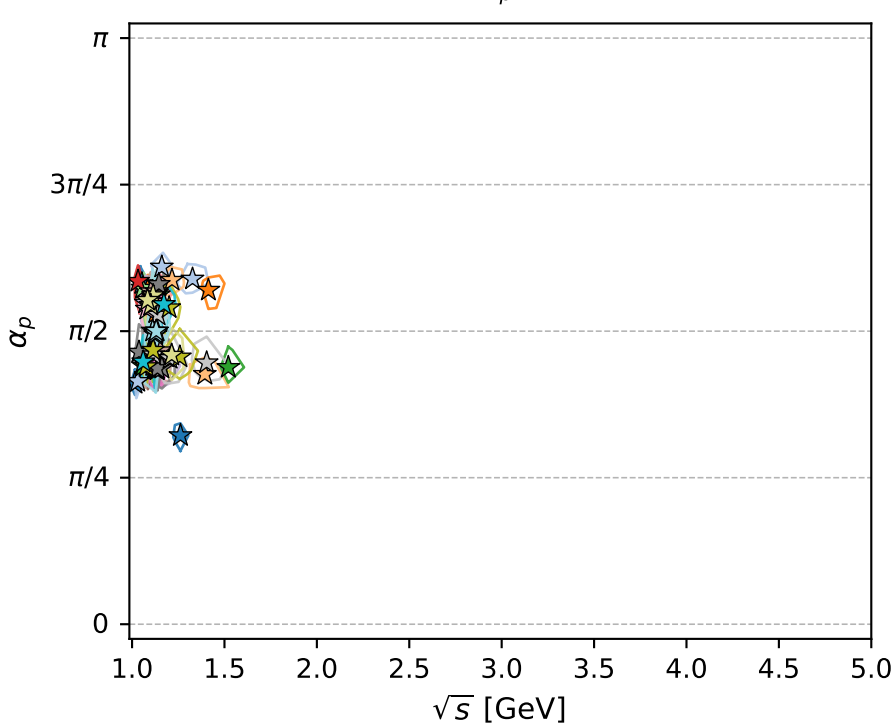
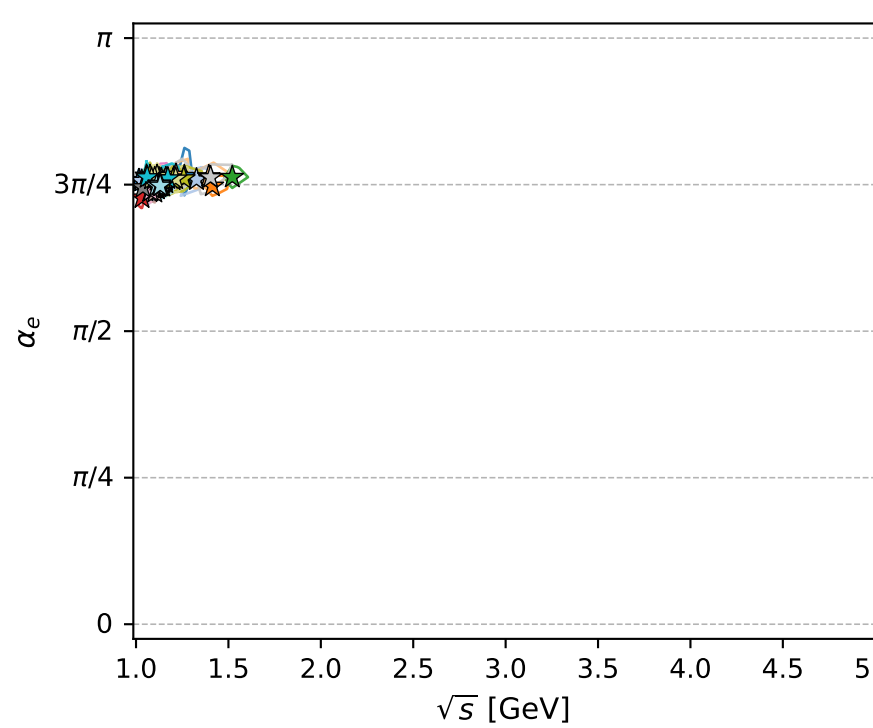
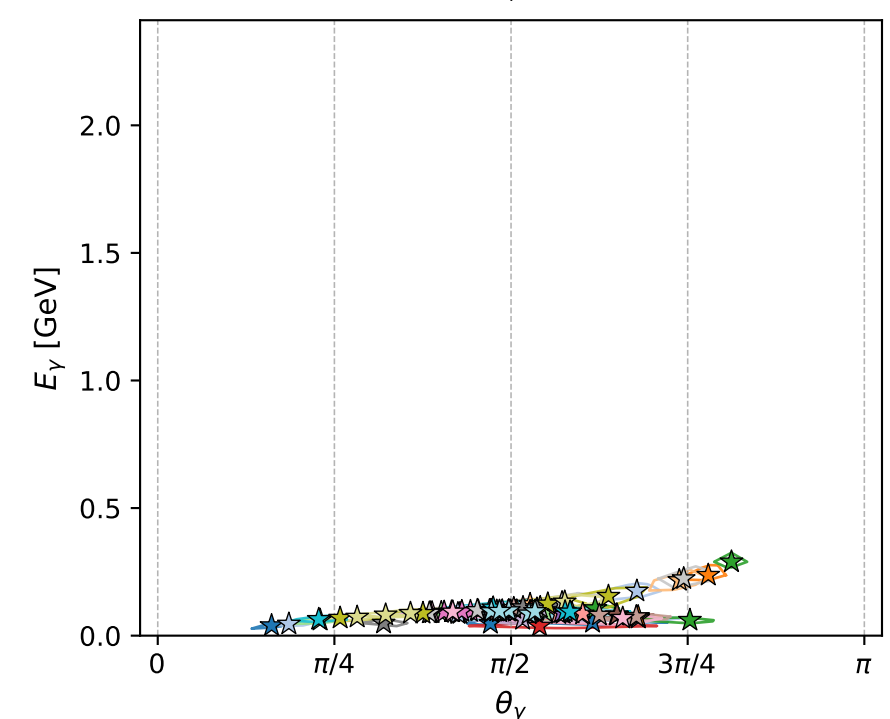
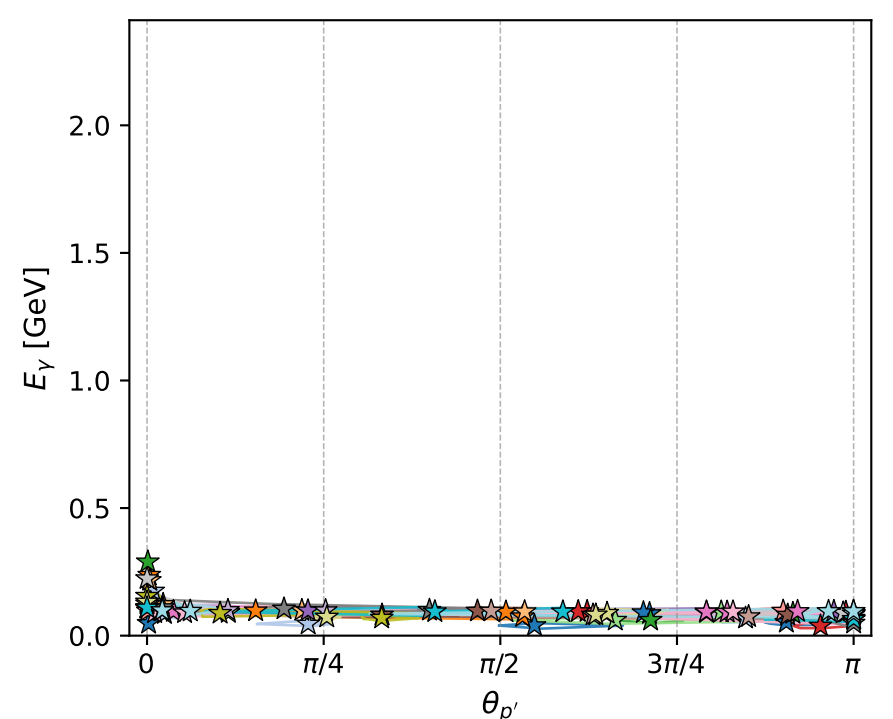
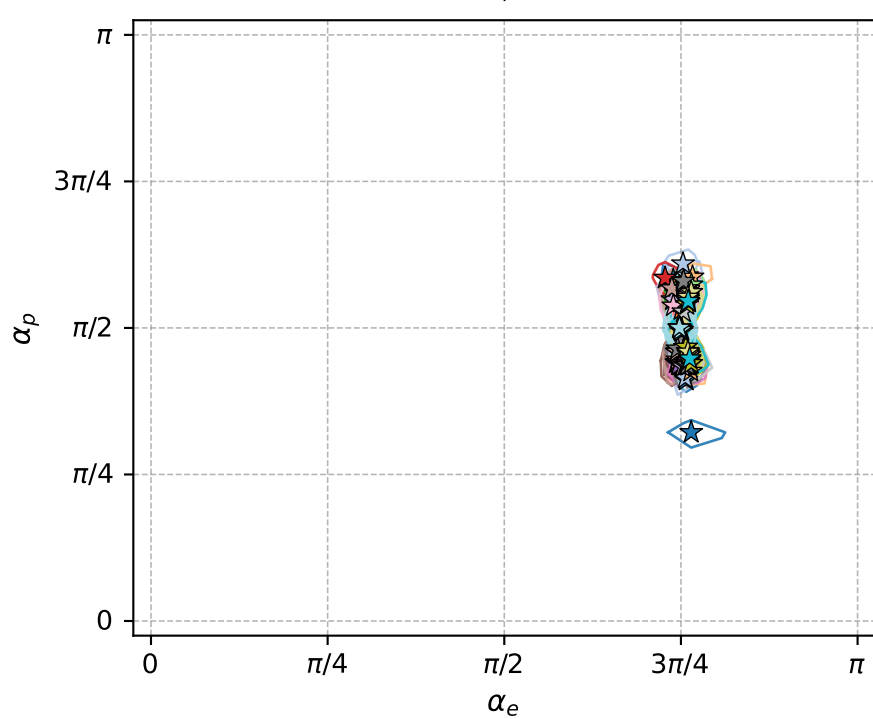
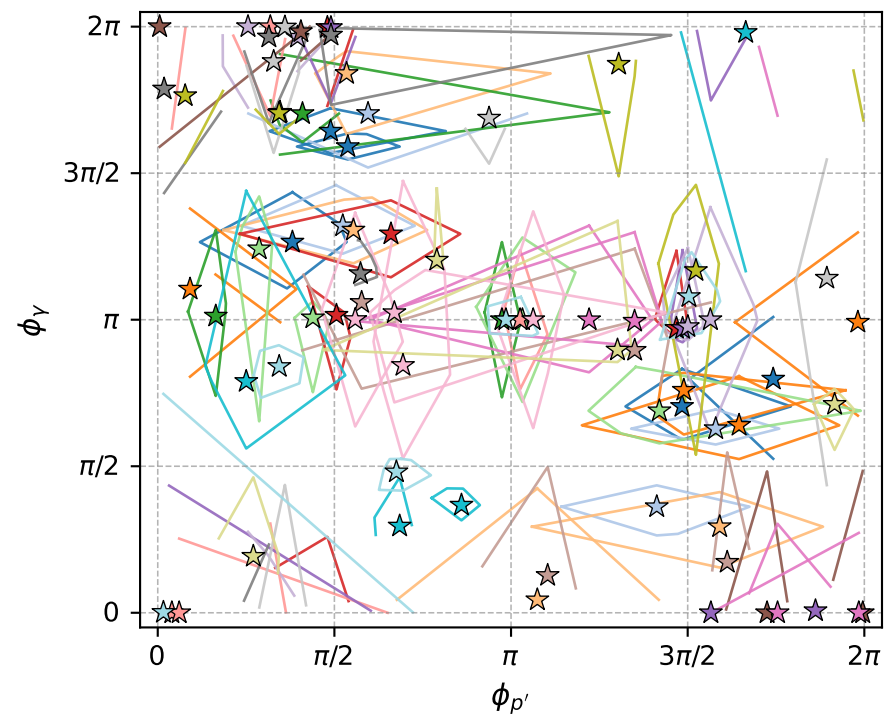
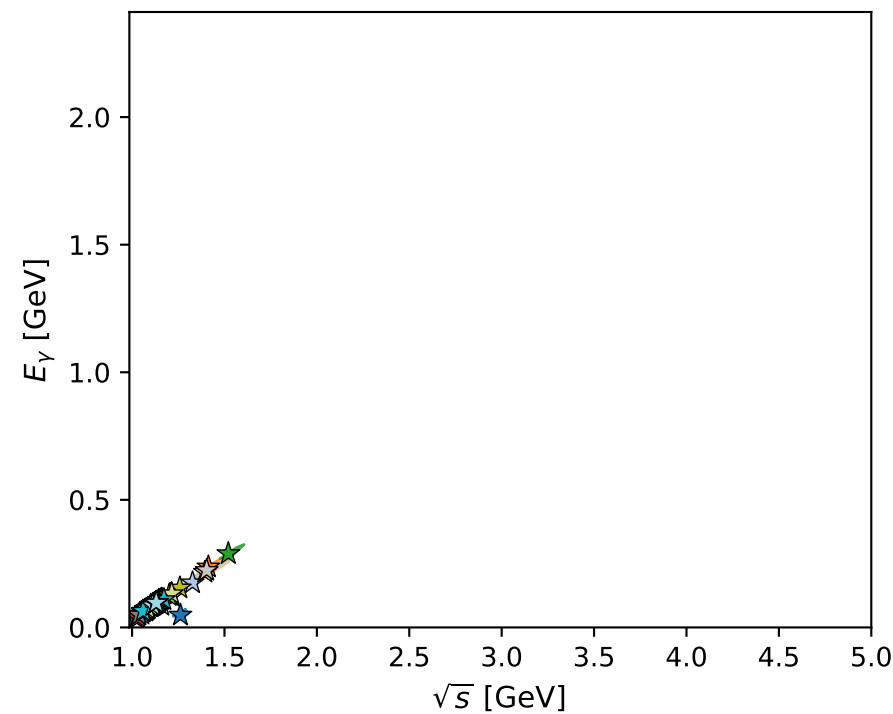
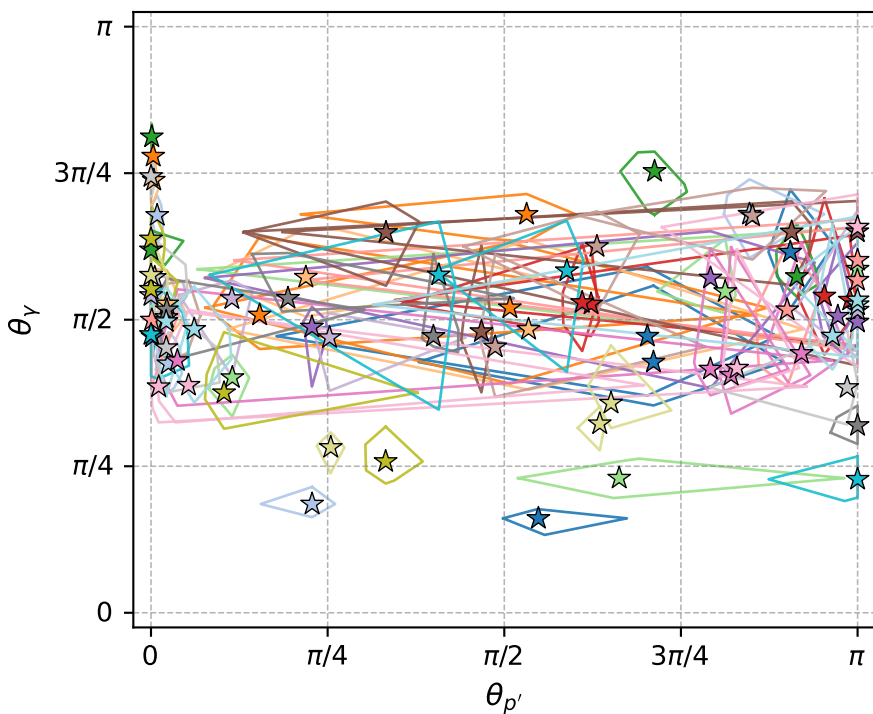


electron: polarization cluster P6; all local-minimum configurations and pairwise projections of their 8D contours



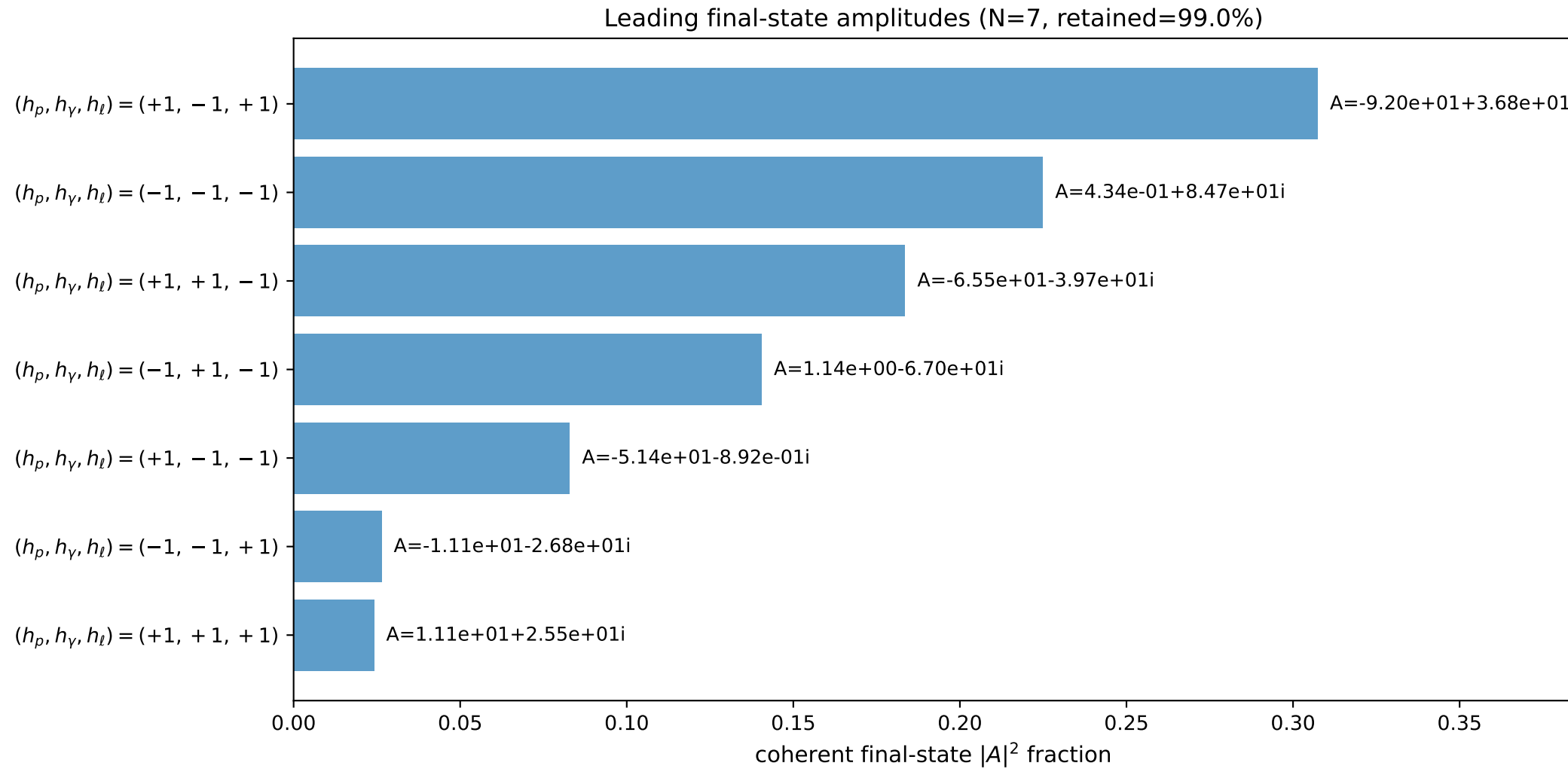
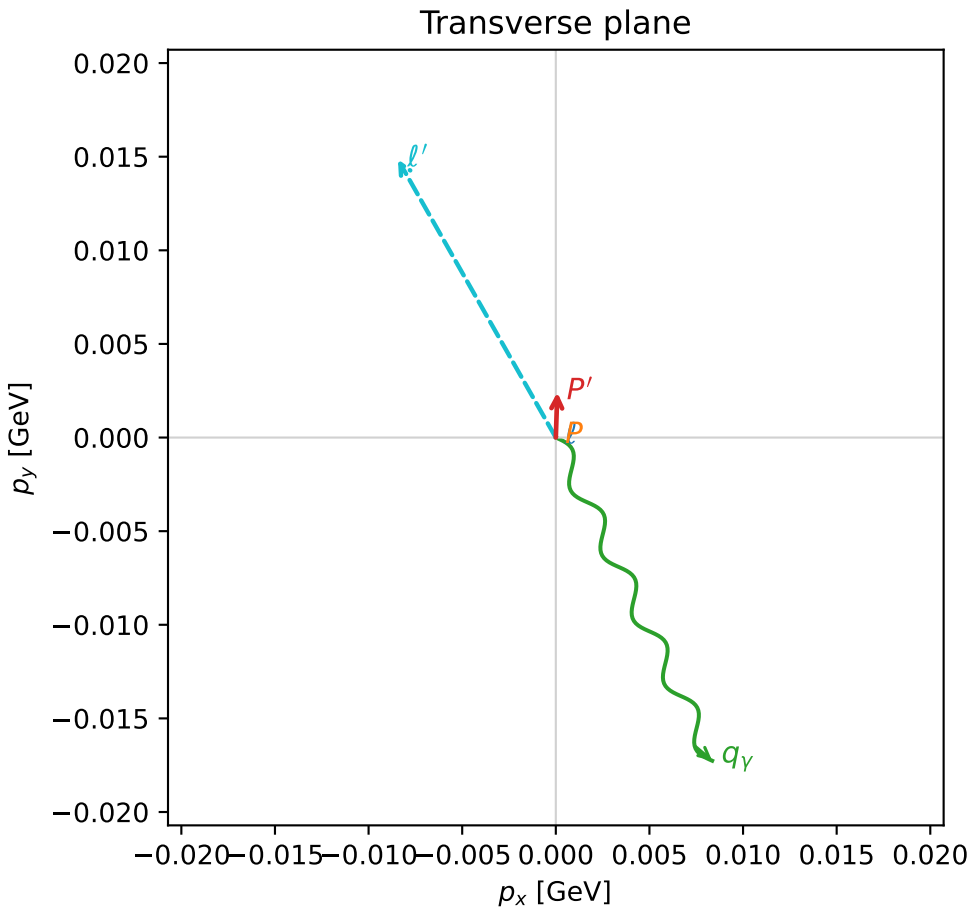
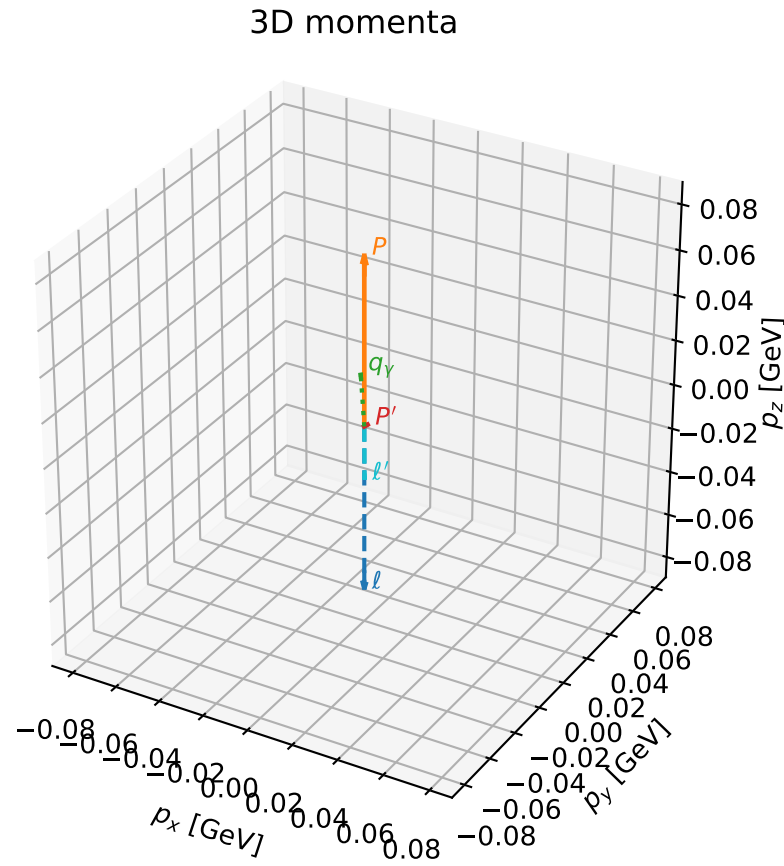
polarization cluster P6: $\alpha_e \sim 3\pi/4, \alpha_\rho \sim \pi/2$
 82 local minima in this polarization cluster
 contour delta=0.05
 configured 8D samples/minimum=256
 D_W range=0.0002647773 to 0.02634278
 global-minimum offsets=0.0002483445 to 0.02632635

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

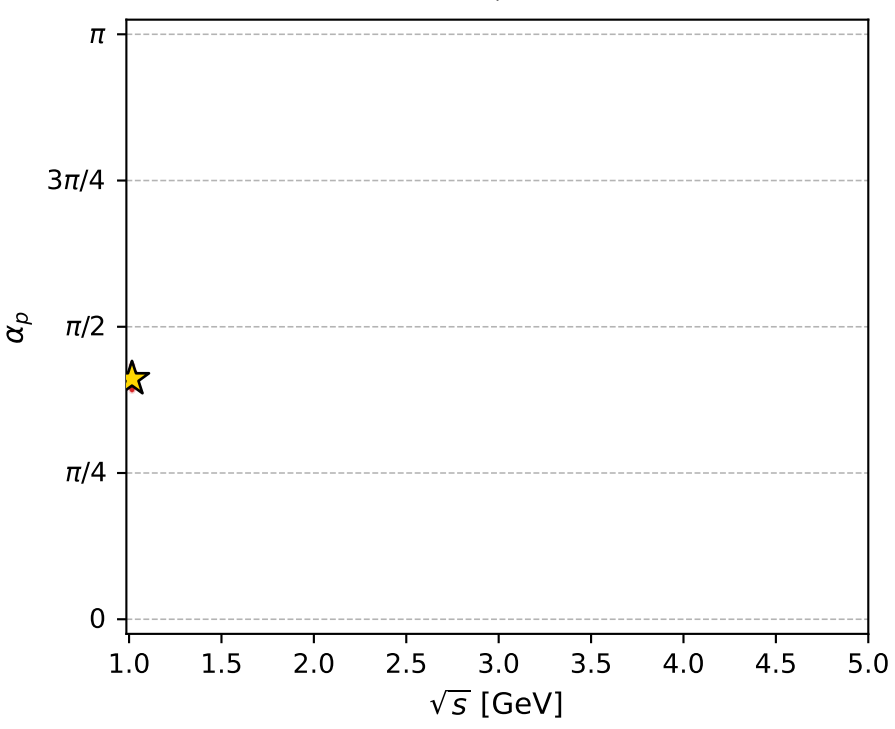
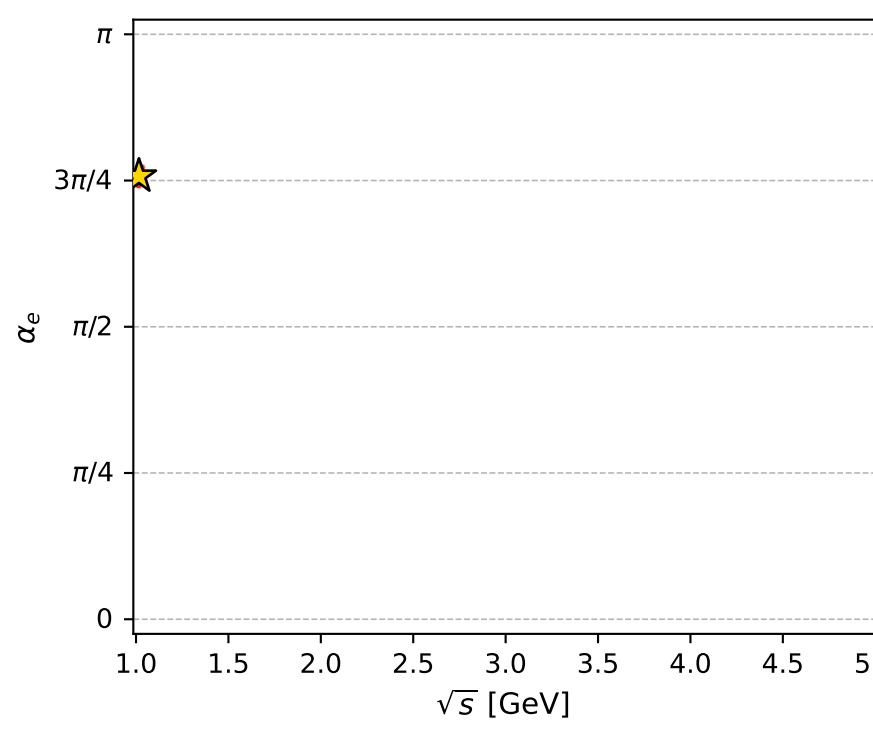
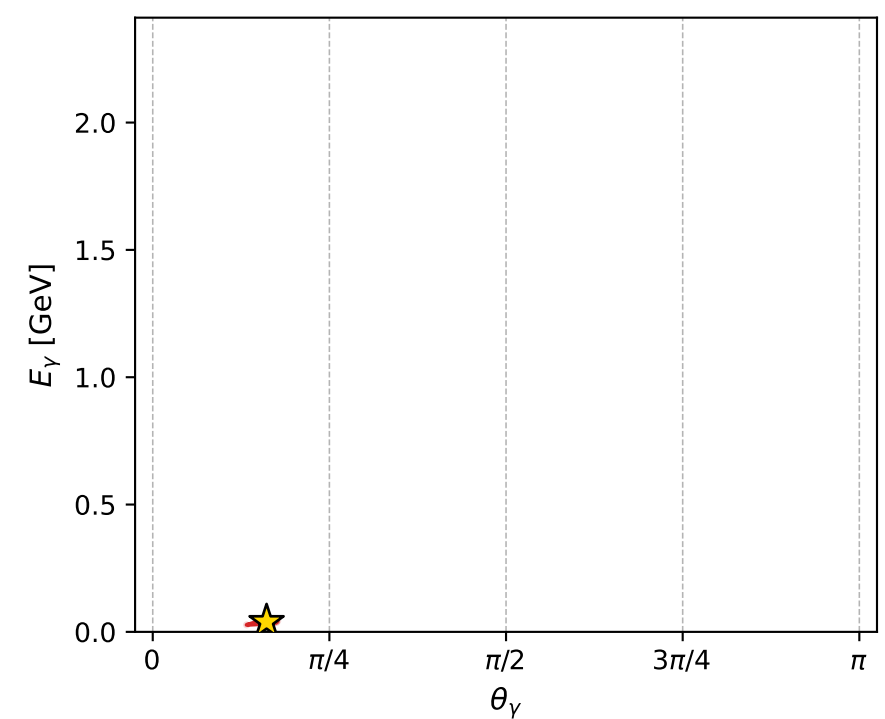
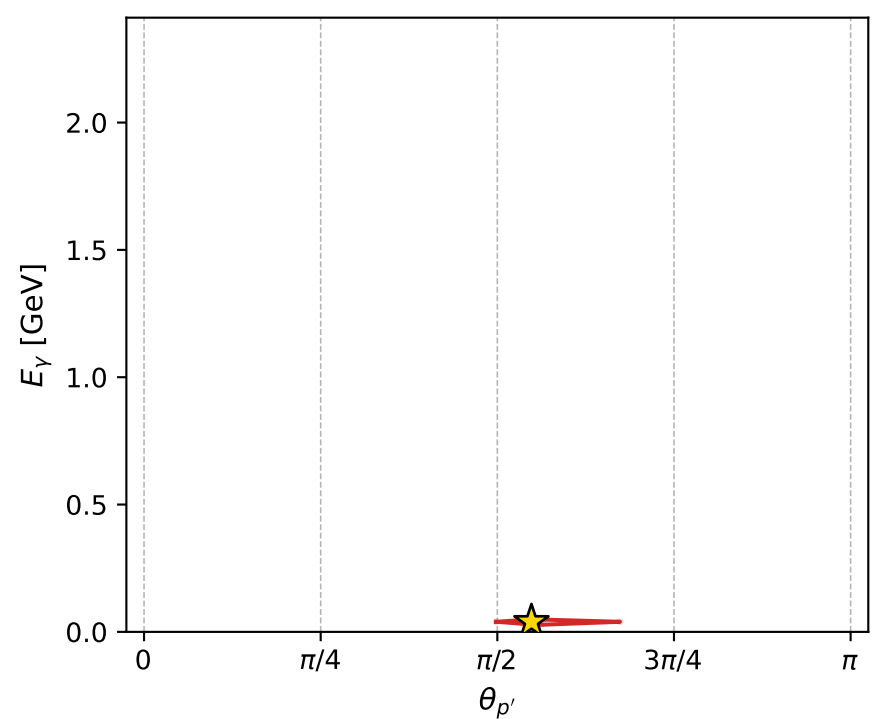
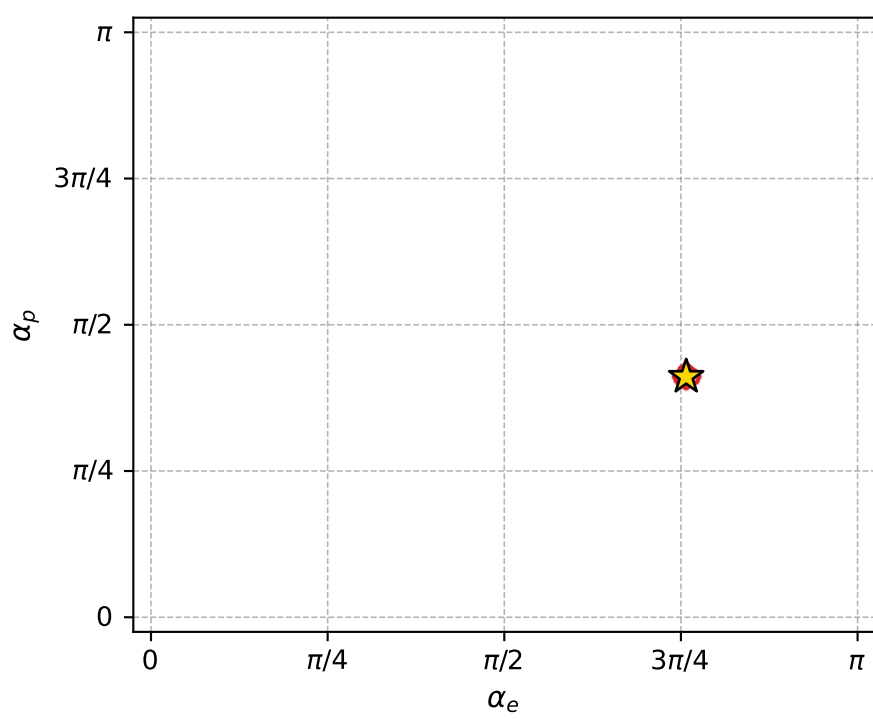
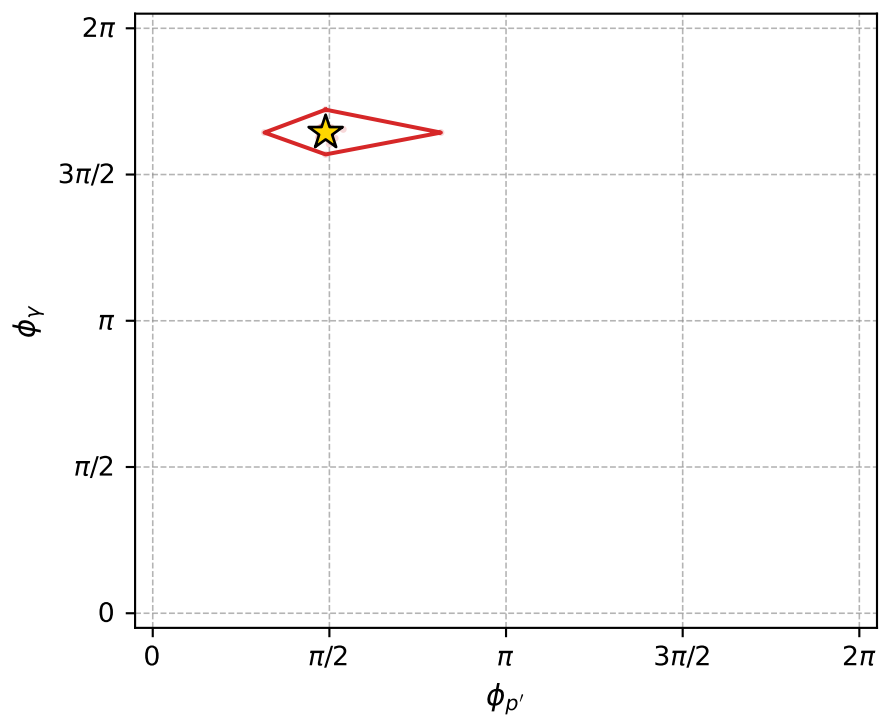
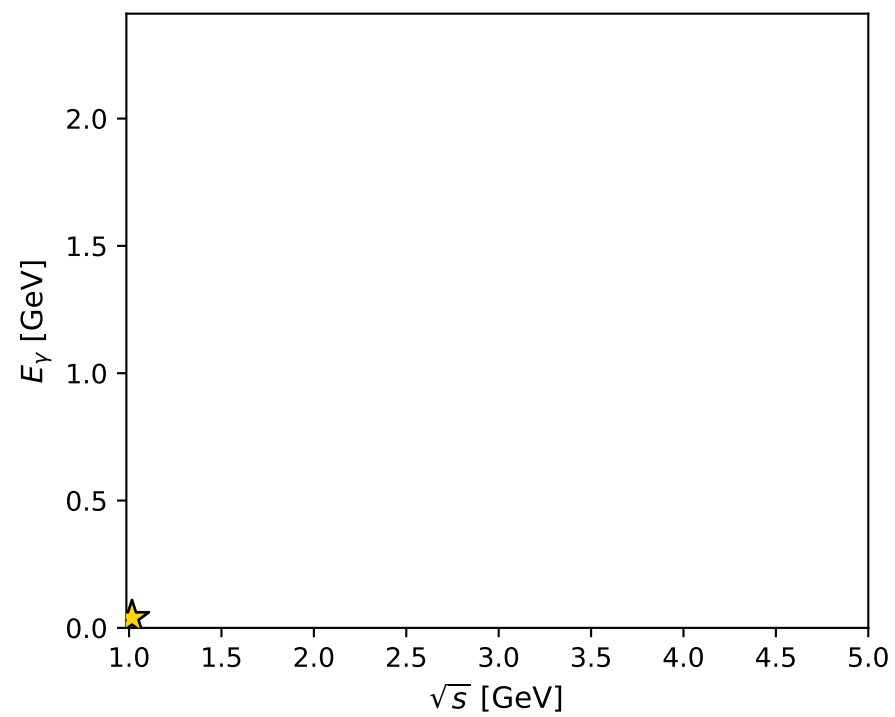
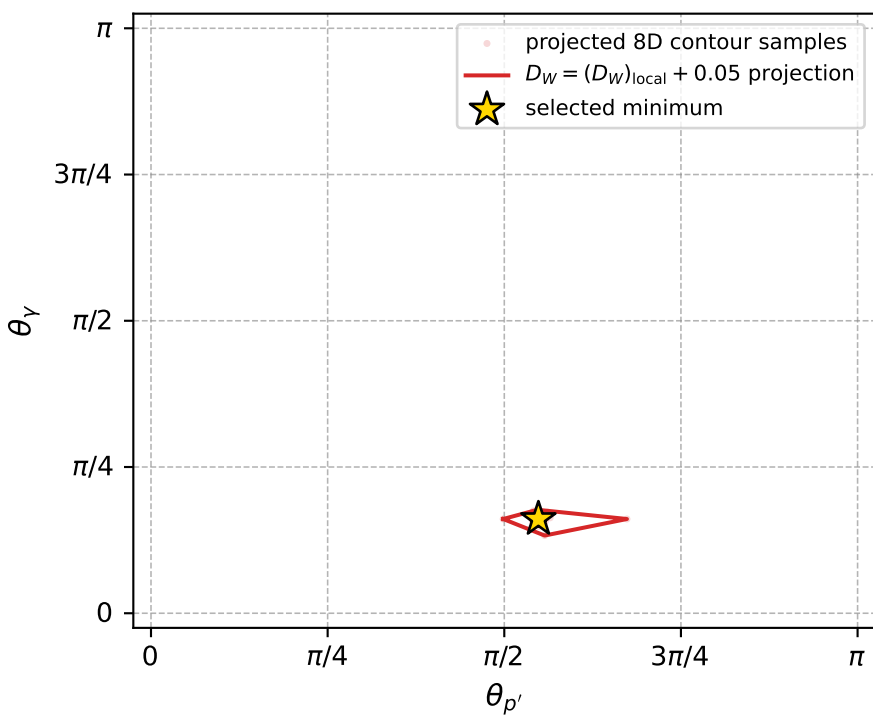
dw_mixing_angles_polarization_06_local_minimum_106 D_W=0.000264777
 region: polarization_cluster_06_local_minimum_106
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3795908$ rad
 incoming lepton: $-0.723455|+> +0.690371|->$
 proton mixing angle: $\alpha_p=1.2914226$ rad
 incoming proton: $0.275754|+> +0.961228|->$

$s=1.03187$, $\sqrt{s}=1.01581$, $|\vec{P}|=0.0748268$, $|\vec{P}'|=0.00244155$
 $\theta_{p'}=1.72249$, $\theta_V=0.506124$, $\phi_{p'}=1.53732$, $\phi_V=5.16301$
 $E_V=0.0395531$, $\theta_{\ell'}=2.67881$, $\phi_{\ell'}=2.08724$
 $Q^2=0.000602218$, $x_B=0.00803906$, $t=-0.00565136$, $W^2=0.954153$, $y=0.492767$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.0748$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0748$
 P $E= 0.9410$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0748$
 ℓ' $E= 0.0383$ $p_x= -0.0084$ $p_y= 0.0148$ $p_z= -0.0342$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0024$ $p_z= -0.0004$
 q_V $E= 0.0396$ $p_x= 0.0084$ $p_y= -0.0173$ $p_z= 0.0346$



electron: polarization cluster P6, local minimum 106; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0002647773$
 $D_W \text{ contour} = 0.050264777$
 above global minimum = 0.00024834446
 8D contour samples = 256

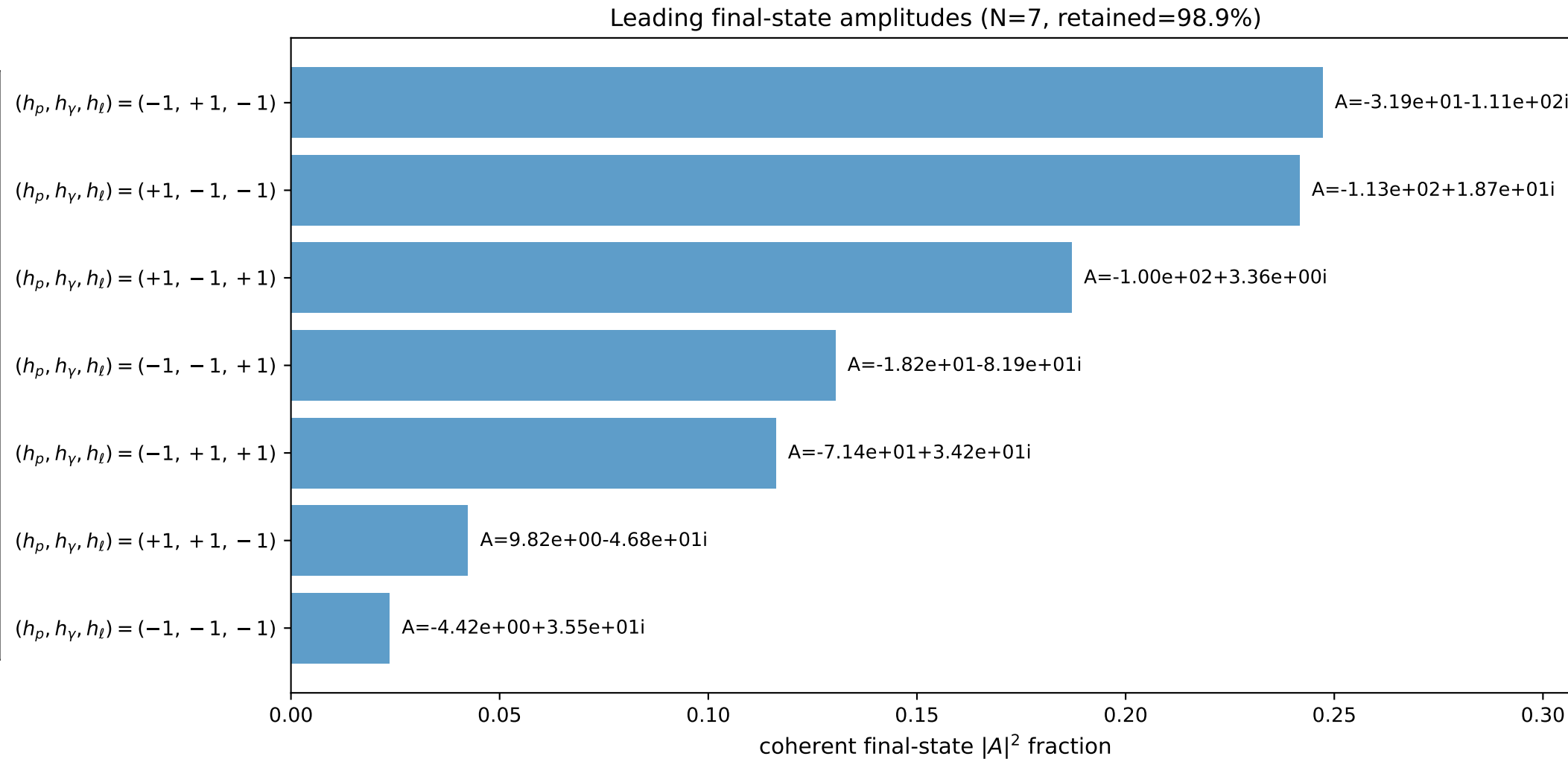
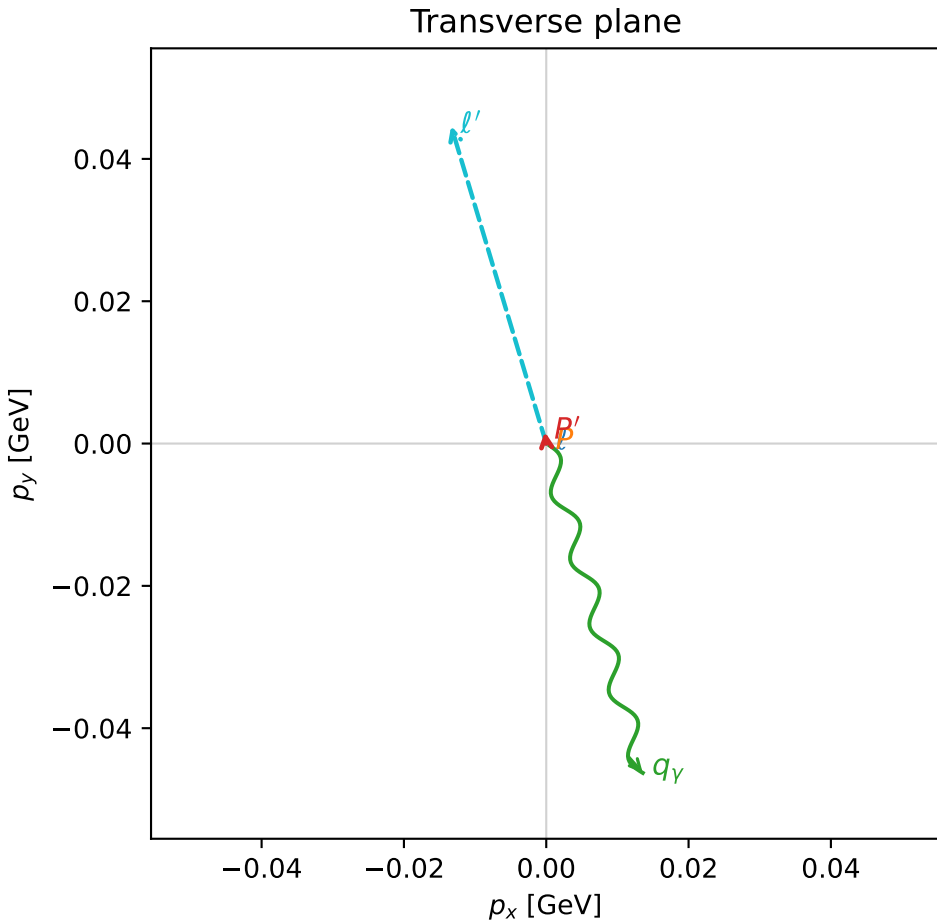
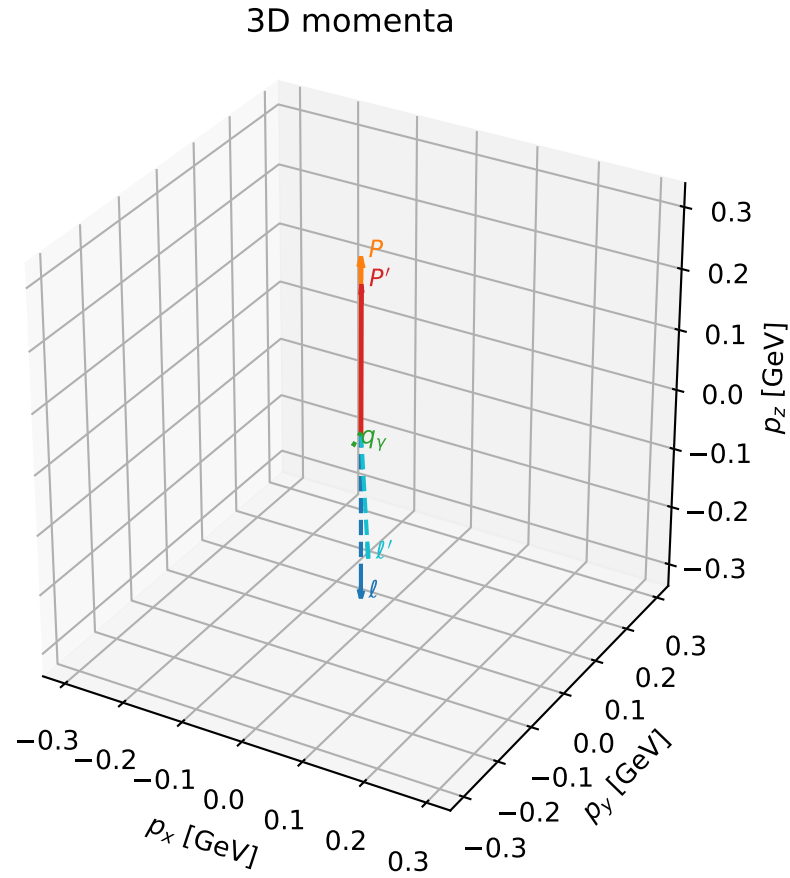
selected phase-space point:
 $\text{sqrt}_s = 1.015808$
 $\text{theta}_p_{\text{out}} = 1.722493$
 $\text{theta}_\gamma_{\text{out}} = 0.5061242$
 $q_{\text{out}} = 0.0395531$
 $\text{phi}_p_{\text{out}} = 1.537318$
 $\text{phi}_\gamma_{\text{out}} = 5.163007$
 $\alpha_e = 2.379591$
 $\alpha_p = 1.291423$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

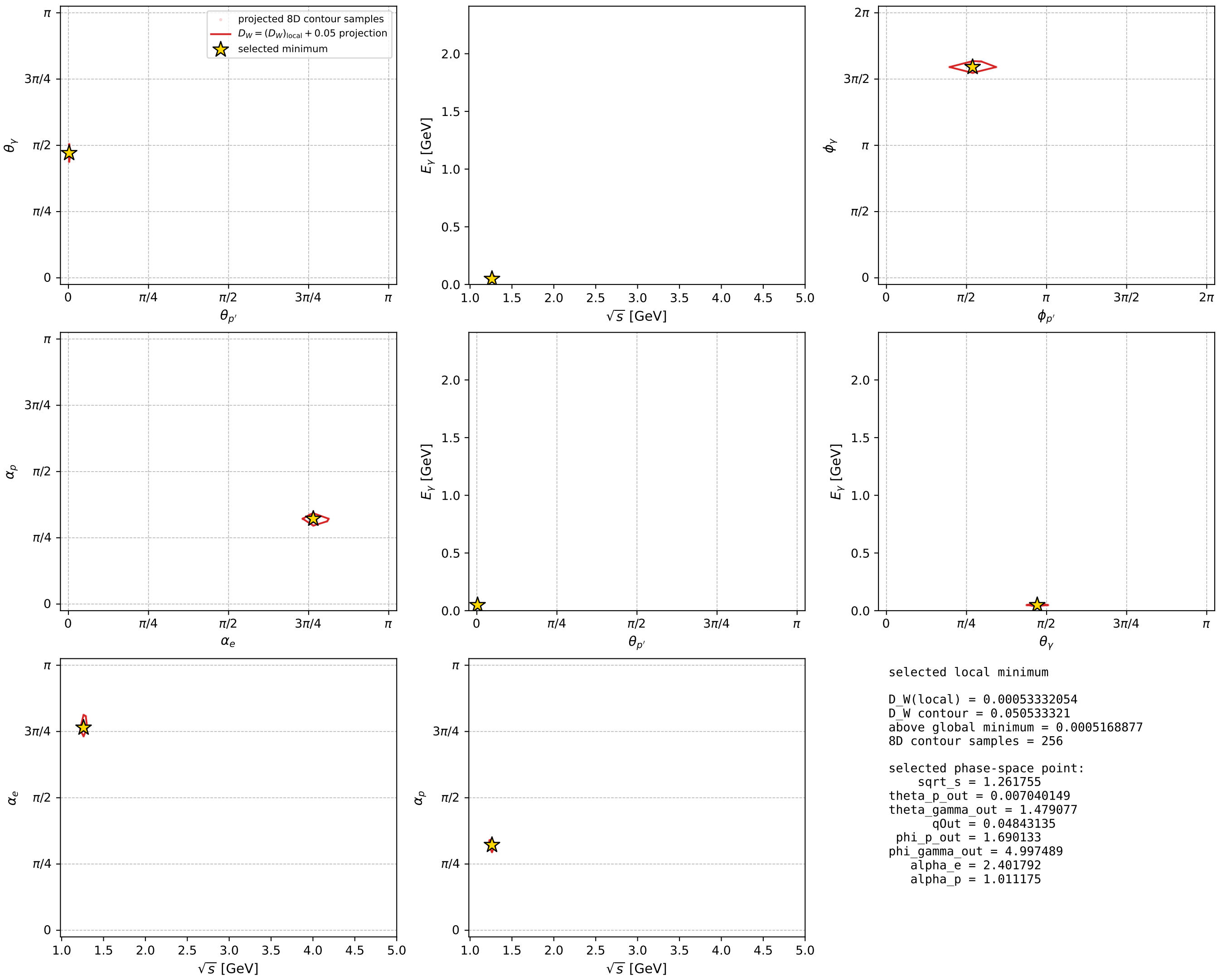
dw_mixing_angles_polarization_06_local_minimum_217 D_W=0.000533321
 region: polarization_cluster_06_local_minimum_217
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.4017919$ rad
 incoming lepton: $-0.738603|+> +0.674141|->$
 proton mixing angle: $\alpha_p=1.0111751$ rad
 incoming proton: $0.530865|+> +0.847456|->$

$s=1.59203$, $\sqrt{s}=1.26176$, $|\vec{P}|=0.282219$, $|\vec{P}'|=0.23697$
 $\theta_{P'}=0.00704015$, $\theta_V=1.47908$, $\phi_{P'}=1.69013$, $\phi_V=4.99749$
 $E_V=0.0484313$, $\theta_{\ell'}=2.95096$, $\phi_{\ell'}=1.86181$
 $Q^2=0.00251374$, $x_B=0.0266619$, $t=-0.00190519$, $W^2=0.971612$, $y=0.132385$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2822$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2822$
 P $E= 0.9795$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2822$
 ℓ' $E= 0.2459$ $p_x= -0.0134$ $p_y= 0.0446$ $p_z= -0.2414$
 P' $E= 0.9675$ $p_x= -0.0002$ $p_y= 0.0017$ $p_z= 0.2370$
 q_V $E= 0.0484$ $p_x= 0.0136$ $p_y= -0.0463$ $p_z= 0.0044$



electron: polarization cluster P6, local minimum 217; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

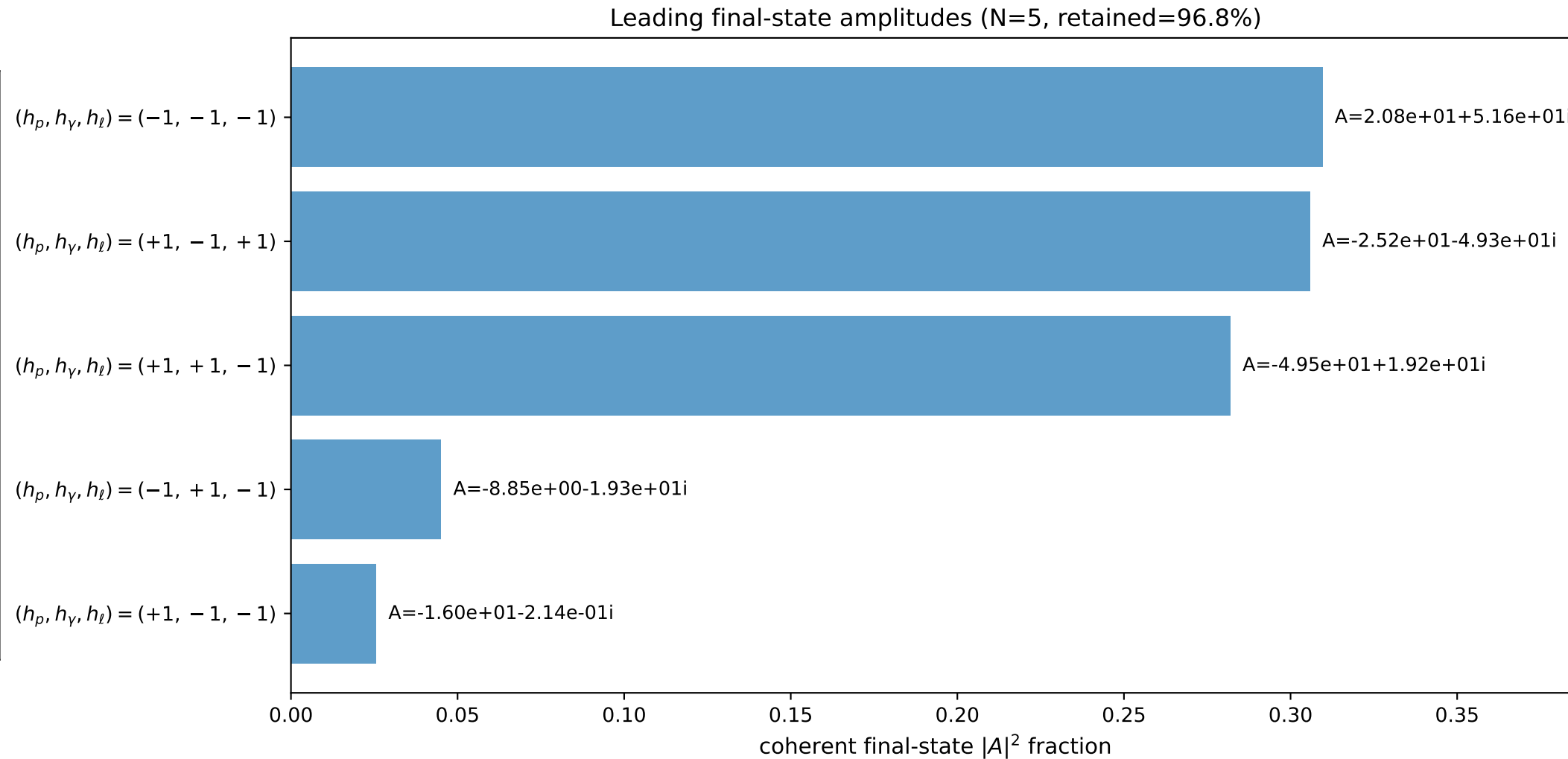
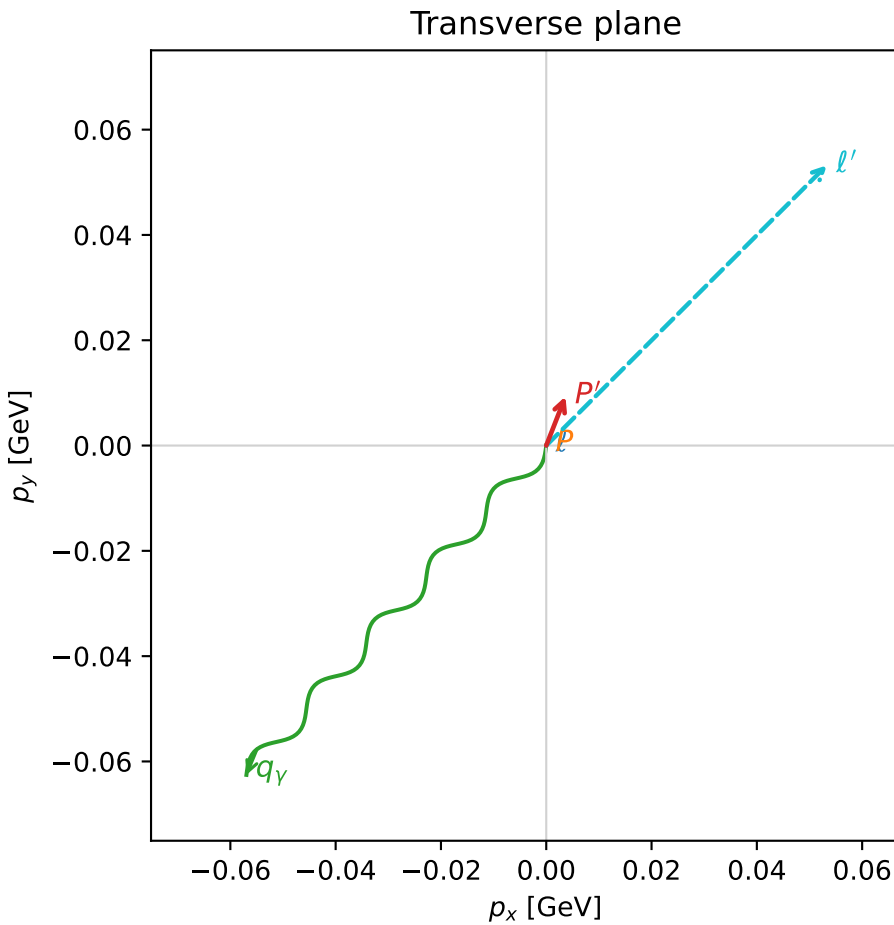
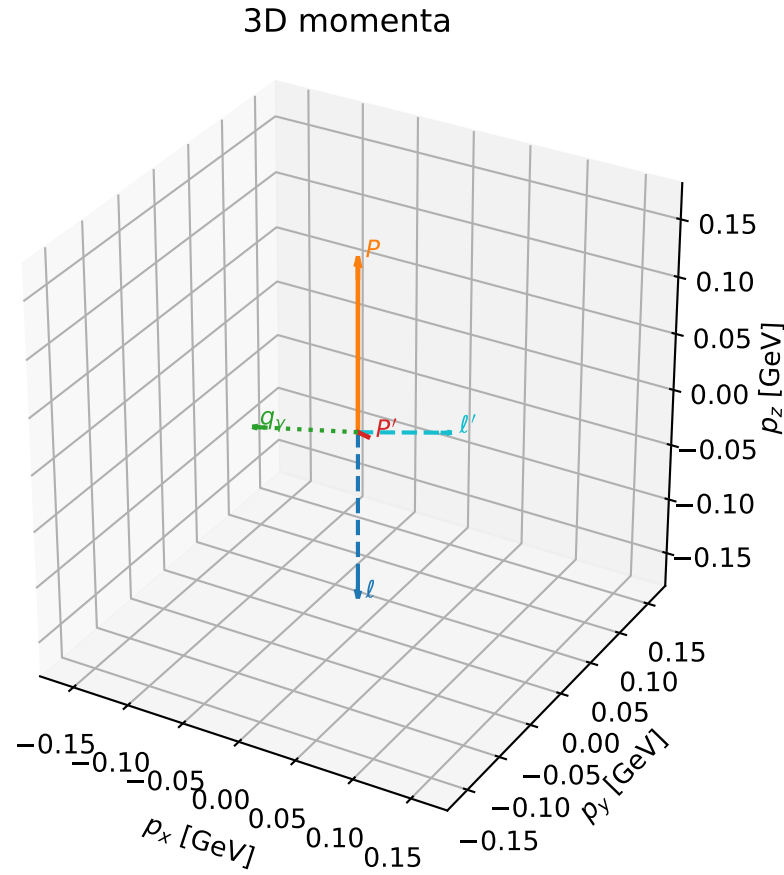


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

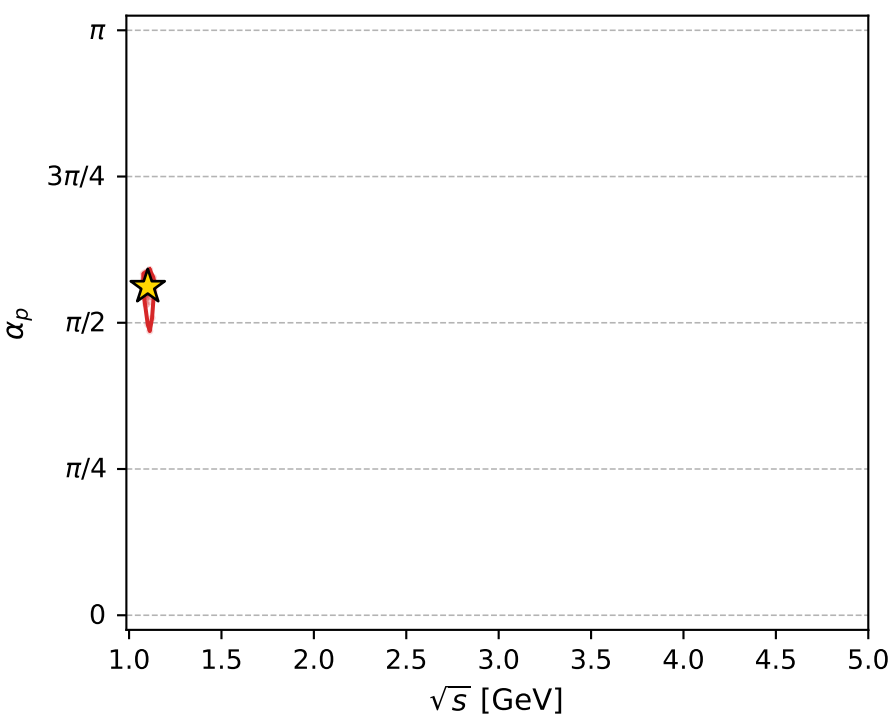
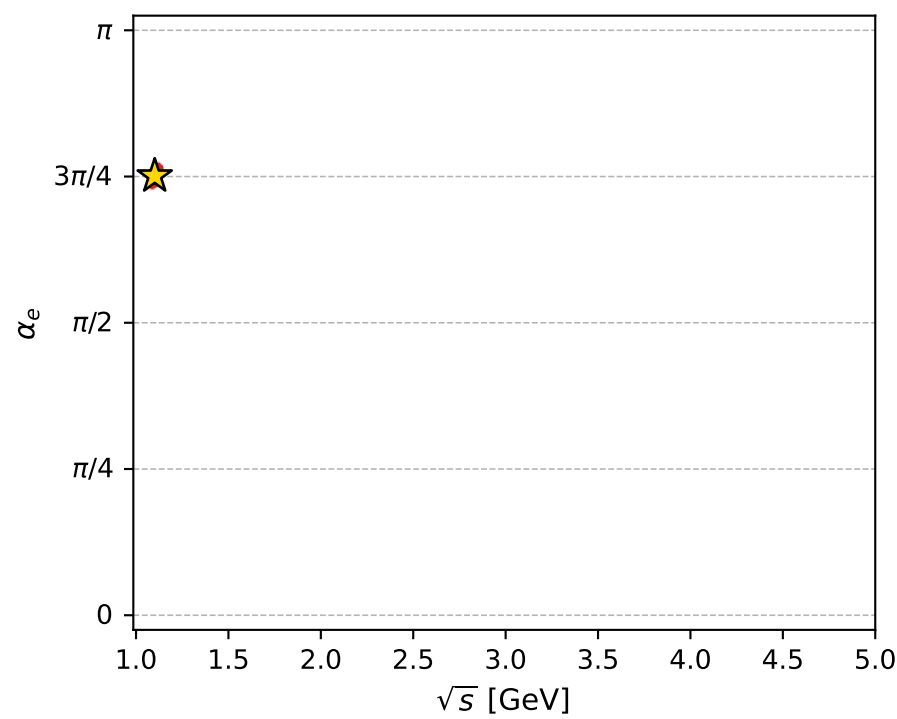
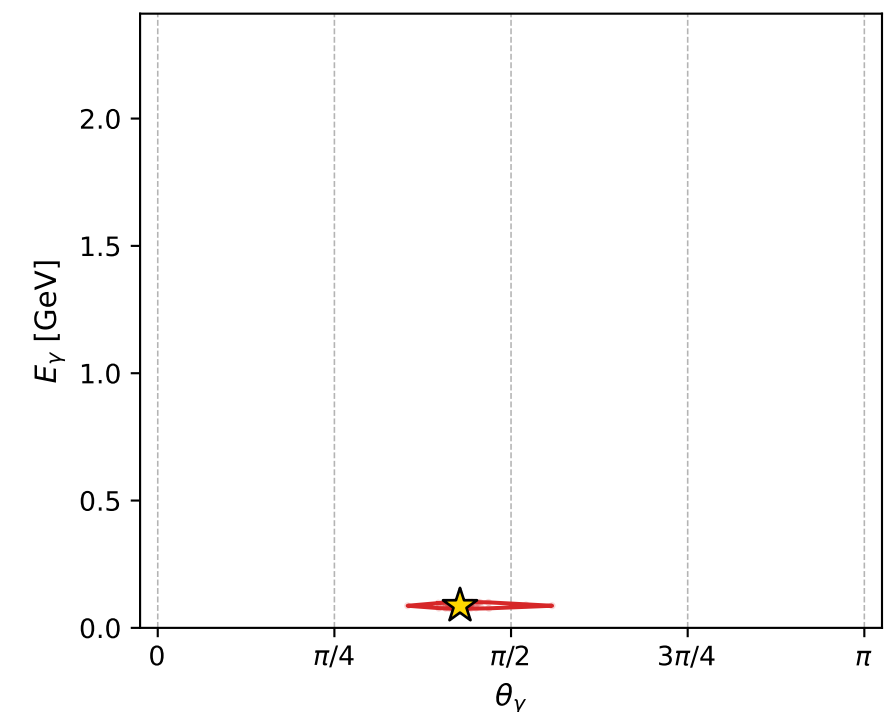
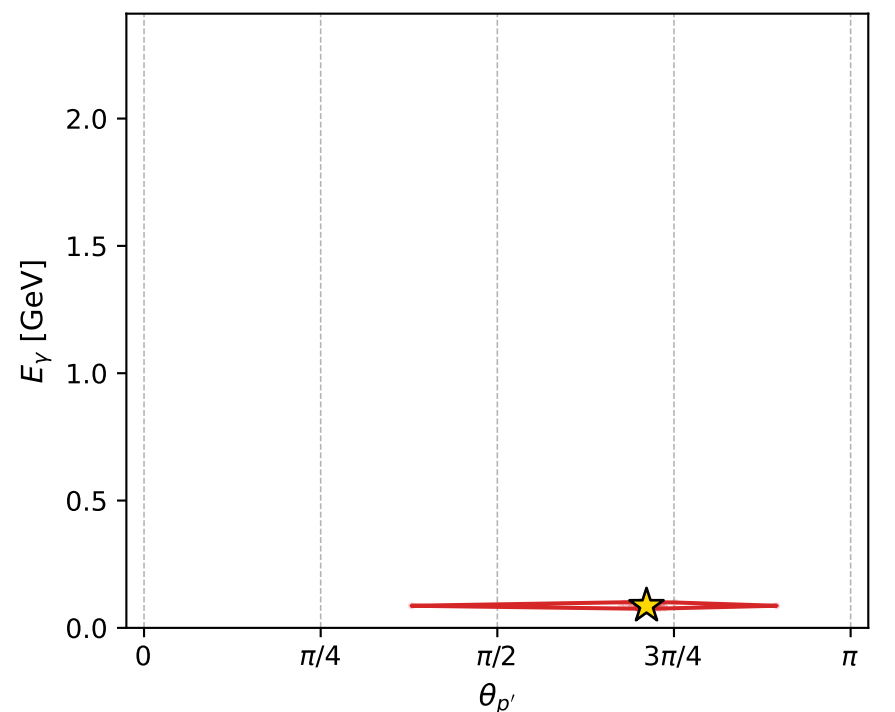
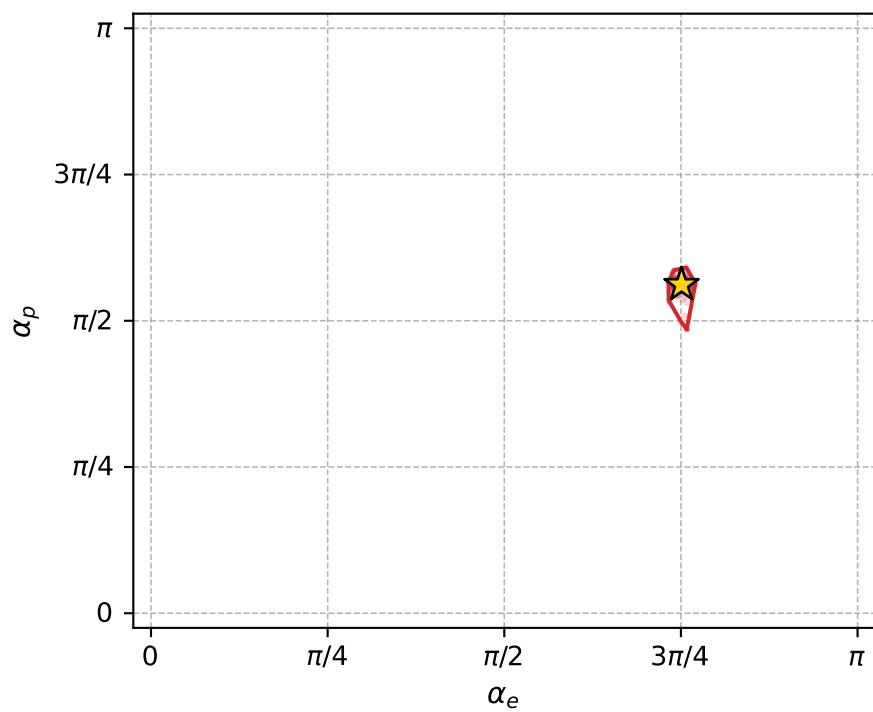
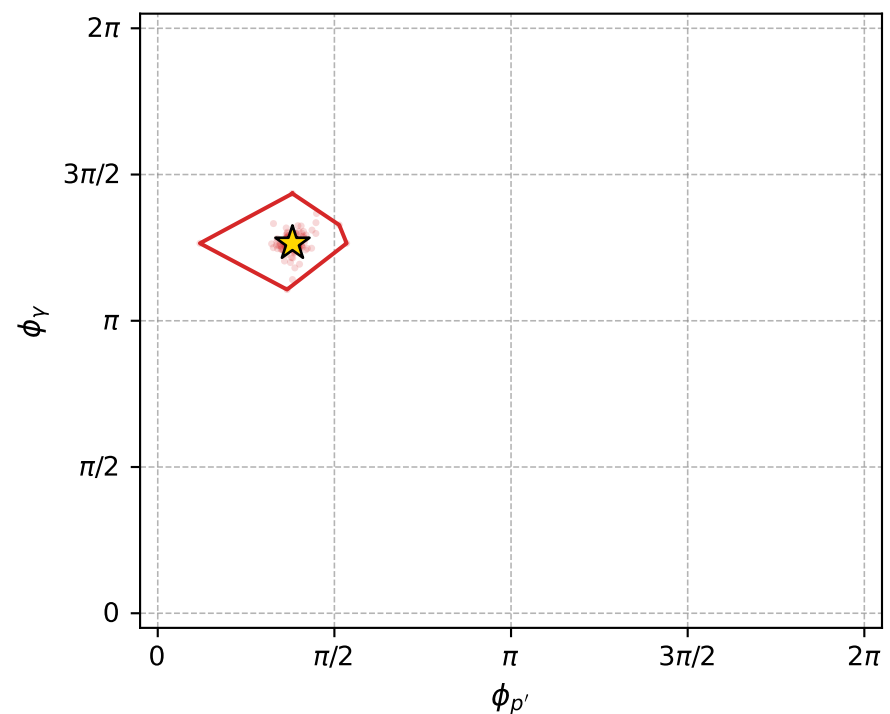
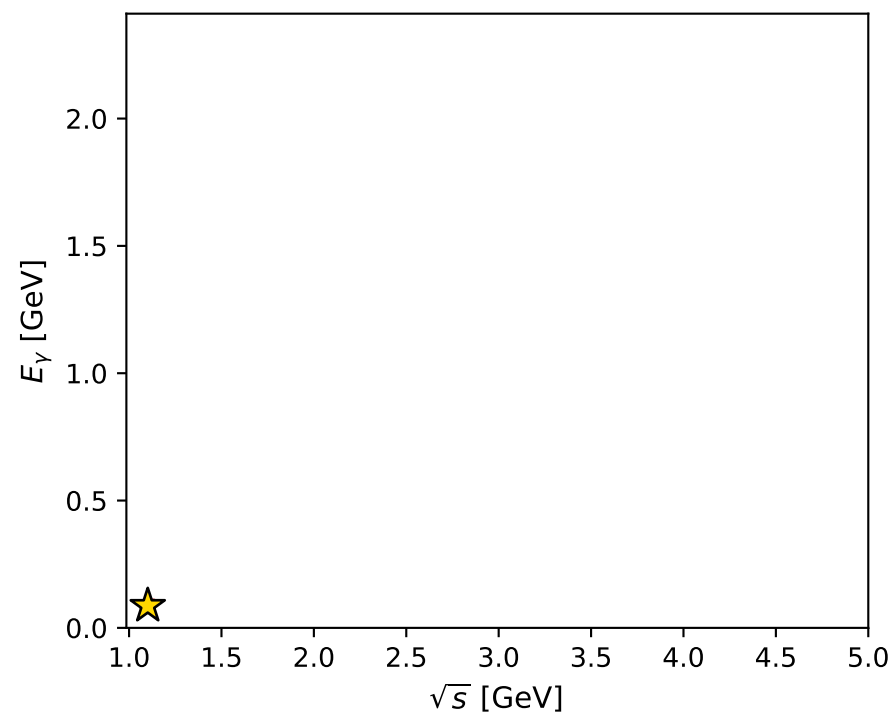
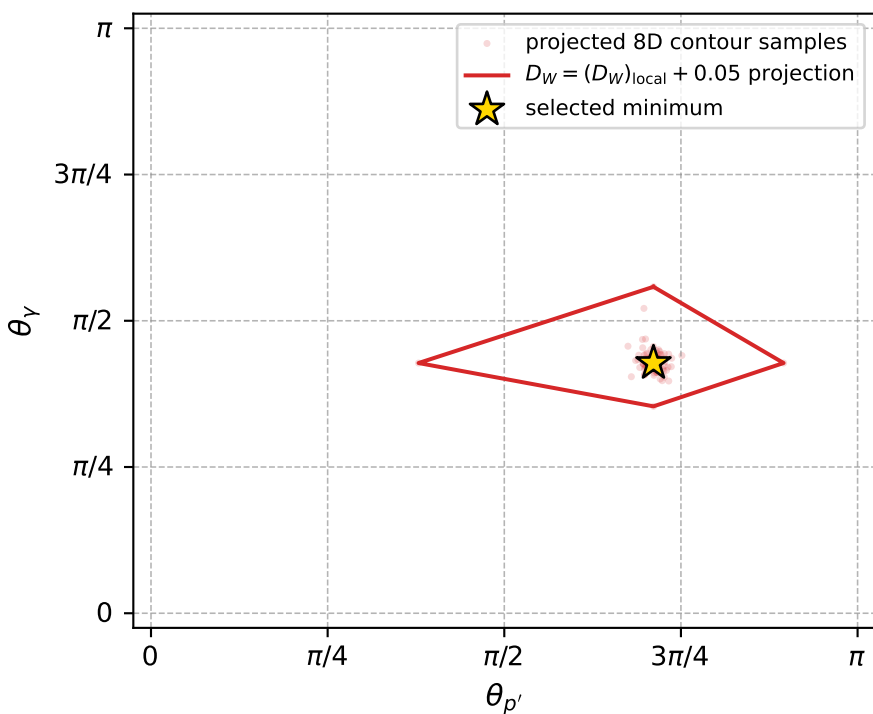
dw_mixing_angles_polarization_06_local_minimum_271 D_W=0.000694337
 region: polarization_cluster_06_local_minimum_271
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3588113$ rad
 incoming lepton: $-0.708955|+> +0.705254|->$
 proton mixing angle: $\alpha_p=1.764368$ rad
 incoming proton: $-0.192365|+> +0.981323|->$

$s=1.2125$, $\sqrt{s}=1.10113$, $|\vec{P}|=0.151049$, $|\vec{P}'|=0.012708$
 $\theta_{P'}=2.23412$, $\theta_V=1.34385$, $\phi_{P'}=1.19769$, $\phi_V=3.97387$
 $E_V=0.0868174$, $\theta_{\ell'}=1.72501$, $\phi_{\ell'}=0.784755$
 $Q^2=0.0194914$, $x_B=0.10578$, $t=-0.0251972$, $W^2=1.04462$, $y=0.553923$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1510$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1510$
 P $E= 0.9501$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1510$
 ℓ' $E= 0.0762$ $p_x= 0.0533$ $p_y= 0.0532$ $p_z= -0.0117$
 P' $E= 0.9381$ $p_x= 0.0036$ $p_y= 0.0093$ $p_z= -0.0078$
 q_V $E= 0.0868$ $p_x= -0.0569$ $p_y= -0.0626$ $p_z= 0.0195$



electron: polarization cluster P6, local minimum 271; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.00069433749$
 $D_W \text{ contour} = 0.050694337$
 above global minimum = 0.00067790465
 8D contour samples = 256

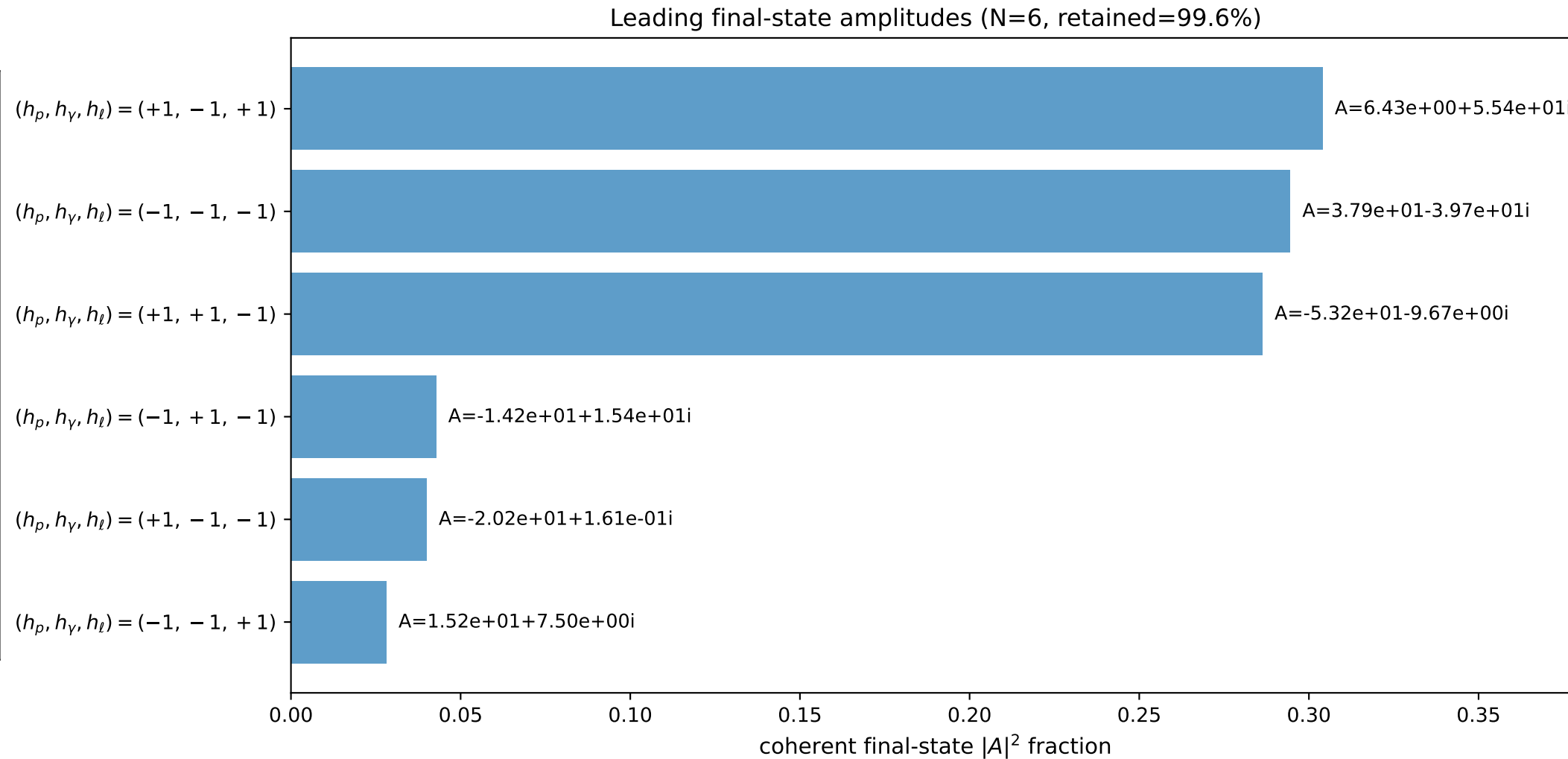
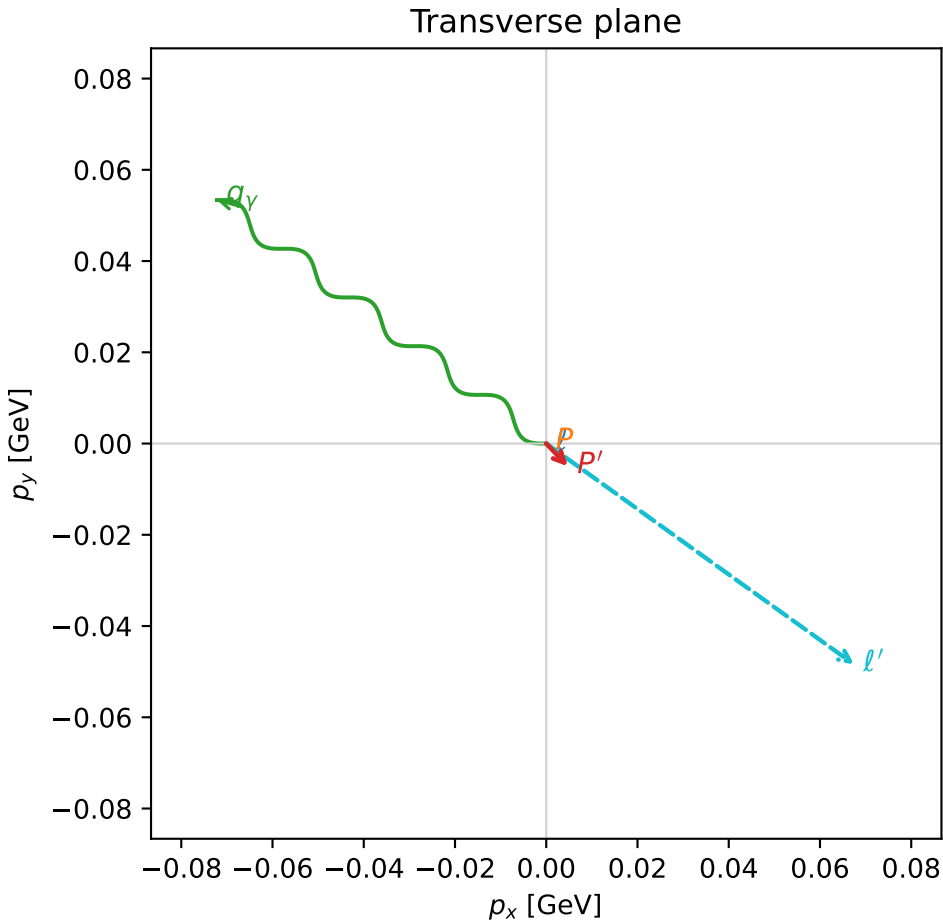
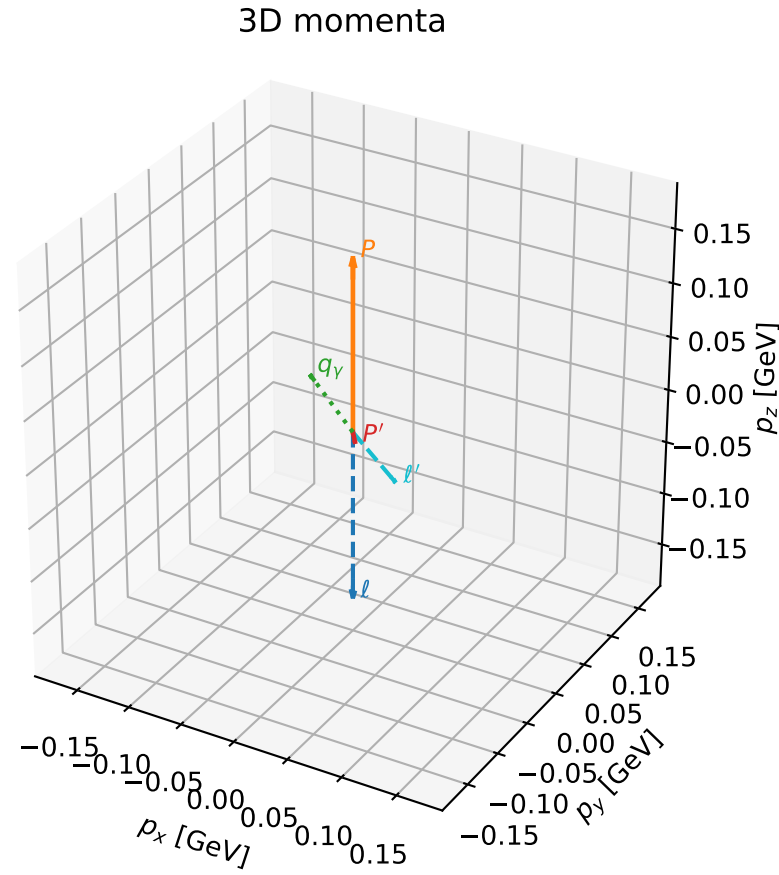
selected phase-space point:
 $\text{sqrt_s} = 1.101134$
 $\text{theta_p_out} = 2.234117$
 $\text{theta_gamma_out} = 1.343852$
 $\text{qOut} = 0.08681744$
 $\text{phi_p_out} = 1.197691$
 $\text{phi_gamma_out} = 3.973868$
 $\text{alpha_e} = 2.358811$
 $\text{alpha_p} = 1.764368$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

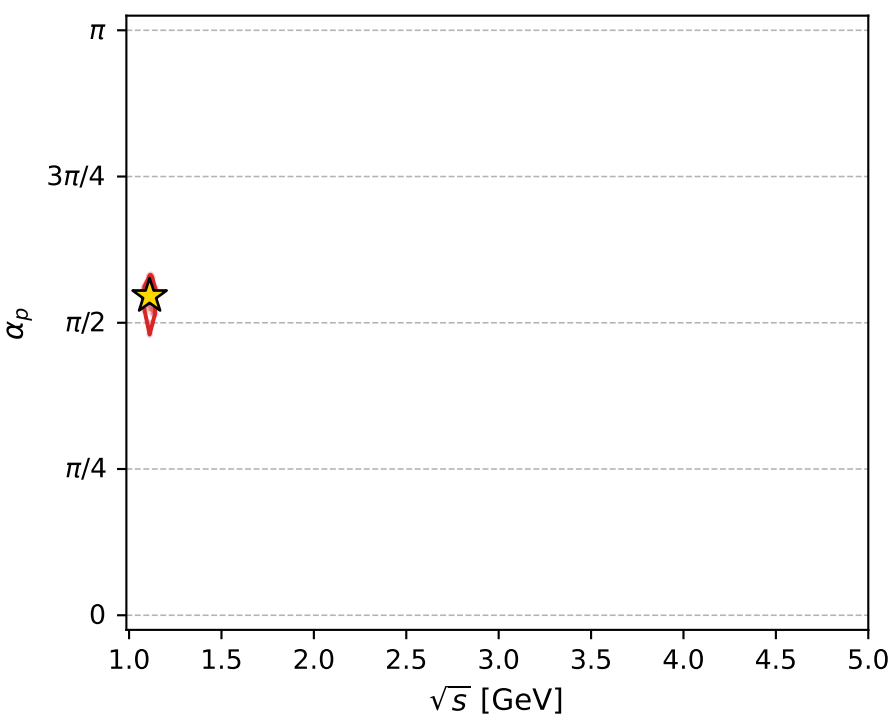
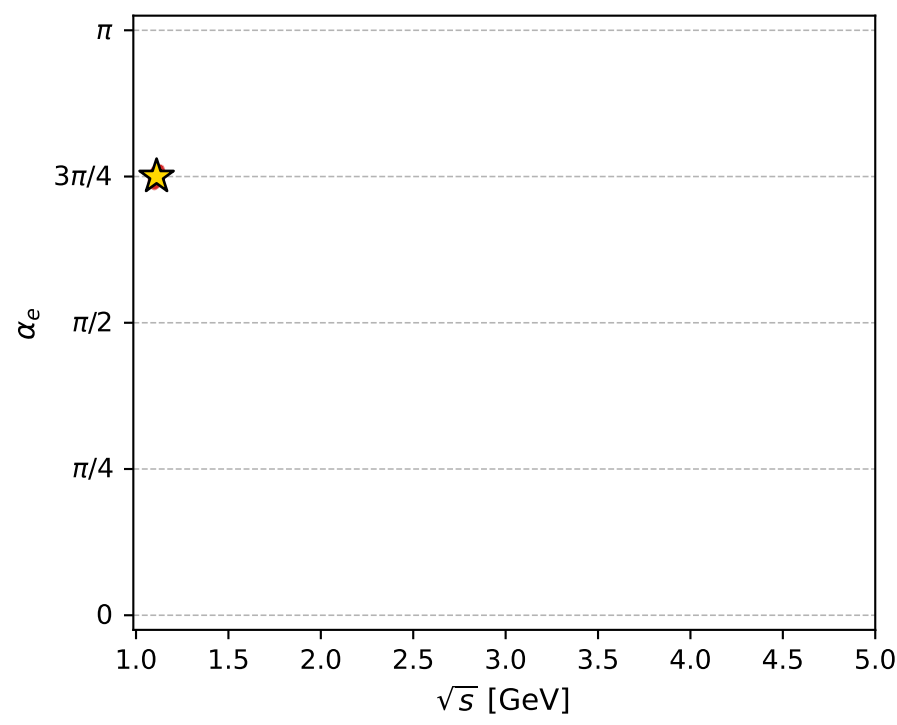
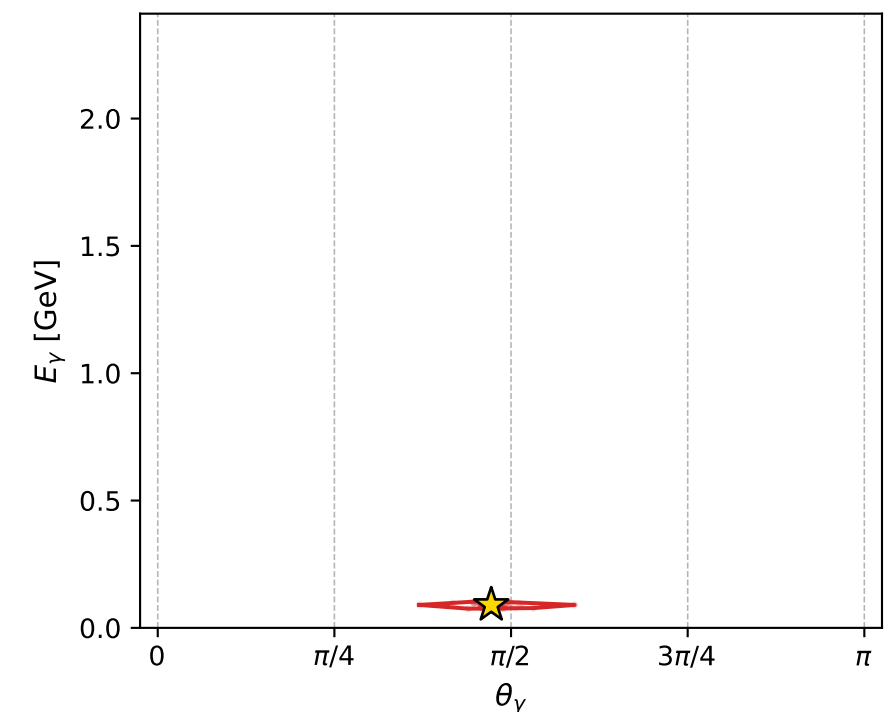
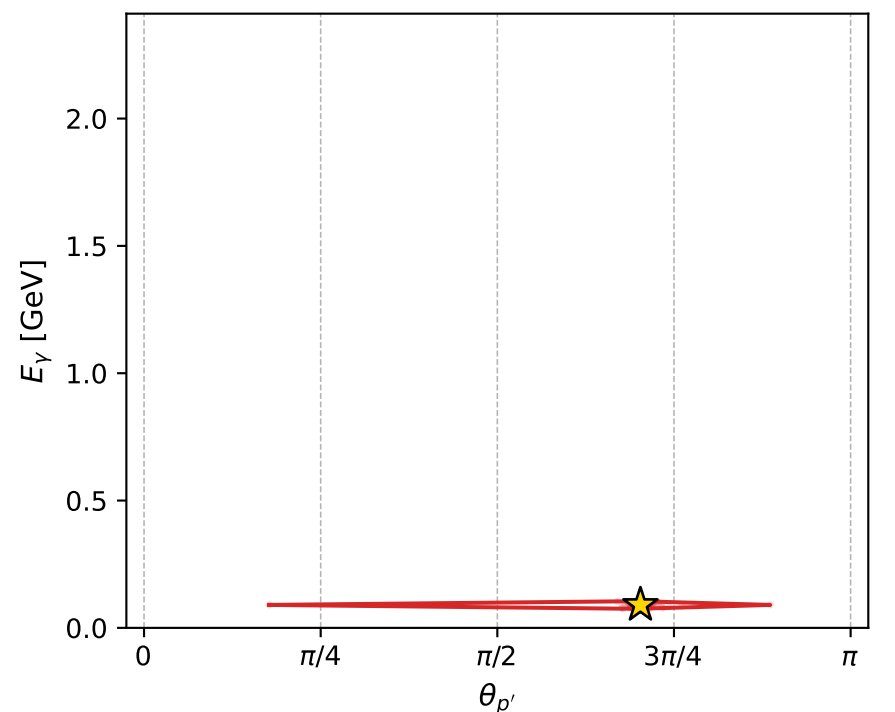
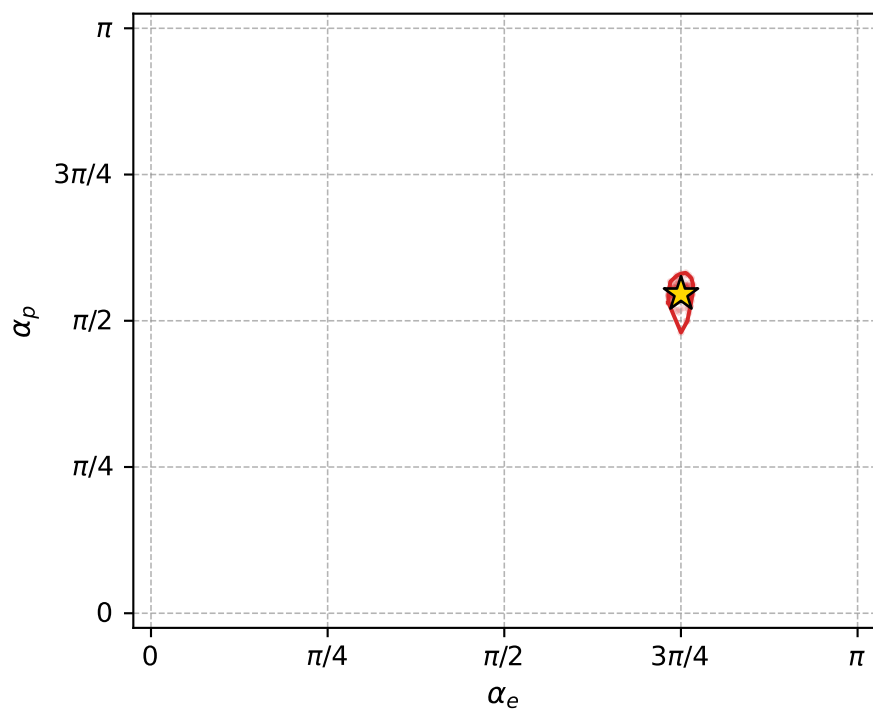
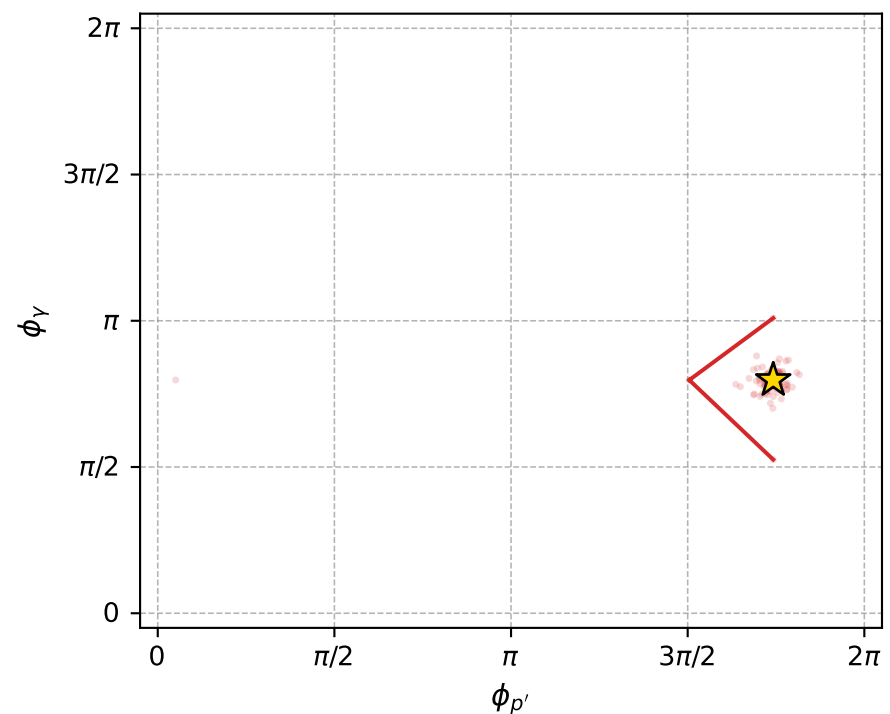
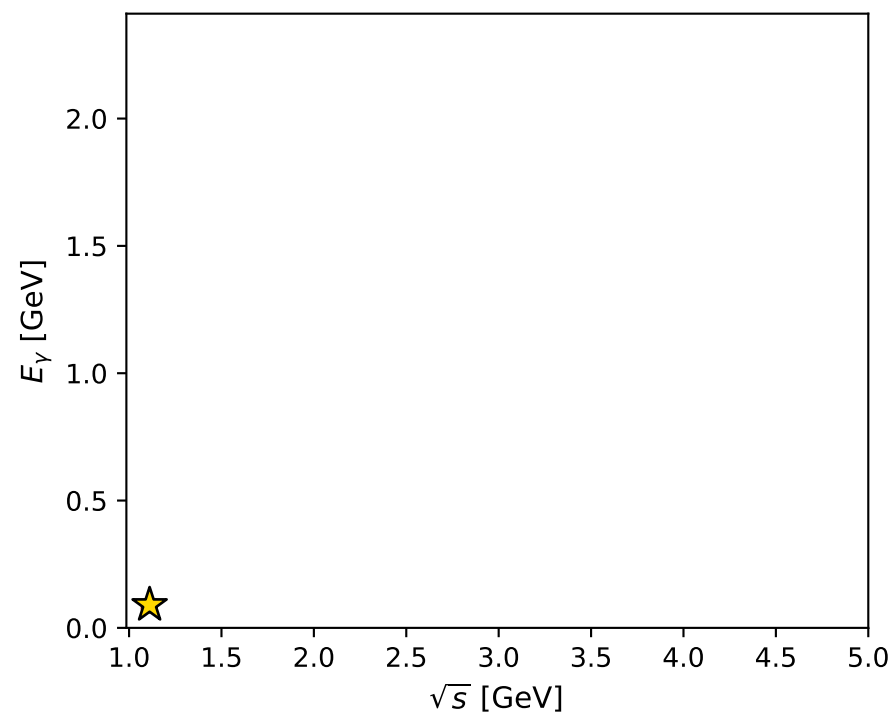
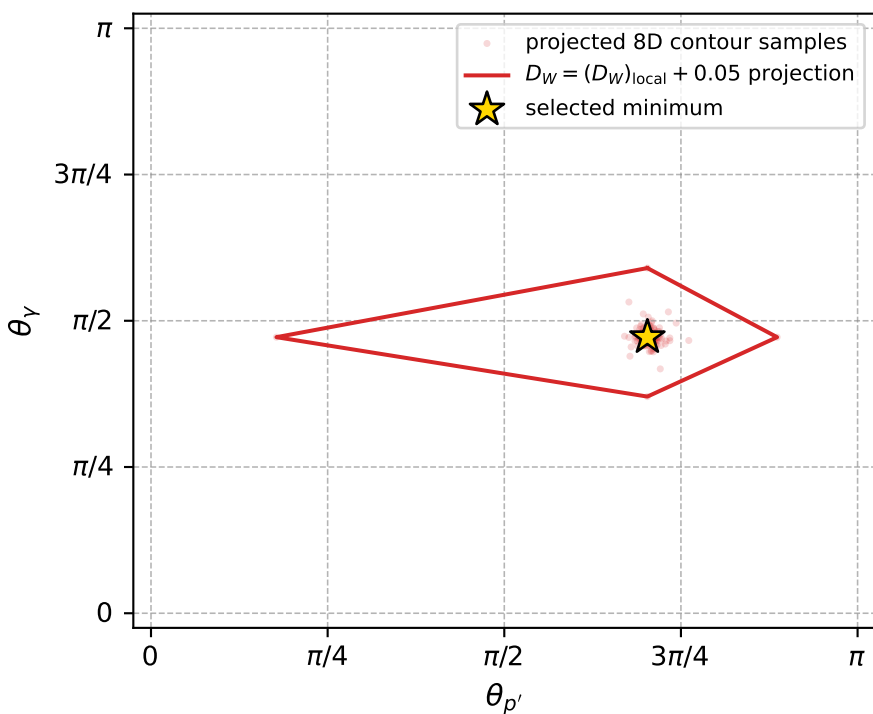
dw_mixing_angles_polarization_06_local_minimum_273 D_W=0.00070773
 region: polarization_cluster_06_local_minimum_273
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3564827$ rad
 incoming lepton: $-0.707311|+\rangle +0.706903|-\rangle$
 proton mixing angle: $\alpha_p=1.7143834$ rad
 incoming proton: $-0.143094|+\rangle +0.989709|-\rangle$

$s=1.23482$, $\sqrt{s}=1.11122$, $|\vec{P}|=0.159721$, $|\vec{P}'|=0.00854883$
 $\theta_{P'}=2.20703$, $\theta_Y=1.48269$, $\phi_{P'}=5.47294$, $\phi_Y=2.50498$
 $E_Y=0.0901237$, $\theta_{\ell'}=1.60513$, $\phi_{\ell'}=5.66089$
 $Q^2=0.0256218$, $x_B=0.130725$, $t=-0.0270252$, $W^2=1.05022$, $y=0.552149$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1597$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1597$
 P $E= 0.9515$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1597$
 ℓ' $E= 0.0831$ $p_x= 0.0674$ $p_y= -0.0484$ $p_z= -0.0029$
 P' $E= 0.9380$ $p_x= 0.0047$ $p_y= -0.0050$ $p_z= -0.0051$
 q_Y $E= 0.0901$ $p_x= -0.0722$ $p_y= 0.0534$ $p_z= 0.0079$



electron: polarization cluster P6, local minimum 273; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.00070773024$
 $D_W \text{ contour} = 0.05070773$
 above global minimum = 0.0006912974
 8D contour samples = 256

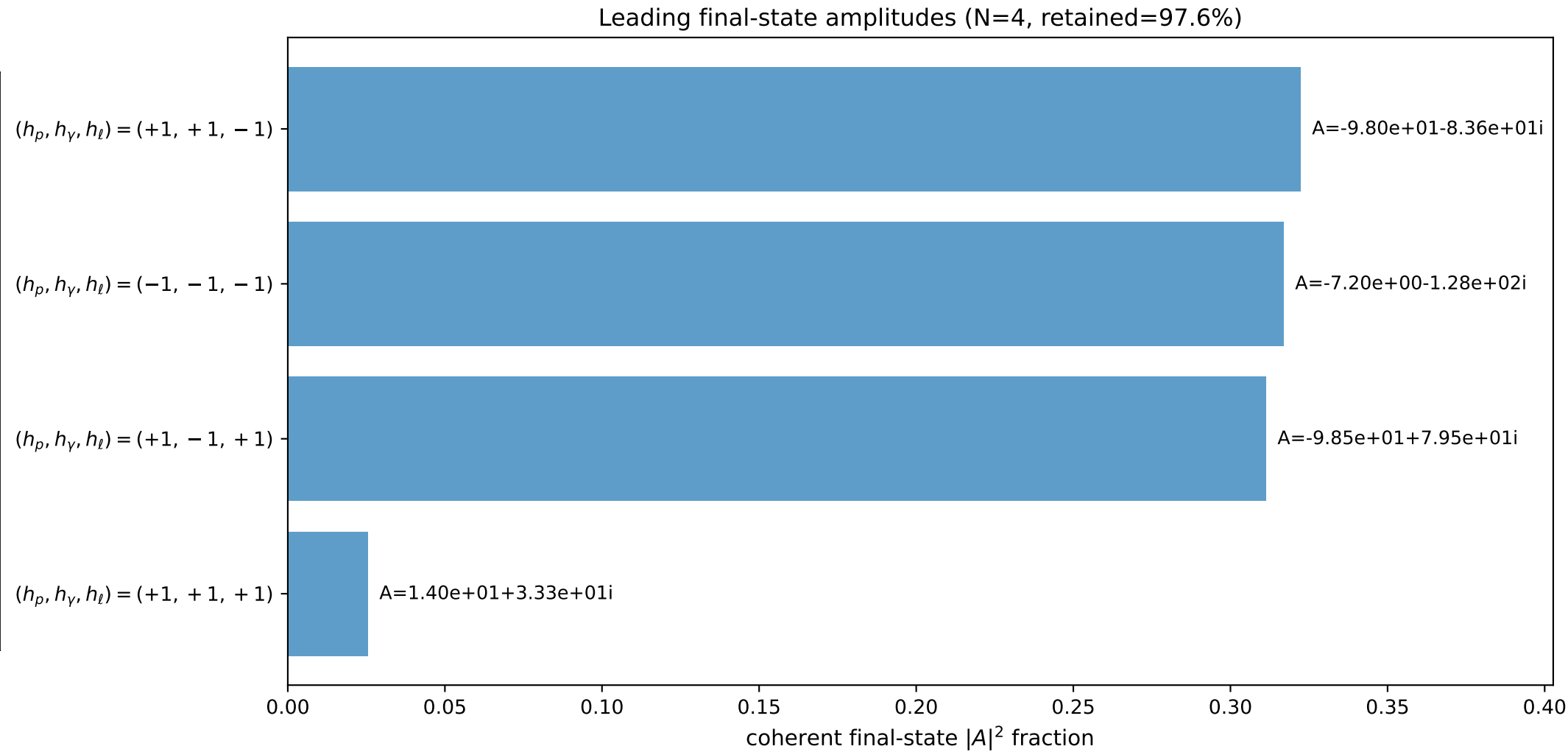
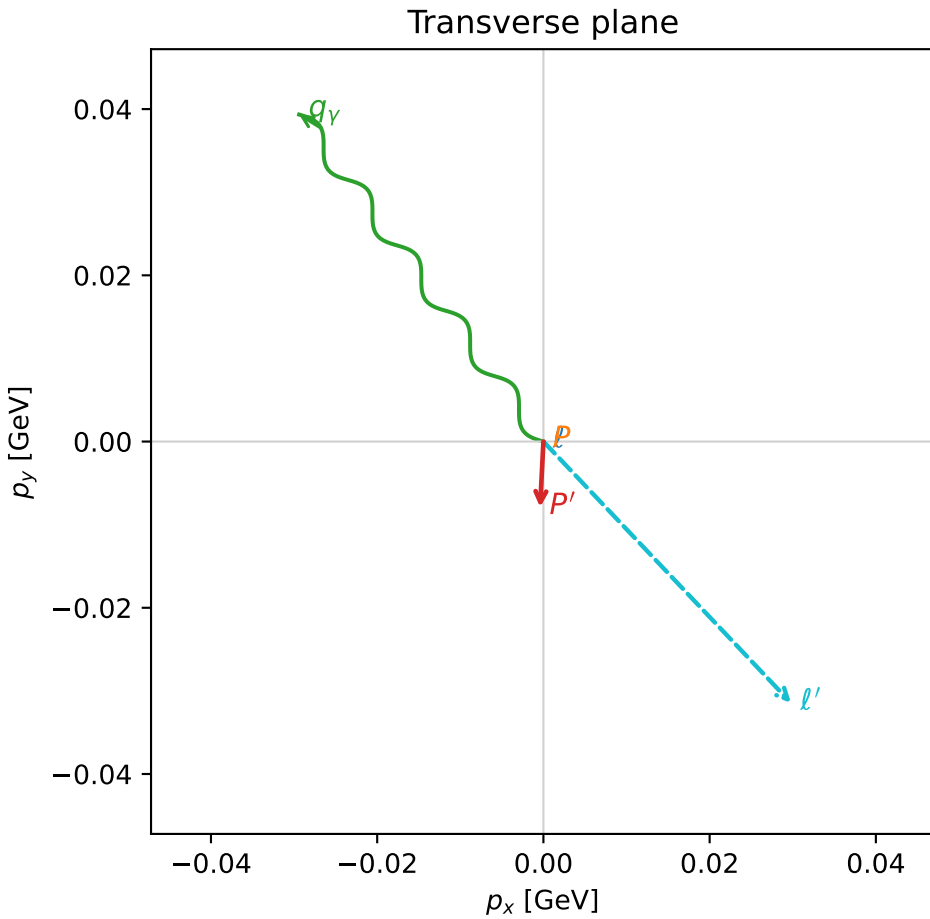
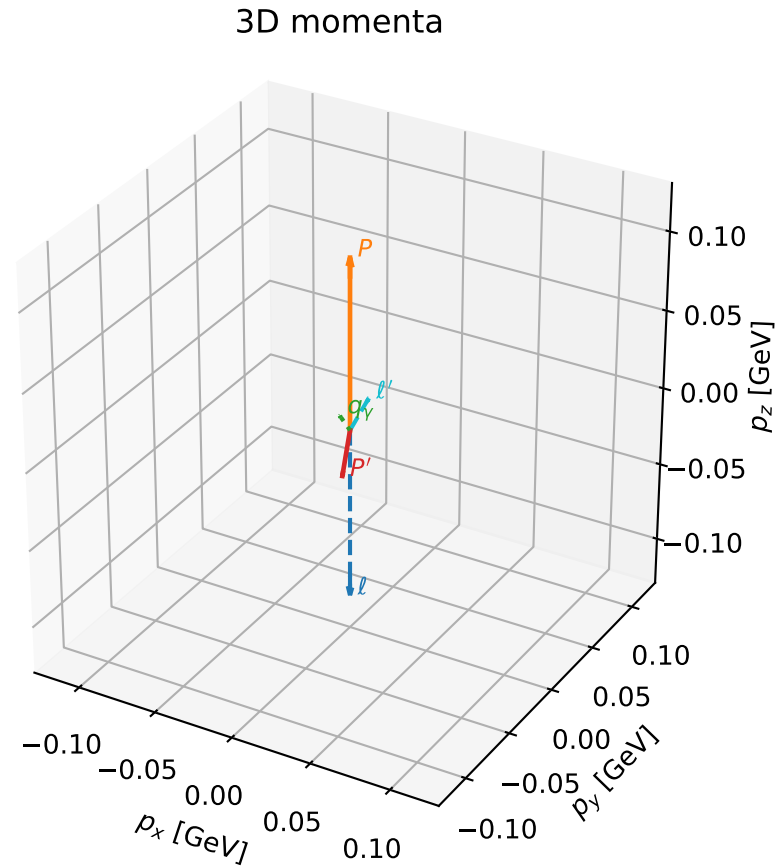
selected phase-space point:
 $\text{sqrt_s} = 1.111223$
 $\text{theta_p_out} = 2.207032$
 $\text{theta_gamma_out} = 1.482686$
 $\text{q0ut} = 0.09012373$
 $\text{phi_p_out} = 5.472943$
 $\text{phi_gamma_out} = 2.504981$
 $\text{alpha_e} = 2.356483$
 $\text{alpha_p} = 1.714383$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

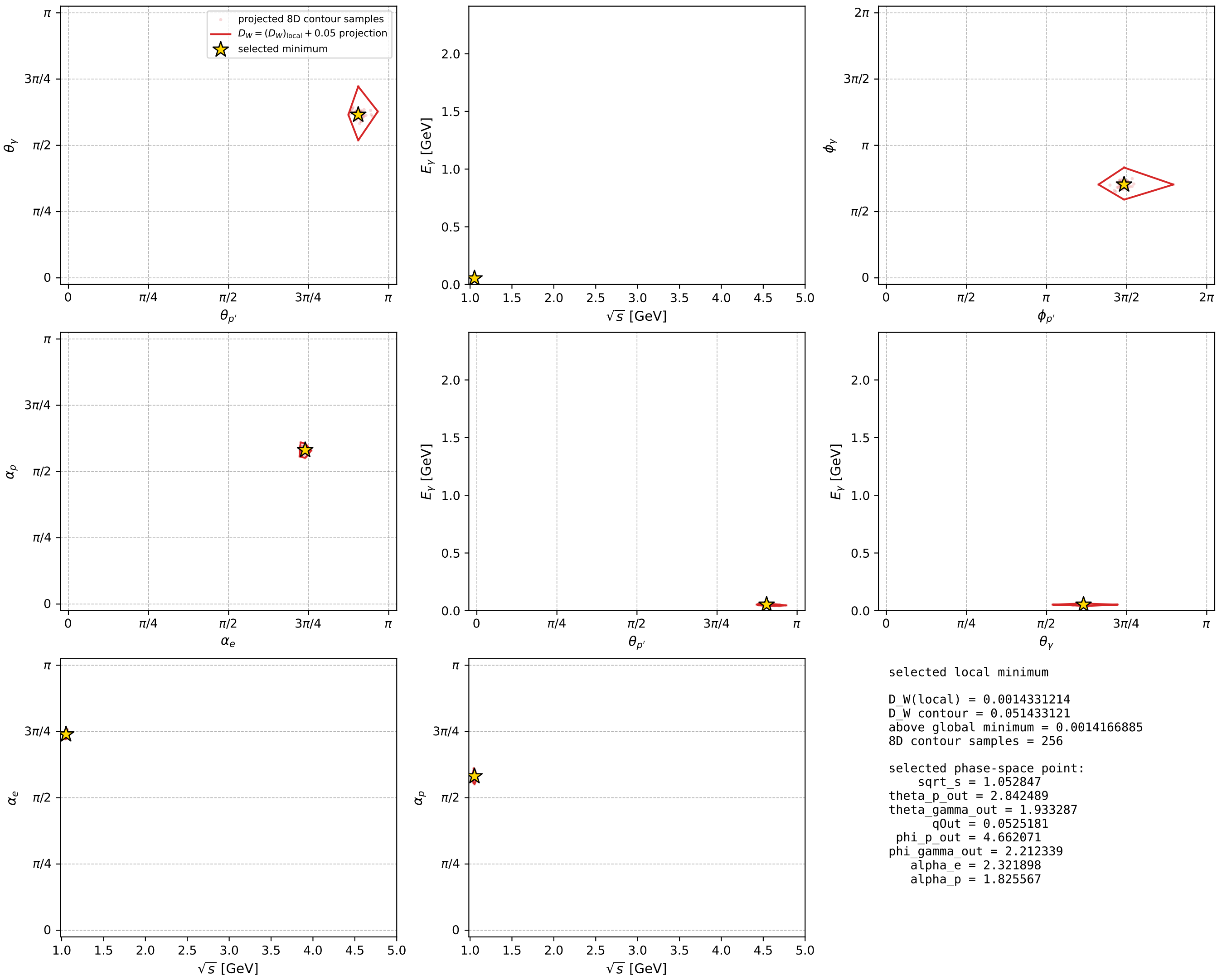
dw_mixing_angles_polarization_06_local_minimum_414 D_W=0.00143312
 region: polarization_cluster_06_local_minimum_414
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3218983$ rad
 incoming lepton: $-0.682445|+> +0.730937|->$
 proton mixing angle: $\alpha_p=1.8255674$ rad
 incoming proton: $-0.252024|+> +0.967721|->$

$s=1.10849$, $\sqrt{s}=1.05285$, $|\vec{P}|=0.108582$, $|\vec{P}'|=0.0268574$
 $\theta_{p'}=2.84249$, $\theta_Y=1.93329$, $\phi_{p'}=4.66207$, $\phi_Y=2.21234$
 $E_Y=0.0525181$, $\theta_{\ell'}=0.774192$, $\phi_{\ell'}=5.47078$
 $Q^2=0.0230696$, $x_B=0.190223$, $t=-0.0180504$, $W^2=0.978051$, $y=0.530419$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1086$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1086$
 P $E= 0.9443$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1086$
 ℓ' $E= 0.0619$ $p_x= 0.0298$ $p_y= -0.0314$ $p_z= 0.0443$
 P' $E= 0.9384$ $p_x= -0.0004$ $p_y= -0.0079$ $p_z= -0.0257$
 q_Y $E= 0.0525$ $p_x= -0.0294$ $p_y= 0.0393$ $p_z= -0.0186$



electron: polarization cluster P6, local minimum 414; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



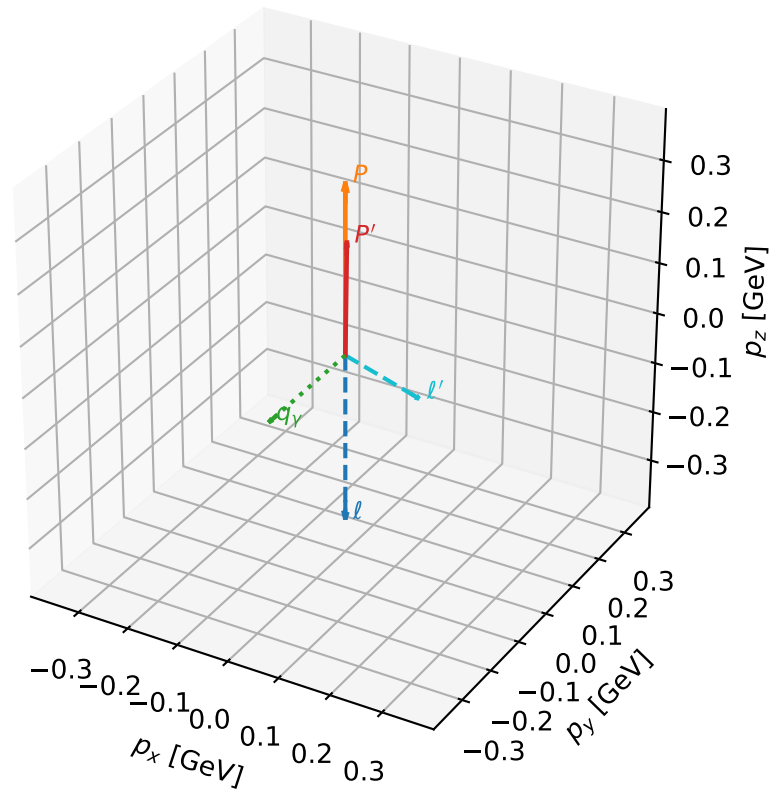
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_463 D_W=0.00198404
region: polarization_cluster_06_local_minimum_463
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3829895$ rad
incoming lepton: $-0.725798|+> +0.687908|->$
proton mixing angle: $\alpha_p=1.8504596$ rad
incoming proton: $-0.276032|+> +0.961148|->$

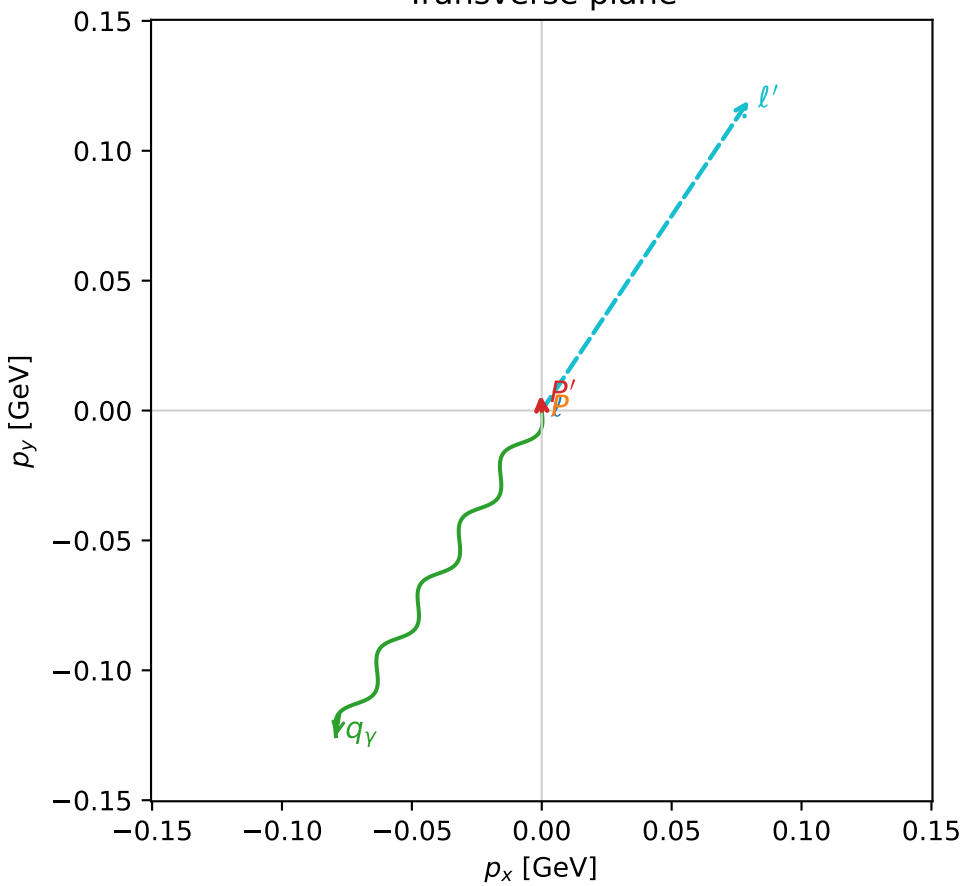
$s=1.76147$, $\sqrt{s}=1.3272$, $|\vec{P}|=0.332137$, $|\vec{P}'|=0.216708$
 $\theta_{P'}=0.0274801$, $\theta_V=2.13021$, $\phi_{P'}=1.64511$, $\phi_V=4.14865$
 $E_V=0.174953$, $\theta_{l'}=2.28239$, $\phi_{l'}=0.982347$
 $Q^2=0.0436849$, $x_B=0.103473$, $t=-0.0123311$, $W^2=1.25835$, $y=0.478871$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.3321$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3321$
 P $E= 0.9951$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3321$
 l' $E= 0.1895$ $p_x= 0.0797$ $p_y= 0.1194$ $p_z= -0.1238$
 P' $E= 0.9627$ $p_x= -0.0004$ $p_y= 0.0059$ $p_z= 0.2166$
 q_Y $E= 0.1750$ $p_x= -0.0792$ $p_y= -0.1253$ $p_z= -0.0928$

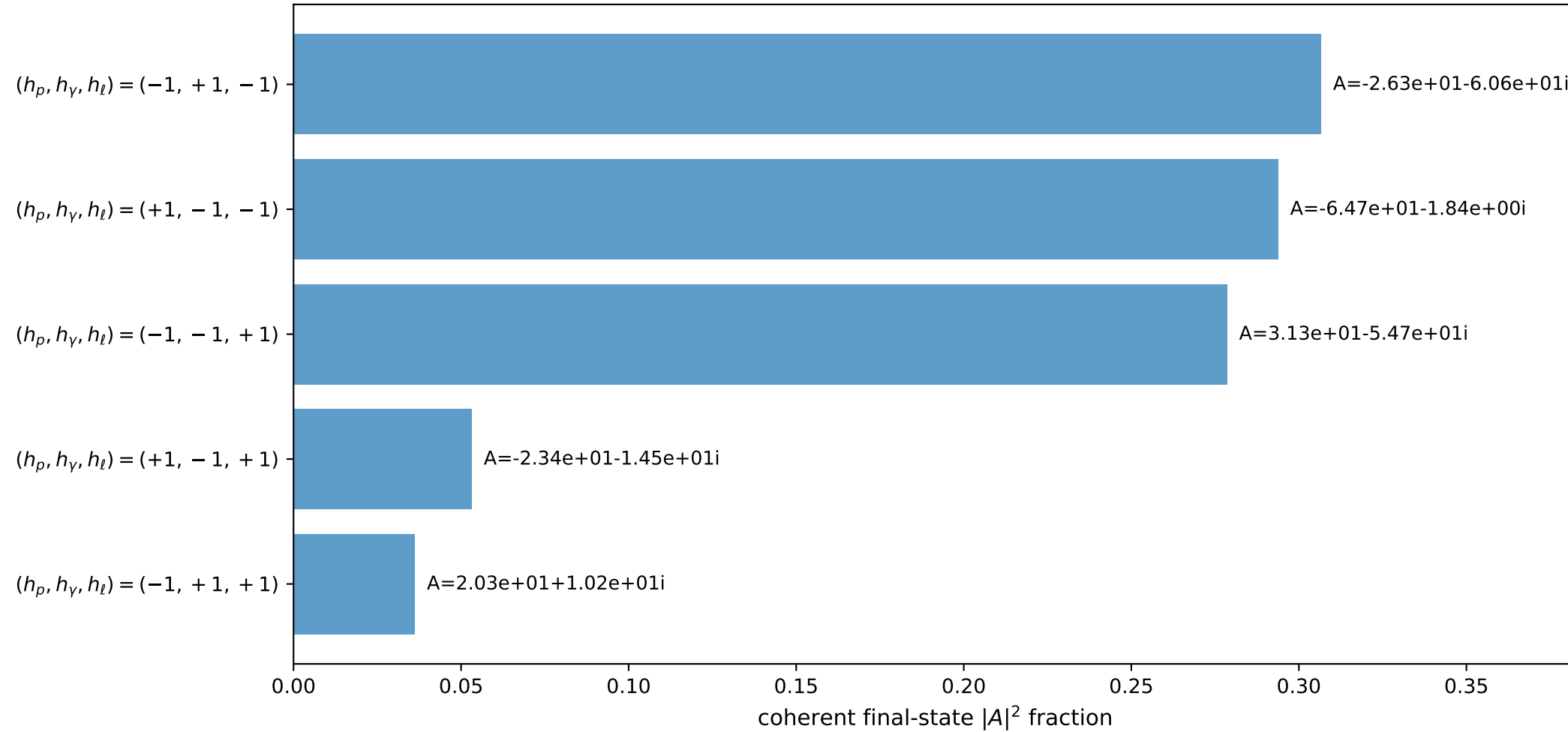
3D momenta



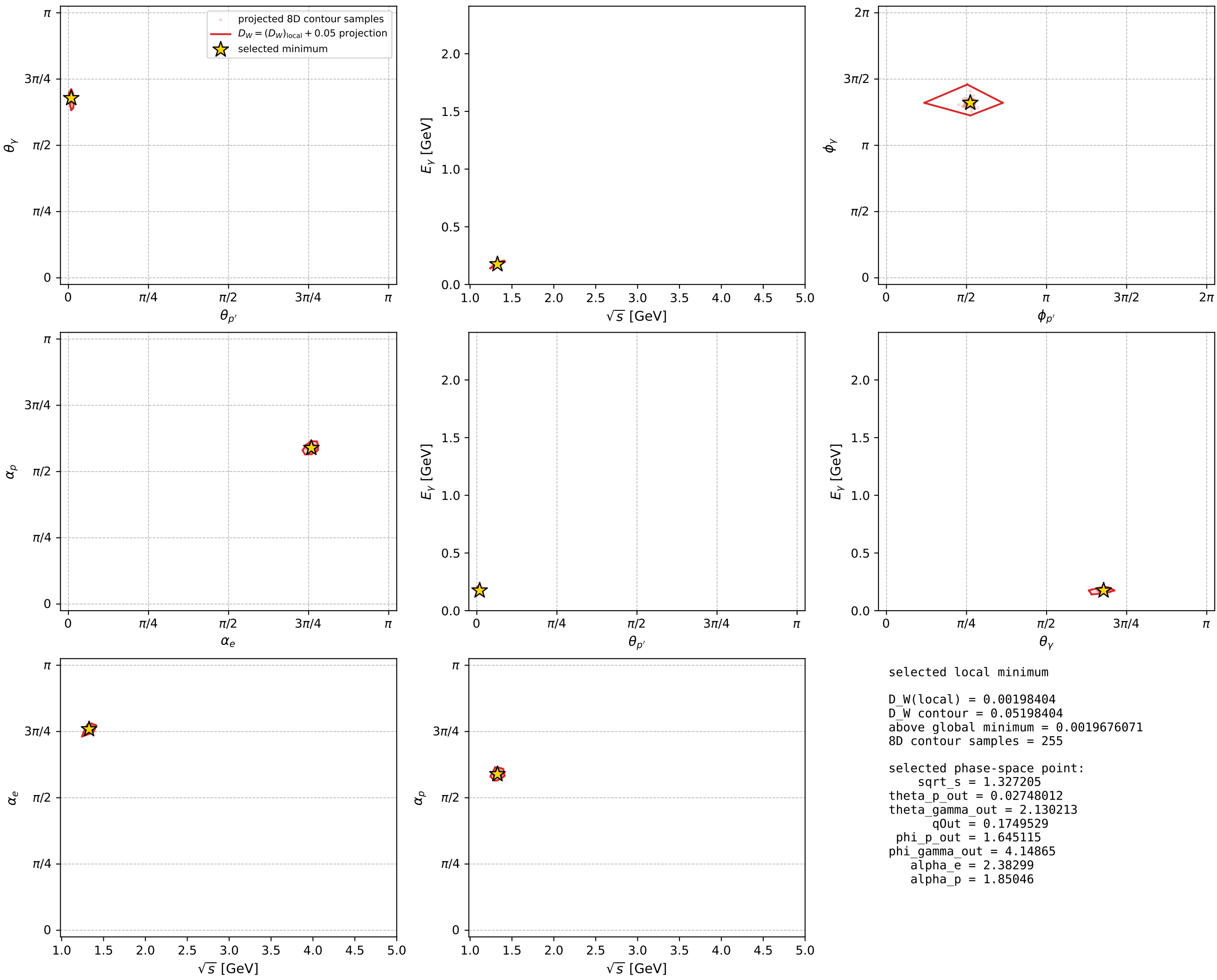
Transverse plane



Leading final-state amplitudes (N=5, retained=96.8%)



electron: polarization cluster P6, local minimum 463; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



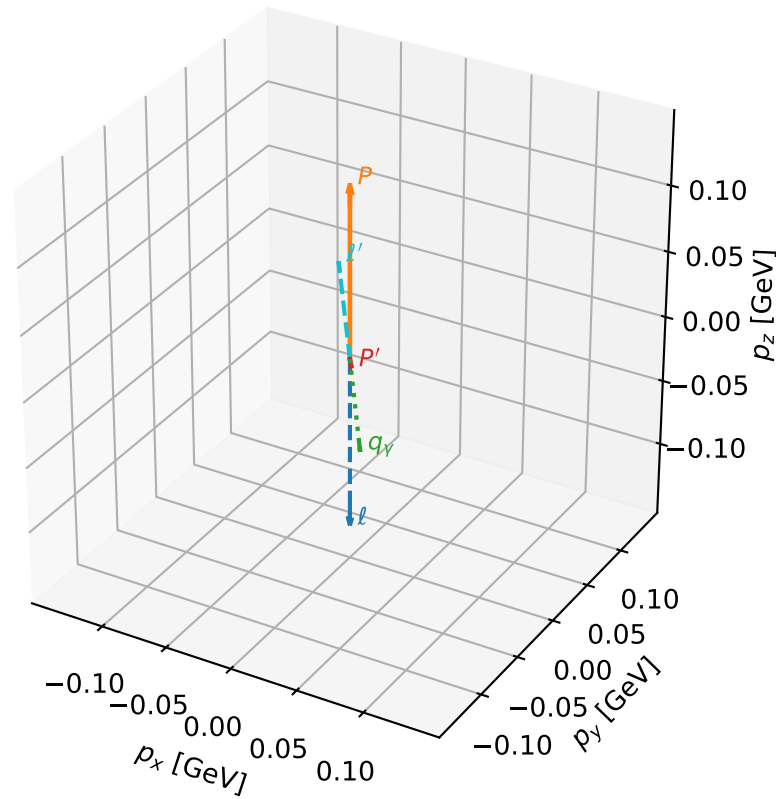
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_509 D_W=0.00259542
 region: polarization_cluster_06_local_minimum_509
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3474721$ rad
 incoming lepton: $-0.700912|+> +0.713247|->$
 proton mixing angle: $\alpha_p=1.3918247$ rad
 incoming proton: $0.178018|+> +0.984027|->$

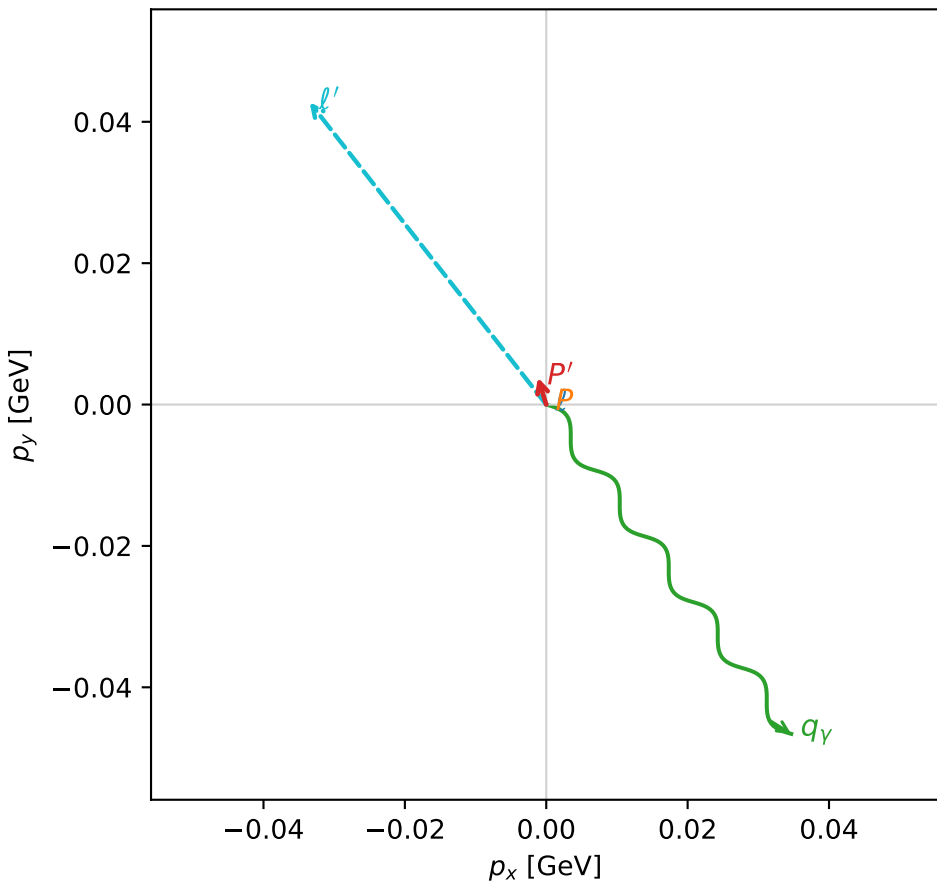
$s=1.16007$, $\sqrt{s}=1.07706$, $|\vec{P}|=0.130085$, $|\vec{P}'|=0.00864853$
 $\theta_{P'}=2.66104$, $\theta_V=2.13648$, $\phi_{P'}=1.87001$, $\phi_V=5.35198$
 $E_V=0.0687822$, $\theta_l=0.884012$, $\phi_{l'}=2.23496$
 $Q^2=0.0298612$, $x_B=0.188073$, $t=-0.0189123$, $W^2=1.00876$, $y=0.566603$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1301$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1301$
 P $E= 0.9470$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1301$
 l' $E= 0.0702$ $p_x= -0.0335$ $p_y= 0.0428$ $p_z= 0.0445$
 P' $E= 0.9380$ $p_x= -0.0012$ $p_y= 0.0038$ $p_z= -0.0077$
 q_V $E= 0.0688$ $p_x= 0.0347$ $p_y= -0.0466$ $p_z= -0.0369$

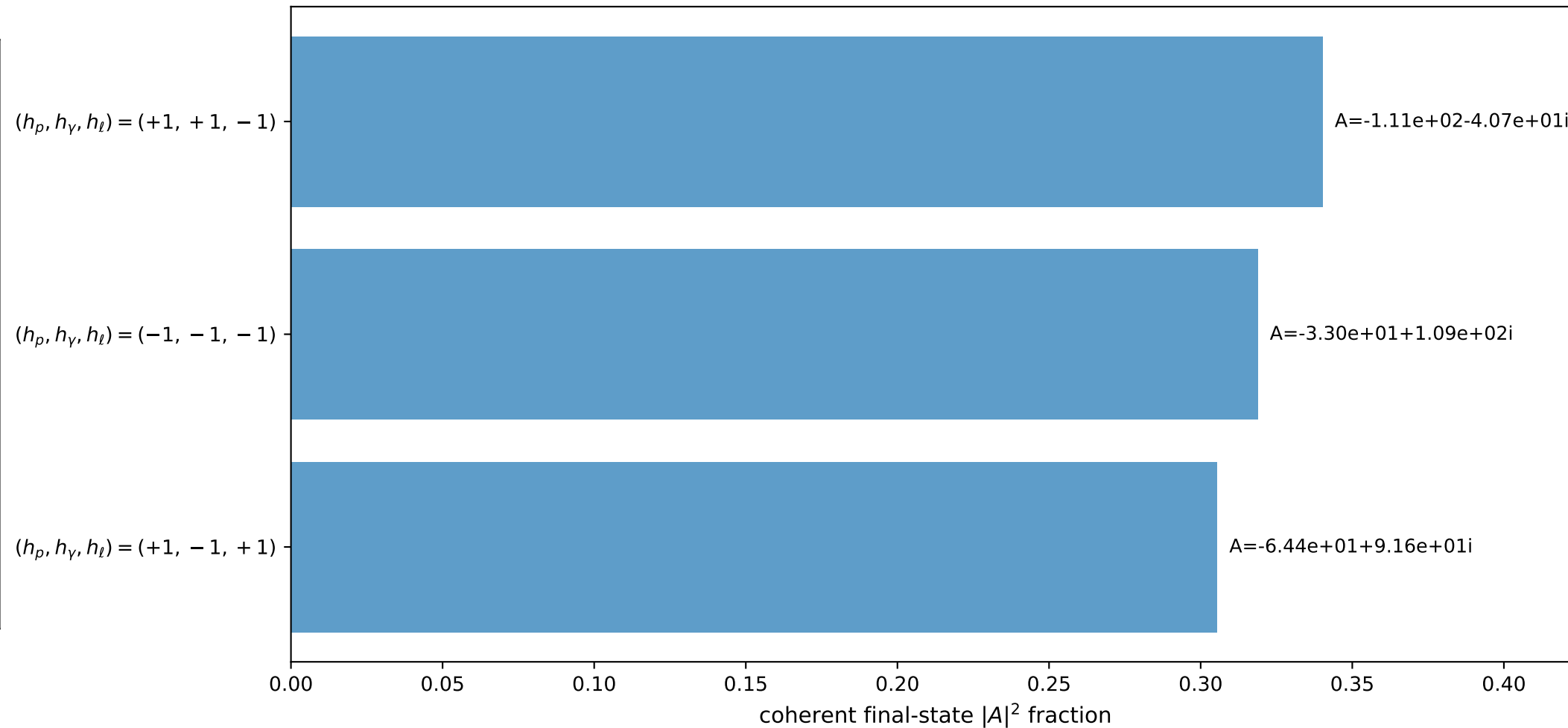
3D momenta



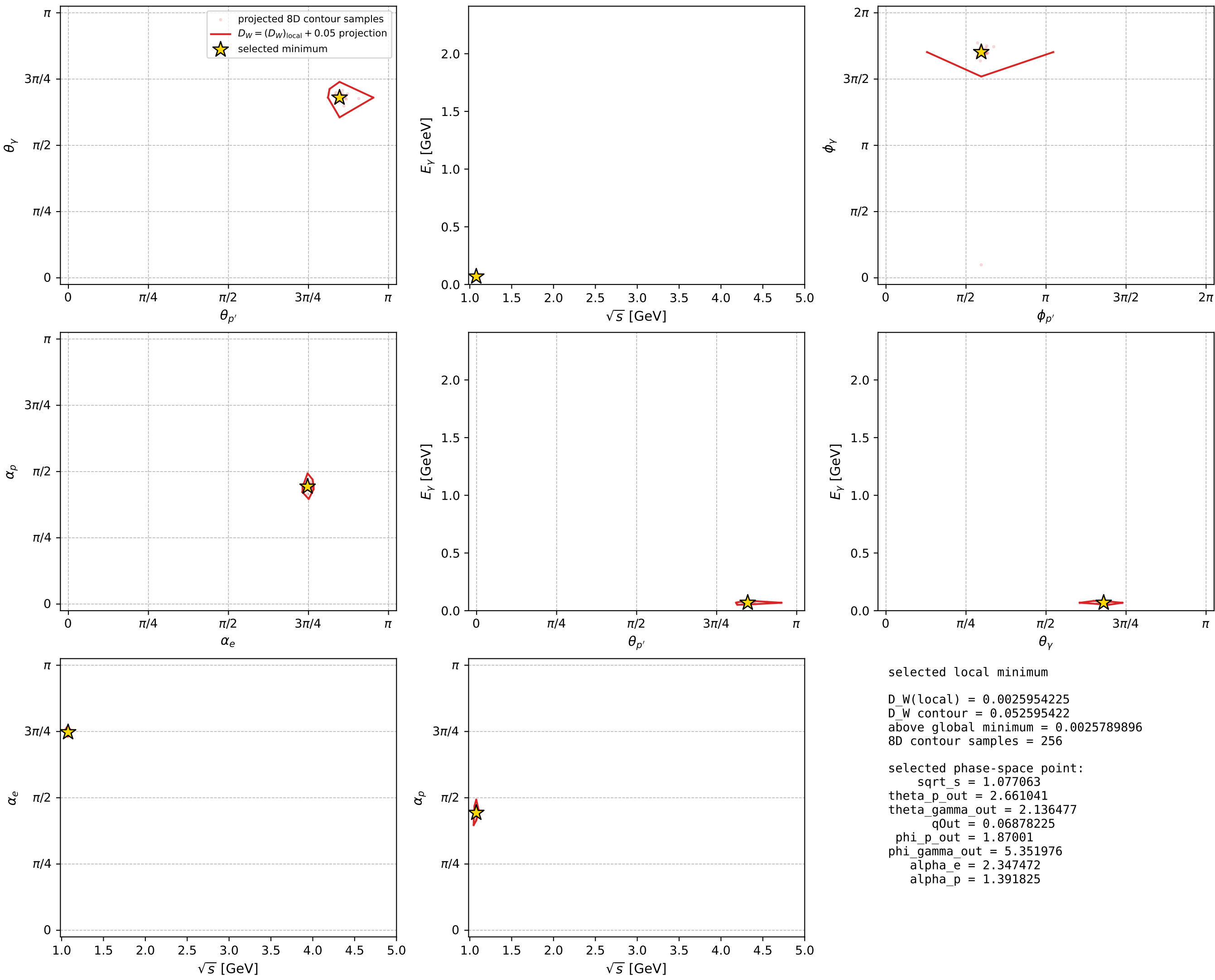
Transverse plane



Leading final-state amplitudes (N=3, retained=96.5%)



electron: polarization cluster P6, local minimum 509; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

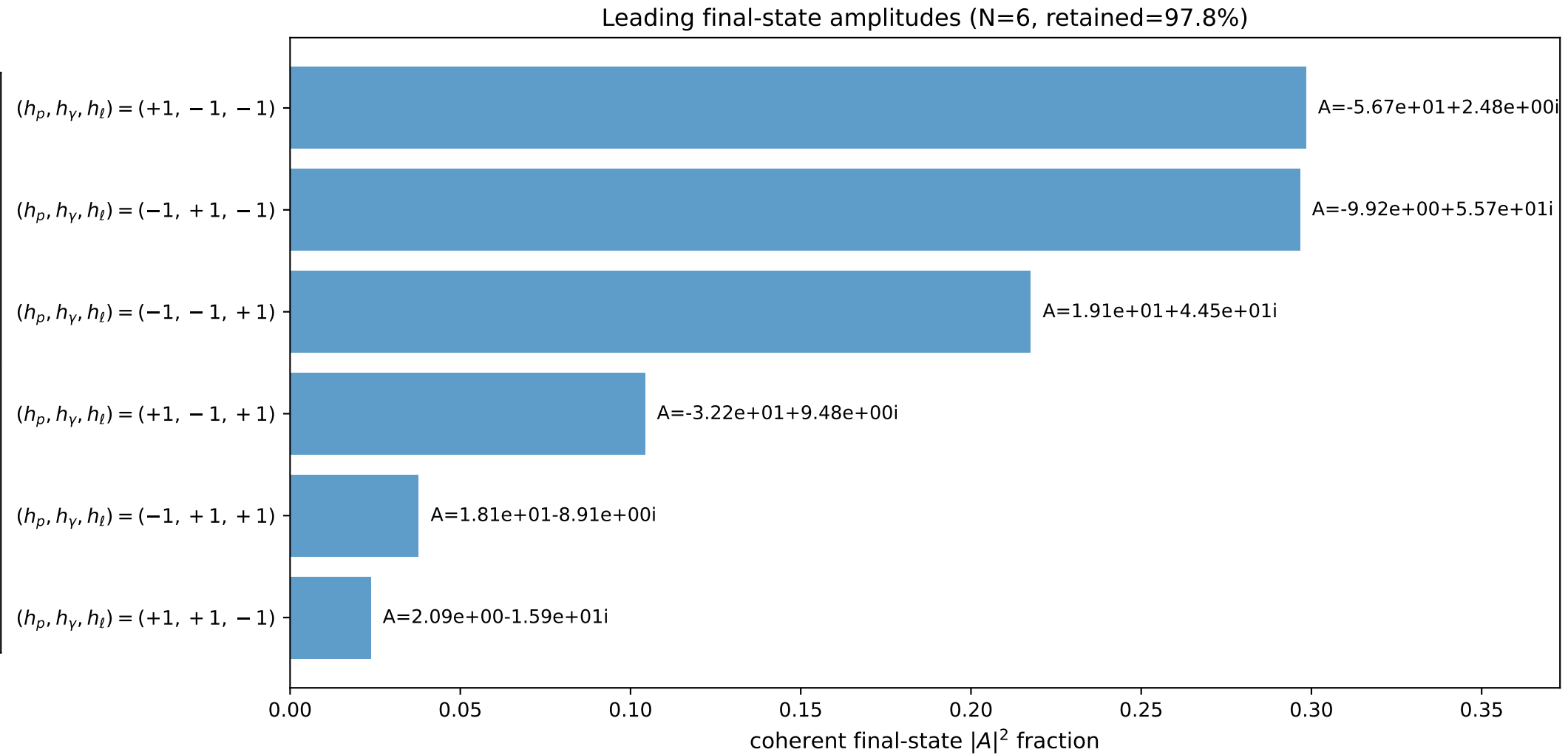
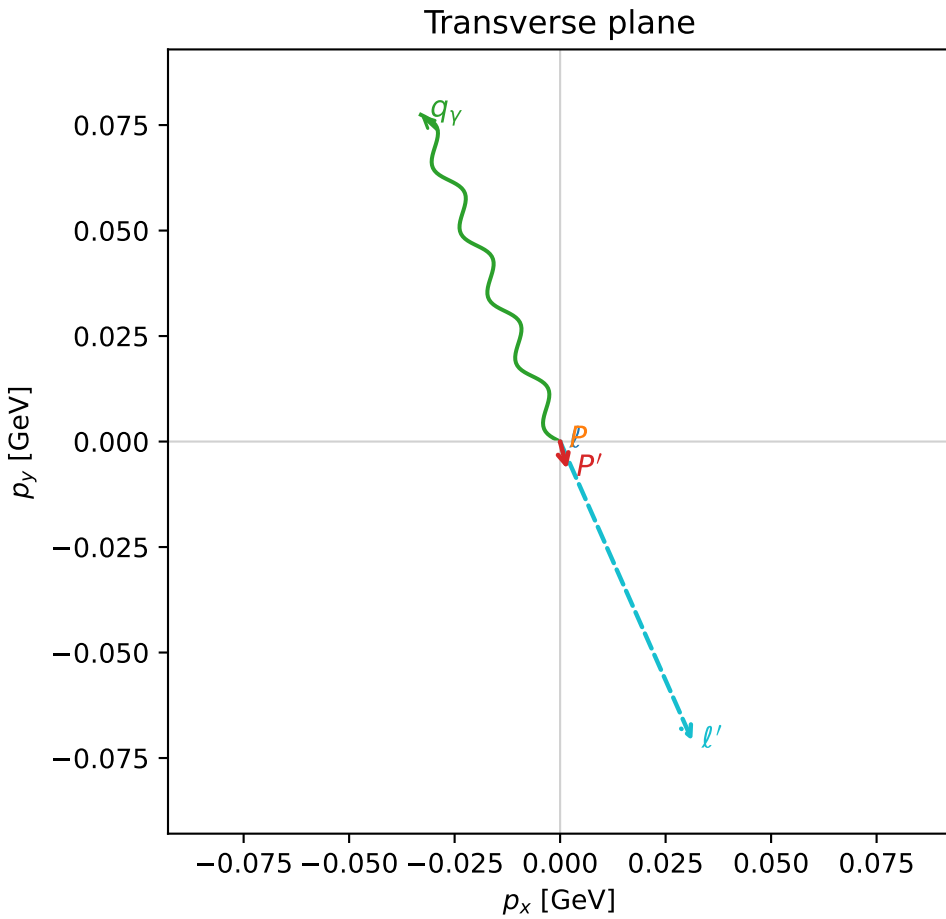
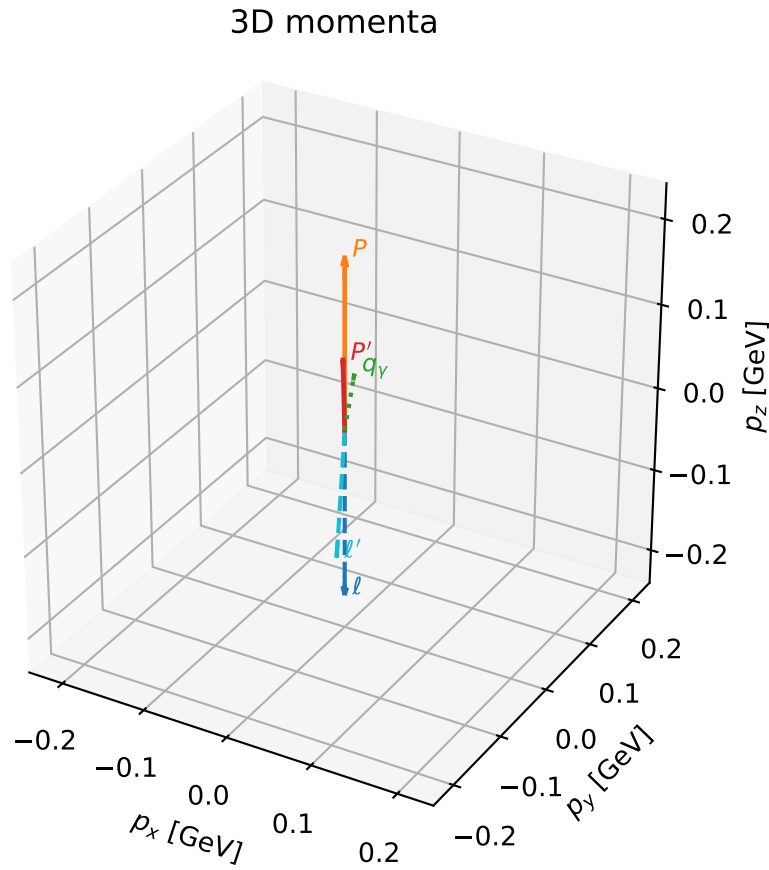


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

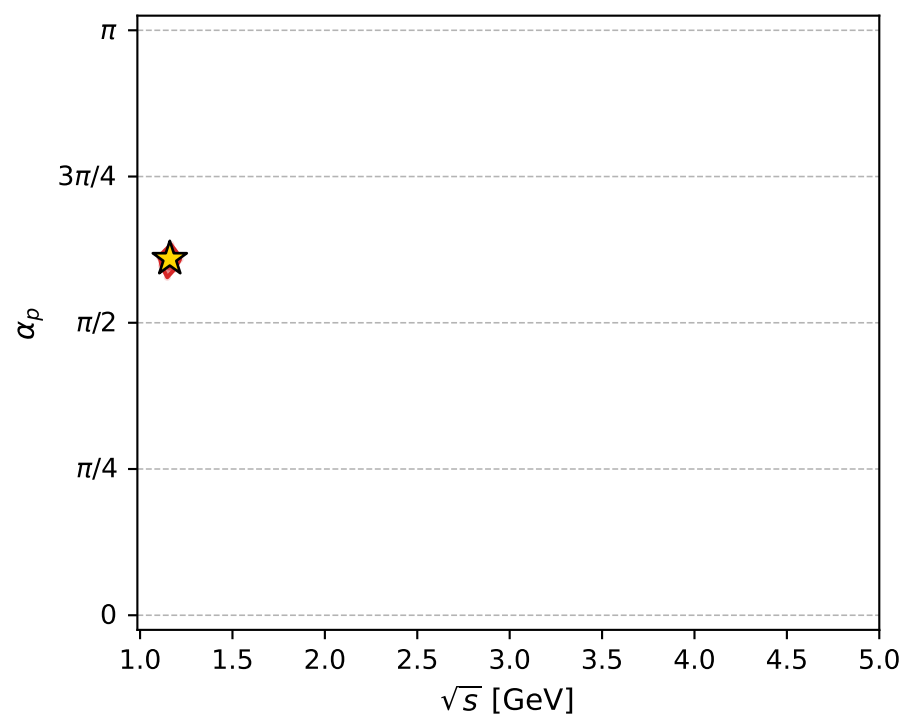
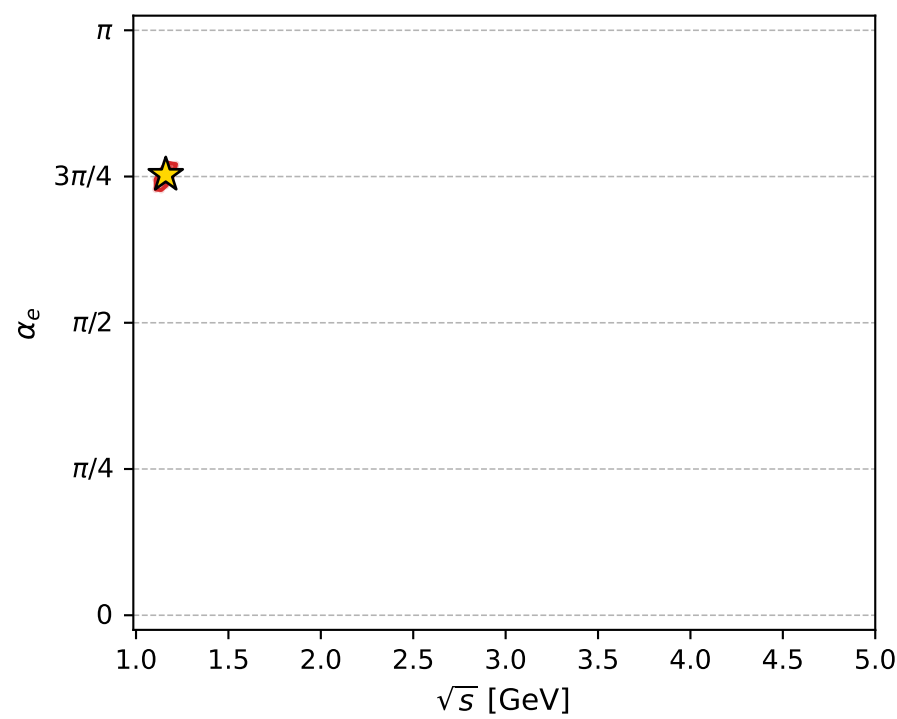
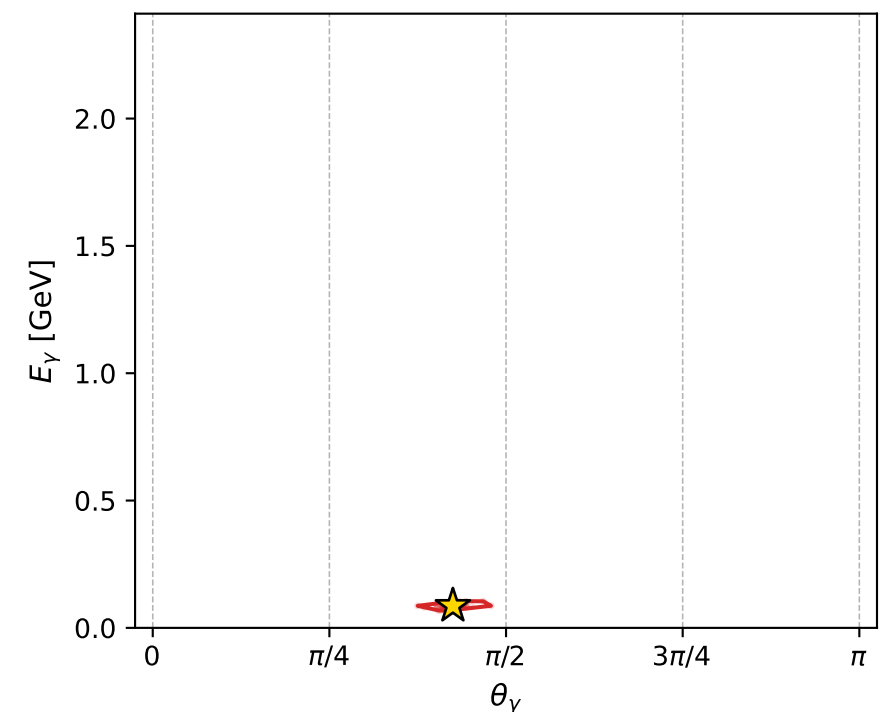
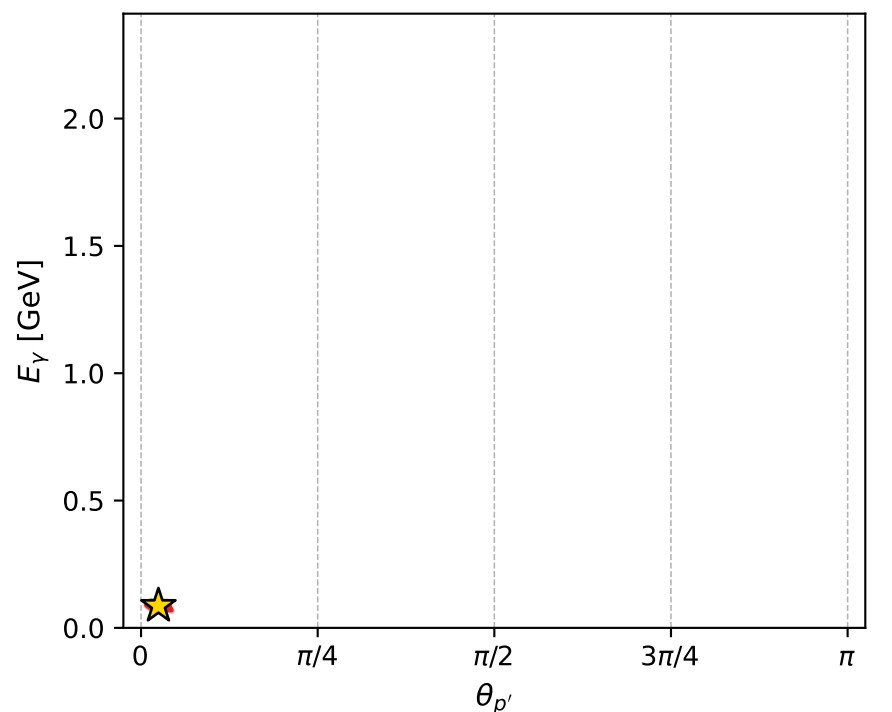
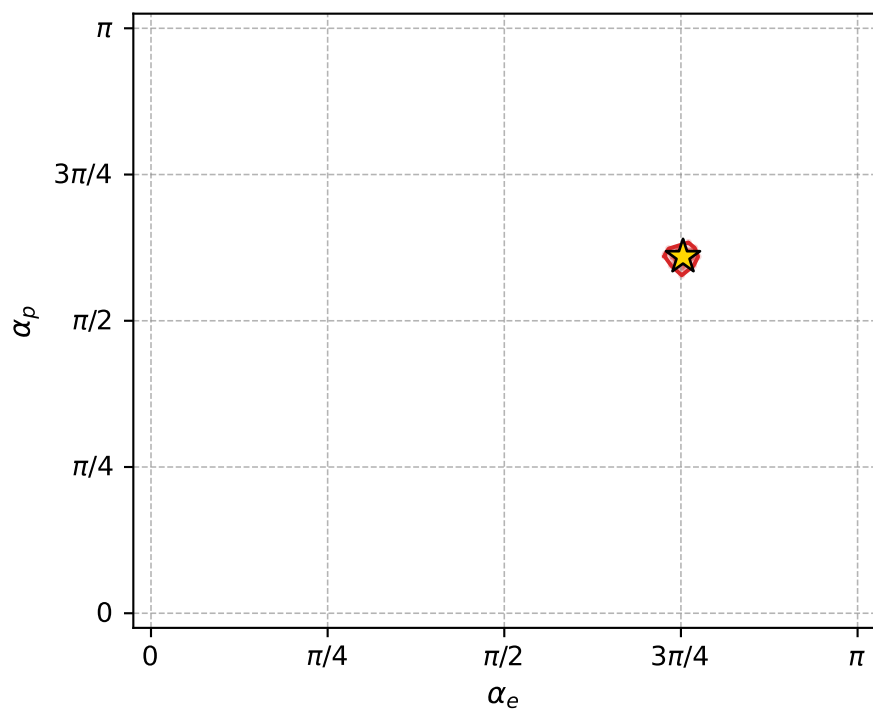
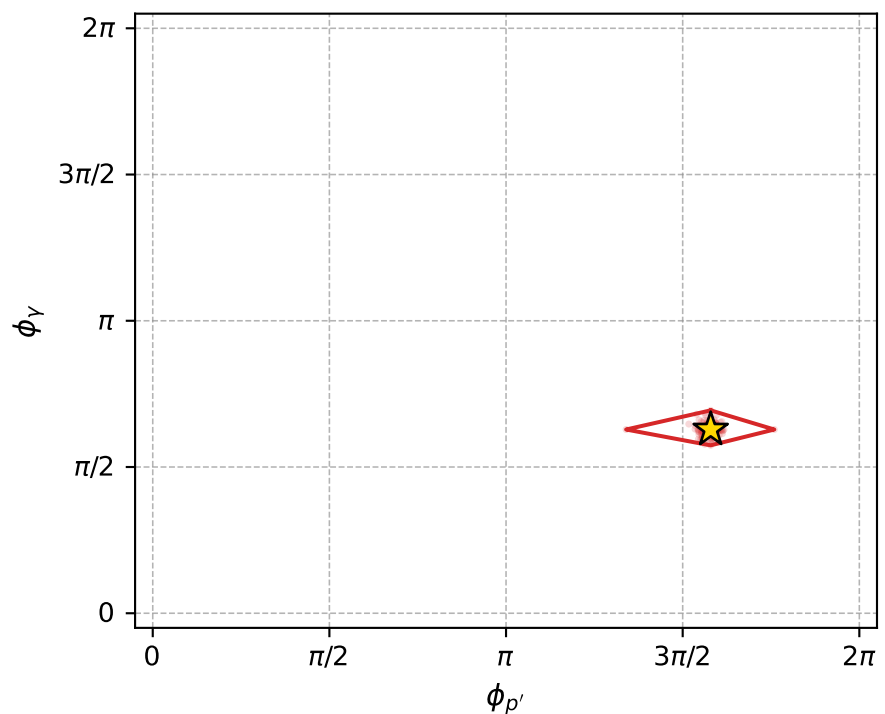
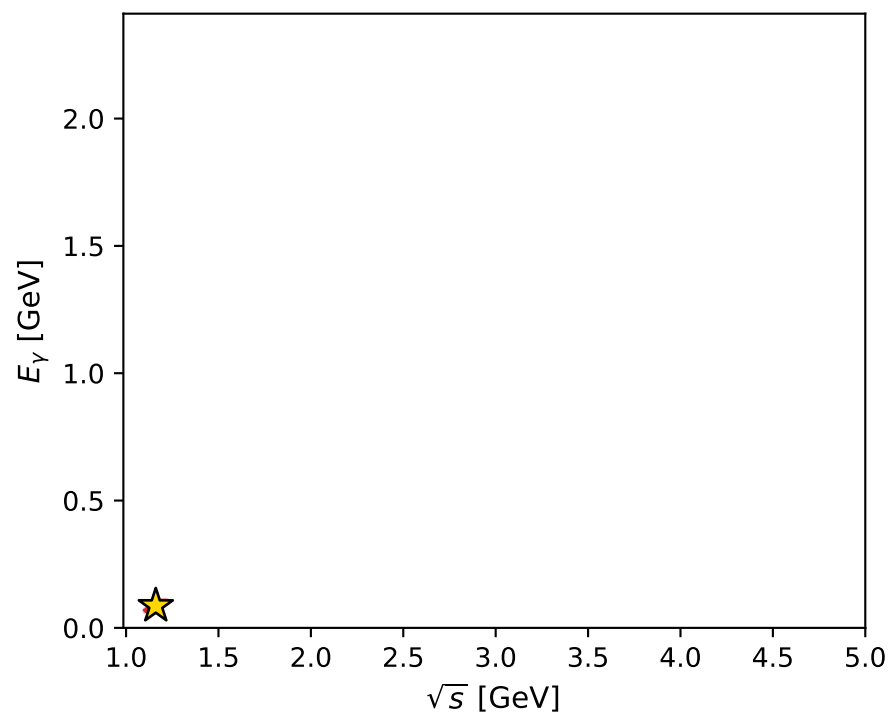
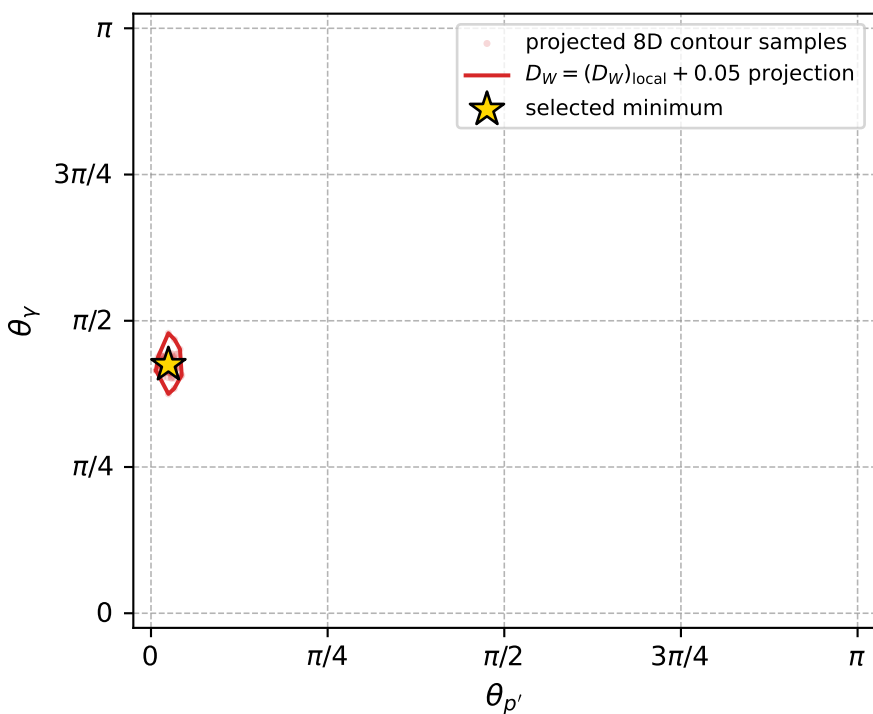
dw_mixing_angles_polarization_06_local_minimum_530 D_W=0.00280974
region: polarization_cluster_06_local_minimum_530
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3654336$ rad
incoming lepton: $-0.71361|+\rangle +0.700544|-\rangle$
proton mixing angle: $\alpha_p=1.9147799$ rad
incoming proton: $-0.33724|+\rangle +0.941419|-\rangle$

$s=1.34767$, $\sqrt{s}=1.16089$, $|\vec{P}|=0.201493$, $|\vec{P}'|=0.0871666$
 $\theta_{p'}=0.0773232$, $\theta_Y=1.33454$, $\phi_{p'}=4.96147$, $\phi_Y=1.97356$
 $E_Y=0.086577$, $\theta_{\ell'}=2.51535$, $\phi_{\ell'}=5.12845$
 $Q^2=0.0101152$, $x_B=0.0592116$, $t=-0.0128742$, $W^2=1.04056$, $y=0.365163$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2015$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2015$
 P $E= 0.9594$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2015$
 ℓ' $E= 0.1323$ $p_x= 0.0313$ $p_y= -0.0709$ $p_z= -0.1072$
 P' $E= 0.9420$ $p_x= 0.0017$ $p_y= -0.0065$ $p_z= 0.0869$
 q_Y $E= 0.0866$ $p_x= -0.0330$ $p_y= 0.0774$ $p_z= 0.0203$



electron: polarization cluster P6, local minimum 530; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0028097386$
 $D_W \text{ contour} = 0.052809739$
 above global minimum = 0.0027933057
 8D contour samples = 256

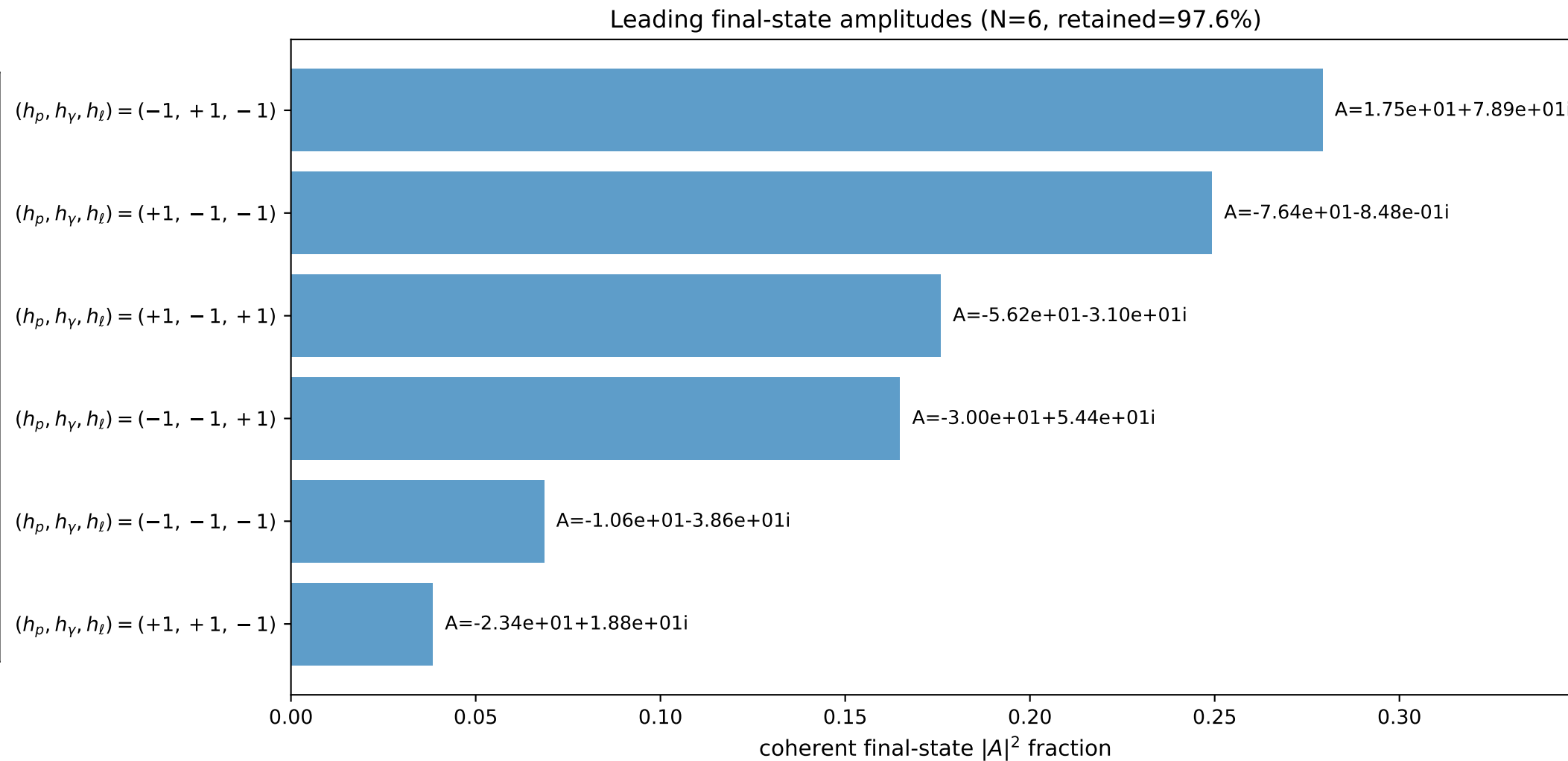
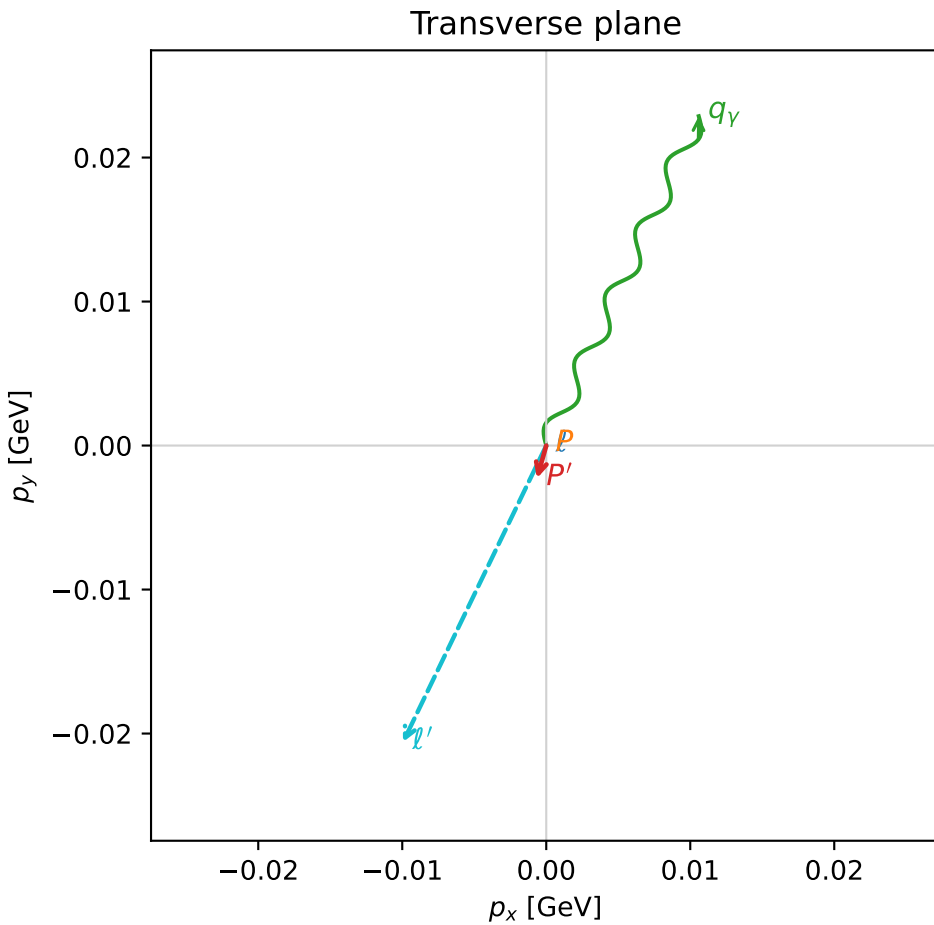
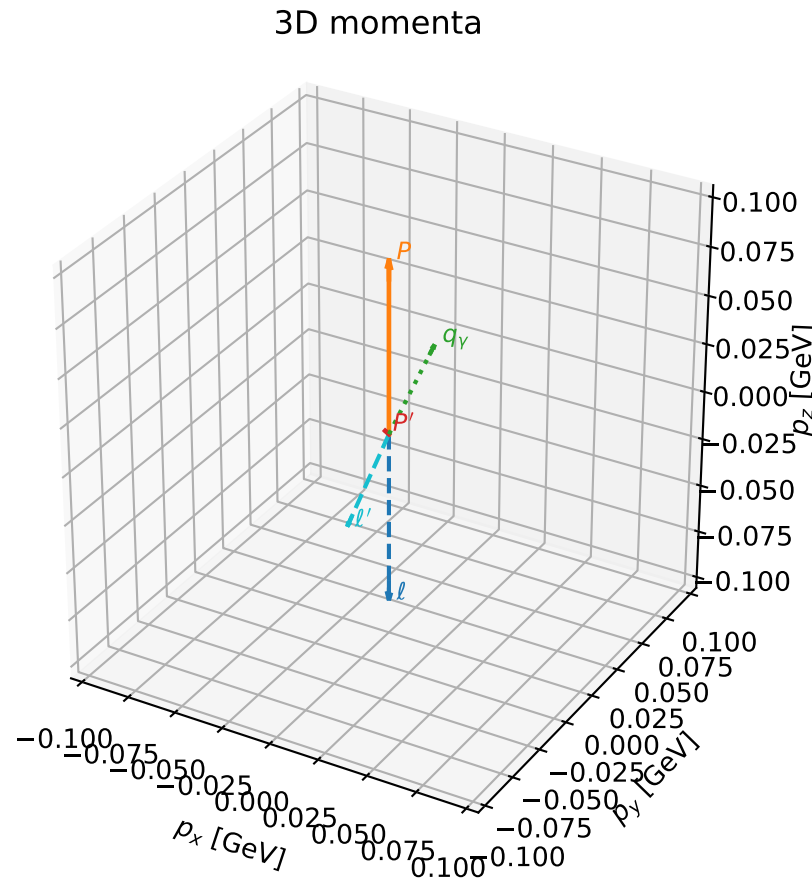
selected phase-space point:
 $\text{sqrt_s} = 1.16089$
 $\text{theta_p_out} = 0.07732316$
 $\text{theta_gamma_out} = 1.334542$
 $\text{qOut} = 0.08657701$
 $\text{phi_p_out} = 4.961468$
 $\text{phi_gamma_out} = 1.973563$
 $\text{alpha_e} = 2.365434$
 $\text{alpha_p} = 1.91478$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

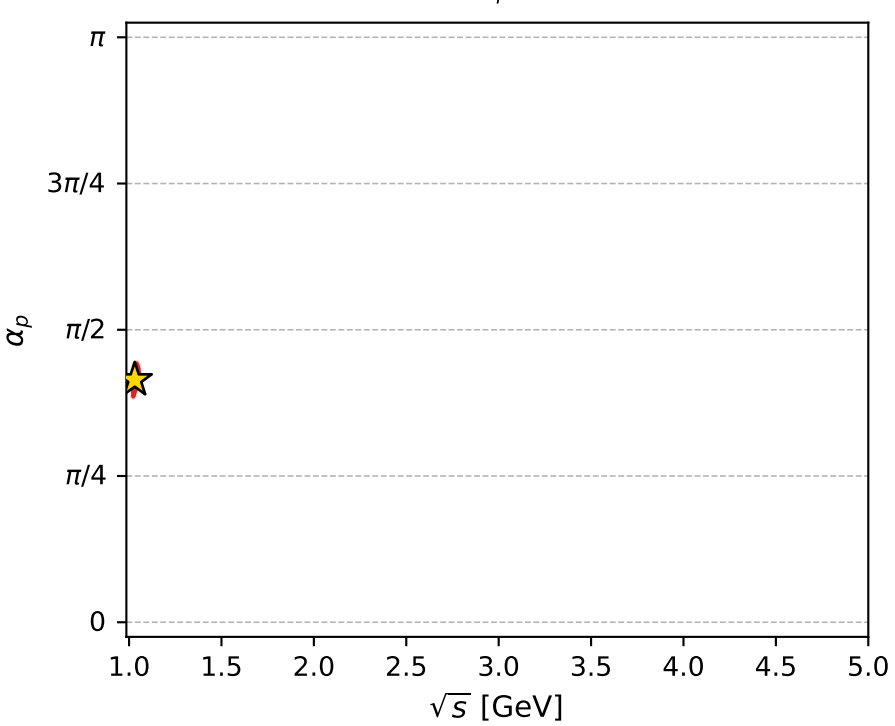
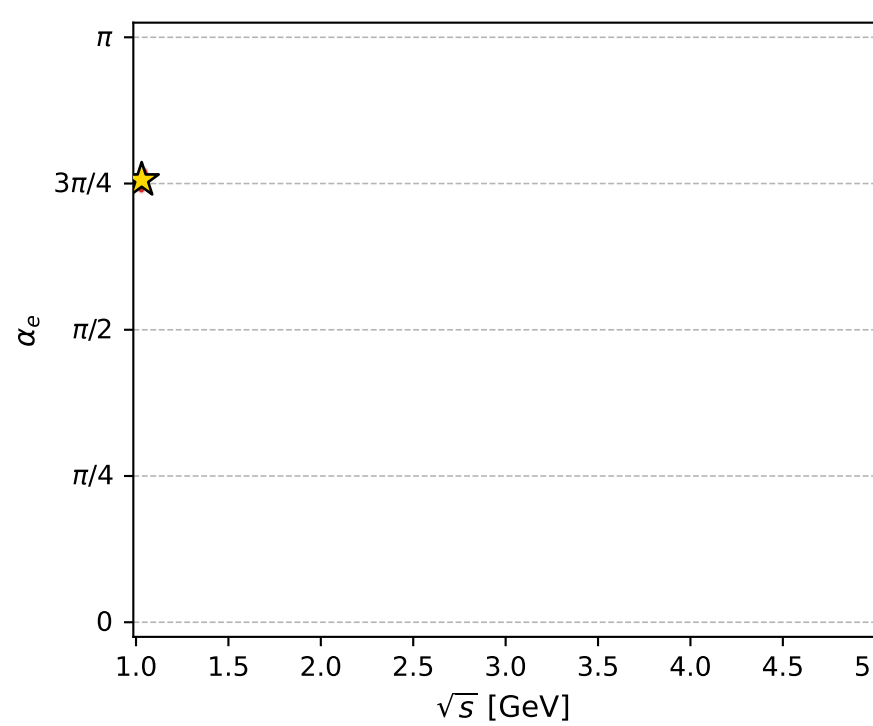
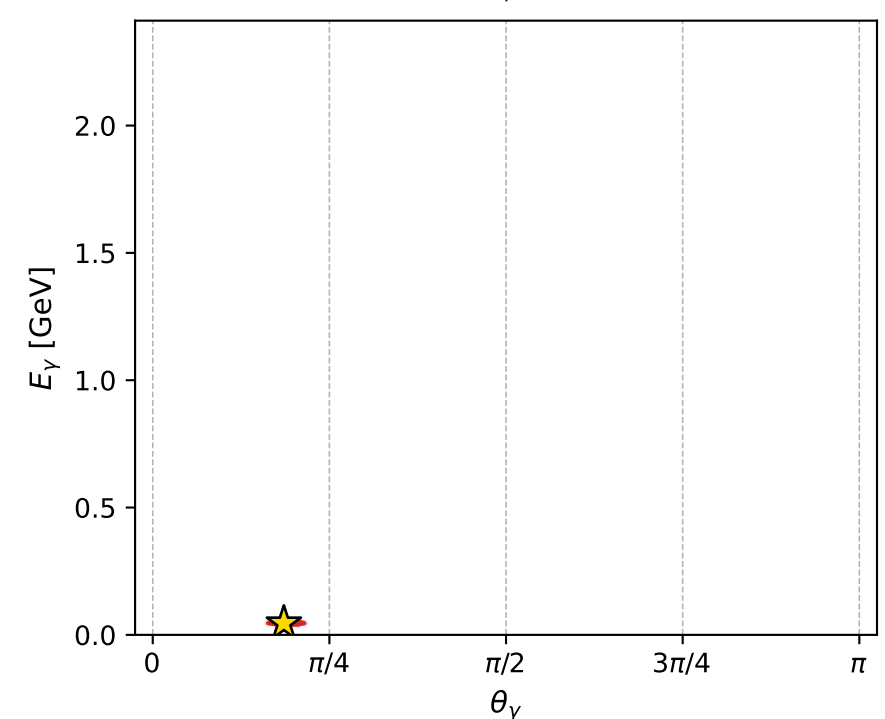
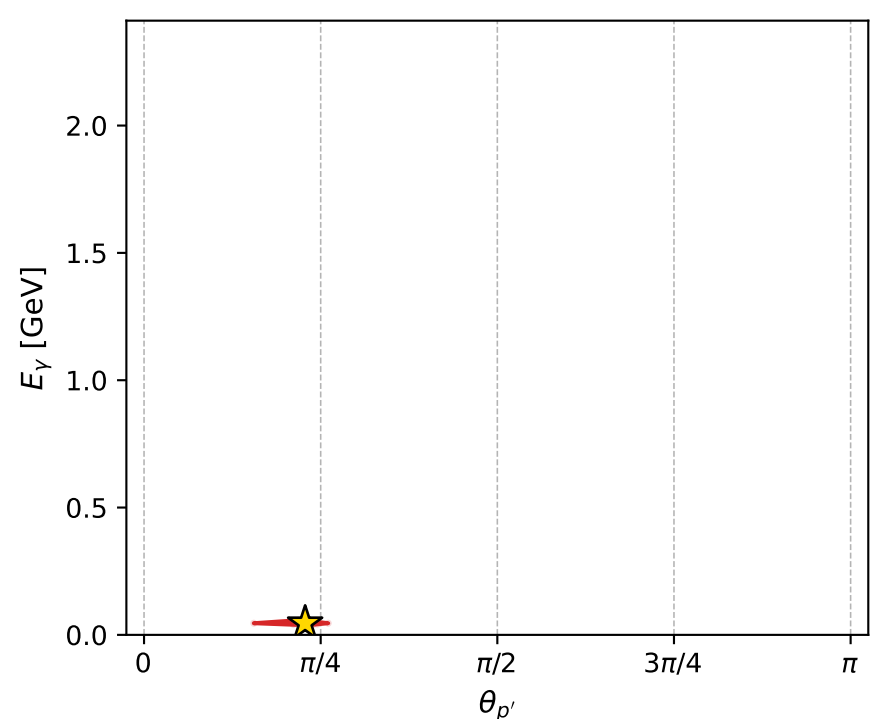
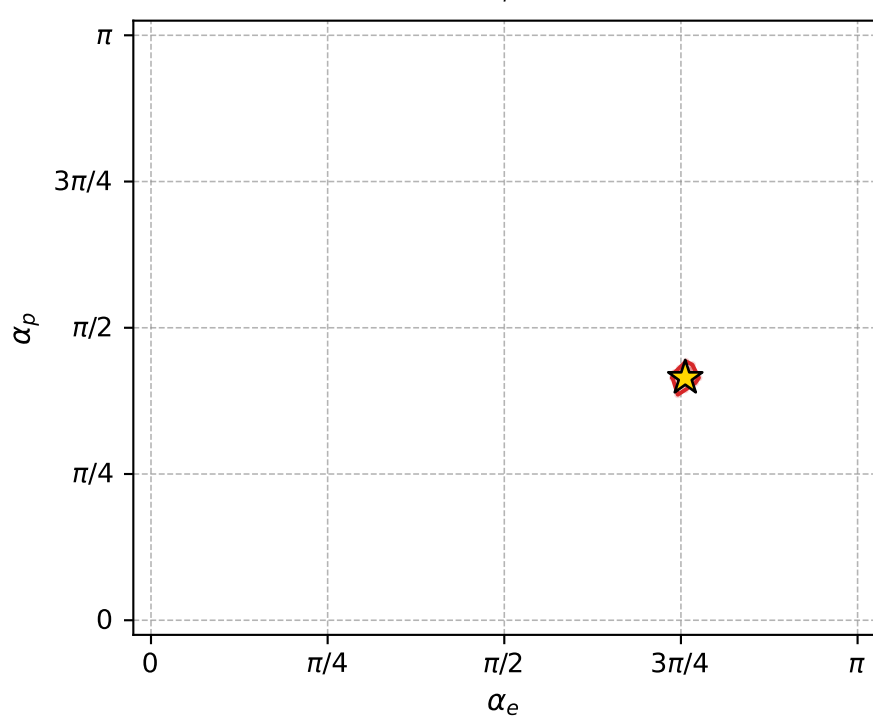
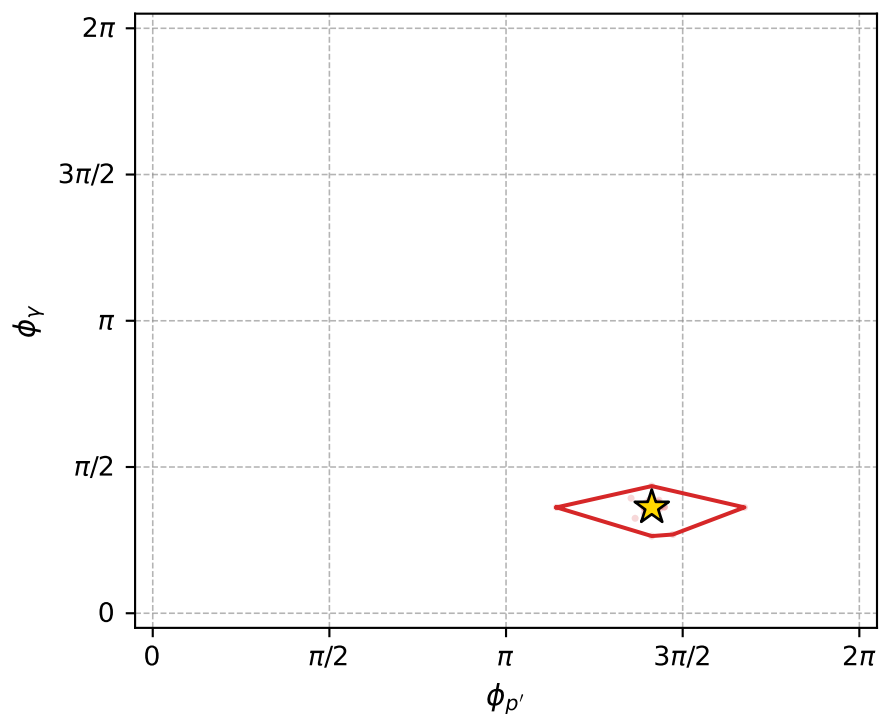
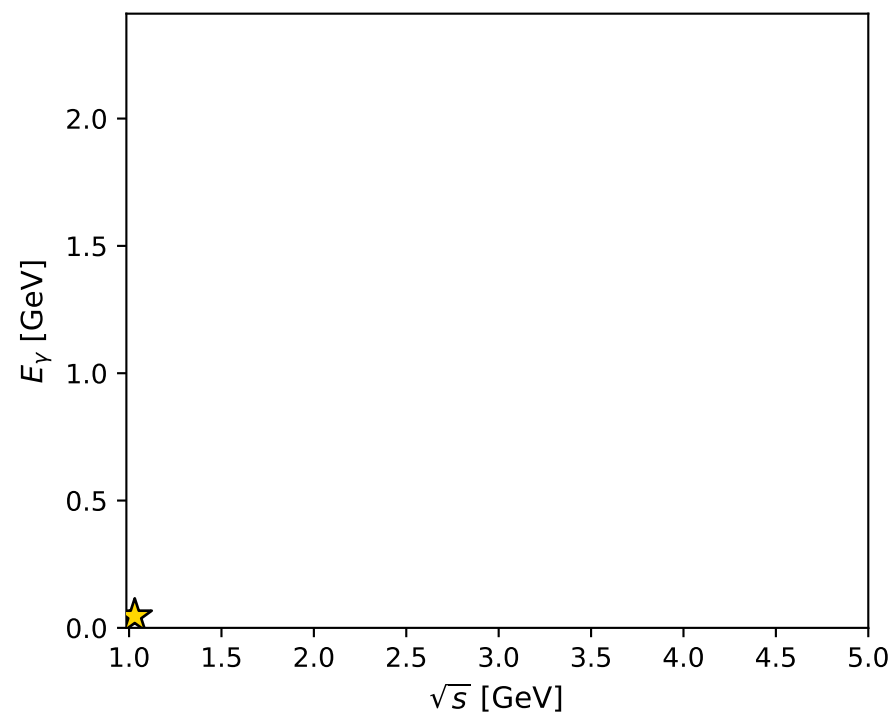
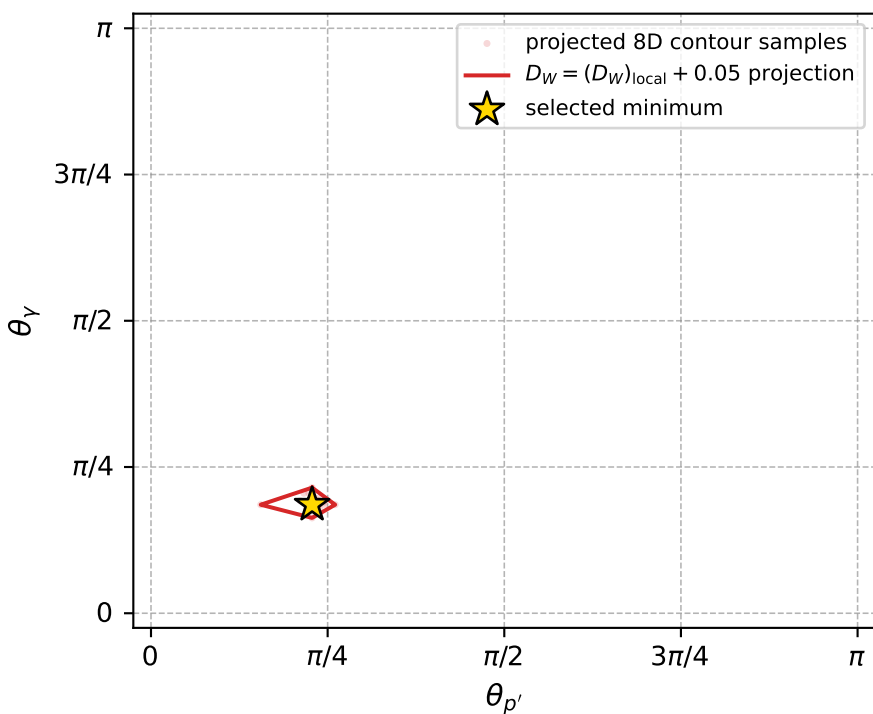
dw_mixing_angles_polarization_06_local_minimum_543 D_W=0.0031388
 region: polarization_cluster_06_local_minimum_543
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3754345$ rad
 incoming lepton: $-0.72058|+\rangle + 0.693372|-\rangle$
 proton mixing angle: $\alpha_p=1.3018382$ rad
 incoming proton: $0.265727|+\rangle + 0.964048|-\rangle$

$s=1.06231$, $\sqrt{s}=1.03068$, $|\vec{P}|=0.0885142$, $|\vec{P}'|=0.00359954$
 $\theta_{P'}=0.716259$, $\theta_Y=0.58298$, $\phi_{P'}=4.43737$, $\phi_Y=1.13719$
 $E_Y=0.0457816$, $\theta_{\ell'}=2.63205$, $\phi_{\ell'}=4.26246$
 $Q^2=0.00105464$, $x_B=0.012143$, $t=-0.00734979$, $W^2=0.965641$, $y=0.475997$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.0885$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0885$
 P $E= 0.9422$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0885$
 ℓ' $E= 0.0469$ $p_x= -0.0099$ $p_y= -0.0206$ $p_z= -0.0409$
 P' $E= 0.9380$ $p_x= -0.0006$ $p_y= -0.0023$ $p_z= 0.0027$
 q_Y $E= 0.0458$ $p_x= 0.0106$ $p_y= 0.0229$ $p_z= 0.0382$



electron: polarization cluster P6, local minimum 543; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0031388042$
 $D_W \text{ contour} = 0.053138804$
 above global minimum = 0.0031223713
 8D contour samples = 256

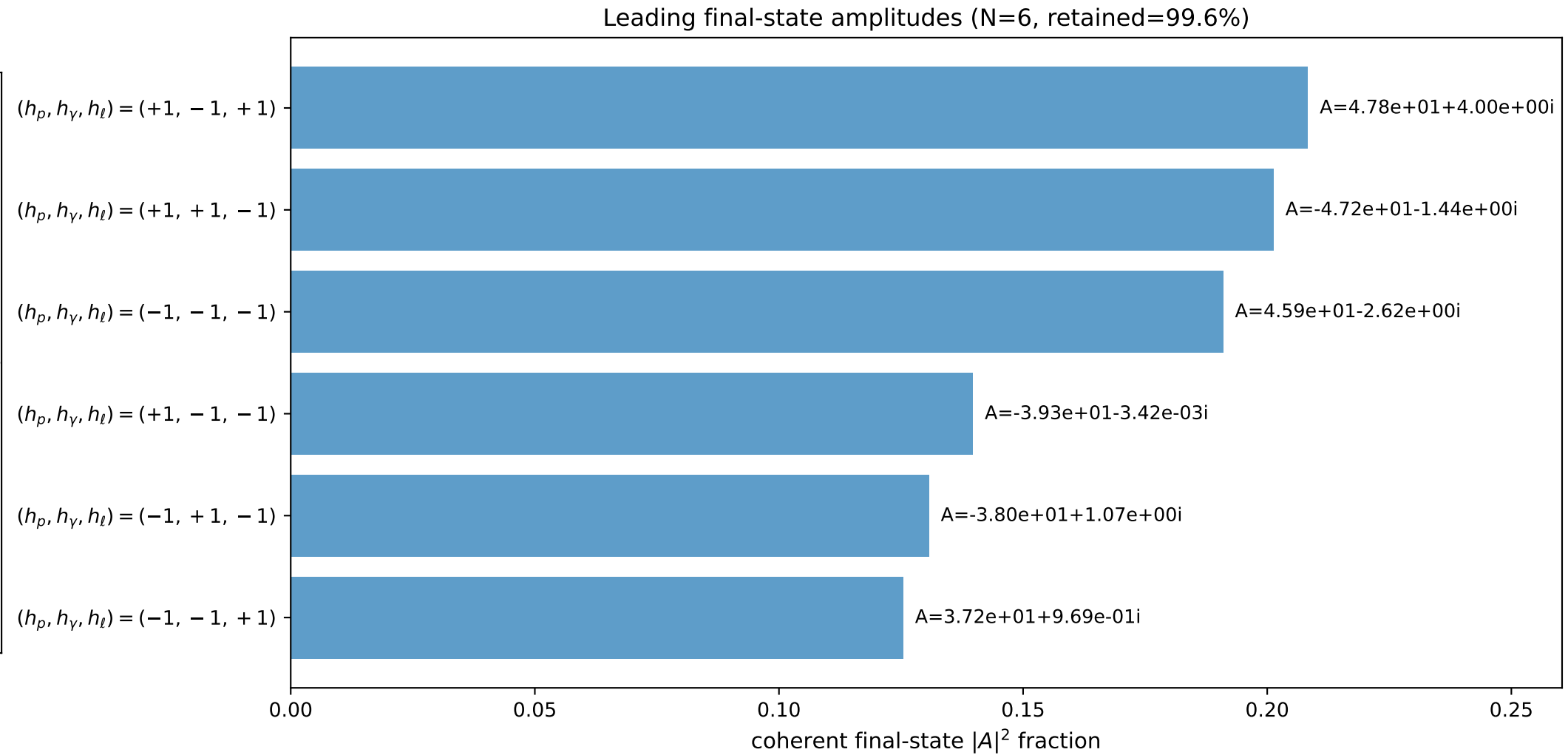
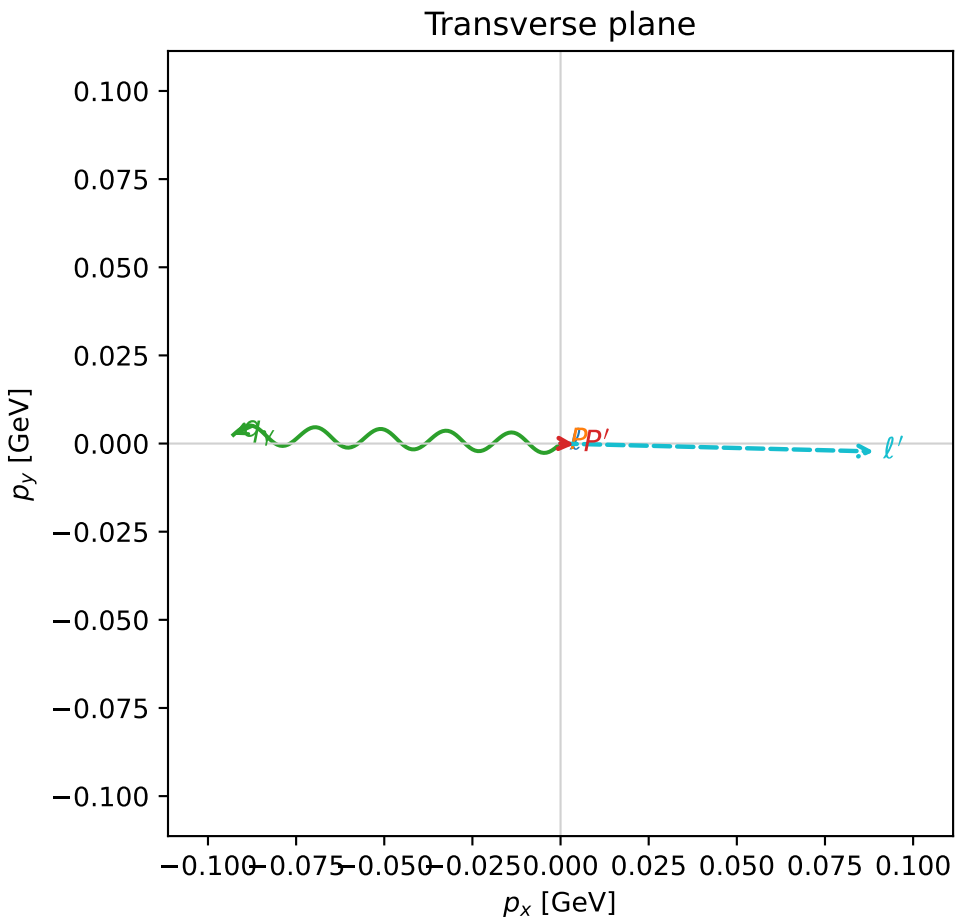
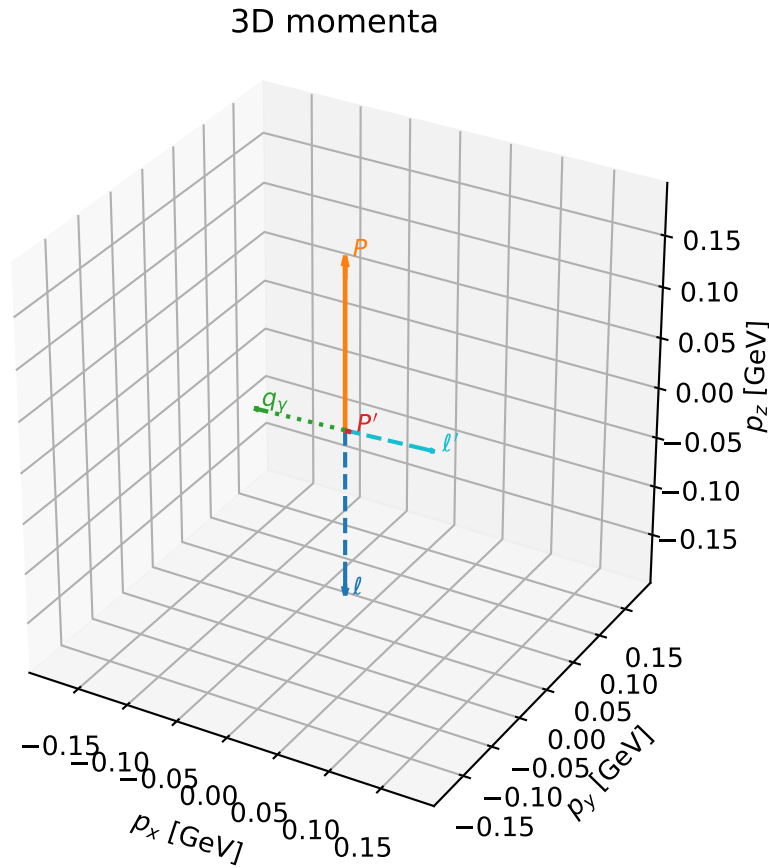
selected phase-space point:
 $\text{sqrt_s} = 1.030683$
 $\text{theta_p_out} = 0.716259$
 $\text{theta_gamma_out} = 0.5829805$
 $\text{qOut} = 0.0457816$
 $\text{phi_p_out} = 4.437368$
 $\text{phi_gamma_out} = 1.137189$
 $\text{alpha_e} = 2.375435$
 $\text{alpha_p} = 1.301838$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

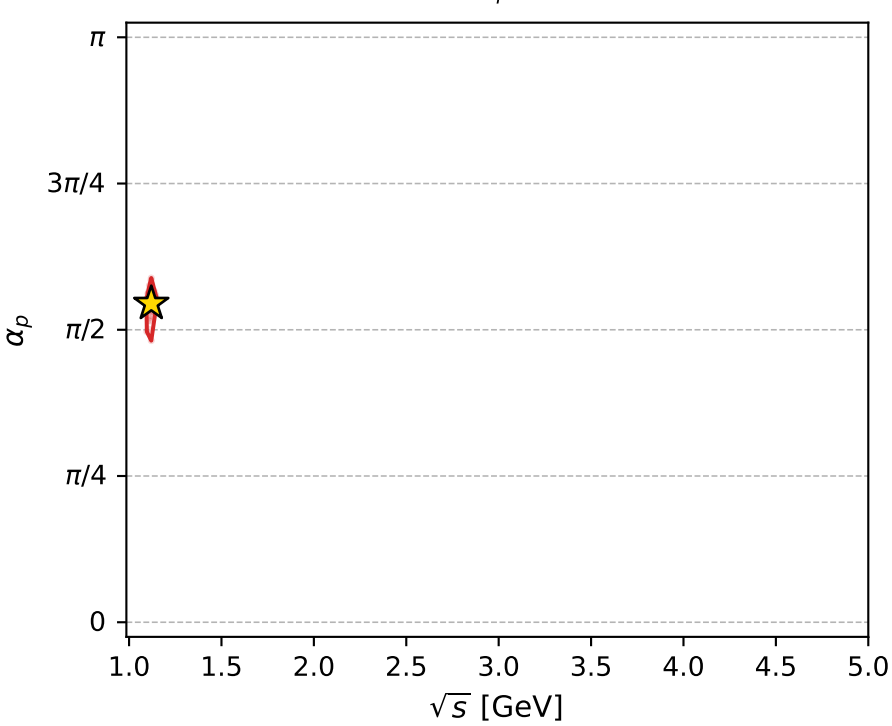
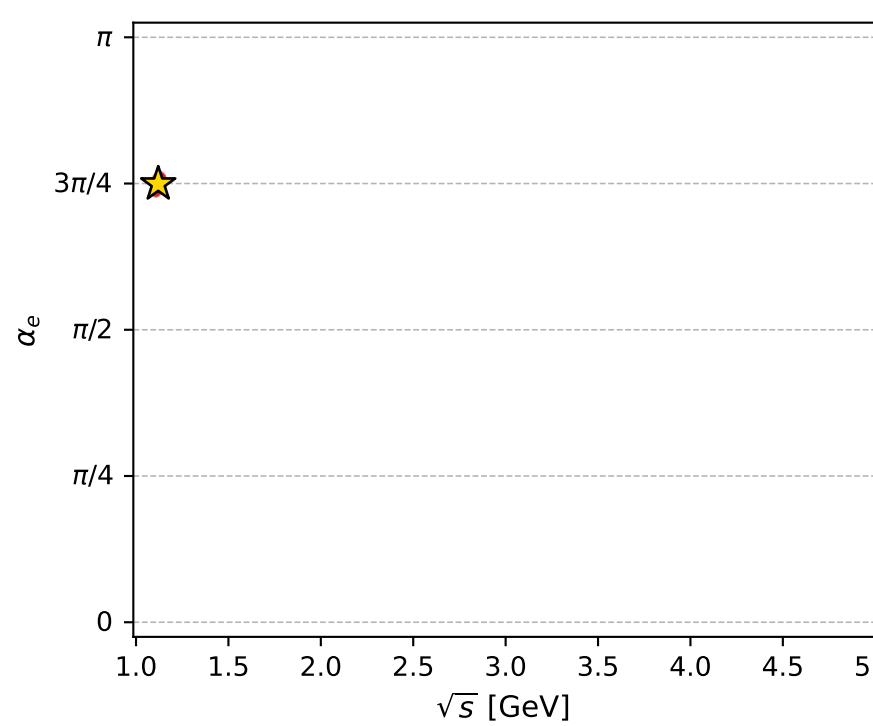
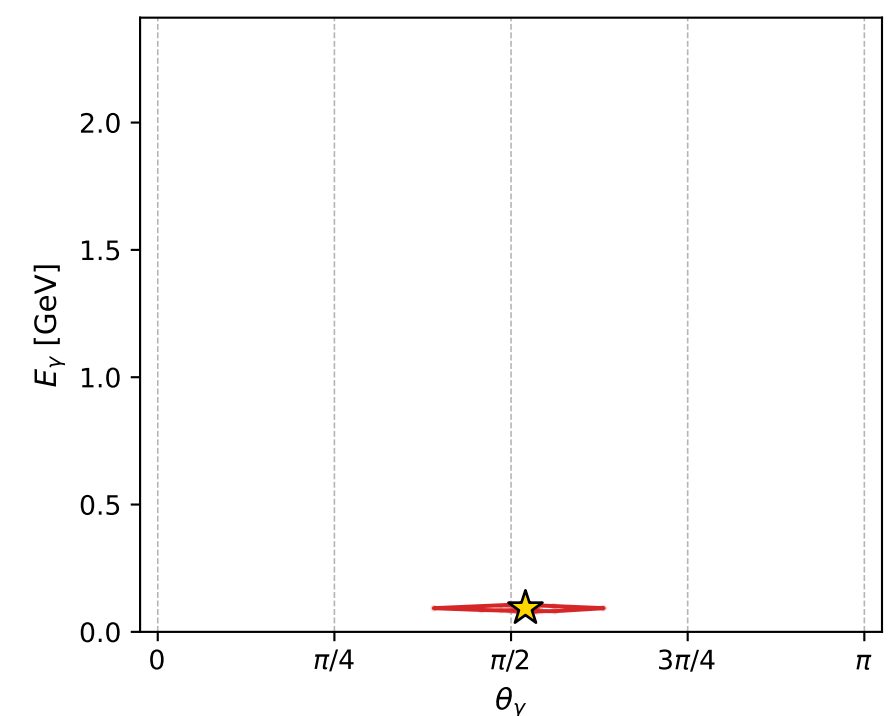
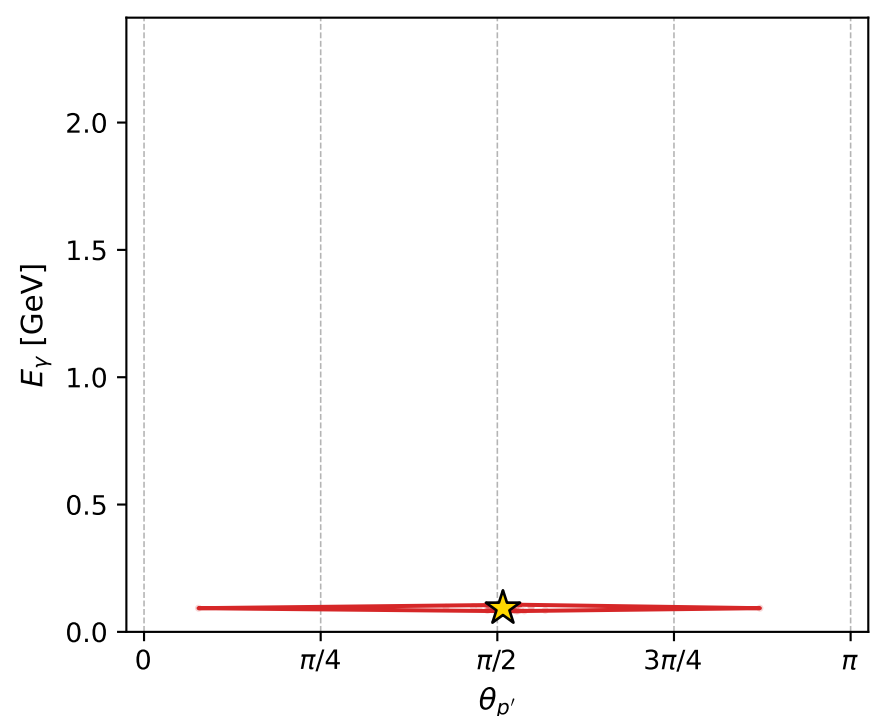
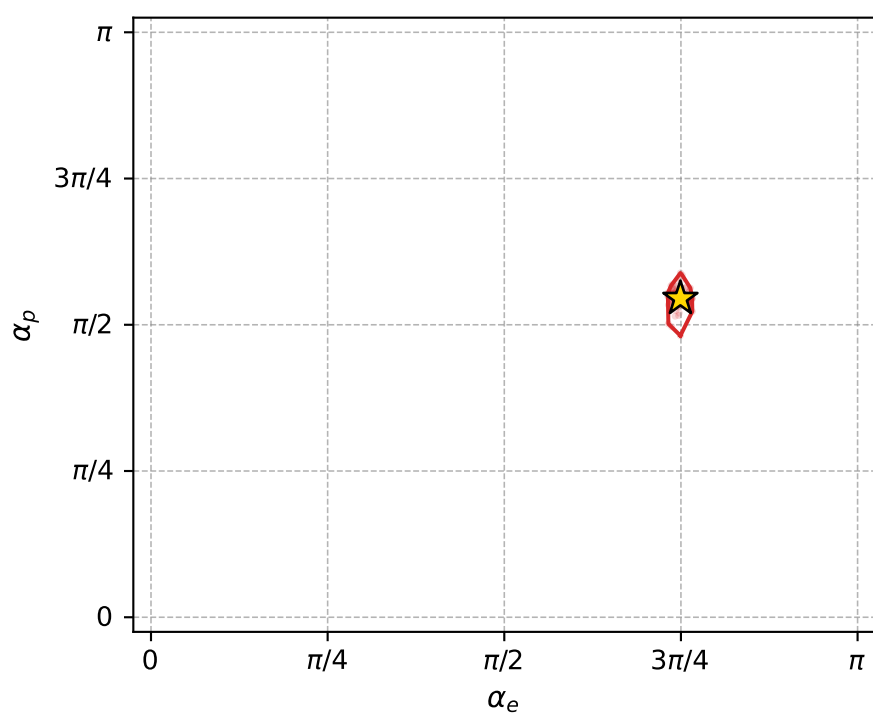
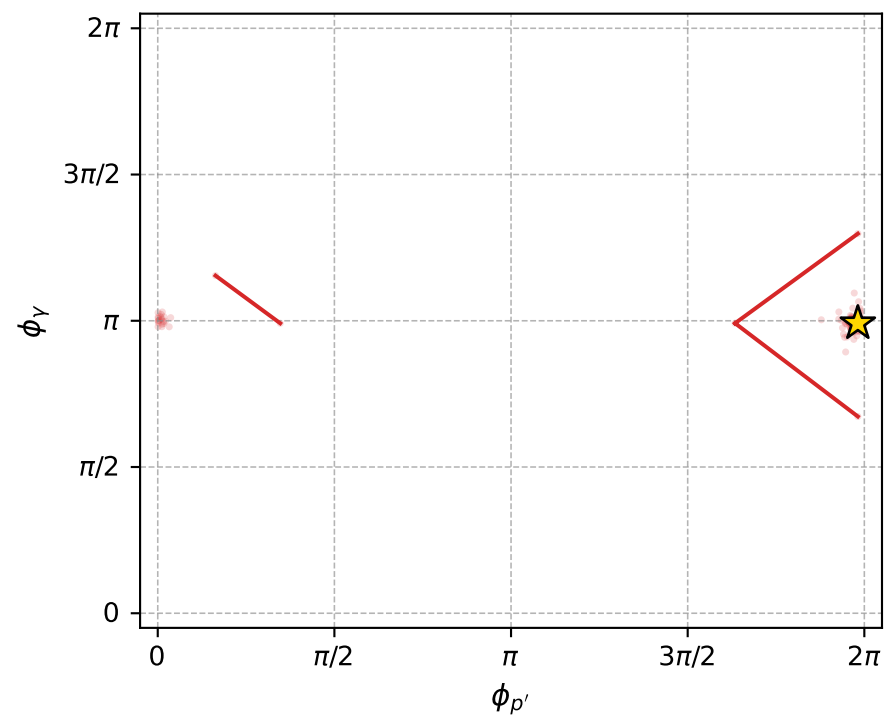
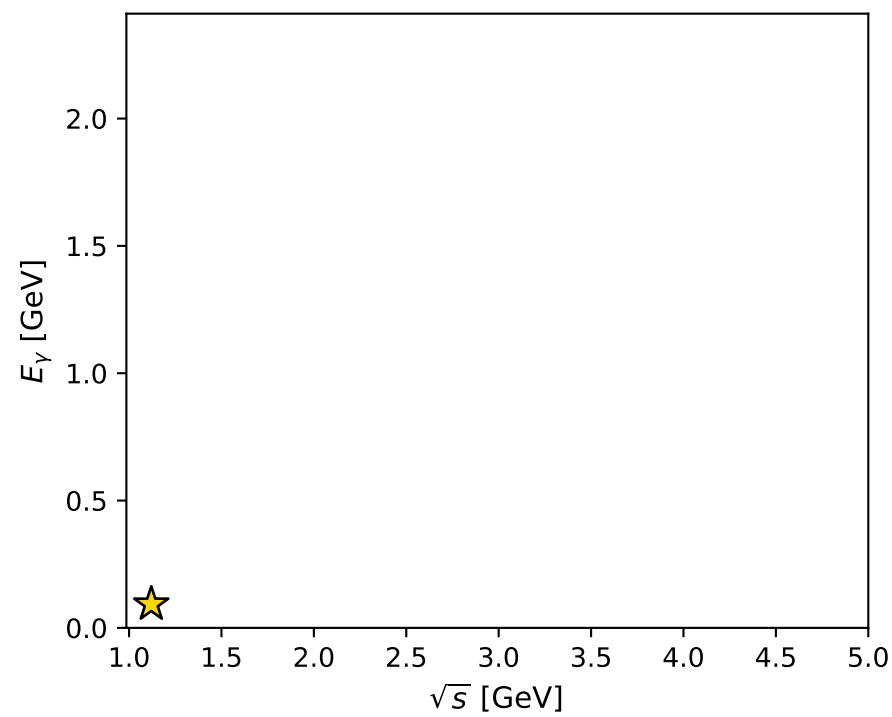
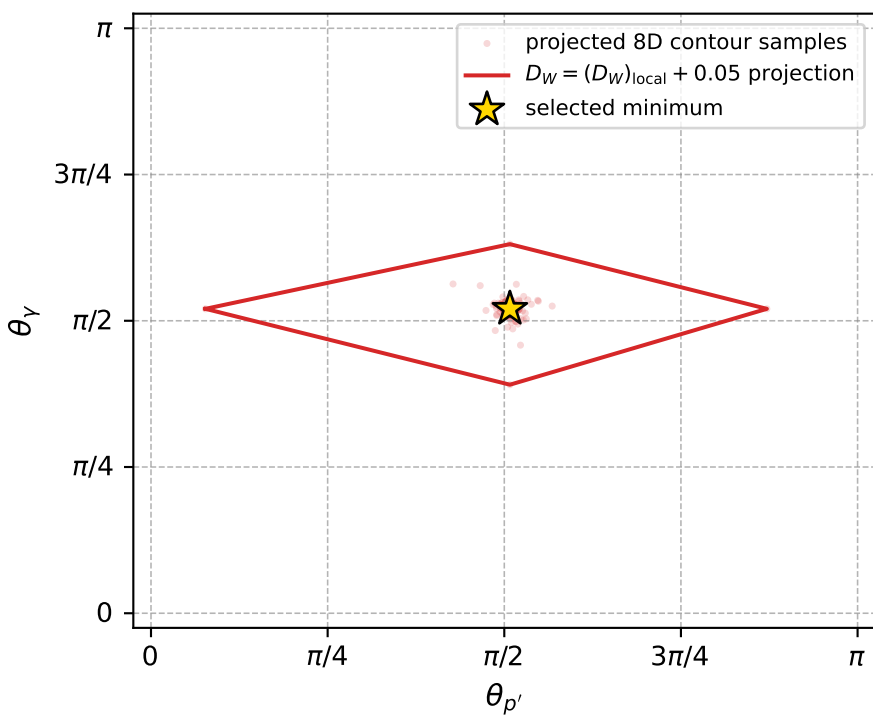
dw_mixing_angles_polarization_06_local_minimum_547 D_W=0.00316272
 region: polarization_cluster_06_local_minimum_547
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3538094$ rad
 incoming lepton: $-0.705418|+> +0.708791|->$
 proton mixing angle: $\alpha_p=1.7117874$ rad
 incoming proton: $-0.140524|+> +0.990077|->$

$s=1.2547$, $\sqrt{s}=1.12013$, $|\vec{P}|=0.167324$, $|\vec{P}'|=0.00390855$
 $\theta_{p'}=1.59575$, $\theta_Y=1.63488$, $\phi_{p'}=6.2256$, $\phi_Y=3.11492$
 $E_Y=0.0930066$, $\theta_{\ell'}=1.50281$, $\phi_{\ell'}=6.25787$
 $Q^2=0.0318487$, $x_B=0.153818$, $t=-0.0278264$, $W^2=1.05505$, $y=0.552363$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1673$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1673$
 P $E= 0.9528$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1673$
 ℓ' $E= 0.0891$ $p_x= 0.0889$ $p_y= -0.0023$ $p_z= 0.0061$
 P' $E= 0.9380$ $p_x= 0.0039$ $p_y= -0.0002$ $p_z= -0.0001$
 q_Y $E= 0.0930$ $p_x= -0.0928$ $p_y= 0.0025$ $p_z= -0.0060$



electron: polarization cluster P6, local minimum 547; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0031627241$
 $D_W \text{ contour} = 0.053162724$
 above global minimum = 0.0031462912
 8D contour samples = 256

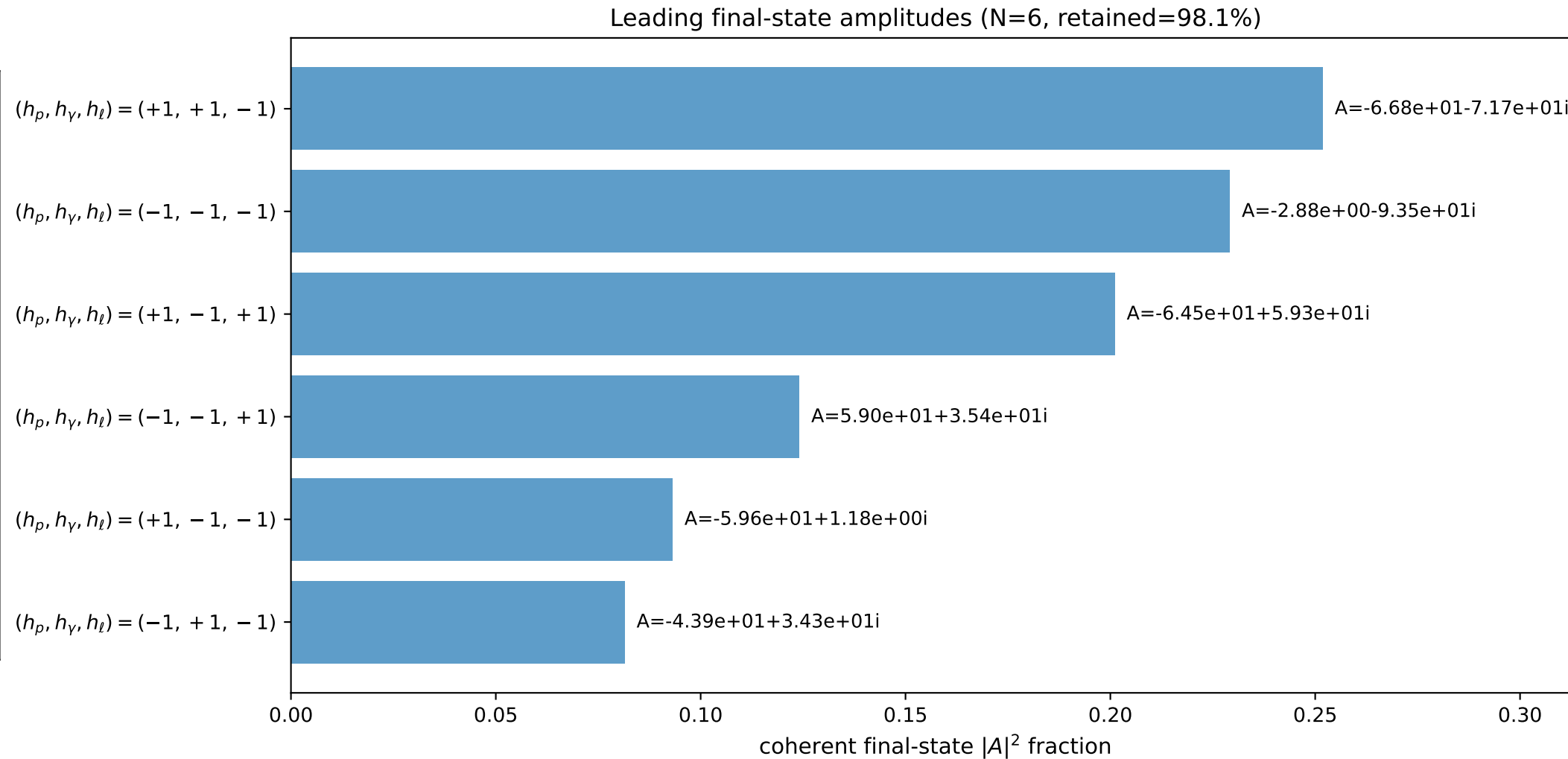
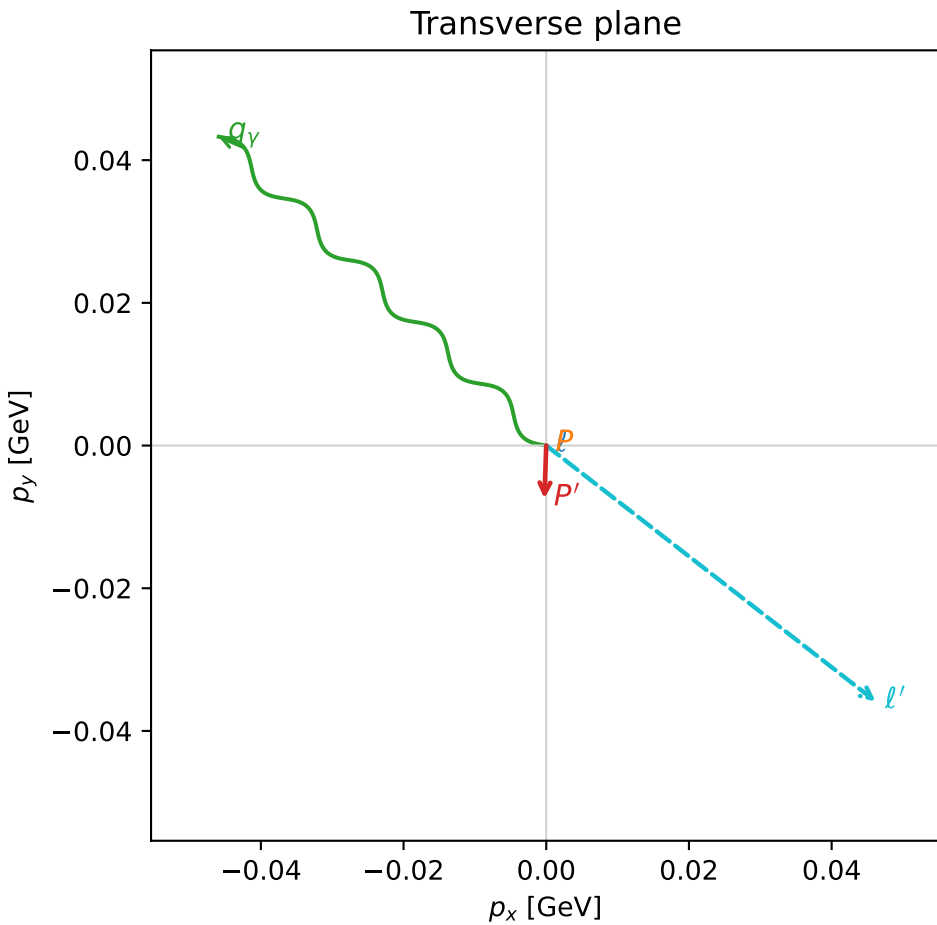
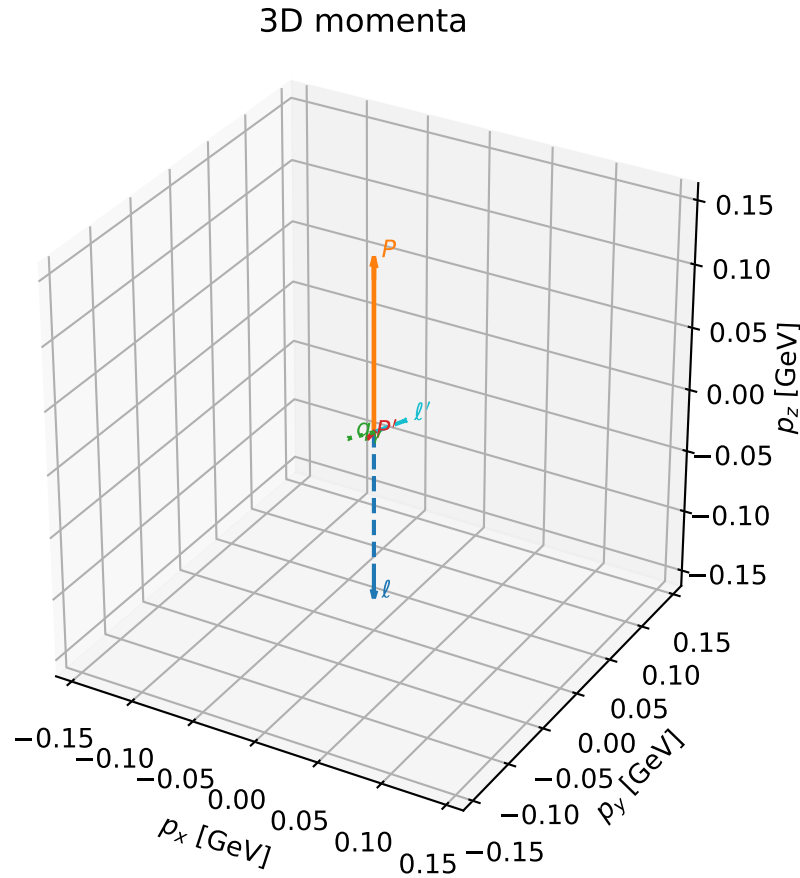
selected phase-space point:
 $\text{sqrt_s} = 1.120132$
 $\text{theta_p_out} = 1.595749$
 $\text{theta_gamma_out} = 1.634884$
 $\text{qOut} = 0.09300662$
 $\text{phi_p_out} = 6.225599$
 $\text{phi_gamma_out} = 3.114919$
 $\text{alpha_e} = 2.353809$
 $\text{alpha_p} = 1.711787$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

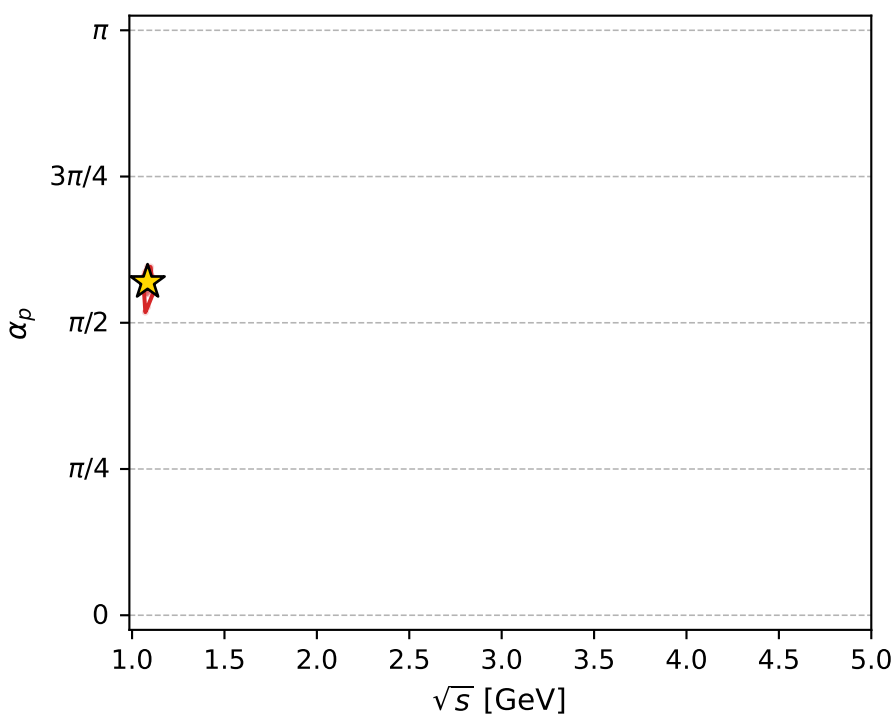
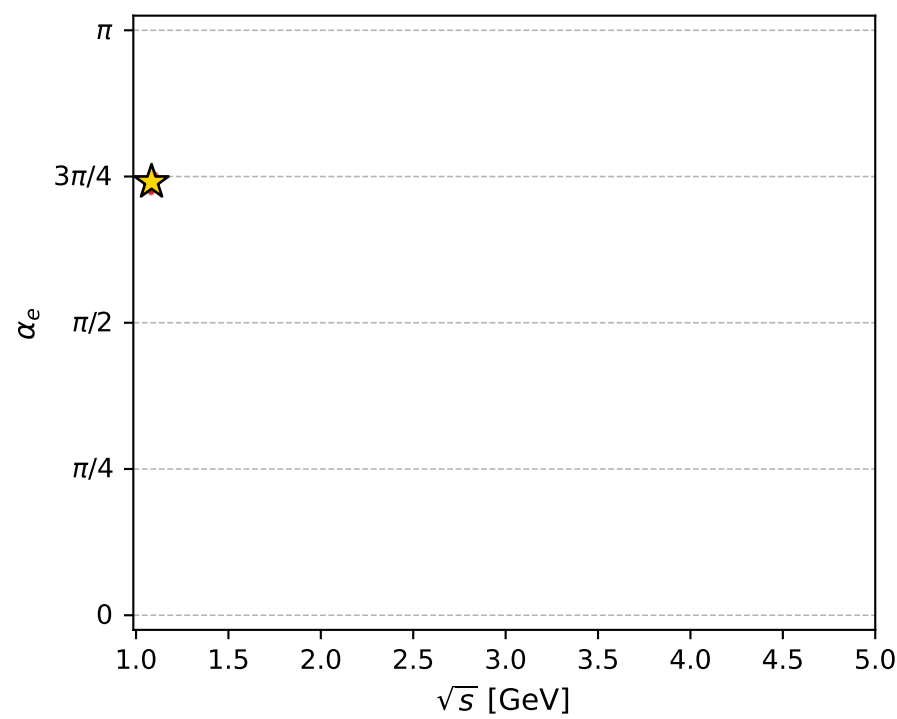
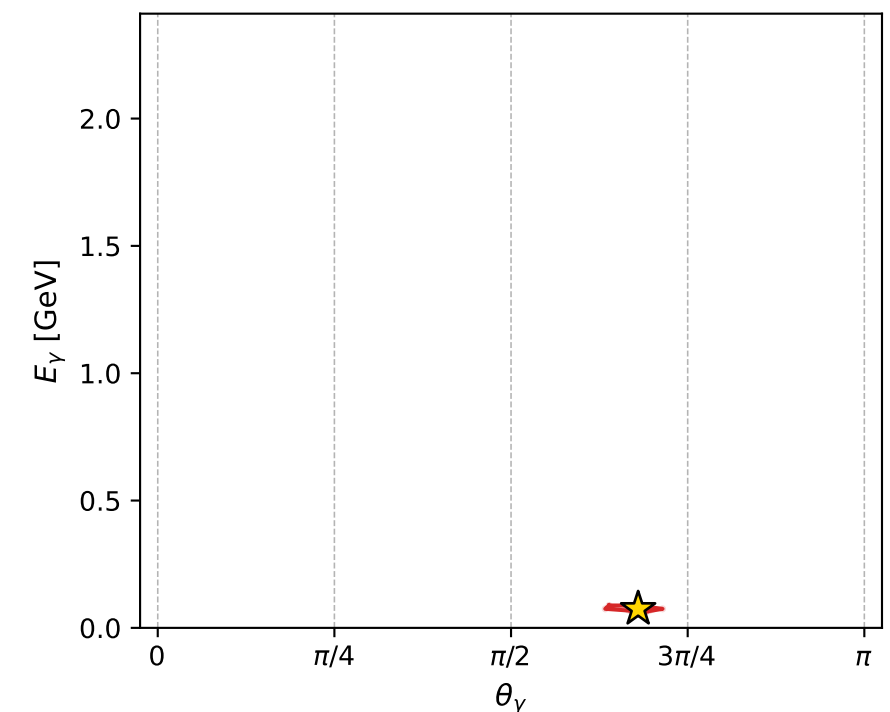
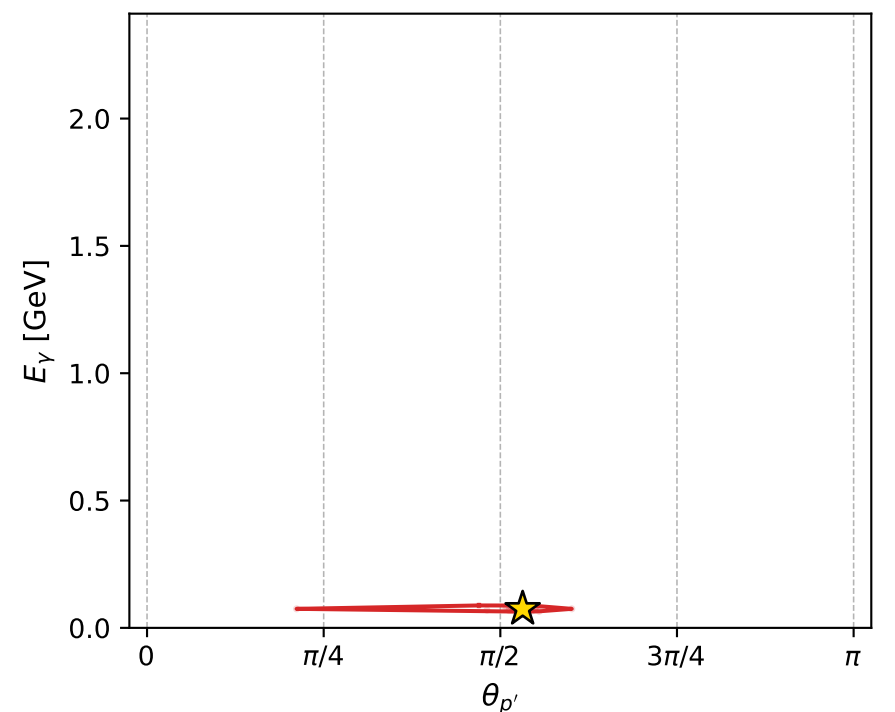
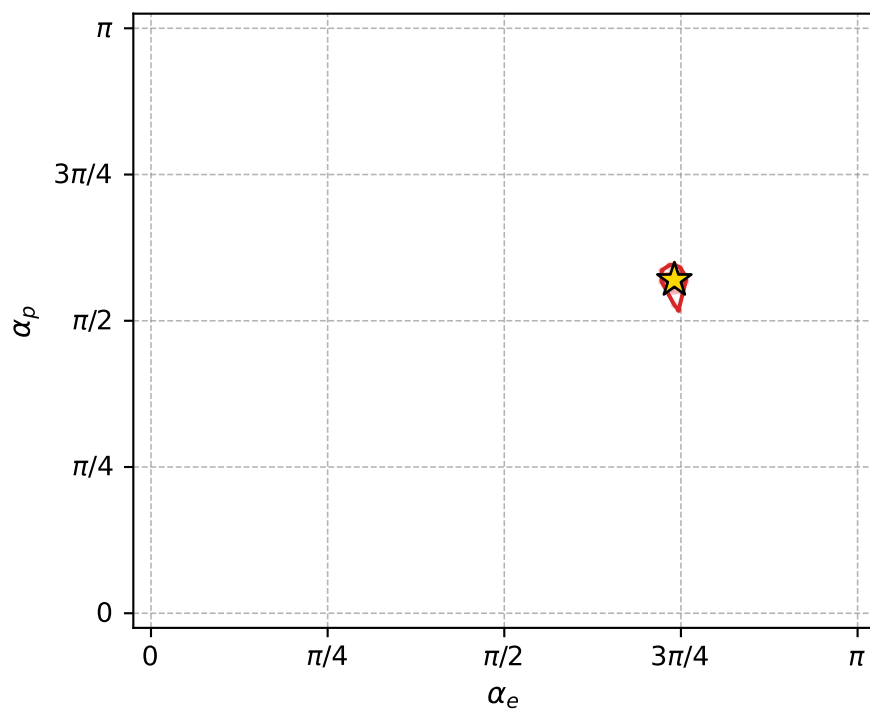
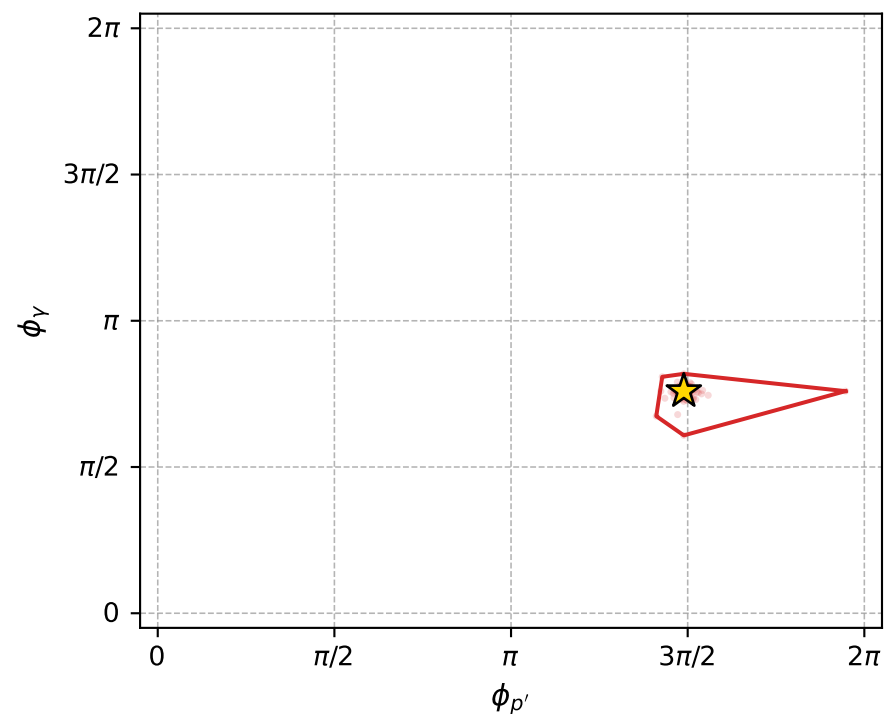
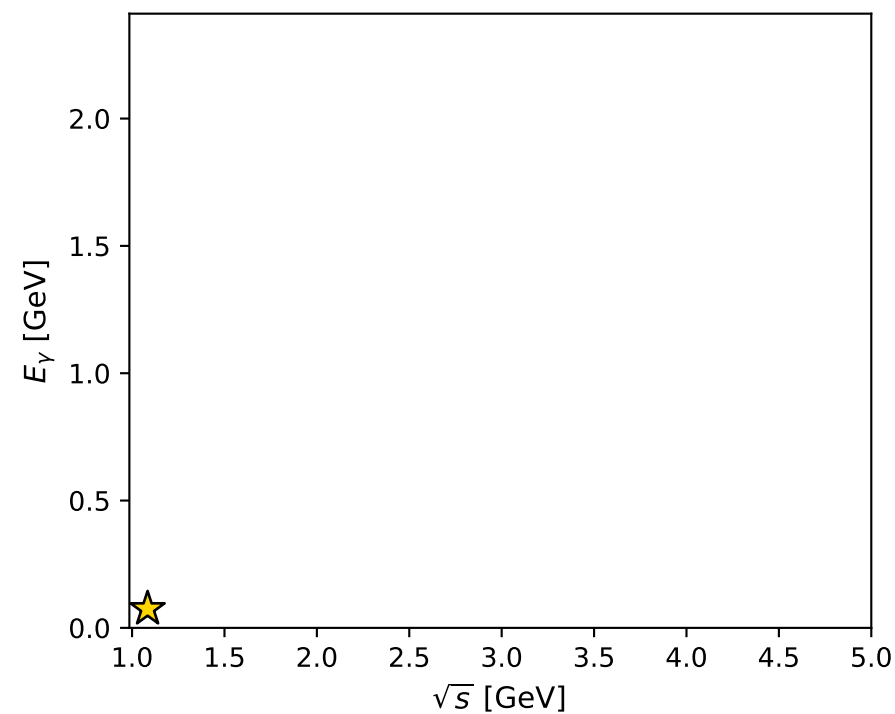
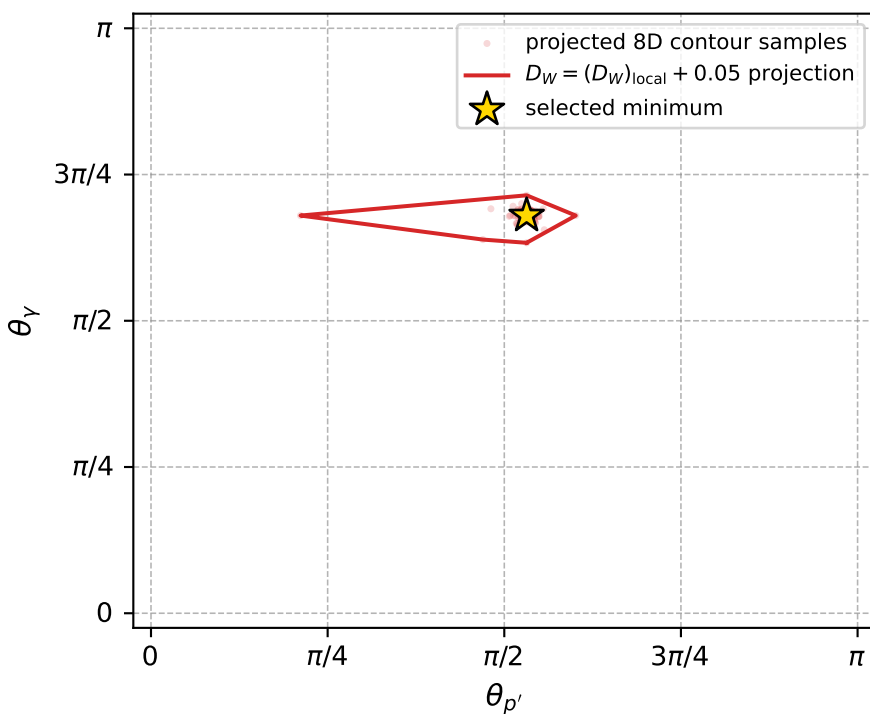
dw_mixing_angles_polarization_06_local_minimum_580 D_W=0.00369861
 region: polarization_cluster_06_local_minimum_580
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3269676$ rad
 incoming lepton: $-0.686141|+\rangle +0.727468|-\rangle$
 proton mixing angle: $\alpha_p=1.7897825$ rad
 incoming proton: $-0.21724|+\rangle +0.976118|-\rangle$

$s=1.17501$, $\sqrt{s}=1.08398$, $|\vec{P}|=0.136147$, $|\vec{P}'|=0.00747269$
 $\theta_{P'}=1.67007$, $\theta_V=2.13584$, $\phi_{P'}=4.678$, $\phi_V=2.38562$
 $E_V=0.0747017$, $\theta_{\ell'}=0.962046$, $\phi_{\ell'}=5.62287$
 $Q^2=0.0304925$, $x_B=0.178113$, $t=-0.0186974$, $W^2=1.02055$, $y=0.580014$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1361$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1361$
 P $E= 0.9478$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1361$
 ℓ' $E= 0.0712$ $p_x= 0.0462$ $p_y= -0.0358$ $p_z= 0.0407$
 P' $E= 0.9380$ $p_x= -0.0003$ $p_y= -0.0074$ $p_z= -0.0007$
 q_Y $E= 0.0747$ $p_x= -0.0459$ $p_y= 0.0433$ $p_z= -0.0400$



electron: polarization cluster P6, local minimum 580; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0036986094$
 $D_W \text{ contour} = 0.053698609$
 above global minimum = 0.0036821765
 8D contour samples = 256

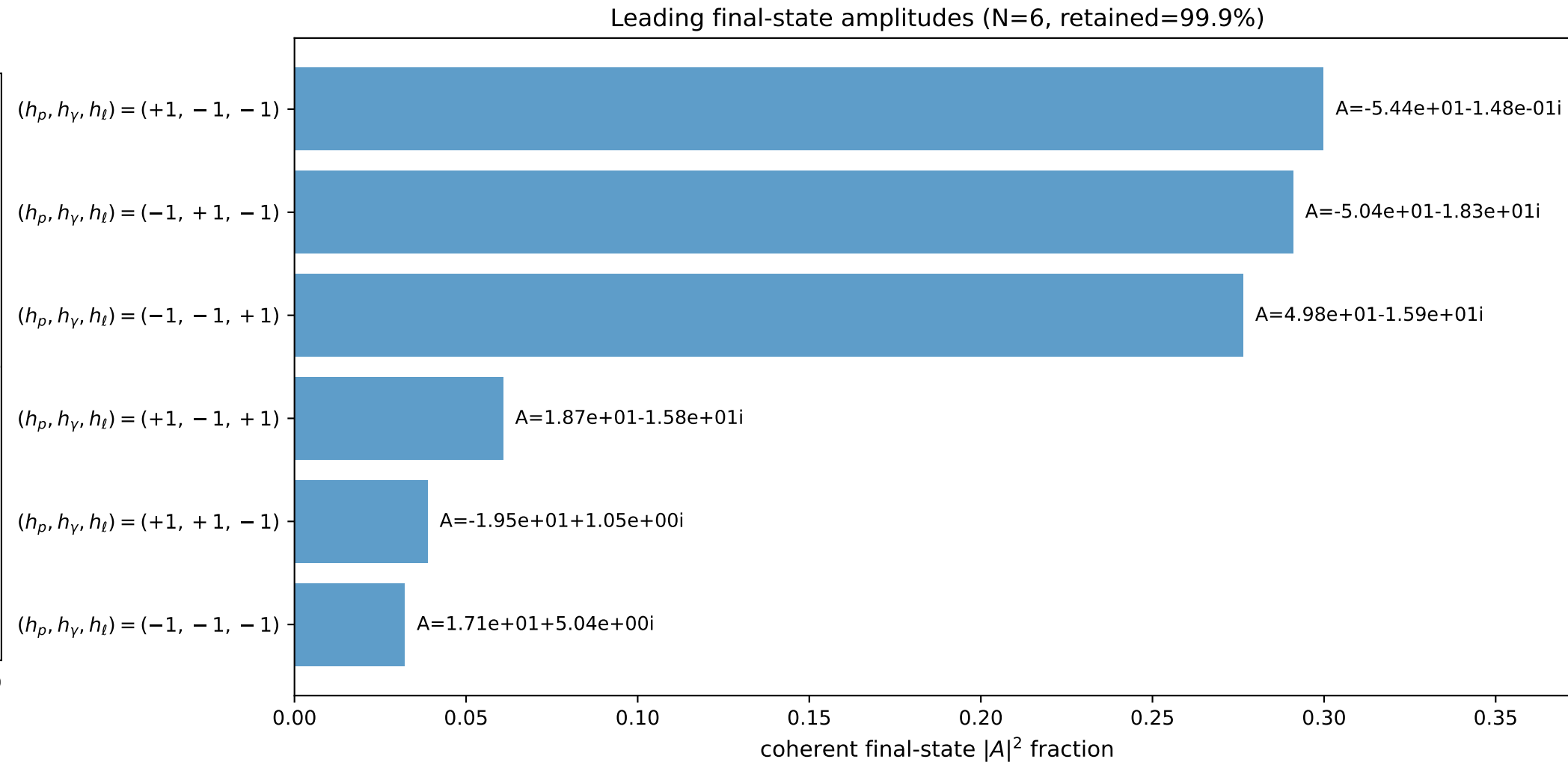
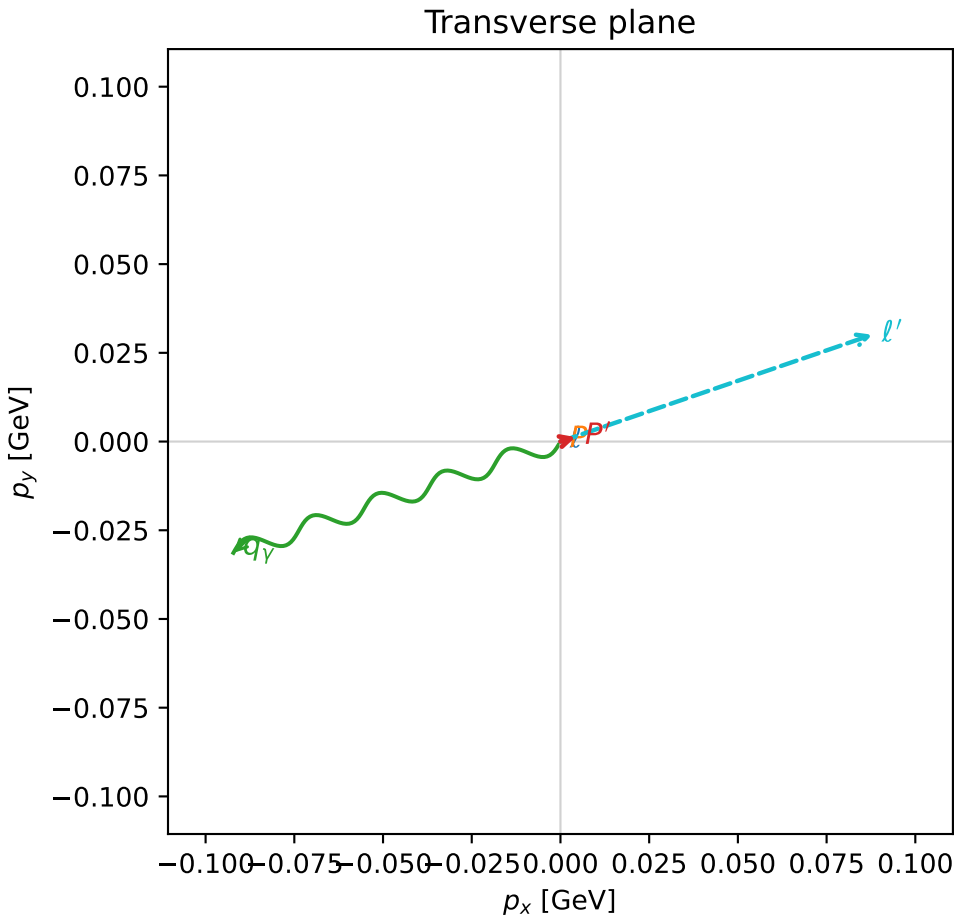
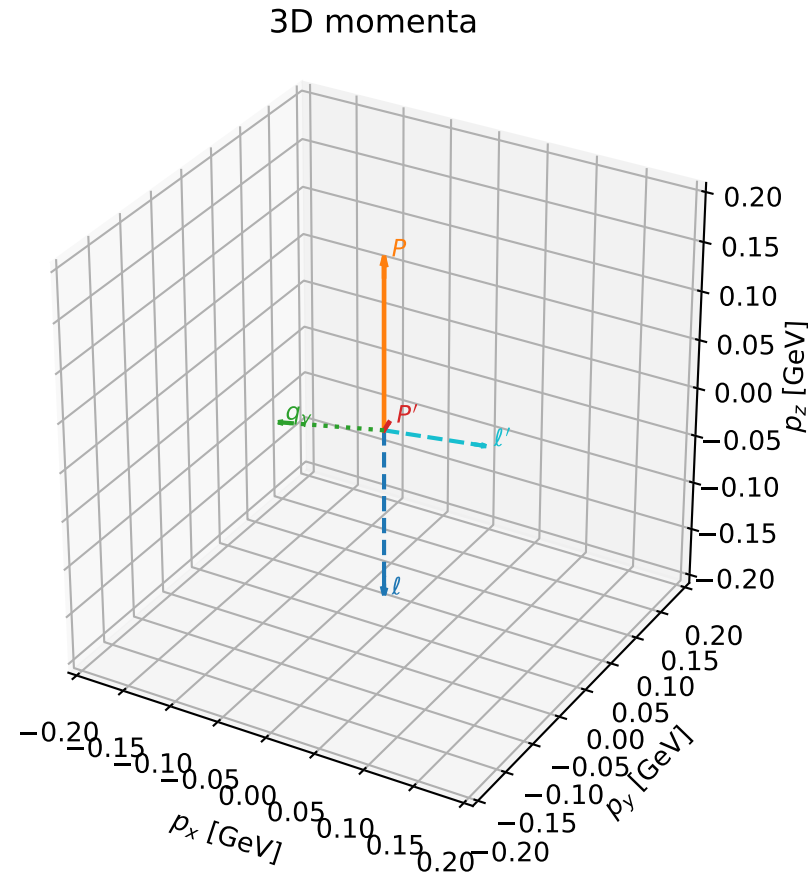
selected phase-space point:
 $\text{sqrt}_s = 1.083977$
 $\text{theta}_p_{\text{out}} = 1.670073$
 $\text{theta}_\gamma_{\text{out}} = 2.135843$
 $q_{\text{out}} = 0.07470168$
 $\text{phi}_p_{\text{out}} = 4.677997$
 $\text{phi}_\gamma_{\text{out}} = 2.385615$
 $\alpha_e = 2.326968$
 $\alpha_p = 1.789783$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

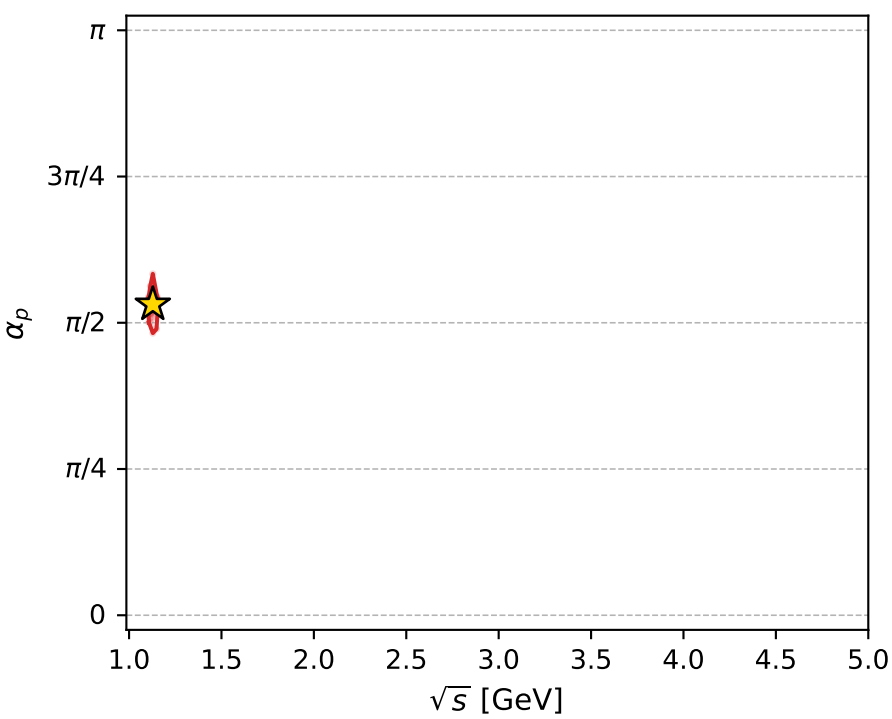
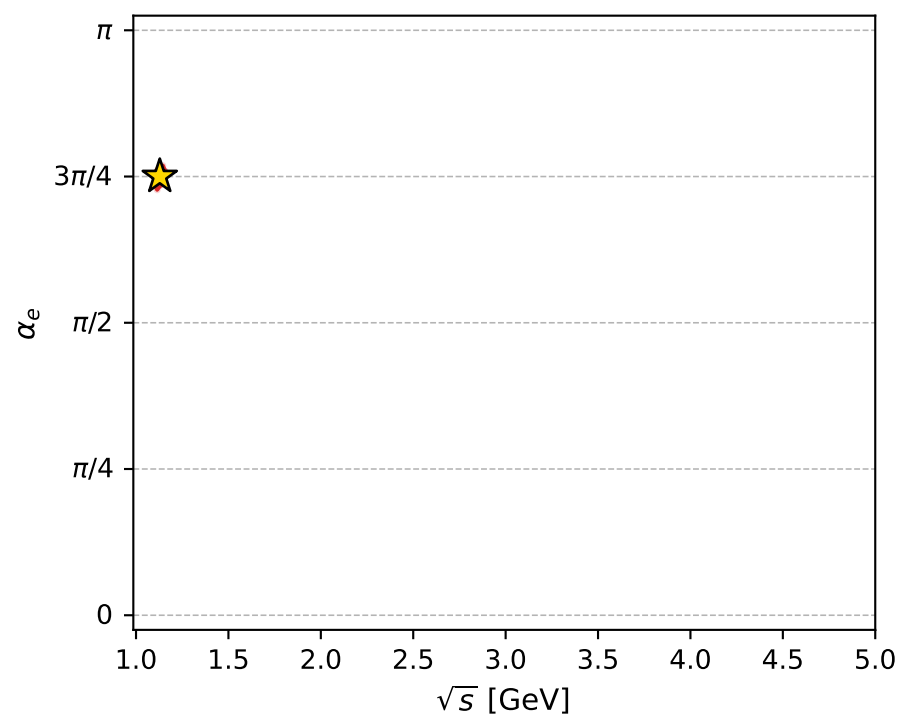
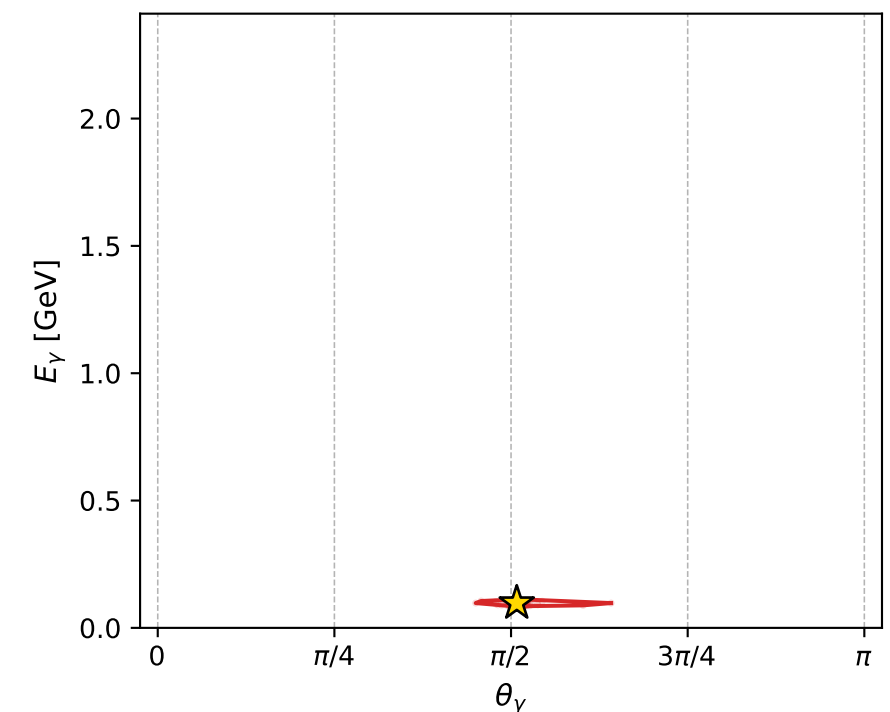
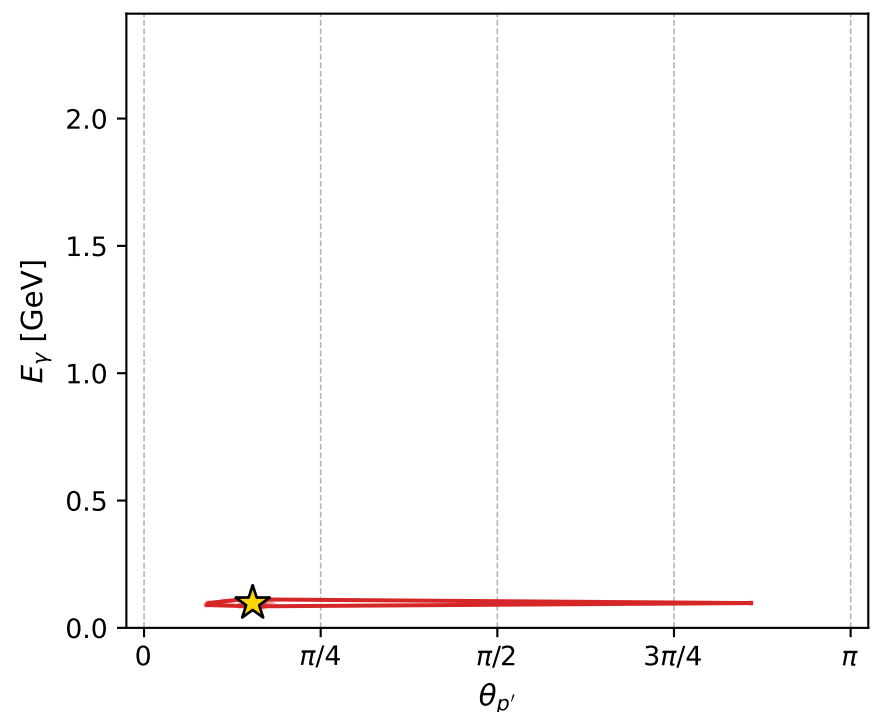
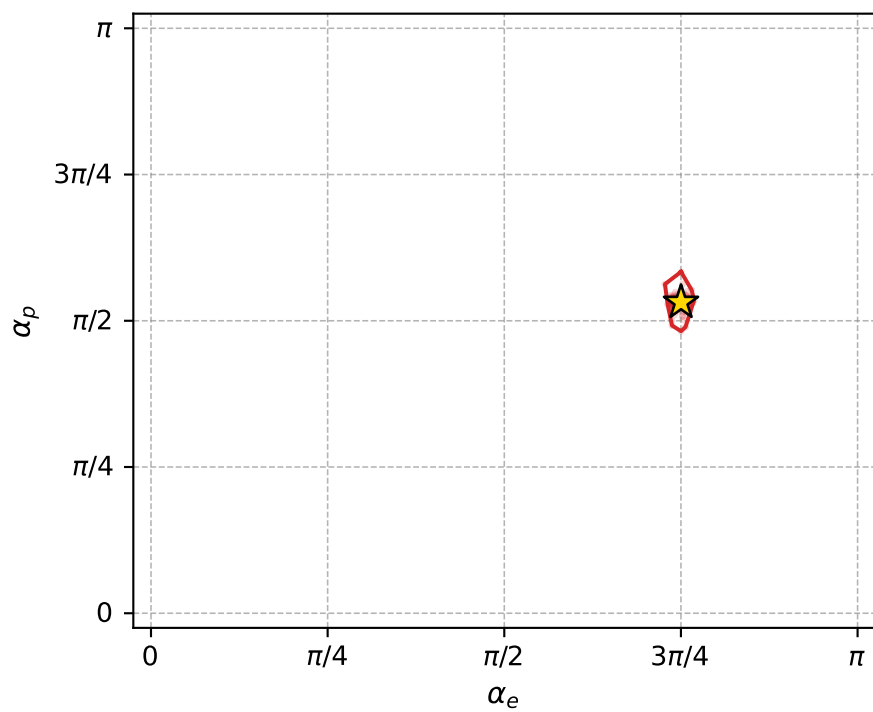
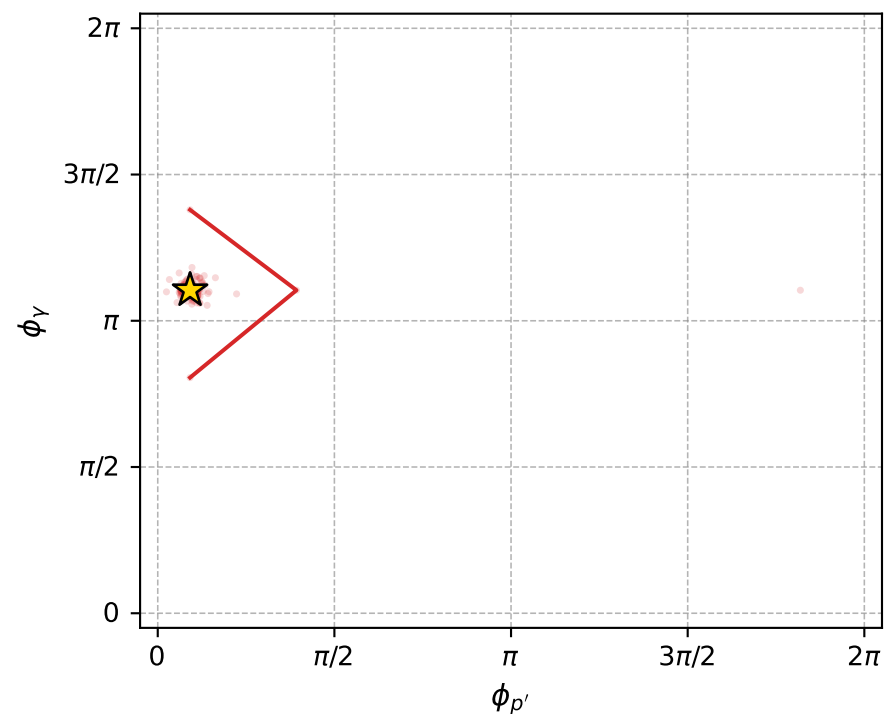
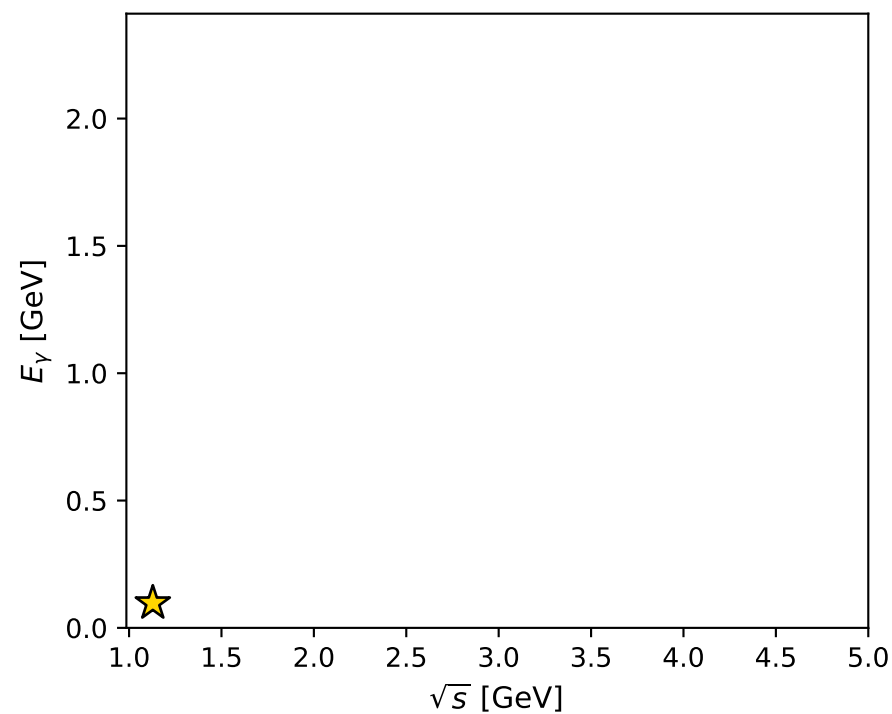
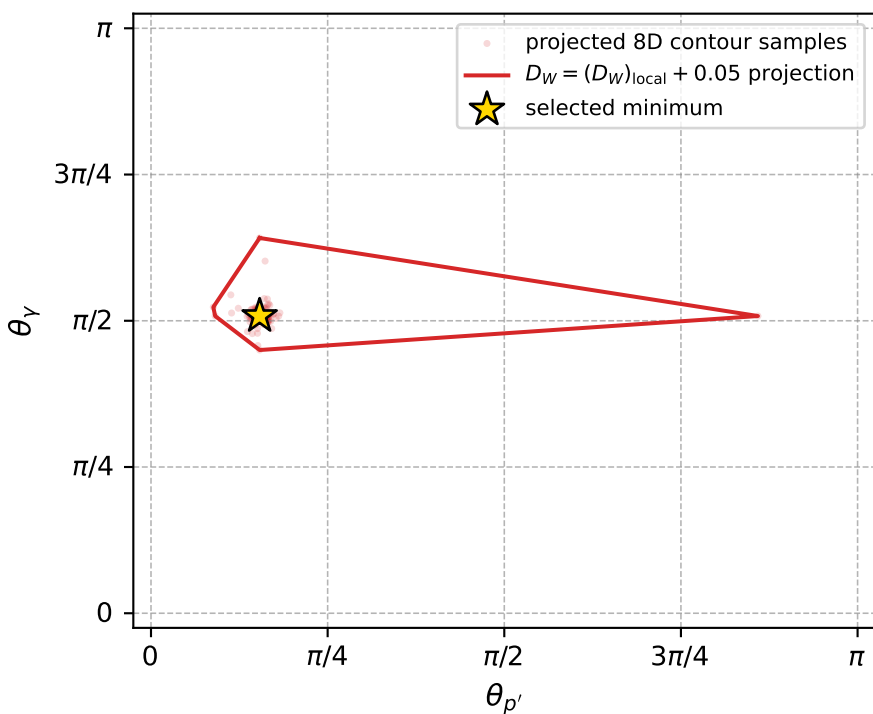
dw_mixing_angles_polarization_06_local_minimum_610 D_W=0.0043094
 region: polarization_cluster_06_local_minimum_610
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3566246$ rad
 incoming lepton: $-0.707411|+> +0.706803|->$
 proton mixing angle: $\alpha_p=1.6695932$ rad
 incoming proton: $-0.0986363|+> +0.995124|->$

$s=1.2736$, $\sqrt{s}=1.12854$, $|\vec{P}|=0.174454$, $|\vec{P}'|=0.00957612$
 $\theta_{p'}=0.48302$, $\theta_Y=1.59625$, $\phi_{p'}=0.287288$, $\phi_Y=3.4696$
 $E_Y=0.0973858$, $\theta_{\ell'}=1.6353$, $\phi_{\ell'}=0.329959$
 $Q^2=0.0303909$, $x_B=0.142011$, $t=-0.0273099$, $W^2=1.06346$, $y=0.543489$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1745$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1745$
 P $E= 0.9541$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1745$
 ℓ' $E= 0.0931$ $p_x= 0.0879$ $p_y= 0.0301$ $p_z= -0.0060$
 P' $E= 0.9380$ $p_x= 0.0043$ $p_y= 0.0013$ $p_z= 0.0085$
 q_Y $E= 0.0974$ $p_x= -0.0922$ $p_y= -0.0314$ $p_z= -0.0025$



electron: polarization cluster P6, local minimum 610; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0043093982$
 $D_W \text{ contour} = 0.054309398$
 above global minimum = 0.0042929654
 8D contour samples = 256

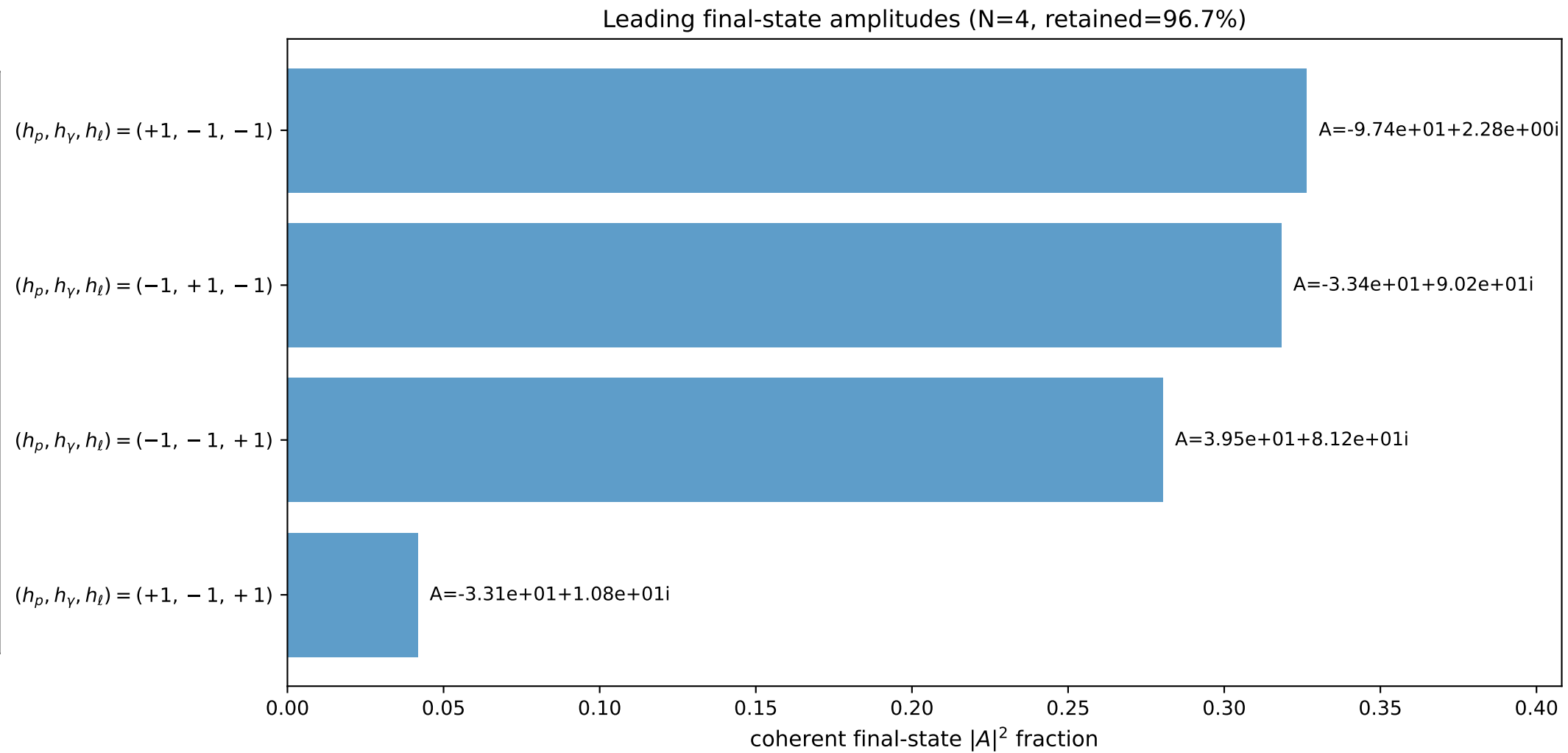
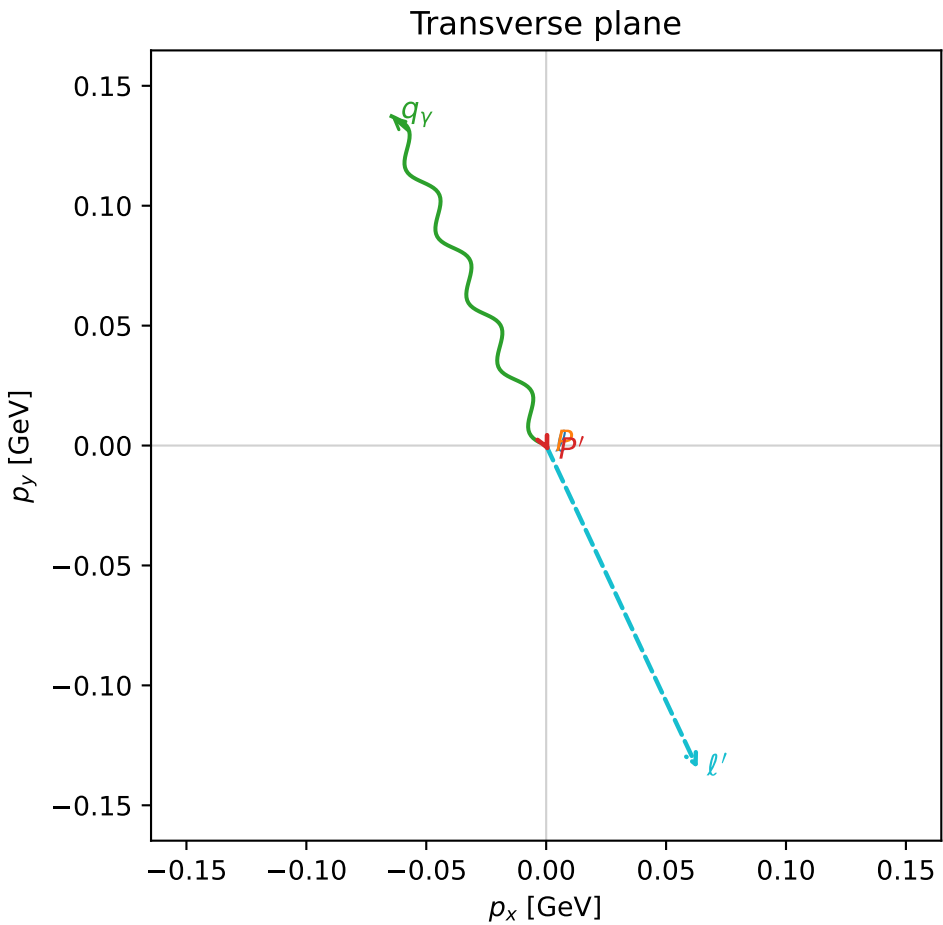
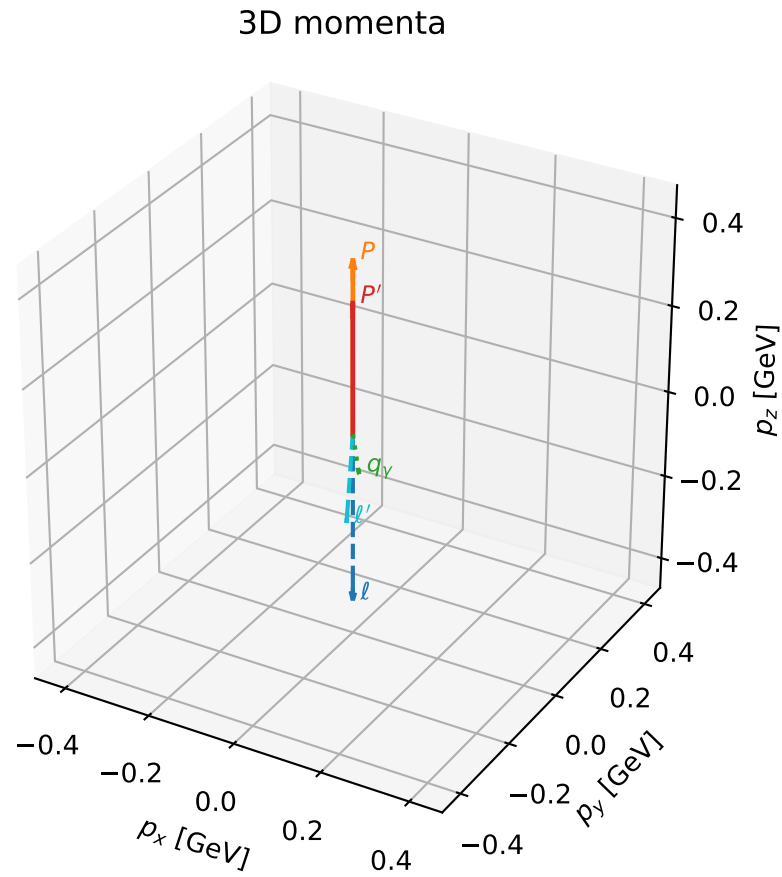
selected phase-space point:
 $\text{sqrt_s} = 1.12854$
 $\text{theta_p_out} = 0.4830201$
 $\text{theta_gamma_out} = 1.596253$
 $\text{q0ut} = 0.09738577$
 $\text{phi_p_out} = 0.2872884$
 $\text{phi_gamma_out} = 3.469603$
 $\text{alpha_e} = 2.356625$
 $\text{alpha_p} = 1.669593$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

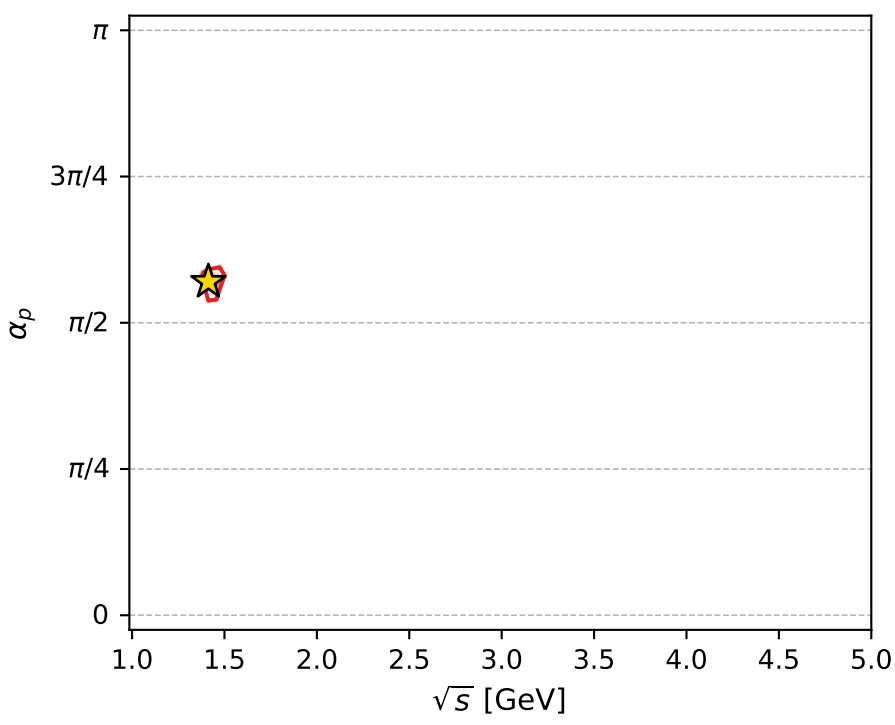
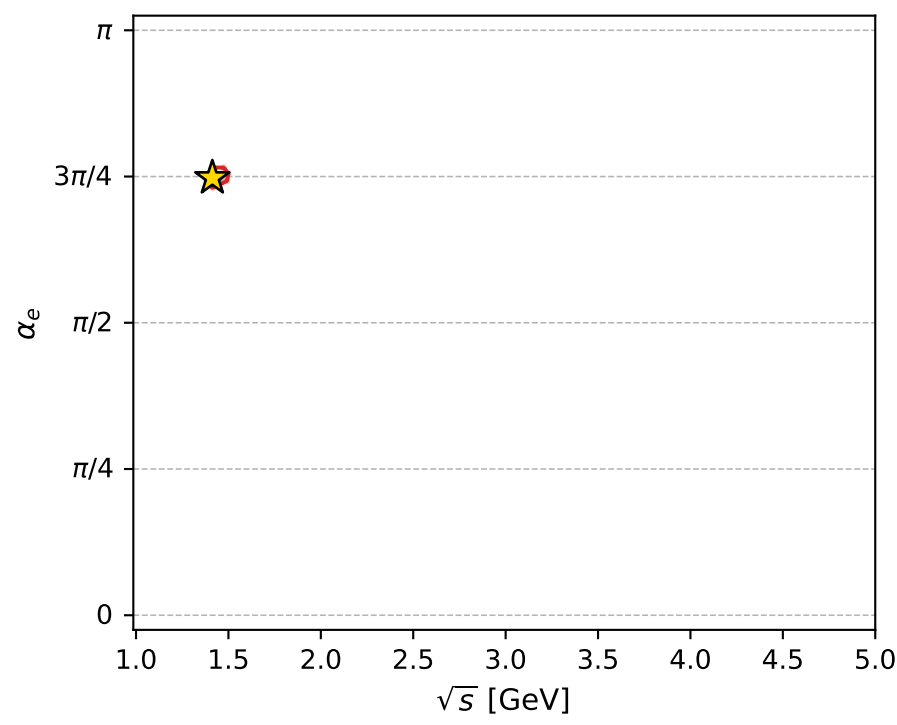
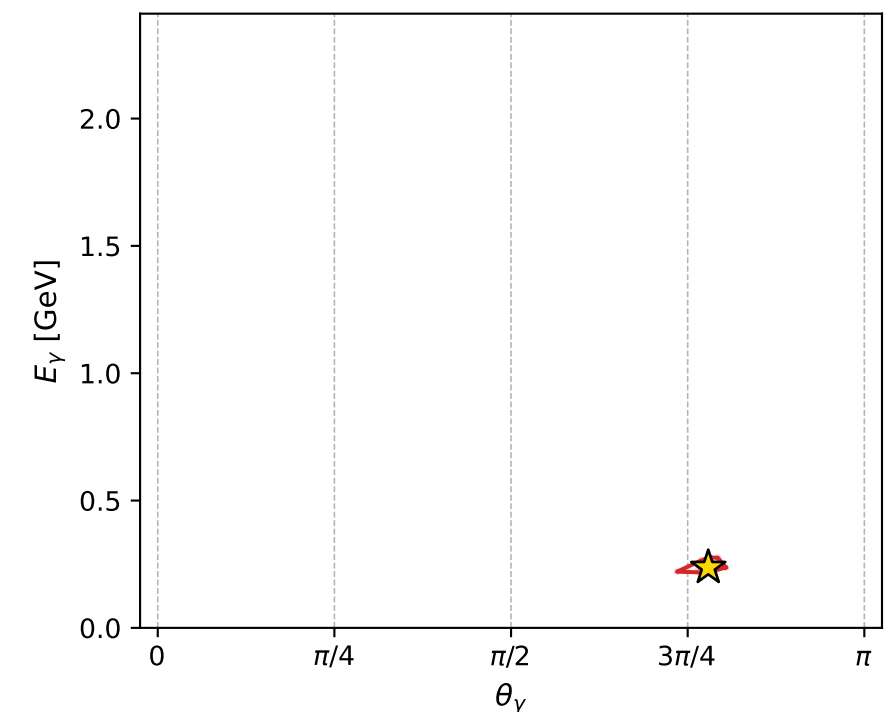
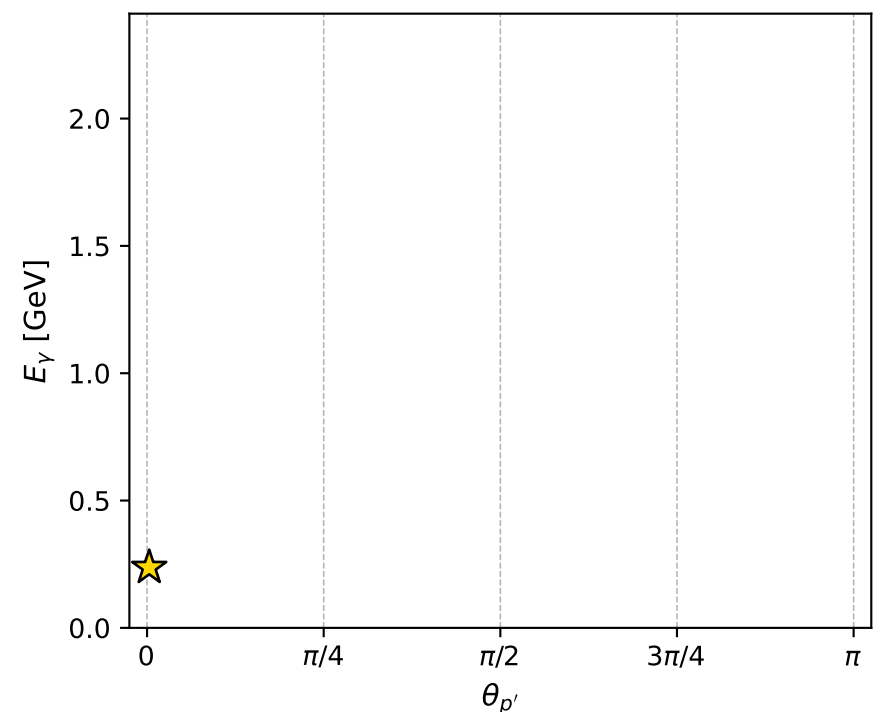
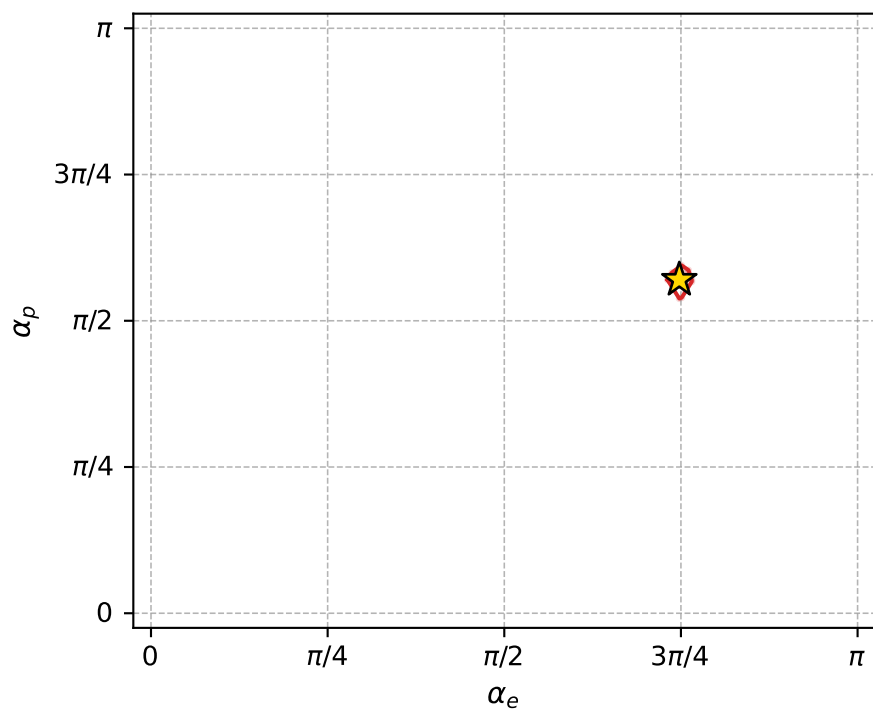
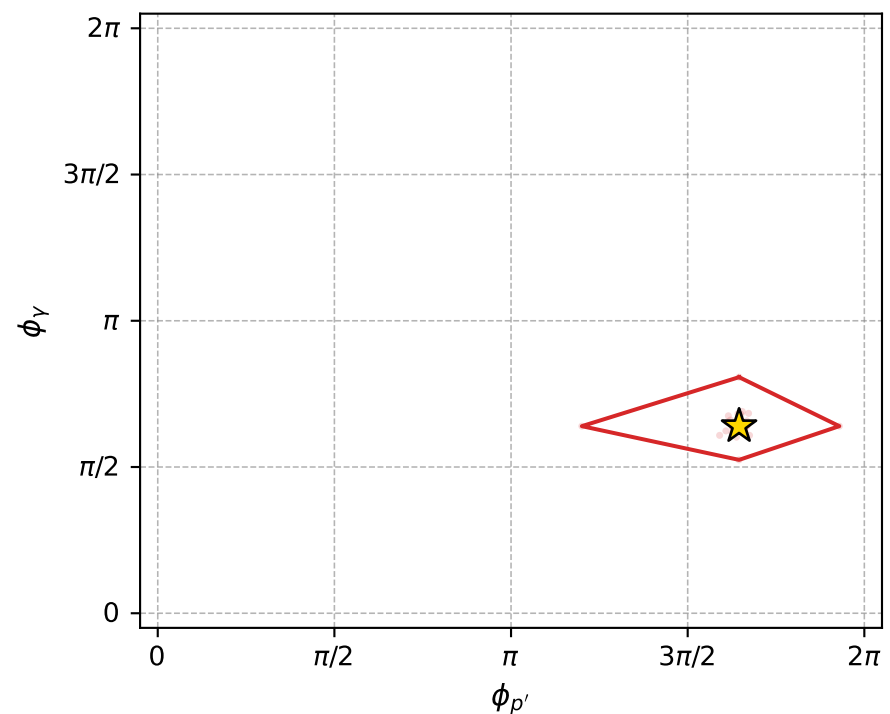
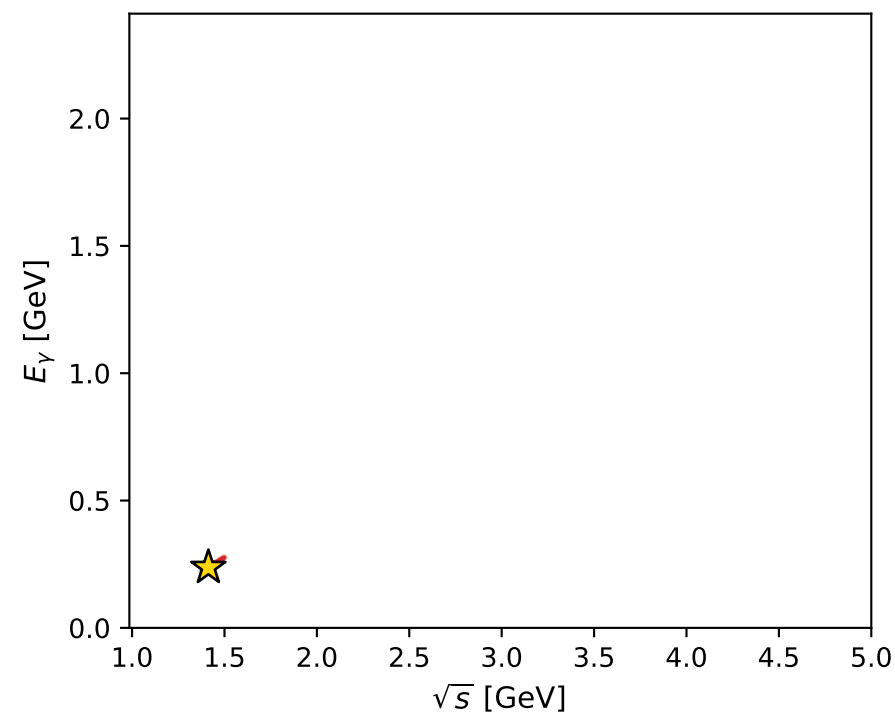
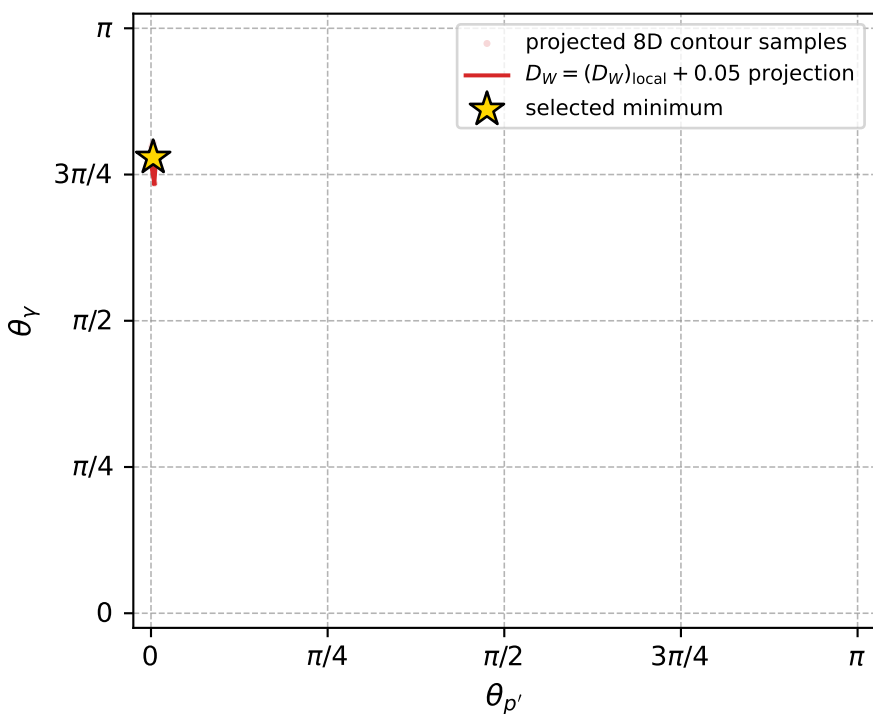
dw_mixing_angles_polarization_06_local_minimum_613 D_W=0.0045002
region: polarization_cluster_06_local_minimum_613
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3489925$ rad
incoming lepton: $-0.701996|+\rangle +0.712181|-\rangle$
proton mixing angle: $\alpha_p=1.7907971$ rad
incoming proton: $-0.21823|+\rangle +0.975897|-\rangle$

$s=1.99647$, $\sqrt{s}=1.41297$, $|\vec{P}|=0.395137$, $|\vec{P}'|=0.301443$
 $\theta_{P'}=0.00969286$, $\theta_Y=2.44765$, $\phi_{P'}=5.16936$, $\phi_Y=2.00965$
 $E_Y=0.237146$, $\theta_{\ell'}=2.24606$, $\phi_{\ell'}=5.15089$
 $Q^2=0.0564617$, $x_B=0.0889801$, $t=-0.00772813$, $W^2=1.45793$, $y=0.568266$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3951$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3951$
 P $E= 1.0178$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3951$
 ℓ' $E= 0.1906$ $p_x= 0.0632$ $p_y= -0.1347$ $p_z= -0.1191$
 P' $E= 0.9852$ $p_x= 0.0013$ $p_y= -0.0026$ $p_z= 0.3014$
 q_Y $E= 0.2371$ $p_x= -0.0644$ $p_y= 0.1373$ $p_z= -0.1823$



electron: polarization cluster P6, local minimum 613; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0045002001$

$D_W \text{ contour} = 0.0545002$

above global minimum = 0.0044837672

8D contour samples = 251

selected phase-space point:

$\text{sqrt_s} = 1.412967$

$\text{theta_p_out} = 0.009692859$

$\text{theta_gamma_out} = 2.44765$

$\text{qOut} = 0.2371457$

$\text{phi_p_out} = 5.169357$

$\text{phi_gamma_out} = 2.009653$

$\text{alpha_e} = 2.348993$

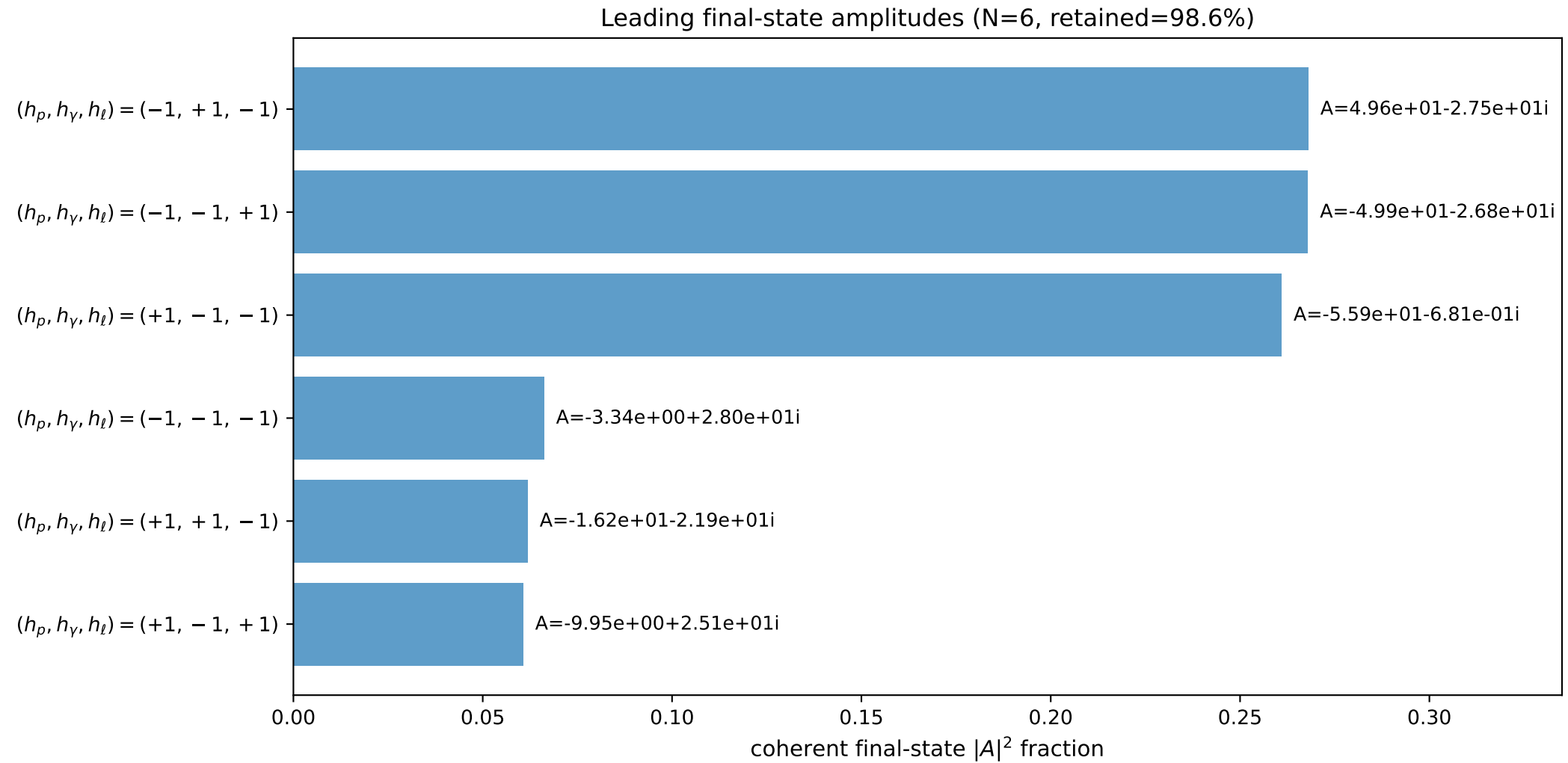
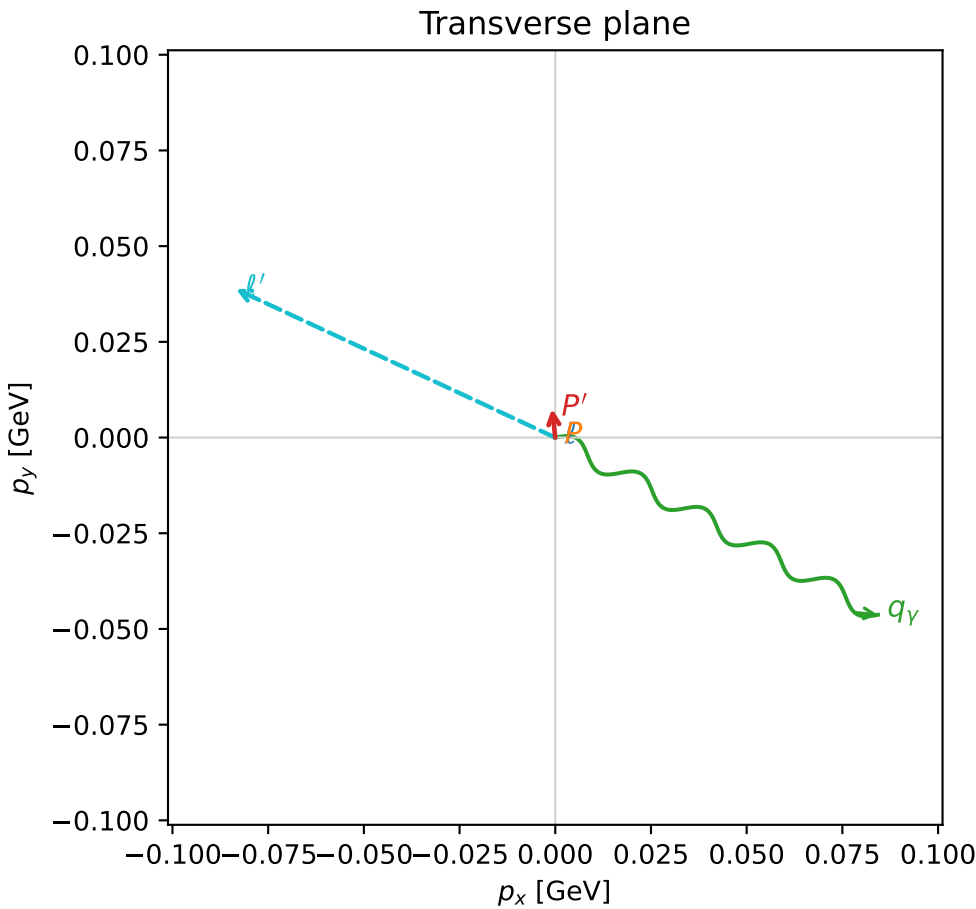
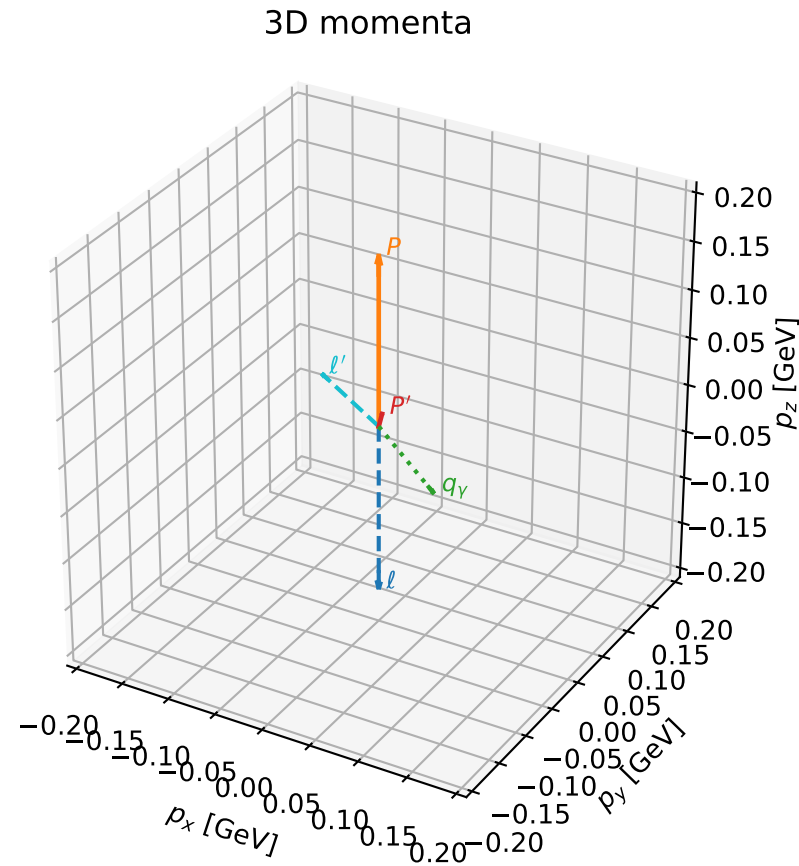
$\text{alpha_p} = 1.790797$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

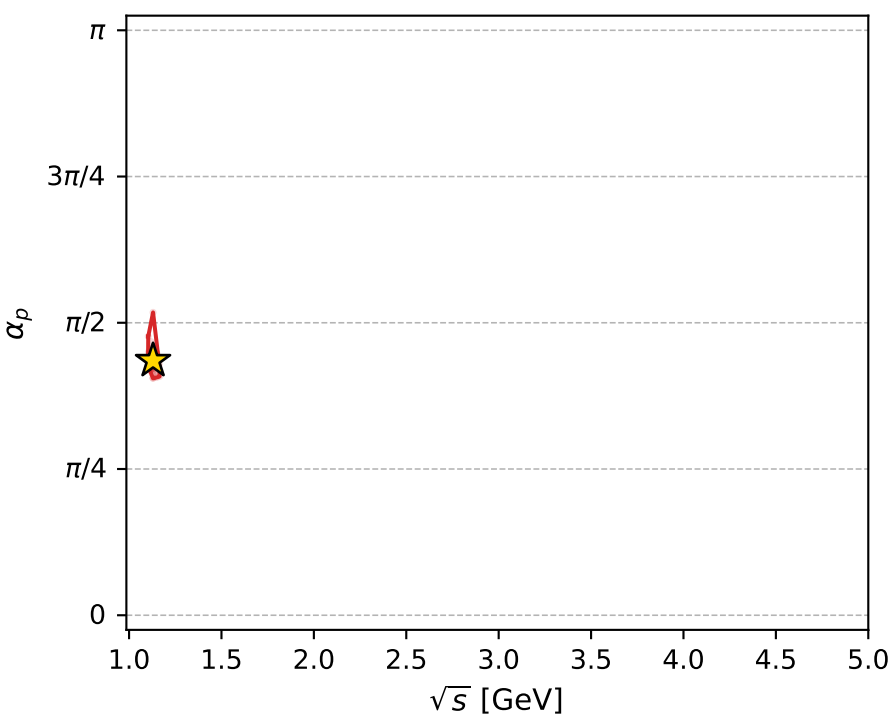
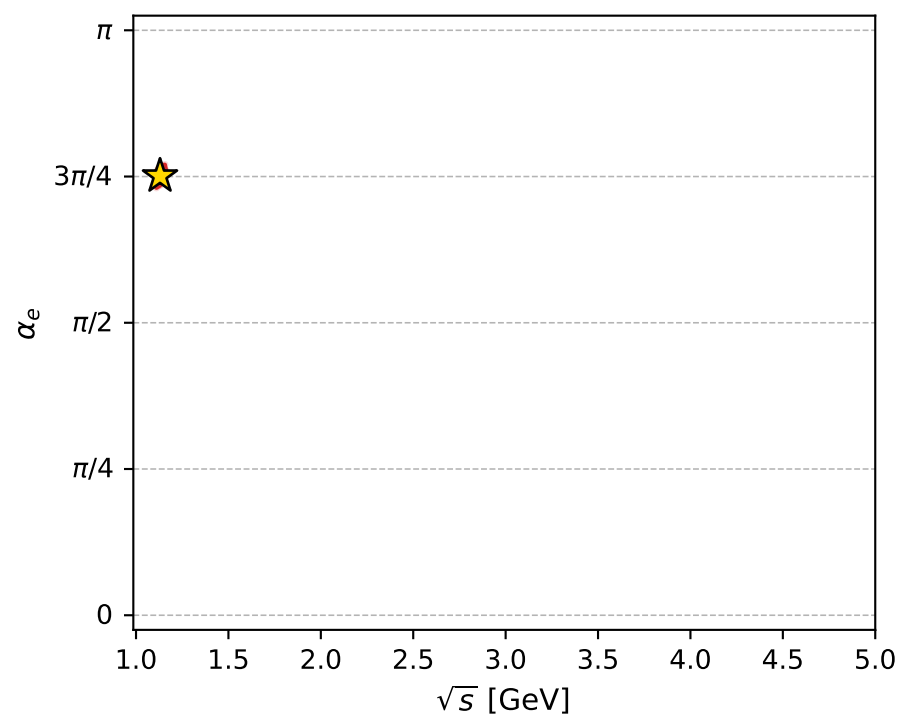
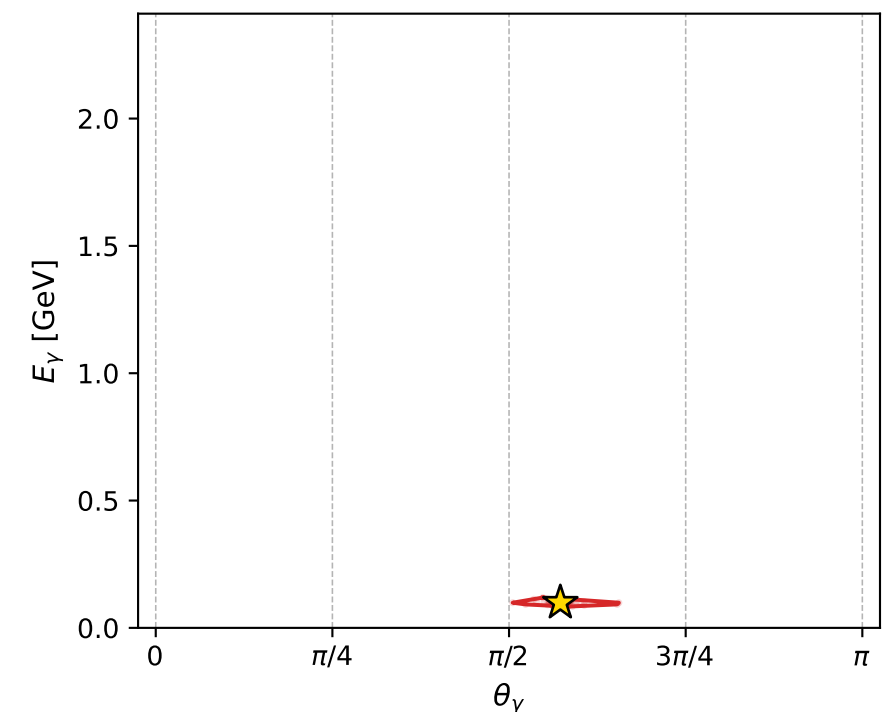
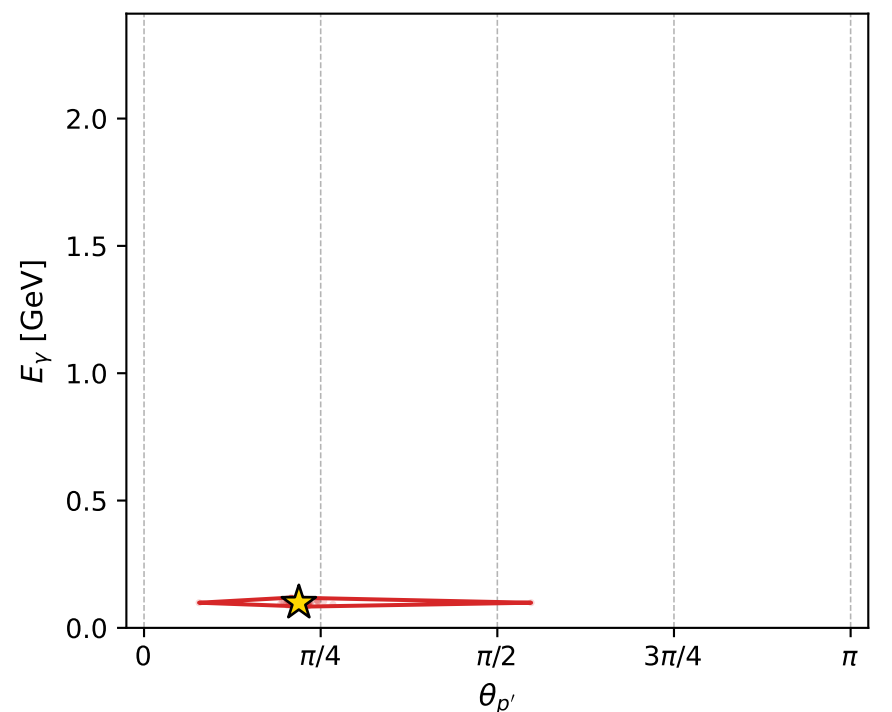
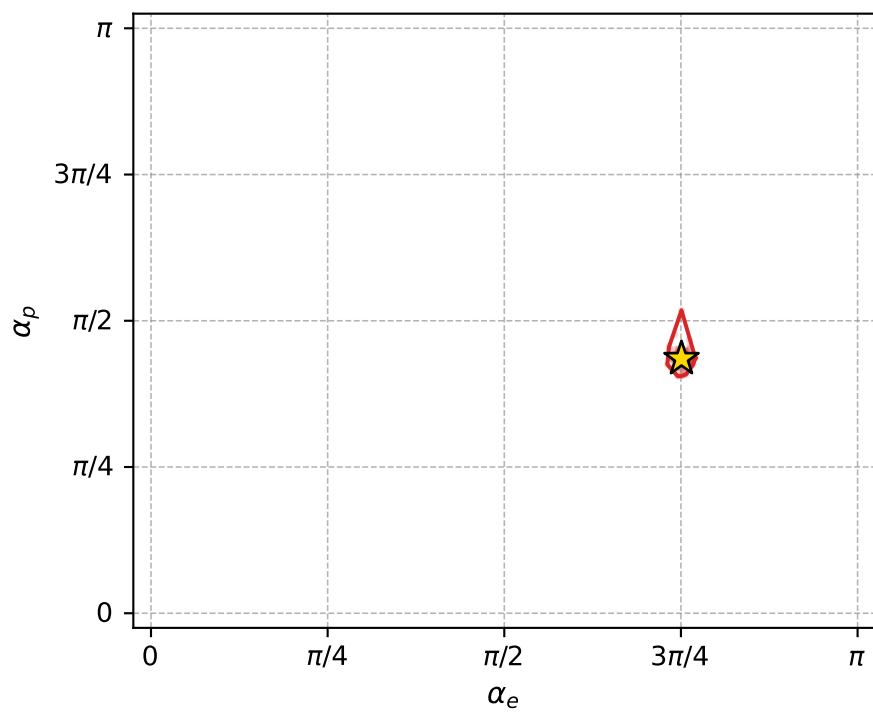
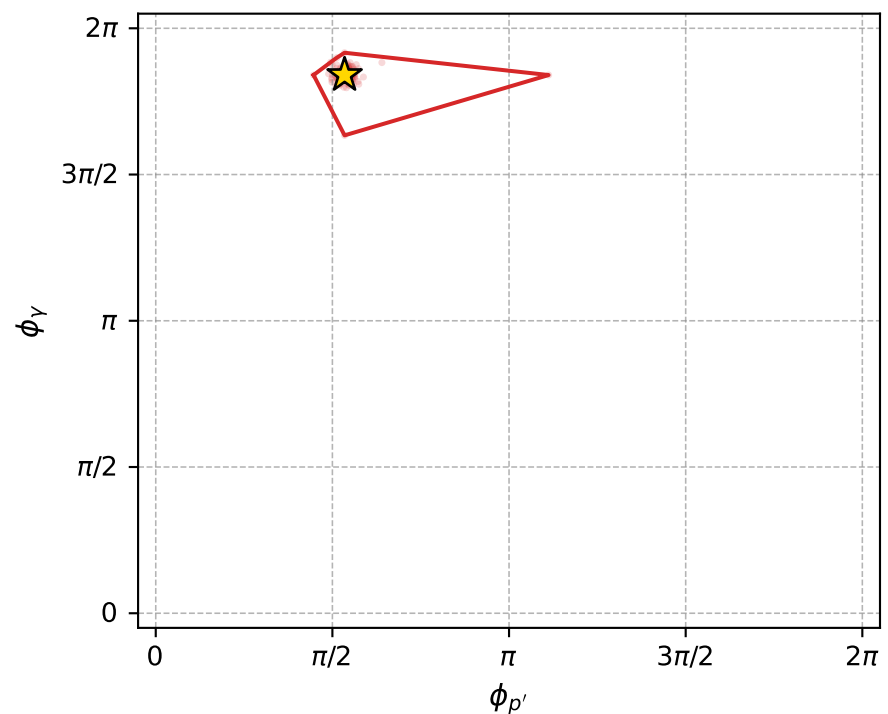
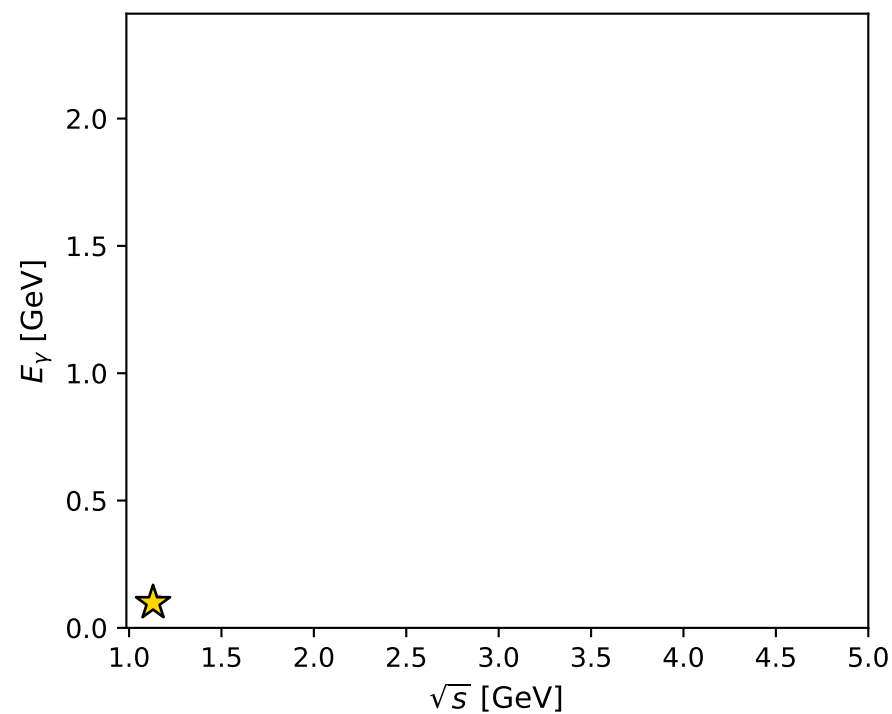
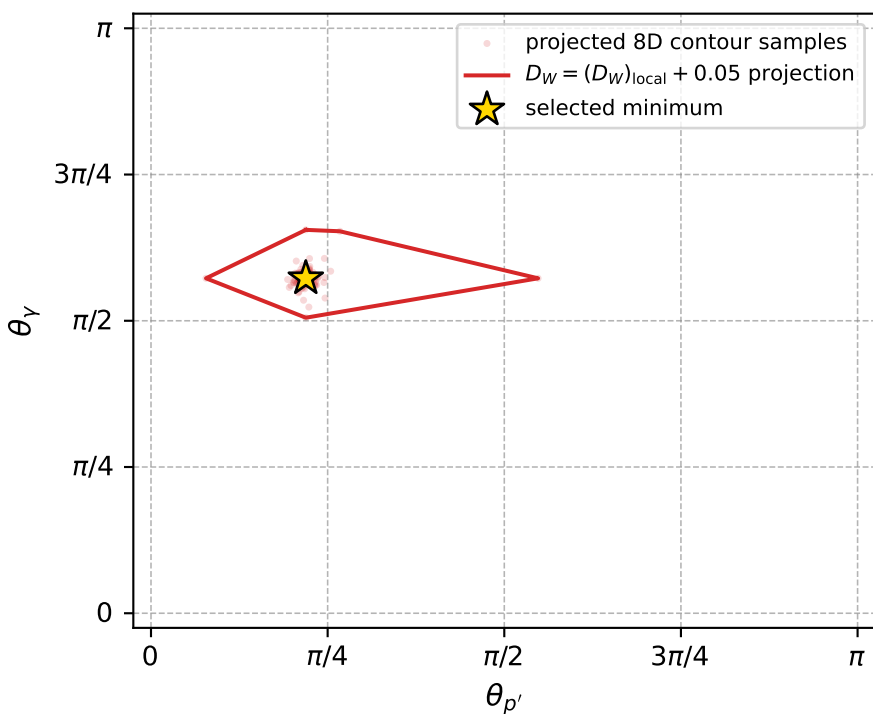
dw_mixing_angles_polarization_06_local_minimum_643 D_W=0.00536493
 region: polarization_cluster_06_local_minimum_643
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3586723$ rad
 incoming lepton: $-0.708857|+> +0.705353|->$
 proton mixing angle: $\alpha_p=1.3673414$ rad
 incoming proton: $0.202054|+> +0.979374|->$

$s=1.27646$, $\sqrt{s}=1.1298$, $|\vec{P}|=0.175522$, $|\vec{P}'|=0.011898$
 $\theta_{p'}=0.688348$, $\theta_V=1.79882$, $\phi_{p'}=1.67917$, $\phi_V=5.78097$
 $E_V=0.0987382$, $\theta_{l'}=1.42911$, $\phi_{l'}=2.7067$
 $Q^2=0.037253$, $x_B=0.166498$, $t=-0.0274614$, $W^2=1.06634$, $y=0.564138$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1755$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1755$
 P $E= 0.9543$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1755$
 l' $E= 0.0930$ $p_x= -0.0835$ $p_y= 0.0388$ $p_z= 0.0131$
 P' $E= 0.9381$ $p_x= -0.0008$ $p_y= 0.0075$ $p_z= 0.0092$
 q_V $E= 0.0987$ $p_x= 0.0843$ $p_y= -0.0463$ $p_z= -0.0223$



electron: polarization cluster P6, local minimum 643; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0053649343$
 $D_W \text{ contour} = 0.055364934$
 above global minimum = 0.0053485015
 8D contour samples = 256

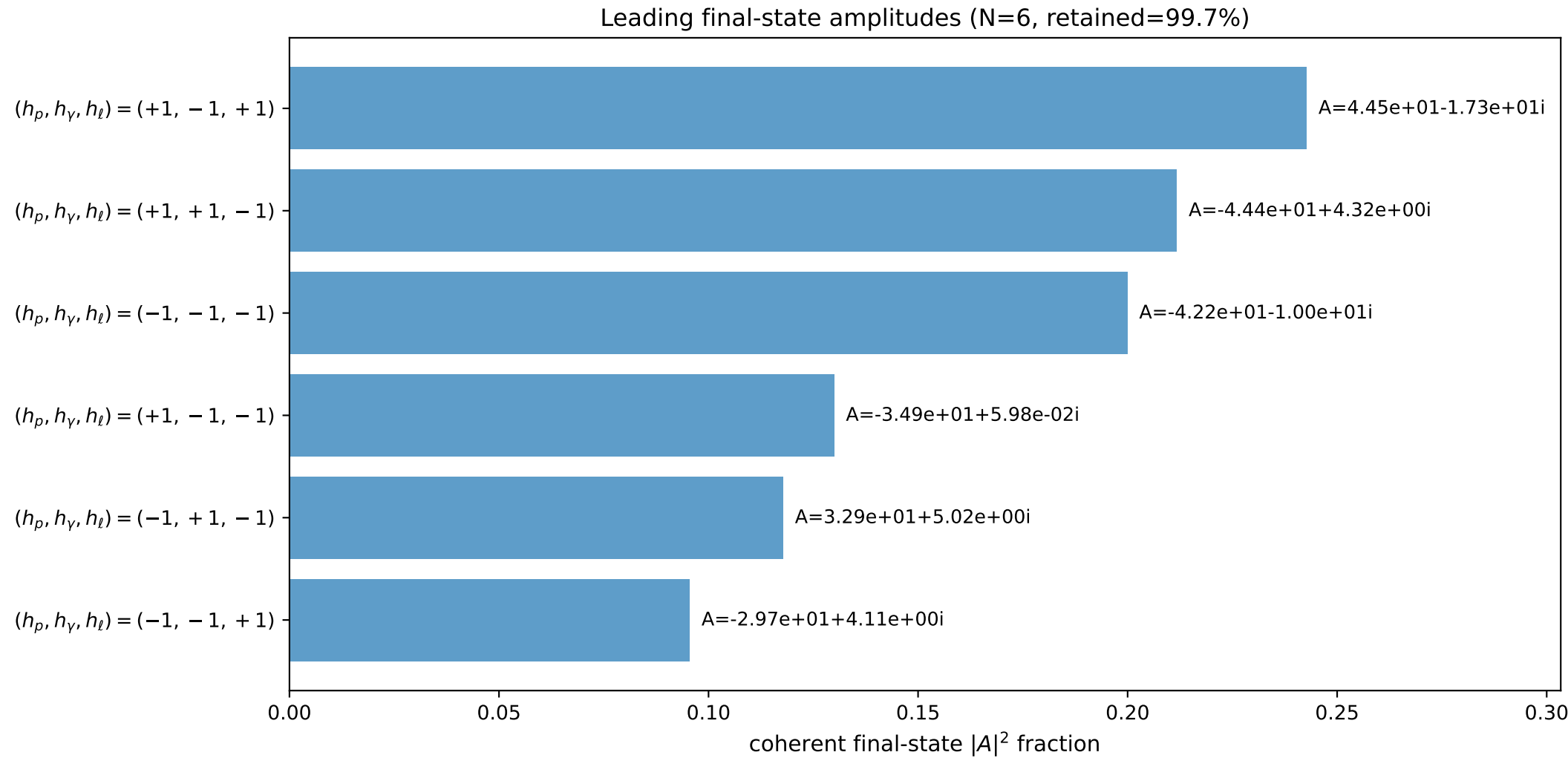
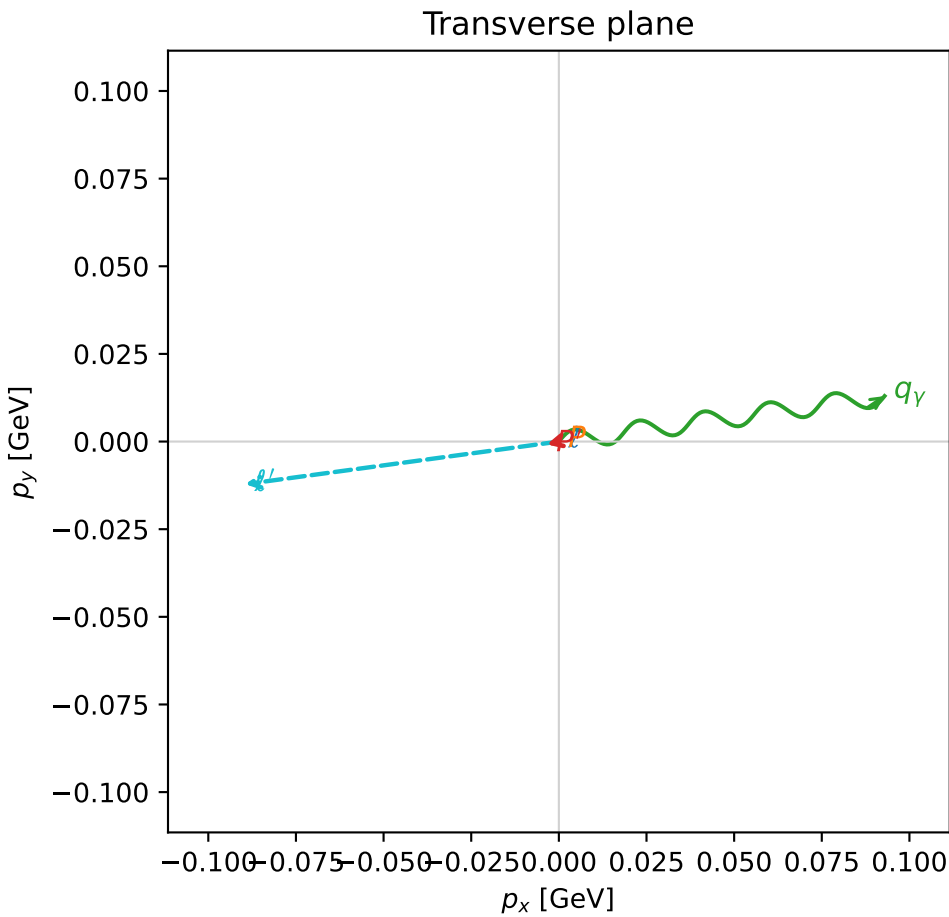
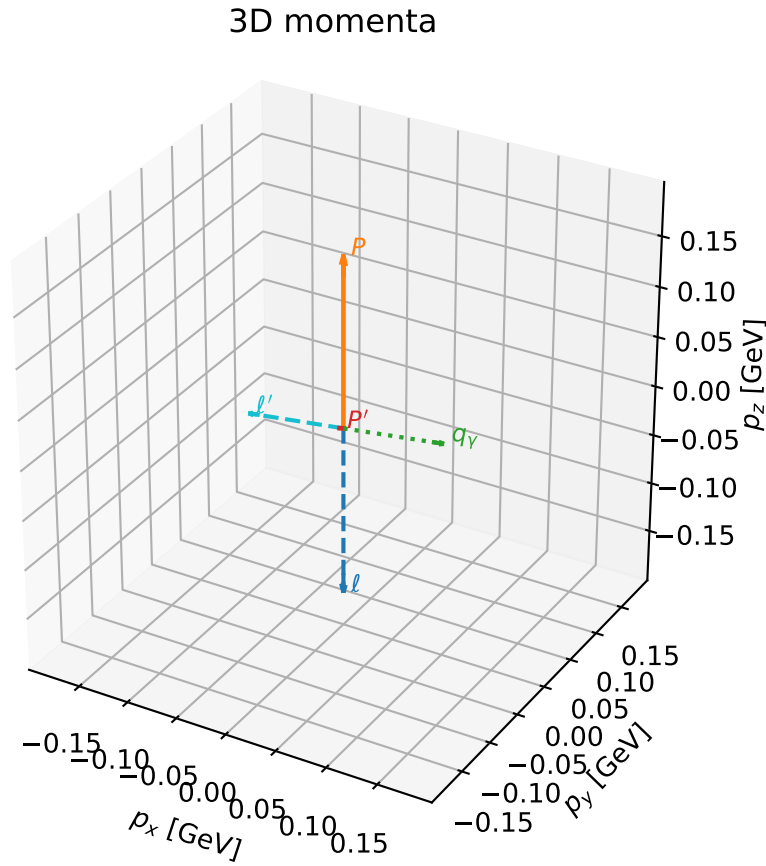
selected phase-space point:
 $\text{sqrt_s} = 1.129804$
 $\text{theta_p_out} = 0.688348$
 $\text{theta_gamma_out} = 1.798818$
 $\text{qOut} = 0.09873819$
 $\text{phi_p_out} = 1.679172$
 $\text{phi_gamma_out} = 5.78097$
 $\text{alpha_e} = 2.358672$
 $\text{alpha_p} = 1.367341$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

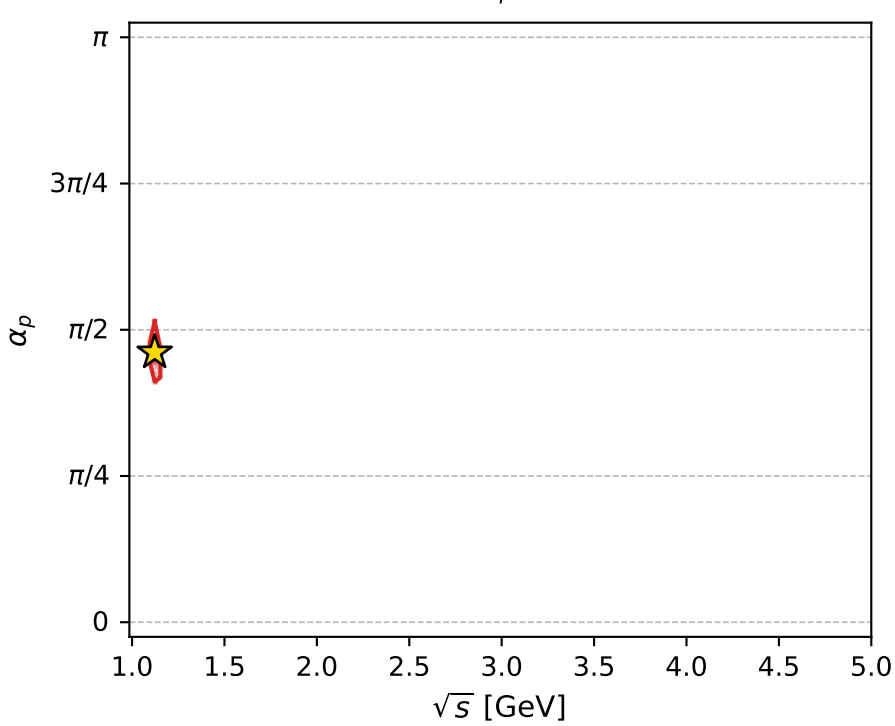
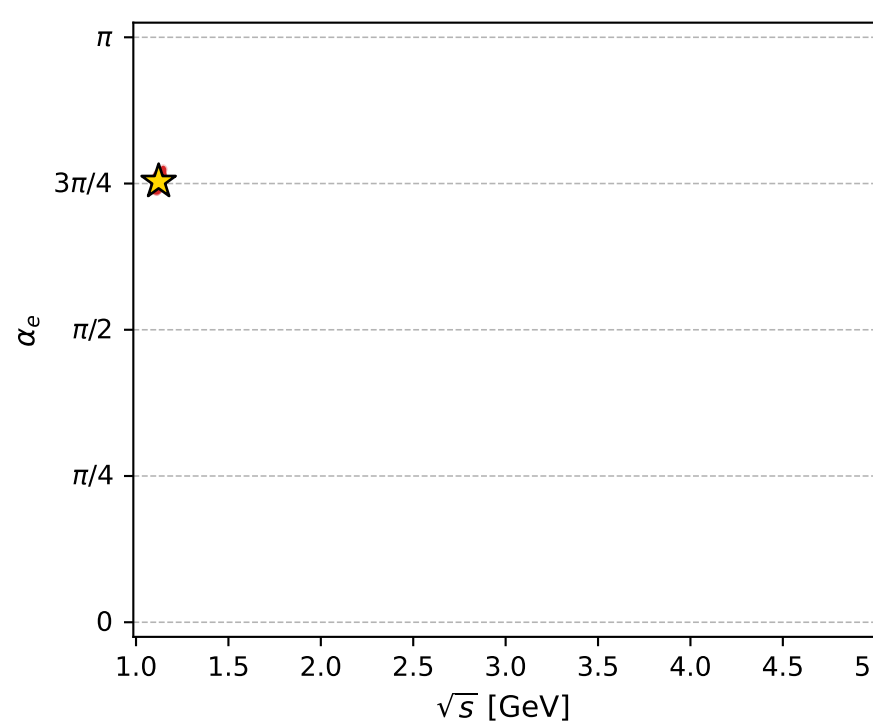
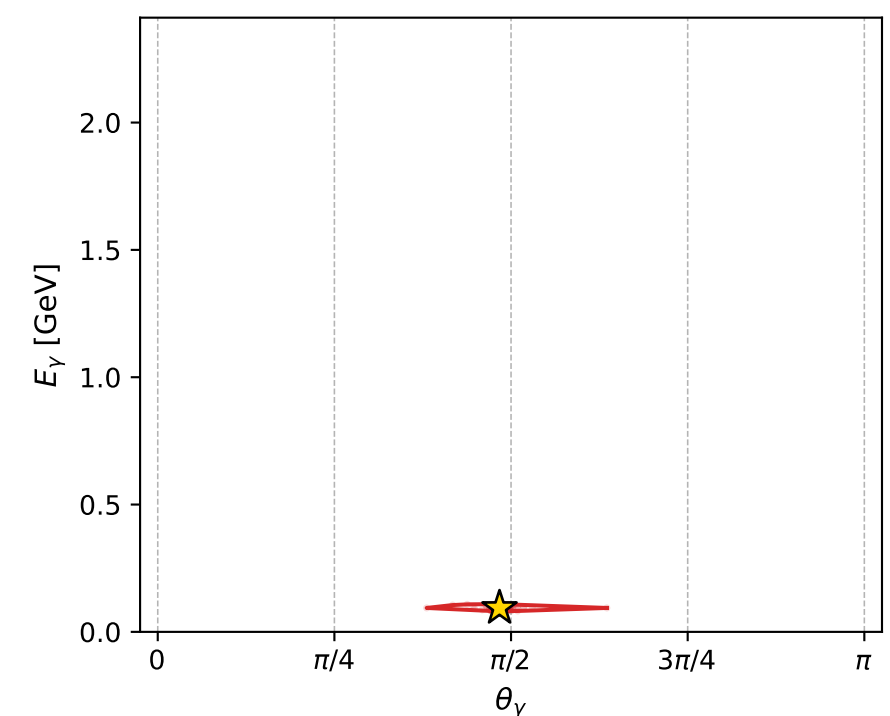
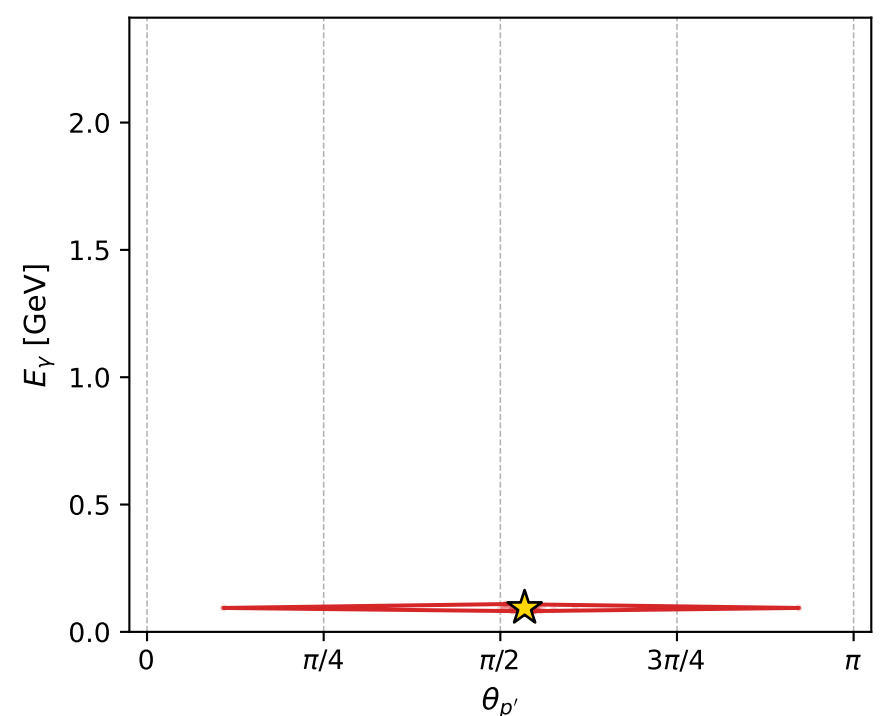
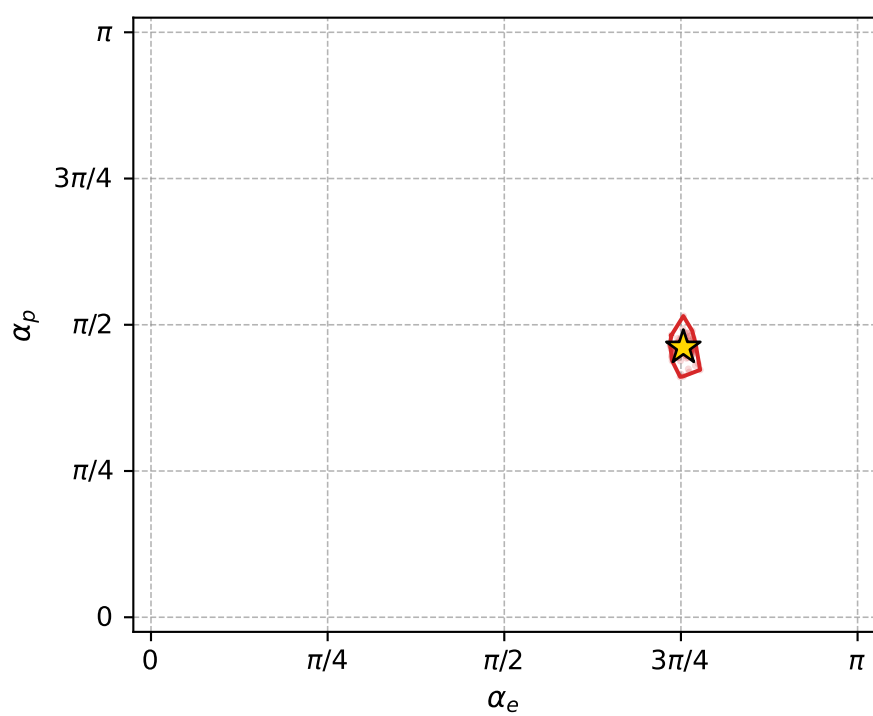
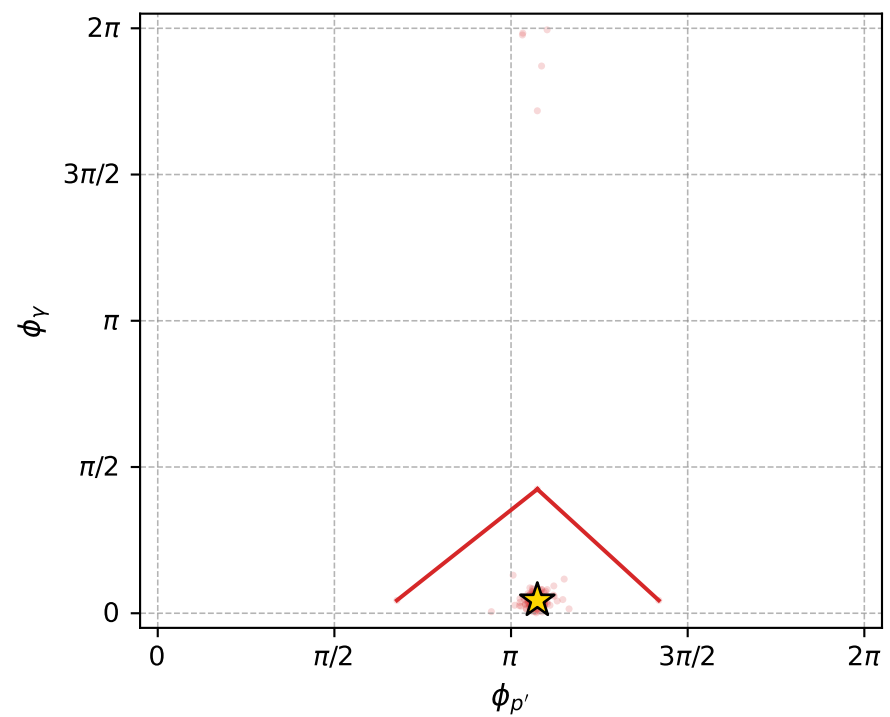
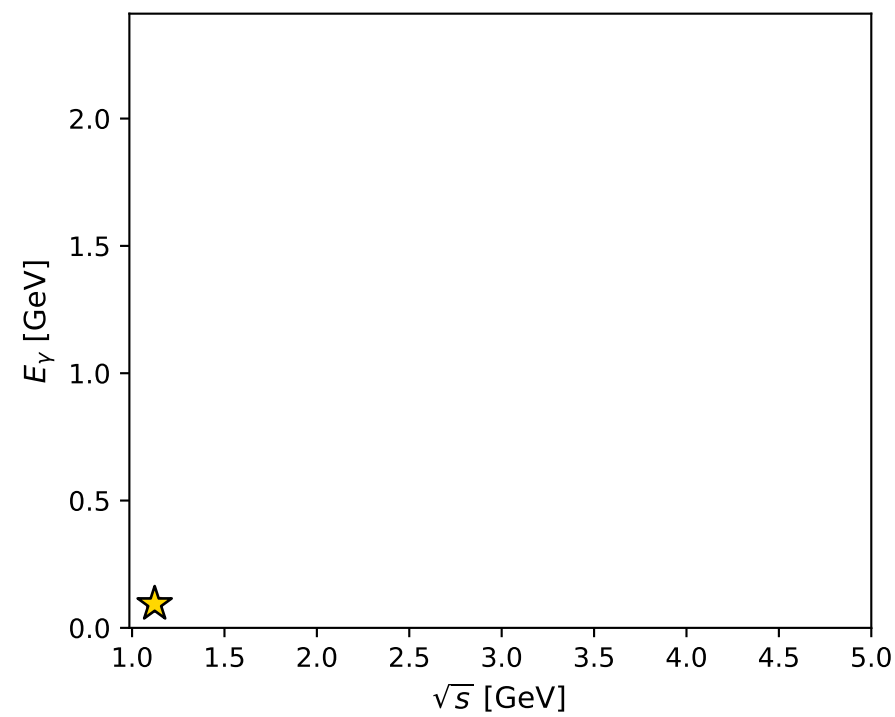
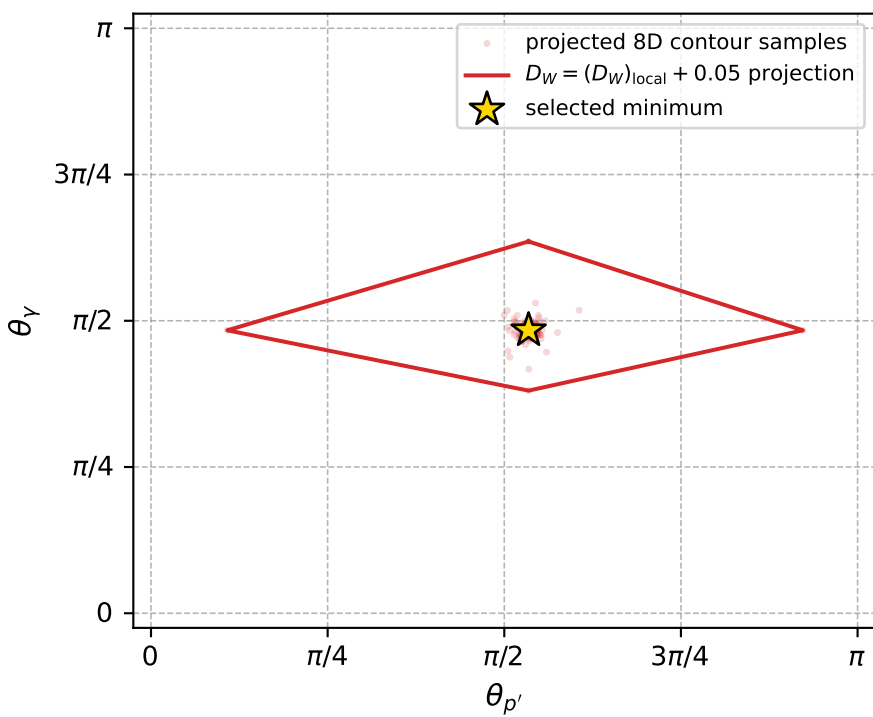
dw_mixing_angles_polarization_06_local_minimum_653 D_W=0.00559009
region: polarization_cluster_06_local_minimum_653
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3669172$ rad
incoming lepton: $-0.714648|+> +0.699484|->$
proton mixing angle: $\alpha_p=1.4491886$ rad
incoming proton: $0.121308|+> +0.992615|->$

$s=1.25969$, $\sqrt{s}=1.12236$, $|\vec{P}|=0.169215$, $|\vec{P}'|=0.00354994$
 $\theta_{p'}=1.67879$, $\theta_\gamma=1.51957$, $\phi_{p'}=3.37541$, $\phi_\gamma=0.138751$
 $E_\gamma=0.0939382$, $\theta_{l'}=1.61979$, $\phi_{l'}=3.27663$
 $Q^2=0.0290994$, $x_B=0.141265$, $t=-0.0285469$, $W^2=1.05674$, $y=0.54231$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1692$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1692$
 P $E= 0.9531$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1692$
 l' $E= 0.0904$ $p_x= -0.0895$ $p_y= -0.0122$ $p_z= -0.0044$
 P' $E= 0.9380$ $p_x= -0.0034$ $p_y= -0.0008$ $p_z= -0.0004$
 q_γ $E= 0.0939$ $p_x= 0.0929$ $p_y= 0.0130$ $p_z= 0.0048$



electron: polarization cluster P6, local minimum 653; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0055900946$
 $D_W \text{ contour} = 0.055590095$
 above global minimum = 0.0055736618
 8D contour samples = 256

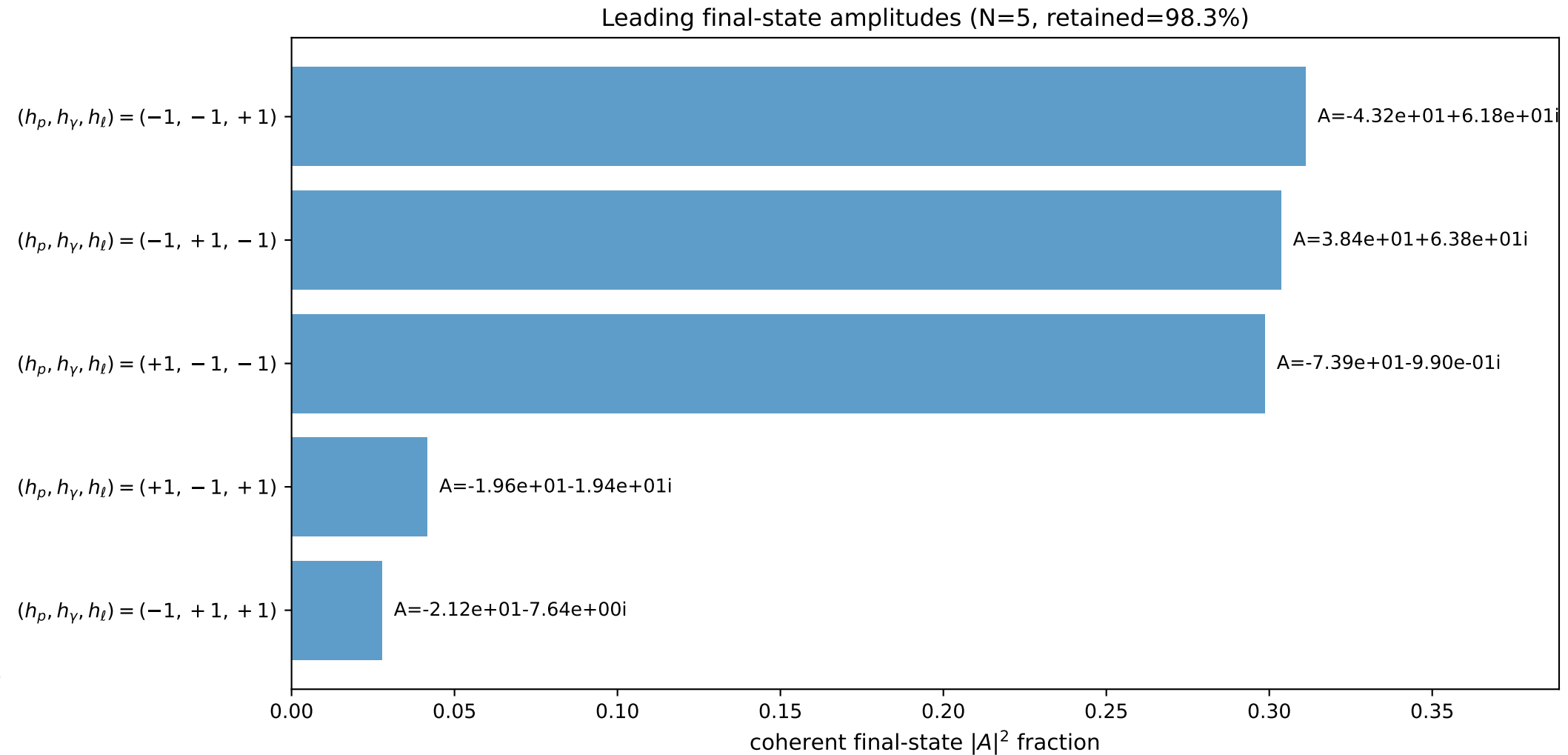
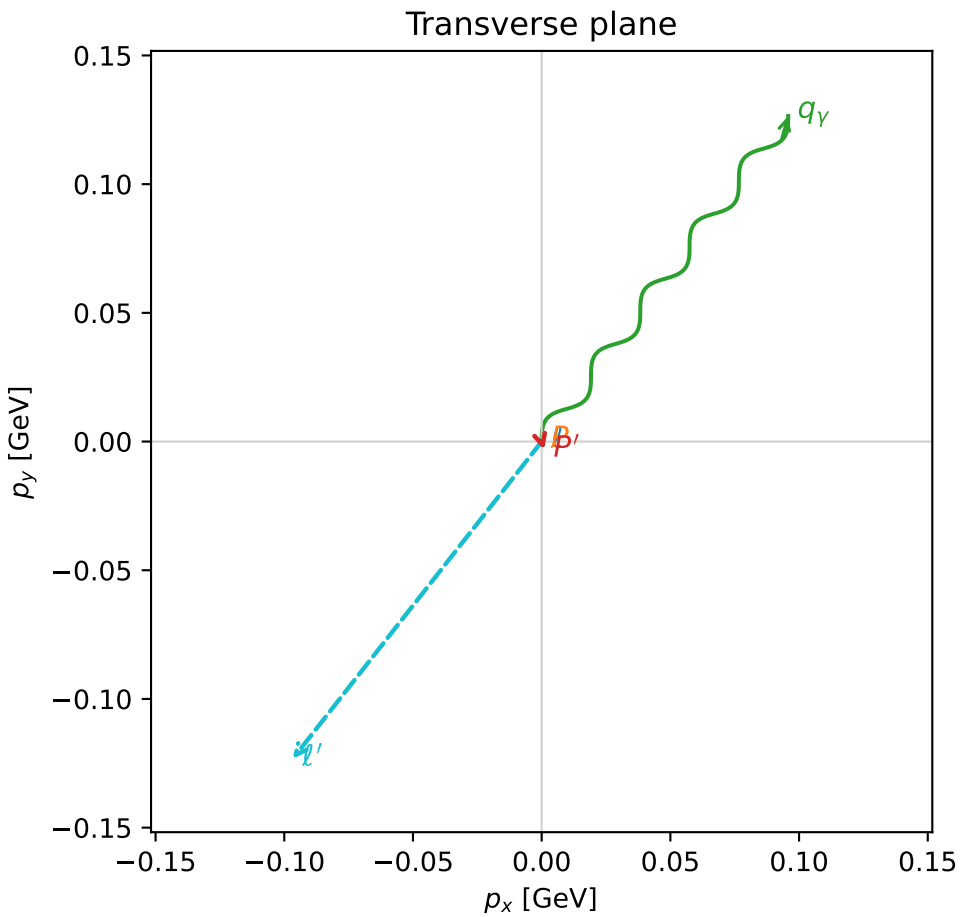
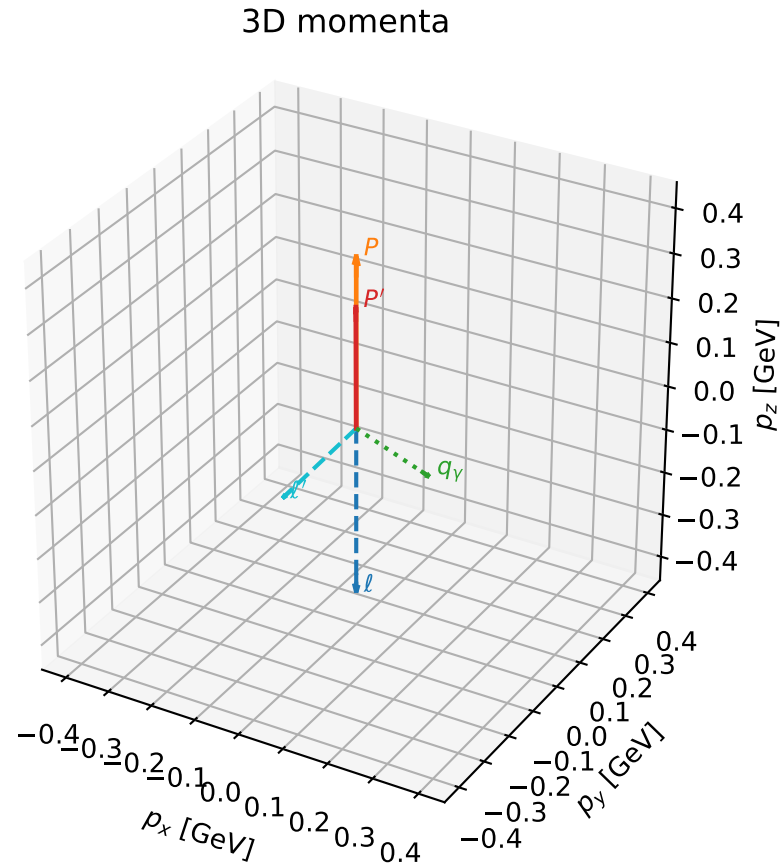
selected phase-space point:
 $\text{sqrt_s} = 1.122357$
 $\text{theta_p_out} = 1.678788$
 $\text{theta_gamma_out} = 1.519568$
 $\text{q0ut} = 0.0939382$
 $\text{phi_p_out} = 3.375406$
 $\text{phi_gamma_out} = 0.1387507$
 $\text{alpha_e} = 2.366917$
 $\text{alpha_p} = 1.449189$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

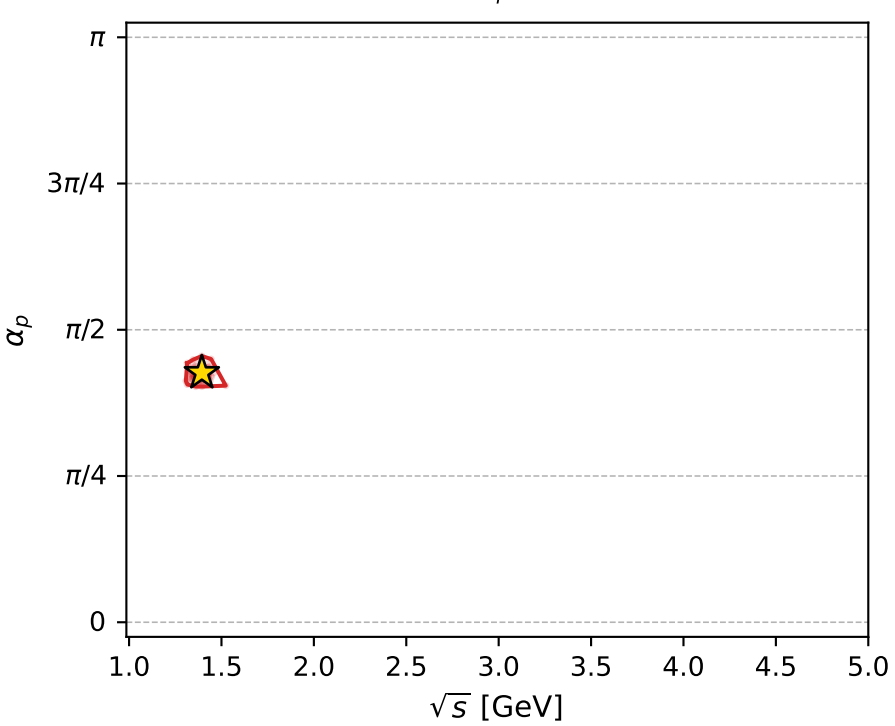
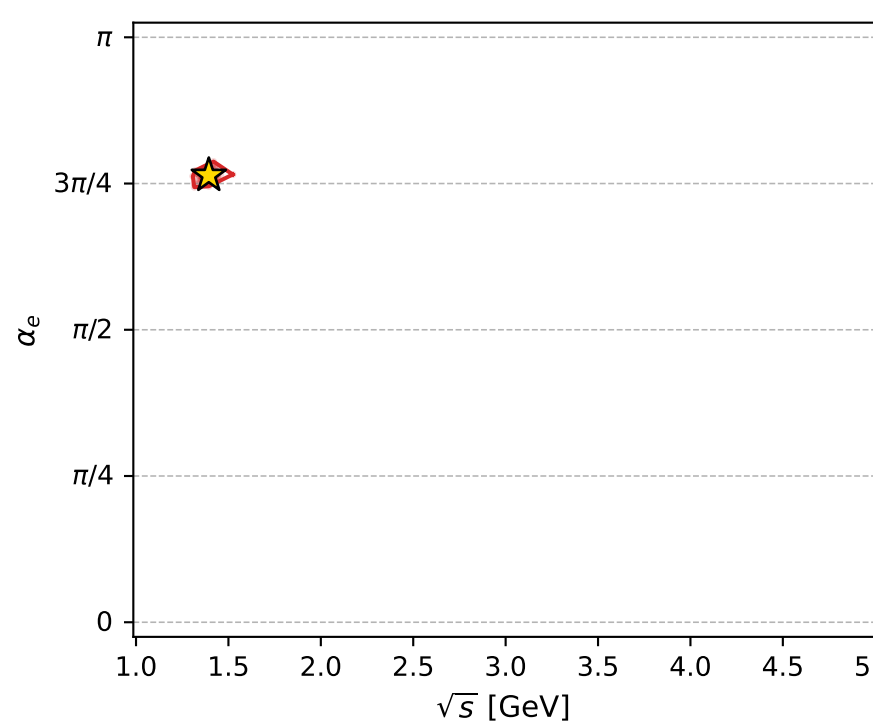
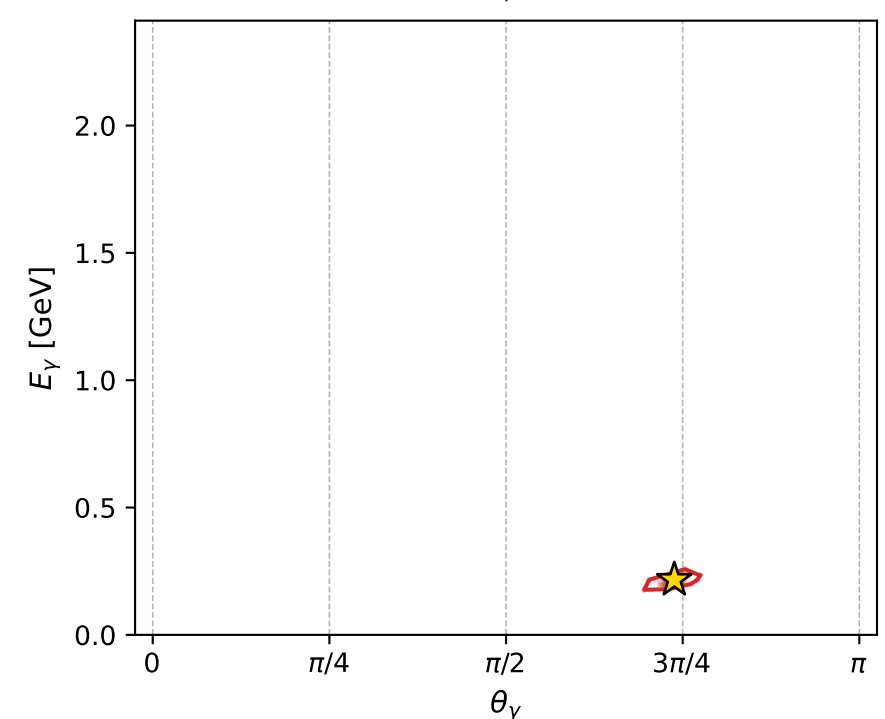
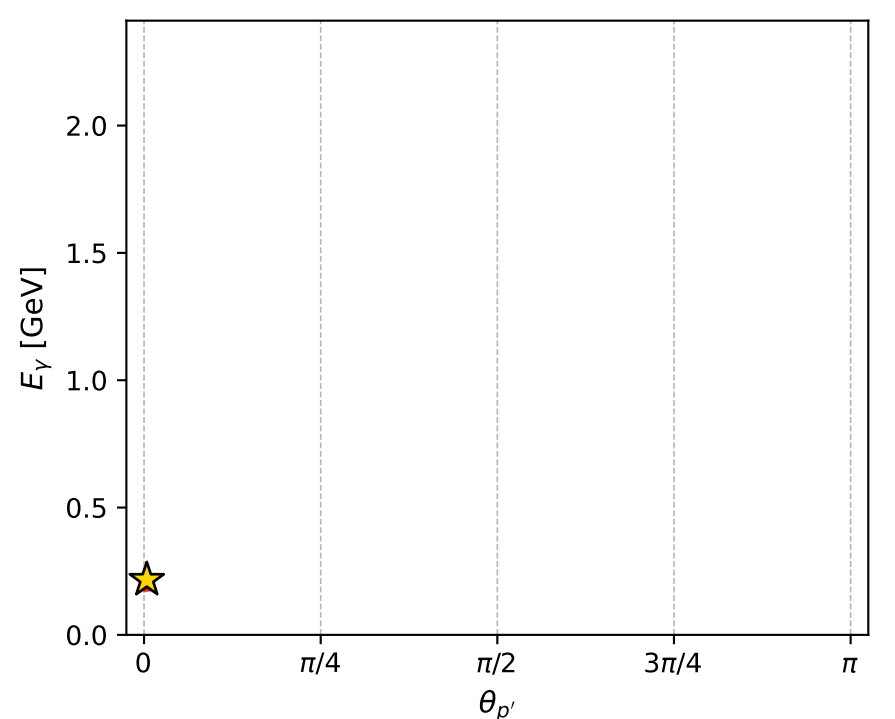
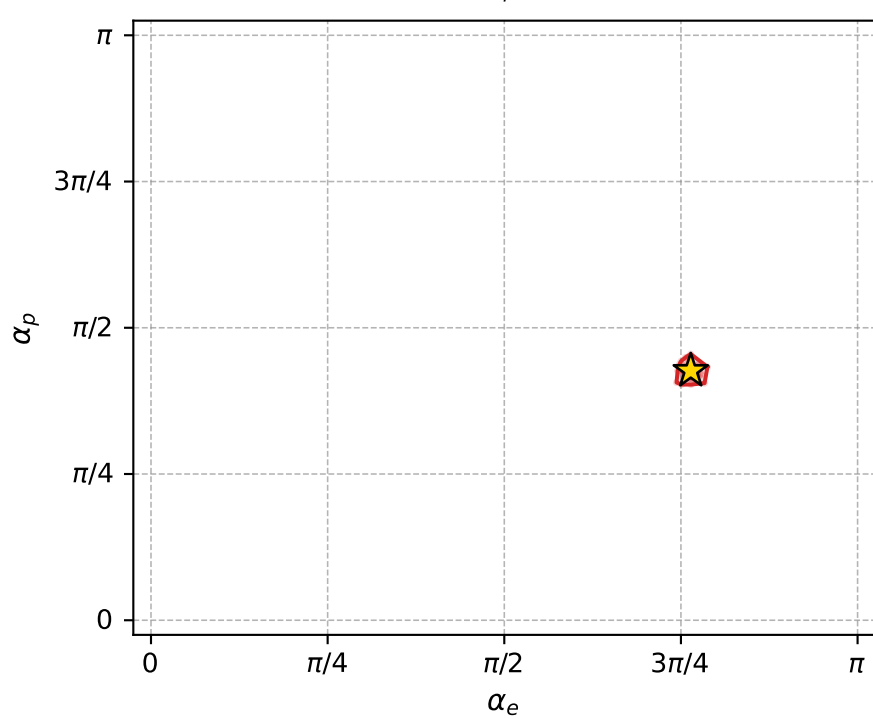
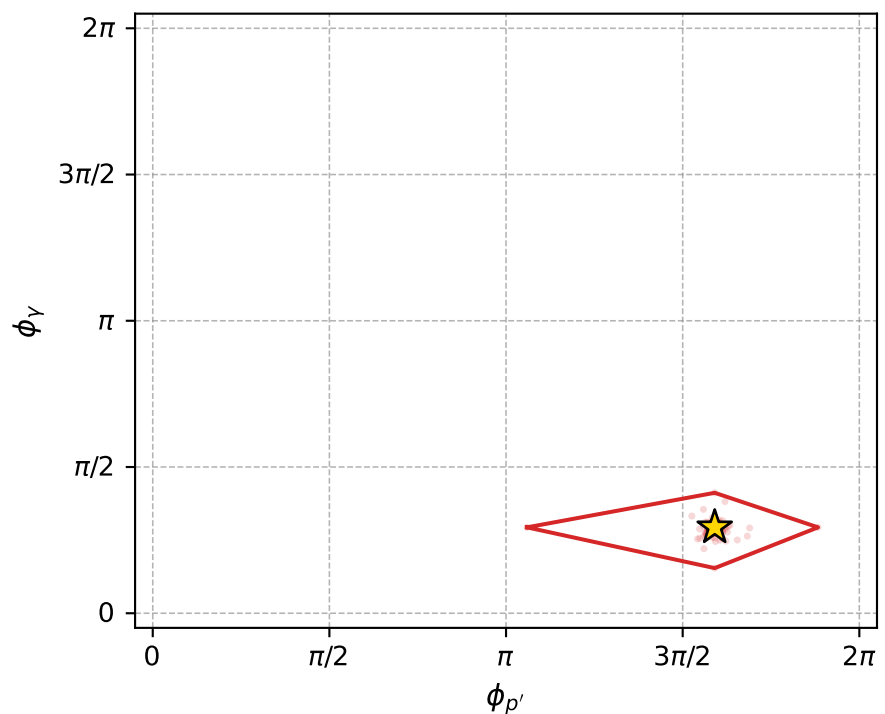
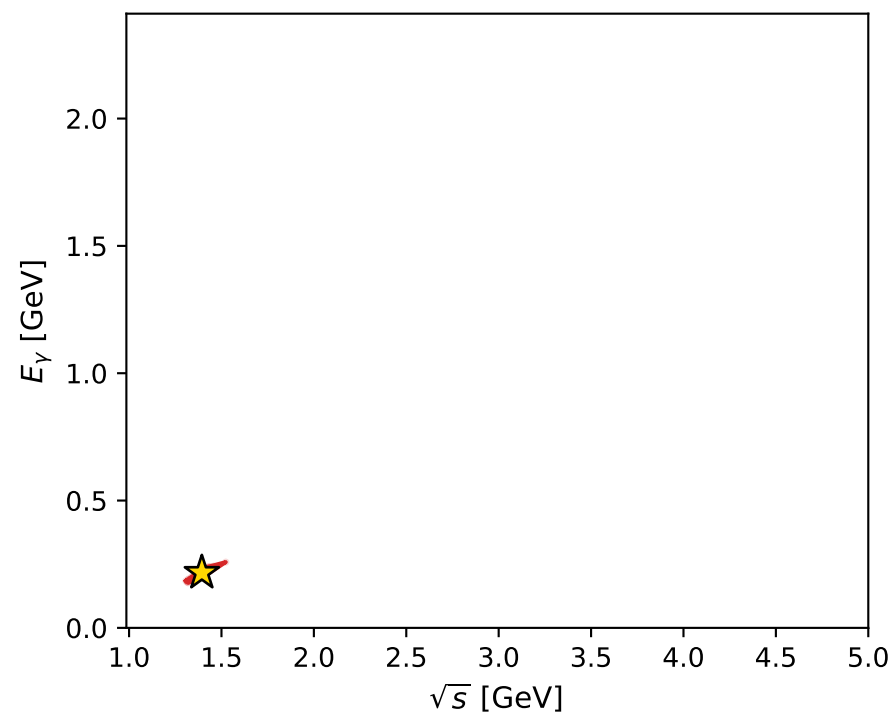
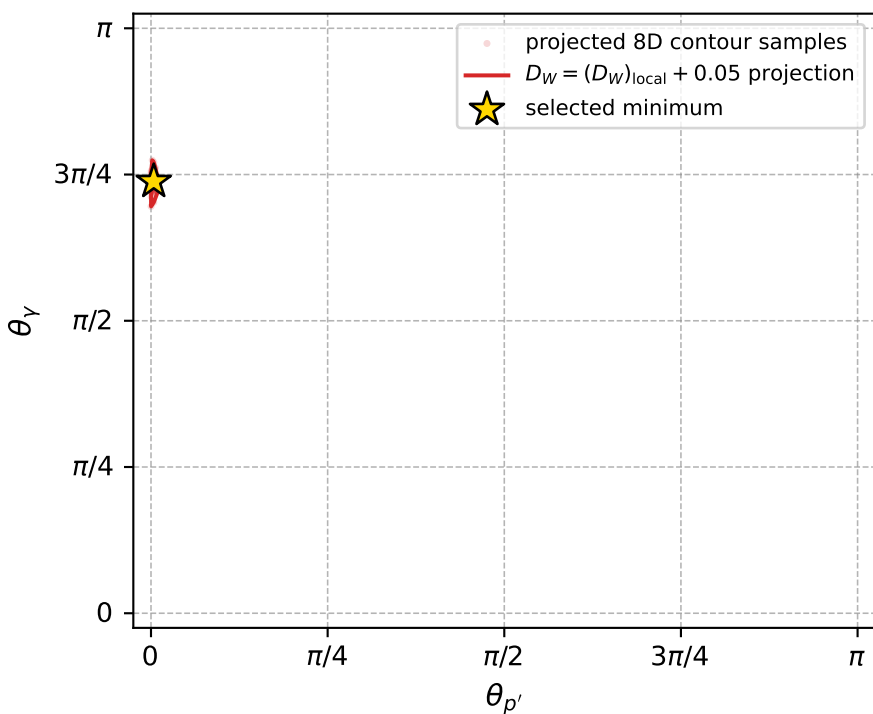
dw_mixing_angles_polarization_06_local_minimum_654 D_W=0.00561805
 region: polarization_cluster_06_local_minimum_654
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3998893$ rad
 incoming lepton: $-0.737319|+\rangle +0.675545|-\rangle$
 proton mixing angle: $\alpha_p=1.3396292$ rad
 incoming proton: $0.229114|+\rangle +0.9734|-\rangle$

$s=1.94292$, $\sqrt{s}=1.39389$, $|\vec{P}|=0.381335$, $|\vec{P}'|=0.272594$
 $\theta_{P'}=0.0122803$, $\theta_V=2.31885$, $\phi_{P'}=4.99713$, $\phi_V=0.922634$
 $E_V=0.216427$, $\theta_{\ell'}=2.24561$, $\phi_{\ell'}=4.04706$
 $Q^2=0.0574249$, $x_B=0.102339$, $t=-0.0105625$, $W^2=1.38354$, $y=0.527832$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.3813$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.3813$
 P $E=1.0126$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.3813$
 ℓ' $E=0.2007$ $p_x=-0.0967$ $p_y=-0.1233$ $p_z=-0.1254$
 P' $E=0.9768$ $p_x=0.0009$ $p_y=-0.0032$ $p_z=0.2726$
 q_V $E=0.2164$ $p_x=0.0958$ $p_y=0.1265$ $p_z=-0.1472$



electron: polarization cluster P6, local minimum 654; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0056180451$
 $D_W \text{ contour} = 0.055618045$
 above global minimum = 0.0056016122
 8D contour samples = 204

selected phase-space point:
 $\text{sqrt_s} = 1.393887$
 $\text{theta_p_out} = 0.01228033$
 $\text{theta_gamma_out} = 2.318845$
 $\text{q0out} = 0.2164267$
 $\text{phi_p_out} = 4.997129$
 $\text{phi_gamma_out} = 0.9226339$
 $\text{alpha_e} = 2.399889$
 $\text{alpha_p} = 1.339629$

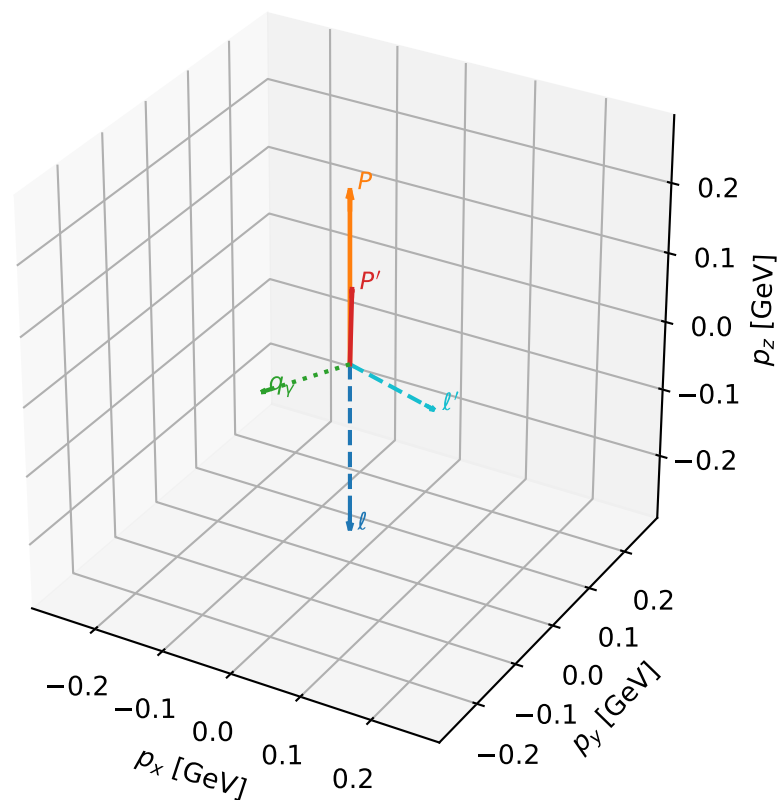
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_678 D_W=0.00630653
 region: polarization_cluster_06_local_minimum_678
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.4077902$ rad
 incoming lepton: $-0.742633|+> +0.669698|->$
 proton mixing angle: $\alpha_p=1.8442141$ rad
 incoming proton: $-0.270024|+> +0.962854|->$

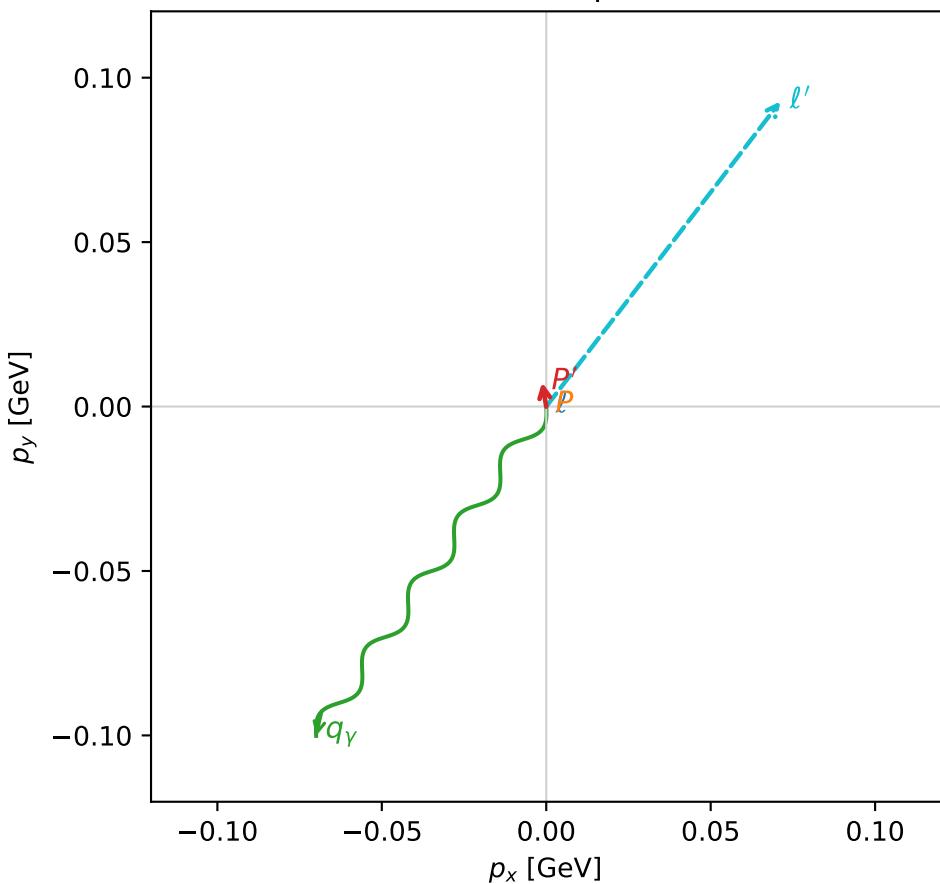
$s=1.47872$, $\sqrt{s}=1.21603$, $|\vec{P}|=0.246242$, $|\vec{P}'|=0.104001$
 $\theta_{P'}=0.0716767$, $\theta_V=1.65548$, $\phi_{P'}=1.74014$, $\phi_V=4.10256$
 $E_V=0.122605$, $\theta_{\ell'}=2.24437$, $\phi_{\ell'}=0.916219$
 $Q^2=0.0277313$, $x_B=0.105605$, $t=-0.0196863$, $W^2=1.11471$, $y=0.438478$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2462$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2462$
 P $E= 0.9698$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2462$
 ℓ' $E= 0.1497$ $p_x= 0.0712$ $p_y= 0.0928$ $p_z= -0.0934$
 P' $E= 0.9437$ $p_x= -0.0013$ $p_y= 0.0073$ $p_z= 0.1037$
 q_Y $E= 0.1226$ $p_x= -0.0700$ $p_y= -0.1001$ $p_z= -0.0104$

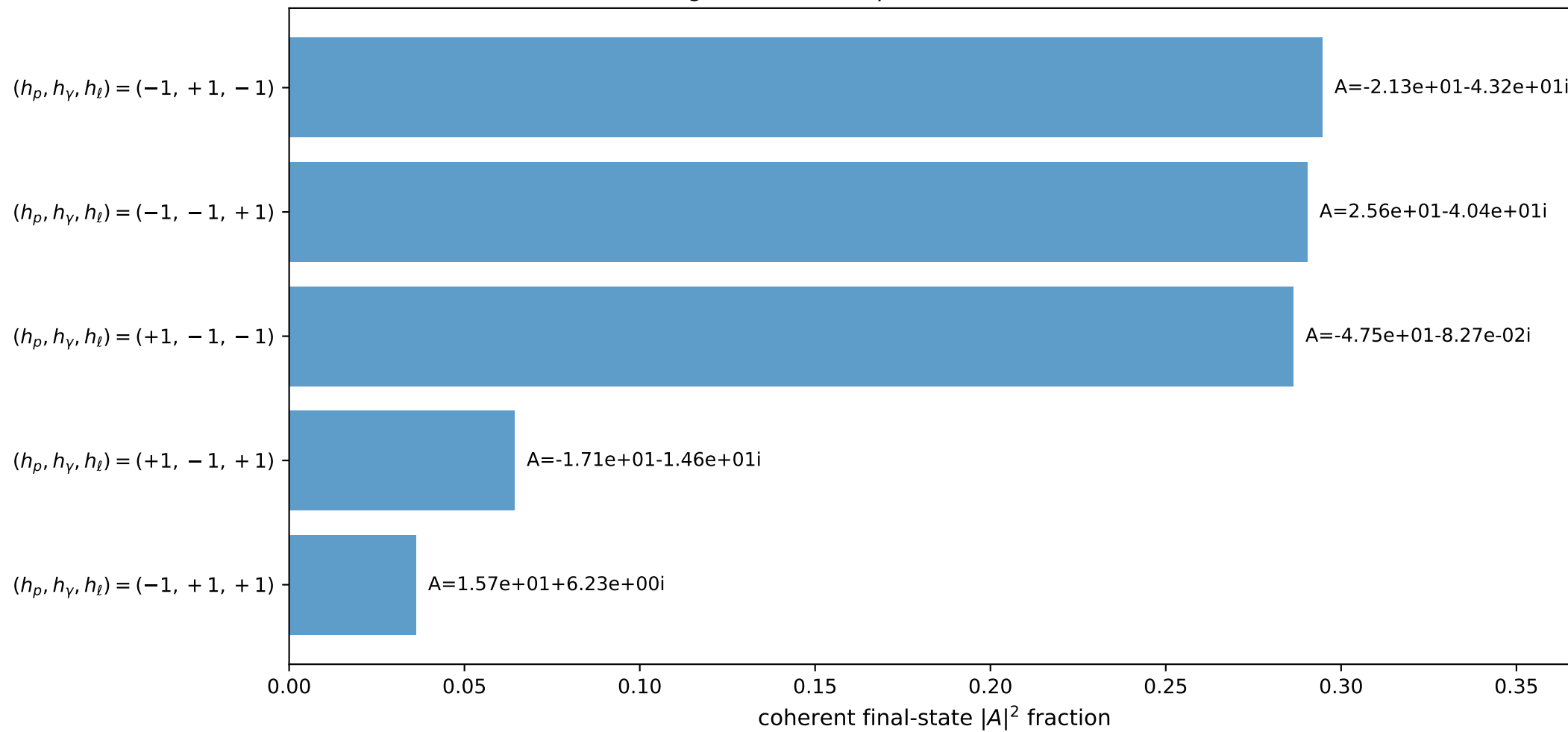
3D momenta



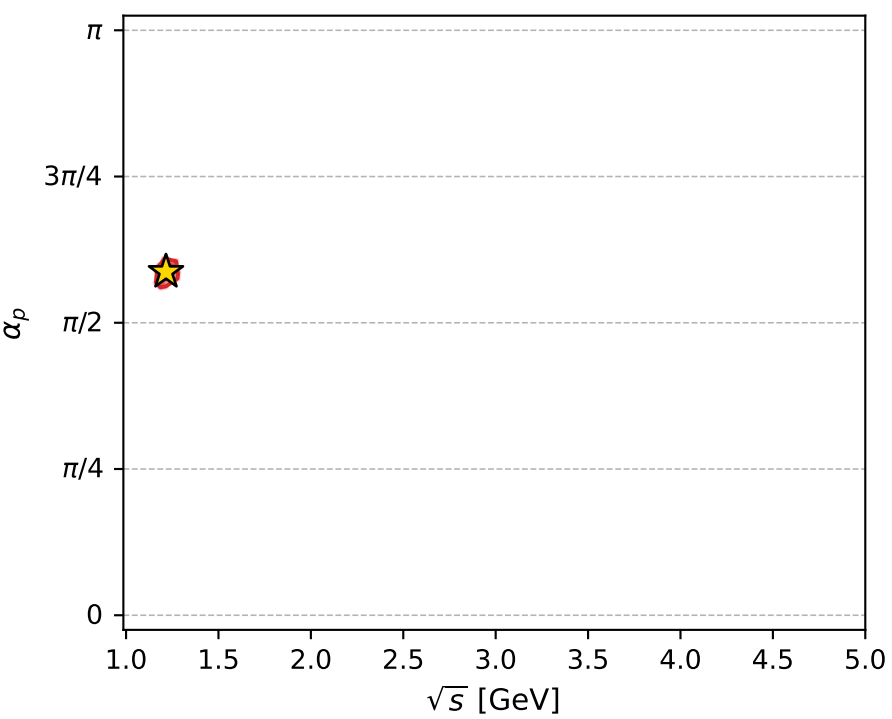
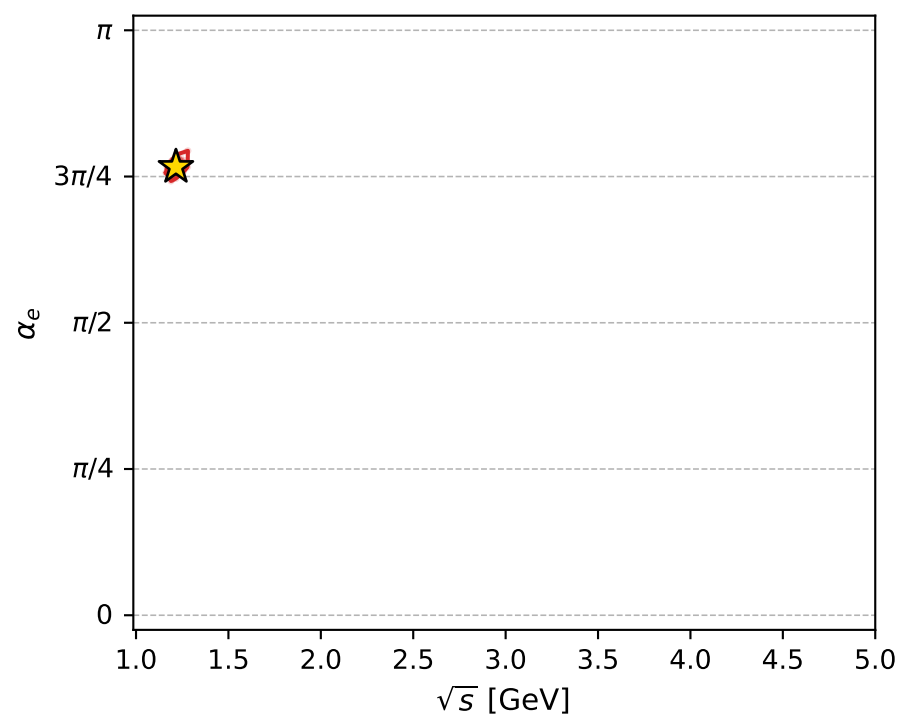
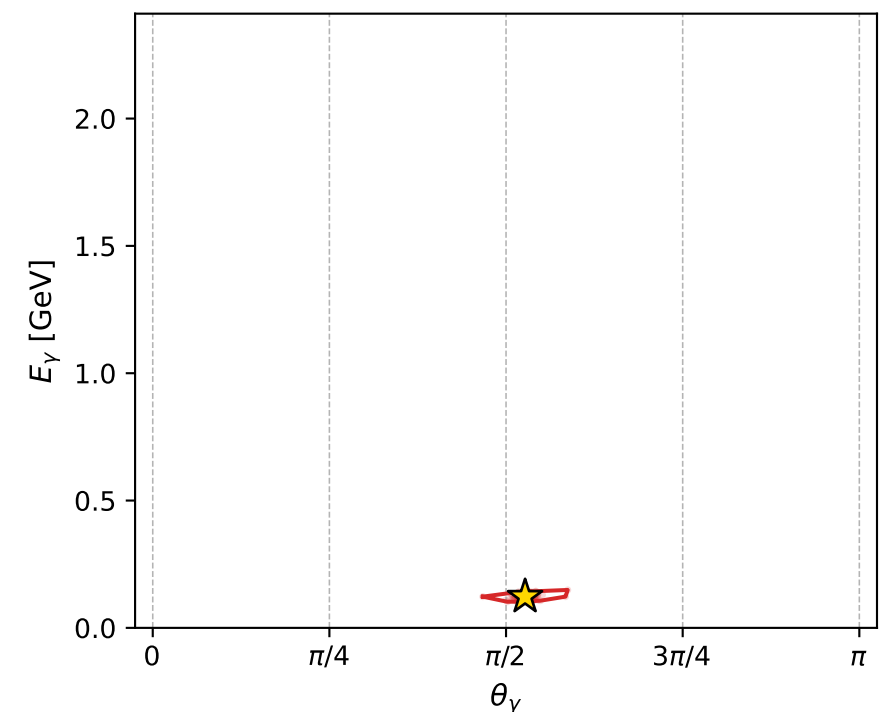
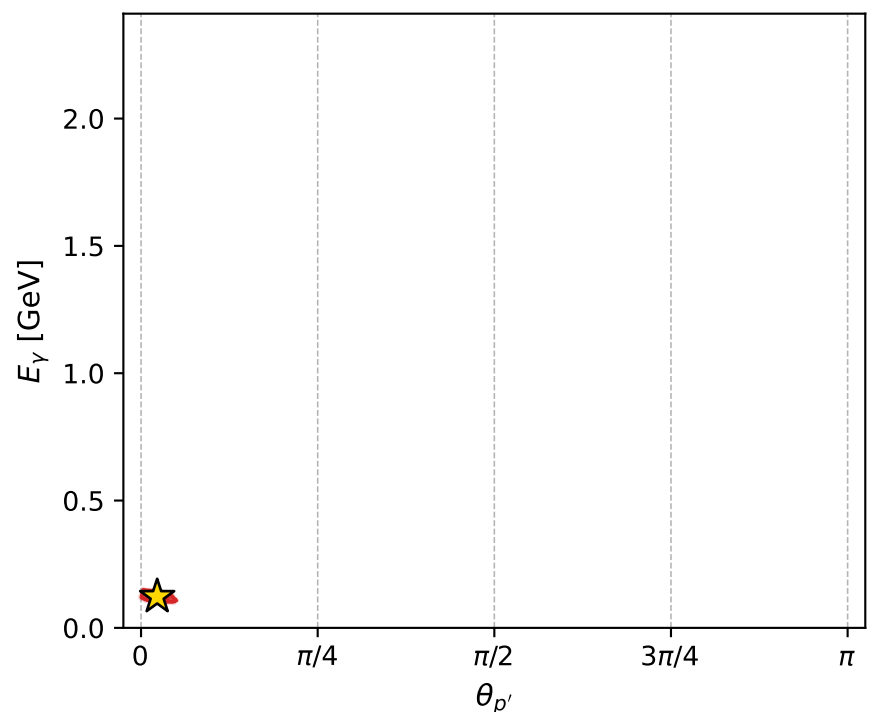
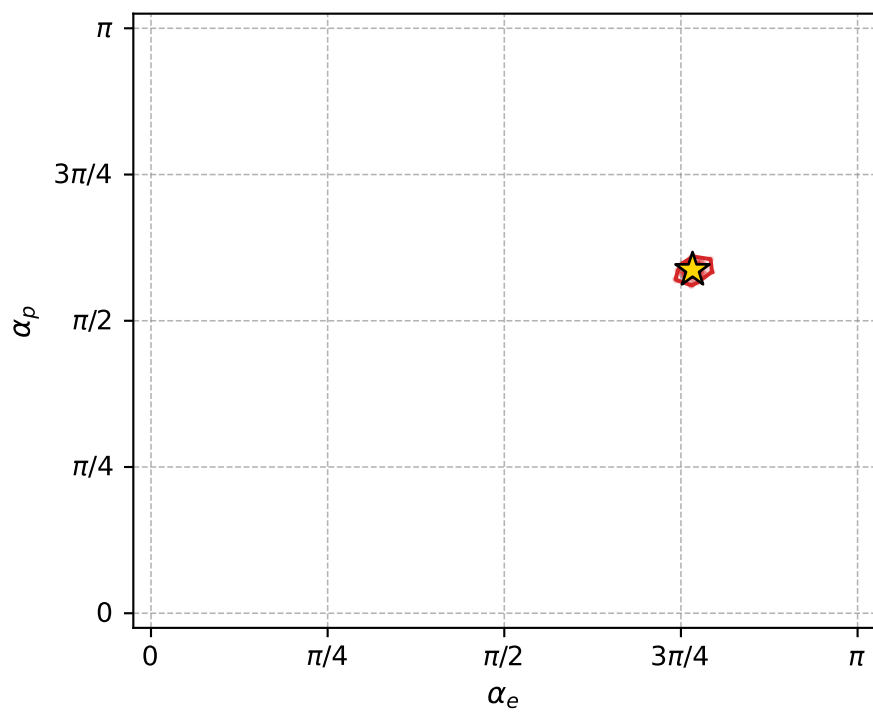
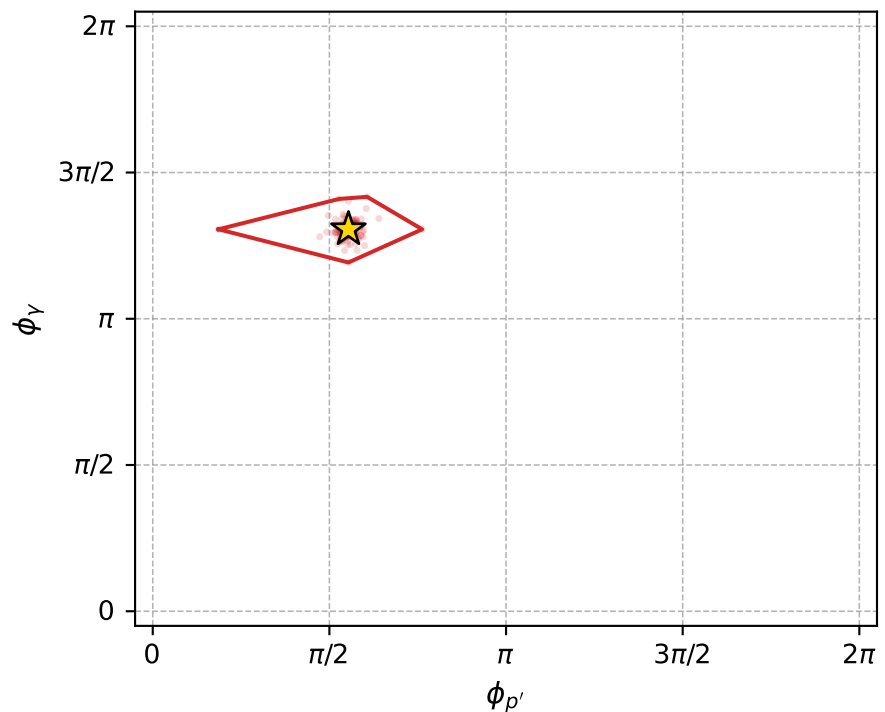
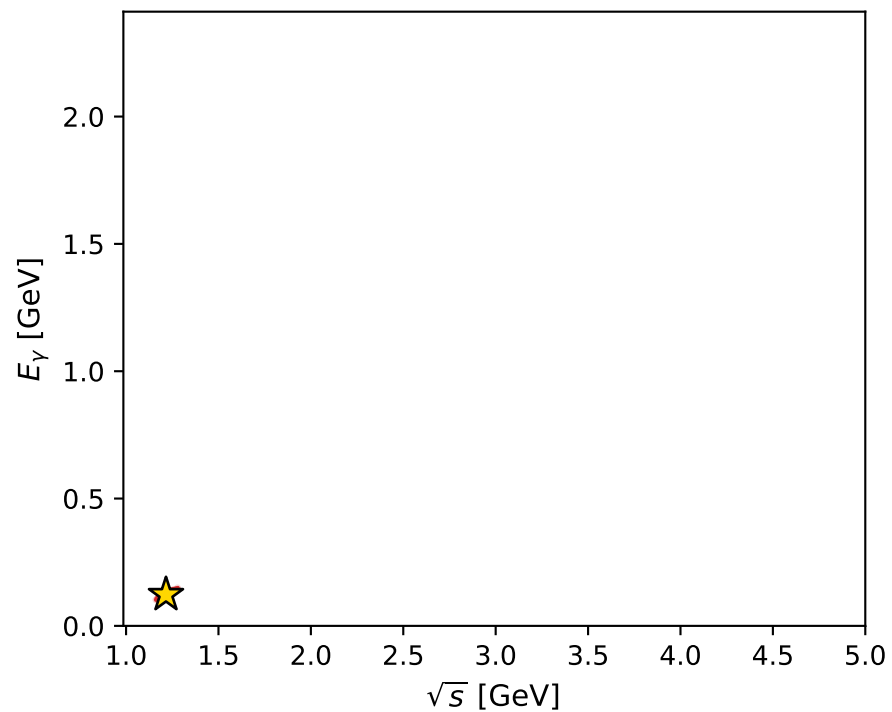
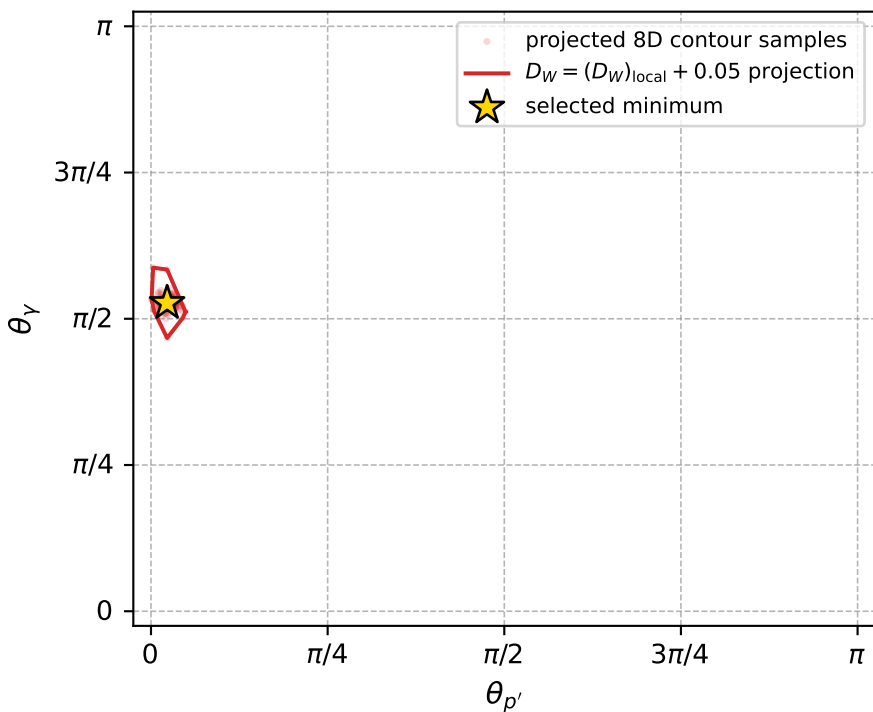
Transverse plane



Leading final-state amplitudes (N=5, retained=97.2%)



electron: polarization cluster P6, local minimum 678; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0063065276$
 $D_W \text{ contour} = 0.056306528$
 above global minimum = 0.0062900948
 8D contour samples = 256

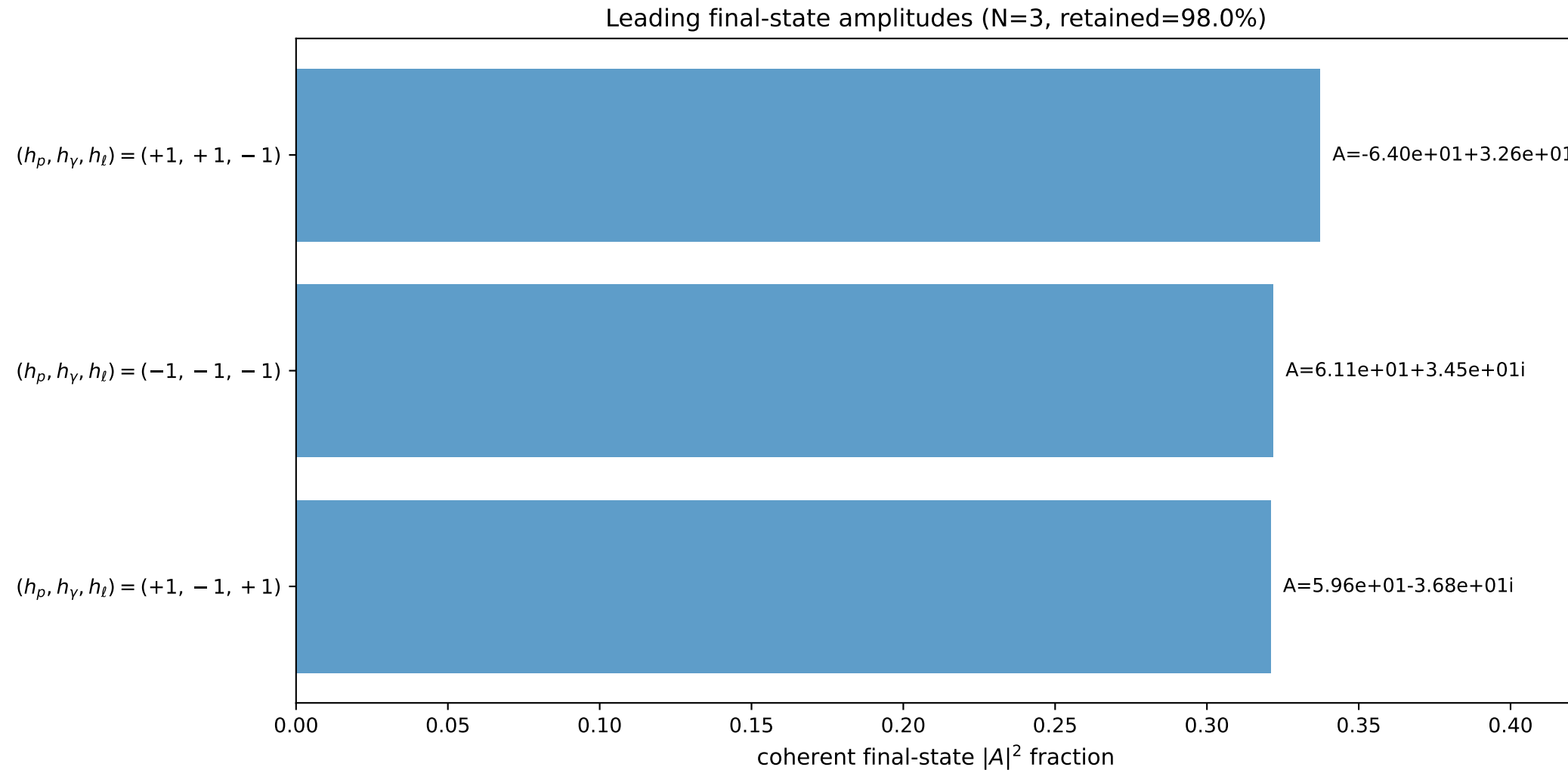
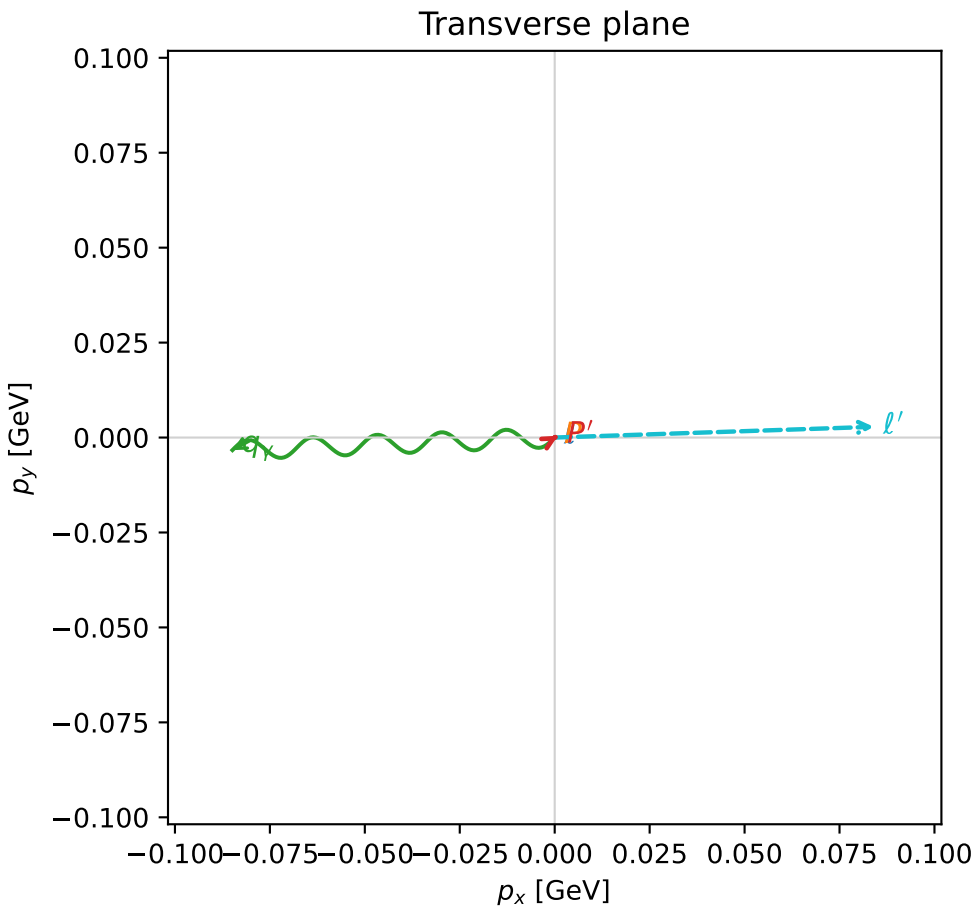
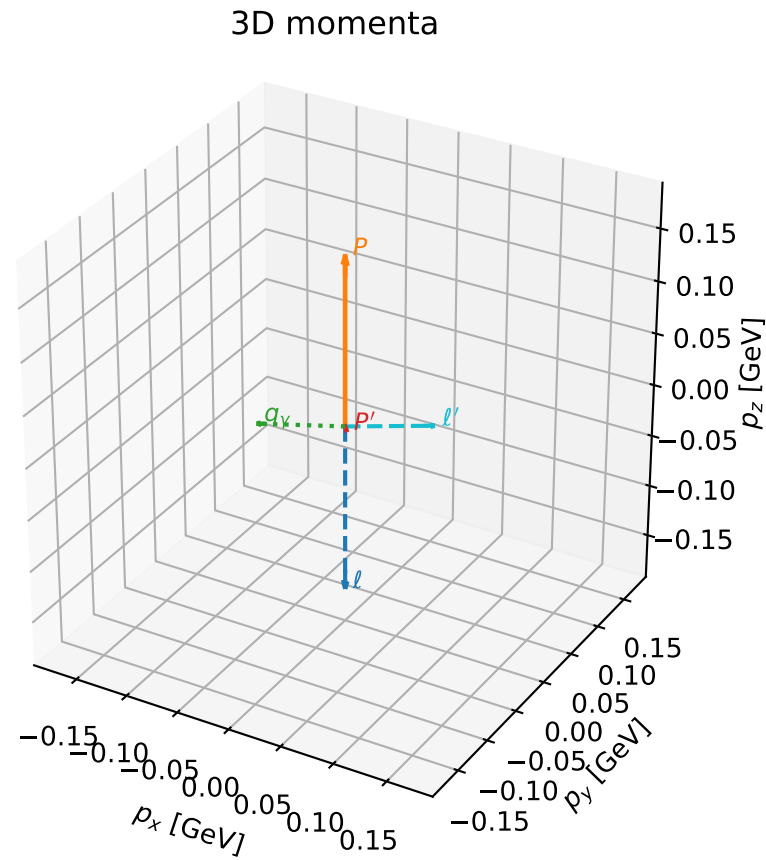
selected phase-space point:
 $\text{sqrt_s} = 1.216026$
 $\text{theta_p_out} = 0.07167669$
 $\text{theta_gamma_out} = 1.655481$
 $\text{qOut} = 0.1226046$
 $\text{phi_p_out} = 1.740142$
 $\text{phi_gamma_out} = 4.102565$
 $\text{alpha_e} = 2.40779$
 $\text{alpha_p} = 1.844214$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

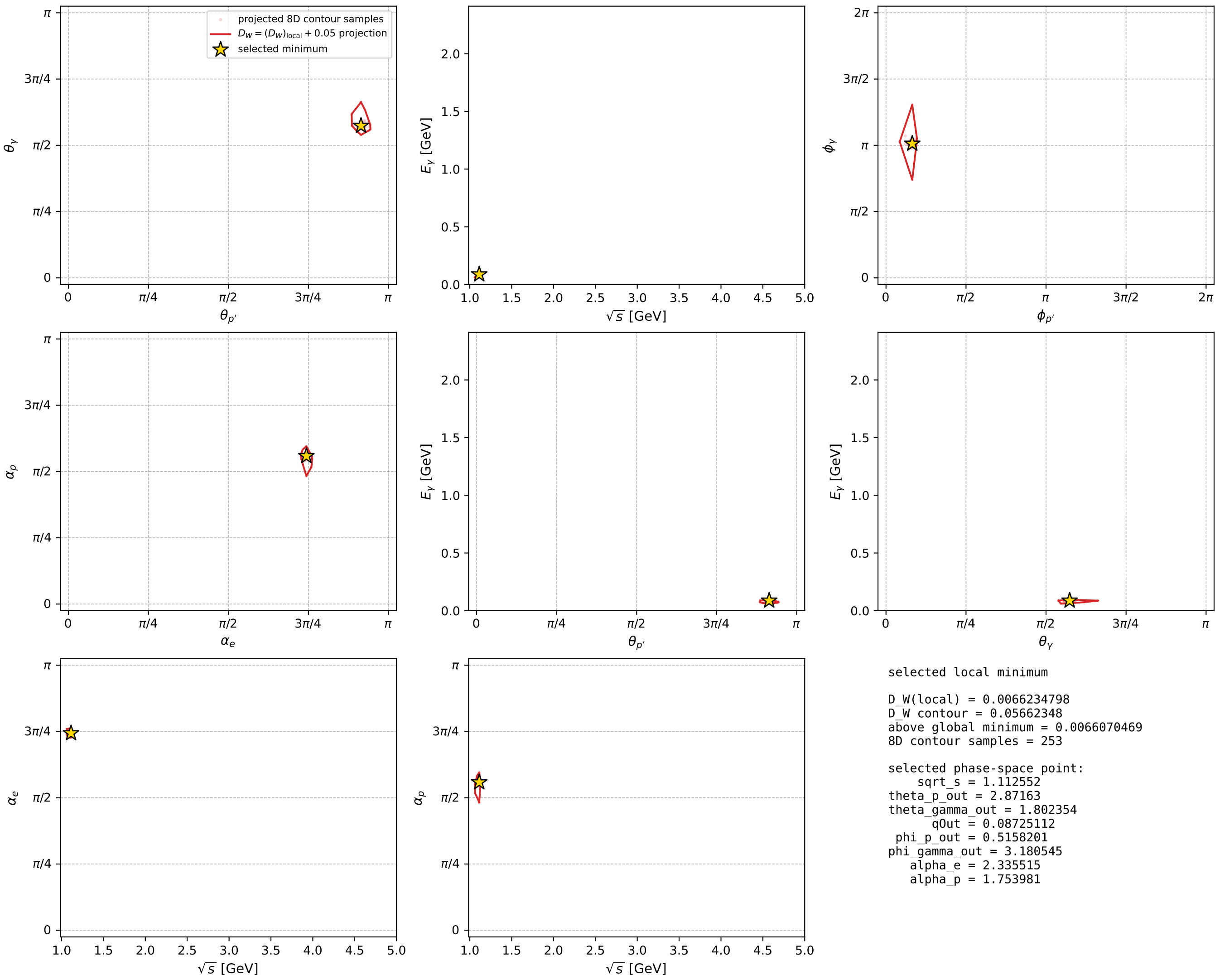
dw_mixing_angles_polarization_06_local_minimum_687 D_W=0.00662348
 region: polarization_cluster_06_local_minimum_687
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3355152$ rad
 incoming lepton: $-0.692334|+> +0.721577|->$
 proton mixing angle: $\alpha_p=1.7539808$ rad
 incoming proton: $-0.182162|+> +0.983269|->$

$s=1.23777$, $\sqrt{s}=1.11255$, $|\vec{P}|=0.160858$, $|\vec{P}'|=0.003618$
 $\theta_{P'}=2.87163$, $\theta_Y=1.80235$, $\phi_{P'}=0.51582$, $\phi_Y=3.18054$
 $E_Y=0.0872511$, $\theta_{\ell'}=1.2981$, $\phi_{\ell'}=0.0336836$
 $Q^2=0.035647$, $x_B=0.178828$, $t=-0.0268229$, $W^2=1.04353$, $y=0.556922$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1609$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1609$
 P $E= 0.9517$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1609$
 ℓ' $E= 0.0873$ $p_x= 0.0840$ $p_y= 0.0028$ $p_z= 0.0235$
 P' $E= 0.9380$ $p_x= 0.0008$ $p_y= 0.0005$ $p_z= -0.0035$
 q_Y $E= 0.0873$ $p_x= -0.0849$ $p_y= -0.0033$ $p_z= -0.0200$



electron: polarization cluster P6, local minimum 687; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



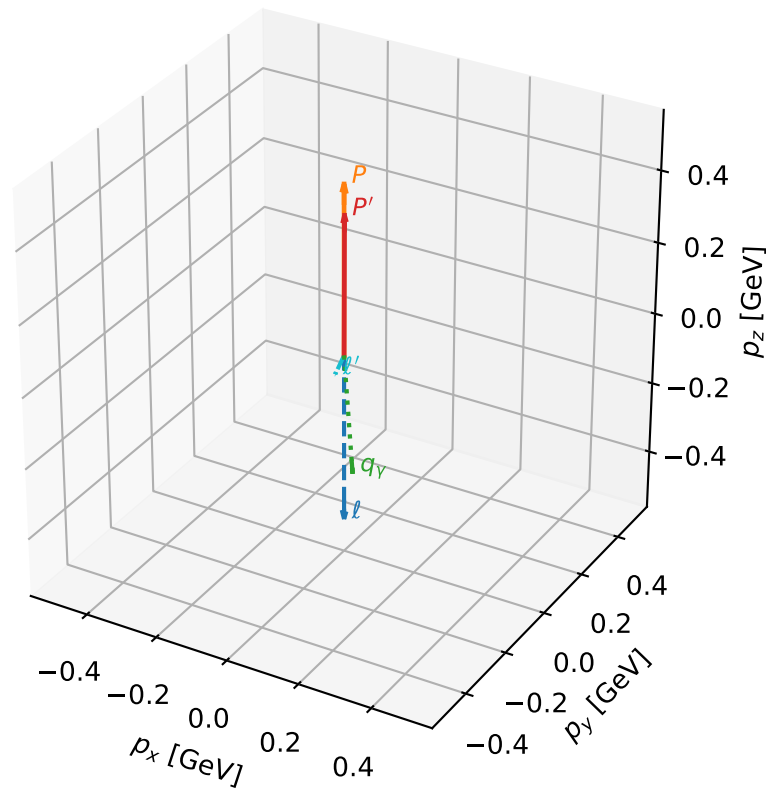
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_690 D_W=0.00683264
 region: polarization_cluster_06_local_minimum_690
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3964457$ rad
 incoming lepton: $-0.734988|+> +0.67808|->$
 proton mixing angle: $\alpha_p=1.3770046$ rad
 incoming proton: $0.192581|+> +0.981281|->$

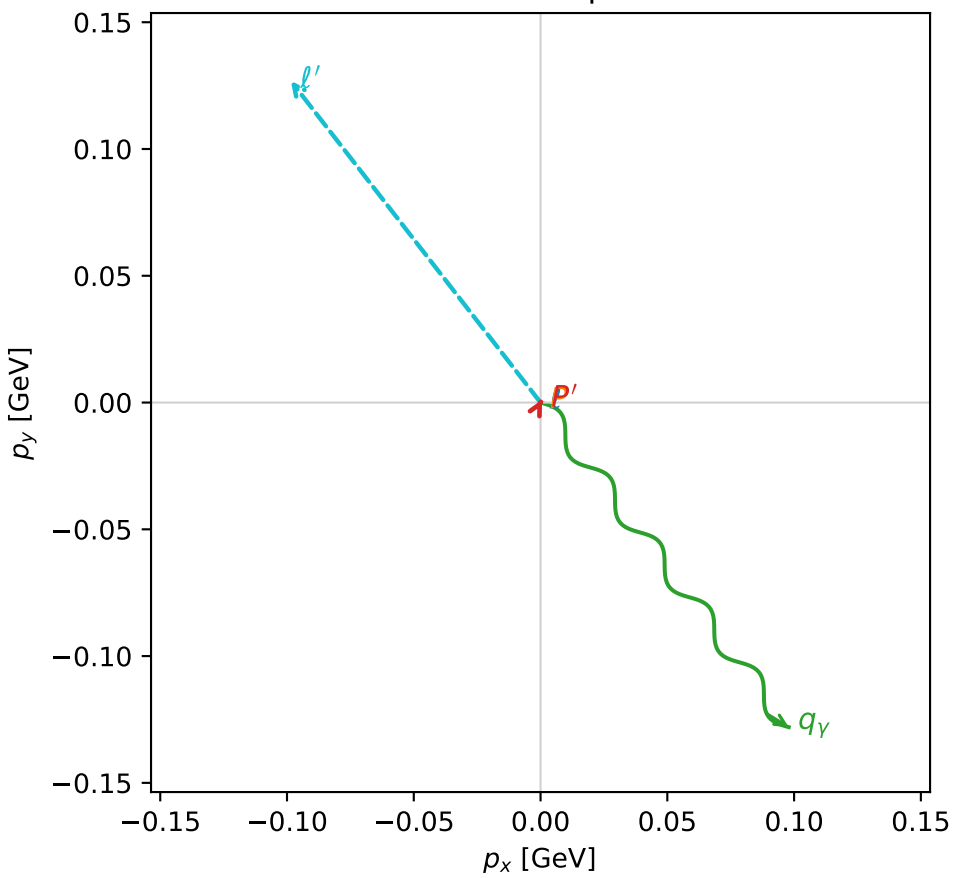
$s=2.31316$, $\sqrt{s}=1.52091$, $|\vec{P}|=0.471204$, $|\vec{P}'|=0.386384$
 $\theta_{p'}=0.00339625$, $\theta_Y=2.55087$, $\phi_{p'}=1.08743$, $\phi_Y=5.36517$
 $E_Y=0.289348$, $\theta_{l'}=2.30881$, $\phi_{l'}=2.23099$
 $Q^2=0.066938$, $x_B=0.0796986$, $t=-0.00595473$, $W^2=1.6528$, $y=0.585977$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.4712$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4712$
 P $E= 1.0497$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4712$
 l' $E= 0.2171$ $p_x= -0.0985$ $p_y= 0.1269$ $p_z= -0.1461$
 P' $E= 1.0145$ $p_x= 0.0006$ $p_y= 0.0012$ $p_z= 0.3864$
 q_Y $E= 0.2893$ $p_x= 0.0979$ $p_y= -0.1280$ $p_z= -0.2403$

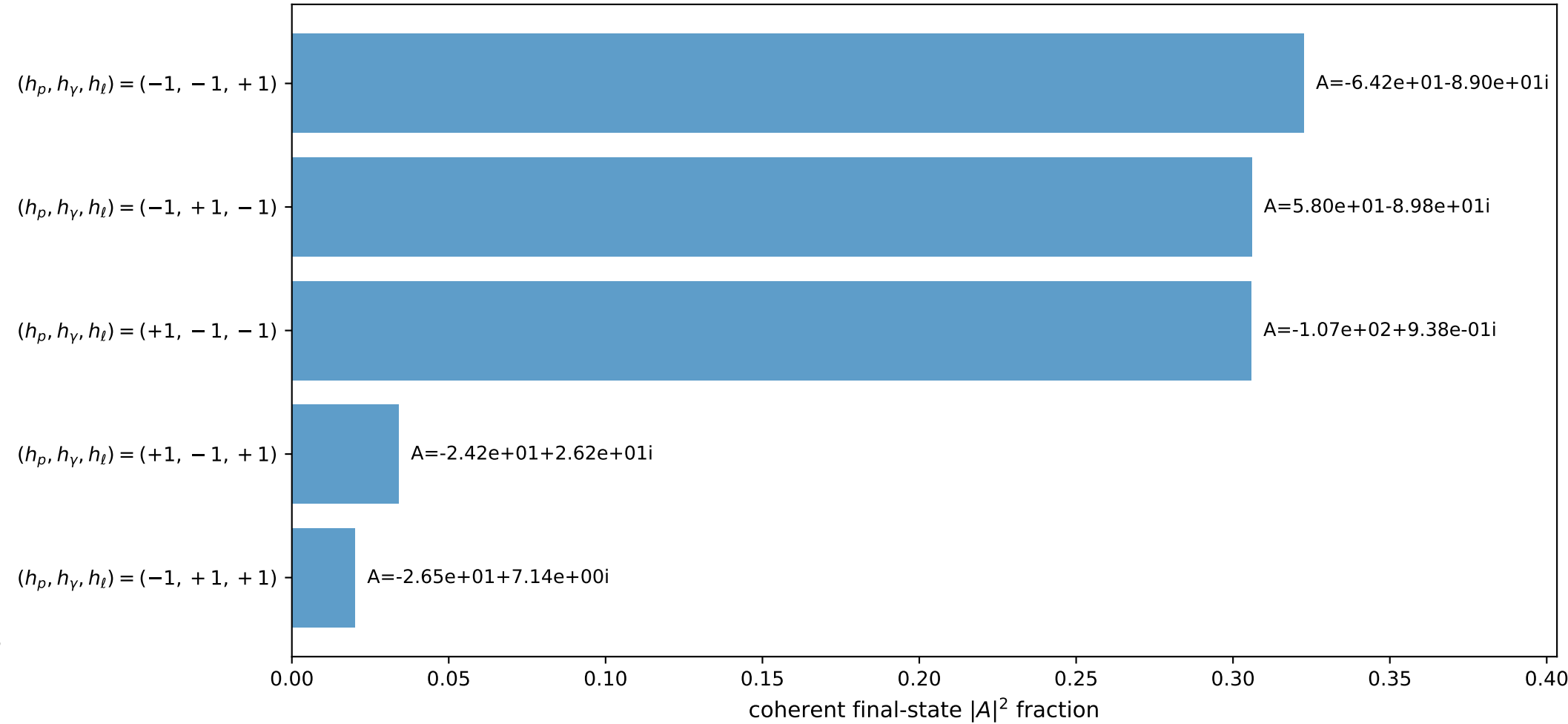
3D momenta



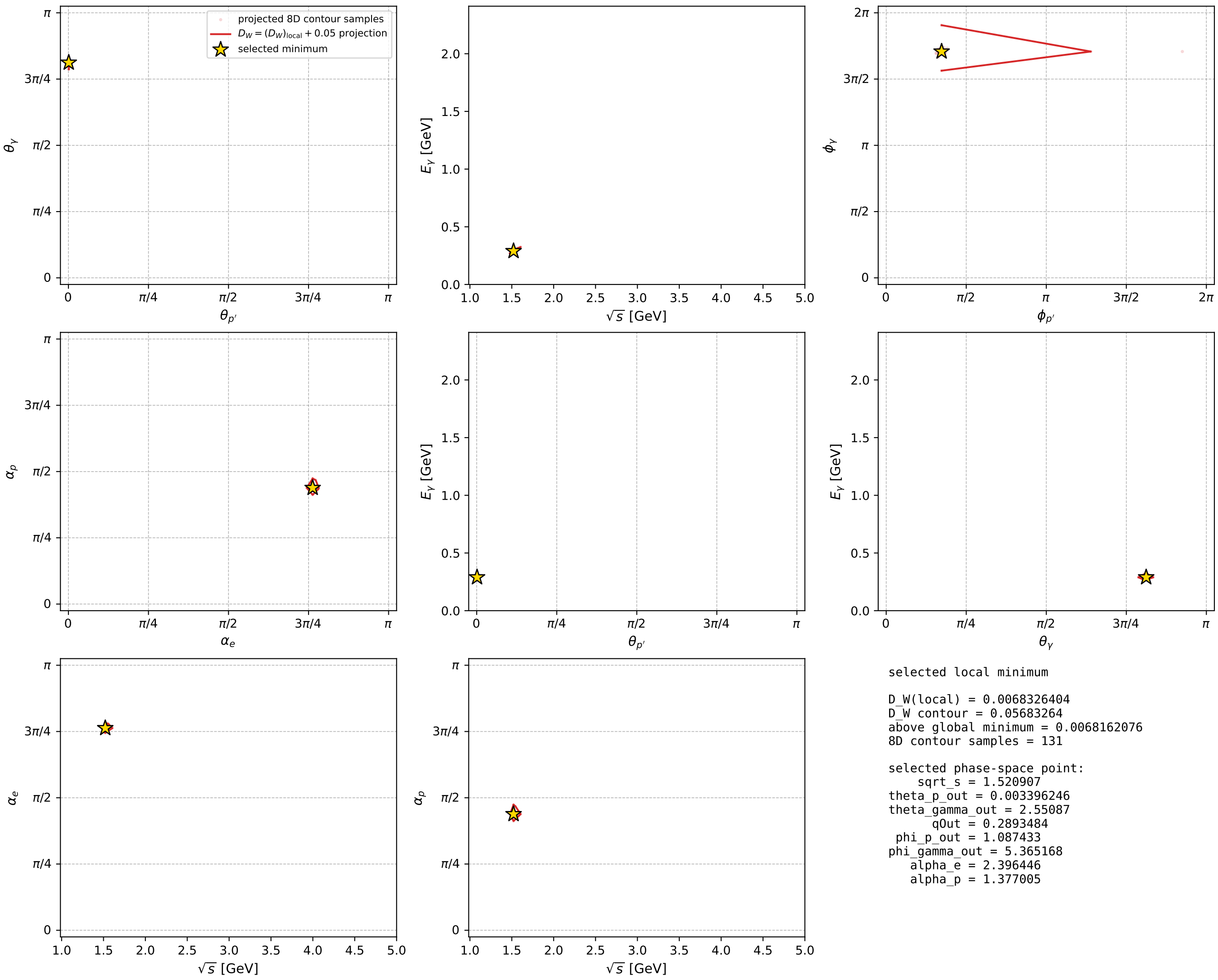
Transverse plane



Leading final-state amplitudes (N=5, retained=98.9%)



electron: polarization cluster P6, local minimum 690; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

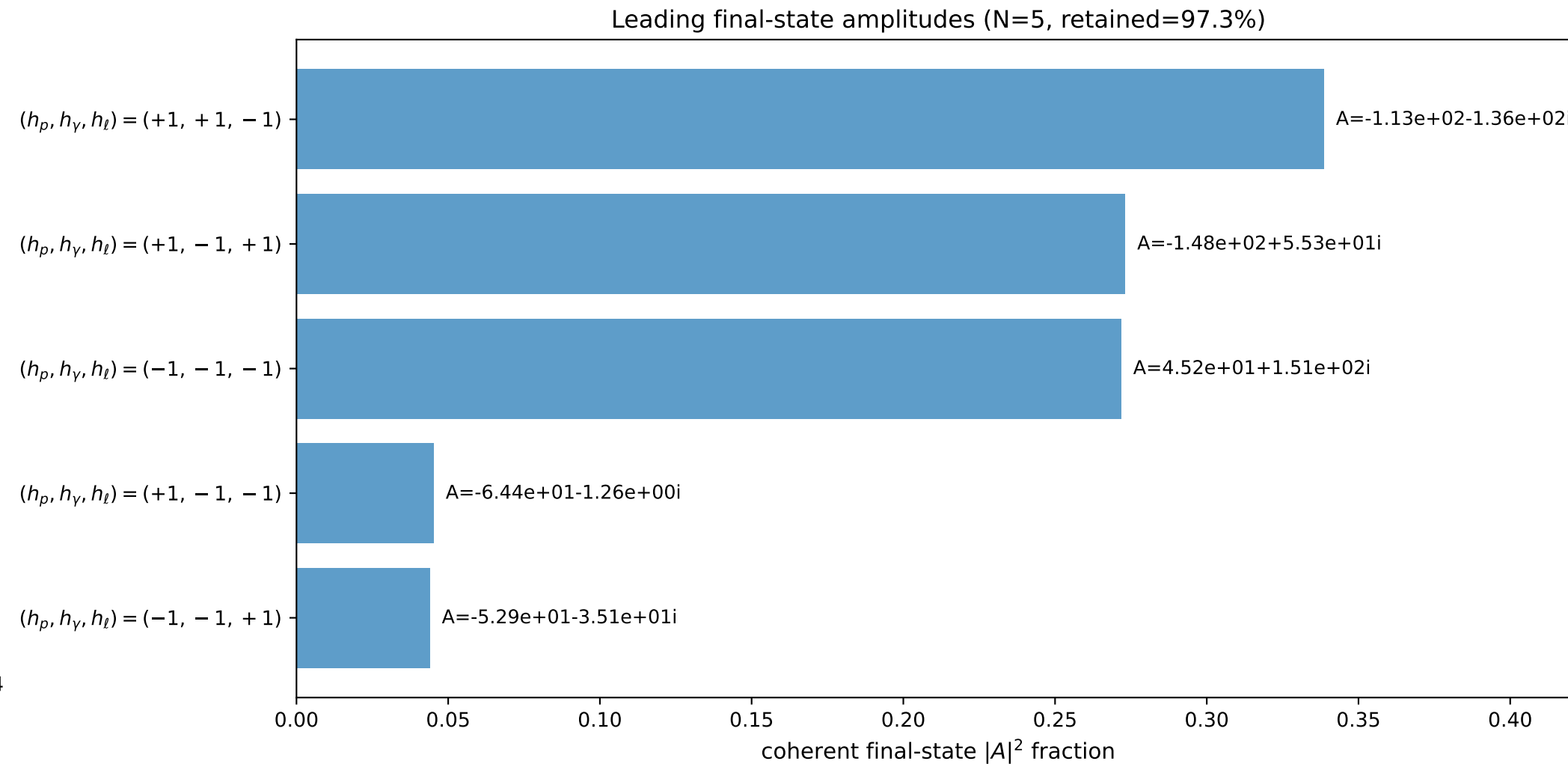
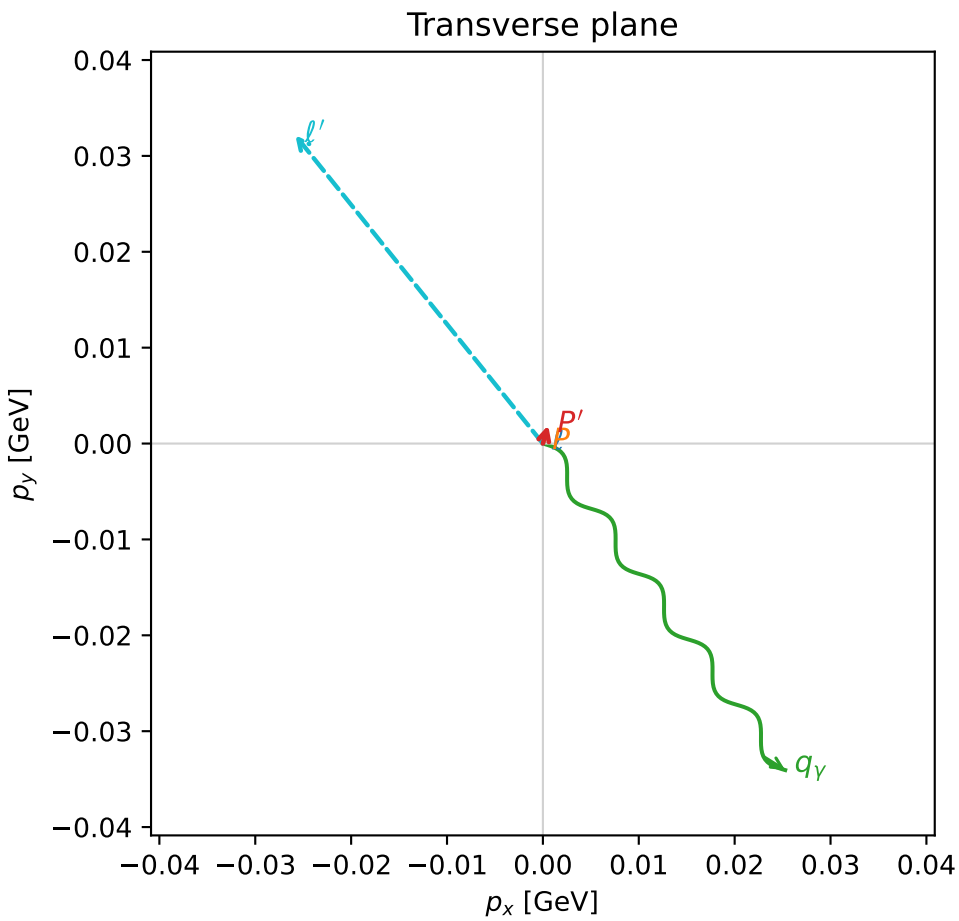
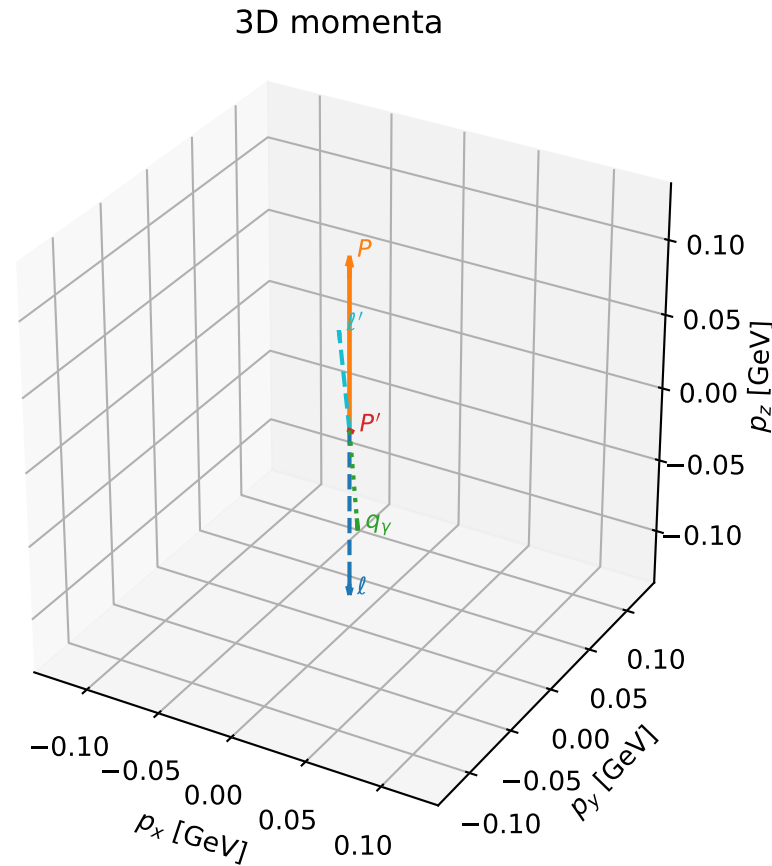


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

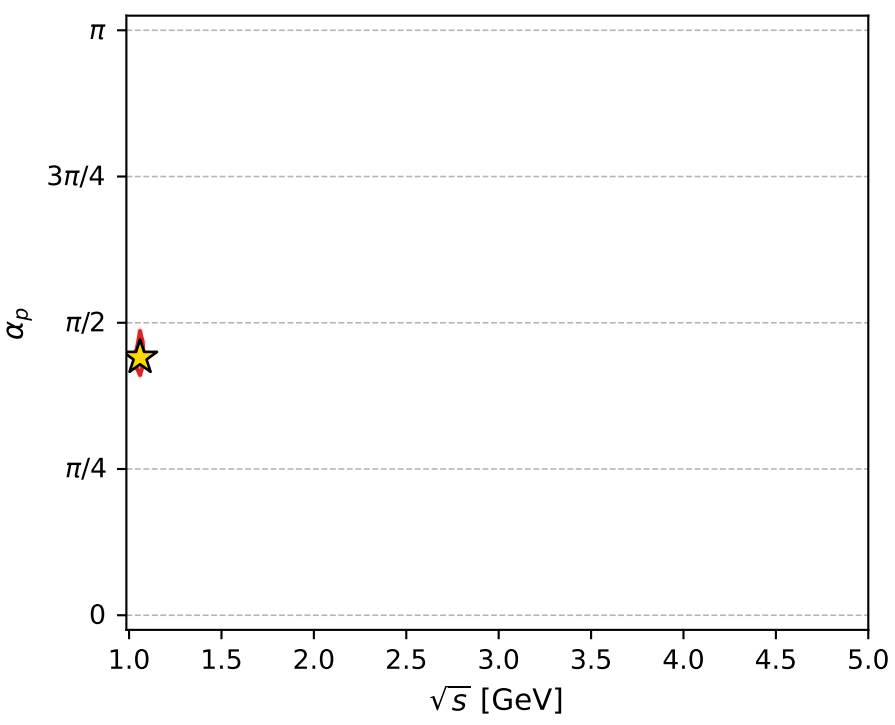
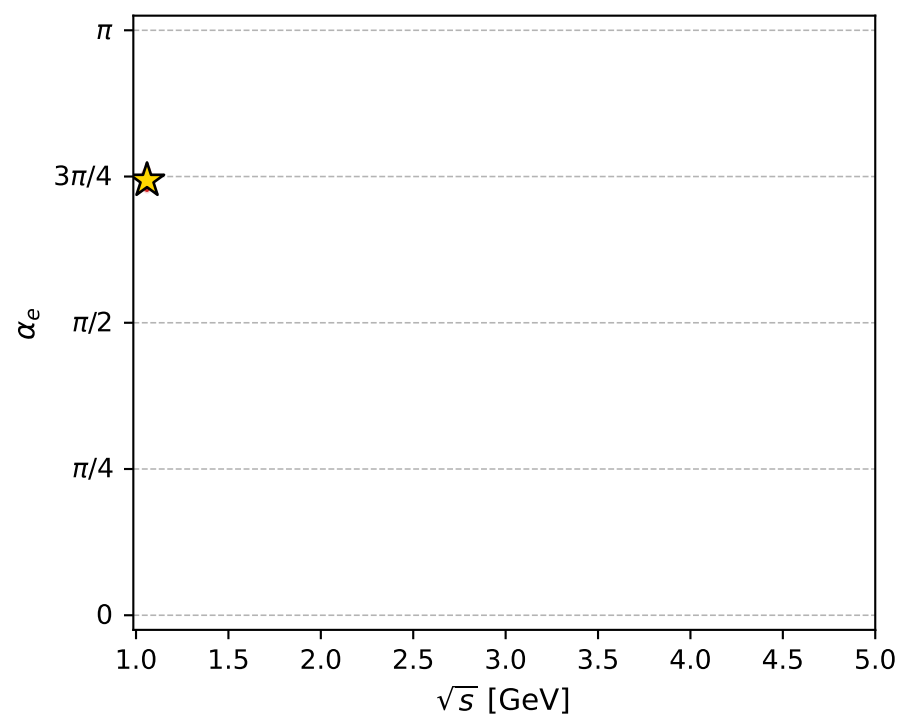
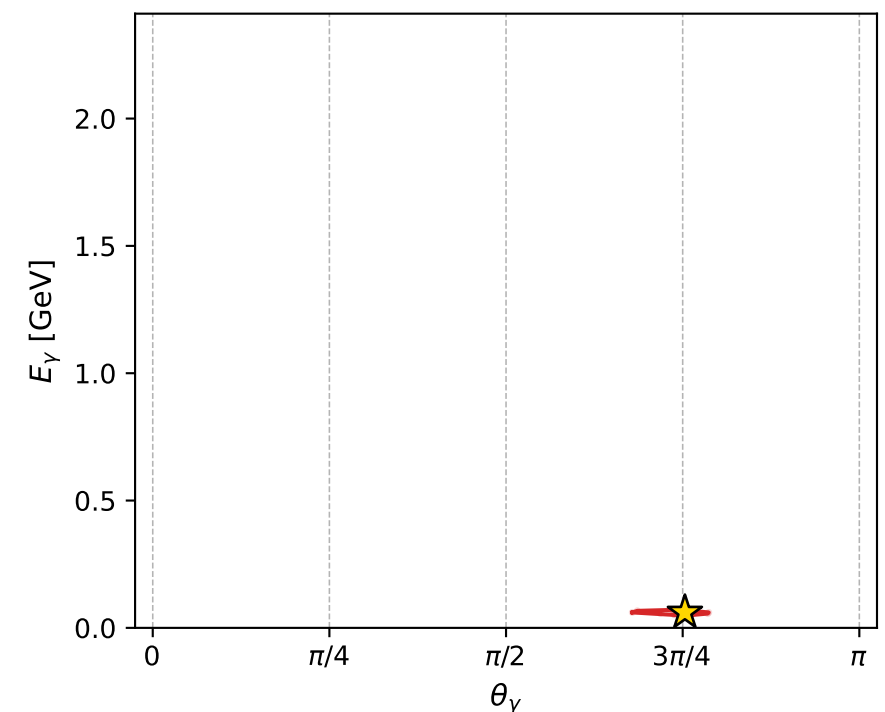
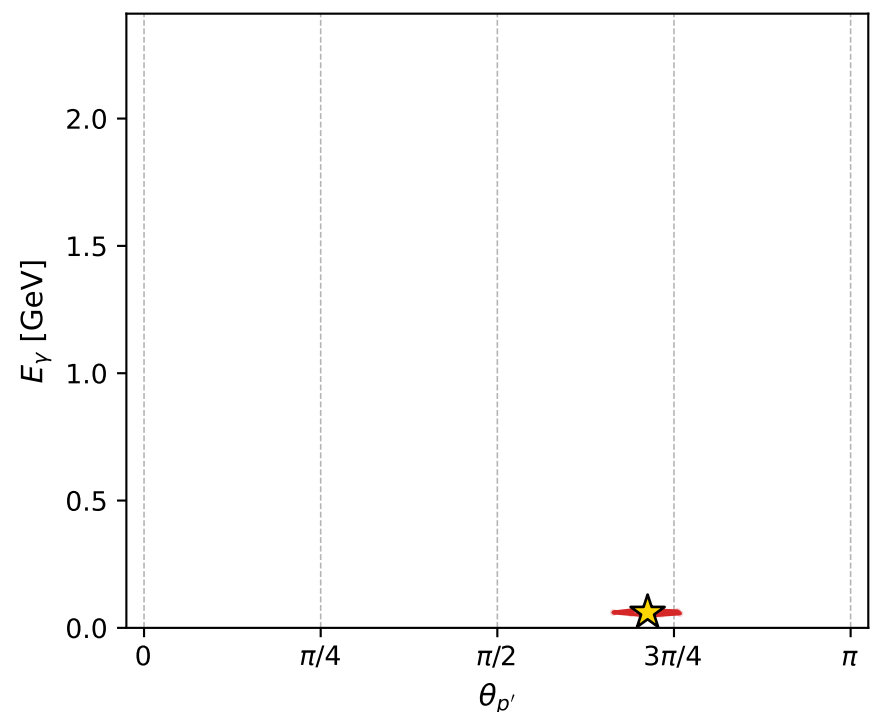
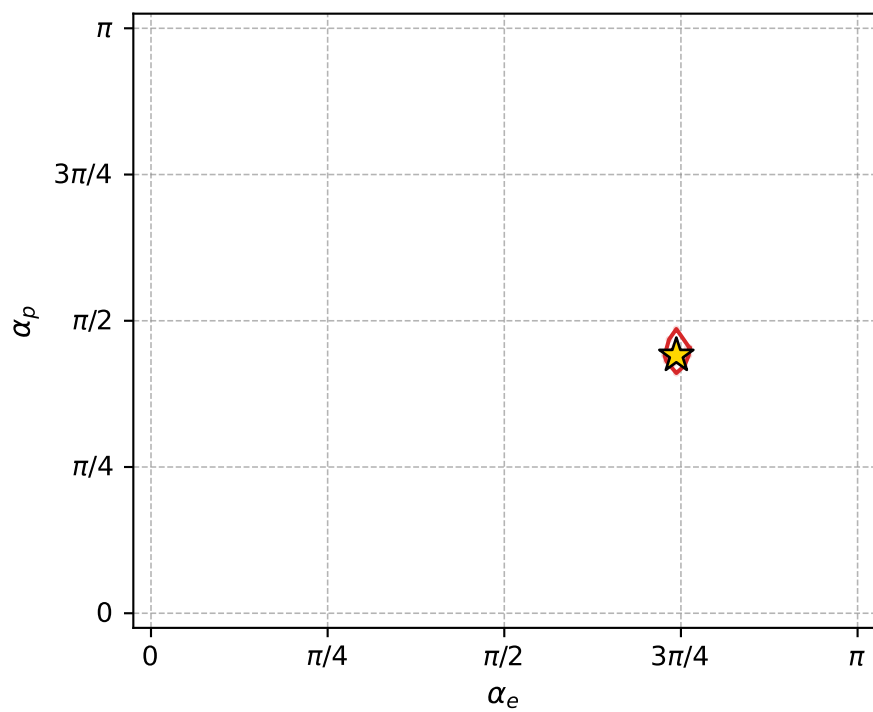
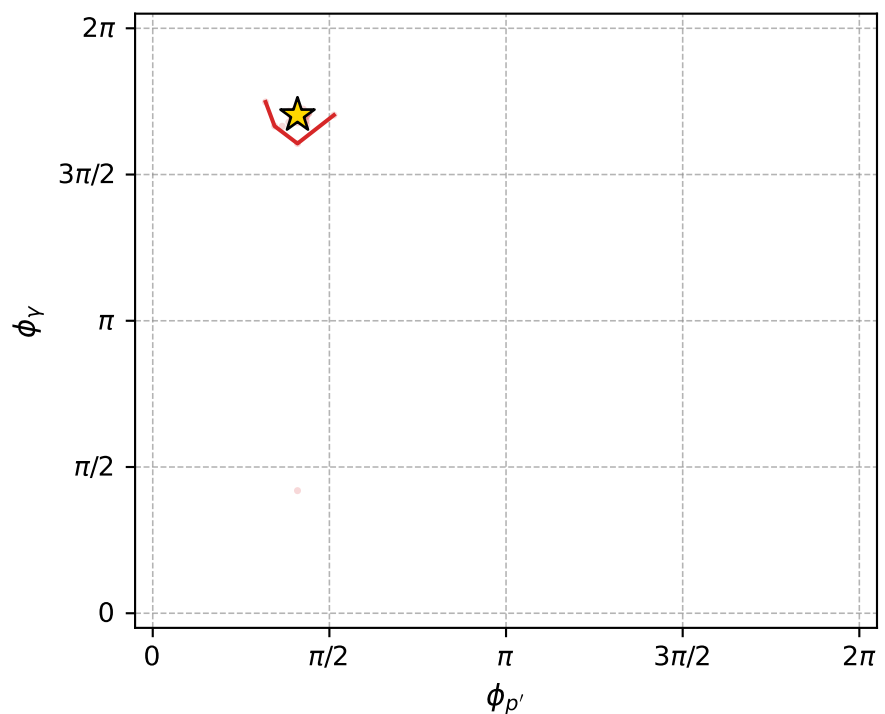
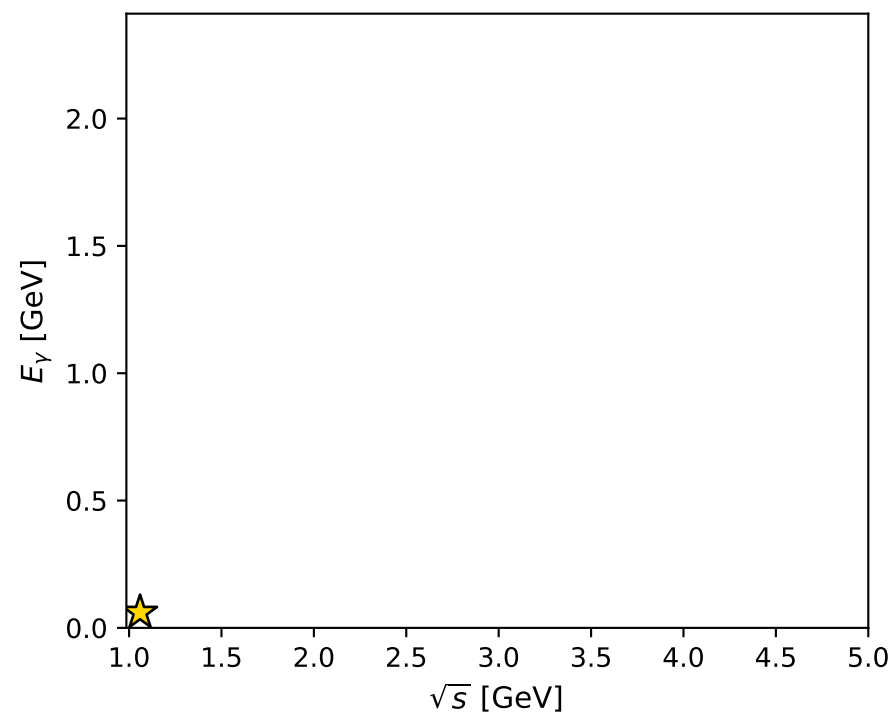
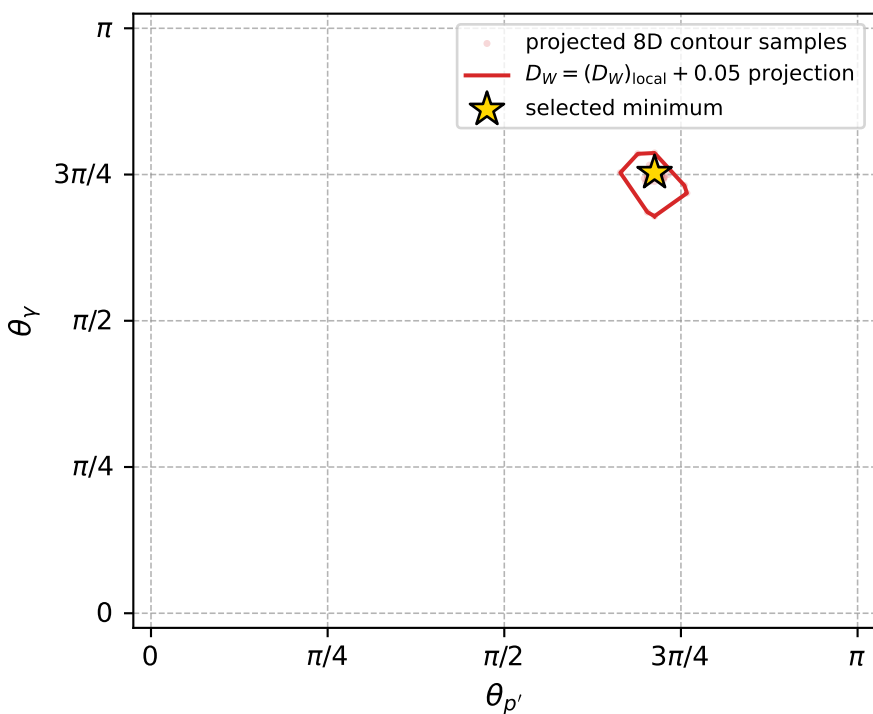
dw_mixing_angles_polarization_06_local_minimum_697 D_W=0.00709047
 region: polarization_cluster_06_local_minimum_697
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3357721$ rad
 incoming lepton: $-0.69252|+> +0.721399|->$
 proton mixing angle: $\alpha_p=1.3847329$ rad
 incoming proton: $0.184992|+> +0.98274|->$

$s=1.12252$, $\sqrt{s}=1.05949$, $|\vec{P}|=0.114526$, $|\vec{P}'|=0.00252257$
 $\theta_{P'}=2.23892$, $\theta_Y=2.36606$, $\phi_{P'}=1.2871$, $\phi_Y=5.35078$
 $E_Y=0.0605786$, $\theta_{l'}=0.743891$, $\phi_{l'}=2.24746$
 $Q^2=0.0242172$, $x_B=0.175704$, $t=-0.013432$, $W^2=0.993456$, $y=0.567945$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1145$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1145$
 P $E= 0.9450$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1145$
 l' $E= 0.0609$ $p_x= -0.0258$ $p_y= 0.0322$ $p_z= 0.0448$
 P' $E= 0.9380$ $p_x= 0.0006$ $p_y= 0.0019$ $p_z= -0.0016$
 q_Y $E= 0.0606$ $p_x= 0.0253$ $p_y= -0.0341$ $p_z= -0.0433$



electron: polarization cluster P6, local minimum 697; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0070904672$
 $D_W \text{ contour} = 0.057090467$
 above global minimum = 0.0070740343
 8D contour samples = 254

selected phase-space point:
 $\text{sqrt_s} = 1.059493$
 $\text{theta_p_out} = 2.238924$
 $\text{theta_gamma_out} = 2.366062$
 $\text{qOut} = 0.06057857$
 $\text{phi_p_out} = 1.287098$
 $\text{phi_gamma_out} = 5.350784$
 $\text{alpha_e} = 2.335772$
 $\text{alpha_p} = 1.384733$

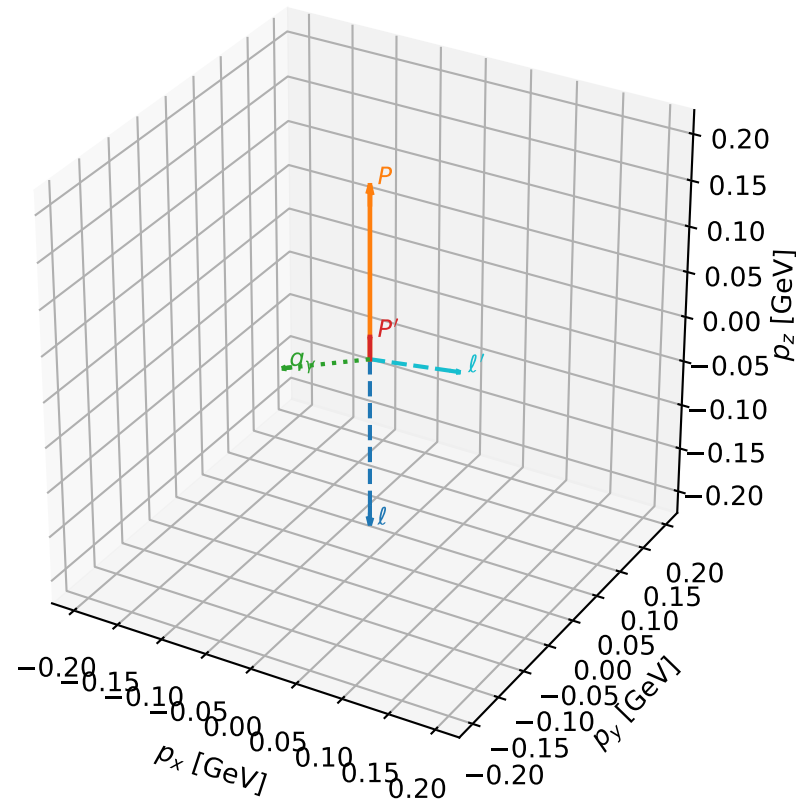
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_705 D_W=0.00745435
 region: polarization_cluster_06_local_minimum_705
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3477982$ rad
 incoming lepton: $-0.701145|+> +0.713019|->$
 proton mixing angle: $\alpha_p=1.7659151$ rad
 incoming proton: $-0.193883|+> +0.981025|->$

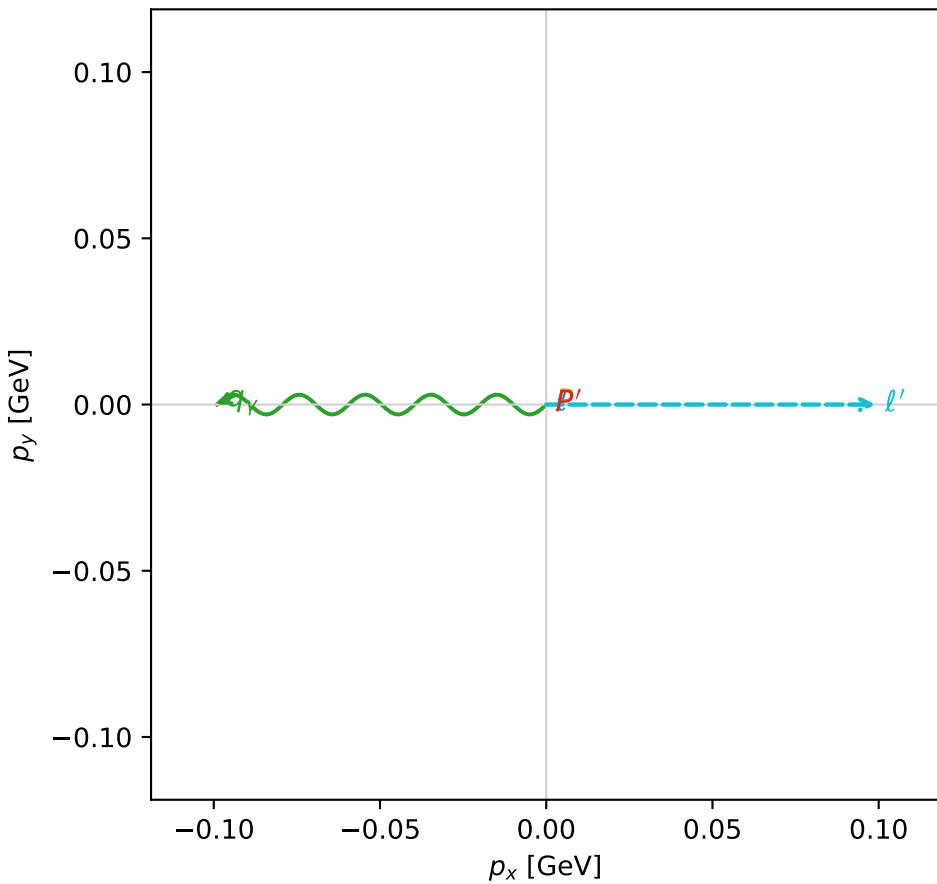
$s=1.31067$, $\sqrt{s}=1.14485$, $|\vec{P}|=0.18816$, $|\vec{P}'|=0.0245489$
 $\theta_{P'}=0$, $\theta_V=1.94491$, $\phi_{P'}=3.06061$, $\phi_V=3.14158$
 $E_V=0.106426$, $\theta_{\ell'}=1.427$, $\phi_{\ell'}=6.28317$
 $Q^2=0.0430666$, $x_B=0.175998$, $t=-0.0264312$, $W^2=1.08148$, $y=0.567974$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1882$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1882$
 P $E= 0.9567$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1882$
 ℓ' $E= 0.1001$ $p_x= 0.0991$ $p_y= -0.0000$ $p_z= 0.0143$
 P' $E= 0.9383$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0245$
 q_Y $E= 0.1064$ $p_x= -0.0991$ $p_y= 0.0000$ $p_z= -0.0389$

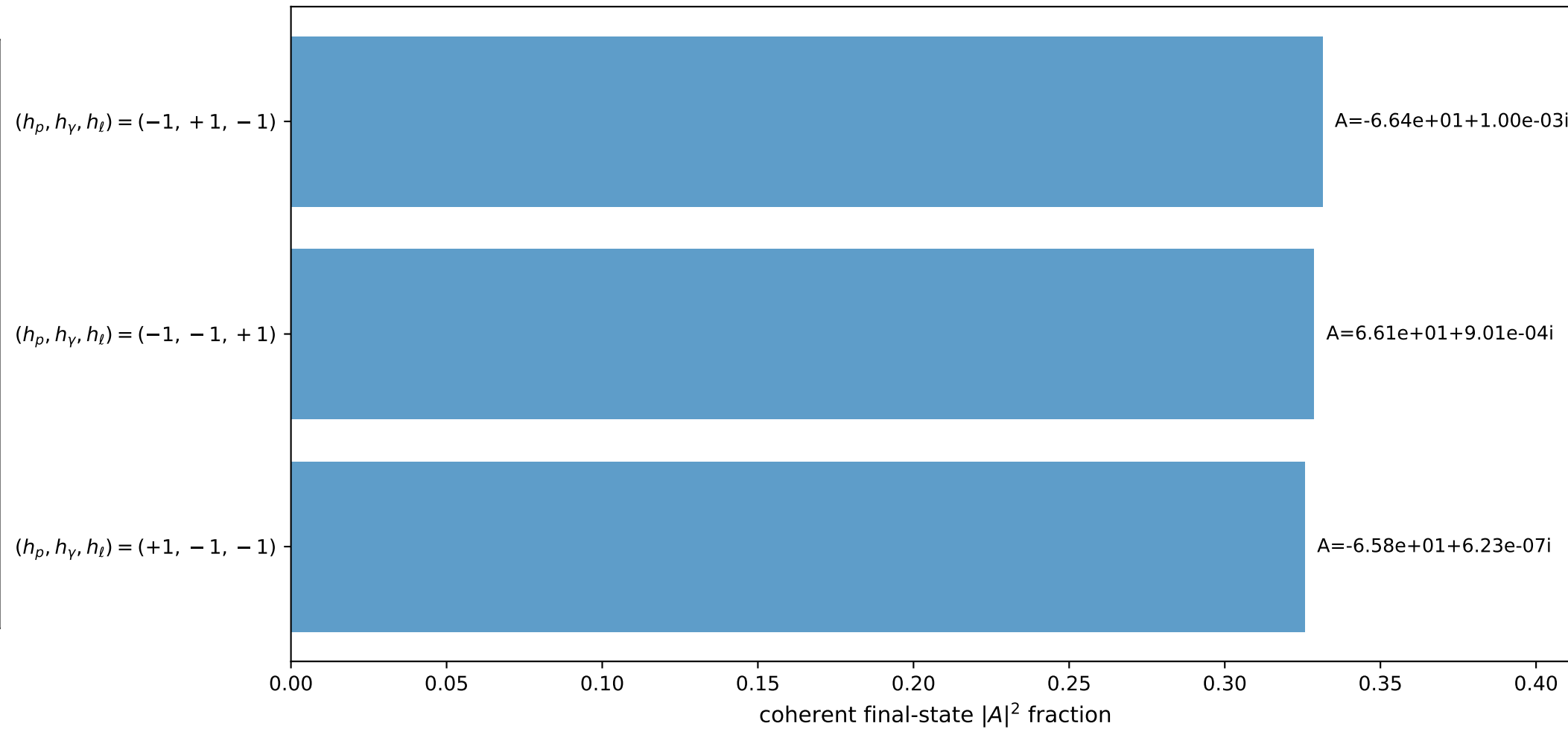
3D momenta



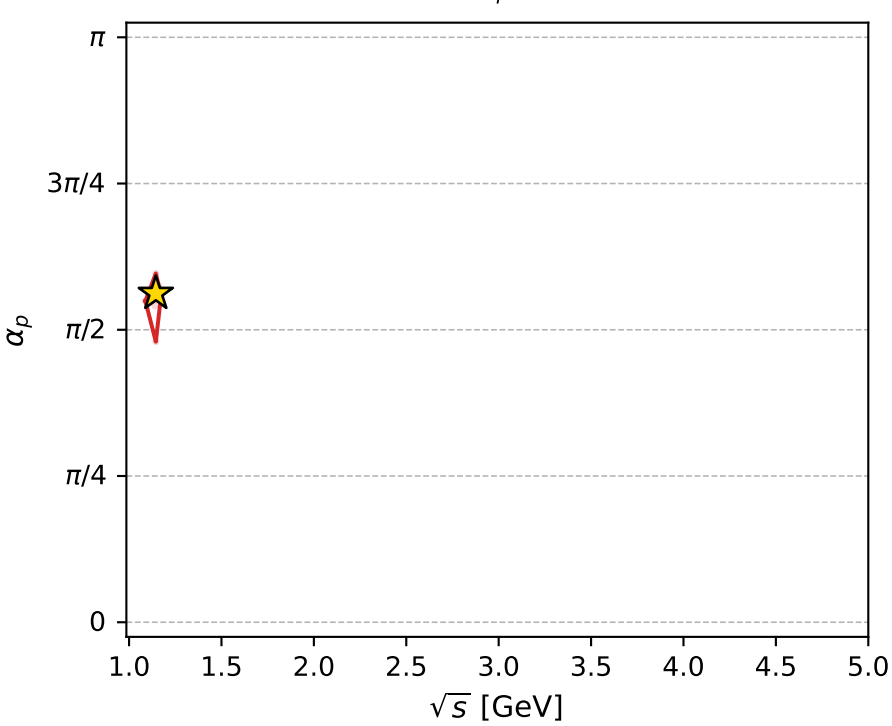
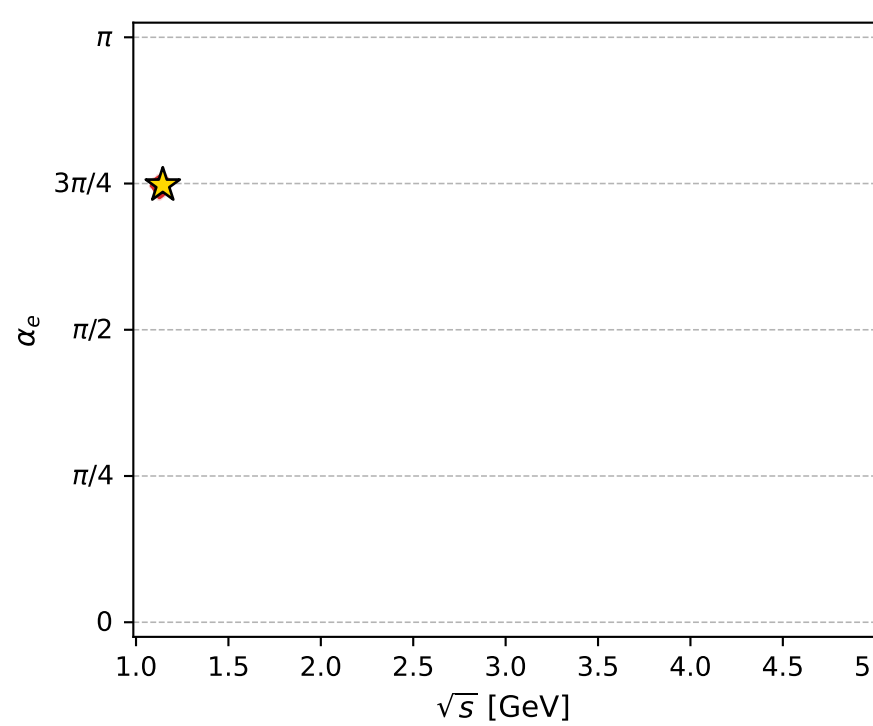
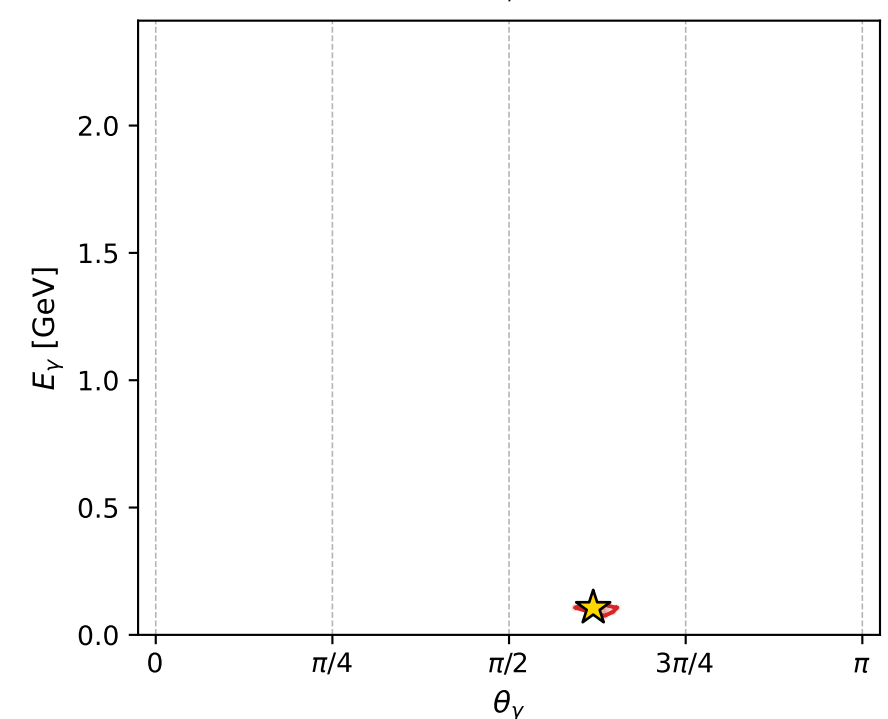
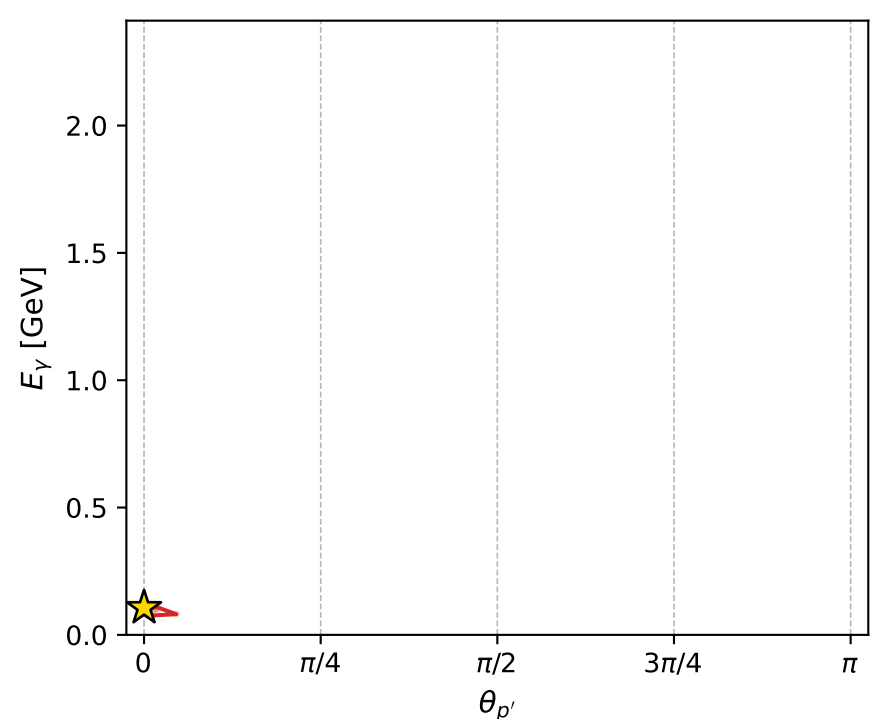
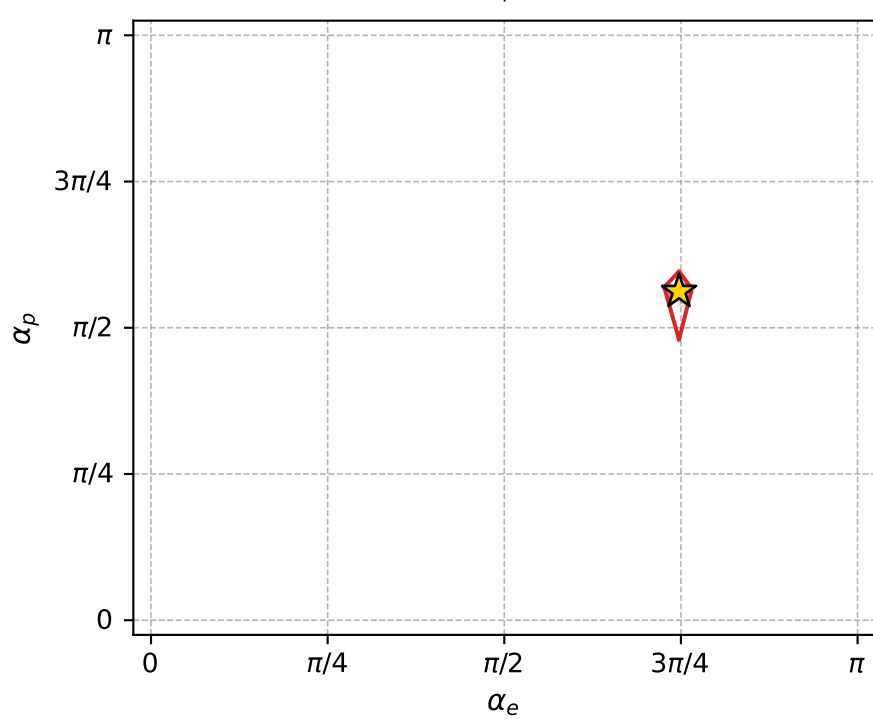
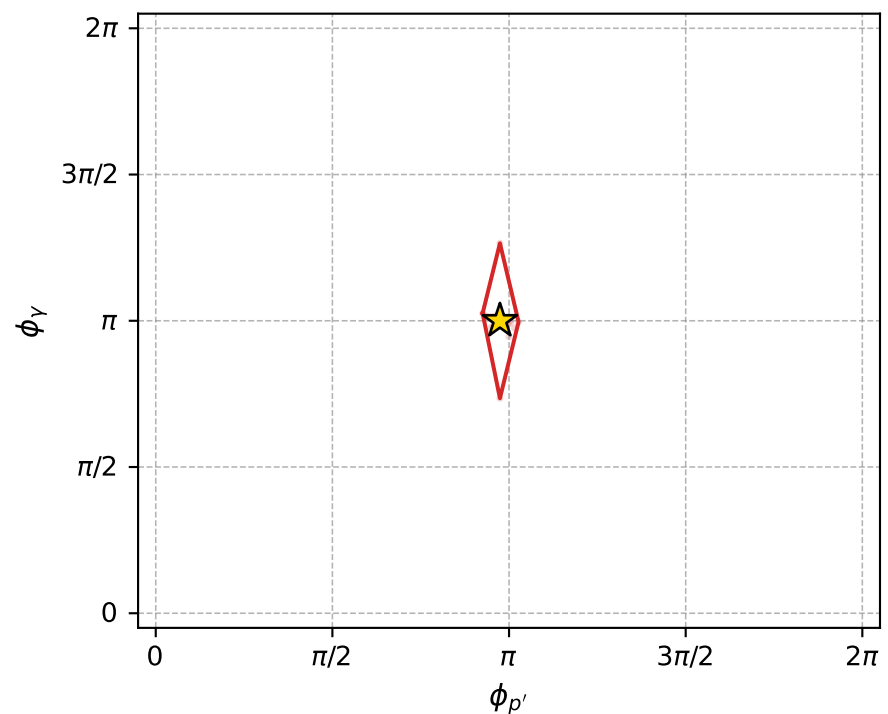
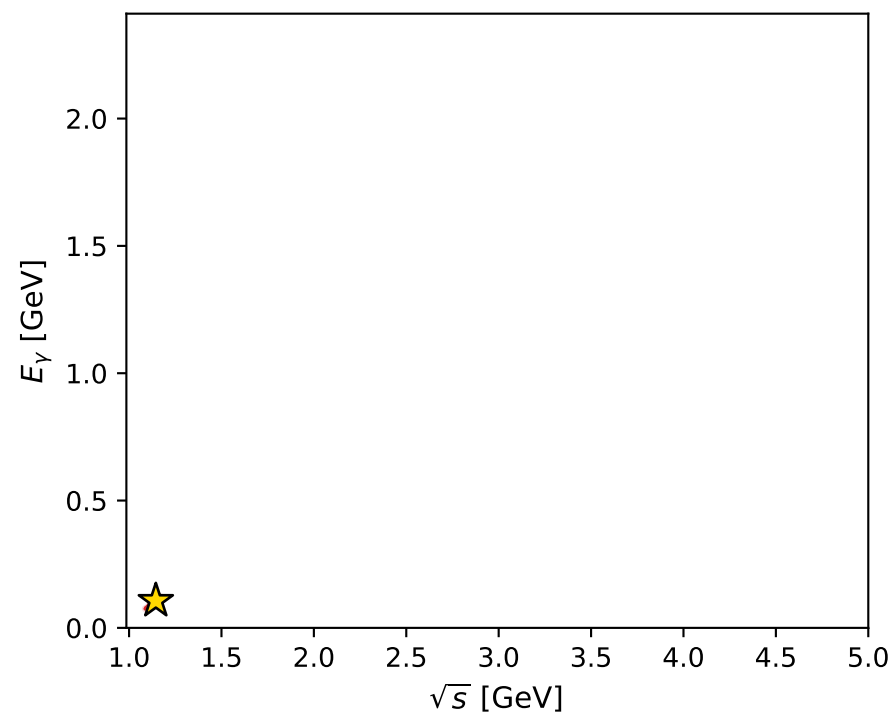
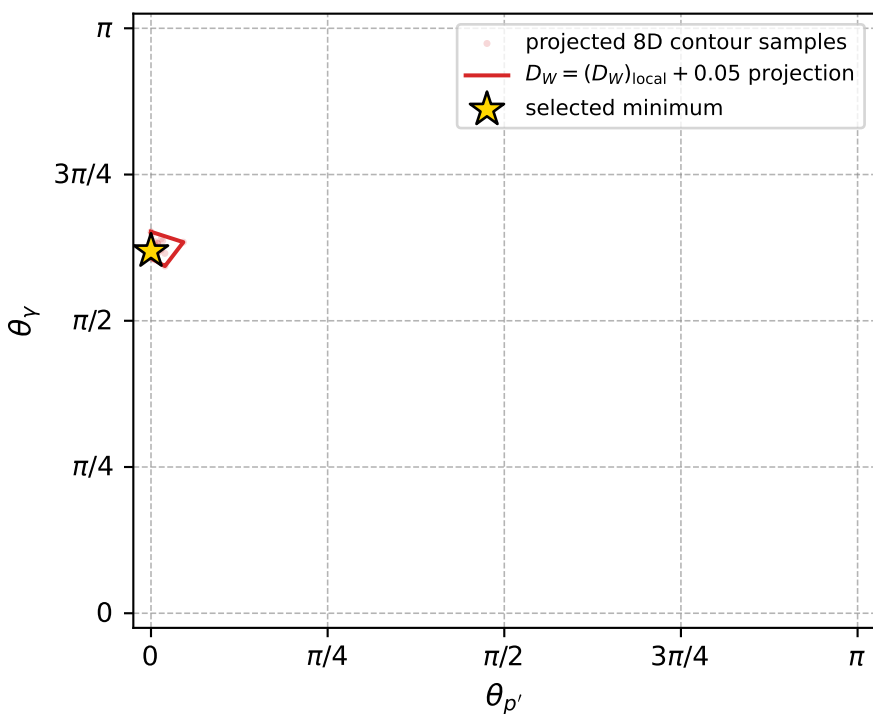
Transverse plane



Leading final-state amplitudes (N=3, retained=98.6%)



electron: polarization cluster P6, local minimum 705; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0074543486$
 $D_W \text{ contour} = 0.057454349$
 above global minimum = 0.0074379158
 8D contour samples = 128

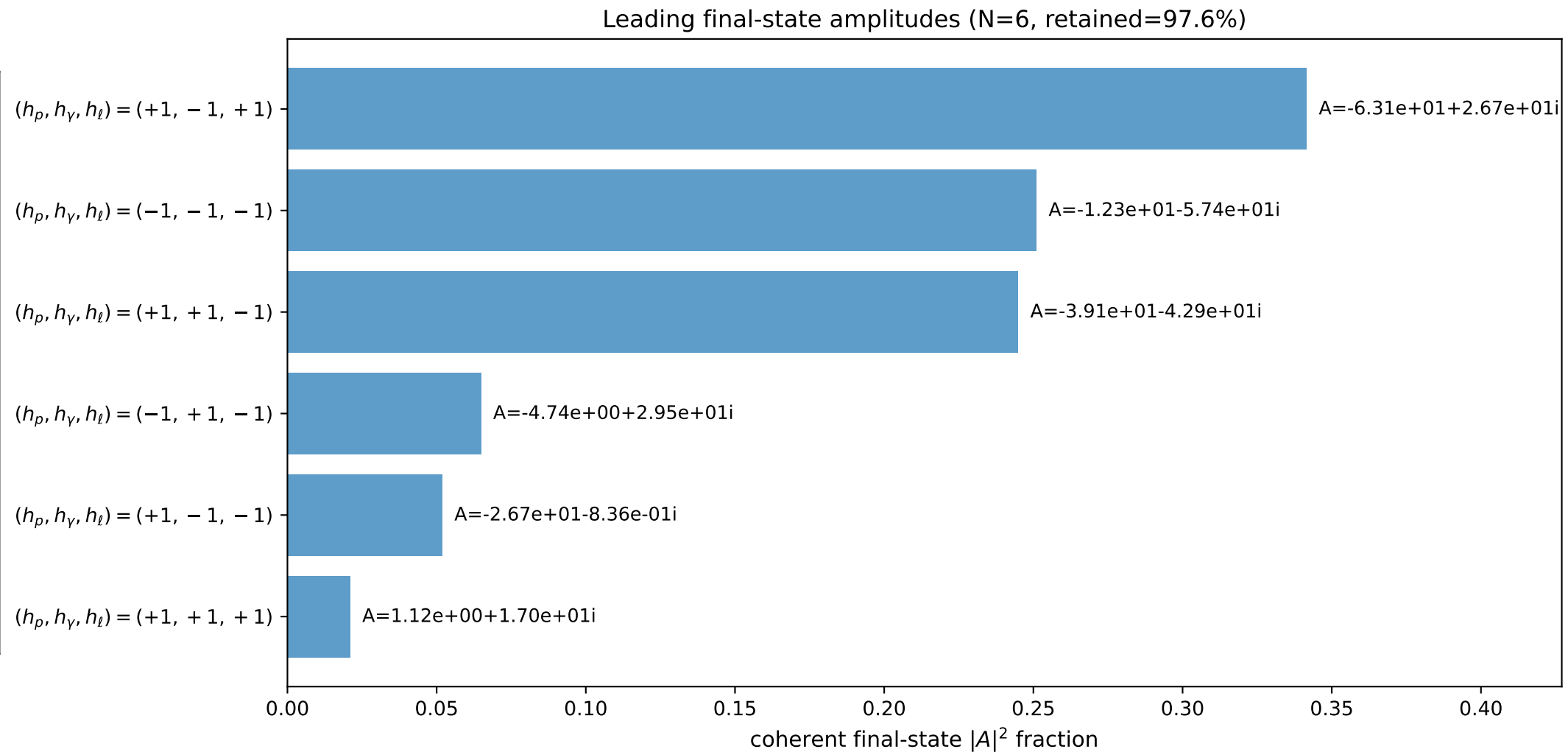
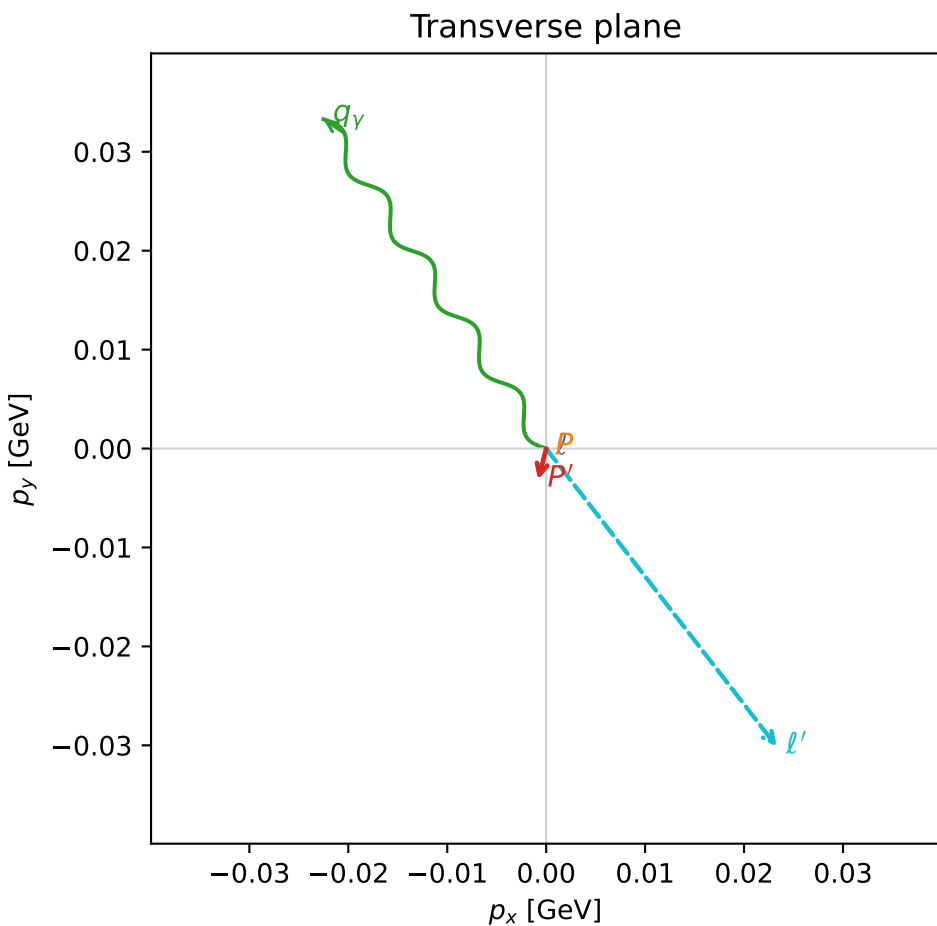
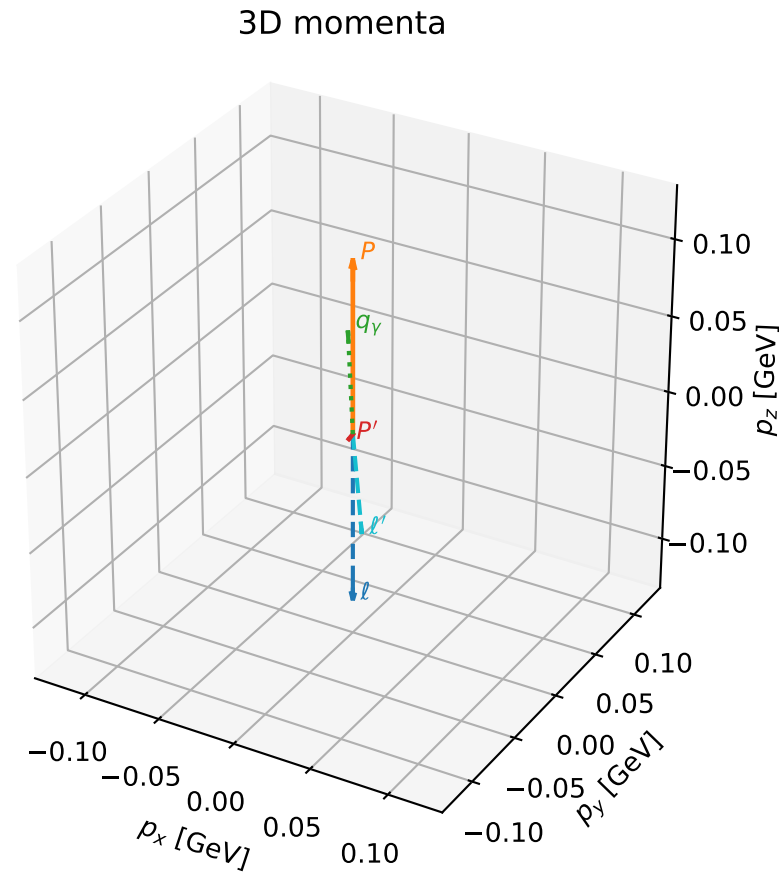
selected phase-space point:
 $\text{sqrt_s} = 1.144846$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 1.944905$
 $\text{q0ut} = 0.1064259$
 $\text{phi_p_out} = 3.06061$
 $\text{phi_gamma_out} = 3.141578$
 $\text{alpha_e} = 2.347798$
 $\text{alpha_p} = 1.765915$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

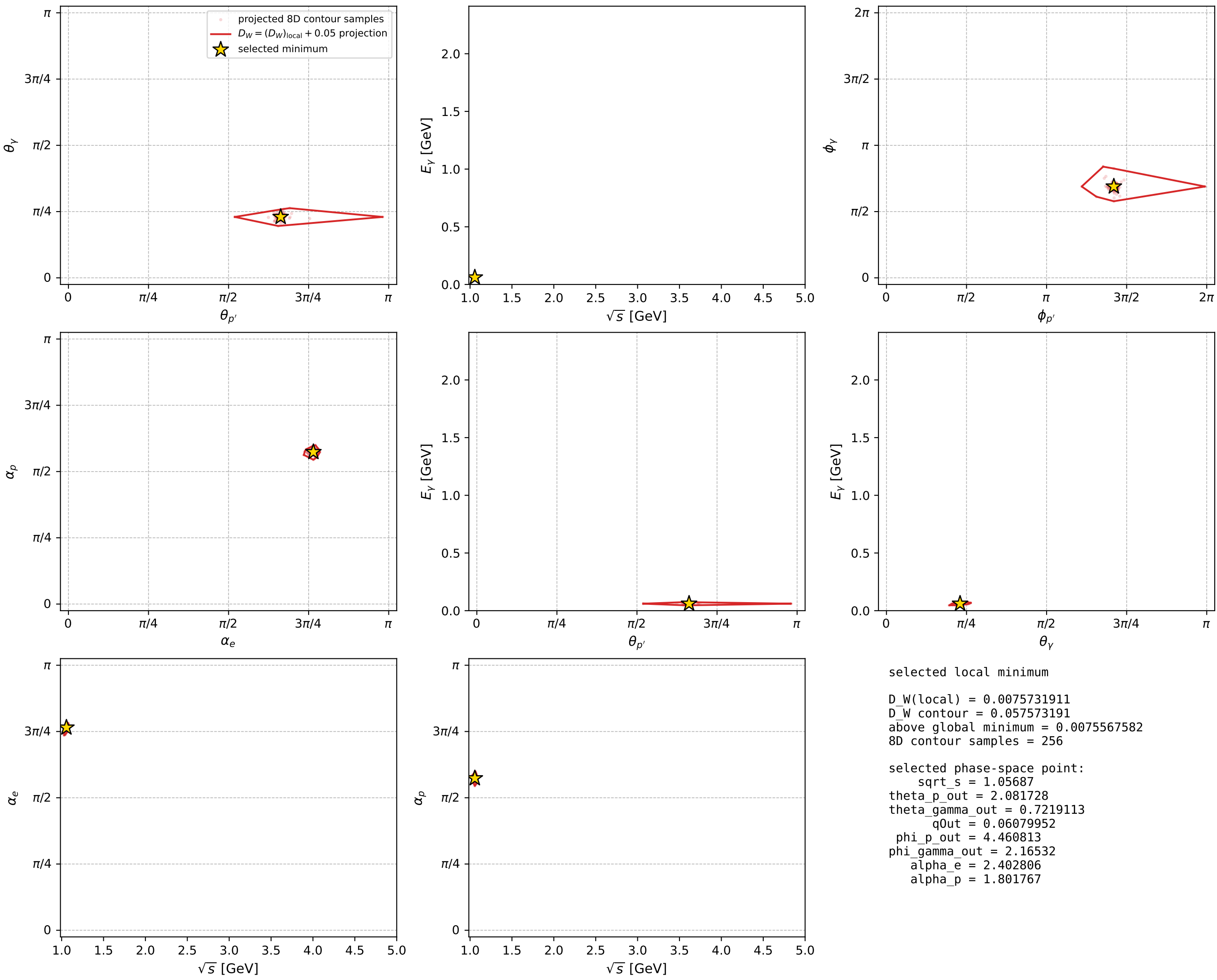
dw_mixing_angles_polarization_06_local_minimum_708 D_W=0.00757319
 region: polarization_cluster_06_local_minimum_708
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.4028059$ rad
 incoming lepton: $-0.739286|+> +0.673391|->$
 proton mixing angle: $\alpha_p=1.8017668$ rad
 incoming proton: $-0.228922|+> +0.973445|->$

$s=1.11697$, $\sqrt{s}=1.05687$, $|\vec{P}|=0.112184$, $|\vec{P}'|=0.0037257$
 $\theta_{P'}=2.08173$, $\theta_V=0.721911$, $\phi_{P'}=4.46081$, $\phi_V=2.16532$
 $E_V=0.0607995$, $\theta_{\ell'}=2.42579$, $\phi_{\ell'}=5.37082$
 $Q^2=0.00319739$, $x_B=0.0271892$, $t=-0.0129634$, $W^2=0.994244$, $y=0.49592$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1122$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1122$
 P $E= 0.9447$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1122$
 ℓ' $E= 0.0581$ $p_x= 0.0233$ $p_y= -0.0301$ $p_z= -0.0438$
 P' $E= 0.9380$ $p_x= -0.0008$ $p_y= -0.0031$ $p_z= -0.0018$
 q_V $E= 0.0608$ $p_x= -0.0225$ $p_y= 0.0333$ $p_z= 0.0456$



electron: polarization cluster P6, local minimum 708; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



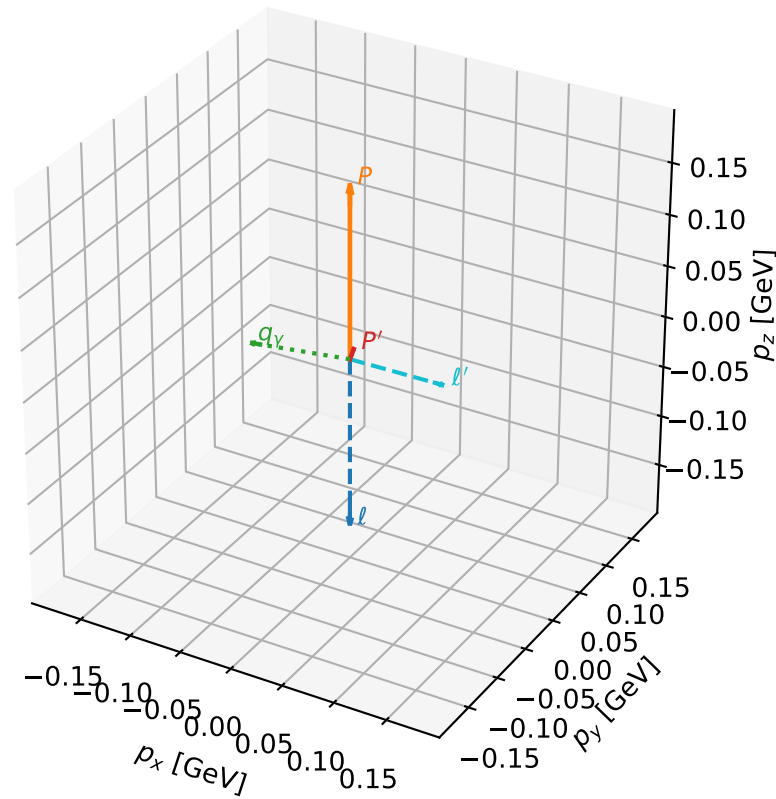
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_710 D_W=0.00772056
region: polarization_cluster_06_local_minimum_710
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3727624$ rad
incoming lepton: $-0.718724|+> +0.695295|->$
proton mixing angle: $\alpha_p=1.7245629$ rad
incoming proton: $-0.153161|+> +0.988201|->$

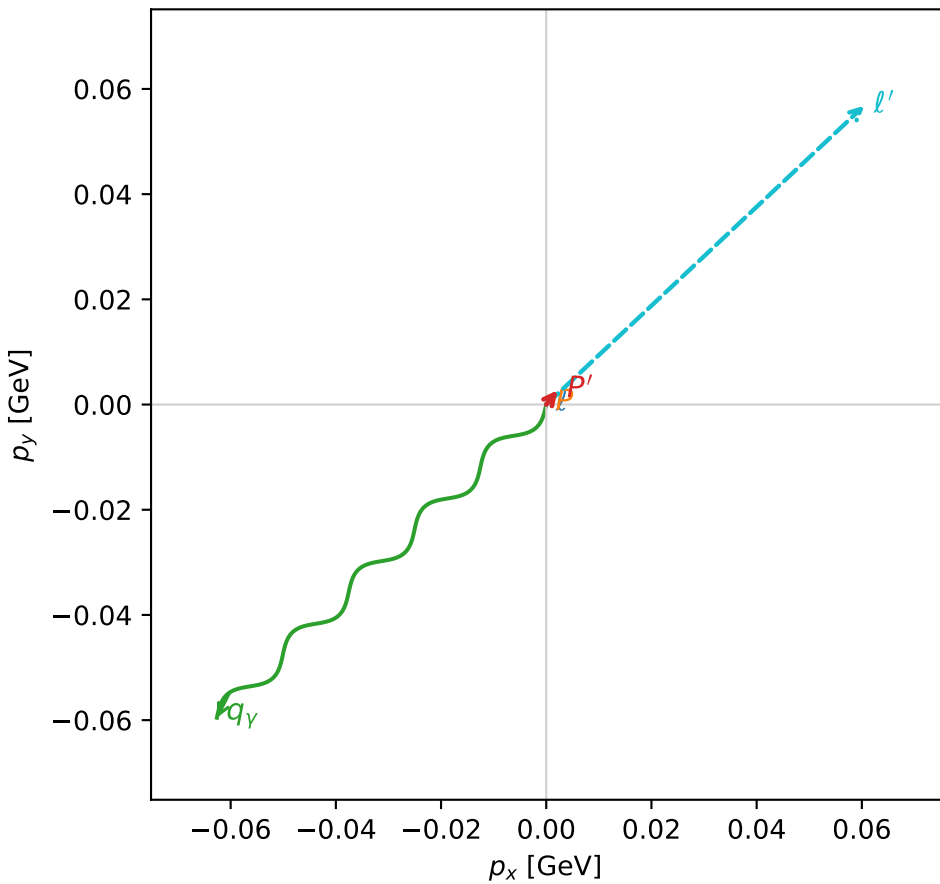
$s=1.25344$, $\sqrt{s}=1.11957$, $|\vec{P}|=0.166846$, $|\vec{P}'|=0.00979936$
 $\theta_{P'}=0.361556$, $\theta_Y=1.26062$, $\phi_{P'}=0.902018$, $\phi_Y=3.90182$
 $E_Y=0.0907291$, $\theta_{\ell'}=1.98885$, $\phi_{\ell'}=0.754318$
 $Q^2=0.017996$, $x_B=0.0955718$, $t=-0.0246597$, $W^2=1.05015$, $y=0.504021$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1668$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1668$
 P $E= 0.9527$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1668$
 ℓ' $E= 0.0908$ $p_x= 0.0605$ $p_y= 0.0568$ $p_z= -0.0369$
 P' $E= 0.9381$ $p_x= 0.0021$ $p_y= 0.0027$ $p_z= 0.0092$
 q_Y $E= 0.0907$ $p_x= -0.0626$ $p_y= -0.0595$ $p_z= 0.0277$

3D momenta



Transverse plane



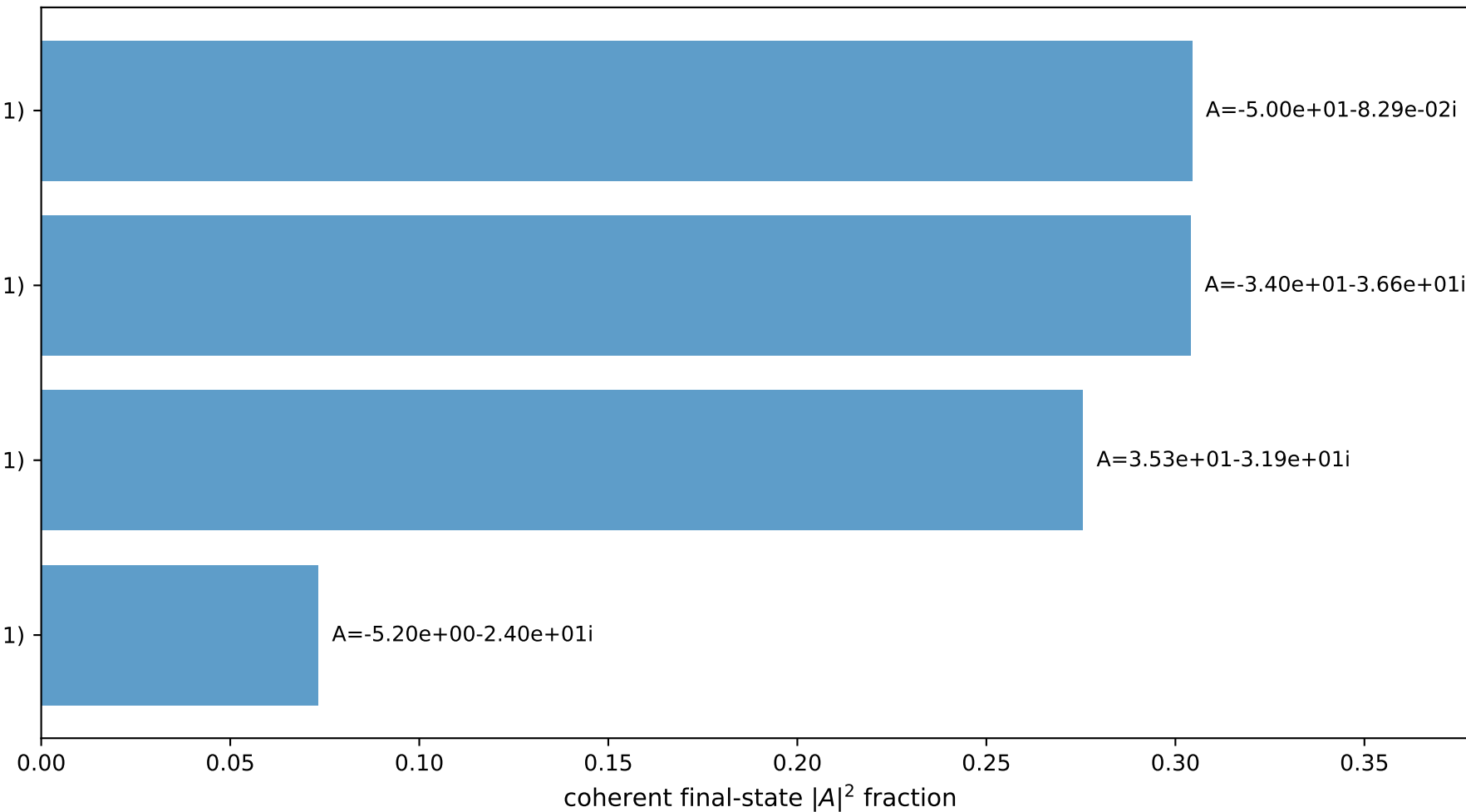
$(h_p, h_Y, h_{\ell}) = (+1, -1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

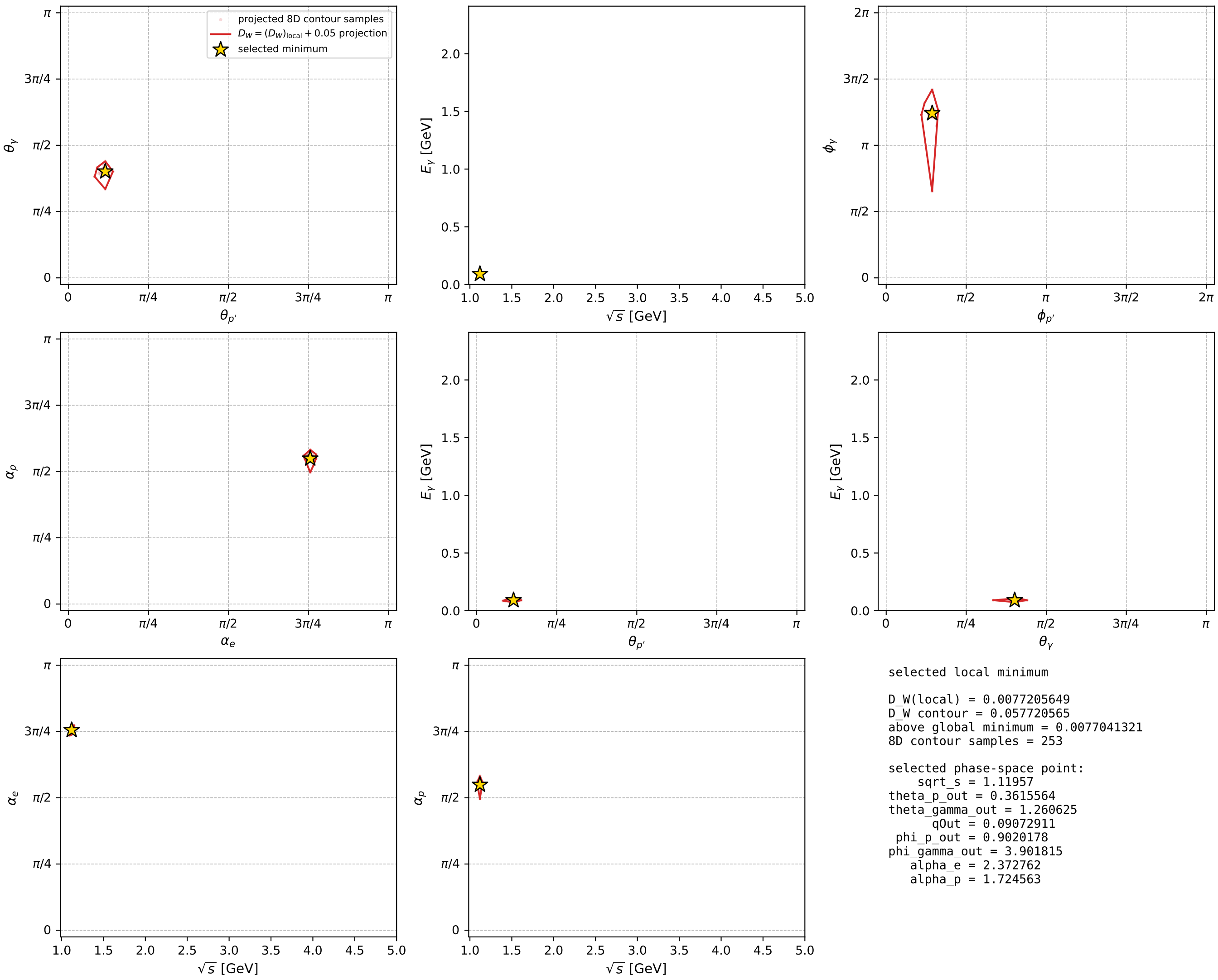
$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$

Leading final-state amplitudes (N=4, retained=95.8%)



electron: polarization cluster P6, local minimum 710; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

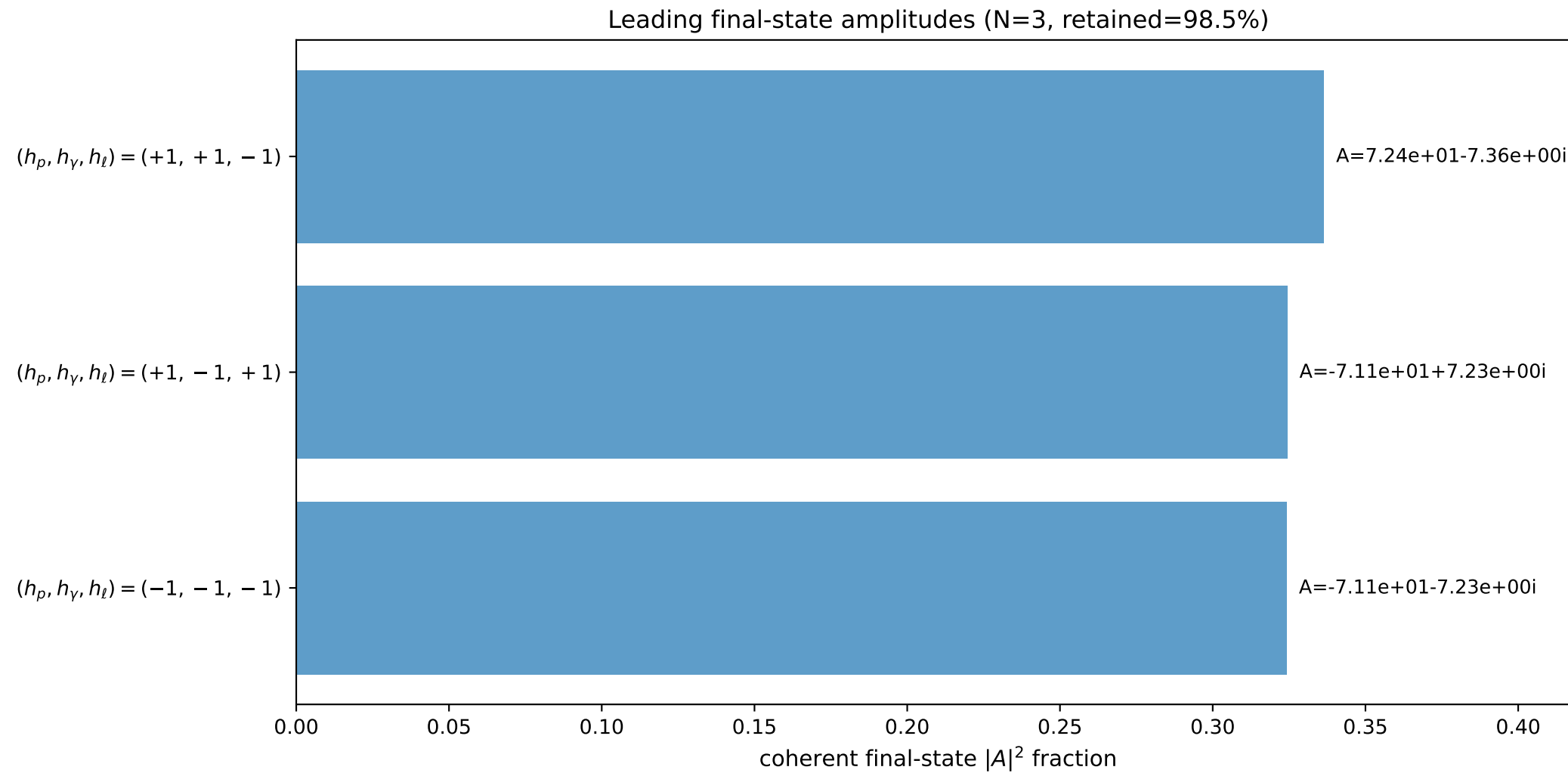
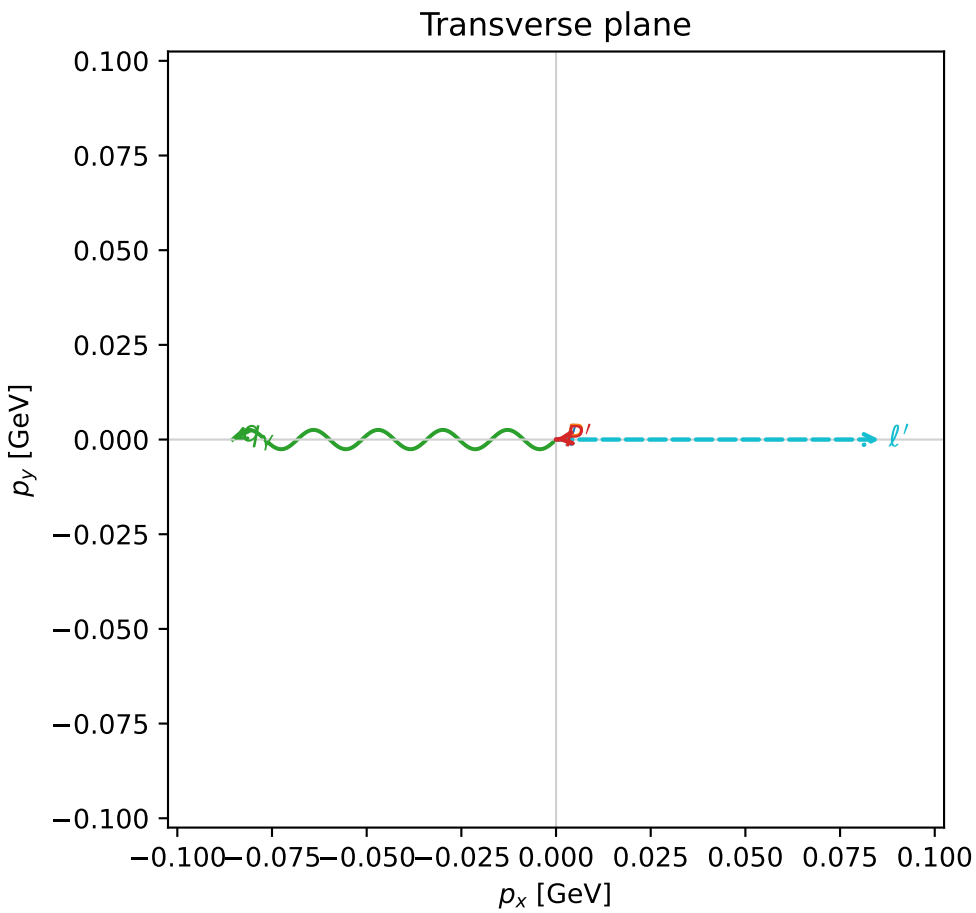
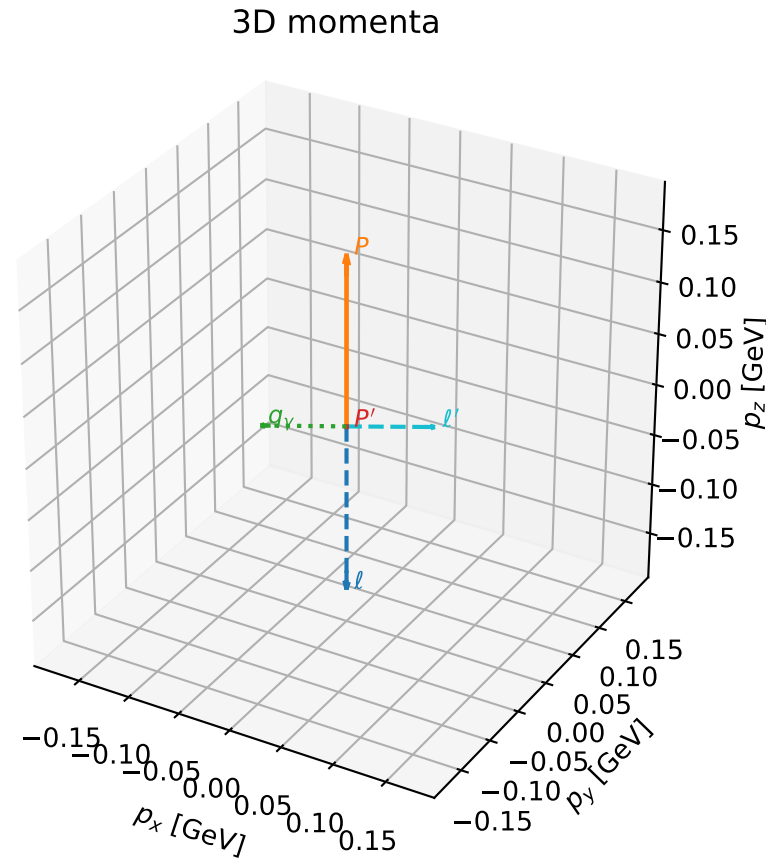


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

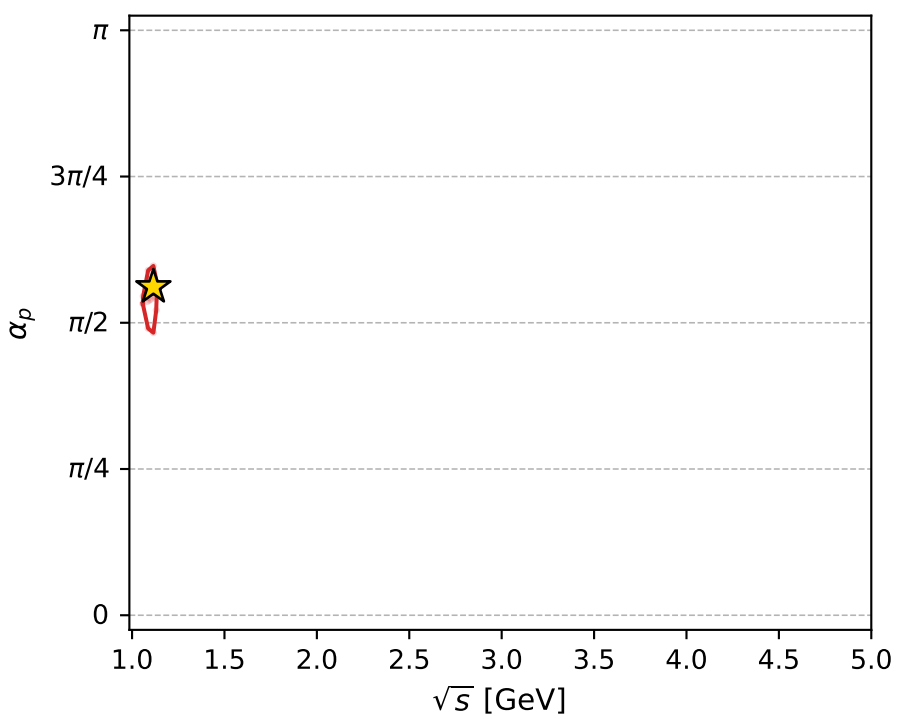
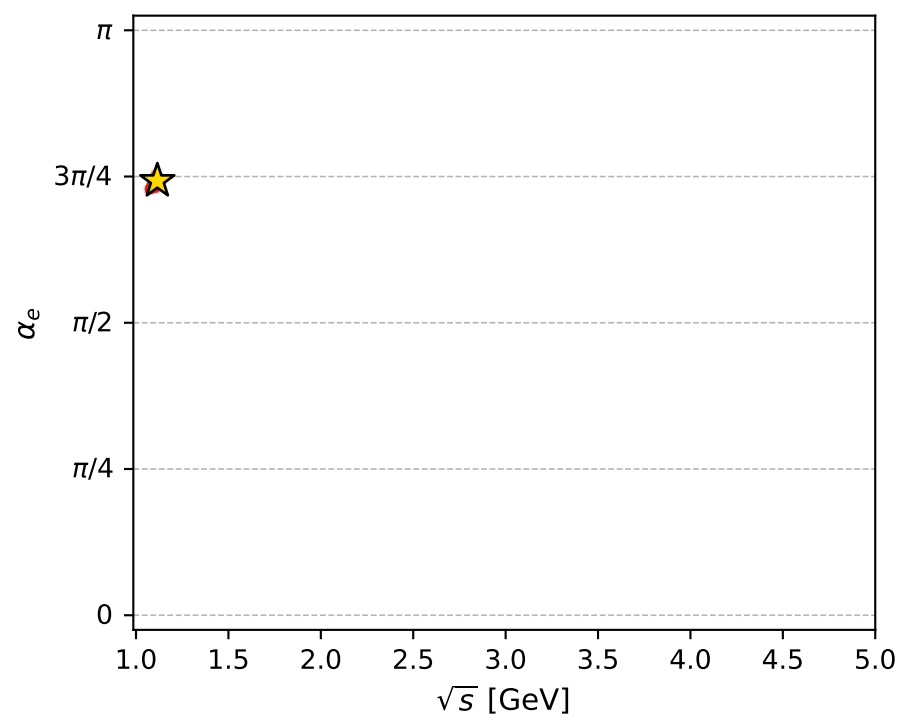
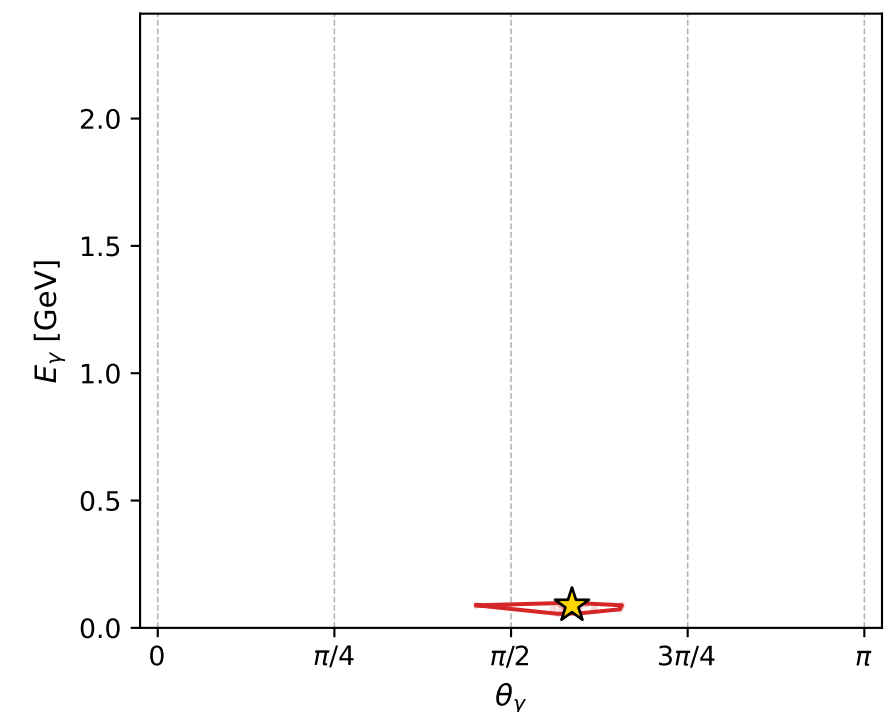
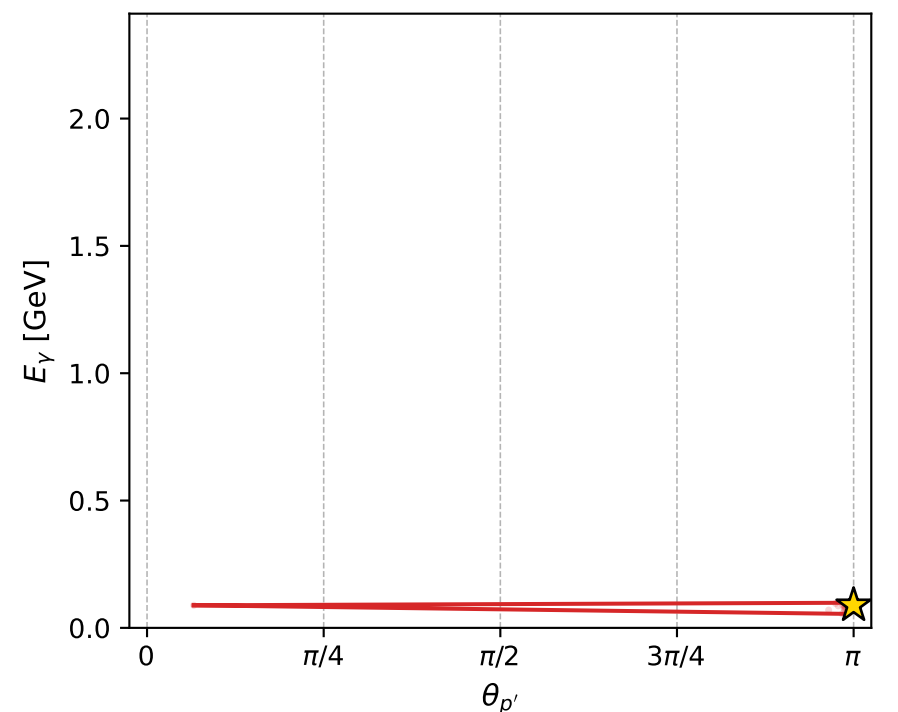
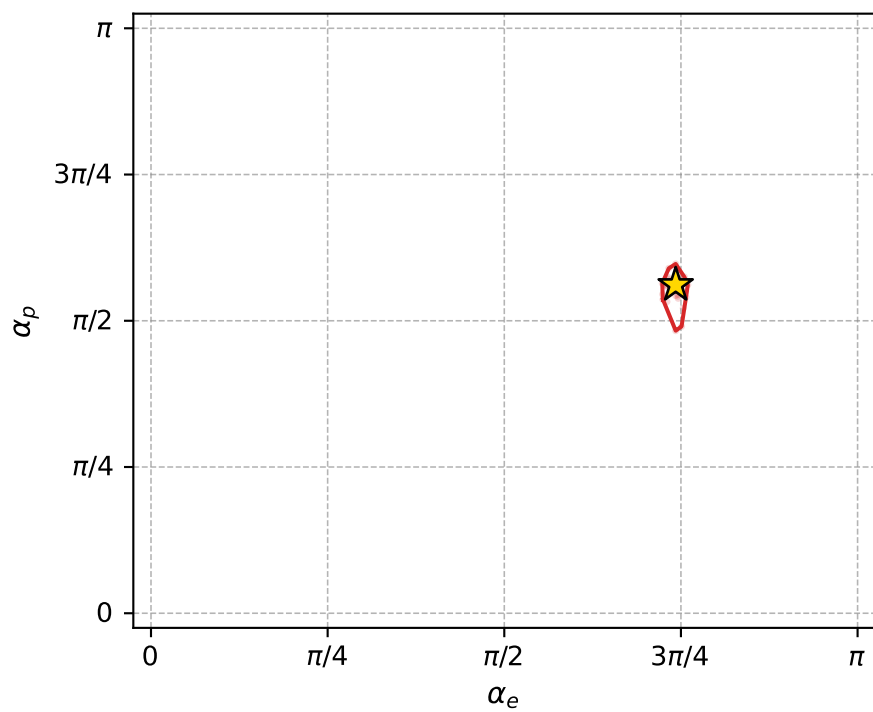
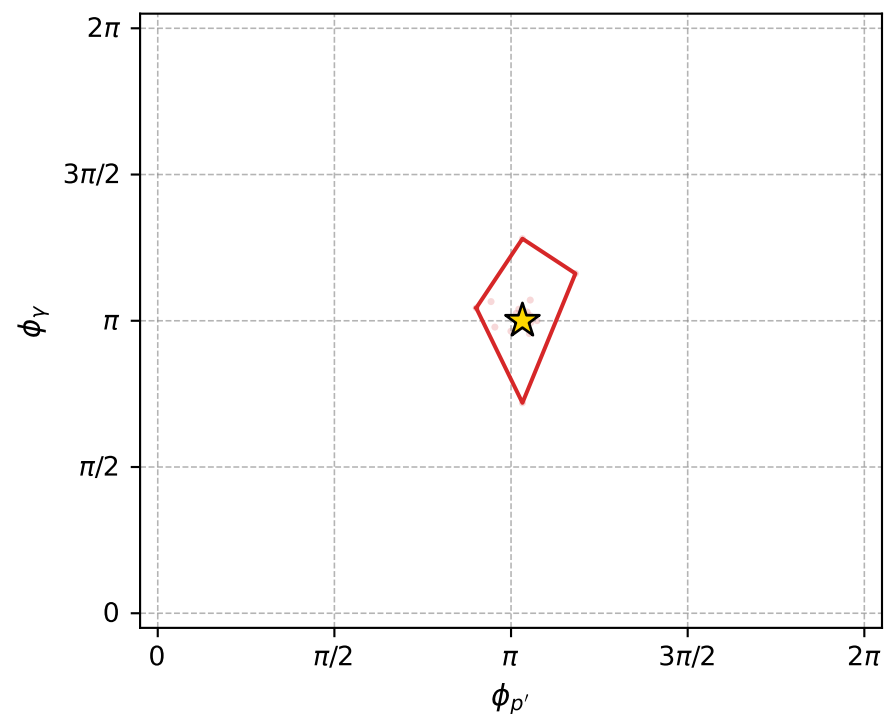
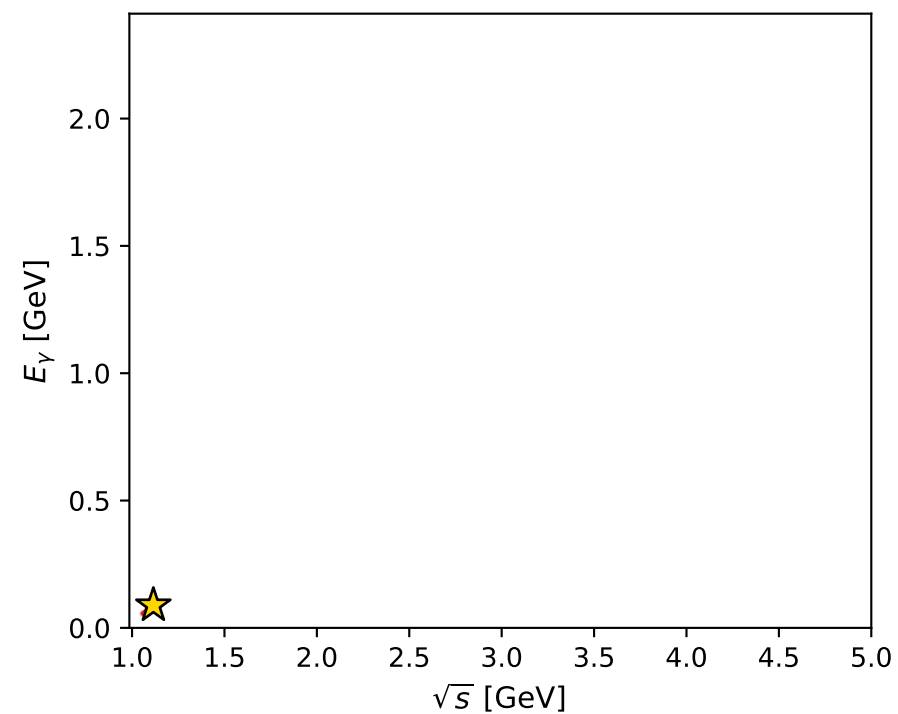
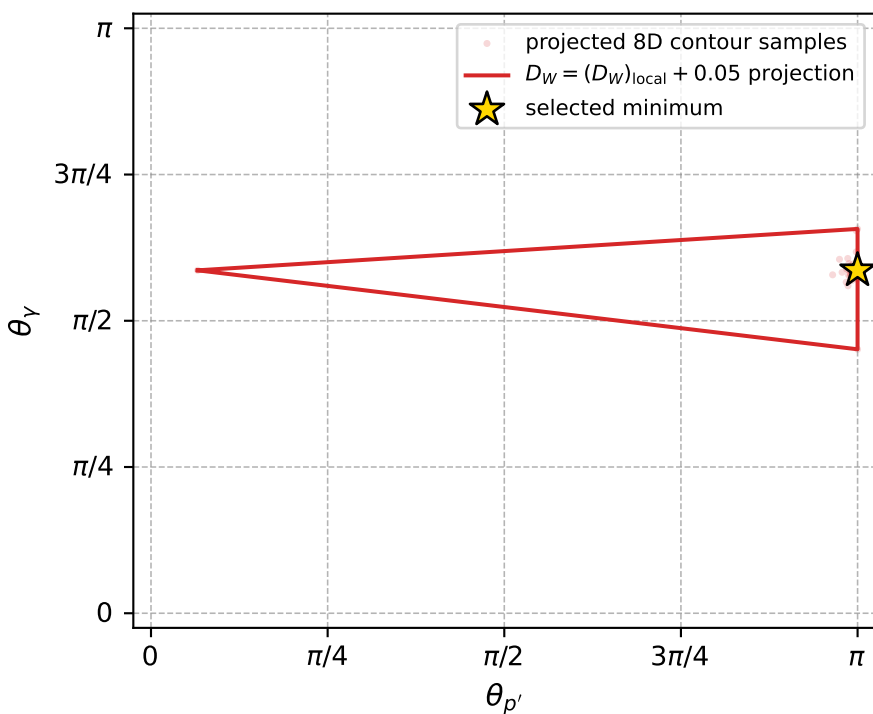
dw_mixing_angles_polarization_06_local_minimum_715 D_W=0.00790999
 region: polarization_cluster_06_local_minimum_715
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3328713$ rad
 incoming lepton: $-0.690424|+> +0.723405|->$
 proton mixing angle: $\alpha_p=1.763399$ rad
 incoming proton: $-0.191414|+> +0.981509|->$

$s=1.24396$, $\sqrt{s}=1.11533$, $|\vec{P}|=0.163233$, $|\vec{P}'|=0.000348371$
 $\theta_{P'}=3.14148$, $\theta_Y=1.8417$, $\phi_{P'}=3.24294$, $\phi_Y=3.14161$
 $E_Y=0.0886176$, $\theta_{\ell'}=1.29611$, $\phi_{\ell'}=1.5997e-05$
 $Q^2=0.0368169$, $x_B=0.181322$, $t=-0.02656$, $W^2=1.04607$, $y=0.55764$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1632$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1632$
 P $E= 0.9521$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1632$
 ℓ' $E= 0.0887$ $p_x= 0.0854$ $p_y= 0.0000$ $p_z= 0.0241$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0003$
 q_Y $E= 0.0886$ $p_x= -0.0854$ $p_y= -0.0000$ $p_z= -0.0237$



electron: polarization cluster P6, local minimum 715; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0079099913$
 $D_W \text{ contour} = 0.057909991$
 above global minimum = 0.0078935585
 8D contour samples = 132

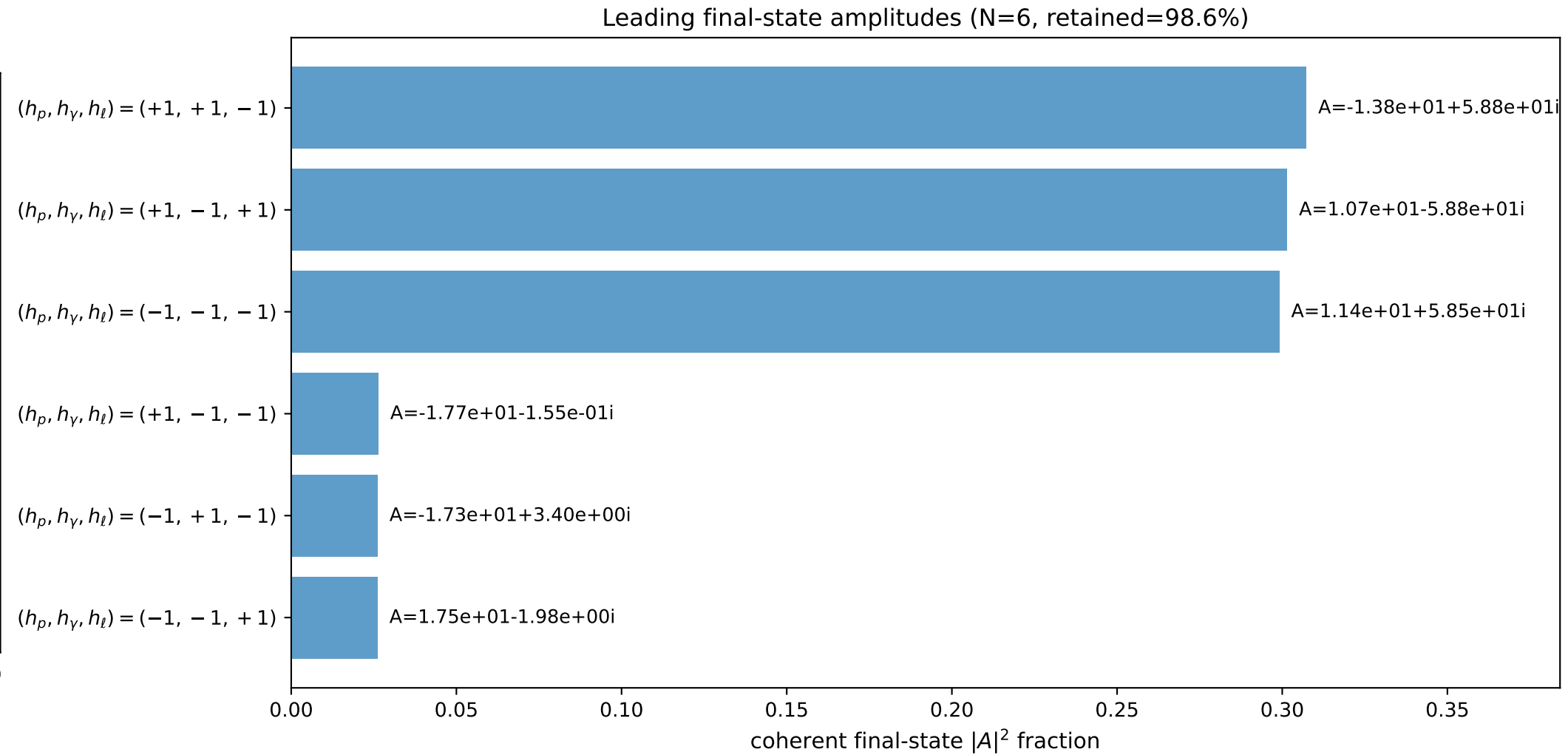
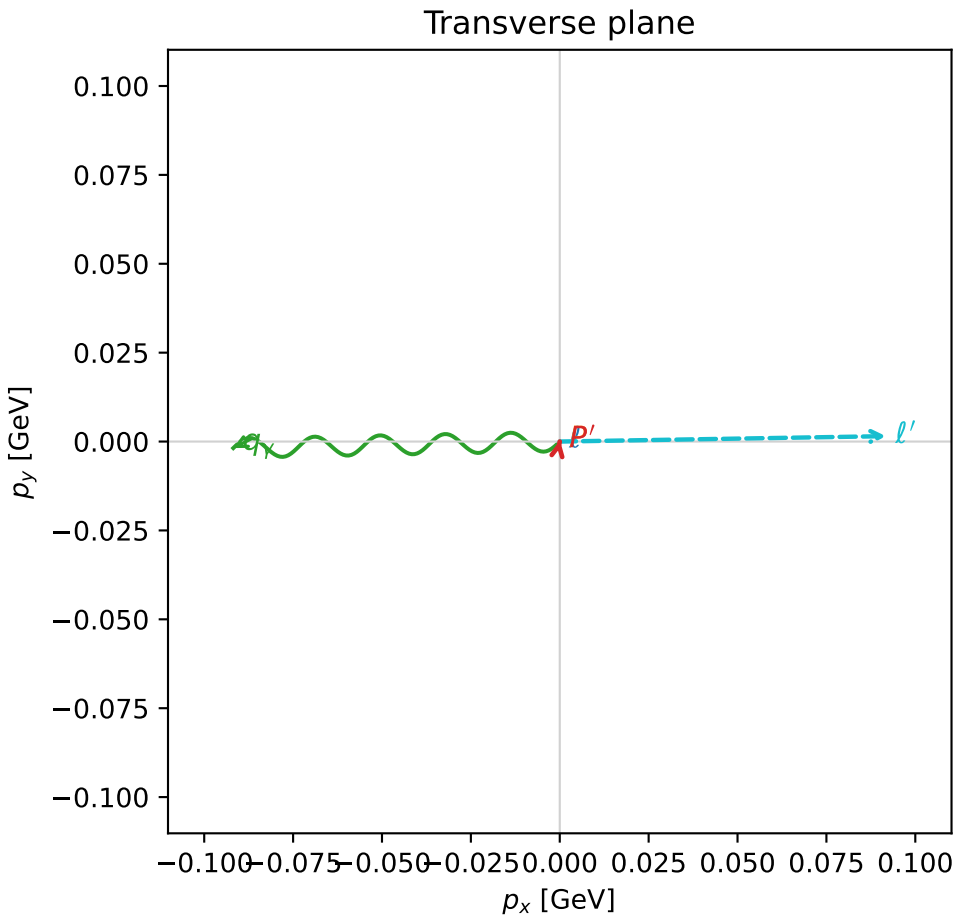
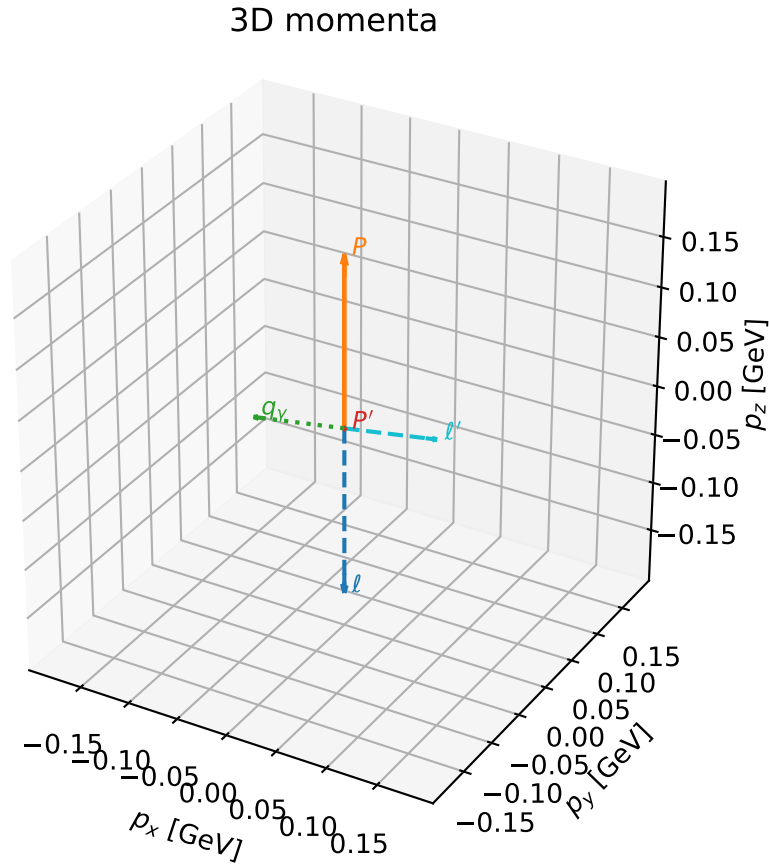
selected phase-space point:
 $\text{sqrt_s} = 1.115331$
 $\text{theta_p_out} = 3.141478$
 $\text{theta_gamma_out} = 1.841697$
 $\text{qOut} = 0.0886176$
 $\text{phi_p_out} = 3.242943$
 $\text{phi_gamma_out} = 3.141609$
 $\text{alpha_e} = 2.332871$
 $\text{alpha_p} = 1.763399$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

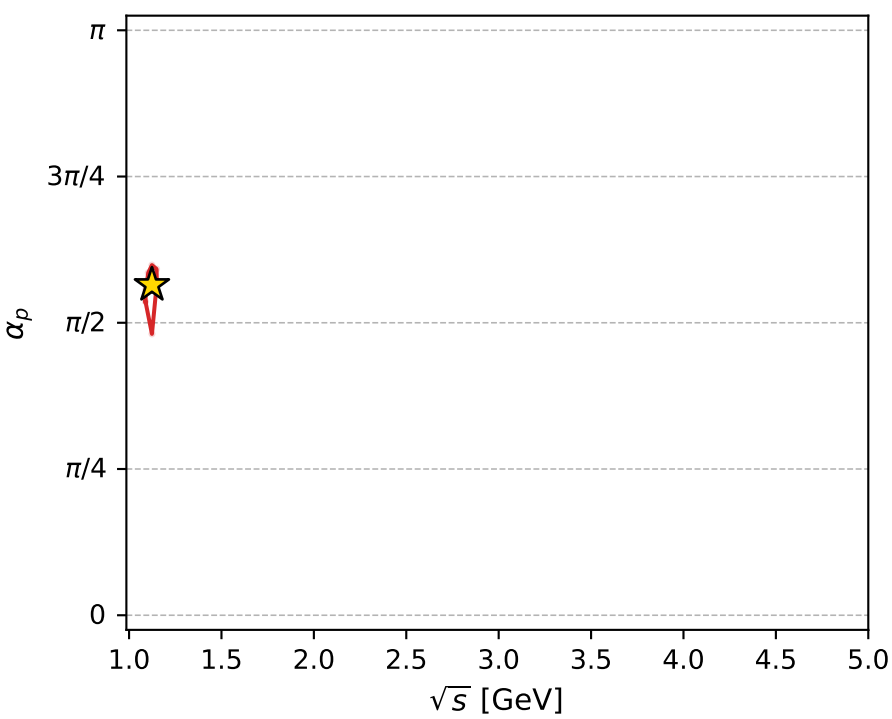
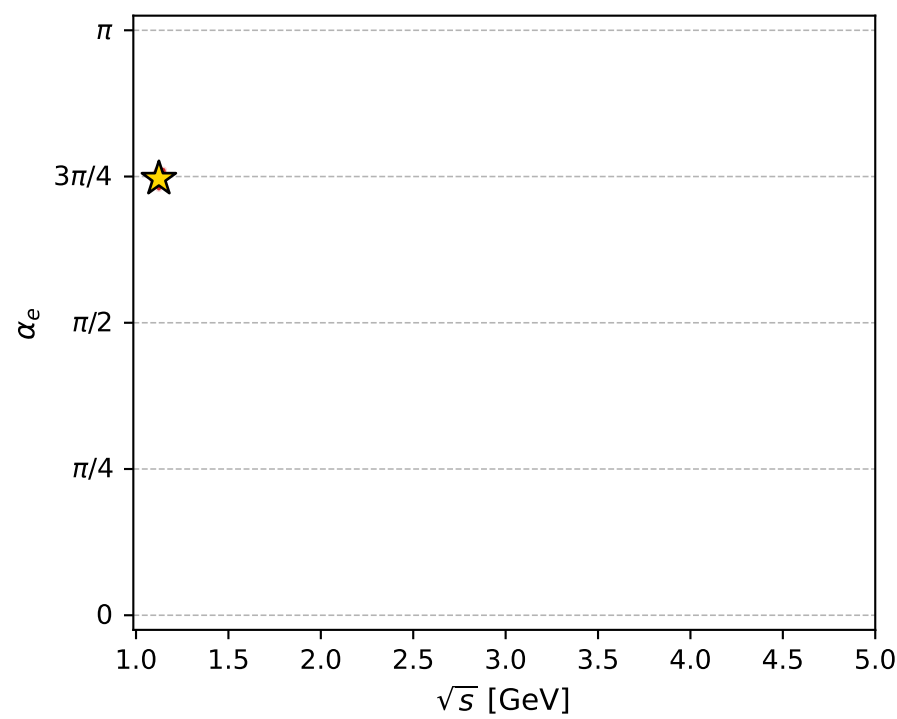
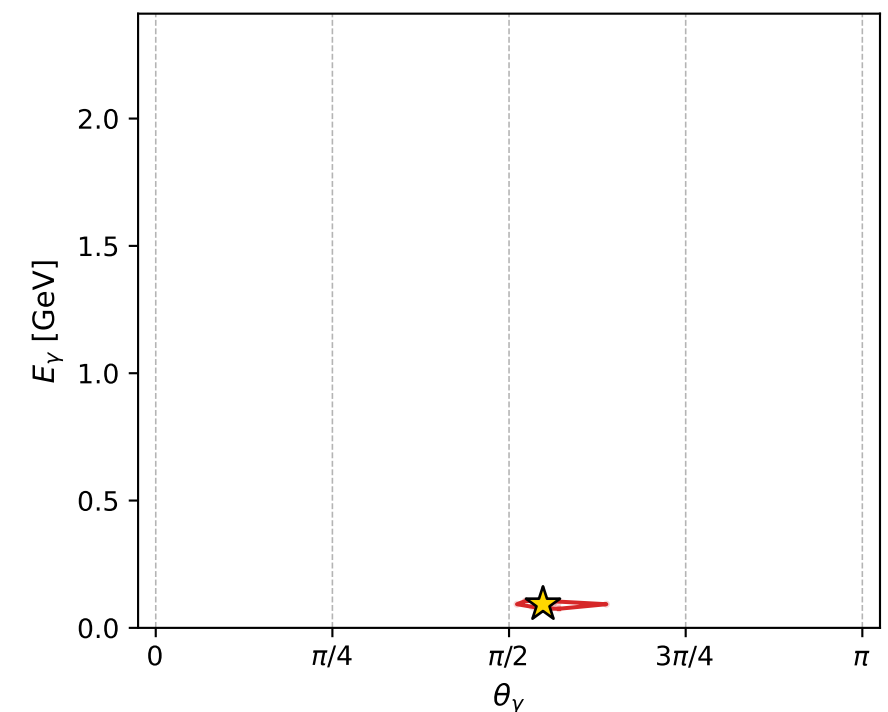
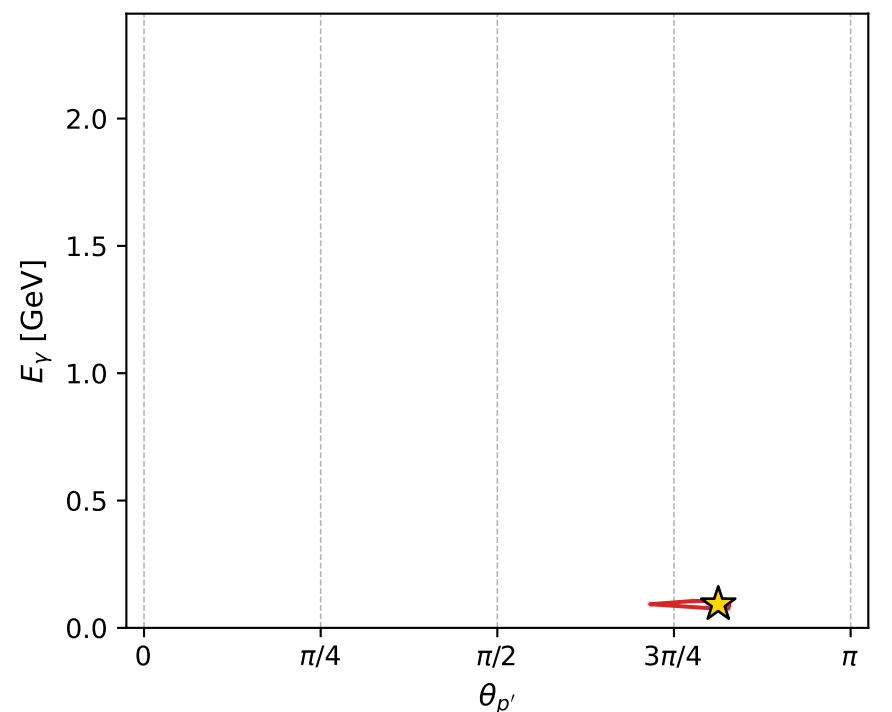
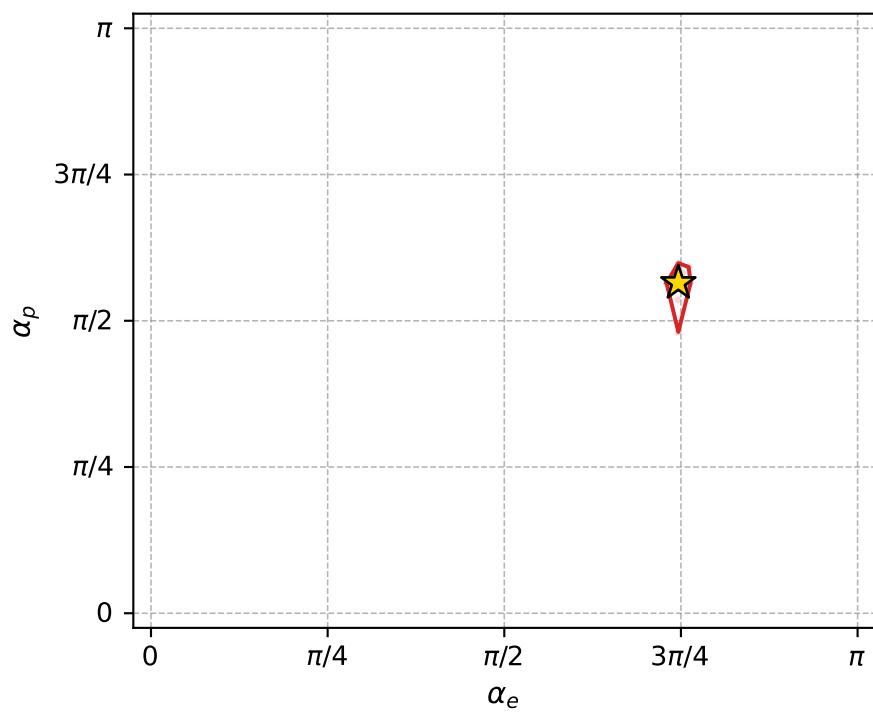
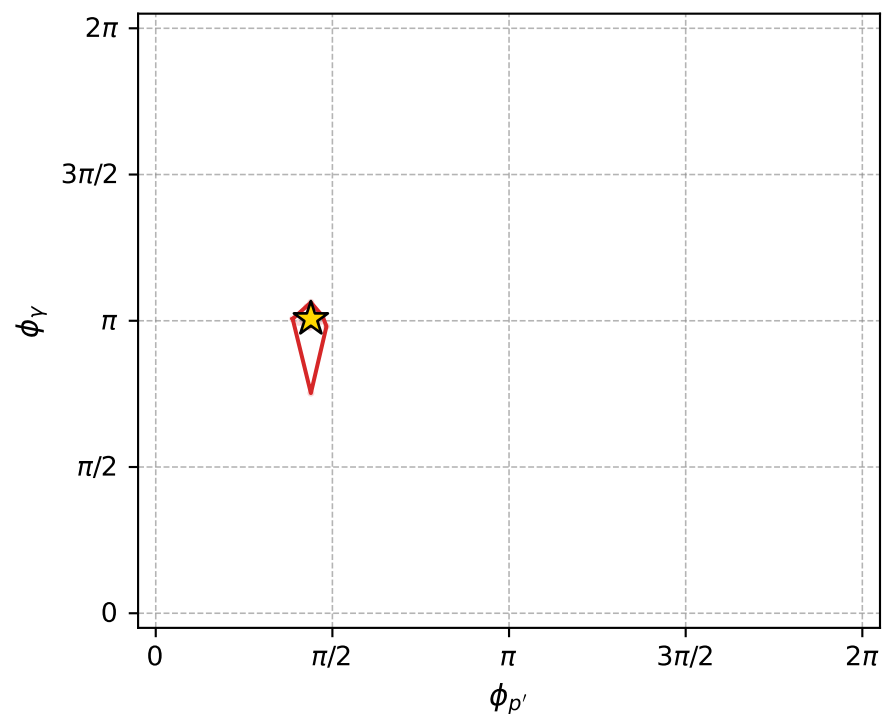
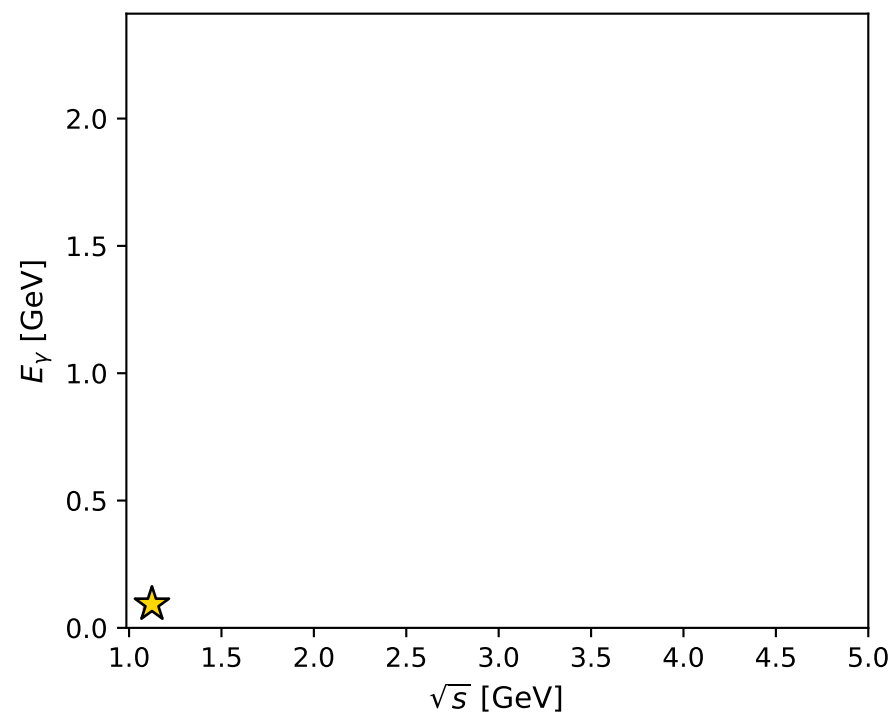
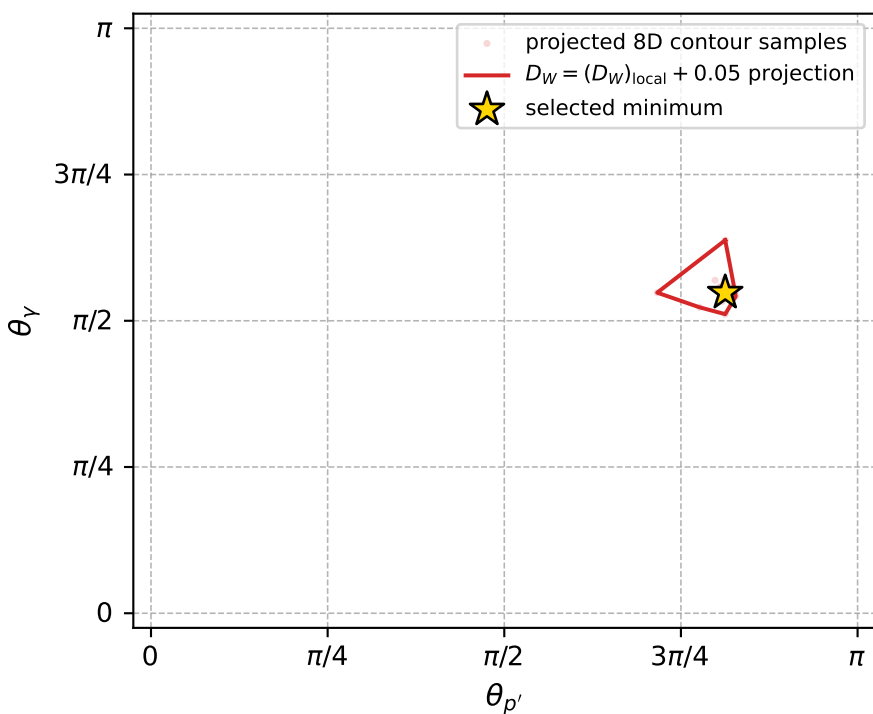
dw_mixing_angles_polarization_06_local_minimum_720 D_W=0.00802906
 region: polarization_cluster_06_local_minimum_720
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3446561$ rad
 incoming lepton: $-0.698901|+\rangle +0.715218|-\rangle$
 proton mixing angle: $\alpha_p=1.7740381$ rad
 incoming proton: $-0.201845|+\rangle +0.979417|-\rangle$

$s=1.26294$, $\sqrt{s}=1.12381$, $|\vec{P}|=0.170445$, $|\vec{P}'|=0.000588971$
 $\theta_{p'}=2.55303$, $\theta_Y=1.72176$, $\phi_{p'}=1.37954$, $\phi_Y=3.16161$
 $E_Y=0.0928981$, $\theta_{\ell'}=1.41451$, $\phi_{\ell'}=0.0165307$
 $Q^2=0.0366002$, $x_B=0.173563$, $t=-0.0289828$, $W^2=1.05412$, $y=0.550454$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1704$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1704$
 P $E=0.9534$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1704$
 ℓ' $E=0.0929$ $p_x=0.0918$ $p_y=0.0015$ $p_z=0.0145$
 P' $E=0.9380$ $p_x=0.0001$ $p_y=0.0003$ $p_z=-0.0005$
 q_Y $E=0.0929$ $p_x=-0.0918$ $p_y=-0.0018$ $p_z=-0.0140$



electron: polarization cluster P6, local minimum 720; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0080290566$
 $D_W \text{ contour} = 0.058029057$
 above global minimum = 0.0080126238
 8D contour samples = 254

selected phase-space point:
 $\text{sqrt_s} = 1.123805$
 $\text{theta_p_out} = 2.553029$
 $\text{theta_gamma_out} = 1.721758$
 $\text{qOut} = 0.09289807$
 $\text{phi_p_out} = 1.379536$
 $\text{phi_gamma_out} = 3.161607$
 $\text{alpha_e} = 2.344656$
 $\text{alpha_p} = 1.774038$

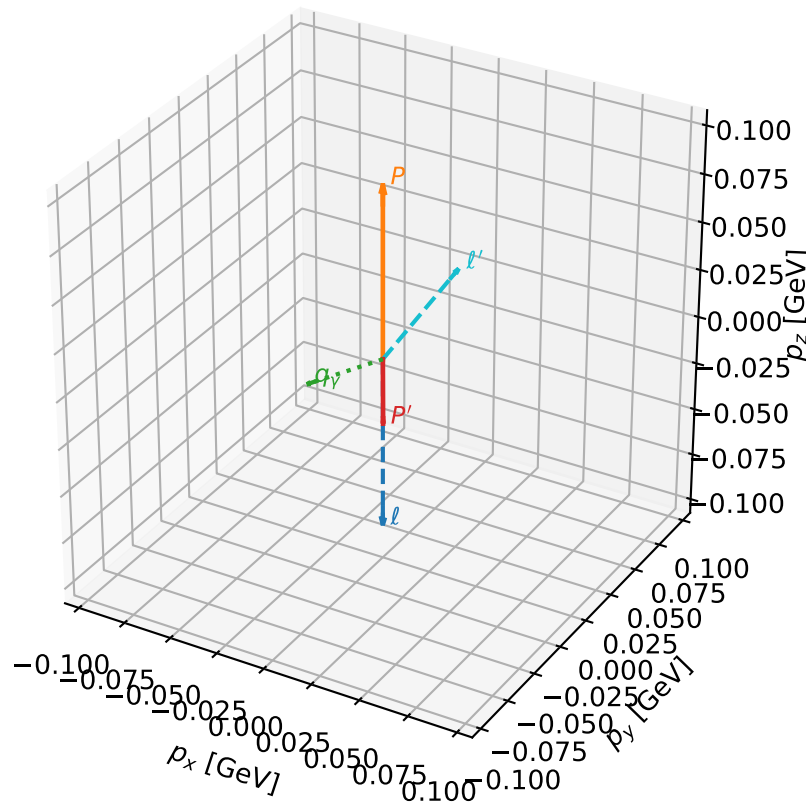
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_734 D_W=0.00847351
 region: polarization_cluster_06_local_minimum_734
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.285969$ rad
 incoming lepton: $-0.655748|+> +0.75498|->$
 proton mixing angle: $\alpha_p=1.8384725$ rad
 incoming proton: $-0.264491|+> +0.964388|->$

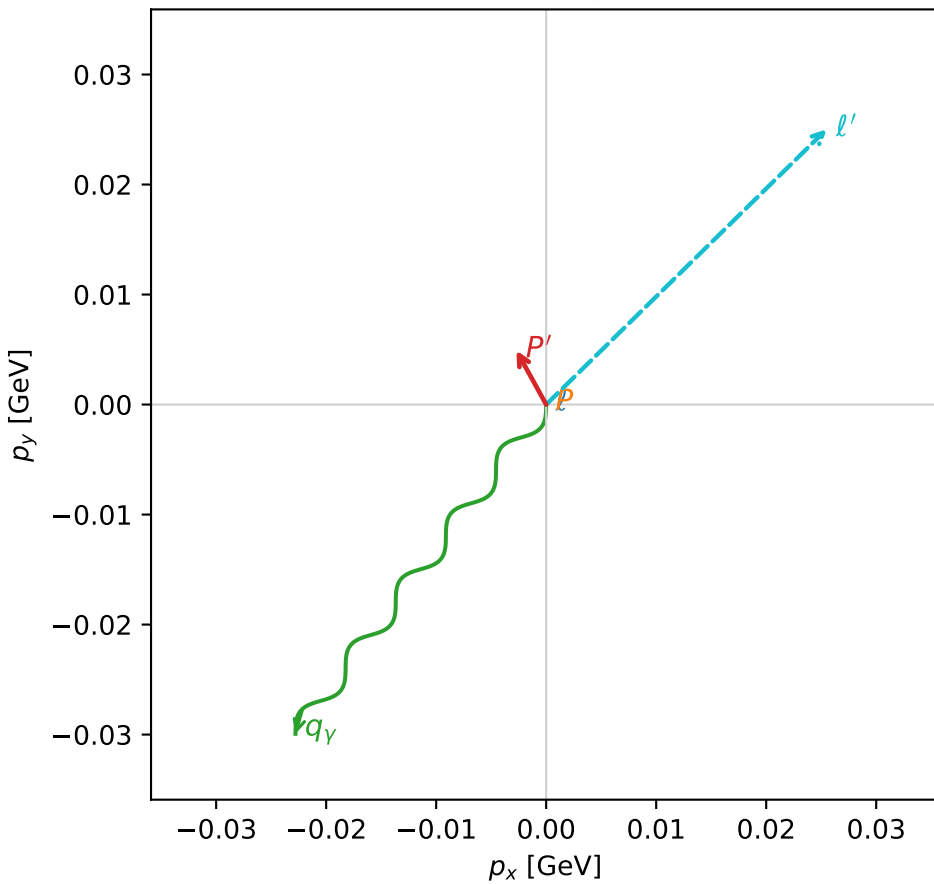
$s=1.06553$, $\sqrt{s}=1.03225$, $|\vec{P}|=0.0899427$, $|\vec{P}'|=0.0381481$
 $\theta_{P'}=2.99407$, $\theta_V=1.69772$, $\phi_{P'}=2.07381$, $\phi_V=4.06164$
 $E_V=0.0379255$, $\theta_{\ell'}=0.698522$, $\phi_{\ell'}=0.77602$
 $Q^2=0.0176428$, $x_B=0.198998$, $t=-0.0163203$, $W^2=0.95086$, $y=0.477458$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.0899$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0899$
 P $E= 0.9423$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0899$
 ℓ' $E= 0.0555$ $p_x= 0.0255$ $p_y= 0.0250$ $p_z= 0.0425$
 P' $E= 0.9388$ $p_x= -0.0027$ $p_y= 0.0049$ $p_z= -0.0377$
 q_V $E= 0.0379$ $p_x= -0.0228$ $p_y= -0.0299$ $p_z= -0.0048$

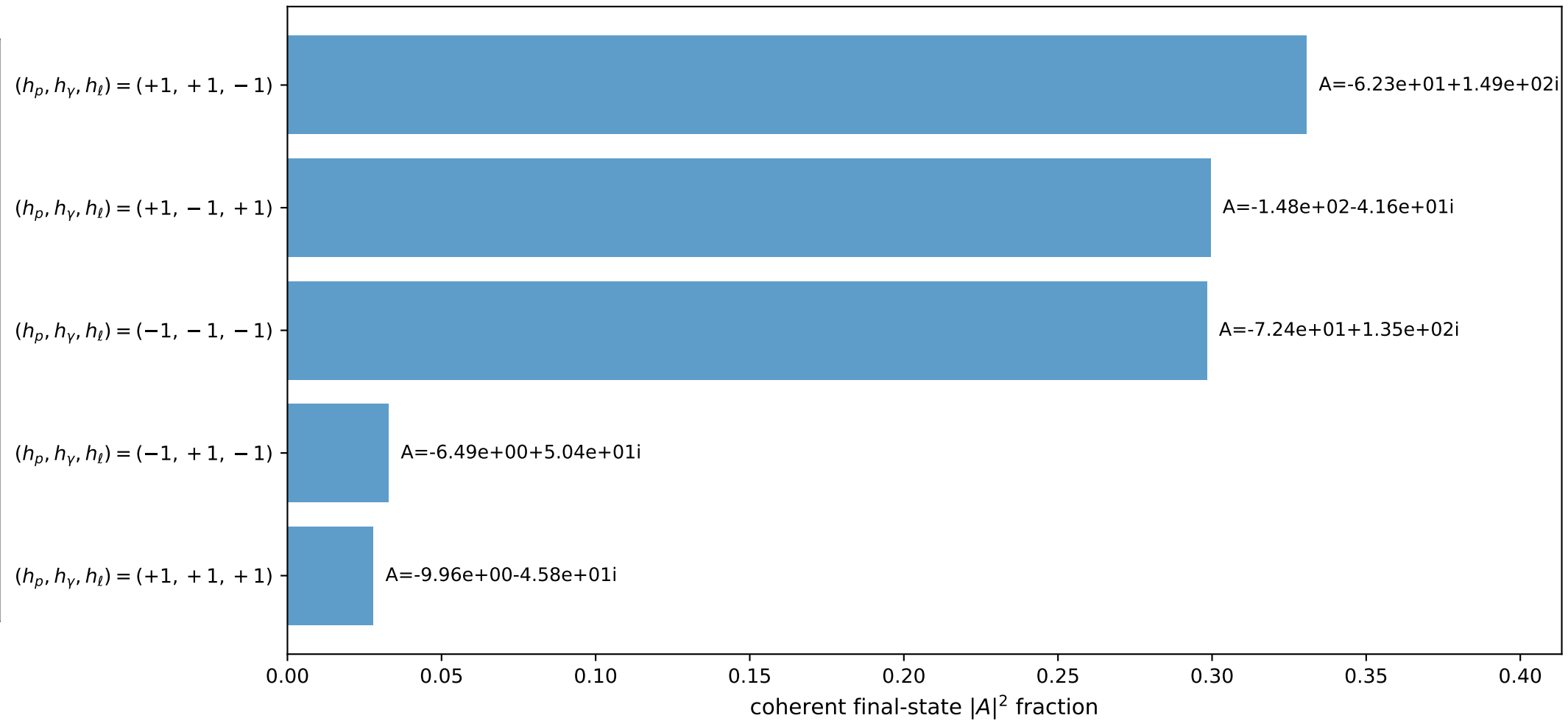
3D momenta



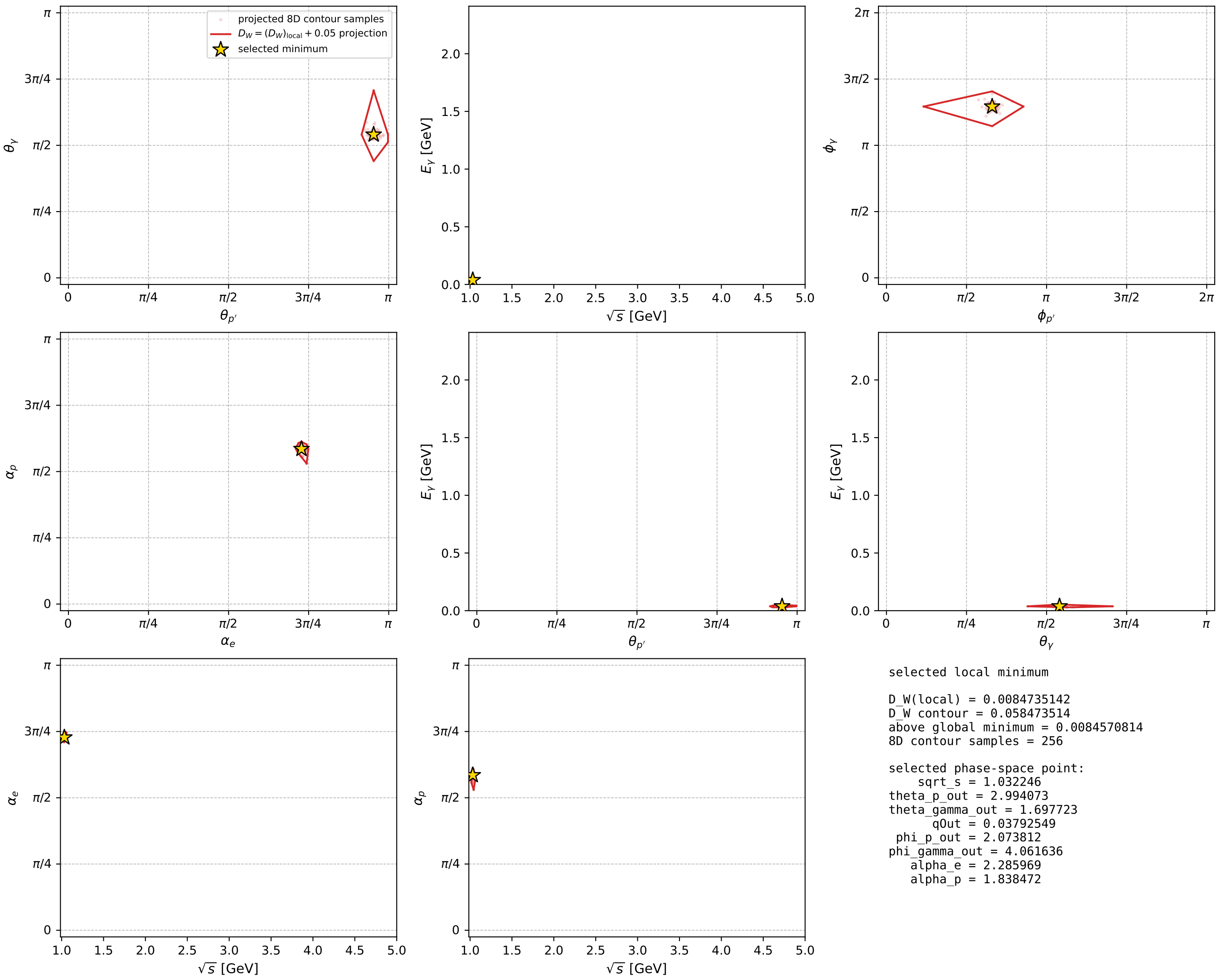
Transverse plane



Leading final-state amplitudes (N=5, retained=99.0%)



electron: polarization cluster P6, local minimum 734; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

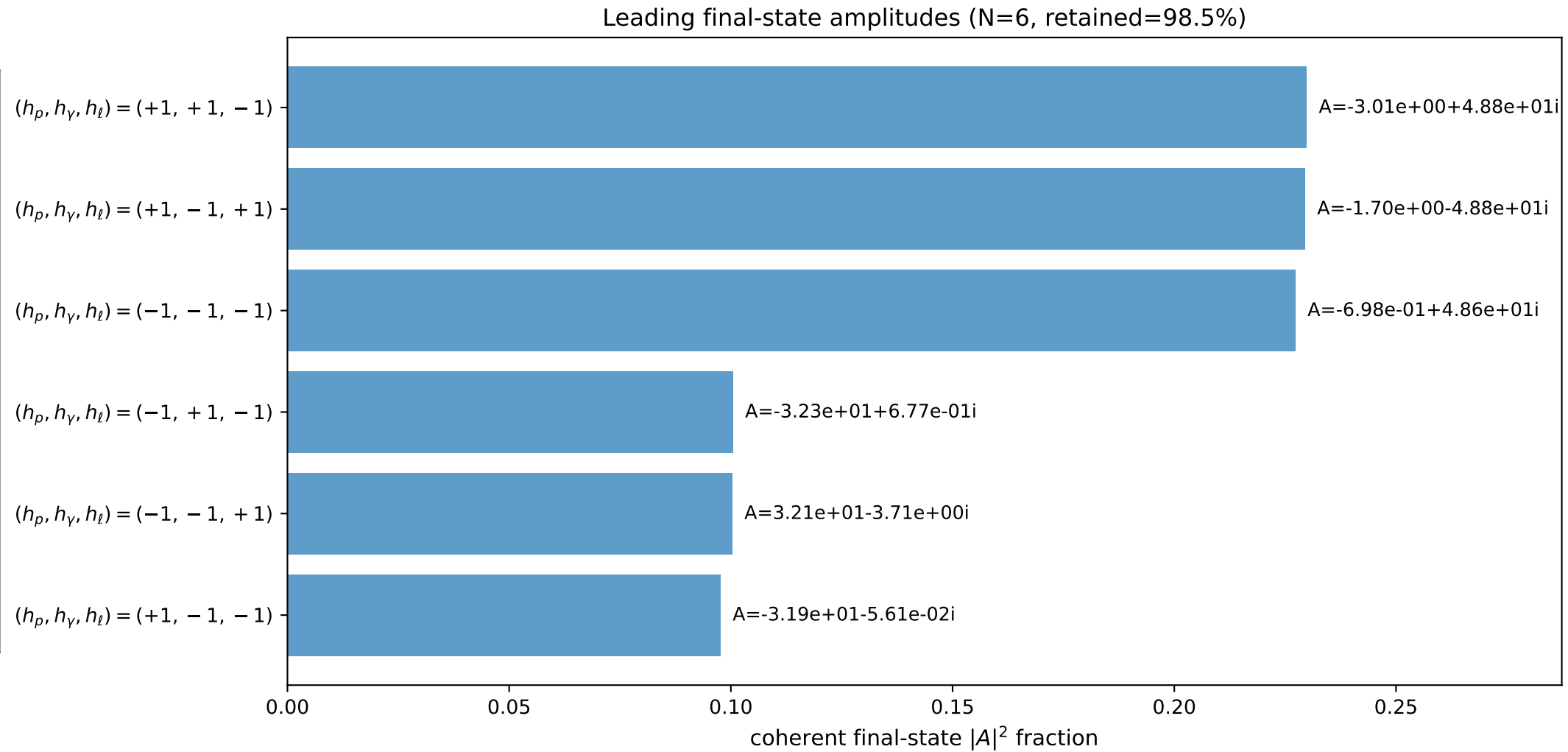
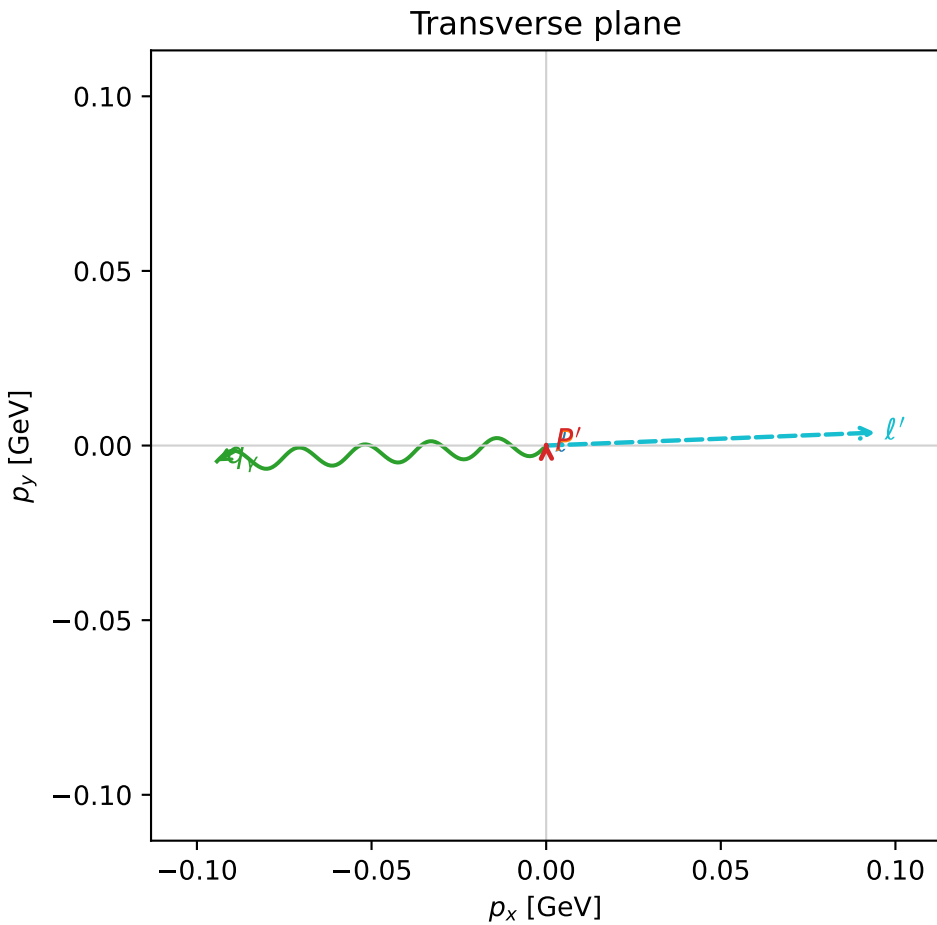
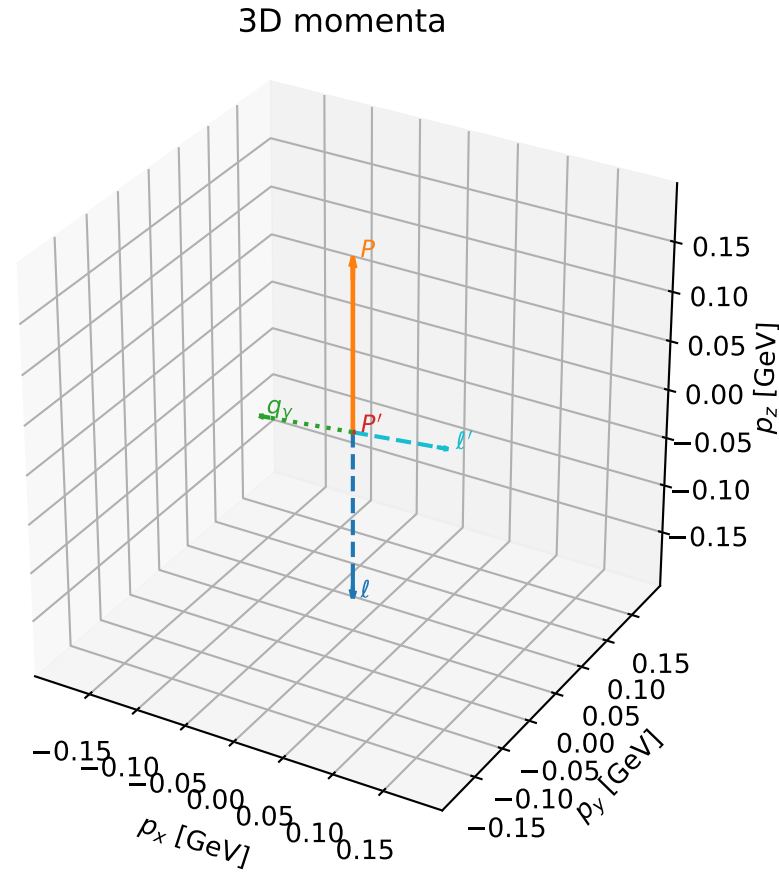


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

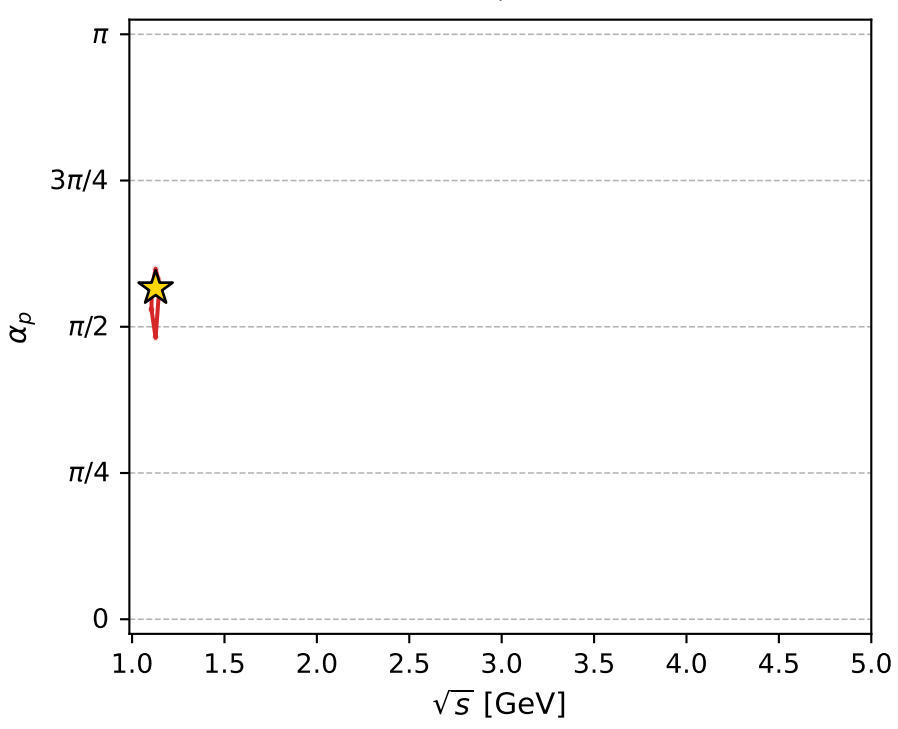
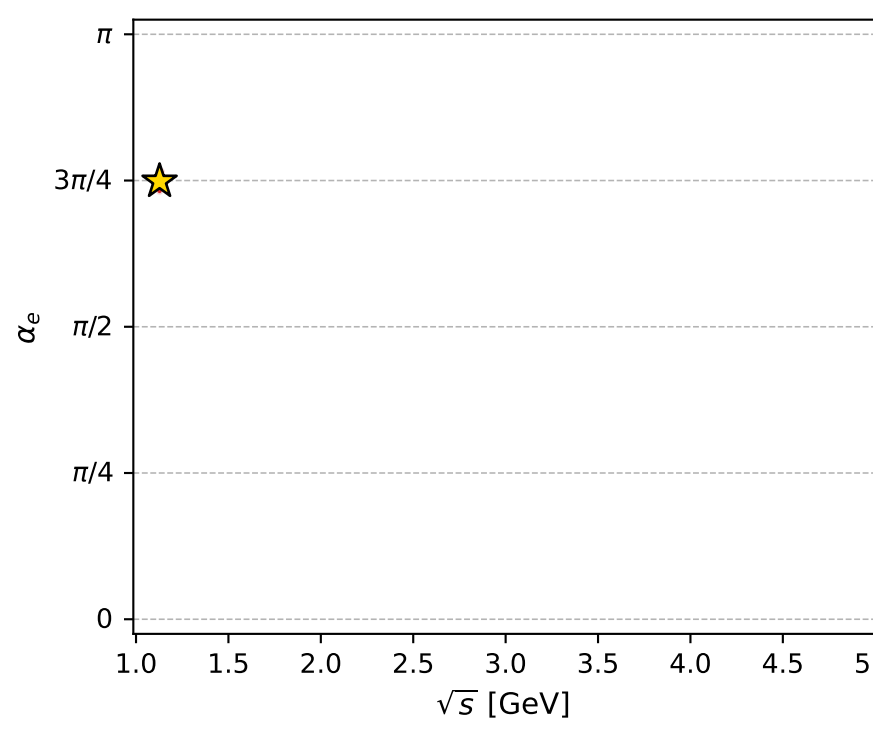
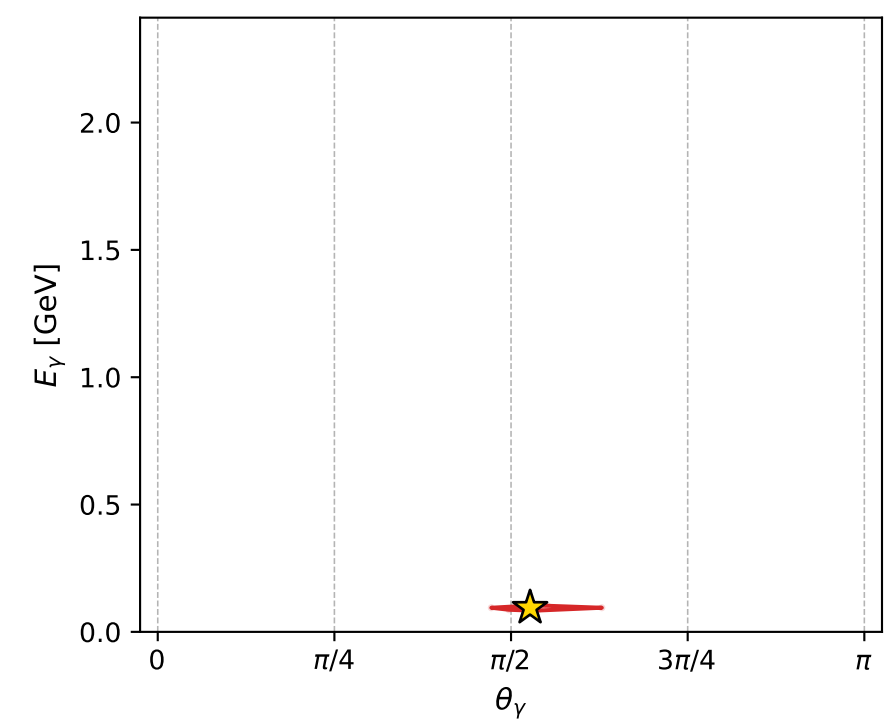
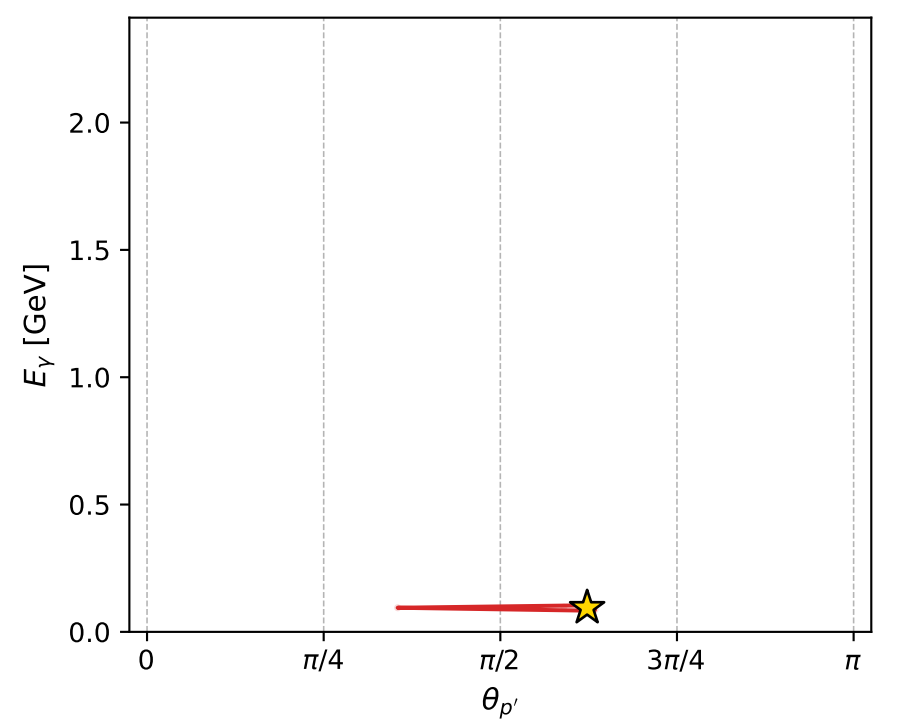
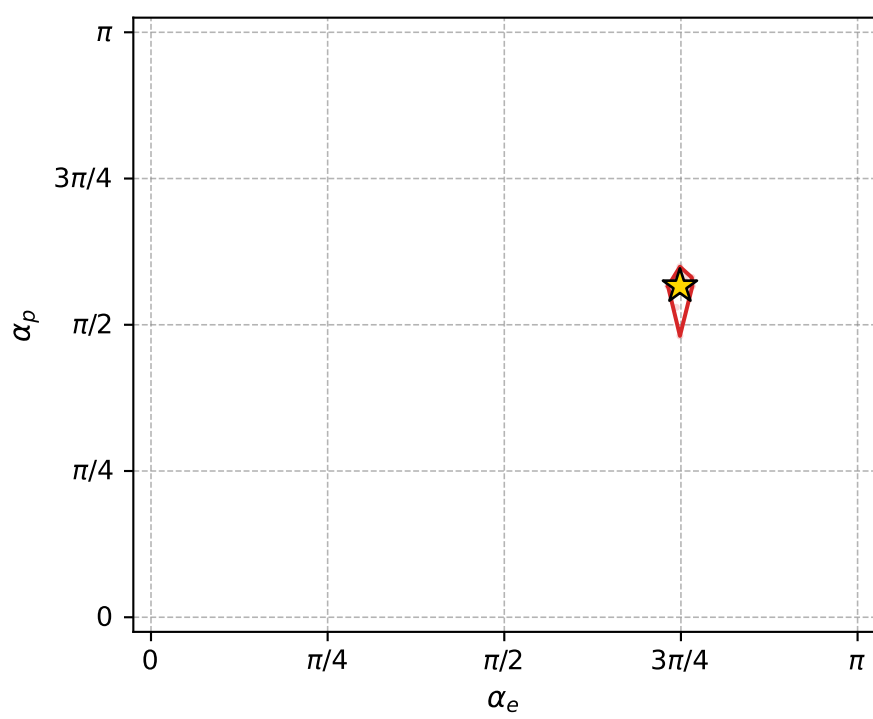
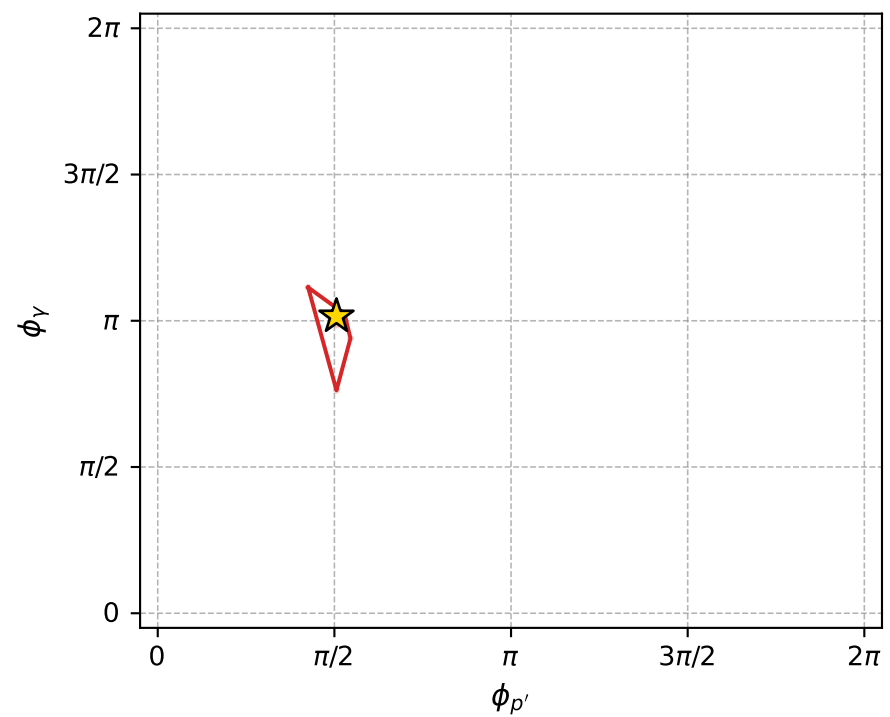
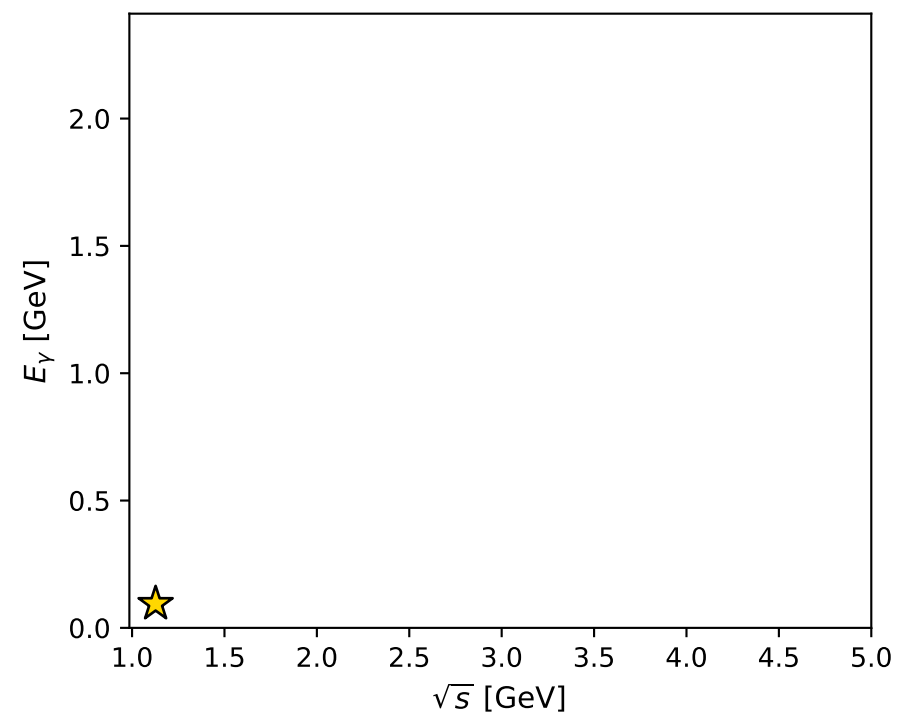
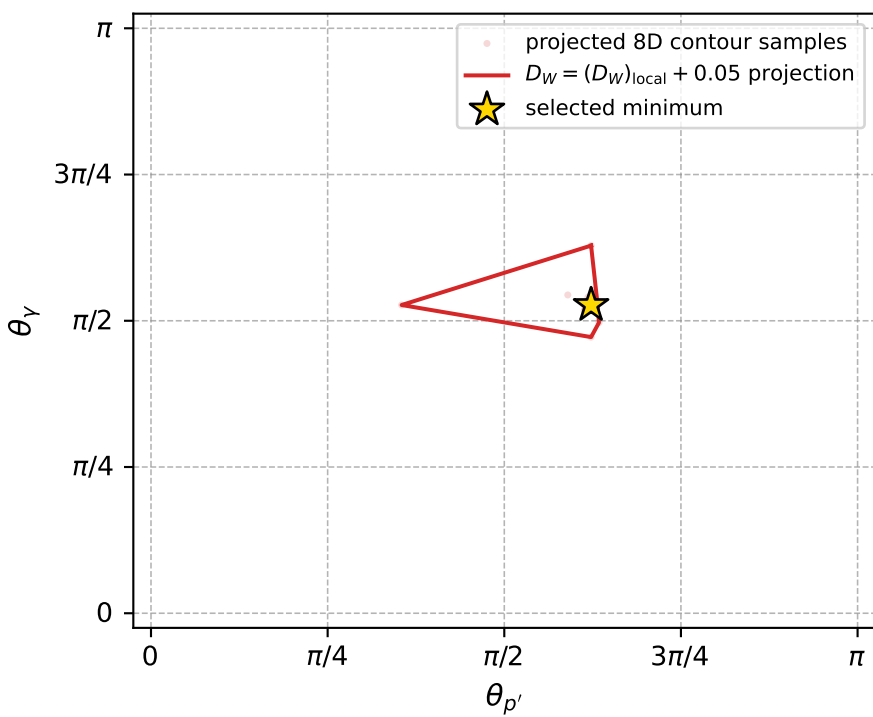
dw_mixing_angles_polarization_06_local_minimum_737 D_W=0.00851272
 region: polarization_cluster_06_local_minimum_737
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3517716$ rad
 incoming lepton: $-0.703972|+> +0.710227|->$
 proton mixing angle: $\alpha_p=1.778068$ rad
 incoming proton: $-0.205791|+> +0.978596|->$

$s=1.27112$, $\sqrt{s}=1.12744$, $|\vec{P}|=0.173522$, $|\vec{P}'|=0.000874221$
 $\theta_{P'}=1.9567$, $\theta_V=1.65541$, $\phi_{P'}=1.58924$, $\phi_V=3.18946$
 $E_V=0.0947139$, $\theta_{\ell'}=1.4827$, $\phi_{\ell'}=0.0392903$
 $Q^2=0.0357649$, $x_B=0.167558$, $t=-0.0299715$, $W^2=1.05753$, $y=0.545523$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1735$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1735$
 P $E= 0.9539$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1735$
 ℓ' $E= 0.0947$ $p_x= 0.0943$ $p_y= 0.0037$ $p_z= 0.0083$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0008$ $p_z= -0.0003$
 q_V $E= 0.0947$ $p_x= -0.0943$ $p_y= -0.0045$ $p_z= -0.0080$



electron: polarization cluster P6, local minimum 737; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.008512719$
 $D_W \text{ contour} = 0.058512719$
 above global minimum = 0.0084962862
 8D contour samples = 254

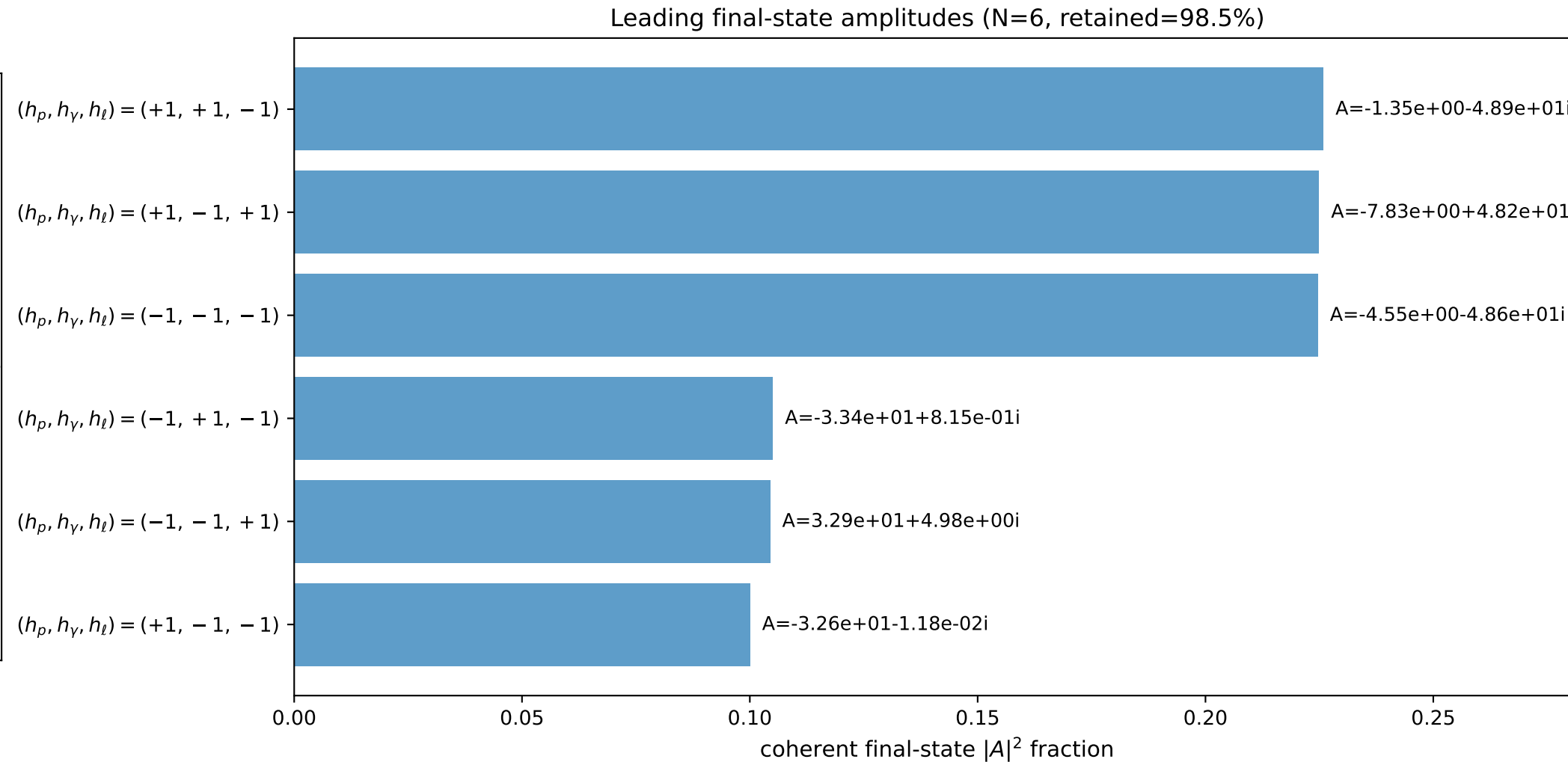
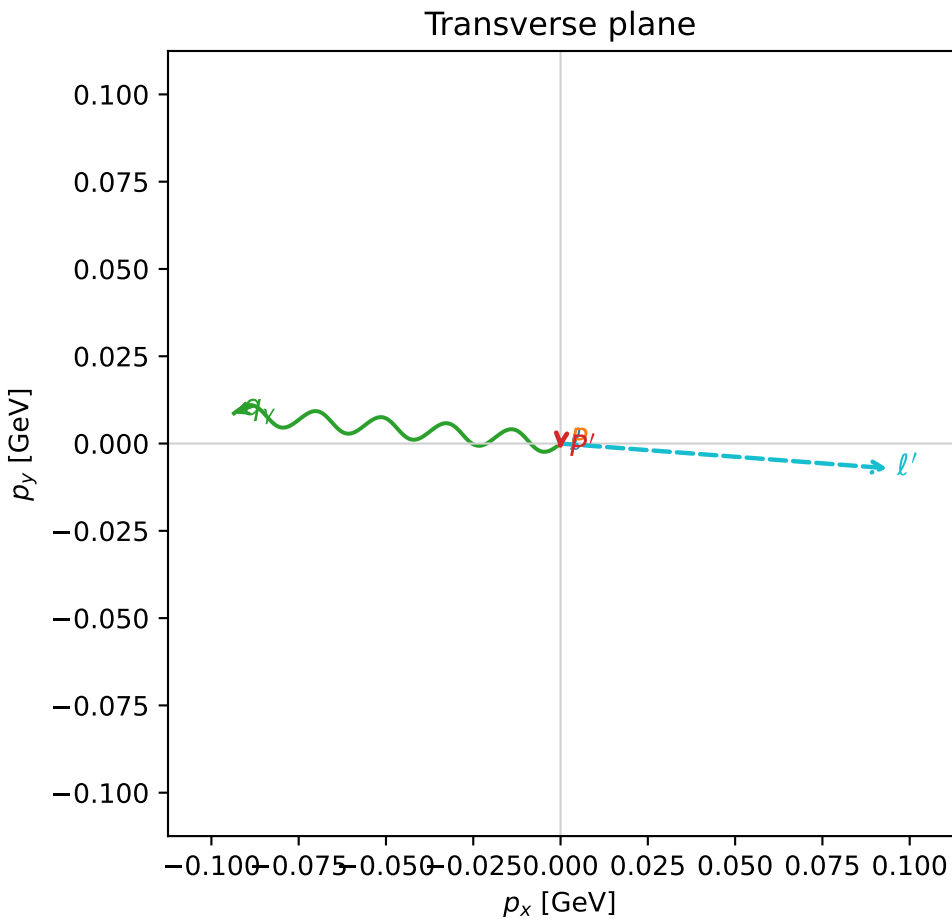
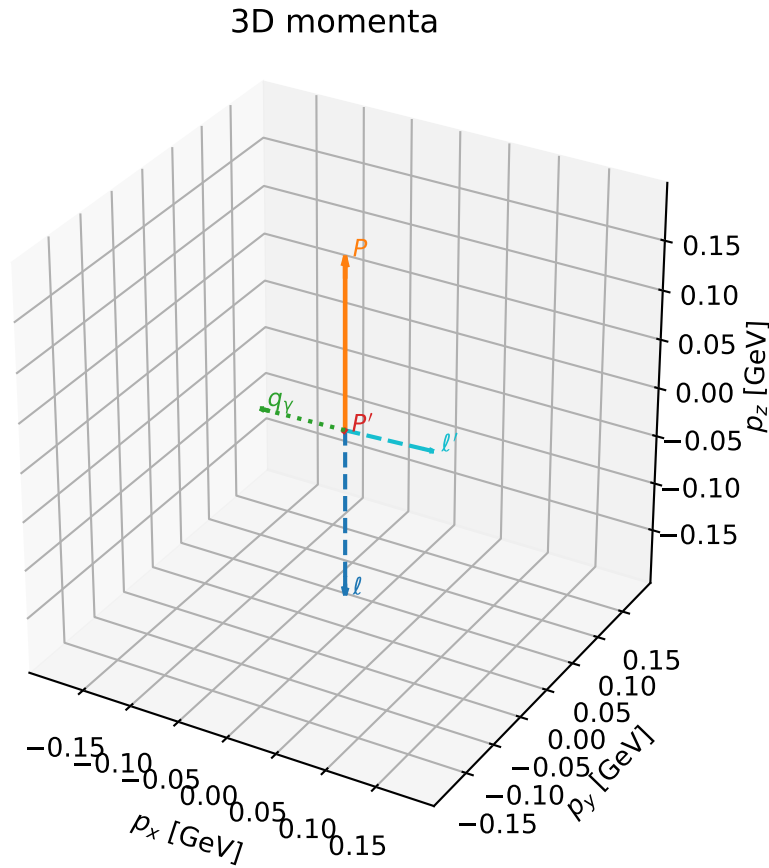
selected phase-space point:
 $\text{sqrt_s} = 1.127438$
 $\text{theta_p_out} = 1.956698$
 $\text{theta_gamma_out} = 1.655408$
 $\text{qOut} = 0.0947139$
 $\text{phi_p_out} = 1.589244$
 $\text{phi_gamma_out} = 3.189463$
 $\text{alpha_e} = 2.351772$
 $\text{alpha_p} = 1.778068$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

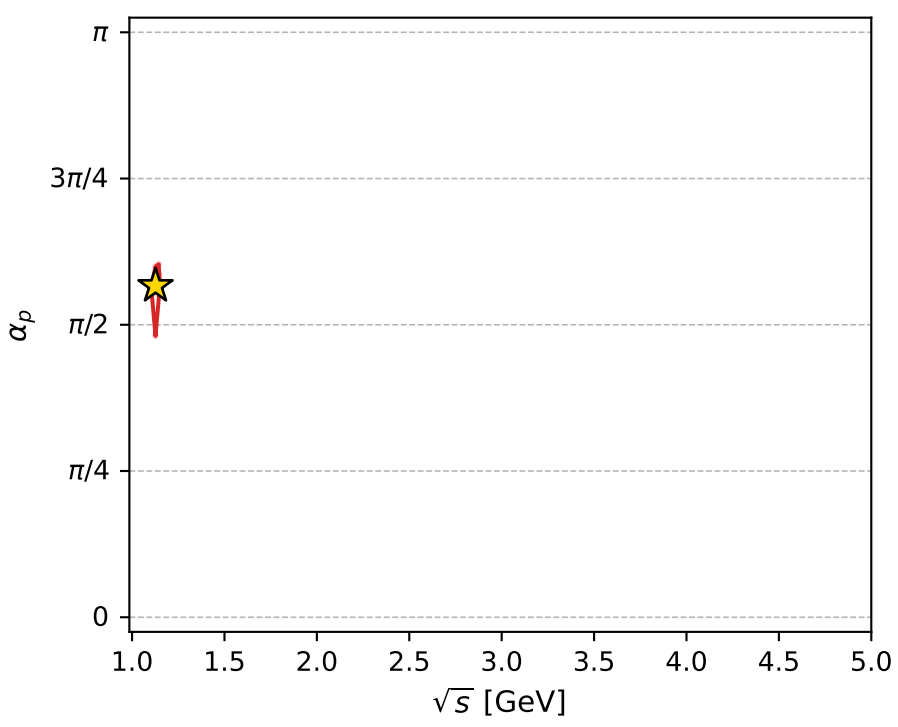
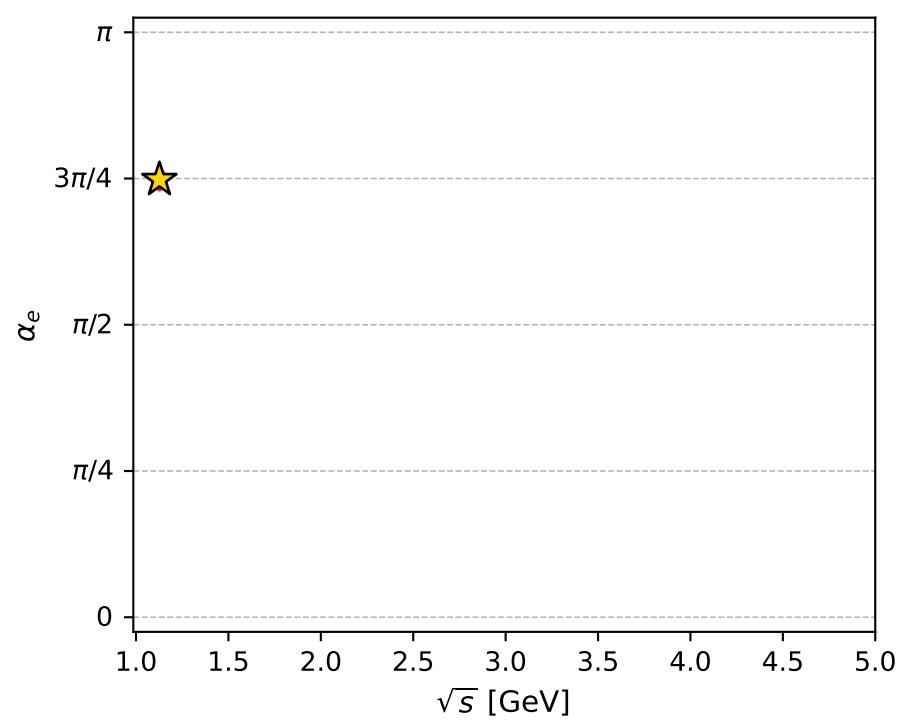
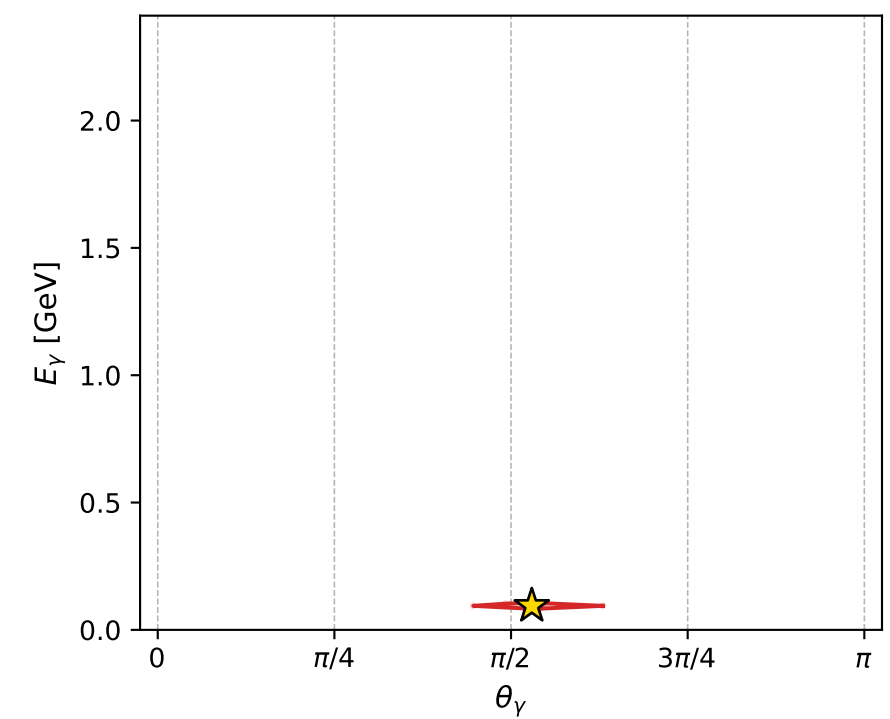
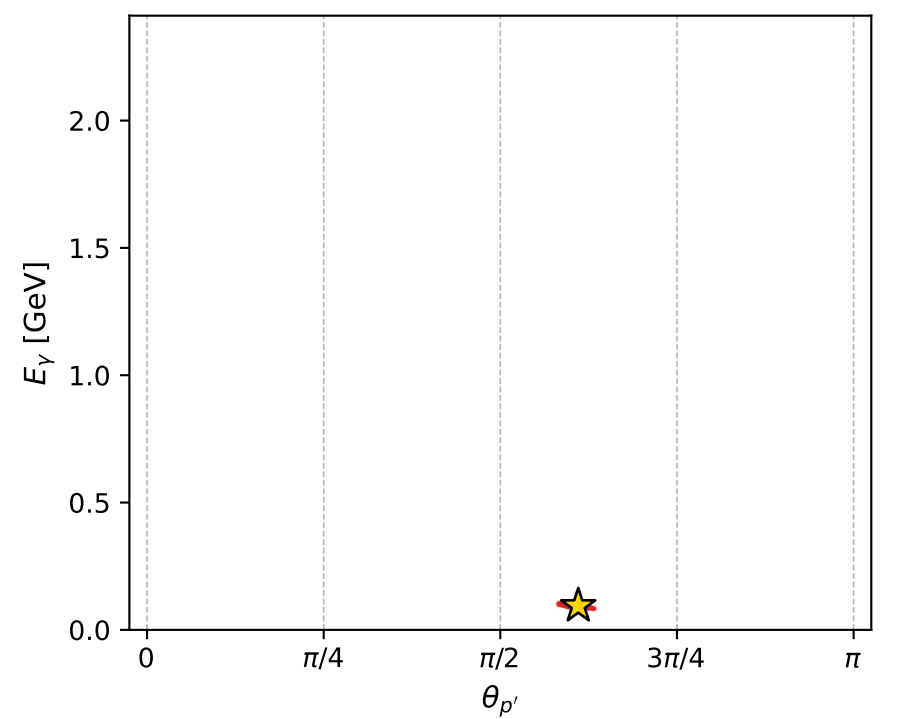
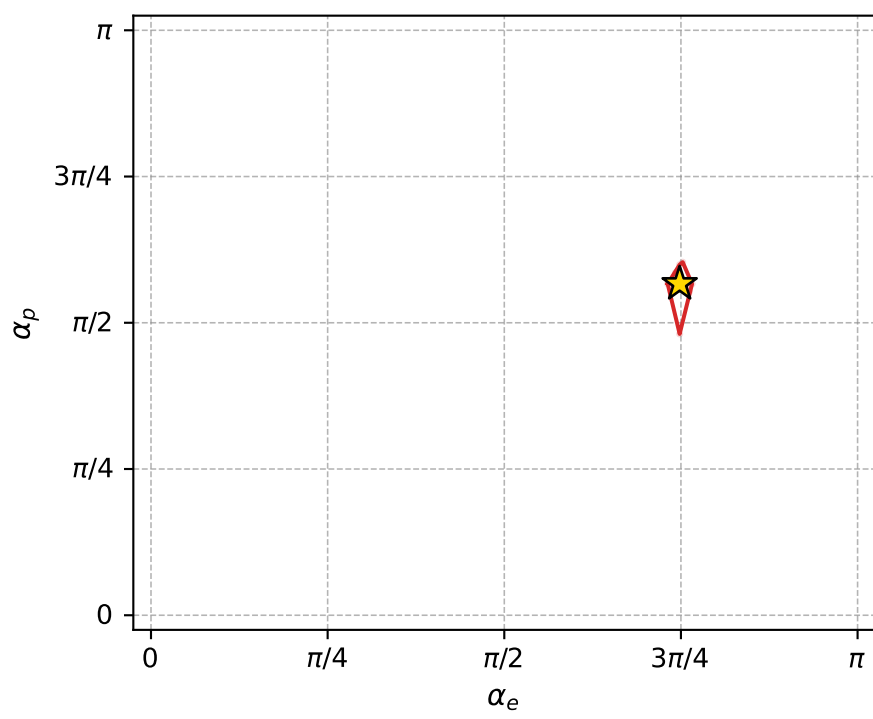
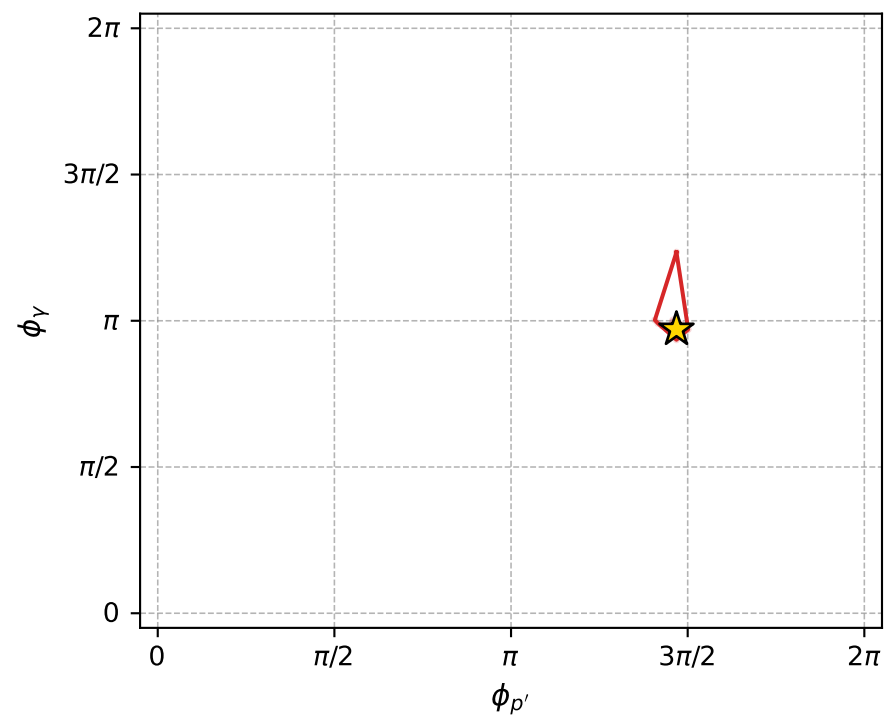
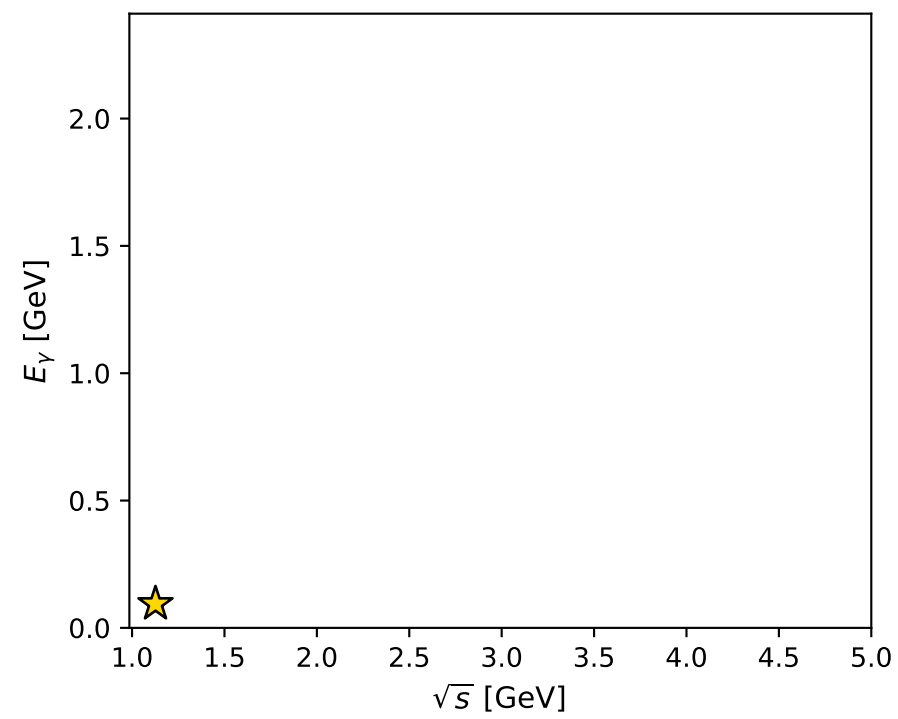
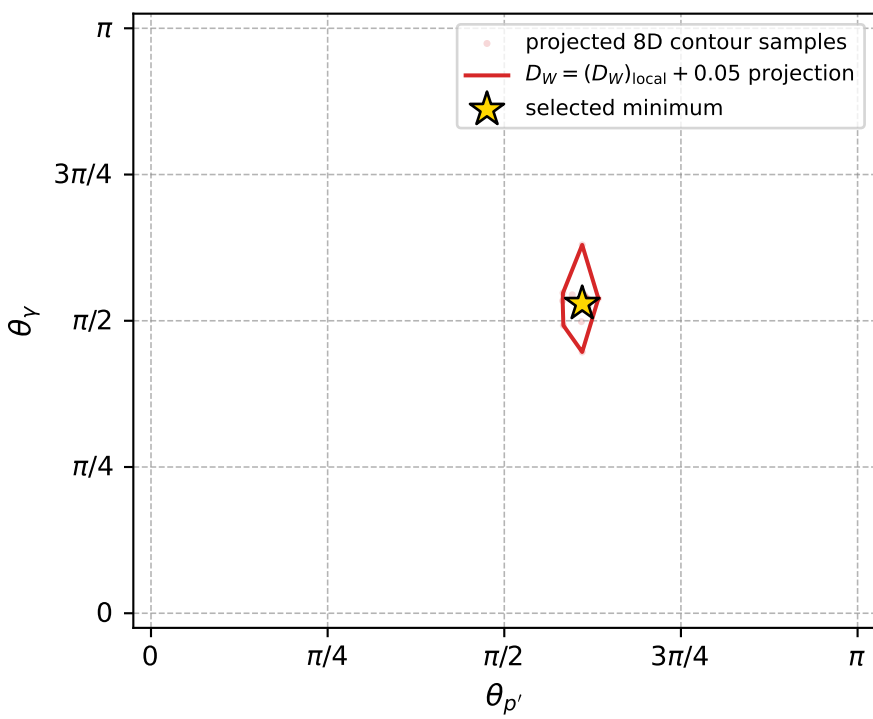
dw_mixing_angles_polarization_06_local_minimum_742 D_W=0.00864219
 region: polarization_cluster_06_local_minimum_742
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3504977$ rad
 incoming lepton: $-0.703067|+> +0.711124|->$
 proton mixing angle: $\alpha_p=1.7798265$ rad
 incoming proton: $-0.207511|+> +0.978233|->$

$s=1.26957$, $\sqrt{s}=1.12675$, $|\vec{P}|=0.172943$, $|\vec{P}'|=0.00168659$
 $\theta_{p'}=1.91717$, $\theta_Y=1.66317$, $\phi_{p'}=4.61255$, $\phi_Y=3.04933$
 $E_Y=0.0943357$, $\theta_{\ell'}=1.47241$, $\phi_{\ell'}=6.20781$
 $Q^2=0.0358651$, $x_B=0.168518$, $t=-0.0298604$, $W^2=1.05681$, $y=0.546087$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1729$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1729$
 P $E= 0.9538$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1729$
 ℓ' $E= 0.0944$ $p_x= 0.0937$ $p_y= -0.0071$ $p_z= 0.0093$
 P' $E= 0.9380$ $p_x= -0.0002$ $p_y= -0.0016$ $p_z= -0.0006$
 q_Y $E= 0.0943$ $p_x= -0.0935$ $p_y= 0.0087$ $p_z= -0.0087$



electron: polarization cluster P6, local minimum 742; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0086421913$
 $D_W \text{ contour} = 0.058642191$
 above global minimum = 0.0086257585
 8D contour samples = 253

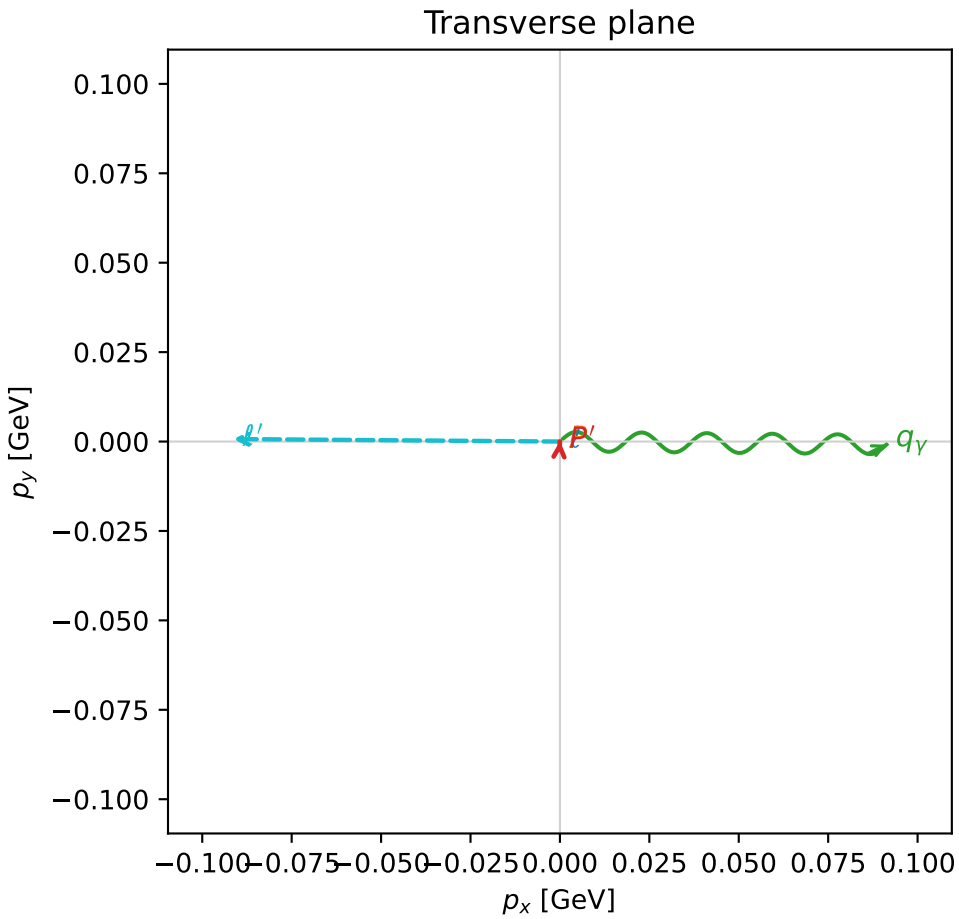
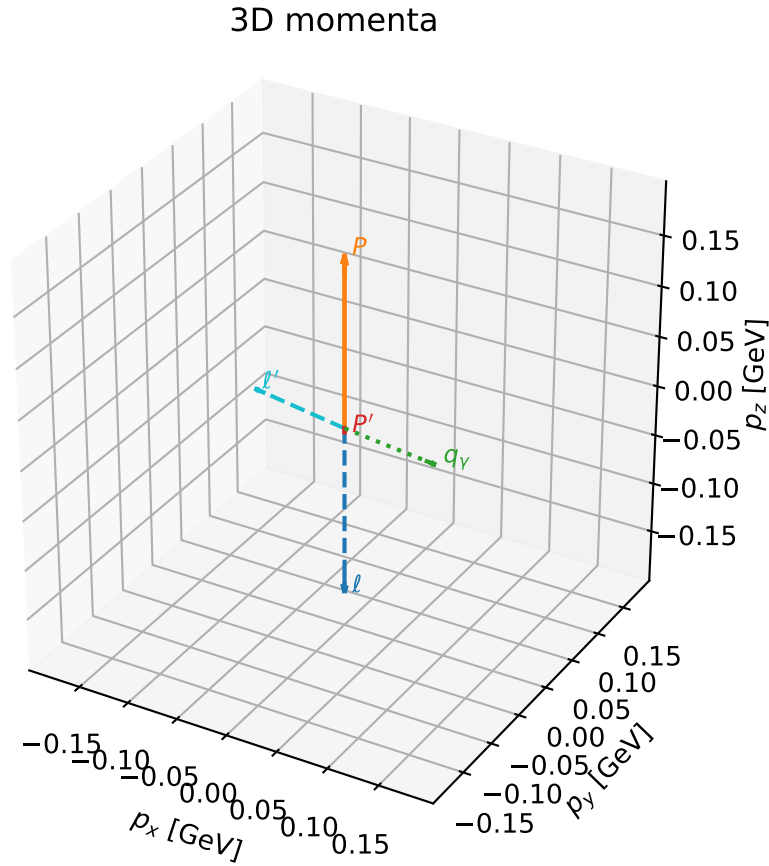
selected phase-space point:
 $\text{sqrt_s} = 1.126754$
 $\text{theta_p_out} = 1.917174$
 $\text{theta_gamma_out} = 1.663172$
 $\text{q0out} = 0.0943357$
 $\text{phi_p_out} = 4.61255$
 $\text{phi_gamma_out} = 3.04933$
 $\text{alpha_e} = 2.350498$
 $\text{alpha_p} = 1.779826$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_759 D_W=0.00935482
 region: polarization_cluster_06_local_minimum_759
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3545656$ rad
 incoming lepton: $-0.705954|+> +0.708258|->$
 proton mixing angle: $\alpha_p=1.3737025$ rad
 incoming proton: $0.19582|+> +0.98064|->$

$s=1.25922$, $\sqrt{s}=1.12215$, $|\vec{P}|=0.169039$, $|\vec{P}'|=0.00397886$
 $\theta_{P'}=3.10698$, $\theta_Y=1.67588$, $\phi_{P'}=1.50837$, $\phi_Y=6.27405$
 $E_Y=0.091815$, $\theta_{\ell'}=1.42287$, $\phi_{\ell'}=3.13397$
 $Q^2=0.0358132$, $x_B=0.172194$, $t=-0.0297062$, $W^2=1.05201$, $y=0.548223$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1690$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1690$
 P $E=0.9531$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1690$
 ℓ' $E=0.0923$ $p_x=-0.0913$ $p_y=0.0007$ $p_z=0.0136$
 P' $E=0.9380$ $p_x=0.0000$ $p_y=0.0001$ $p_z=-0.0040$
 q_Y $E=0.0918$ $p_x=0.0913$ $p_y=-0.0008$ $p_z=-0.0096$

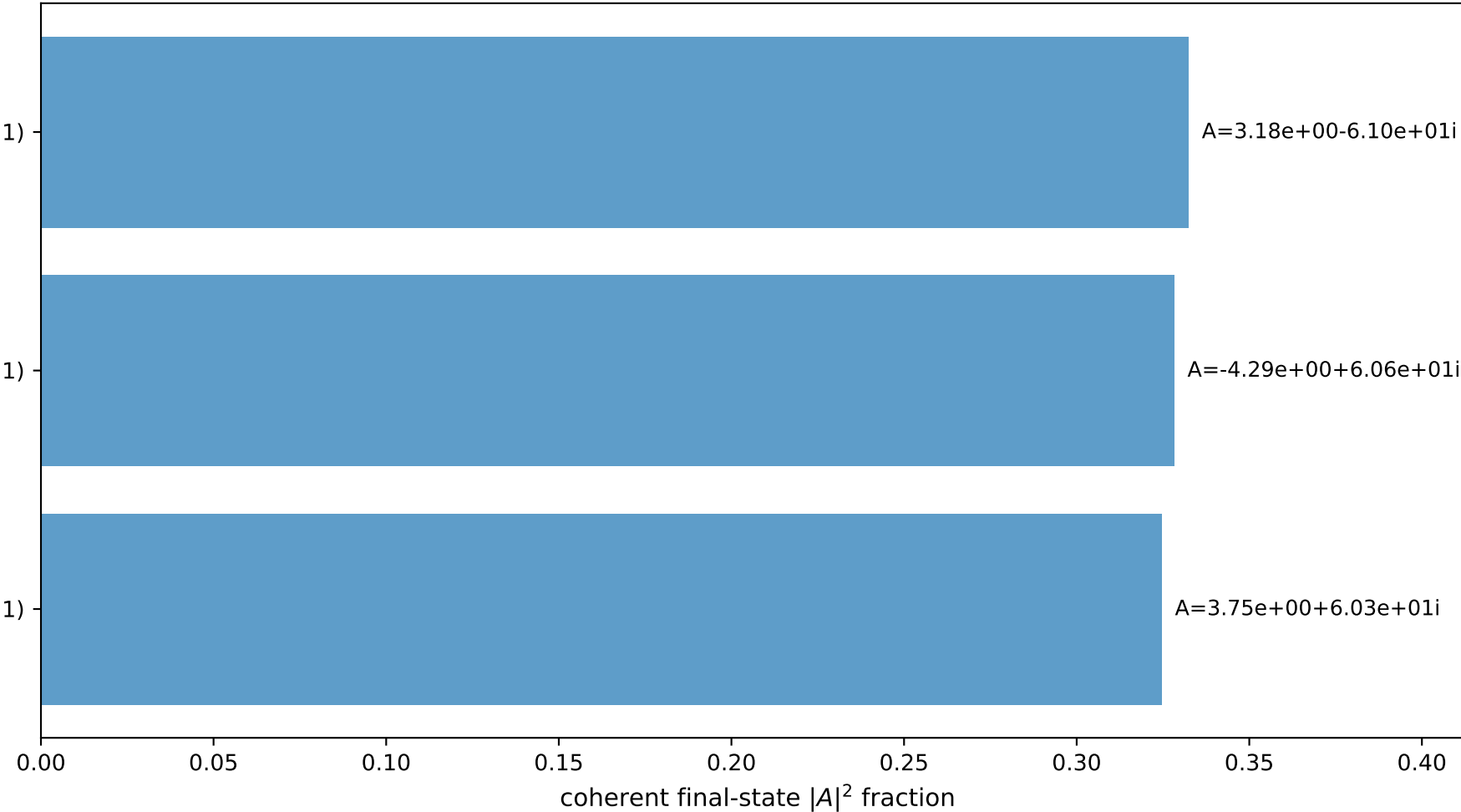


$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

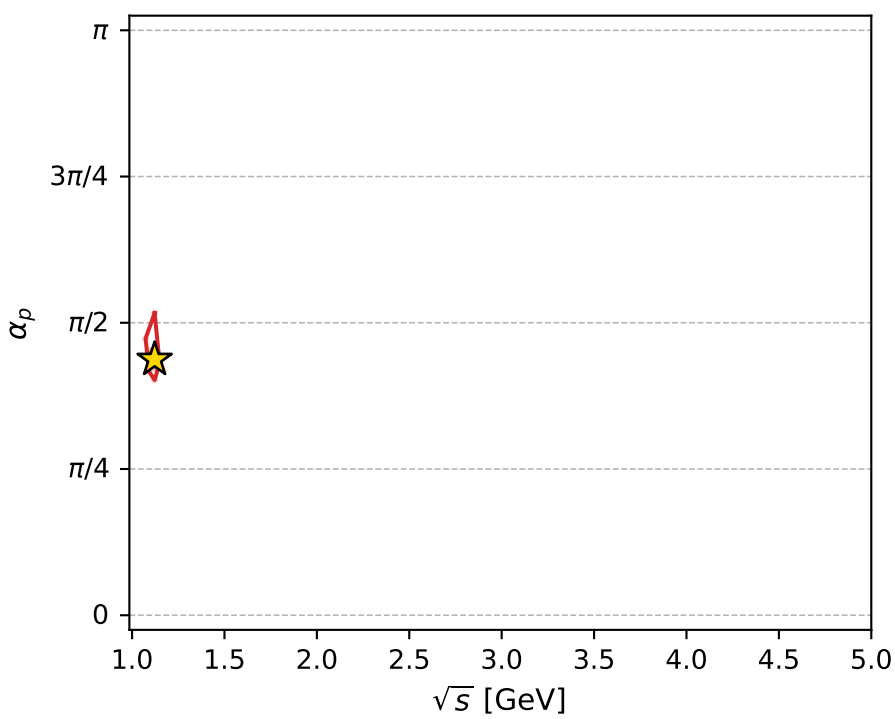
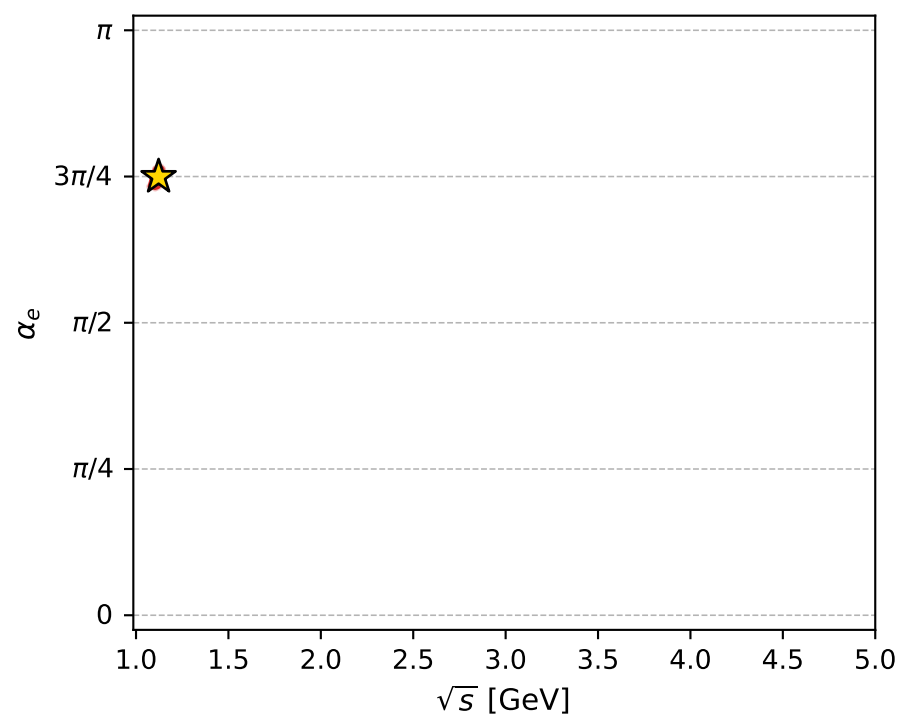
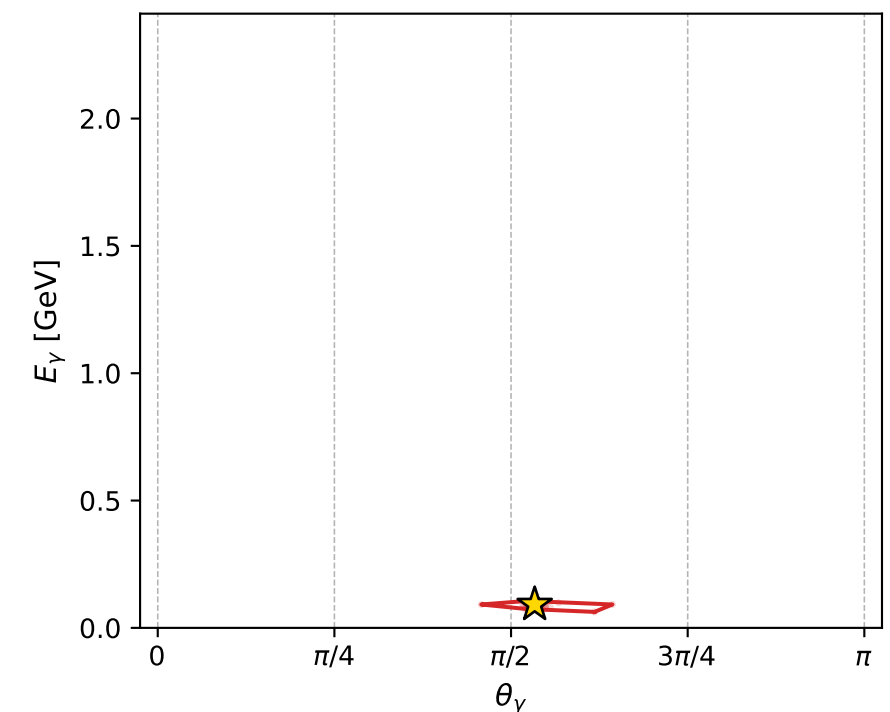
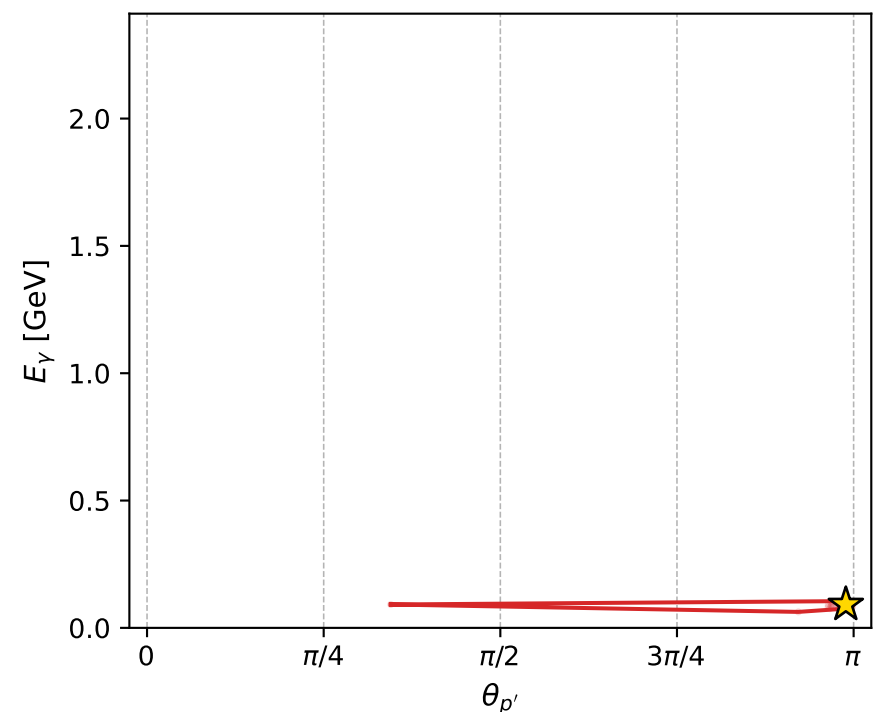
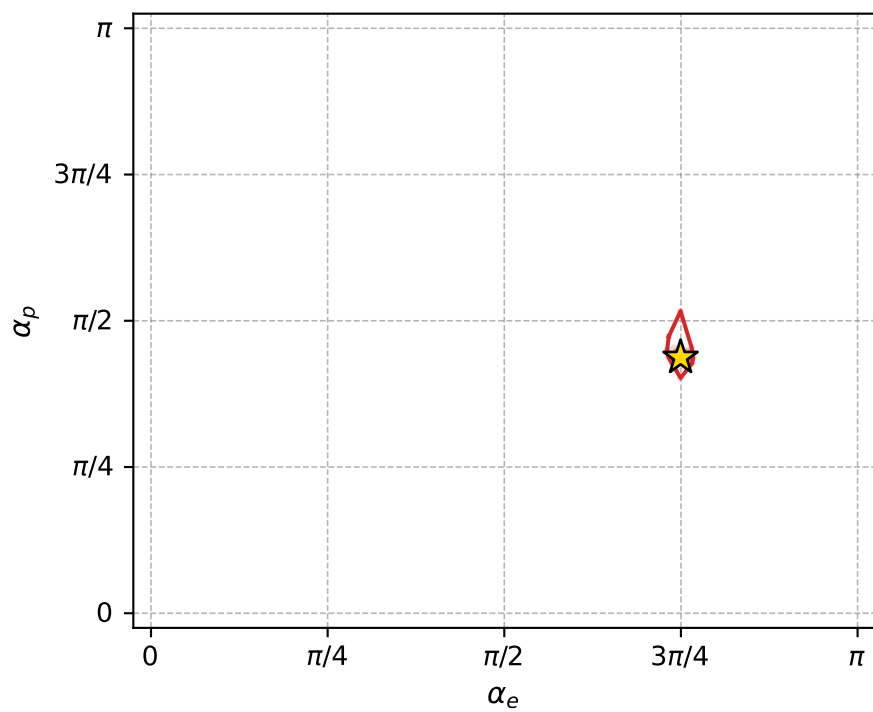
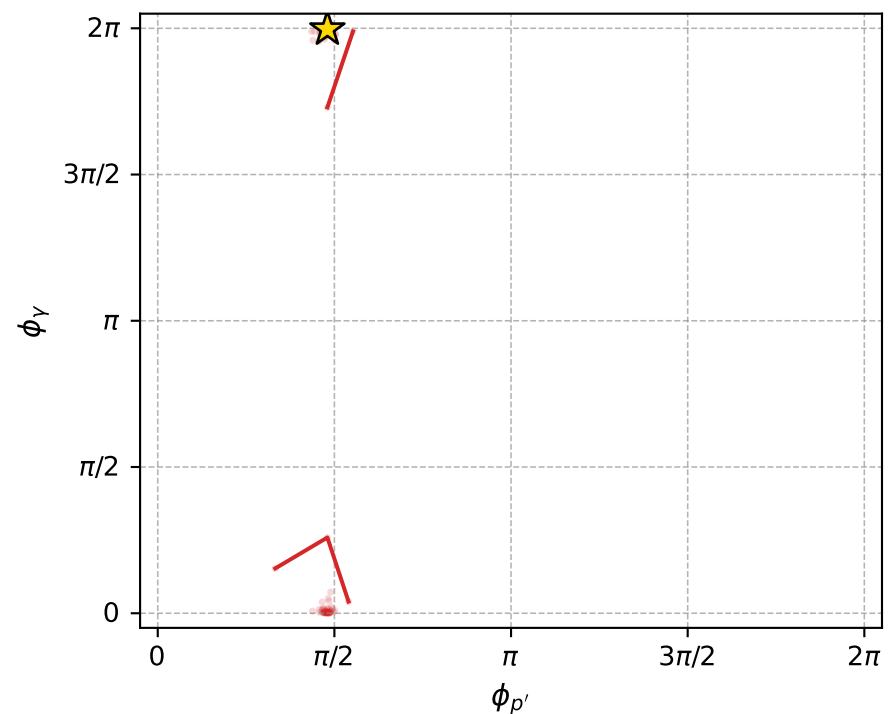
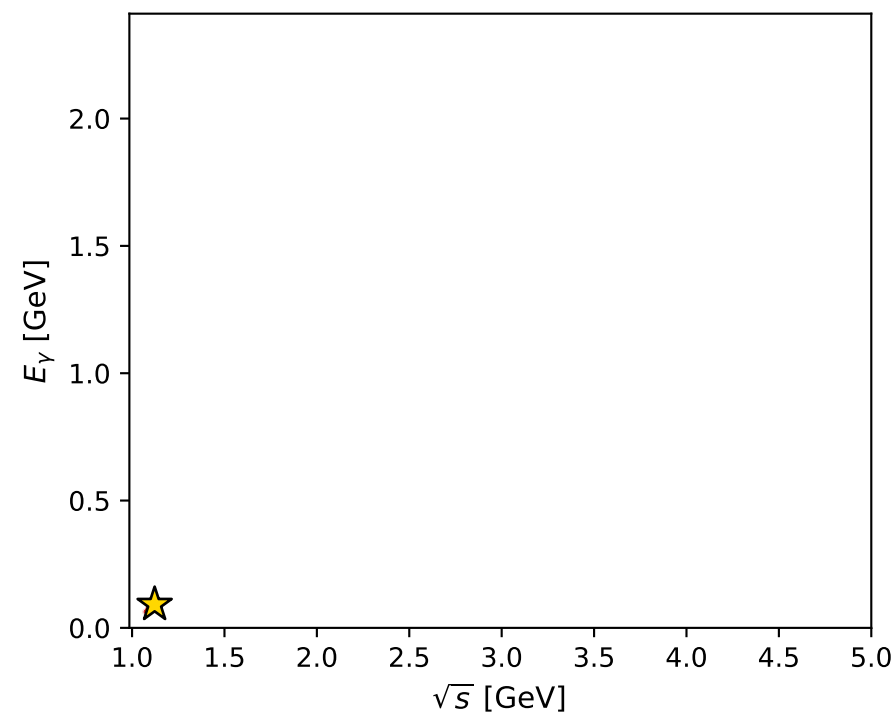
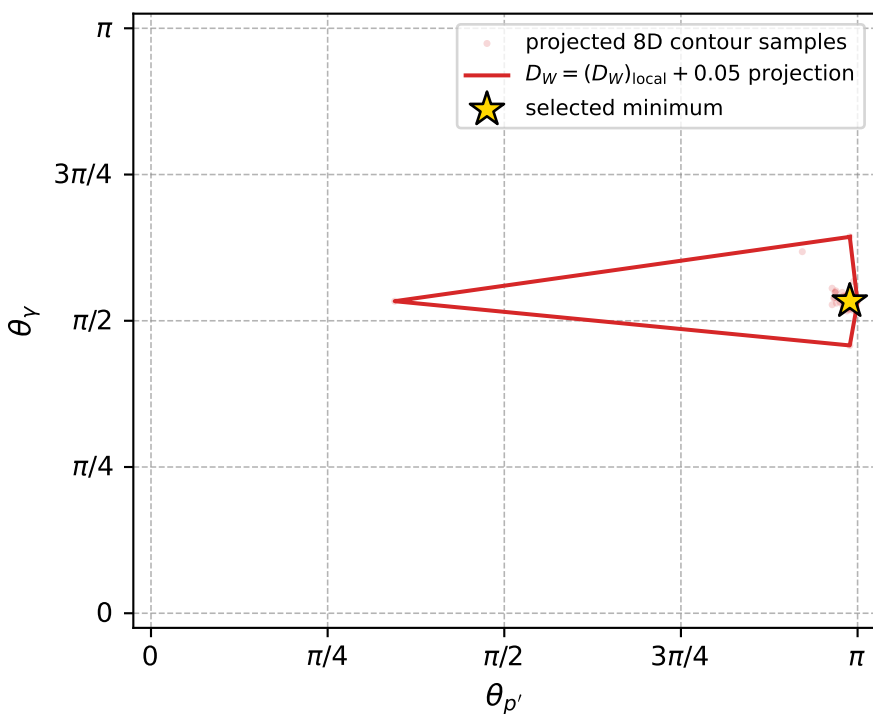
$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, -1)$

Leading final-state amplitudes (N=3, retained=98.5%)



electron: polarization cluster P6, local minimum 759; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0093548213$
 $D_W \text{ contour} = 0.059354821$
 above global minimum = 0.0093383885
 8D contour samples = 239

selected phase-space point:
 $\text{sqrt}_s = 1.122149$
 $\text{theta}_p_{\text{out}} = 3.106979$
 $\text{theta}_\gamma_{\text{out}} = 1.675884$
 $q_{\text{out}} = 0.09181504$
 $\text{phi}_p_{\text{out}} = 1.508373$
 $\text{phi}_\gamma_{\text{out}} = 6.274054$
 $\alpha_e = 2.354566$
 $\alpha_p = 1.373703$

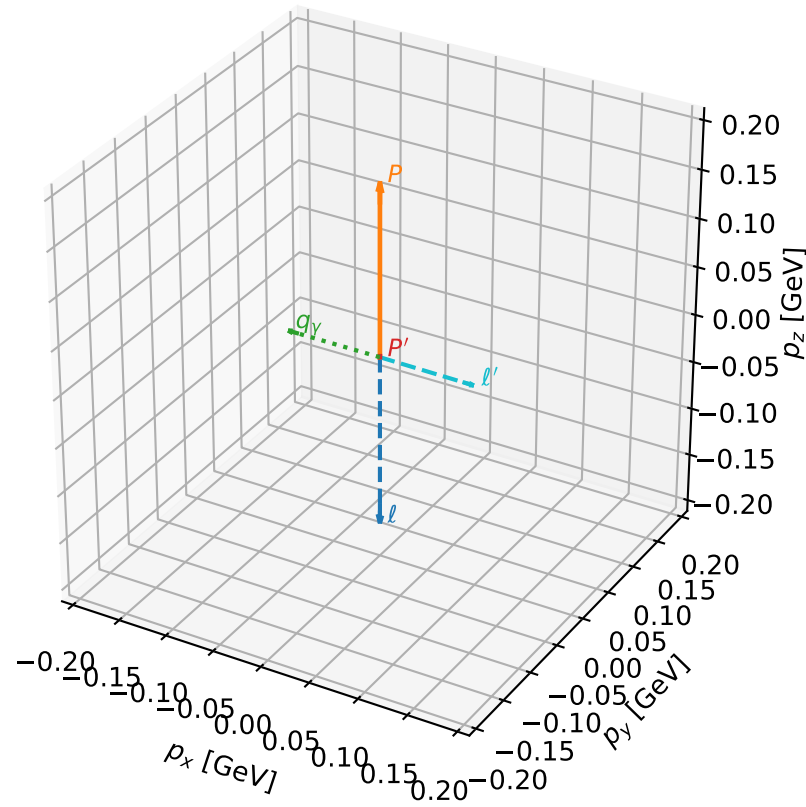
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_762 D_W=0.00938309
 region: polarization_cluster_06_local_minimum_762
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.362513$ rad
 incoming lepton: $-0.71156|+\rangle + 0.702625|-\rangle$
 proton mixing angle: $\alpha_p=1.7776891$ rad
 incoming proton: $-0.20542|+\rangle + 0.978674|-\rangle$

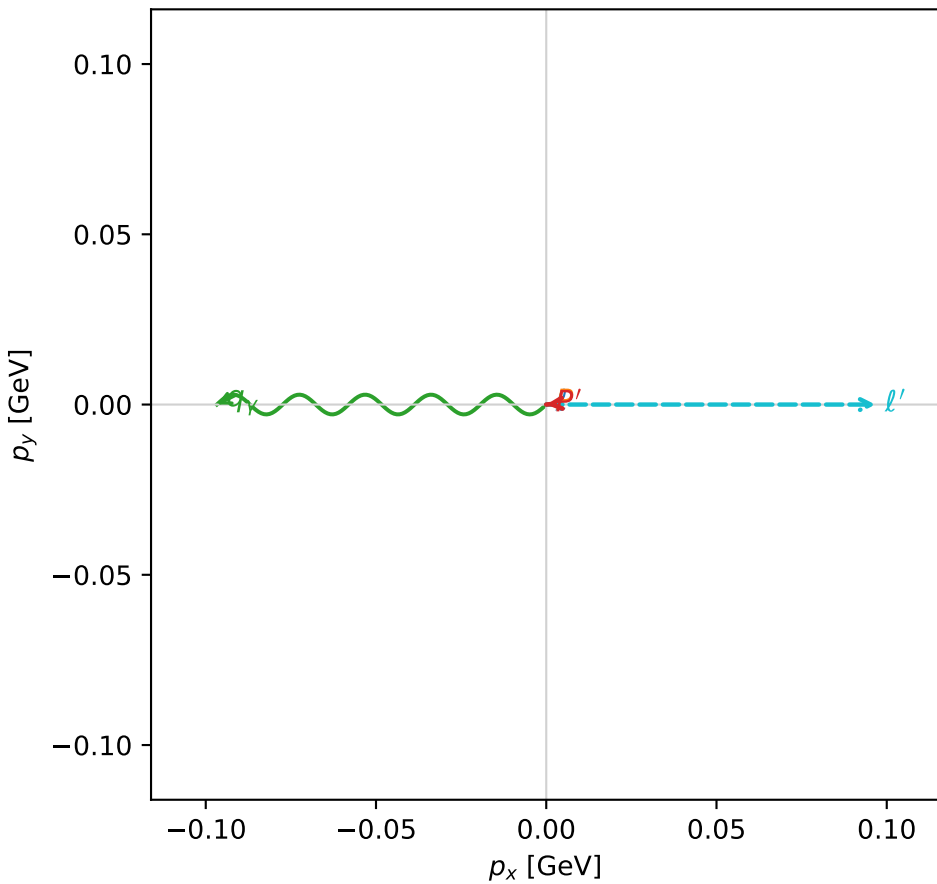
$s=1.28018$, $\sqrt{s}=1.13145$, $|\vec{P}|=0.176913$, $|\vec{P}'|=0.00103867$
 $\theta_{P'}=0.00343516$, $\theta_Y=1.56915$, $\phi_{P'}=3.22195$, $\phi_Y=3.14155$
 $E_Y=0.0967195$, $\theta_{\ell'}=1.58318$, $\phi_{\ell'}=6.28314$
 $Q^2=0.033802$, $x_B=0.157038$, $t=-0.0306584$, $W^2=1.06129$, $y=0.537662$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1769$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1769$
 P $E= 0.9545$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1769$
 ℓ' $E= 0.0967$ $p_x= 0.0967$ $p_y= -0.0000$ $p_z= -0.0012$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0010$
 q_Y $E= 0.0967$ $p_x= -0.0967$ $p_y= 0.0000$ $p_z= 0.0002$

3D momenta



Transverse plane

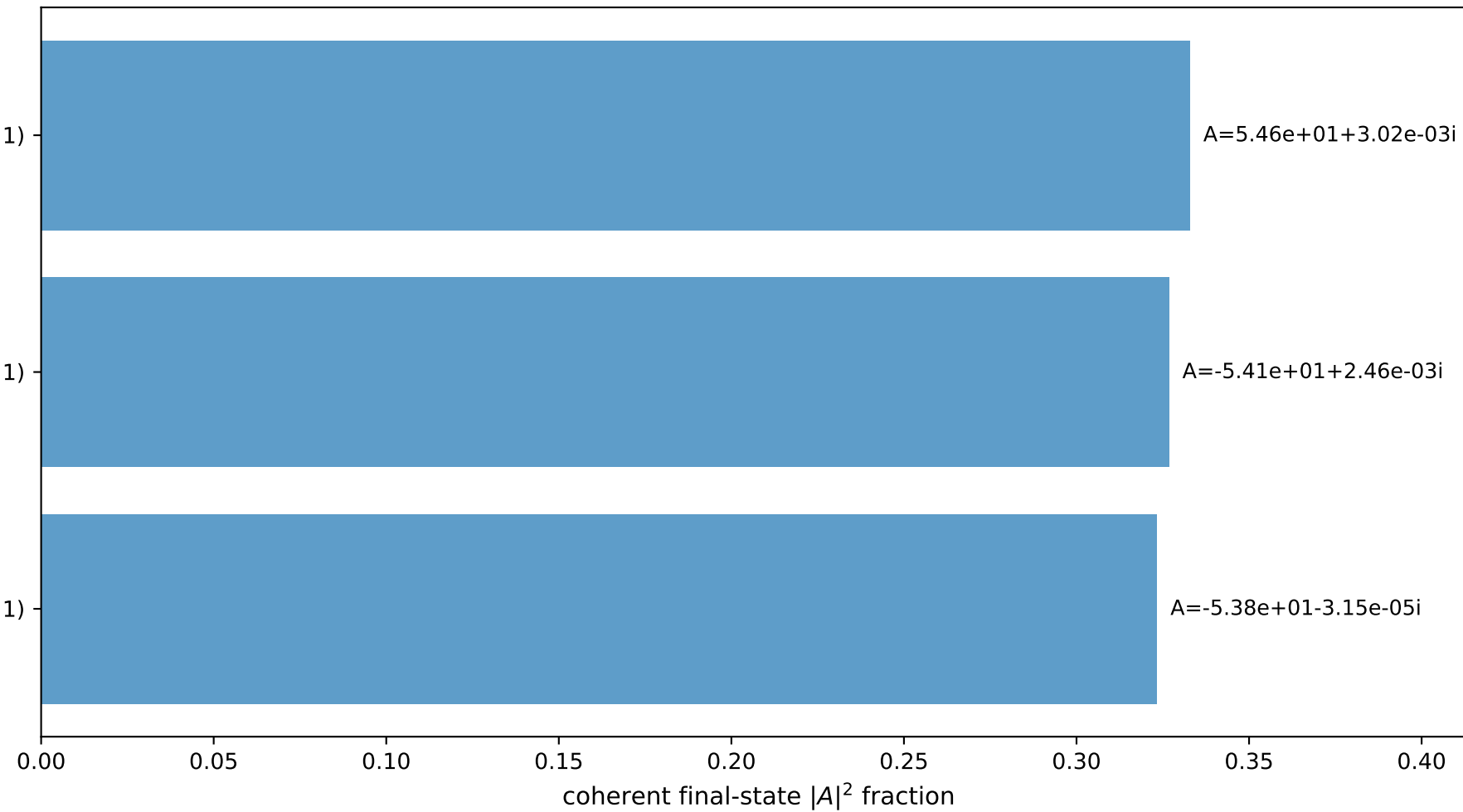


$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

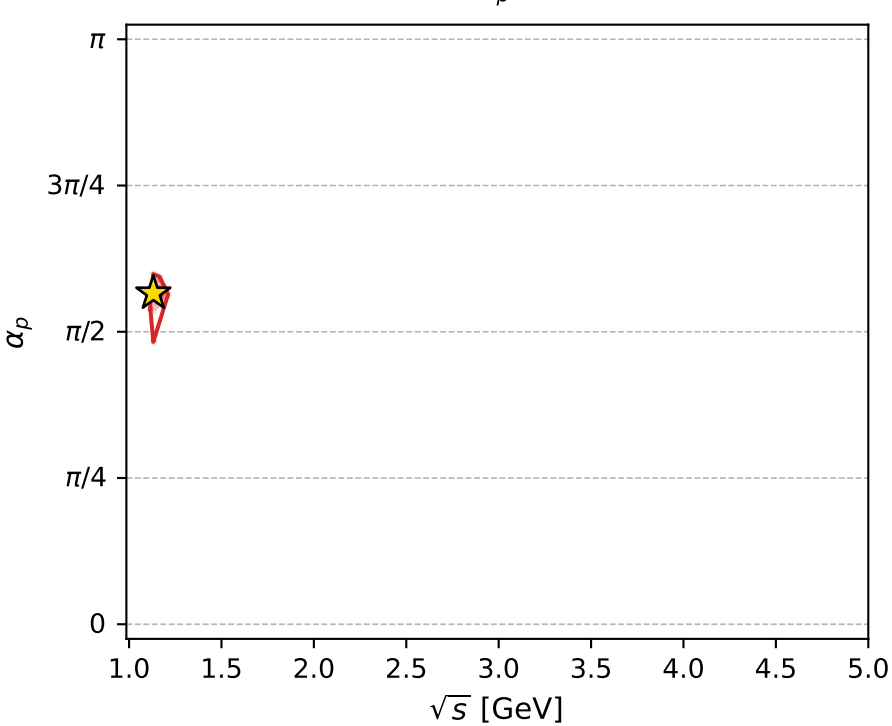
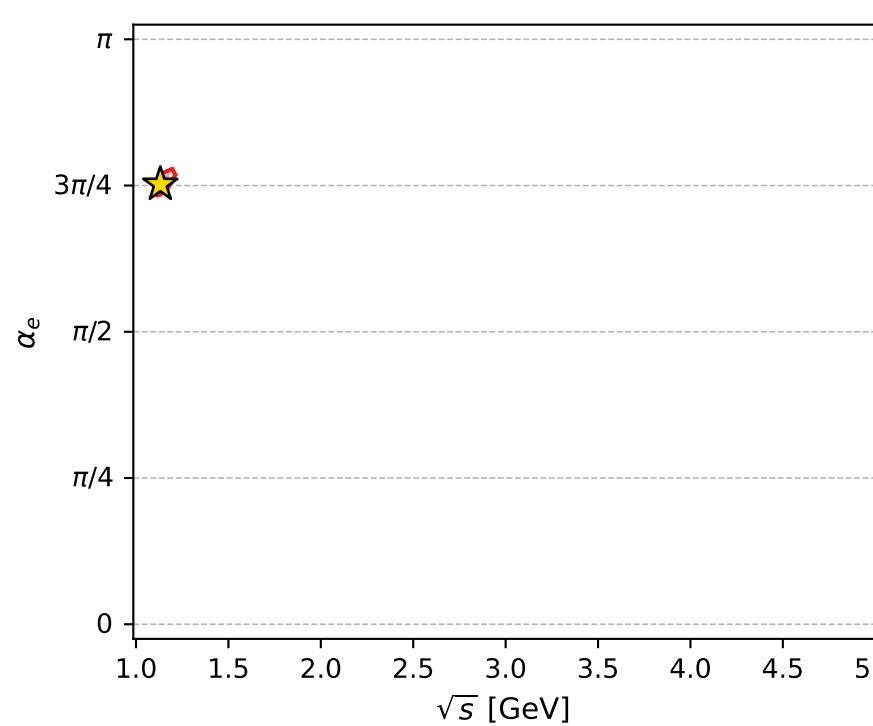
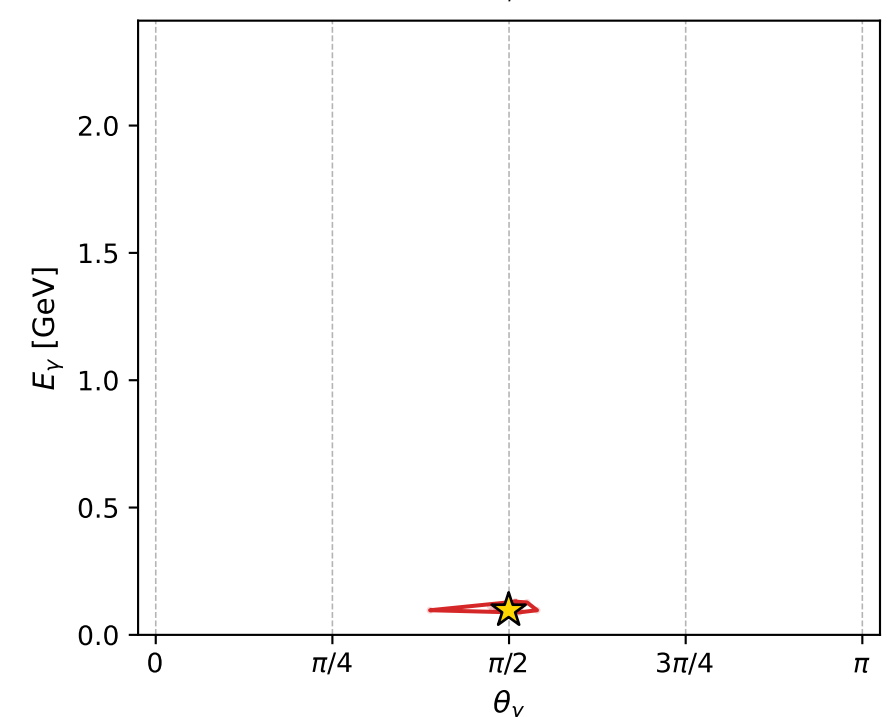
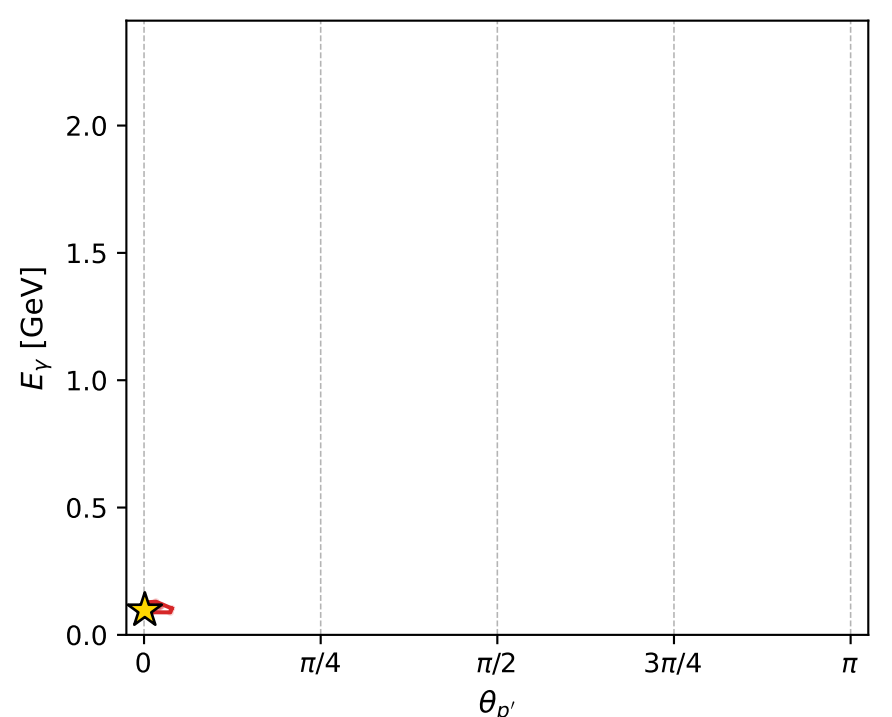
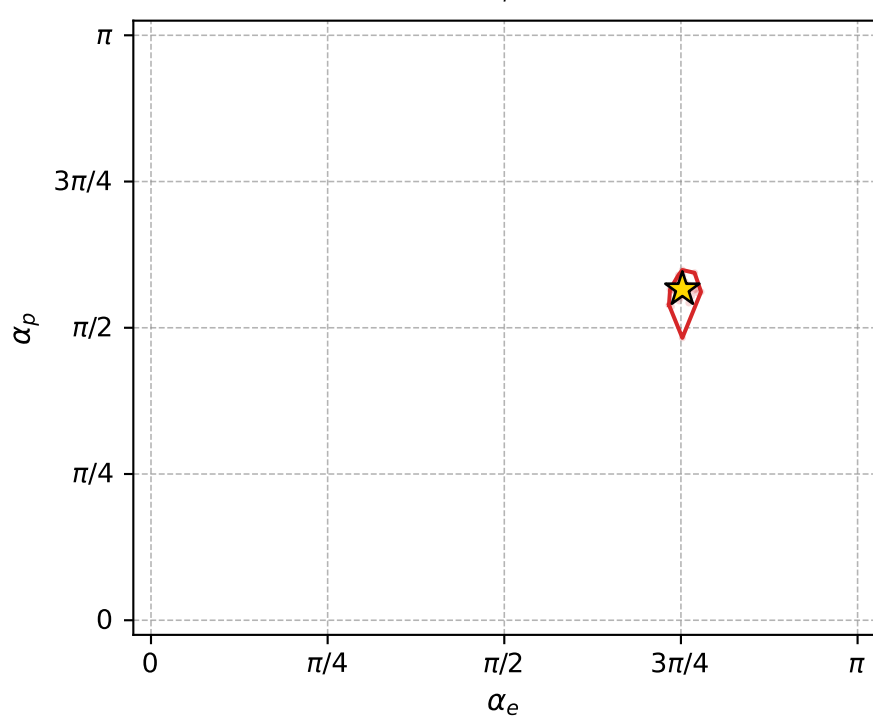
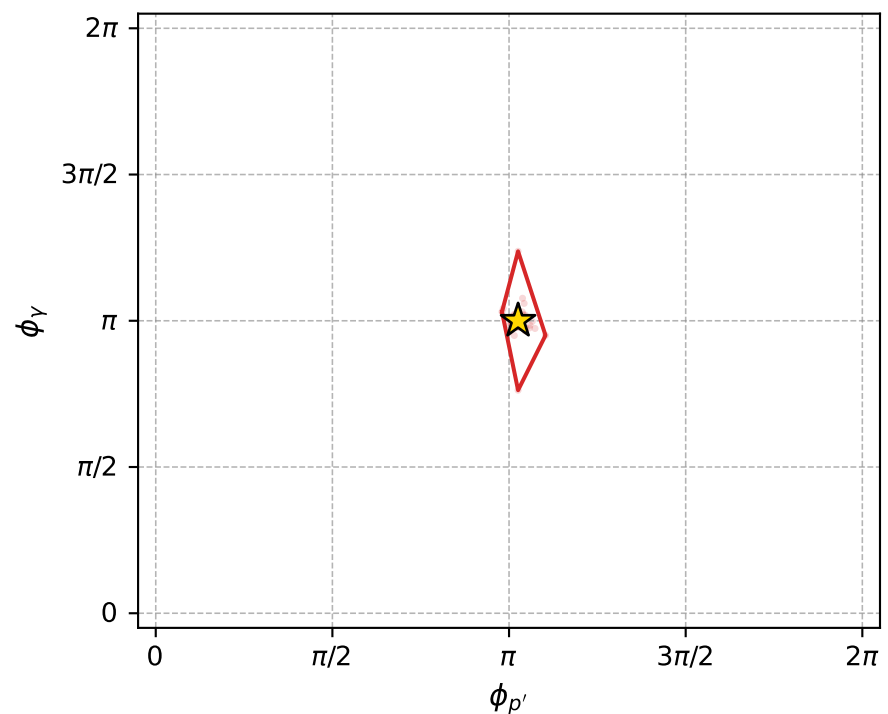
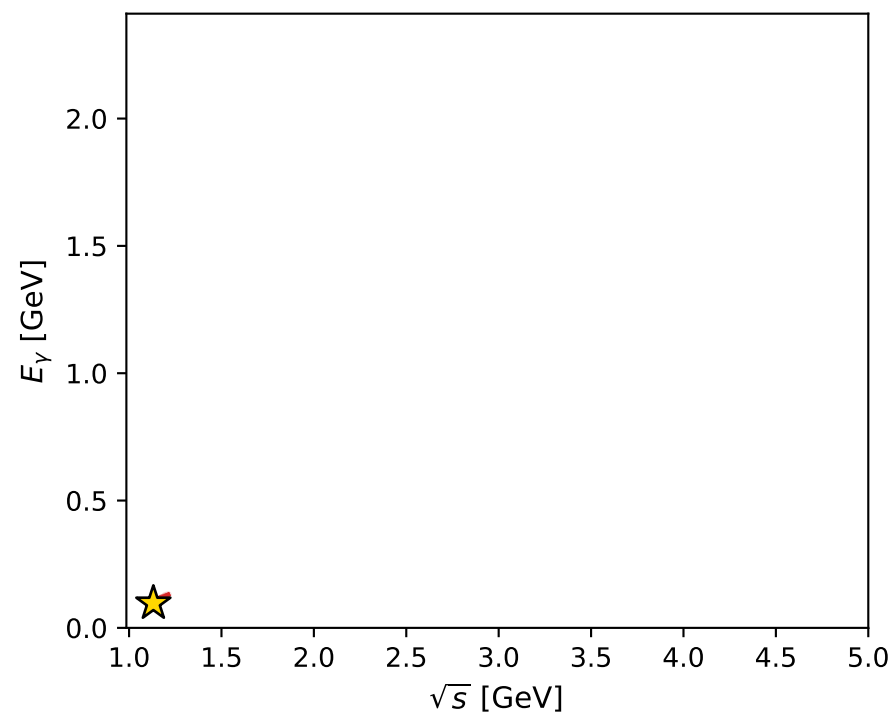
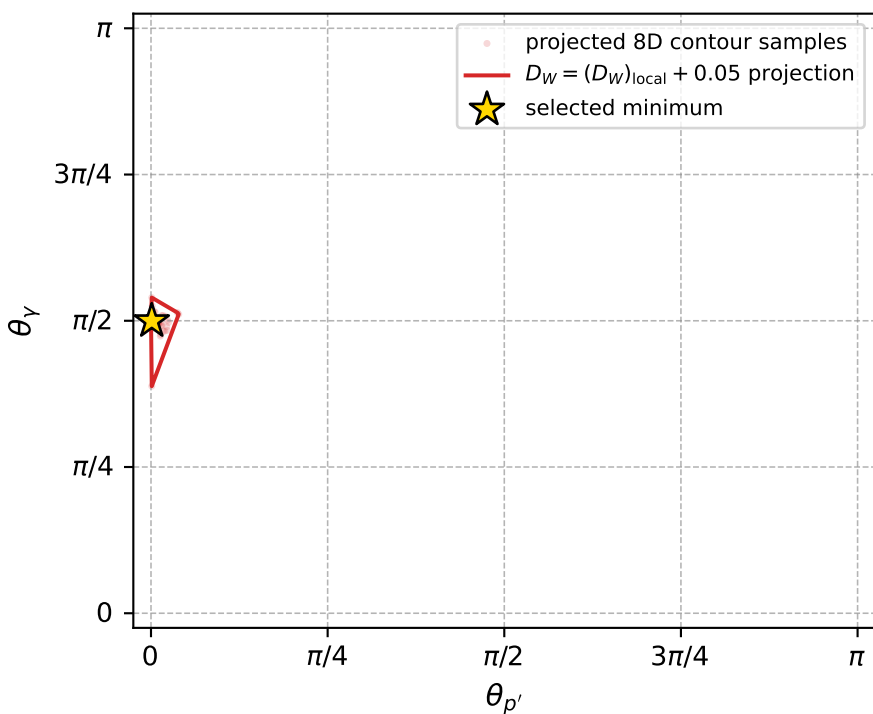
$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (+1, -1, -1)$

Leading final-state amplitudes (N=3, retained=98.3%)



electron: polarization cluster P6, local minimum 762; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.009383093$
 $D_W \text{ contour} = 0.059383093$
 above global minimum = 0.0093666601
 8D contour samples = 188

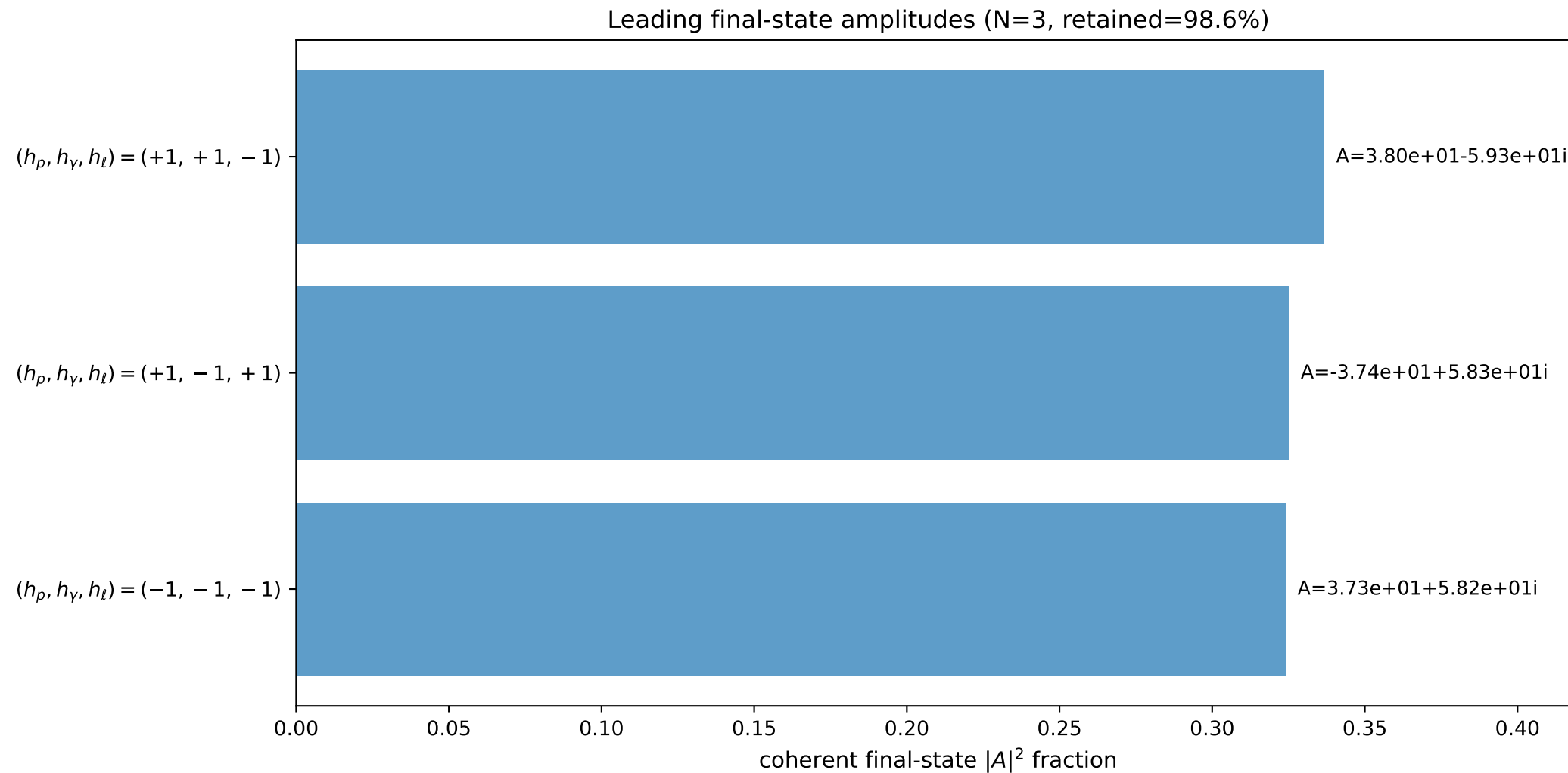
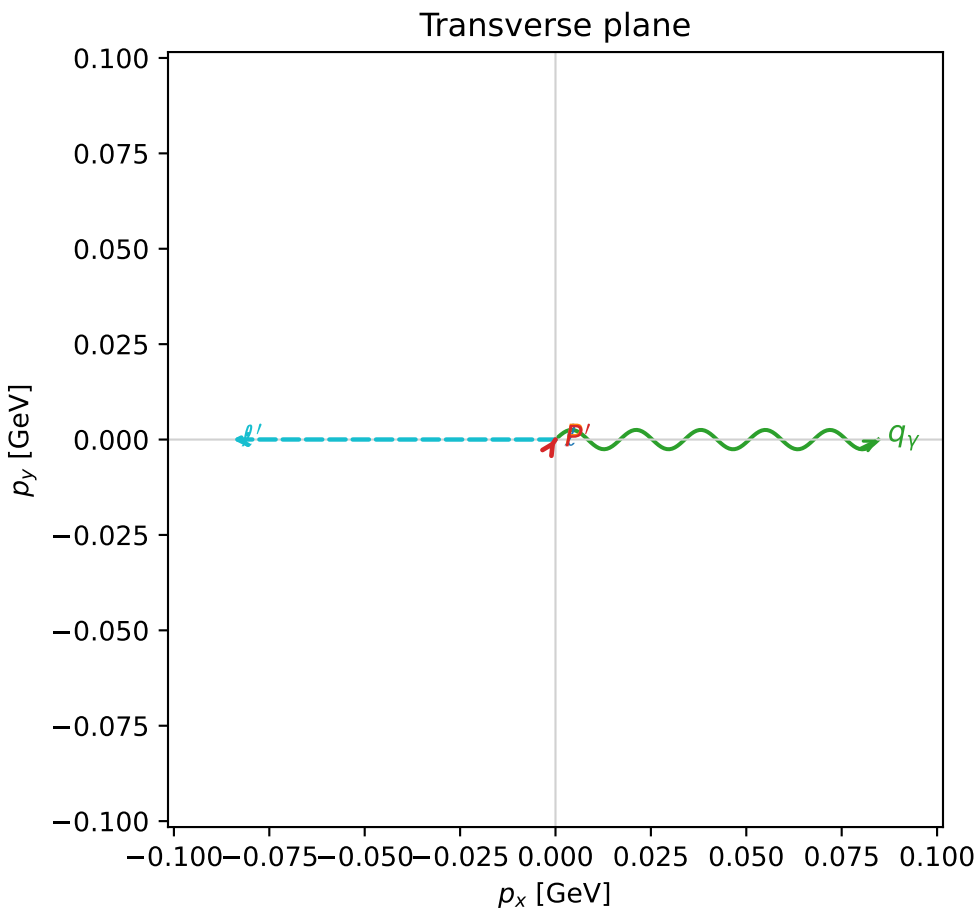
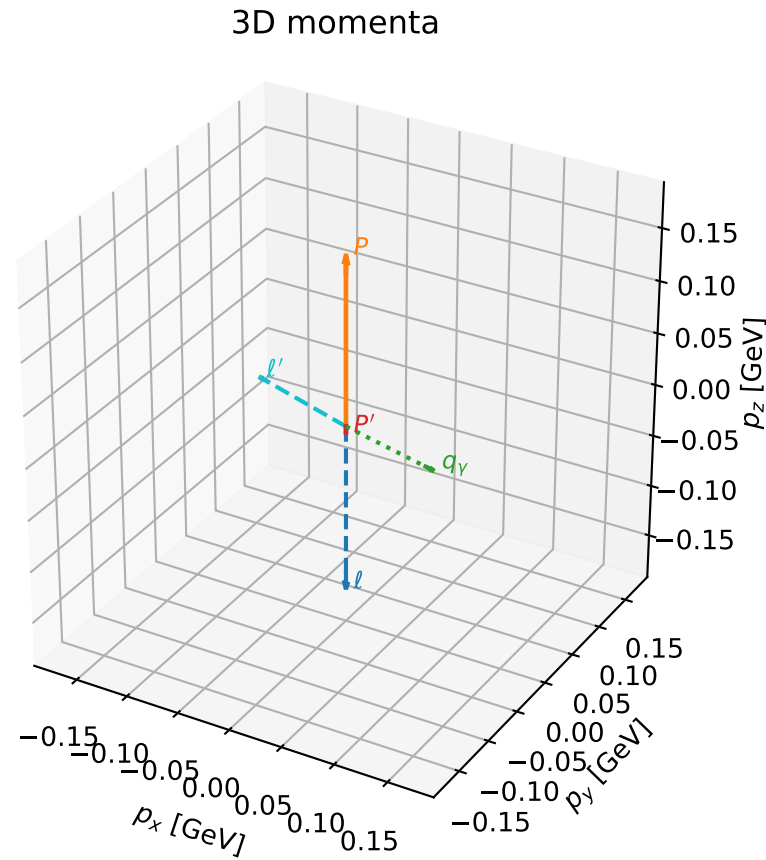
selected phase-space point:
 $\text{sqrt_s} = 1.131452$
 $\text{theta_p_out} = 0.003435164$
 $\text{theta_gamma_out} = 1.569148$
 $\text{qOut} = 0.09671949$
 $\text{phi_p_out} = 3.221949$
 $\text{phi_gamma_out} = 3.141546$
 $\text{alpha_e} = 2.362513$
 $\text{alpha_p} = 1.777689$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

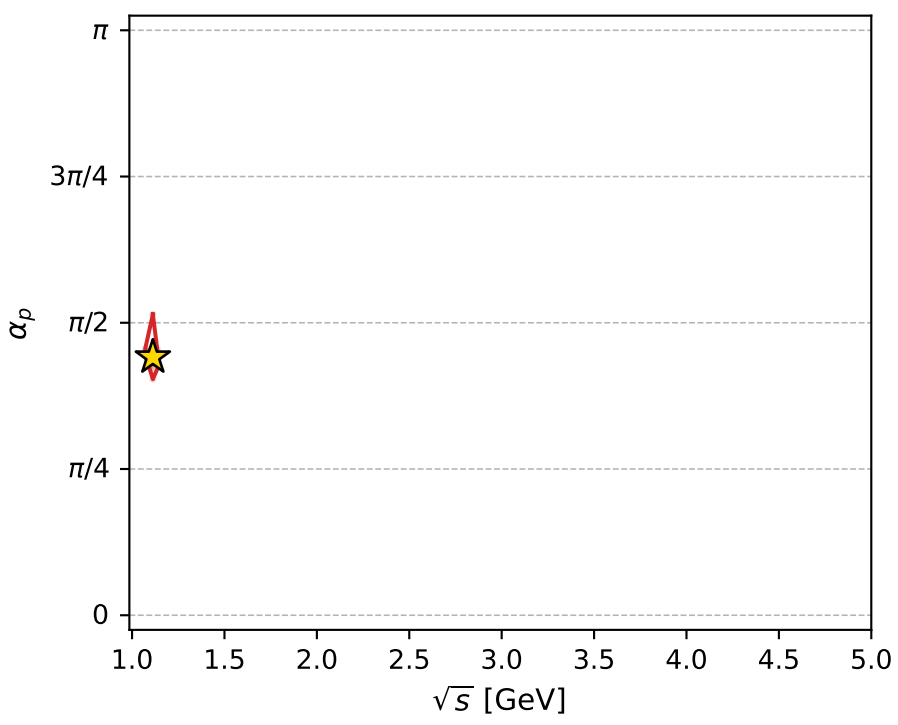
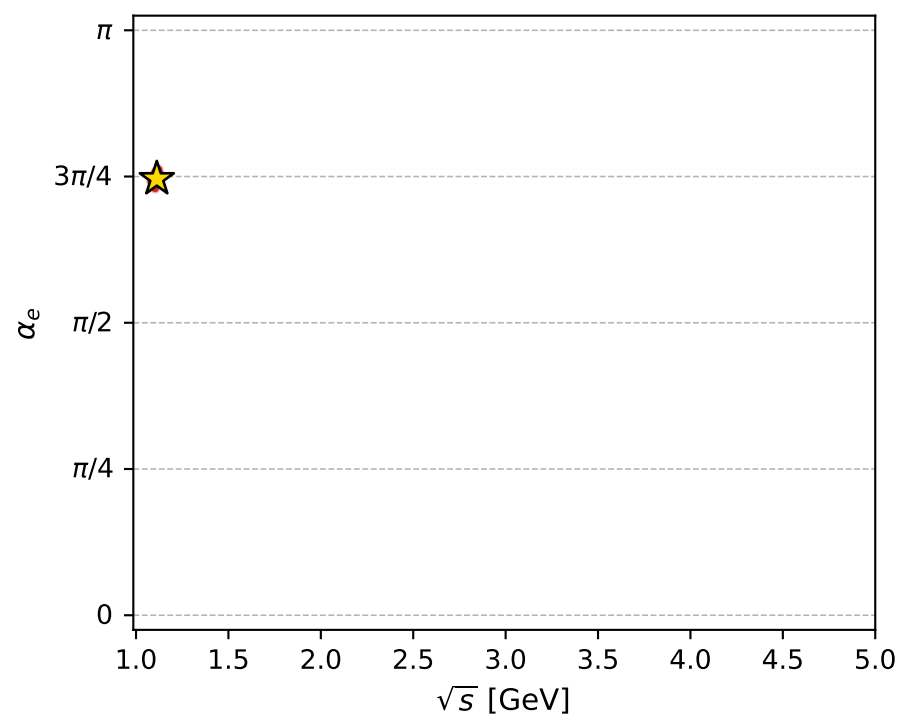
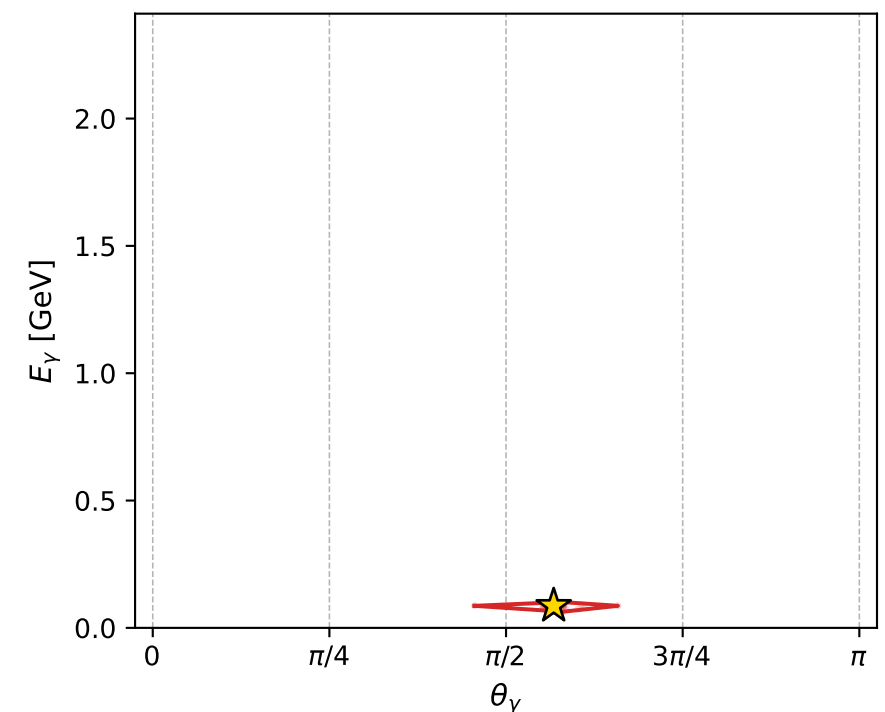
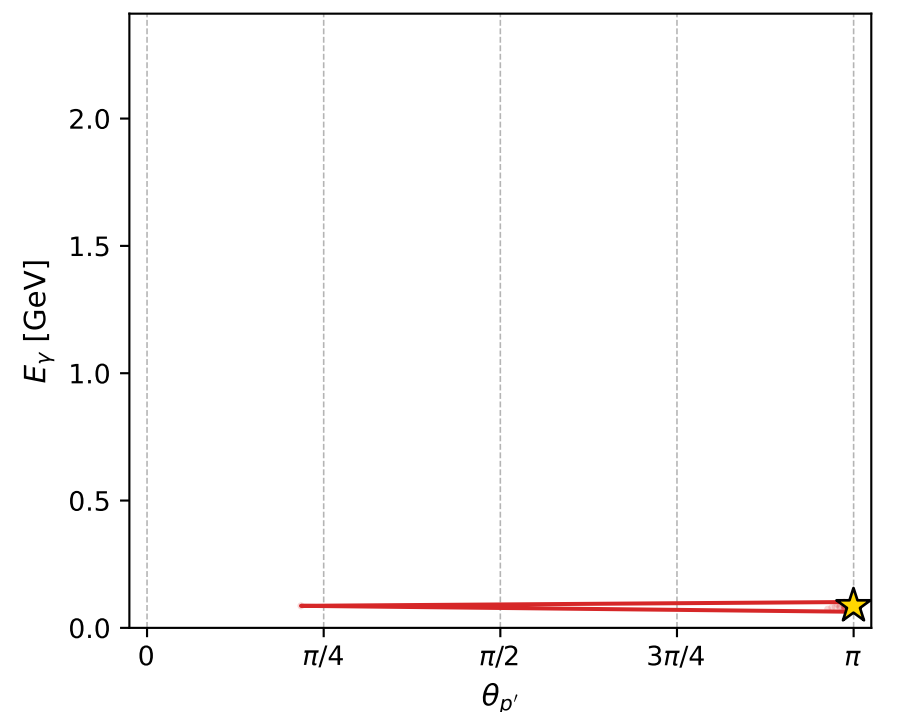
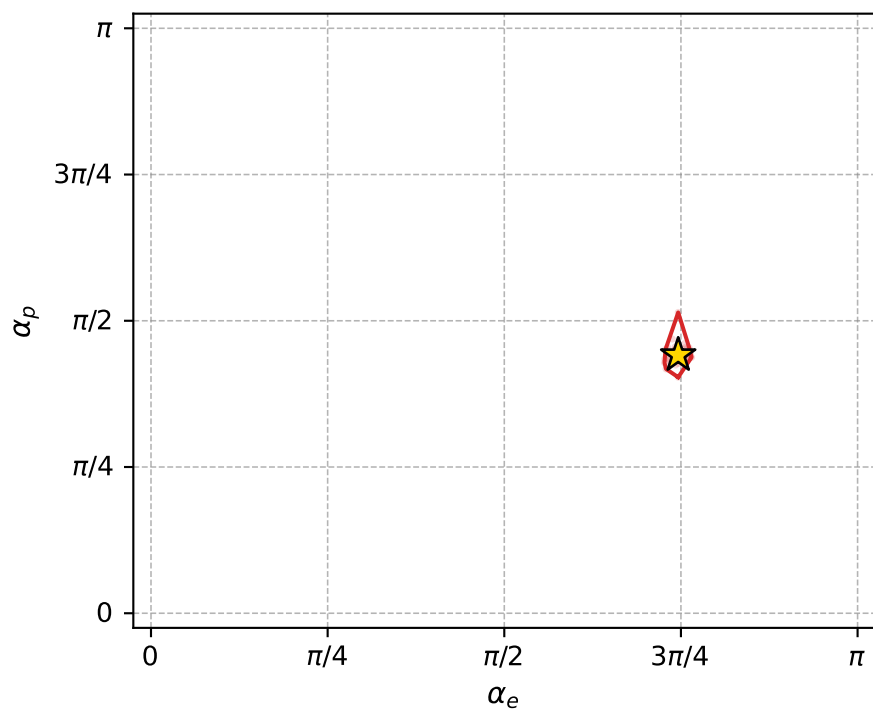
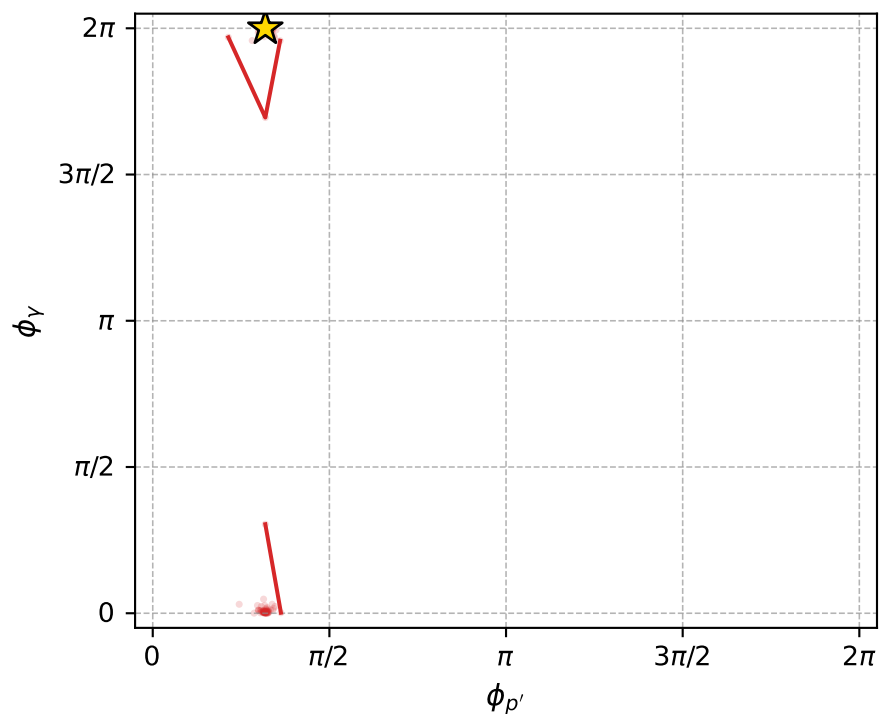
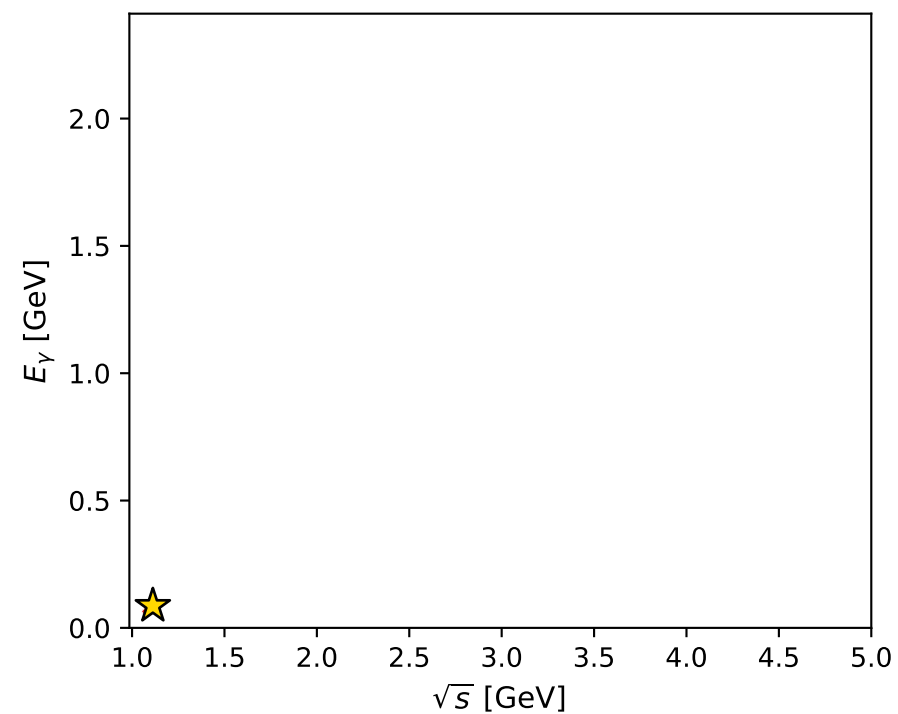
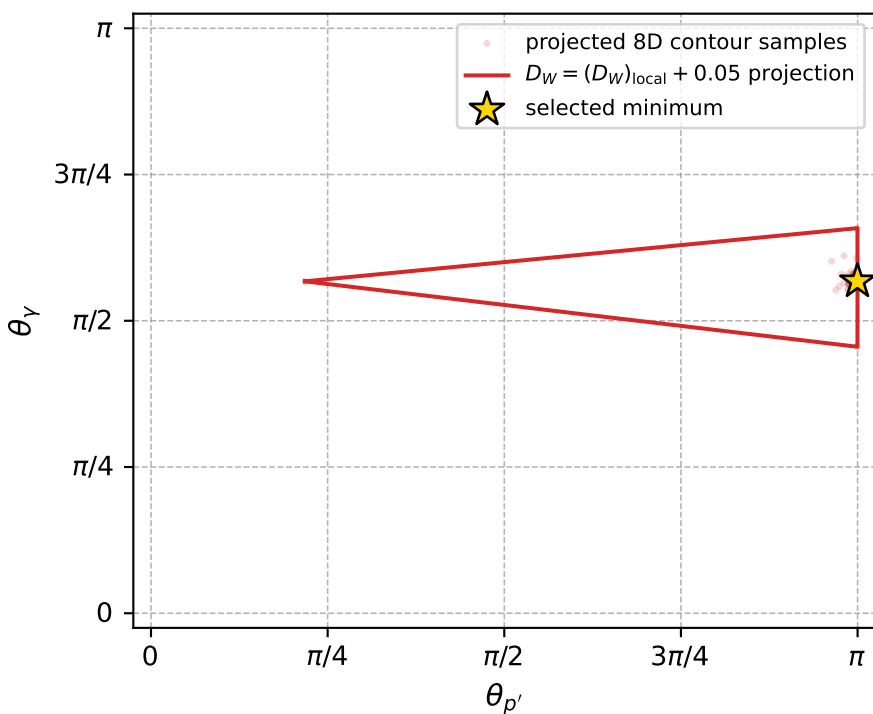
dw_mixing_angles_polarization_06_local_minimum_765 D_W=0.00940191
 region: polarization_cluster_06_local_minimum_765
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3440441$ rad
 incoming lepton: $-0.698463|+> +0.715646|->$
 proton mixing angle: $\alpha_p=1.3862429$ rad
 incoming proton: $0.183508|+> +0.983018|->$

$s=1.23772$, $\sqrt{s}=1.11253$, $|\vec{P}|=0.16084$, $|\vec{P}'|=0.00576713$
 $\theta_{p'}=3.14098$, $\theta_V=1.7826$, $\phi_{p'}=1.00097$, $\phi_V=6.28294$
 $E_V=0.086558$, $\theta_{l'}=1.29485$, $\phi_{l'}=3.14138$
 $Q^2=0.0360014$, $x_B=0.181663$, $t=-0.0275709$, $W^2=1.04202$, $y=0.553754$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1608$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1608$
 P $E= 0.9517$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1608$
 l' $E= 0.0880$ $p_x= -0.0846$ $p_y= 0.0000$ $p_z= 0.0240$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0058$
 q_V $E= 0.0866$ $p_x= 0.0846$ $p_y= -0.0000$ $p_z= -0.0182$



electron: polarization cluster P6, local minimum 765; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.009401907$
 $D_W \text{ contour} = 0.059401907$
 above global minimum = 0.0093854741
 8D contour samples = 148

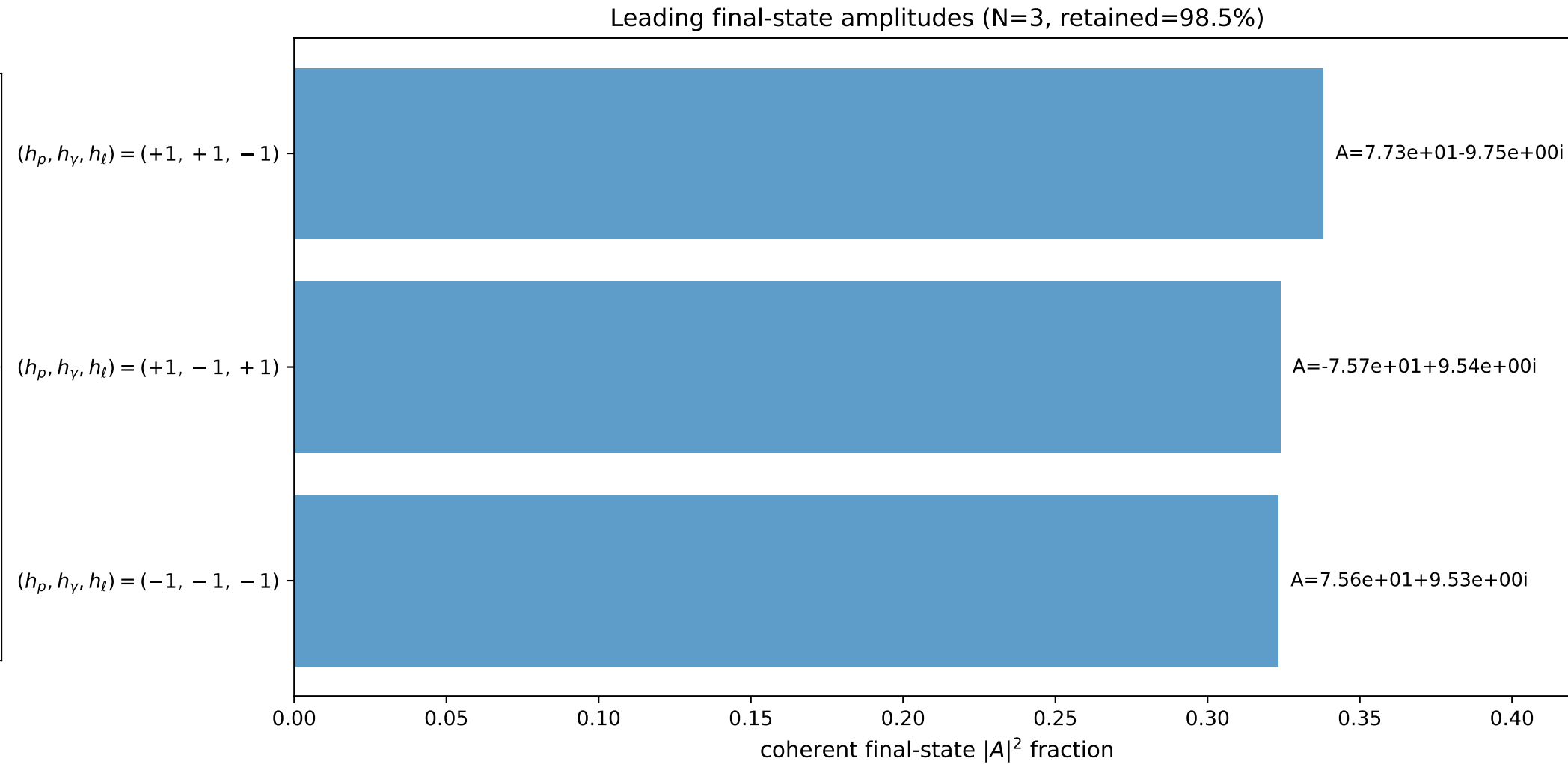
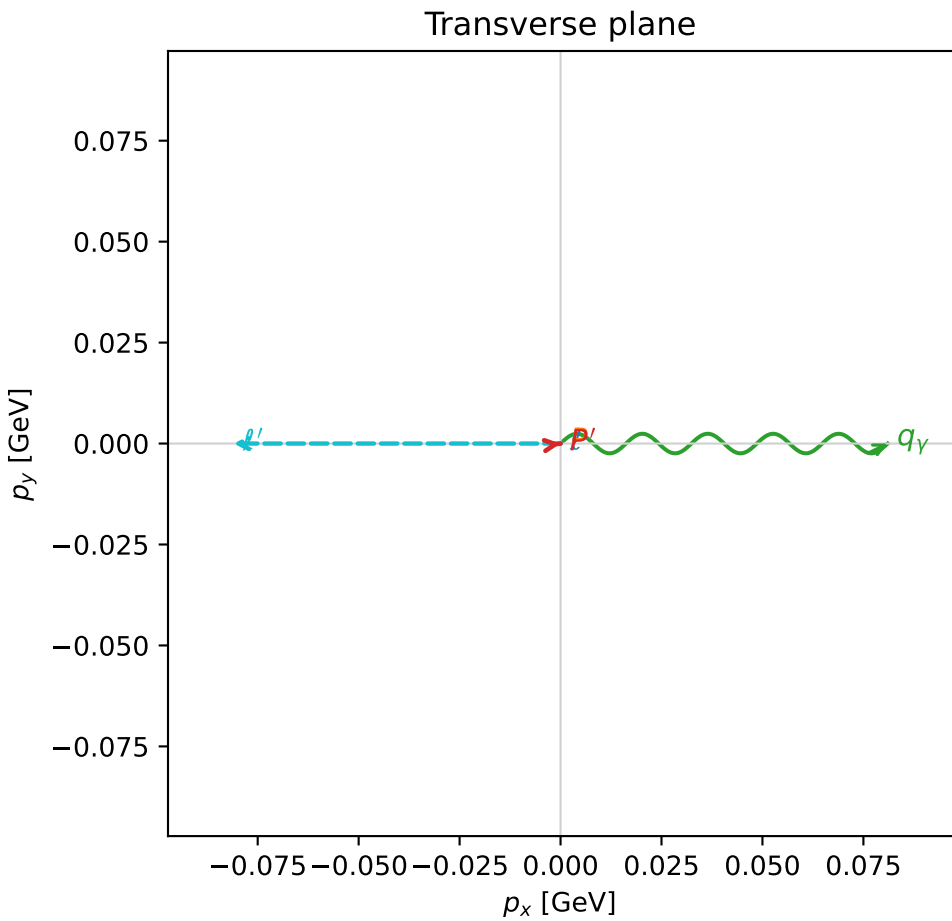
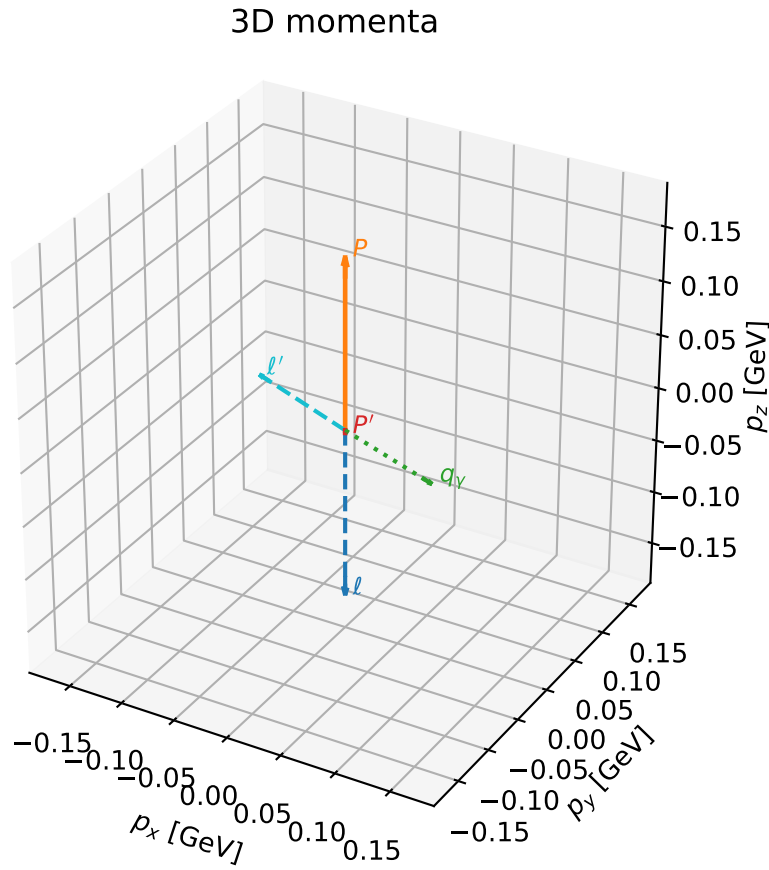
selected phase-space point:
 $\text{sqrt_s} = 1.11253$
 $\text{theta_p_out} = 3.140979$
 $\text{theta_gamma_out} = 1.782598$
 $\text{qOut} = 0.08655799$
 $\text{phi_p_out} = 1.000972$
 $\text{phi_gamma_out} = 6.282937$
 $\text{alpha_e} = 2.344044$
 $\text{alpha_p} = 1.386243$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

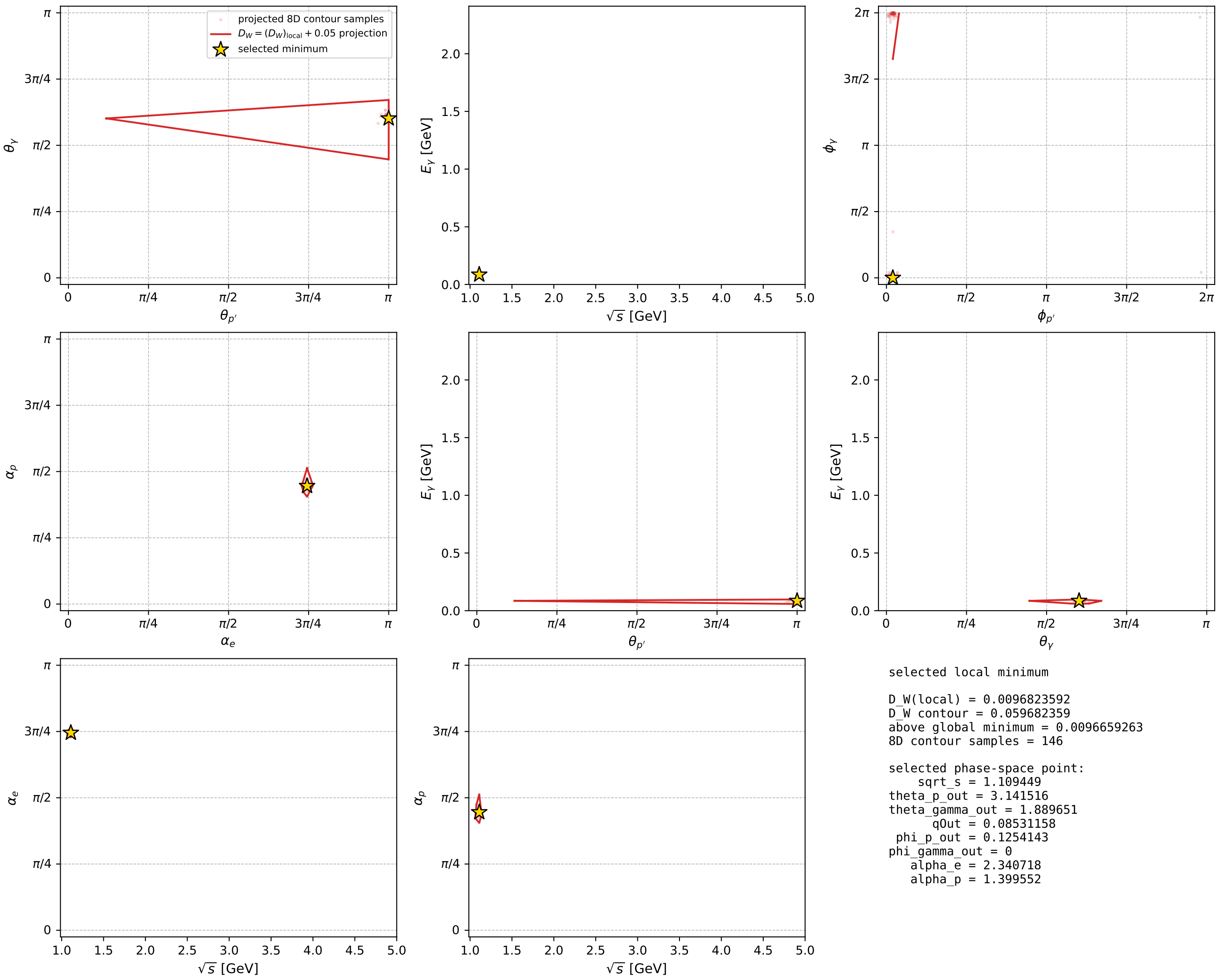
dw_mixing_angles_polarization_06_local_minimum_777 D_W=0.00968236
 region: polarization_cluster_06_local_minimum_777
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3407176$ rad
 incoming lepton: $-0.696079|+> +0.717965|->$
 proton mixing angle: $\alpha_p=1.3995518$ rad
 incoming proton: $0.170409|+> +0.985373|->$

$s=1.23088$, $\sqrt{s}=1.10945$, $|\vec{P}|=0.158201$, $|\vec{P}'|=0.00251337$
 $\theta_{p'}=3.14152$, $\theta_Y=1.88965$, $\phi_{p'}=0.125414$, $\phi_Y=0$
 $E_Y=0.0853116$, $\theta_{\ell'}=1.22423$, $\phi_{\ell'}=3.14159$
 $Q^2=0.0365096$, $x_B=0.185875$, $t=-0.0256537$, $W^2=1.03975$, $y=0.559549$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1582$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1582$
 P $E= 0.9512$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1582$
 ℓ' $E= 0.0861$ $p_x= -0.0810$ $p_y= -0.0000$ $p_z= 0.0293$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0025$
 q_Y $E= 0.0853$ $p_x= 0.0810$ $p_y= 0.0000$ $p_z= -0.0267$



electron: polarization cluster P6, local minimum 777; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



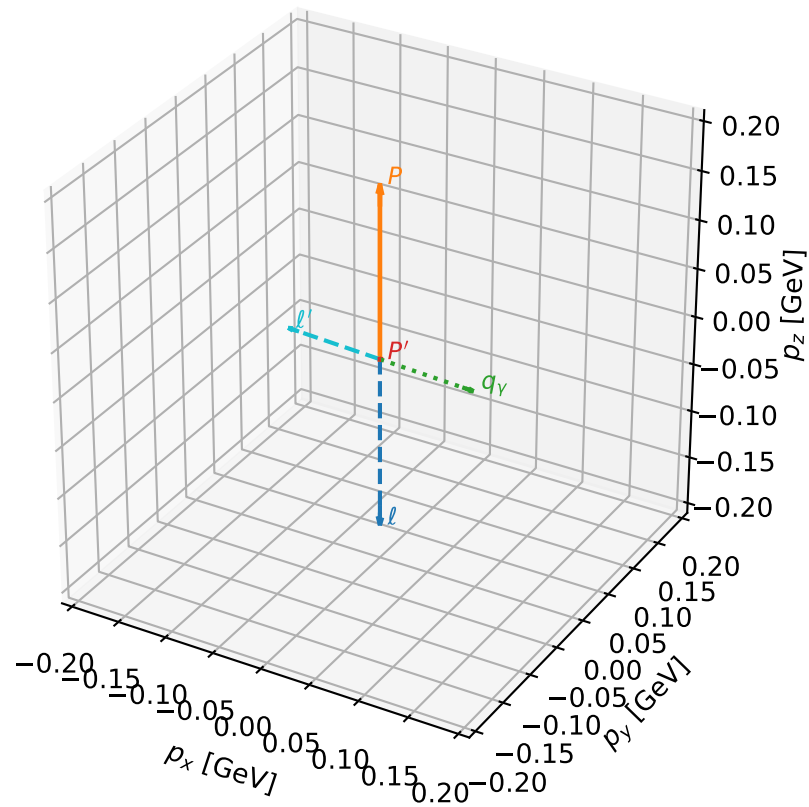
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_782 D_W=0.00984539
 region: polarization_cluster_06_local_minimum_782
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3645136$ rad
 incoming lepton: $-0.712965|+> +0.7012|->$
 proton mixing angle: $\alpha_p=1.3707723$ rad
 incoming proton: $0.198693|+> +0.980062|->$

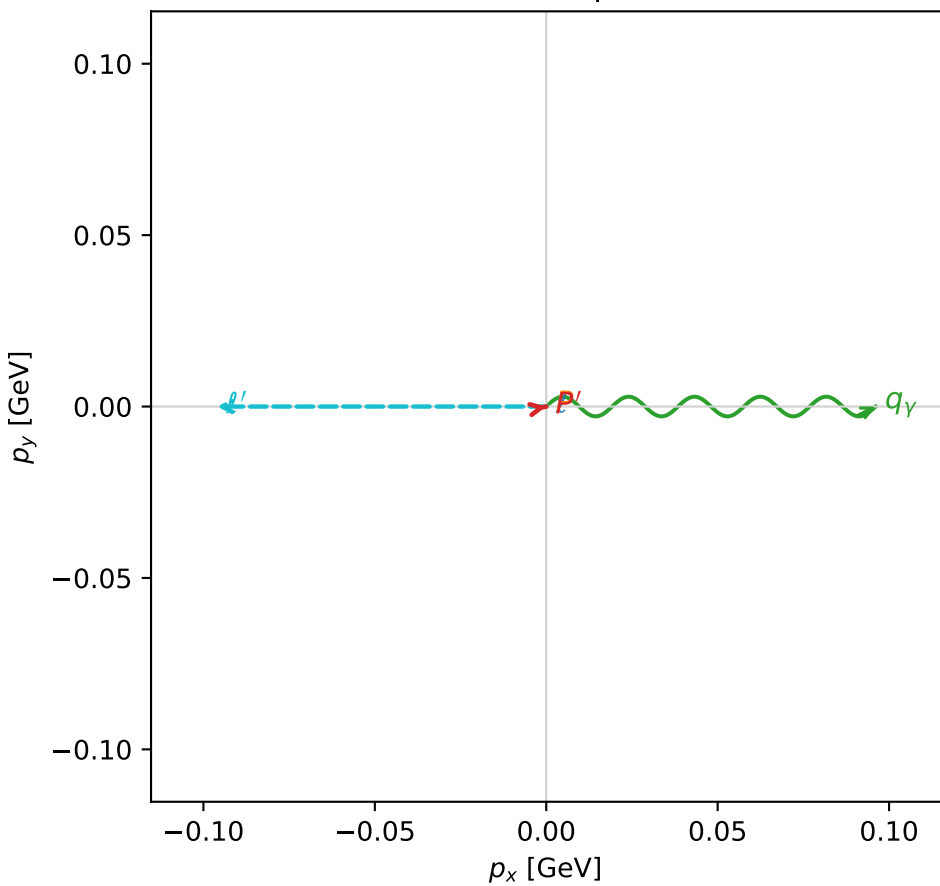
$s=1.27783$, $\sqrt{s}=1.13041$, $|\vec{P}|=0.176036$, $|\vec{P}'|=4.9131e-08$
 $\theta_{P'}=2.82852$, $\theta_V=1.62411$, $\phi_{P'}=0.190647$, $\phi_V=2.23335e-05$
 $E_V=0.0962054$, $\theta_{\ell'}=1.51749$, $\phi_{\ell'}=3.14162$
 $Q^2=0.0356761$, $x_B=0.165047$, $t=-0.0307205$, $W^2=1.06033$, $y=0.543126$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1760$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1760$
 P $E= 0.9544$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1760$
 ℓ' $E= 0.0962$ $p_x= -0.0961$ $p_y= -0.0000$ $p_z= 0.0051$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_V $E= 0.0962$ $p_x= 0.0961$ $p_y= 0.0000$ $p_z= -0.0051$

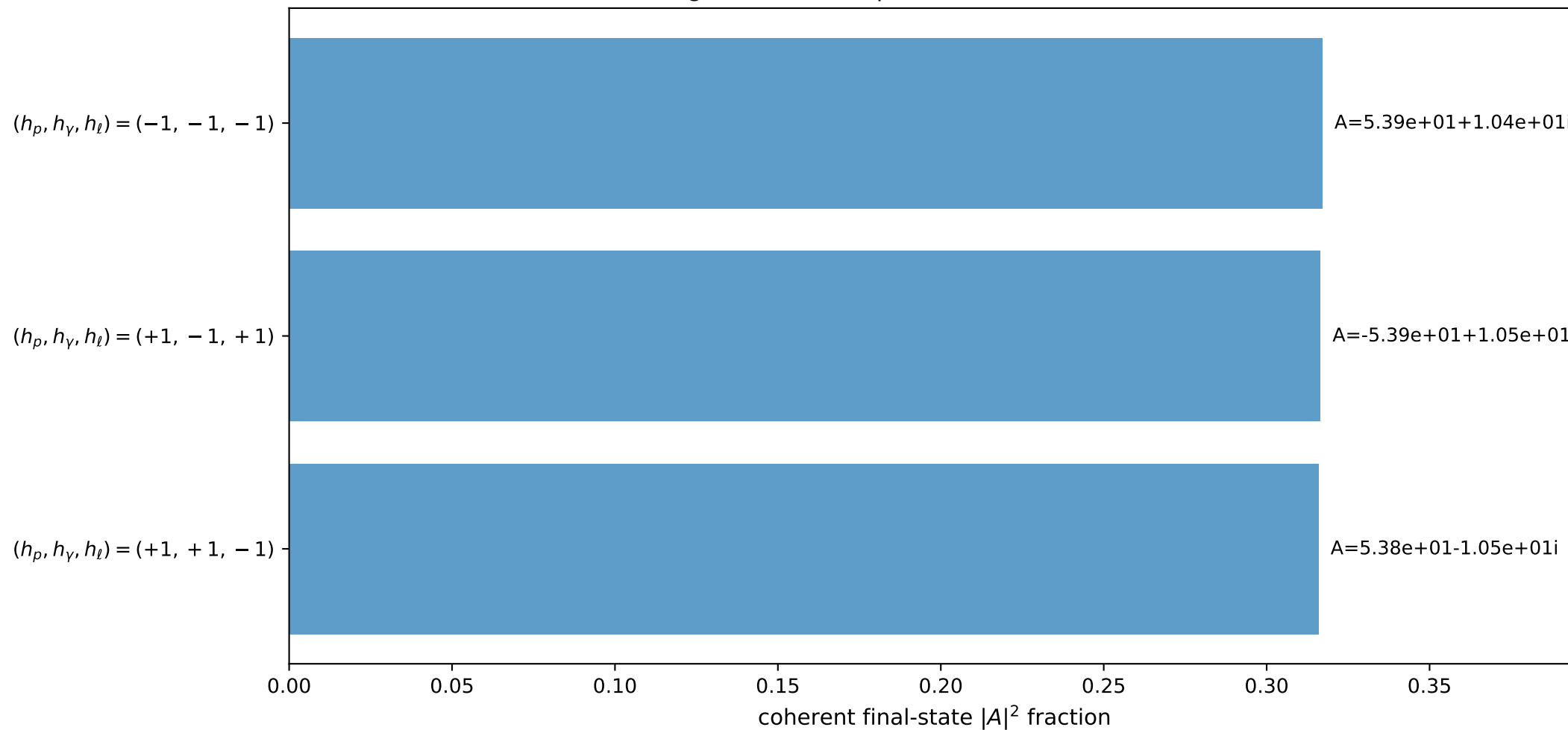
3D momenta



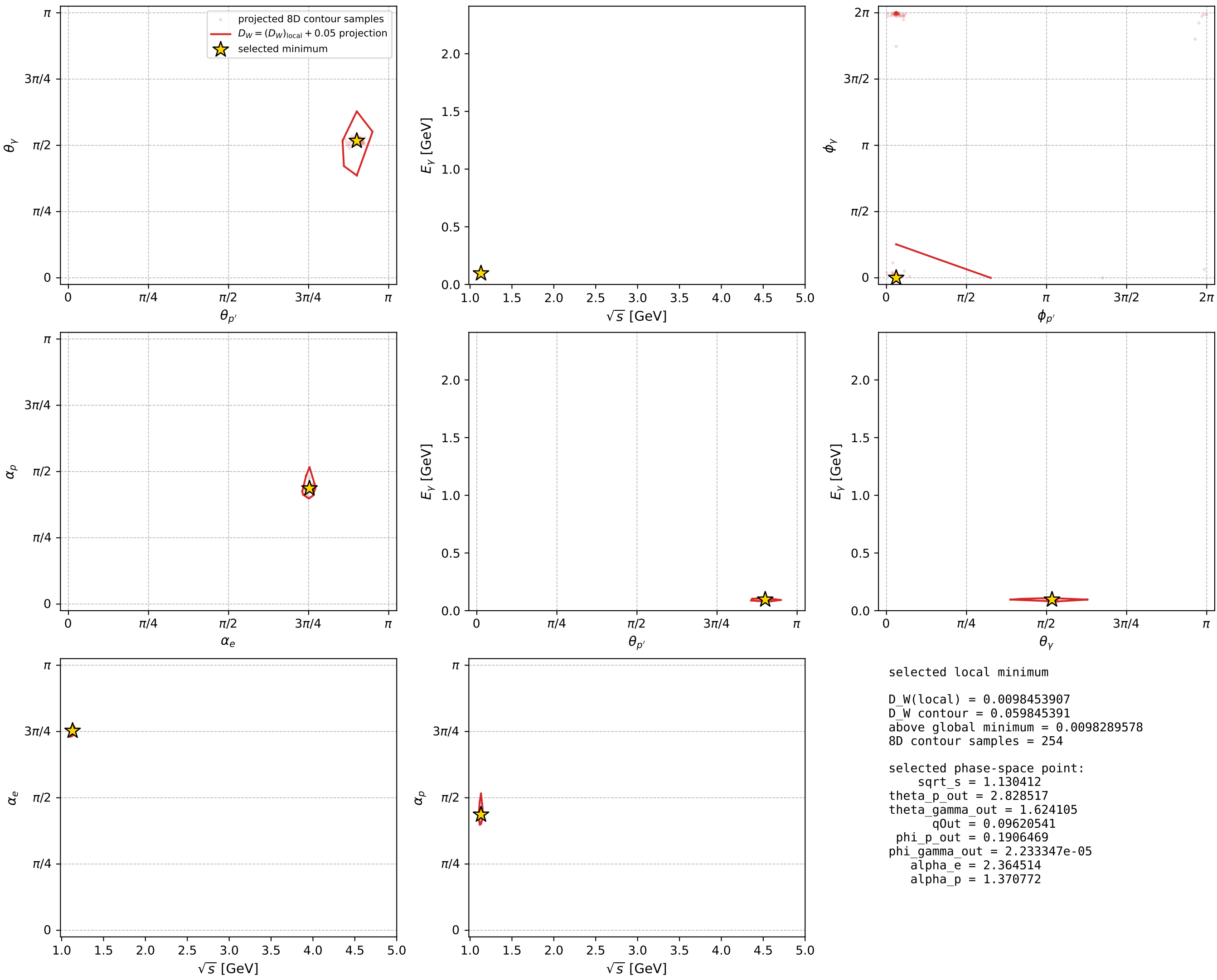
Transverse plane



Leading final-state amplitudes (N=3, retained=95.0%)



electron: polarization cluster P6, local minimum 782; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



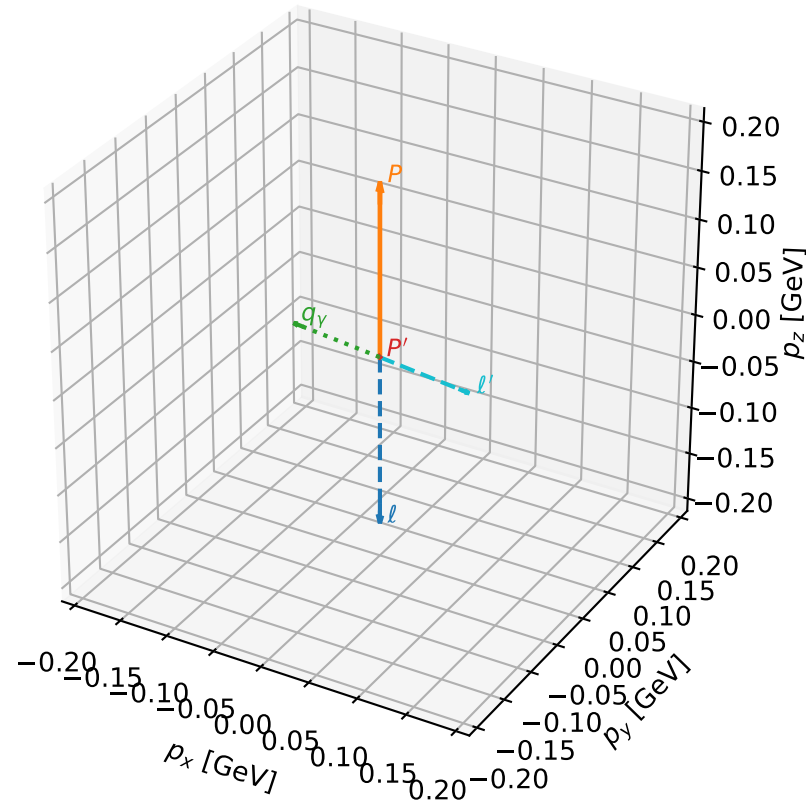
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_784 D_W=0.00991473
 region: polarization_cluster_06_local_minimum_784
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3682459$ rad
 incoming lepton: $-0.715577|+> +0.698534|->$
 proton mixing angle: $\alpha_p=1.7778209$ rad
 incoming proton: $-0.205549|+> +0.978647|->$

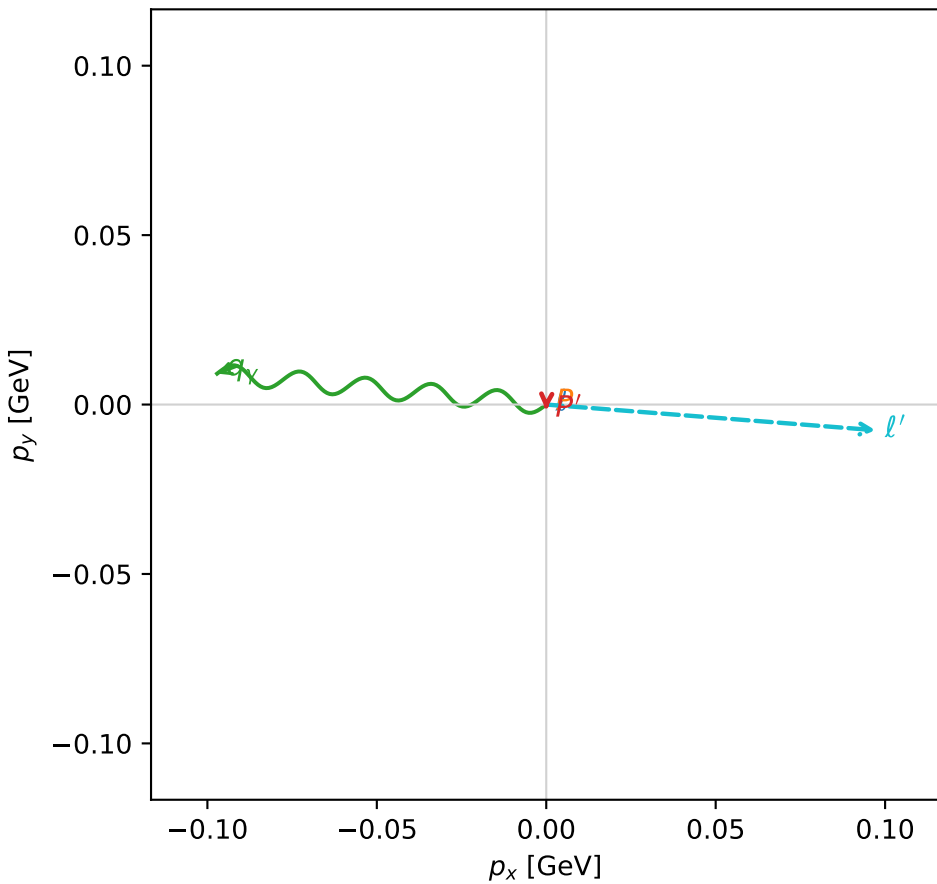
$s=1.28432$, $\sqrt{s}=1.13328$, $|\vec{P}|=0.178454$, $|\vec{P}'|=0.00236397$
 $\theta_{P'}=0.715431$, $\theta_V=1.5316$, $\phi_{P'}=4.66693$, $\phi_V=3.04788$
 $E_V=0.0976257$, $\theta_{\ell'}=1.62827$, $\phi_{\ell'}=6.20536$
 $Q^2=0.0328499$, $x_B=0.152085$, $t=-0.0309317$, $W^2=1.06299$, $y=0.534015$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1785$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1785$
 P $E= 0.9548$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1785$
 ℓ' $E= 0.0977$ $p_x= 0.0972$ $p_y= -0.0076$ $p_z= -0.0056$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0015$ $p_z= 0.0018$
 q_Y $E= 0.0976$ $p_x= -0.0971$ $p_y= 0.0091$ $p_z= 0.0038$

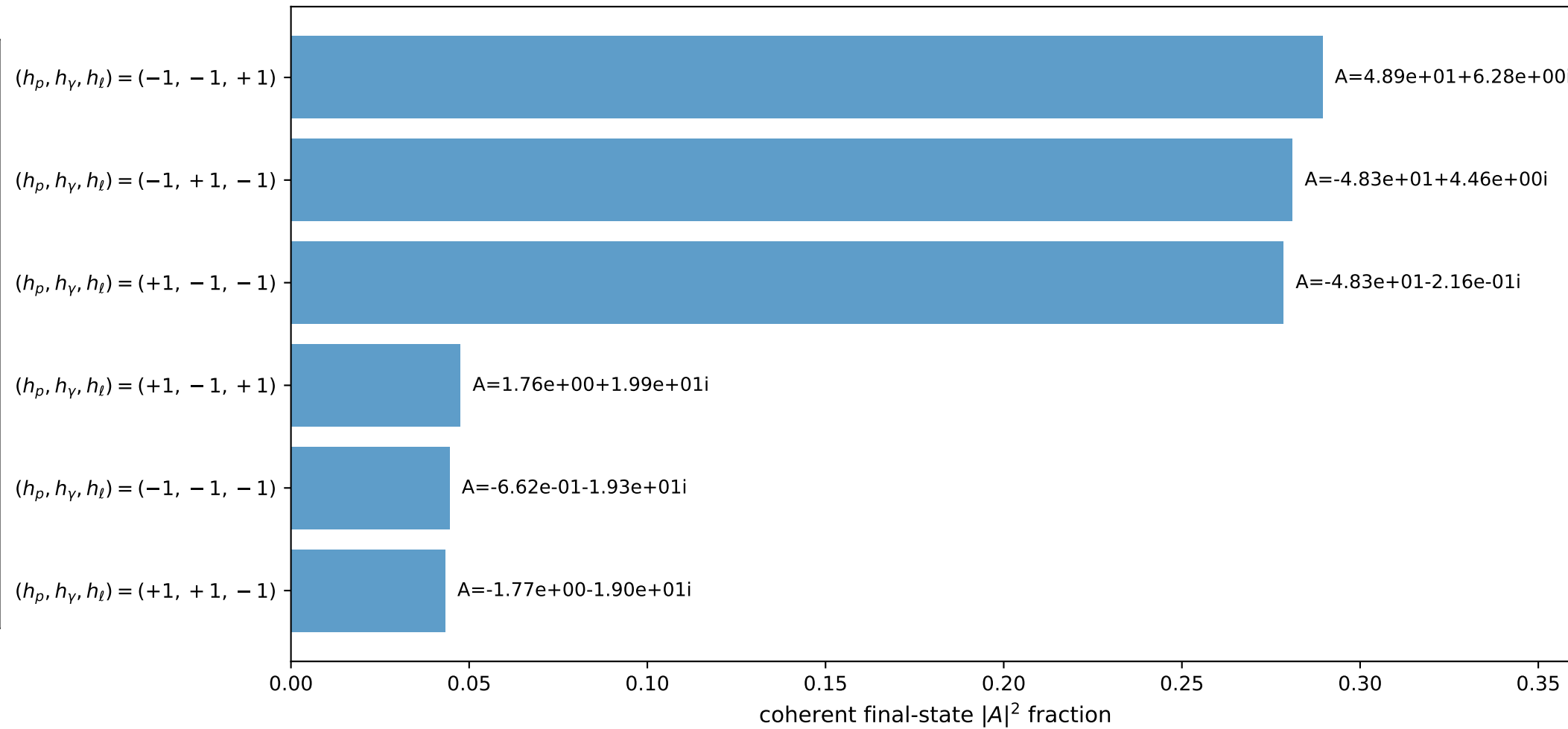
3D momenta



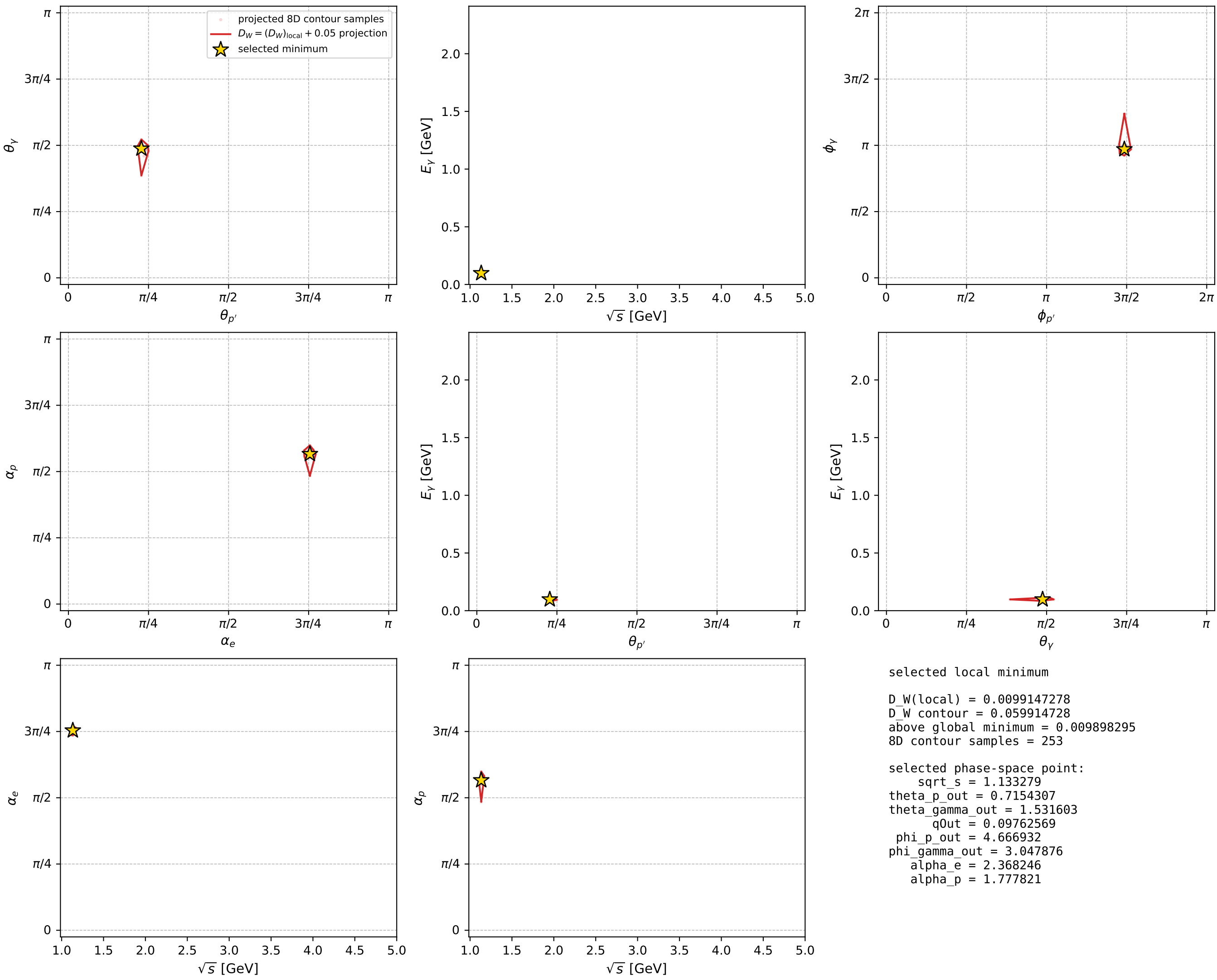
Transverse plane



Leading final-state amplitudes (N=6, retained=98.5%)



electron: polarization cluster P6, local minimum 784; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



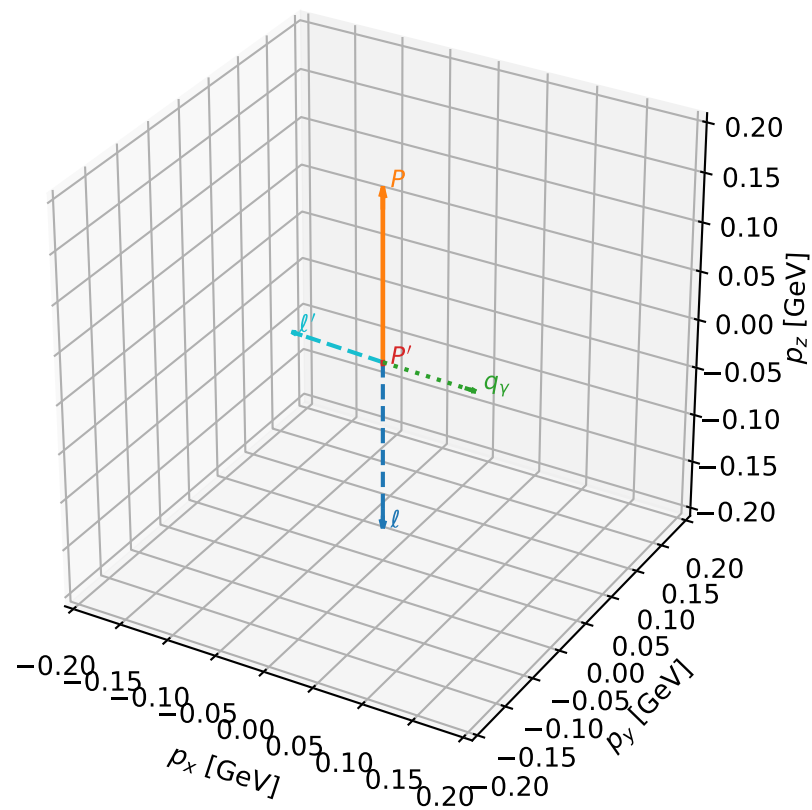
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_786 D_W=0.0099536
 region: polarization_cluster_06_local_minimum_786
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3653262$ rad
 incoming lepton: $-0.713534|+\rangle +0.70062|-\rangle$
 proton mixing angle: $\alpha_p=1.3705843$ rad
 incoming proton: $0.198877|+\rangle +0.980024|-\rangle$

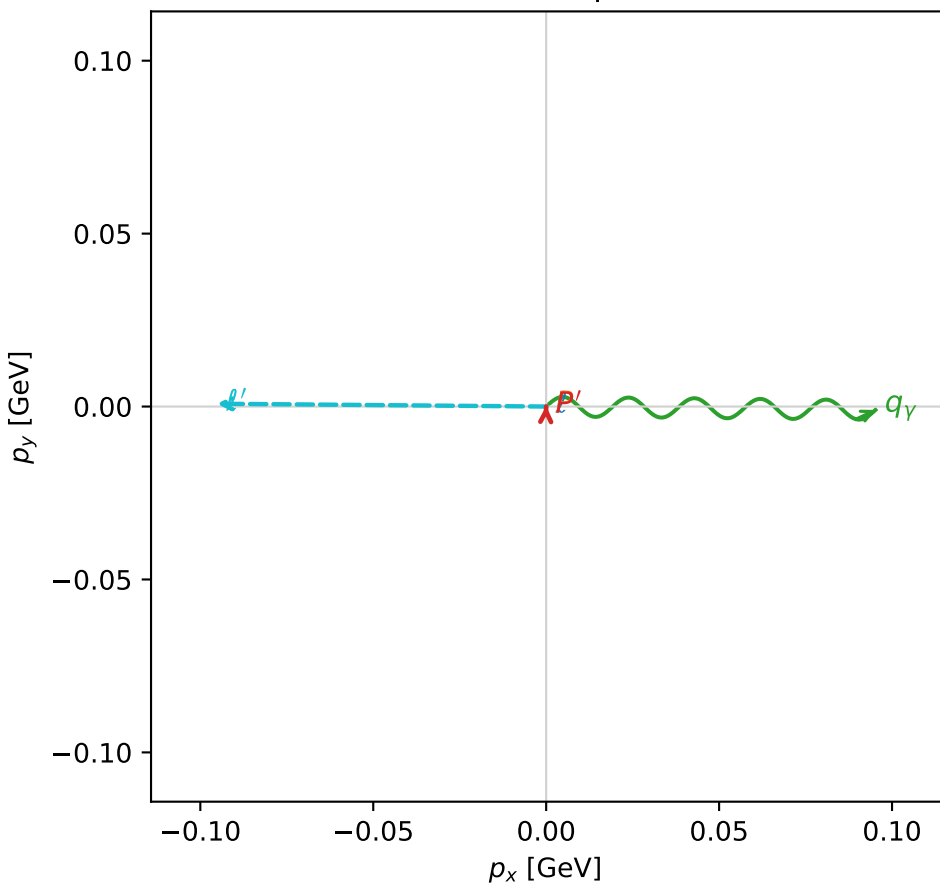
$s=1.27353$, $\sqrt{s}=1.12851$, $|\vec{P}|=0.174427$, $|\vec{P}'|=0.00184551$
 $\theta_{P'}=3.05371$, $\theta_V=1.58959$, $\phi_{P'}=1.53975$, $\phi_V=6.27289$
 $E_V=0.0952242$, $\theta_{\ell'}=1.53271$, $\phi_{\ell'}=3.133$
 $Q^2=0.0345044$, $x_B=0.161888$, $t=-0.0308108$, $W^2=1.05848$, $y=0.541393$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1744$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1744$
 P $E=0.9541$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1744$
 ℓ' $E=0.0953$ $p_x=-0.0952$ $p_y=0.0008$ $p_z=0.0036$
 P' $E=0.9380$ $p_x=0.0000$ $p_y=0.0002$ $p_z=-0.0018$
 q_V $E=0.0952$ $p_x=0.0952$ $p_y=-0.0010$ $p_z=-0.0018$

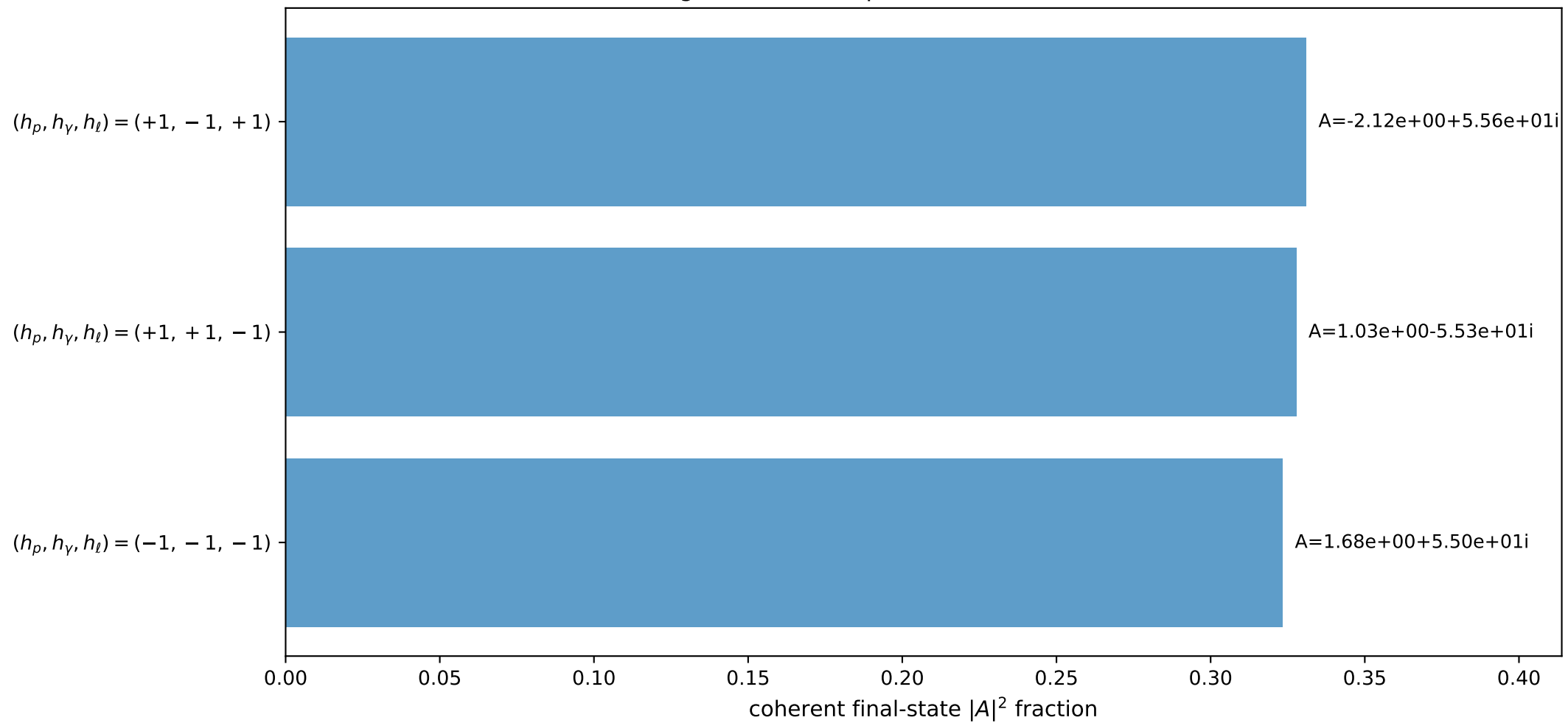
3D momenta



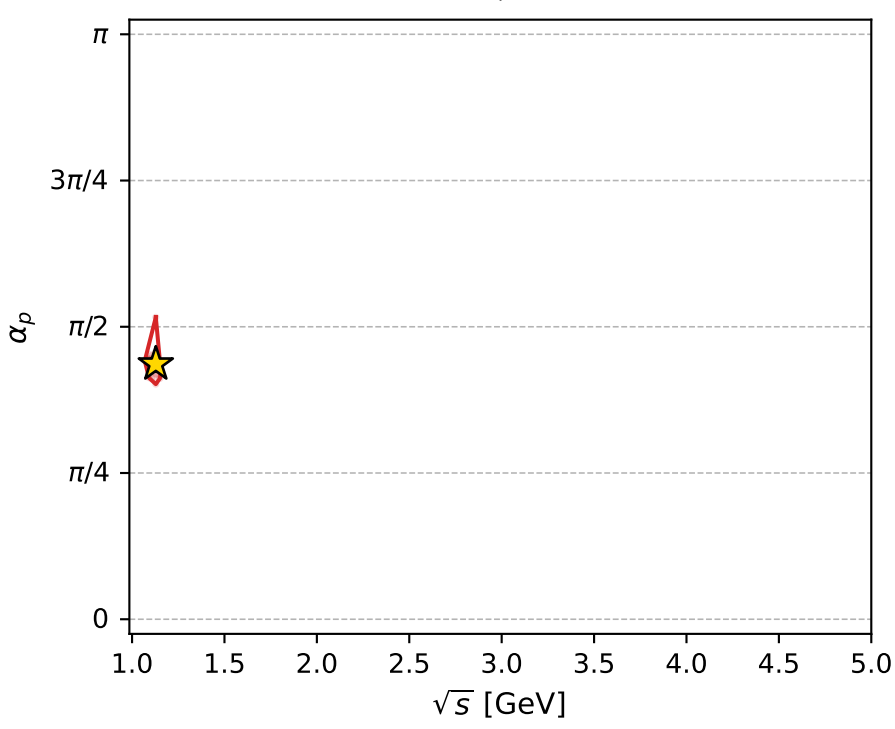
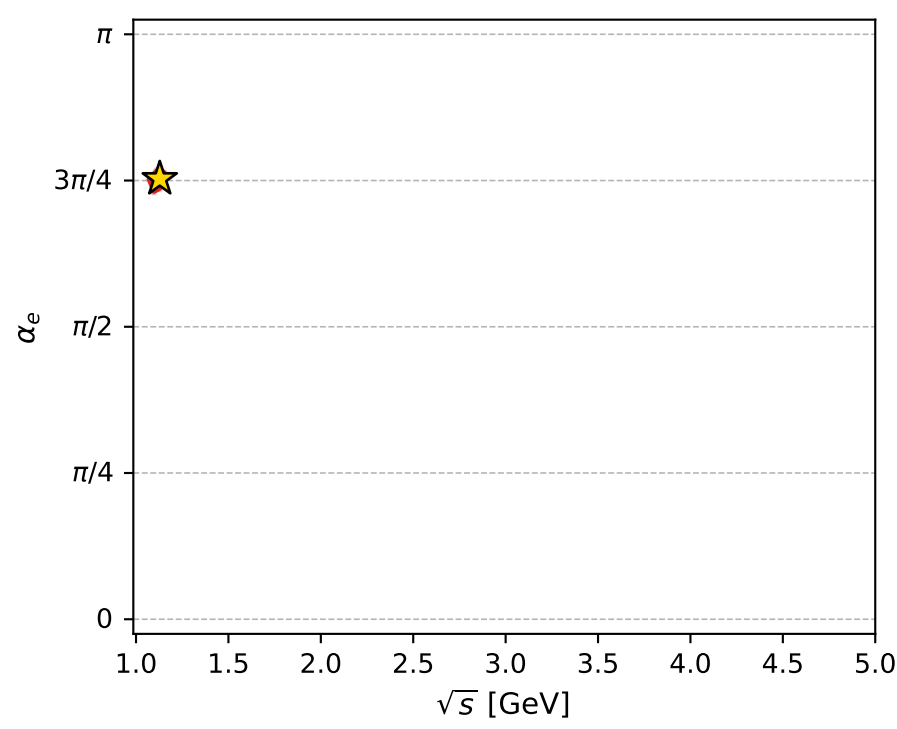
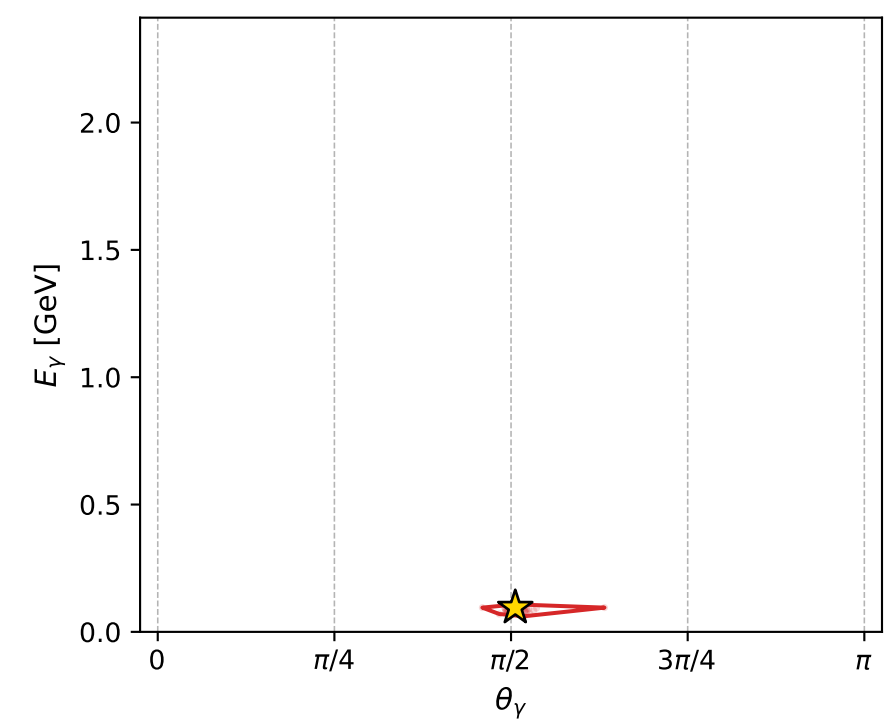
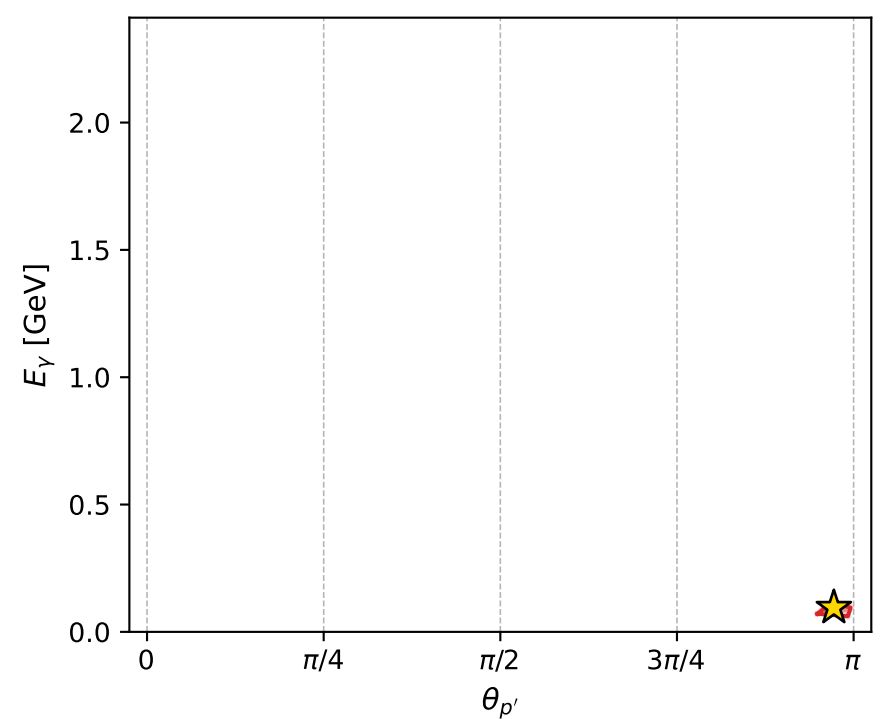
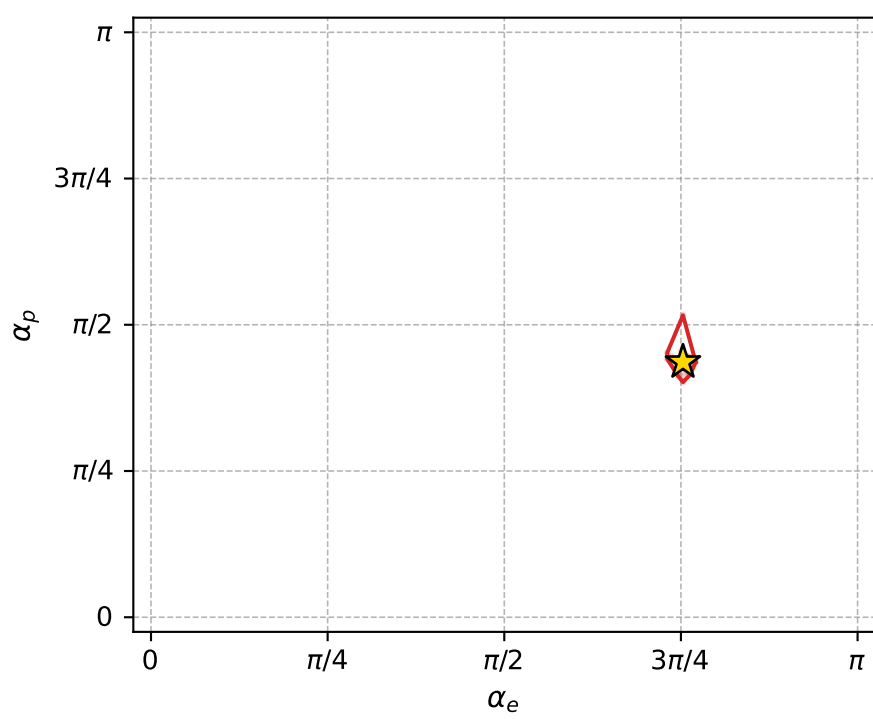
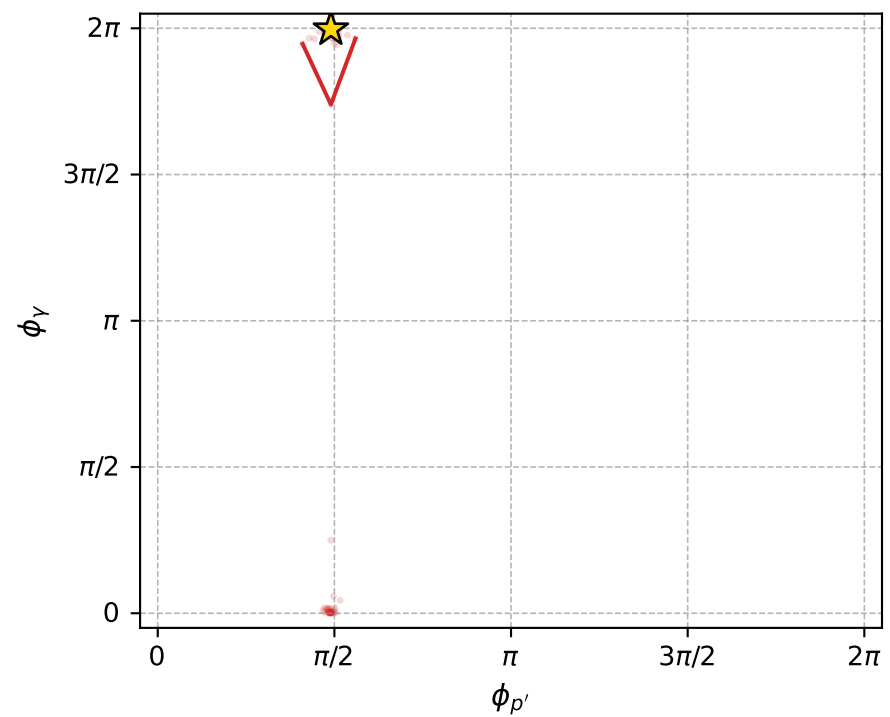
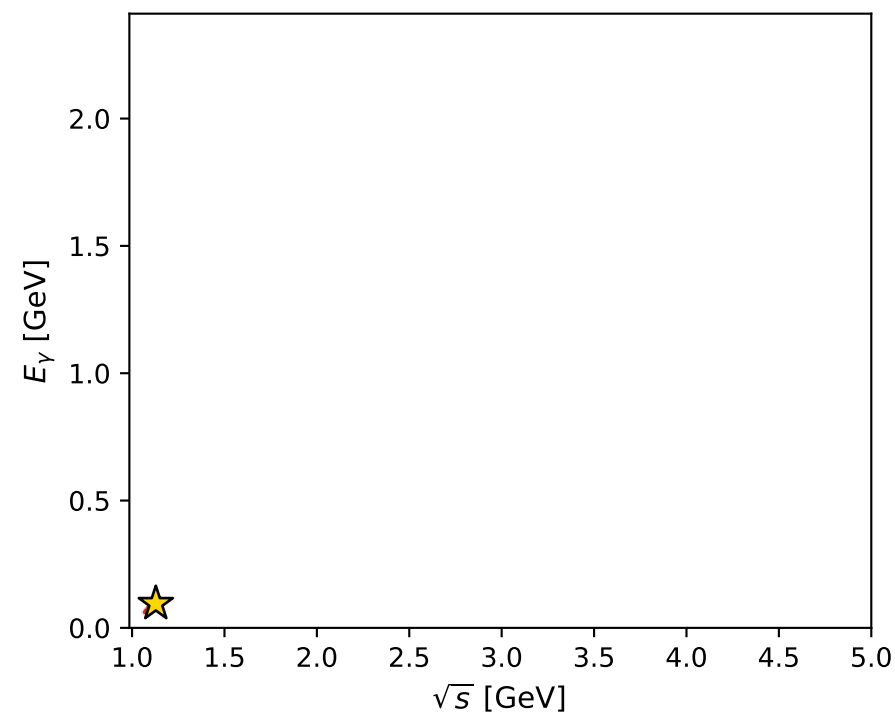
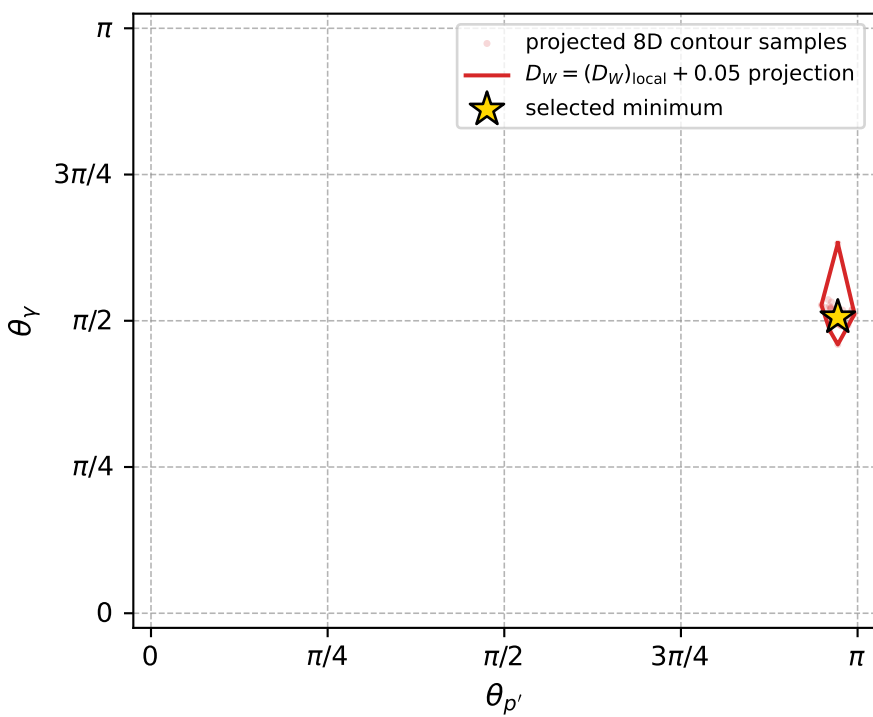
Transverse plane



Leading final-state amplitudes (N=3, retained=98.2%)



electron: polarization cluster P6, local minimum 786; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0099535973$
 $D_W \text{ contour} = 0.059953597$
 above global minimum = 0.0099371645
 8D contour samples = 250

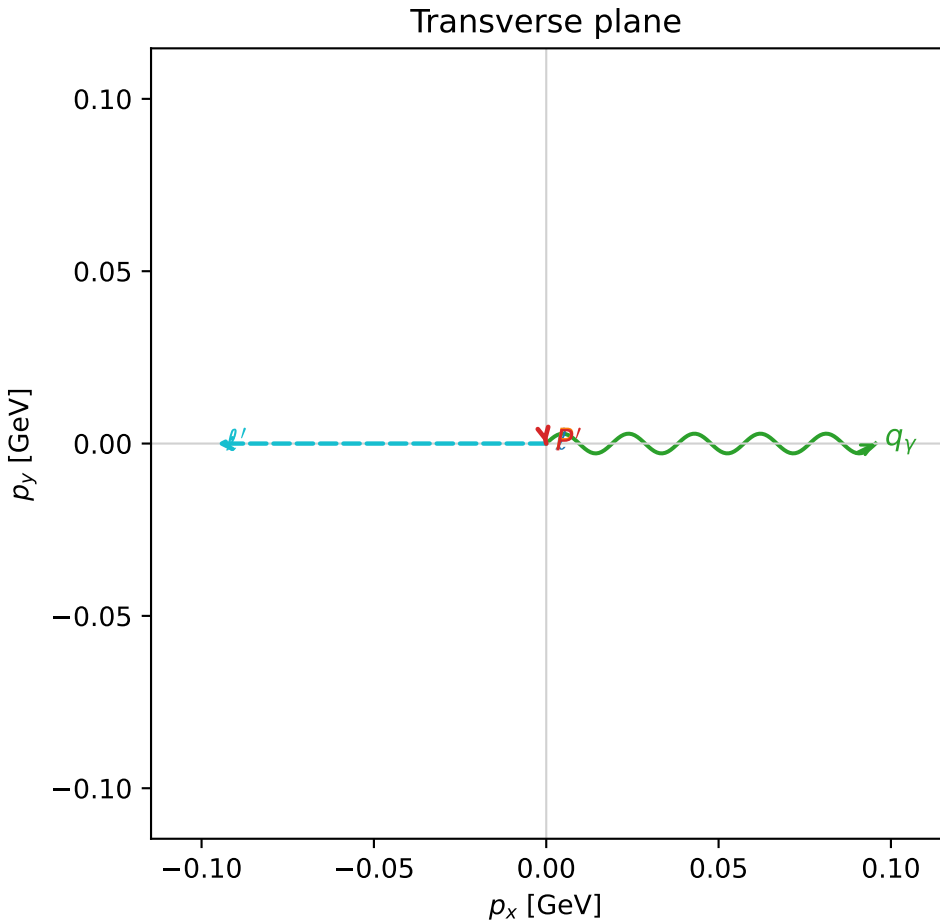
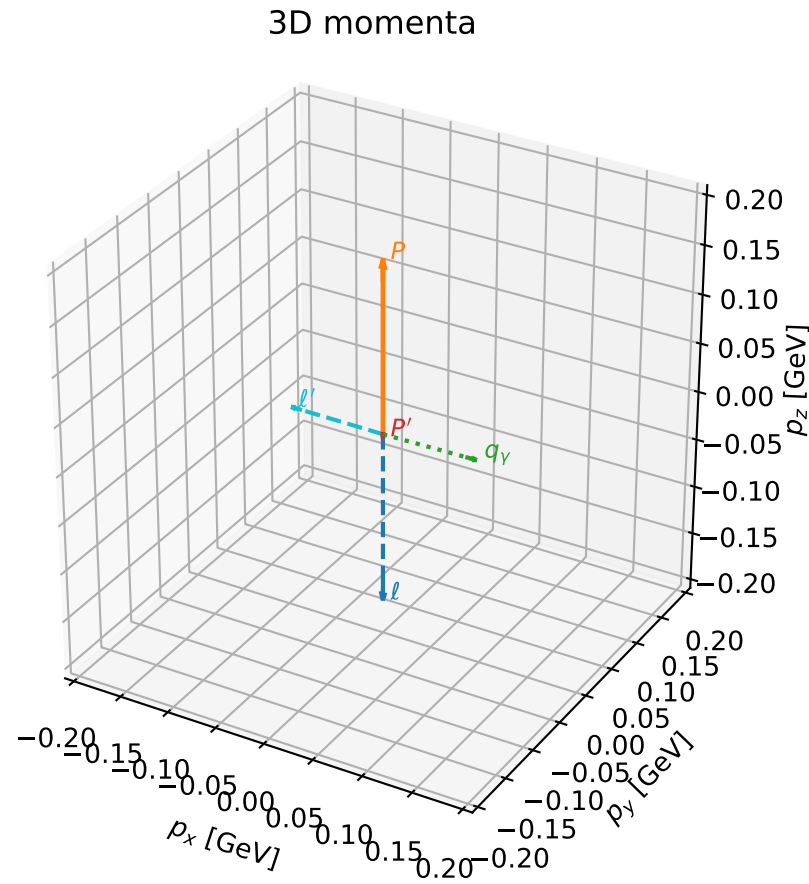
selected phase-space point:
 $\text{sqrt_s} = 1.128507$
 $\text{theta_p_out} = 3.053705$
 $\text{theta_gamma_out} = 1.589591$
 $\text{qOut} = 0.09522416$
 $\text{phi_p_out} = 1.539748$
 $\text{phi_gamma_out} = 6.272892$
 $\text{alpha_e} = 2.365326$
 $\text{alpha_p} = 1.370584$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_795 D_W=0.0102153
 region: polarization_cluster_06_local_minimum_795
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3684745$ rad
 incoming lepton: $-0.715737|+> +0.69837|->$
 proton mixing angle: $\alpha_p=1.3705715$ rad
 incoming proton: $0.19889|+> +0.980022|->$

$s=1.2749$, $\sqrt{s}=1.12911$, $|\vec{P}|=0.174939$, $|\vec{P}'|=0.00210552$
 $\theta_{P'}=3.14096$, $\theta_V=1.55855$, $\phi_{P'}=4.91864$, $\phi_V=0.000108903$
 $E_V=0.095556$, $\theta_{\ell'}=1.56101$, $\phi_{\ell'}=3.14169$
 $Q^2=0.0337594$, $x_B=0.158474$, $t=-0.0310832$, $W^2=1.05911$, $y=0.539239$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1749$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1749$
 P $E= 0.9542$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1749$
 ℓ' $E= 0.0956$ $p_x= -0.0955$ $p_y= -0.0000$ $p_z= 0.0009$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0021$
 q_V $E= 0.0956$ $p_x= 0.0955$ $p_y= 0.0000$ $p_z= 0.0012$

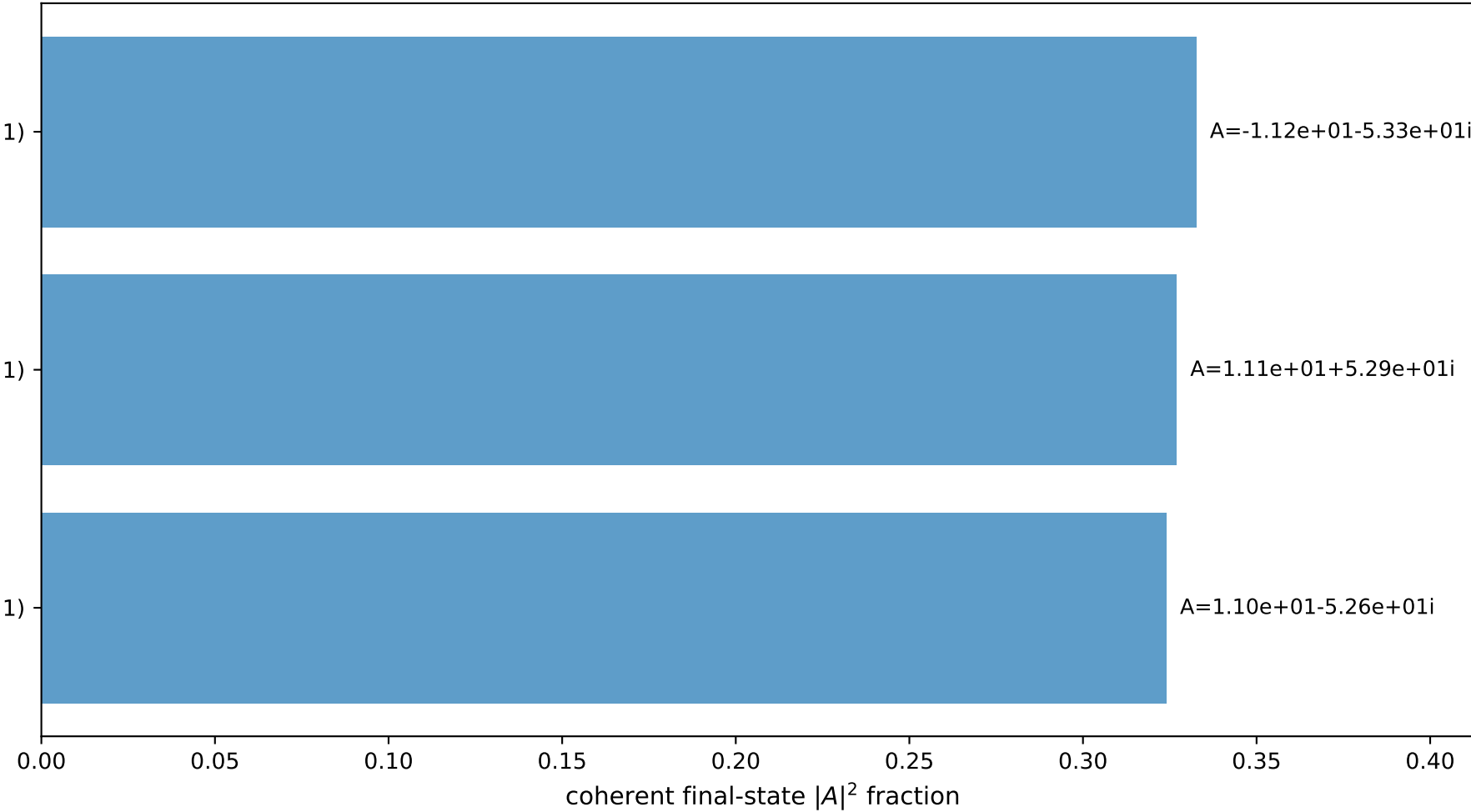


$(h_p, h_V, h_\ell) = (+1, -1, +1)$

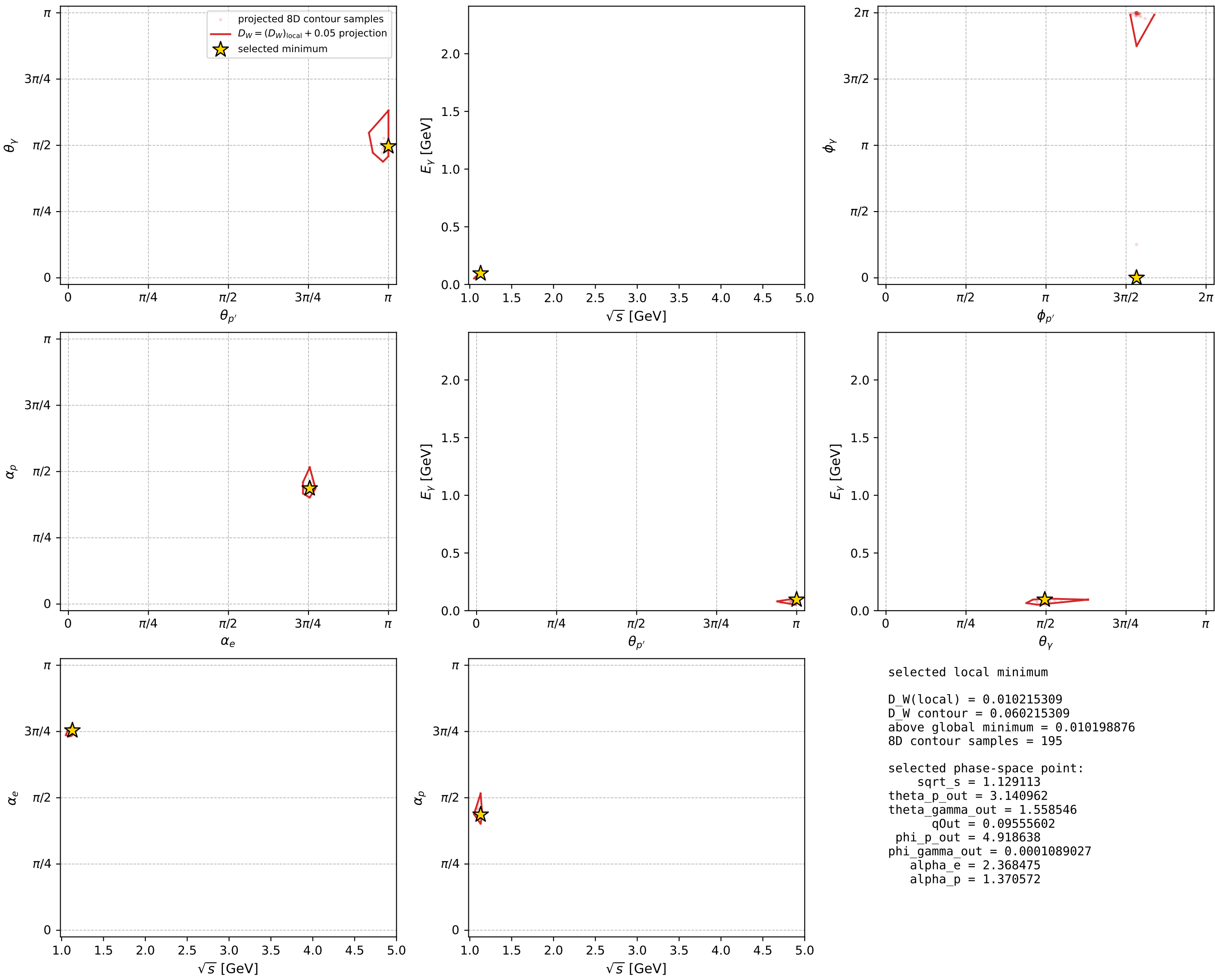
$(h_p, h_V, h_\ell) = (+1, +1, -1)$

$(h_p, h_V, h_\ell) = (-1, -1, -1)$

Leading final-state amplitudes (N=3, retained=98.4%)



electron: polarization cluster P6, local minimum 795; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

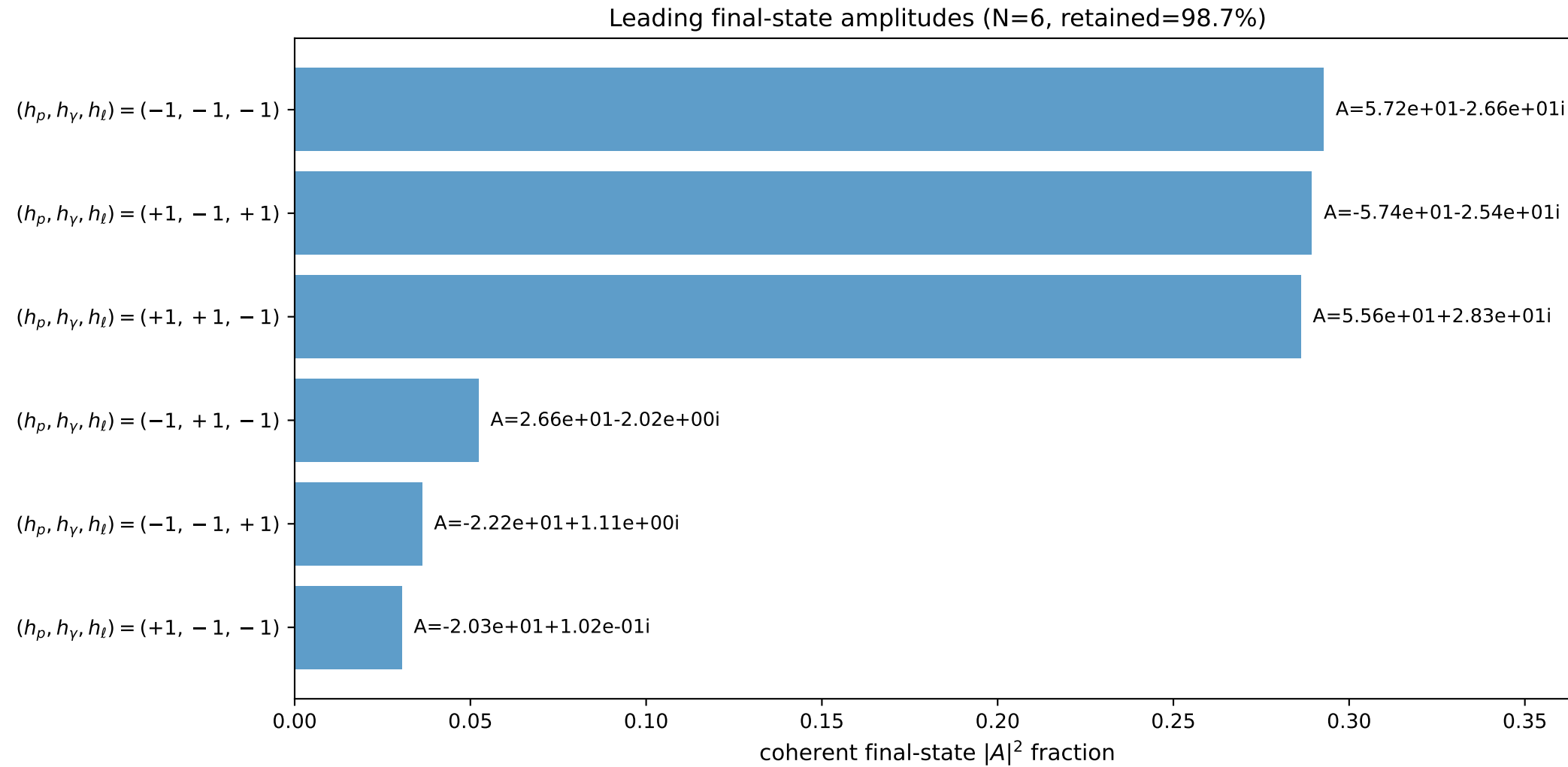
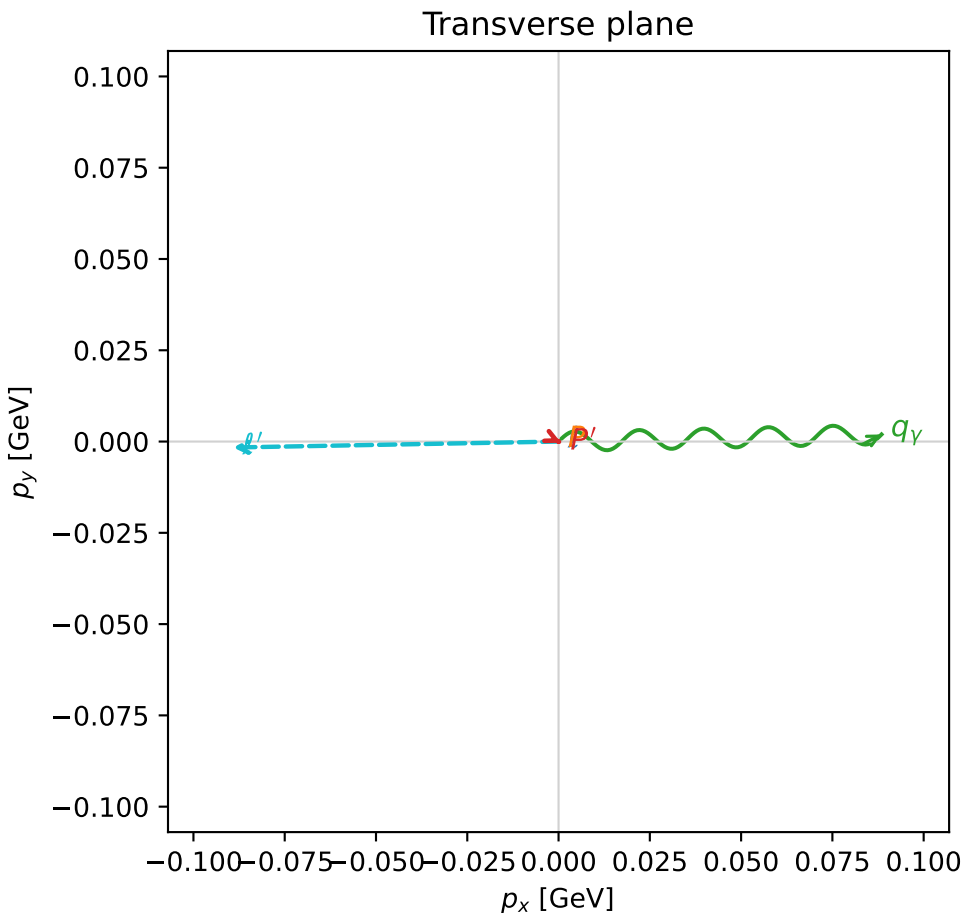
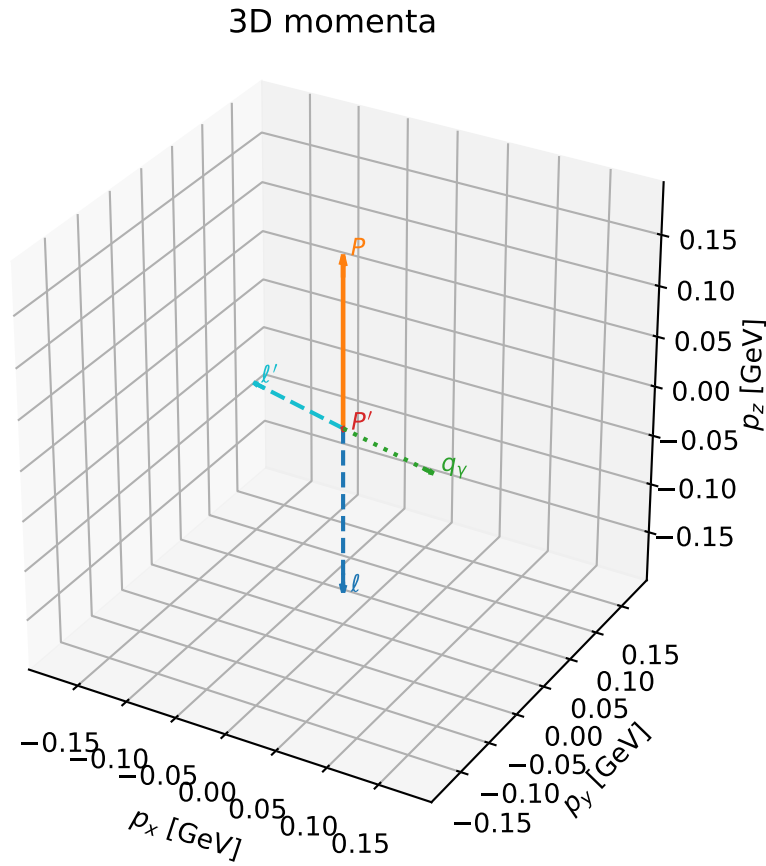


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

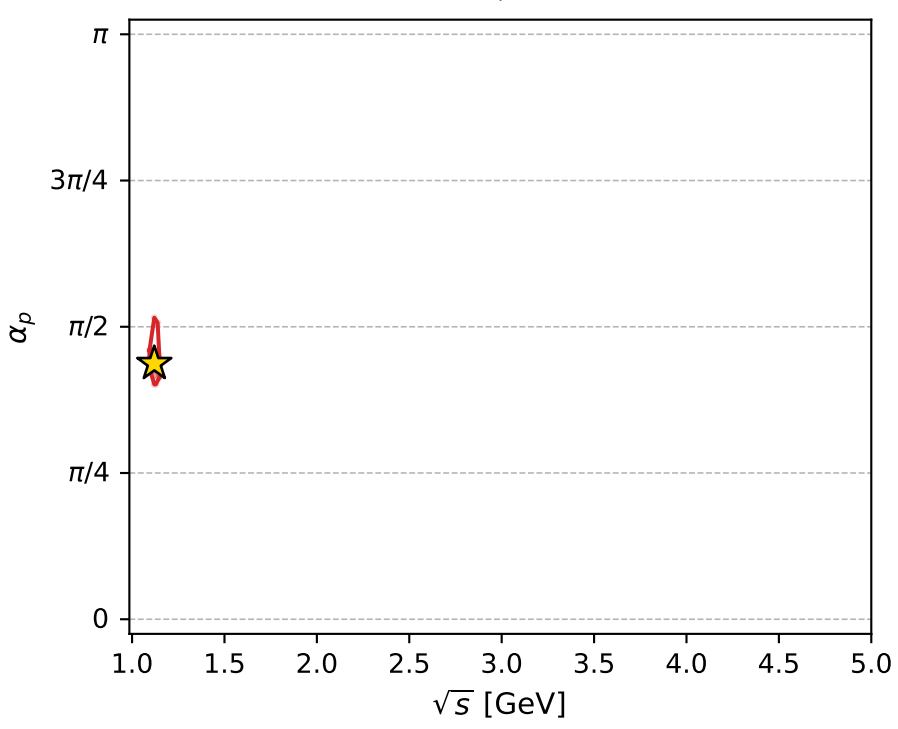
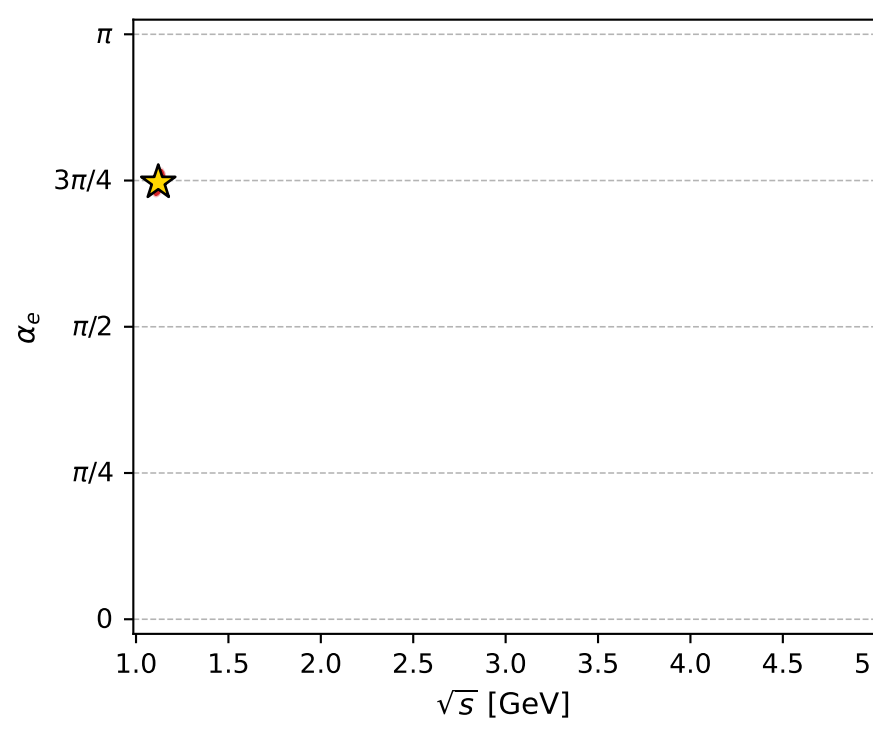
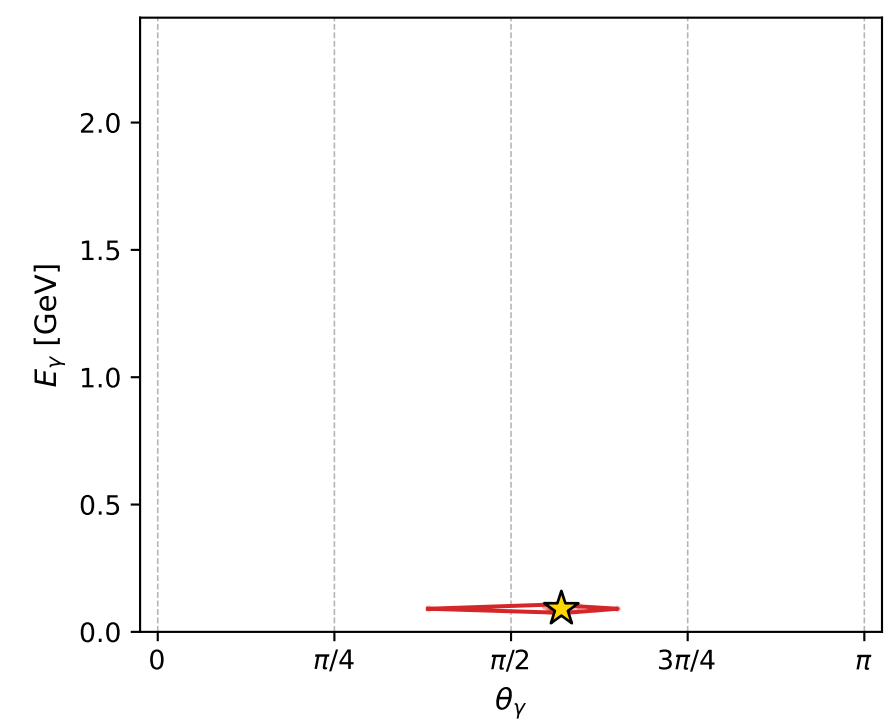
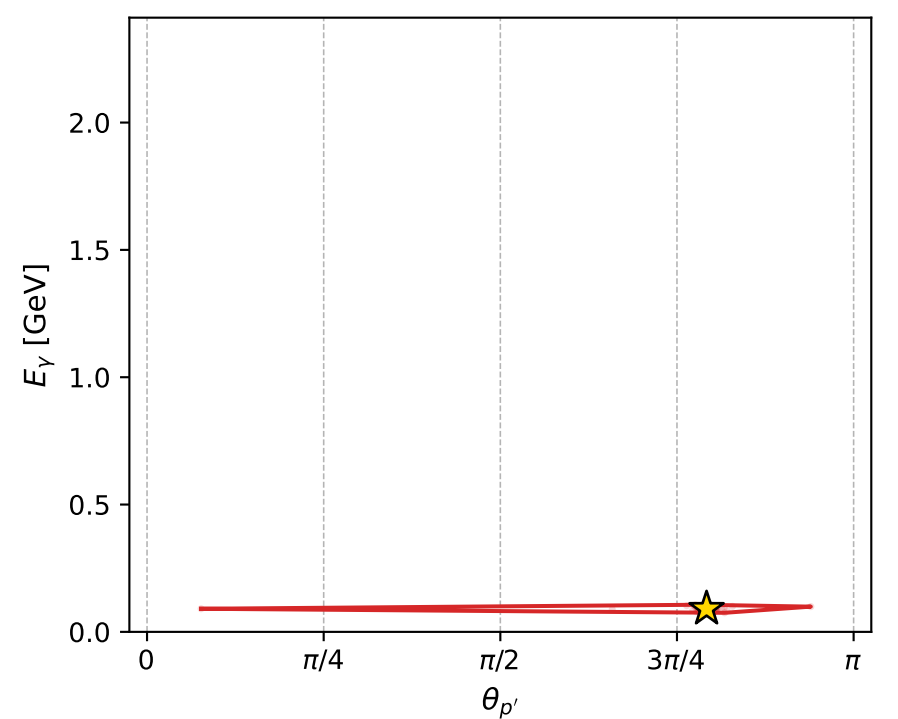
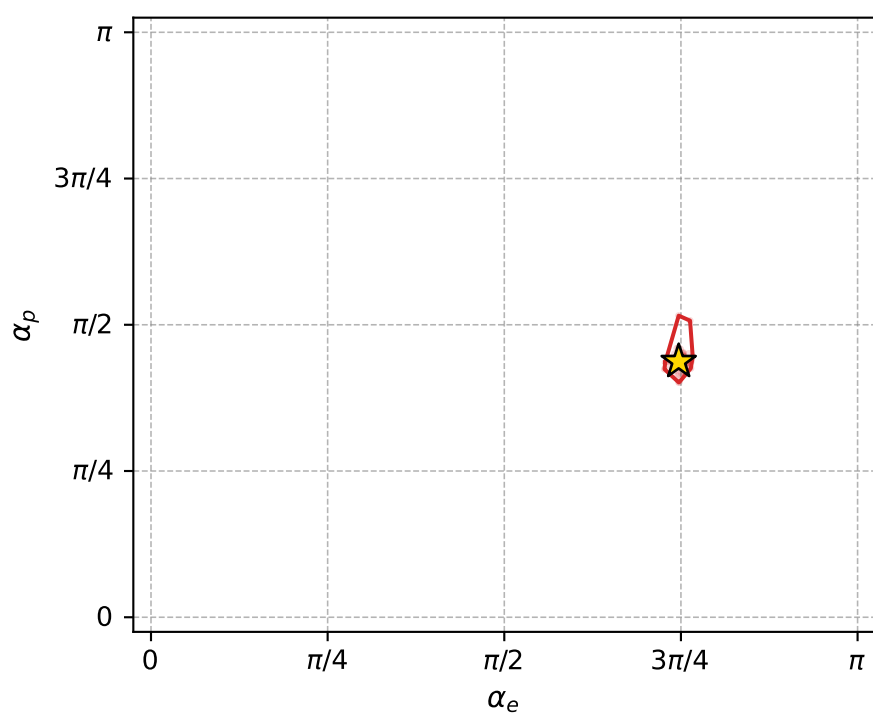
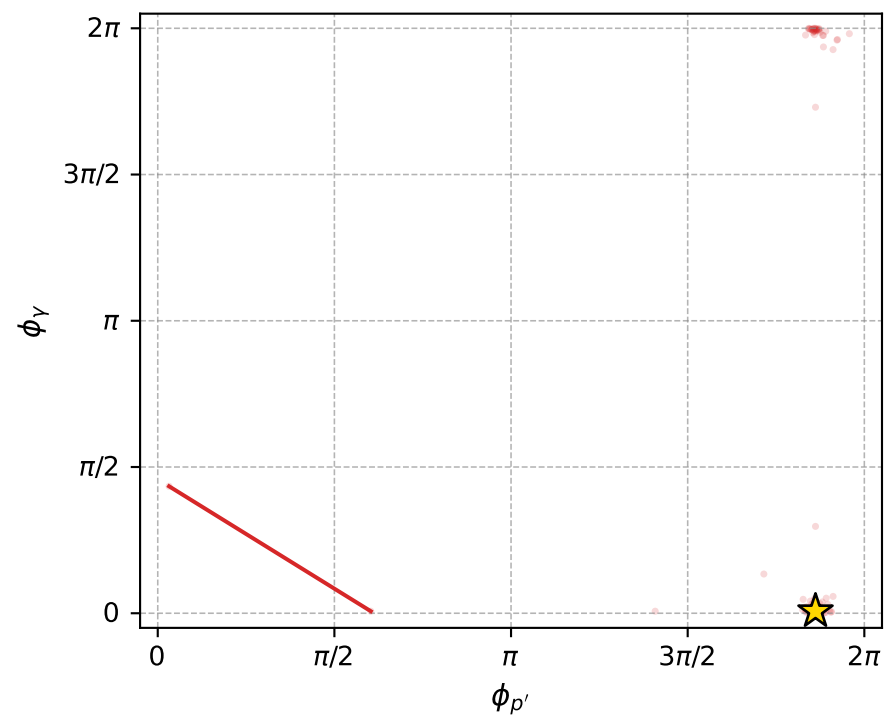
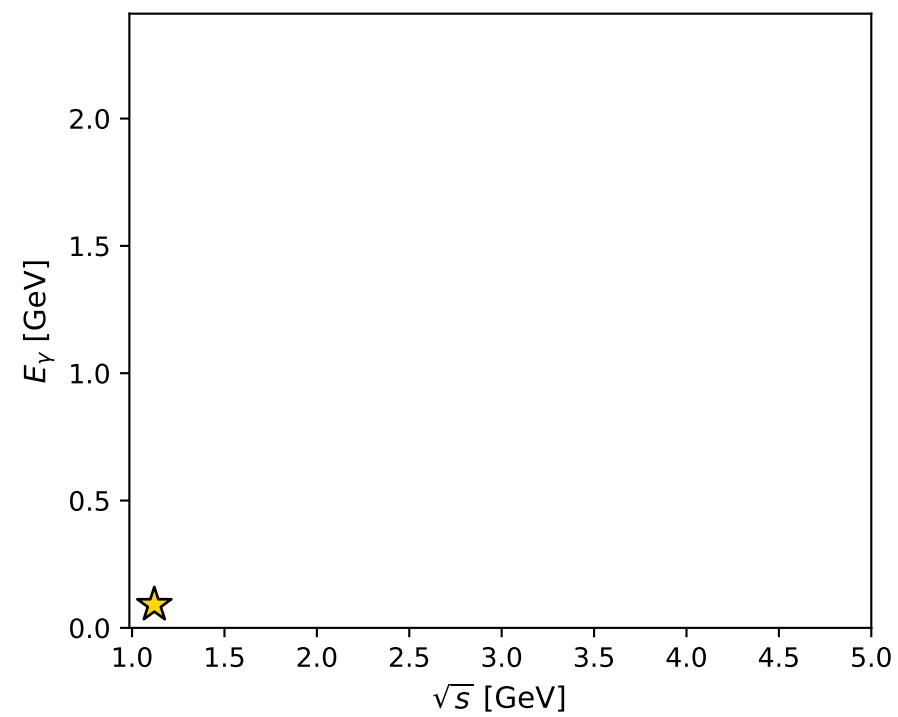
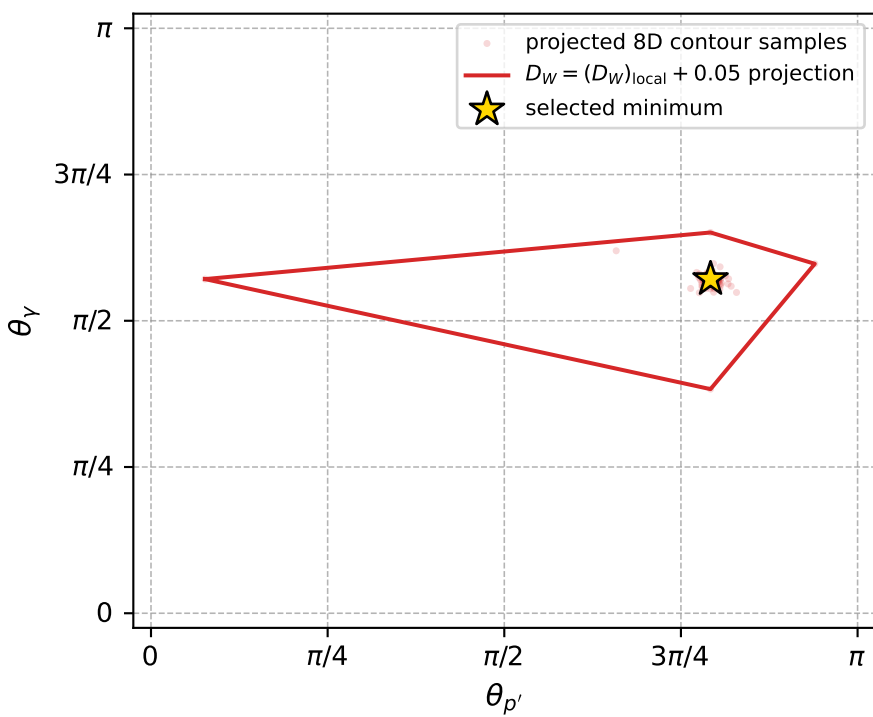
dw_mixing_angles_polarization_06_local_minimum_803 D_W=0.0104478
 region: polarization_cluster_06_local_minimum_803
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3462122$ rad
 incoming lepton: $-0.700013|+\rangle +0.71413|-\rangle$
 proton mixing angle: $\alpha_p=1.3732709$ rad
 incoming proton: $0.196244|+\rangle +0.980555|-\rangle$

$s=1.25522$, $\sqrt{s}=1.12037$, $|\vec{P}|=0.167524$, $|\vec{P}'|=0.00123948$
 $\theta_{P'}=2.48776$, $\theta_Y=1.79459$, $\phi_{P'}=5.84888$, $\phi_Y=0.0219861$
 $E_Y=0.0907412$, $\theta_{\ell'}=1.33817$, $\phi_{\ell'}=3.15985$
 $Q^2=0.0377753$, $x_B=0.181746$, $t=-0.0281752$, $W^2=1.04992$, $y=0.553699$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1675$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1675$
 P $E= 0.9528$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1675$
 ℓ' $E= 0.0916$ $p_x= -0.0891$ $p_y= -0.0016$ $p_z= 0.0211$
 P' $E= 0.9380$ $p_x= 0.0007$ $p_y= -0.0003$ $p_z= -0.0010$
 q_Y $E= 0.0907$ $p_x= 0.0885$ $p_y= 0.0019$ $p_z= -0.0201$



electron: polarization cluster P6, local minimum 803; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.010447771$
 $D_W \text{ contour} = 0.060447771$
 above global minimum = 0.010431338
 8D contour samples = 255

selected phase-space point:
 $\text{sqrt_s} = 1.120367$
 $\text{theta_p_out} = 2.487756$
 $\text{theta_gamma_out} = 1.794589$
 $\text{qOut} = 0.09074124$
 $\text{phi_p_out} = 5.848877$
 $\text{phi_gamma_out} = 0.02198615$
 $\text{alpha_e} = 2.346212$
 $\text{alpha_p} = 1.373271$

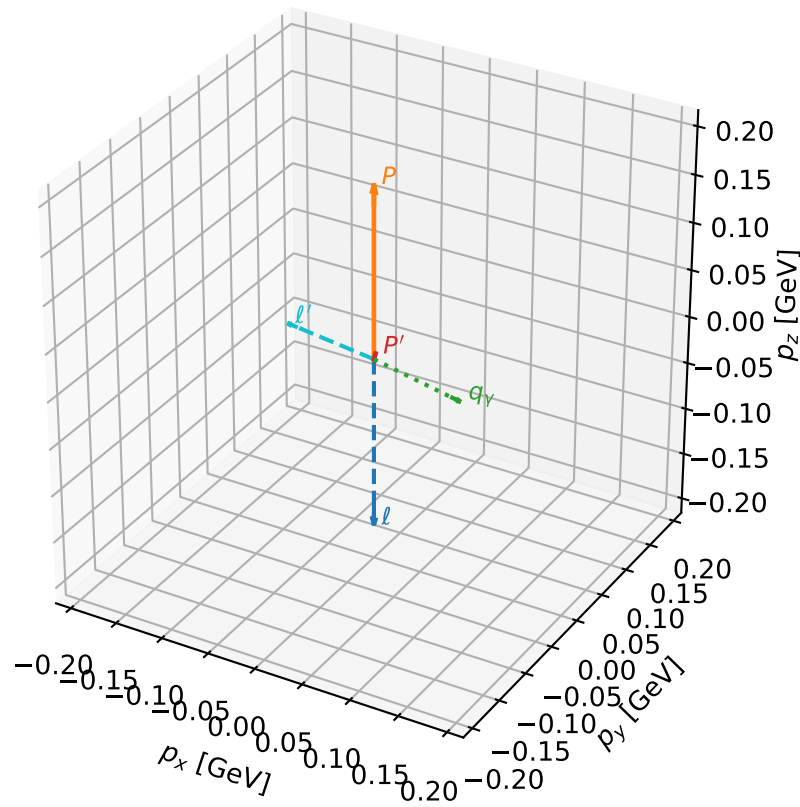
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_812 D_W=0.0107861
 region: polarization_cluster_06_local_minimum_812
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3638088$ rad
 incoming lepton: $-0.71247|+\rangle +0.701702|-\rangle$
 proton mixing angle: $\alpha_p=1.3635838$ rad
 incoming proton: $0.205733|+\rangle +0.978608|-\rangle$

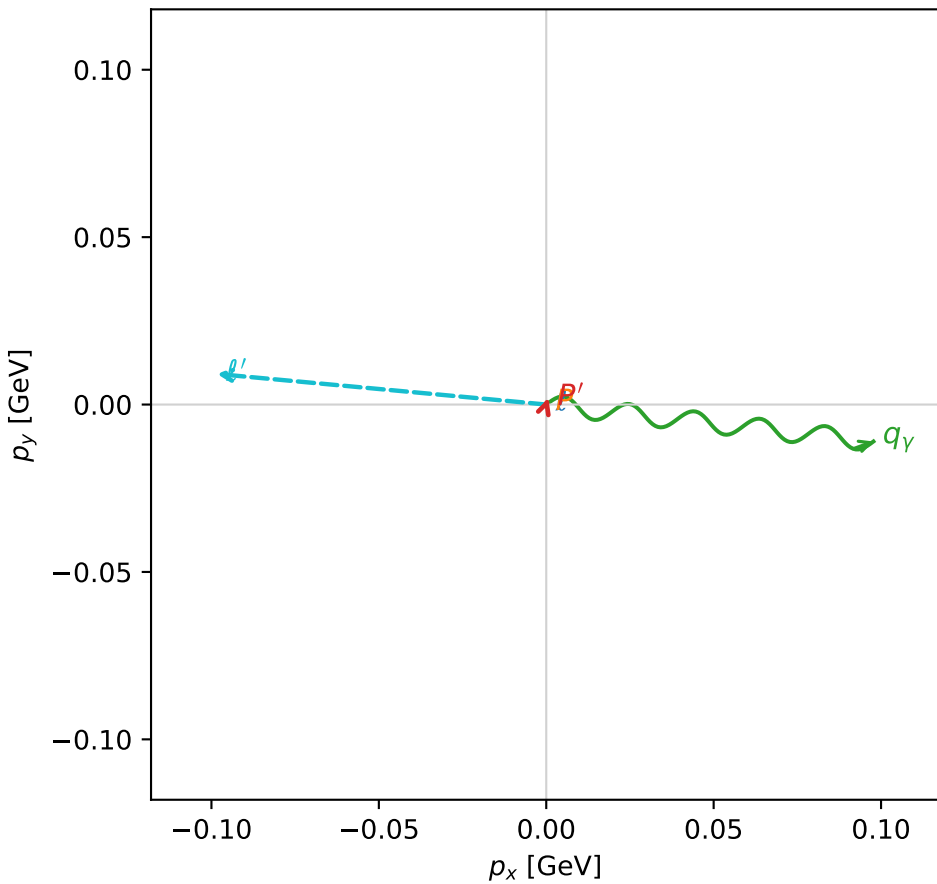
$s=1.29042$, $\sqrt{s}=1.13597$, $|\vec{P}|=0.180716$, $|\vec{P}'|=0.00566642$
 $\theta_{P'}=0.359605$, $\theta_Y=1.68413$, $\phi_{P'}=1.24282$, $\phi_Y=6.1708$
 $E_Y=0.0989809$, $\theta_{\ell'}=1.51125$, $\phi_{\ell'}=3.04893$
 $Q^2=0.0378986$, $x_B=0.169474$, $t=-0.0304764$, $W^2=1.06557$, $y=0.544662$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1807$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1807$
 P $E= 0.9552$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1807$
 ℓ' $E= 0.0990$ $p_x= -0.0984$ $p_y= 0.0091$ $p_z= 0.0059$
 P' $E= 0.9380$ $p_x= 0.0006$ $p_y= 0.0019$ $p_z= 0.0053$
 q_Y $E= 0.0990$ $p_x= 0.0977$ $p_y= -0.0110$ $p_z= -0.0112$

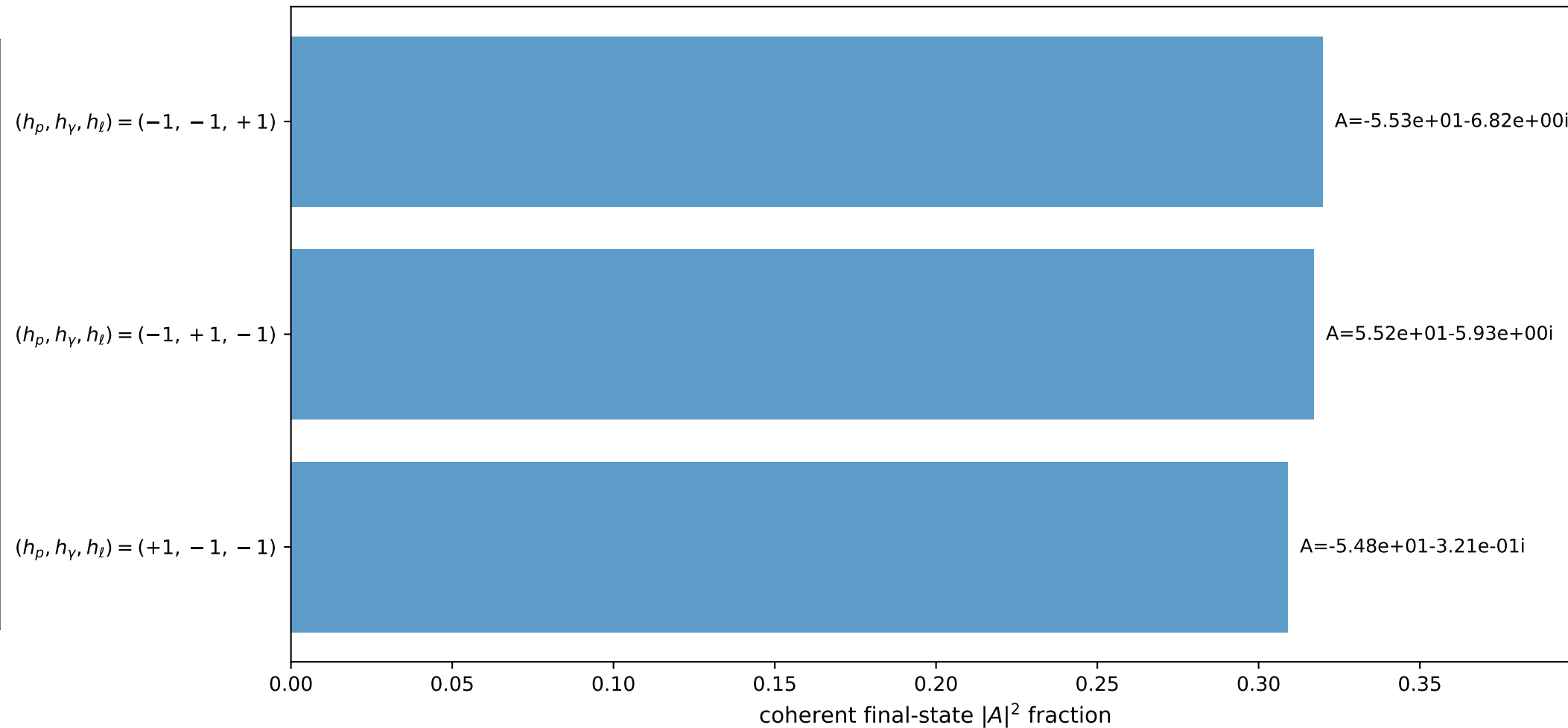
3D momenta



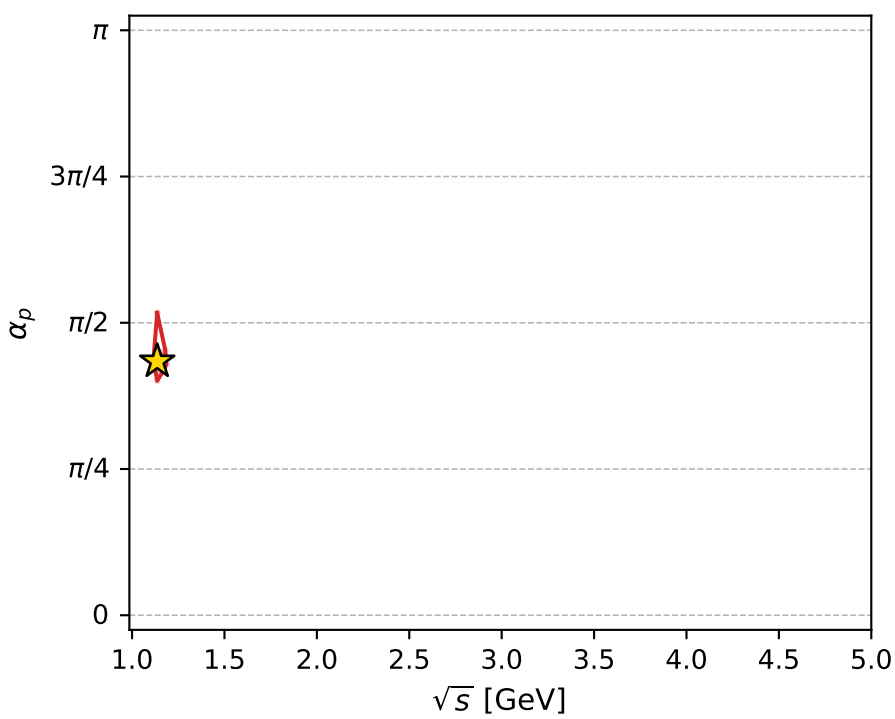
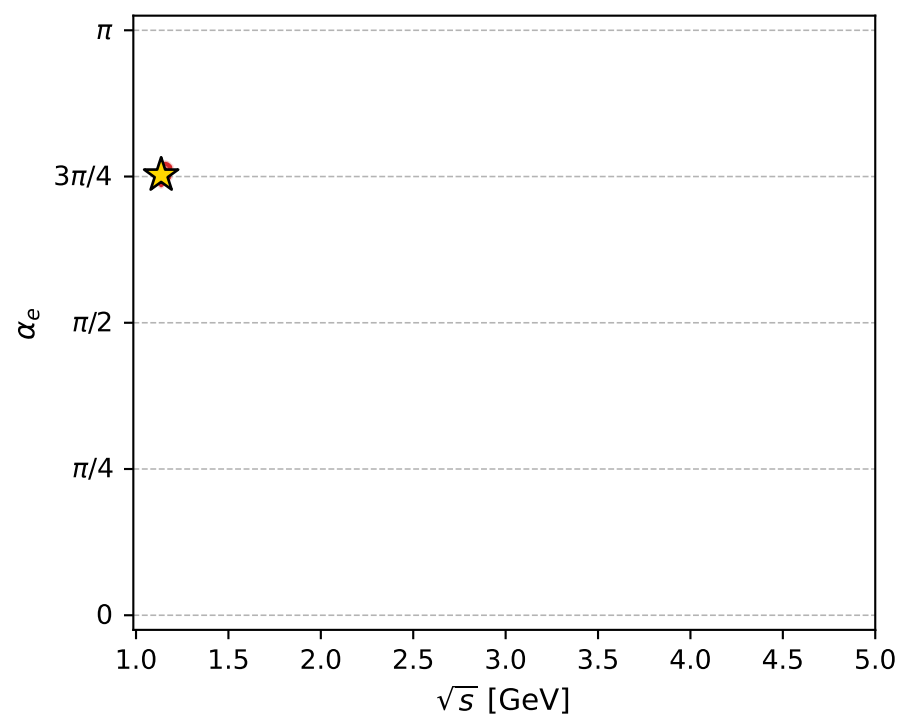
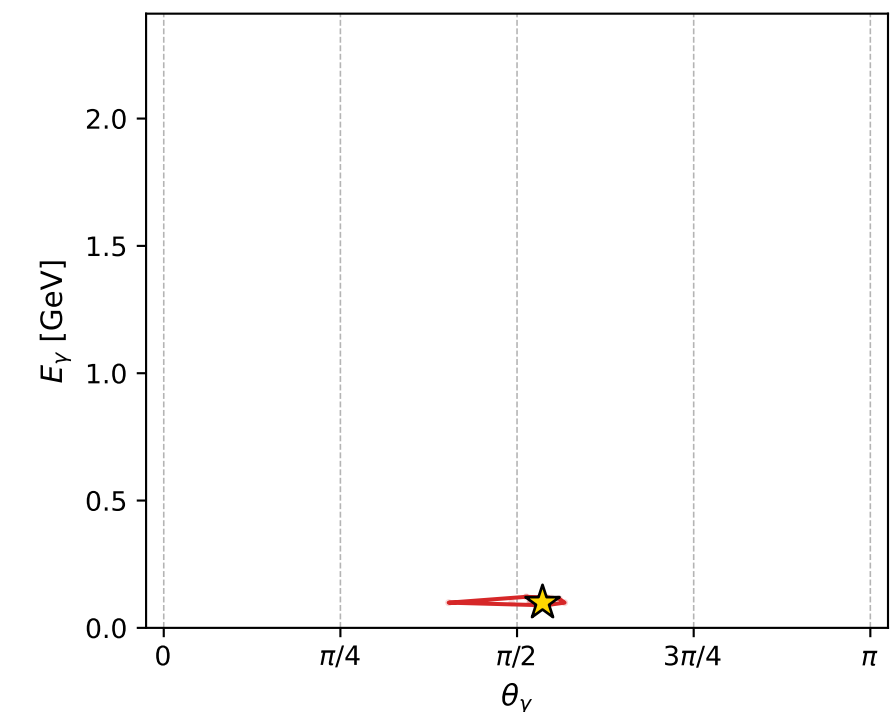
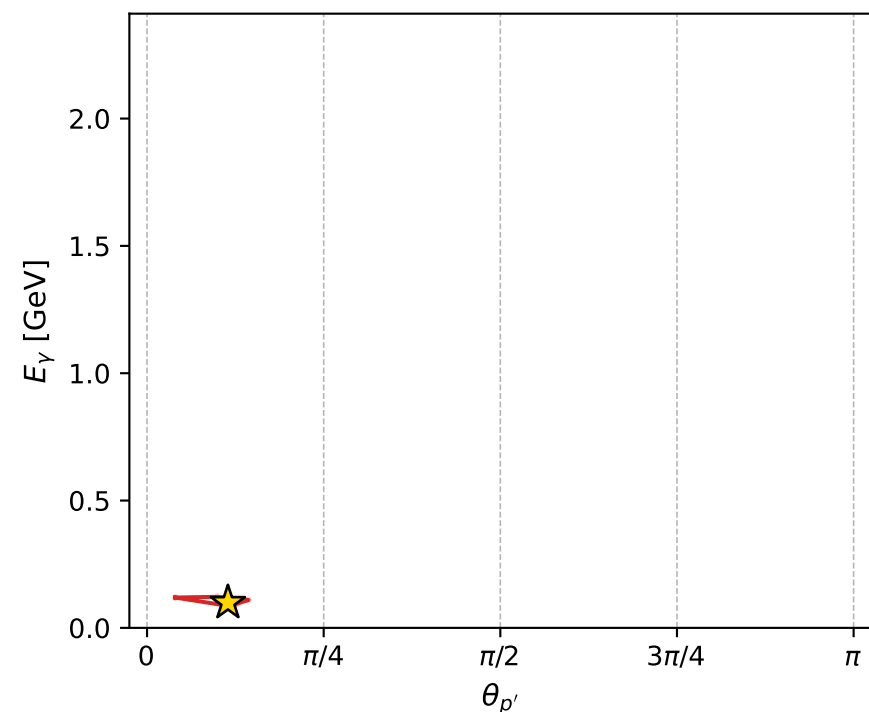
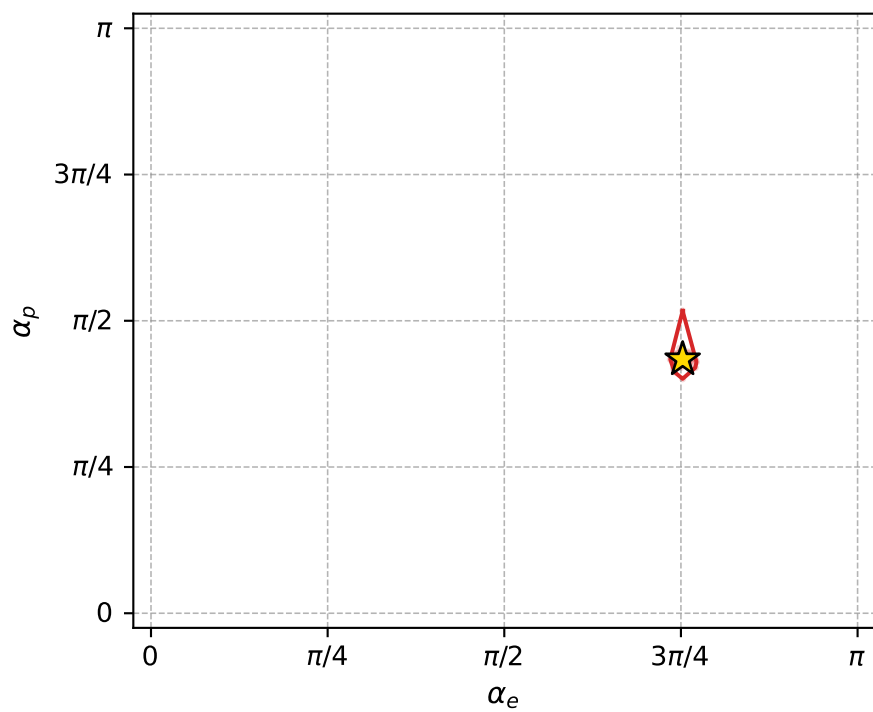
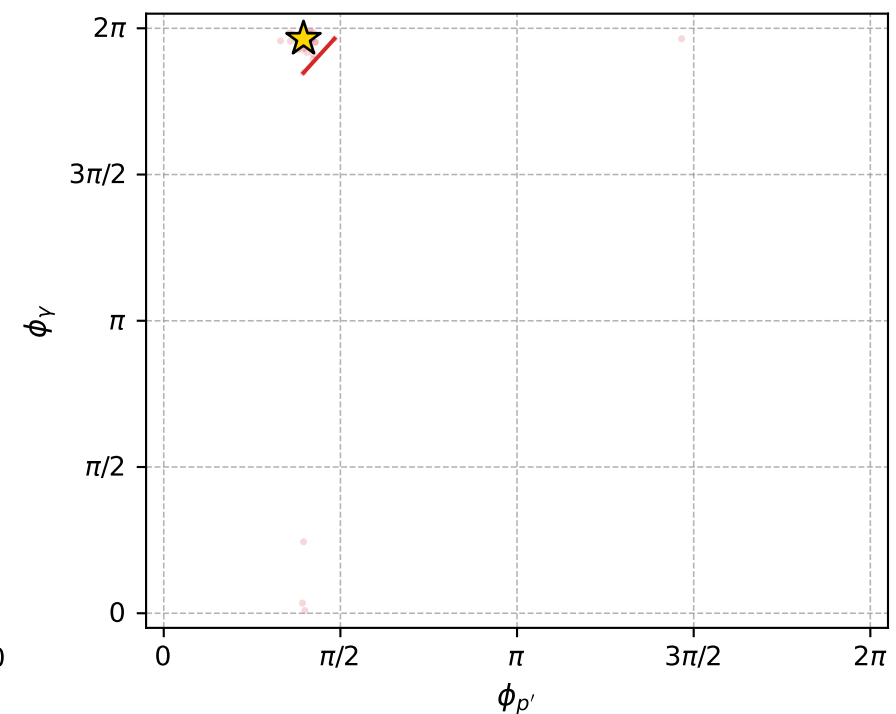
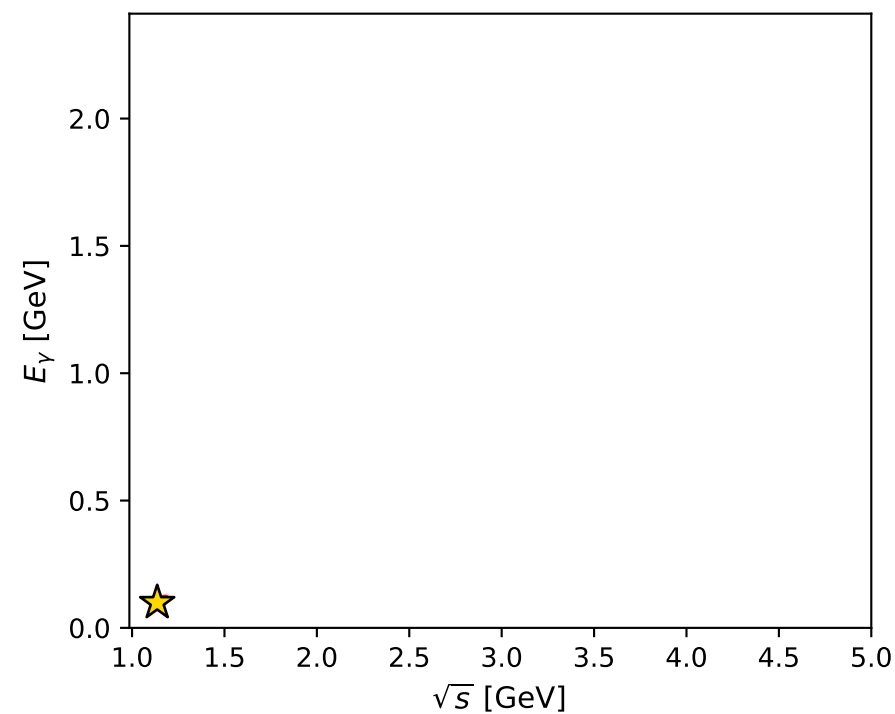
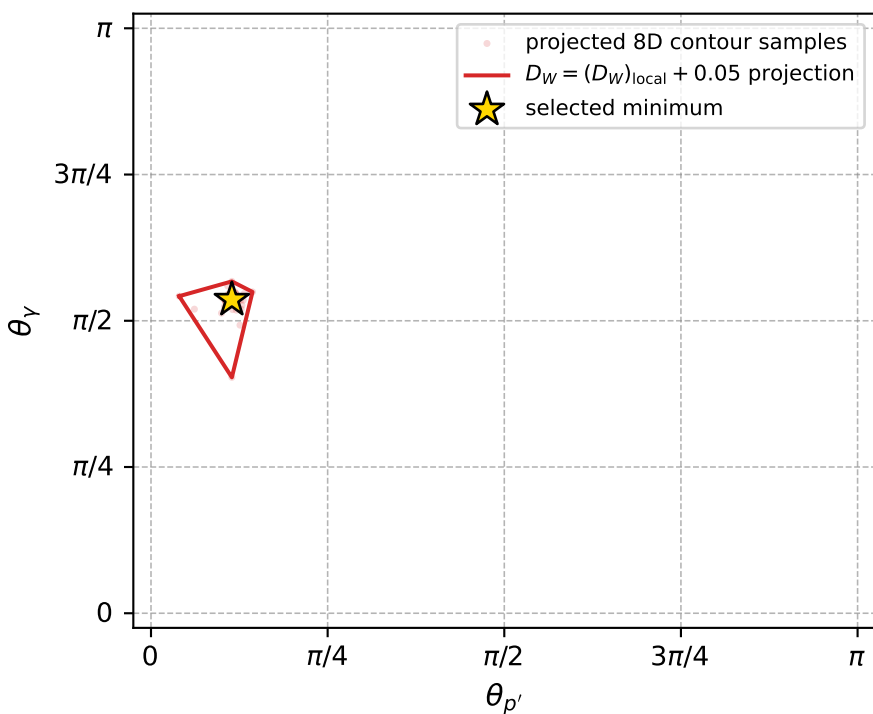
Transverse plane



Leading final-state amplitudes (N=3, retained=94.6%)



electron: polarization cluster P6, local minimum 812; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.010786106$
 $D_W \text{ contour} = 0.060786106$
 above global minimum = 0.010769673
 8D contour samples = 254

selected phase-space point:
 $\text{sqrt_s} = 1.135967$
 $\text{theta_p_out} = 0.3596054$
 $\text{theta_gamma_out} = 1.684125$
 $\text{qOut} = 0.09898094$
 $\text{phi_p_out} = 1.242821$
 $\text{phi_gamma_out} = 6.170801$
 $\text{alpha_e} = 2.363809$
 $\text{alpha_p} = 1.363584$

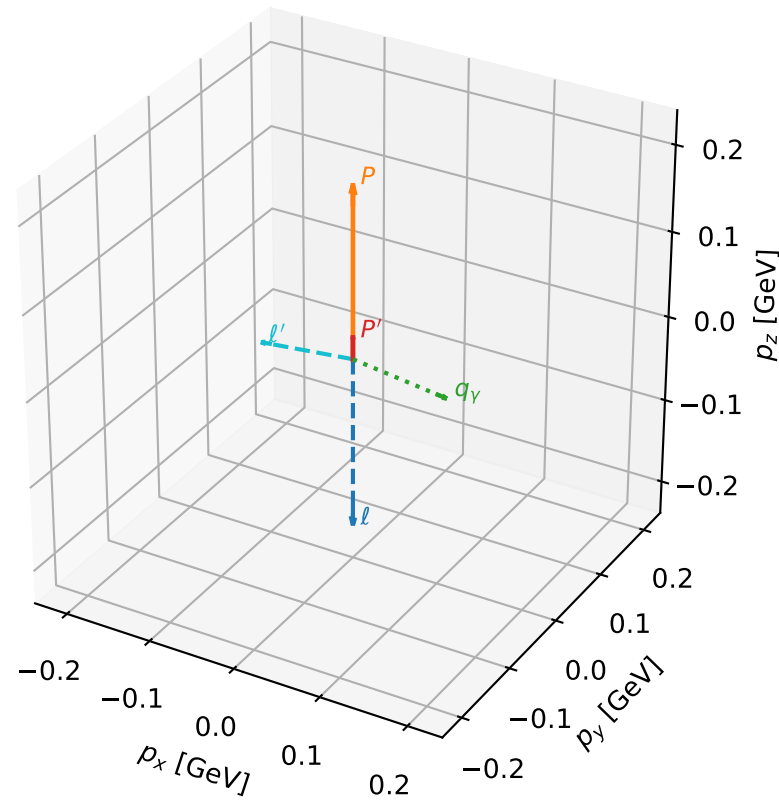
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_831 D_W=0.0114378
 region: polarization_cluster_06_local_minimum_831
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3807655$ rad
 incoming lepton: $-0.724266|+> +0.689521|->$
 proton mixing angle: $\alpha_p=1.3809873$ rad
 incoming proton: $0.188671|+> +0.98204|->$

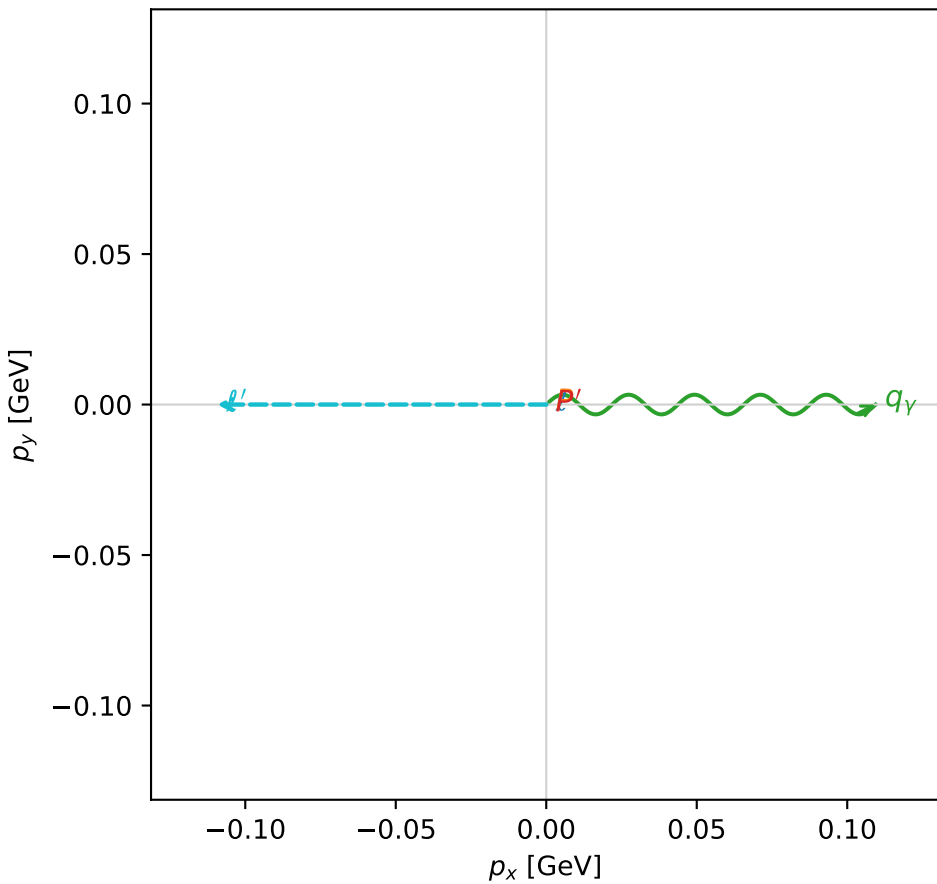
$s=1.34258$, $\sqrt{s}=1.1587$, $|\vec{P}|=0.199679$, $|\vec{P}'|=0.0255397$
 $\theta_{P'}=0$, $\theta_V=1.70042$, $\phi_{P'}=0.802418$, $\phi_V=6.28318$
 $E_V=0.110347$, $\theta_{\ell'}=1.67348$, $\phi_{\ell'}=3.14158$
 $Q^2=0.0394268$, $x_B=0.159465$, $t=-0.0298971$, $W^2=1.08766$, $y=0.53431$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1997$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1997$
 P $E= 0.9590$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1997$
 ℓ' $E= 0.1100$ $p_x= -0.1094$ $p_y= 0.0000$ $p_z= -0.0113$
 P' $E= 0.9383$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0255$
 q_V $E= 0.1103$ $p_x= 0.1094$ $p_y= -0.0000$ $p_z= -0.0143$

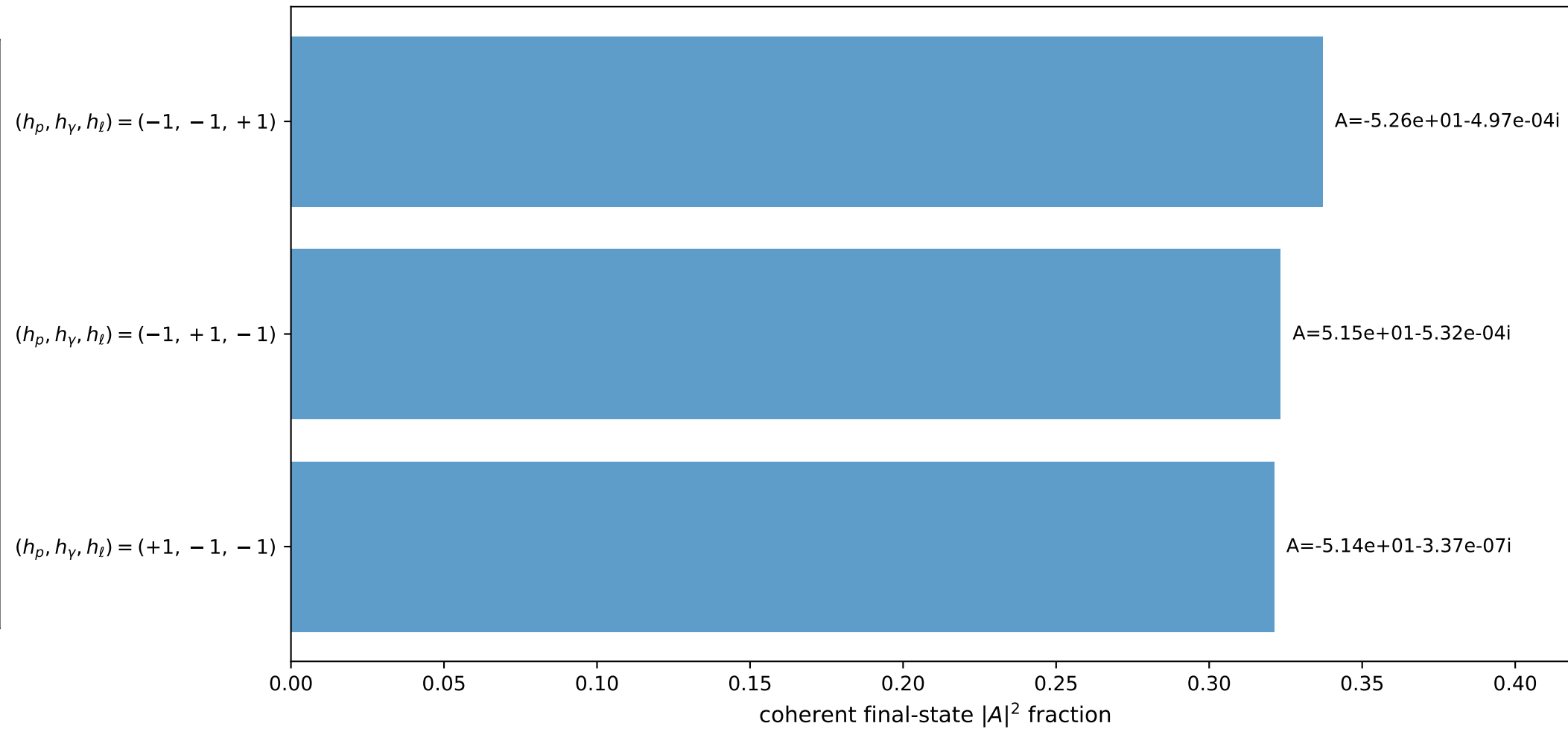
3D momenta



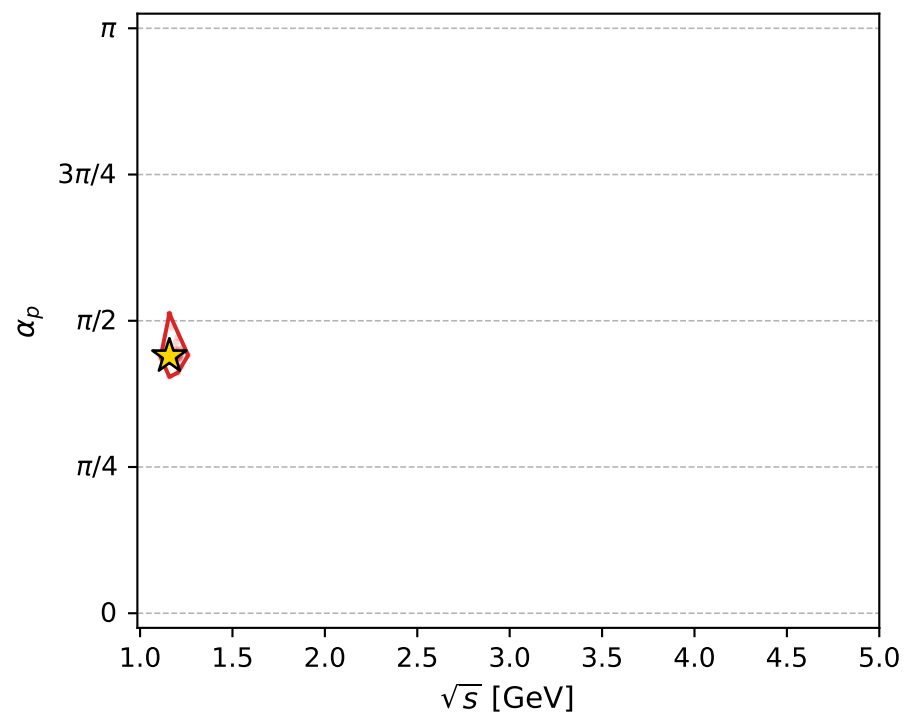
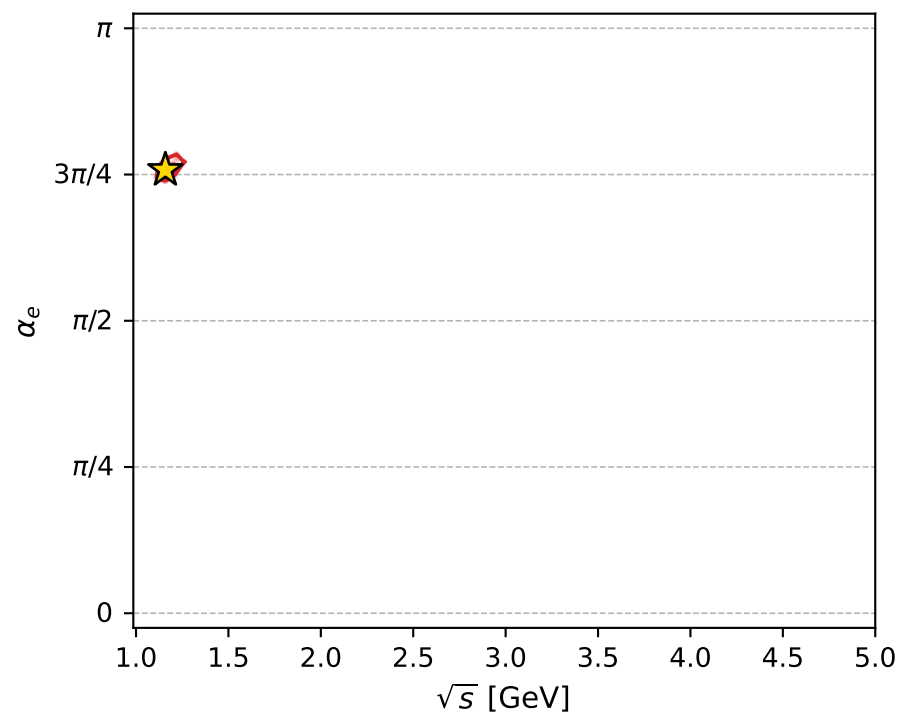
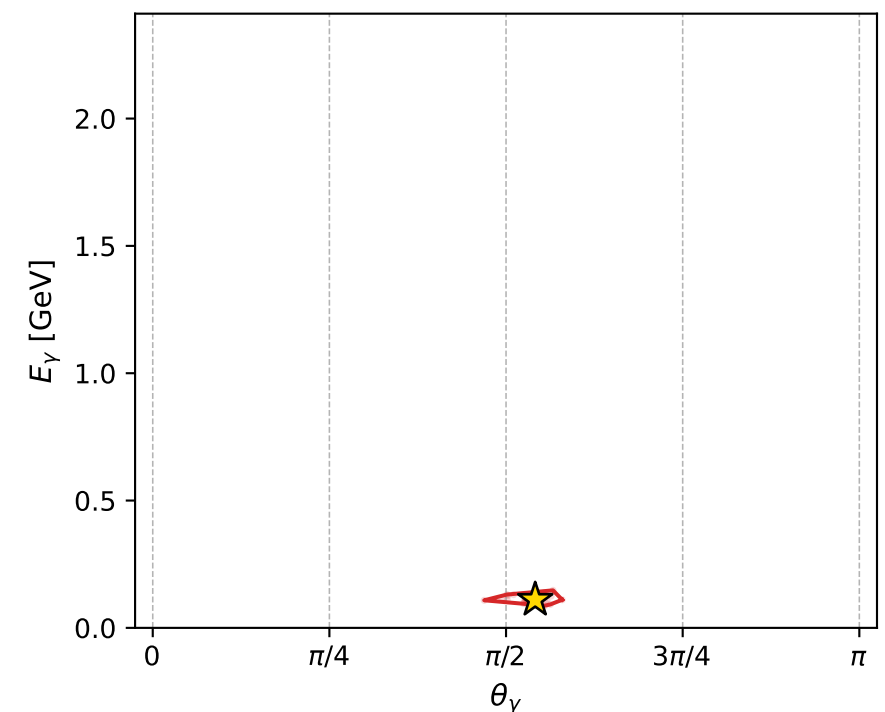
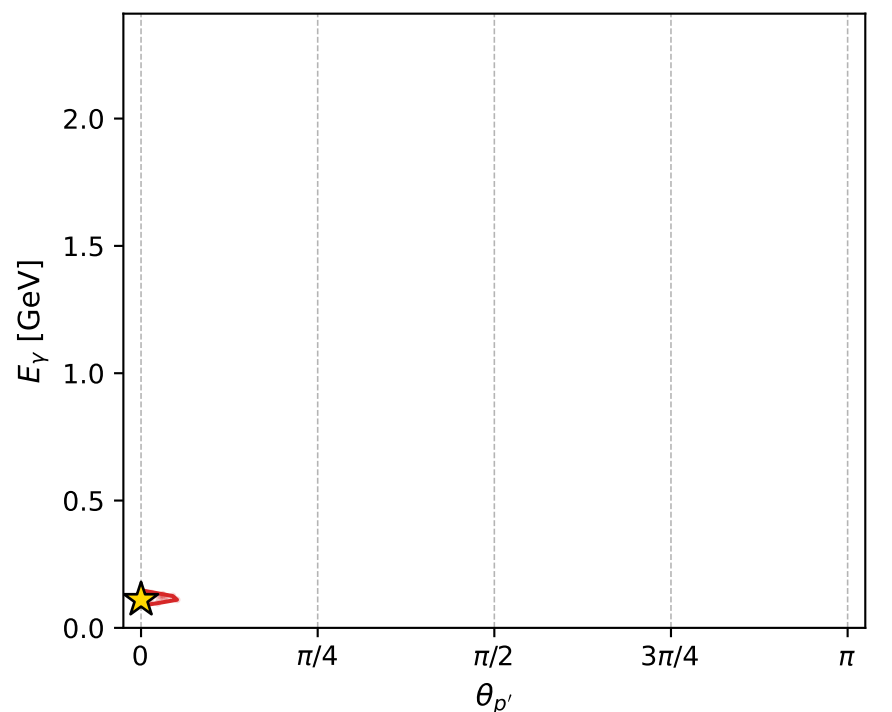
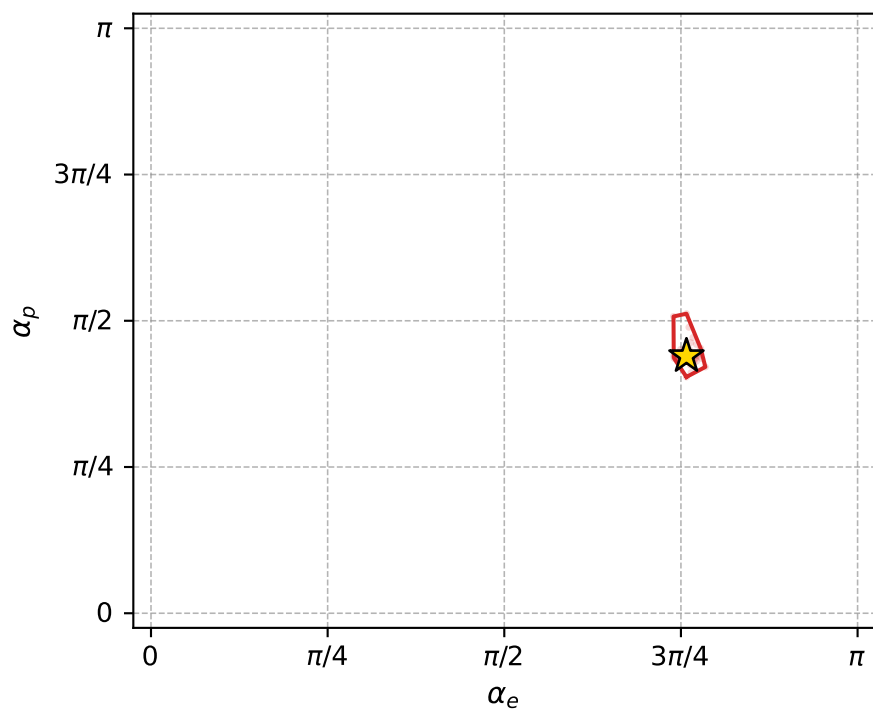
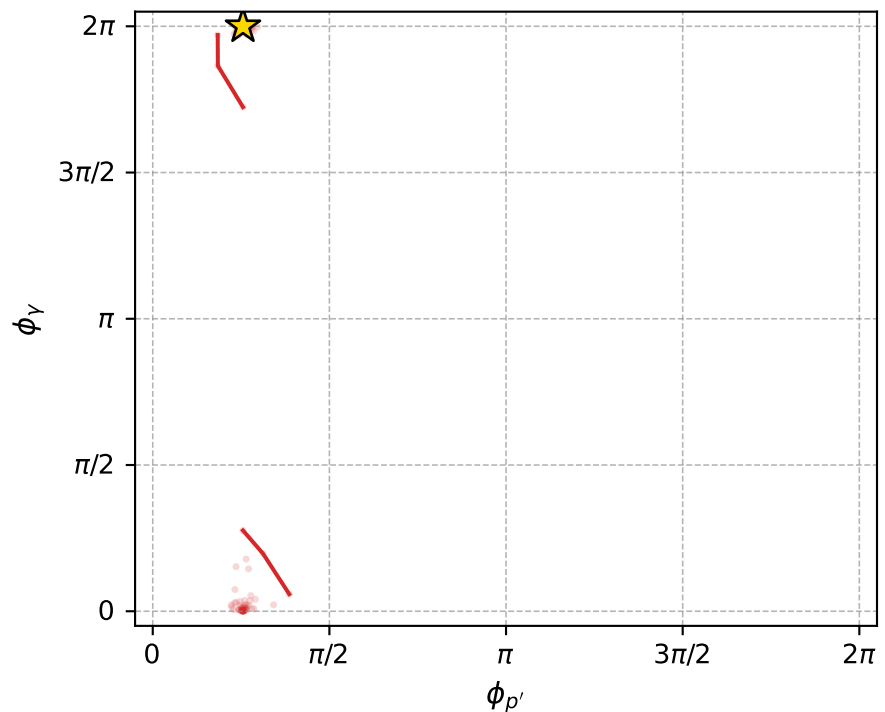
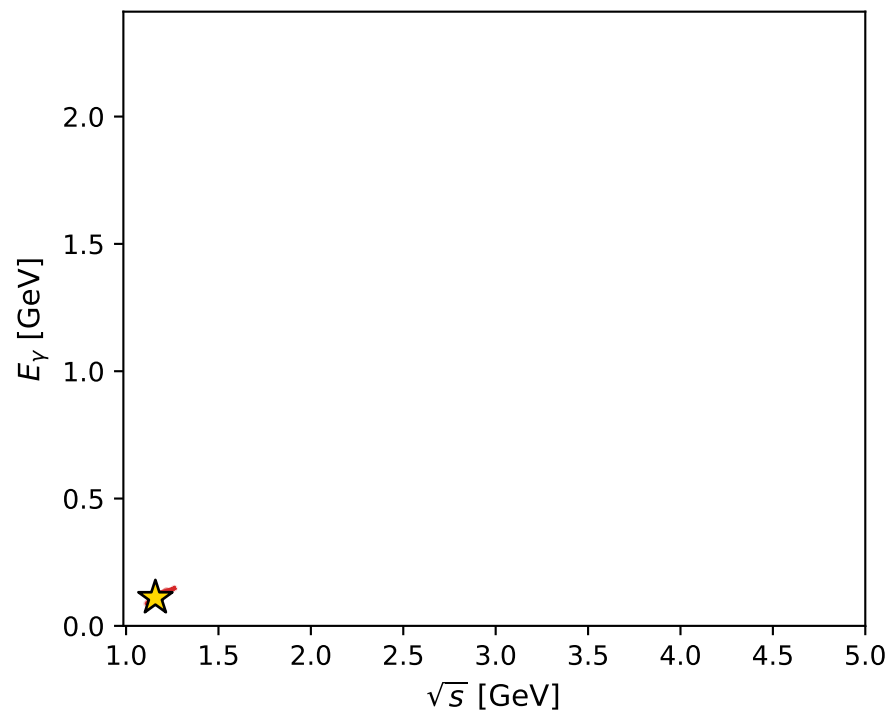
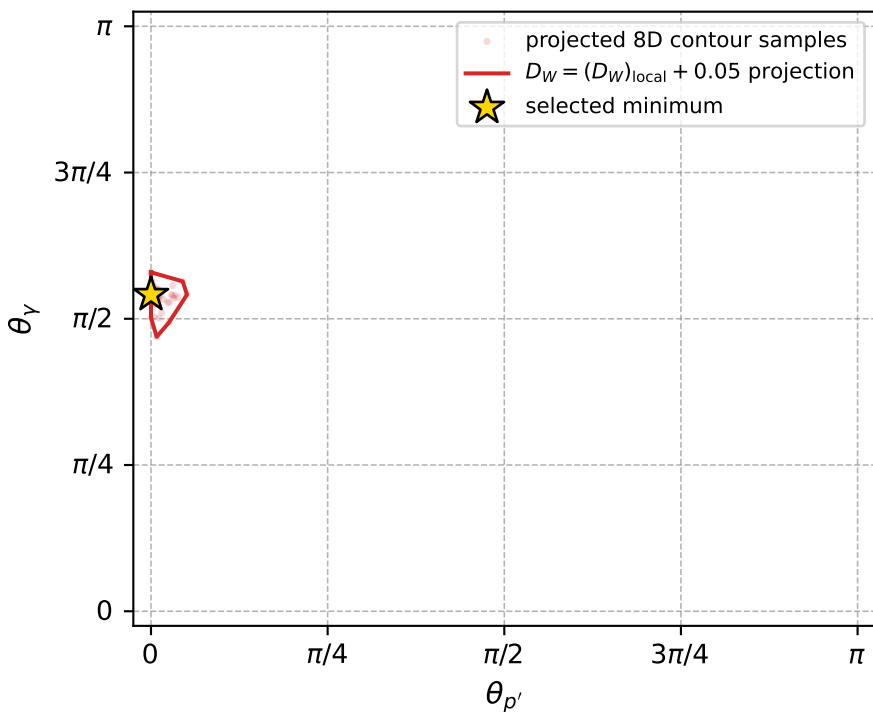
Transverse plane



Leading final-state amplitudes (N=3, retained=98.2%)



electron: polarization cluster P6, local minimum 831; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.011437784$
 $D_W \text{ contour} = 0.061437784$
 above global minimum = 0.011421351
 8D contour samples = 119

selected phase-space point:
 $\text{sqrt_s} = 1.158697$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 1.700423$
 $\text{qOut} = 0.1103474$
 $\text{phi_p_out} = 0.8024177$
 $\text{phi_gamma_out} = 6.283176$
 $\text{alpha_e} = 2.380766$
 $\text{alpha_p} = 1.380987$

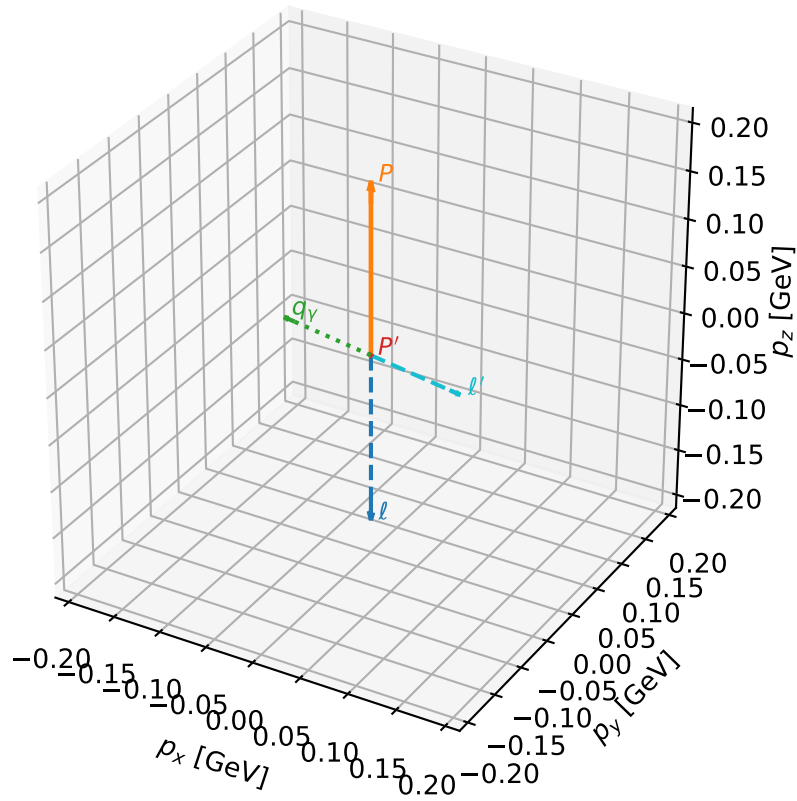
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_832 D_W=0.0114399
 region: polarization_cluster_06_local_minimum_832
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3759727$ rad
 incoming lepton: $-0.720953|+> +0.692984|->$
 proton mixing angle: $\alpha_p=1.7749962$ rad
 incoming proton: $-0.202784|+> +0.979224|->$

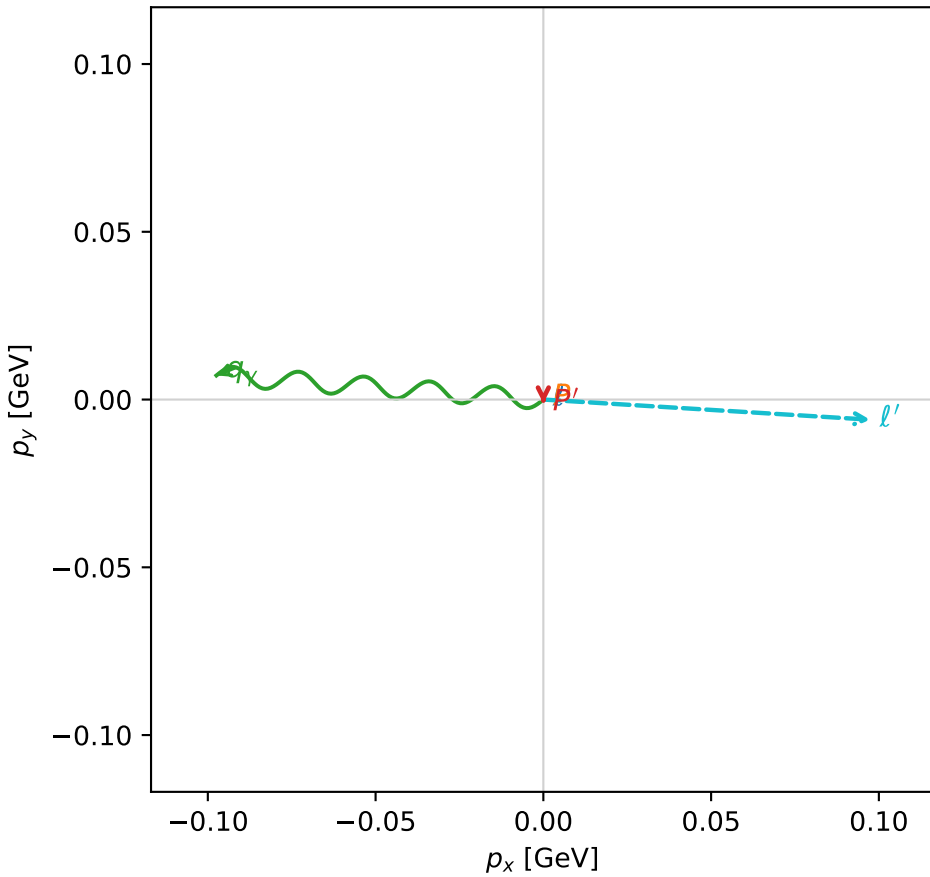
$s=1.28673$, $\sqrt{s}=1.13434$, $|\vec{P}|=0.179348$, $|\vec{P}'|=0.00164204$
 $\theta_{p'}=0.793957$, $\theta_Y=1.47513$, $\phi_{p'}=4.71386$, $\phi_Y=3.06804$
 $E_Y=0.0981507$, $\theta_{\ell'}=1.67821$, $\phi_{\ell'}=6.2216$
 $Q^2=0.0314435$, $x_B=0.145862$, $t=-0.0314668$, $W^2=1.06397$, $y=0.529807$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1793$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1793$
 P $E= 0.9550$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1793$
 ℓ' $E= 0.0982$ $p_x= 0.0974$ $p_y= -0.0060$ $p_z= -0.0105$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0012$ $p_z= 0.0012$
 q_Y $E= 0.0982$ $p_x= -0.0974$ $p_y= 0.0072$ $p_z= 0.0094$

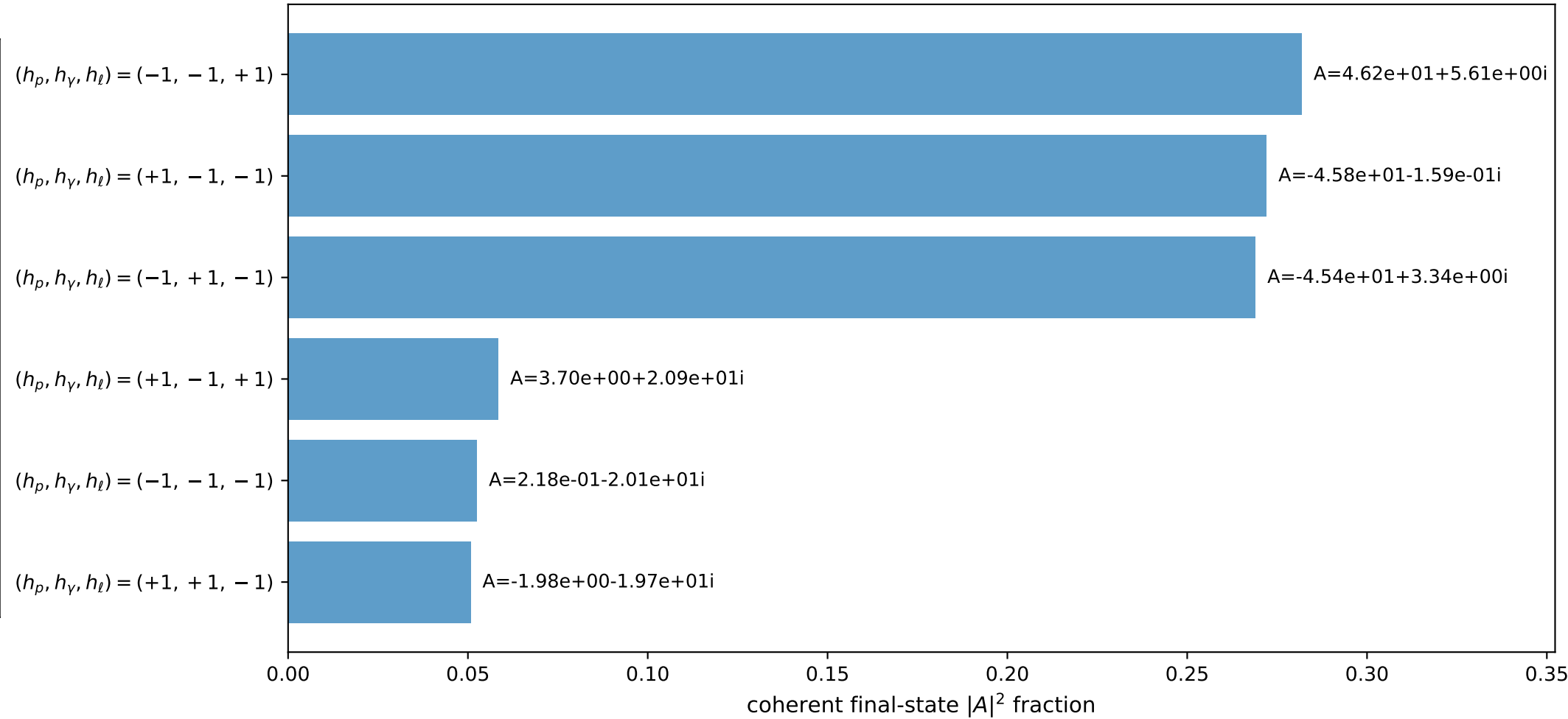
3D momenta



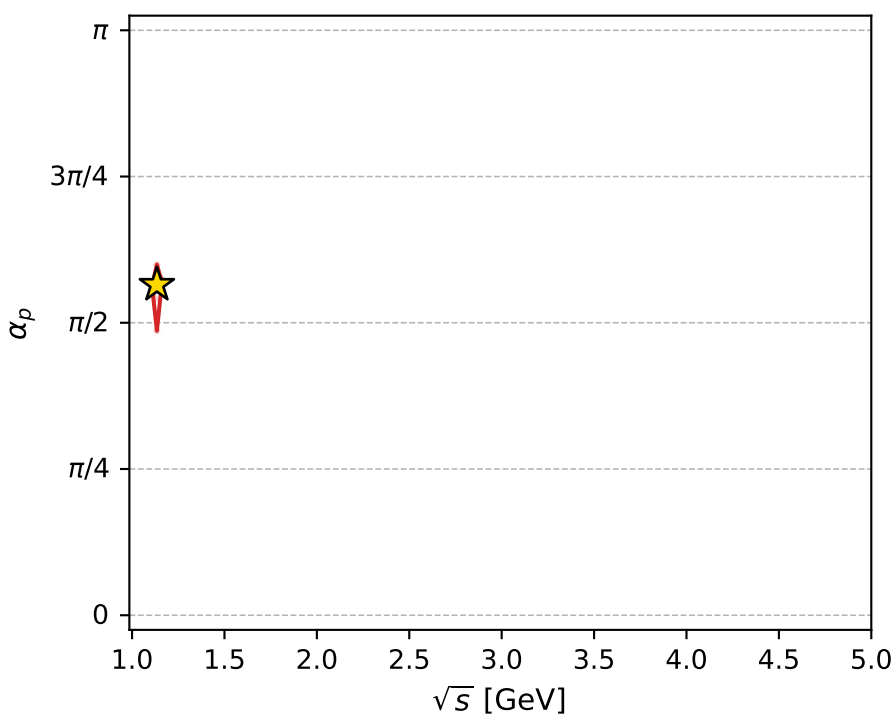
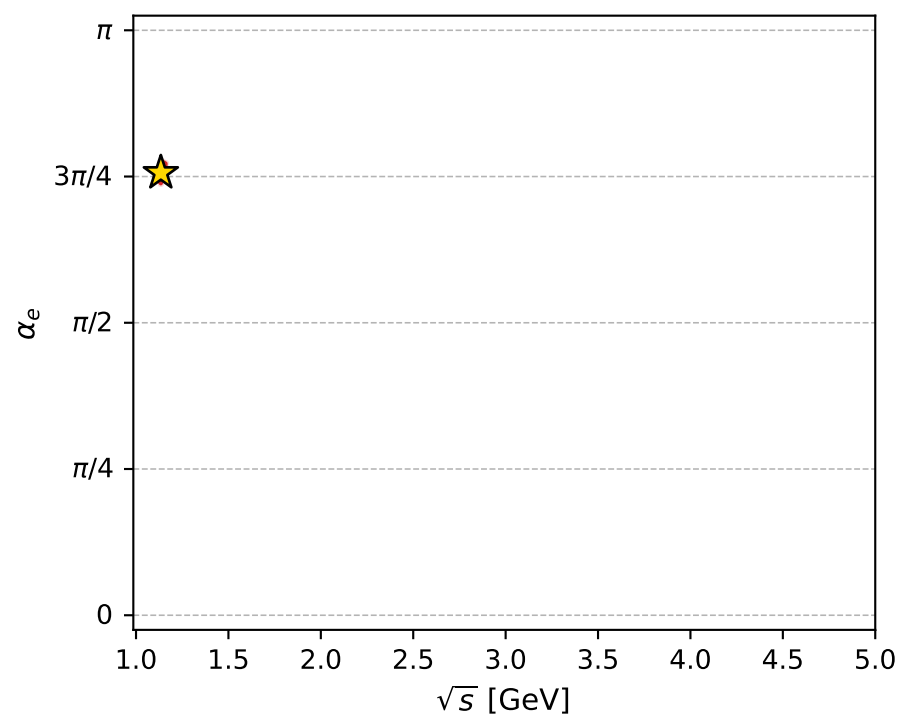
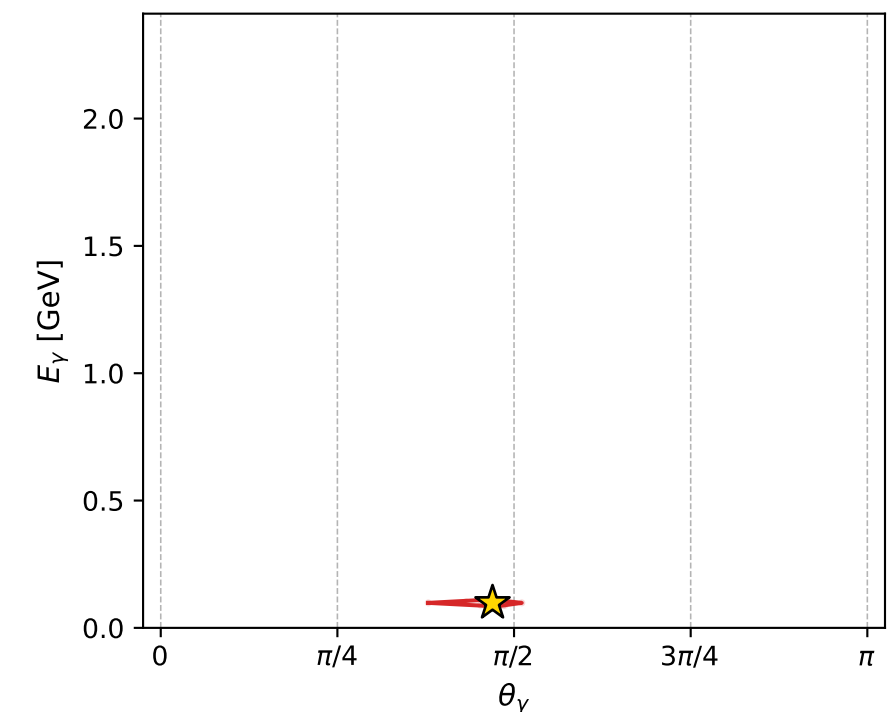
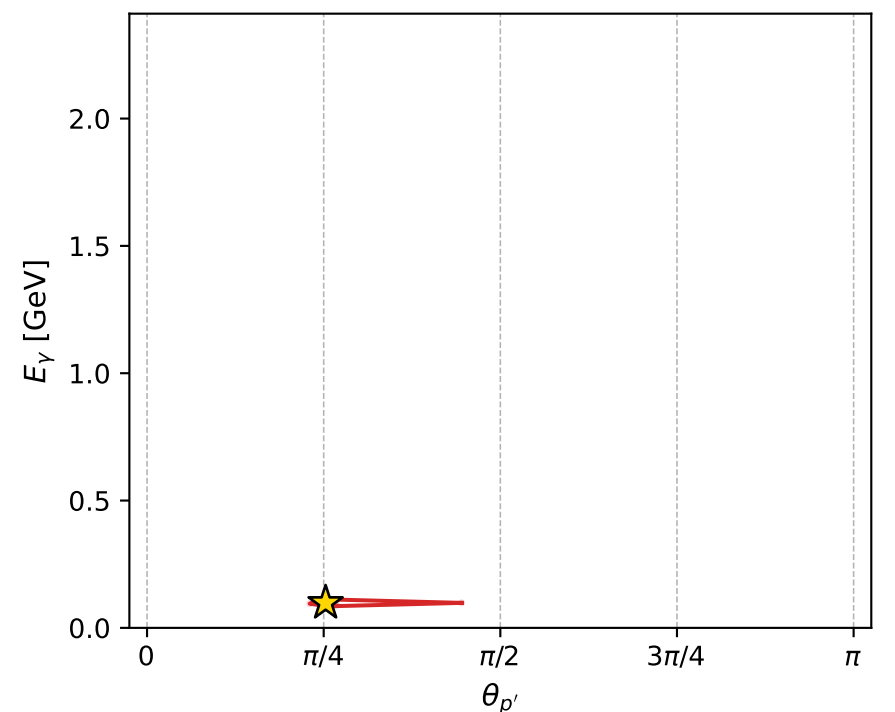
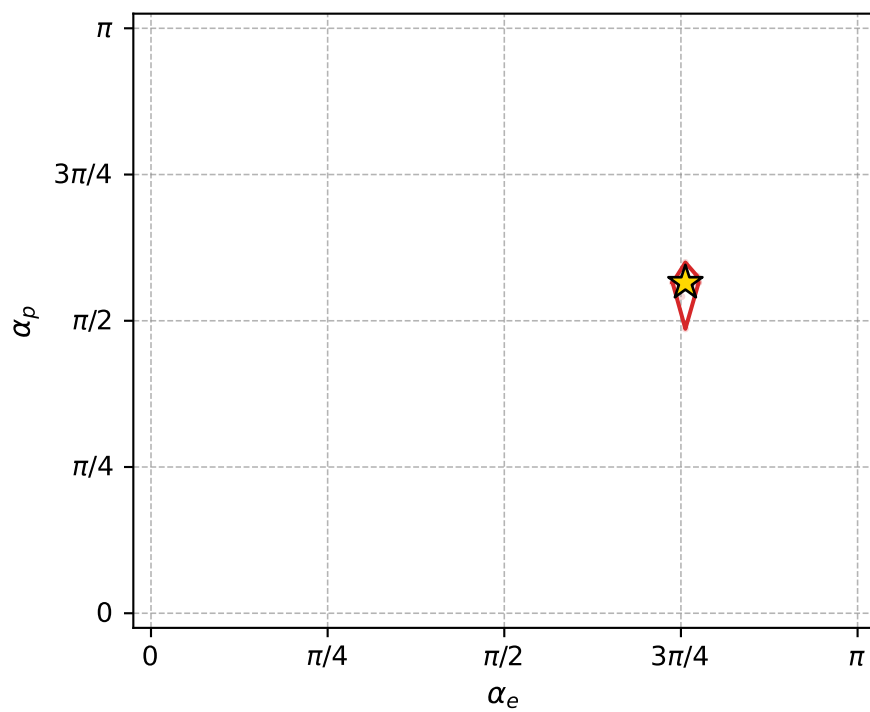
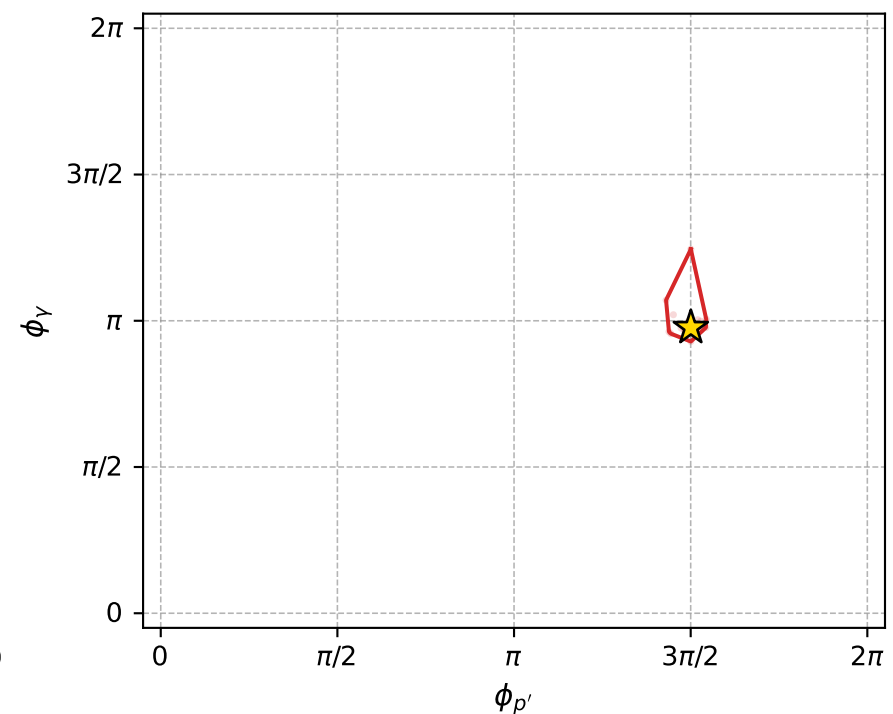
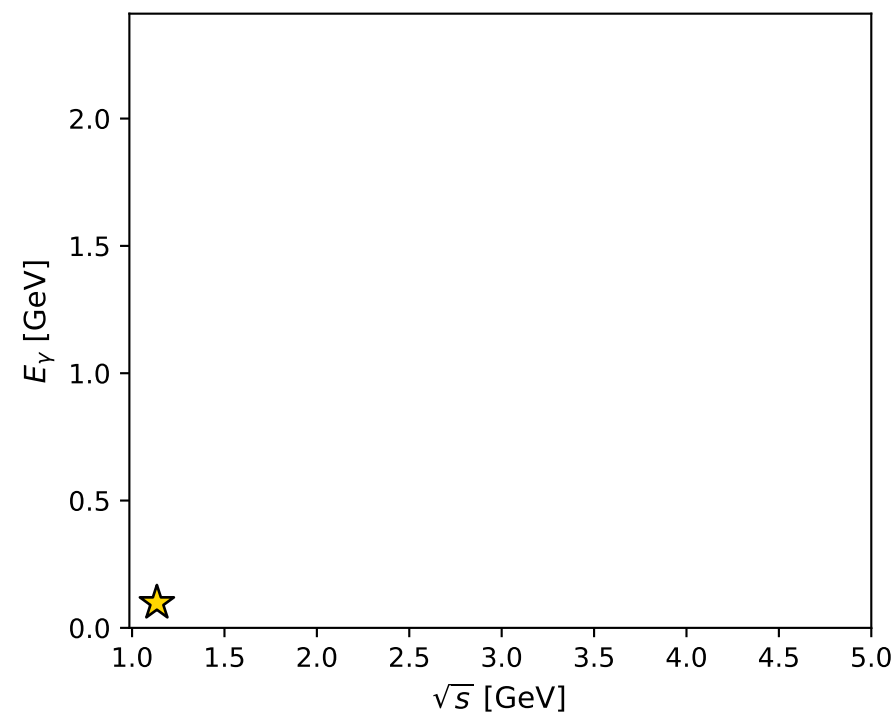
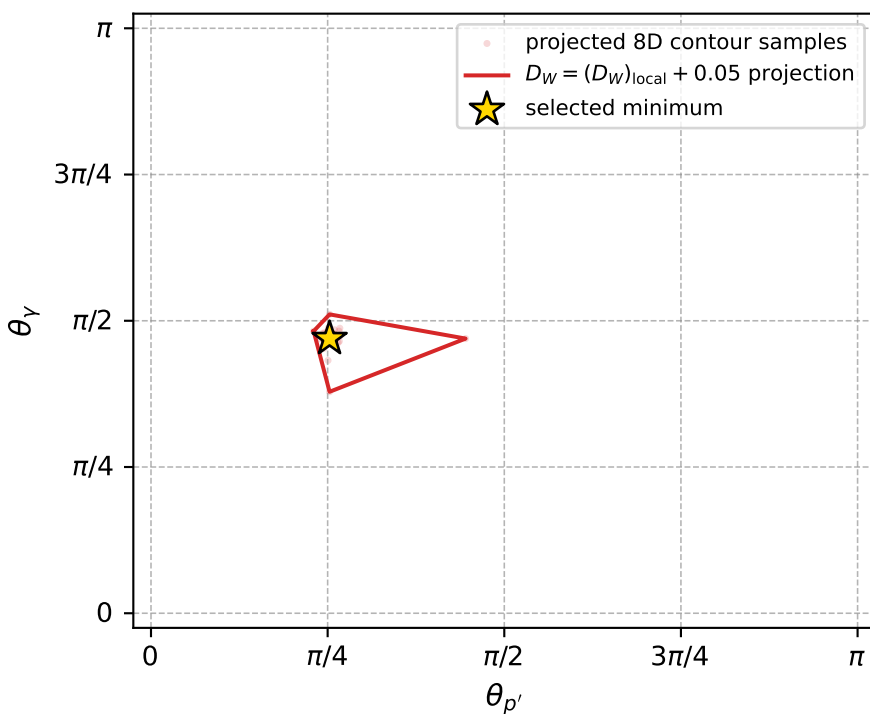
Transverse plane



Leading final-state amplitudes (N=6, retained=98.5%)



electron: polarization cluster P6, local minimum 832; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.011439902$
 $D_W \text{ contour} = 0.061439902$
 above global minimum = 0.011423469
 8D contour samples = 254

selected phase-space point:

$\text{sqrt_s} = 1.134341$
 $\text{theta_p_out} = 0.793957$
 $\text{theta_gamma_out} = 1.47513$
 $\text{qOut} = 0.09815072$
 $\text{phi_p_out} = 4.713864$
 $\text{phi_gamma_out} = 3.068043$
 $\text{alpha_e} = 2.375973$
 $\text{alpha_p} = 1.774996$

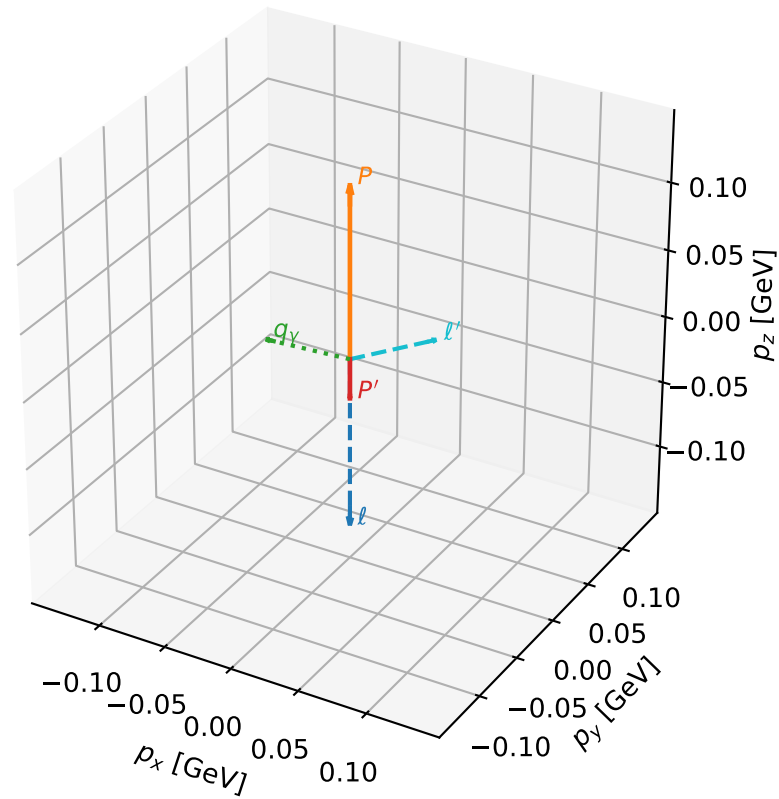
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_840 D_W=0.0116403
region: polarization_cluster_06_local_minimum_840
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3185035$ rad
incoming lepton: $-0.679959|+> +0.73325|->$
proton mixing angle: $\alpha_p=1.7221857$ rad
incoming proton: $-0.150812|+> +0.988562|->$

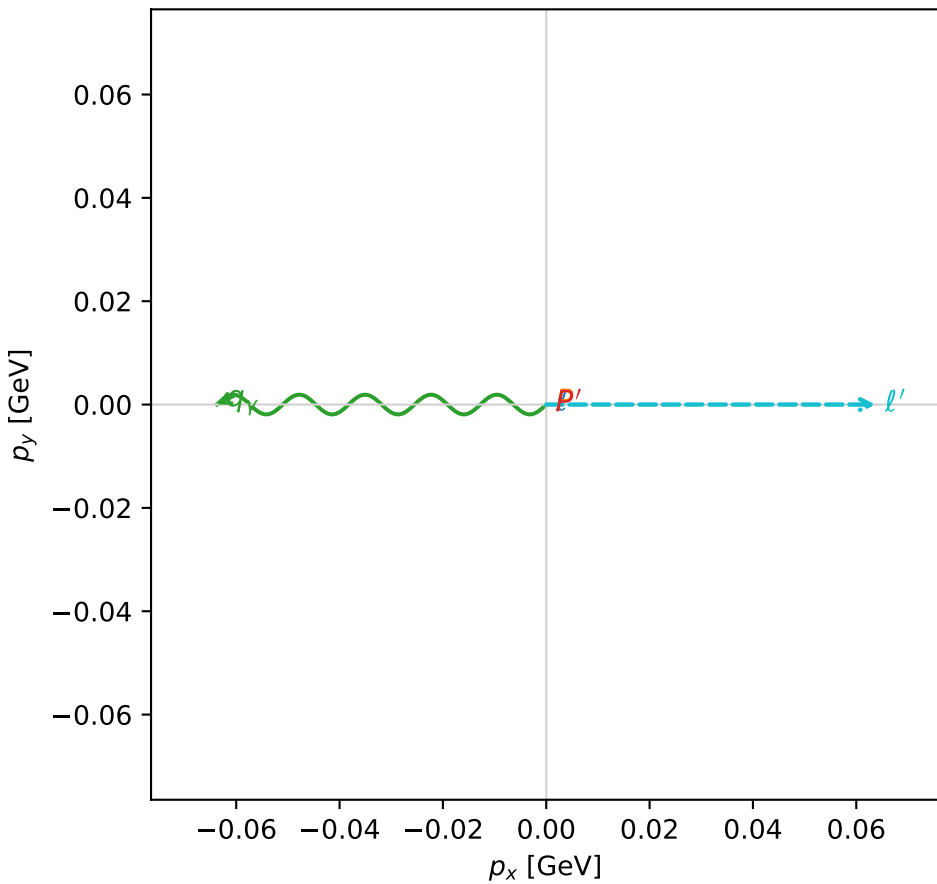
$s=1.15333$, $\sqrt{s}=1.07393$, $|\vec{P}|=0.12733$, $|\vec{P}'|=0.0294481$
 $\theta_{P'}=3.14159$, $\theta_V=1.62221$, $\phi_{P'}=4.91412$, $\phi_V=3.14161$
 $E_V=0.0638214$, $\theta_{\ell'}=1.09642$, $\phi_{\ell'}=1.27813e-05$
 $Q^2=0.0265808$, $x_B=0.181842$, $t=-0.0245132$, $W^2=0.999439$, $y=0.534482$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1273$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1273$
 P $E= 0.9466$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1273$
 ℓ' $E= 0.0717$ $p_x= 0.0637$ $p_y= 0.0000$ $p_z= 0.0327$
 P' $E= 0.9385$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0294$
 q_Y $E= 0.0638$ $p_x= -0.0637$ $p_y= -0.0000$ $p_z= -0.0033$

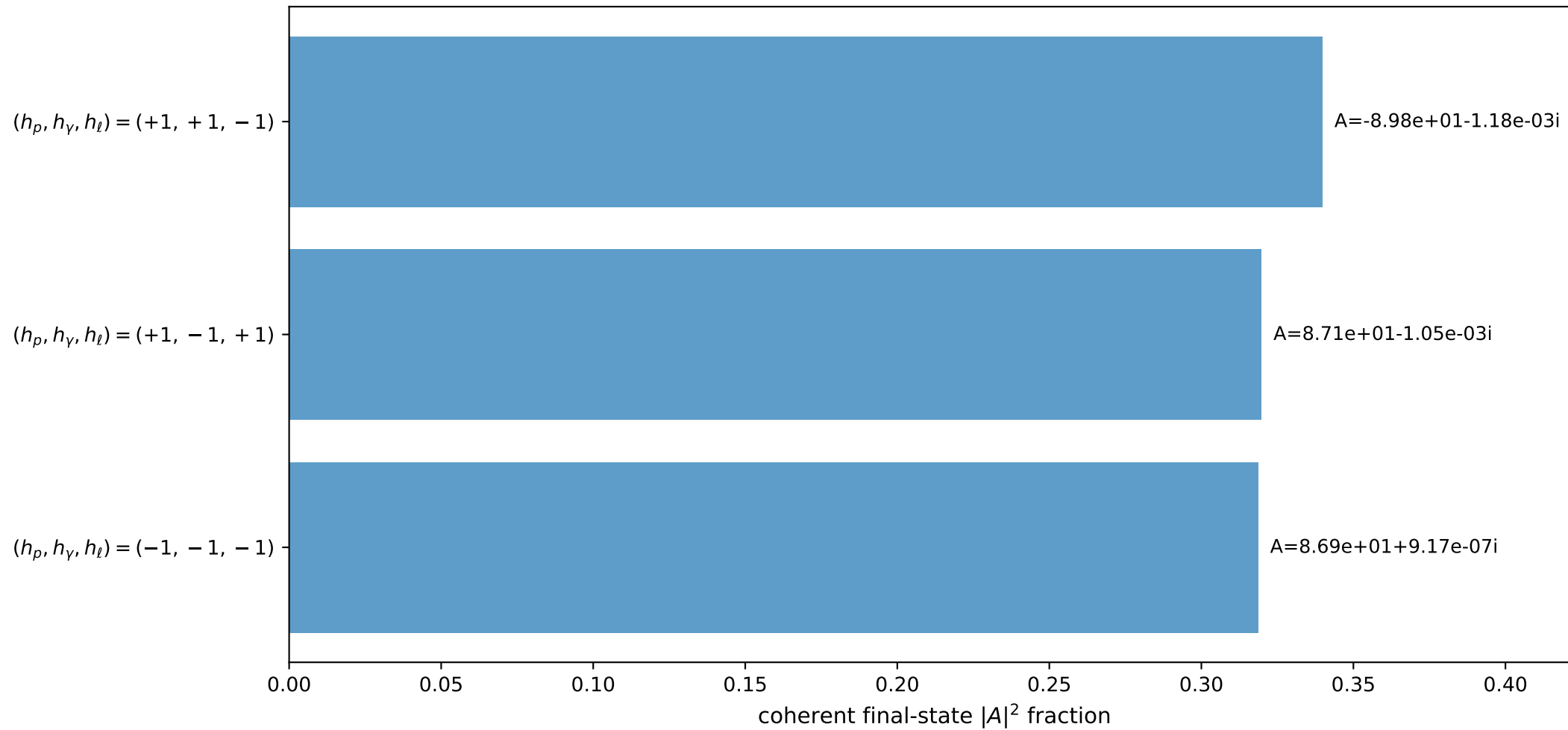
3D momenta



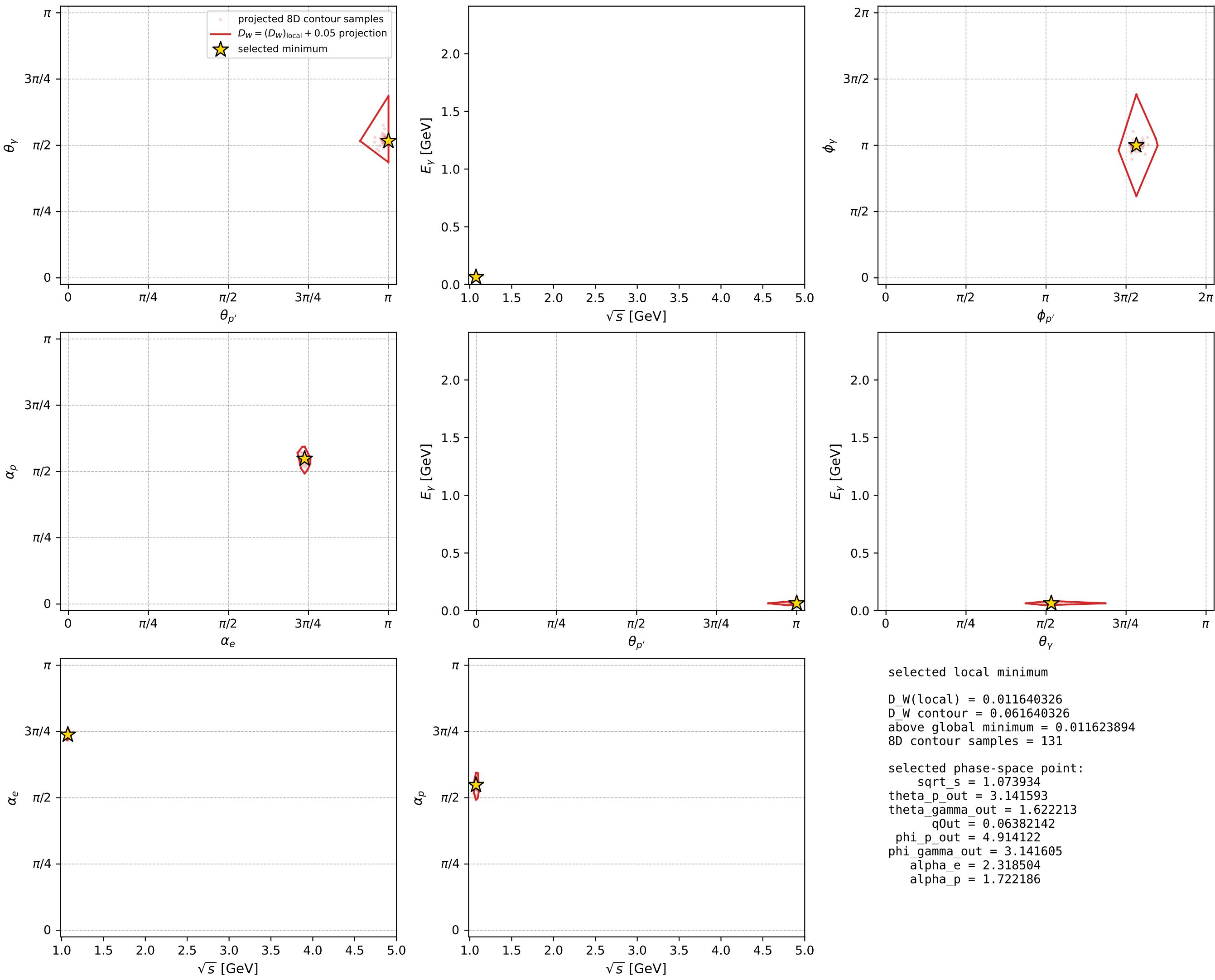
Transverse plane



Leading final-state amplitudes (N=3, retained=97.8%)



electron: polarization cluster P6, local minimum 840; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



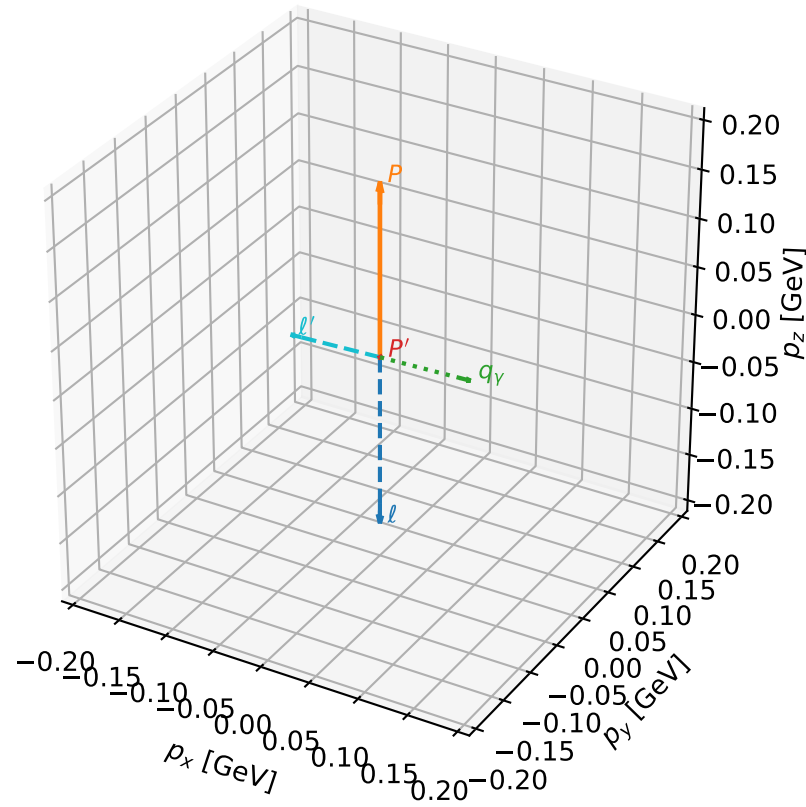
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_842 D_W=0.0116742
 region: polarization_cluster_06_local_minimum_842
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.374465$ rad
 incoming lepton: $-0.719907|+> +0.69407|->$
 proton mixing angle: $\alpha_p=1.3705897$ rad
 incoming proton: $0.198872|+> +0.980026|->$

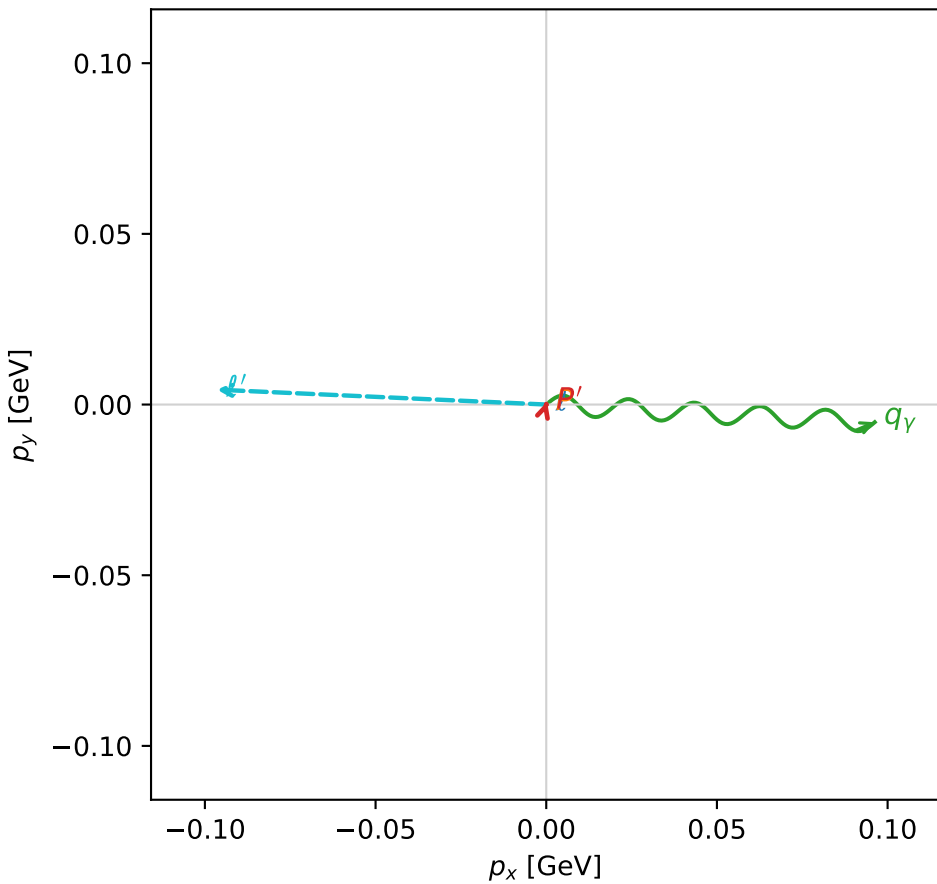
$s=1.27994$, $\sqrt{s}=1.13134$, $|\vec{P}|=0.176823$, $|\vec{P}'|=0.000857698$
 $\theta_{P'}=1.46915$, $\theta_Y=1.50719$, $\phi_{P'}=1.27279$, $\phi_Y=6.22947$
 $E_Y=0.096564$, $\theta_{\ell'}=1.63516$, $\phi_{\ell'}=3.09645$
 $Q^2=0.0320242$, $x_B=0.150251$, $t=-0.0309634$, $W^2=1.06096$, $y=0.532715$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1768$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1768$
 P $E= 0.9545$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1768$
 ℓ' $E= 0.0968$ $p_x= -0.0965$ $p_y= 0.0044$ $p_z= -0.0062$
 P' $E= 0.9380$ $p_x= 0.0003$ $p_y= 0.0008$ $p_z= 0.0001$
 q_Y $E= 0.0966$ $p_x= 0.0962$ $p_y= -0.0052$ $p_z= 0.0061$

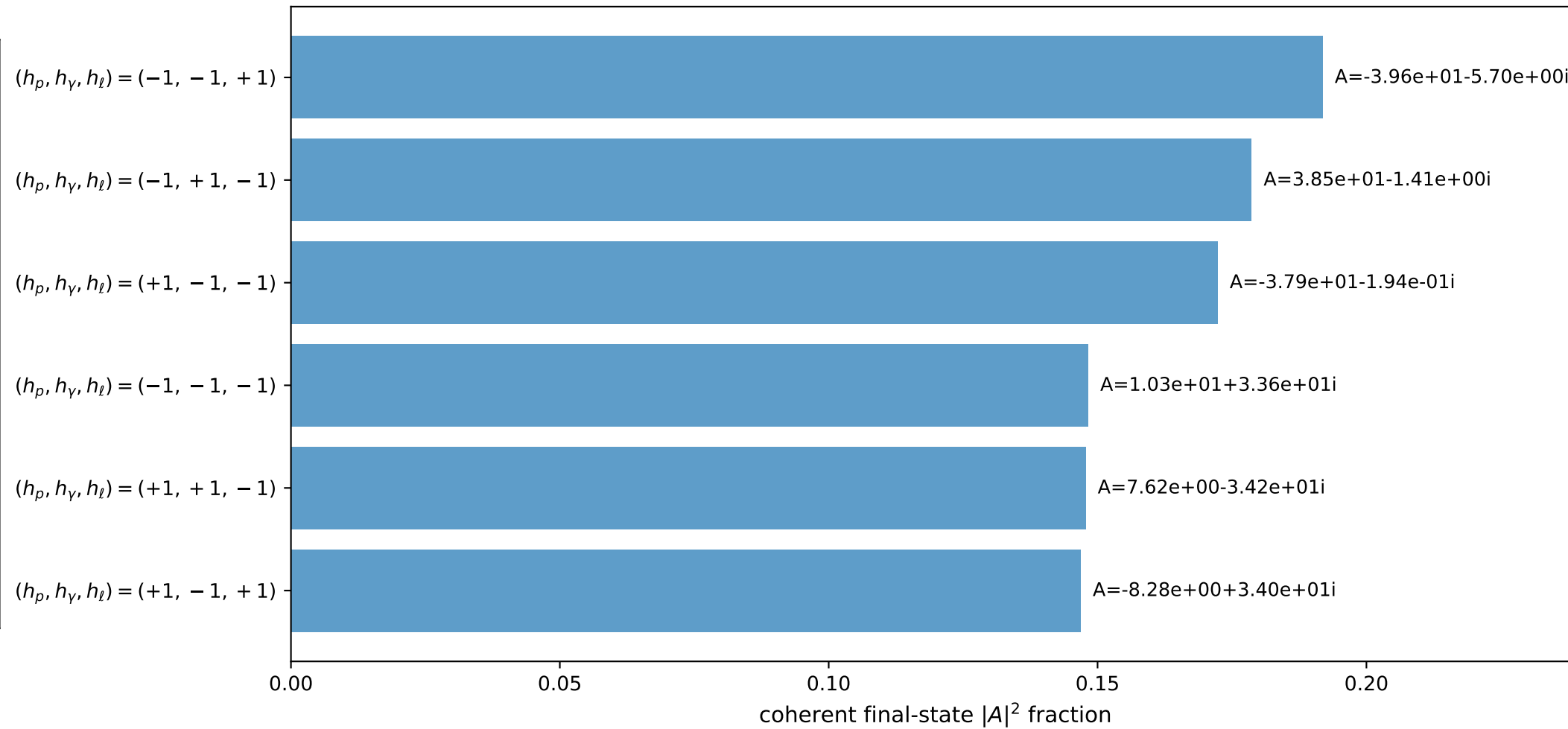
3D momenta



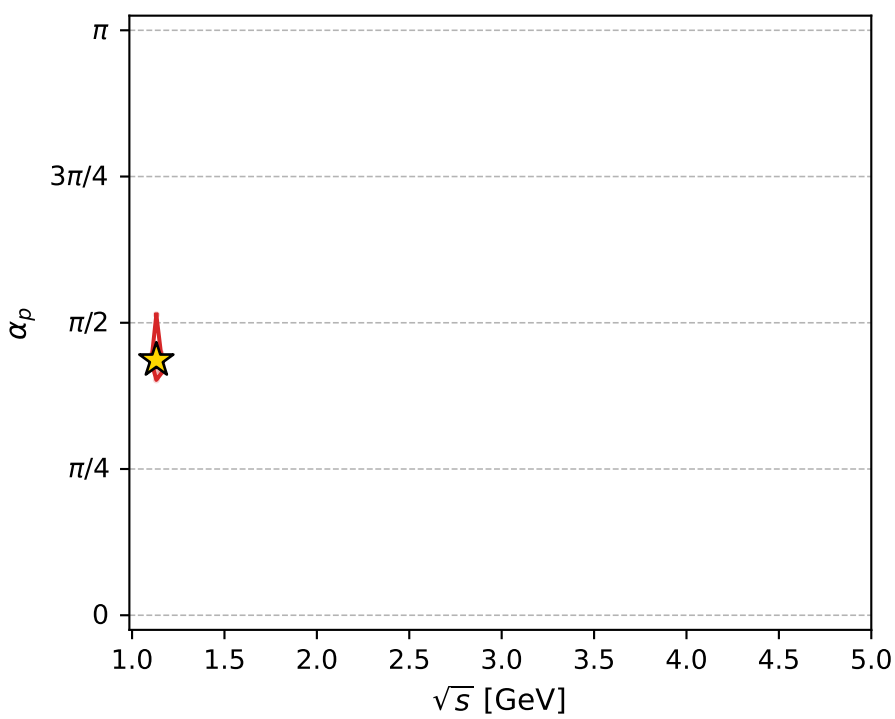
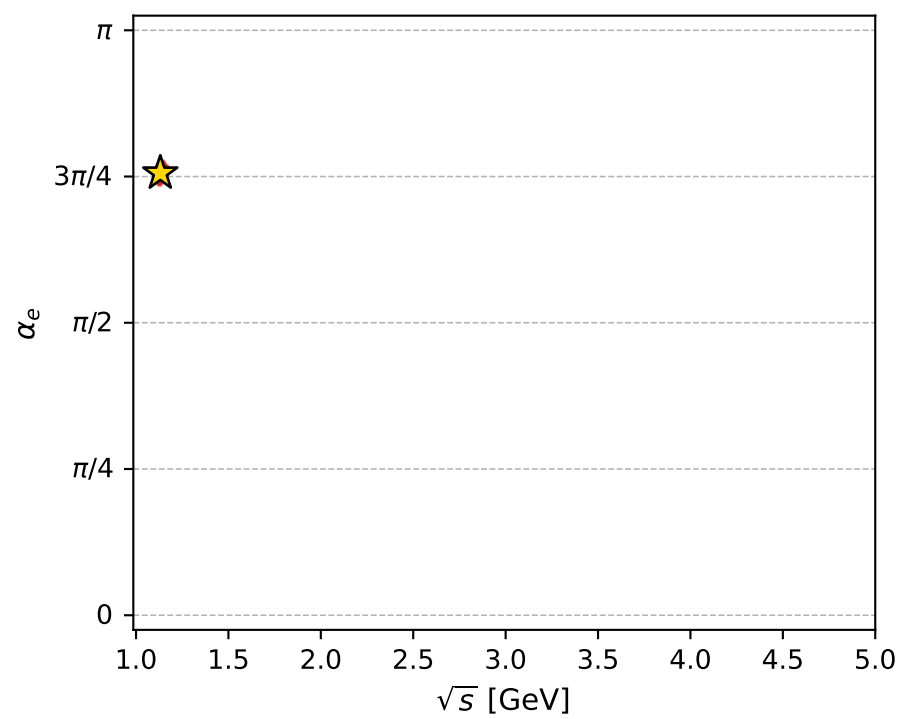
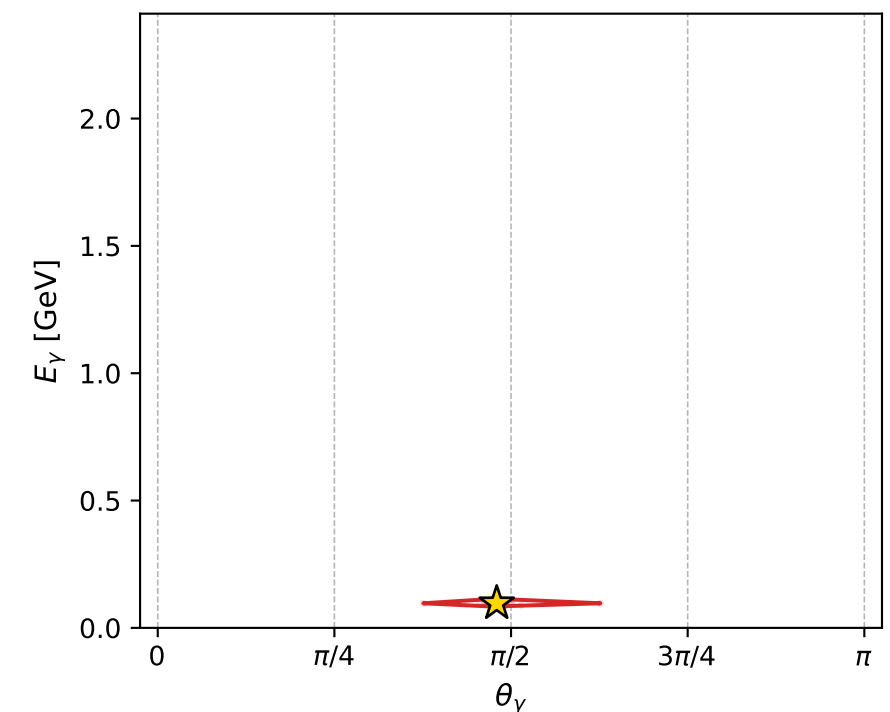
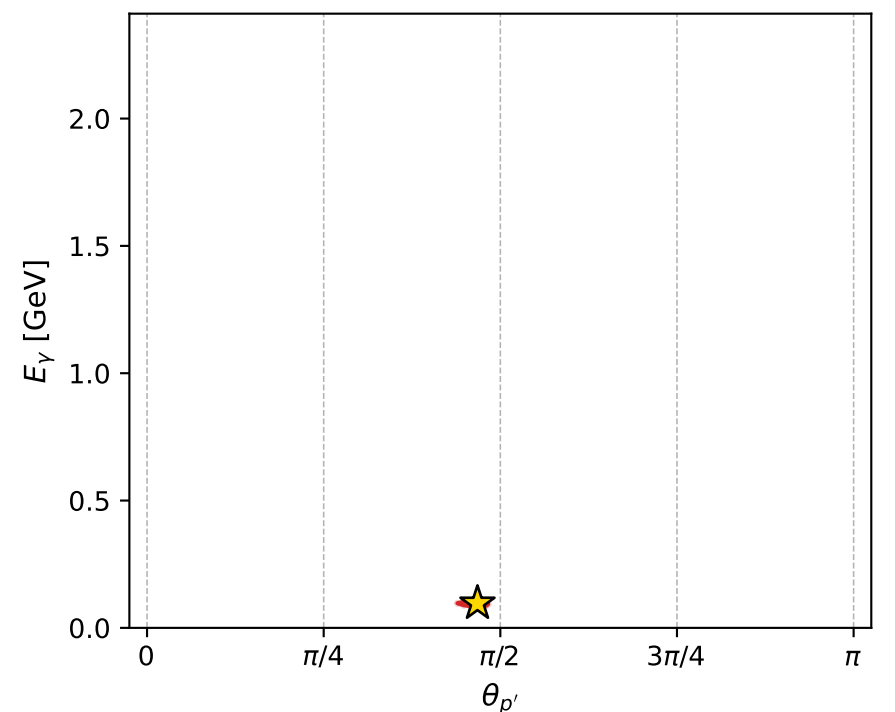
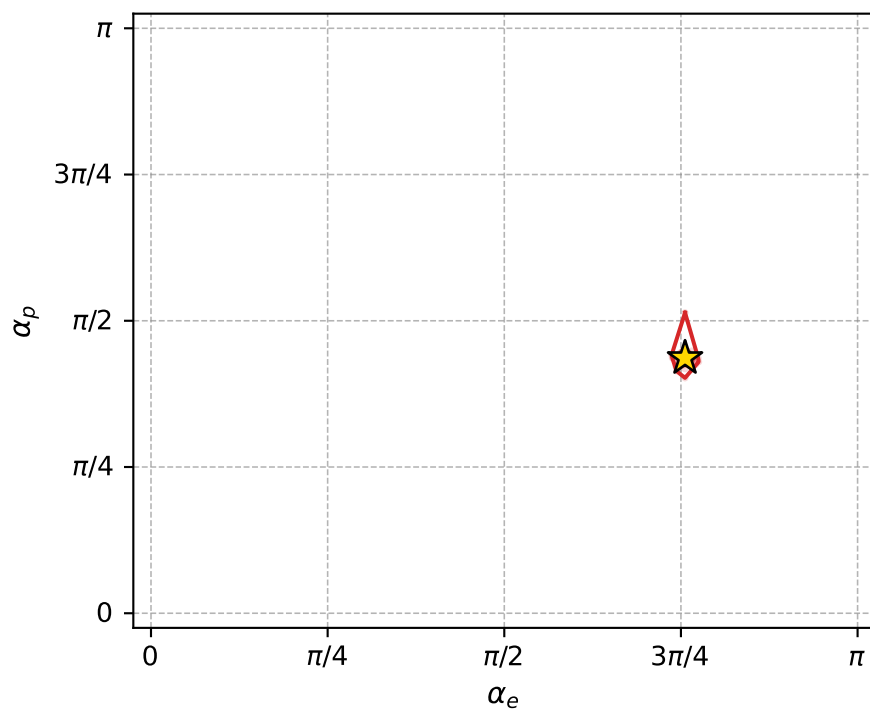
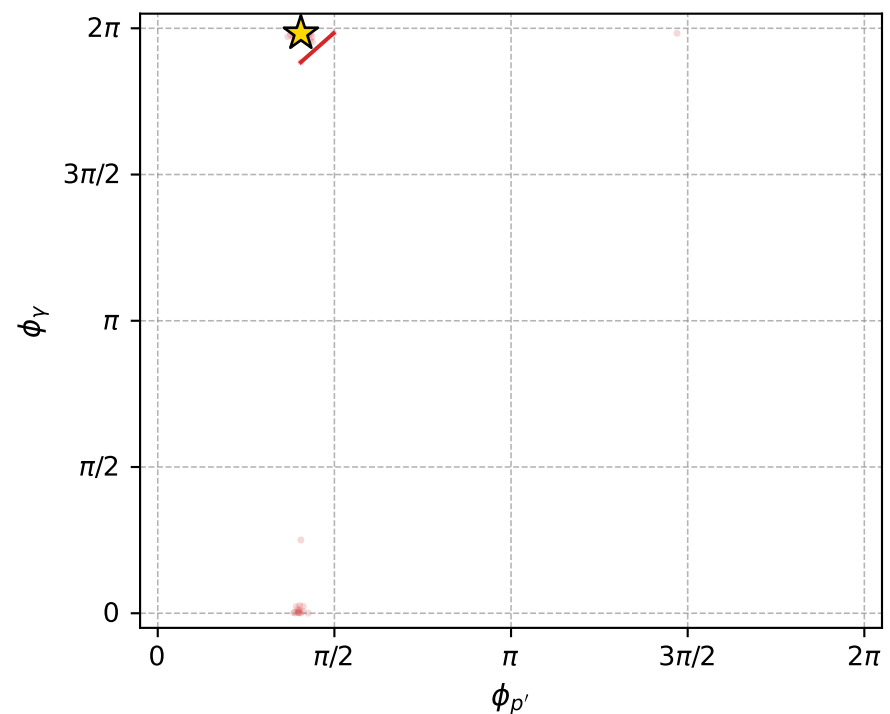
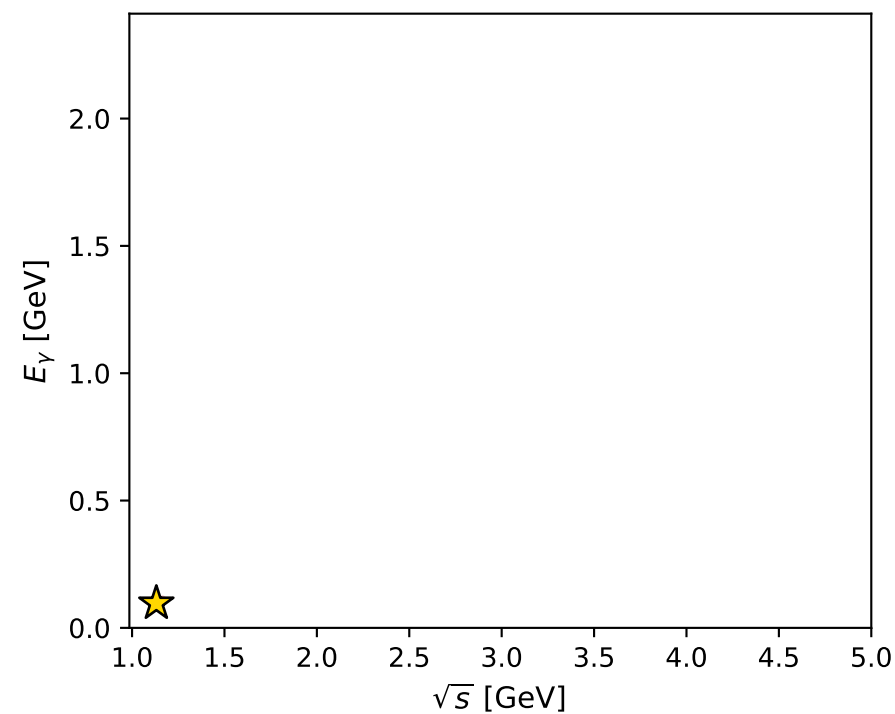
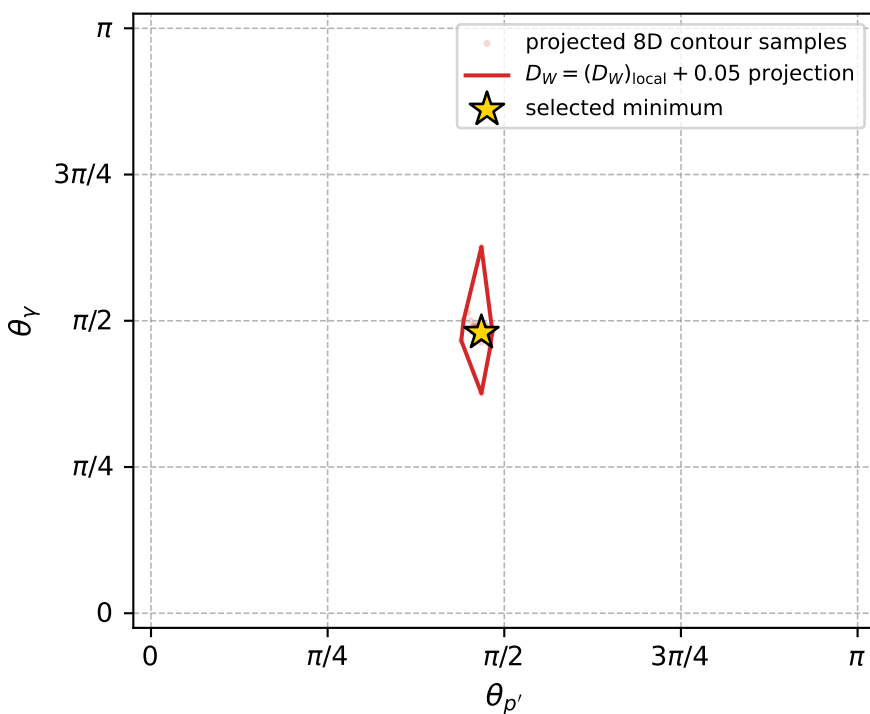
Transverse plane



Leading final-state amplitudes (N=6, retained=98.6%)



electron: polarization cluster P6, local minimum 842; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.011674152$
 $D_W \text{ contour} = 0.061674152$
 above global minimum = 0.01165772
 8D contour samples = 254

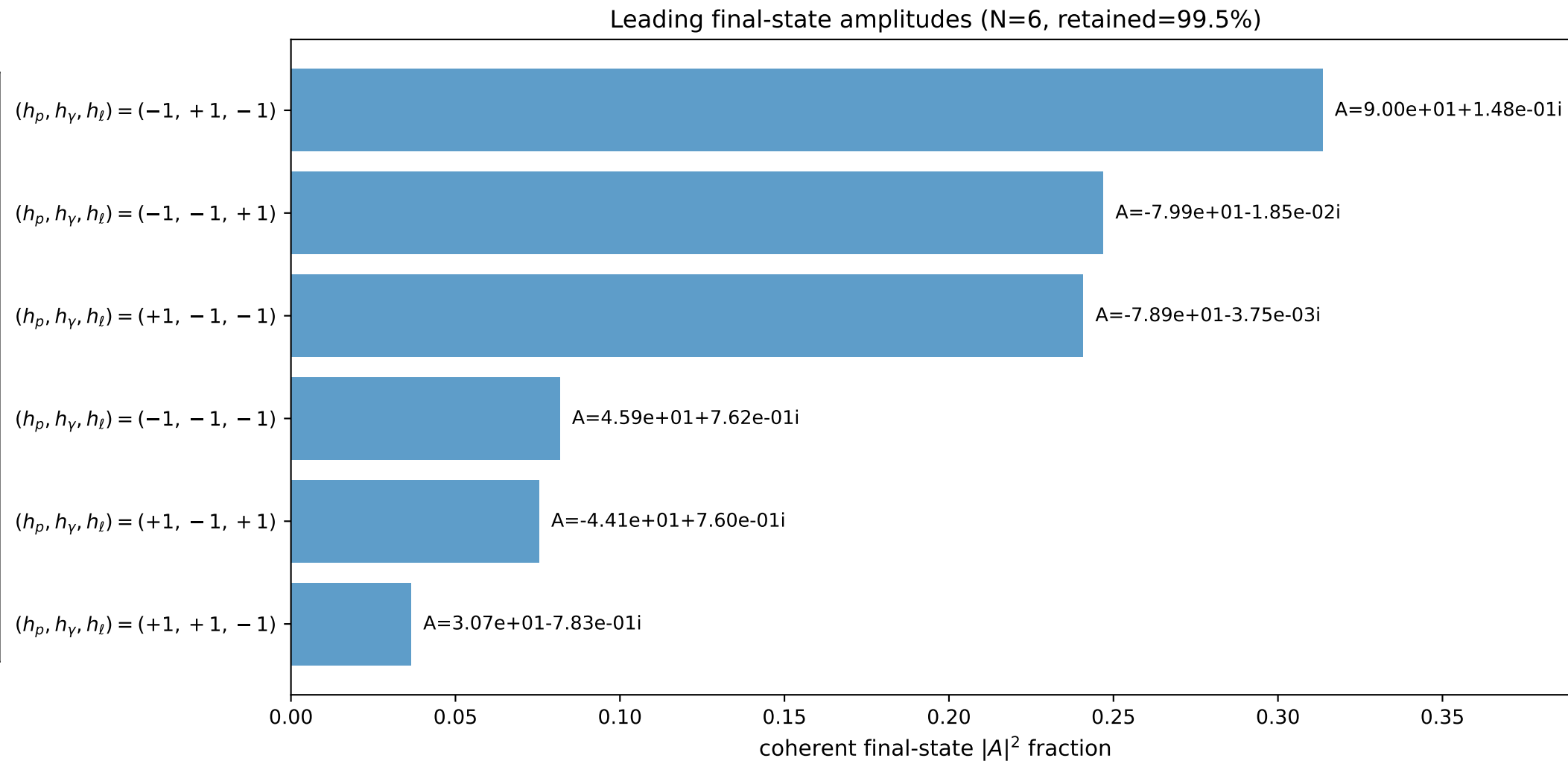
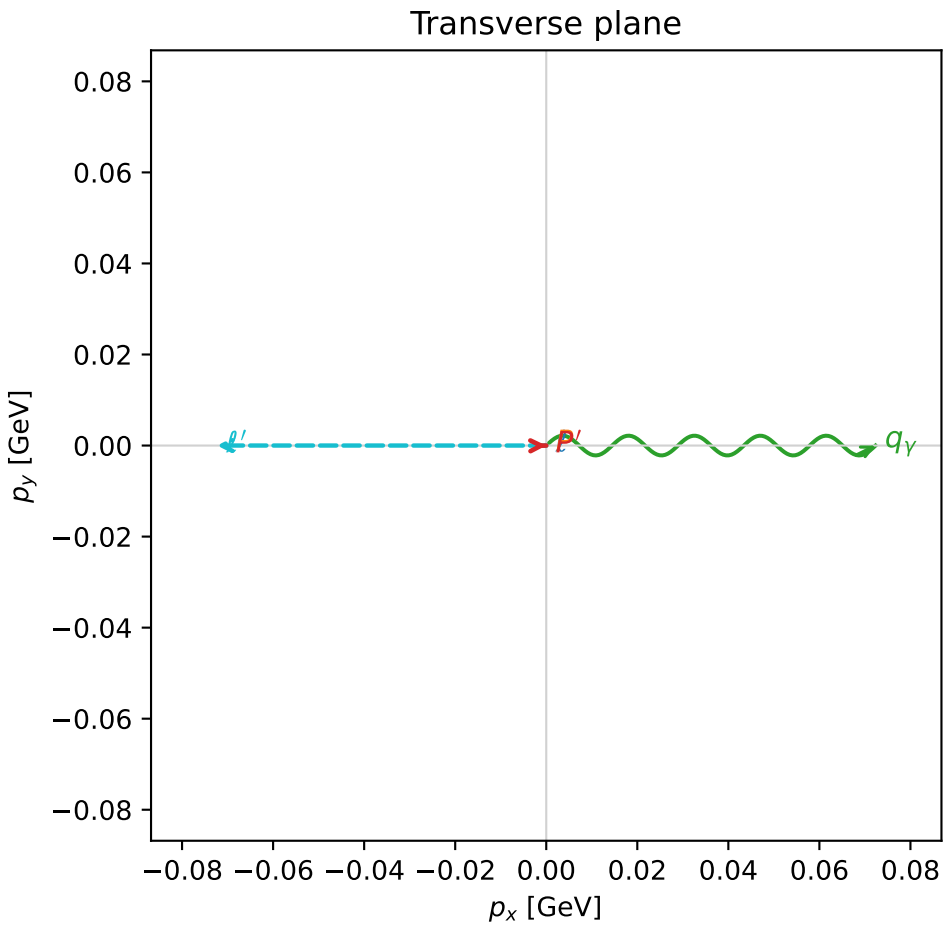
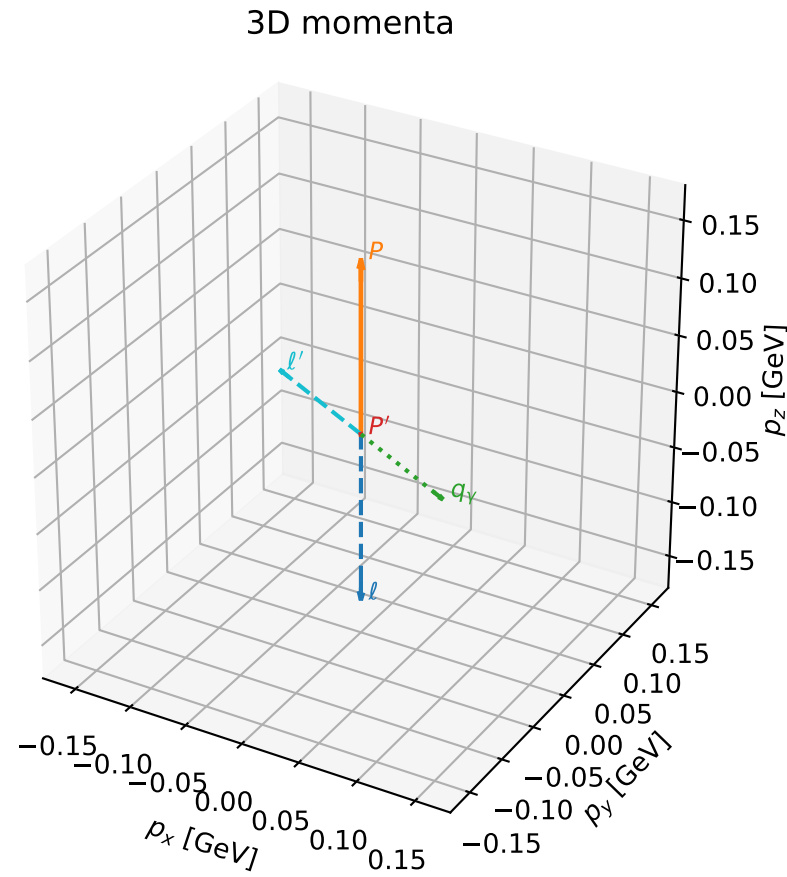
selected phase-space point:
 $\text{sqrt_s} = 1.131345$
 $\text{theta_p_out} = 1.469146$
 $\text{theta_gamma_out} = 1.507189$
 $\text{qOut} = 0.09656402$
 $\text{phi_p_out} = 1.272791$
 $\text{phi_gamma_out} = 6.229472$
 $\text{alpha_e} = 2.374465$
 $\text{alpha_p} = 1.37059$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

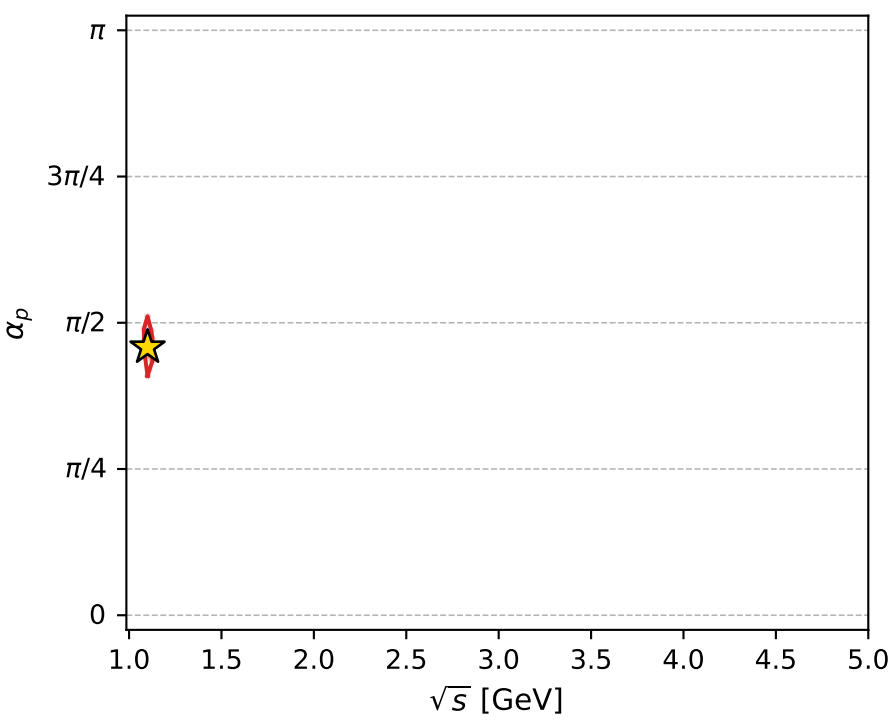
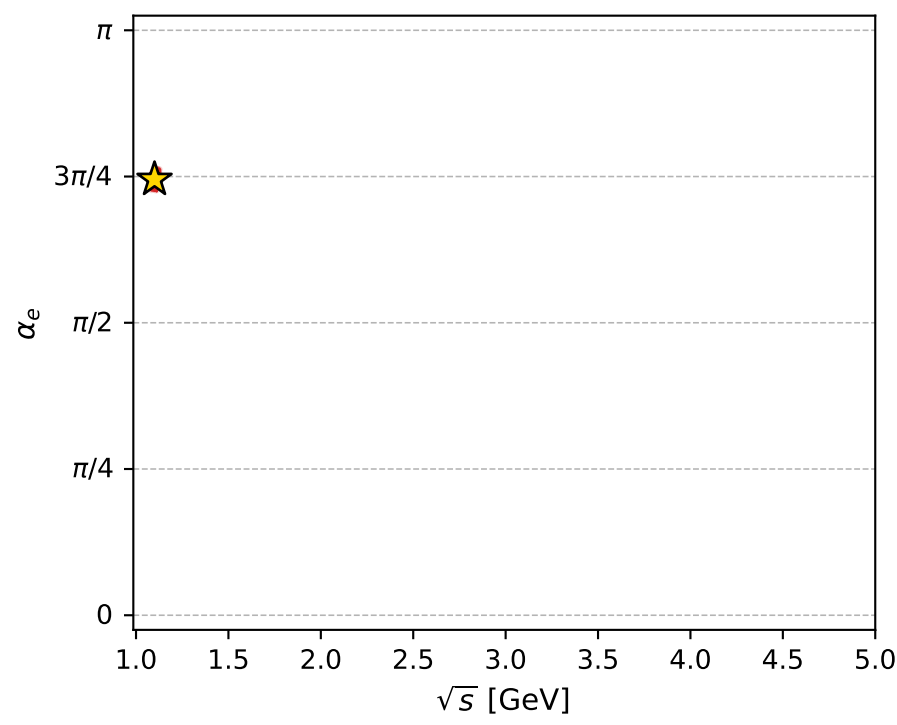
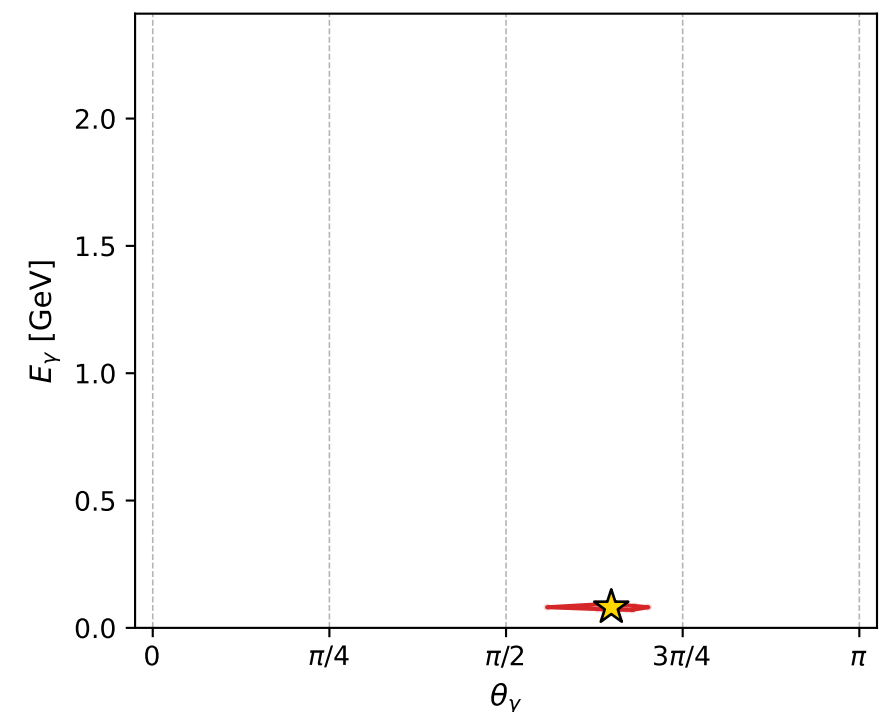
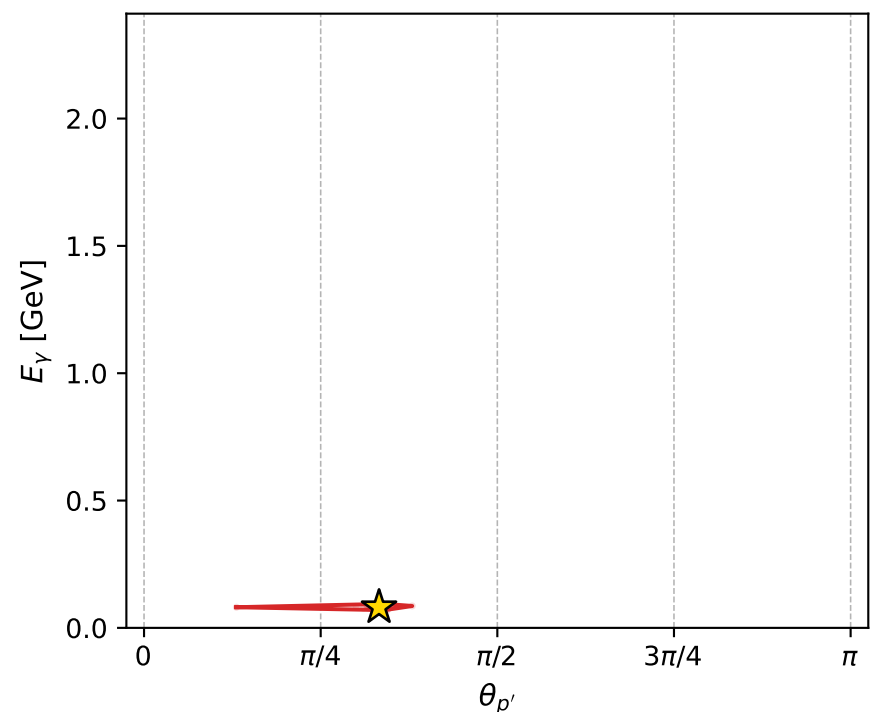
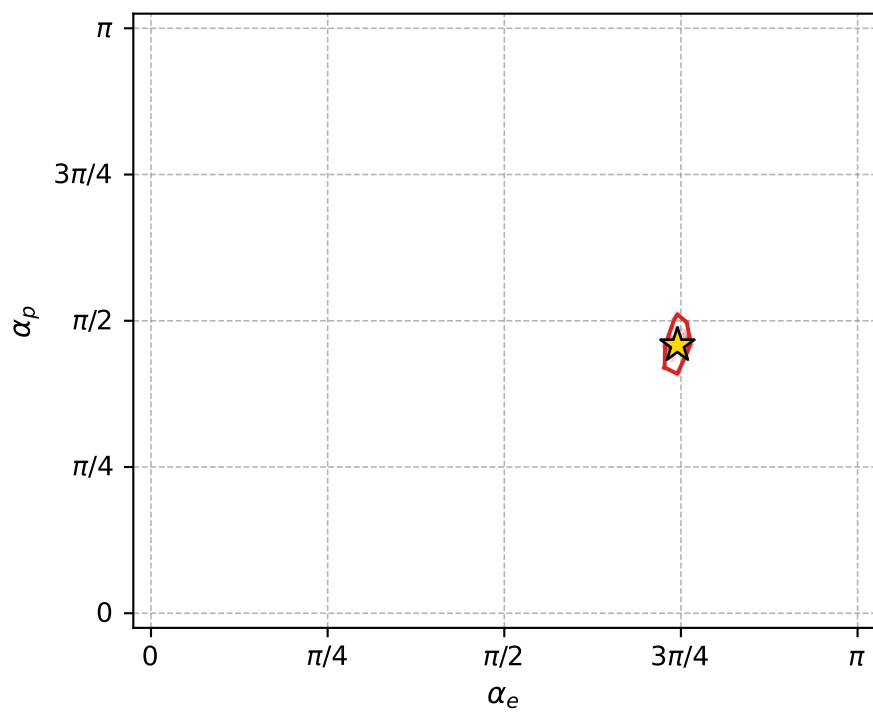
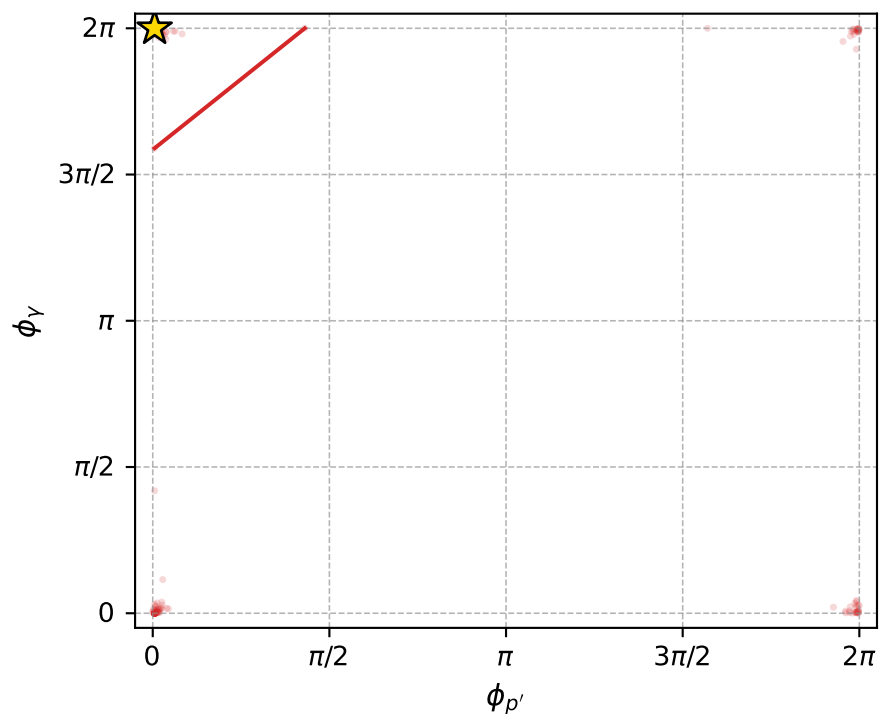
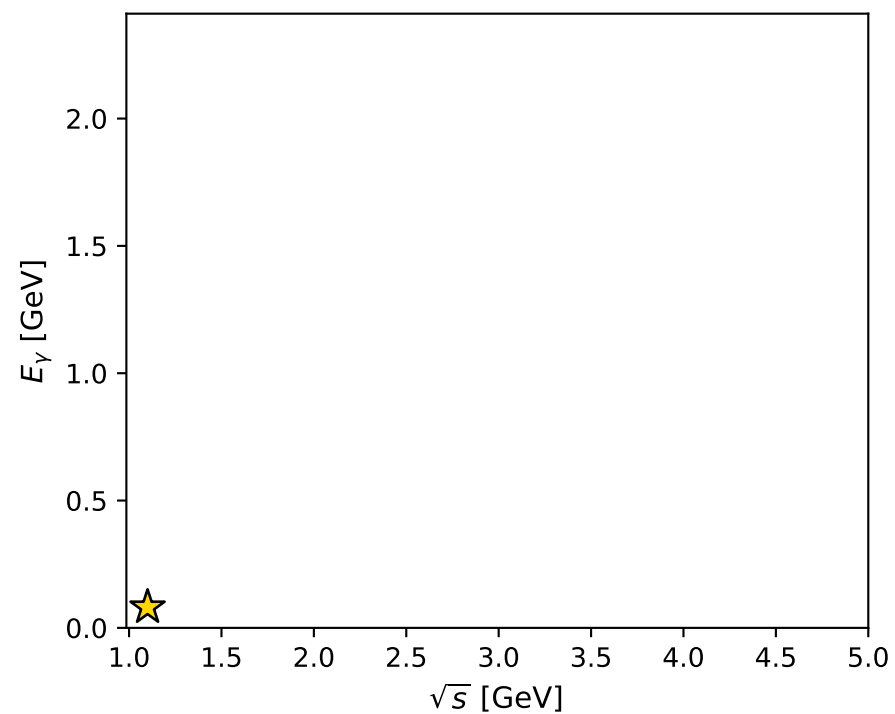
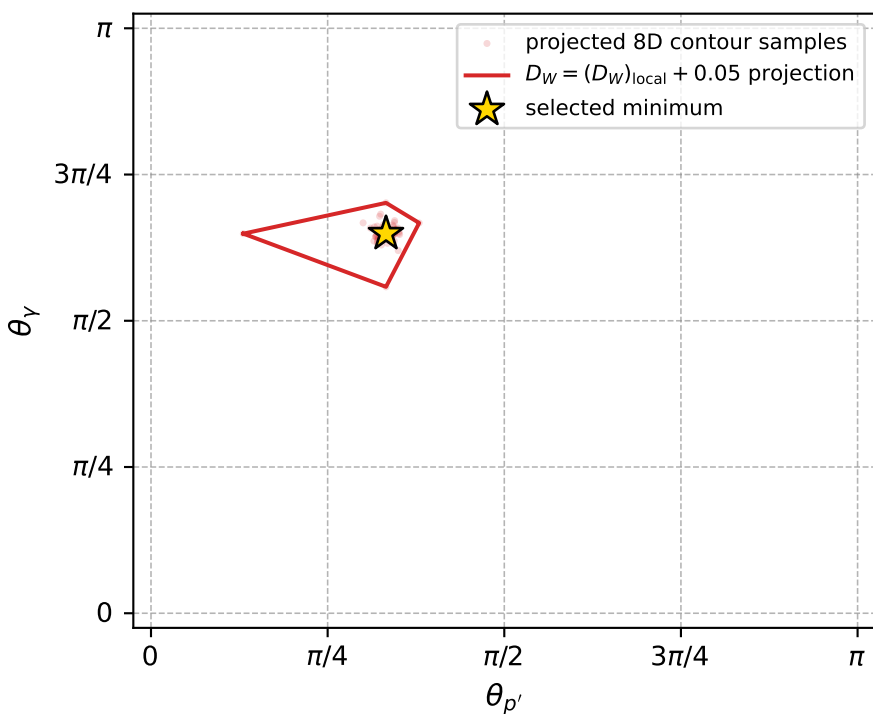
dw_mixing_angles_polarization_06_local_minimum_843 D_W=0.0117295
 region: polarization_cluster_06_local_minimum_843
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3405129$ rad
 incoming lepton: $-0.695932|+> +0.718108|->$
 proton mixing angle: $\alpha_p=1.4394801$ rad
 incoming proton: $0.130939|+> +0.99139|->$

$s=1.21021$, $\sqrt{s}=1.10009$, $|\vec{P}|=0.150151$, $|\vec{P}'|=1.37214e-05$
 $\theta_{P'}=1.04489$, $\theta_V=2.03867$, $\phi_{P'}=0.016731$, $\phi_V=6.28312$
 $E_V=0.0810425$, $\theta_{\ell'}=1.10307$, $\phi_{\ell'}=3.14153$
 $Q^2=0.0353133$, $x_B=0.188491$, $t=-0.0224008$, $W^2=1.03188$, $y=0.567097$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1502$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1502$
 P $E= 0.9499$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1502$
 ℓ' $E= 0.0811$ $p_x= -0.0723$ $p_y= 0.0000$ $p_z= 0.0365$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0000$
 q_V $E= 0.0810$ $p_x= 0.0723$ $p_y= -0.0000$ $p_z= -0.0365$



electron: polarization cluster P6, local minimum 843; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.011729528$
 $D_W \text{ contour} = 0.061729528$
 above global minimum = 0.011713096
 8D contour samples = 255

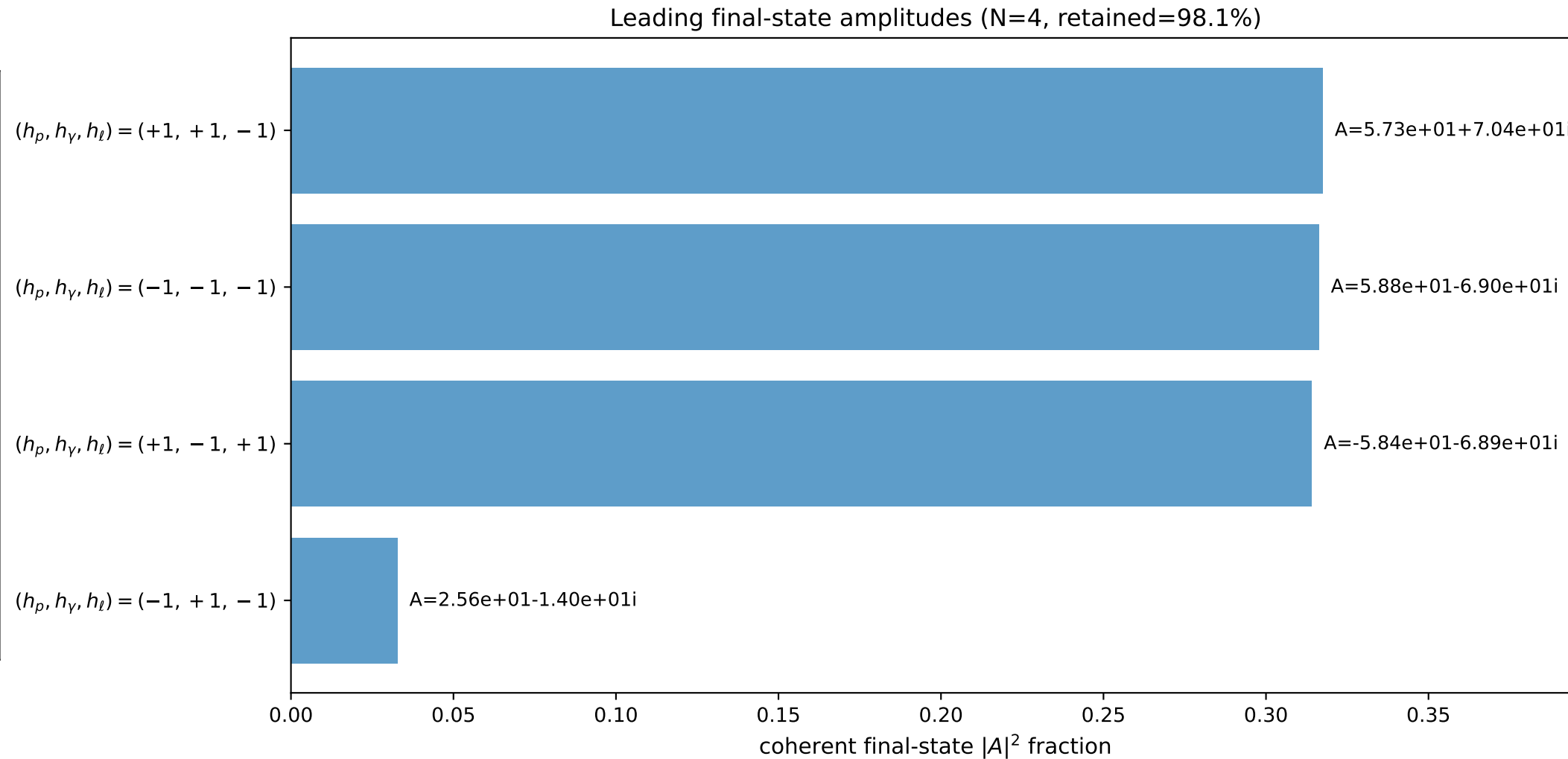
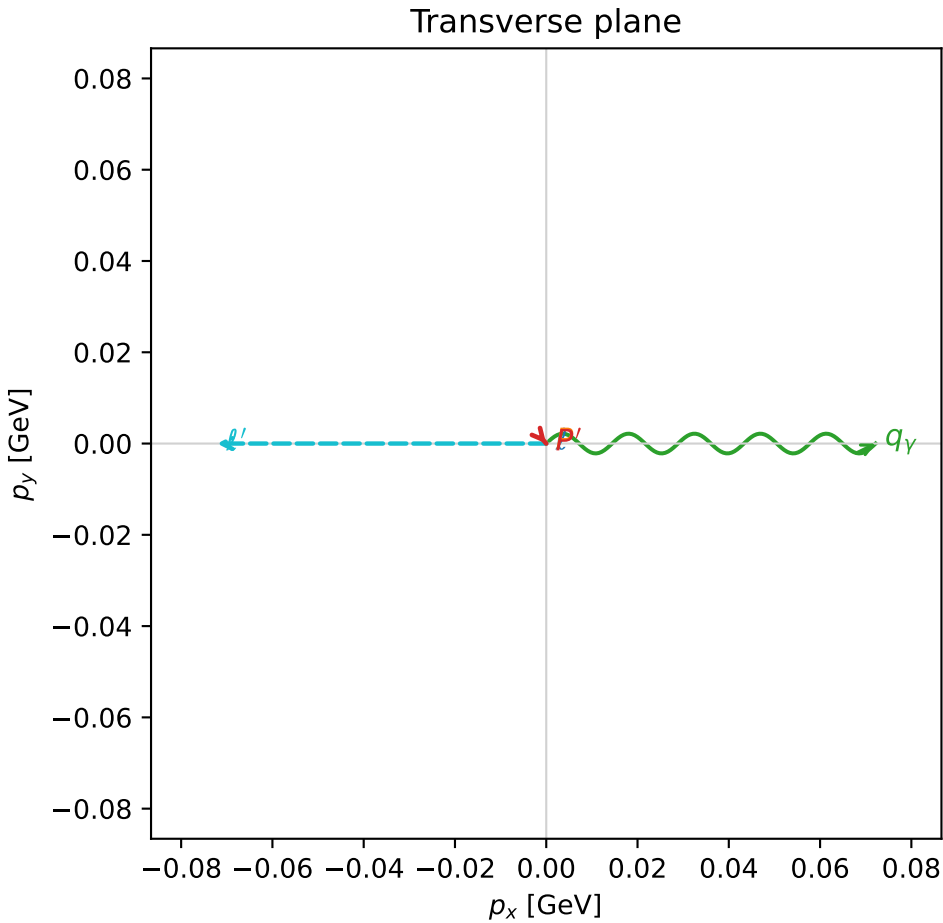
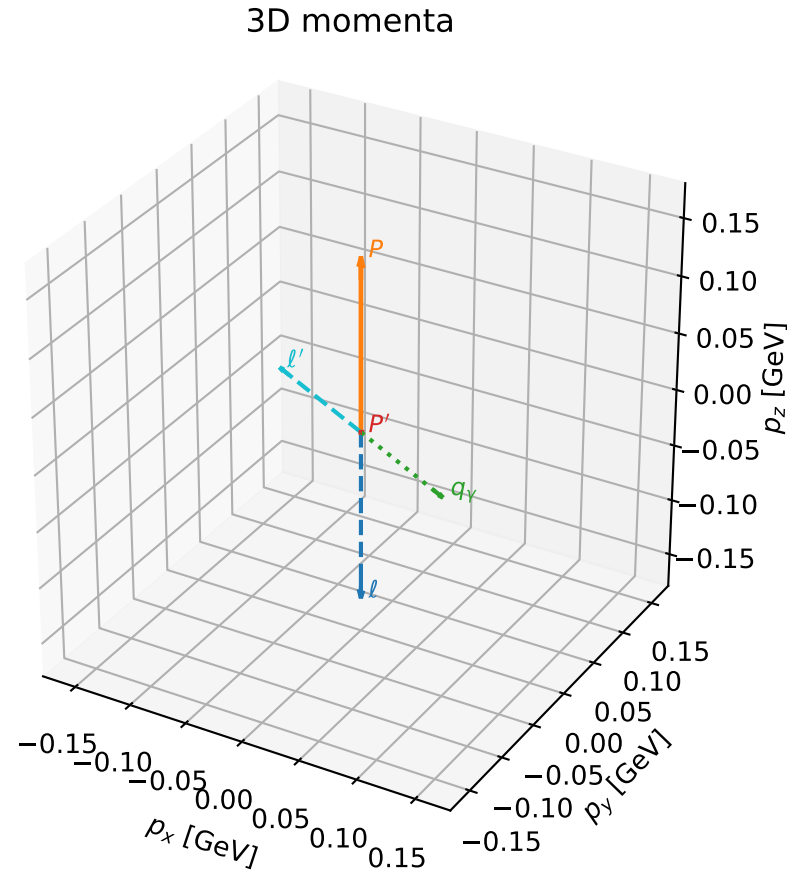
selected phase-space point:
 $\text{sqrt_s} = 1.100094$
 $\text{theta_p_out} = 1.044891$
 $\text{theta_gamma_out} = 2.038665$
 $\text{qOut} = 0.08104247$
 $\text{phi_p_out} = 0.01673104$
 $\text{phi_gamma_out} = 6.283123$
 $\text{alpha_e} = 2.340513$
 $\text{alpha_p} = 1.43948$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

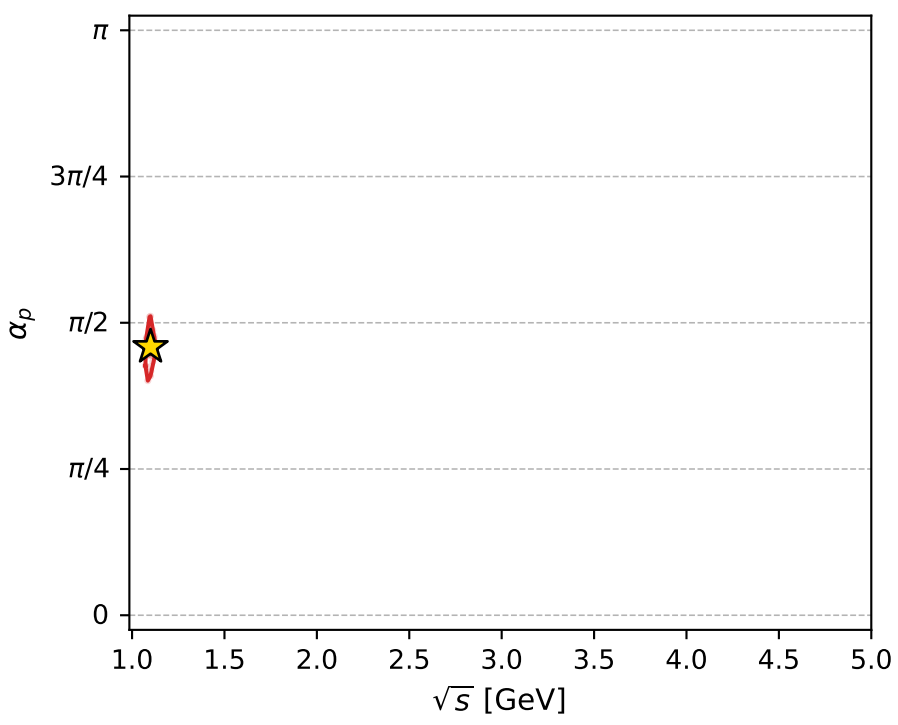
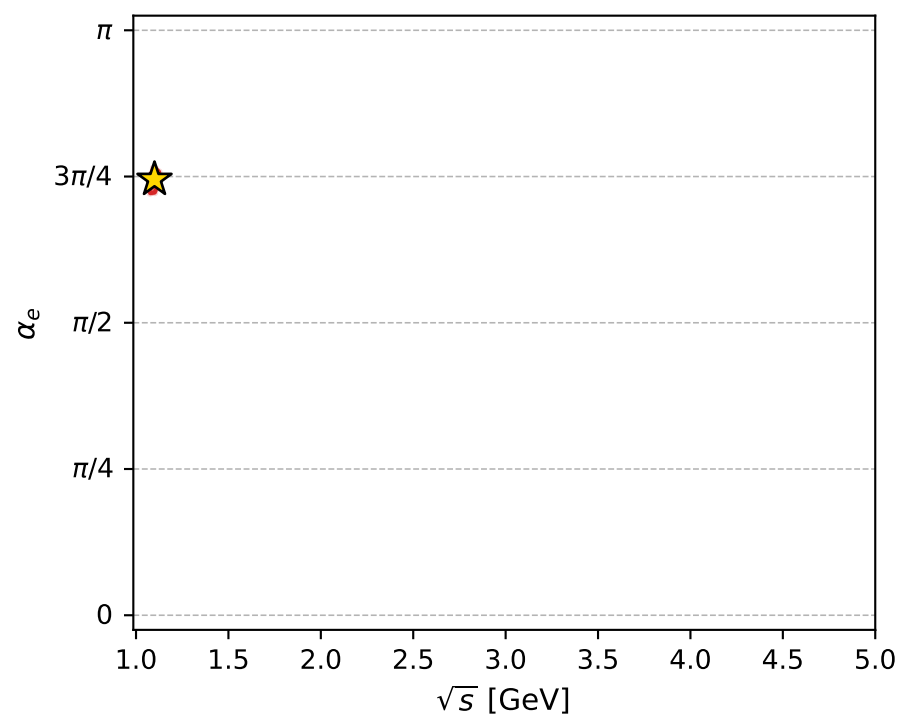
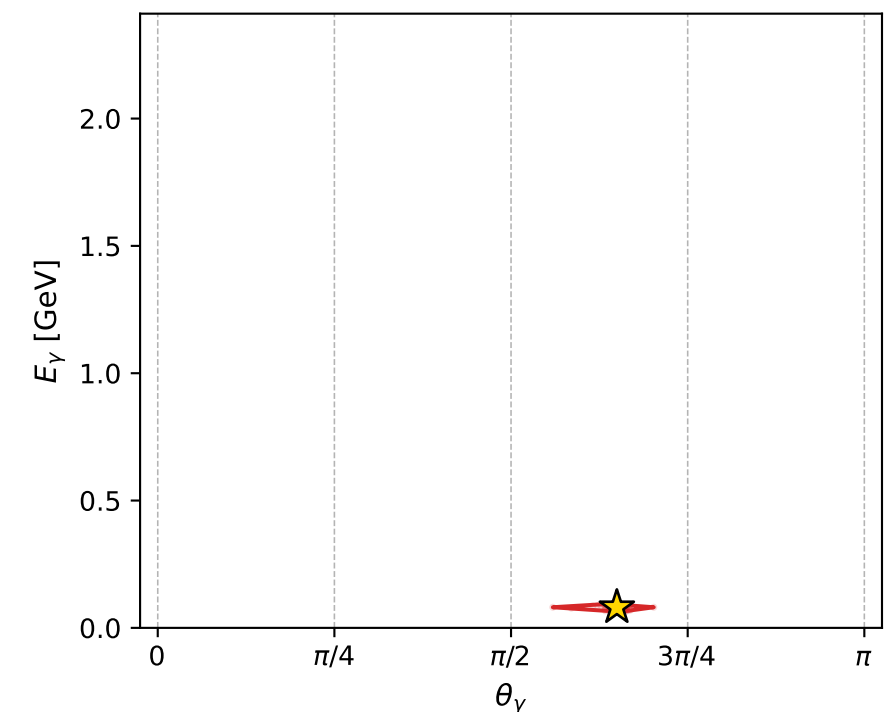
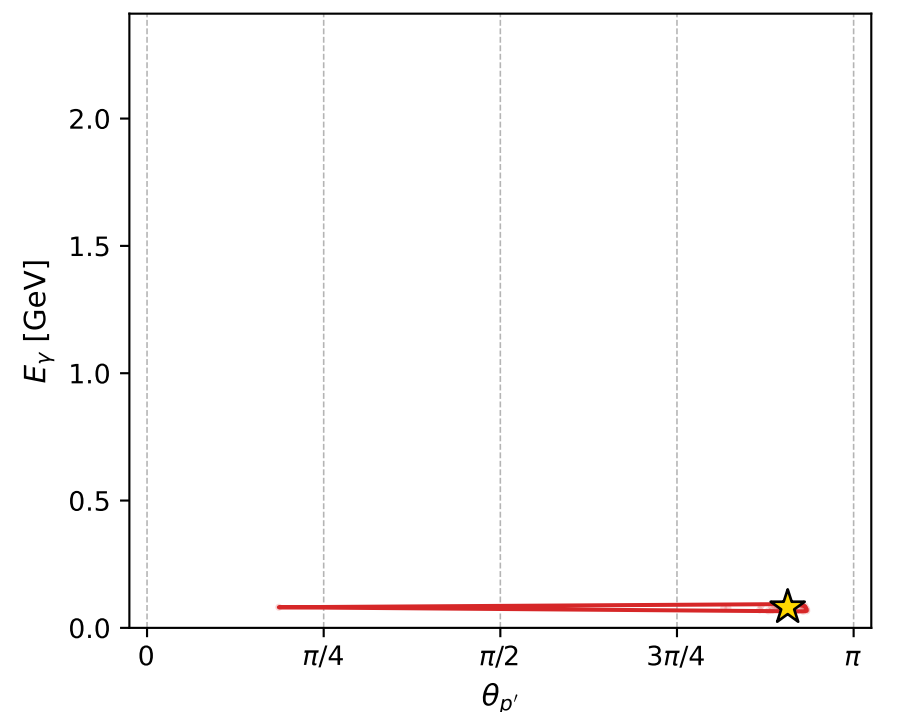
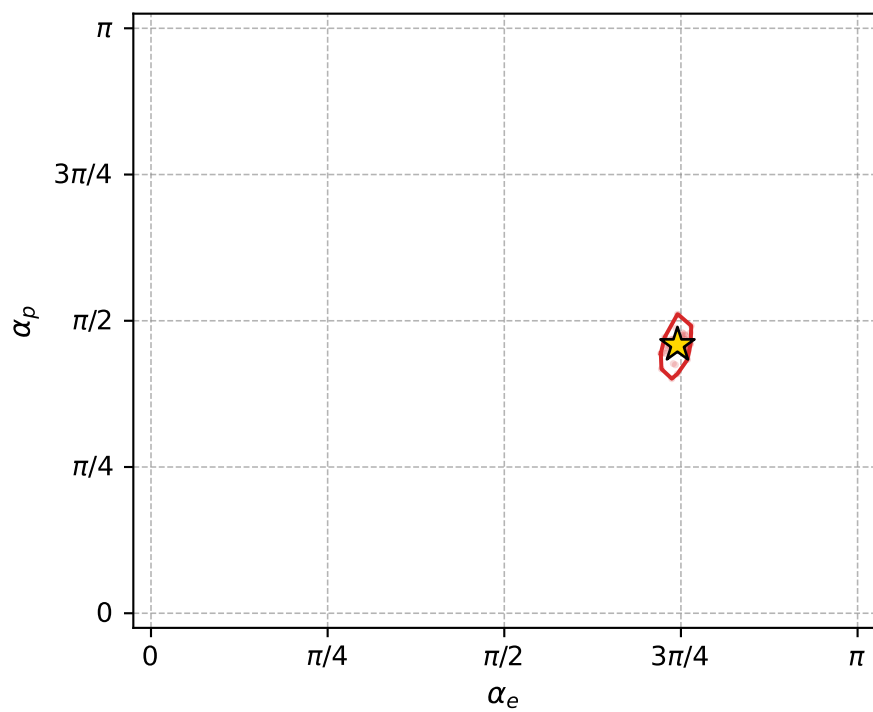
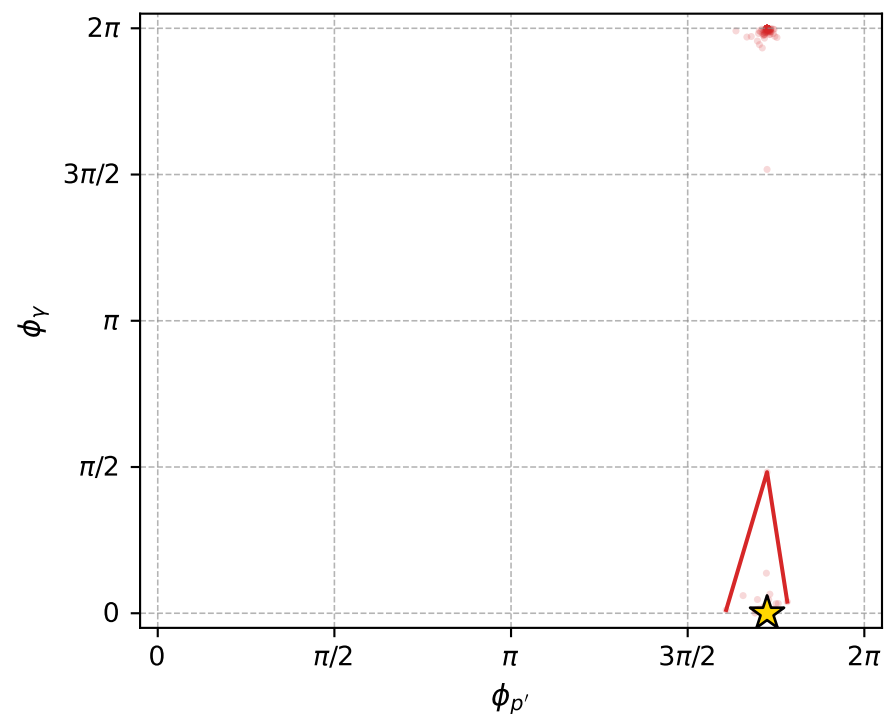
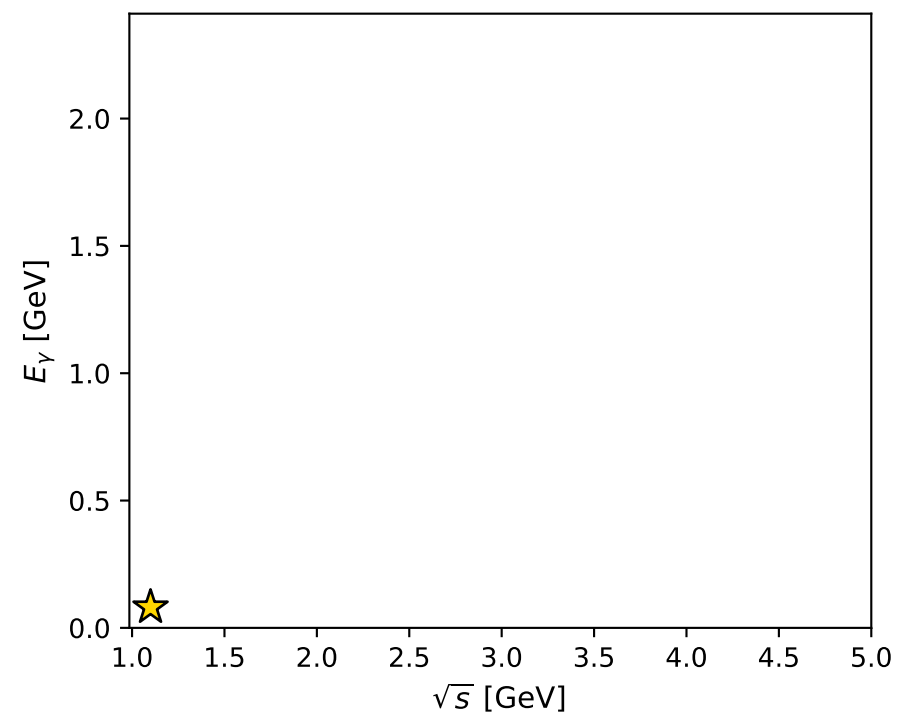
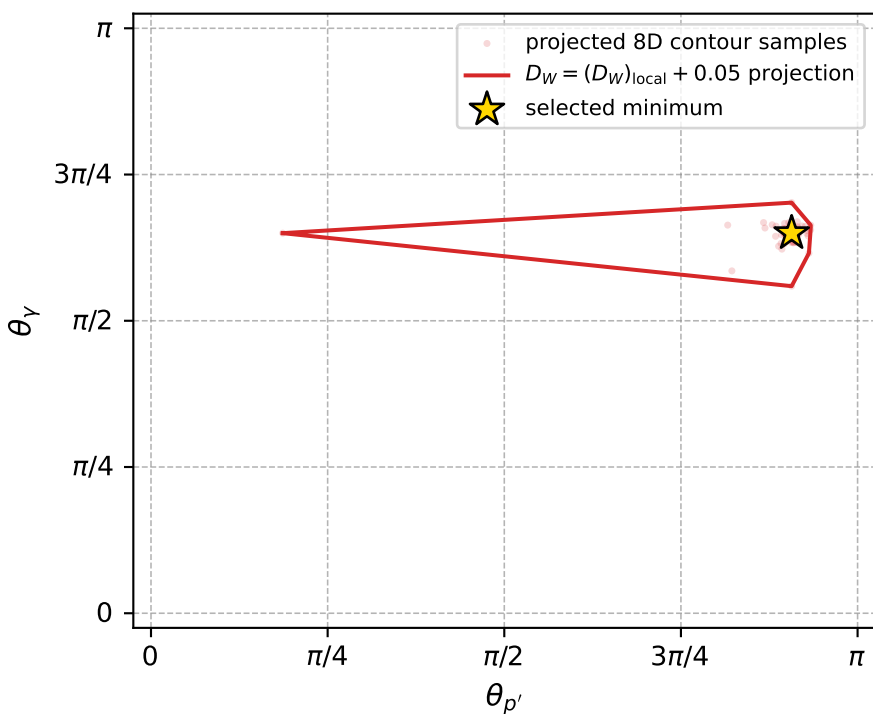
dw_mixing_angles_polarization_06_local_minimum_845 D_W=0.0118195
 region: polarization_cluster_06_local_minimum_845
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3407015$ rad
 incoming lepton: $-0.696067|+> +0.717977|->$
 proton mixing angle: $\alpha_p=1.4413352$ rad
 incoming proton: $0.1291|+> +0.991632|->$

$s=1.20983$, $\sqrt{s}=1.09992$, $|\vec{P}|=0.150005$, $|\vec{P}'|=5.62371e-08$
 $\theta_{P'}=2.84844$, $\theta_V=2.04132$, $\phi_{P'}=5.41822$, $\phi_V=1.61383e-05$
 $E_V=0.0809613$, $\theta_{\ell'}=1.10027$, $\phi_{\ell'}=3.14161$
 $Q^2=0.0353008$, $x_B=0.188589$, $t=-0.0223594$, $W^2=1.03173$, $y=0.567244$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1500$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1500$
 P $E= 0.9499$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1500$
 ℓ' $E= 0.0810$ $p_x= -0.0722$ $p_y= -0.0000$ $p_z= 0.0367$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_V $E= 0.0810$ $p_x= 0.0722$ $p_y= 0.0000$ $p_z= -0.0367$



electron: polarization cluster P6, local minimum 845; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.011819458$
 $D_W \text{ contour} = 0.061819458$
 above global minimum = 0.011803025
 8D contour samples = 253

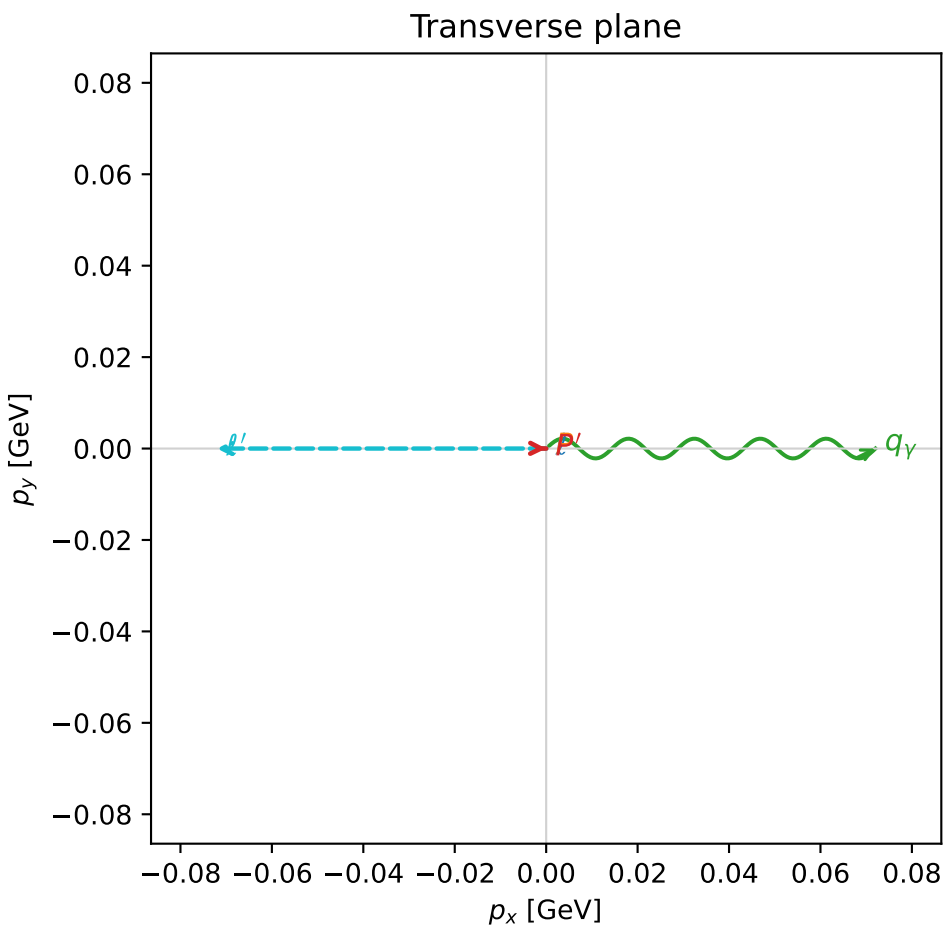
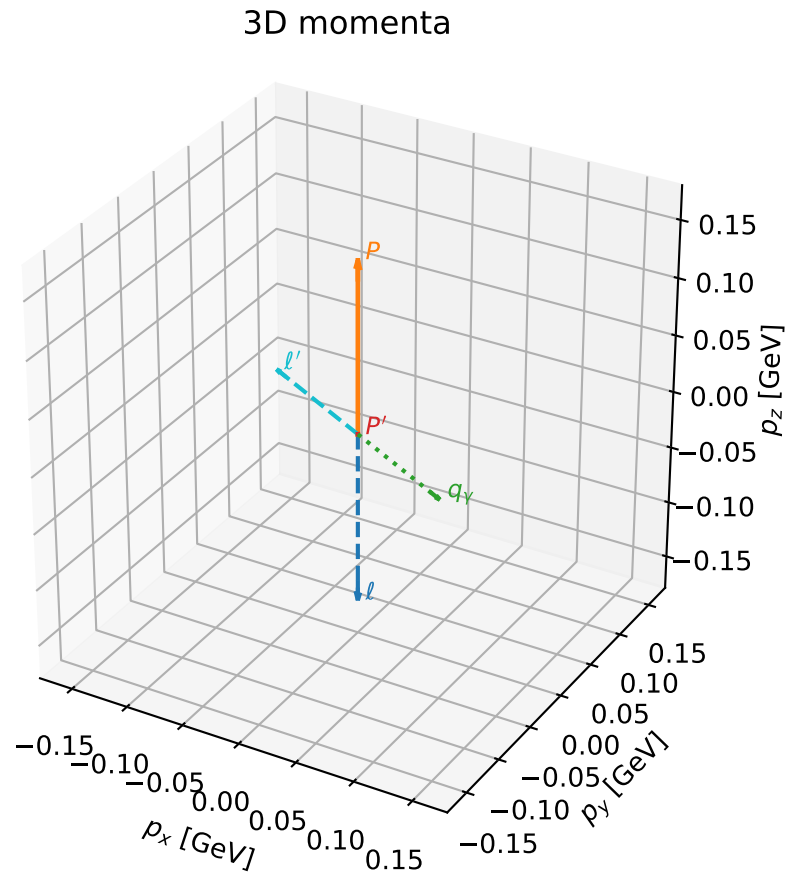
selected phase-space point:
 $\text{sqrt_s} = 1.099924$
 $\text{theta_p_out} = 2.848439$
 $\text{theta_gamma_out} = 2.041317$
 $\text{q0ut} = 0.08096131$
 $\text{phi_p_out} = 5.418215$
 $\text{phi_gamma_out} = 1.613826\text{e-}05$
 $\text{alpha_e} = 2.340702$
 $\text{alpha_p} = 1.441335$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_847 D_W=0.0118716
 region: polarization_cluster_06_local_minimum_847
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3406718$ rad
 incoming lepton: $-0.696046|+\rangle +0.717997|-\rangle$
 proton mixing angle: $\alpha_p=1.4421737$ rad
 incoming proton: $0.128268|+\rangle +0.99174|-\rangle$

$s=1.20934$, $\sqrt{s}=1.0997$, $|\vec{P}|=0.149809$, $|\vec{P}'|=8.16231e-06$
 $\theta_{P'}=3.14003$, $\theta_V=2.04218$, $\phi_{P'}=6.26908$, $\phi_V=8.62237e-06$
 $E_V=0.0808463$, $\theta_{\ell'}=1.09932$, $\phi_{\ell'}=3.1416$
 $Q^2=0.035227$, $x_B=0.188486$, $t=-0.022304$, $W^2=1.03151$, $y=0.56722$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1498$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1498$
 P $E= 0.9499$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1498$
 ℓ' $E= 0.0809$ $p_x= -0.0720$ $p_y= -0.0000$ $p_z= 0.0367$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_V $E= 0.0808$ $p_x= 0.0720$ $p_y= 0.0000$ $p_z= -0.0367$

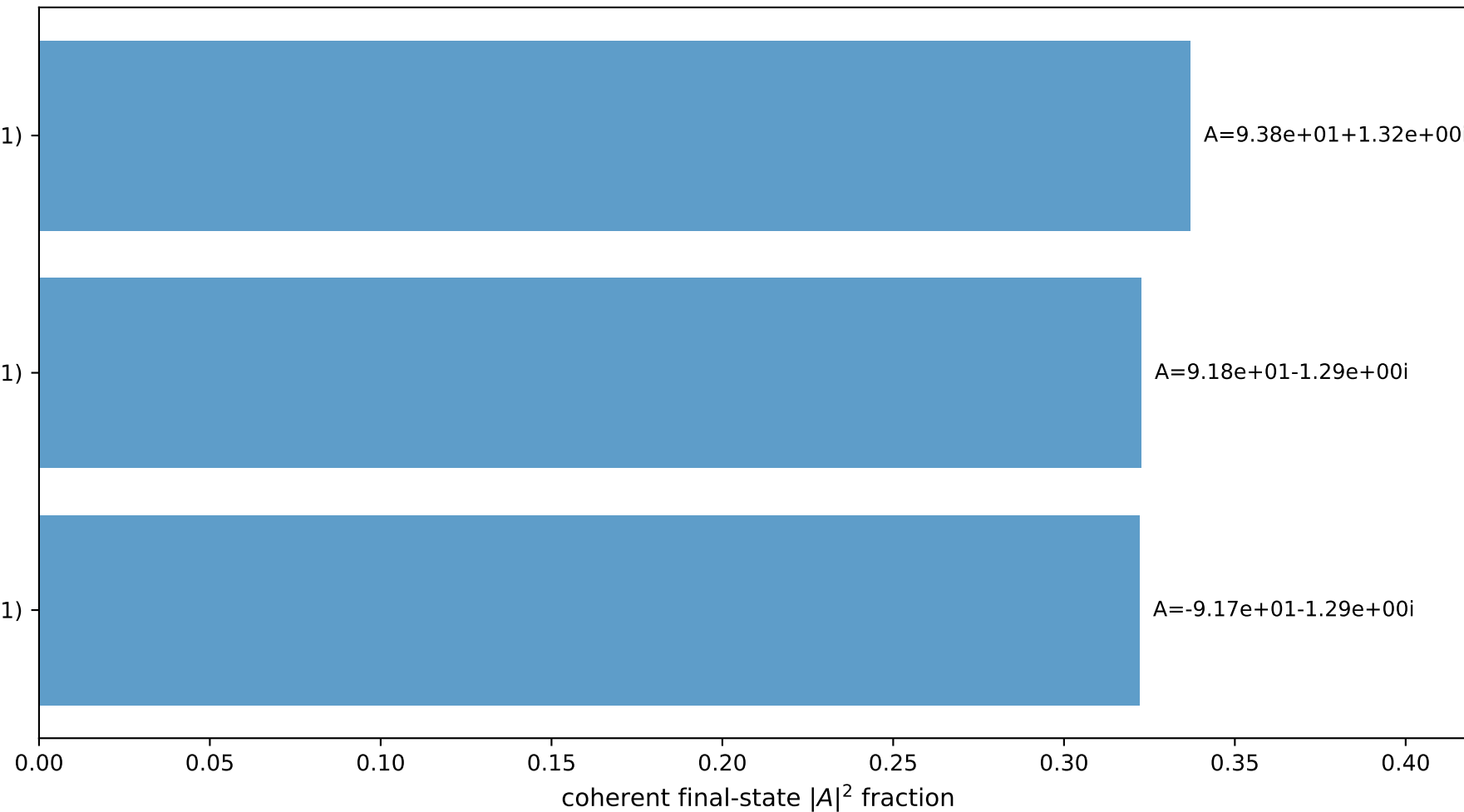


$(h_p, h_V, h_\ell) = (+1, +1, -1)$

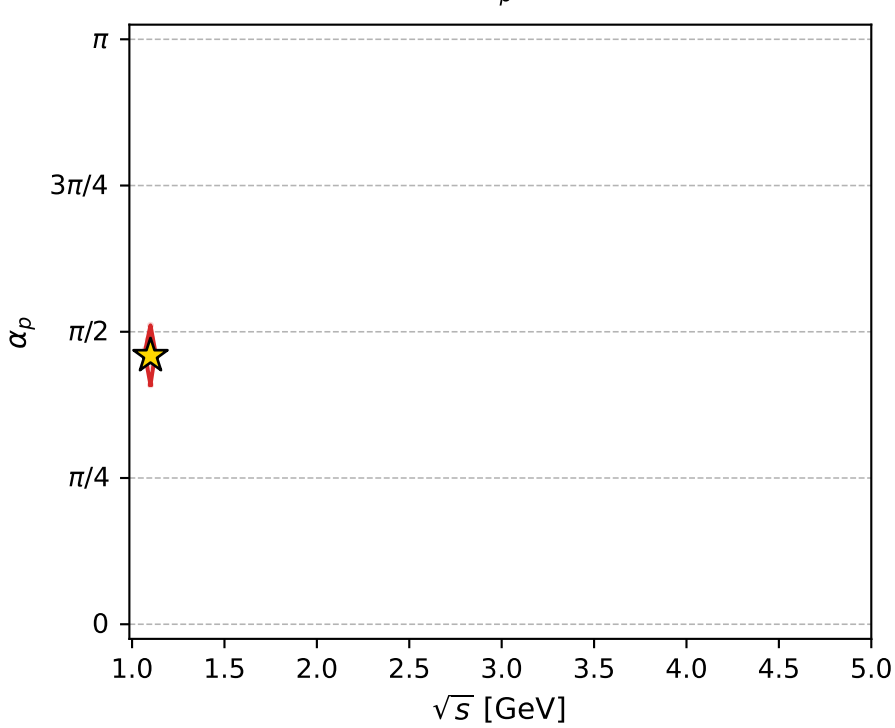
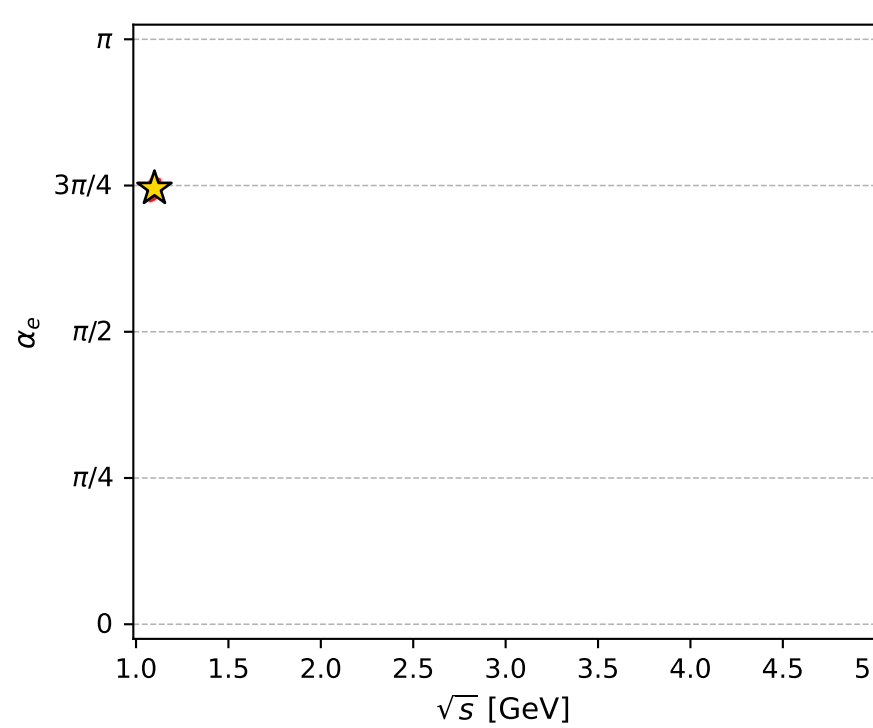
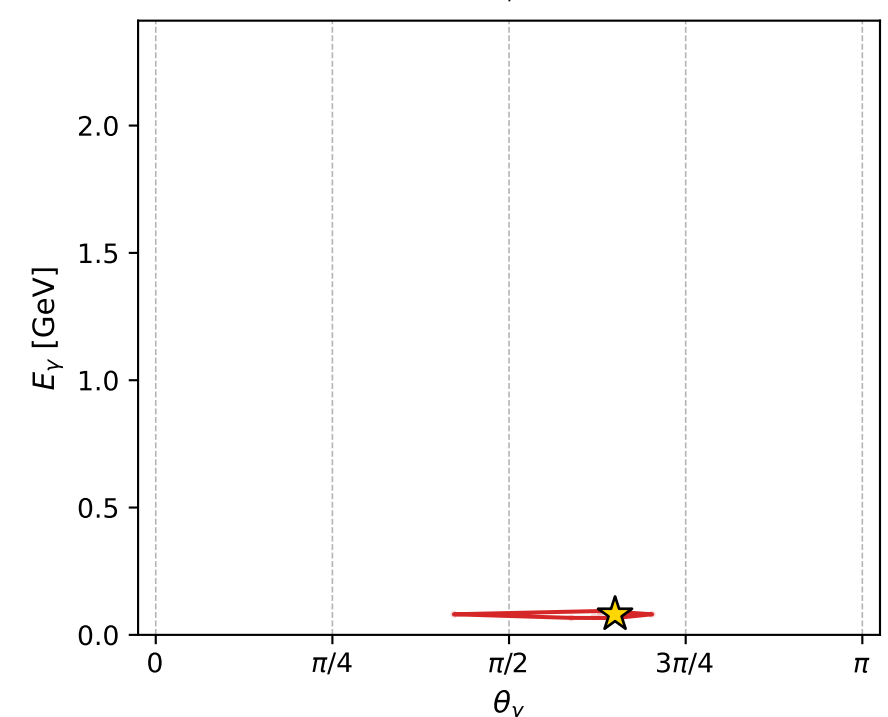
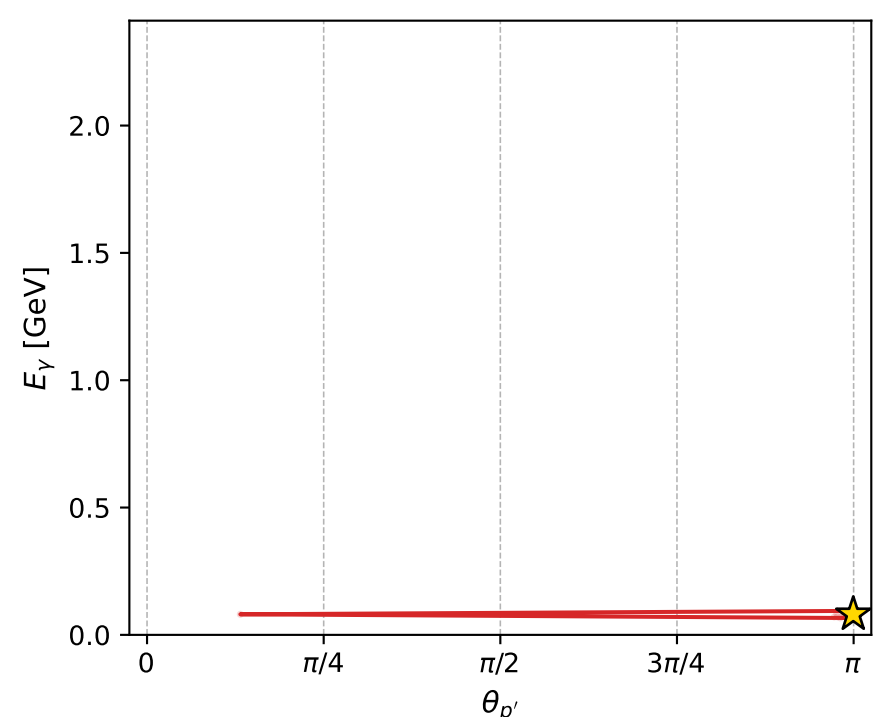
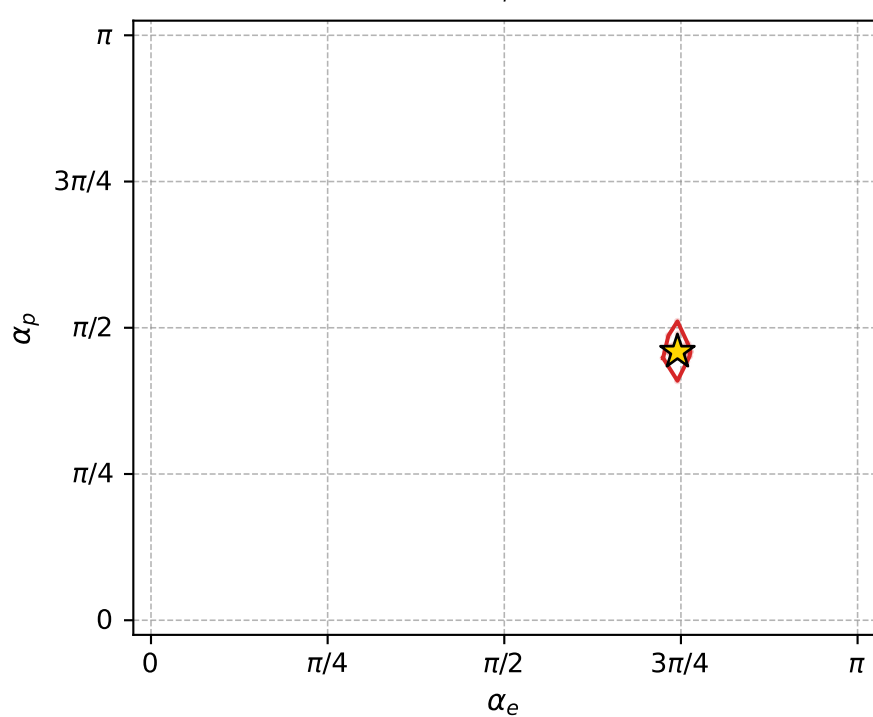
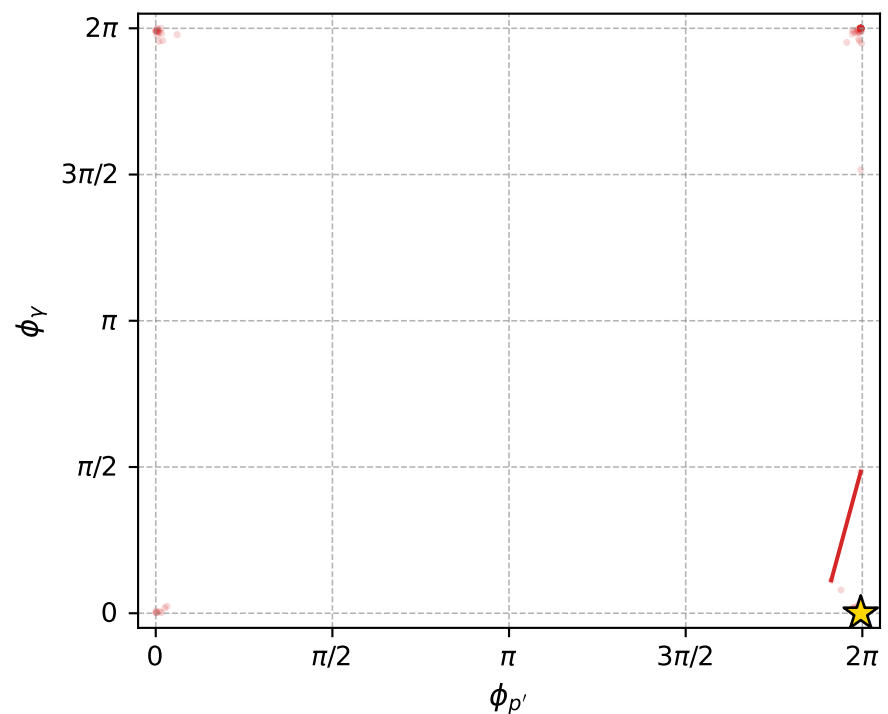
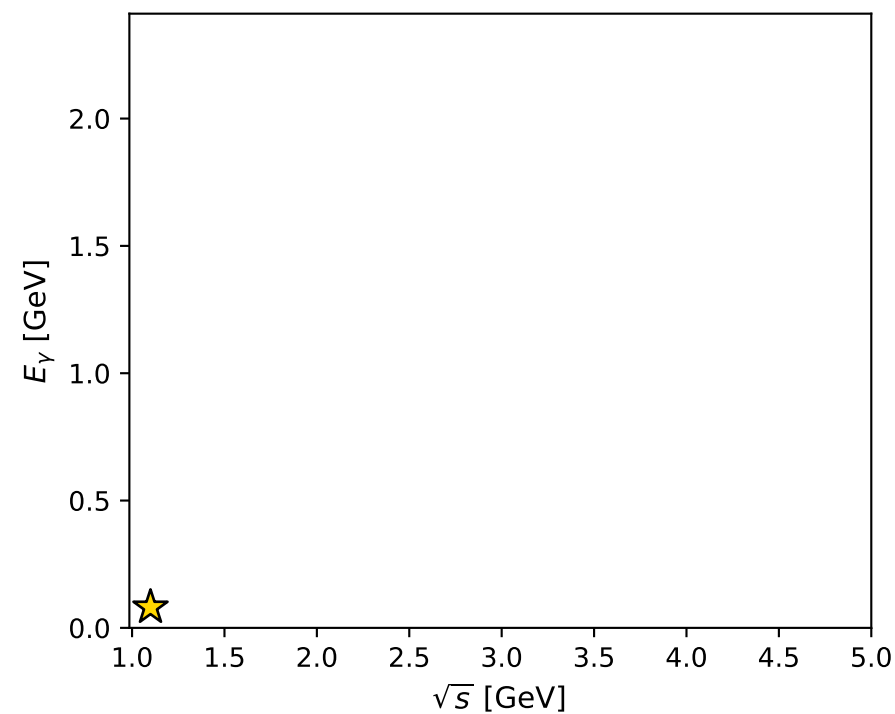
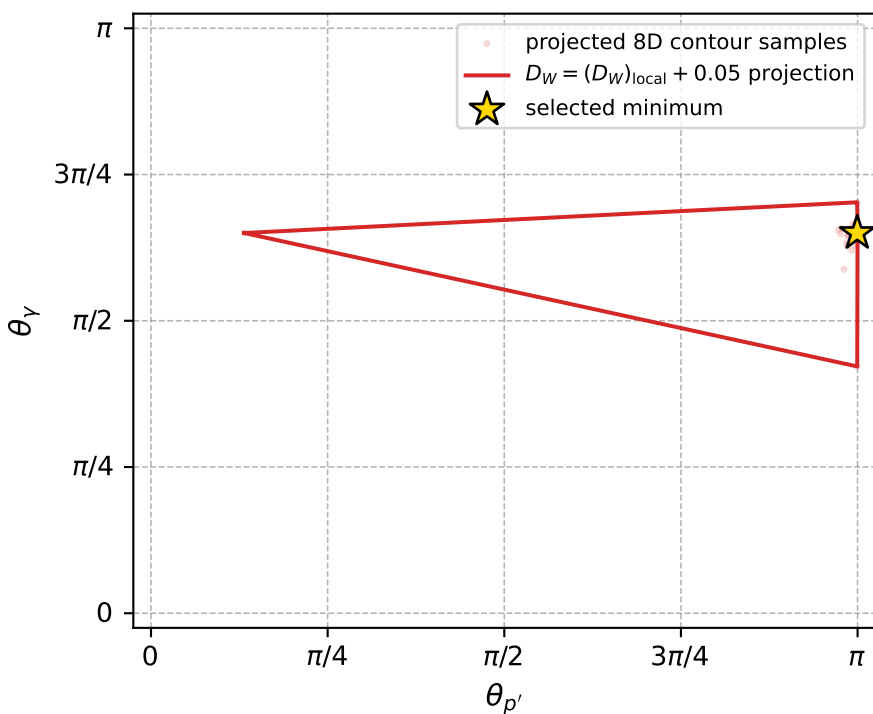
$(h_p, h_V, h_\ell) = (-1, -1, -1)$

$(h_p, h_V, h_\ell) = (+1, -1, +1)$

Leading final-state amplitudes (N=3, retained=98.2%)



electron: polarization cluster P6, local minimum 847; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.011871564$
 $D_W \text{ contour} = 0.061871564$
 above global minimum = 0.011855132
 8D contour samples = 200

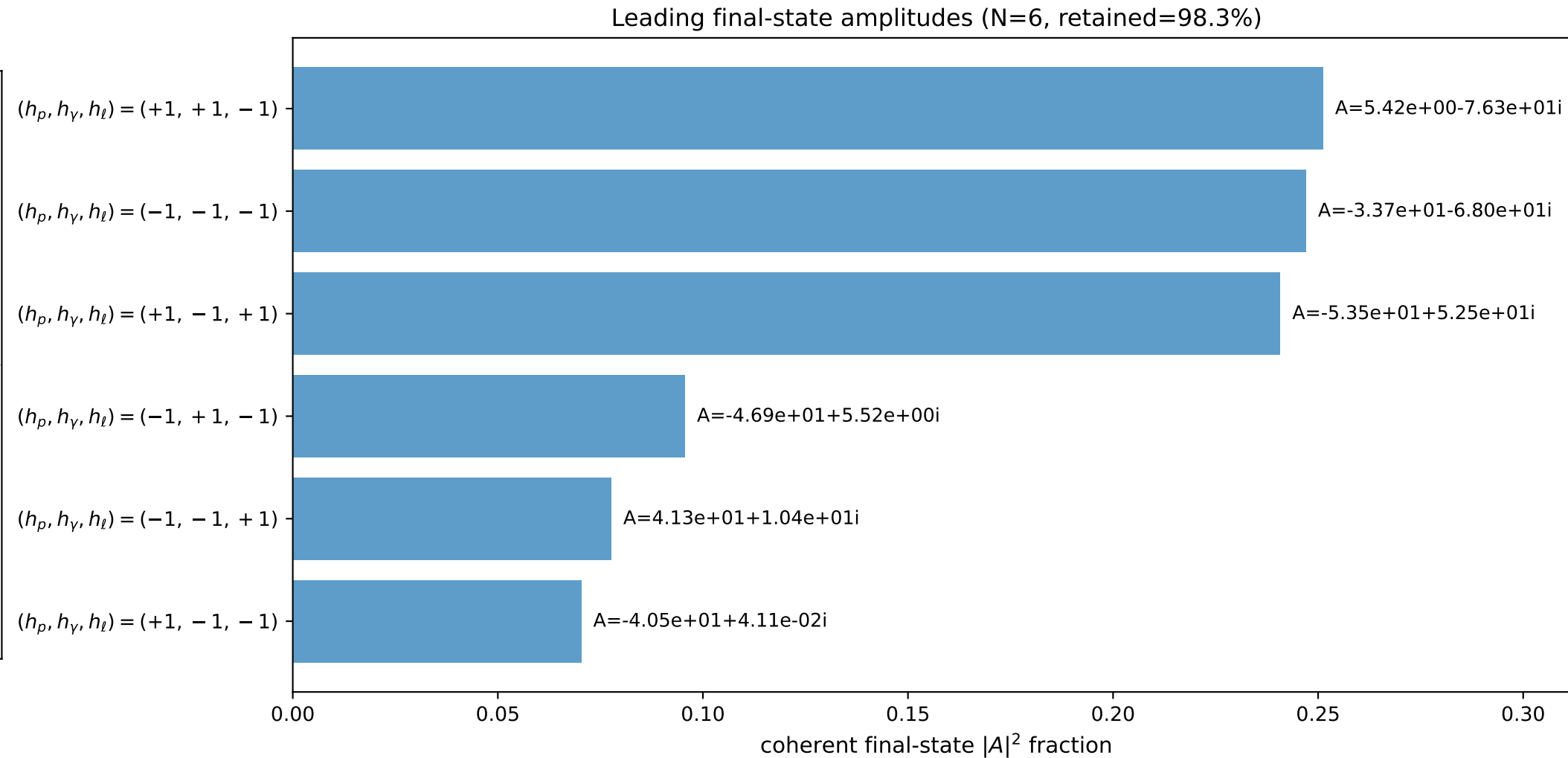
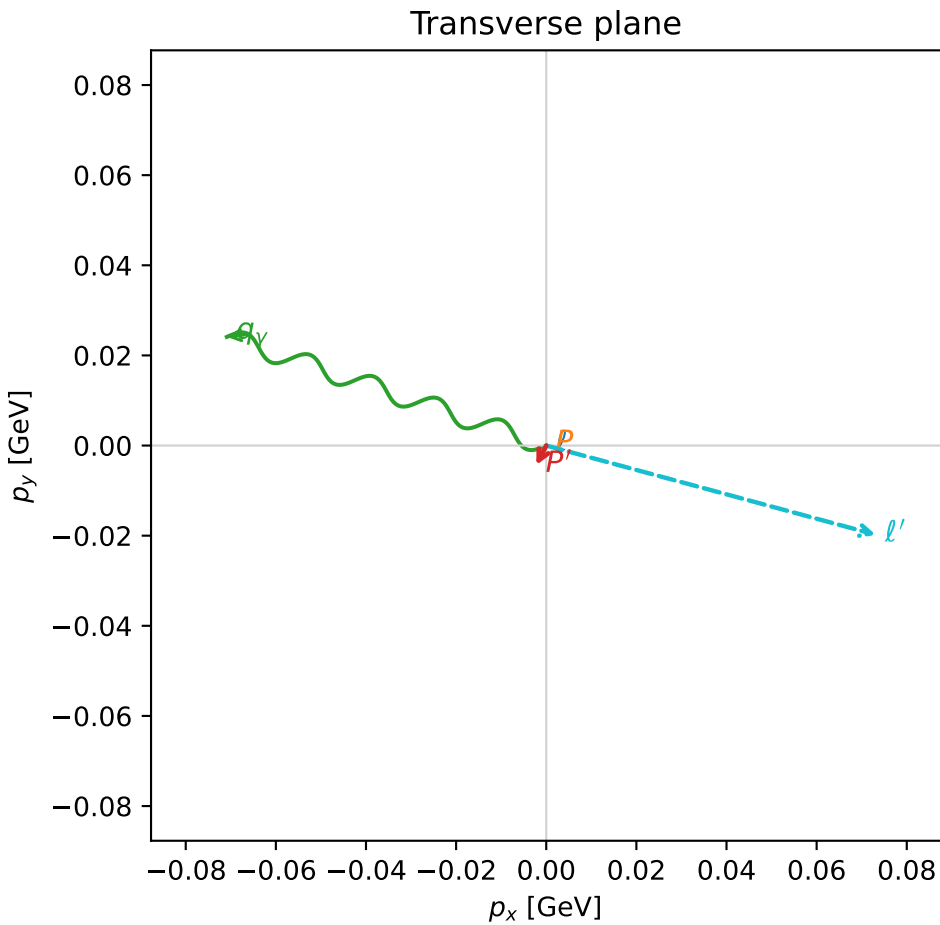
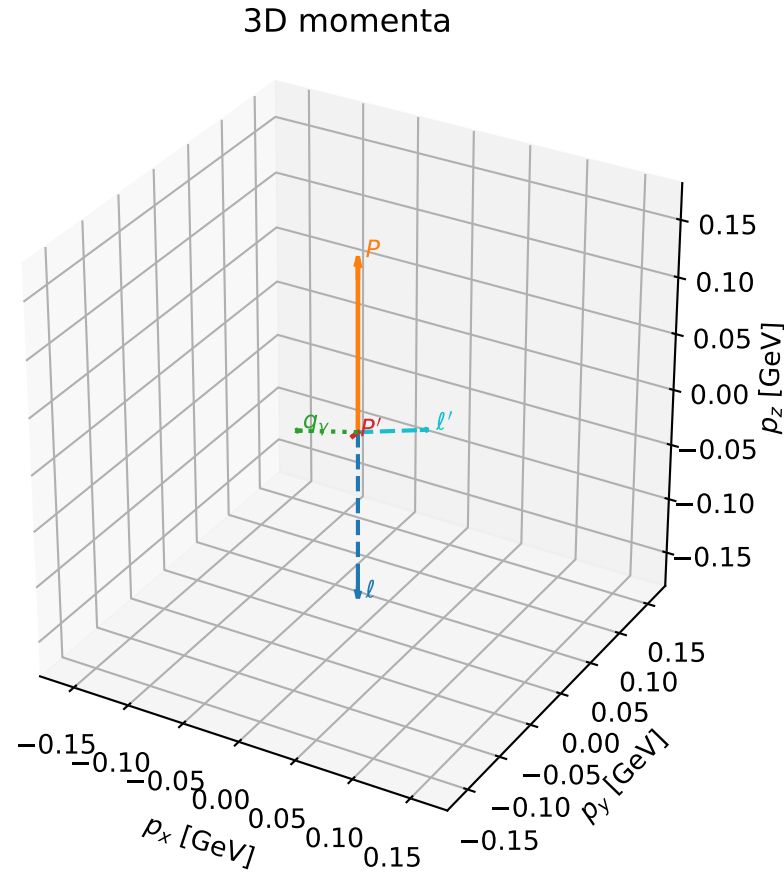
selected phase-space point:
 $\text{sqrt}_s = 1.099698$
 $\text{theta}_p_{\text{out}} = 3.140034$
 $\text{theta}_\gamma_{\text{out}} = 2.042184$
 $q_{\text{out}} = 0.08084635$
 $\text{phi}_p_{\text{out}} = 6.269078$
 $\text{phi}_\gamma_{\text{out}} = 8.62237\text{e-}06$
 $\alpha_e = 2.340672$
 $\alpha_p = 1.442174$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

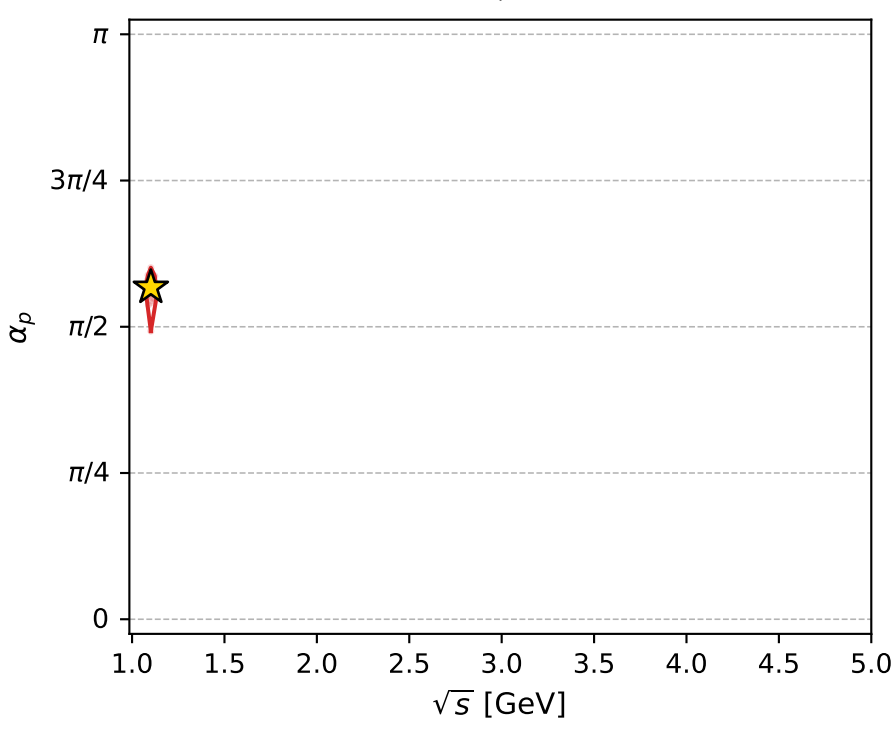
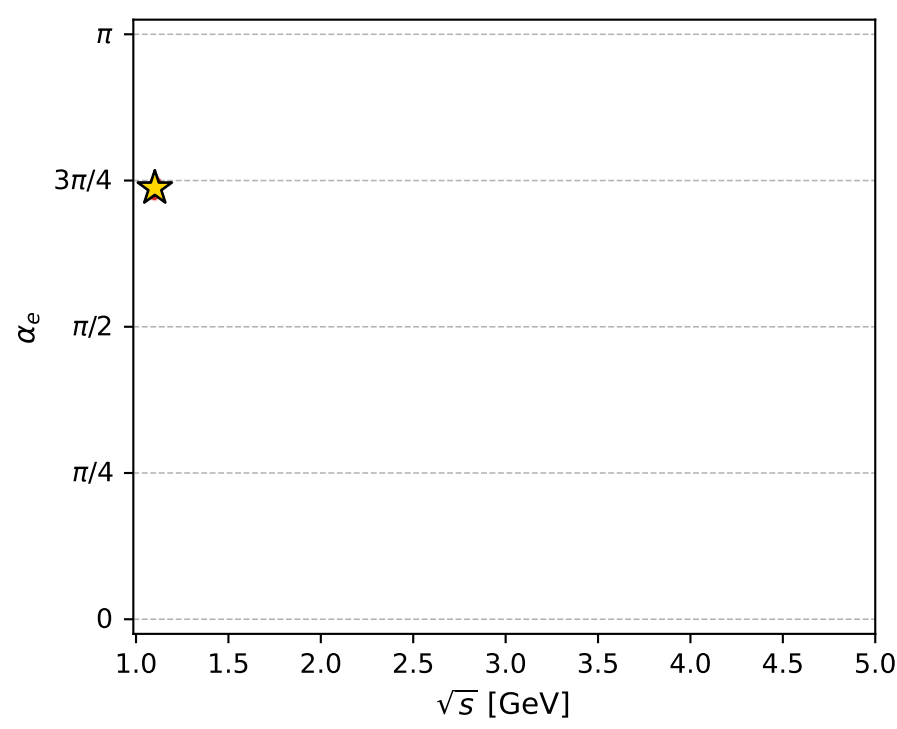
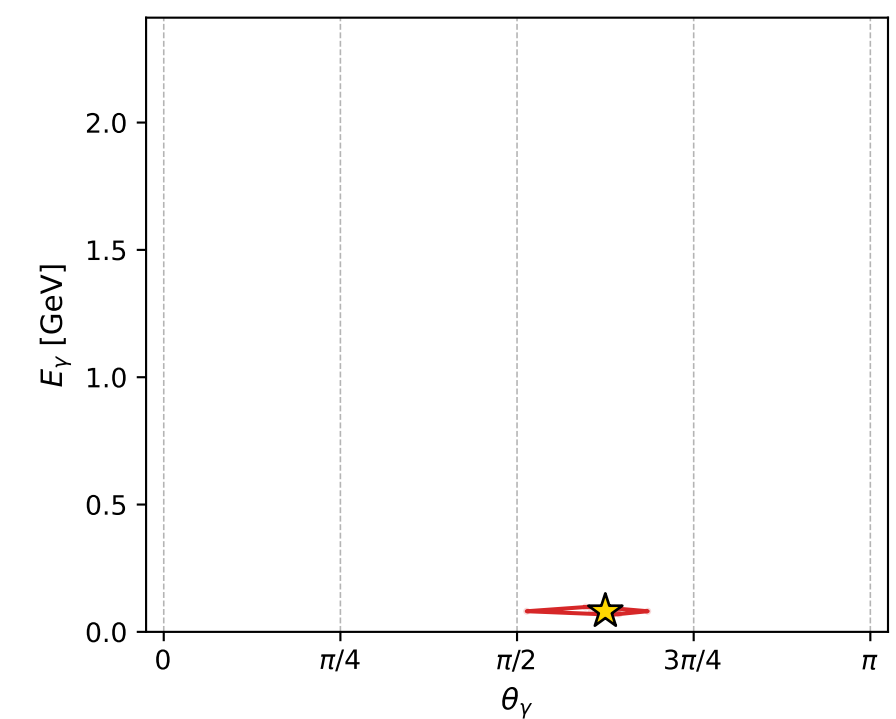
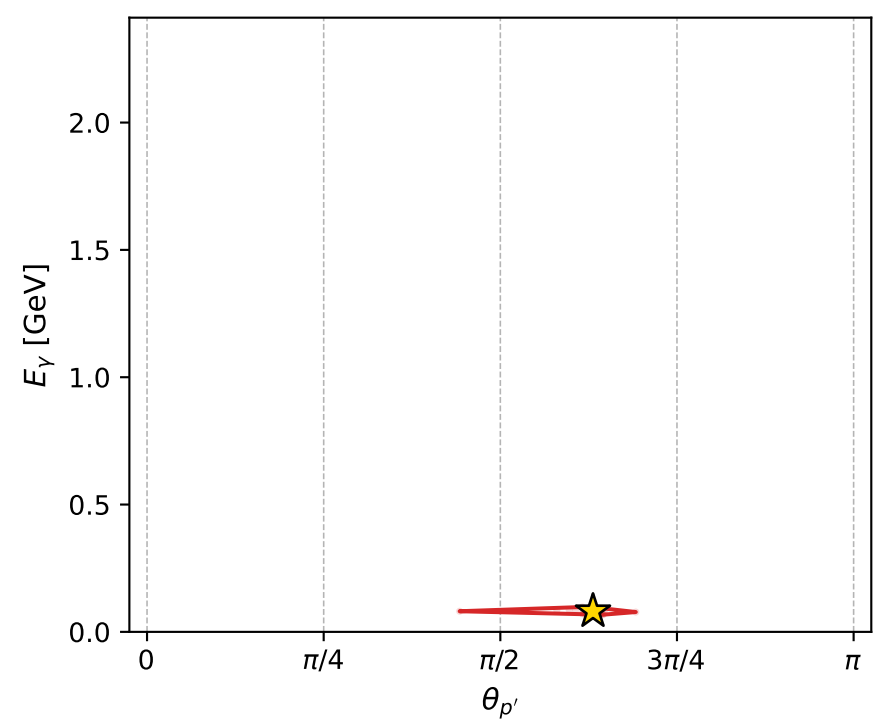
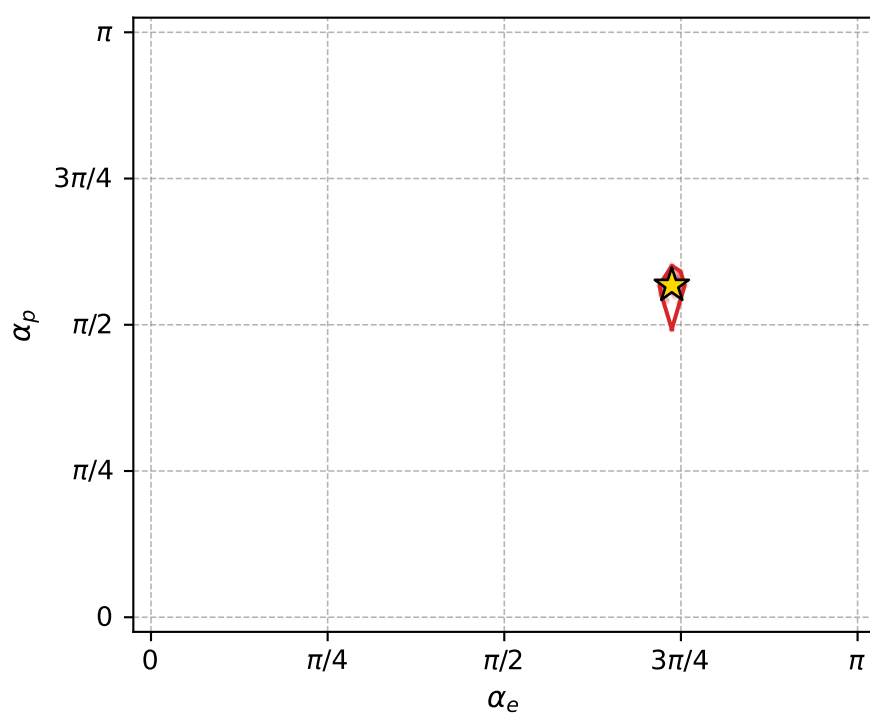
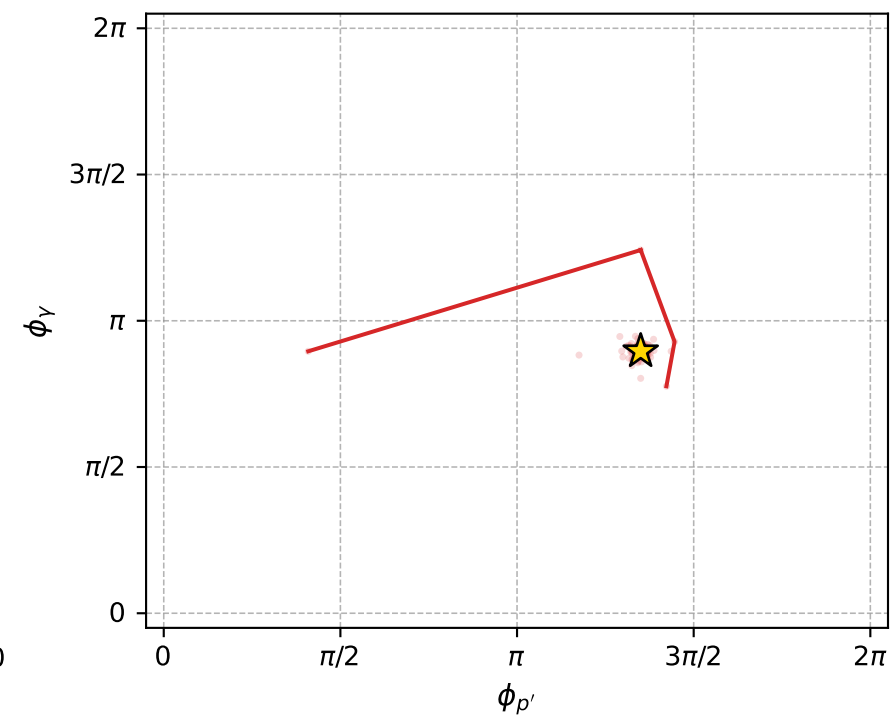
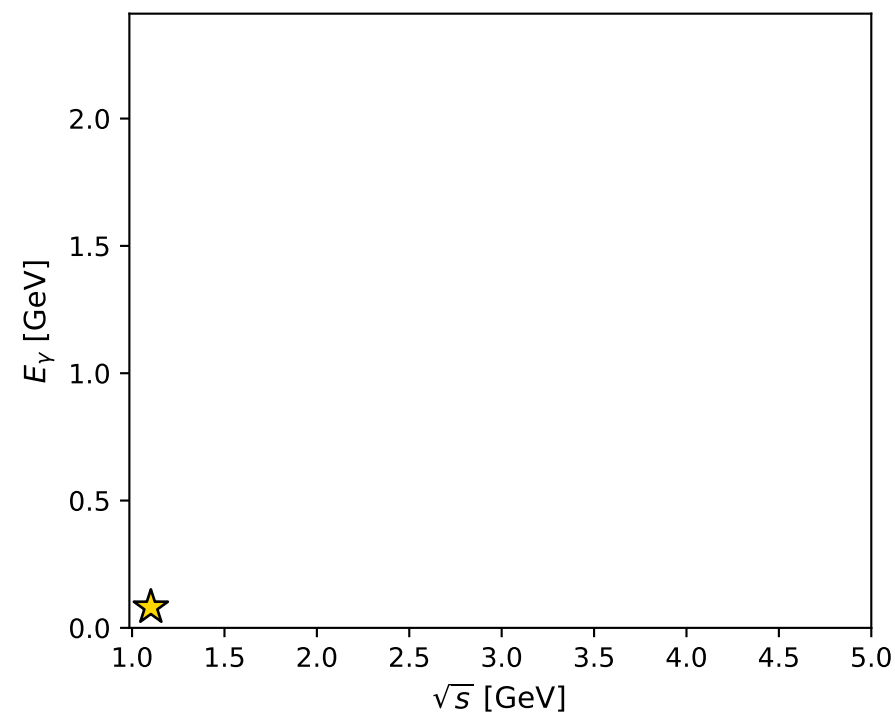
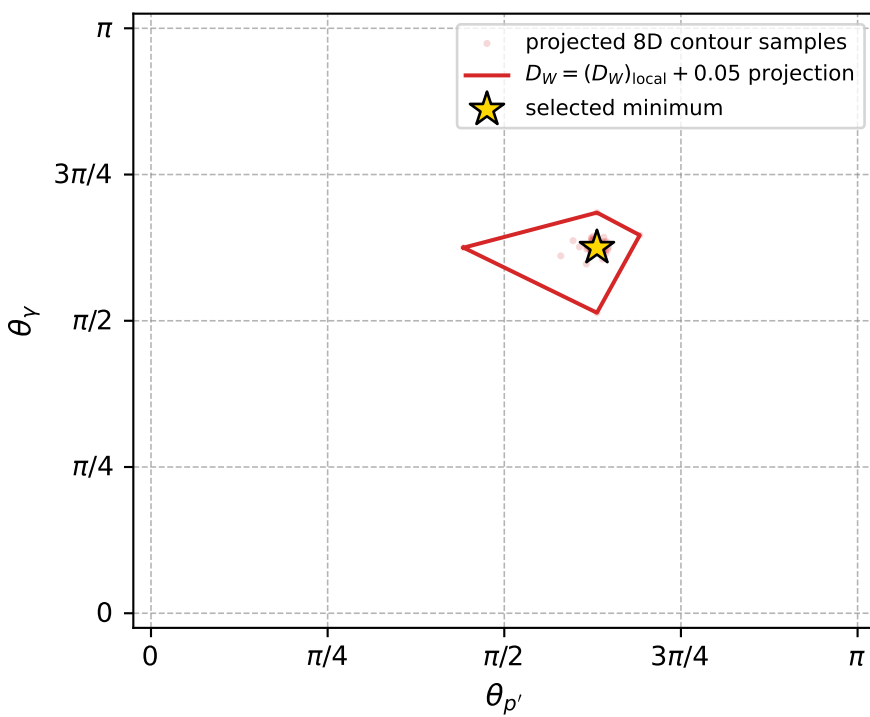
dw_mixing_angles_polarization_06_local_minimum_852 D_W=0.0120068
 region: polarization_cluster_06_local_minimum_852
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.315682$ rad
 incoming lepton: $-0.677888|+> +0.735165|->$
 proton mixing angle: $\alpha_p=1.7822703$ rad
 incoming proton: $-0.209901|+> +0.977723|->$

$s=1.2137$, $\sqrt{s}=1.10168$, $|\vec{P}|=0.15152$, $|\vec{P}'|=0.00530609$
 $\theta_{P'}=1.98279$, $\theta_V=1.96335$, $\phi_{P'}=4.24111$, $\phi_V=2.81374$
 $E_V=0.0810254$, $\theta_{\ell'}=1.15841$, $\phi_{\ell'}=6.01894$
 $Q^2=0.0350798$, $x_B=0.187743$, $t=-0.0234829$, $W^2=1.03161$, $y=0.559675$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1515$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1515$
 P $E= 0.9502$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1515$
 ℓ' $E= 0.0826$ $p_x= 0.0731$ $p_y= -0.0198$ $p_z= 0.0331$
 P' $E= 0.9380$ $p_x= -0.0022$ $p_y= -0.0043$ $p_z= -0.0021$
 q_V $E= 0.0810$ $p_x= -0.0709$ $p_y= 0.0241$ $p_z= -0.0310$



electron: polarization cluster P6, local minimum 852; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.012006777$

$D_W \text{ contour} = 0.062006777$

above global minimum = 0.011990344

8D contour samples = 255

selected phase-space point:

$\text{sqrt_s} = 1.10168$

$\text{theta_p_out} = 1.982791$

$\text{theta_gamma_out} = 1.963352$

$\text{qOut} = 0.0810254$

$\text{phi_p_out} = 4.241107$

$\text{phi_gamma_out} = 2.813744$

$\alpha_e = 2.315682$

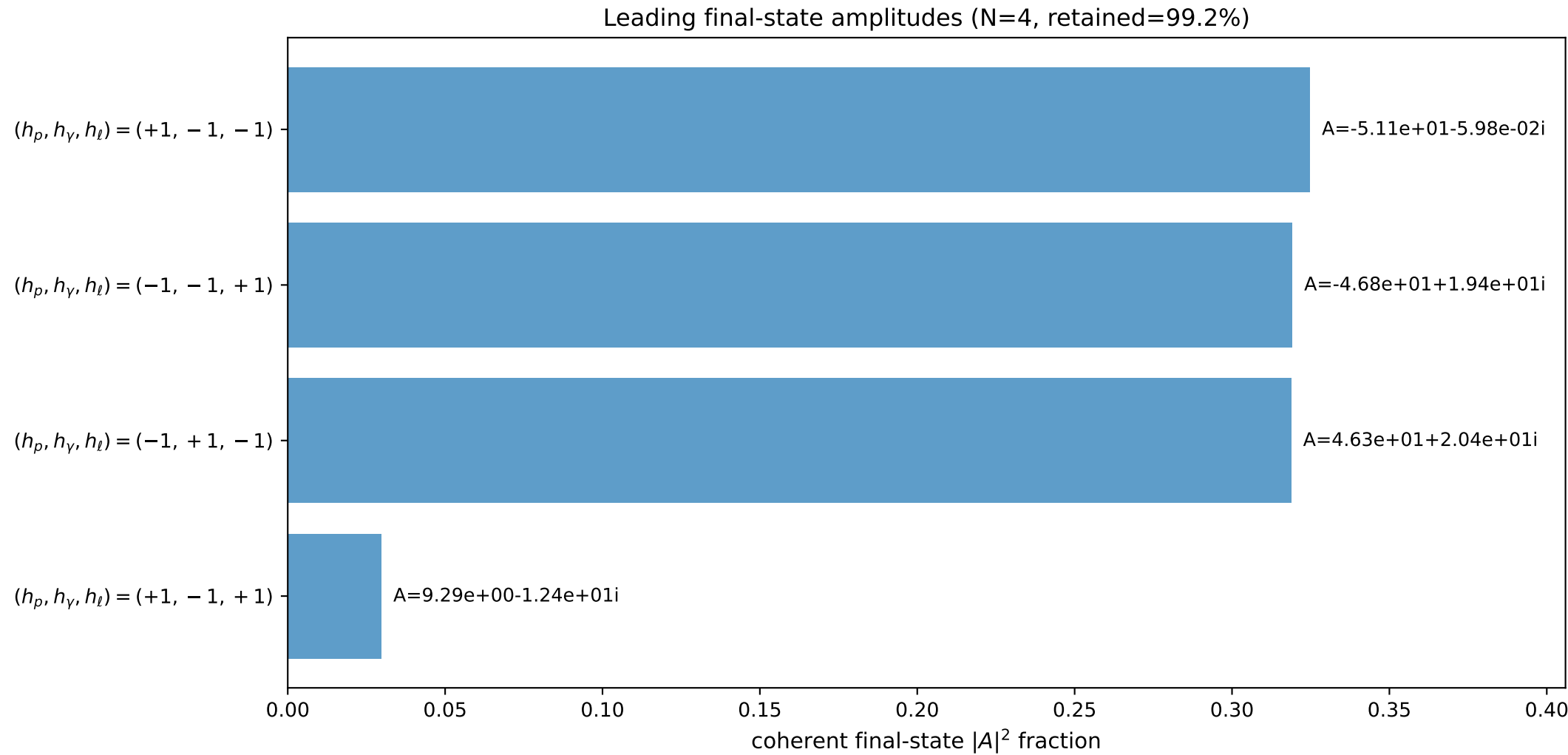
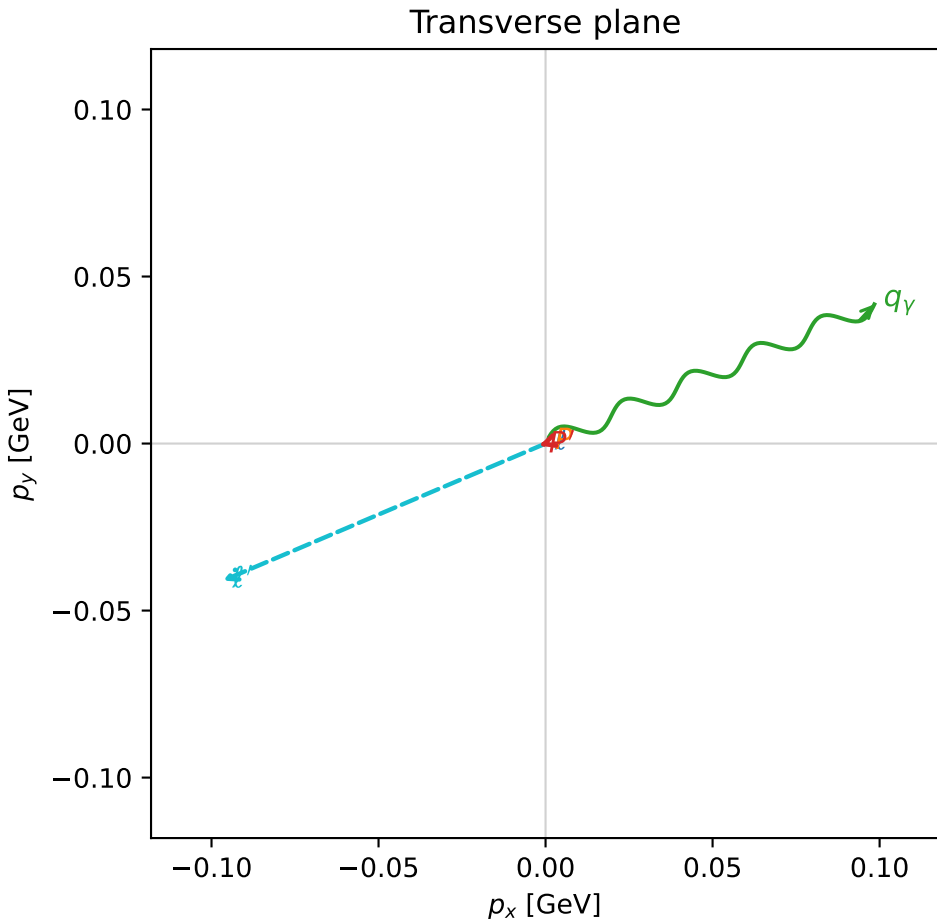
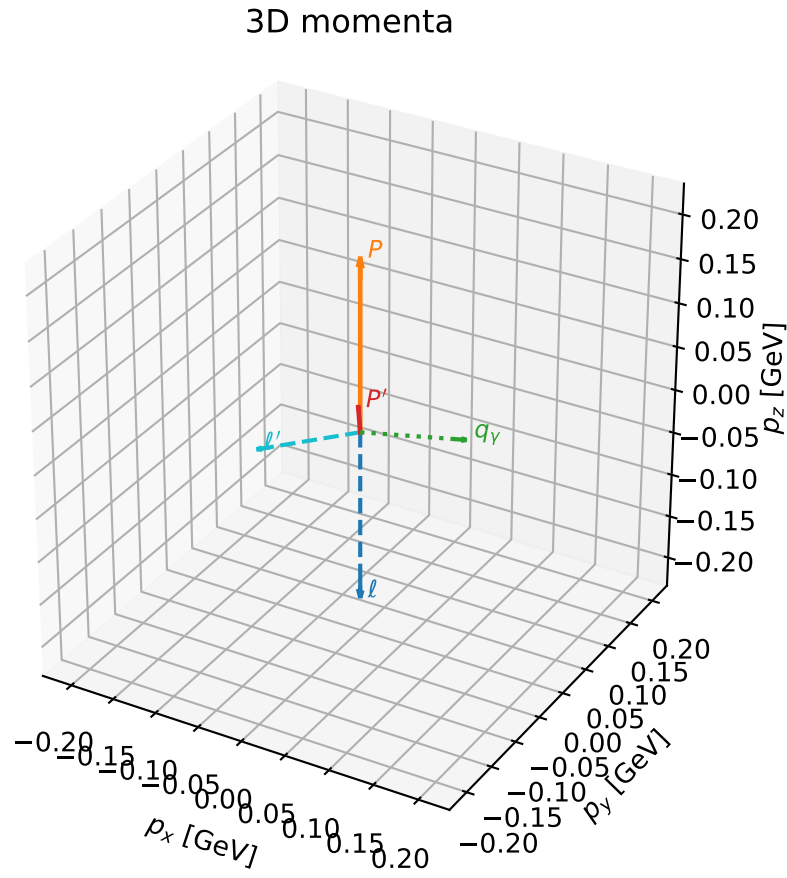
$\alpha_p = 1.78227$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

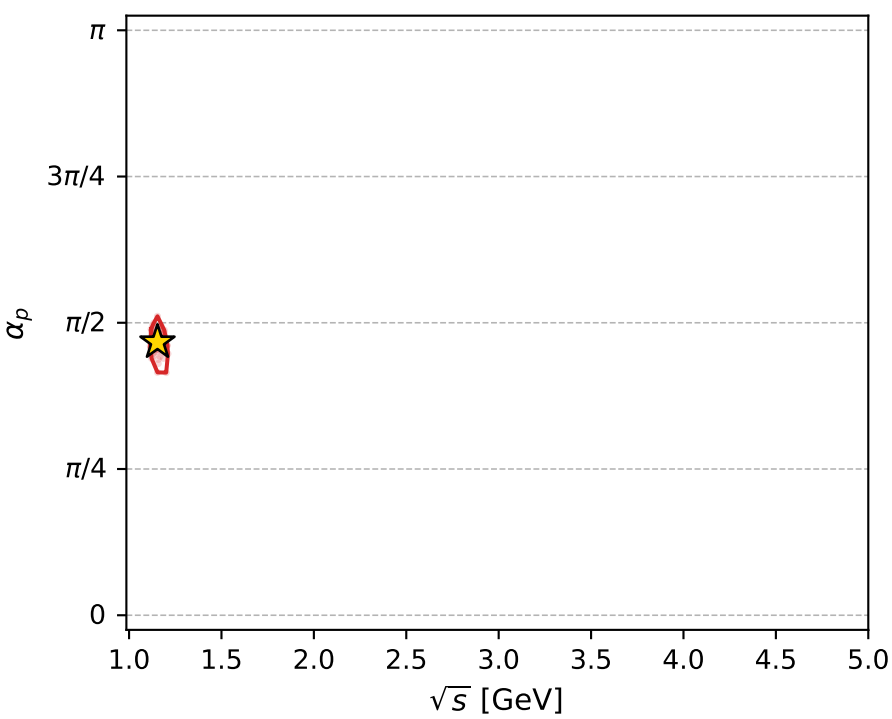
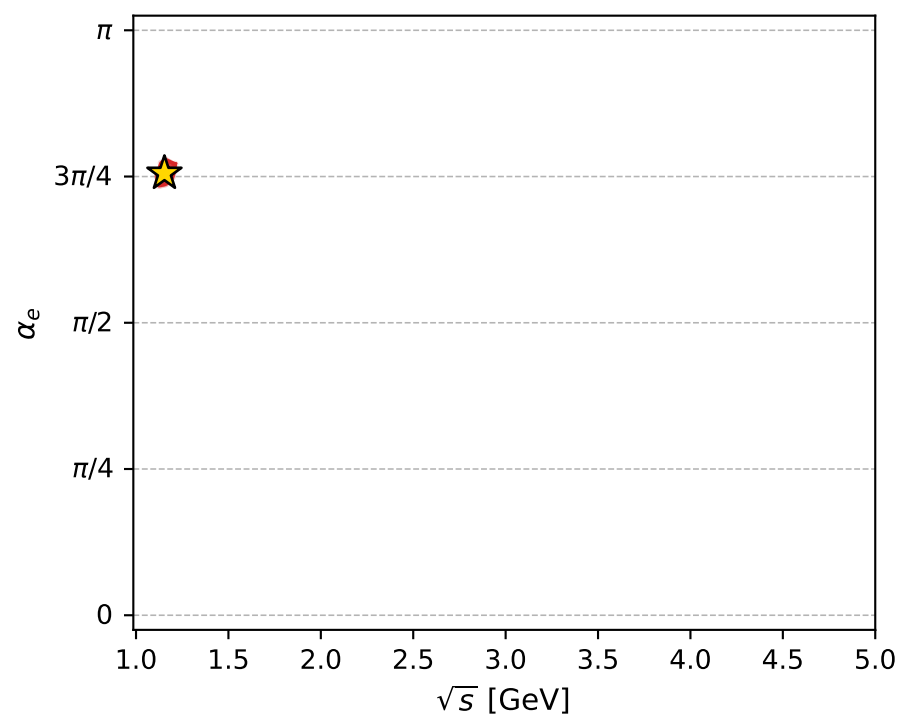
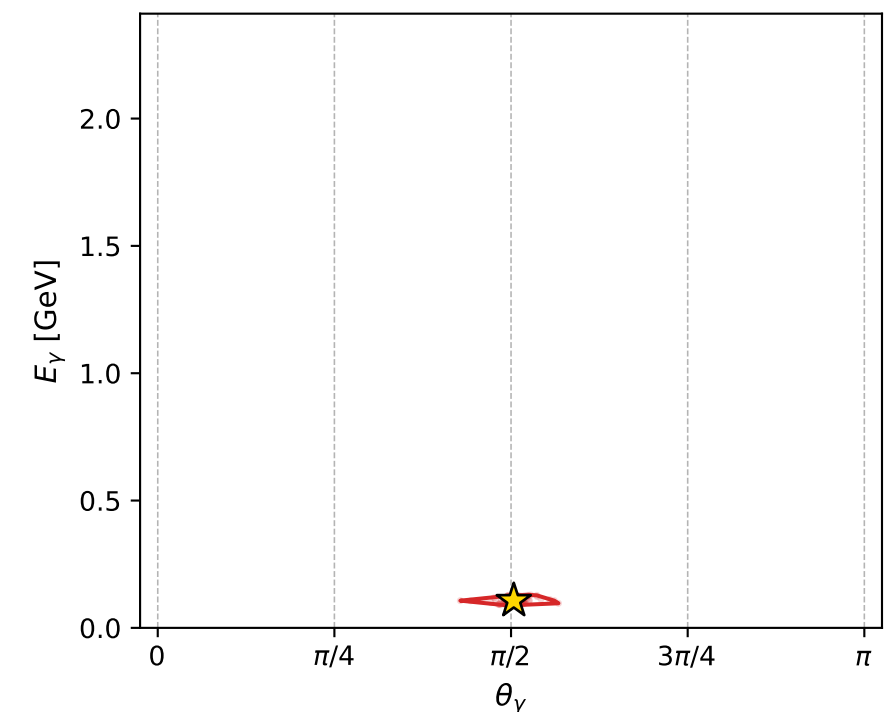
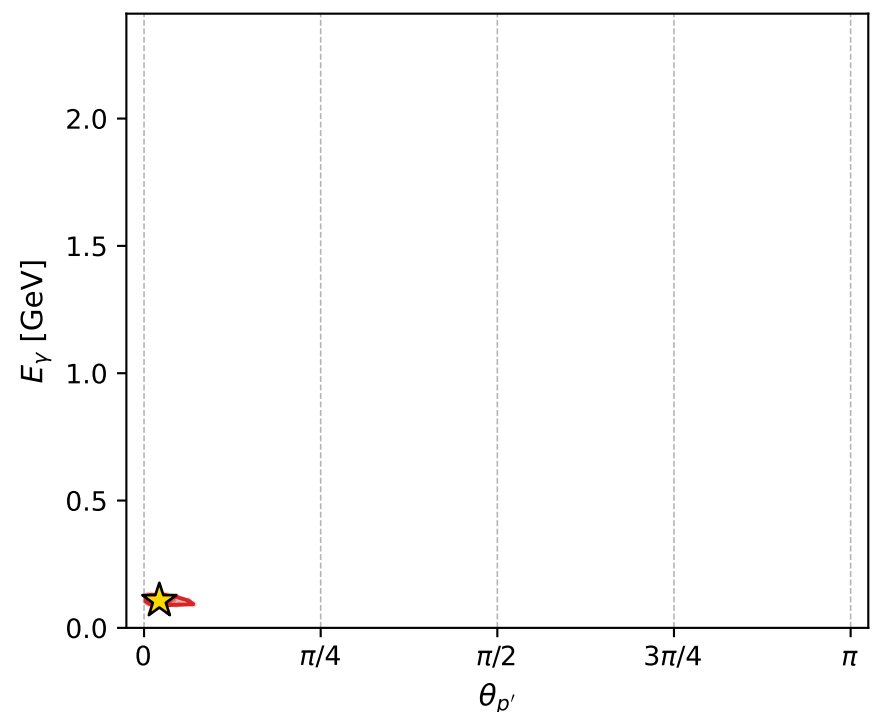
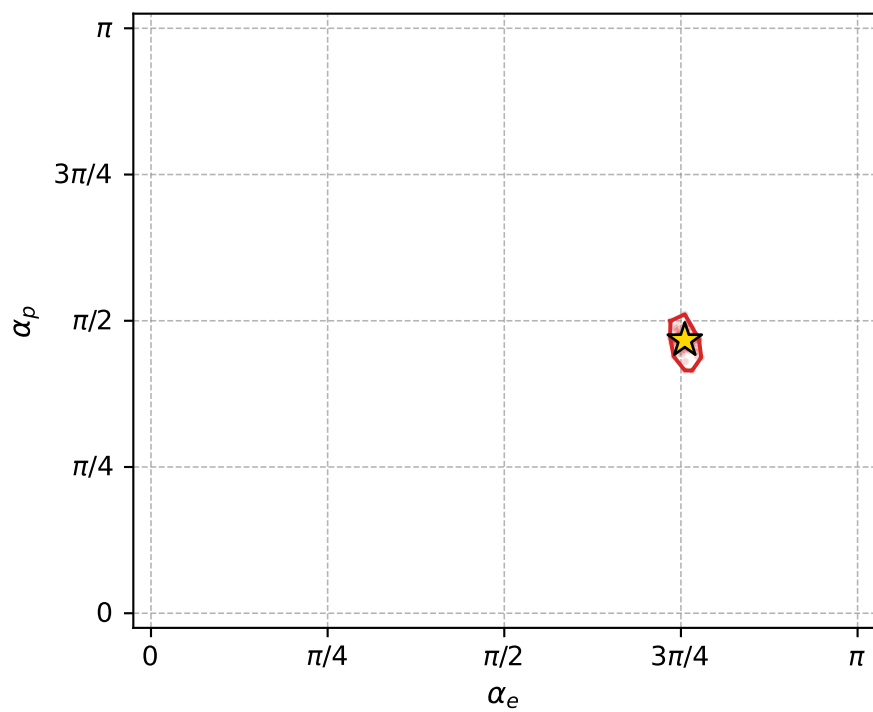
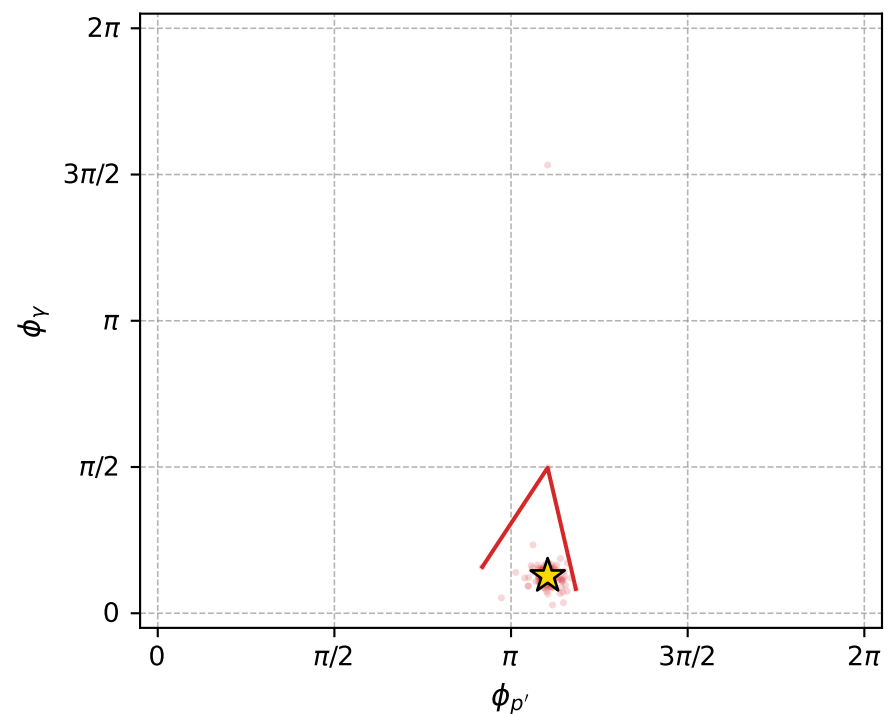
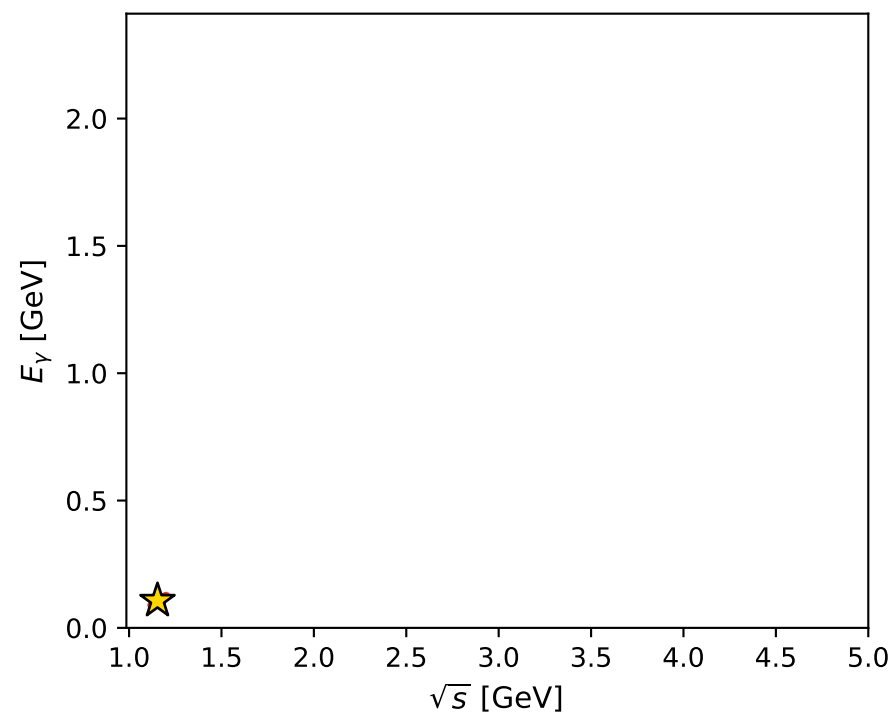
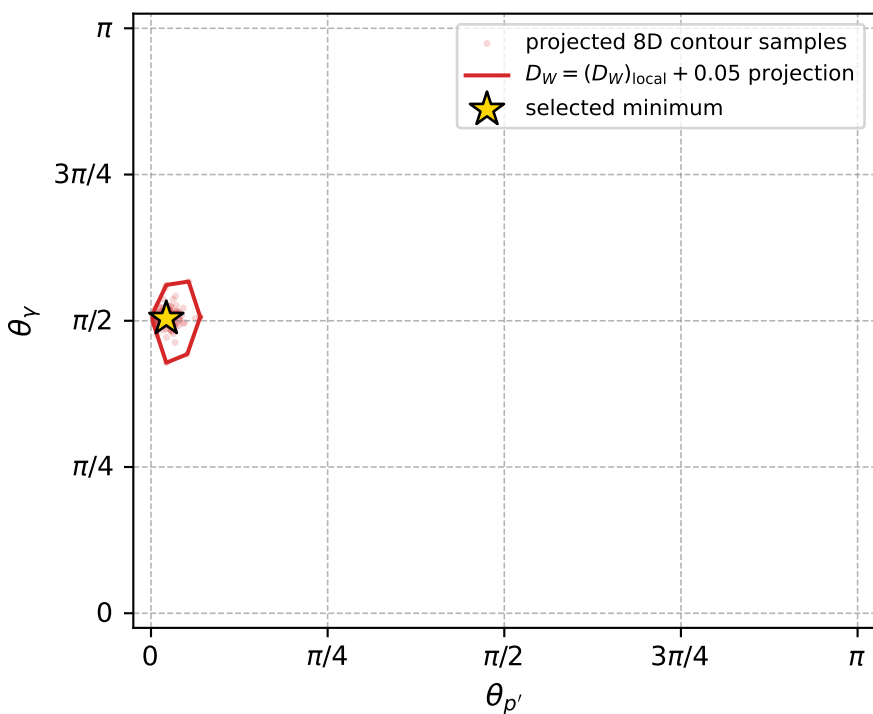
dw_mixing_angles_polarization_06_local_minimum_855 D_W=0.012073
 region: polarization_cluster_06_local_minimum_855
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3734038$ rad
 incoming lepton: $-0.71917|+\rangle +0.694834|-\rangle$
 proton mixing angle: $\alpha_p=1.4663193$ rad
 incoming proton: $0.104287|+\rangle +0.994547|-\rangle$

$s=1.33113$, $\sqrt{s}=1.15374$, $|\vec{P}|=0.195572$, $|\vec{P}'|=0.0286398$
 $\theta_{P'}=0.0680517$, $\theta_Y=1.58294$, $\phi_{P'}=3.46718$, $\phi_Y=0.400475$
 $E_Y=0.106884$, $\theta_{\ell'}=1.8251$, $\phi_{\ell'}=3.54346$
 $Q^2=0.03174$, $x_B=0.136318$, $t=-0.0275029$, $W^2=1.08094$, $y=0.515948$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1956$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1956$
 P $E= 0.9582$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1956$
 ℓ' $E= 0.1084$ $p_x= -0.0966$ $p_y= -0.0410$ $p_z= -0.0273$
 P' $E= 0.9384$ $p_x= -0.0018$ $p_y= -0.0006$ $p_z= 0.0286$
 q_Y $E= 0.1069$ $p_x= 0.0984$ $p_y= 0.0417$ $p_z= -0.0013$



electron: polarization cluster P6, local minimum 855; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.012072981$
 $D_W \text{ contour} = 0.062072981$
 above global minimum = 0.012056549
 8D contour samples = 246

selected phase-space point:
 $\text{sqrt_s} = 1.153744$
 $\text{theta_p_out} = 0.06805175$
 $\text{theta_gamma_out} = 1.582942$
 $\text{qOut} = 0.1068845$
 $\text{phi_p_out} = 3.467181$
 $\text{phi_gamma_out} = 0.4004754$
 $\text{alpha_e} = 2.373404$
 $\text{alpha_p} = 1.466319$

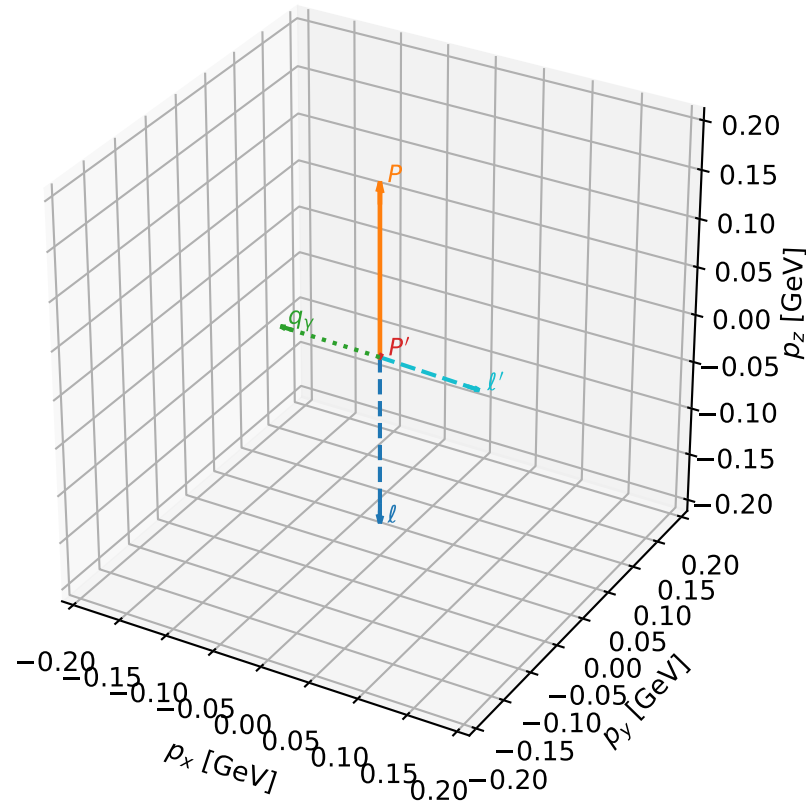
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_862 D_W=0.0122258
 region: polarization_cluster_06_local_minimum_862
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3786691$ rad
 incoming lepton: $-0.722819|+\rangle +0.691038|-\rangle$
 proton mixing angle: $\alpha_p=1.7822548$ rad
 incoming proton: $-0.209886|+\rangle +0.977726|-\rangle$

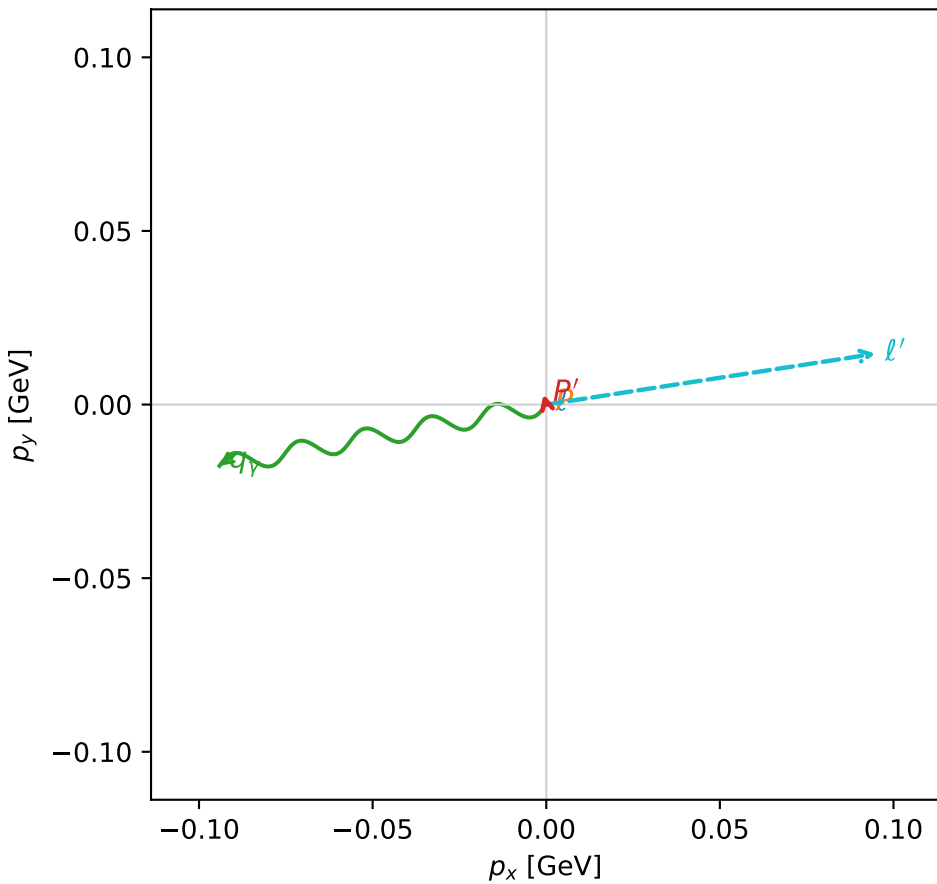
$s=1.28095$, $\sqrt{s}=1.13179$, $|\vec{P}|=0.177197$, $|\vec{P}'|=0.00311273$
 $\theta_{P'}=1.53021$, $\theta_V=1.42511$, $\phi_{P'}=1.81292$, $\phi_V=3.32712$
 $E_V=0.09677$, $\theta_{\ell'}=1.71743$, $\phi_{\ell'}=0.153163$
 $Q^2=0.0293574$, $x_B=0.139226$, $t=-0.0310888$, $W^2=1.06135$, $y=0.525707$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1772$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1772$
 P $E= 0.9546$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1772$
 ℓ' $E= 0.0970$ $p_x= 0.0948$ $p_y= 0.0146$ $p_z= -0.0142$
 P' $E= 0.9380$ $p_x= -0.0007$ $p_y= 0.0030$ $p_z= 0.0001$
 q_Y $E= 0.0968$ $p_x= -0.0941$ $p_y= -0.0177$ $p_z= 0.0140$

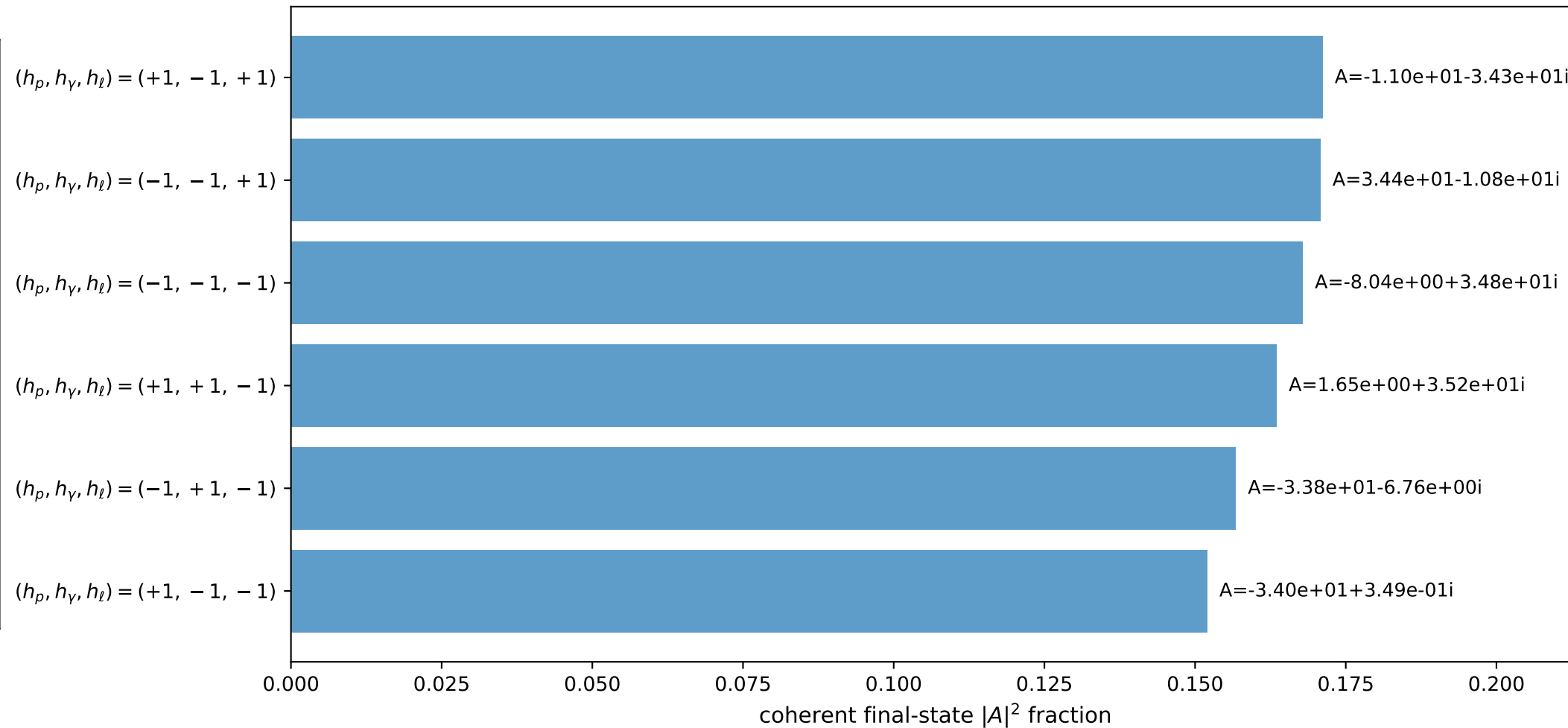
3D momenta



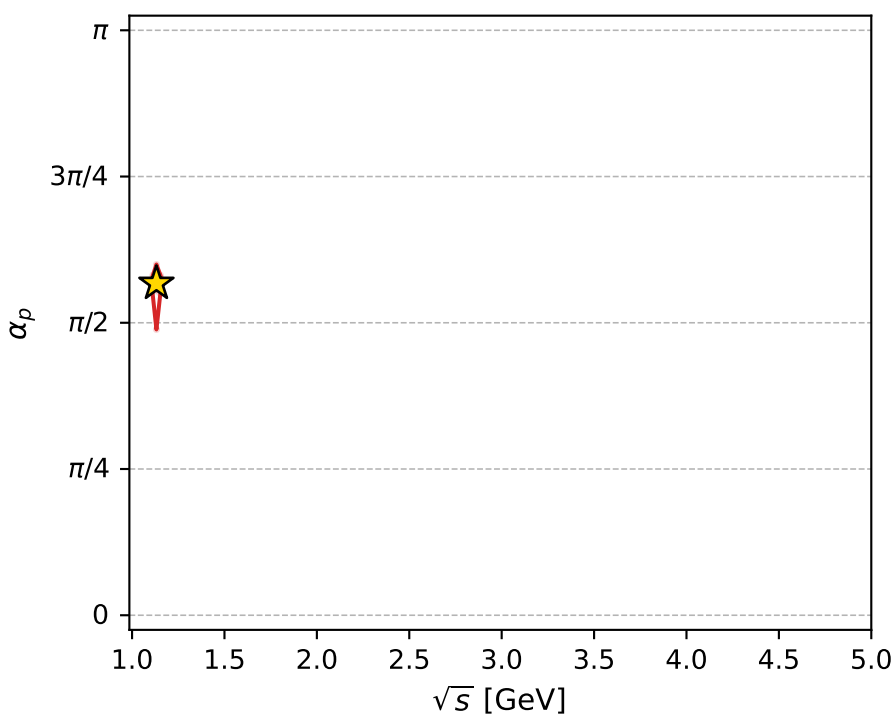
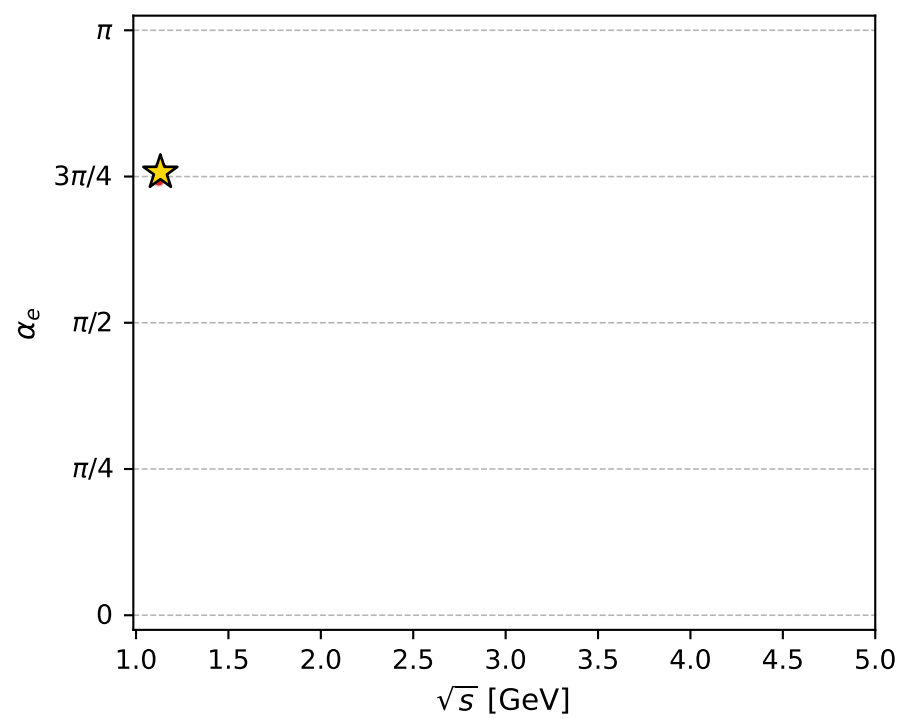
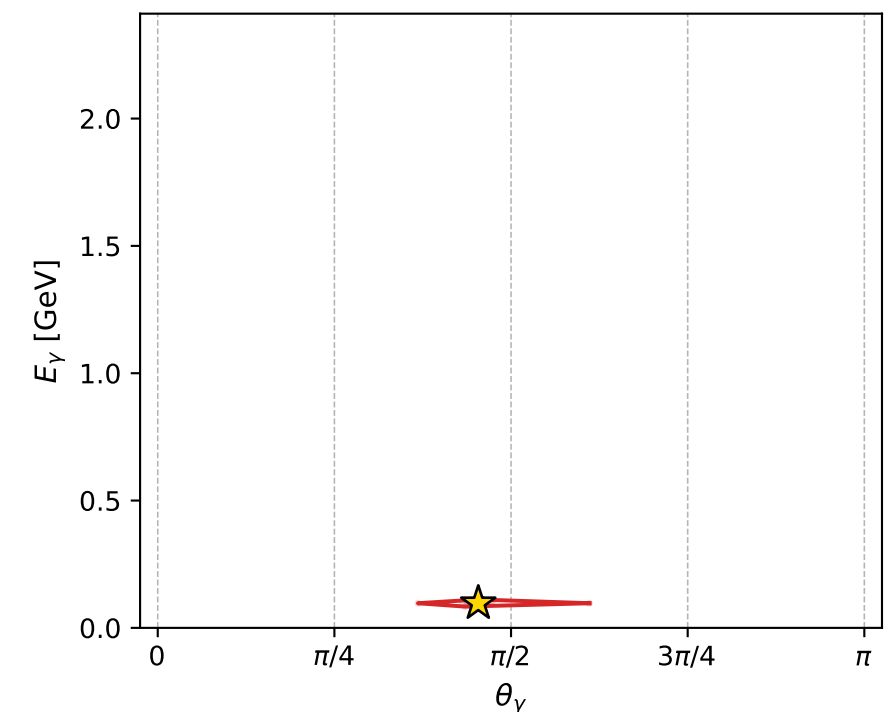
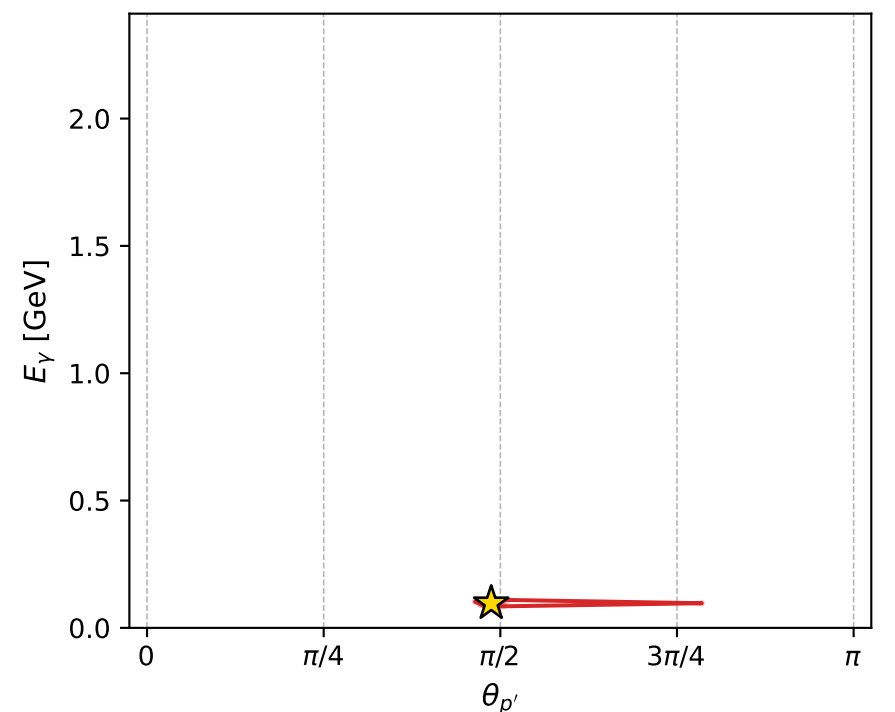
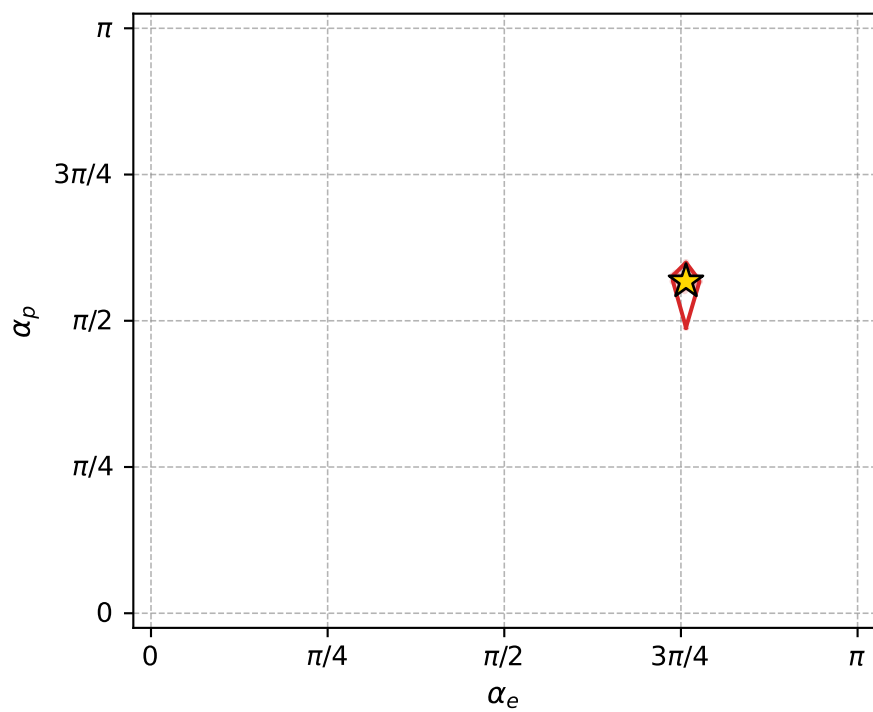
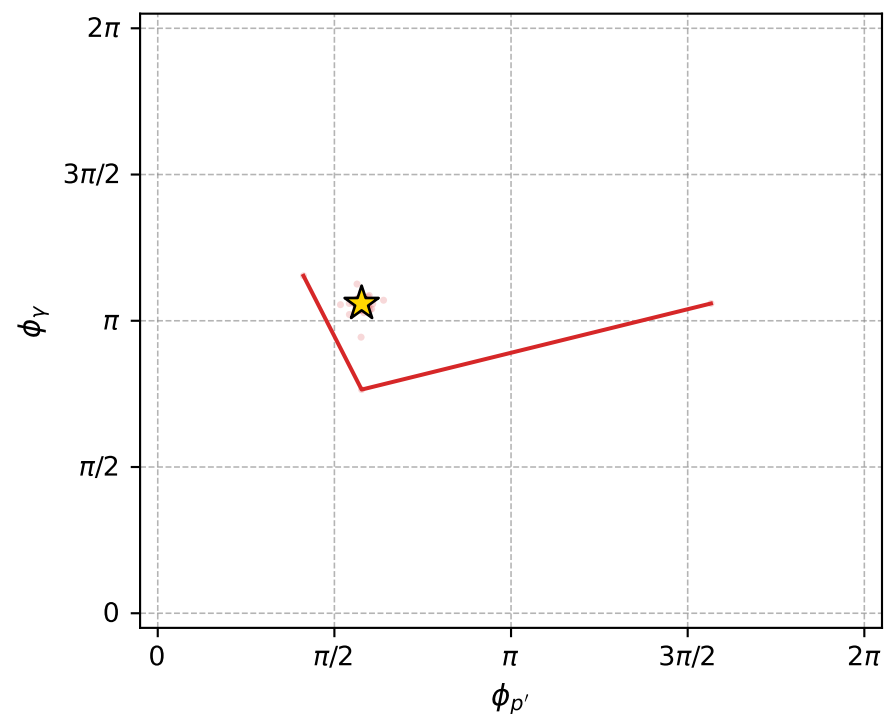
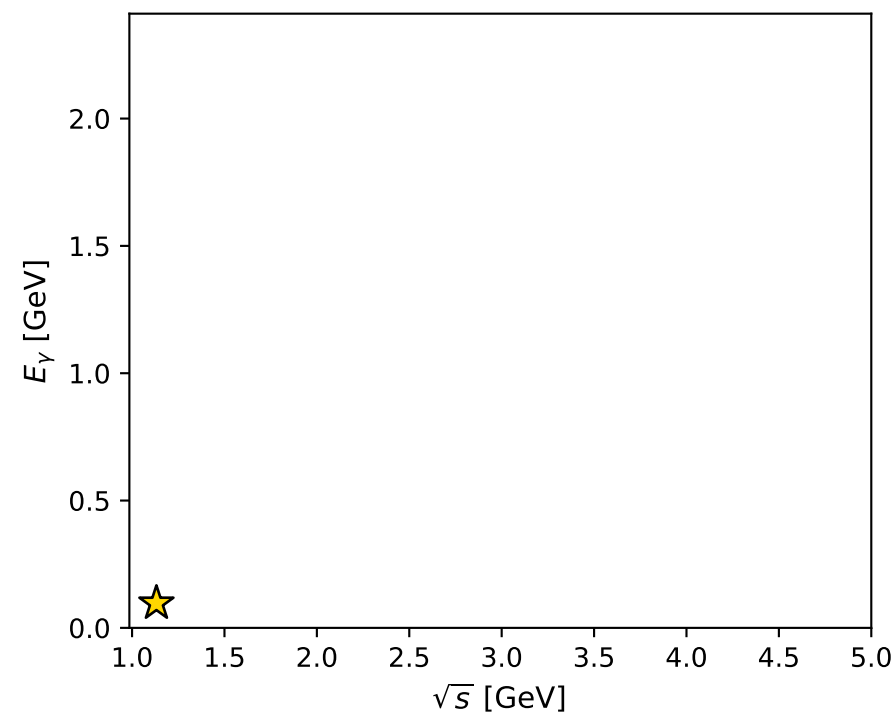
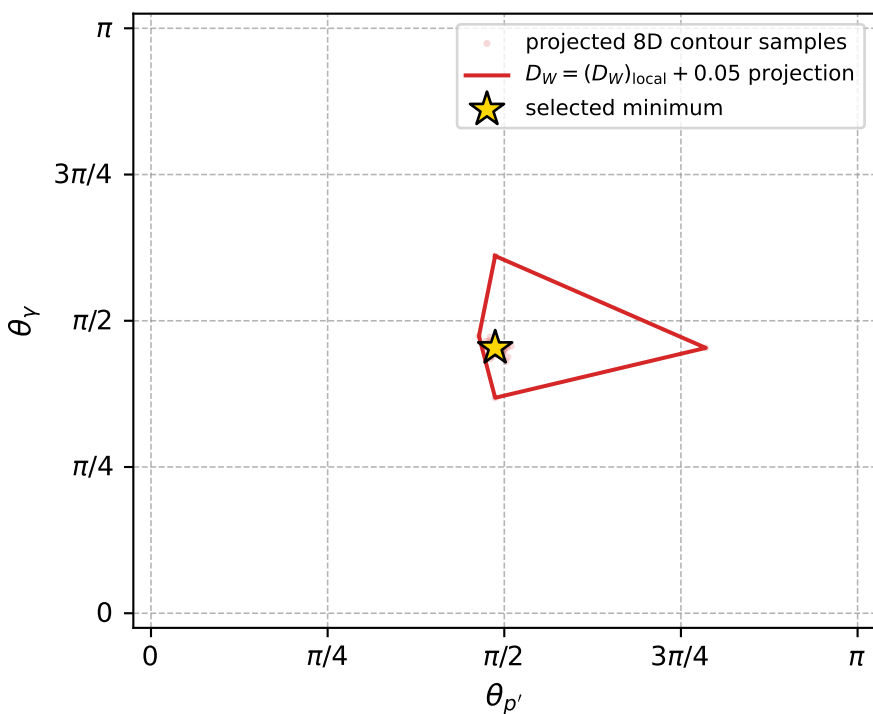
Transverse plane



Leading final-state amplitudes (N=6, retained=98.2%)



electron: polarization cluster P6, local minimum 862; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.012225773$
 $D_W \text{ contour} = 0.062225773$
 above global minimum = 0.01220934
 8D contour samples = 255

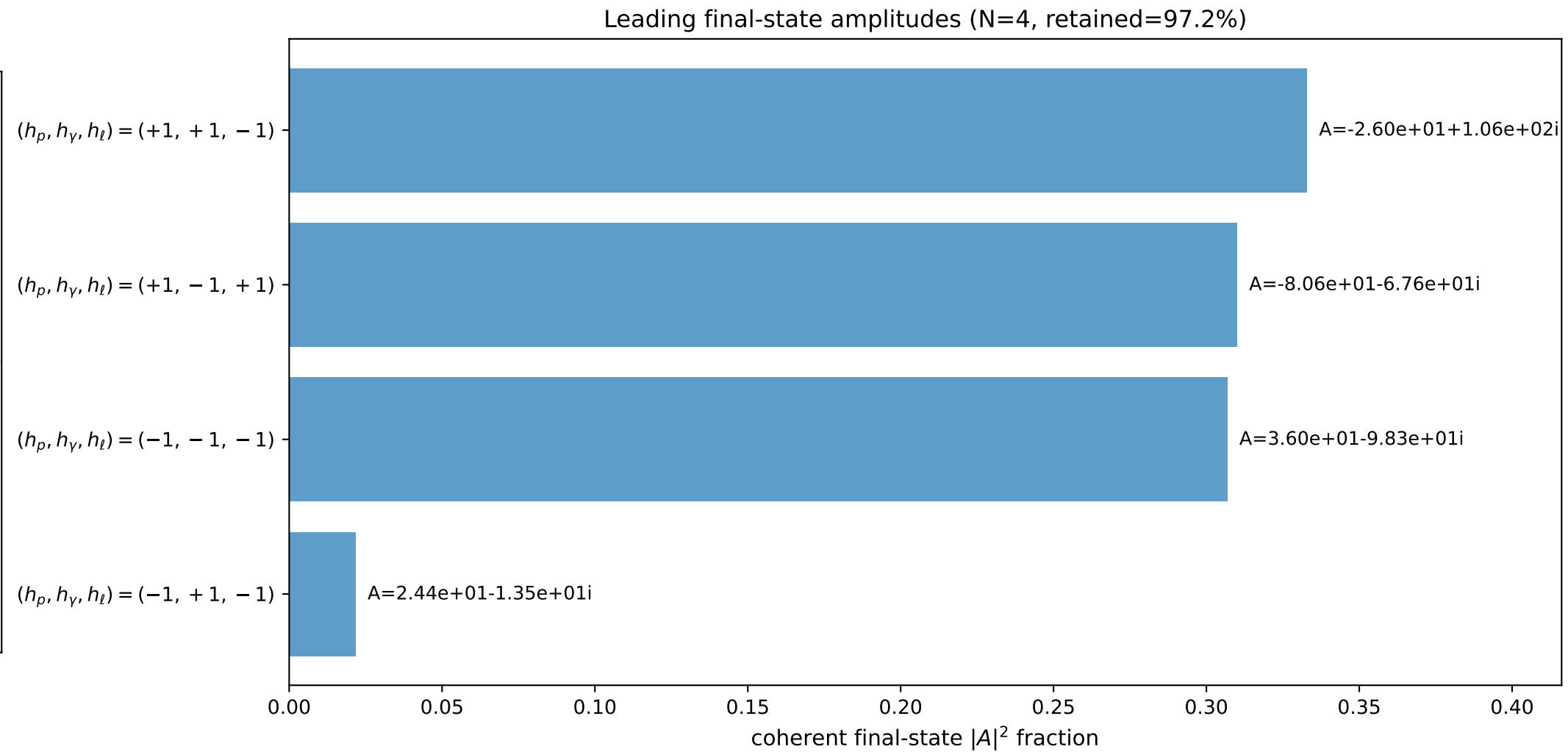
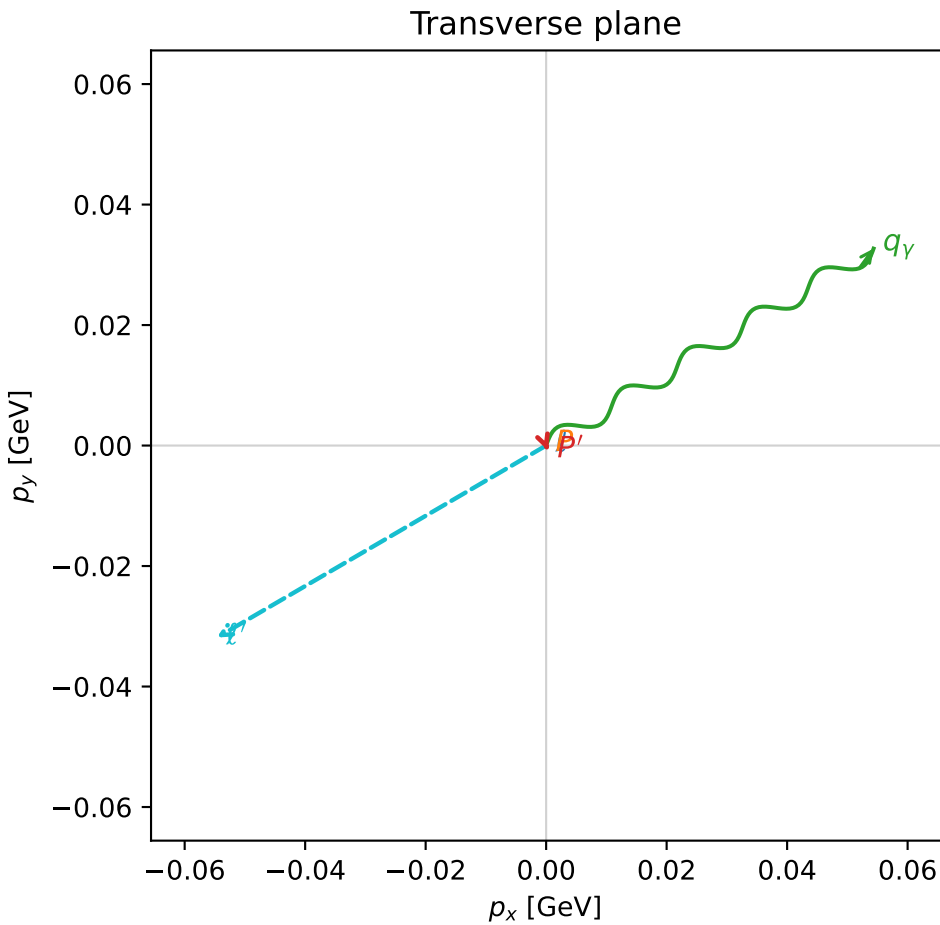
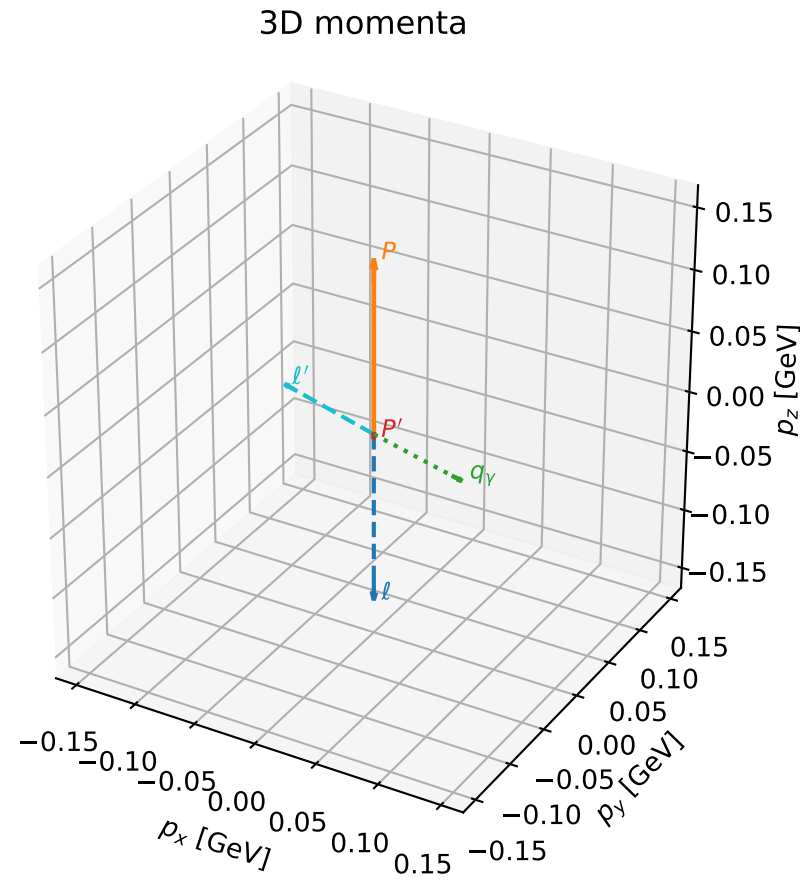
selected phase-space point:
 $\text{sqrt_s} = 1.131789$
 $\text{theta_p_out} = 1.530213$
 $\text{theta_gamma_out} = 1.425113$
 $\text{qOut} = 0.09677$
 $\text{phi_p_out} = 1.812924$
 $\text{phi_gamma_out} = 3.327117$
 $\text{alpha_e} = 2.378669$
 $\text{alpha_p} = 1.782255$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

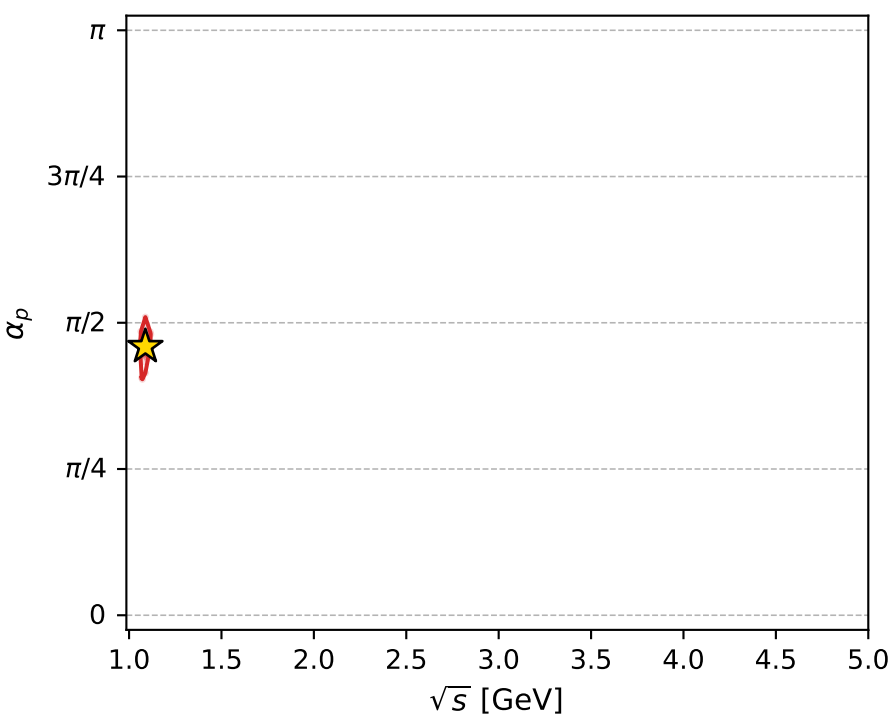
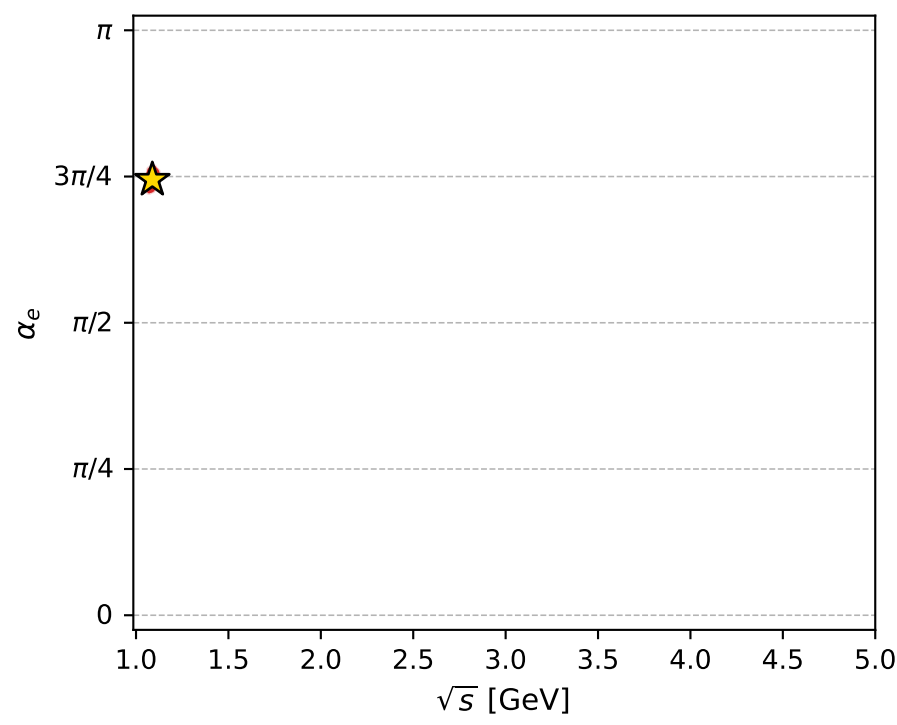
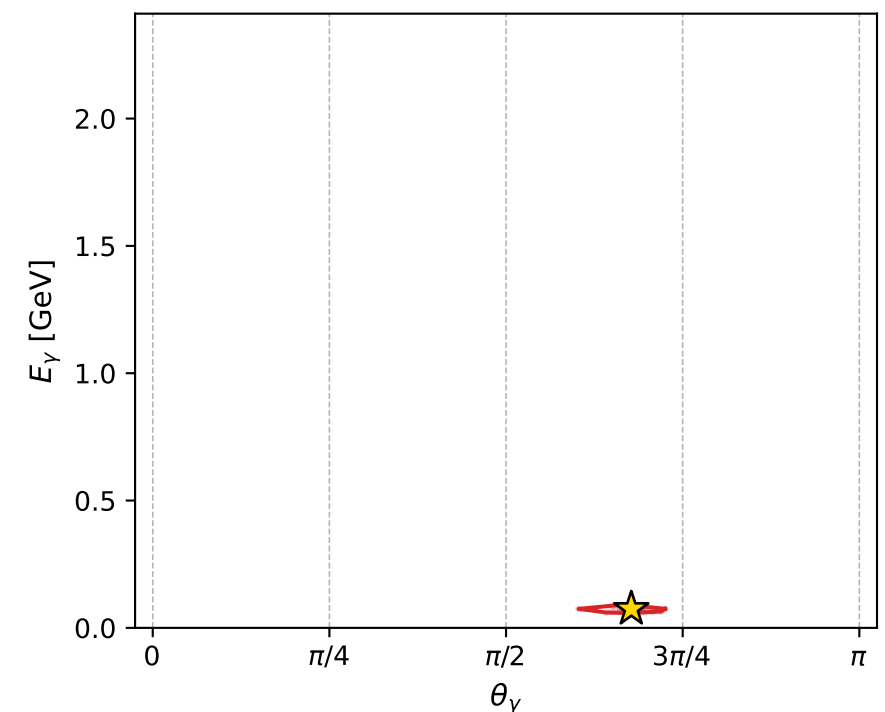
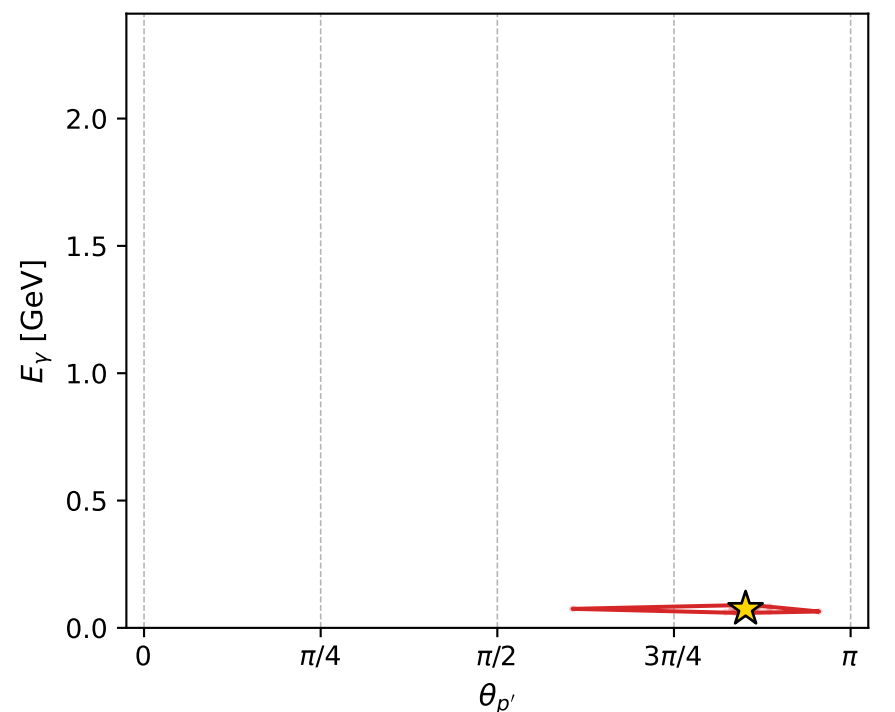
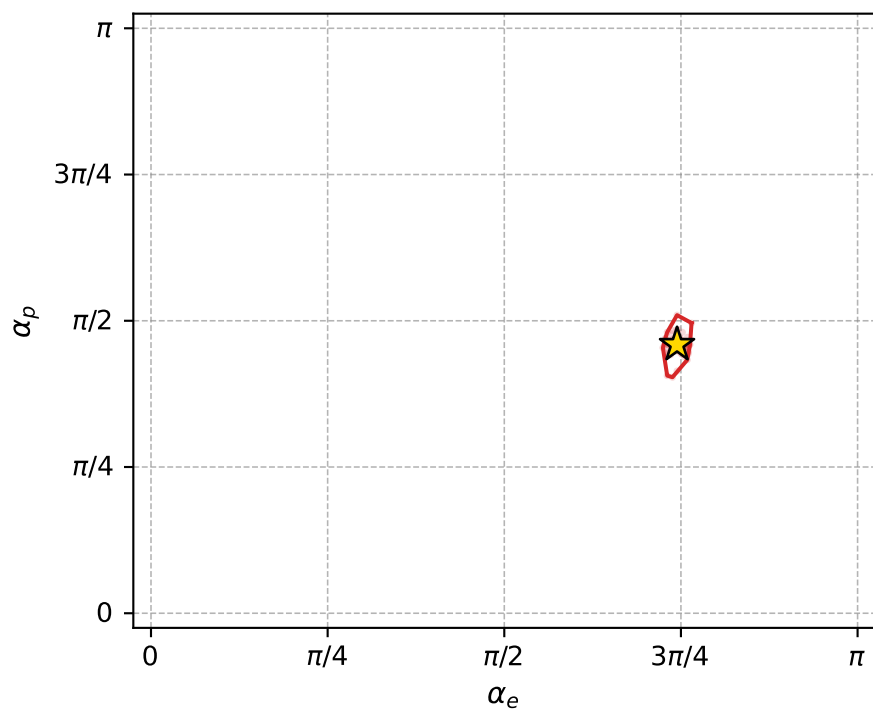
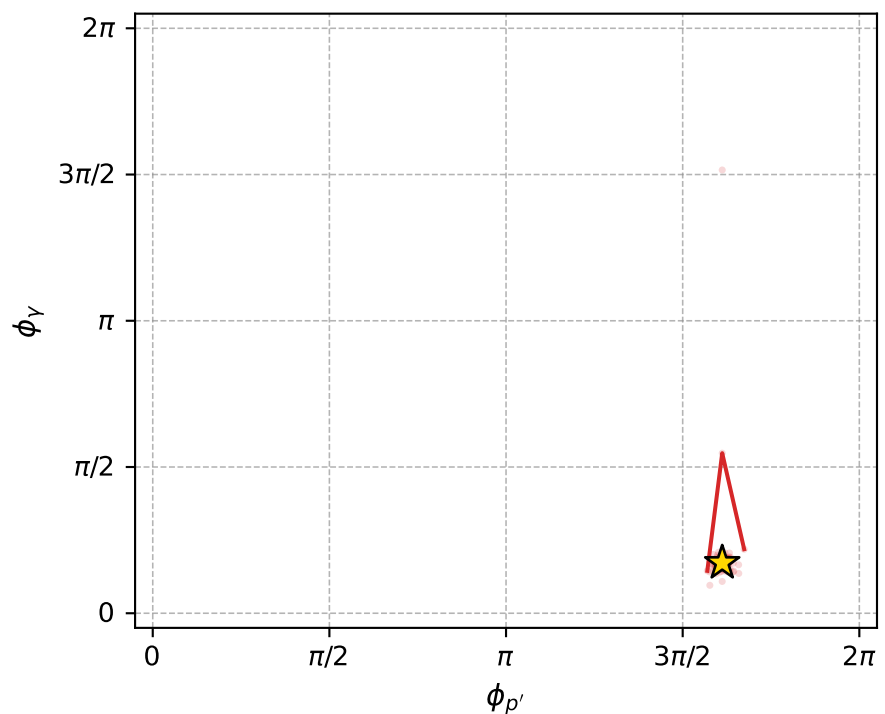
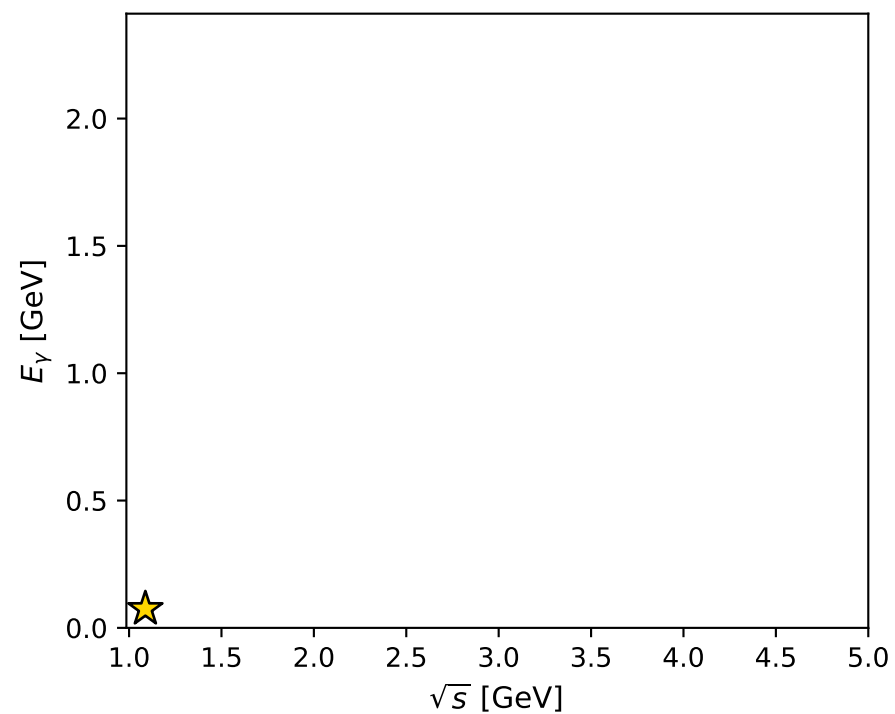
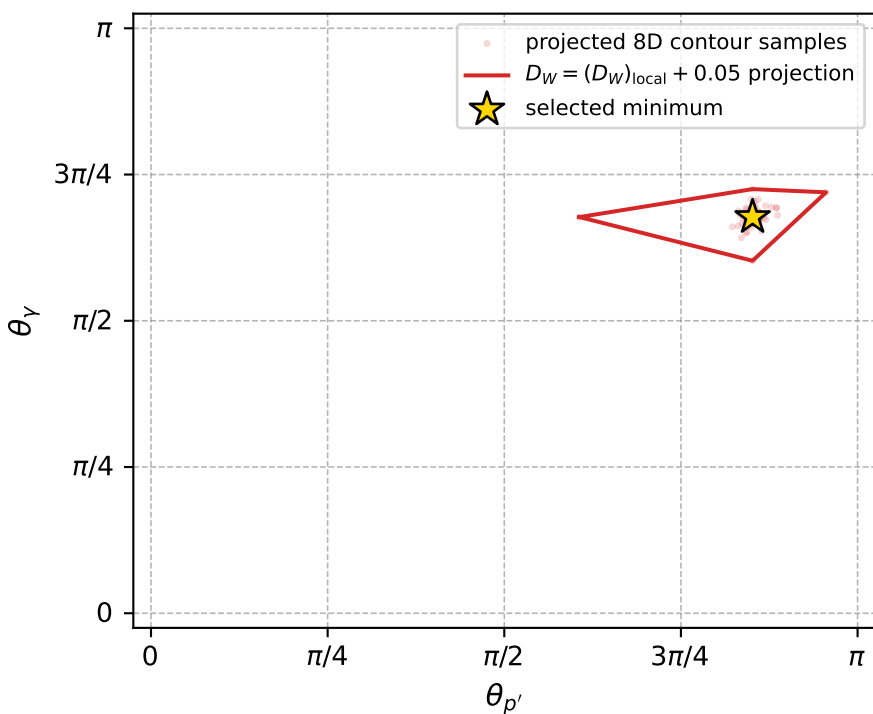
dw_mixing_angles_polarization_06_local_minimum_875 D_W=0.0126931
 region: polarization_cluster_06_local_minimum_875
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3389046$ rad
 incoming lepton: $-0.694776|+\rangle +0.719226|-\rangle$
 proton mixing angle: $\alpha_p=1.4425622$ rad
 incoming proton: $0.127883|+\rangle +0.991789|-\rangle$

$s=1.18423$, $\sqrt{s}=1.08823$, $|\vec{P}|=0.139856$, $|\vec{P}'|=0.00193654$
 $\theta_{P'}=2.67465$, $\theta_V=2.1277$, $\phi_{P'}=5.06392$, $\phi_V=0.54147$
 $E_V=0.0747133$, $\theta_{\ell'}=0.993335$, $\phi_{\ell'}=3.66953$
 $Q^2=0.0326503$, $x_B=0.189061$, $t=-0.0199396$, $W^2=1.01989$, $y=0.567354$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1399$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1399$
 P $E= 0.9484$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1399$
 ℓ' $E= 0.0755$ $p_x= -0.0547$ $p_y= -0.0319$ $p_z= 0.0412$
 P' $E= 0.9380$ $p_x= 0.0003$ $p_y= -0.0008$ $p_z= -0.0017$
 q_Y $E= 0.0747$ $p_x= 0.0544$ $p_y= 0.0327$ $p_z= -0.0395$



electron: polarization cluster P6, local minimum 875; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.012693082$
 $D_W \text{ contour} = 0.062693082$
 above global minimum = 0.012676649
 8D contour samples = 253

selected phase-space point:
 $\text{sqrt_s} = 1.088225$
 $\text{theta_p_out} = 2.674653$
 $\text{theta_gamma_out} = 2.127704$
 $\text{q0out} = 0.07471328$
 $\text{phi_p_out} = 5.06392$
 $\text{phi_gamma_out} = 0.5414696$
 $\text{alpha_e} = 2.338905$
 $\text{alpha_p} = 1.442562$

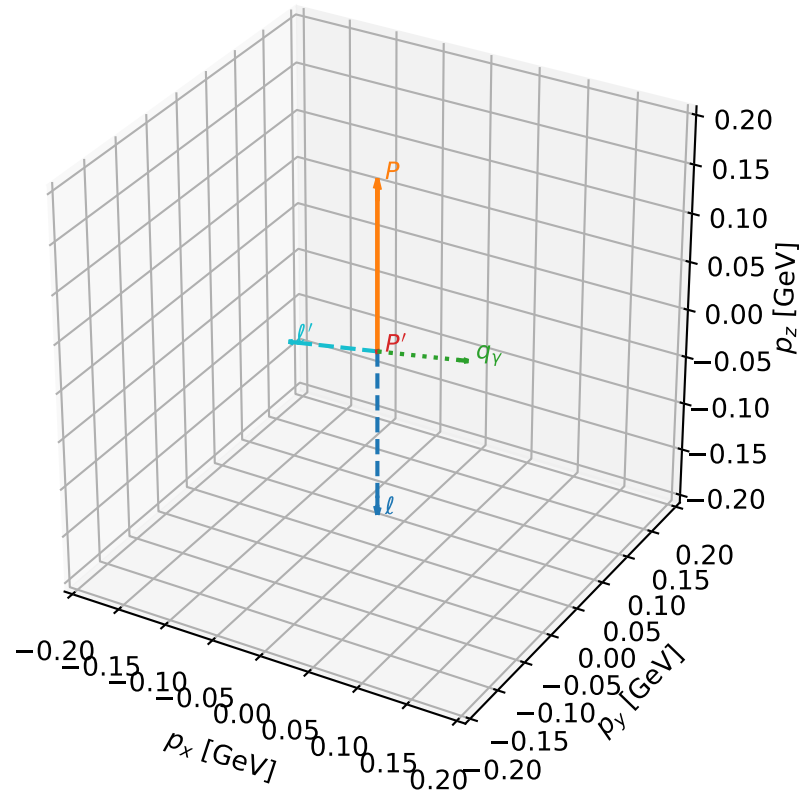
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_886 D_W=0.0130868
region: polarization_cluster_06_local_minimum_886
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3833876$ rad
incoming lepton: $-0.726071|+\rangle +0.687619|-\rangle$
proton mixing angle: $\alpha_p=1.3838509$ rad
incoming proton: $0.185858|+\rangle +0.982577|-\rangle$

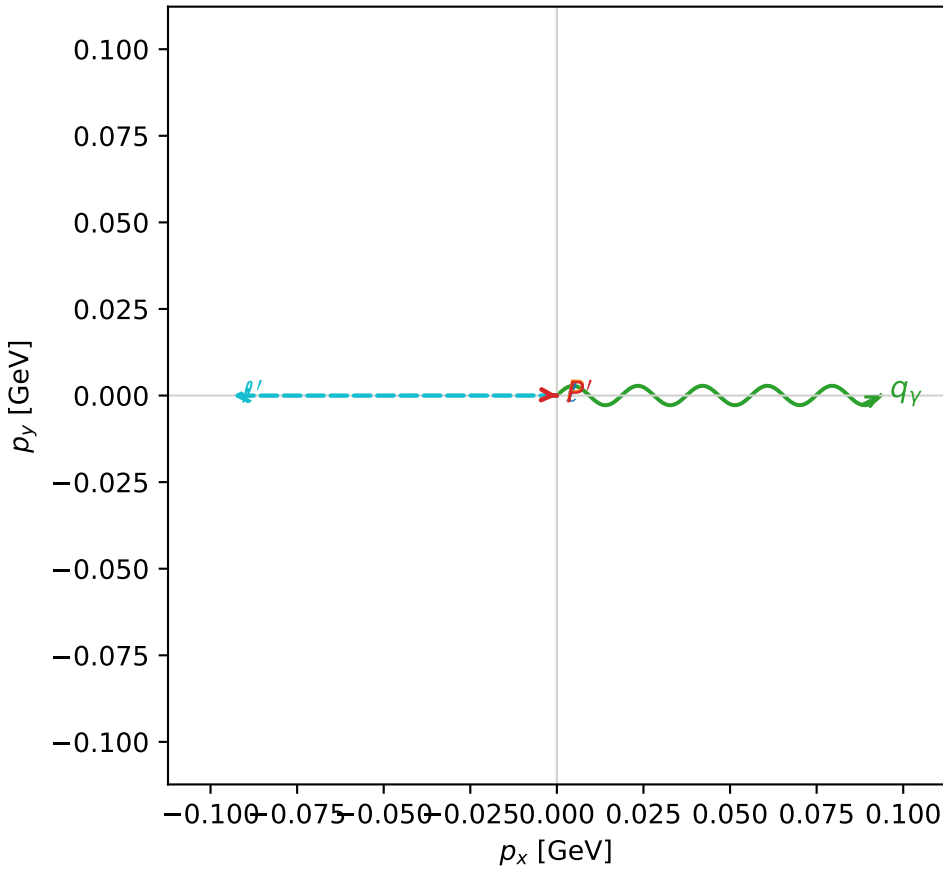
$s=1.27262$, $\sqrt{s}=1.1281$, $|\vec{P}|=0.174086$, $|\vec{P}'|=0.000297533$
 $\theta_{P'}=2.89231$, $\theta_Y=1.39033$, $\phi_{P'}=6.23383$, $\phi_Y=0.000295168$
 $E_Y=0.0950414$, $\theta_{\ell'}=1.74814$, $\phi_{\ell'}=3.14185$
 $Q^2=0.0272592$, $x_B=0.132614$, $t=-0.03015$, $W^2=1.05814$, $y=0.523333$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1741$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1741$
 P $E= 0.9540$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1741$
 ℓ' $E= 0.0951$ $p_x= -0.0936$ $p_y= -0.0000$ $p_z= -0.0168$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= -0.0003$
 q_Y $E= 0.0950$ $p_x= 0.0935$ $p_y= 0.0000$ $p_z= 0.0171$

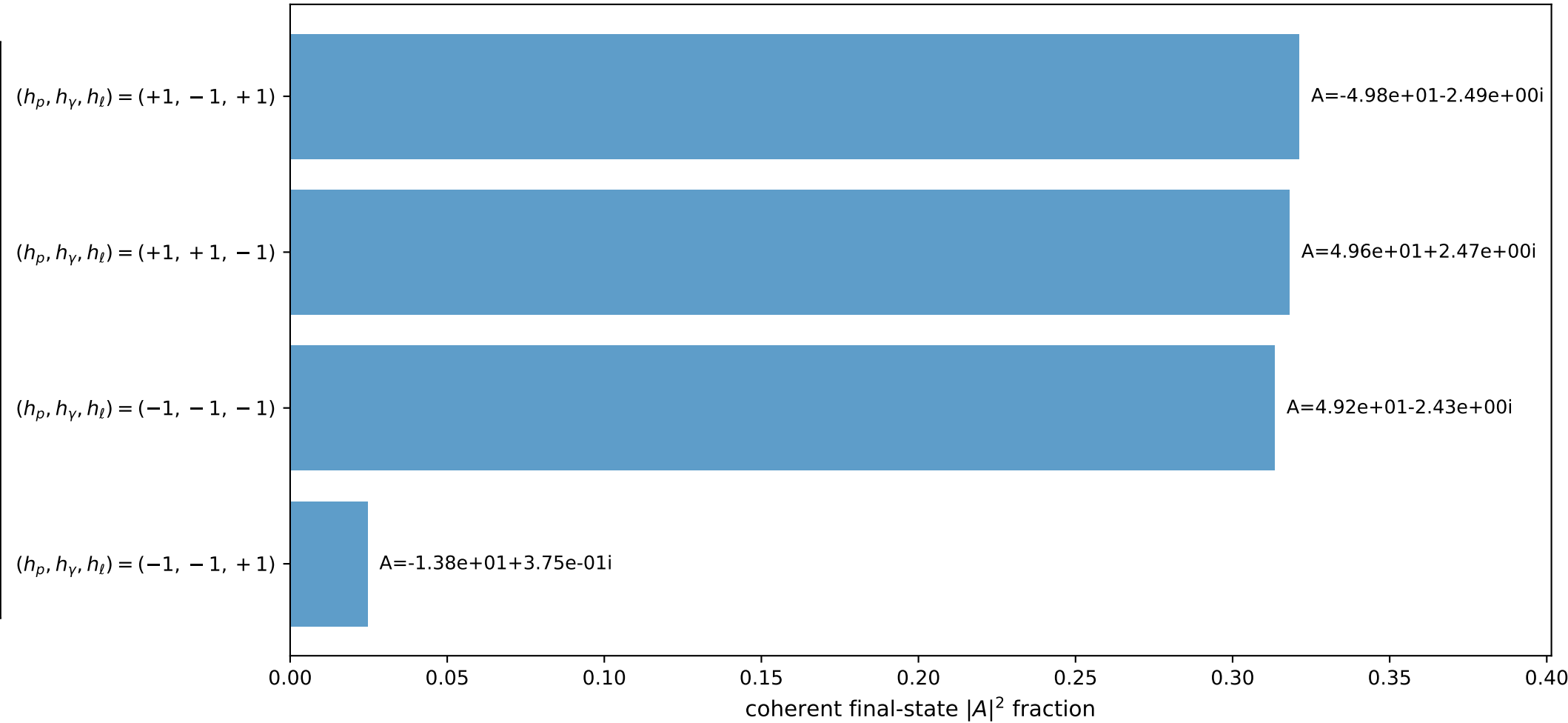
3D momenta



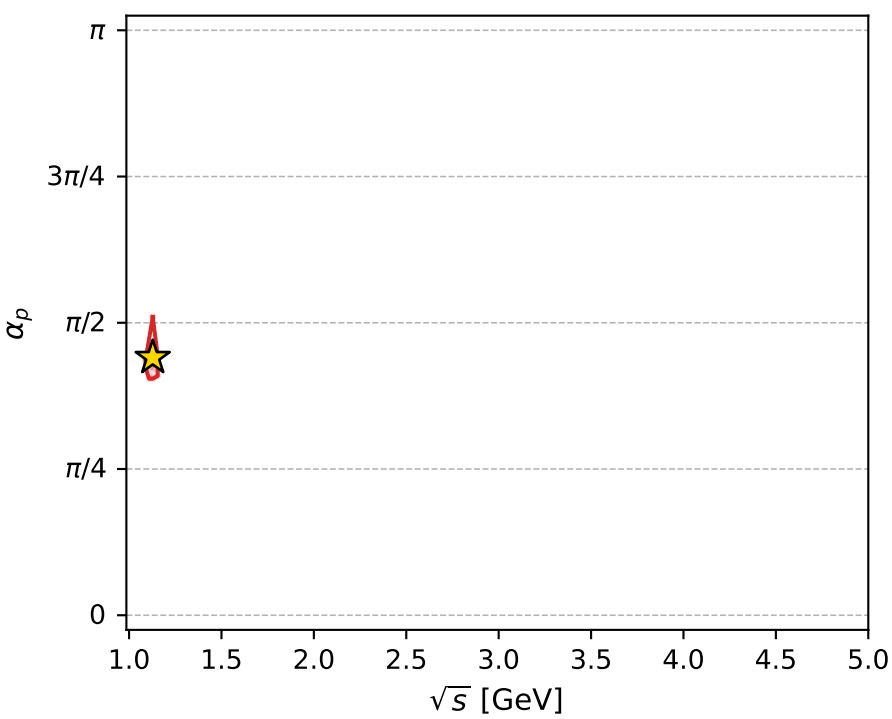
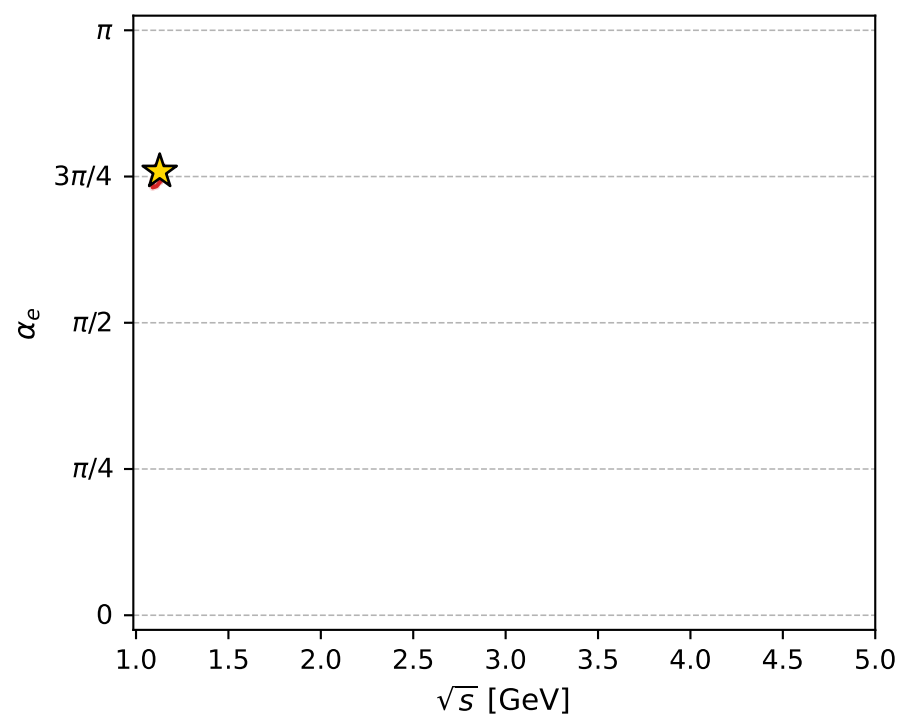
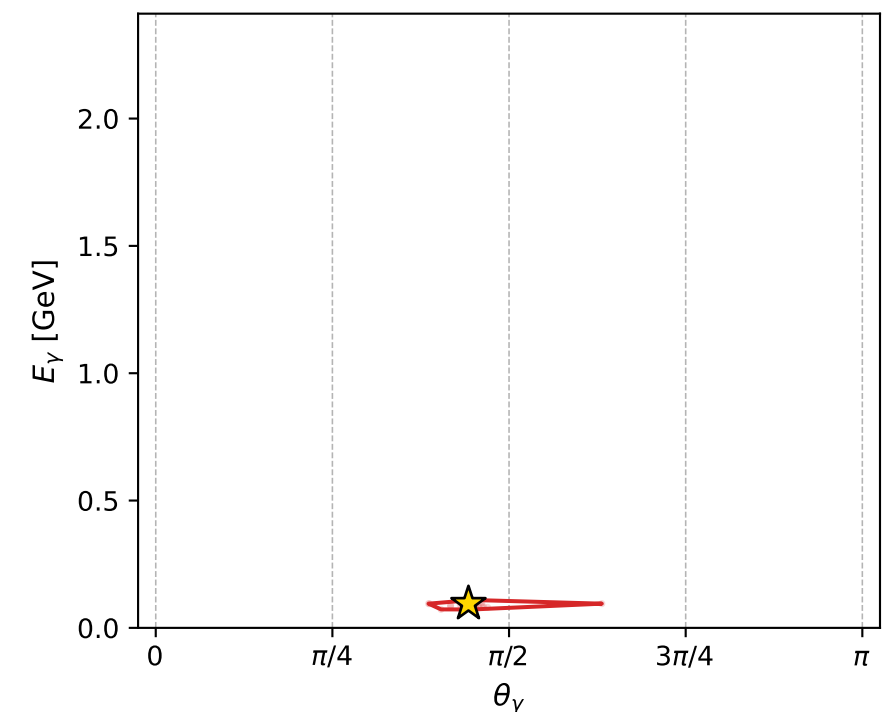
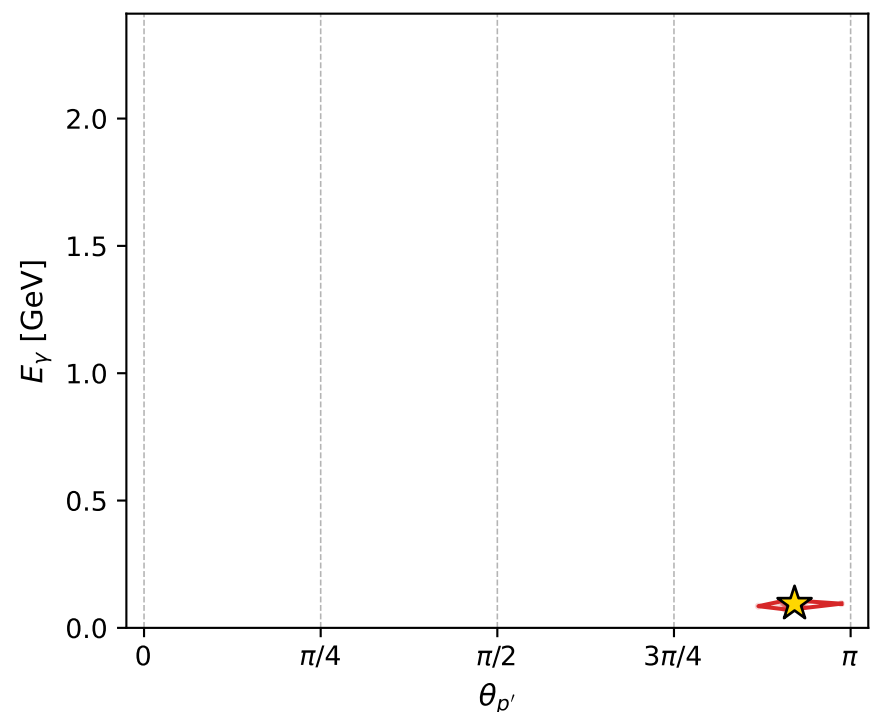
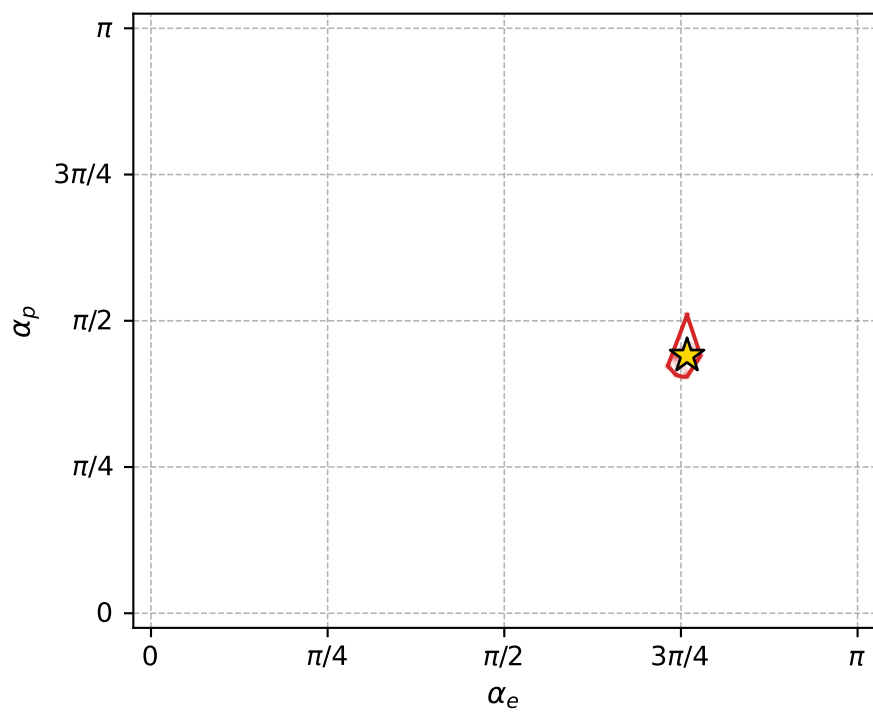
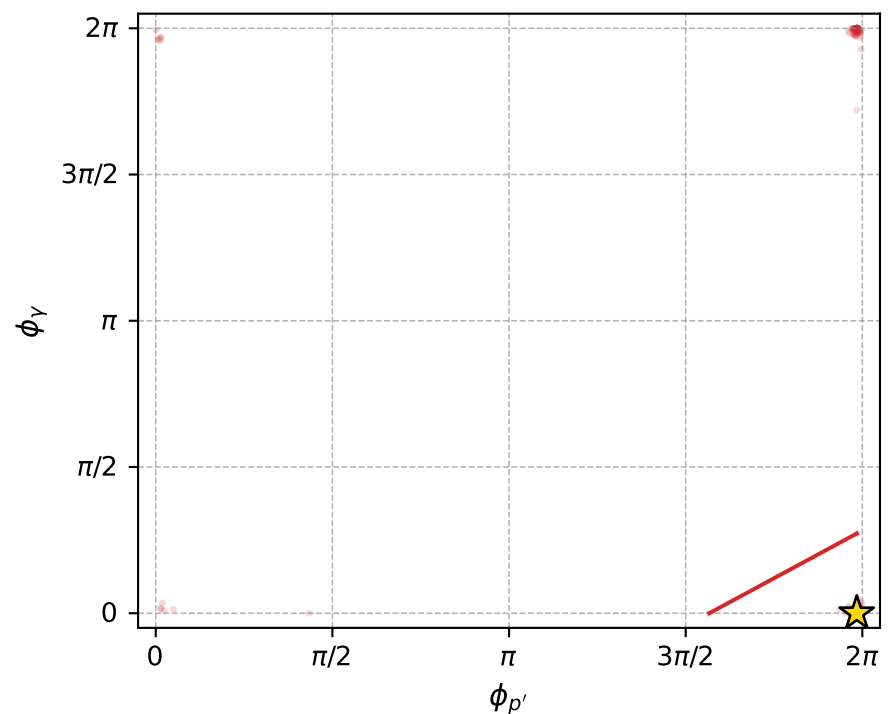
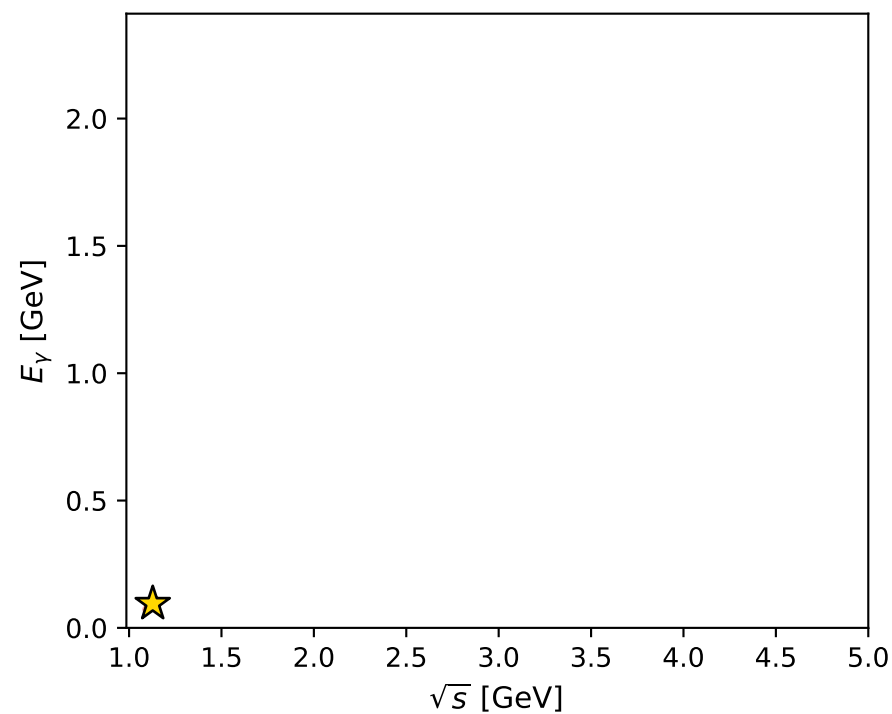
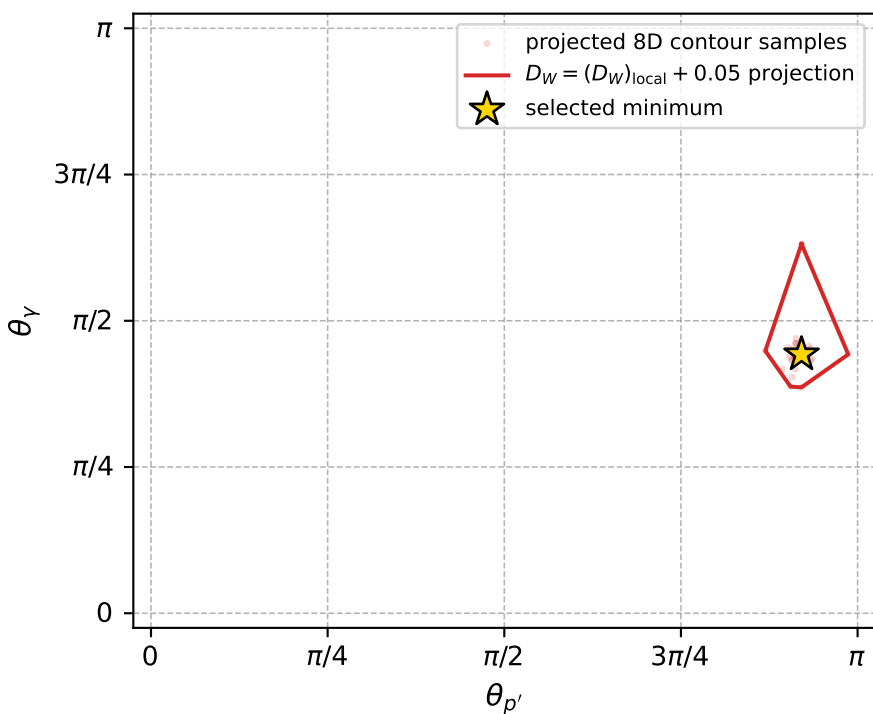
Transverse plane



Leading final-state amplitudes (N=4, retained=97.7%)



electron: polarization cluster P6, local minimum 886; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.013086794$
 $D_W \text{ contour} = 0.063086794$
 above global minimum = 0.013070362
 8D contour samples = 255

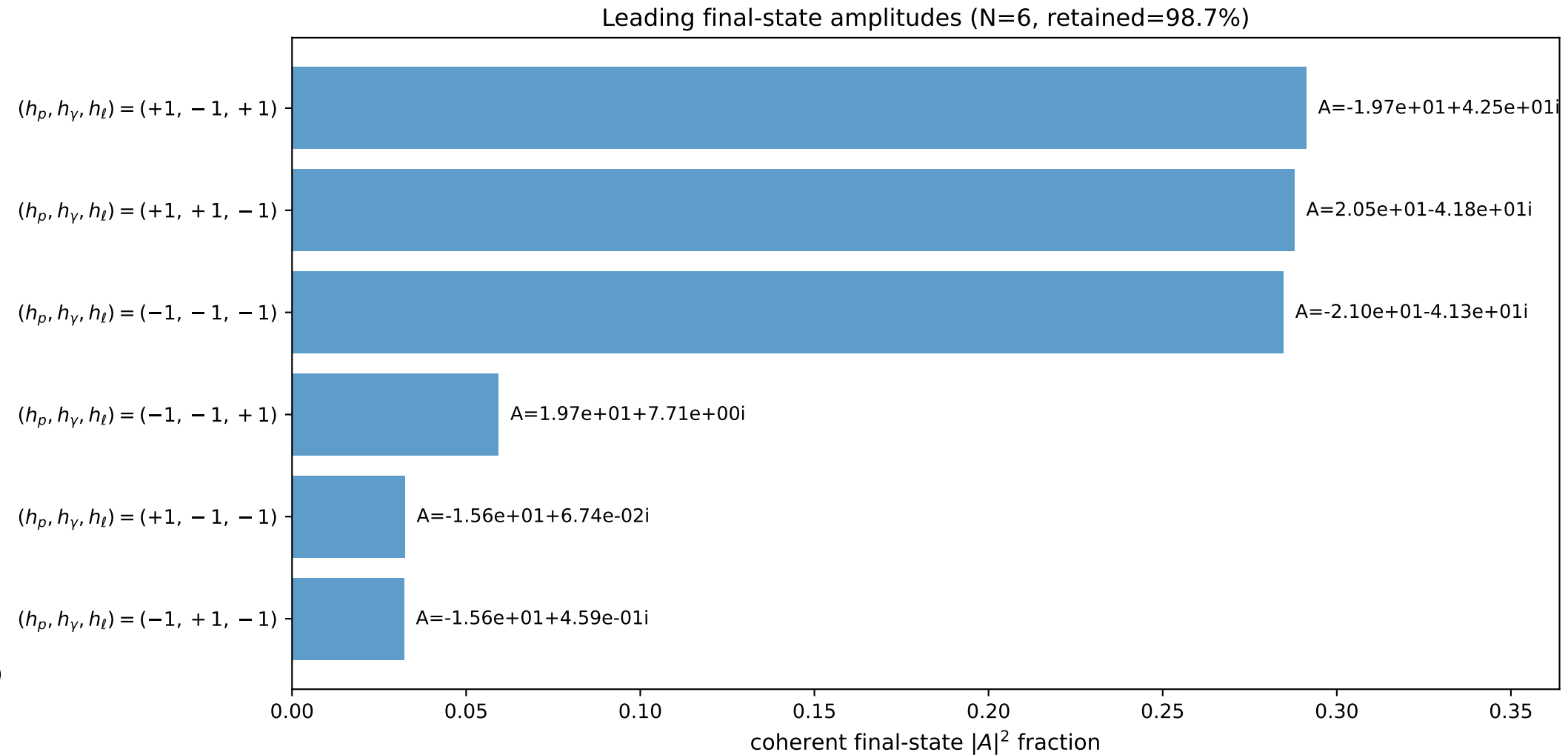
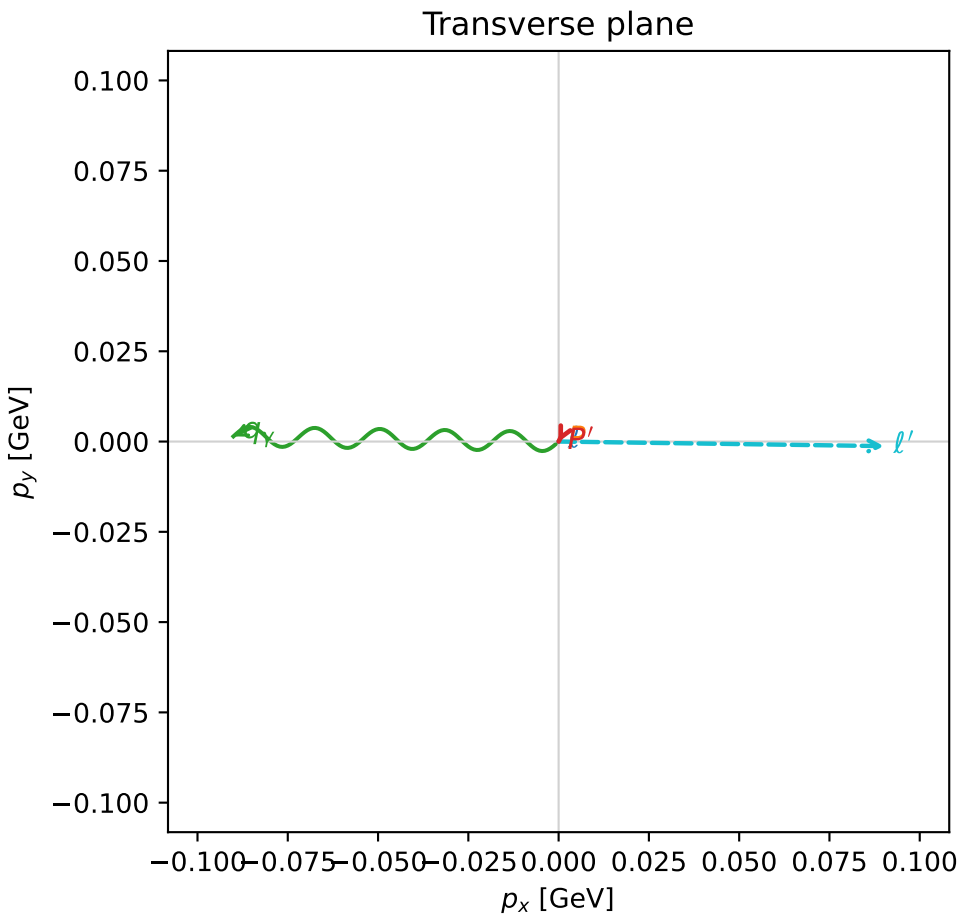
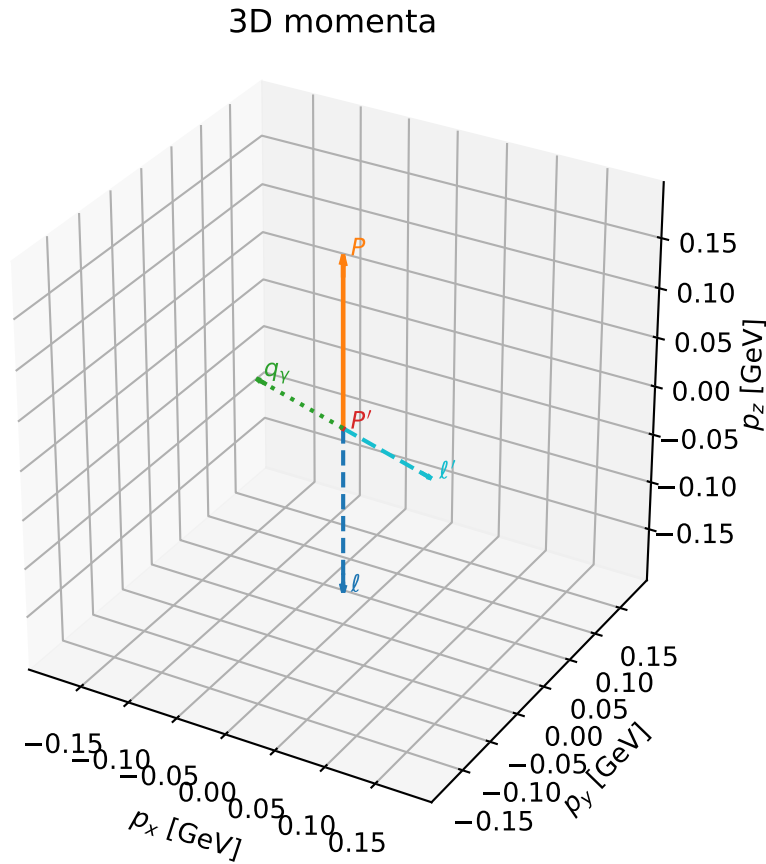
selected phase-space point:
 $\text{sqrt_s} = 1.128105$
 $\text{theta_p_out} = 2.892313$
 $\text{theta_gamma_out} = 1.390331$
 $\text{qOut} = 0.09504136$
 $\text{phi_p_out} = 6.233826$
 $\text{phi_gamma_out} = 0.0002951683$
 $\text{alpha_e} = 2.383388$
 $\text{alpha_p} = 1.383851$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

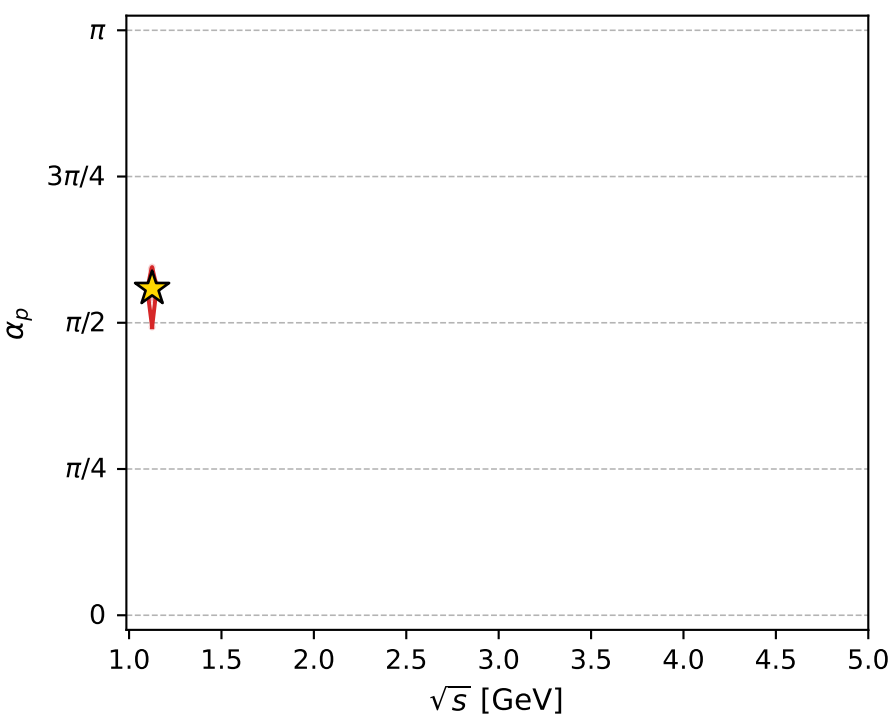
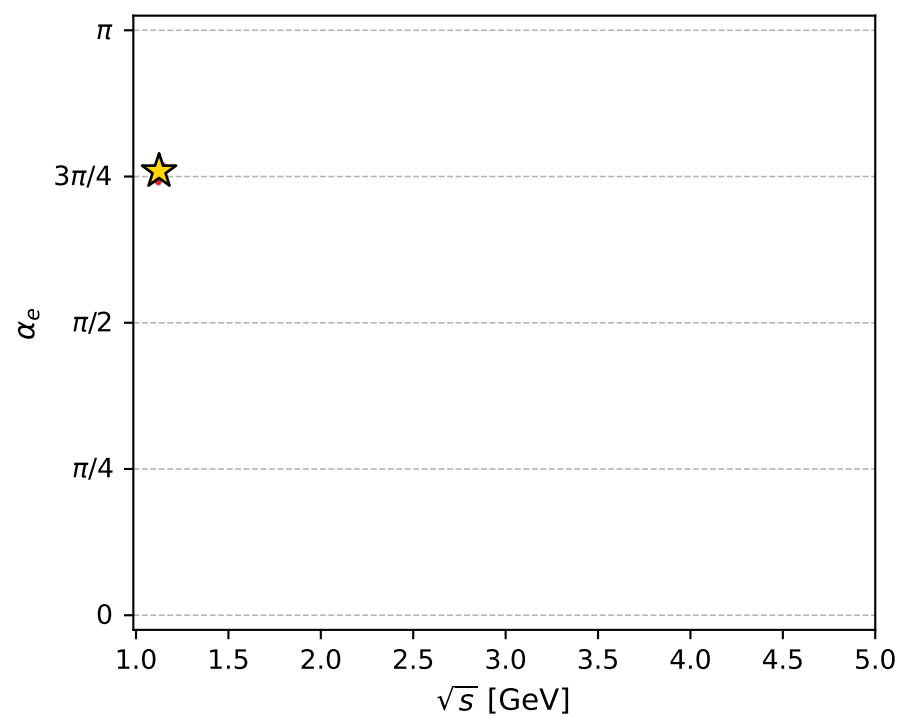
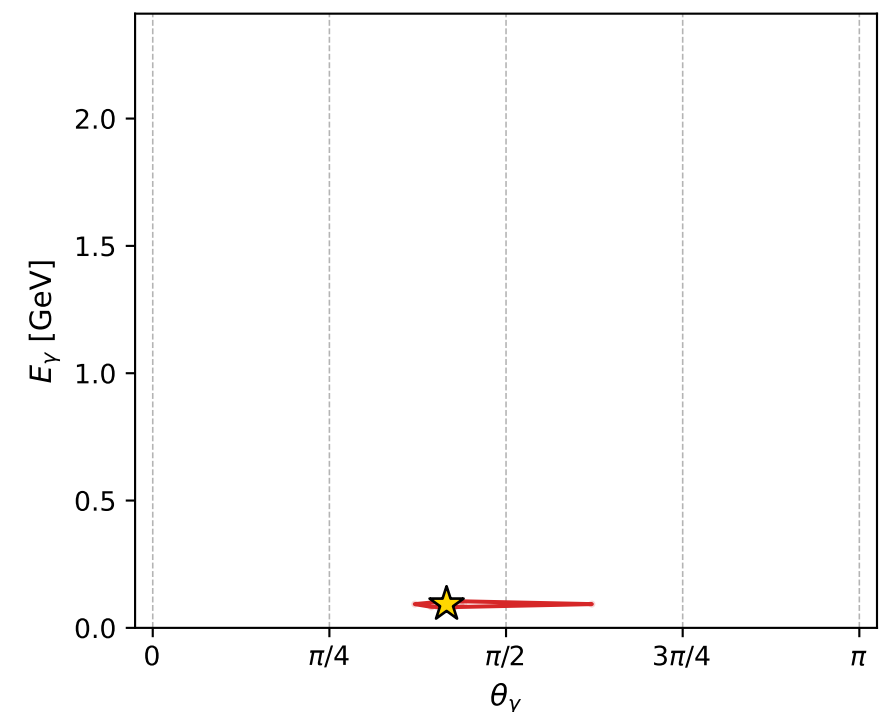
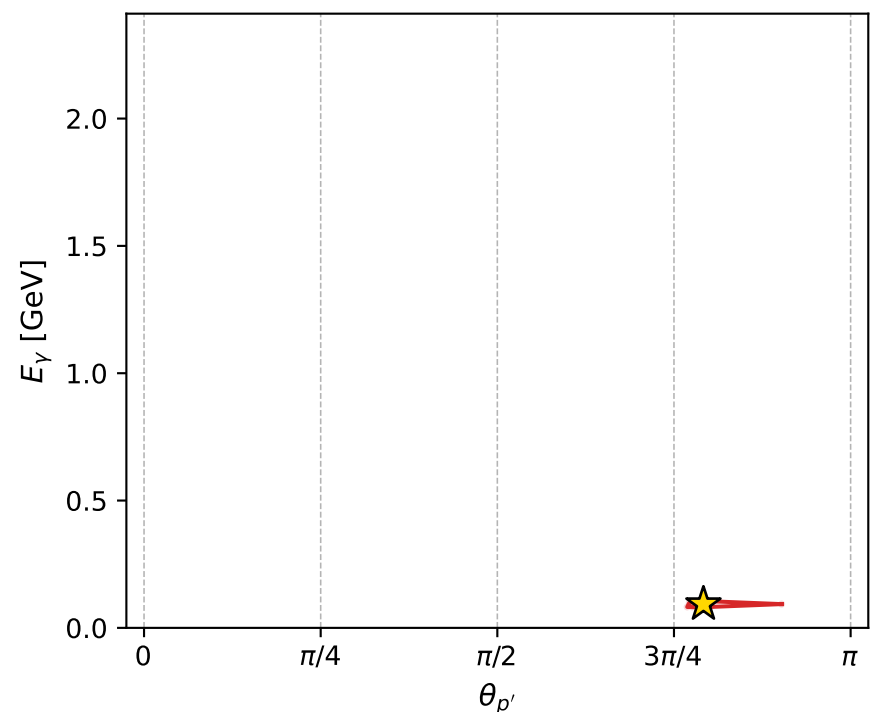
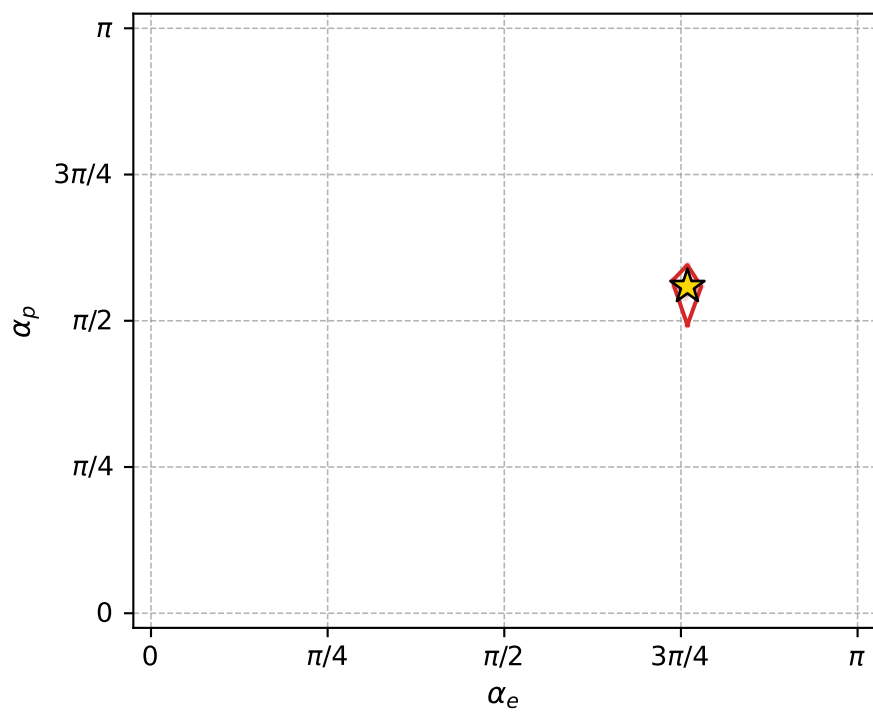
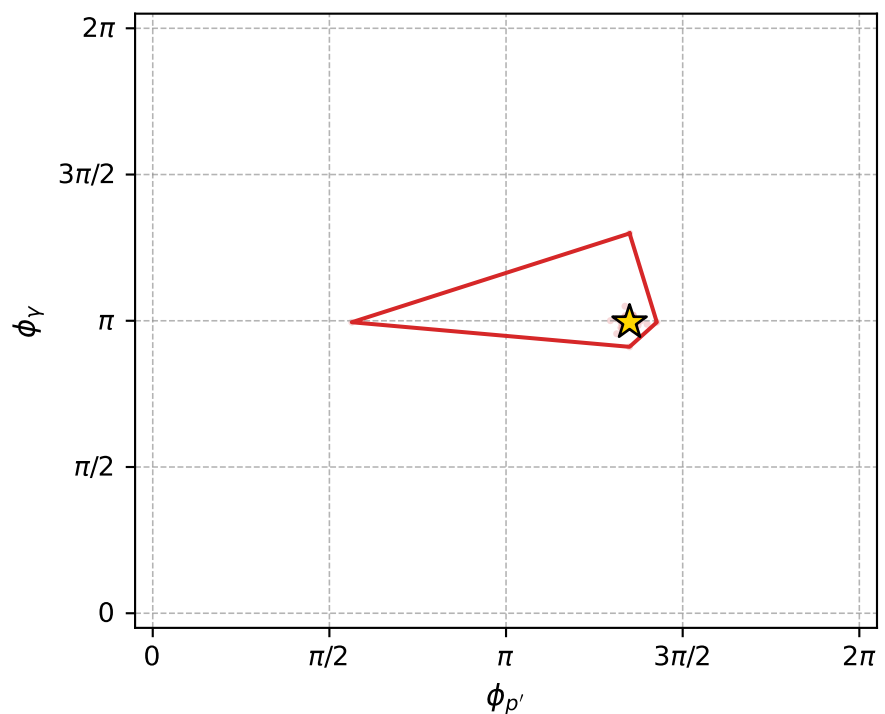
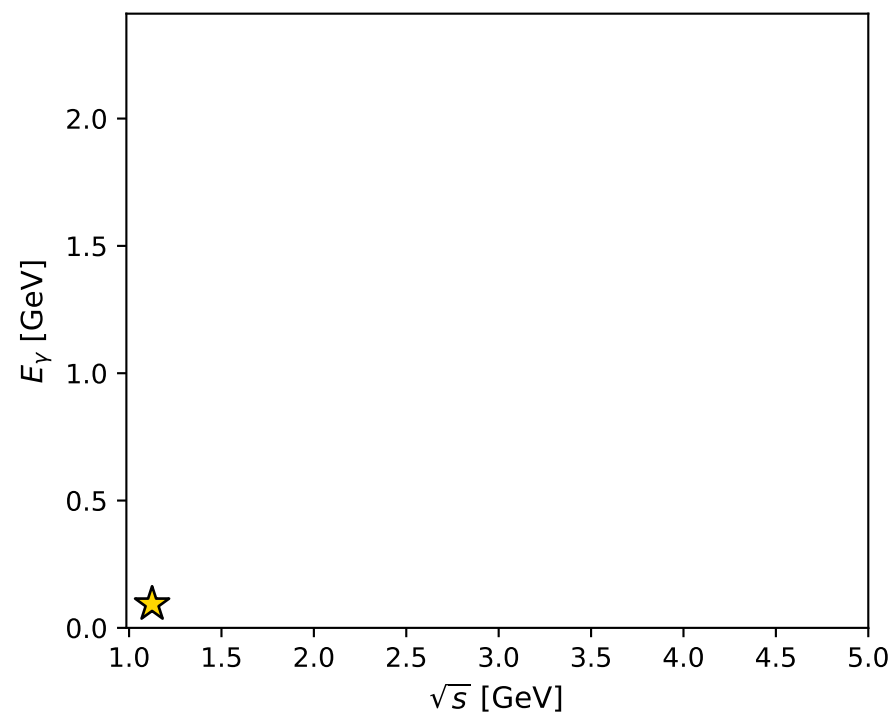
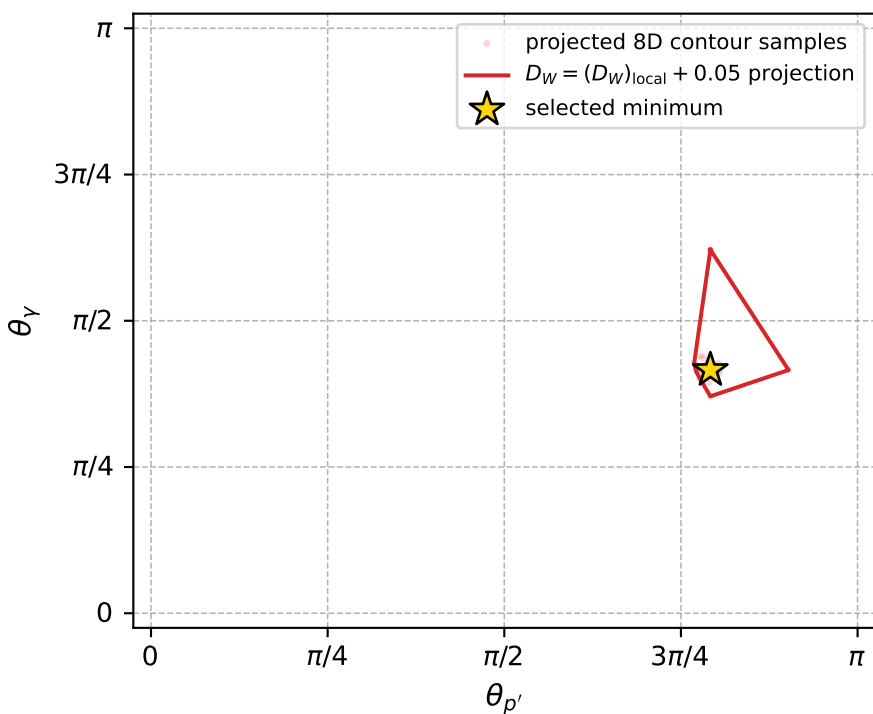
dw_mixing_angles_polarization_06_local_minimum_887 D_W=0.0131829
 region: polarization_cluster_06_local_minimum_887
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.385056$ rad
 incoming lepton: $-0.727218|+\rangle + 0.686407|-\rangle$
 proton mixing angle: $\alpha_p=1.7547248$ rad
 incoming proton: $-0.182893|+\rangle + 0.983133|-\rangle$

$s=1.2648$, $\sqrt{s}=1.12464$, $|\vec{P}|=0.171148$, $|\vec{P}'|=0.000315788$
 $\theta_{P'}=2.48717$, $\theta_V=1.30635$, $\phi_{P'}=4.24002$, $\phi_V=3.12567$
 $E_V=0.0933084$, $\theta_{\ell'}=1.83242$, $\phi_{\ell'}=6.26917$
 $Q^2=0.0236826$, $x_B=0.119172$, $t=-0.0291377$, $W^2=1.05489$, $y=0.516226$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1711$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1711$
 P $E= 0.9535$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1711$
 ℓ' $E= 0.0933$ $p_x= 0.0901$ $p_y= -0.0013$ $p_z= -0.0241$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0002$ $p_z= -0.0003$
 q_V $E= 0.0933$ $p_x= -0.0901$ $p_y= 0.0014$ $p_z= 0.0244$



electron: polarization cluster P6, local minimum 887; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.013182936$
 $D_W \text{ contour} = 0.063182936$
 above global minimum = 0.013166503
 8D contour samples = 255

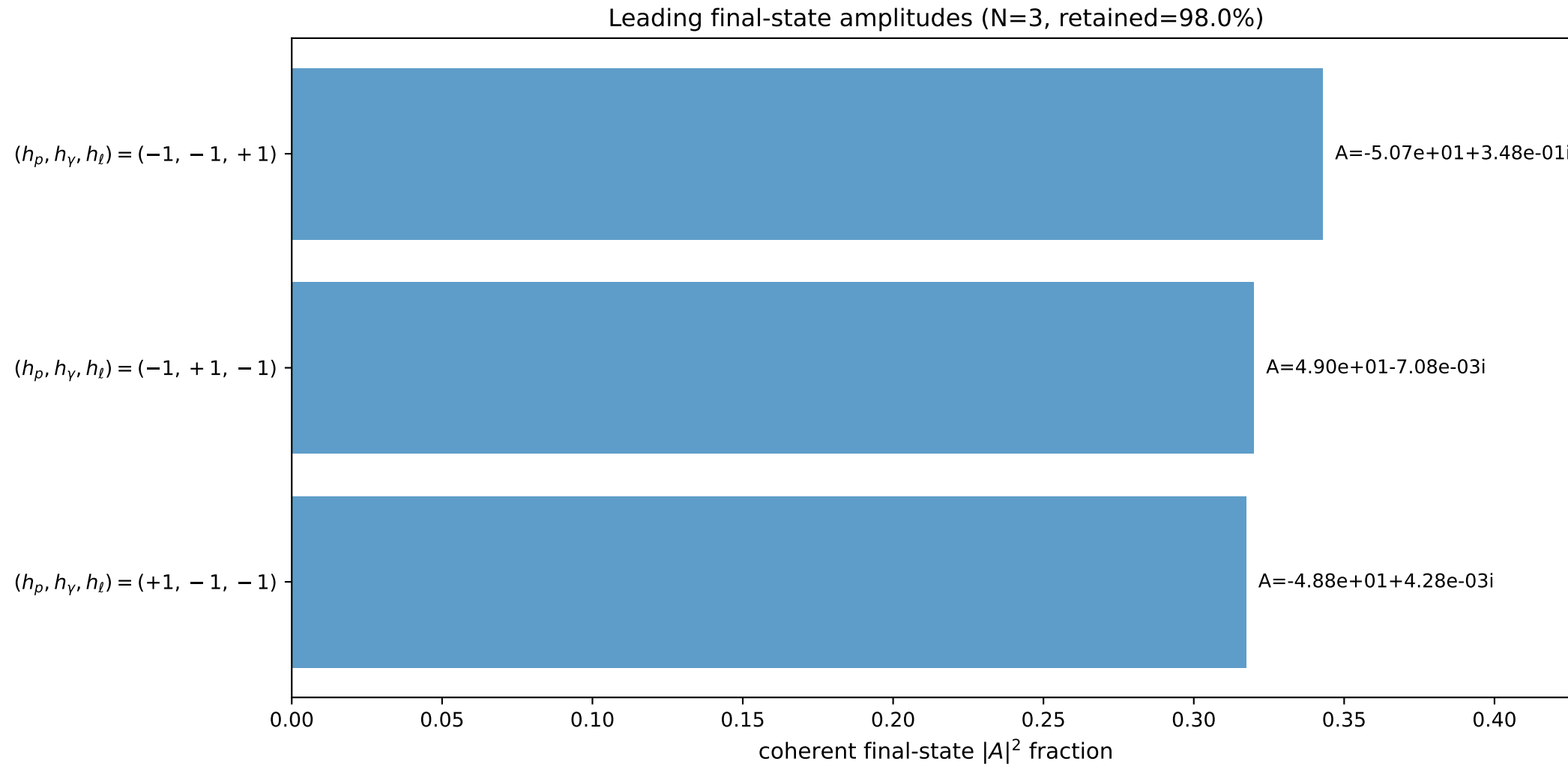
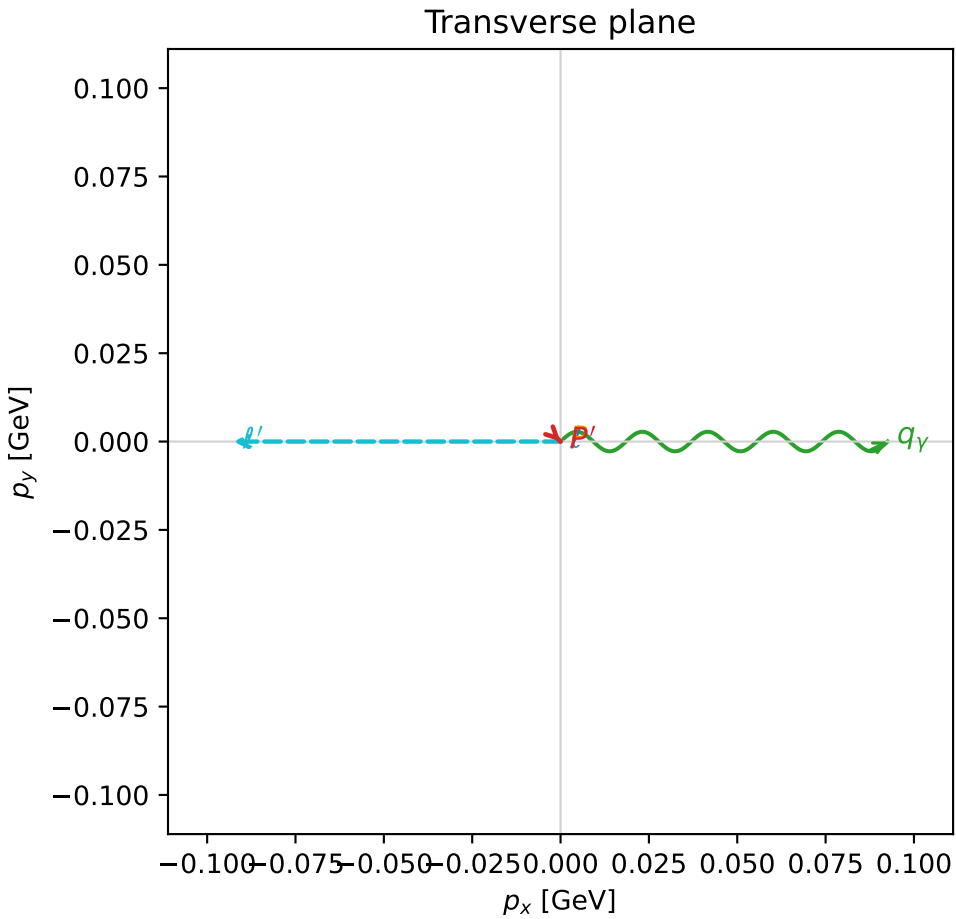
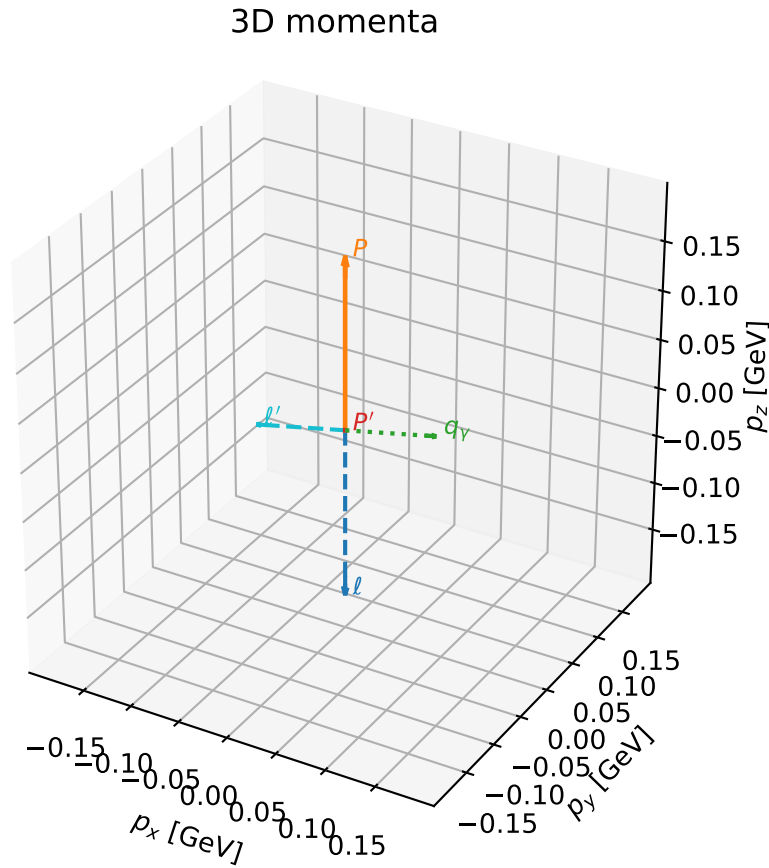
selected phase-space point:
 $\text{sqrt_s} = 1.124635$
 $\text{theta_p_out} = 2.48717$
 $\text{theta_gamma_out} = 1.306348$
 $\text{qOut} = 0.09330838$
 $\text{phi_p_out} = 4.24002$
 $\text{phi_gamma_out} = 3.125667$
 $\text{alpha_e} = 2.385056$
 $\text{alpha_p} = 1.754725$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

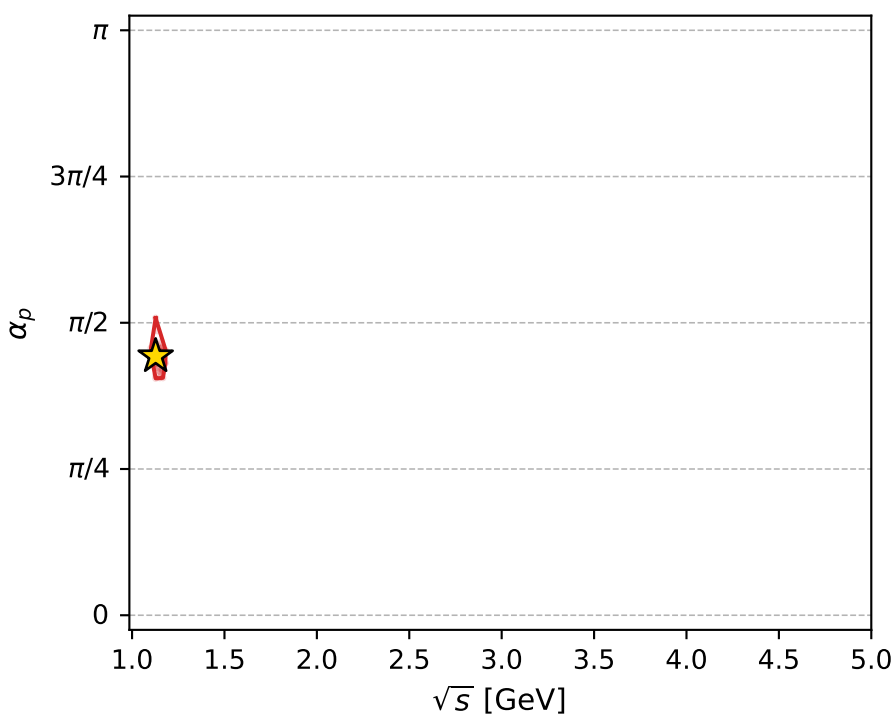
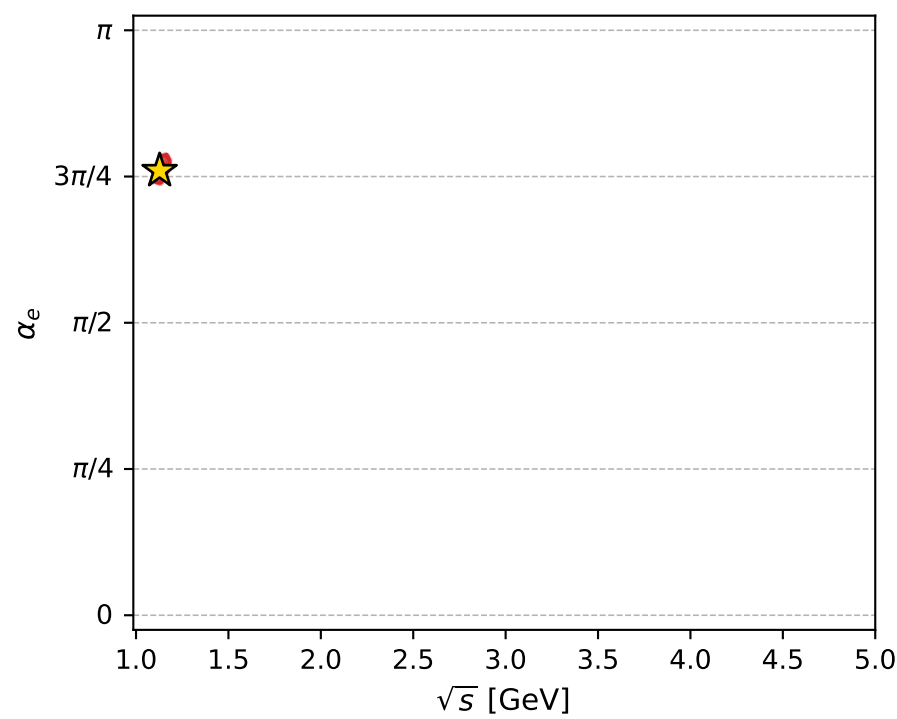
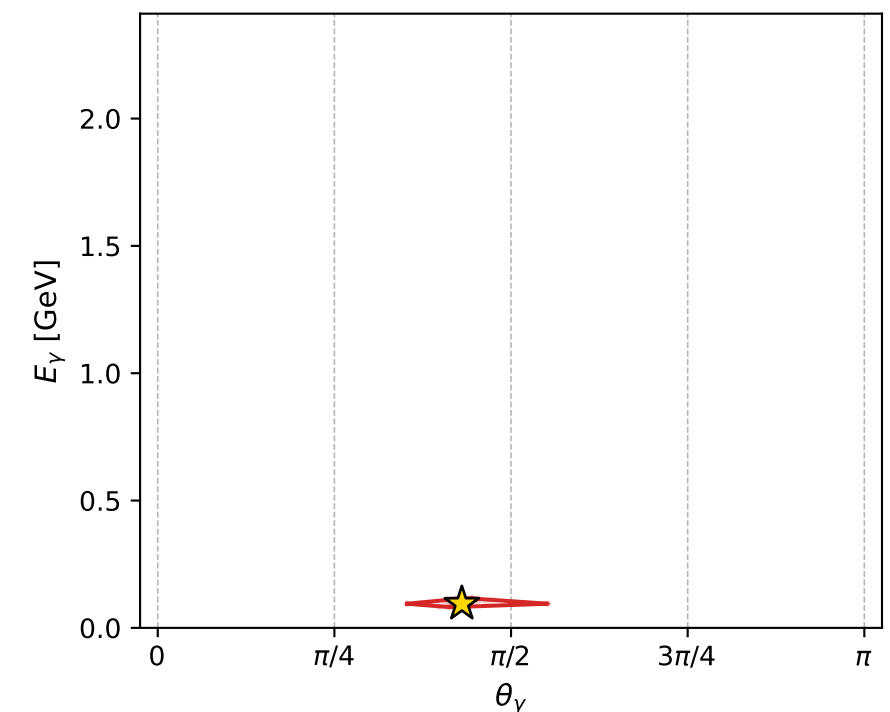
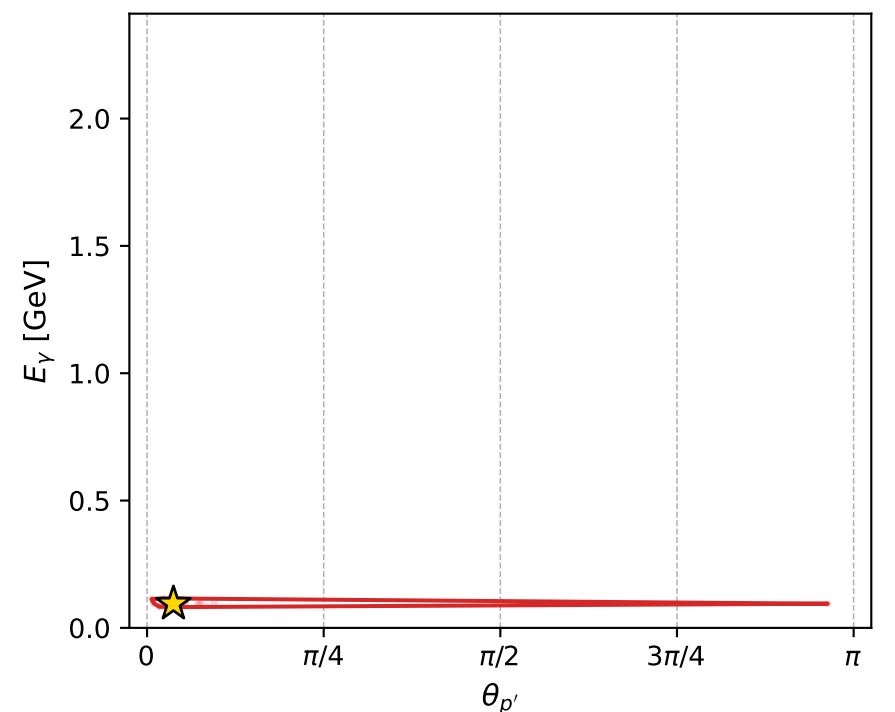
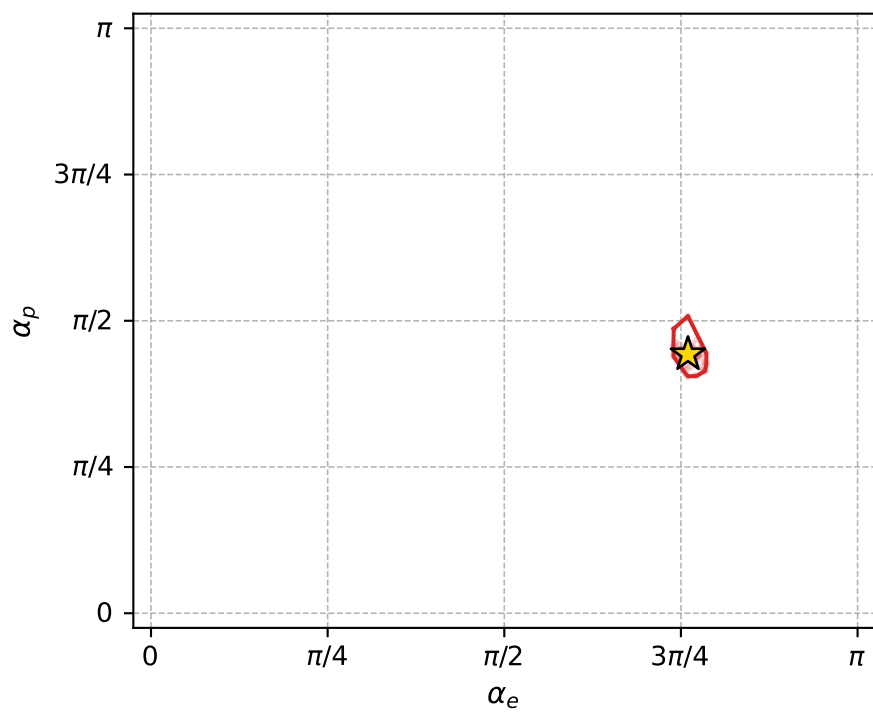
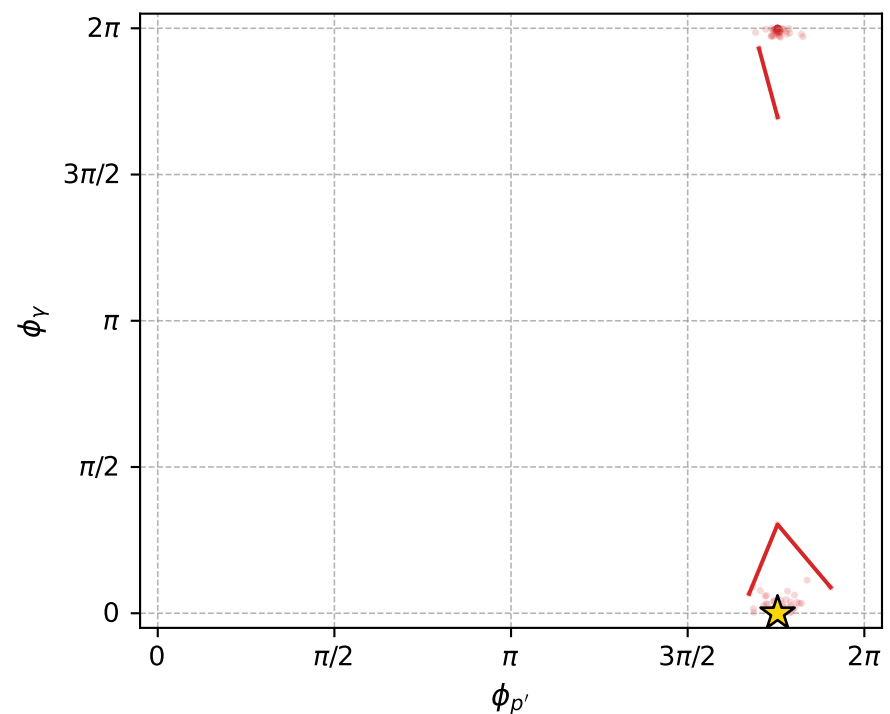
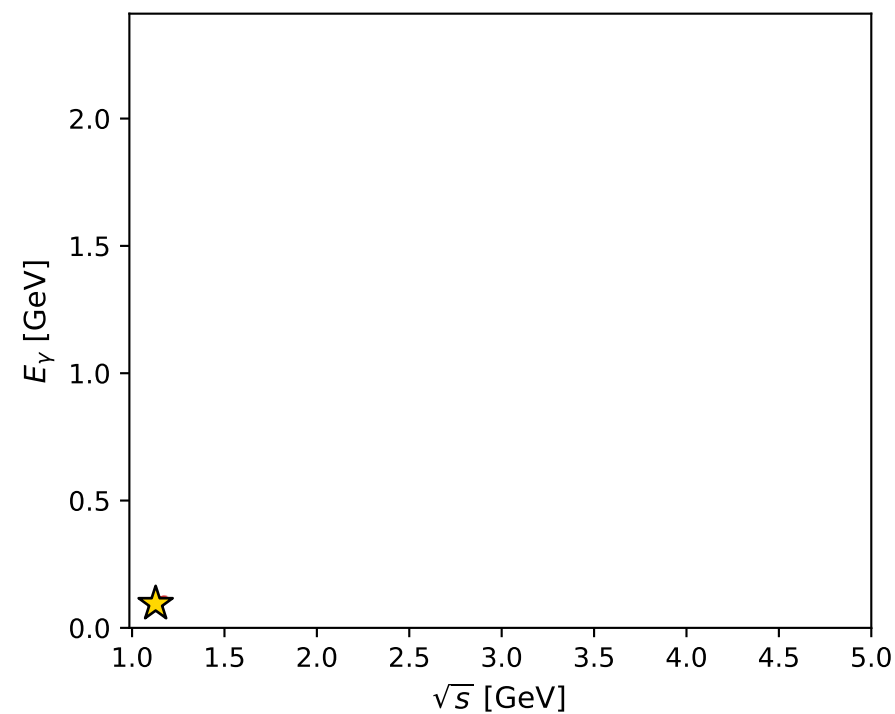
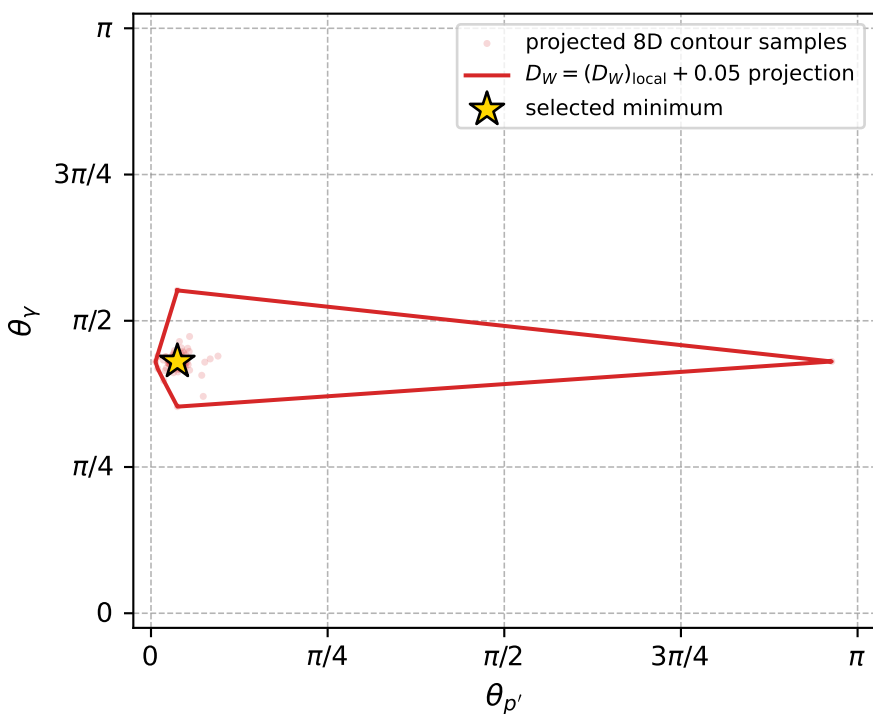
dw_mixing_angles_polarization_06_local_minimum_890 D_W=0.0133474
 region: polarization_cluster_06_local_minimum_890
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3874645$ rad
 incoming lepton: $-0.728869|+> +0.684653|->$
 proton mixing angle: $\alpha_p=1.3908398$ rad
 incoming proton: $0.178987|+> +0.983851|->$

$s=1.2716$, $\sqrt{s}=1.12765$, $|\vec{P}|=0.173702$, $|\vec{P}'|=8.55738e-06$
 $\theta_{p'}=0.117307$, $\theta_Y=1.3521$, $\phi_{p'}=5.51155$, $\phi_Y=8.8004e-05$
 $E_Y=0.0948232$, $\theta_{\ell'}=1.78958$, $\phi_{\ell'}=3.14167$
 $Q^2=0.0257928$, $x_B=0.126634$, $t=-0.029915$, $W^2=1.05773$, $y=0.519924$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1737$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1737$
 P $E= 0.9539$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1737$
 ℓ' $E= 0.0948$ $p_x= -0.0926$ $p_y= -0.0000$ $p_z= -0.0206$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0000$
 q_Y $E= 0.0948$ $p_x= 0.0926$ $p_y= 0.0000$ $p_z= 0.0206$



electron: polarization cluster P6, local minimum 890; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.013347384$
 $D_W \text{ contour} = 0.063347384$
 above global minimum = 0.013330951
 8D contour samples = 253

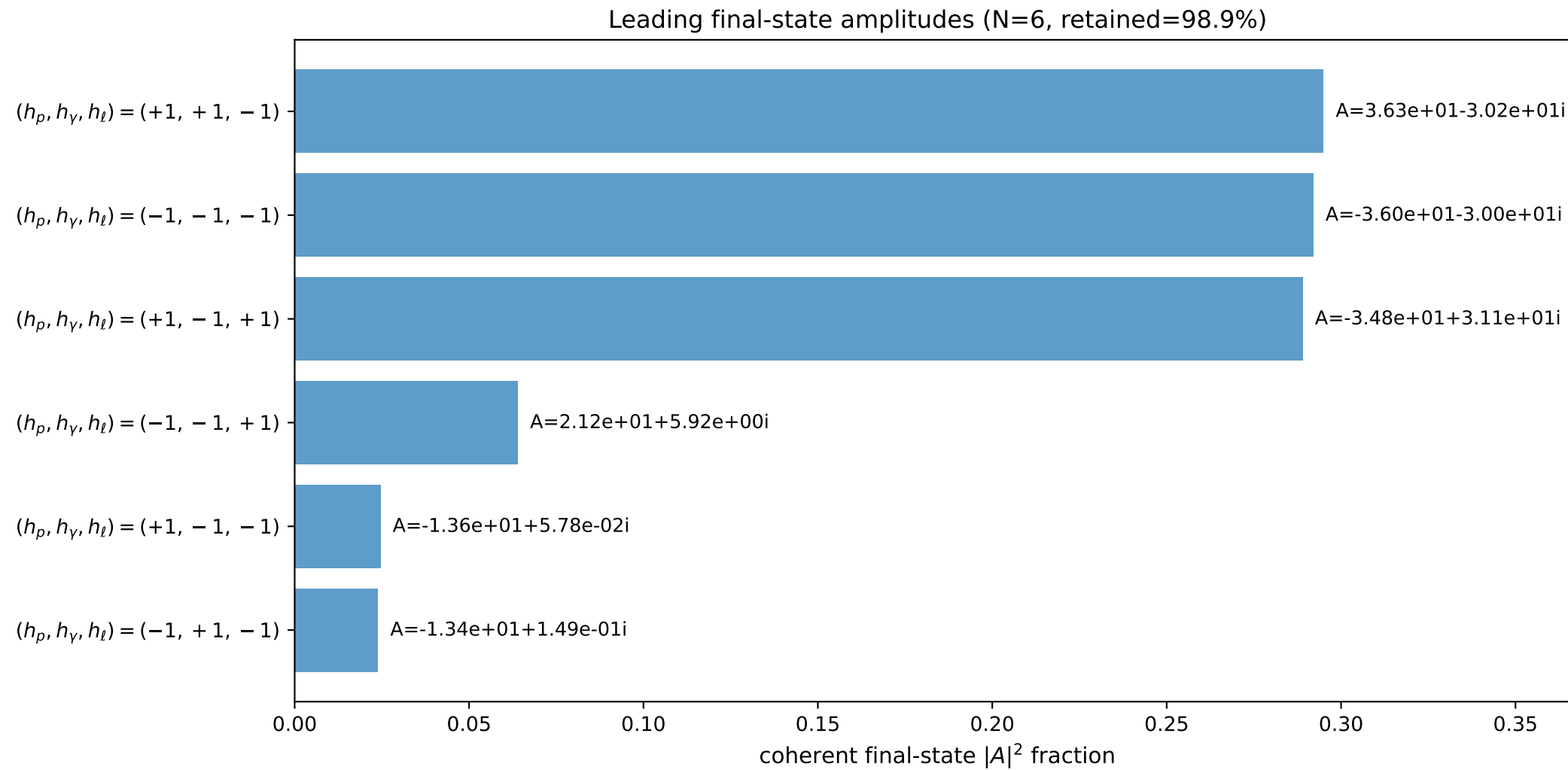
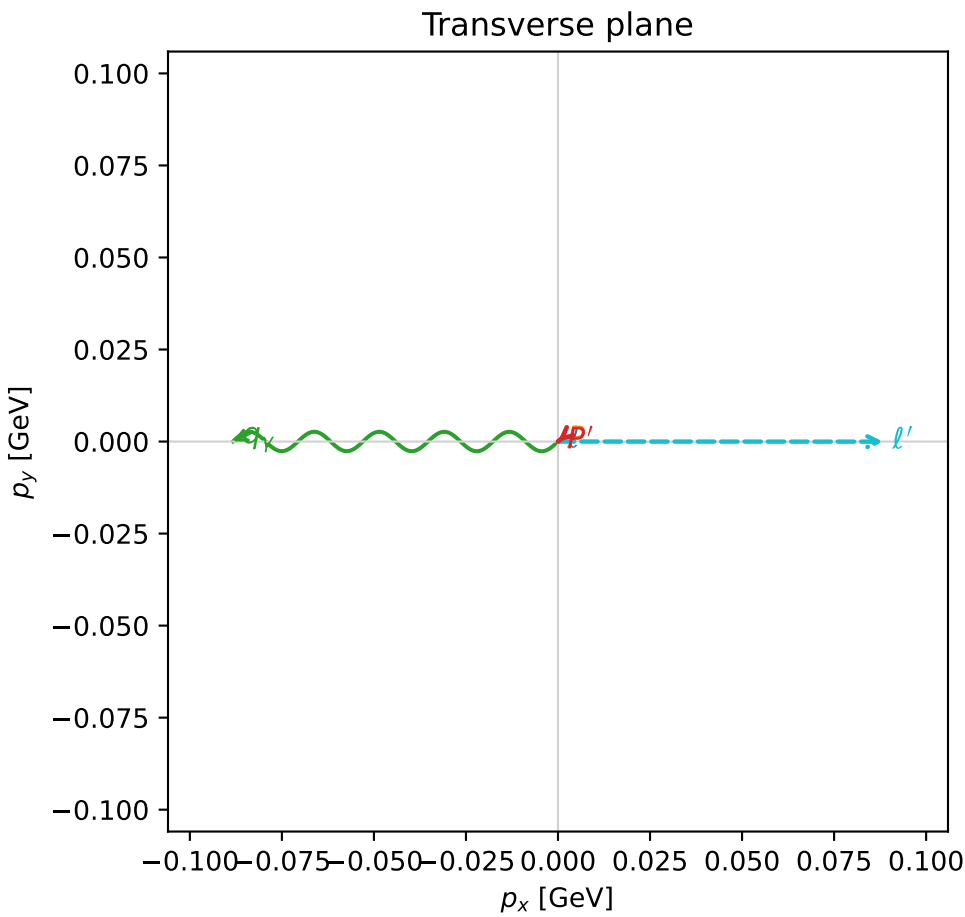
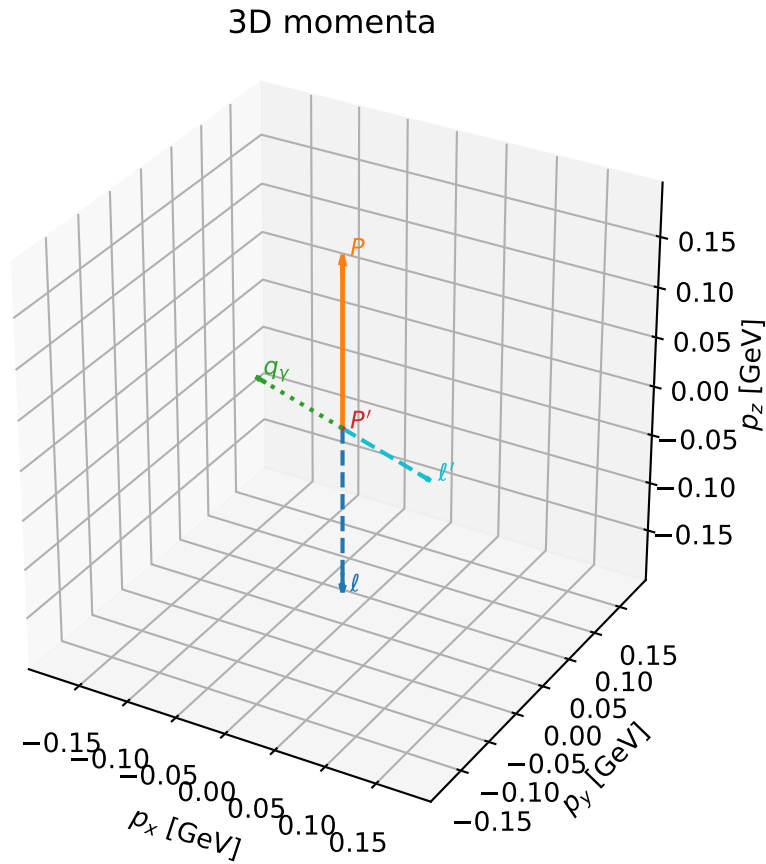
selected phase-space point:
 $\text{sqrt_s} = 1.12765$
 $\text{theta_p_out} = 0.1173065$
 $\text{theta_gamma_out} = 1.352095$
 $\text{qOut} = 0.09482317$
 $\text{phi_p_out} = 5.511549$
 $\text{phi_gamma_out} = 8.800403\text{e-}05$
 $\text{alpha_e} = 2.387464$
 $\text{alpha_p} = 1.39084$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

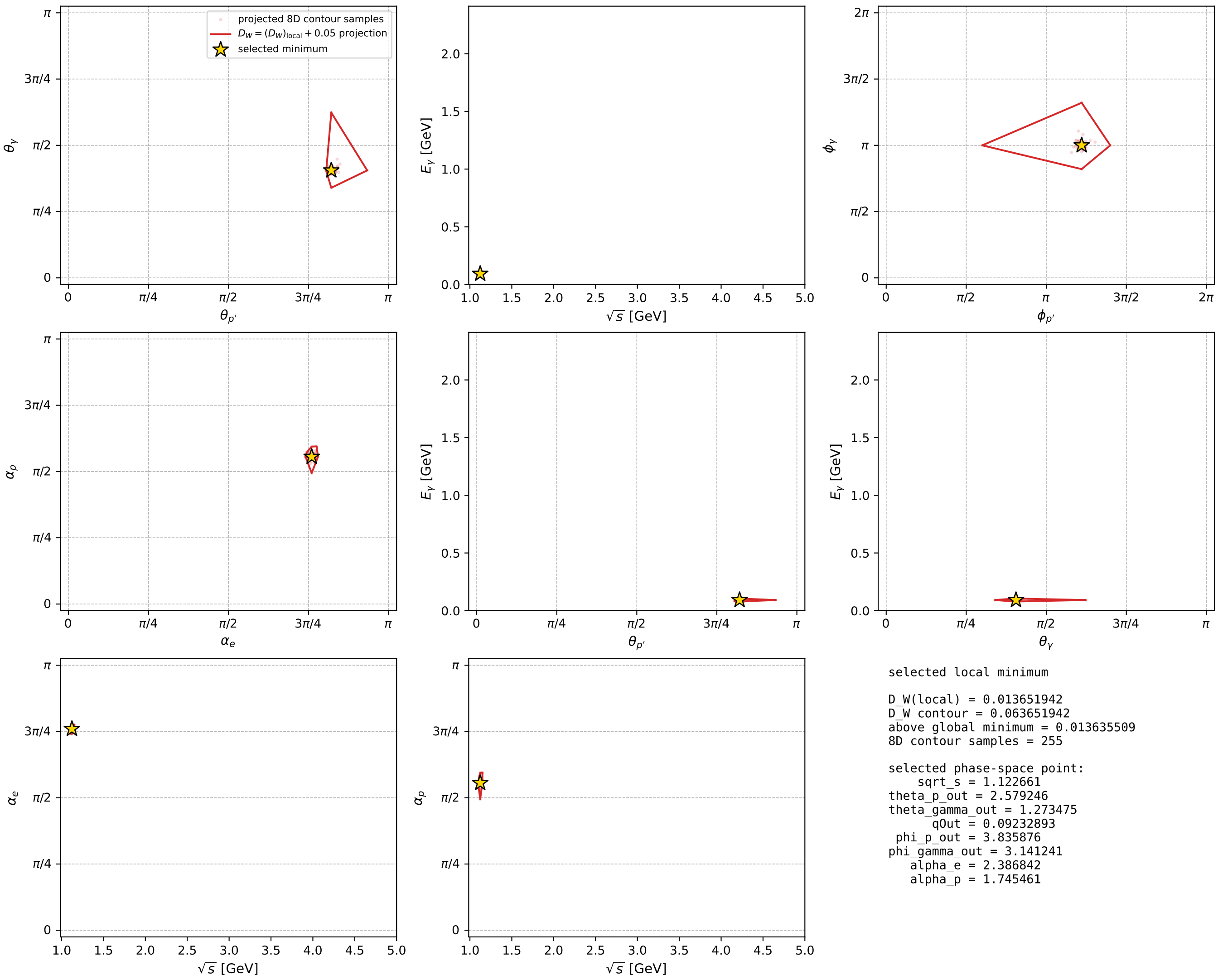
dw_mixing_angles_polarization_06_local_minimum_903 D_W=0.0136519
 region: polarization_cluster_06_local_minimum_903
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3868424$ rad
 incoming lepton: $-0.728443|+> +0.685107|->$
 proton mixing angle: $\alpha_p=1.7454607$ rad
 incoming proton: $-0.173778|+> +0.984785|->$

$s=1.26037$, $\sqrt{s}=1.12266$, $|\vec{P}|=0.169473$, $|\vec{P}'|=9.43393\text{e-}06$
 $\theta_{P'}=2.57925$, $\theta_V=1.27347$, $\phi_{P'}=3.83588$, $\phi_V=3.14124$
 $E_V=0.0923289$, $\theta_{\ell'}=1.86802$, $\phi_{\ell'}=6.28287$
 $Q^2=0.0221297$, $x_B=0.113289$, $t=-0.0284932$, $W^2=1.05305$, $y=0.513343$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1695$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1695$
 P $E= 0.9532$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1695$
 ℓ' $E= 0.0923$ $p_x= 0.0883$ $p_y= -0.0000$ $p_z= -0.0270$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_V $E= 0.0923$ $p_x= -0.0883$ $p_y= 0.0000$ $p_z= 0.0270$



electron: polarization cluster P6, local minimum 903; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

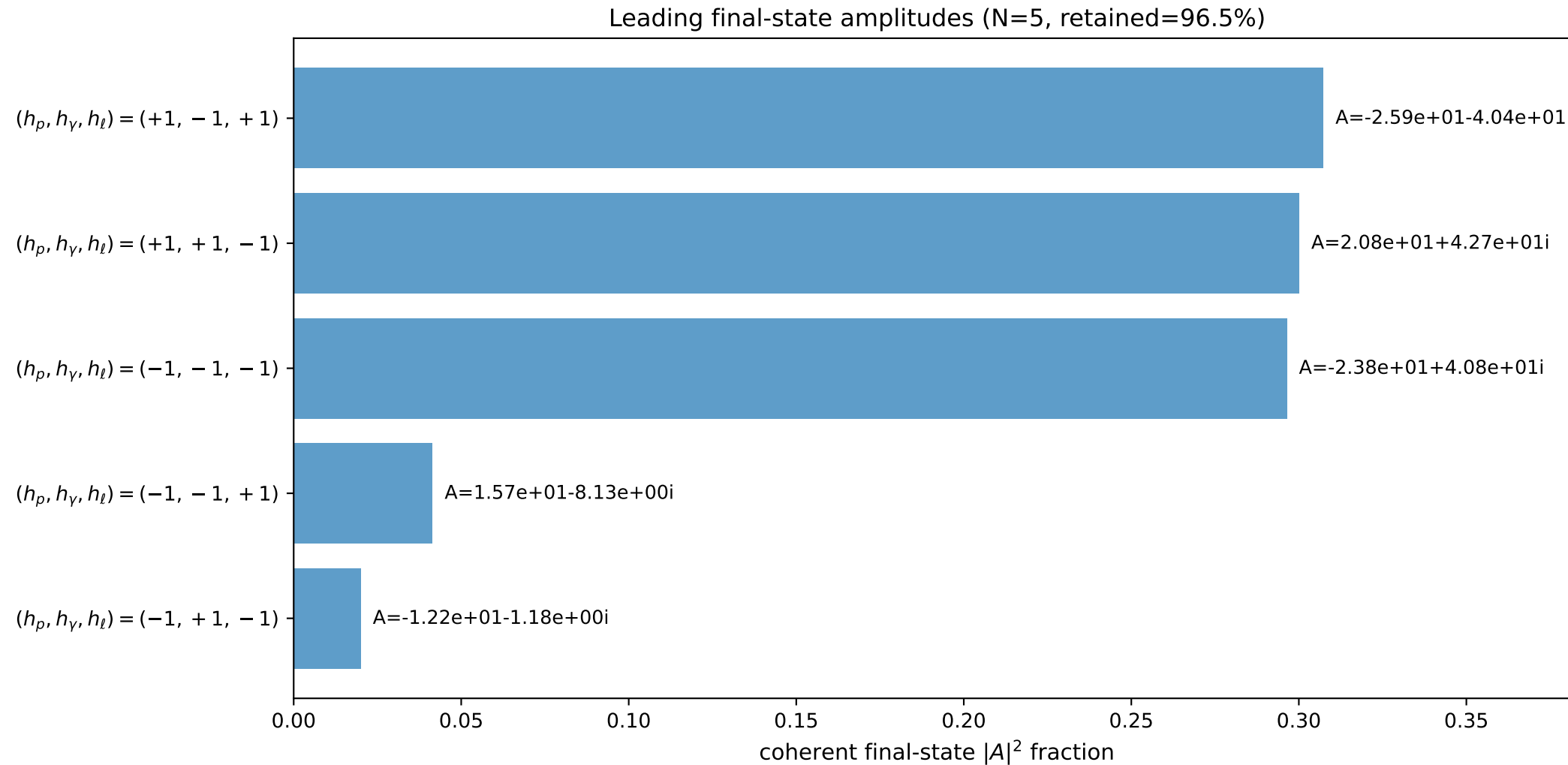
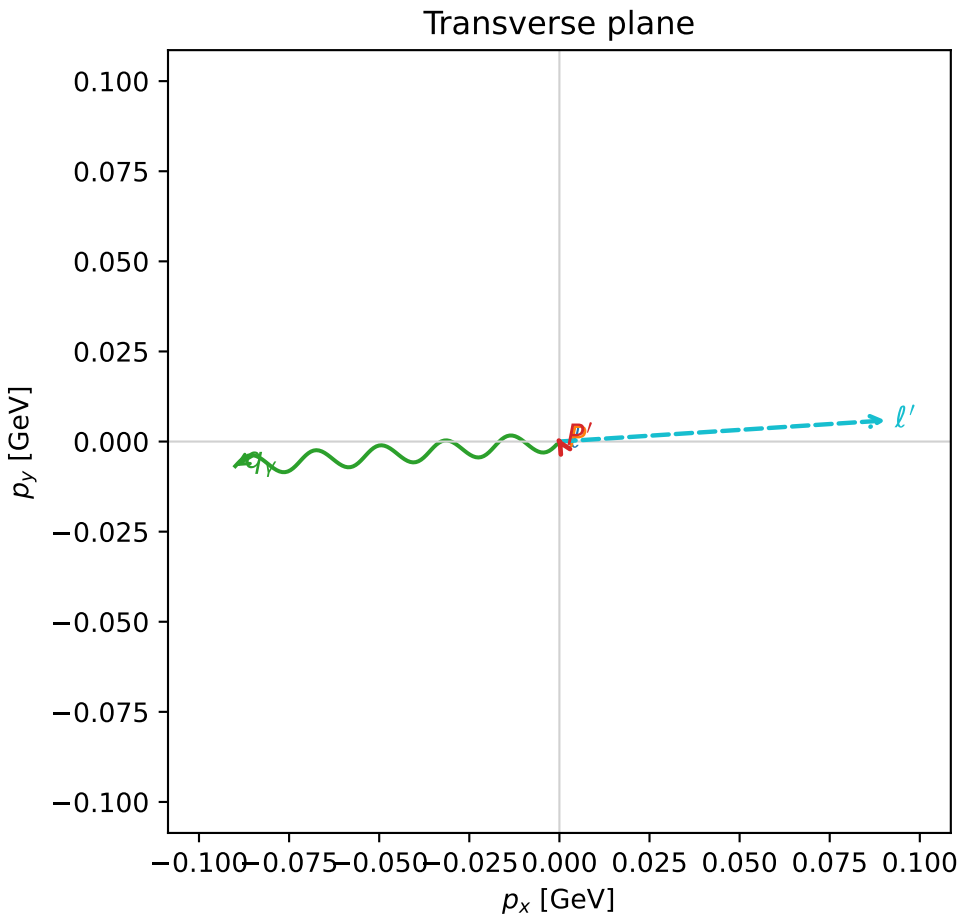
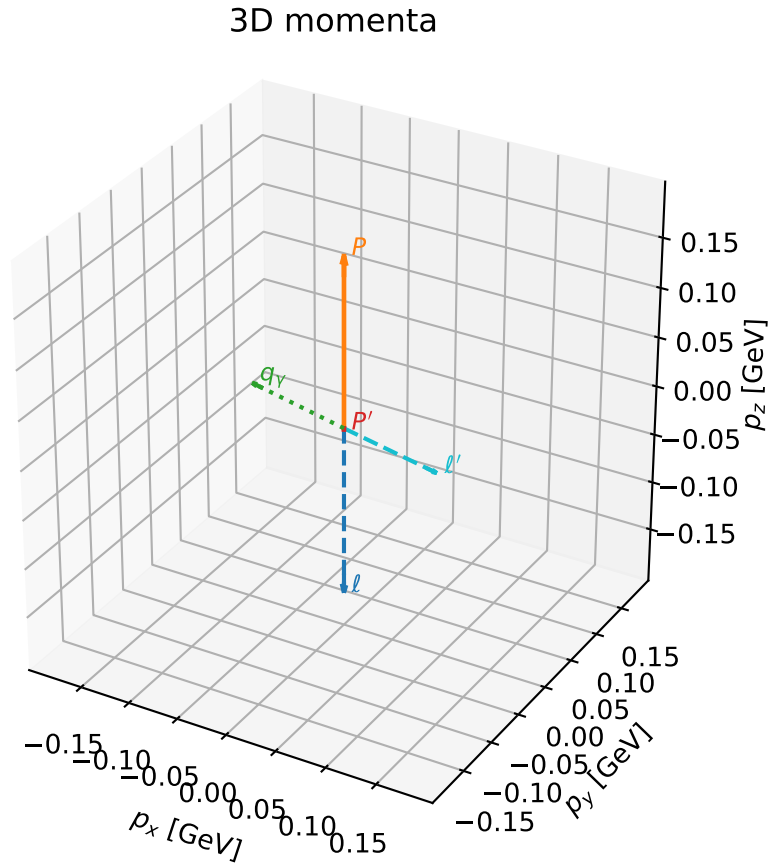


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

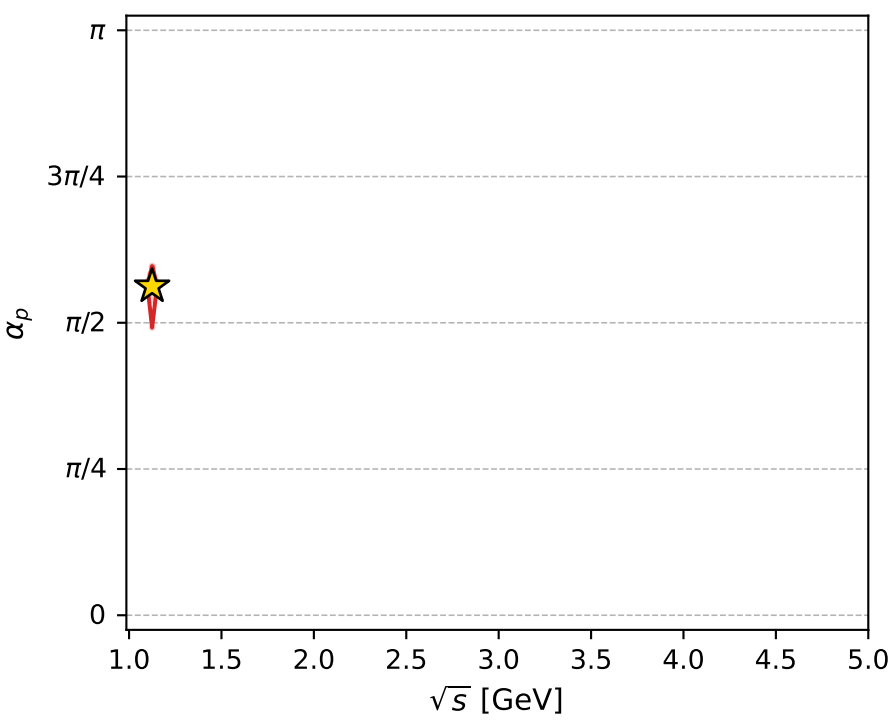
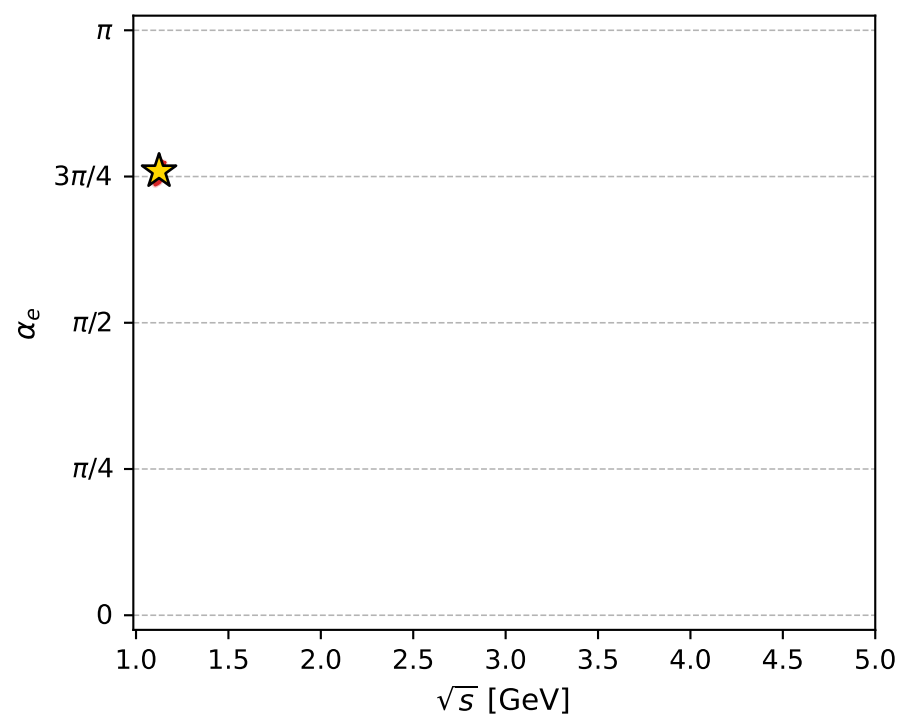
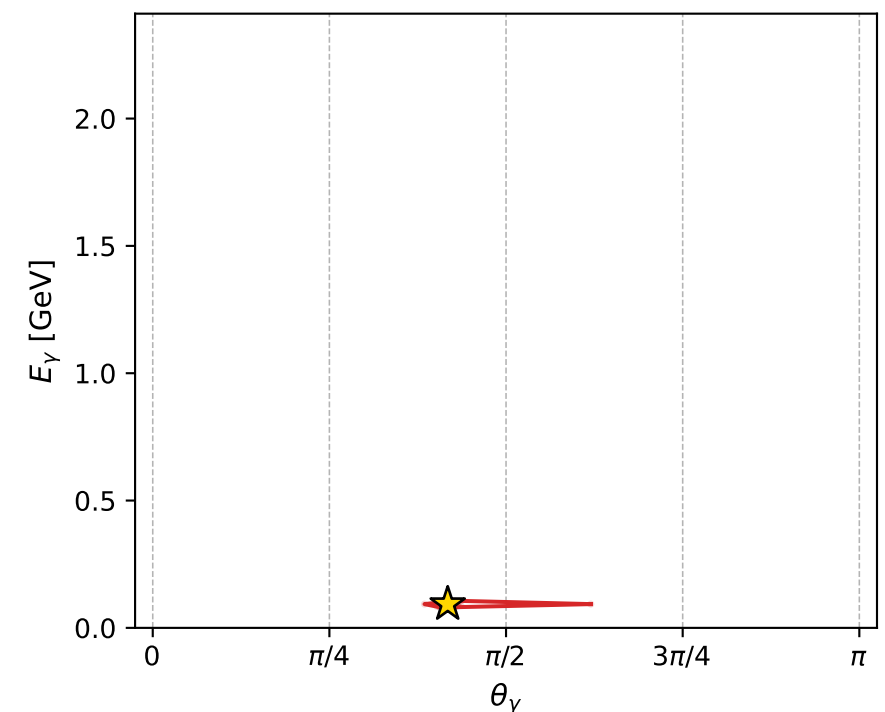
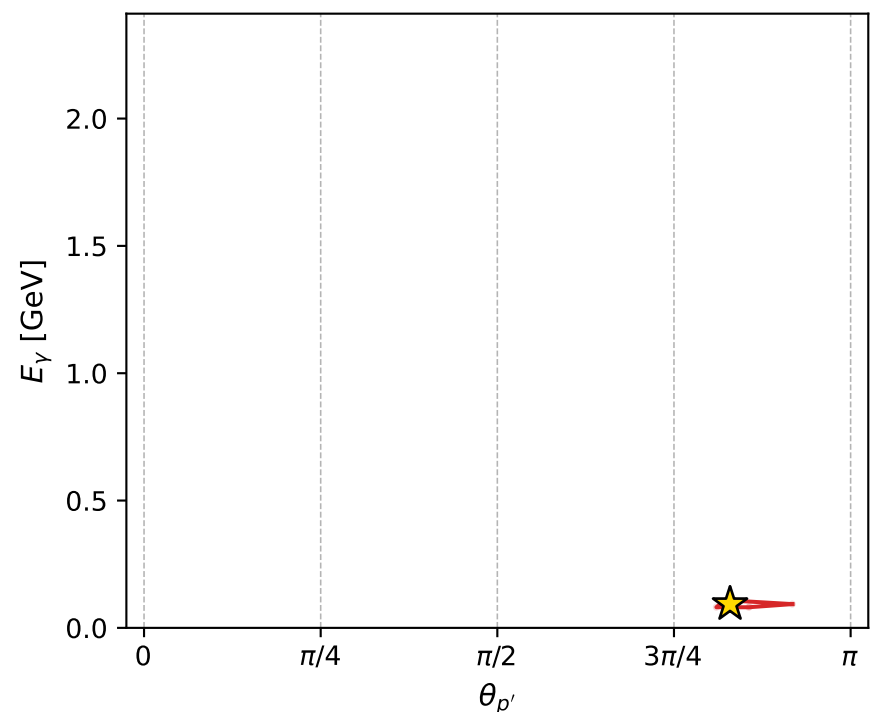
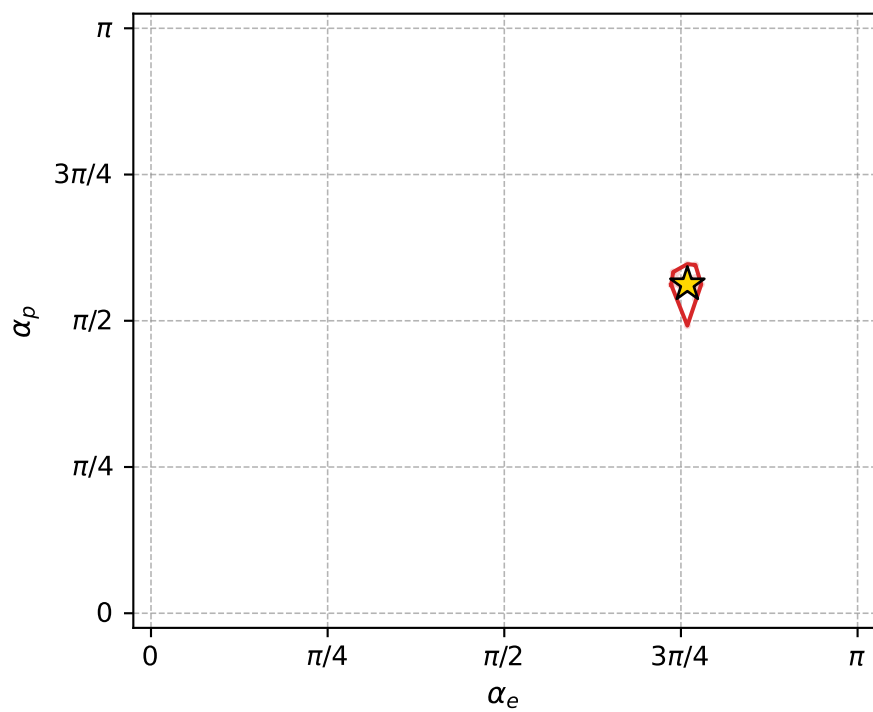
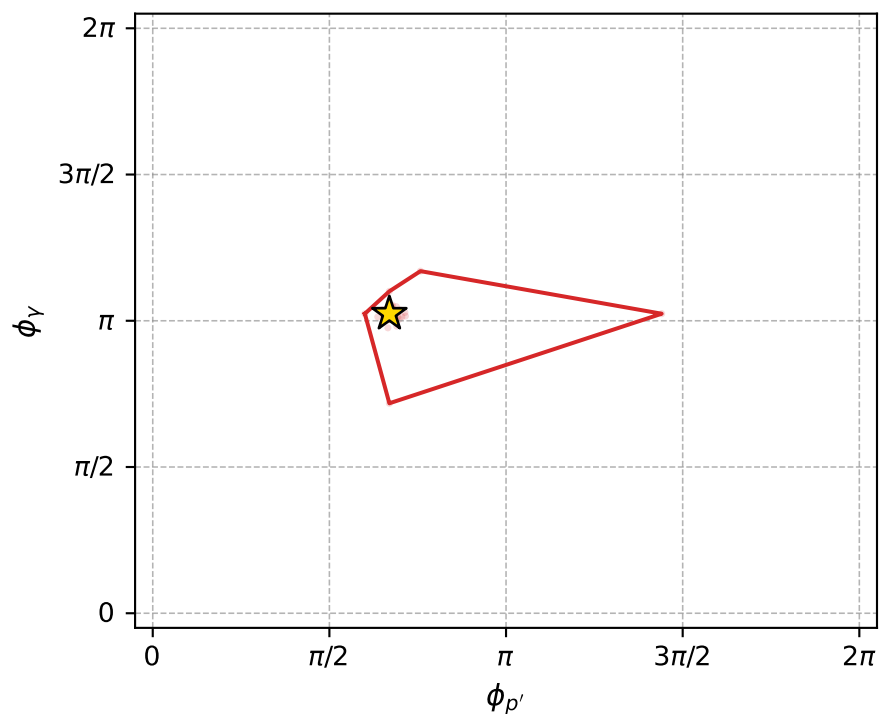
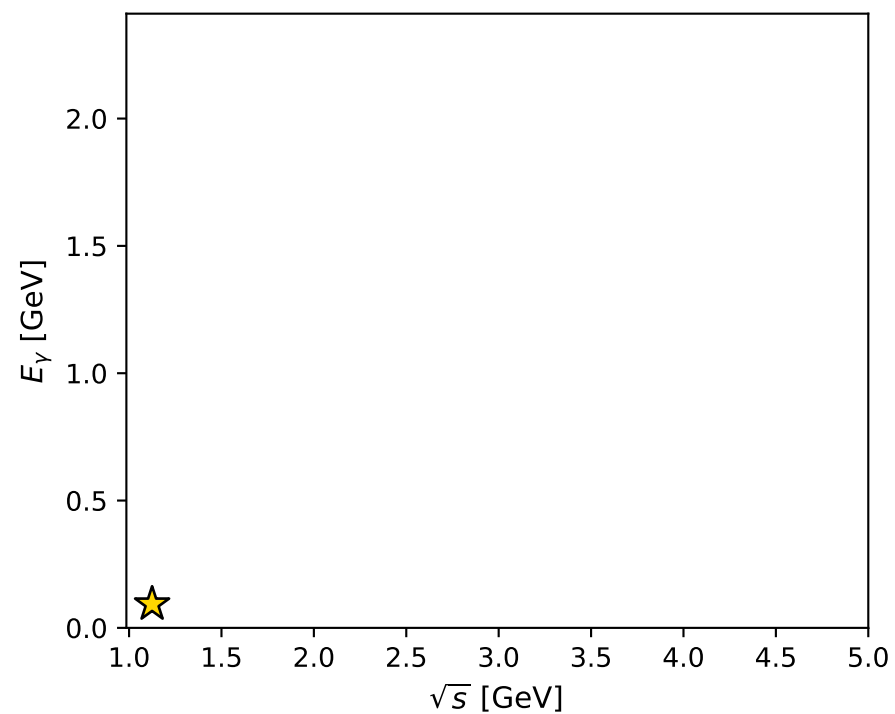
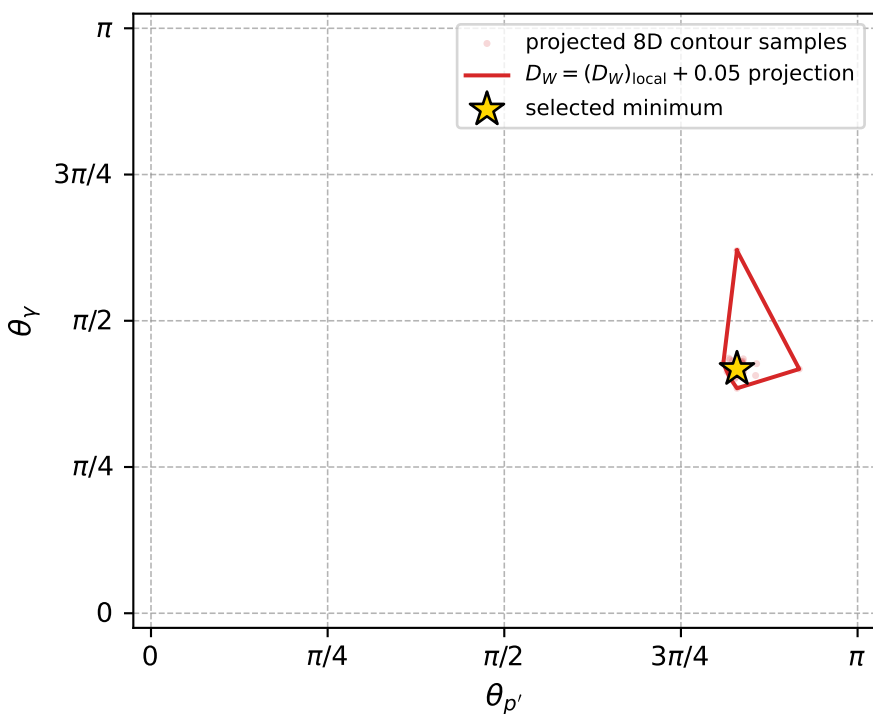
dw_mixing_angles_polarization_06_local_minimum_906 D_W=0.0138562
 region: polarization_cluster_06_local_minimum_906
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3847187$ rad
 incoming lepton: $-0.726986|+\rangle +0.686652|-\rangle$
 proton mixing angle: $\alpha_p=1.766548$ rad
 incoming proton: $-0.194504|+\rangle +0.980902|-\rangle$

$s=1.26483$, $\sqrt{s}=1.12465$, $|\vec{P}|=0.171157$, $|\vec{P}'|=0.00218636$
 $\theta_{P'}=2.6054$, $\theta_V=1.31148$, $\phi_{P'}=2.10402$, $\phi_V=3.21728$
 $E_V=0.0933104$, $\theta_{\ell'}=1.80928$, $\phi_{\ell'}=0.0646343$
 $Q^2=0.0244014$, $x_B=0.122342$, $t=-0.0297031$, $W^2=1.0549$, $y=0.518083$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1712$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1712$
 P $E=0.9535$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1712$
 ℓ' $E=0.0933$ $p_x=0.0905$ $p_y=0.0059$ $p_z=-0.0220$
 P' $E=0.9380$ $p_x=-0.0006$ $p_y=0.0010$ $p_z=-0.0019$
 q_V $E=0.0933$ $p_x=-0.0899$ $p_y=-0.0068$ $p_z=0.0239$



electron: polarization cluster P6, local minimum 906; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.013856238$
 $D_W \text{ contour} = 0.063856238$
 above global minimum = 0.013839805
 8D contour samples = 255

selected phase-space point:
 $\text{sqrt_s} = 1.124645$
 $\text{theta_p_out} = 2.605404$
 $\text{theta_gamma_out} = 1.311476$
 $\text{qOut} = 0.0933104$
 $\text{phi_p_out} = 2.104017$
 $\text{phi_gamma_out} = 3.217276$
 $\text{alpha_e} = 2.384719$
 $\text{alpha_p} = 1.766548$

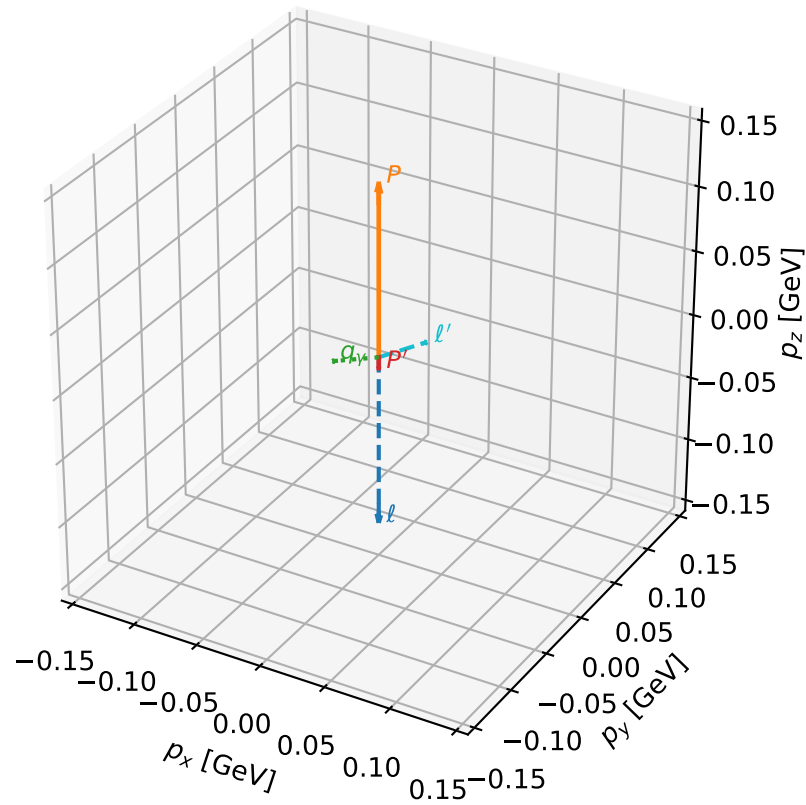
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_913 D_W=0.0141574
 region: polarization_cluster_06_local_minimum_913
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3222921$ rad
 incoming lepton: $-0.682732|+> +0.730668|->$
 proton mixing angle: $\alpha_p=1.6891473$ rad
 incoming proton: $-0.118075|+> +0.993005|->$

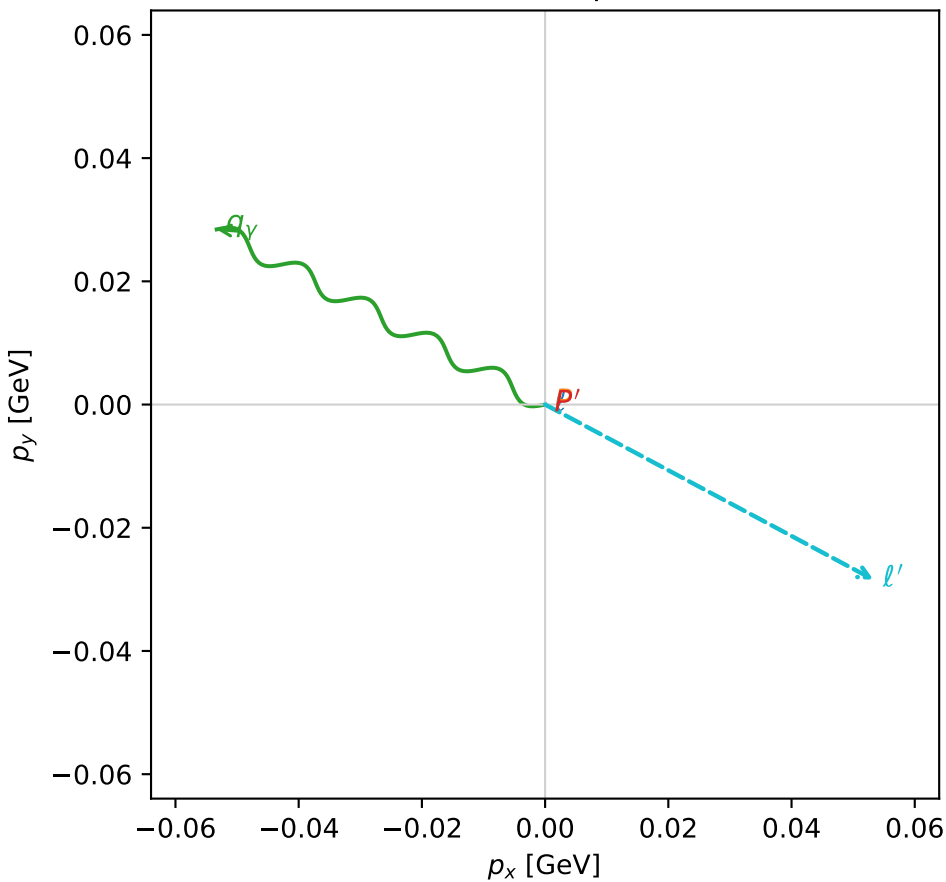
$s=1.16608$, $\sqrt{s}=1.07985$, $|\vec{P}|=0.132535$, $|\vec{P}'|=0.0084374$
 $\theta_{P'}=3.14159$, $\theta_Y=2.0673$, $\phi_{P'}=2.18074$, $\phi_Y=2.6516$
 $E_Y=0.0687084$, $\theta_{\ell'}=0.972642$, $\phi_{\ell'}=5.79319$
 $Q^2=0.0302901$, $x_B=0.190936$, $t=-0.0197872$, $W^2=1.00819$, $y=0.554223$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1325$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1325$
 P $E= 0.9473$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1325$
 ℓ' $E= 0.0731$ $p_x= 0.0533$ $p_y= -0.0284$ $p_z= 0.0412$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0084$
 q_Y $E= 0.0687$ $p_x= -0.0533$ $p_y= 0.0284$ $p_z= -0.0327$

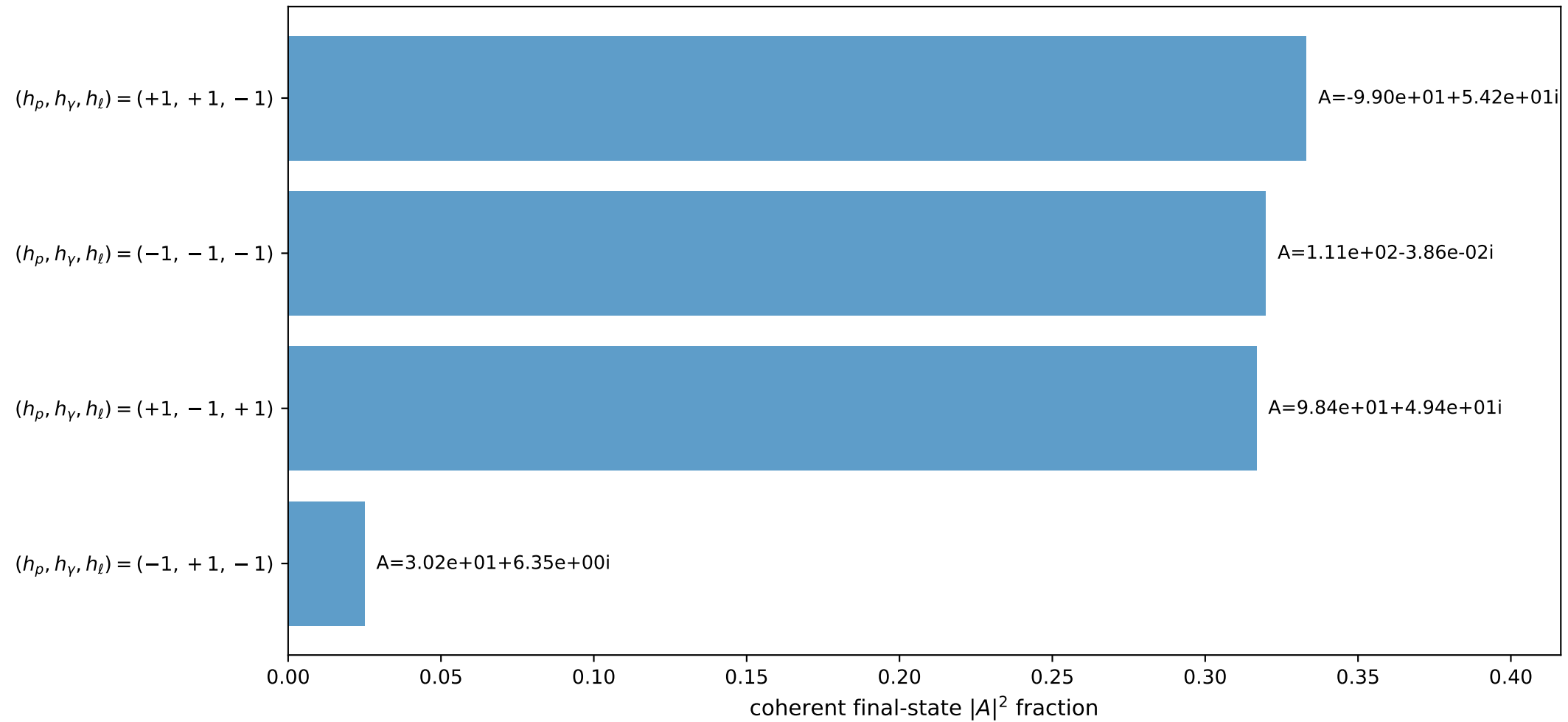
3D momenta



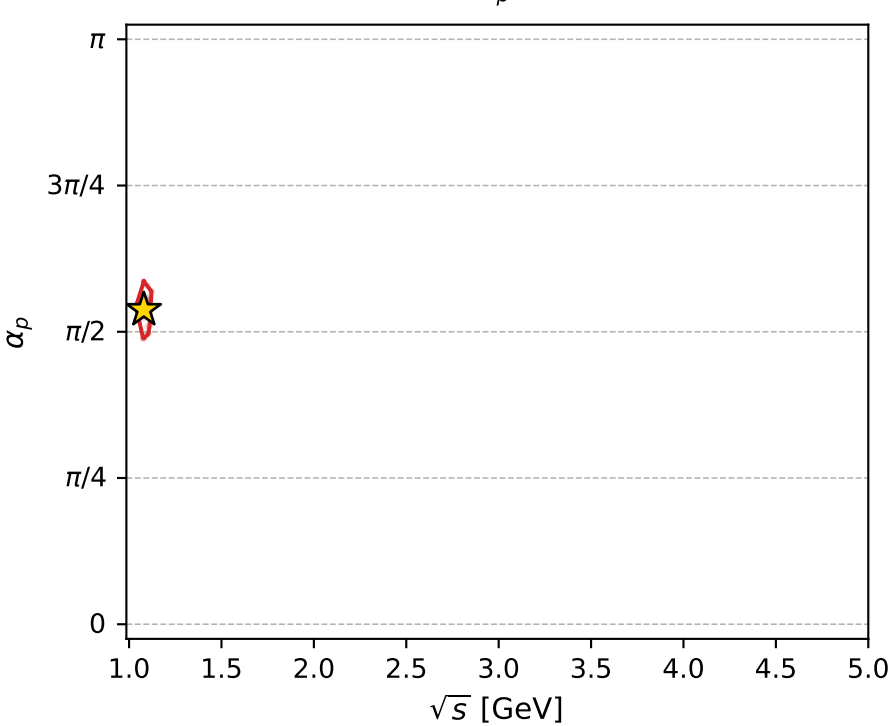
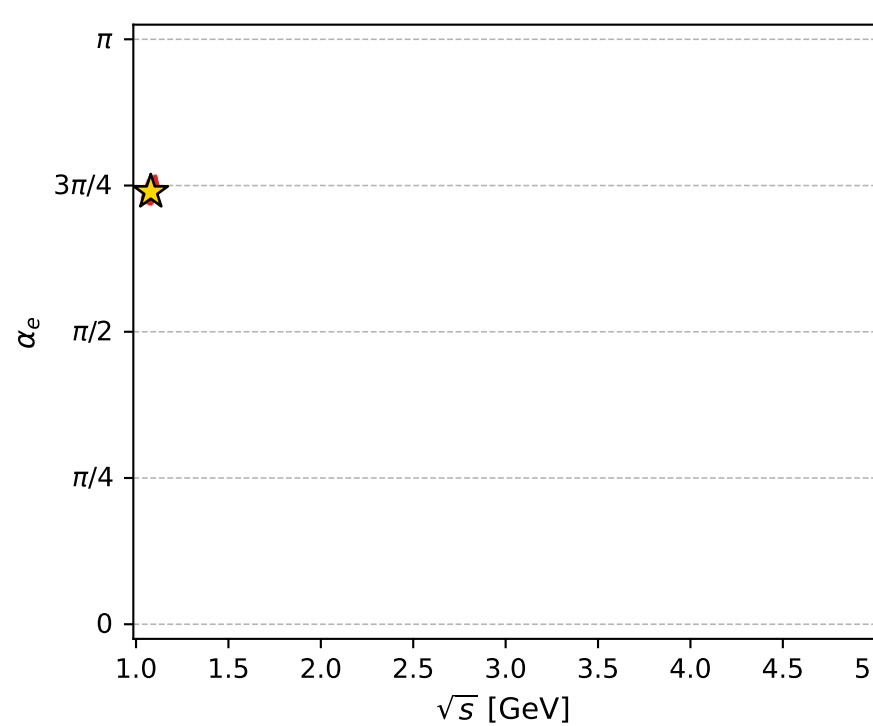
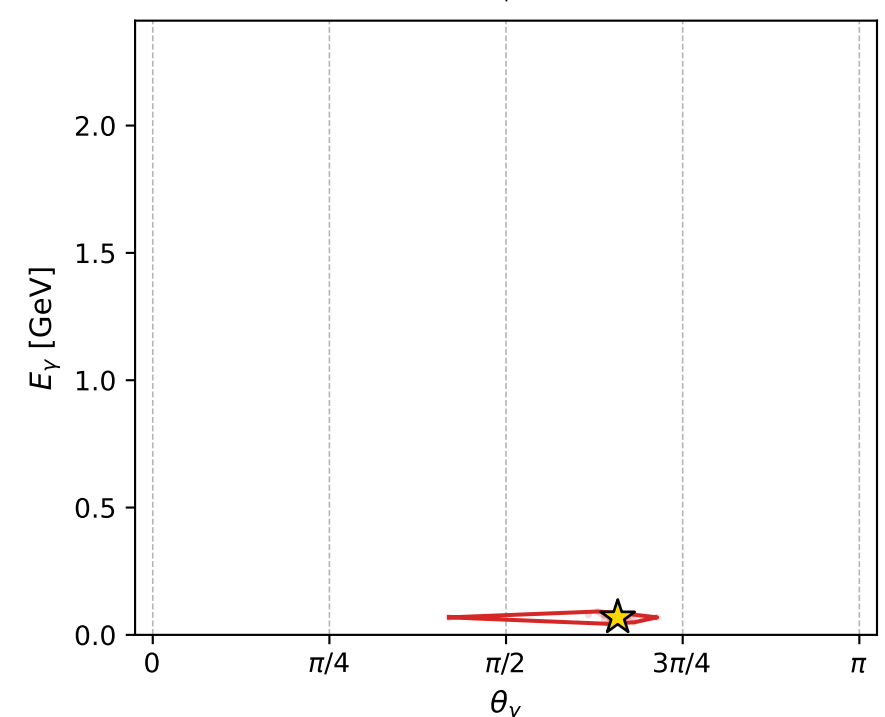
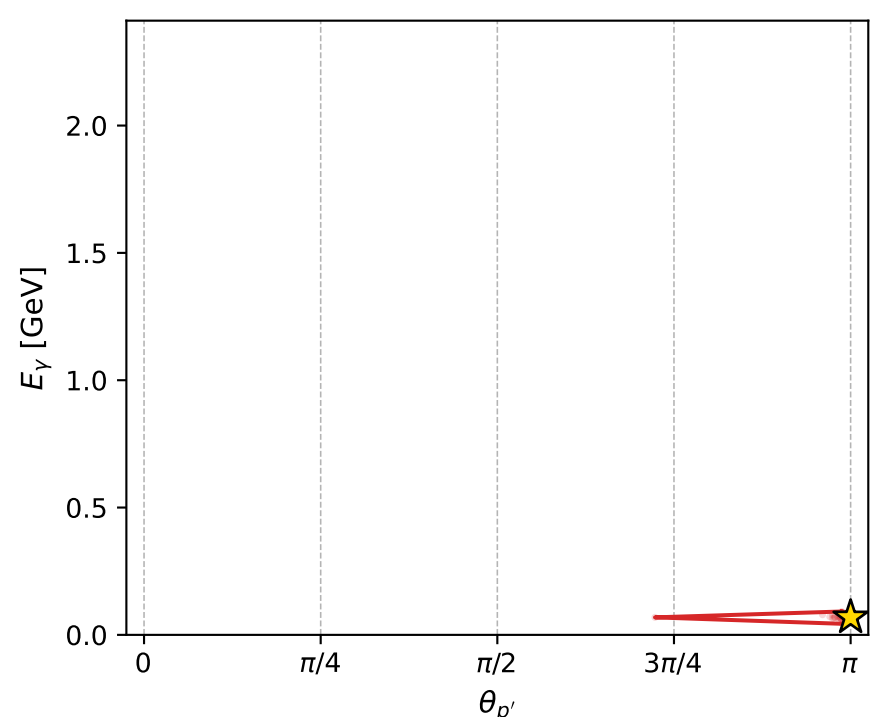
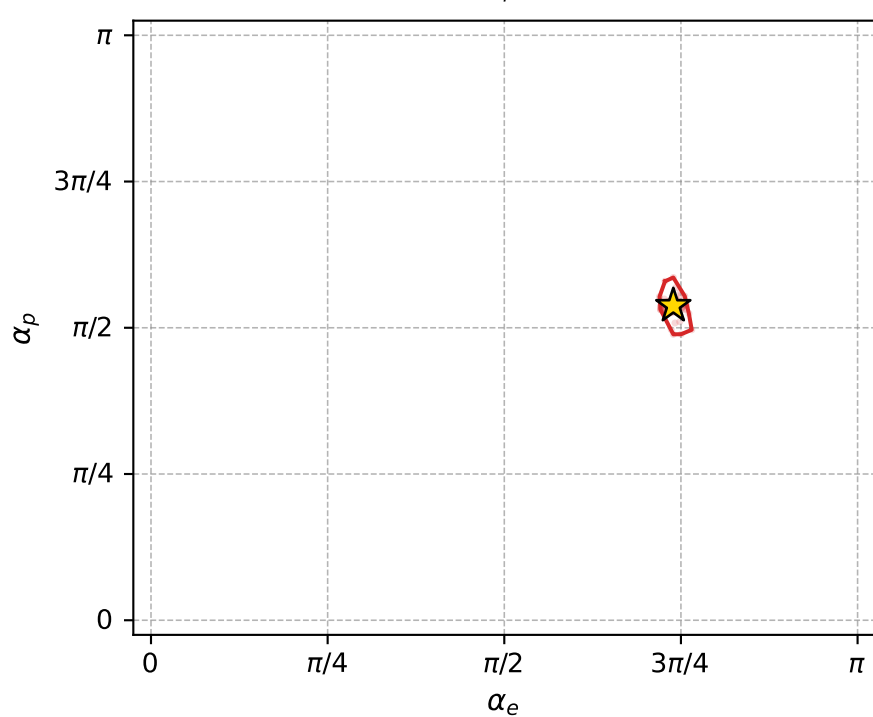
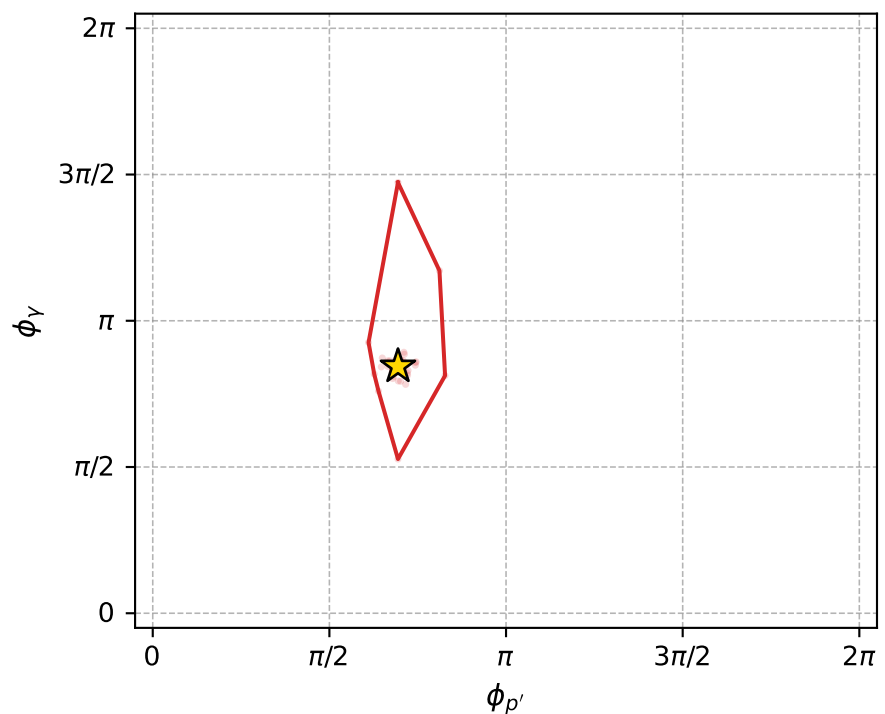
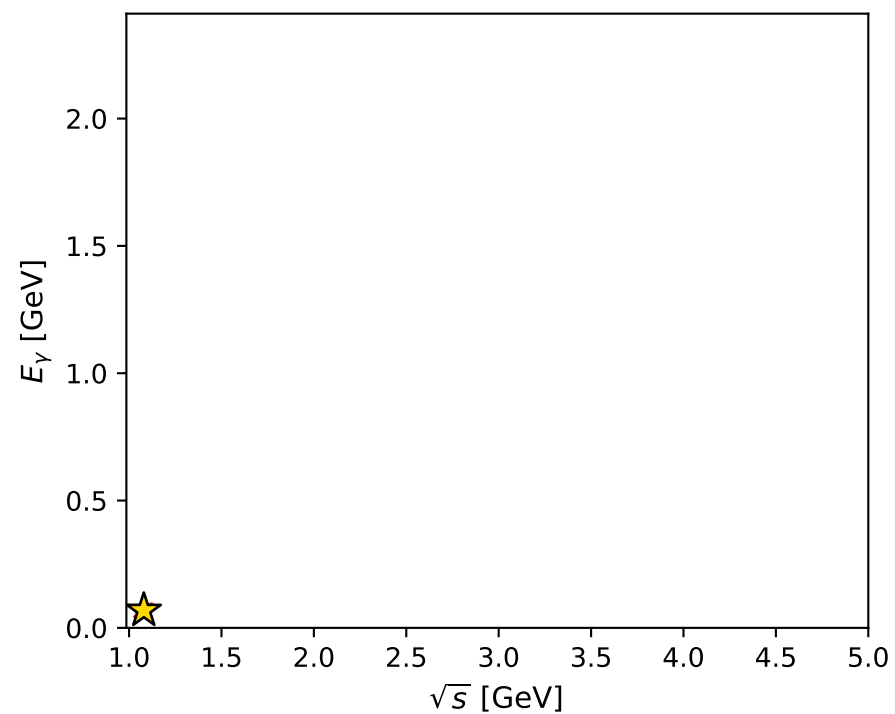
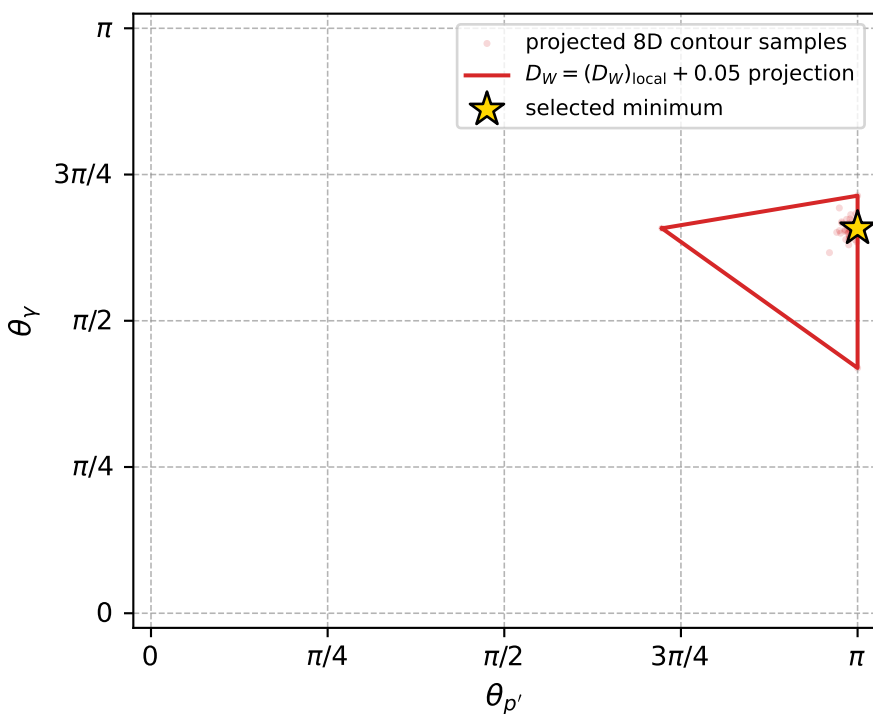
Transverse plane



Leading final-state amplitudes (N=4, retained=99.5%)



electron: polarization cluster P6, local minimum 913; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.014157396$
 $D_W \text{ contour} = 0.064157396$
 above global minimum = 0.014140963
 8D contour samples = 146

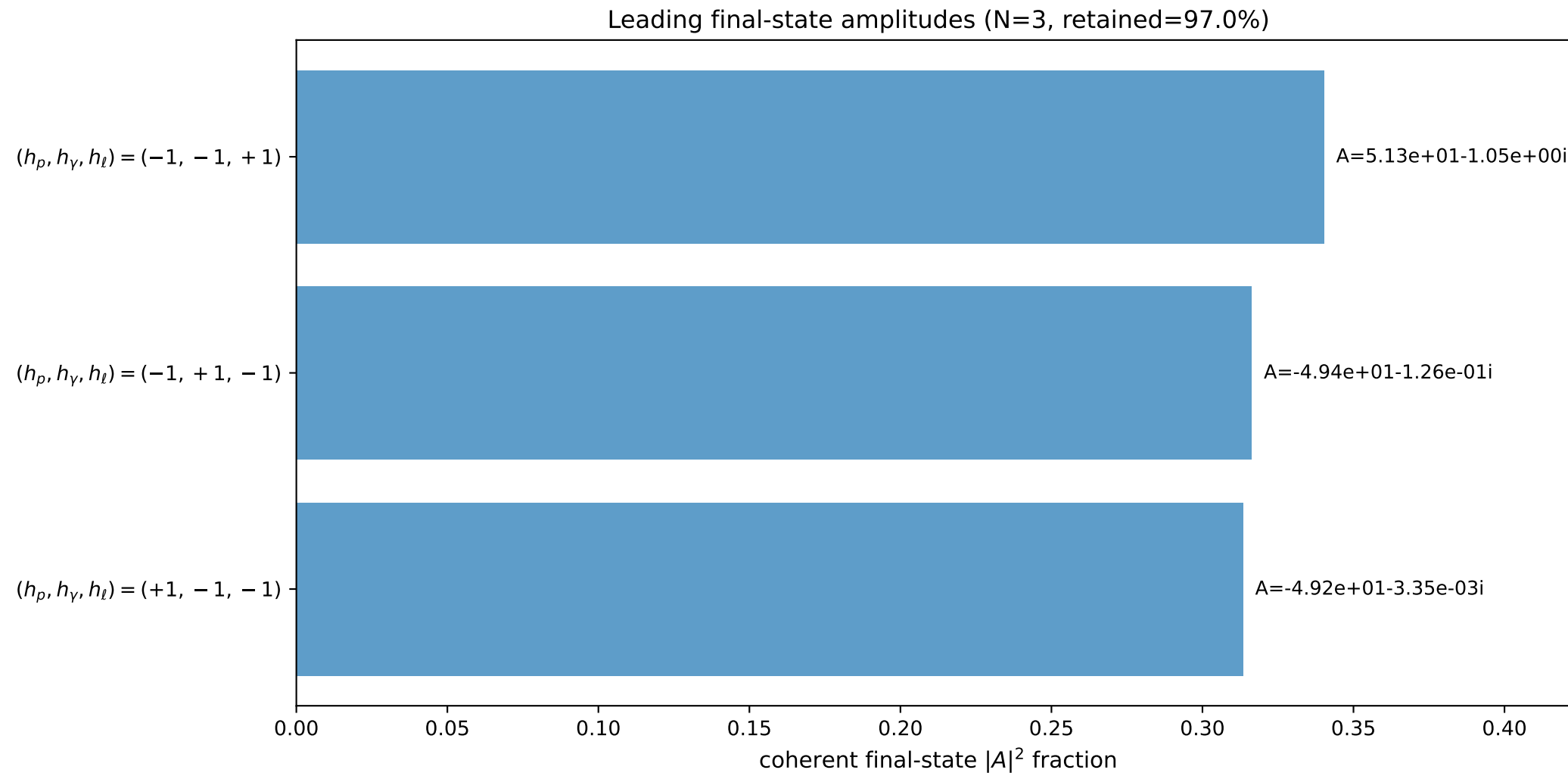
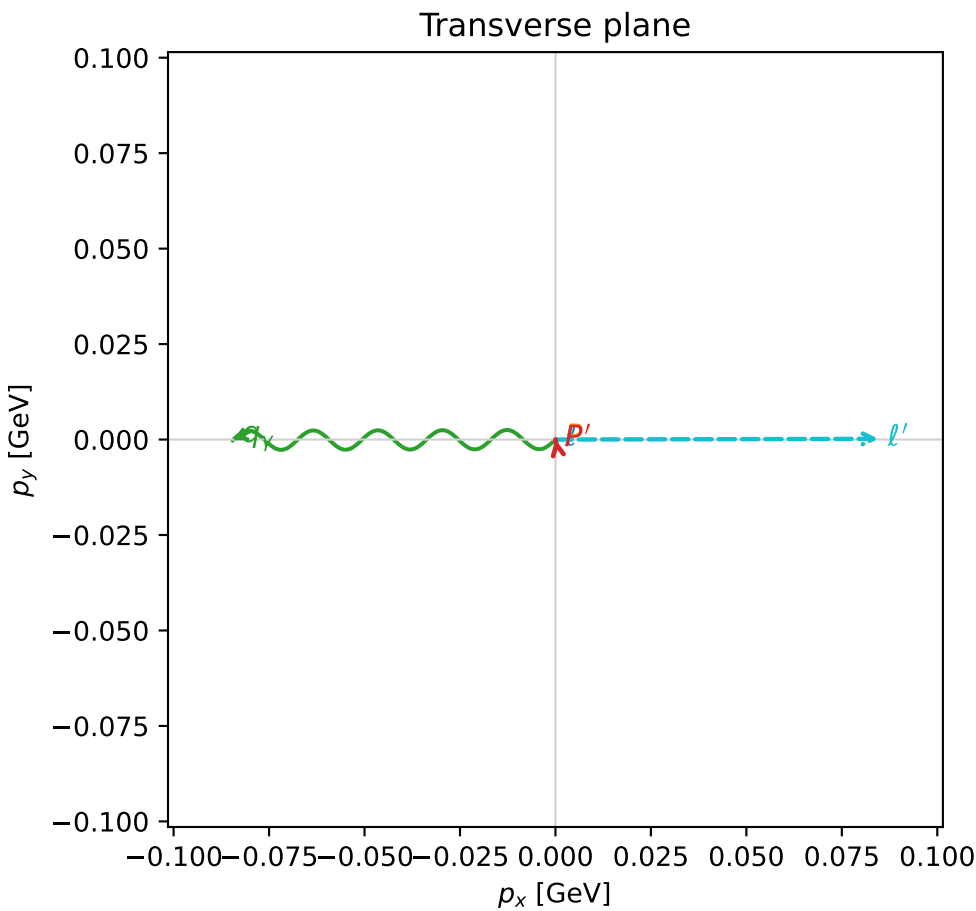
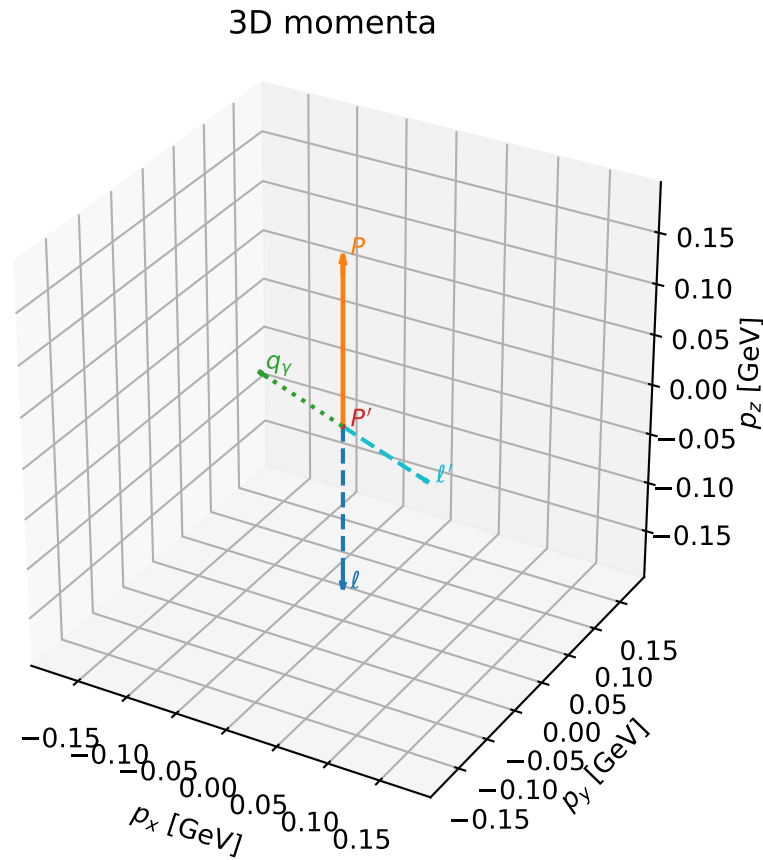
selected phase-space point:
 $\text{sqrt_s} = 1.079853$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 2.067297$
 $\text{qOut} = 0.06870842$
 $\text{phi_p_out} = 2.180737$
 $\text{phi_gamma_out} = 2.651597$
 $\text{alpha_e} = 2.322292$
 $\text{alpha_p} = 1.689147$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

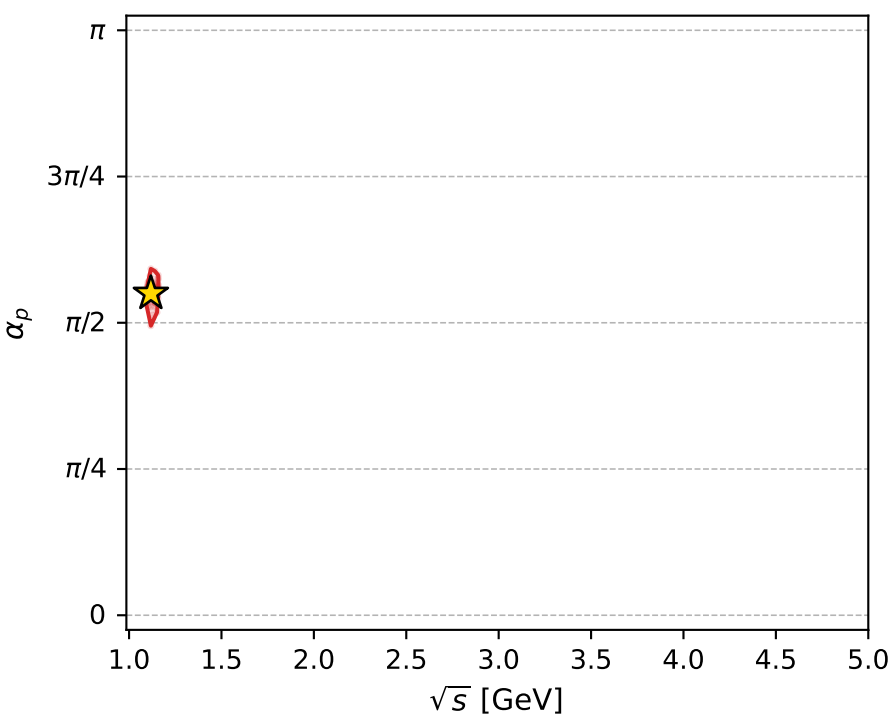
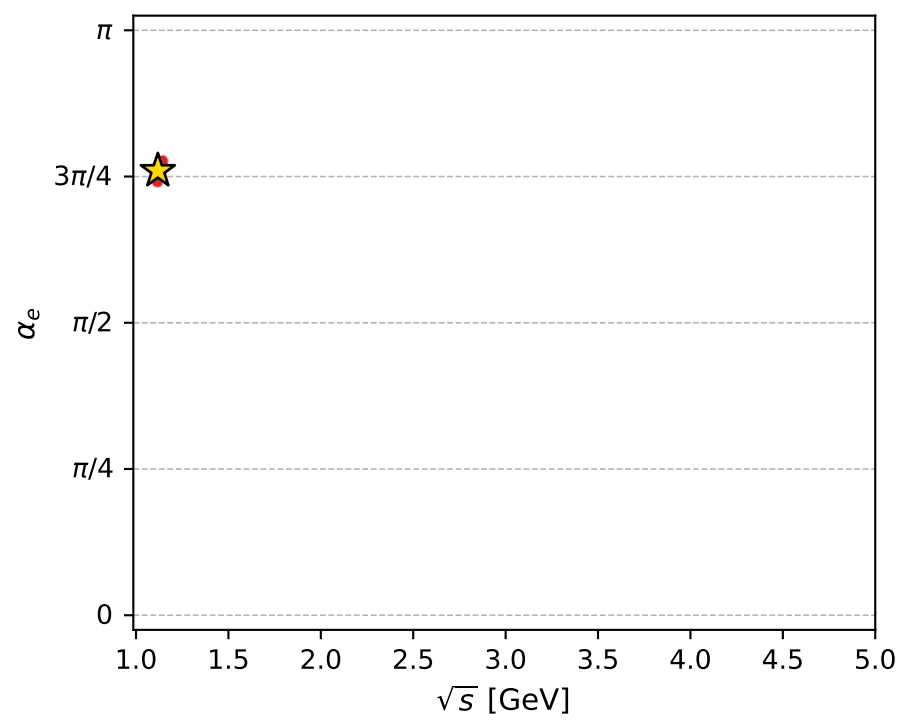
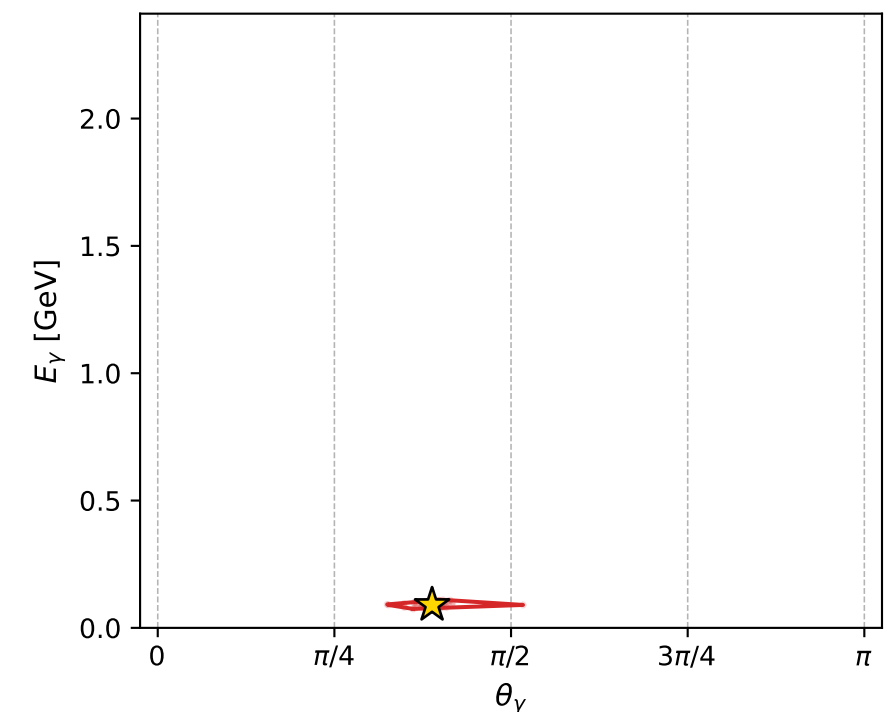
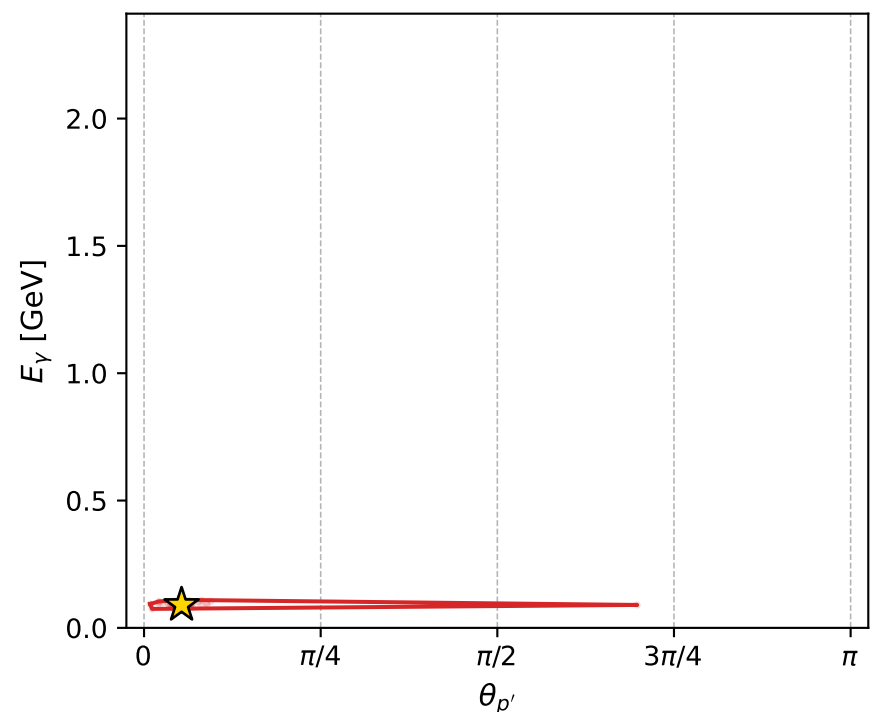
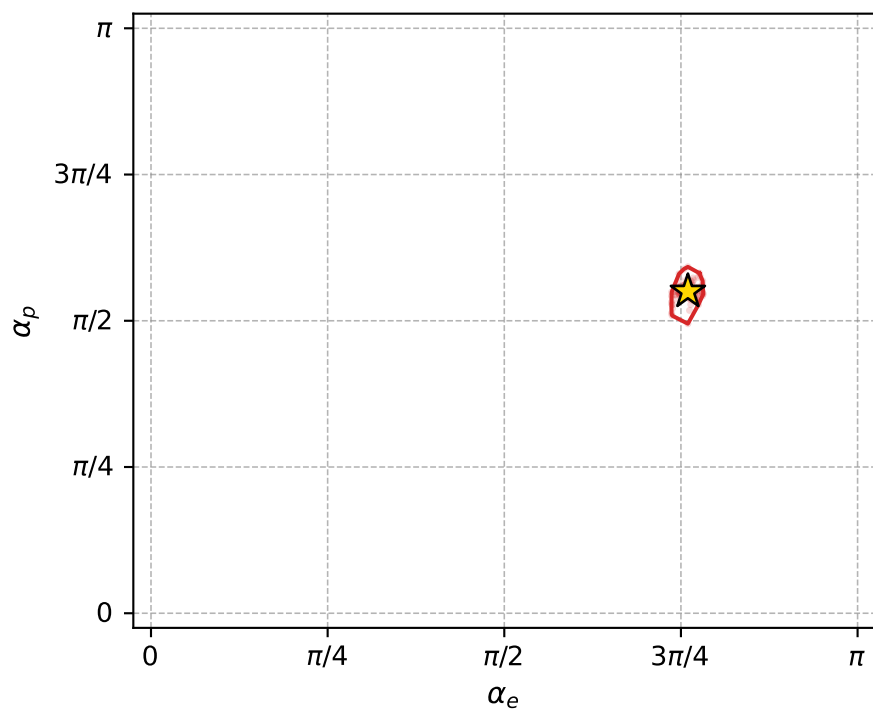
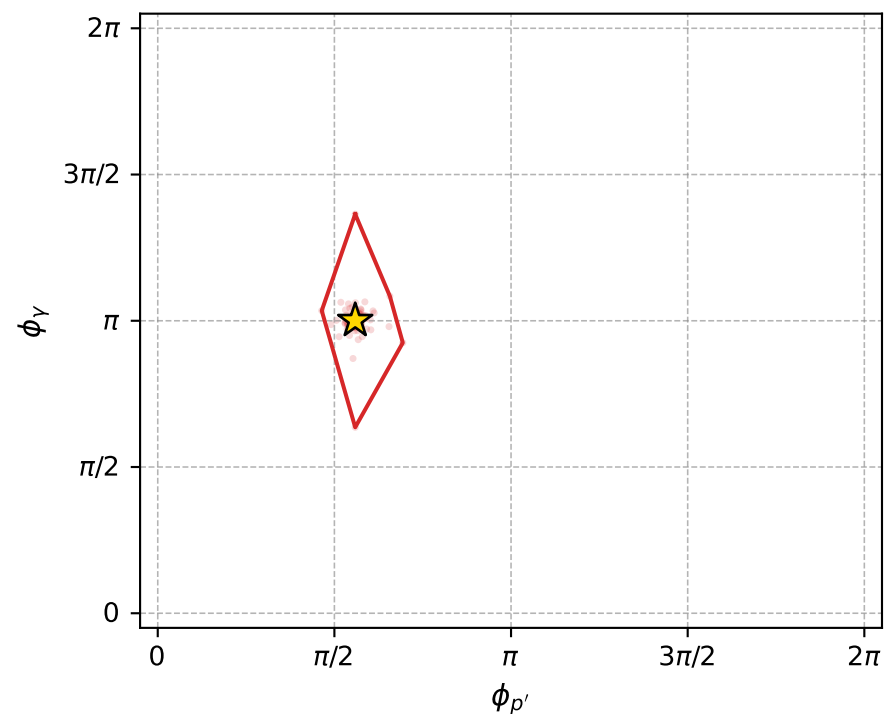
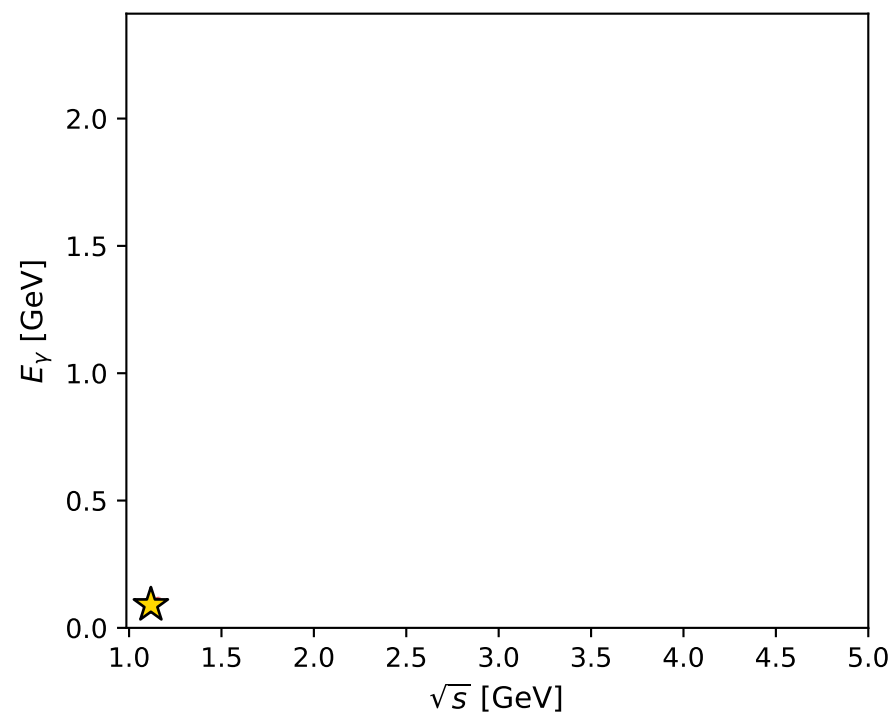
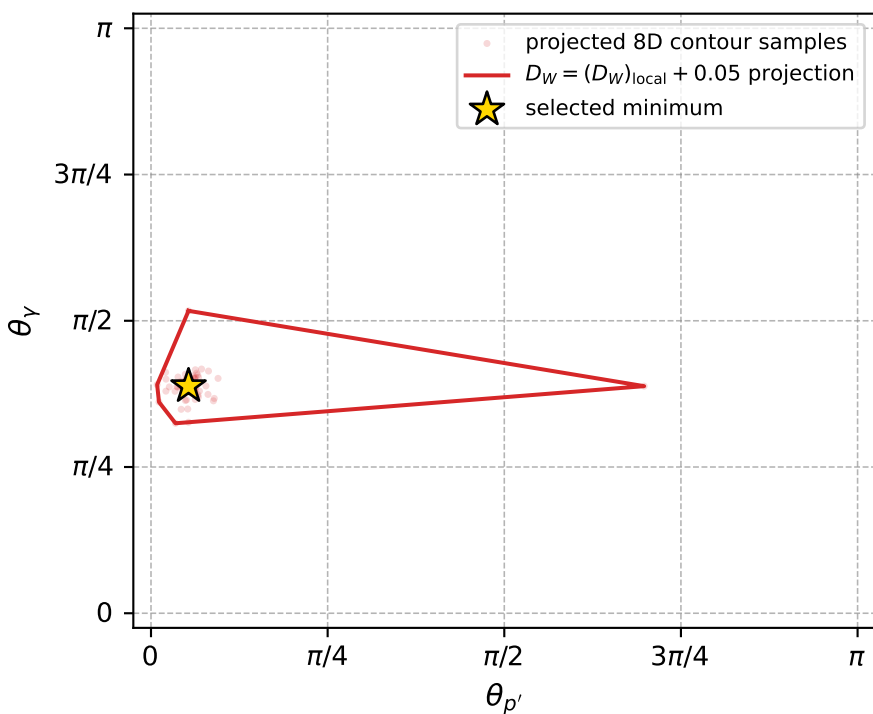
dw_mixing_angles_polarization_06_local_minimum_930 D_W=0.0148952
region: polarization_cluster_06_local_minimum_930
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3871059$ rad
incoming lepton: $-0.728623|+> +0.684915|->$
proton mixing angle: $\alpha_p=1.728226$ rad
incoming proton: $-0.15678|+> +0.987634|->$

$s=1.25021$, $\sqrt{s}=1.11813$, $|\vec{P}|=0.165617$, $|\vec{P}'|=8.61277e-05$
 $\theta_{P'}=0.166897$, $\theta_V=1.21964$, $\phi_{P'}=1.75585$, $\phi_V=3.14396$
 $E_V=0.0900467$, $\theta_{\ell'}=1.92283$, $\phi_{\ell'}=0.00219716$
 $Q^2=0.0195491$, $x_B=0.103724$, $t=-0.0271904$, $W^2=1.04877$, $y=0.508882$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1656$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1656$
 P $E= 0.9525$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1656$
 ℓ' $E= 0.0901$ $p_x= 0.0846$ $p_y= 0.0002$ $p_z= -0.0311$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0001$
 q_V $E= 0.0900$ $p_x= -0.0846$ $p_y= -0.0002$ $p_z= 0.0310$



electron: polarization cluster P6, local minimum 930; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.014895242$
 $D_W \text{ contour} = 0.064895242$
 above global minimum = 0.01487881
 8D contour samples = 253

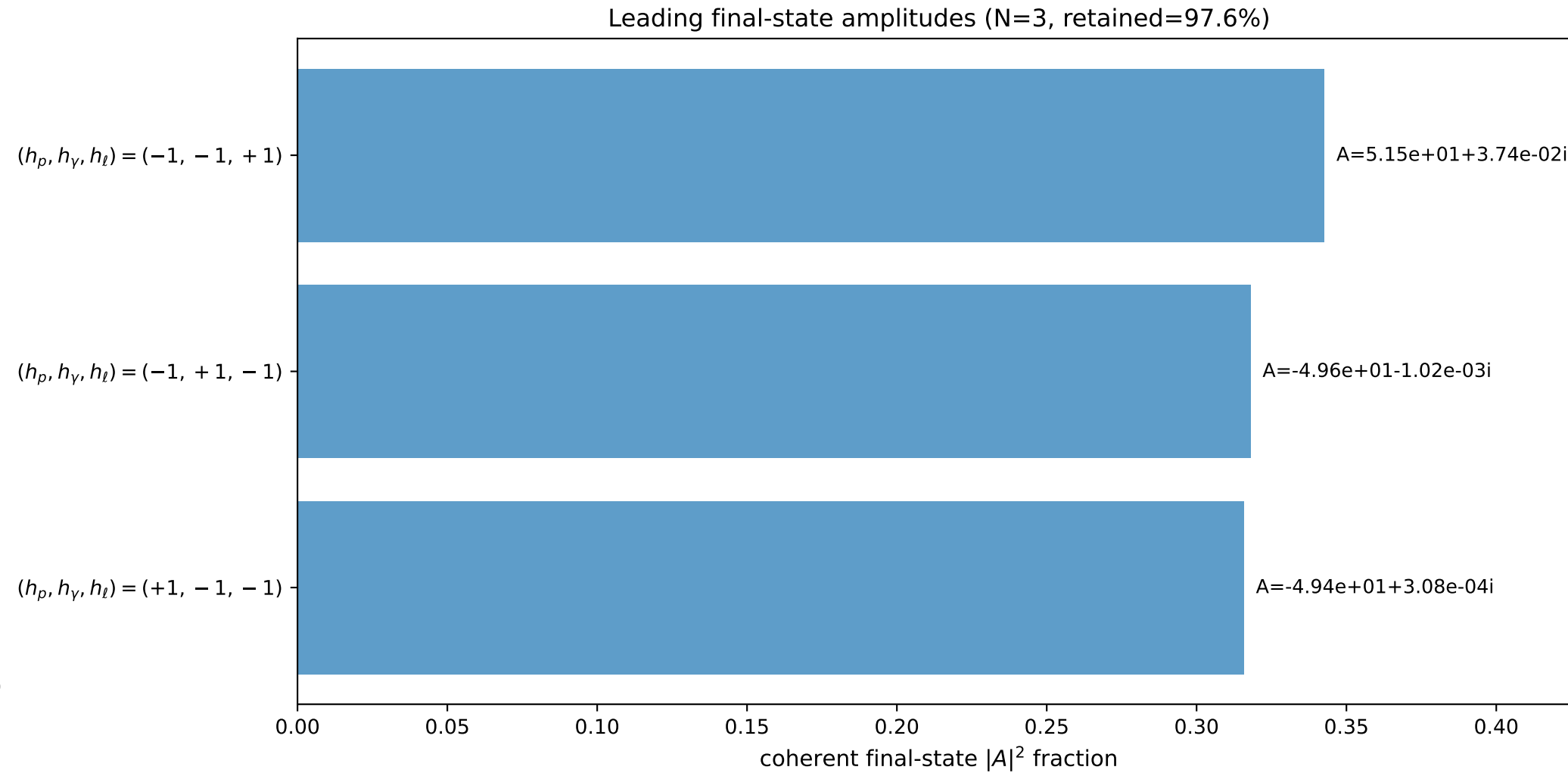
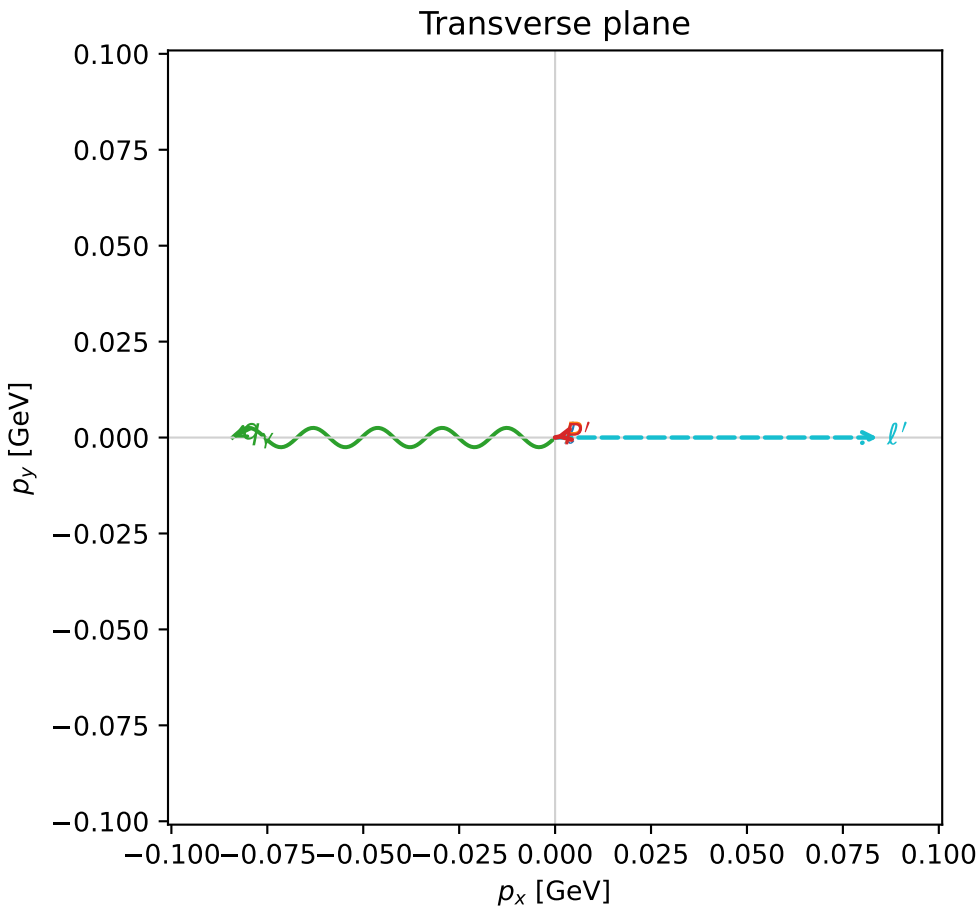
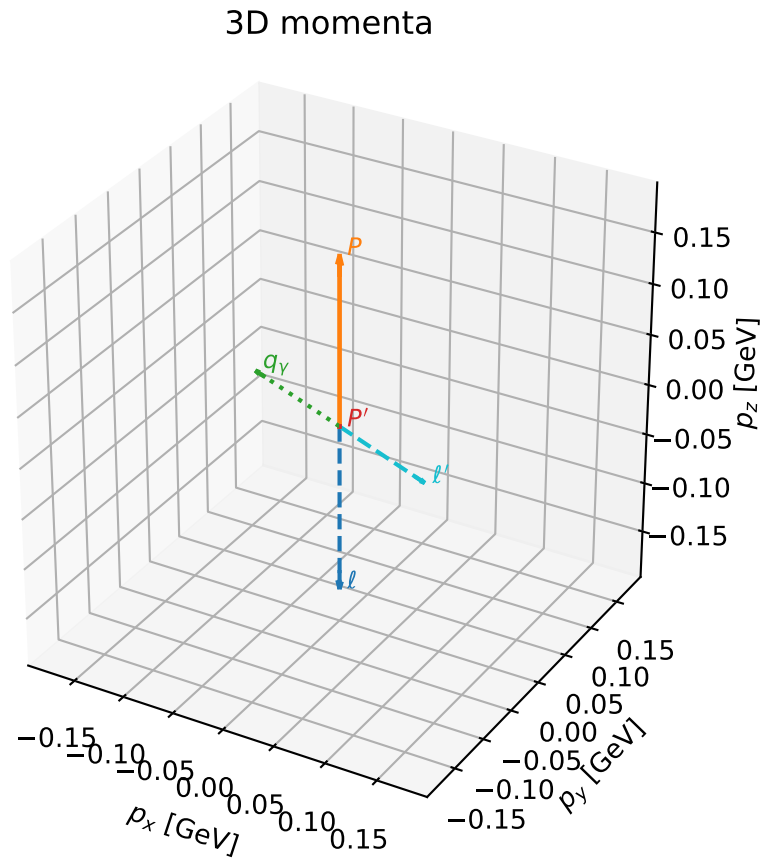
selected phase-space point:
 $\text{sqrt_s} = 1.118127$
 $\text{theta_p_out} = 0.1668971$
 $\text{theta_gamma_out} = 1.219637$
 $\text{q0out} = 0.0900467$
 $\text{phi_p_out} = 1.755846$
 $\text{phi_gamma_out} = 3.143956$
 $\text{alpha_e} = 2.387106$
 $\text{alpha_p} = 1.728226$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

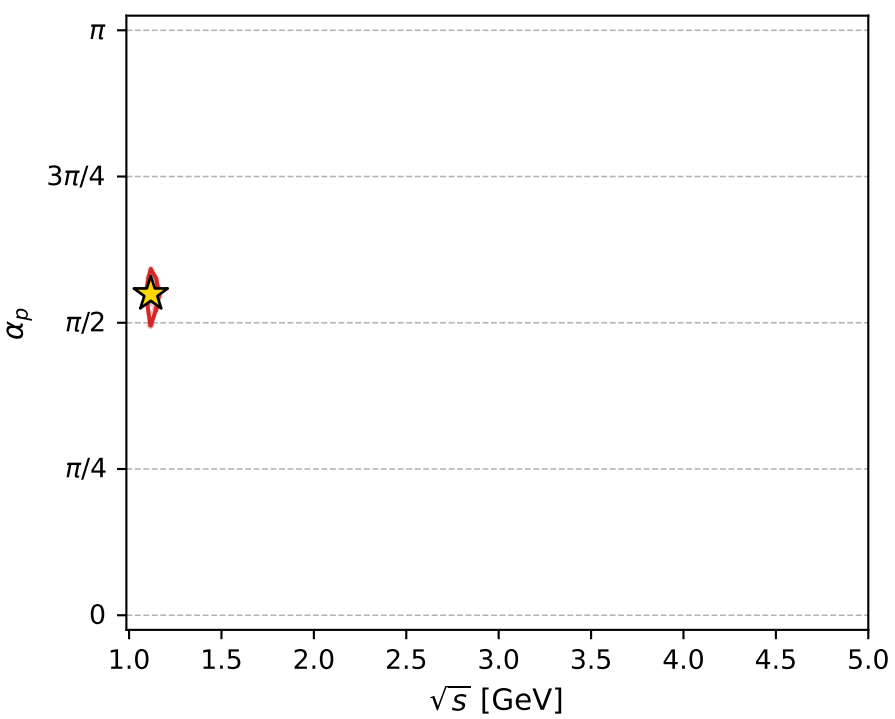
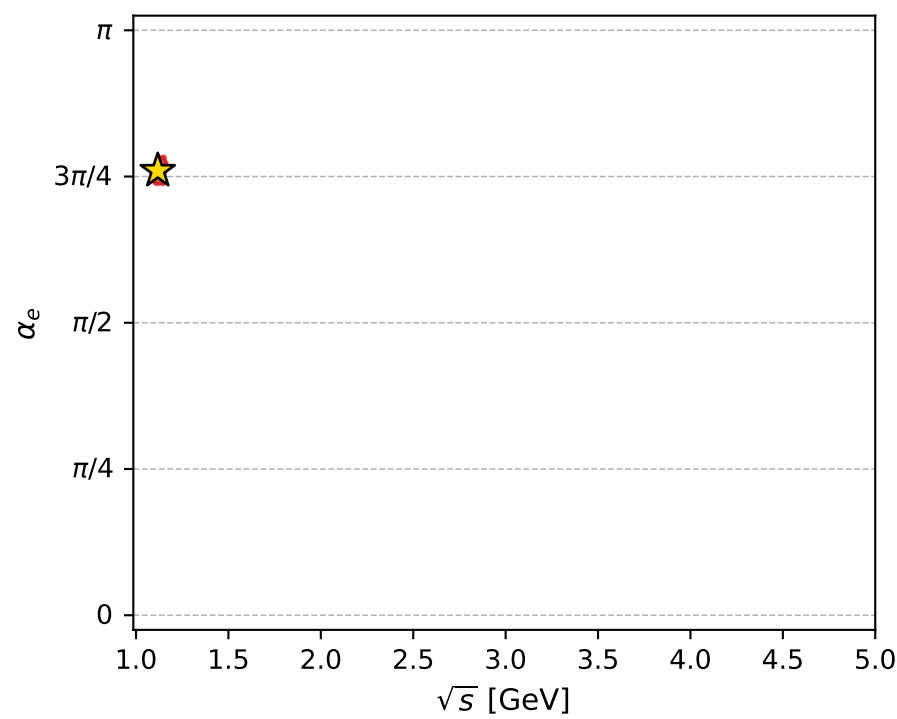
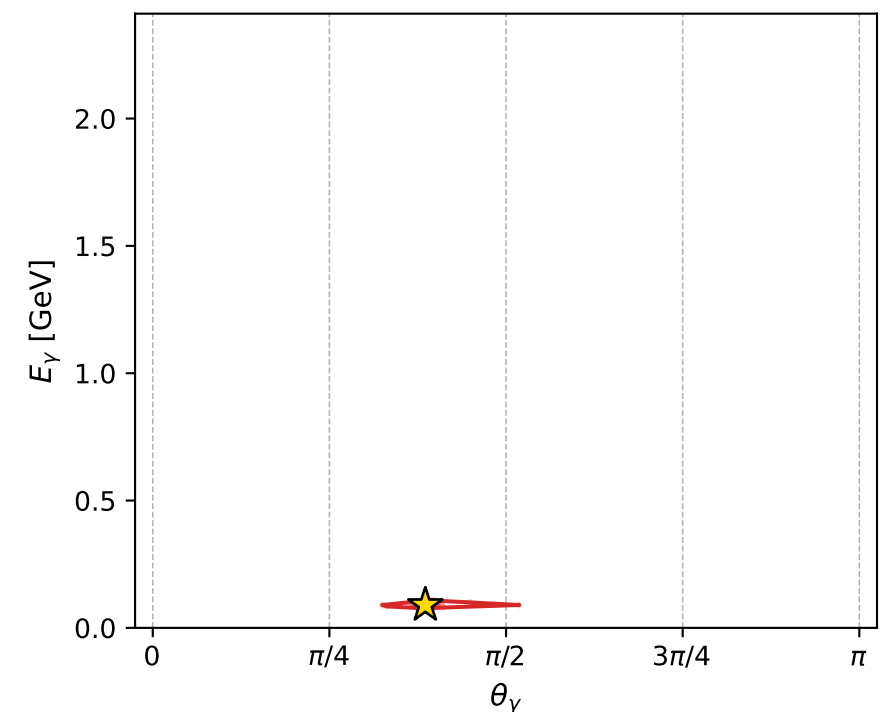
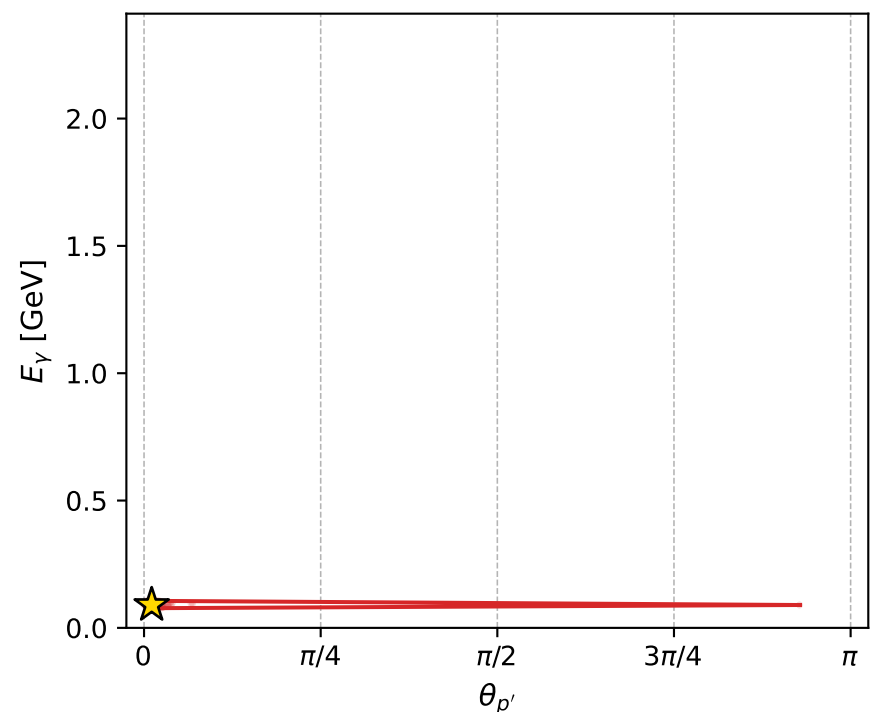
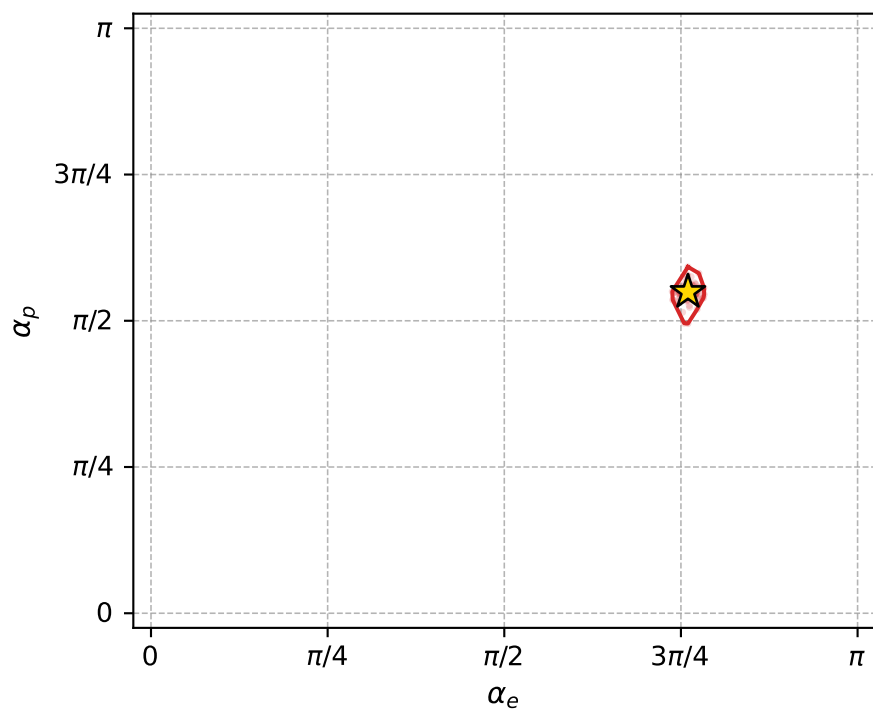
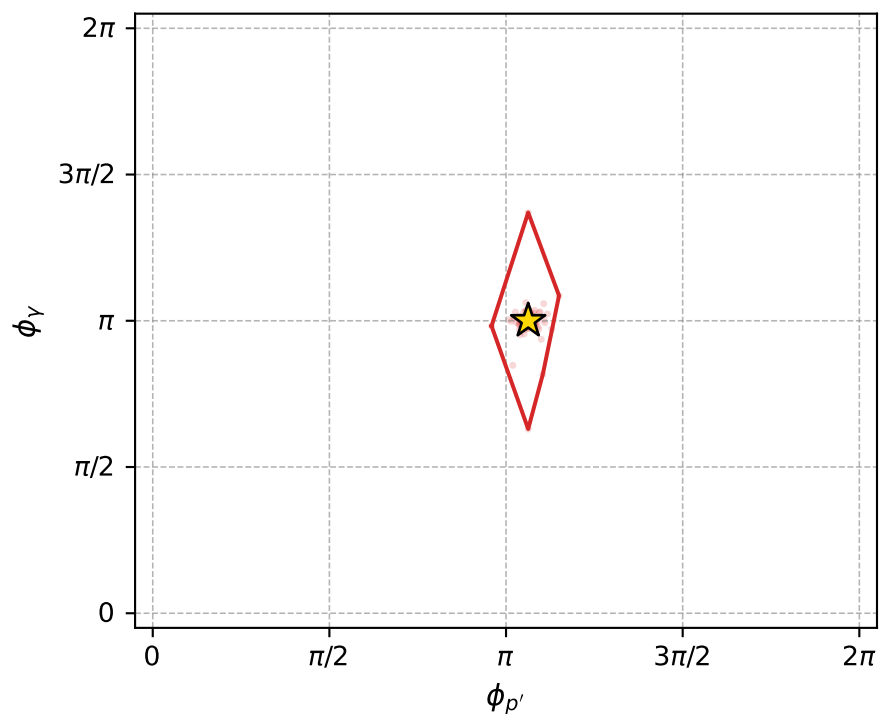
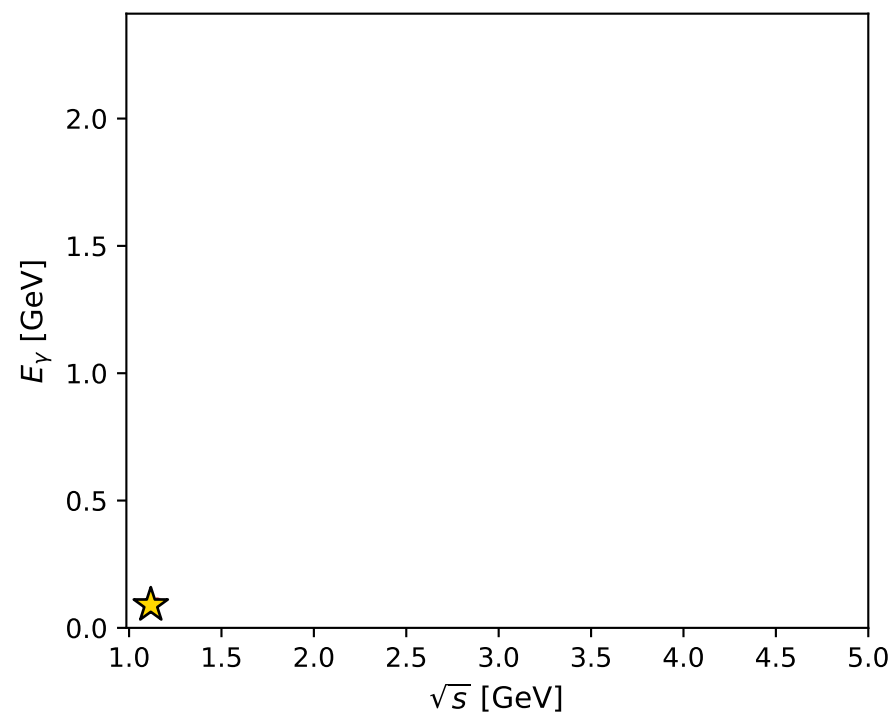
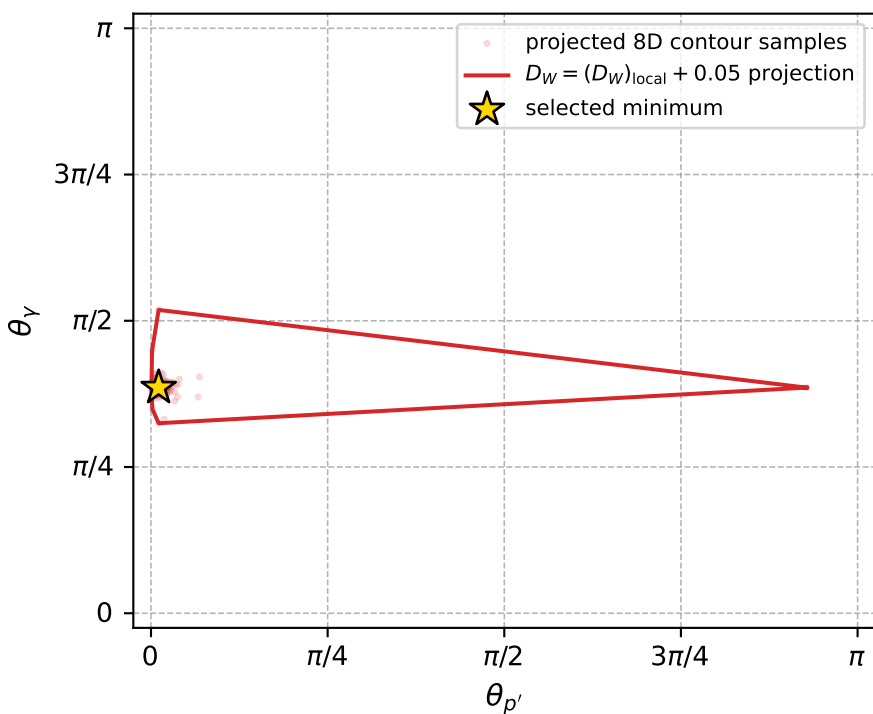
dw_mixing_angles_polarization_06_local_minimum_932 D_W=0.0150586
 region: polarization_cluster_06_local_minimum_932
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3873934$ rad
 incoming lepton: $-0.72882|+\rangle +0.684705|-\rangle$
 proton mixing angle: $\alpha_p=1.7256129$ rad
 incoming proton: $-0.154199|+\rangle +0.98804|-\rangle$

$s=1.249$, $\sqrt{s}=1.11758$, $|\vec{P}|=0.165155$, $|\vec{P}'|=1.2777e-07$
 $\theta_{P'}=0.0340212$, $\theta_V=1.21186$, $\phi_{P'}=3.33858$, $\phi_V=3.14161$
 $E_V=0.0897916$, $\theta_{l'}=1.92973$, $\phi_{l'}=1.74269e-05$
 $Q^2=0.0192406$, $x_B=0.102513$, $t=-0.027068$, $W^2=1.04829$, $y=0.508435$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1652$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1652$
 P $E= 0.9524$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1652$
 l' $E= 0.0898$ $p_x= 0.0841$ $p_y= 0.0000$ $p_z= -0.0315$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0000$
 q_V $E= 0.0898$ $p_x= -0.0841$ $p_y= -0.0000$ $p_z= 0.0315$



electron: polarization cluster P6, local minimum 932; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.015058566$
 $D_W \text{ contour} = 0.065058566$
 above global minimum = 0.015042133
 8D contour samples = 236

selected phase-space point:
 $\text{sqrt_s} = 1.117585$
 $\text{theta_p_out} = 0.03402123$
 $\text{theta_gamma_out} = 1.211859$
 $\text{qOut} = 0.08979161$
 $\text{phi_p_out} = 3.338576$
 $\text{phi_gamma_out} = 3.14161$
 $\text{alpha_e} = 2.387393$
 $\text{alpha_p} = 1.725613$

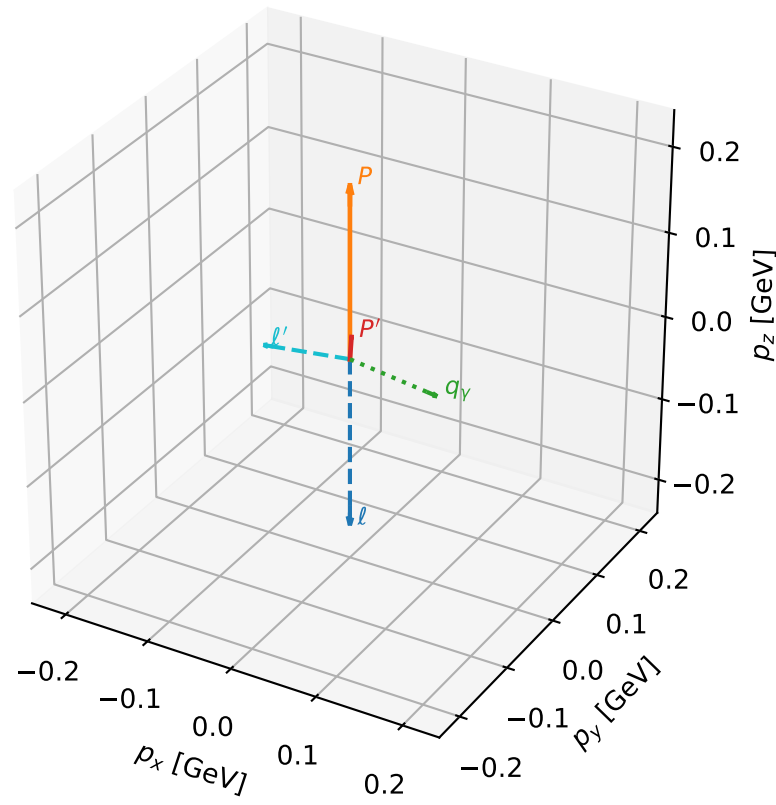
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_933 D_W=0.0151282
 region: polarization_cluster_06_local_minimum_933
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.389794$ rad
 incoming lepton: $-0.730462|+> +0.682954|->$
 proton mixing angle: $\alpha_p=1.3723144$ rad
 incoming proton: $0.197181|+> +0.980367|->$

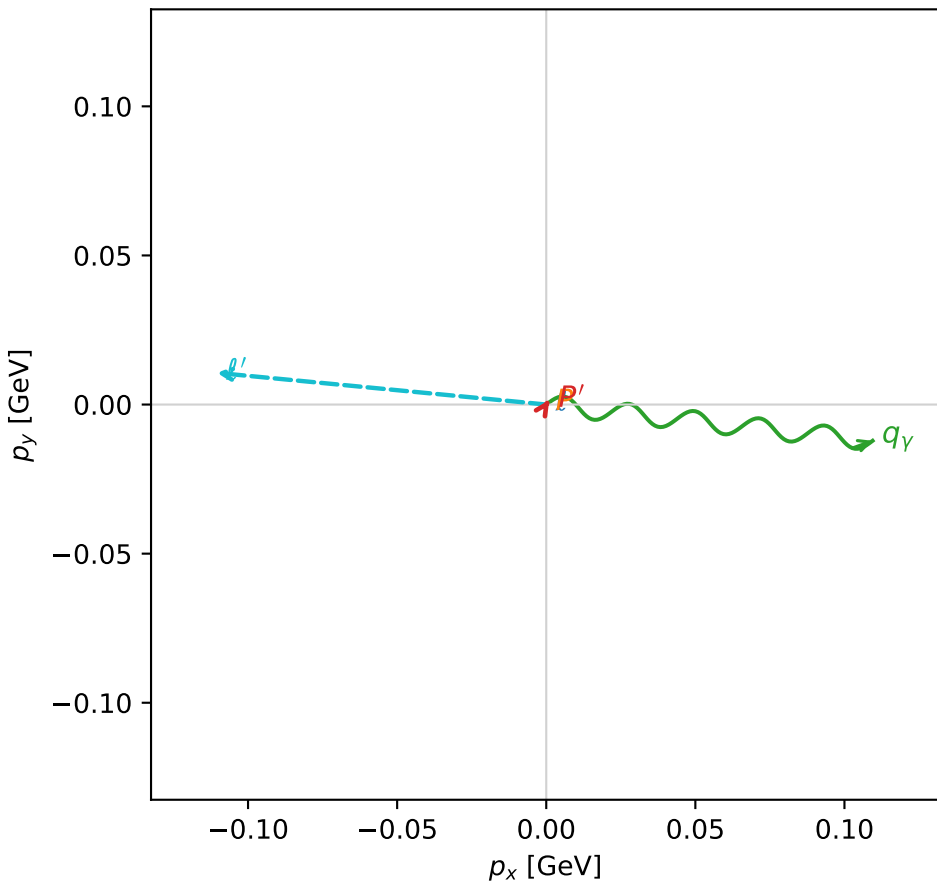
$s=1.34877$, $\sqrt{s}=1.16137$, $|\vec{P}|=0.201885$, $|\vec{P}'|=0.0259237$
 $\theta_{P'}=0.0702731$, $\theta_Y=1.62406$, $\phi_{P'}=0.988205$, $\phi_Y=6.17259$
 $E_Y=0.110269$, $\theta_{\ell'}=1.74905$, $\phi_{\ell'}=3.04561$
 $Q^2=0.0374493$, $x_B=0.153158$, $t=-0.0305422$, $W^2=1.08691$, $y=0.521433$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2019$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2019$
 P $E= 0.9595$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2019$
 ℓ' $E= 0.1127$ $p_x= -0.1104$ $p_y= 0.0106$ $p_z= -0.0200$
 P' $E= 0.9384$ $p_x= 0.0010$ $p_y= 0.0015$ $p_z= 0.0259$
 q_Y $E= 0.1103$ $p_x= 0.1094$ $p_y= -0.0122$ $p_z= -0.0059$

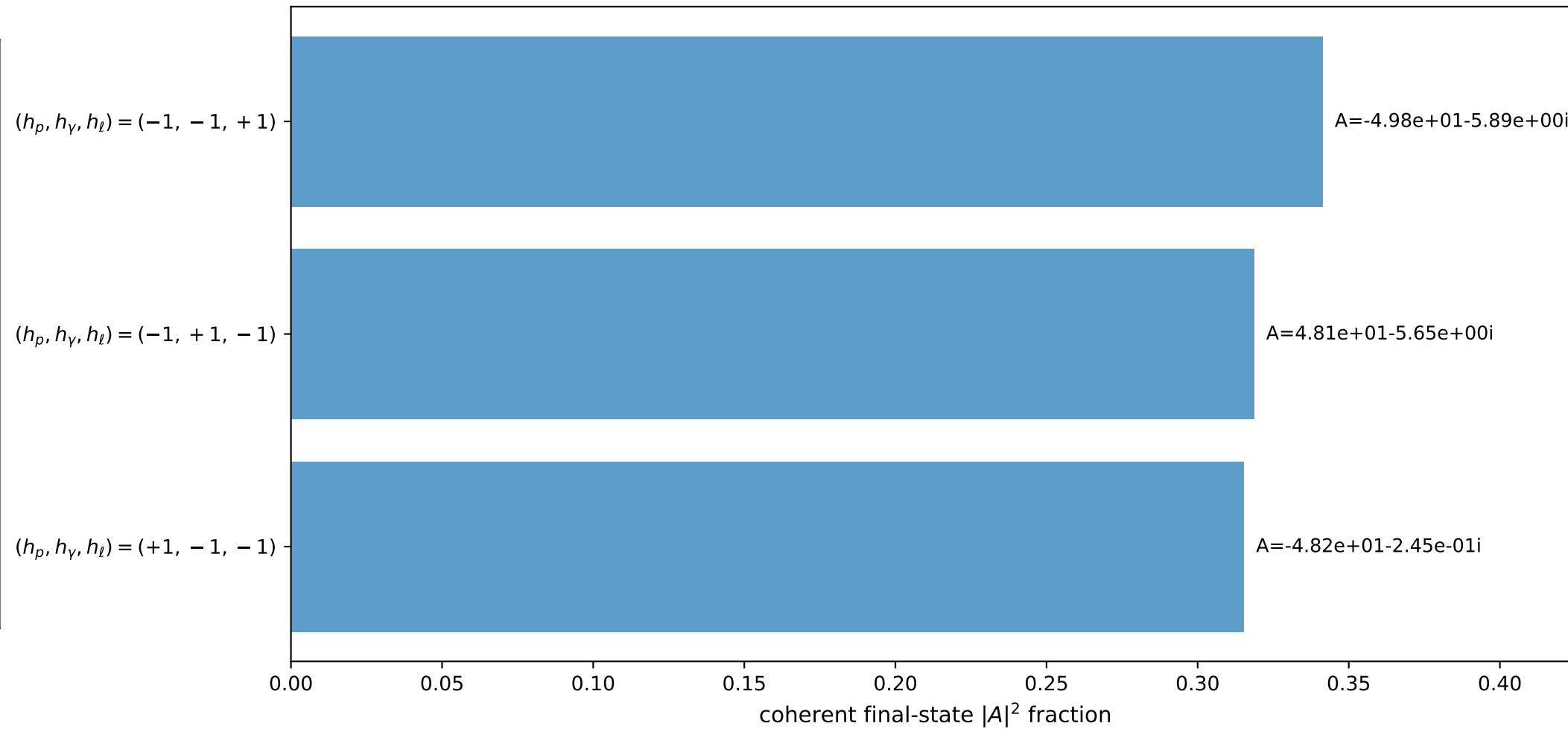
3D momenta



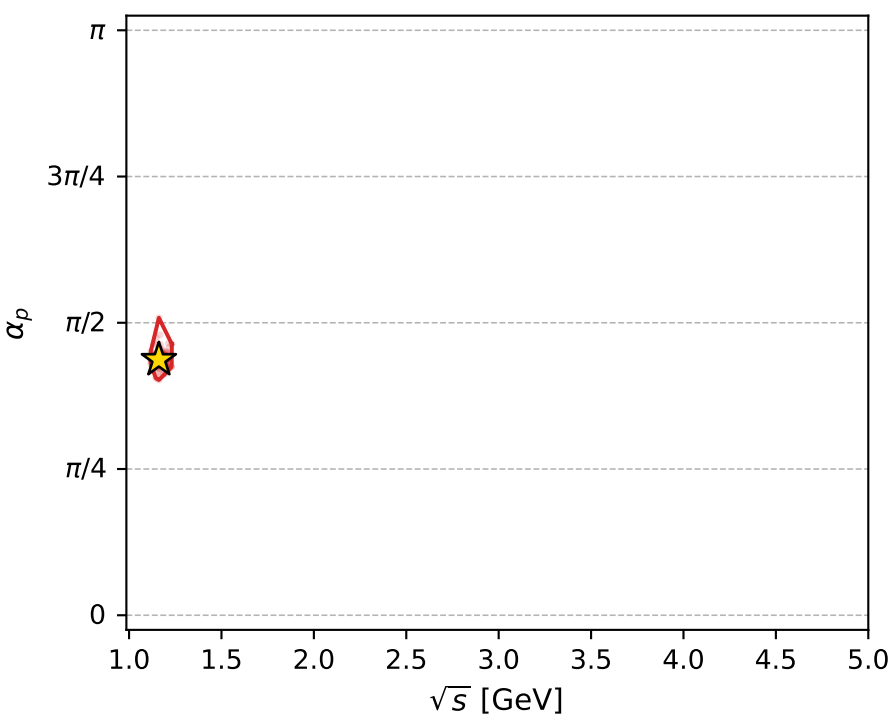
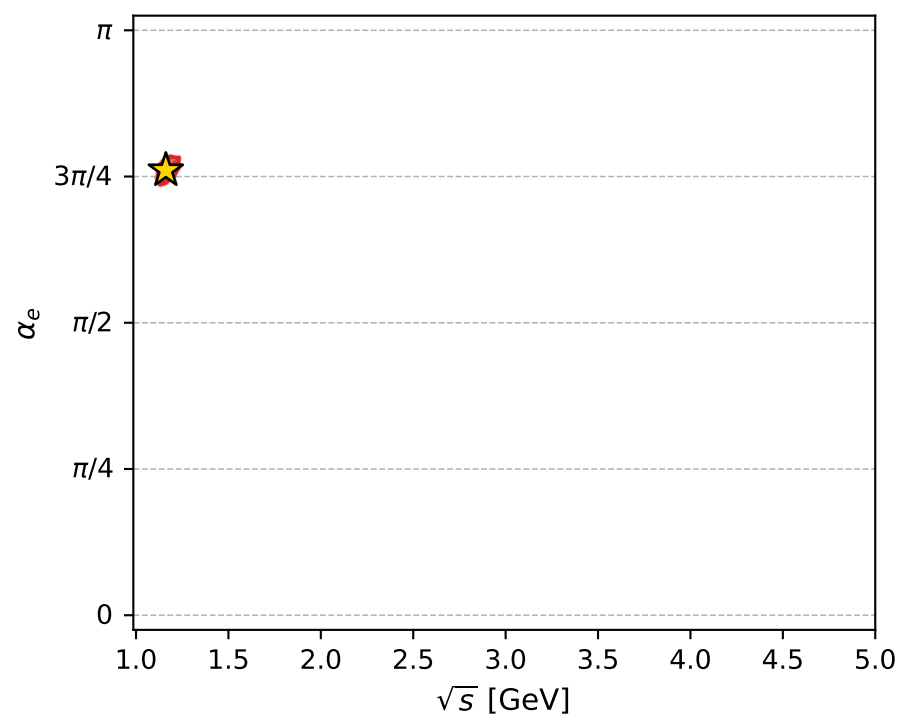
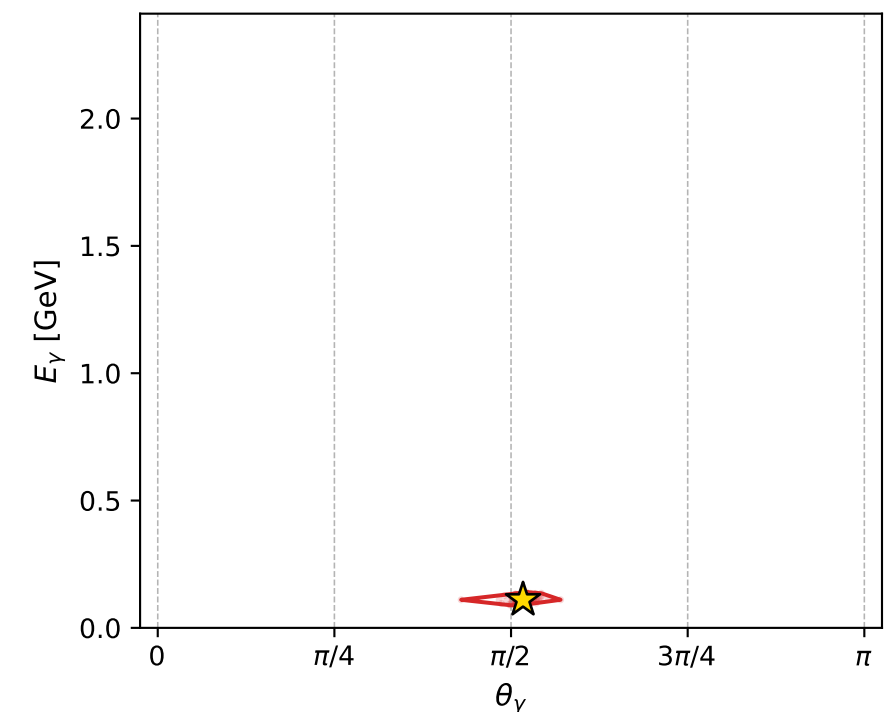
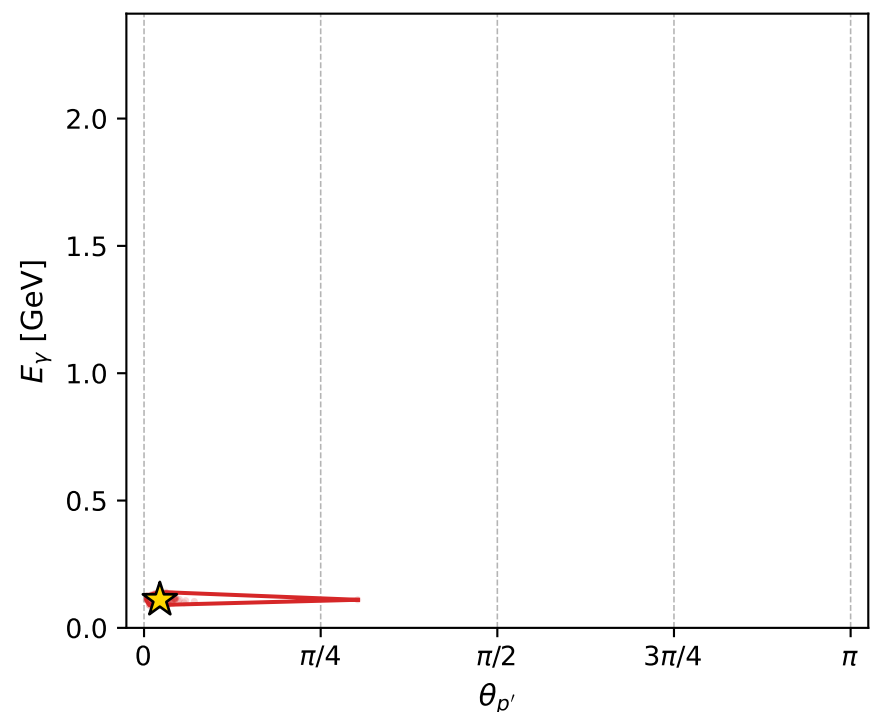
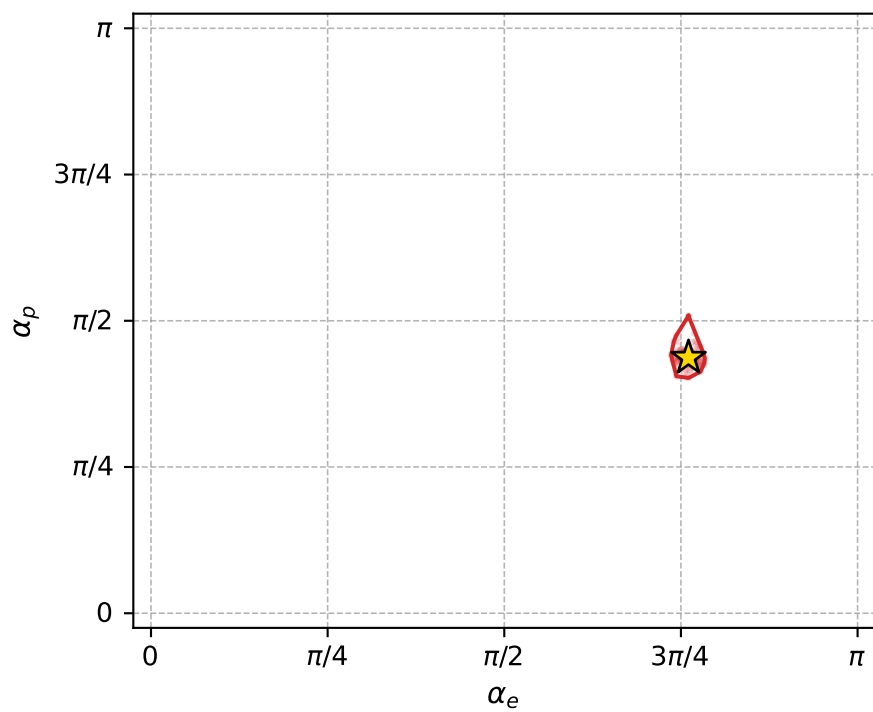
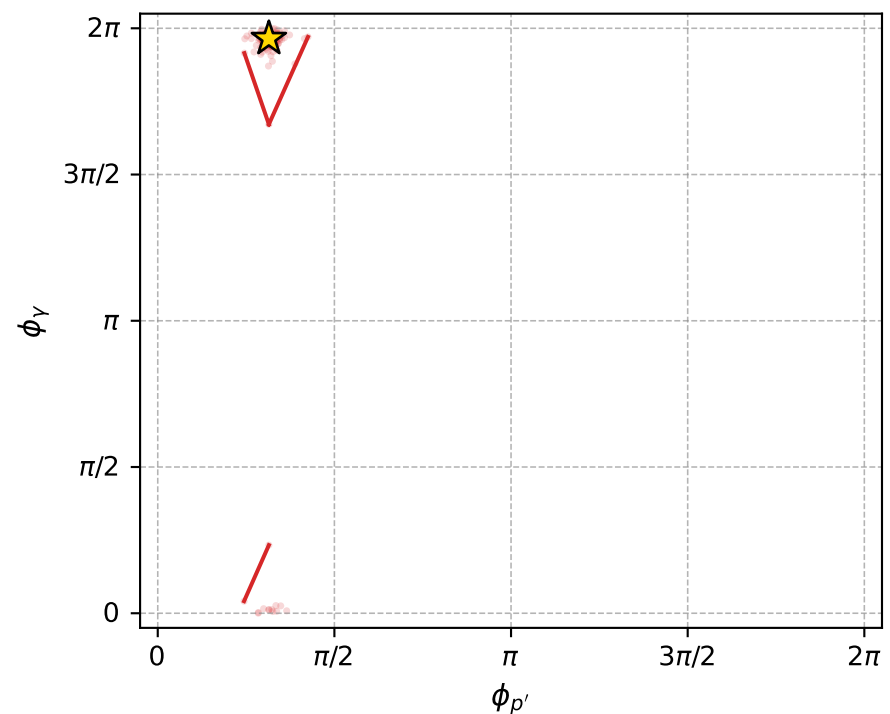
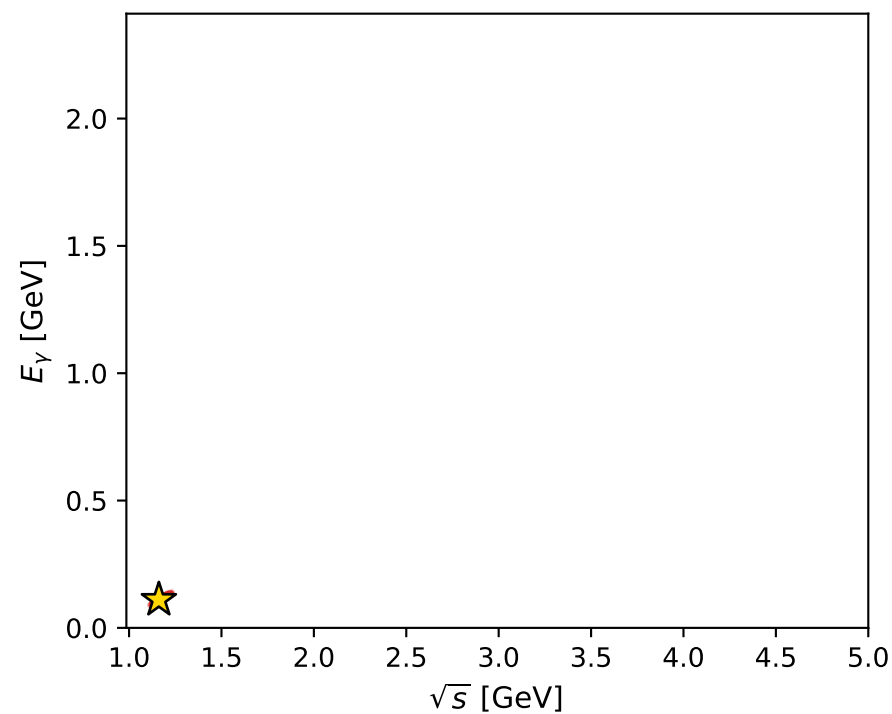
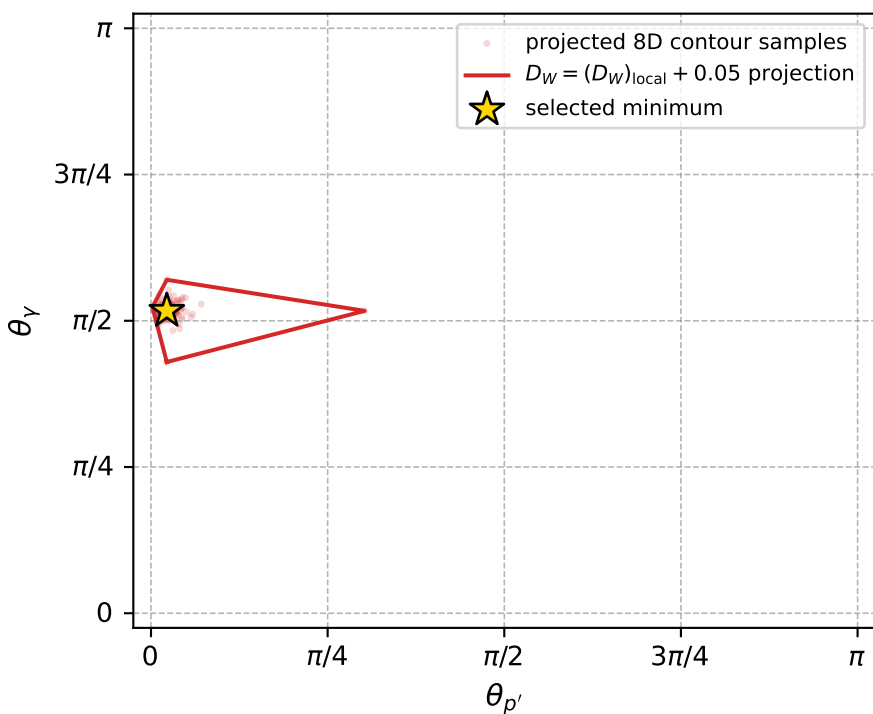
Transverse plane



Leading final-state amplitudes (N=3, retained=97.5%)



electron: polarization cluster P6, local minimum 933; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.015128228$
 $D_W \text{ contour} = 0.065128228$
 above global minimum = 0.015111795
 8D contour samples = 246

selected phase-space point:
 $\text{sqrt_s} = 1.161366$
 $\text{theta_p_out} = 0.07027307$
 $\text{theta_gamma_out} = 1.624059$
 $\text{qOut} = 0.1102687$
 $\text{phi_p_out} = 0.9882045$
 $\text{phi_gamma_out} = 6.172589$
 $\text{alpha_e} = 2.389794$
 $\text{alpha_p} = 1.372314$

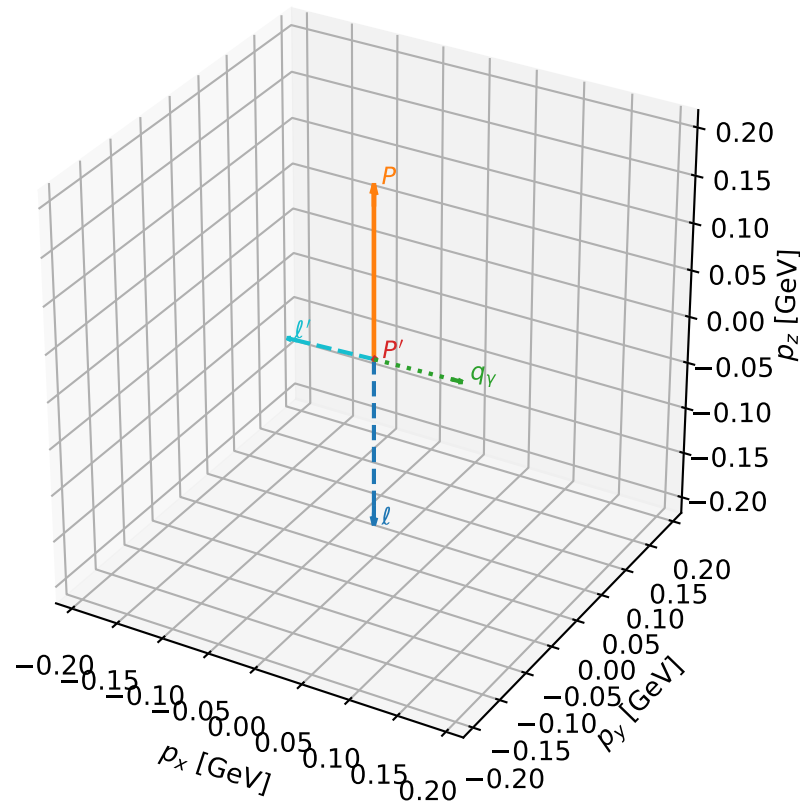
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_939 D_W=0.0153856
 region: polarization_cluster_06_local_minimum_939
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3848476$ rad
 incoming lepton: $-0.727075|+\rangle +0.686558|-\rangle$
 proton mixing angle: $\alpha_p=1.3709056$ rad
 incoming proton: $0.198562|+\rangle +0.980088|-\rangle$

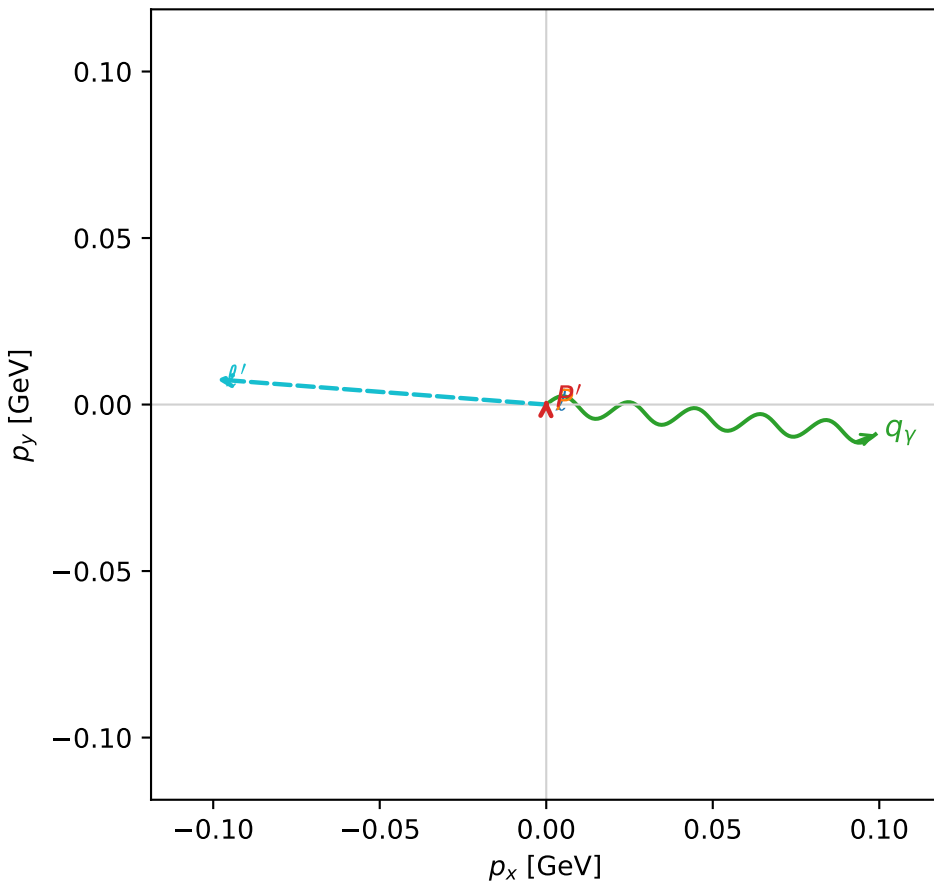
$s=1.29337$, $\sqrt{s}=1.13727$, $|\vec{P}|=0.181808$, $|\vec{P}'|=0.00147494$
 $\theta_{P'}=1.25563$, $\theta_V=1.48186$, $\phi_{P'}=1.54024$, $\phi_V=6.19289$
 $E_V=0.0996475$, $\theta_{\ell'}=1.66437$, $\phi_{\ell'}=3.06542$
 $Q^2=0.0328376$, $x_B=0.149408$, $t=-0.0325853$, $W^2=1.06679$, $y=0.531487$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1818$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1818$
 P $E= 0.9555$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1818$
 ℓ' $E= 0.0996$ $p_x= -0.0989$ $p_y= 0.0075$ $p_z= -0.0093$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0014$ $p_z= 0.0005$
 q_Y $E= 0.0996$ $p_x= 0.0988$ $p_y= -0.0089$ $p_z= 0.0089$

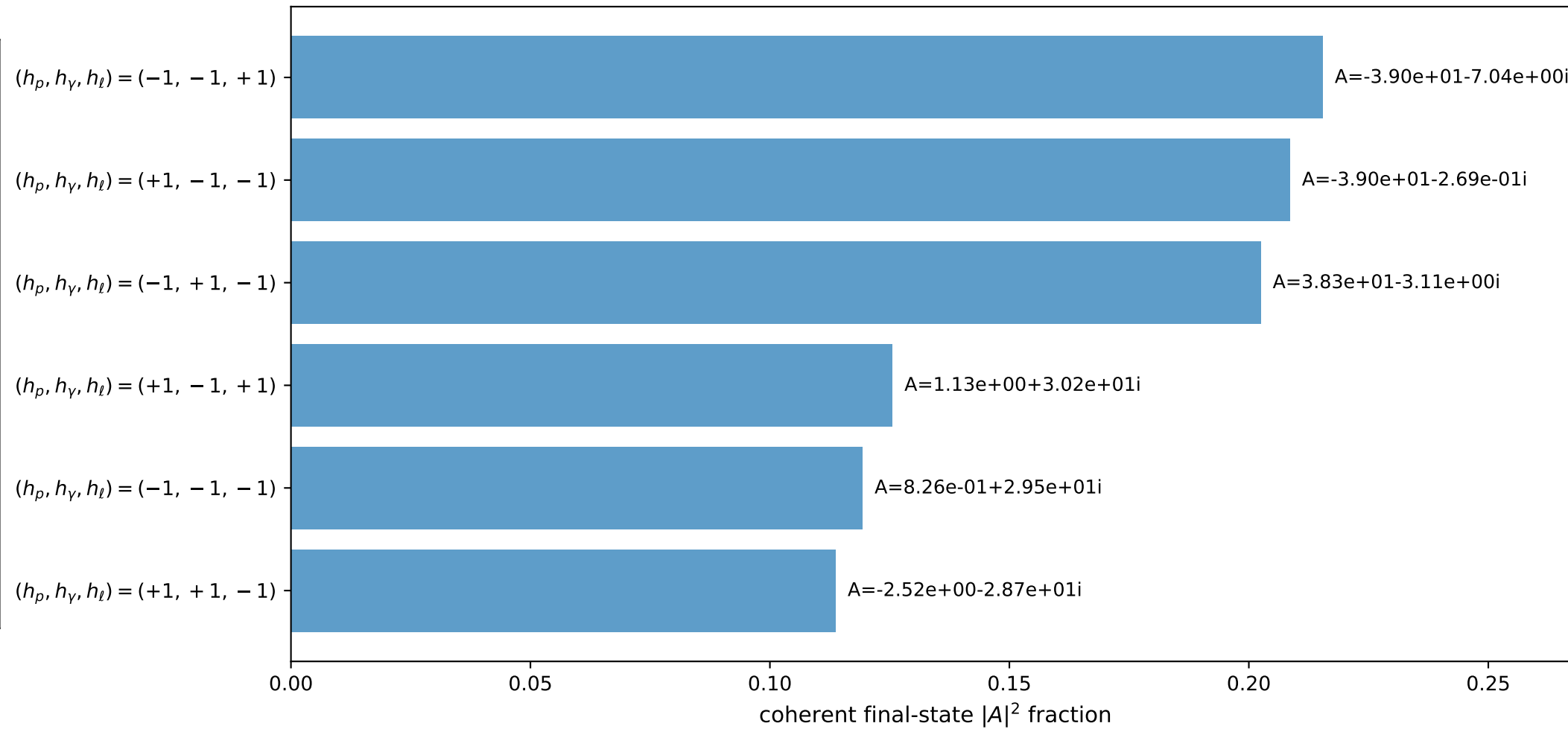
3D momenta



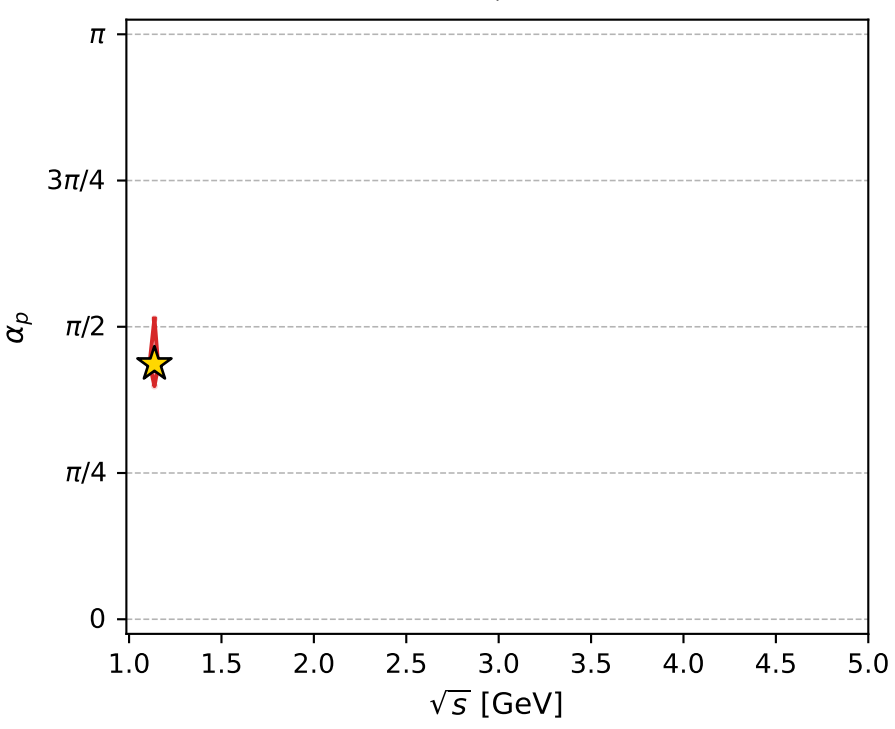
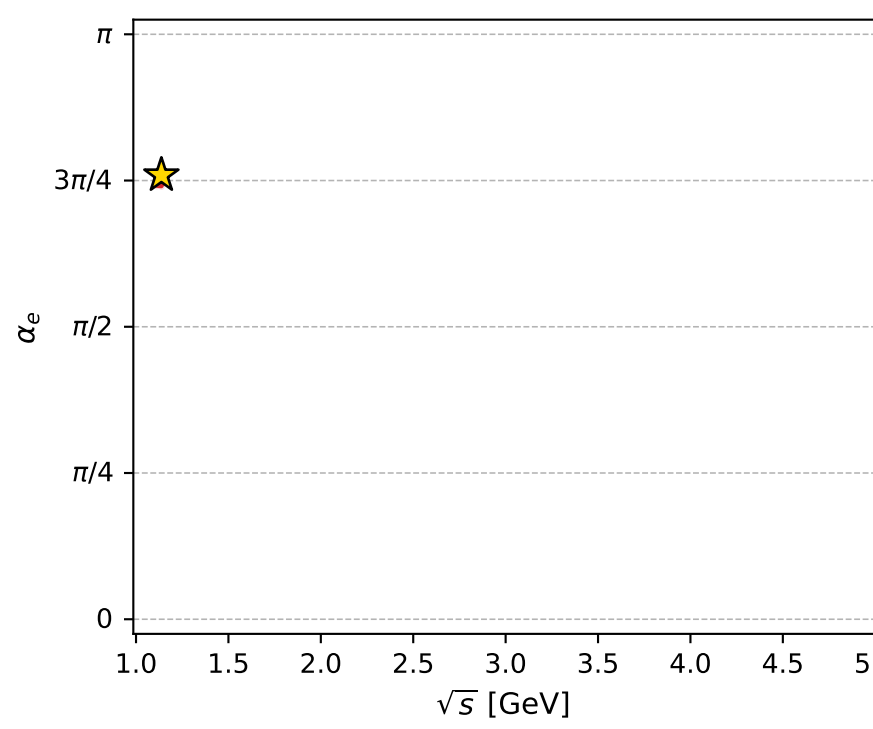
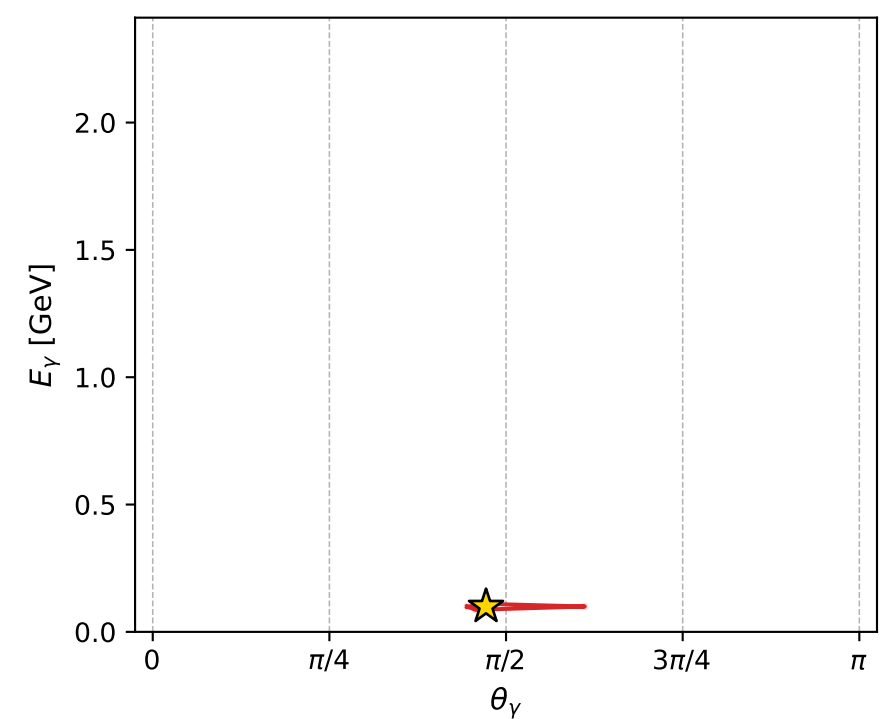
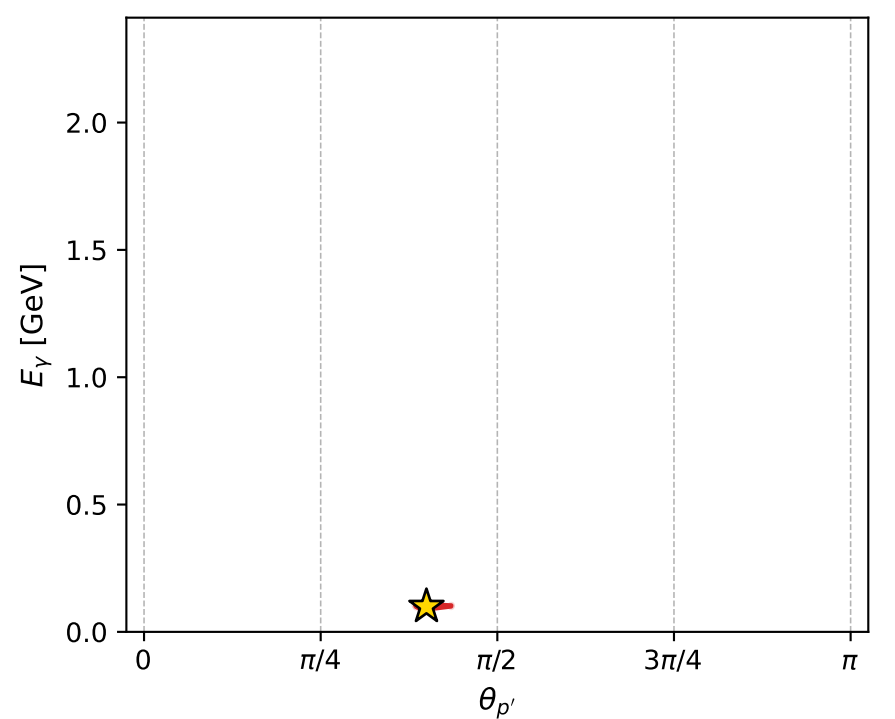
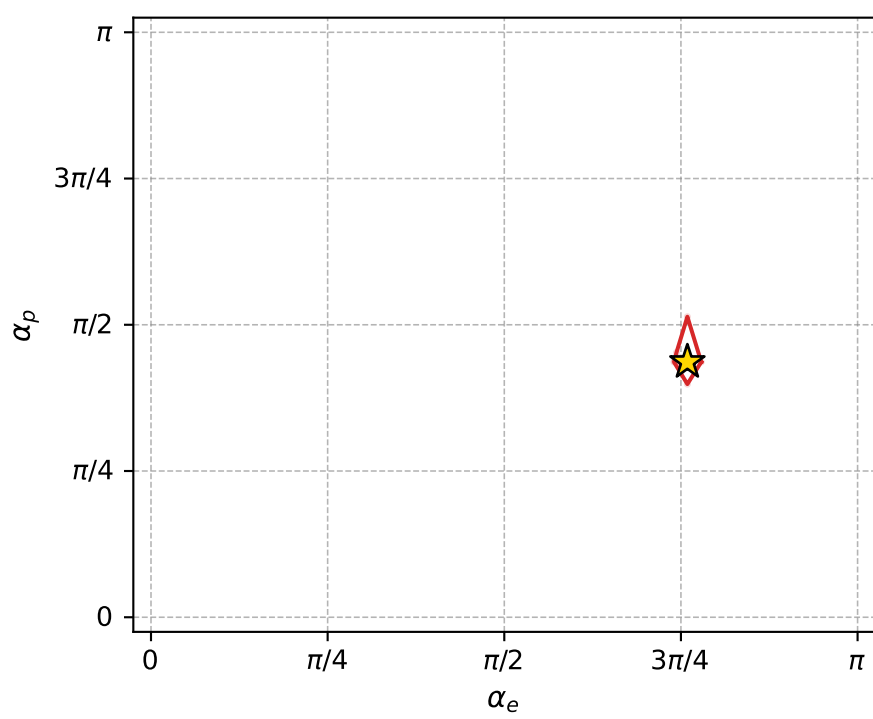
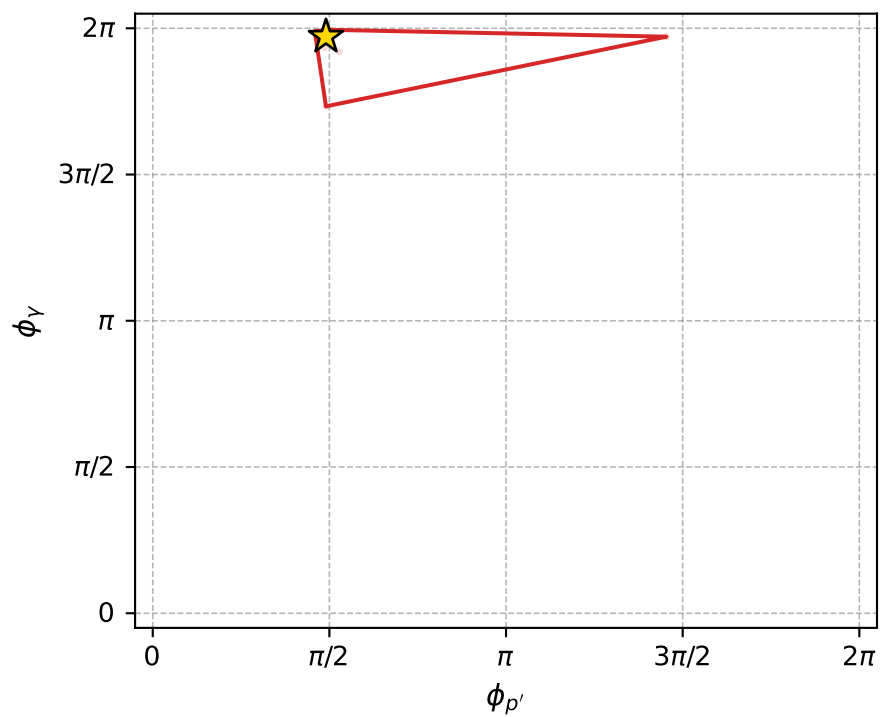
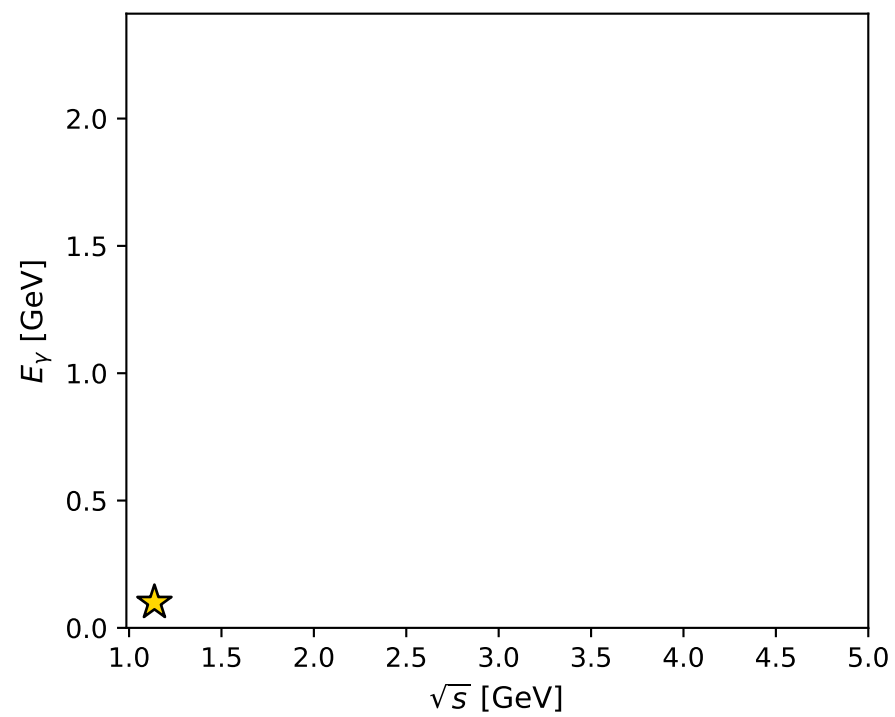
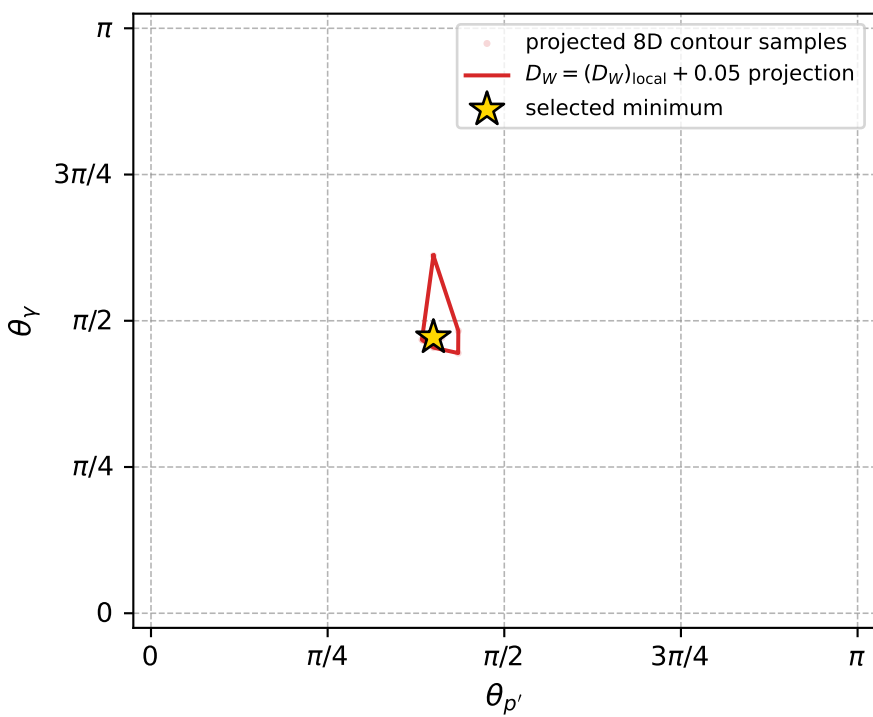
Transverse plane



Leading final-state amplitudes (N=6, retained=98.5%)



electron: polarization cluster P6, local minimum 939; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.015385586$
 $D_W \text{ contour} = 0.065385586$
 above global minimum = 0.015369153
 8D contour samples = 255

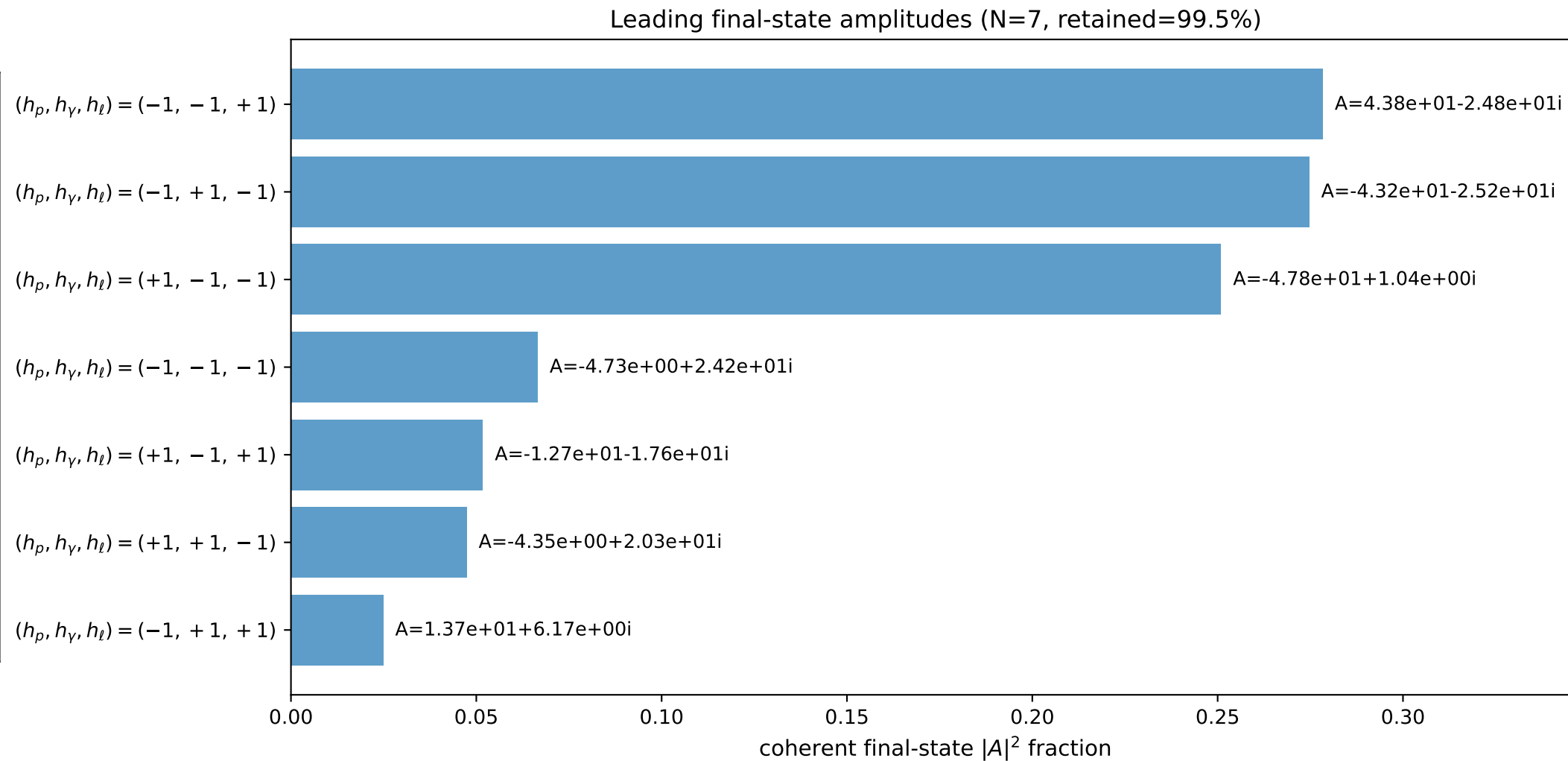
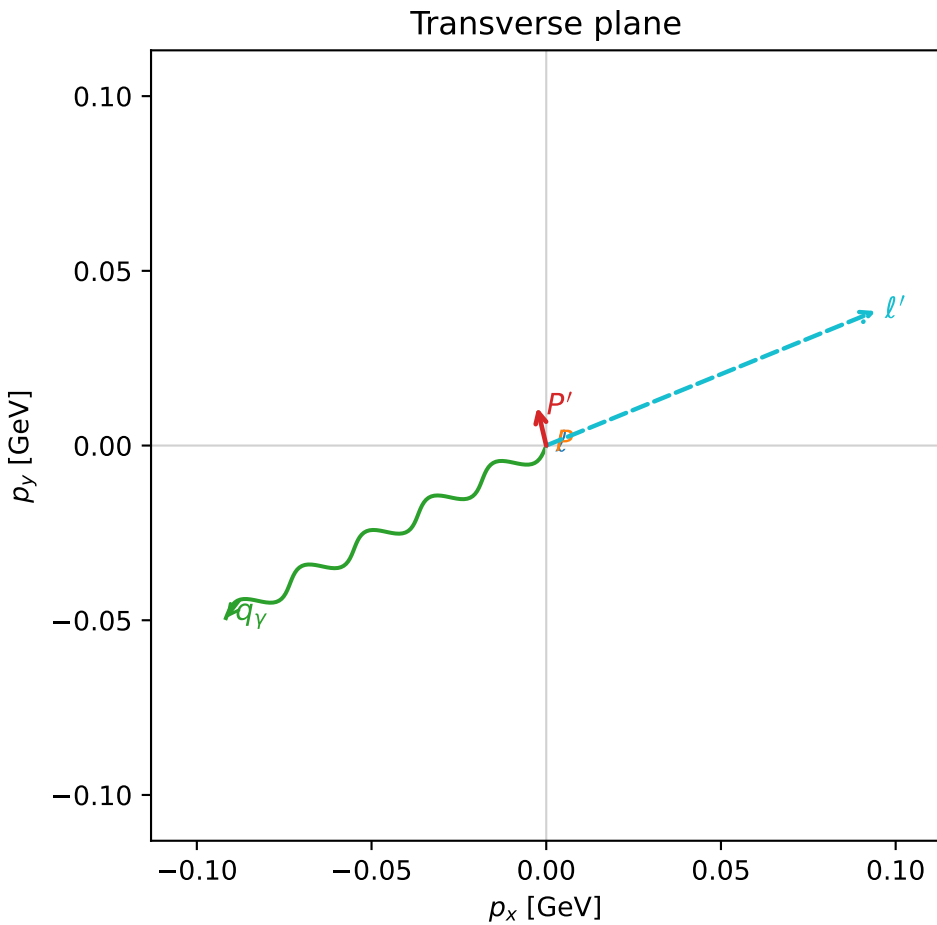
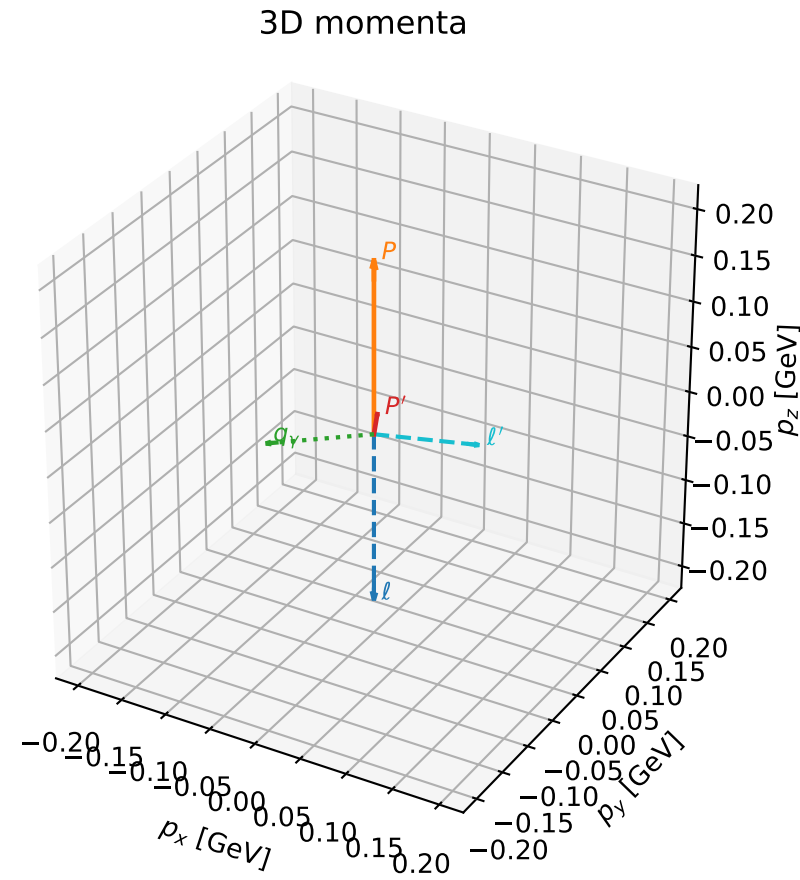
selected phase-space point:
 $\text{sqrt}_s = 1.137266$
 $\text{theta}_p_{\text{out}} = 1.255627$
 $\text{theta}_\gamma_{\text{out}} = 1.481864$
 $q_{\text{out}} = 0.09964747$
 $\text{phi}_p_{\text{out}} = 1.540238$
 $\text{phi}_\gamma_{\text{out}} = 6.192894$
 $\alpha_e = 2.384848$
 $\alpha_p = 1.370906$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

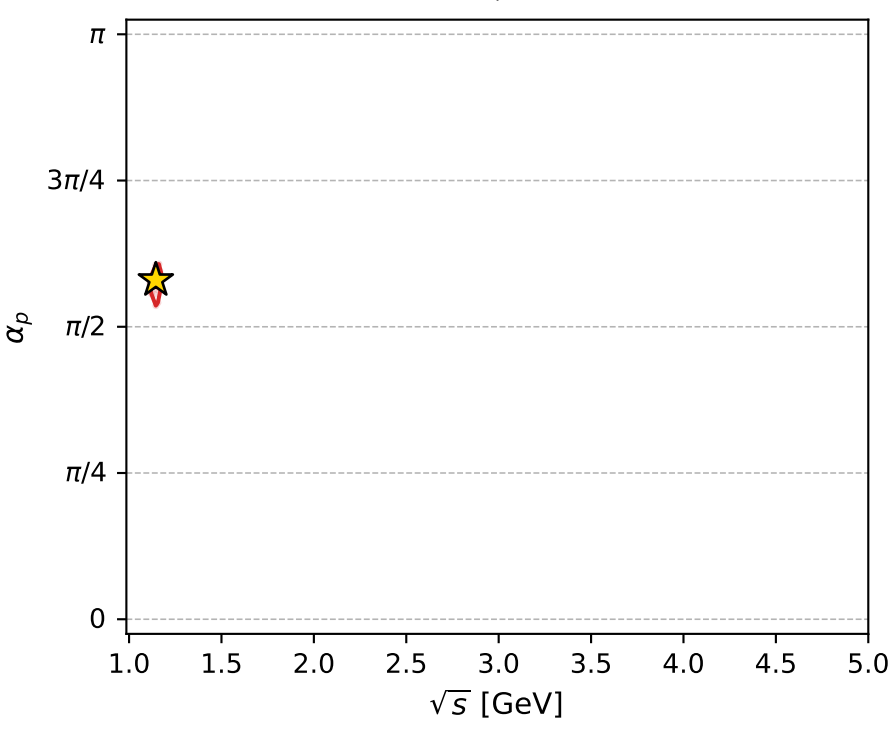
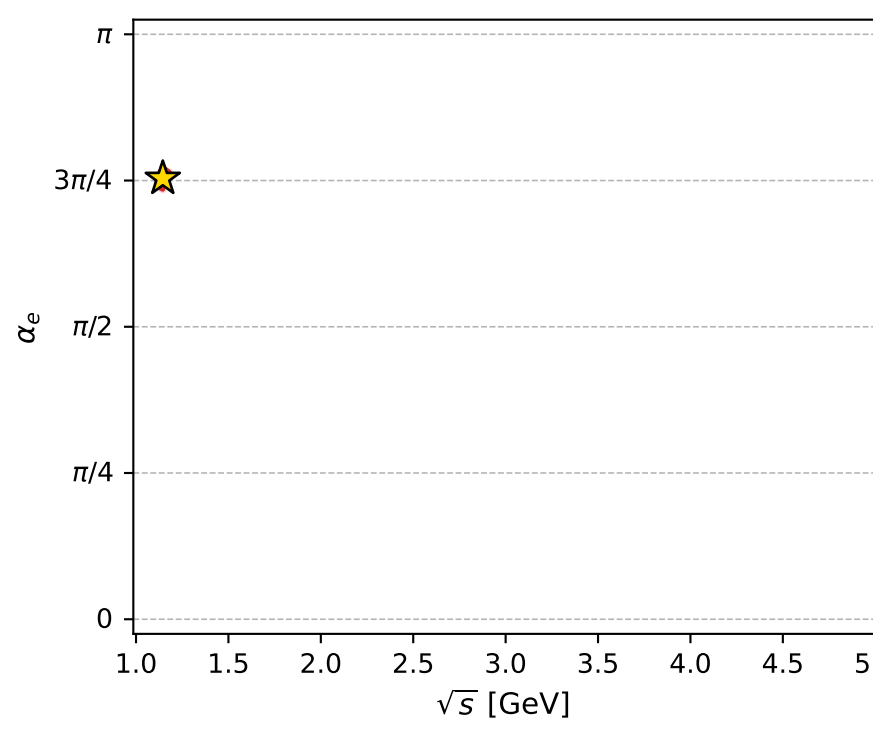
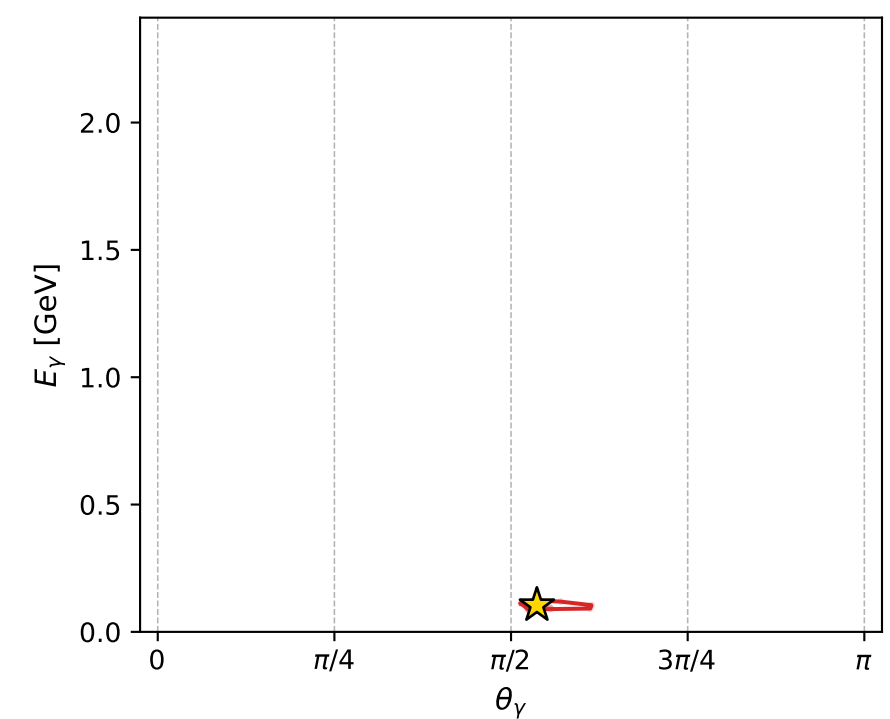
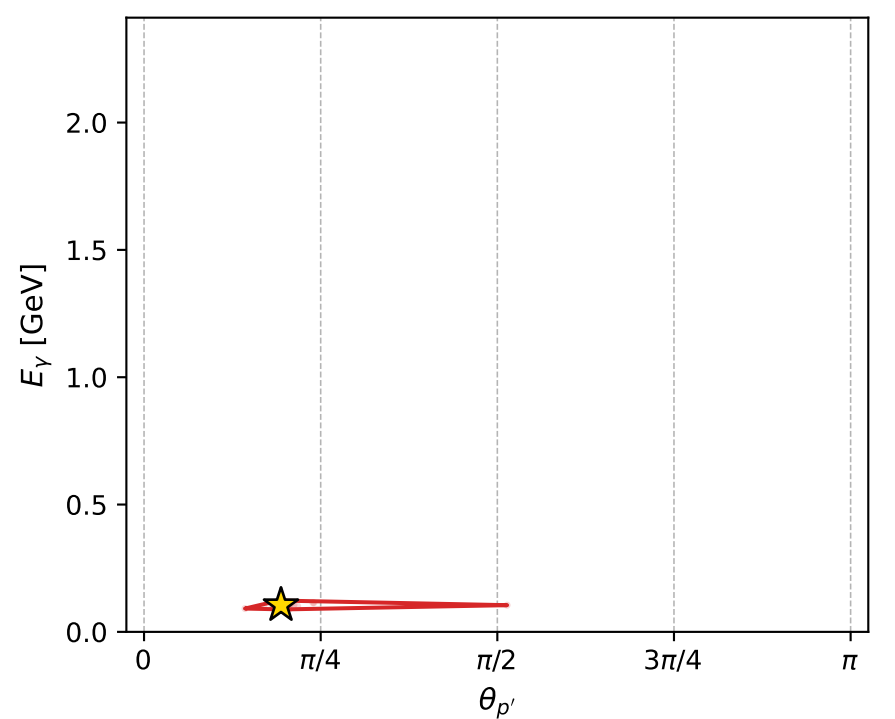
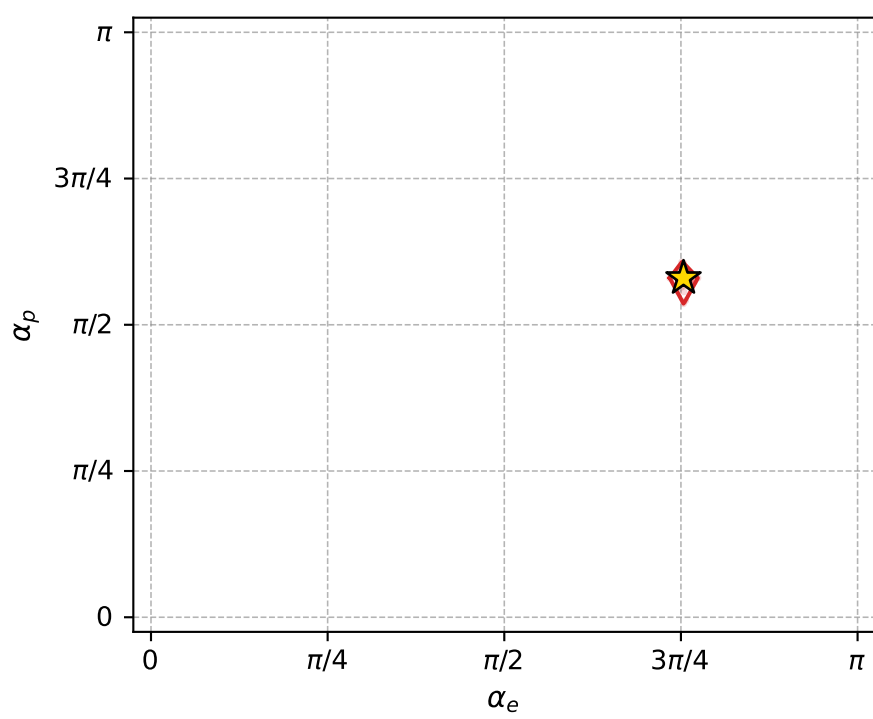
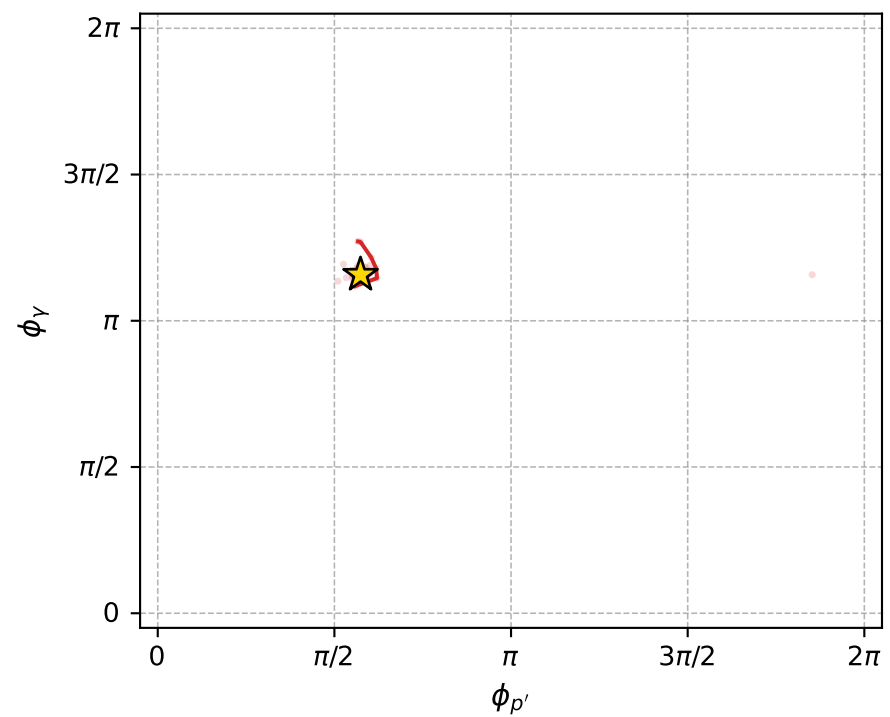
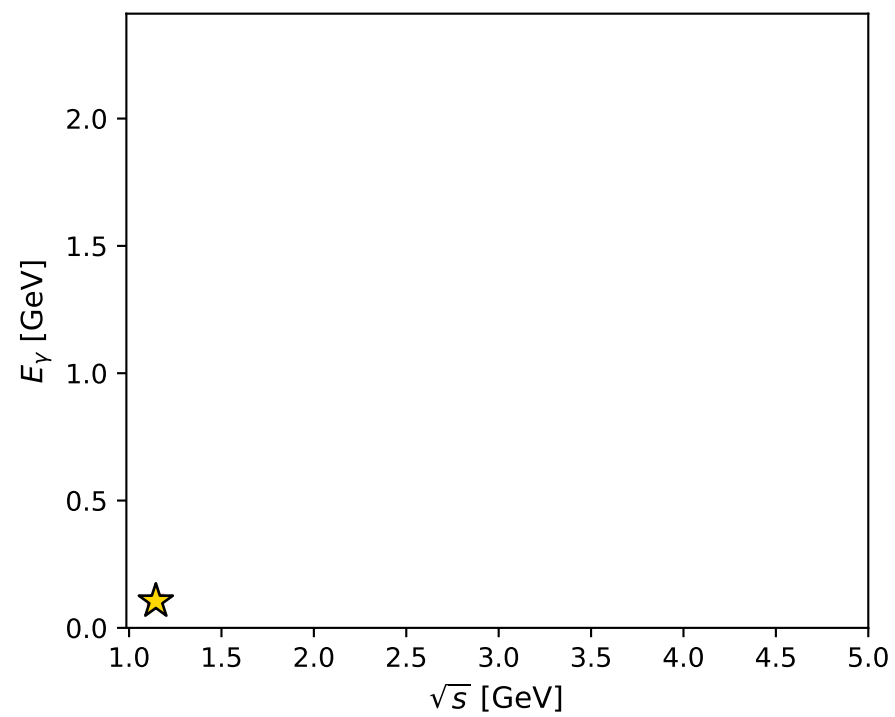
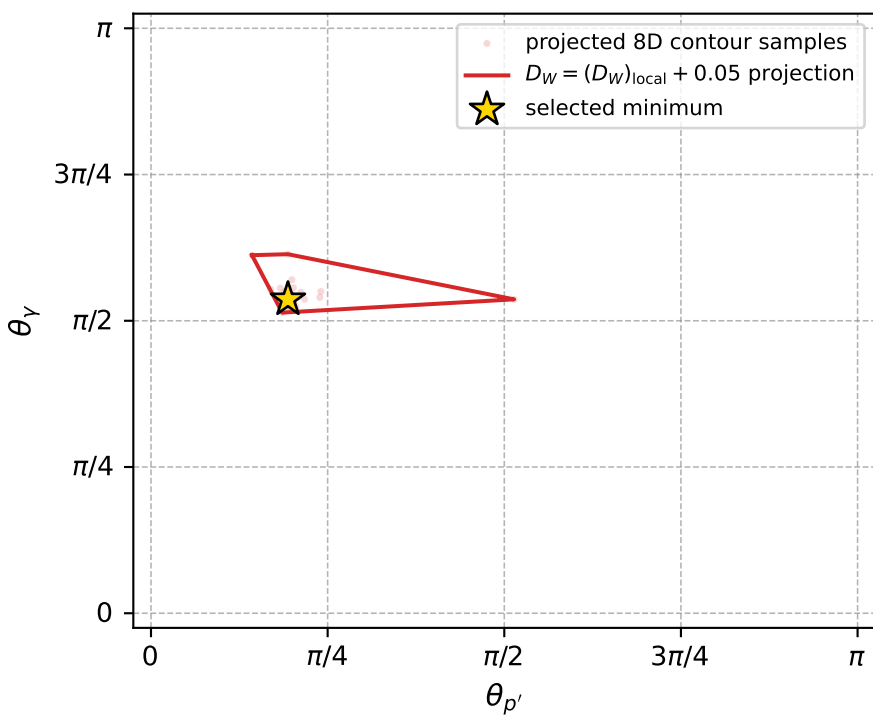
dw_mixing_angles_polarization_06_local_minimum_955 D_W=0.0164171
 region: polarization_cluster_06_local_minimum_955
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3675788$ rad
 incoming lepton: $-0.715111|+\rangle +0.699011|-\rangle$
 proton mixing angle: $\alpha_p=1.8217926$ rad
 incoming proton: $-0.248369|+\rangle +0.968665|-\rangle$

$s=1.31085$, $\sqrt{s}=1.14492$, $|\vec{P}|=0.188224$, $|\vec{P}'|=0.0195707$
 $\theta_{P'}=0.608934$, $\theta_Y=1.68583$, $\phi_{P'}=1.80306$, $\phi_Y=3.63568$
 $E_Y=0.104826$, $\theta_{\ell'}=1.61028$, $\phi_{\ell'}=0.387687$
 $Q^2=0.0368428$, $x_B=0.157092$, $t=-0.029426$, $W^2=1.07753$, $y=0.544147$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1882$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1882$
 P $E= 0.9567$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1882$
 ℓ' $E= 0.1019$ $p_x= 0.0943$ $p_y= 0.0385$ $p_z= -0.0040$
 P' $E= 0.9382$ $p_x= -0.0026$ $p_y= 0.0109$ $p_z= 0.0161$
 q_Y $E= 0.1048$ $p_x= -0.0917$ $p_y= -0.0494$ $p_z= -0.0120$



electron: polarization cluster P6, local minimum 955; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.016417118$
 $D_W \text{ contour} = 0.066417118$
 above global minimum = 0.016400685
 8D contour samples = 256

selected phase-space point:
 $\text{sqrt_s} = 1.144923$
 $\text{theta_p_out} = 0.6089336$
 $\text{theta_gamma_out} = 1.685825$
 $\text{qOut} = 0.1048263$
 $\text{phi_p_out} = 1.803057$
 $\text{phi_gamma_out} = 3.635684$
 $\text{alpha_e} = 2.367579$
 $\text{alpha_p} = 1.821793$

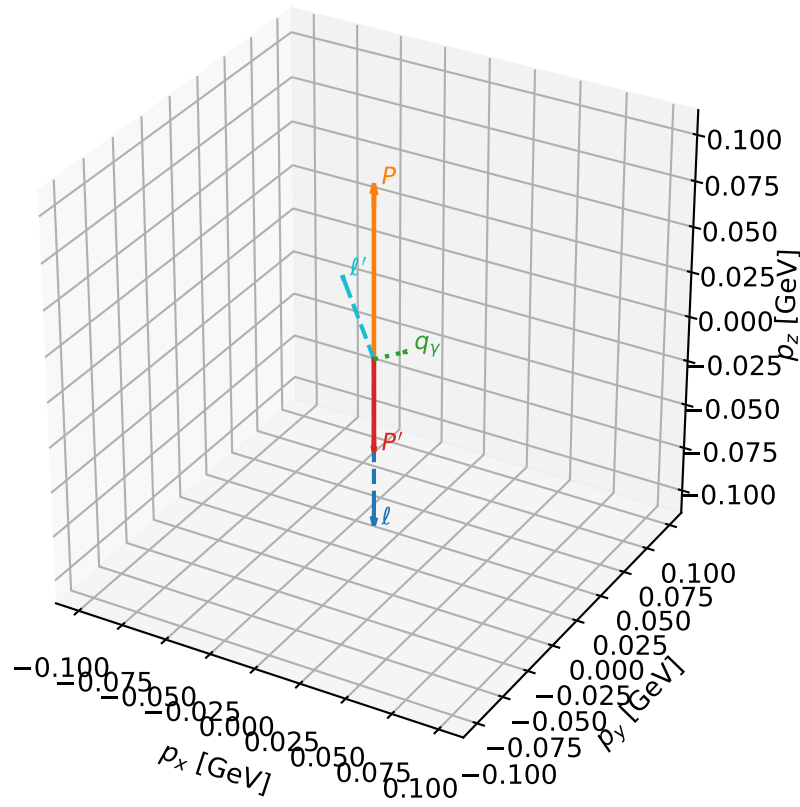
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_966 D_W=0.0167039
 region: polarization_cluster_06_local_minimum_966
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3412481$ rad
 incoming lepton: $-0.696459|+\rangle +0.717596|-\rangle$
 proton mixing angle: $\alpha_p=1.4588753$ rad
 incoming proton: $0.111687|+\rangle +0.993743|-\rangle$

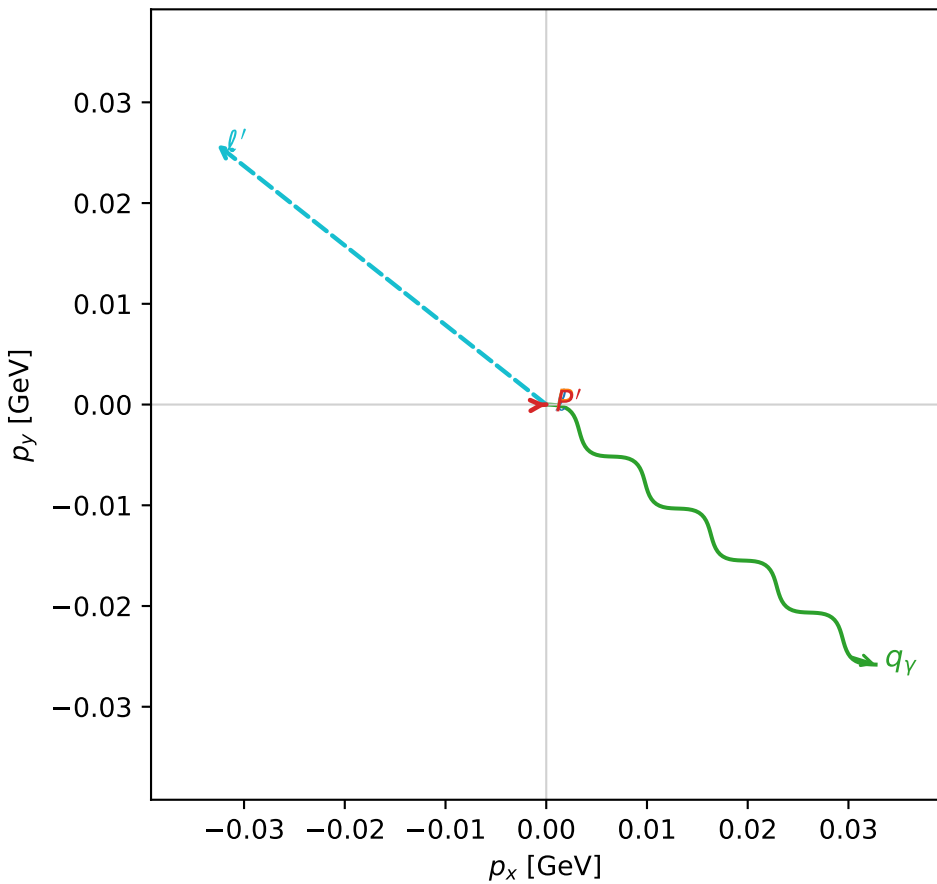
$s=1.07592$, $\sqrt{s}=1.03727$, $|\vec{P}|=0.0945156$, $|\vec{P}'|=0.0513191$
 $\theta_{P'}=3.14159$, $\theta_V=1.00401$, $\phi_{P'}=0.0566132$, $\phi_V=5.61482$
 $E_V=0.0493776$, $\theta_{\ell'}=1.03368$, $\phi_{\ell'}=2.47323$
 $Q^2=0.0138544$, $x_B=0.126701$, $t=-0.0212566$, $W^2=0.975336$, $y=0.557668$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.0945$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0945$
 P $E= 0.9427$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0945$
 ℓ' $E= 0.0485$ $p_x= -0.0327$ $p_y= 0.0258$ $p_z= 0.0248$
 P' $E= 0.9394$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0513$
 q_V $E= 0.0494$ $p_x= 0.0327$ $p_y= -0.0258$ $p_z= 0.0265$

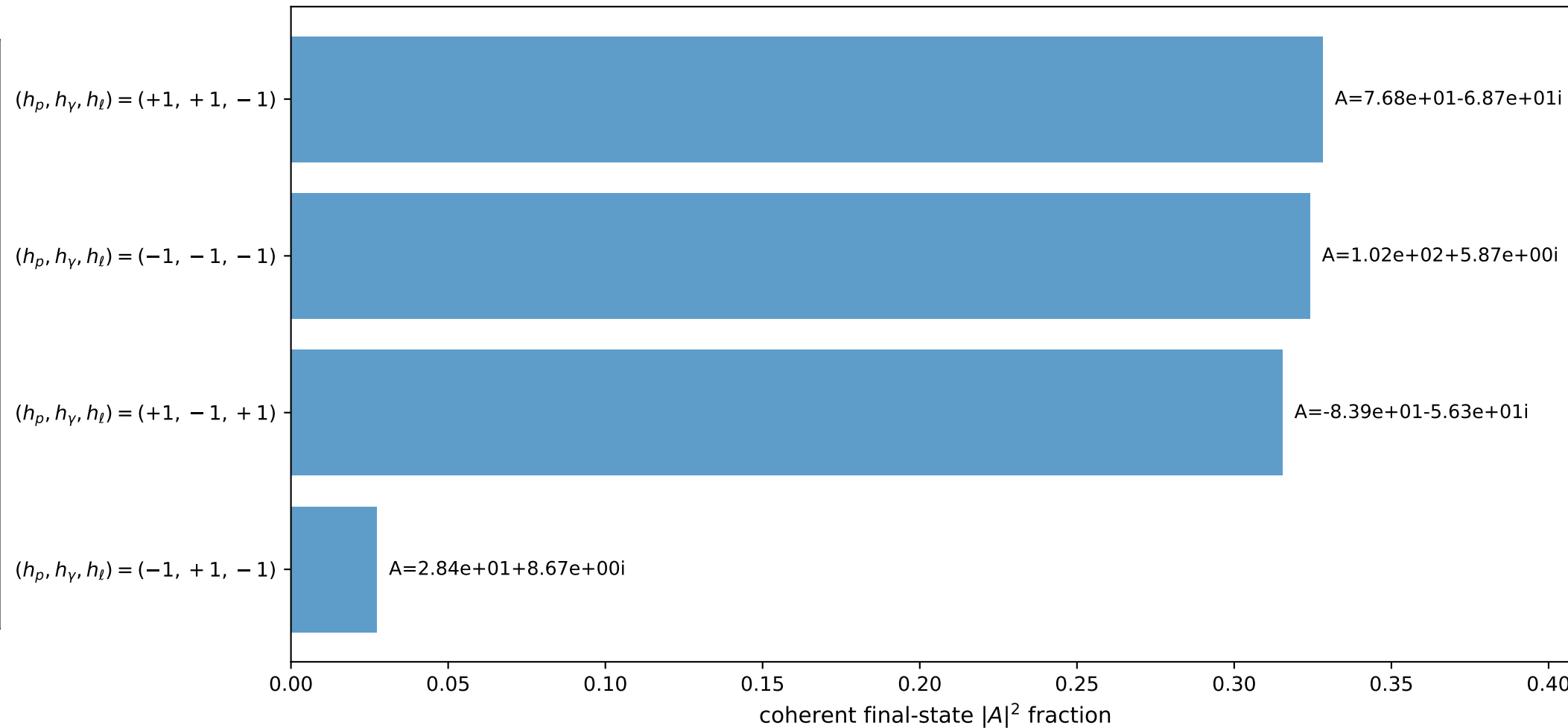
3D momenta



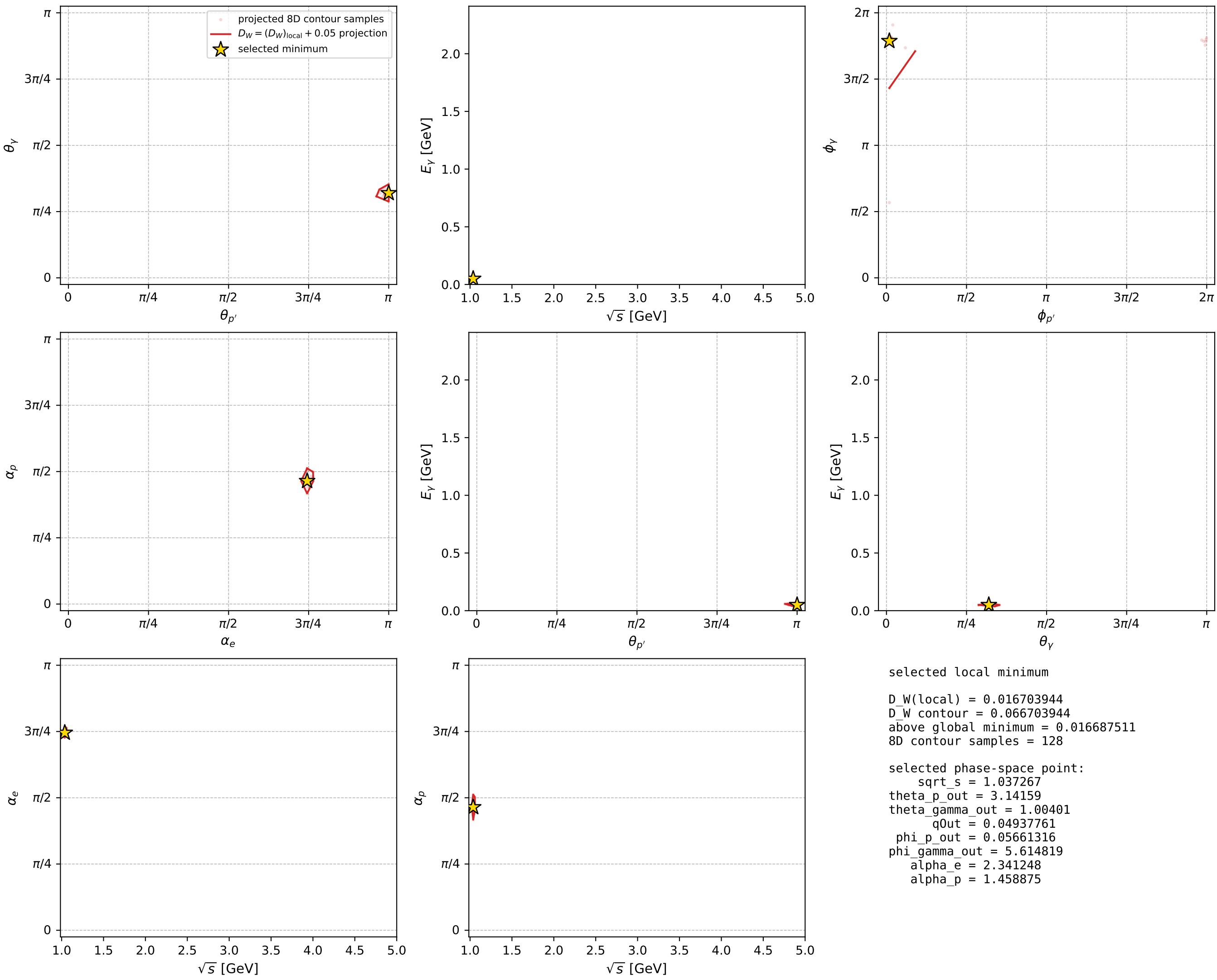
Transverse plane



Leading final-state amplitudes (N=4, retained=99.5%)



electron: polarization cluster P6, local minimum 966; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

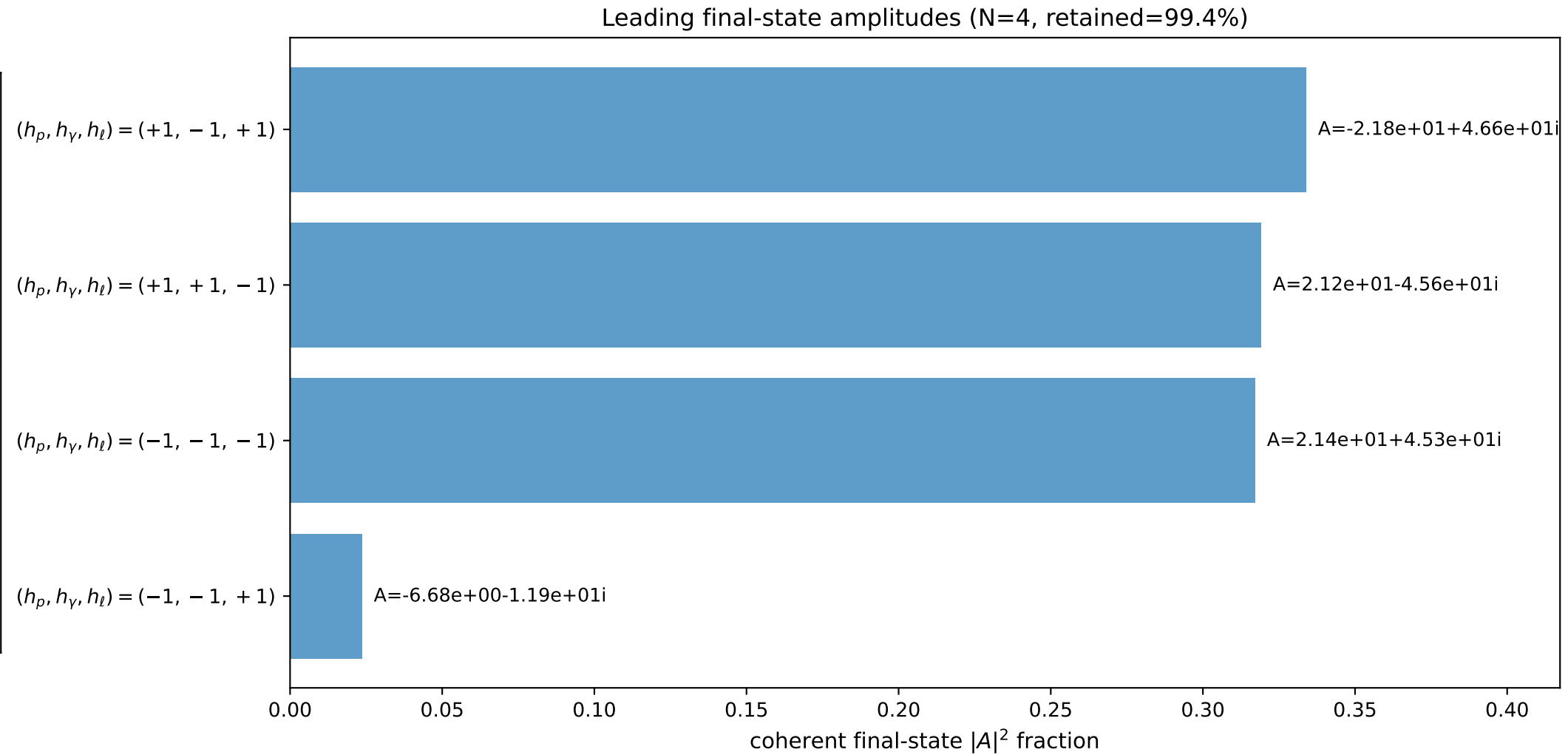
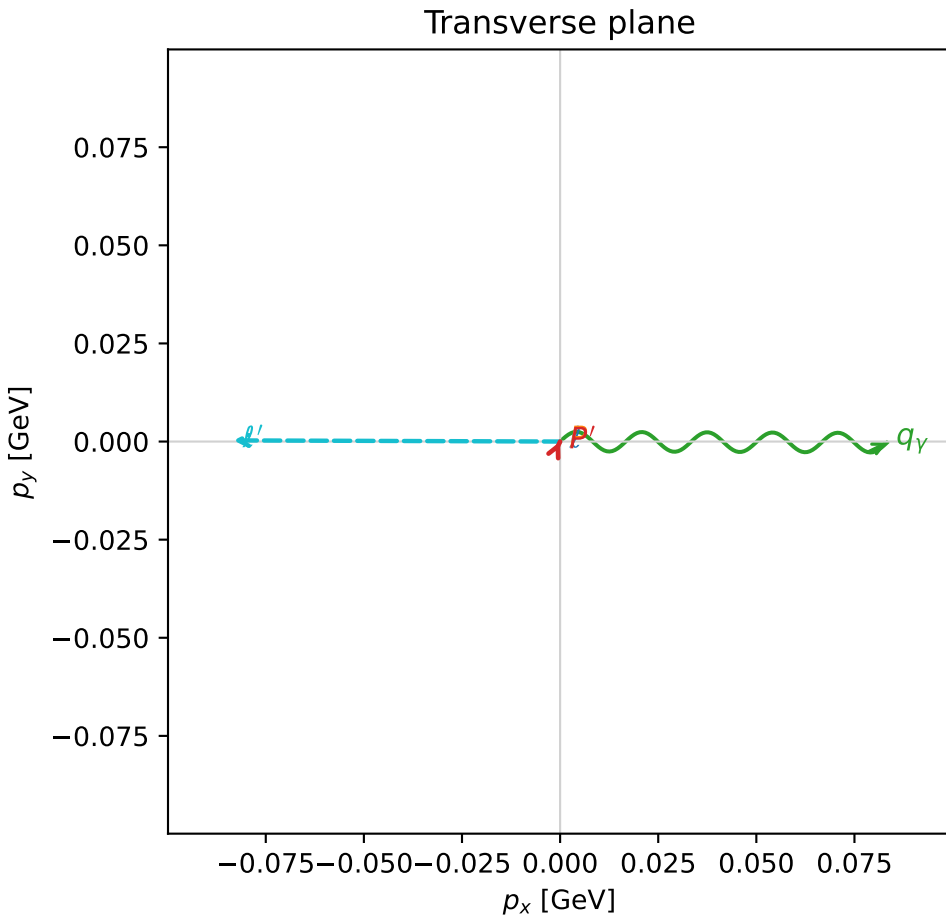
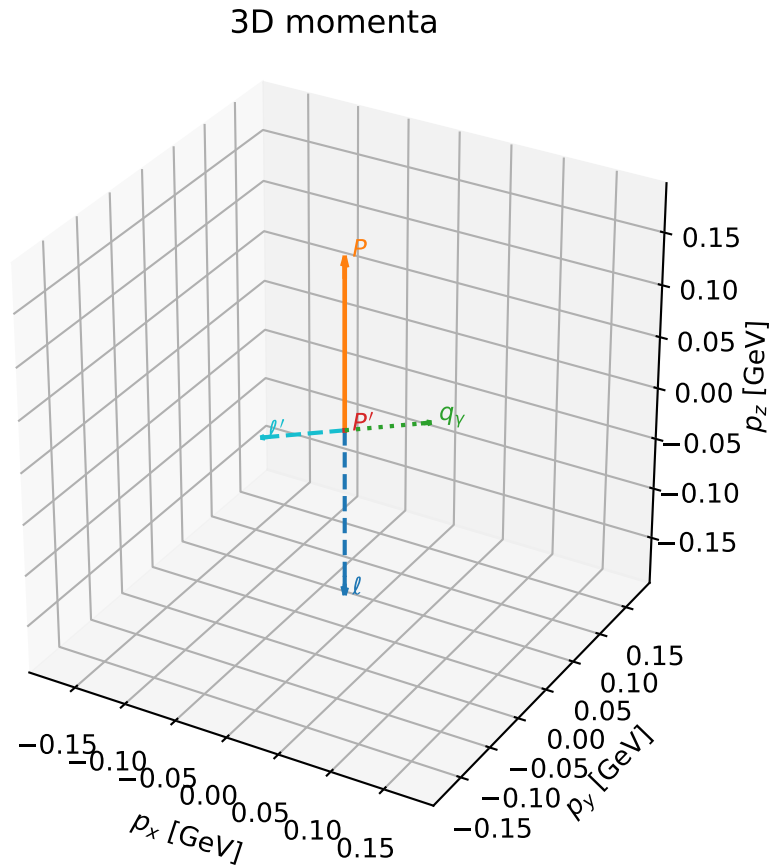


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

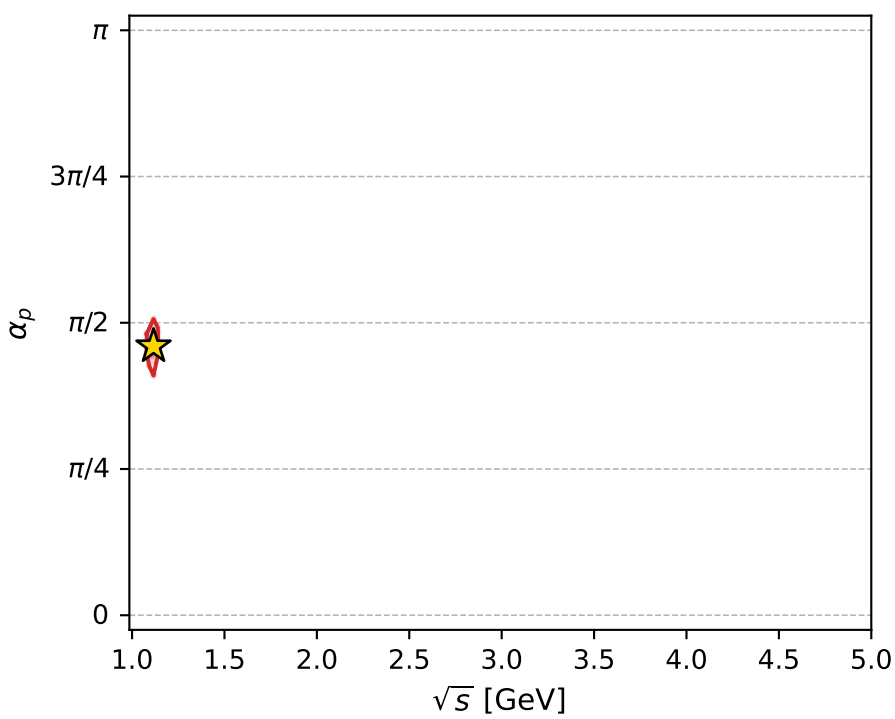
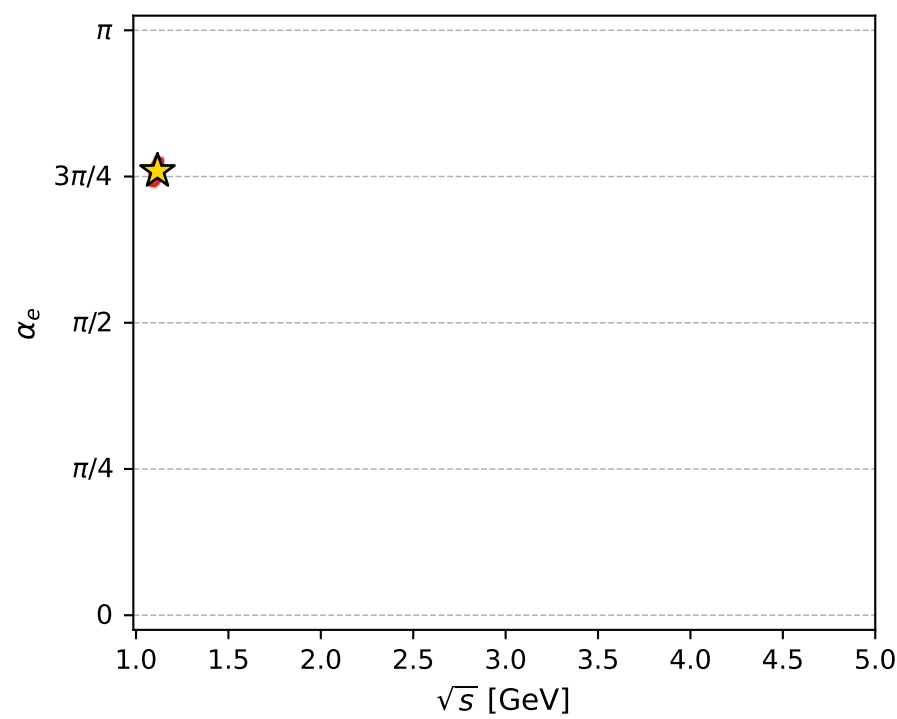
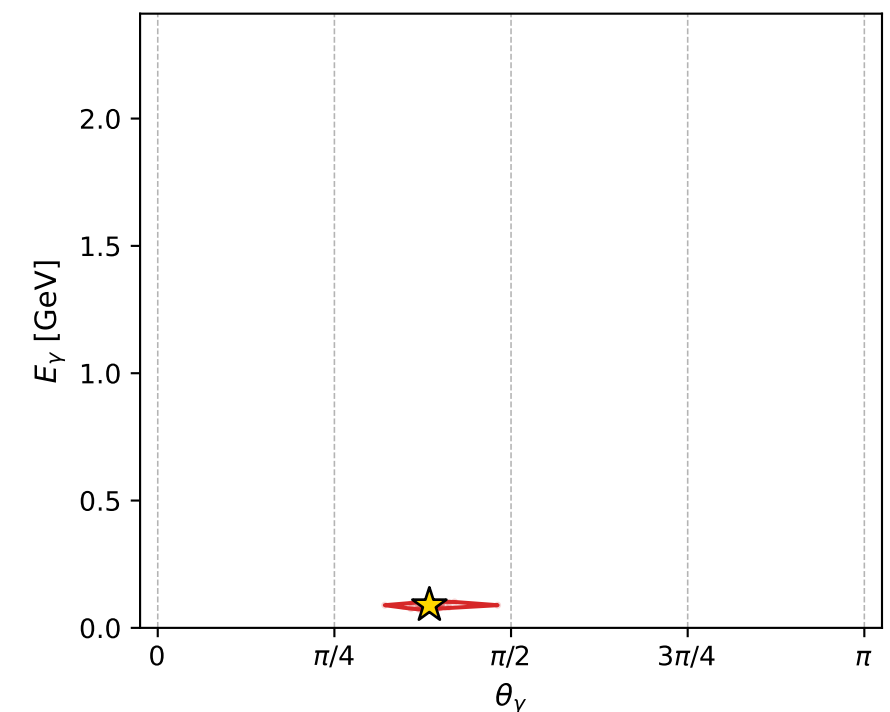
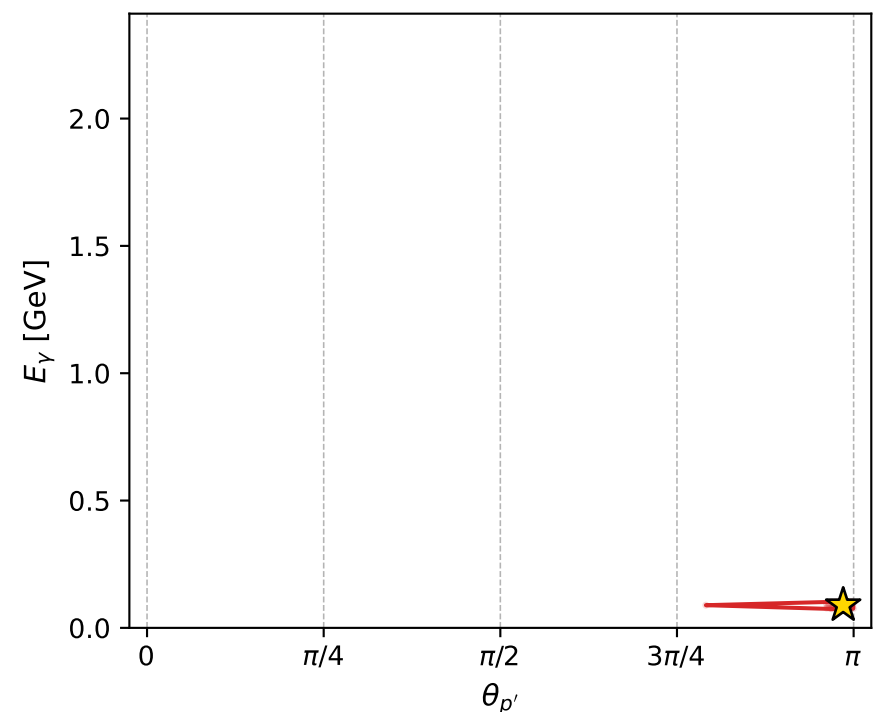
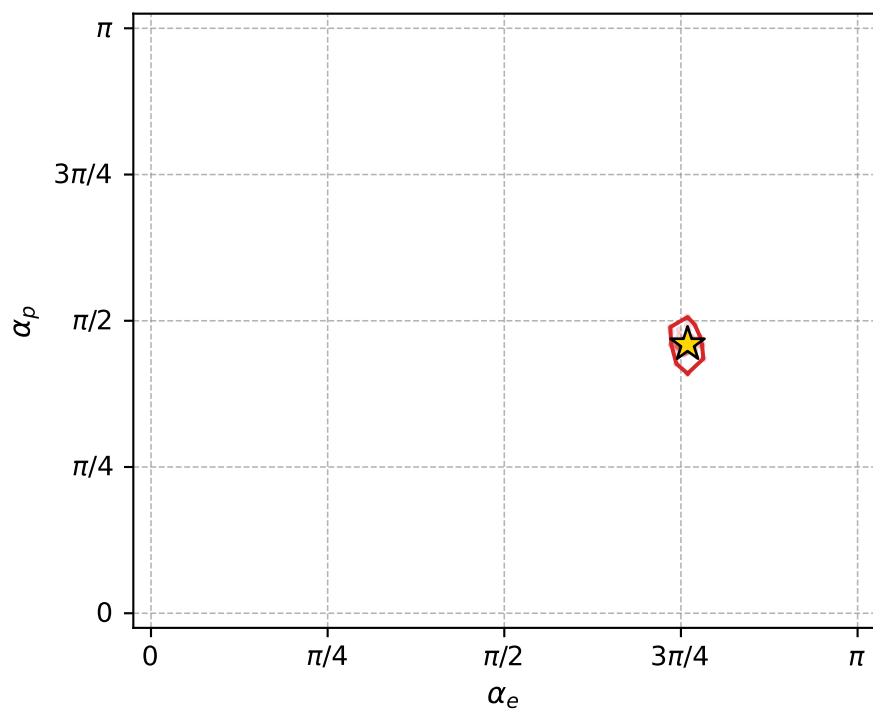
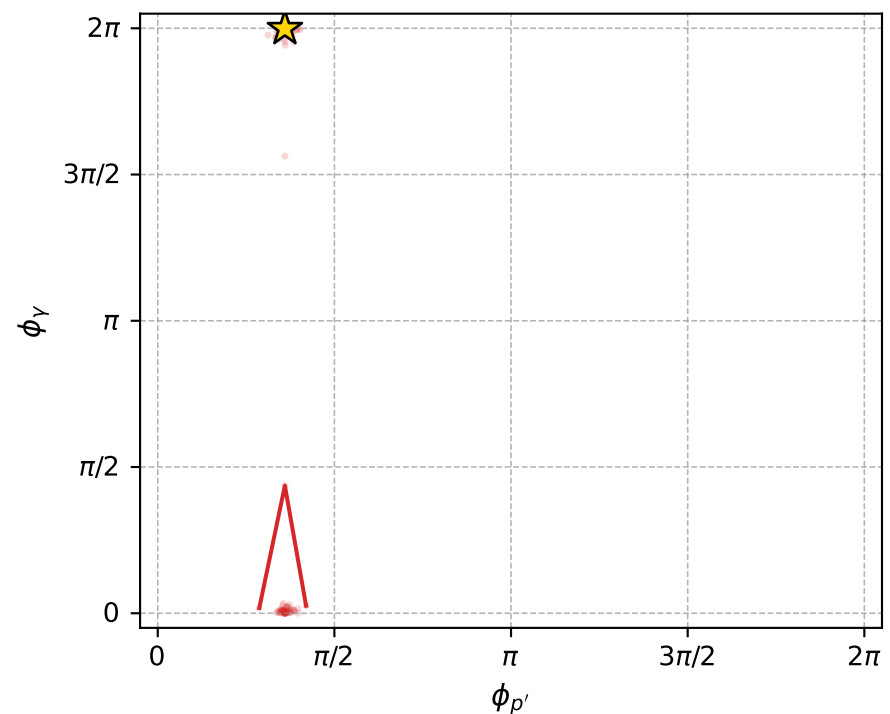
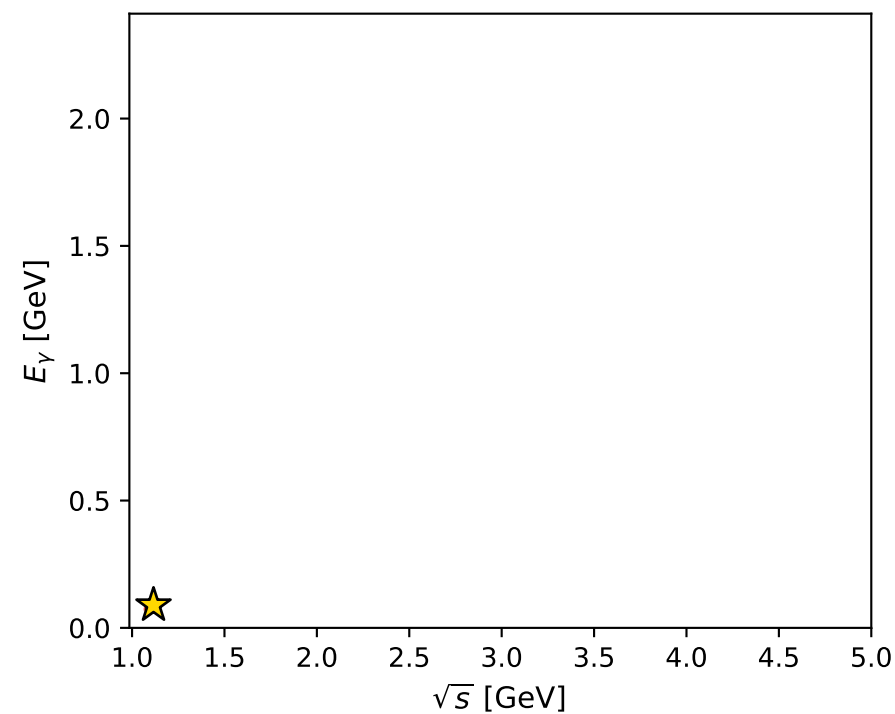
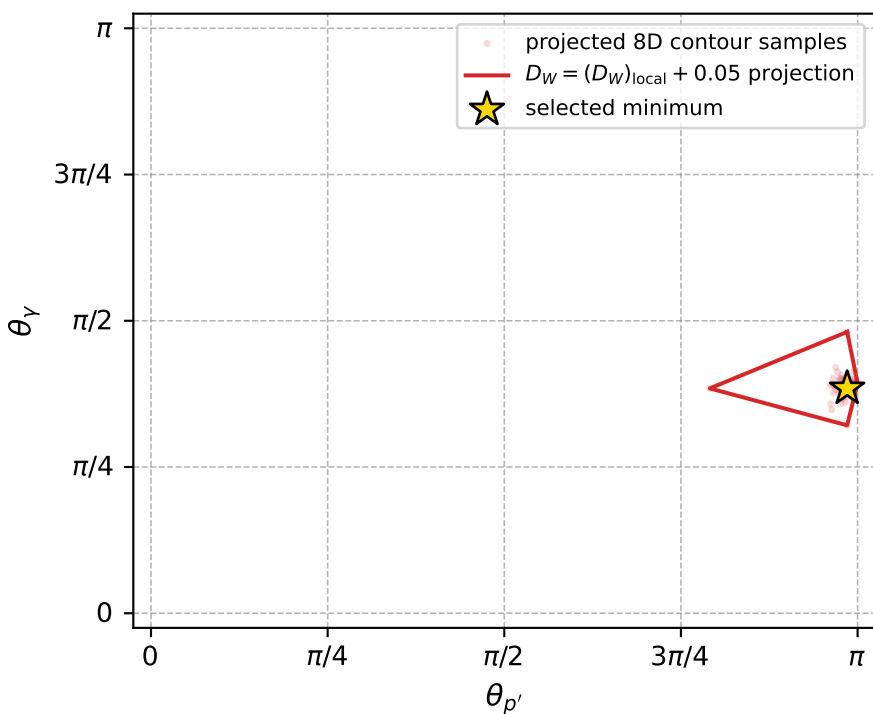
dw_mixing_angles_polarization_06_local_minimum_977 D_W=0.017211
 region: polarization_cluster_06_local_minimum_977
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3857029$ rad
 incoming lepton: $-0.727662|+> +0.685936|->$
 proton mixing angle: $\alpha_p=1.4445675$ rad
 incoming proton: $0.125894|+> +0.992044|->$

$s=1.2456$, $\sqrt{s}=1.11607$, $|\vec{P}|=0.16386$, $|\vec{P}'|=0.000195335$
 $\theta_{p'}=3.09519$, $\theta_Y=1.2077$, $\phi_{p'}=1.13063$, $\phi_Y=6.27958$
 $E_Y=0.0890647$, $\theta_{l'}=1.93182$, $\phi_{l'}=3.13809$
 $Q^2=0.0188641$, $x_B=0.101441$, $t=-0.0267122$, $W^2=1.04694$, $y=0.508427$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1639$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1639$
 P $E= 0.9522$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1639$
 ℓ' $E= 0.0890$ $p_x= -0.0833$ $p_y= 0.0003$ $p_z= -0.0314$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0002$
 q_Y $E= 0.0891$ $p_x= 0.0833$ $p_y= -0.0003$ $p_z= 0.0316$



electron: polarization cluster P6, local minimum 977; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.017211008$
 $D_W \text{ contour} = 0.067211008$
 above global minimum = 0.017194575
 8D contour samples = 241

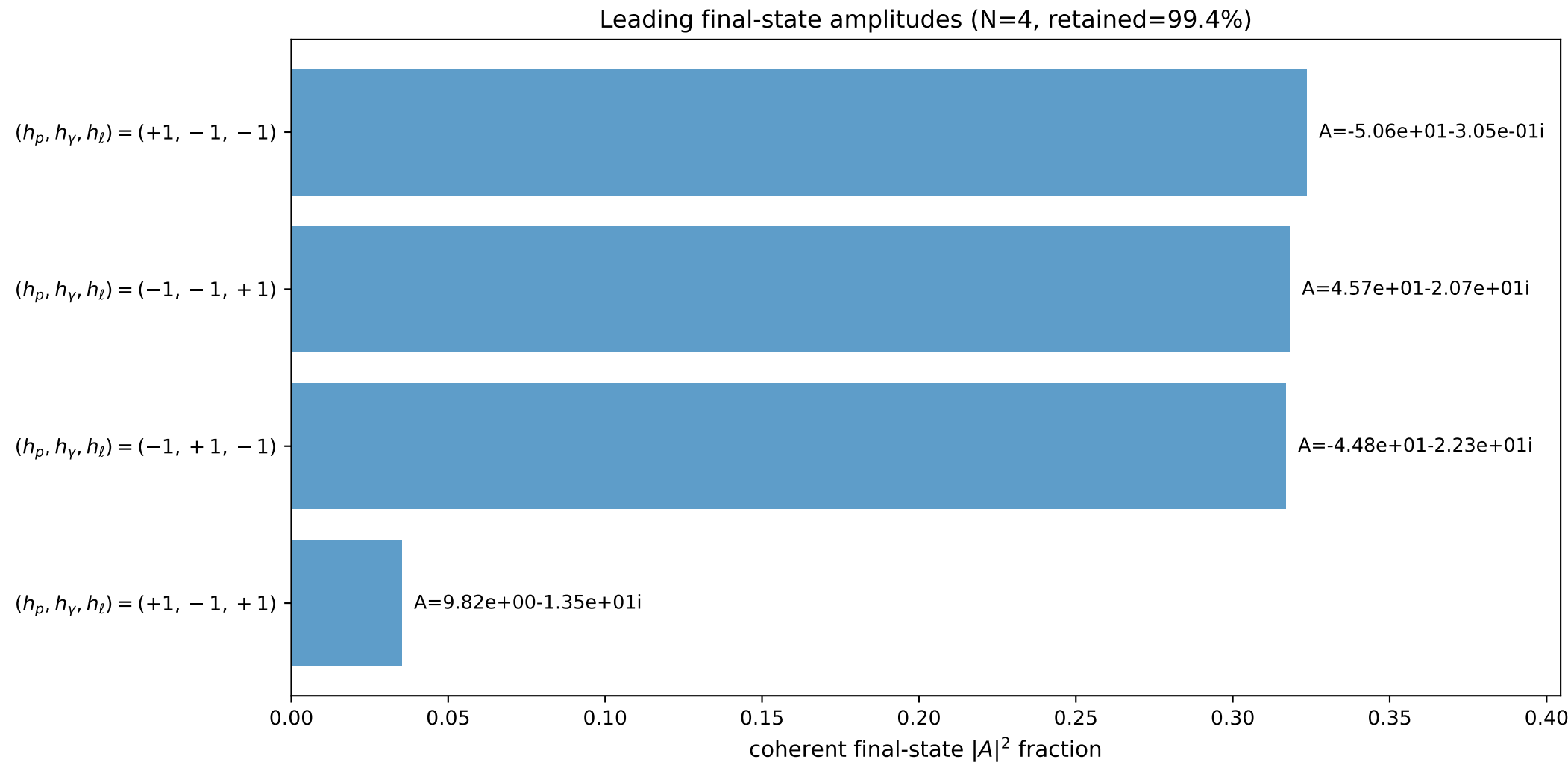
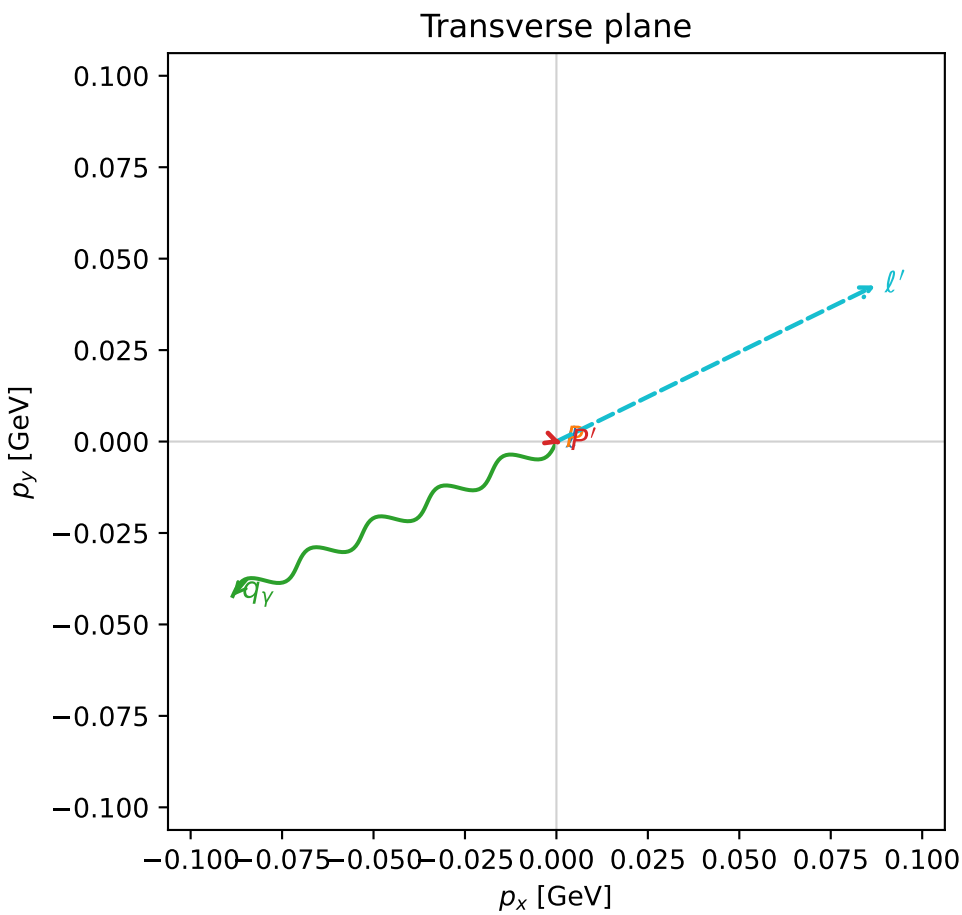
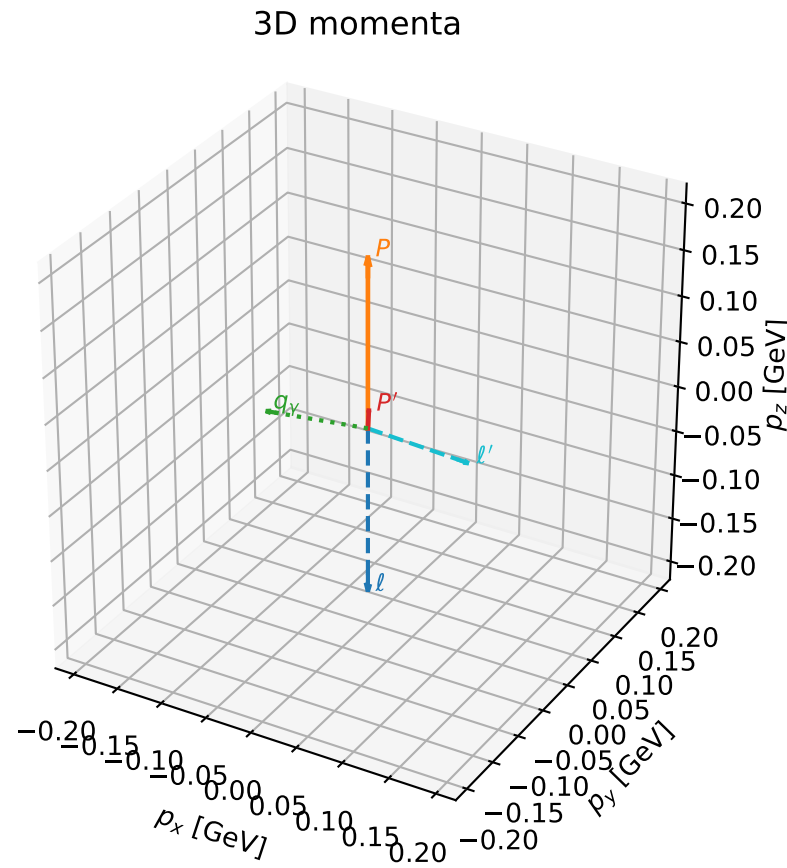
selected phase-space point:
 $\text{sqrt_s} = 1.116065$
 $\text{theta_p_out} = 3.095188$
 $\text{theta_gamma_out} = 1.207703$
 $\text{qOut} = 0.08906471$
 $\text{phi_p_out} = 1.130626$
 $\text{phi_gamma_out} = 6.279579$
 $\text{alpha_e} = 2.385703$
 $\text{alpha_p} = 1.444567$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

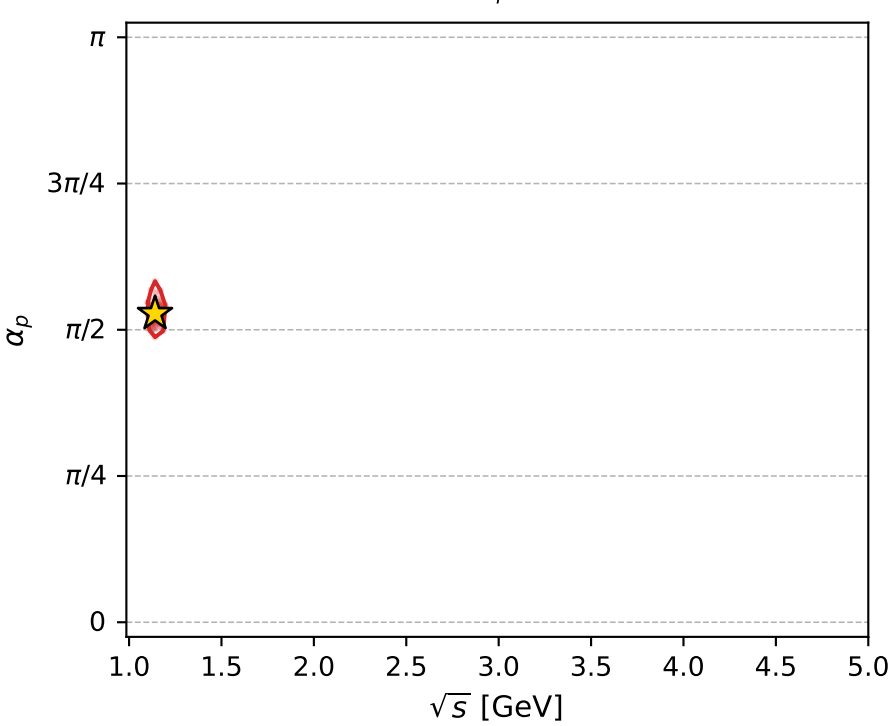
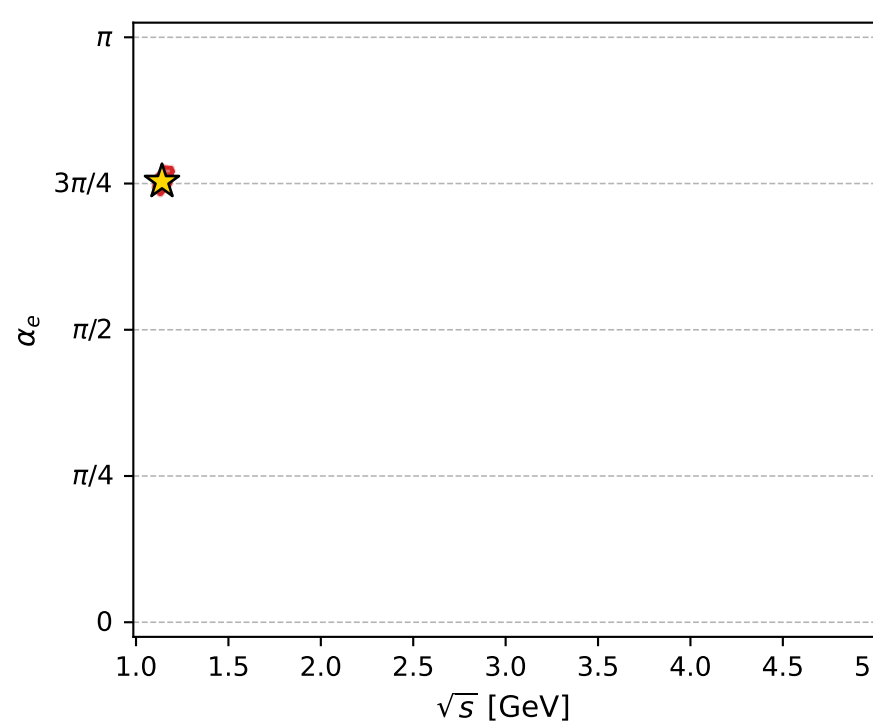
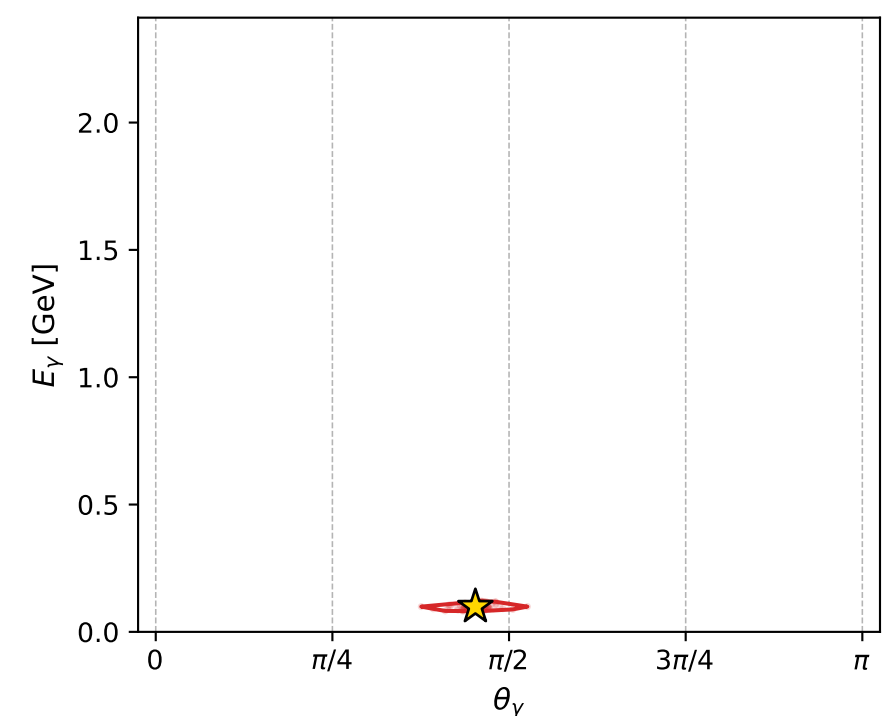
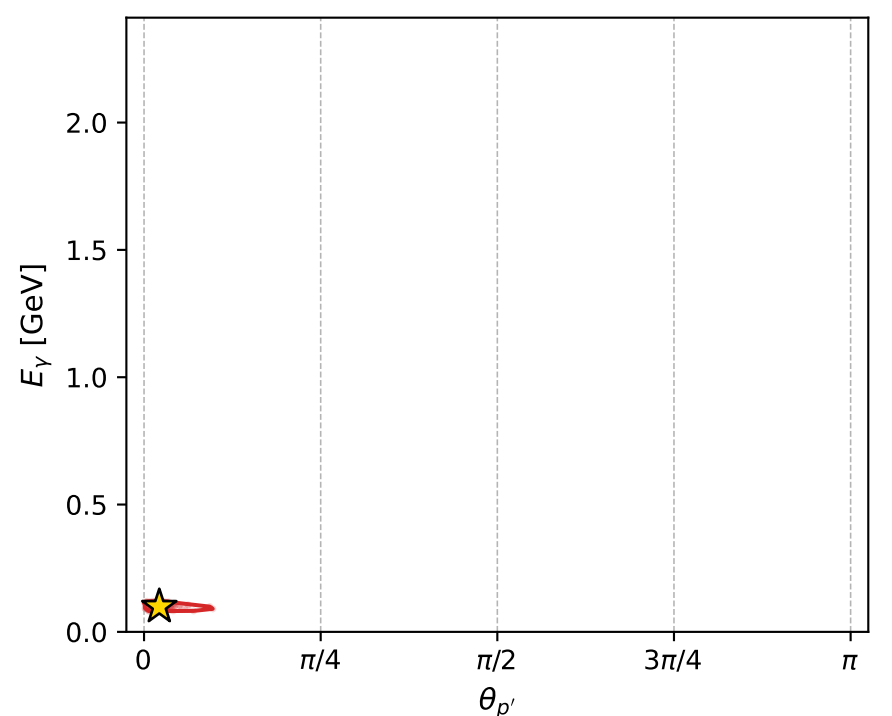
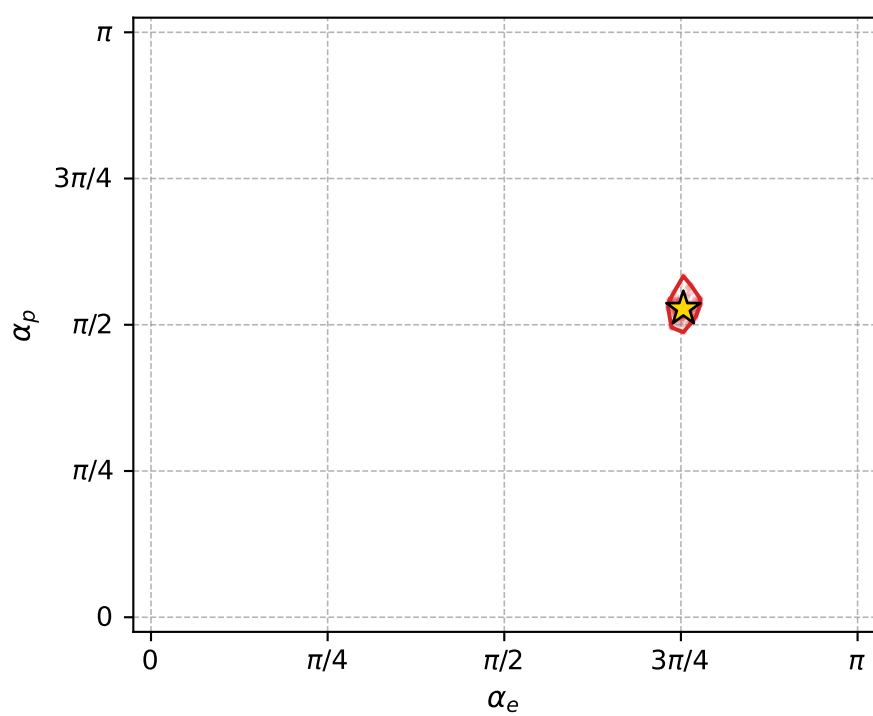
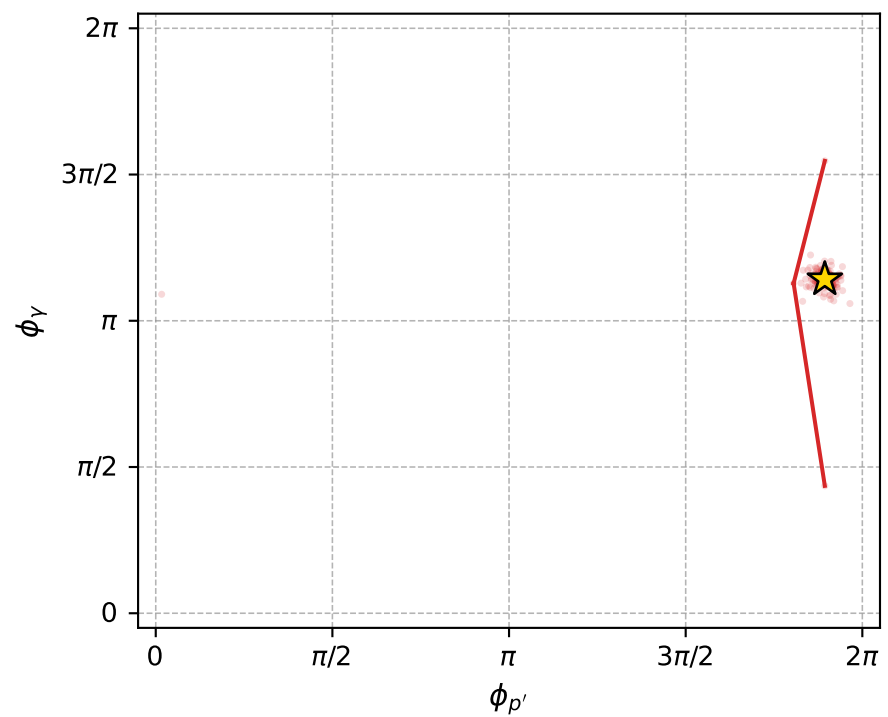
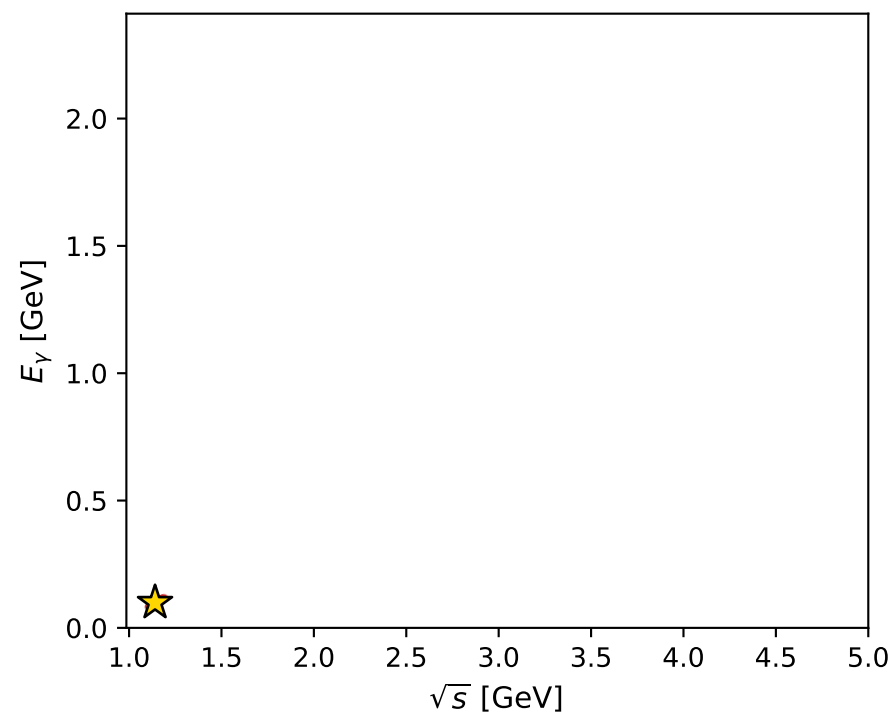
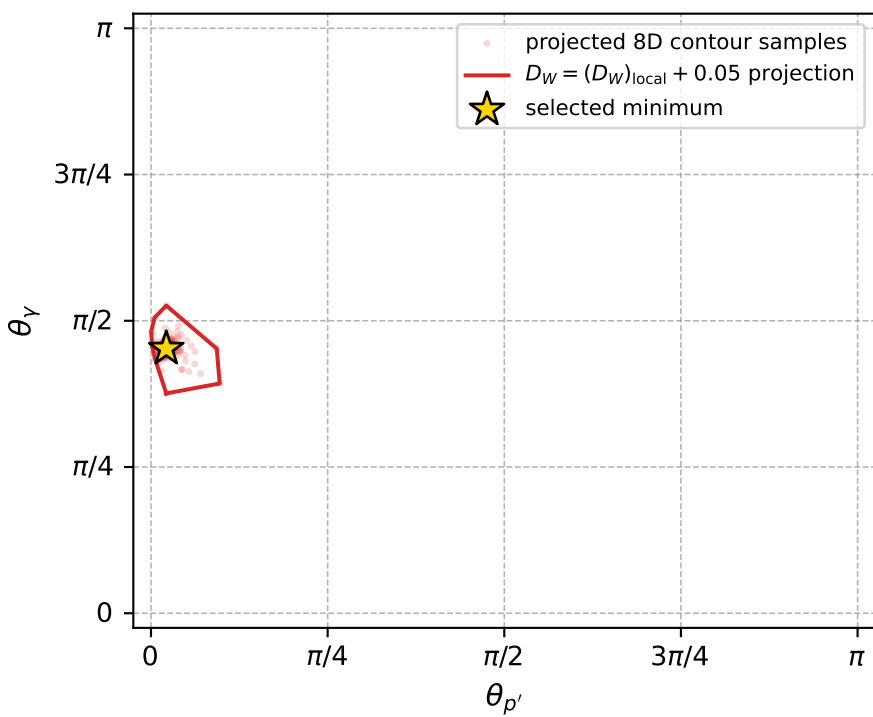
dw_mixing_angles_polarization_06_local_minimum_979 D_W=0.0173198
 region: polarization_cluster_06_local_minimum_979
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.36691$ rad
 incoming lepton: $-0.714643|+\rangle +0.699489|-\rangle$
 proton mixing angle: $\alpha_p=1.6576404$ rad
 incoming proton: $-0.086735|+\rangle +0.996231|-\rangle$

$s=1.30084$, $\sqrt{s}=1.14054$, $|\vec{P}|=0.184557$, $|\vec{P}'|=0.0201758$
 $\theta_{P'}=0.067876$, $\theta_Y=1.42114$, $\phi_{P'}=5.94959$, $\phi_Y=3.58678$
 $E_Y=0.0991571$, $\theta_{\ell'}=1.91603$, $\phi_{\ell'}=0.455092$
 $Q^2=0.0251933$, $x_B=0.119484$, $t=-0.0267228$, $W^2=1.0655$, $y=0.500843$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1846$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1846$
 P $E= 0.9560$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1846$
 ℓ' $E= 0.1032$ $p_x= 0.0872$ $p_y= 0.0427$ $p_z= -0.0349$
 P' $E= 0.9382$ $p_x= 0.0013$ $p_y= -0.0004$ $p_z= 0.0201$
 q_Y $E= 0.0992$ $p_x= -0.0885$ $p_y= -0.0422$ $p_z= 0.0148$



electron: polarization cluster P6, local minimum 979; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.017319848$
 $D_W \text{ contour} = 0.067319848$
 above global minimum = 0.017303415
 8D contour samples = 244

selected phase-space point:
 $\text{sqrt_s} = 1.140542$
 $\text{theta_p_out} = 0.06787604$
 $\text{theta_gamma_out} = 1.421138$
 $\text{qOut} = 0.09915709$
 $\text{phi_p_out} = 5.949588$
 $\text{phi_gamma_out} = 3.586784$
 $\text{alpha_e} = 2.36691$
 $\text{alpha_p} = 1.65764$

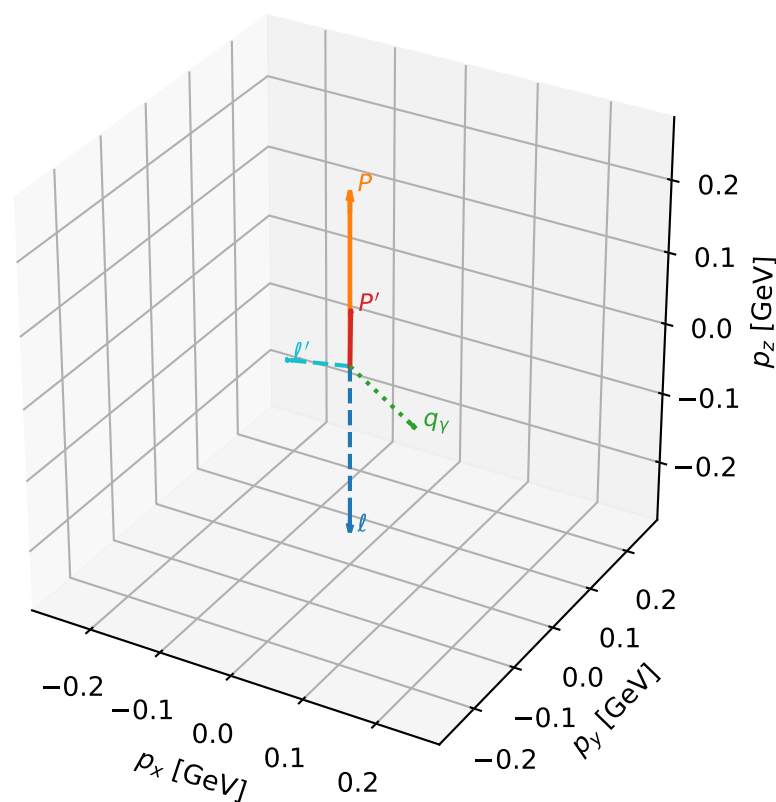
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_997 D_W=0.0181263
 region: polarization_cluster_06_local_minimum_997
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3936653$ rad
 incoming lepton: $-0.7331|+\rangle +0.680121|-\rangle$
 proton mixing angle: $\alpha_p=1.4250893$ rad
 incoming proton: $0.145192|+\rangle +0.989403|-\rangle$

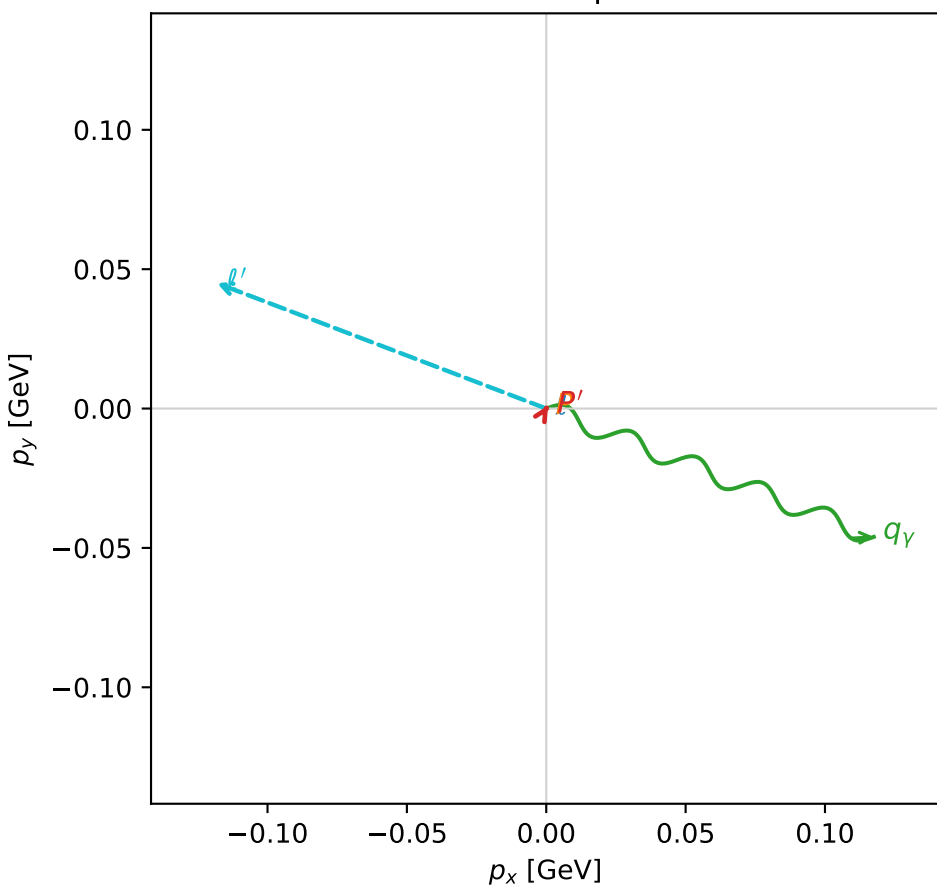
$s=1.45358$, $\sqrt{s}=1.20564$, $|\vec{P}|=0.237937$, $|\vec{P}'|=0.0764869$
 $\theta_{P'}=0.0164515$, $\theta_V=1.79758$, $\phi_{P'}=1.03077$, $\phi_V=5.9097$
 $E_V=0.129528$, $\theta_{\ell'}=1.92917$, $\phi_{\ell'}=2.77792$
 $Q^2=0.0417105$, $x_B=0.143872$, $t=-0.0253637$, $W^2=1.12805$, $y=0.505309$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2379$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2379$
 P $E= 0.9677$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2379$
 ℓ' $E= 0.1350$ $p_x= -0.1182$ $p_y= 0.0450$ $p_z= -0.0474$
 P' $E= 0.9411$ $p_x= 0.0006$ $p_y= 0.0011$ $p_z= 0.0765$
 q_Y $E= 0.1295$ $p_x= 0.1175$ $p_y= -0.0461$ $p_z= -0.0291$

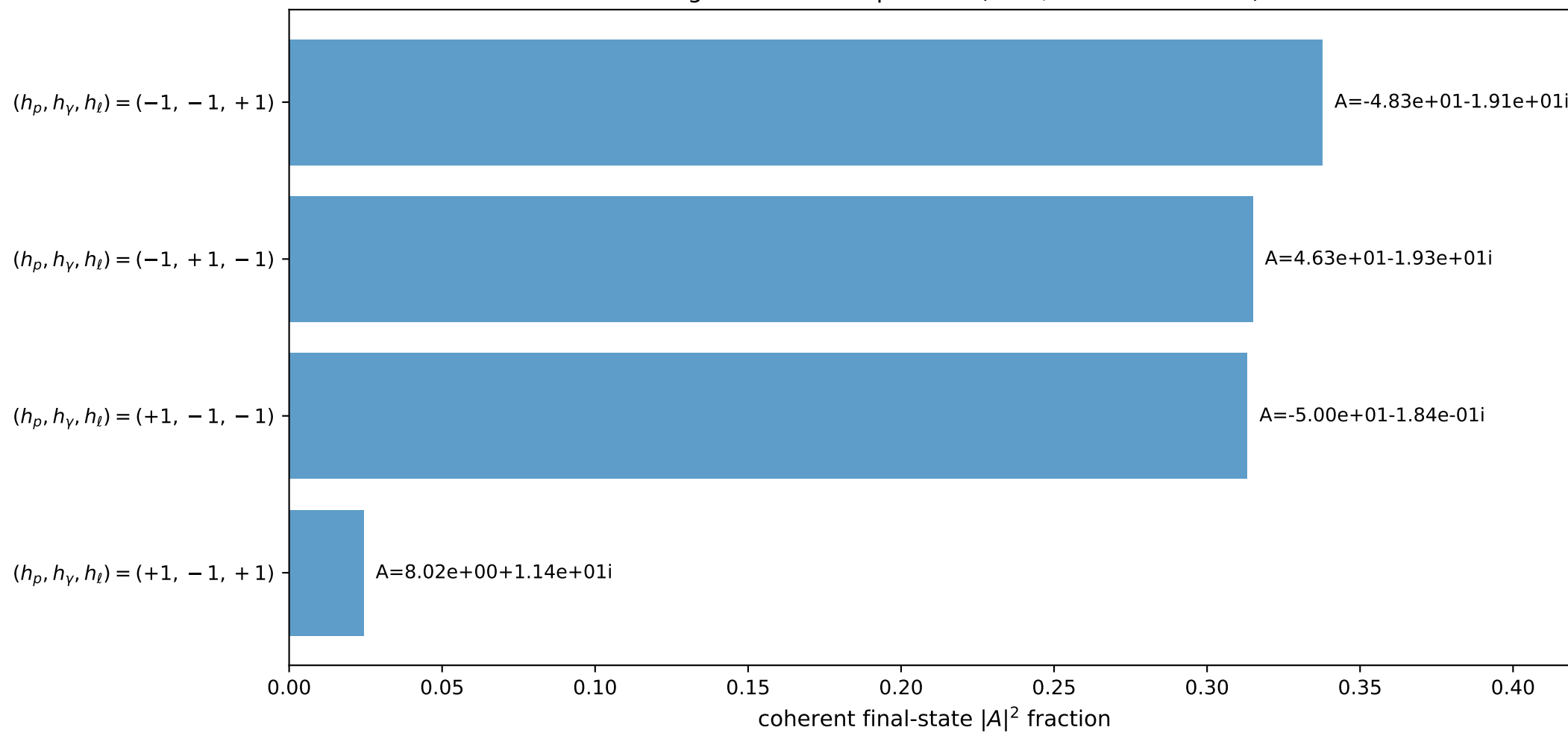
3D momenta



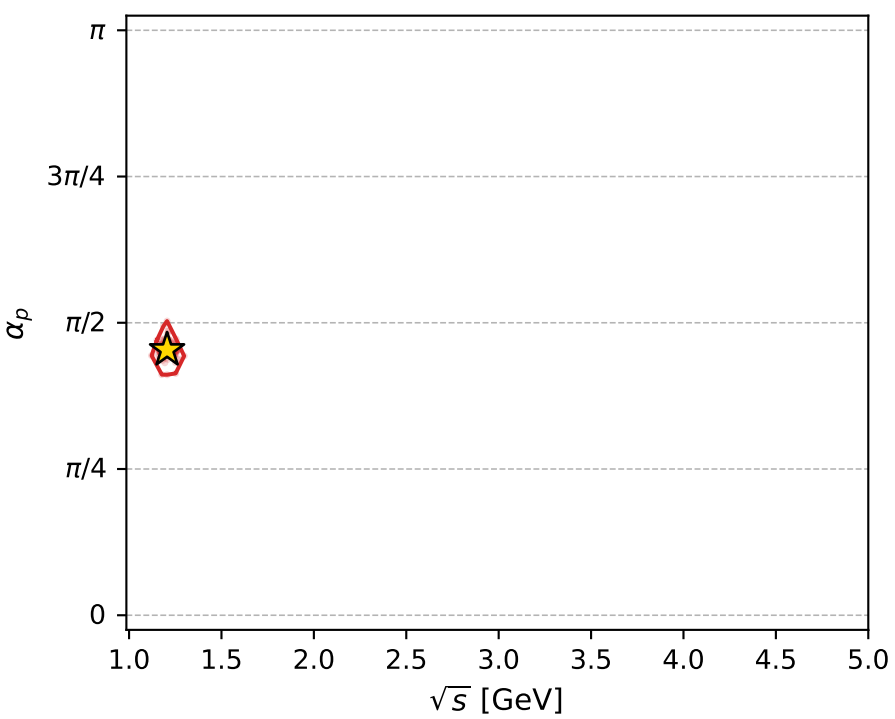
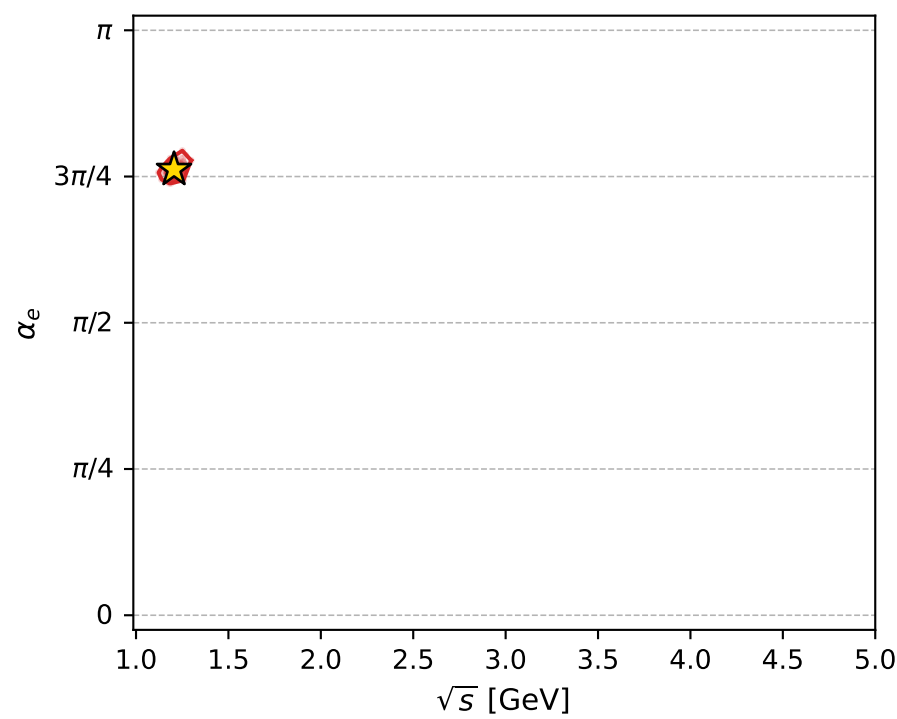
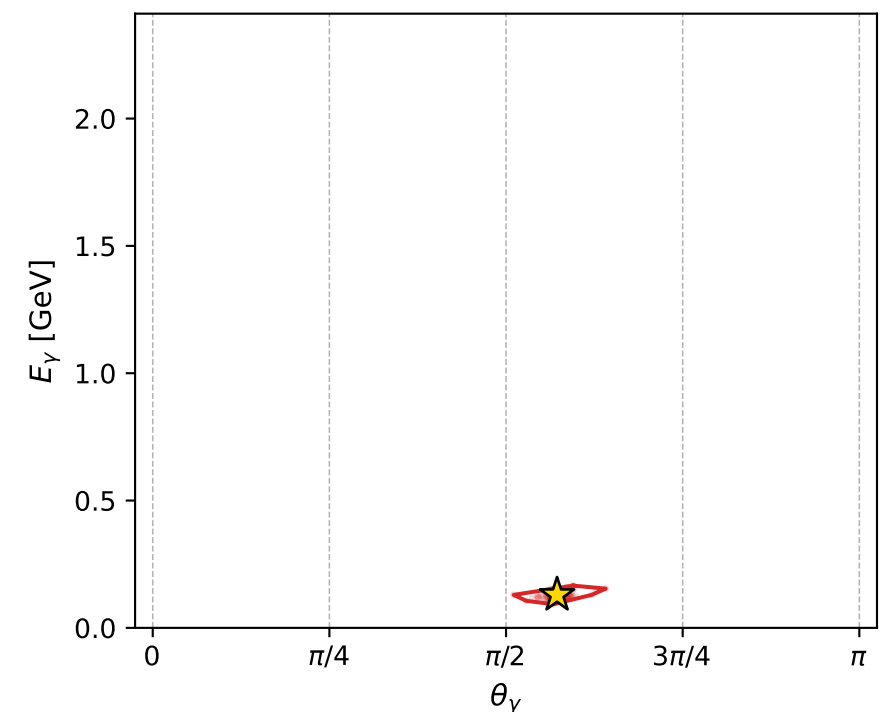
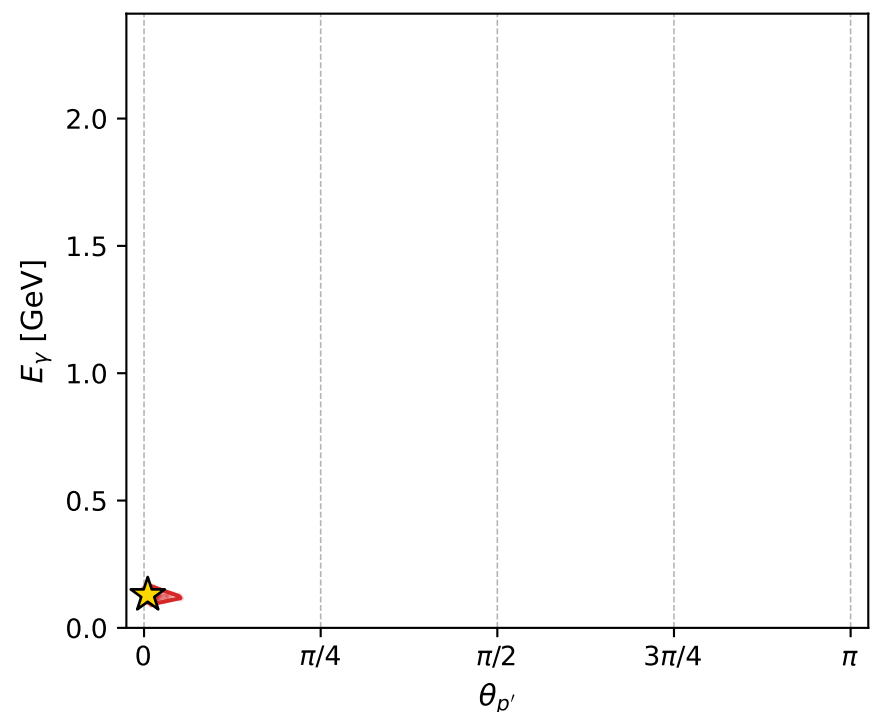
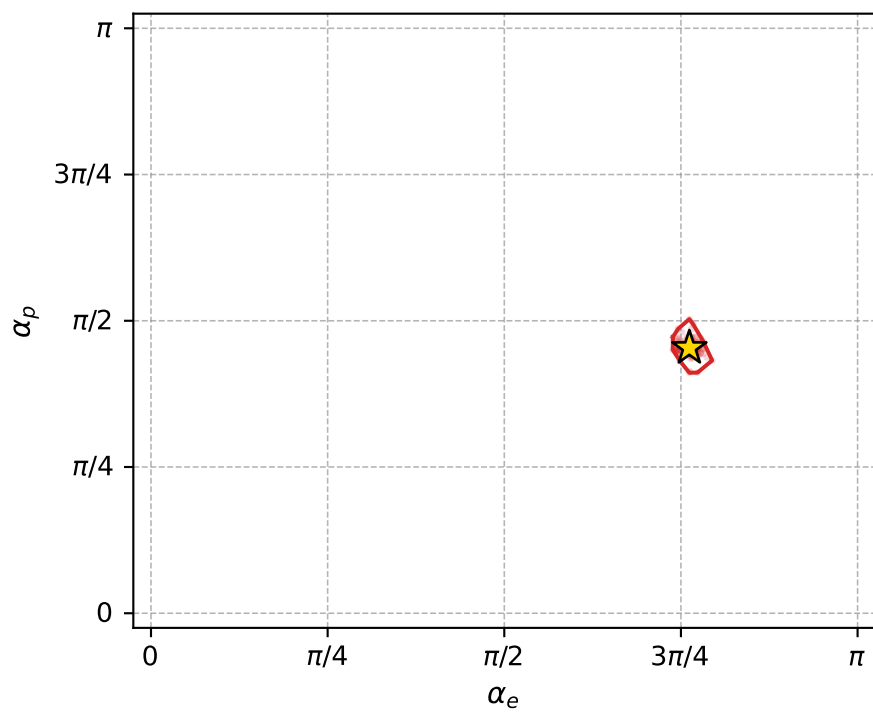
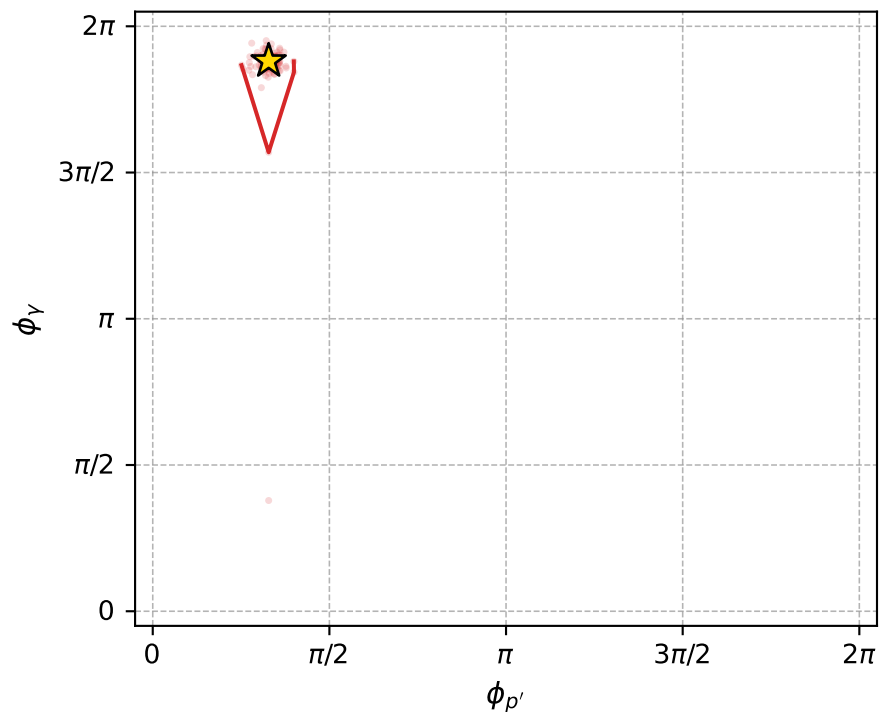
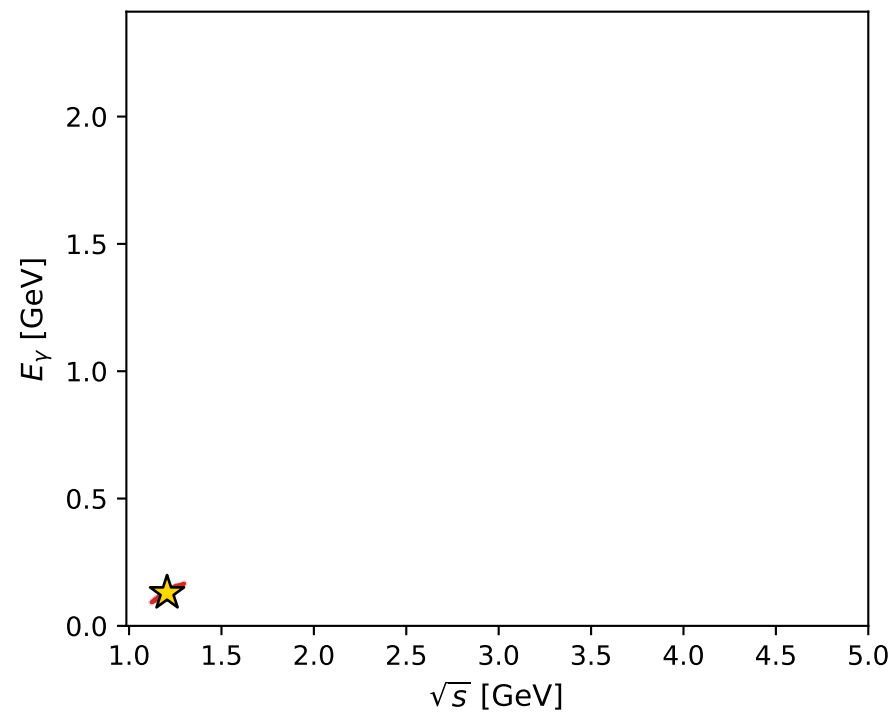
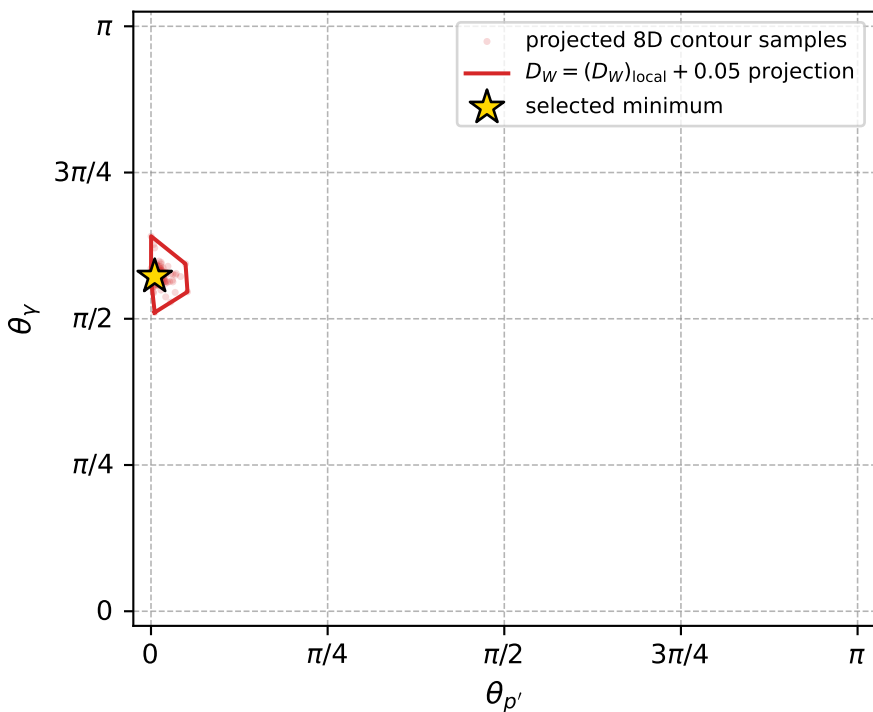
Transverse plane



Leading final-state amplitudes (N=4, retained=99.0%)



electron: polarization cluster P6, local minimum 997; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.018126276$

$D_W \text{ contour} = 0.068126276$

above global minimum = 0.018109843

8D contour samples = 178

selected phase-space point:

$\text{sqrt_s} = 1.205645$

$\text{theta_p_out} = 0.01645151$

$\text{theta_gamma_out} = 1.797585$

$\text{qOut} = 0.1295278$

$\text{phi_p_out} = 1.03077$

$\text{phi_gamma_out} = 5.909696$

$\text{alpha_e} = 2.393665$

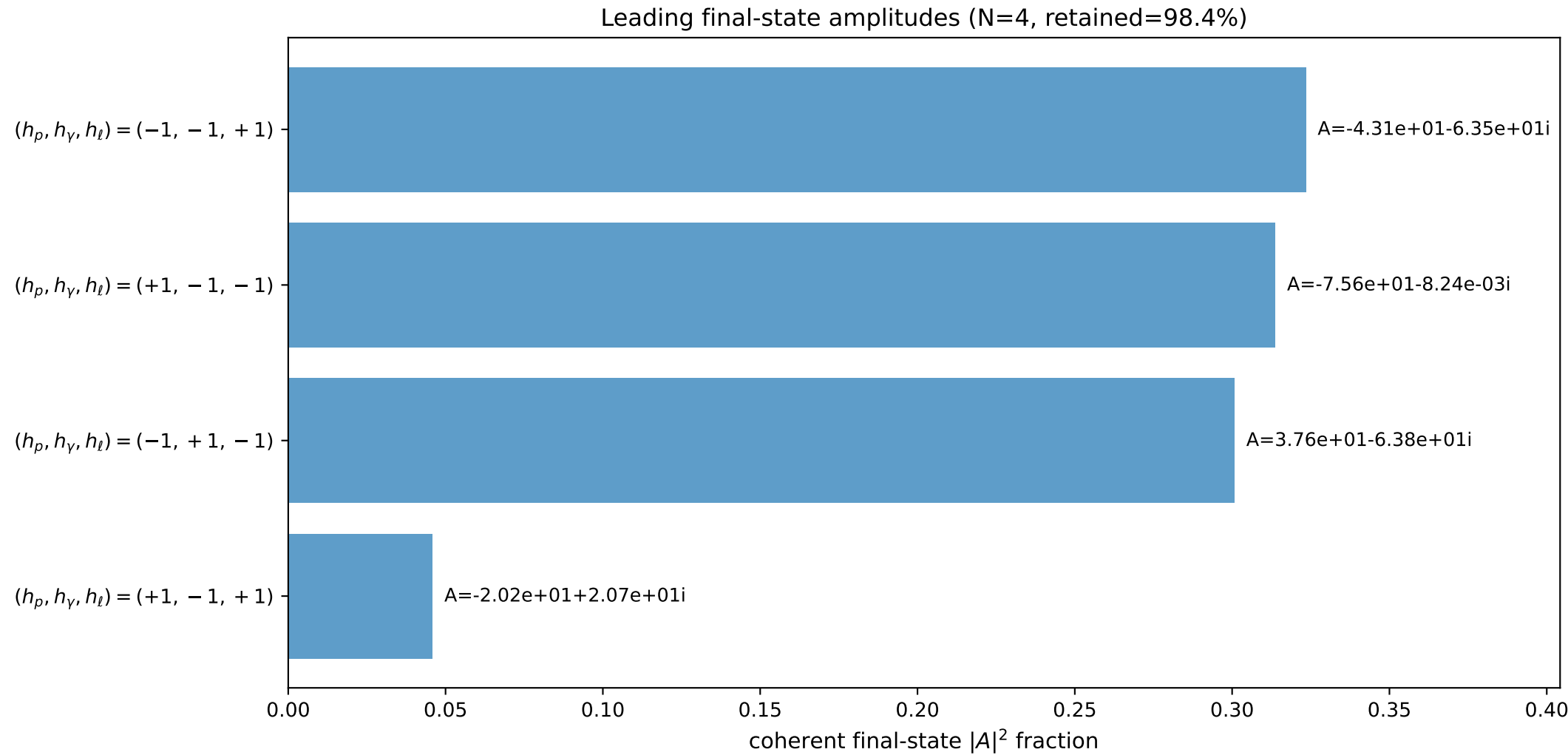
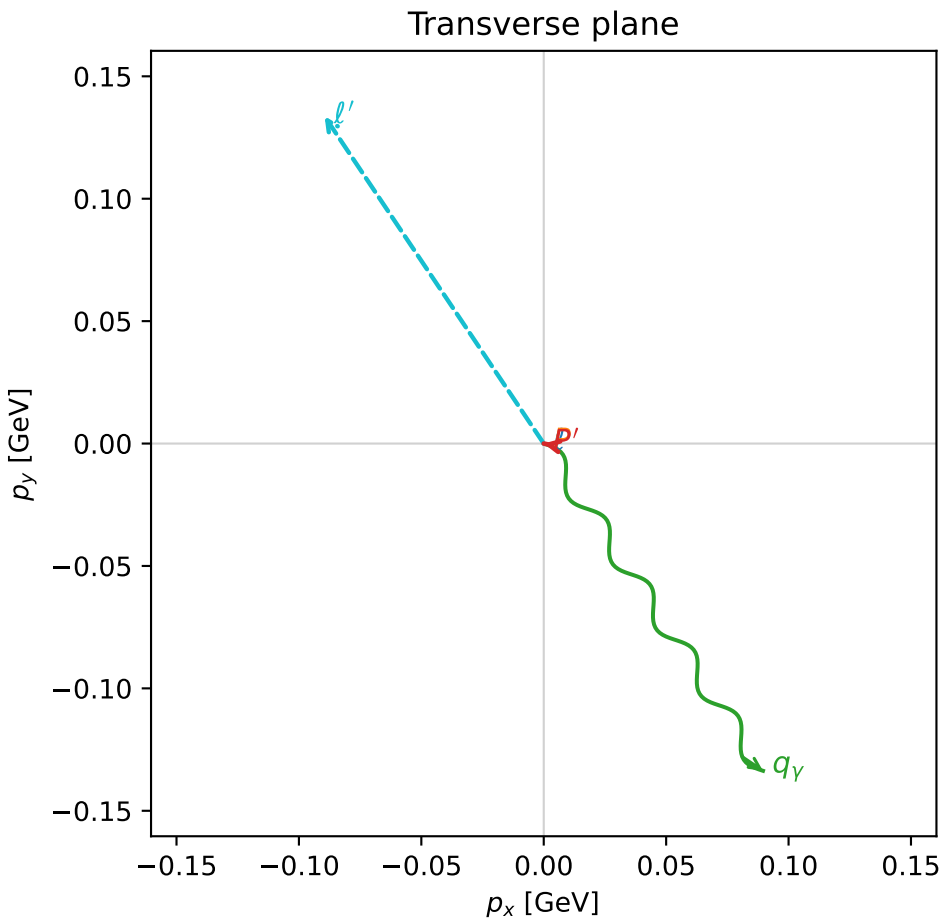
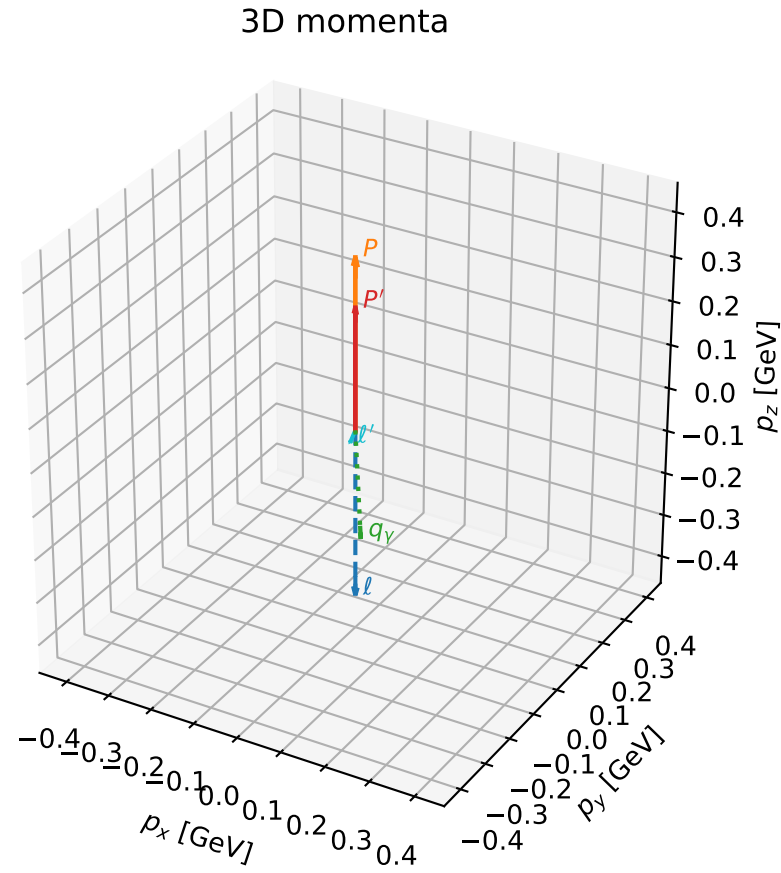
$\text{alpha_p} = 1.425089$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

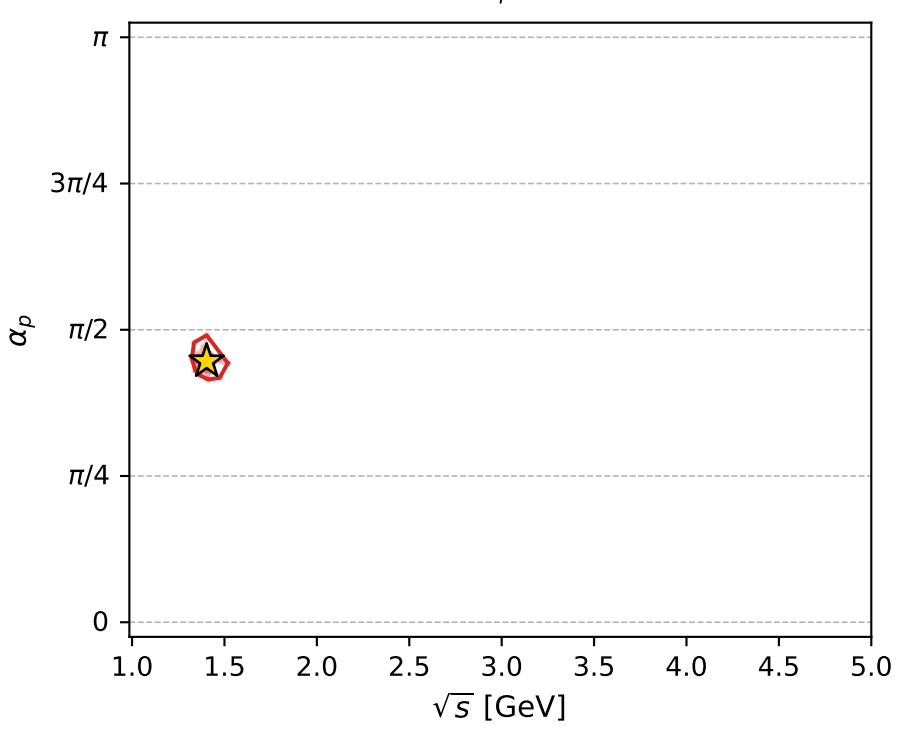
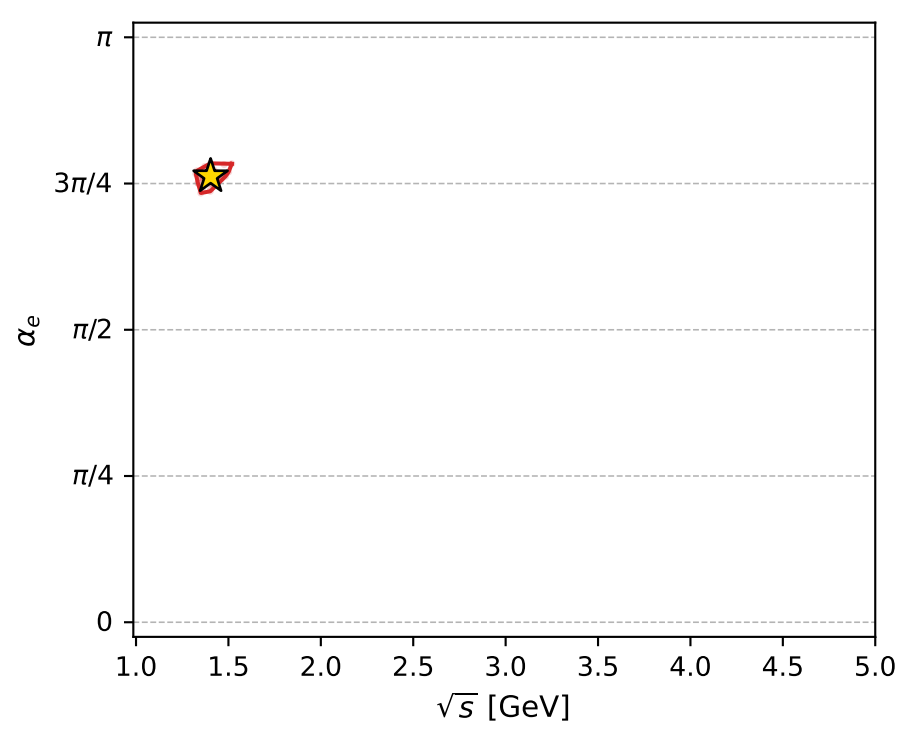
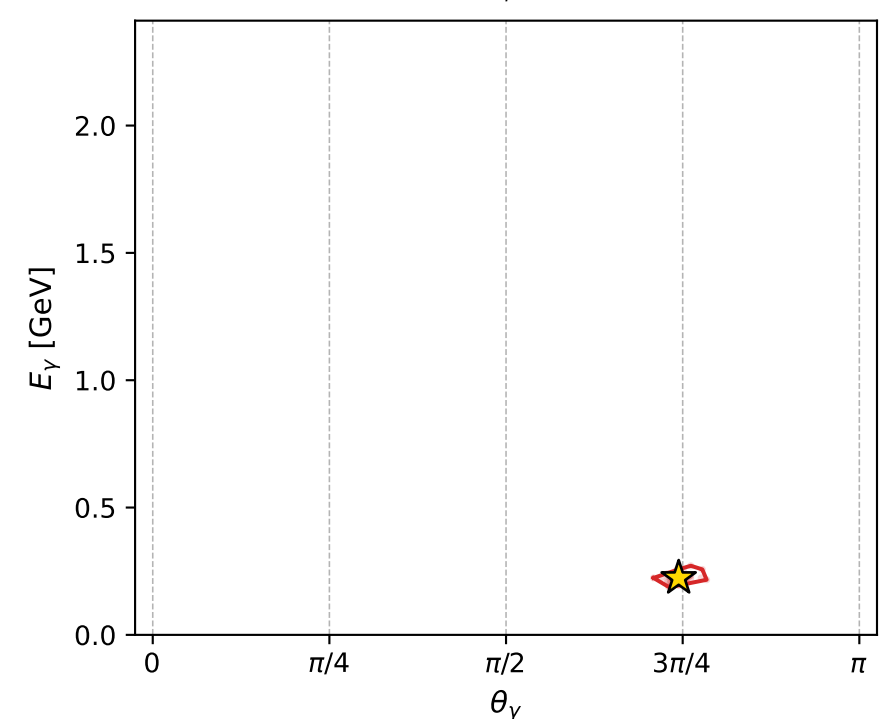
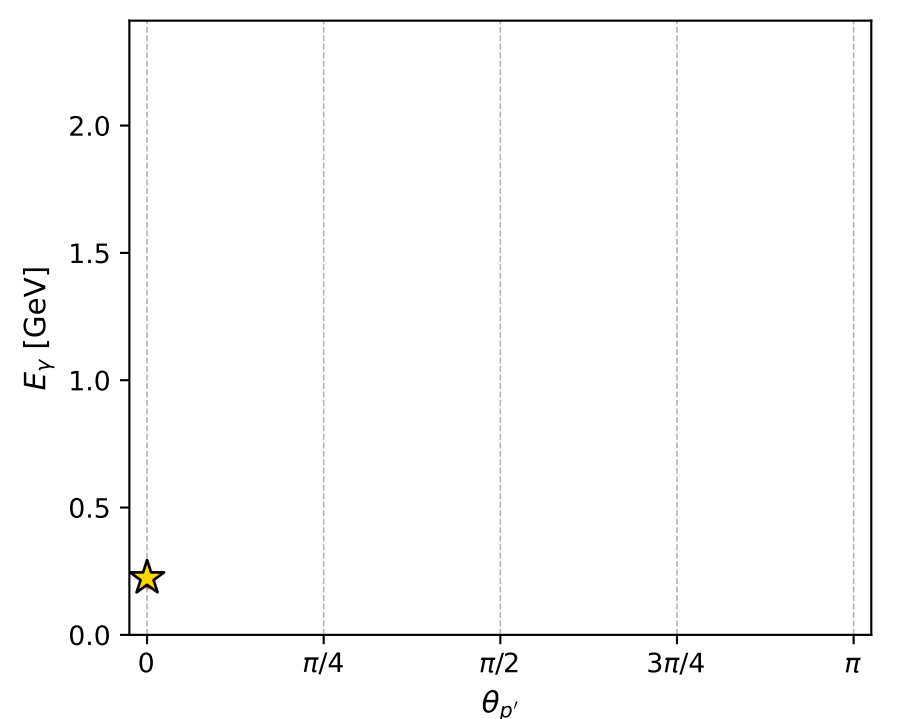
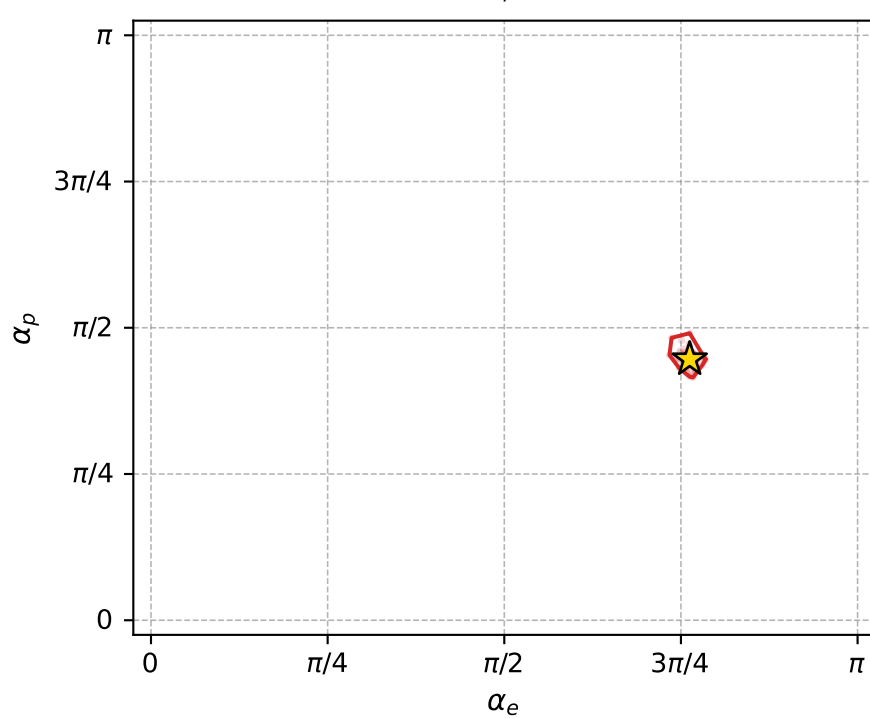
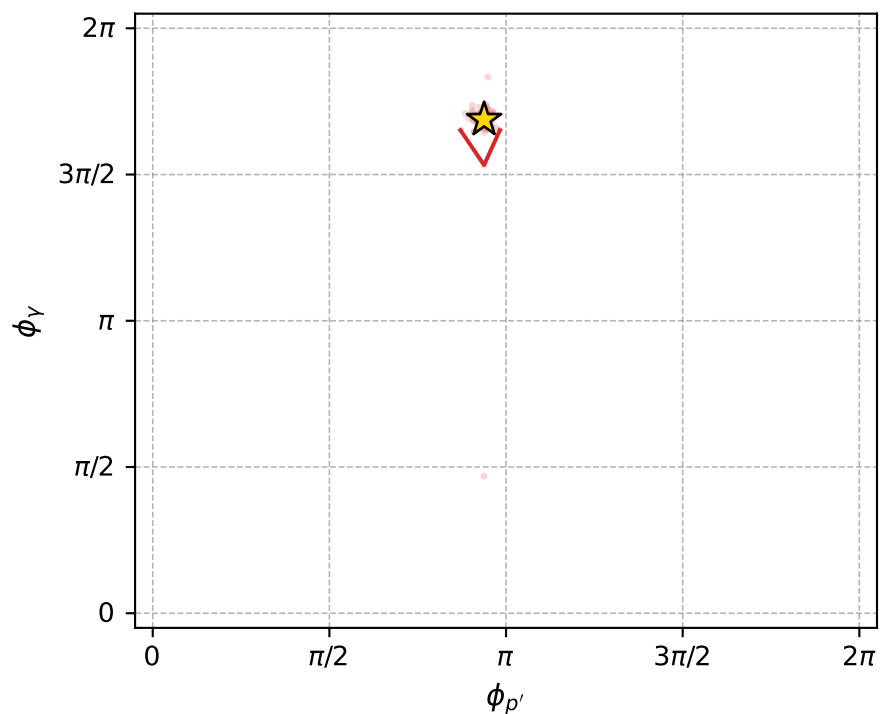
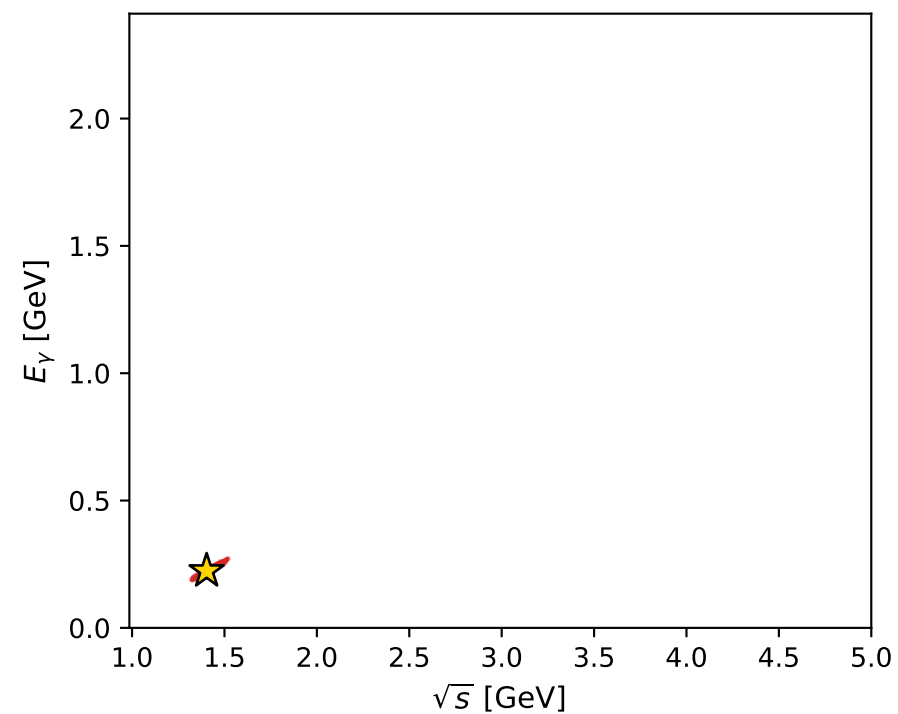
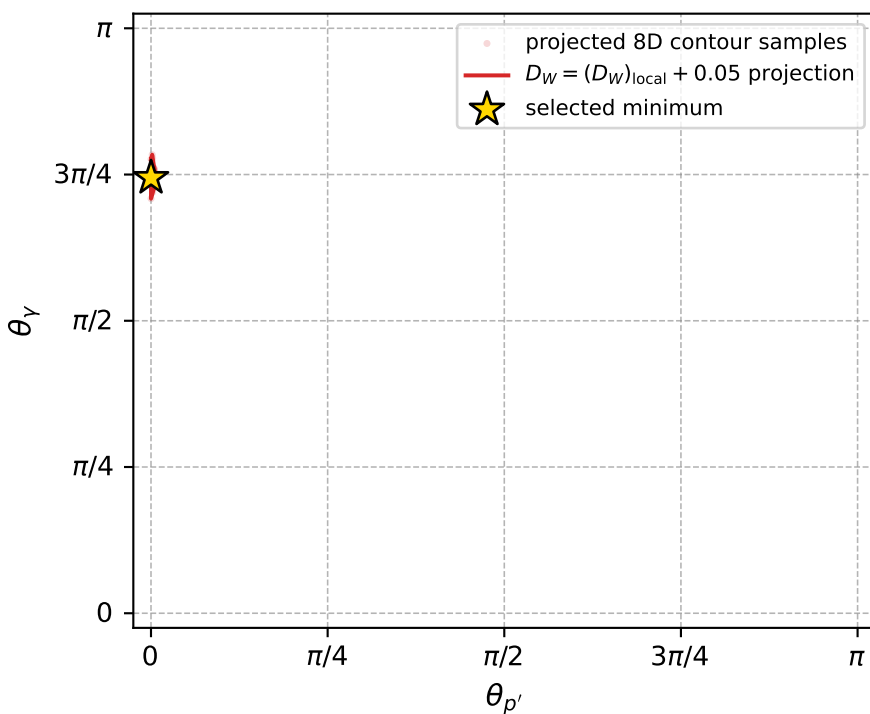
dw_mixing_angles_polarization_06_local_minimum_1008 D_W=0.0188186
region: polarization_cluster_06_local_minimum_1008
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3950954$ rad
incoming lepton: $-0.734072|+> +0.679072|->$
proton mixing angle: $\alpha_p=1.4016538$ rad
incoming proton: $0.168337|+> +0.985729|->$

$s=1.97001$, $\sqrt{s}=1.40357$, $|\vec{P}|=0.388355$, $|\vec{P}'|=0.277217$
 $\theta_{P'}=0.00015683$, $\theta_Y=2.33843$, $\phi_{P'}=2.94637$, $\phi_Y=5.3027$
 $E_Y=0.223606$, $\theta_{\ell'}=2.2194$, $\phi_{\ell'}=2.16092$
 $Q^2=0.0620755$, $x_B=0.106003$, $t=-0.0109746$, $W^2=1.40337$, $y=0.537164$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3884$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3884$
 P $E= 1.0152$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3884$
 ℓ' $E= 0.2019$ $p_x= -0.0895$ $p_y= 0.1337$ $p_z= -0.1219$
 P' $E= 0.9781$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.2772$
 q_Y $E= 0.2236$ $p_x= 0.0896$ $p_y= -0.1337$ $p_z= -0.1553$



electron: polarization cluster P6, local minimum 1008; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.018818614$
 $D_W \text{ contour} = 0.068818614$
 above global minimum = 0.018802181
 8D contour samples = 136

selected phase-space point:

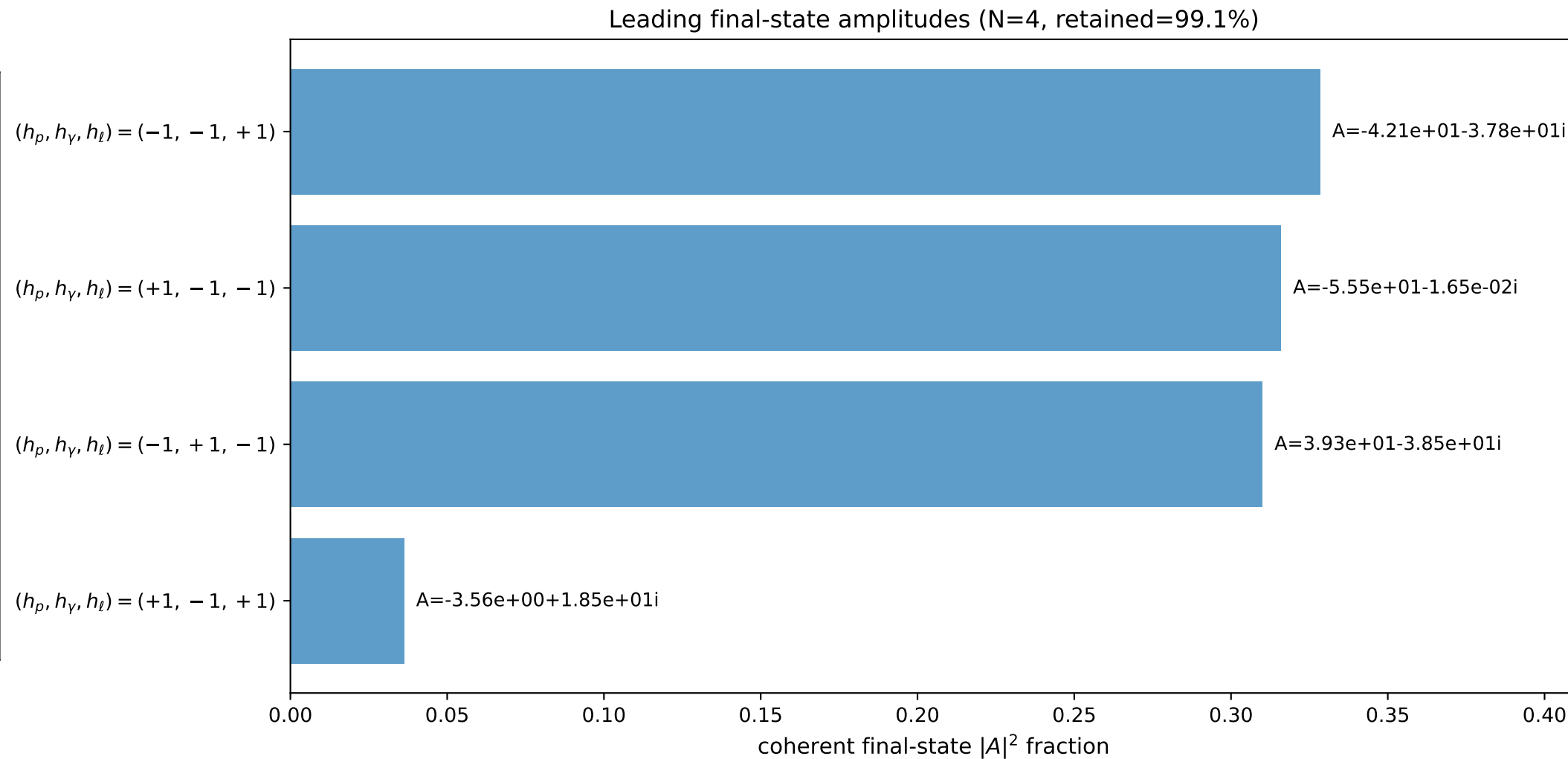
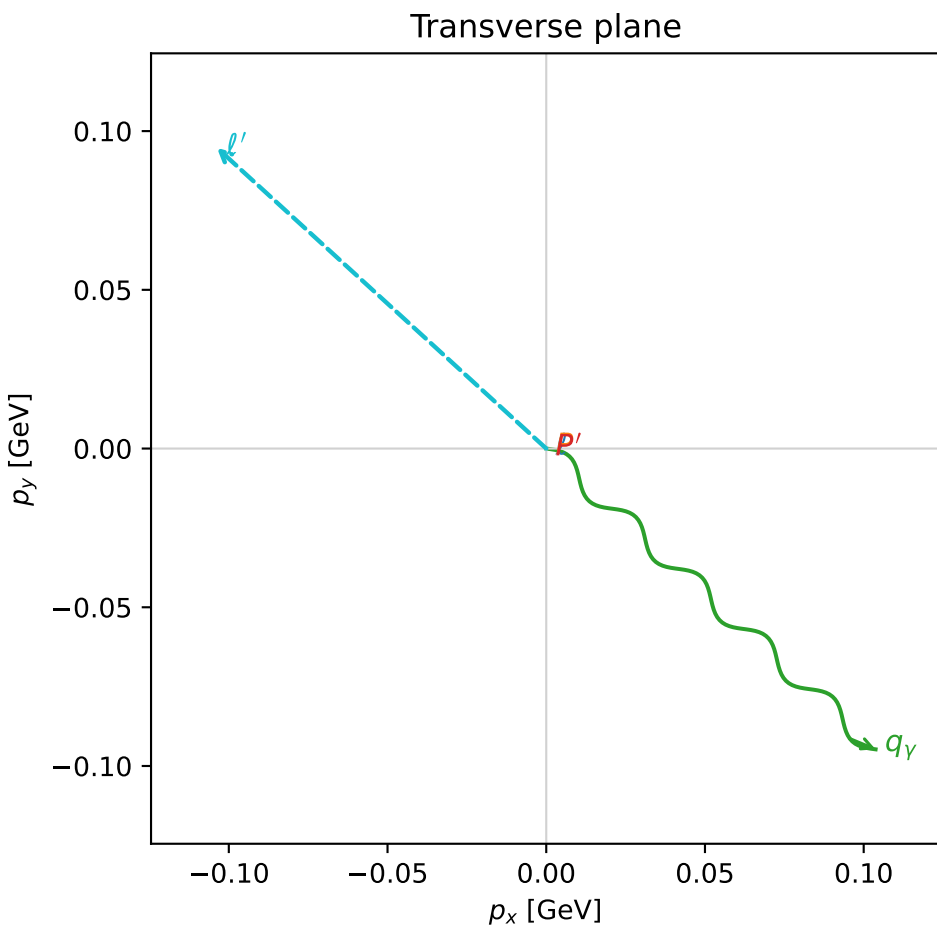
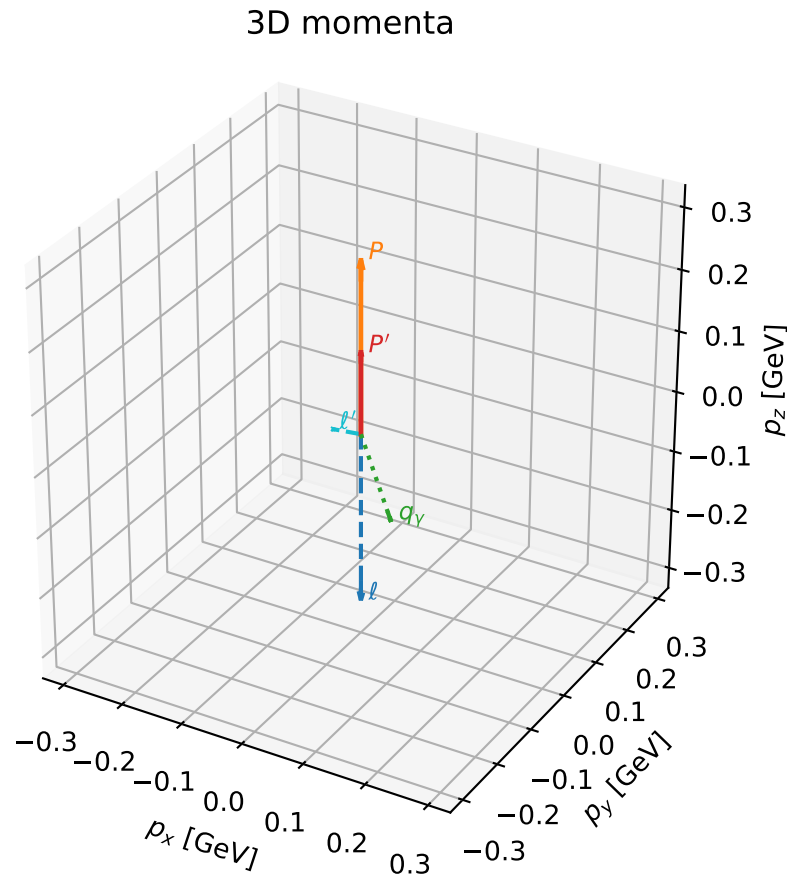
$\text{sqrt_s} = 1.403572$
 $\text{theta_p_out} = 0.0001568302$
 $\text{theta_gamma_out} = 2.338431$
 $\text{q0ut} = 0.2236063$
 $\text{phi_p_out} = 2.94637$
 $\text{phi_gamma_out} = 5.302701$
 $\text{alpha_e} = 2.395095$
 $\text{alpha_p} = 1.401654$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

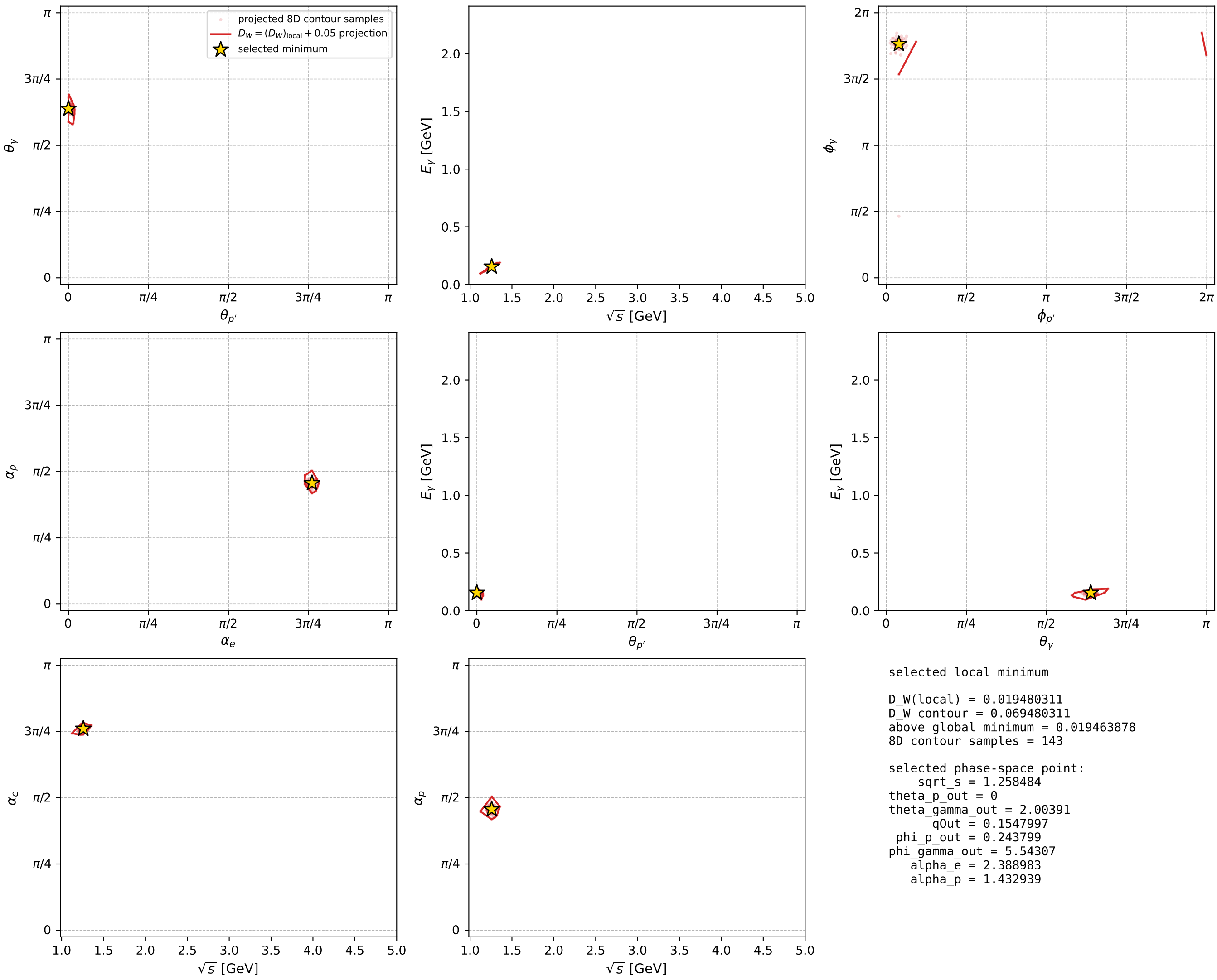
dw_mixing_angles_polarization_06_local_minimum_1019 D_W=0.0194803
region: polarization_cluster_06_local_minimum_1019
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3889829$ rad
incoming lepton: $-0.729907|+> +0.683546|->$
proton mixing angle: $\alpha_p=1.4329385$ rad
incoming proton: $0.137422|+> +0.990513|->$

$s=1.58378$, $\sqrt{s}=1.25848$, $|\vec{P}|=0.279676$, $|\vec{P}'|=0.133332$
 $\theta_{P'}=0$, $\theta_V=2.00391$, $\phi_{P'}=0.243799$, $\phi_V=5.54307$
 $E_V=0.1548$, $\theta_{\ell'}=2.02362$, $\phi_{\ell'}=2.40148$
 $Q^2=0.0491625$, $x_B=0.136634$, $t=-0.0204321$, $W^2=1.19049$, $y=0.511141$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2797$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2797$
 P $E= 0.9788$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2797$
 ℓ' $E= 0.1563$ $p_x= -0.1037$ $p_y= 0.0948$ $p_z= -0.0684$
 P' $E= 0.9474$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1333$
 q_Y $E= 0.1548$ $p_x= 0.1037$ $p_y= -0.0948$ $p_z= -0.0650$



electron: polarization cluster P6, local minimum 1019; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

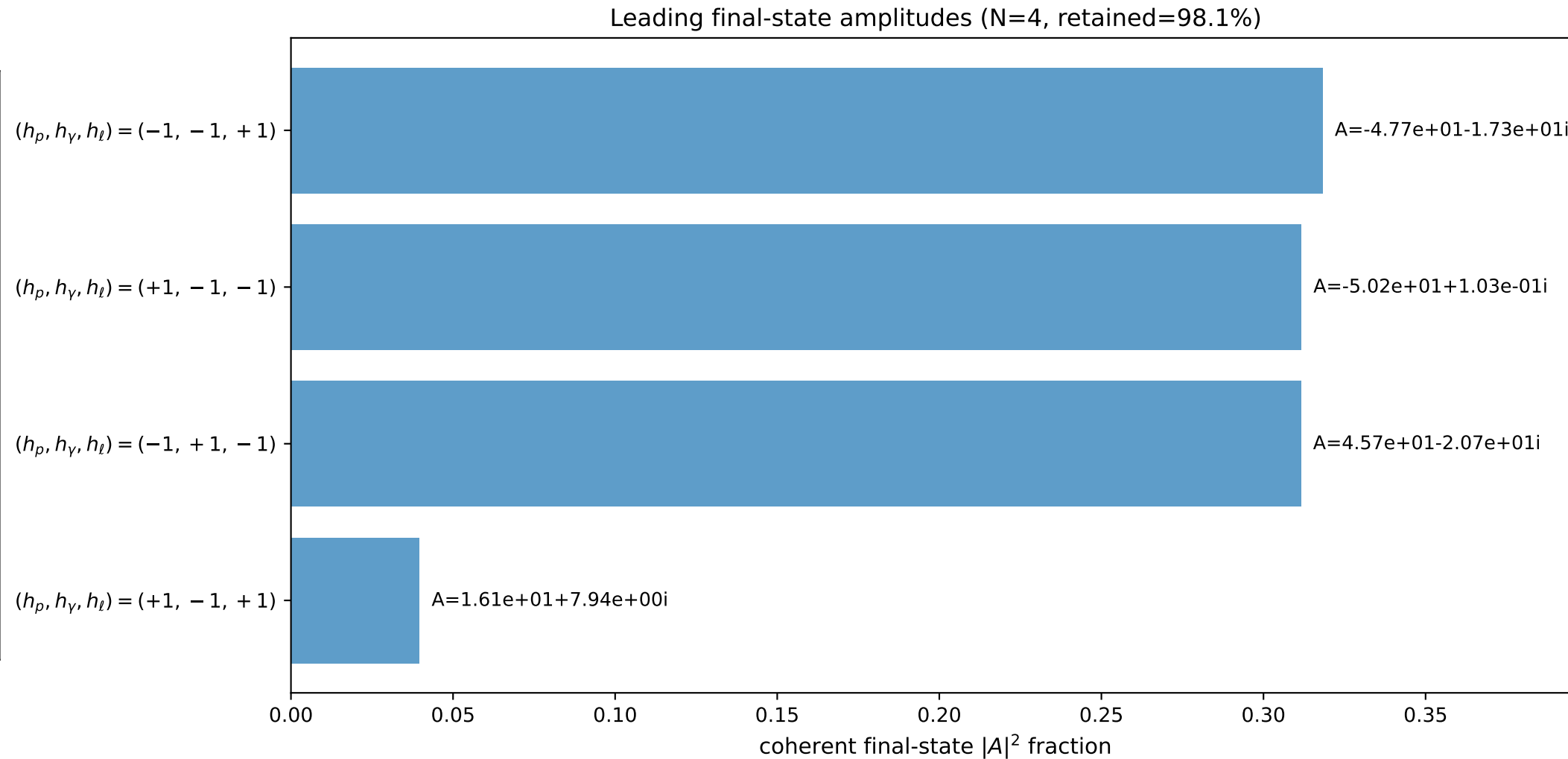
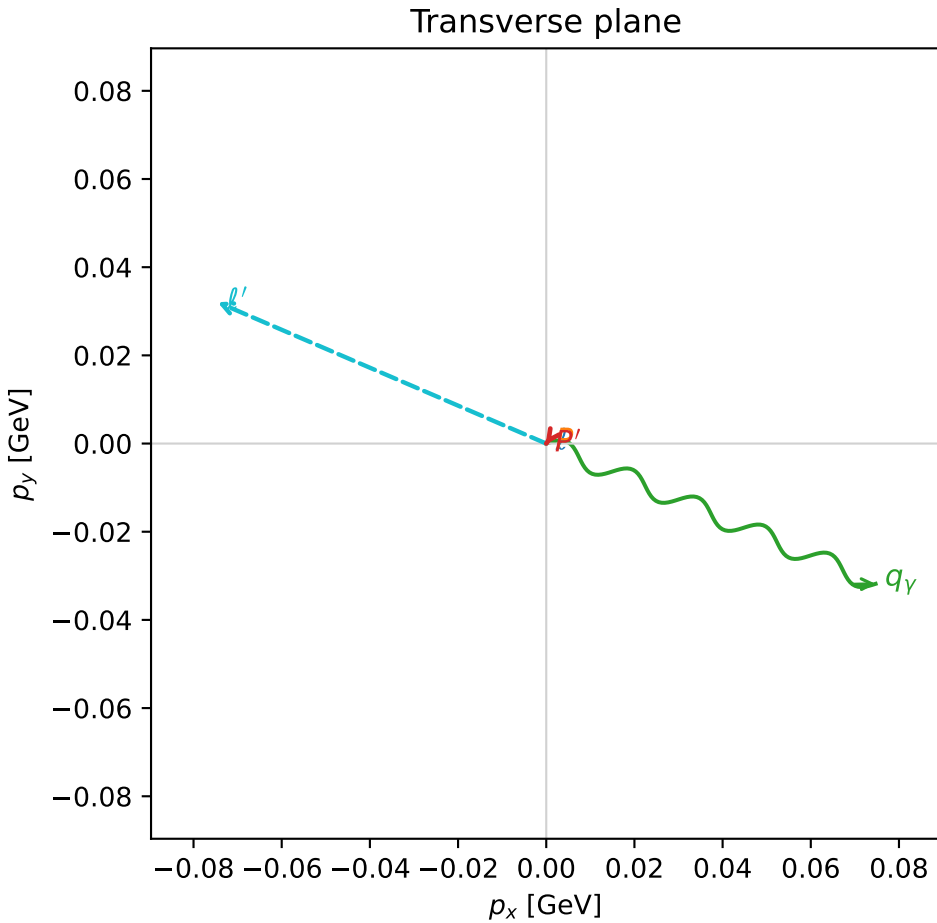
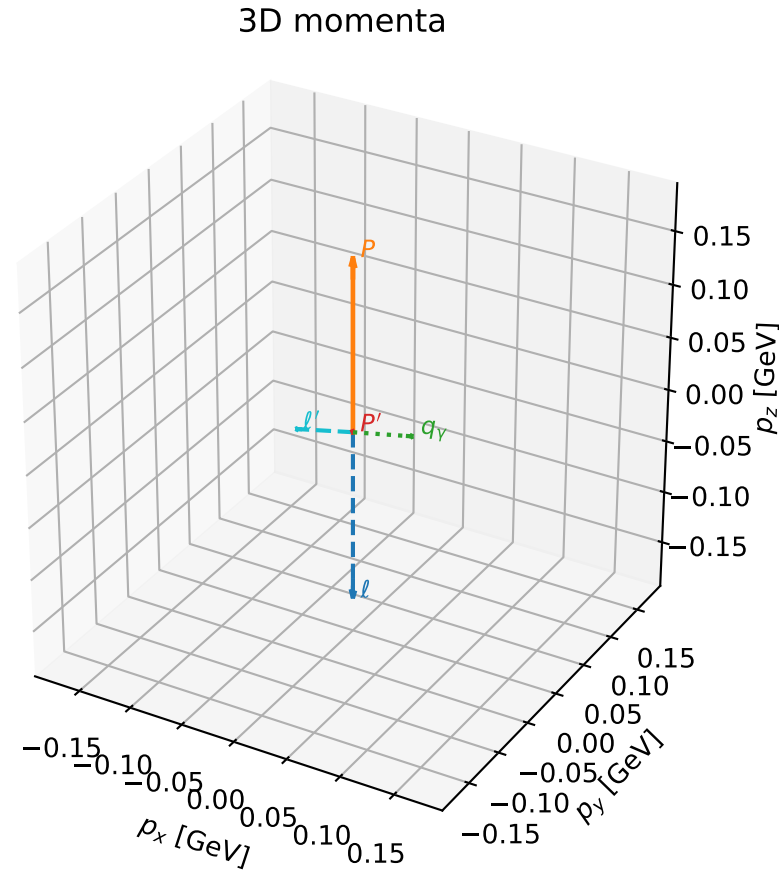


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

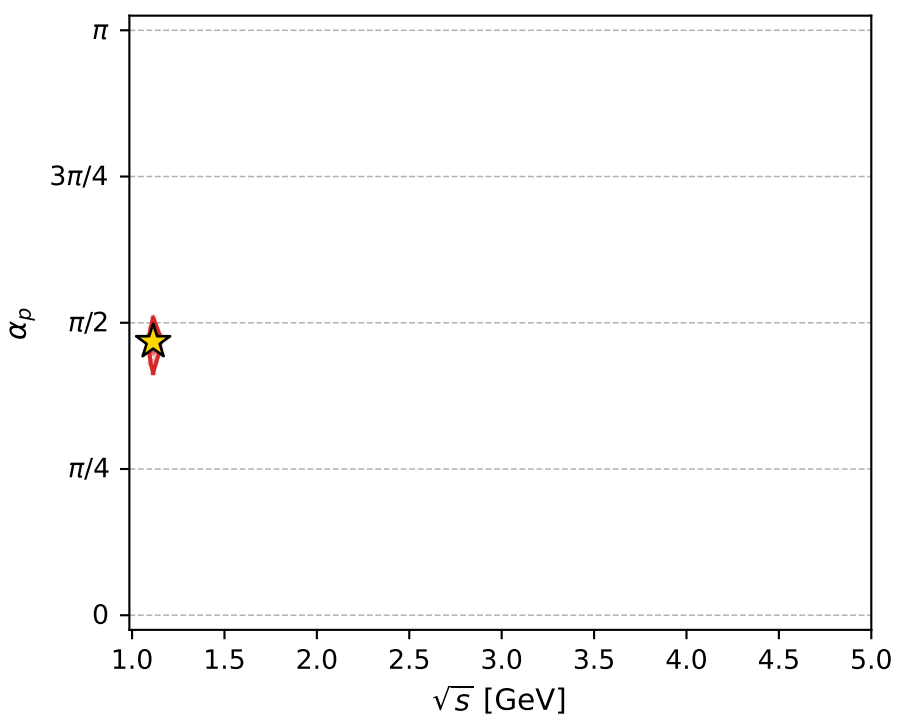
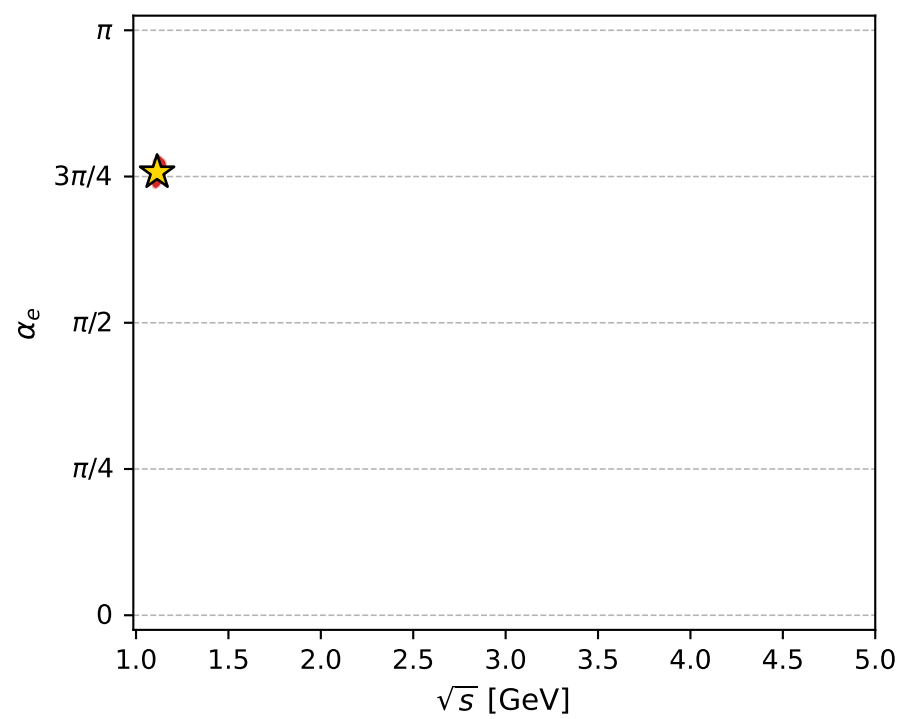
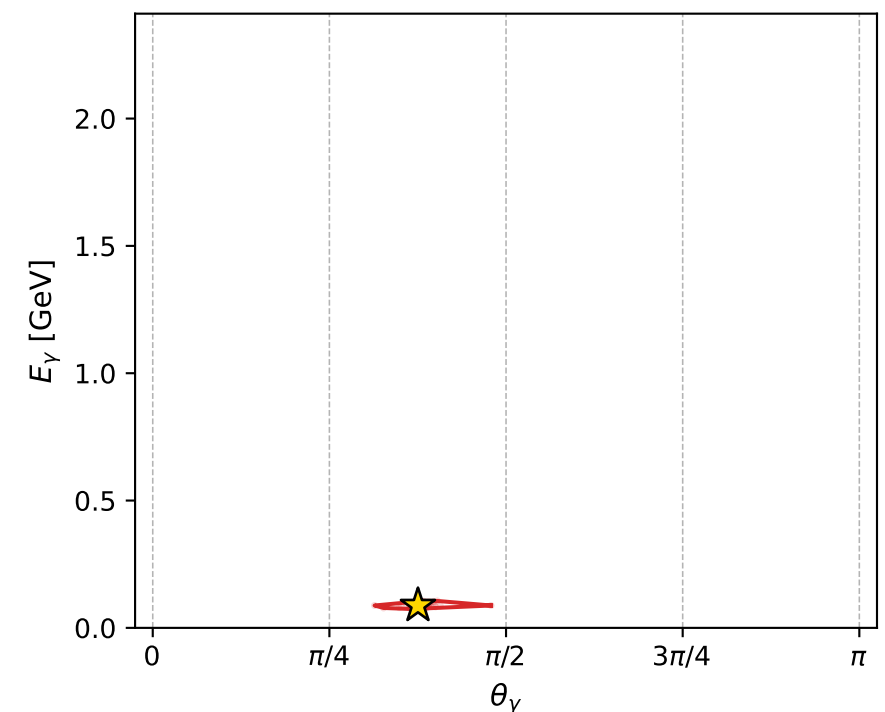
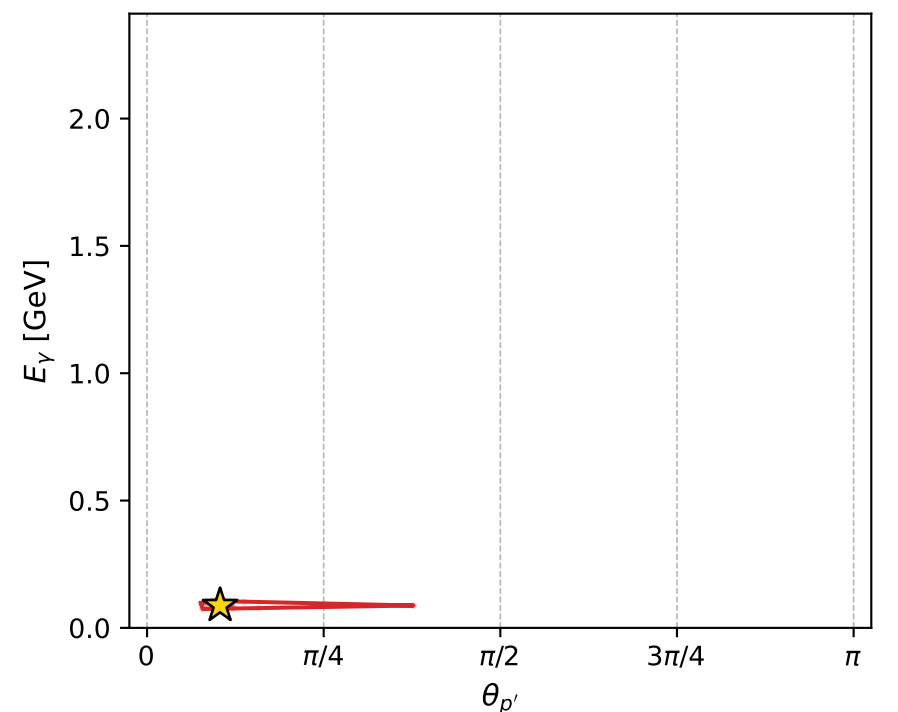
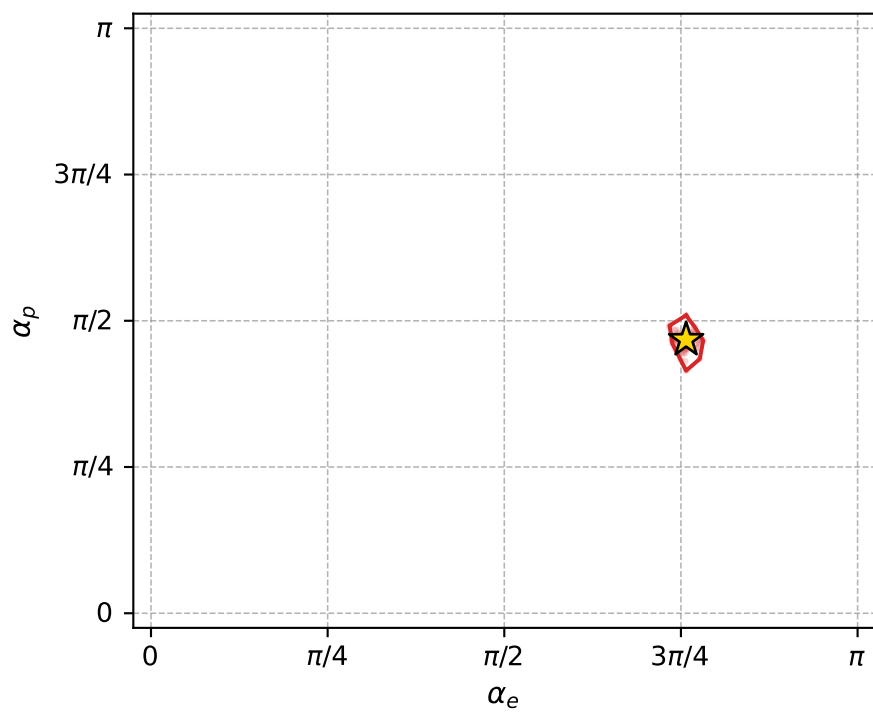
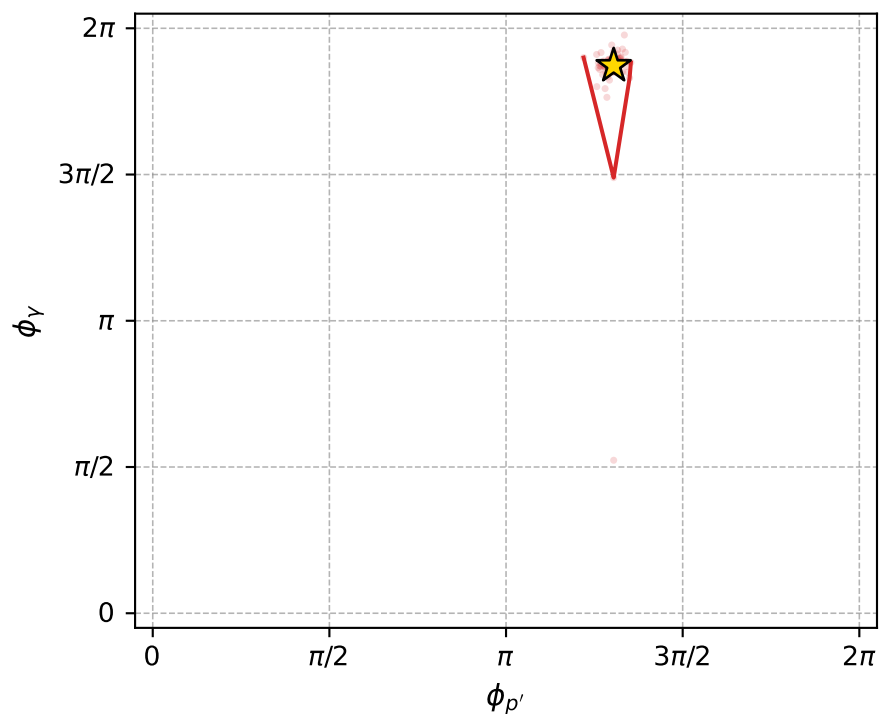
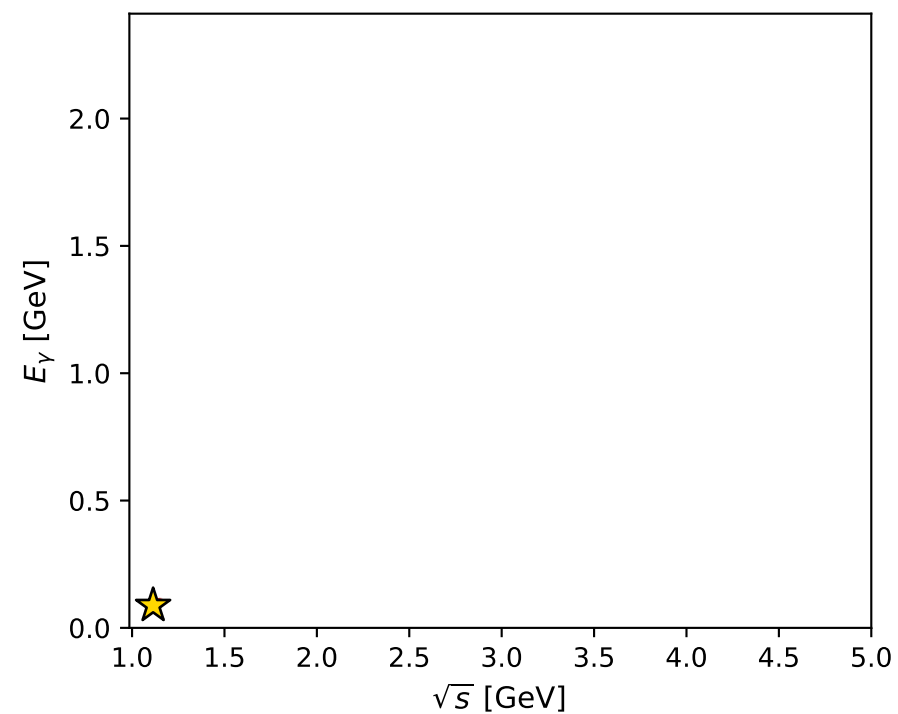
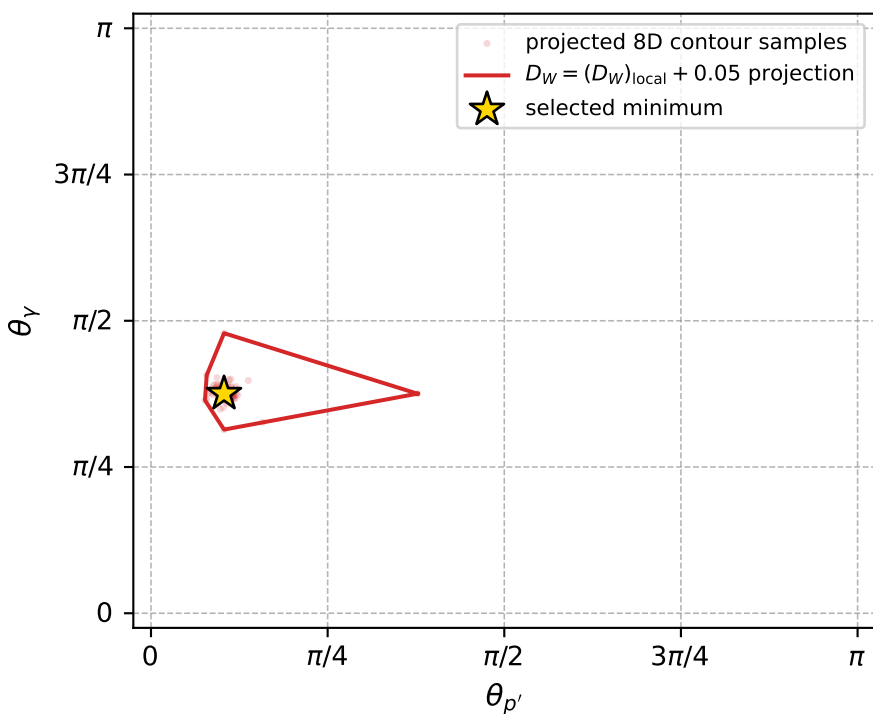
dw_mixing_angles_polarization_06_local_minimum_1021 D_W=0.0195835
region: polarization_cluster_06_local_minimum_1021
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3791285$ rad
incoming lepton: $-0.723136|+> +0.690705|->$
proton mixing angle: $\alpha_p=1.4688565$ rad
incoming proton: $0.101763|+> +0.994809|->$

$s=1.2409$, $\sqrt{s}=1.11396$, $|\vec{P}|=0.162058$, $|\vec{P}'|=0.000883616$
 $\theta_{P'}=0.324755$, $\theta_Y=1.17918$, $\phi_{P'}=4.09872$, $\phi_Y=5.88049$
 $E_Y=0.0878422$, $\theta_{\ell'}=1.97145$, $\phi_{\ell'}=2.7355$
 $Q^2=0.0174201$, $x_B=0.0956278$, $t=-0.0257991$, $W^2=1.04459$, $y=0.504541$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1621$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1621$
 P $E= 0.9519$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1621$
 ℓ' $E= 0.0881$ $p_x= -0.0745$ $p_y= 0.0320$ $p_z= -0.0344$
 P' $E= 0.9380$ $p_x= -0.0002$ $p_y= -0.0002$ $p_z= 0.0008$
 q_Y $E= 0.0878$ $p_x= 0.0747$ $p_y= -0.0318$ $p_z= 0.0335$



electron: polarization cluster P6, local minimum 1021; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.01958353$
 $D_W \text{ contour} = 0.06958353$
 above global minimum = 0.019567097
 8D contour samples = 253

selected phase-space point:
 $\text{sqrt_s} = 1.113955$
 $\text{theta_p_out} = 0.3247546$
 $\text{theta_gamma_out} = 1.179182$
 $\text{qOut} = 0.08784217$
 $\text{phi_p_out} = 4.098722$
 $\text{phi_gamma_out} = 5.880492$
 $\text{alpha_e} = 2.379128$
 $\text{alpha_p} = 1.468857$

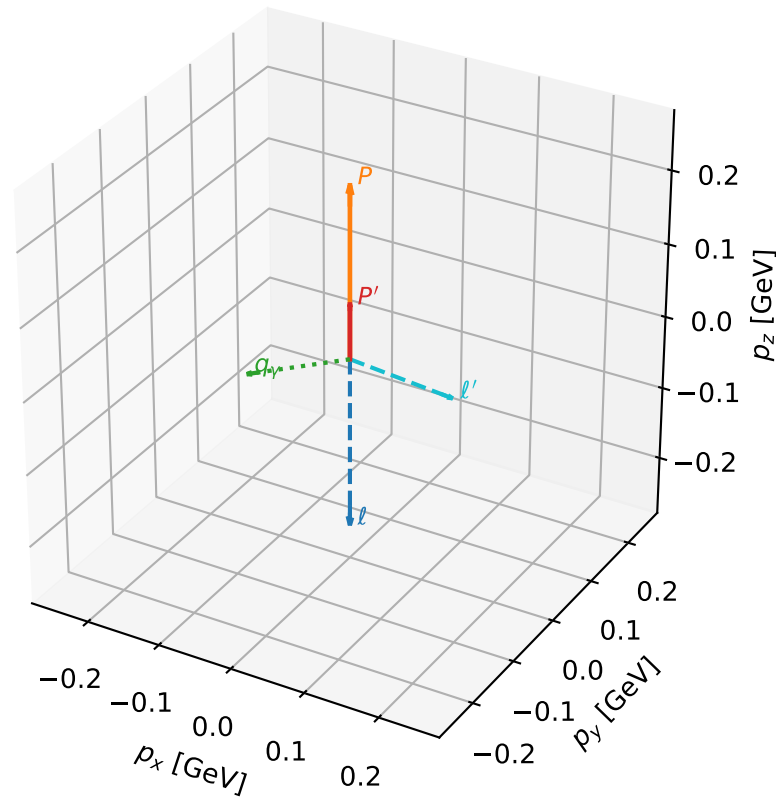
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_1023 D_W=0.0199196
 region: polarization_cluster_06_local_minimum_1023
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3825913$ rad
 incoming lepton: $-0.725524|+\rangle +0.688197|-\rangle$
 proton mixing angle: $\alpha_p=1.6967366$ rad
 incoming proton: $-0.125608|+\rangle +0.99208|-\rangle$

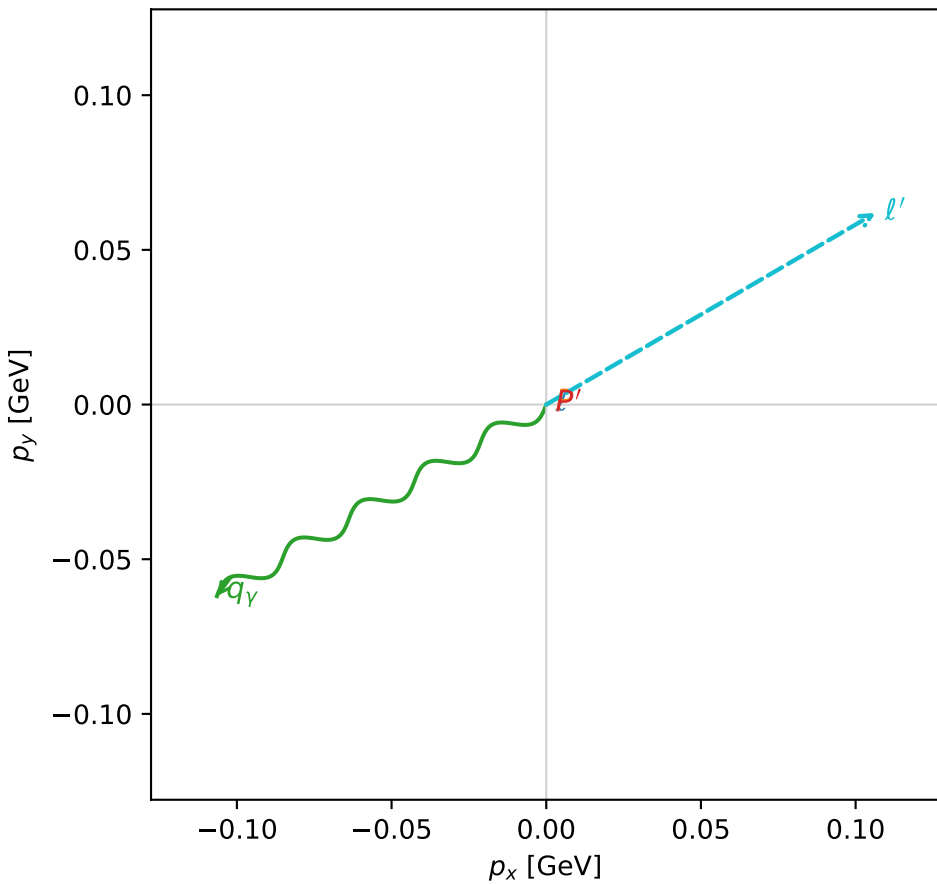
$s=1.44112$, $\sqrt{s}=1.20047$, $|\vec{P}|=0.233773$, $|\vec{P}'|=0.0747506$
 $\theta_{P'}=0$, $\theta_V=1.73417$, $\phi_{P'}=4.78564$, $\phi_V=3.66851$
 $E_V=0.124827$, $\theta_{\ell'}=1.98703$, $\phi_{\ell'}=0.526922$
 $Q^2=0.0375046$, $x_B=0.136153$, $t=-0.0246266$, $W^2=1.1178$, $y=0.490775$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2338$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2338$
 P $E= 0.9667$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2338$
 ℓ' $E= 0.1347$ $p_x= 0.1065$ $p_y= 0.0619$ $p_z= -0.0544$
 P' $E= 0.9410$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0748$
 q_Y $E= 0.1248$ $p_x= -0.1065$ $p_y= -0.0619$ $p_z= -0.0203$

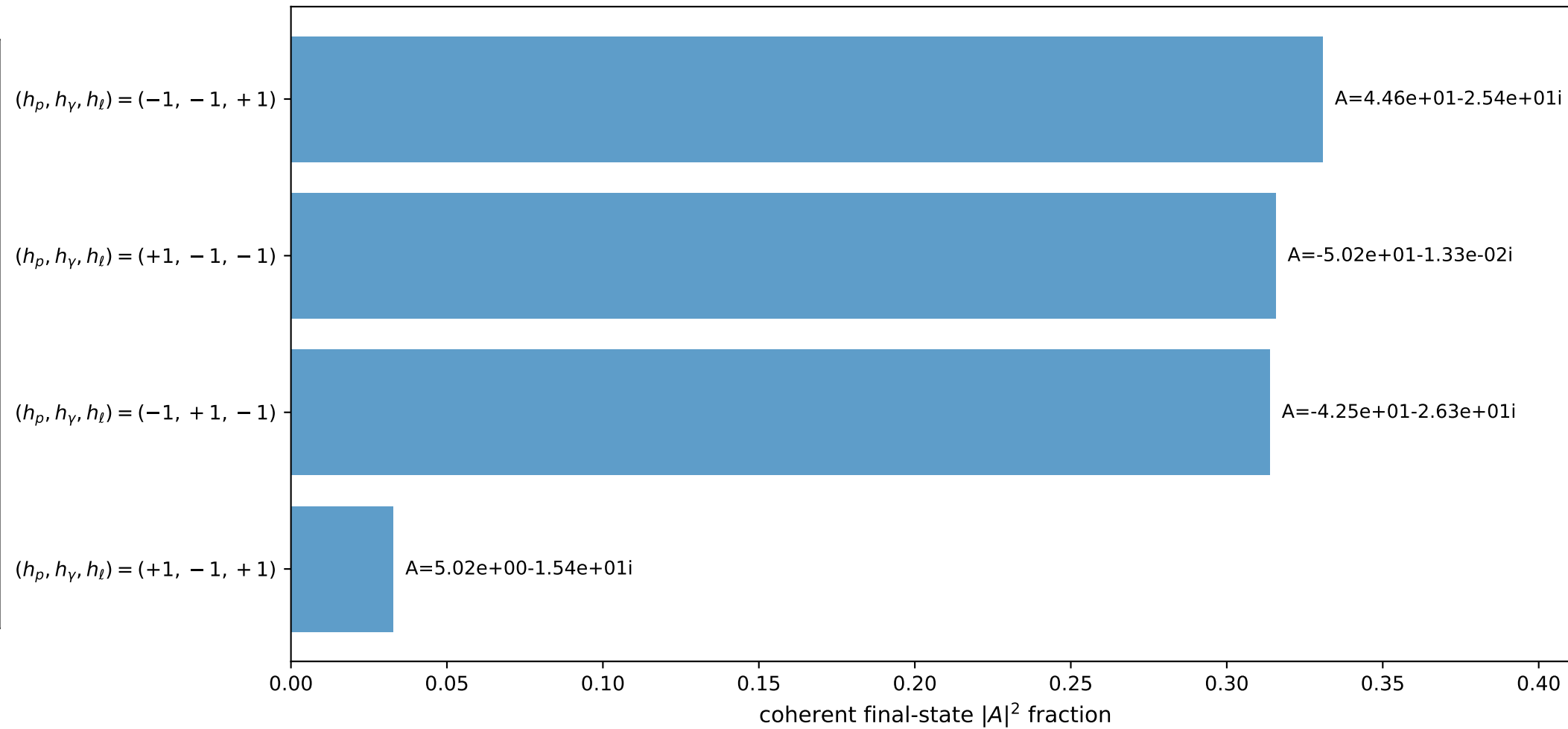
3D momenta



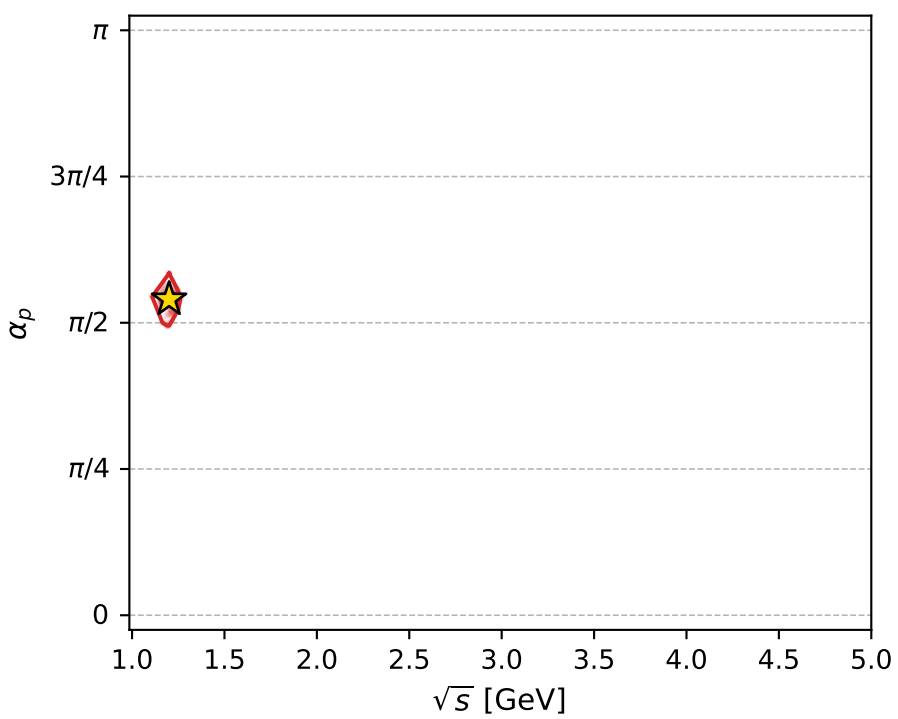
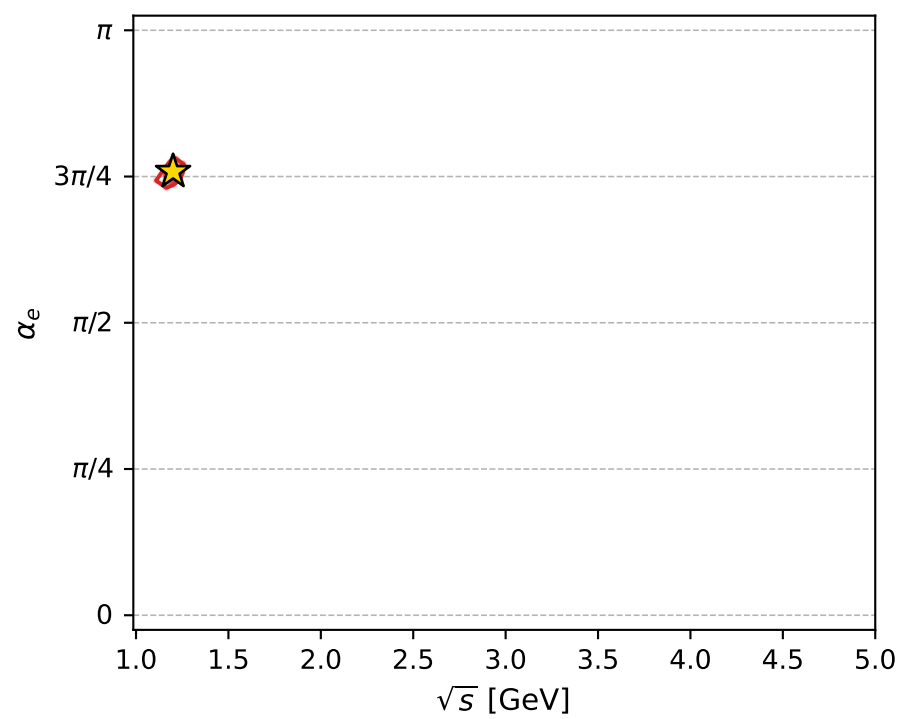
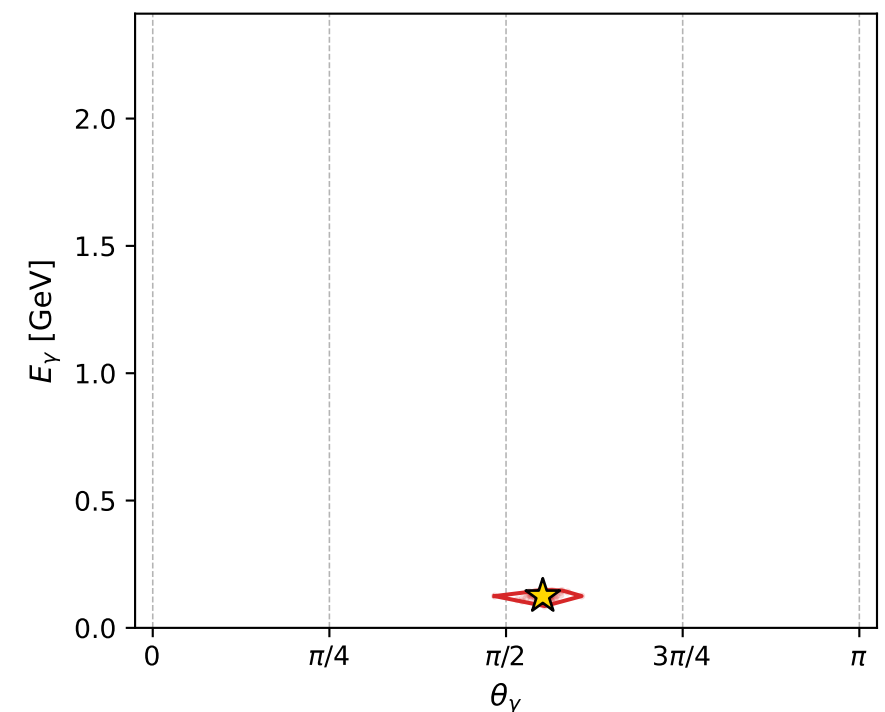
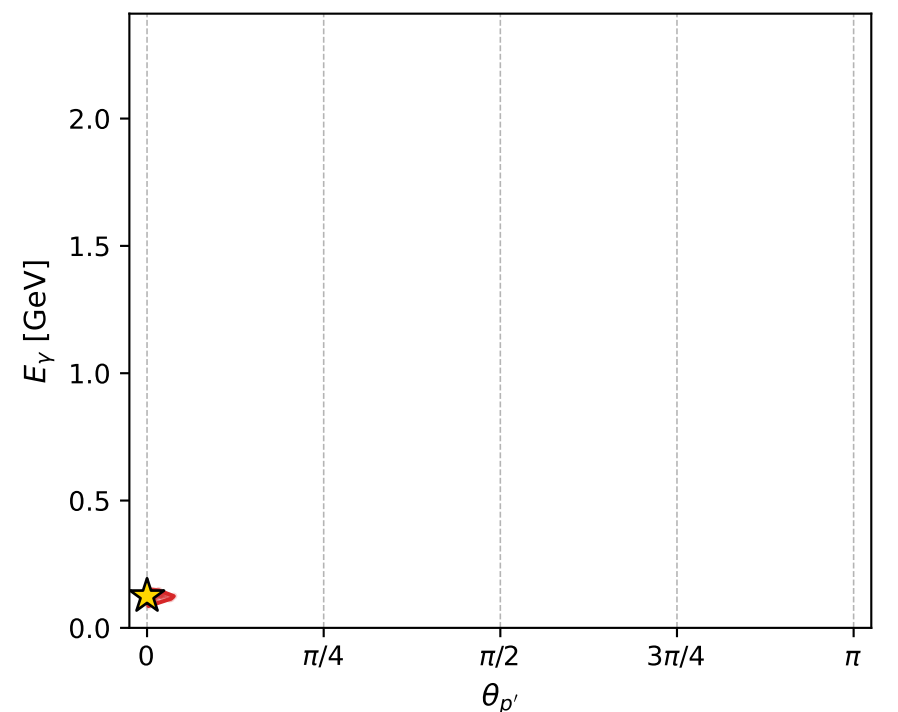
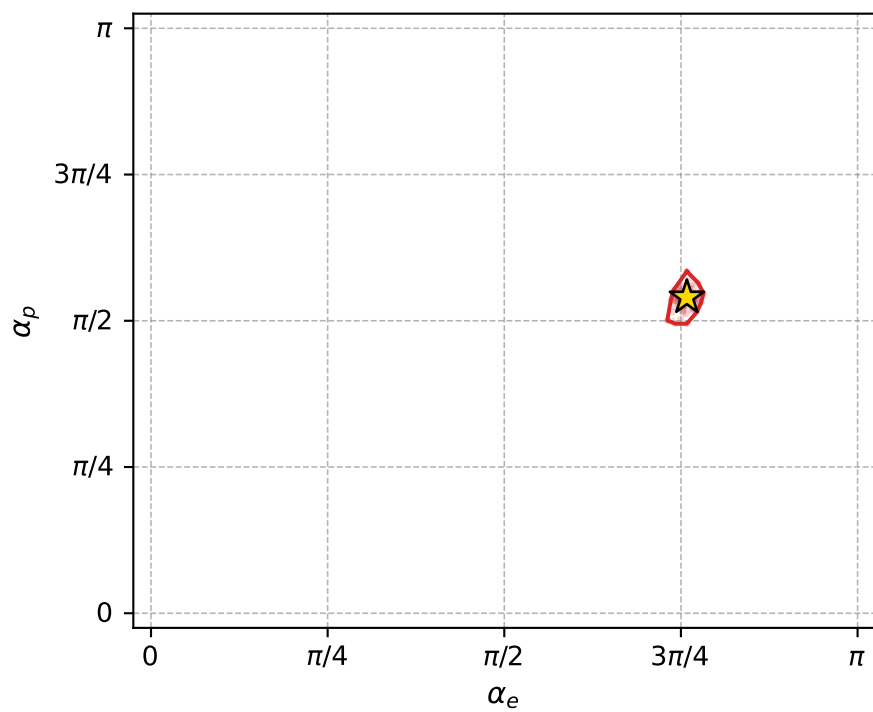
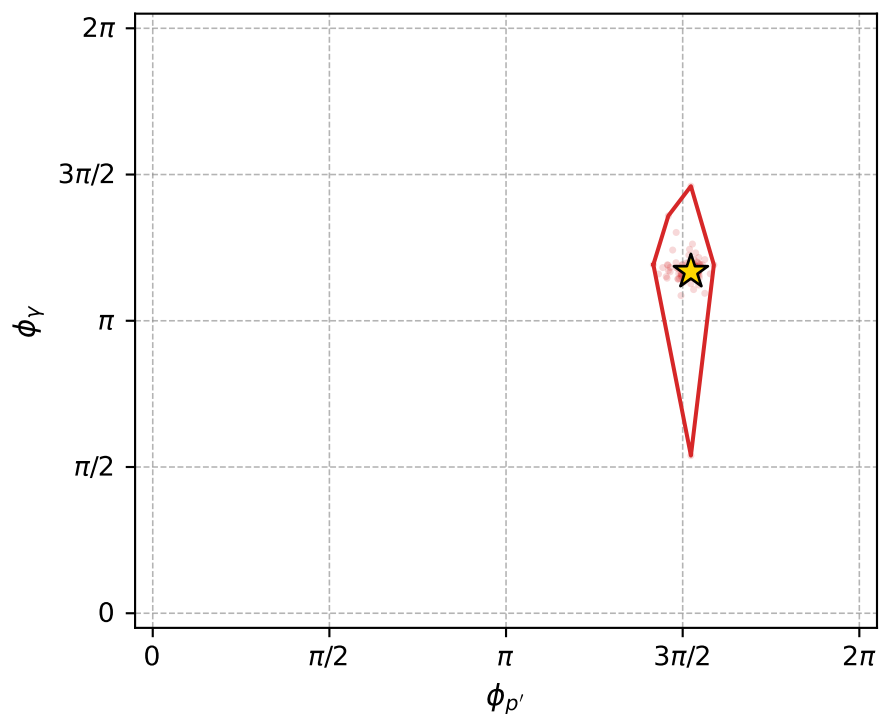
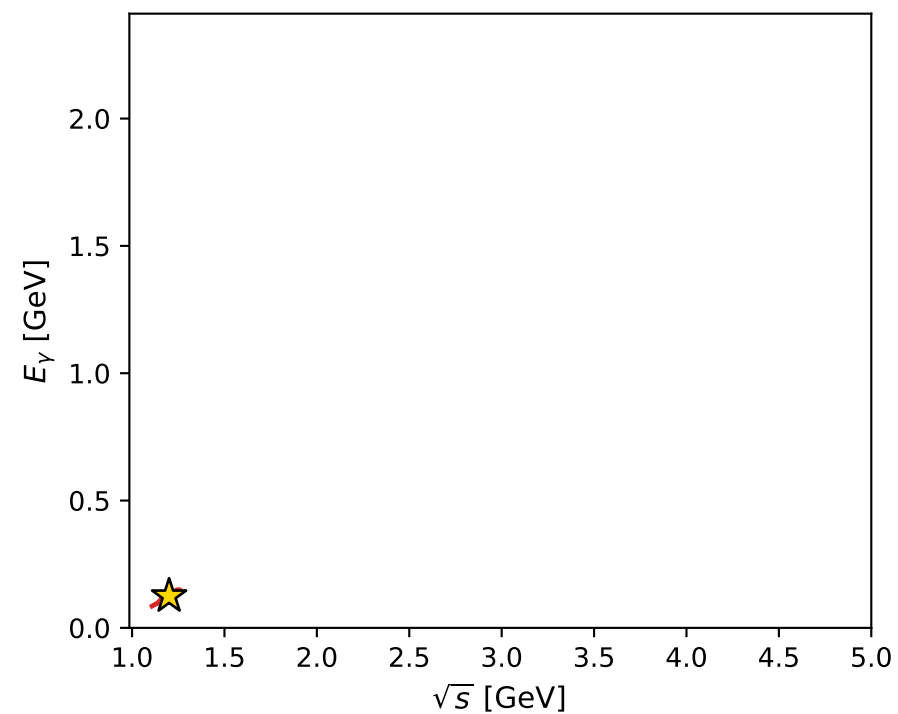
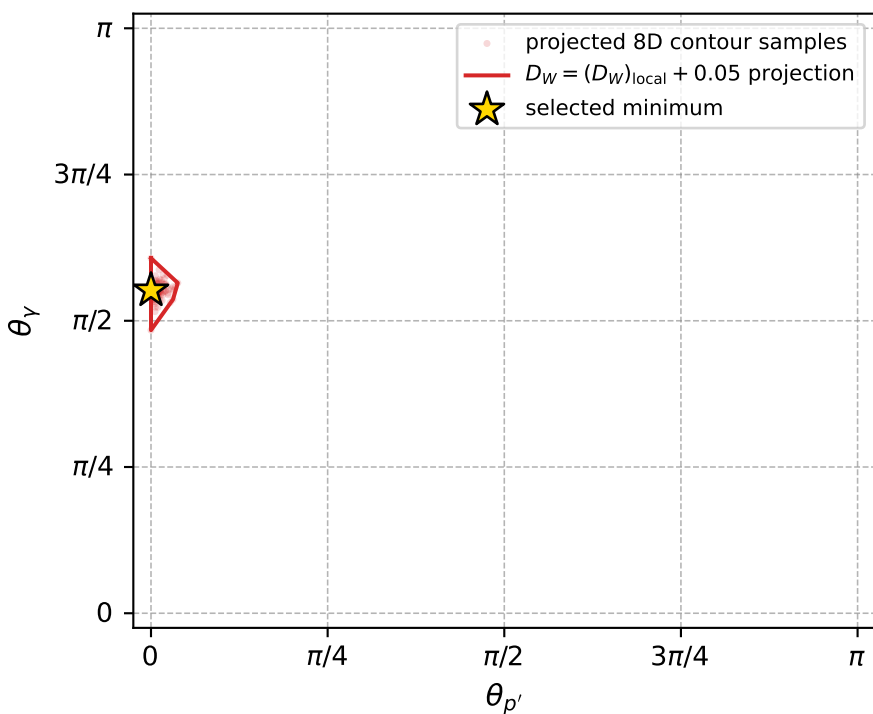
Transverse plane



Leading final-state amplitudes (N=4, retained=99.3%)



electron: polarization cluster P6, local minimum 1023; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.01991957$
 $D_W \text{ contour} = 0.06991957$
 above global minimum = 0.019903138
 8D contour samples = 139

selected phase-space point:
 $\text{sqrt_s} = 1.200465$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 1.734172$
 $\text{qOut} = 0.1248273$
 $\text{phi_p_out} = 4.785636$
 $\text{phi_gamma_out} = 3.668514$
 $\text{alpha_e} = 2.382591$
 $\text{alpha_p} = 1.696737$

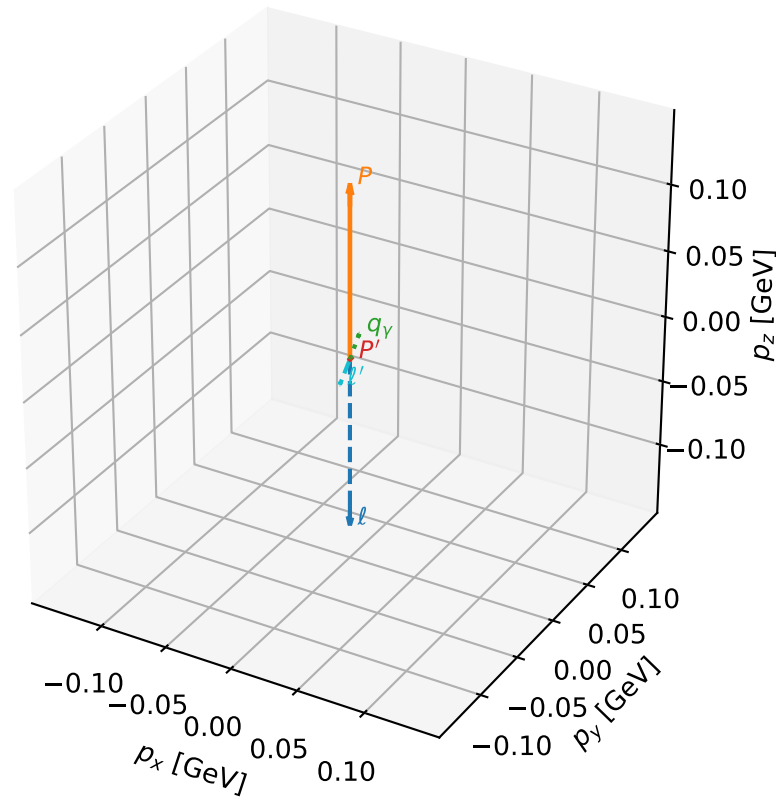
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_1025 D_W=0.0201095
 region: polarization_cluster_06_local_minimum_1025
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.40194$ rad
 incoming lepton: $-0.738703|+> +0.674031|->$
 proton mixing angle: $\alpha_p=1.3813229$ rad
 incoming proton: $0.188342|+> +0.982104|->$

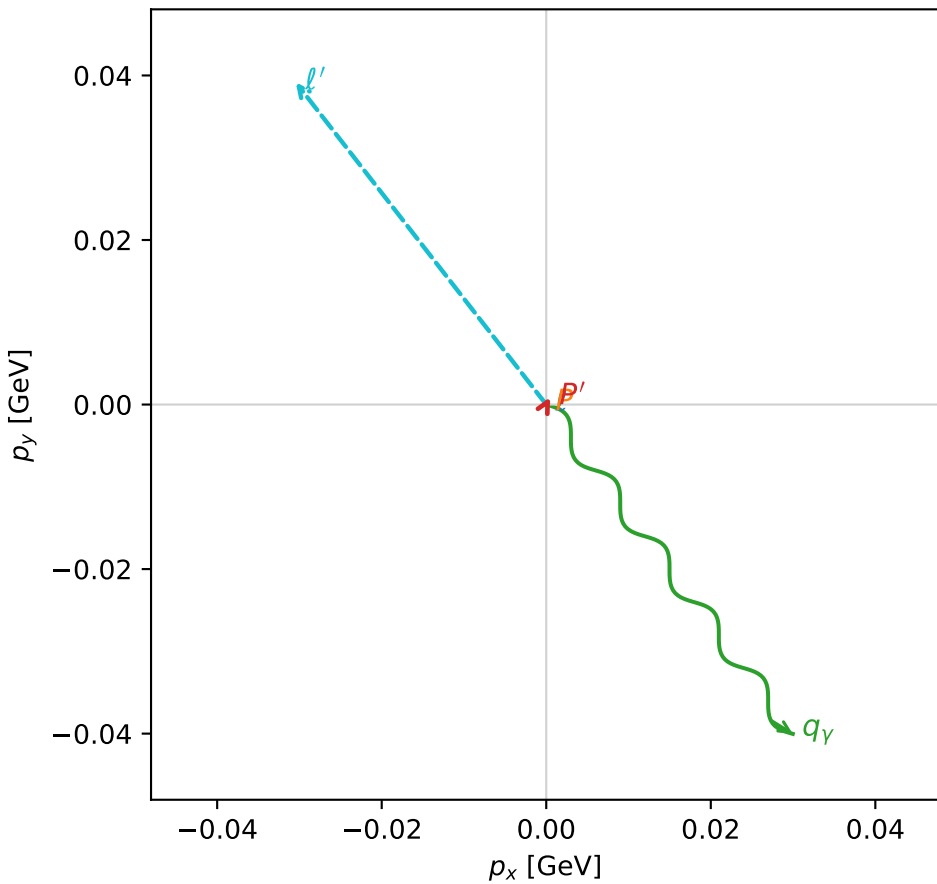
$s=1.15793$, $\sqrt{s}=1.07607$, $|\vec{P}|=0.129212$, $|\vec{P}'|=0.00115369$
 $\theta_{P'}=1.04332$, $\theta_V=0.810925$, $\phi_{P'}=1.07956$, $\phi_V=5.35501$
 $E_V=0.0689825$, $\theta_l=2.34089$, $\phi_l=2.23163$
 $Q^2=0.00542405$, $x_B=0.0402311$, $t=-0.0164687$, $W^2=1.00924$, $y=0.484824$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1292$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1292$
 P $E= 0.9469$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1292$
 ℓ' $E= 0.0691$ $p_x= -0.0304$ $p_y= 0.0392$ $p_z= -0.0481$
 P' $E= 0.9380$ $p_x= 0.0005$ $p_y= 0.0009$ $p_z= 0.0006$
 q_V $E= 0.0690$ $p_x= 0.0300$ $p_y= -0.0400$ $p_z= 0.0475$

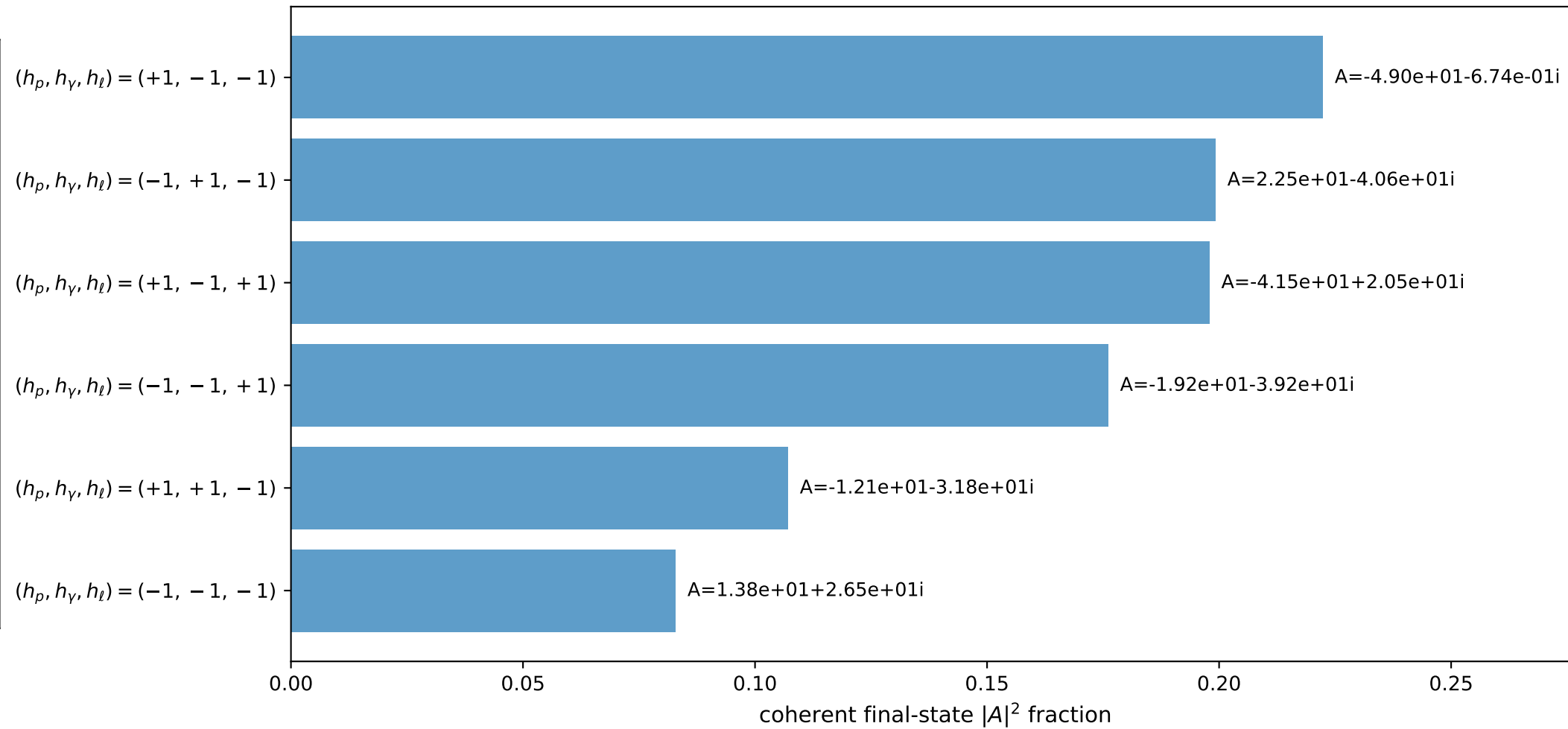
3D momenta



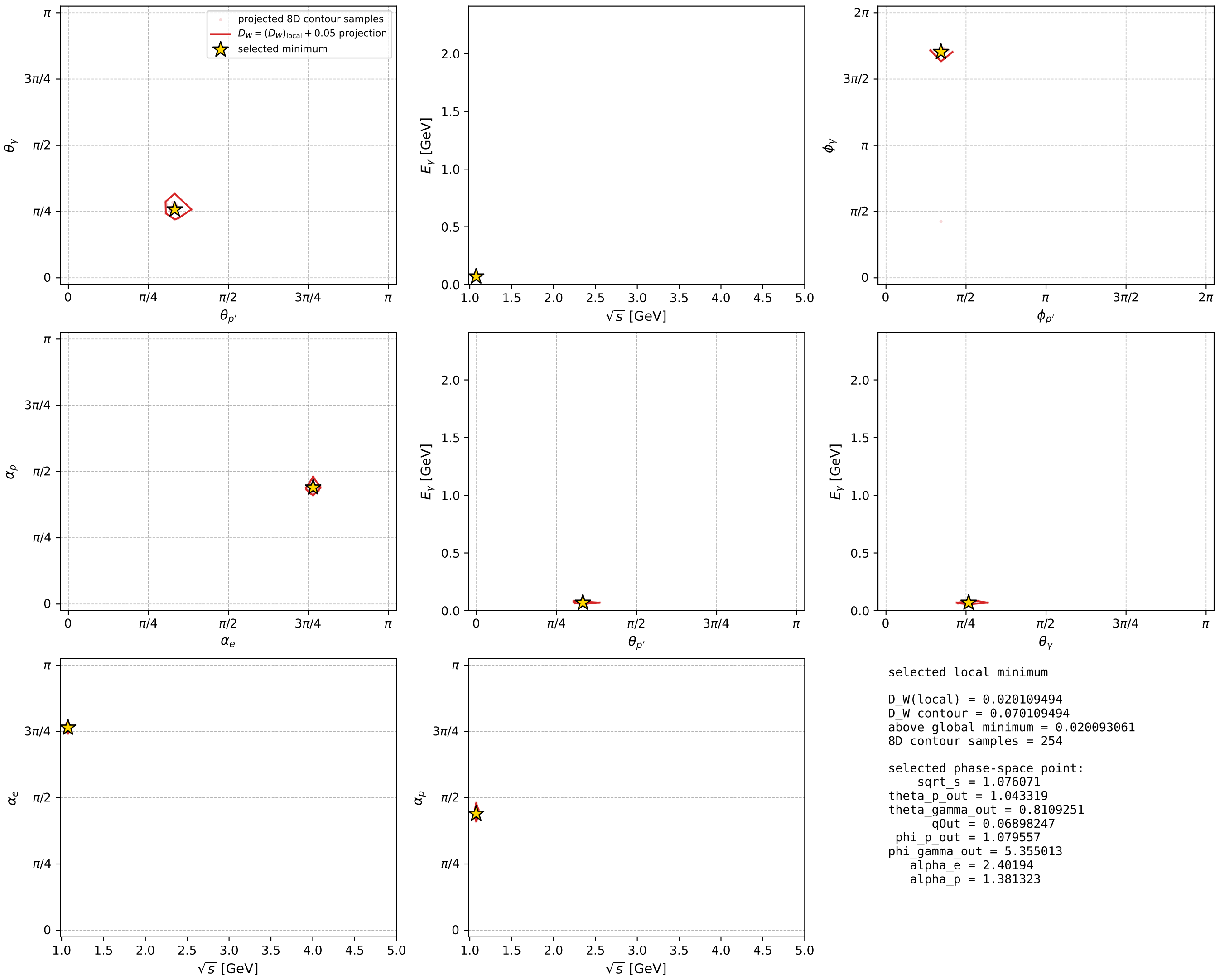
Transverse plane



Leading final-state amplitudes (N=6, retained=98.5%)



electron: polarization cluster P6, local minimum 1025; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

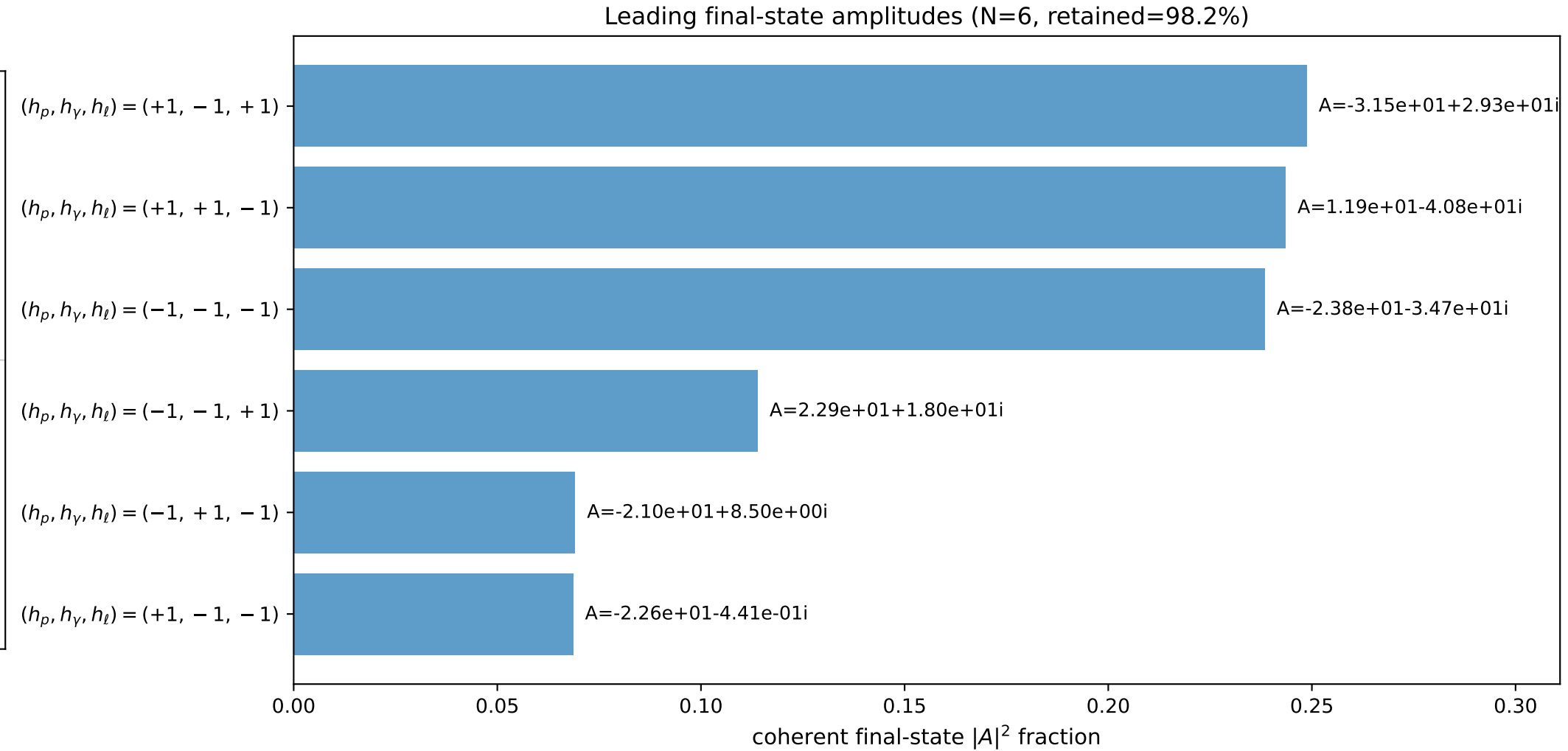
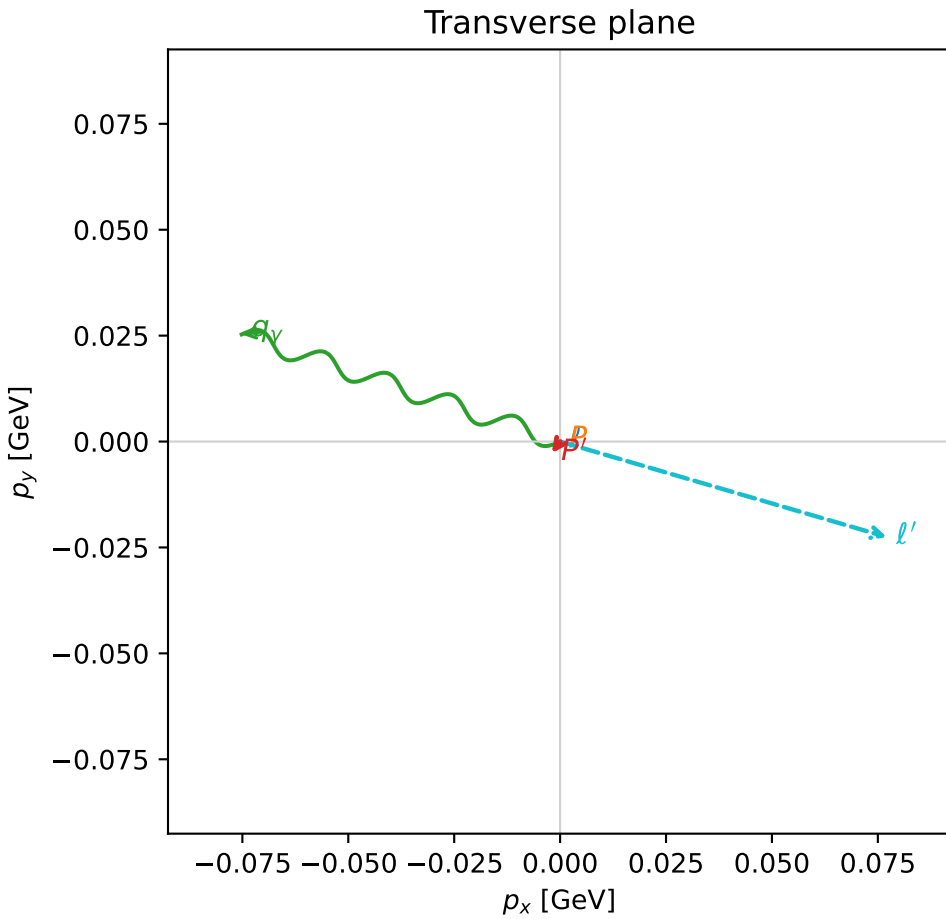
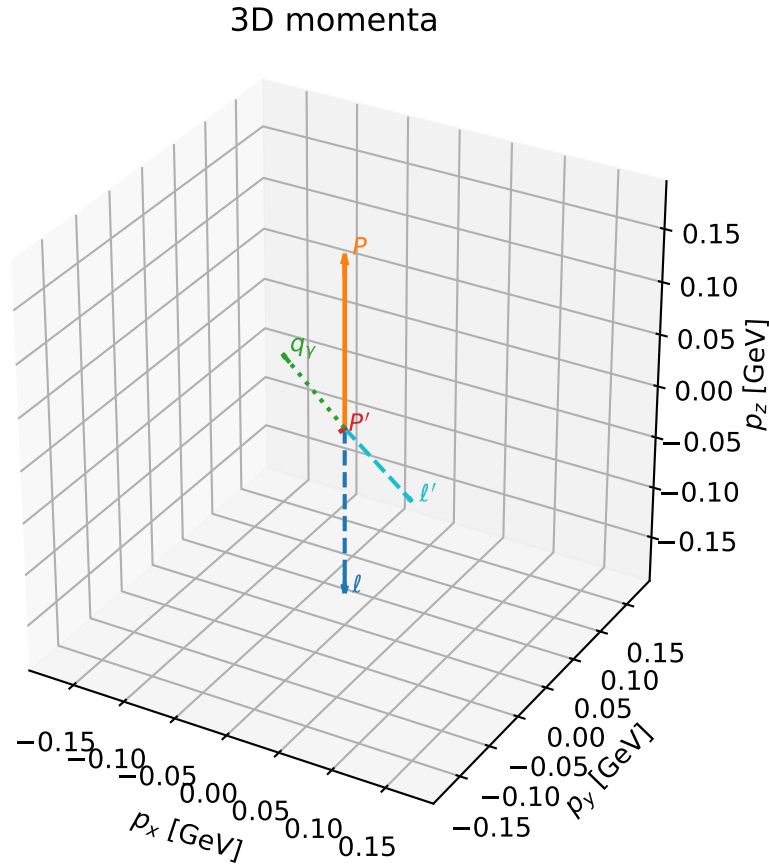


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

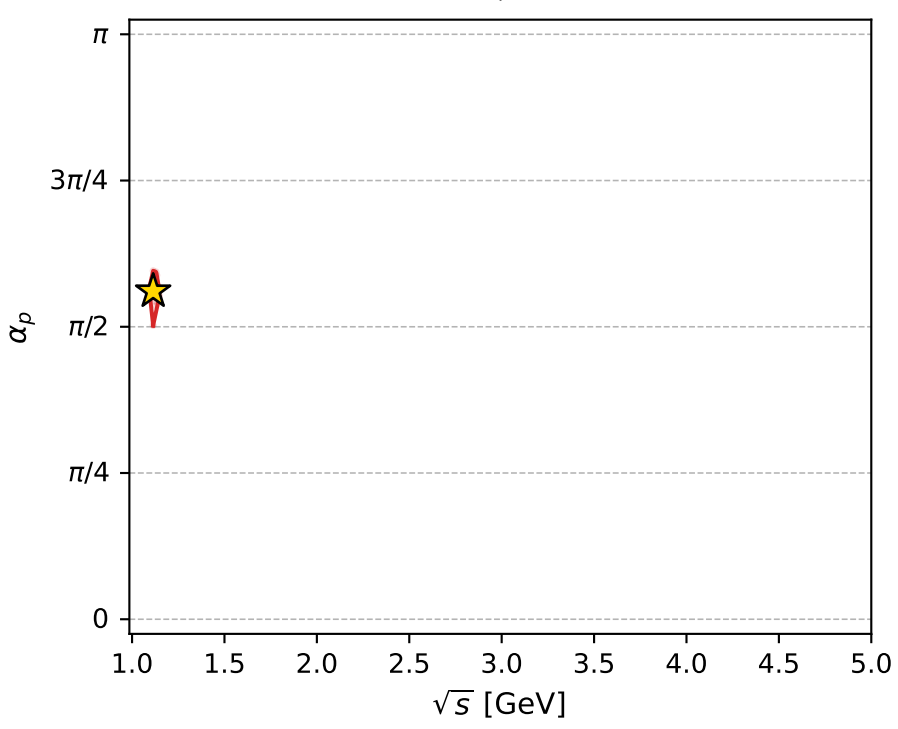
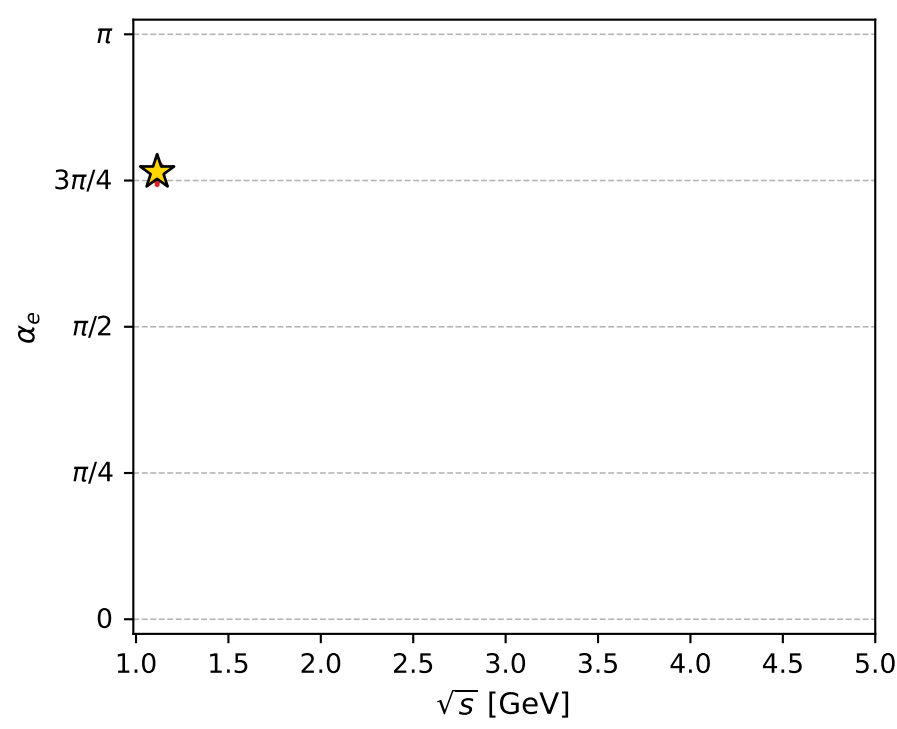
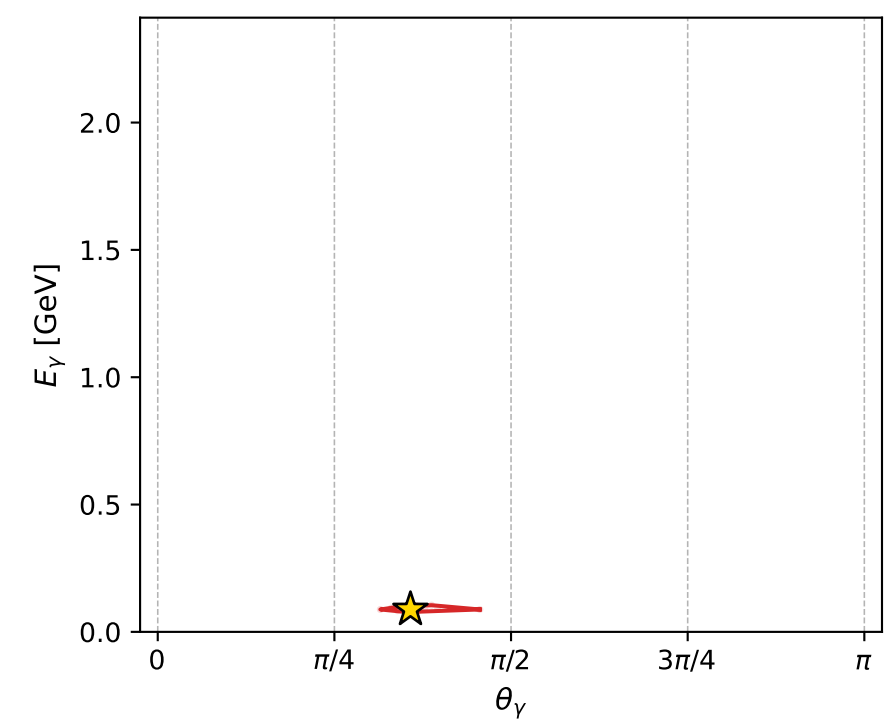
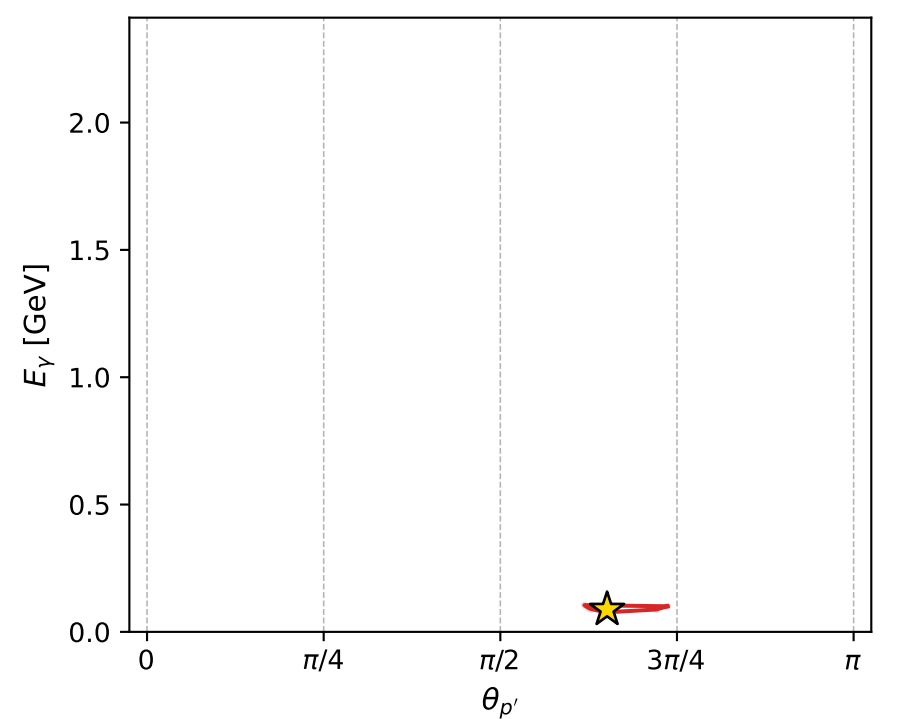
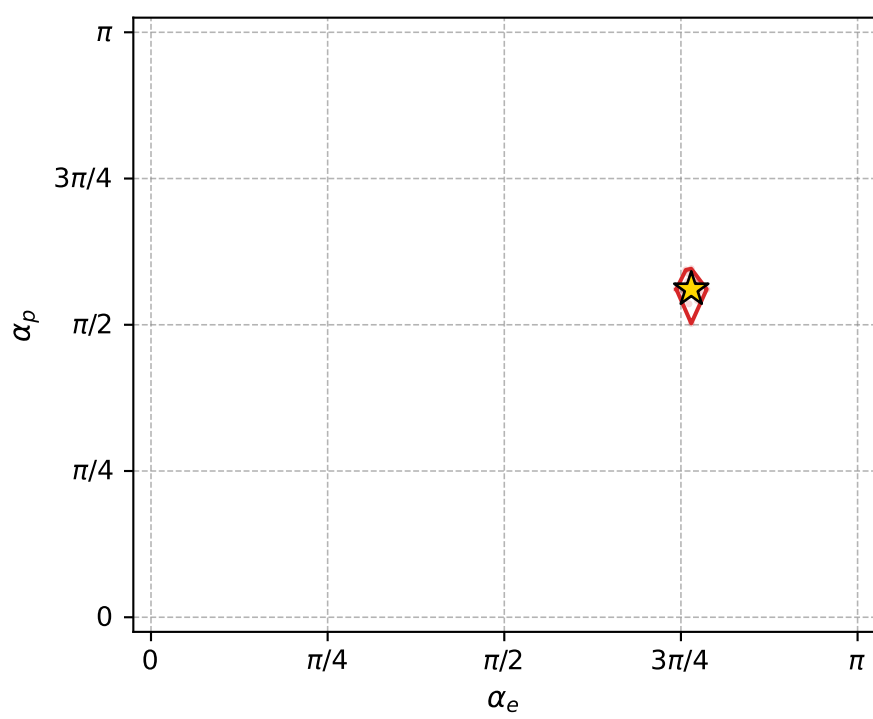
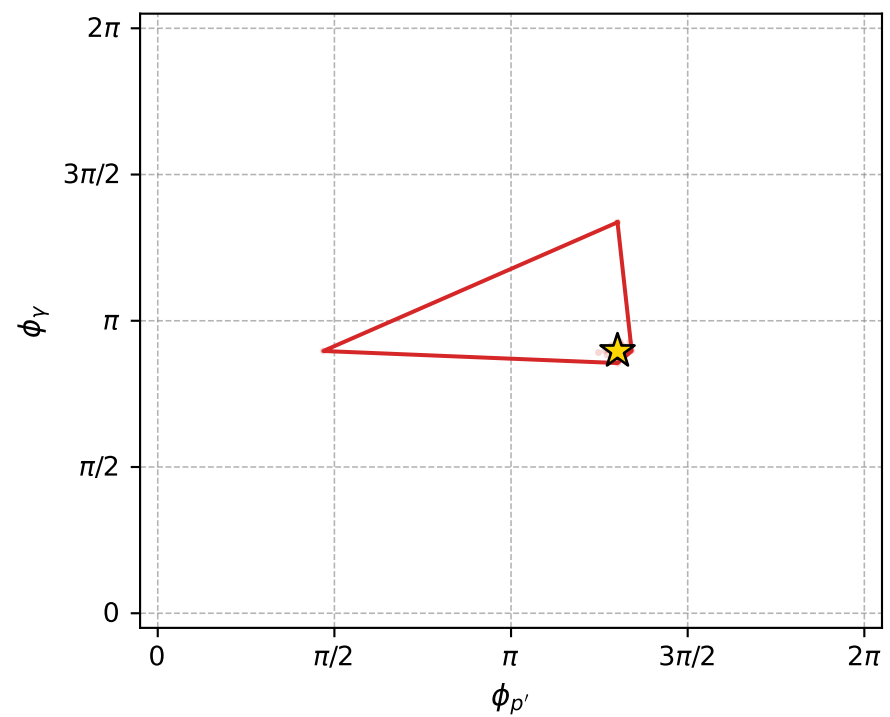
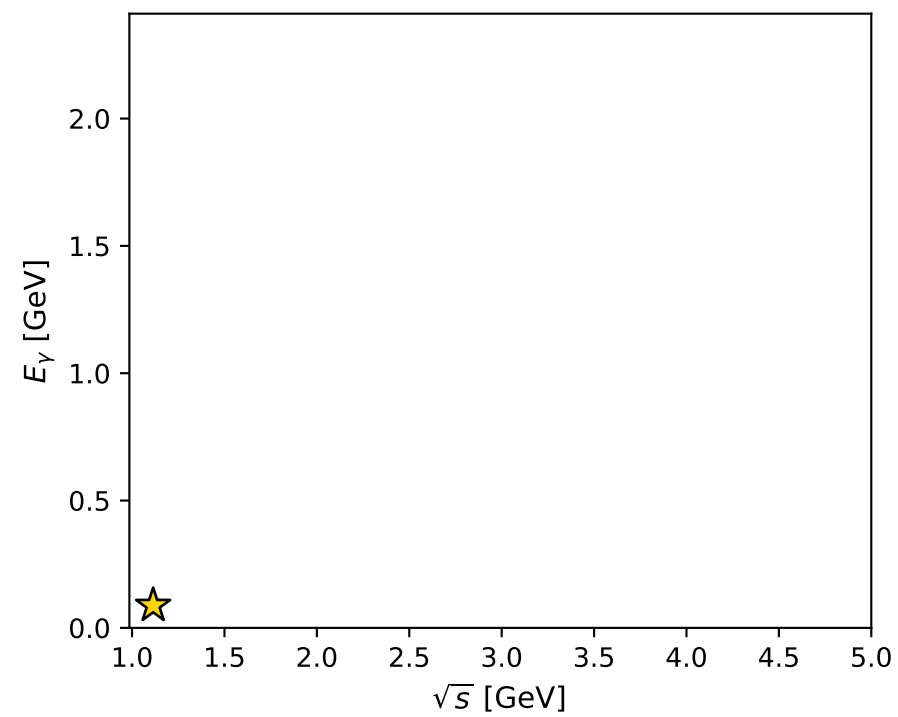
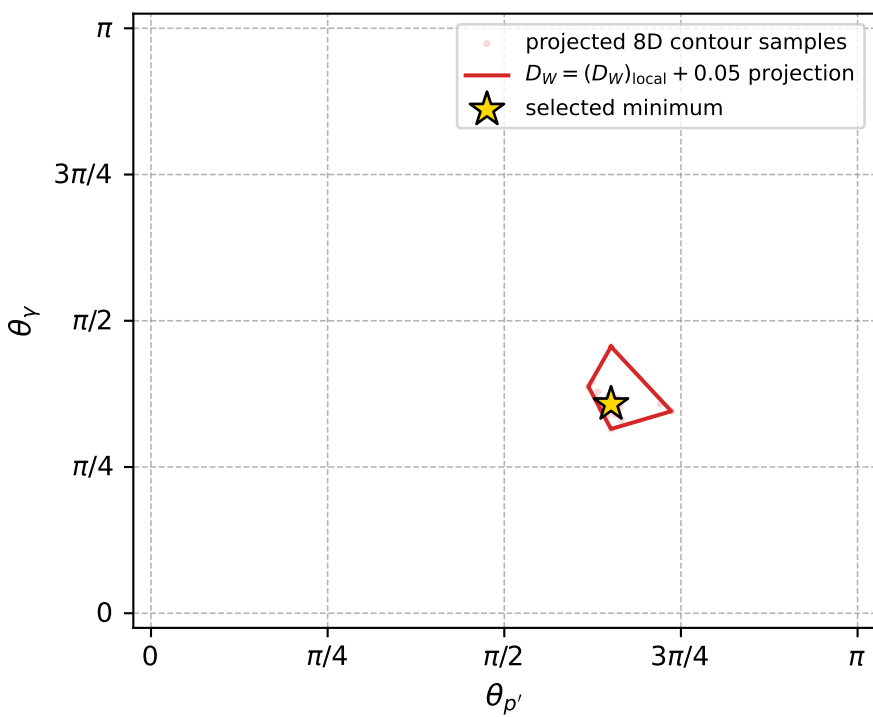
dw_mixing_angles_polarization_06_local_minimum_1026 D_W=0.0201491
 region: polarization_cluster_06_local_minimum_1026
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.4020699$ rad
 incoming lepton: $-0.73879|+\rangle +0.673935|-\rangle$
 proton mixing angle: $\alpha_p=1.7616287$ rad
 incoming proton: $-0.189676|+\rangle +0.981847|-\rangle$

$s=1.24127$, $\sqrt{s}=1.11412$, $|\vec{P}|=0.162201$, $|\vec{P}'|=0.0038079$
 $\theta_{p'}=2.04543$, $\theta_Y=1.12361$, $\phi_{p'}=4.08827$, $\phi_Y=2.81699$
 $E_Y=0.0879412$, $\theta_{\ell'}=1.99496$, $\phi_{\ell'}=5.99887$
 $Q^2=0.0168314$, $x_B=0.0925899$, $t=-0.0266946$, $W^2=1.0448$, $y=0.502967$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1622$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1622$
 P $E= 0.9519$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1622$
 ℓ' $E= 0.0882$ $p_x= 0.0771$ $p_y= -0.0225$ $p_z= -0.0363$
 P' $E= 0.9380$ $p_x= -0.0020$ $p_y= -0.0027$ $p_z= -0.0017$
 q_Y $E= 0.0879$ $p_x= -0.0752$ $p_y= 0.0253$ $p_z= 0.0380$



electron: polarization cluster P6, local minimum 1026; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.020149116$
 $D_W \text{ contour} = 0.070149116$
 above global minimum = 0.020132683
 8D contour samples = 255

selected phase-space point:
 $\text{sqrt_s} = 1.114122$
 $\text{theta_p_out} = 2.045426$
 $\text{theta_gamma_out} = 1.123615$
 $\text{qOut} = 0.08794121$
 $\text{phi_p_out} = 4.08827$
 $\text{phi_gamma_out} = 2.816992$
 $\text{alpha_e} = 2.40207$
 $\text{alpha_p} = 1.761629$

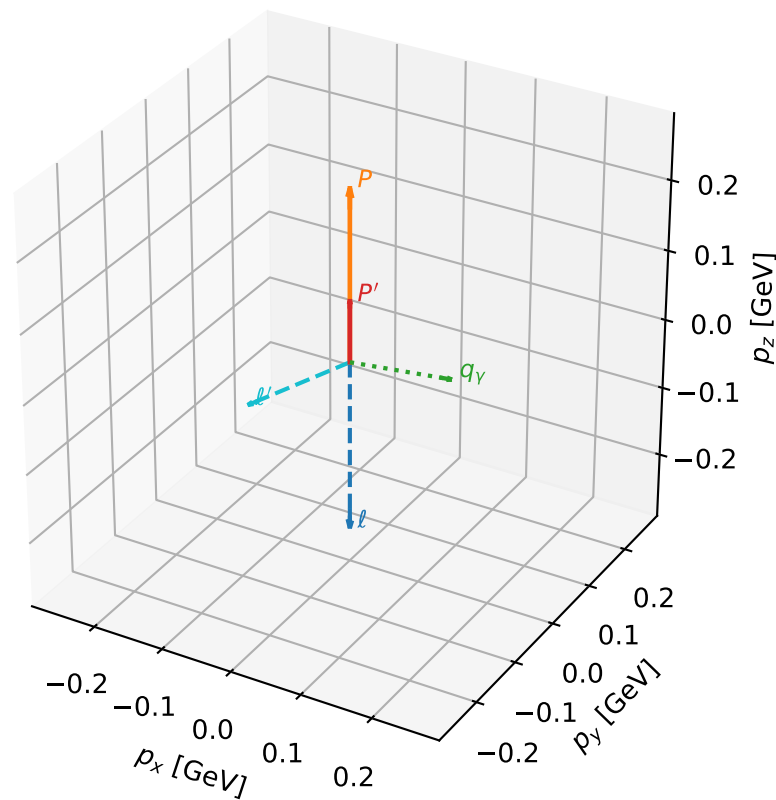
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_1027 D_W=0.0202045
 region: polarization_cluster_06_local_minimum_1027
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3874123$ rad
 incoming lepton: $-0.728833|+\rangle +0.684691|-\rangle$
 proton mixing angle: $\alpha_p=1.442657$ rad
 incoming proton: $0.127789|+\rangle +0.991801|-\rangle$

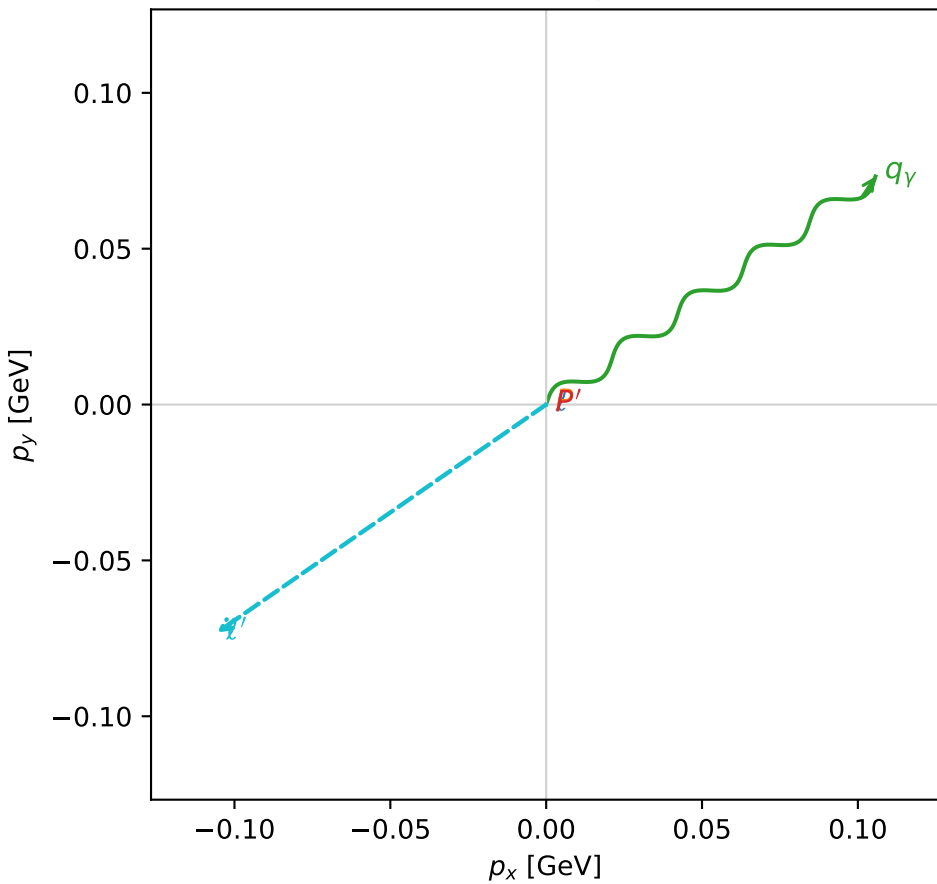
$s=1.47487$, $\sqrt{s}=1.21444$, $|\vec{P}|=0.244979$, $|\vec{P}'|=0.0871895$
 $\theta_{P'}=0$, $\theta_V=1.80946$, $\phi_{P'}=0.849955$, $\phi_V=0.605752$
 $E_V=0.132253$, $\theta_{\ell'}=1.98128$, $\phi_{\ell'}=3.74734$
 $Q^2=0.0412648$, $x_B=0.139459$, $t=-0.0241458$, $W^2=1.13447$, $y=0.497275$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.2450$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.2450$
 P $E=0.9695$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.2450$
 ℓ' $E=0.1401$ $p_x=-0.1056$ $p_y=-0.0732$ $p_z=-0.0559$
 P' $E=0.9420$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.0872$
 q_Y $E=0.1323$ $p_x=0.1056$ $p_y=0.0732$ $p_z=-0.0313$

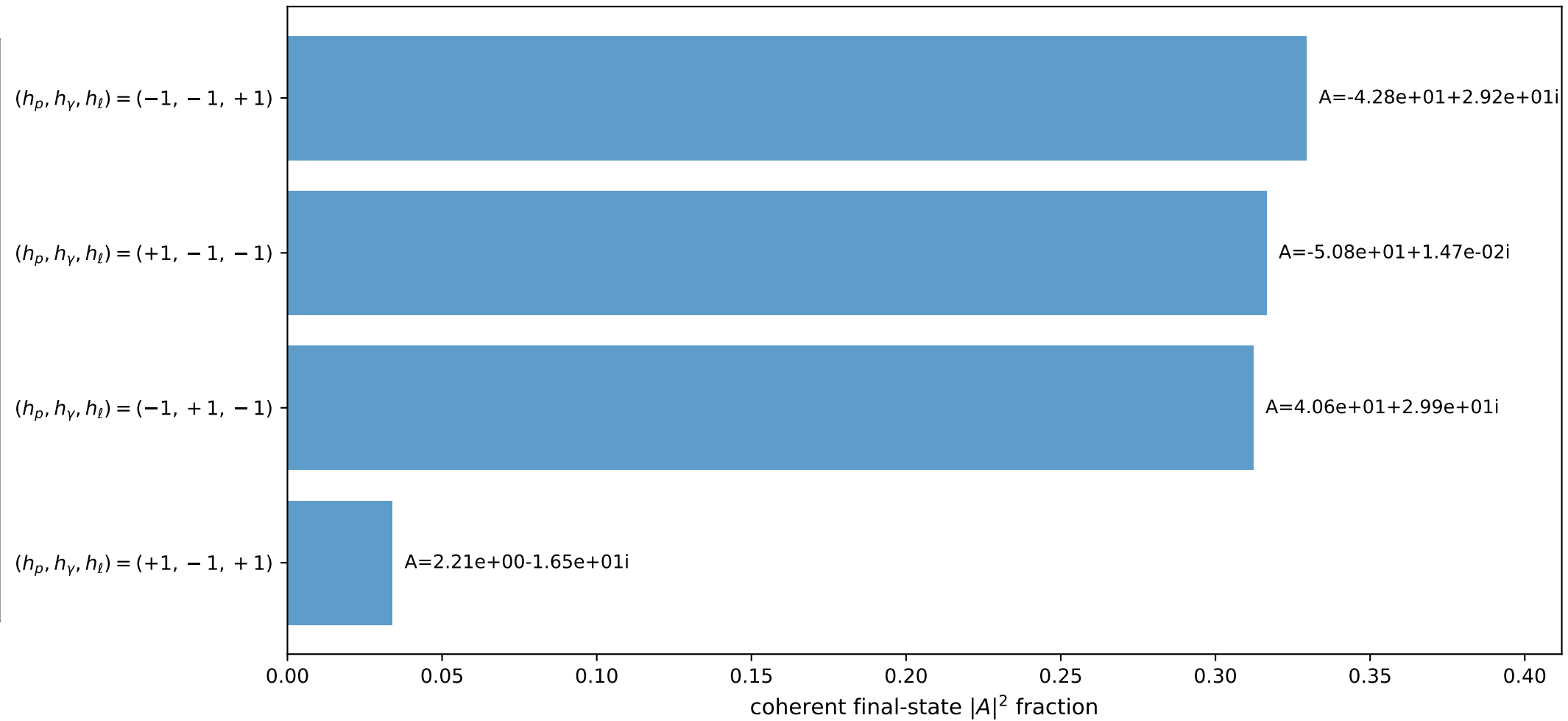
3D momenta



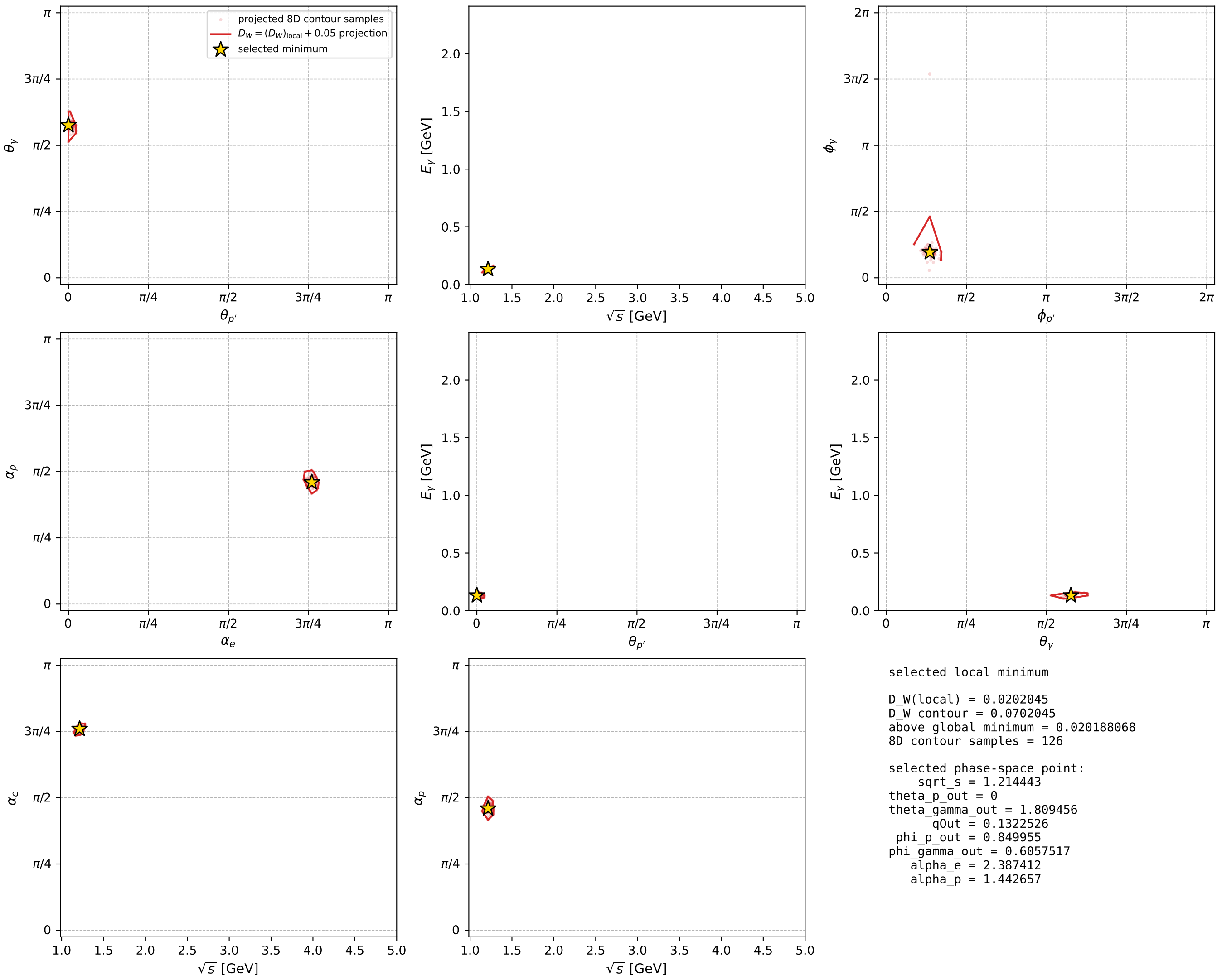
Transverse plane



Leading final-state amplitudes (N=4, retained=99.3%)



electron: polarization cluster P6, local minimum 1027; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

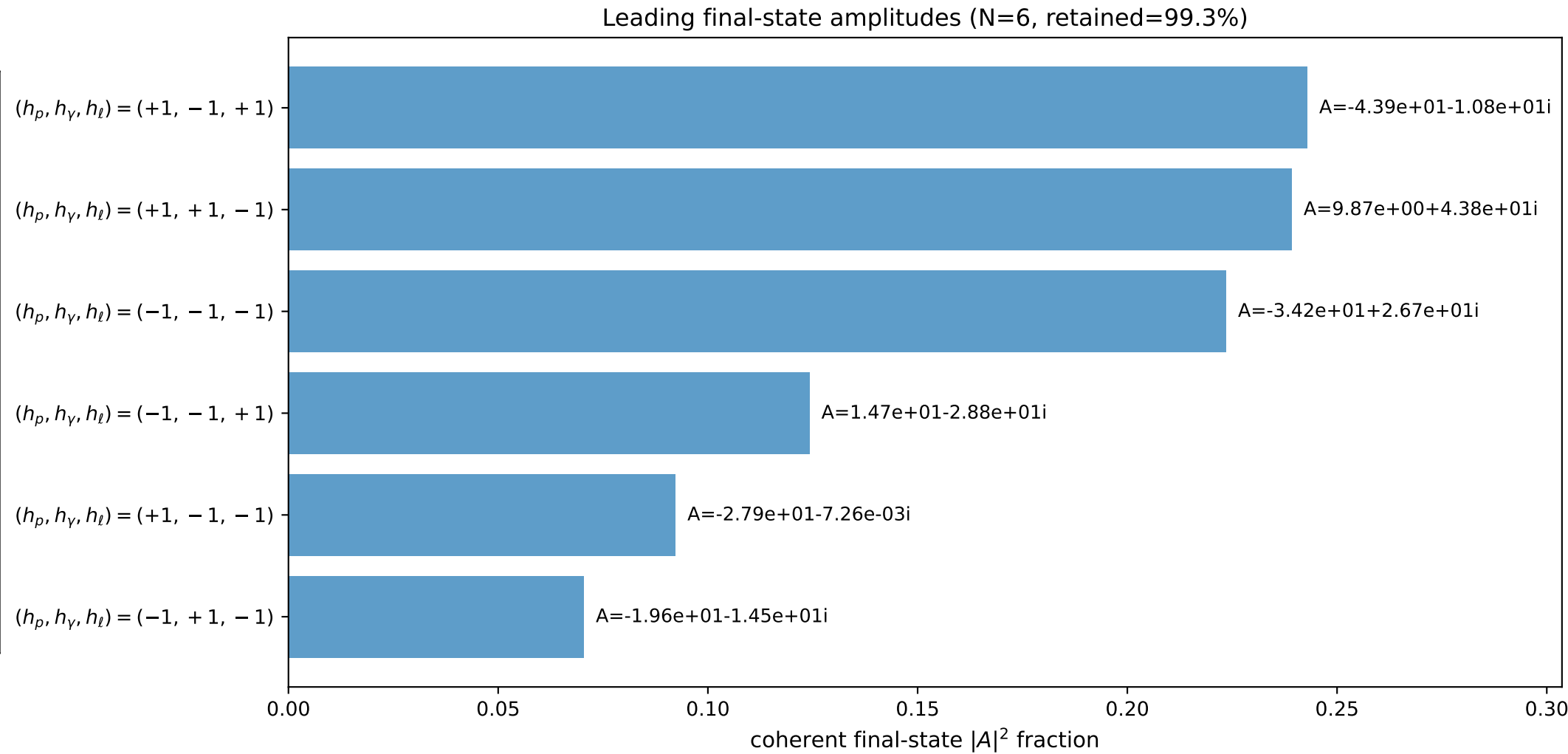
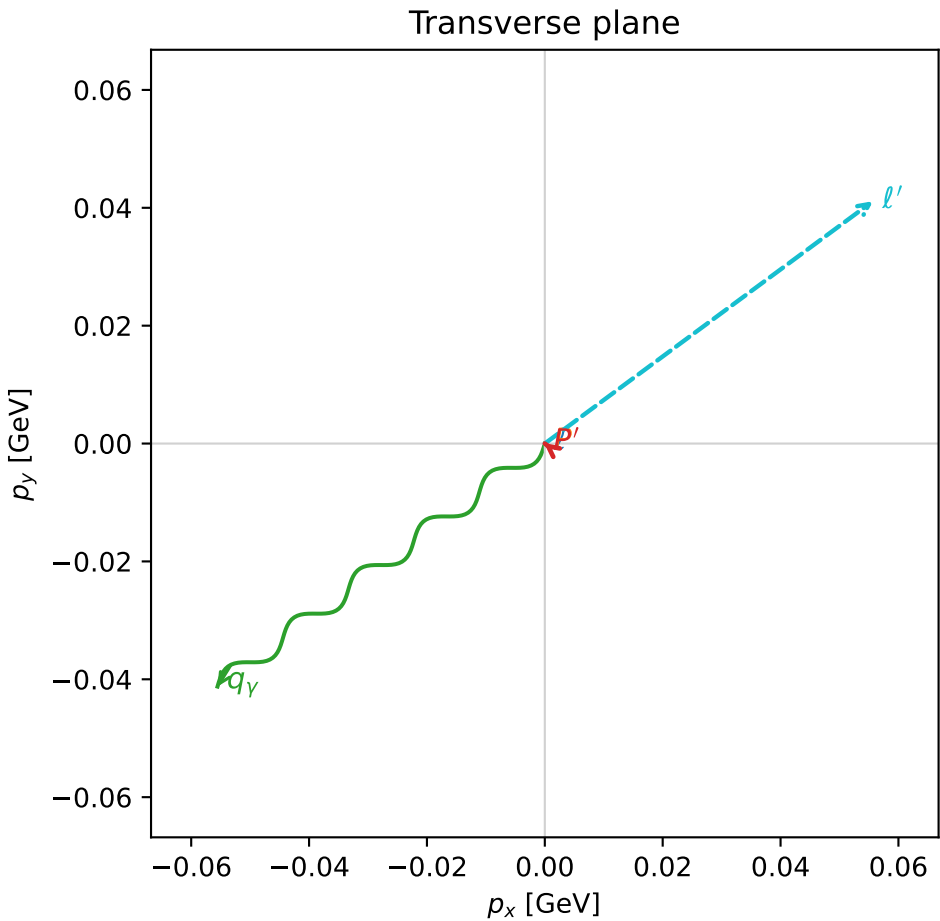
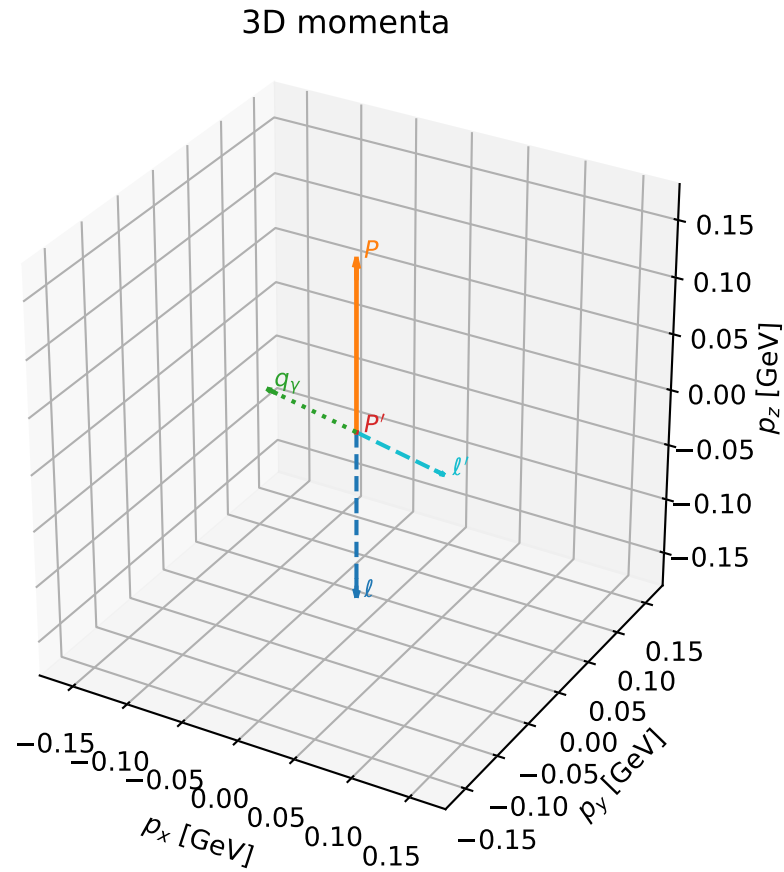


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

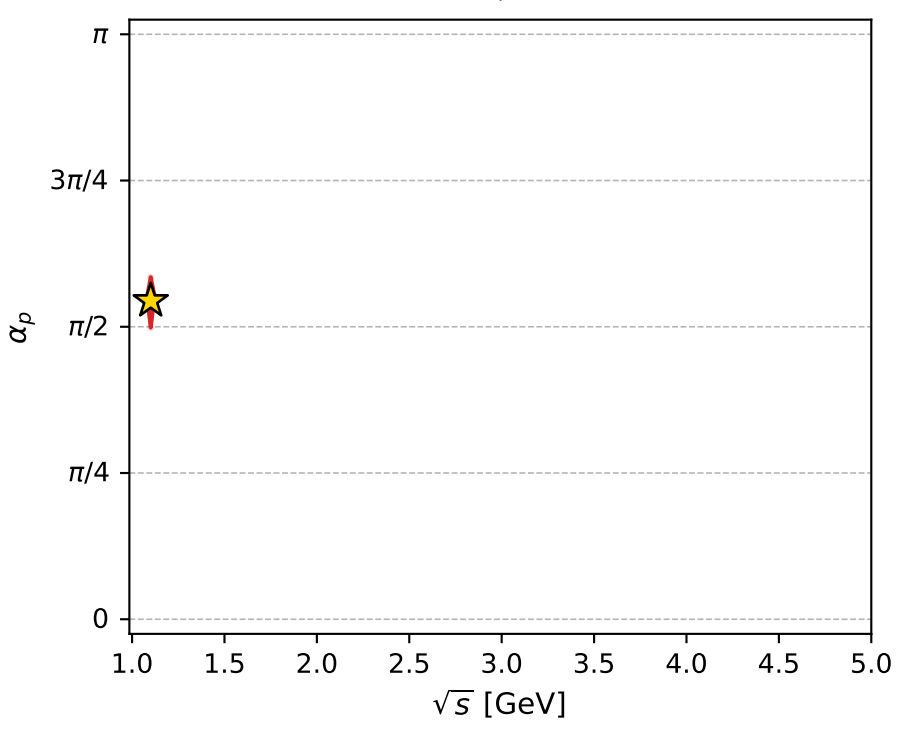
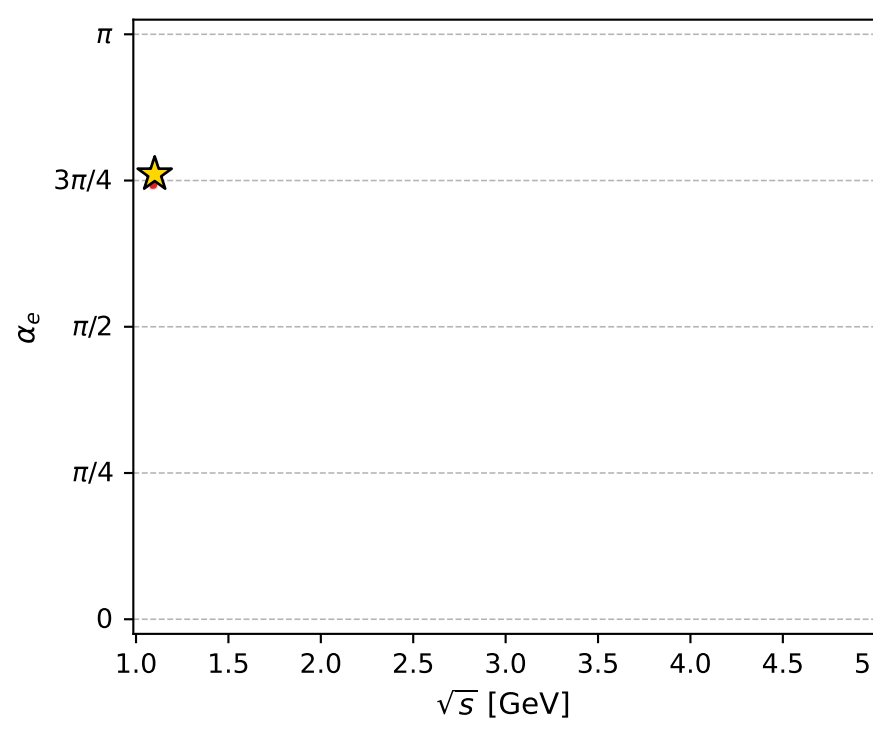
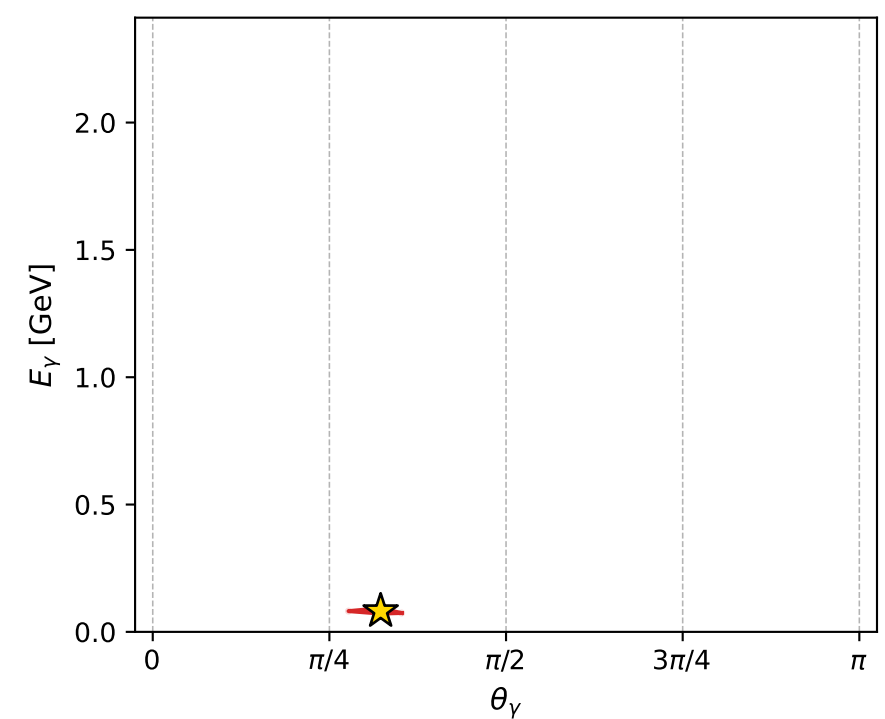
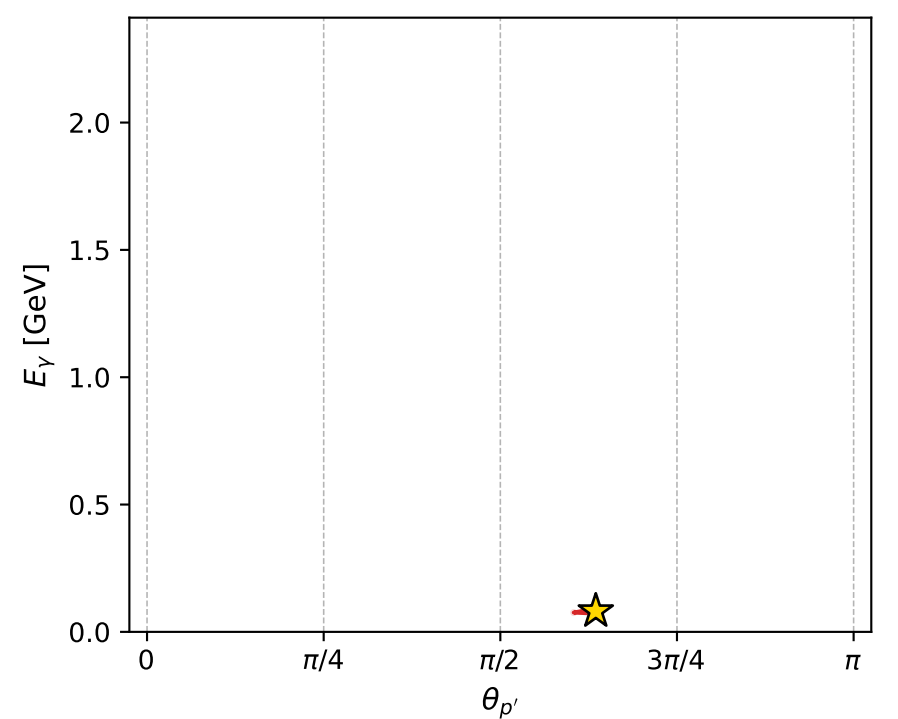
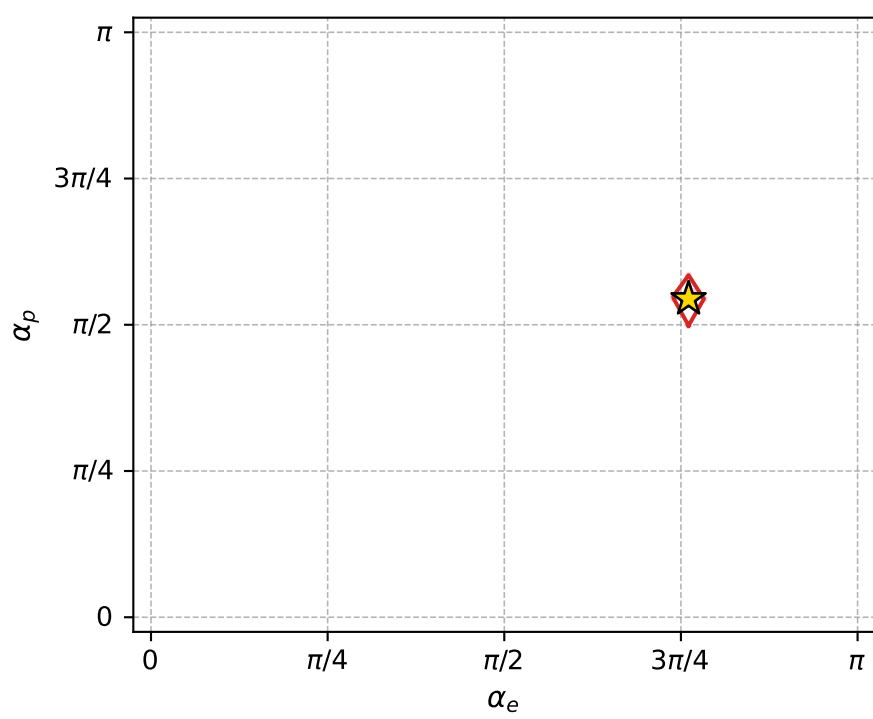
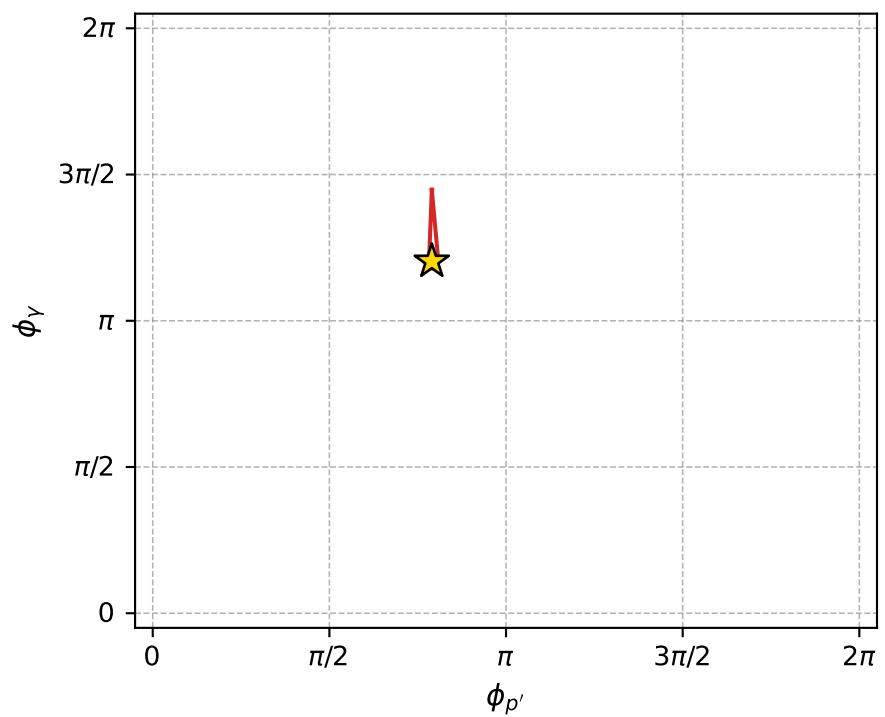
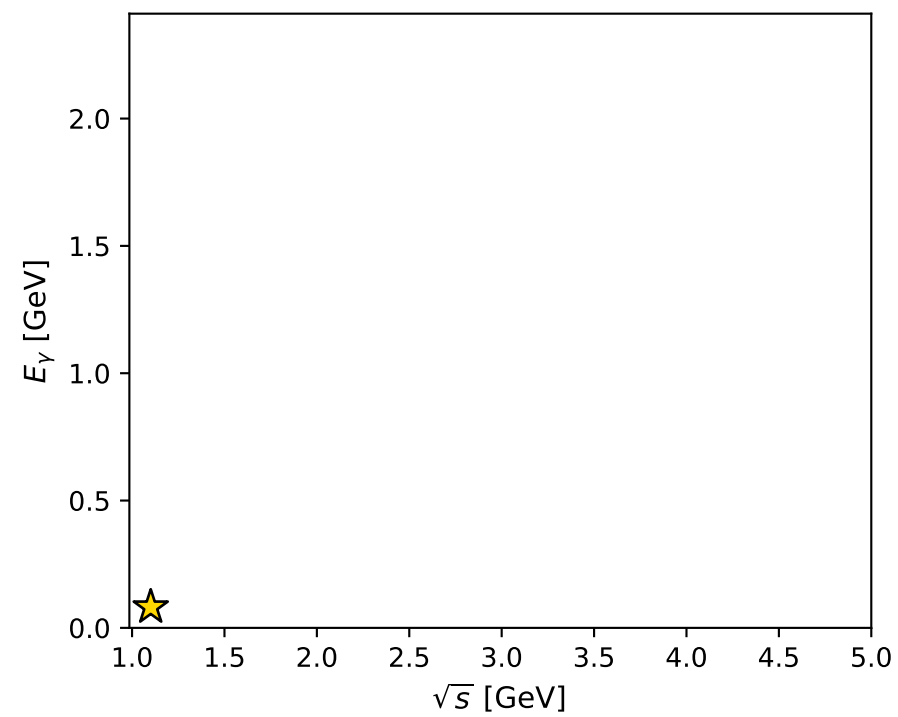
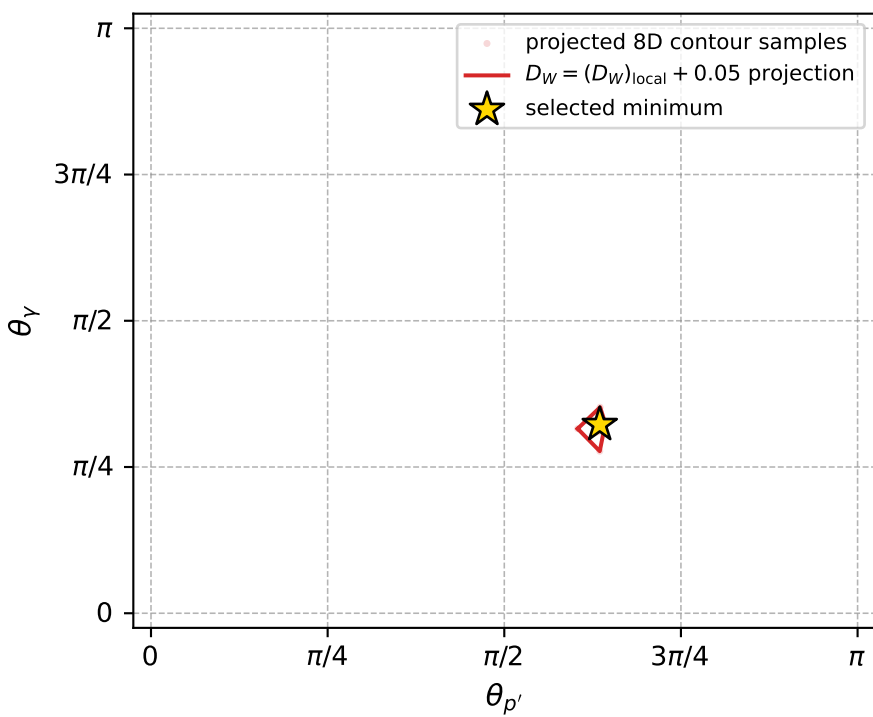
dw_mixing_angles_polarization_06_local_minimum_1030 D_W=0.0203394
 region: polarization_cluster_06_local_minimum_1030
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3901706$ rad
 incoming lepton: $-0.730719|+> +0.682679|->$
 proton mixing angle: $\alpha_p=1.7098586$ rad
 incoming proton: $-0.138614|+> +0.990346|->$

$s=1.21226$, $\sqrt{s}=1.10103$, $|\vec{P}|=0.150957$, $|\vec{P}'|=0.000196886$
 $\theta_{P'}=1.99525$, $\theta_Y=1.01326$, $\phi_{P'}=2.48153$, $\phi_Y=3.78024$
 $E_Y=0.0815139$, $\theta_{\ell'}=2.12717$, $\phi_{\ell'}=0.636148$
 $Q^2=0.0116131$, $x_B=0.0705823$, $t=-0.0226669$, $W^2=1.03276$, $y=0.49496$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1510$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1510$
 P $E= 0.9501$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1510$
 ℓ' $E= 0.0815$ $p_x= 0.0557$ $p_y= 0.0411$ $p_z= -0.0430$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= 0.0001$ $p_z= -0.0001$
 q_Y $E= 0.0815$ $p_x= -0.0555$ $p_y= -0.0412$ $p_z= 0.0431$



electron: polarization cluster P6, local minimum 1030; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.020339448$
 $D_W \text{ contour} = 0.070339448$
 above global minimum = 0.020323015
 8D contour samples = 254

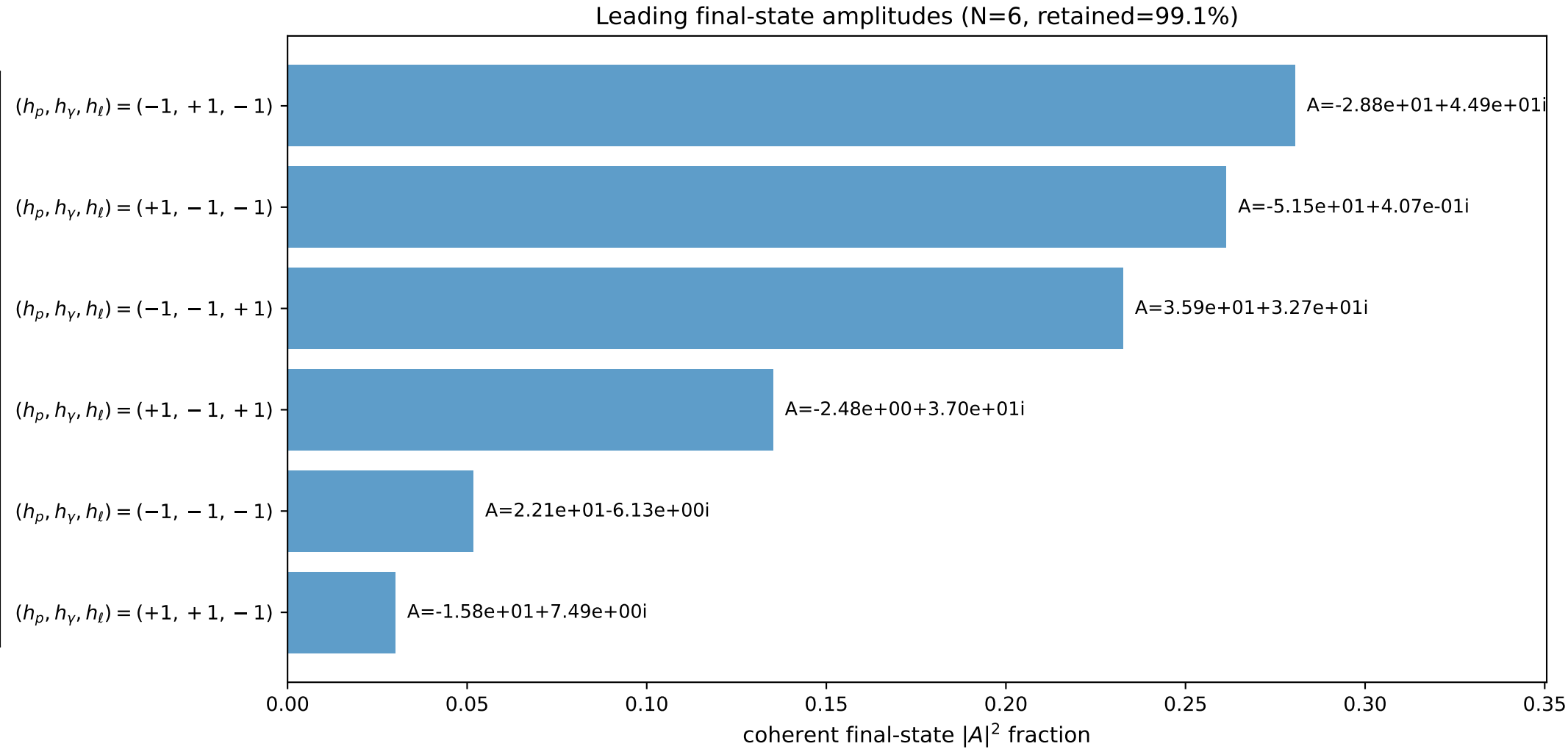
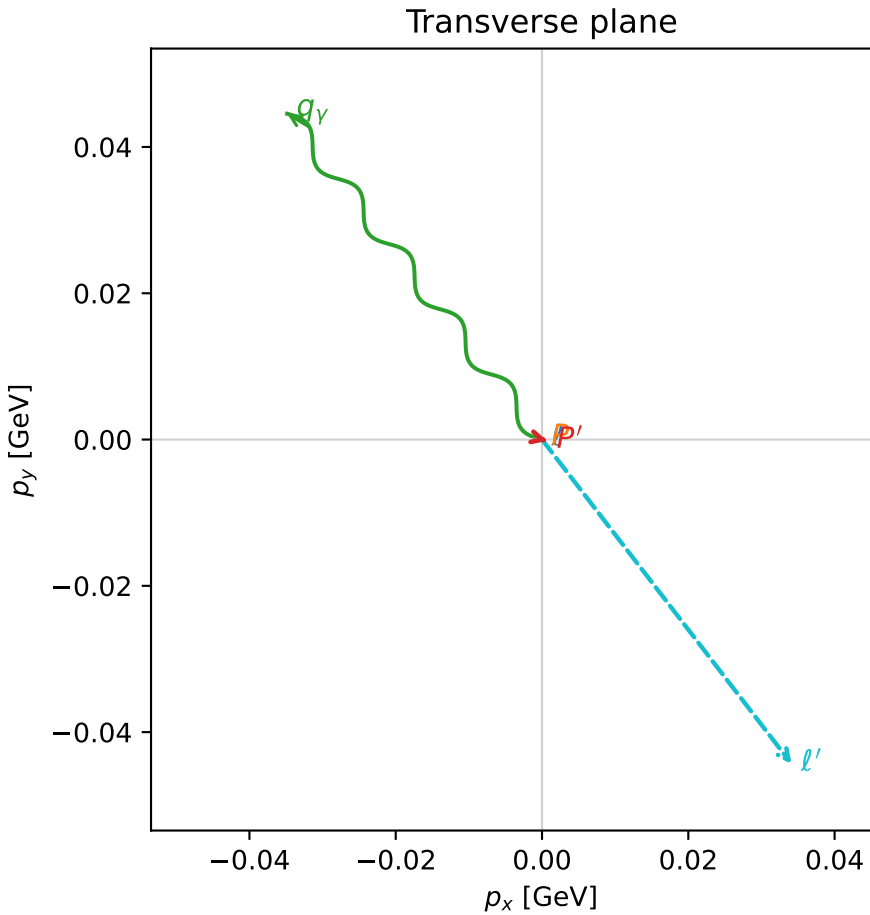
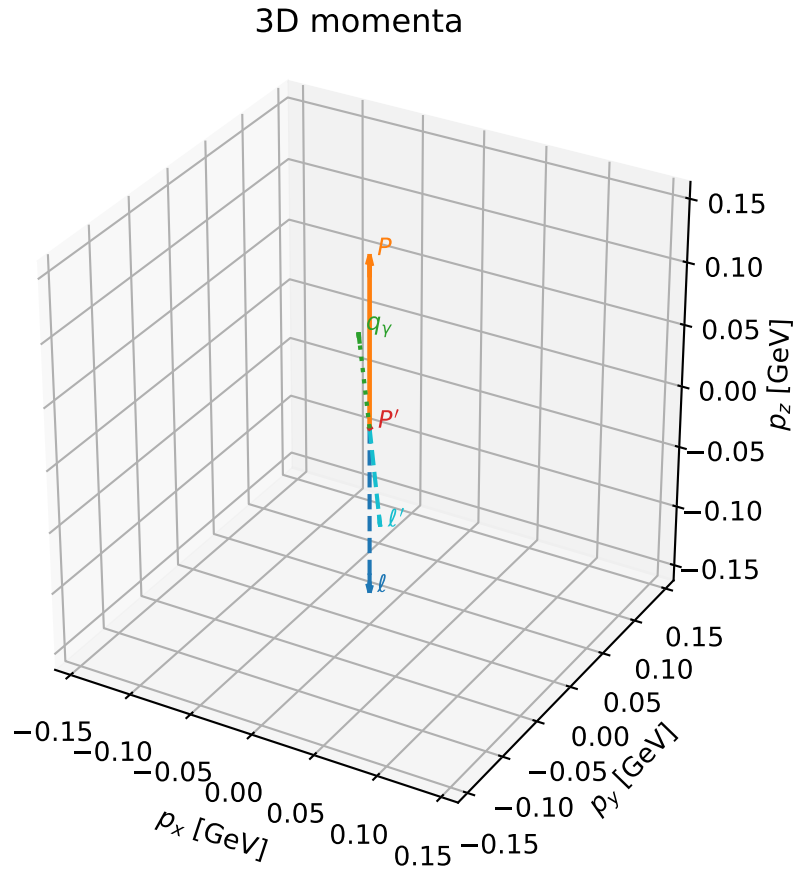
selected phase-space point:
 $\text{sqrt_s} = 1.101028$
 $\text{theta_p_out} = 1.995249$
 $\text{theta_gamma_out} = 1.013264$
 $\text{qOut} = 0.08151387$
 $\text{phi_p_out} = 2.481527$
 $\text{phi_gamma_out} = 3.780237$
 $\text{alpha_e} = 2.390171$
 $\text{alpha_p} = 1.709859$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

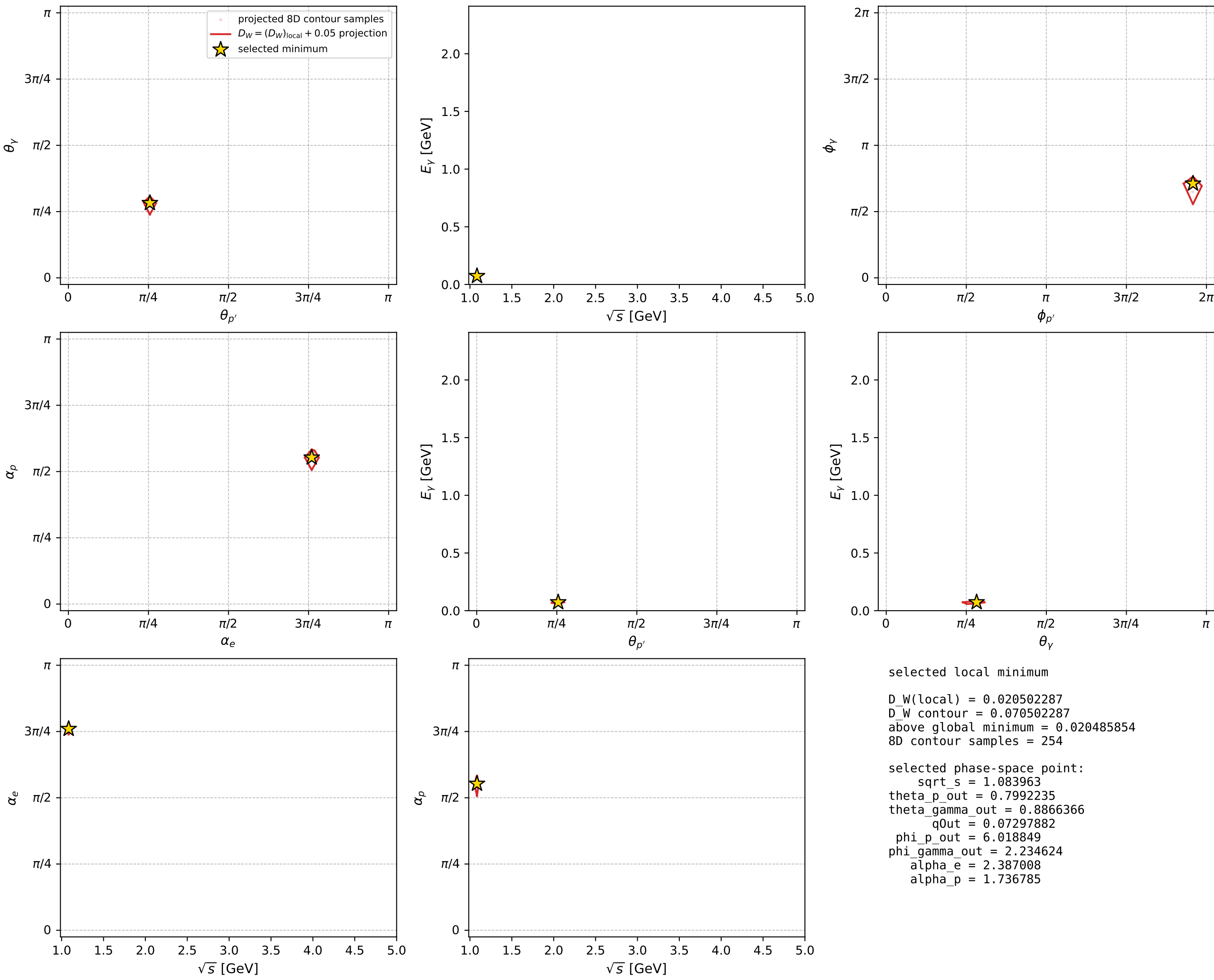
dw_mixing_angles_polarization_06_local_minimum_1032 D_W=0.0205023
 region: polarization_cluster_06_local_minimum_1032
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3870078$ rad
 incoming lepton: $-0.728556|+> +0.684986|->$
 proton mixing angle: $\alpha_p=1.7367854$ rad
 incoming proton: $-0.165228|+> +0.986255|->$

$s=1.17498$, $\sqrt{s}=1.08396$, $|\vec{P}|=0.136135$, $|\vec{P}'|=0.00104328$
 $\theta_{p'}=0.799223$, $\theta_Y=0.886637$, $\phi_{p'}=6.01885$, $\phi_Y=2.23462$
 $E_Y=0.0729788$, $\theta_{\ell'}=2.26785$, $\phi_{\ell'}=5.36821$
 $Q^2=0.00711464$, $x_B=0.0493991$, $t=-0.0182392$, $W^2=1.01675$, $y=0.487997$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1361$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1361$
 P $E= 0.9478$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1361$
 ℓ' $E= 0.0730$ $p_x= 0.0341$ $p_y= -0.0443$ $p_z= -0.0469$
 P' $E= 0.9380$ $p_x= 0.0007$ $p_y= -0.0002$ $p_z= 0.0007$
 q_Y $E= 0.0730$ $p_x= -0.0348$ $p_y= 0.0445$ $p_z= 0.0461$



electron: polarization cluster P6, local minimum 1032; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

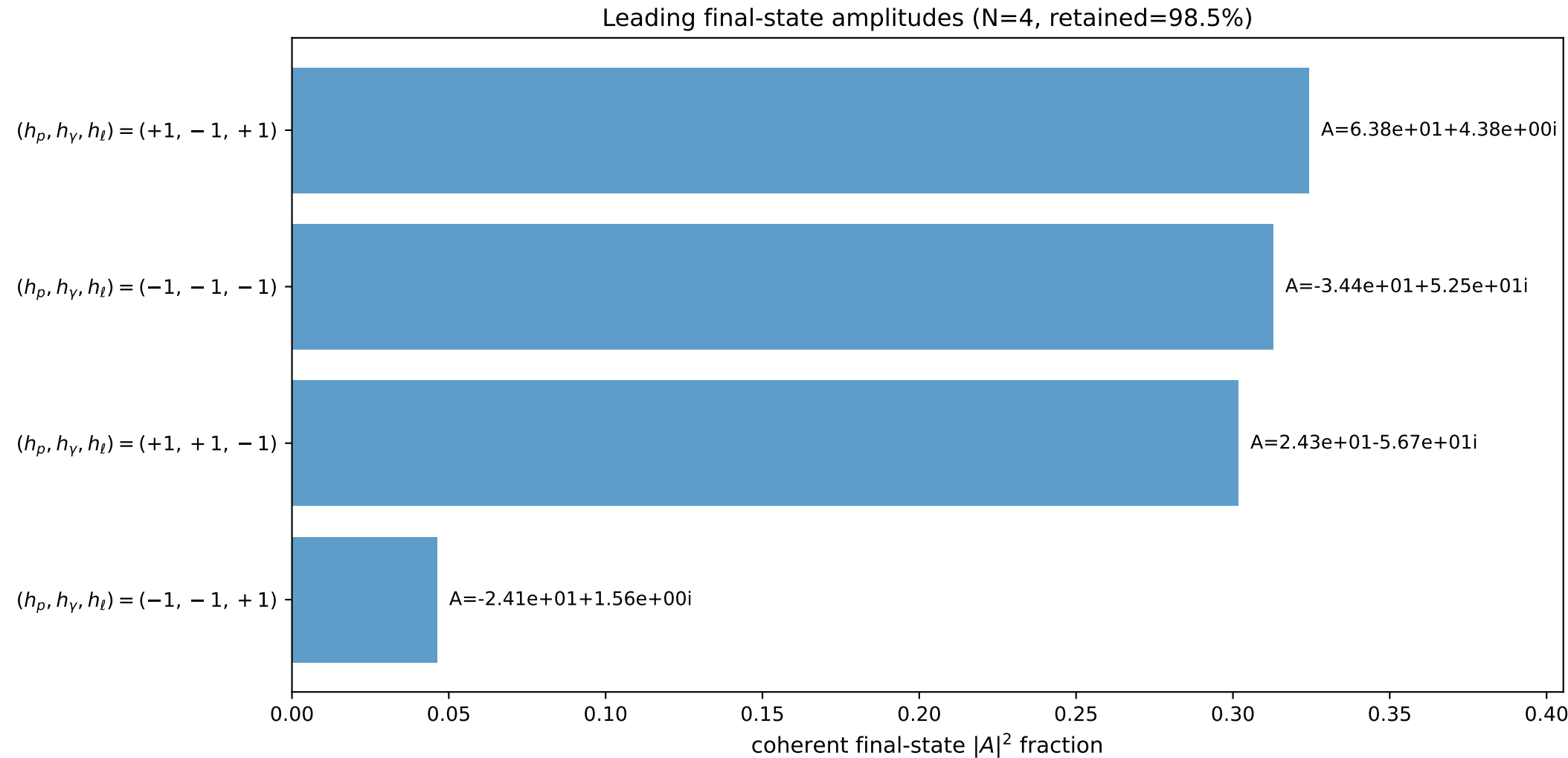
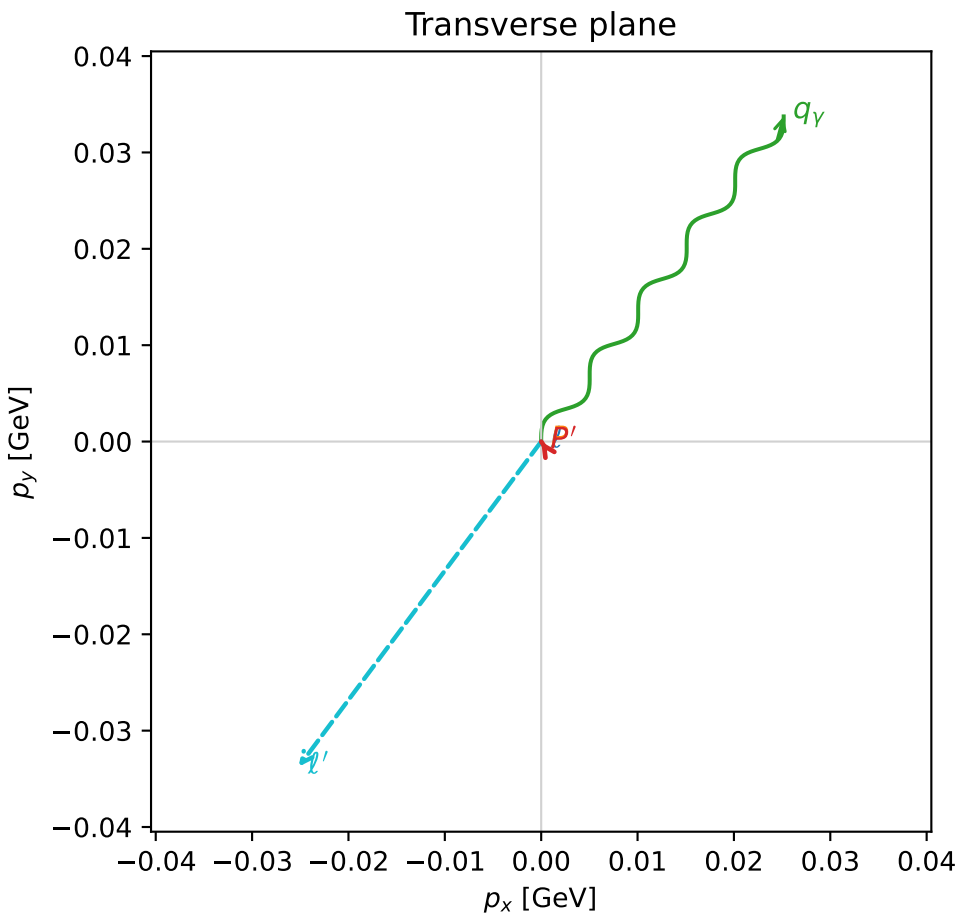
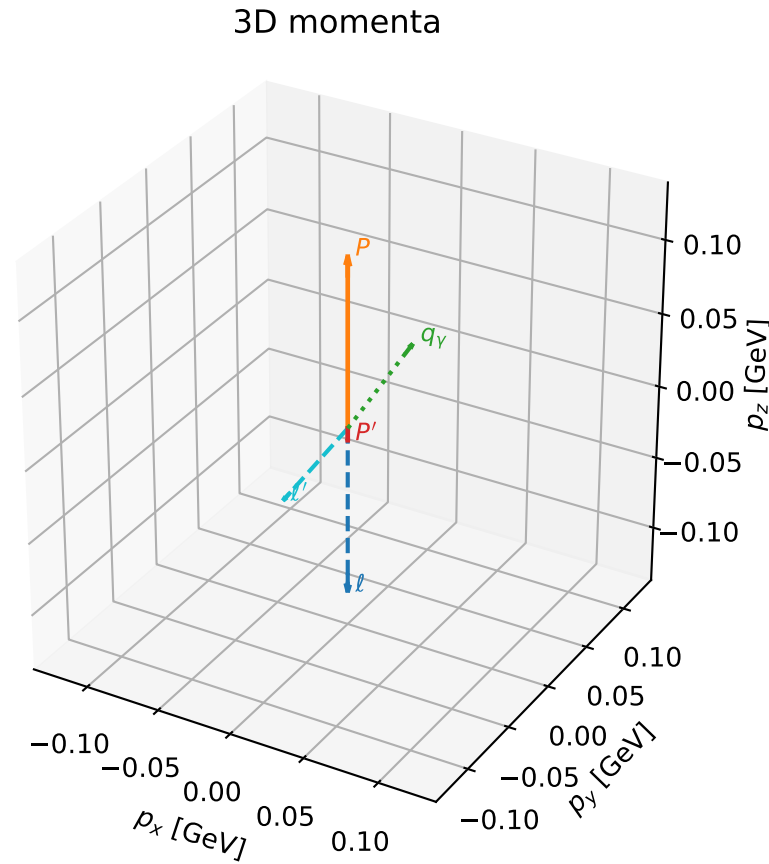


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

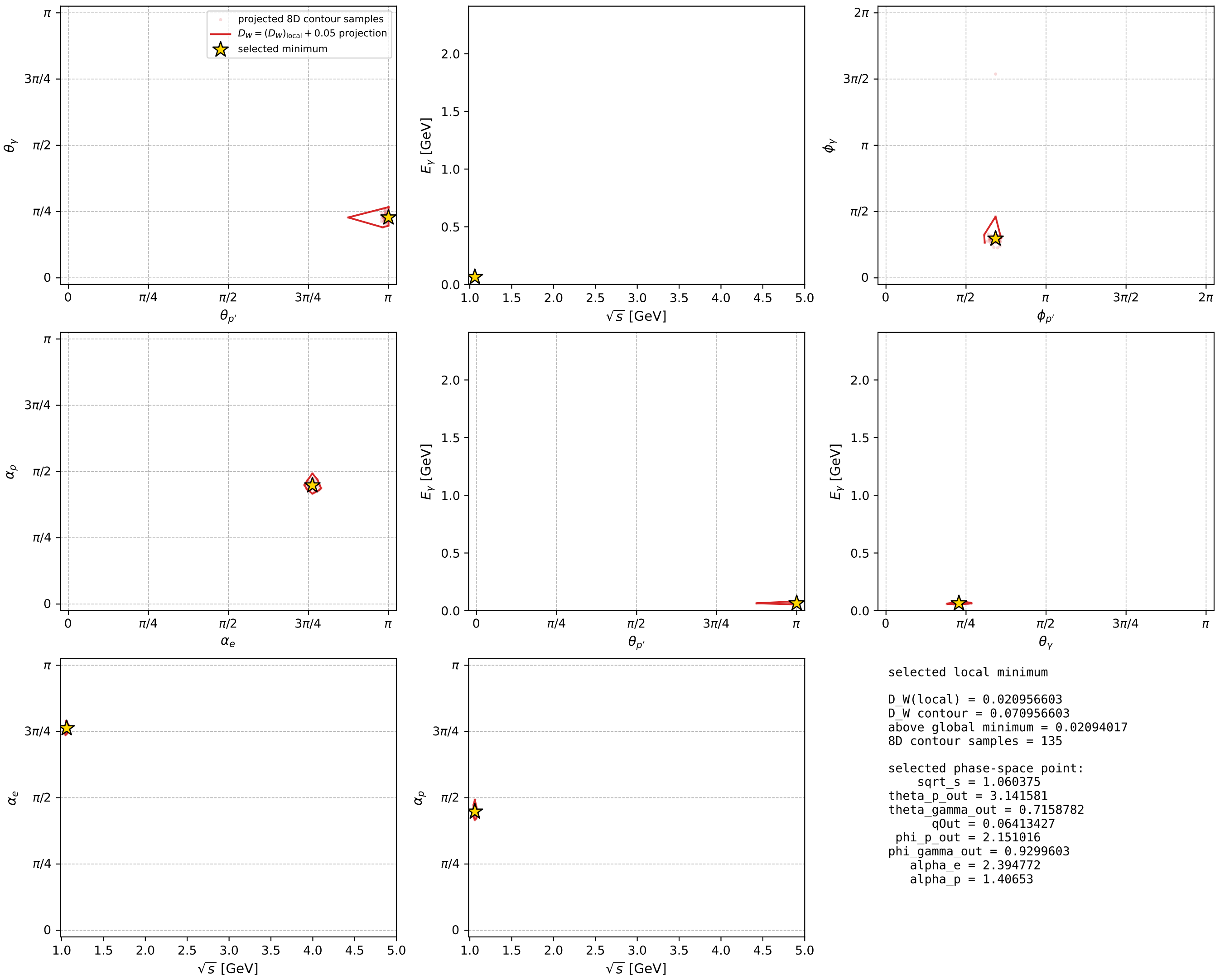
dw_mixing_angles_polarization_06_local_minimum_1039 D_W=0.0209566
 region: polarization_cluster_06_local_minimum_1039
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3947723$ rad
 incoming lepton: $-0.733853|+> +0.679309|->$
 proton mixing angle: $\alpha_p=1.4065296$ rad
 incoming proton: $0.163529|+> +0.986539|->$

$s=1.1244$, $\sqrt{s}=1.06038$, $|\vec{P}|=0.115313$, $|\vec{P}'|=0.00819076$
 $\theta_{p'}=3.14158$, $\theta_Y=0.715878$, $\phi_{p'}=2.15102$, $\phi_Y=0.92996$
 $E_Y=0.0641343$, $\theta_{\ell'}=2.33323$, $\phi_{\ell'}=4.07156$
 $Q^2=0.00415215$, $x_B=0.0331469$, $t=-0.0152038$, $W^2=1.00096$, $y=0.512225$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1153$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1153$
 P $E= 0.9451$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1153$
 ℓ' $E= 0.0582$ $p_x= -0.0252$ $p_y= -0.0337$ $p_z= -0.0402$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0082$
 q_Y $E= 0.0641$ $p_x= 0.0252$ $p_y= 0.0337$ $p_z= 0.0484$



electron: polarization cluster P6, local minimum 1039; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

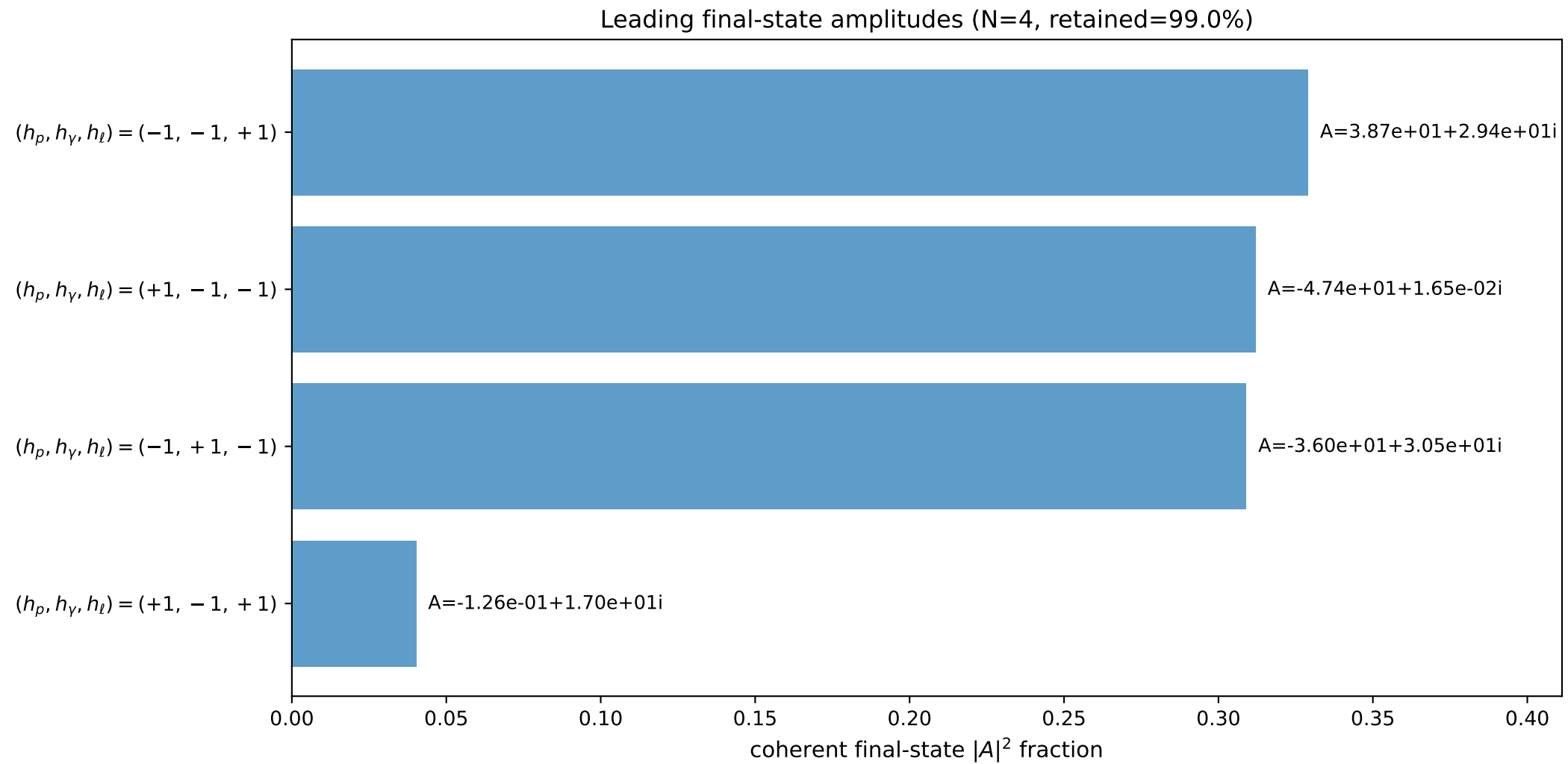
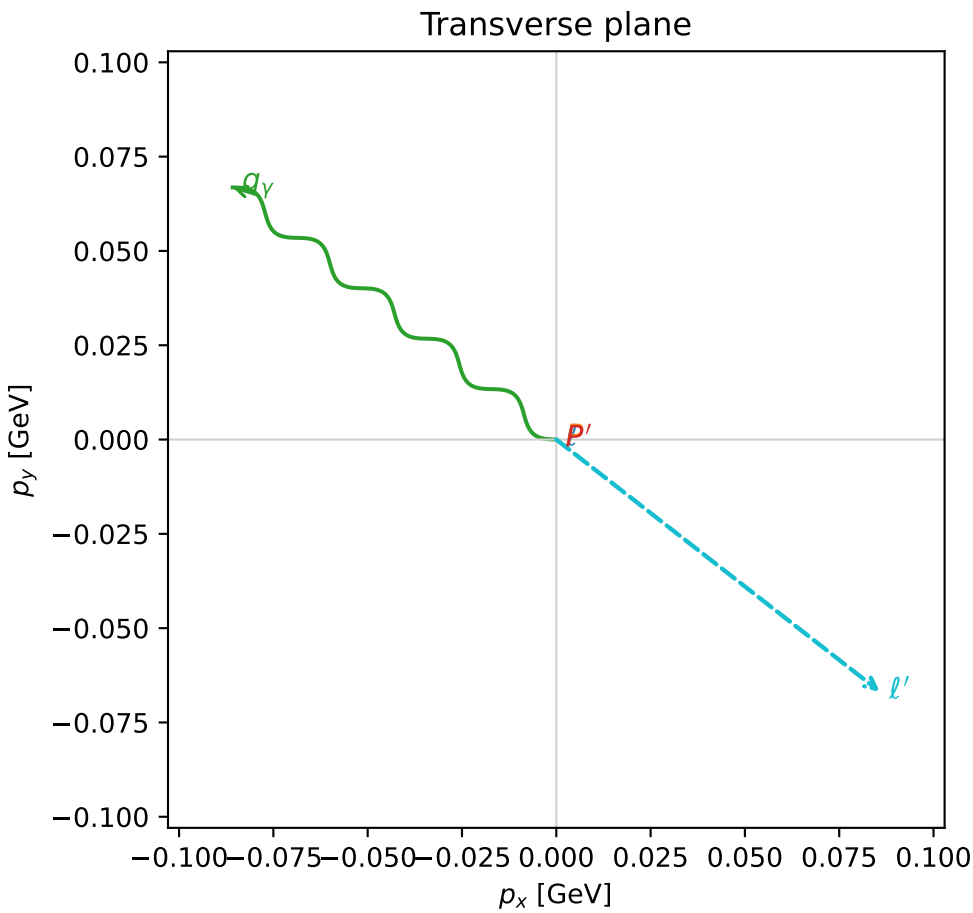
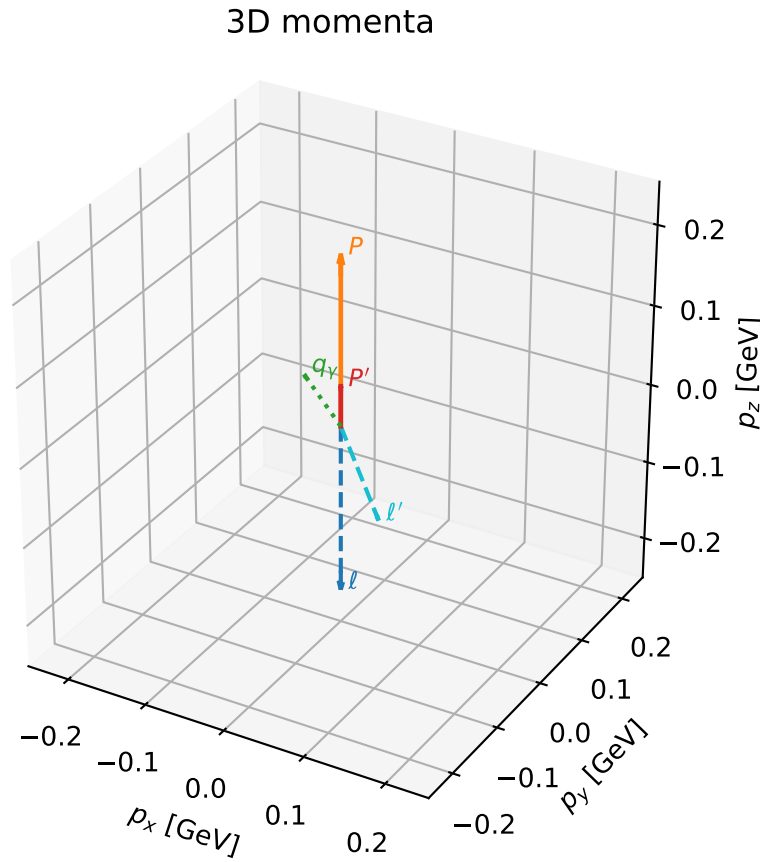


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

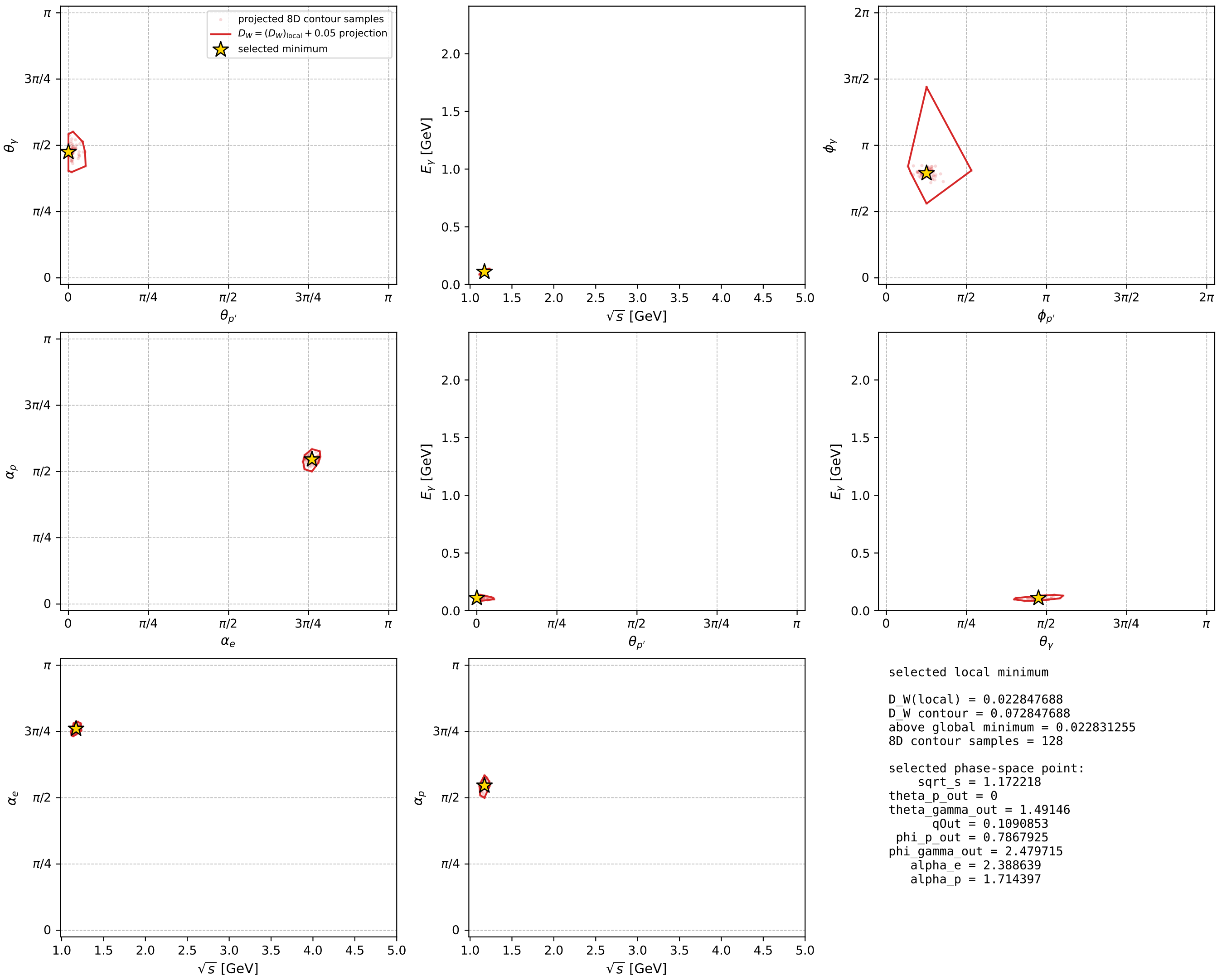
dw_mixing_angles_polarization_06_local_minimum_1051 D_W=0.0228477
region: polarization_cluster_06_local_minimum_1051
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3886388$ rad
incoming lepton: $-0.729672|+> +0.683797|->$
proton mixing angle: $\alpha_p=1.7143968$ rad
incoming proton: $-0.143107|+> +0.989707|->$

$s=1.37409$, $\sqrt{s}=1.17222$, $|\vec{P}|=0.210818$, $|\vec{P}'|=0.0504754$
 $\theta_{P'}=0$, $\theta_Y=1.49146$, $\phi_{P'}=0.786793$, $\phi_Y=2.47971$
 $E_Y=0.109085$, $\theta_{\ell'}=2.06877$, $\phi_{\ell'}=5.62131$
 $Q^2=0.0272605$, $x_B=0.117843$, $t=-0.025224$, $W^2=1.08391$, $y=0.468037$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2108$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2108$
 P $E= 0.9614$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2108$
 ℓ' $E= 0.1238$ $p_x= 0.0858$ $p_y= -0.0668$ $p_z= -0.0591$
 P' $E= 0.9394$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0505$
 q_Y $E= 0.1091$ $p_x= -0.0858$ $p_y= 0.0668$ $p_z= 0.0086$



electron: polarization cluster P6, local minimum 1051; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

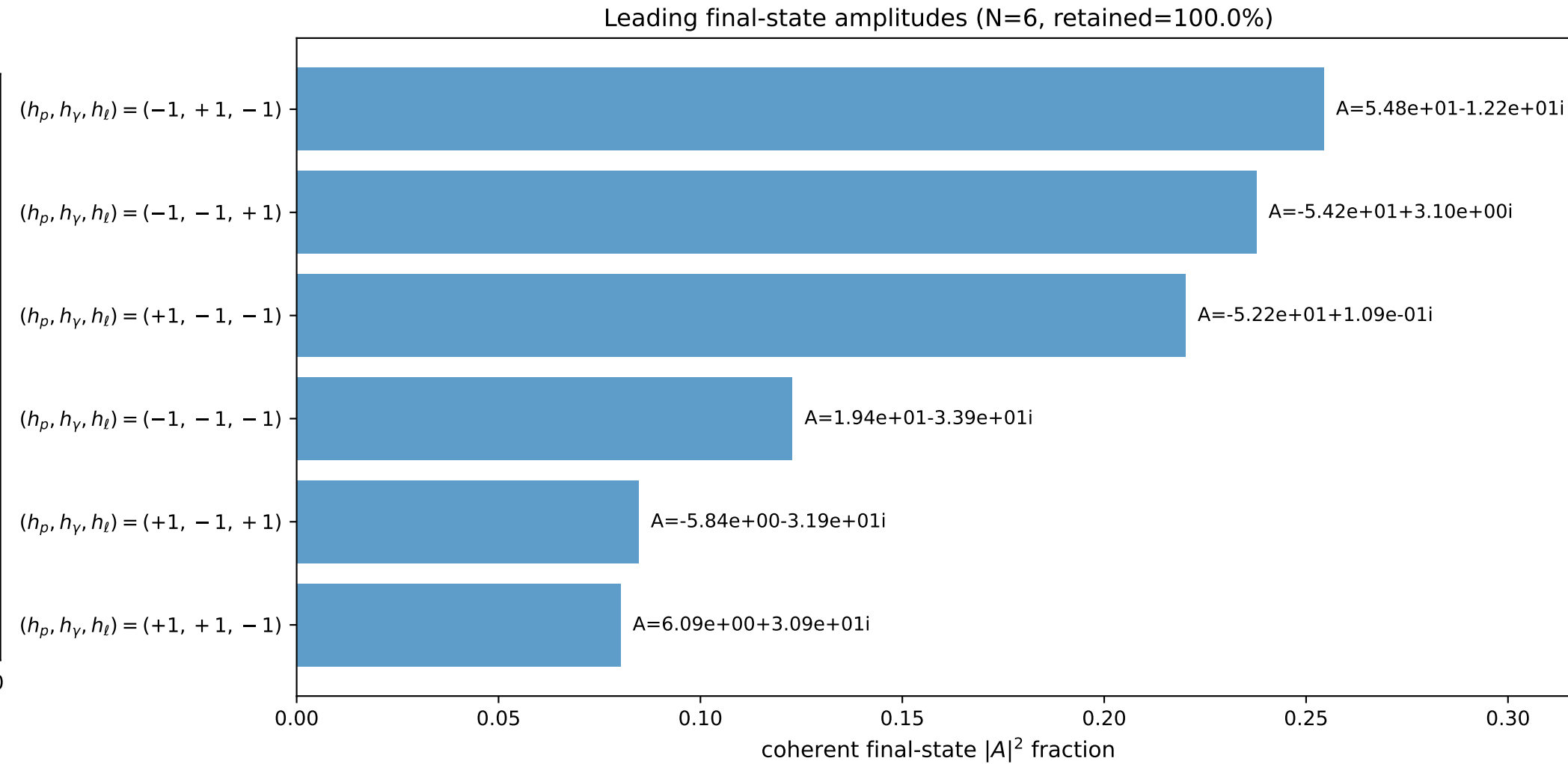
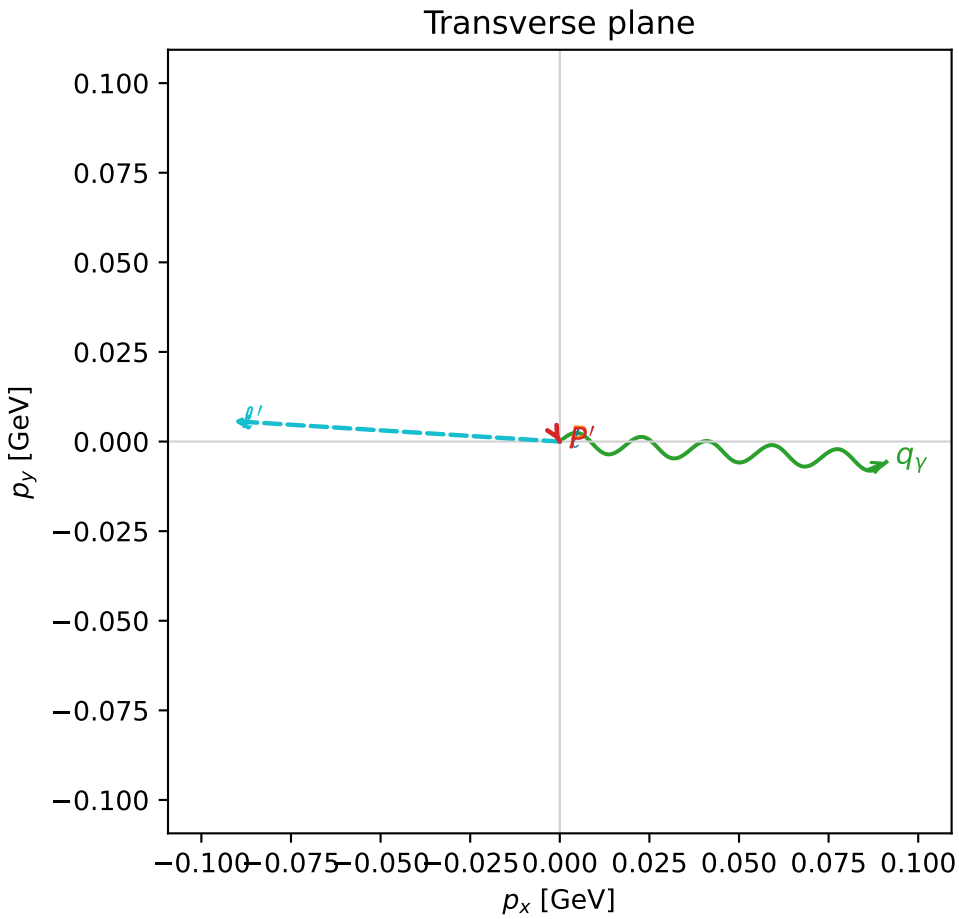
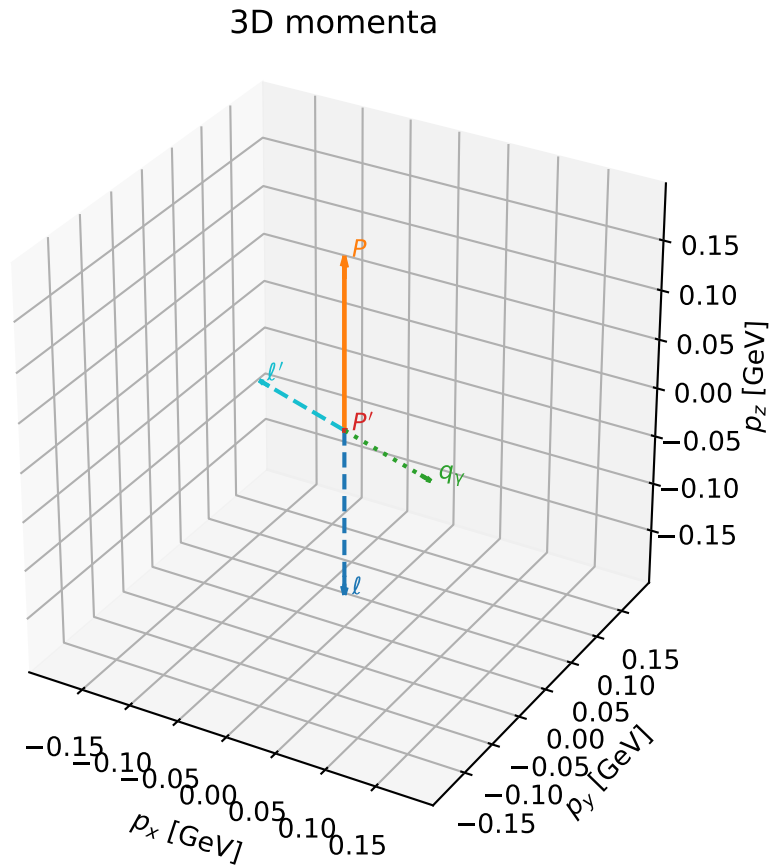


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

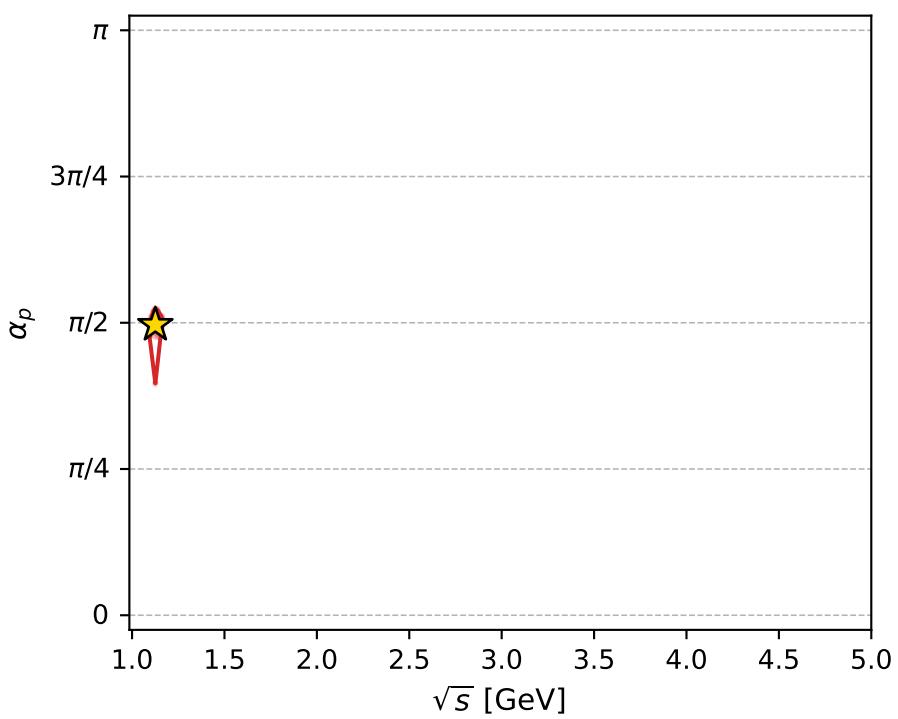
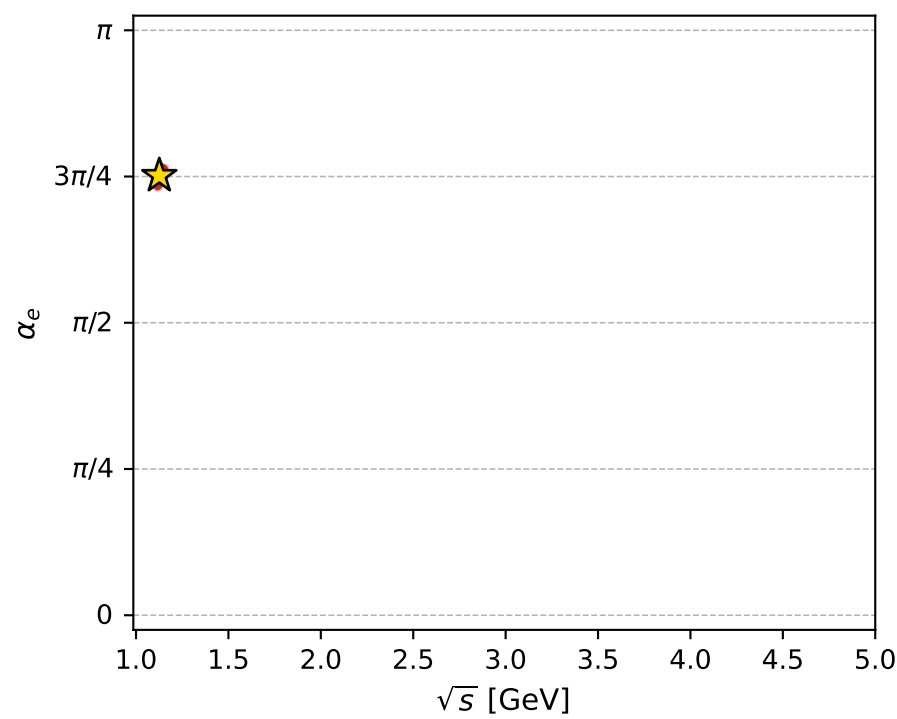
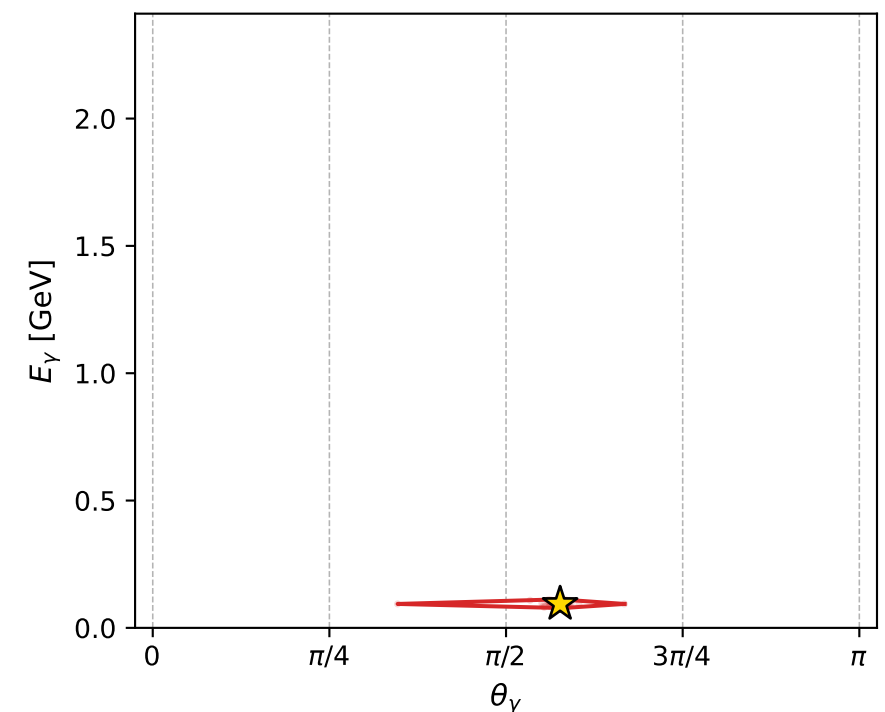
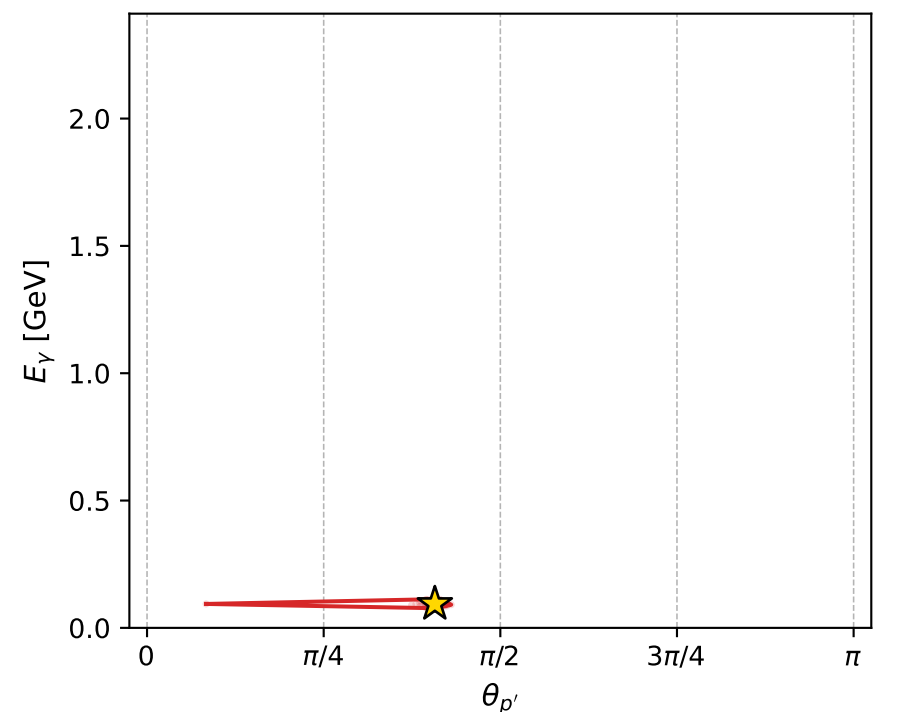
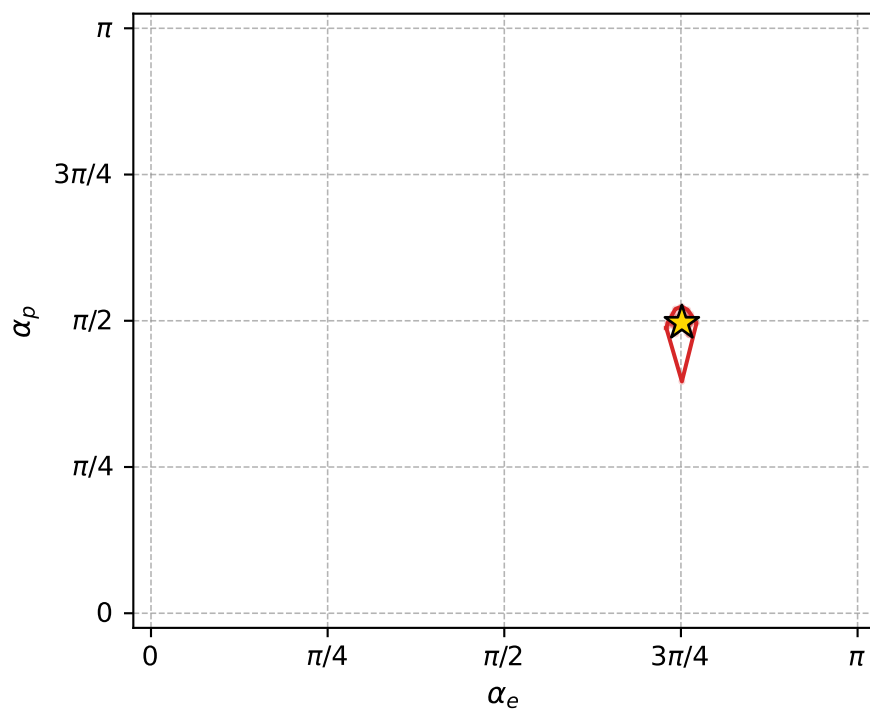
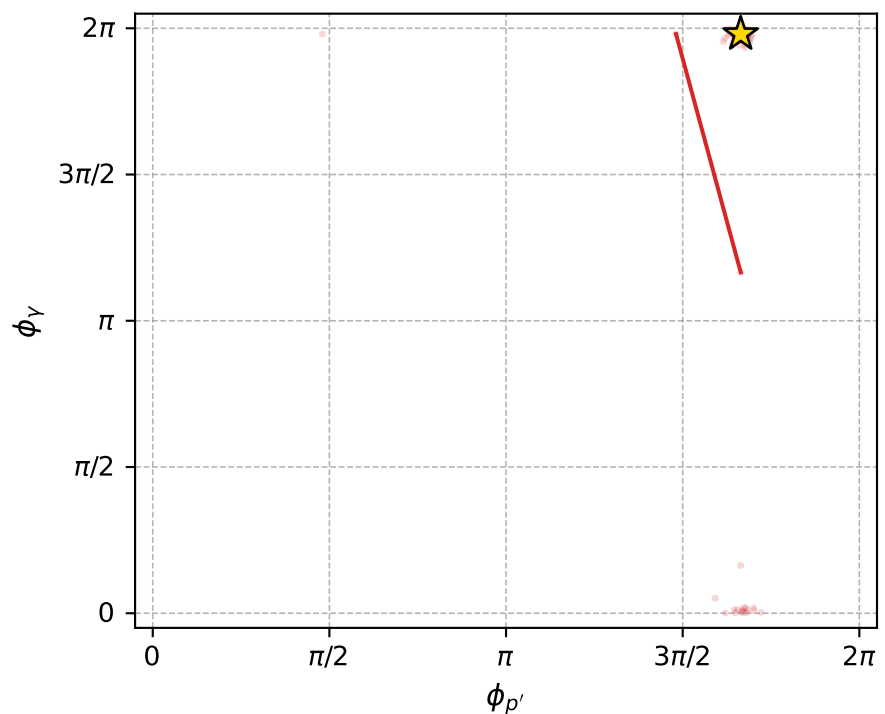
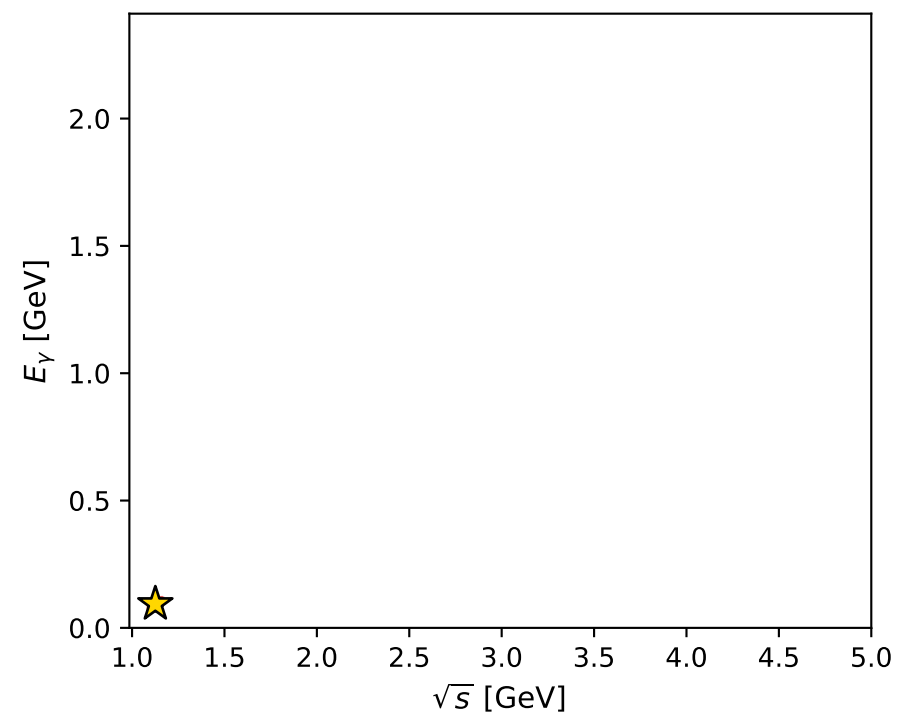
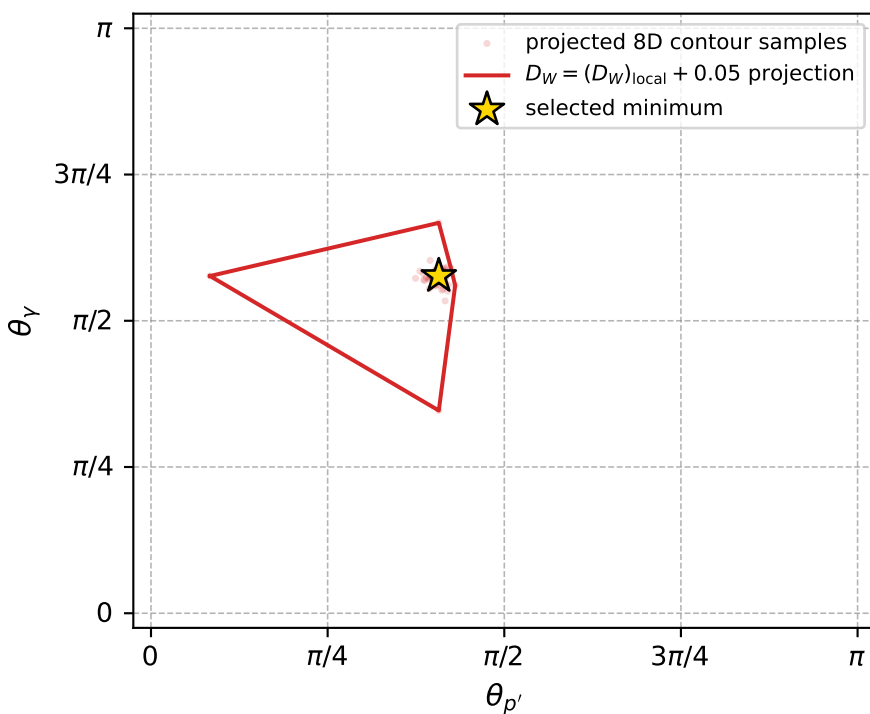
dw_mixing_angles_polarization_06_local_minimum_1069 D_W=0.0245115
 region: polarization_cluster_06_local_minimum_1069
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3604274$ rad
 incoming lepton: $-0.710094|+> +0.704107|->$
 proton mixing angle: $\alpha_p=1.559823$ rad
 incoming proton: $0.0109731|+> +0.99994|->$

$s=1.26776$, $\sqrt{s}=1.12595$, $|\vec{P}|=0.172261$, $|\vec{P}'|=4.75156e-09$
 $\theta_{p'}=1.27963$, $\theta_Y=1.81115$, $\phi_{p'}=5.22847$, $\phi_Y=6.22075$
 $E_Y=0.0939732$, $\theta_{\ell'}=1.33044$, $\phi_{\ell'}=3.07916$
 $Q^2=0.0400829$, $x_B=0.185246$, $t=-0.0294277$, $W^2=1.05614$, $y=0.557795$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1723$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1723$
 P $E= 0.9537$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1723$
 ℓ' $E= 0.0940$ $p_x= -0.0911$ $p_y= 0.0057$ $p_z= 0.0224$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0000$
 q_Y $E= 0.0940$ $p_x= 0.0911$ $p_y= -0.0057$ $p_z= -0.0224$



electron: polarization cluster P6, local minimum 1069; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.024511544$
 $D_W \text{ contour} = 0.074511544$
 above global minimum = 0.024495111
 8D contour samples = 255

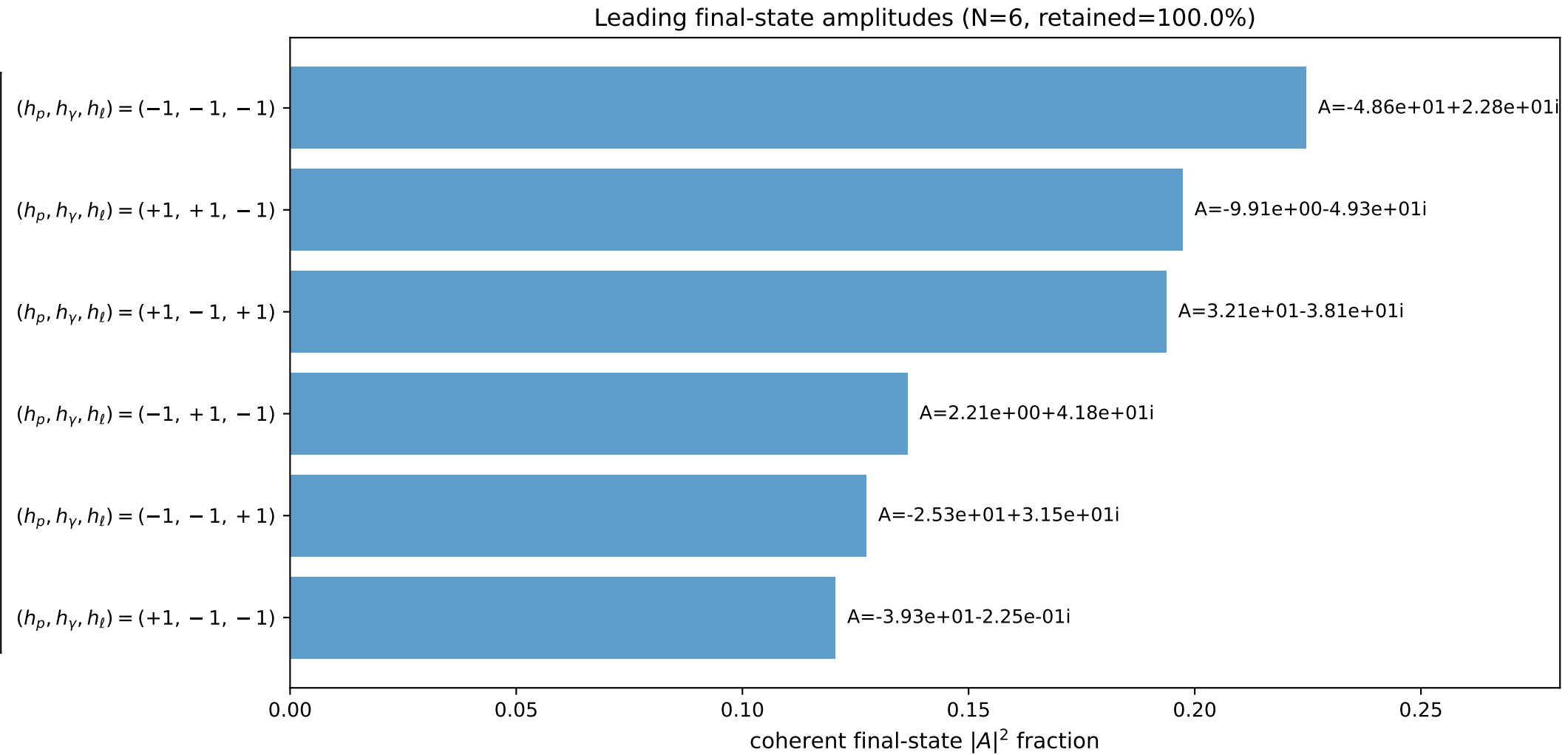
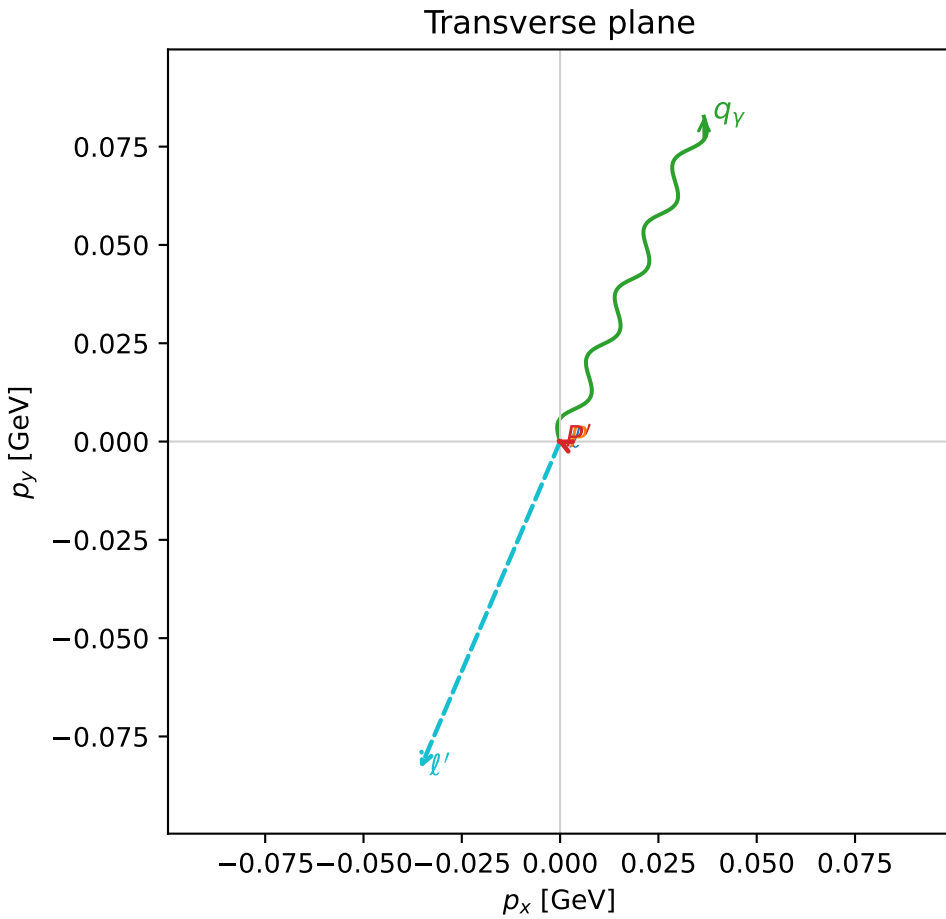
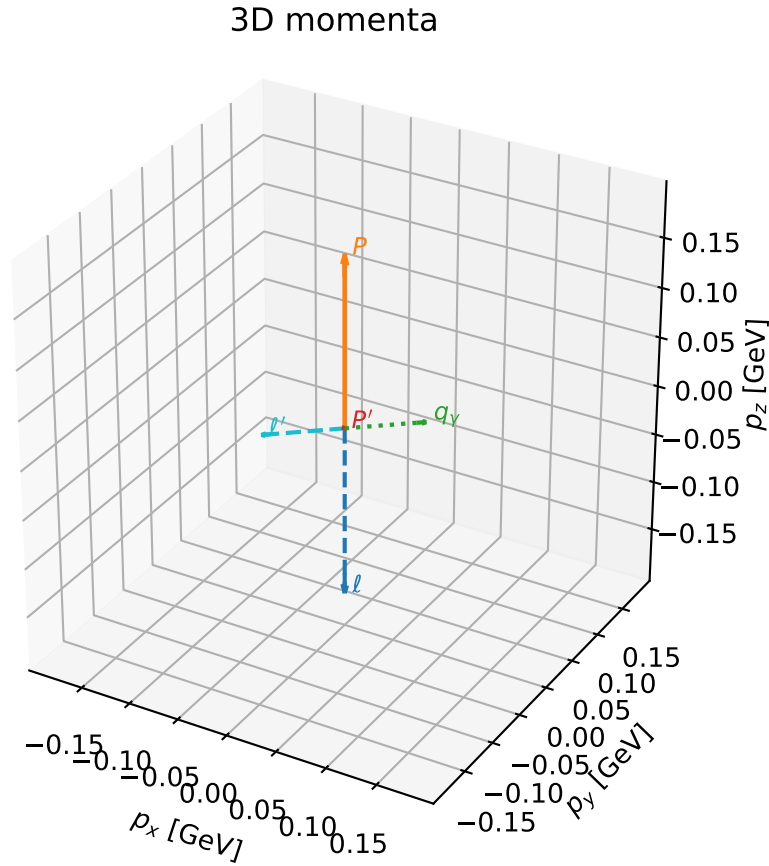
selected phase-space point:
 $\text{sqrt_s} = 1.125948$
 $\text{theta_p_out} = 1.279628$
 $\text{theta_gamma_out} = 1.811154$
 $\text{q0out} = 0.09397322$
 $\text{phi_p_out} = 5.22847$
 $\text{phi_gamma_out} = 6.22075$
 $\text{alpha_e} = 2.360427$
 $\text{alpha_p} = 1.559823$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

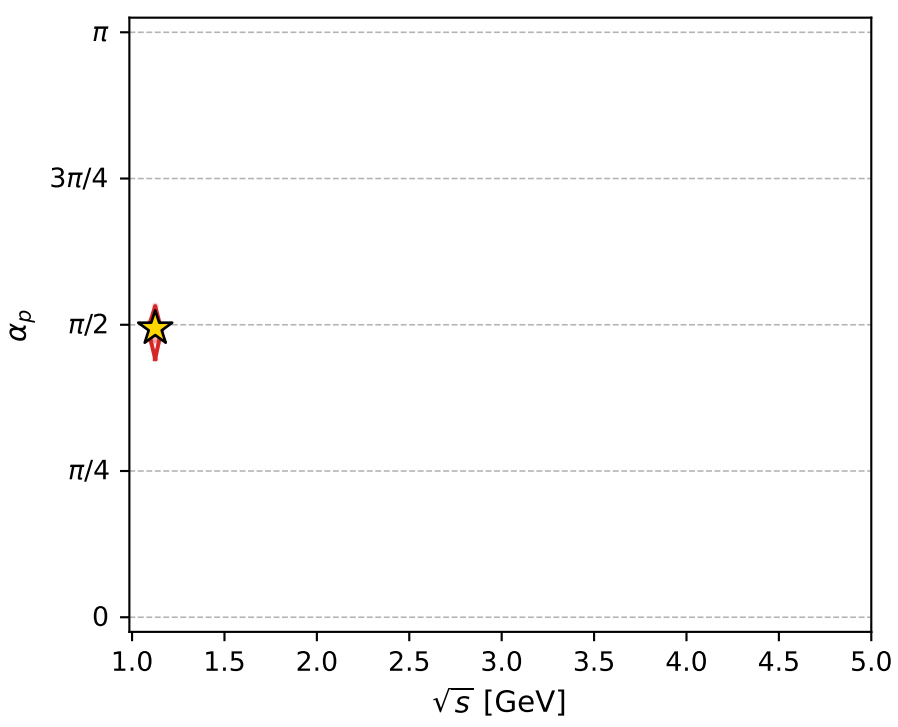
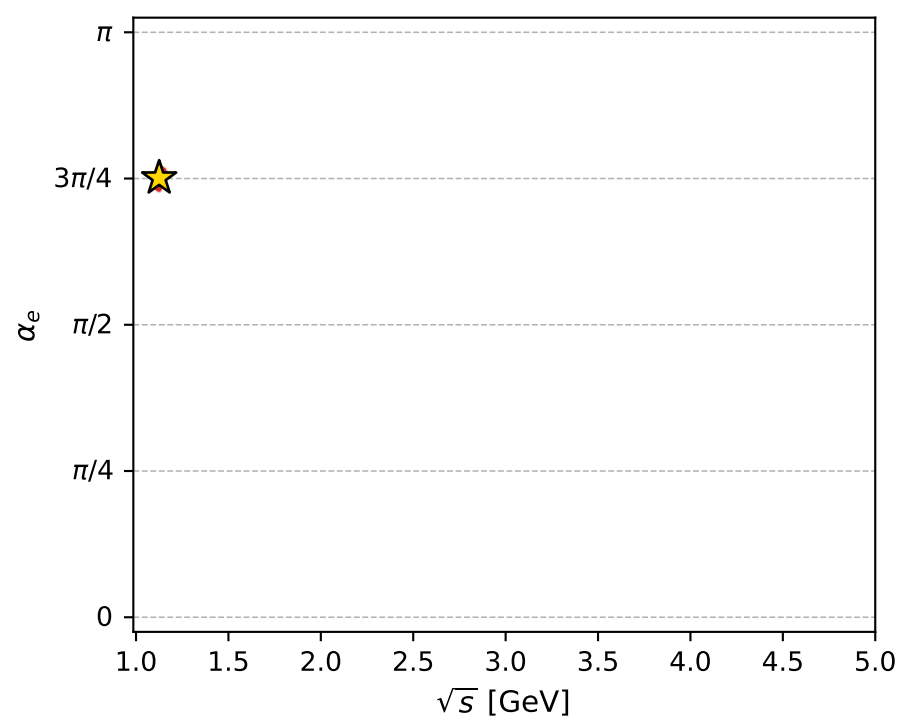
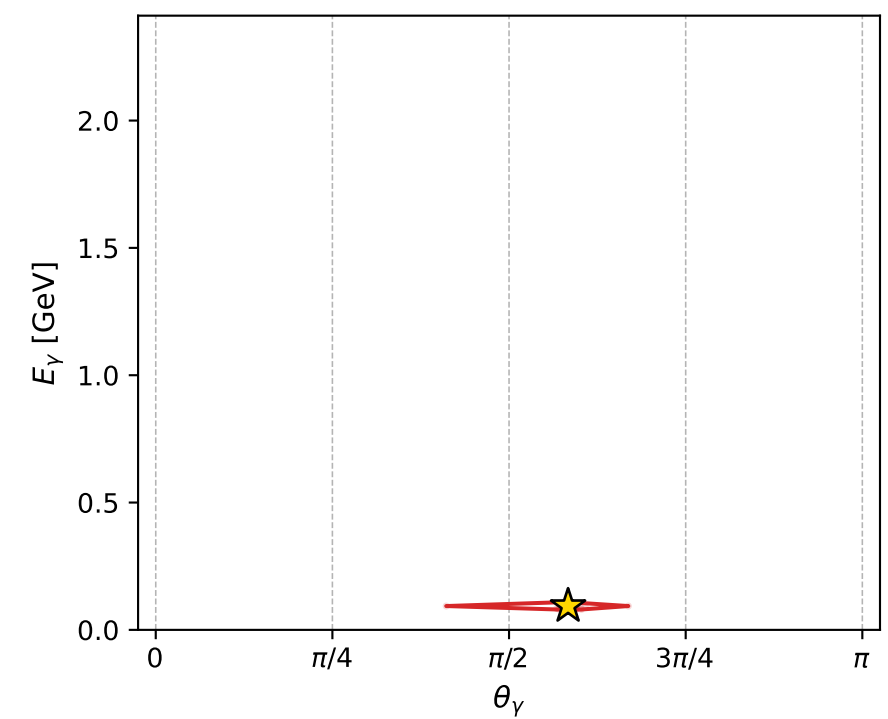
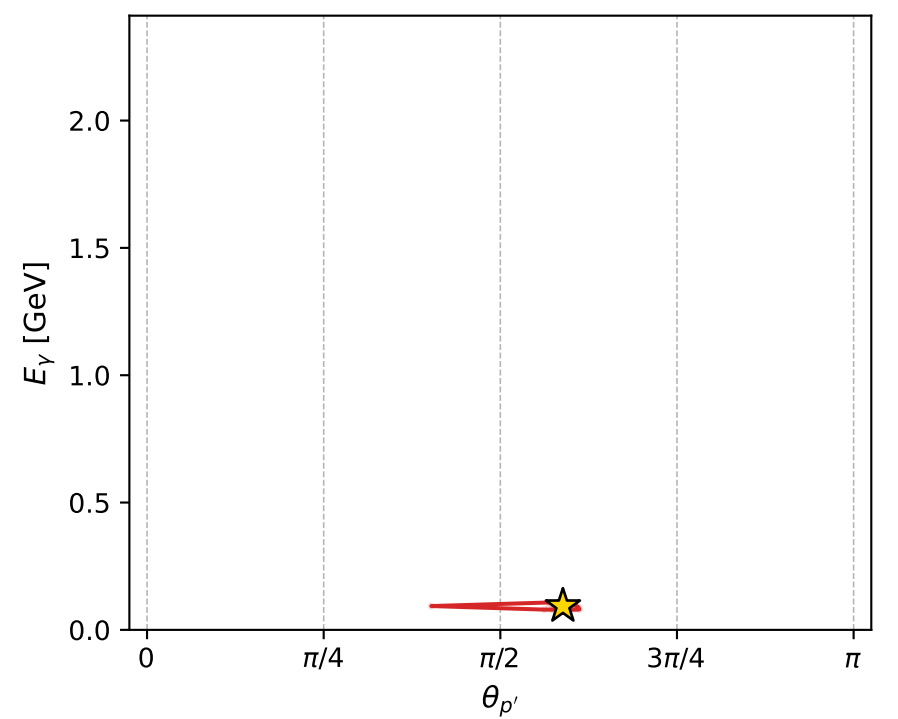
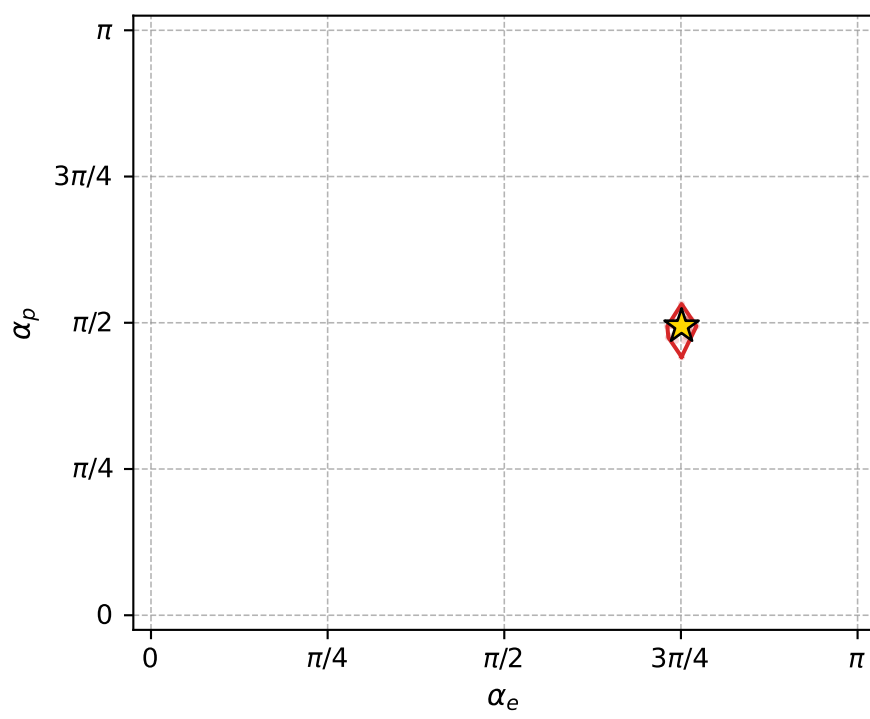
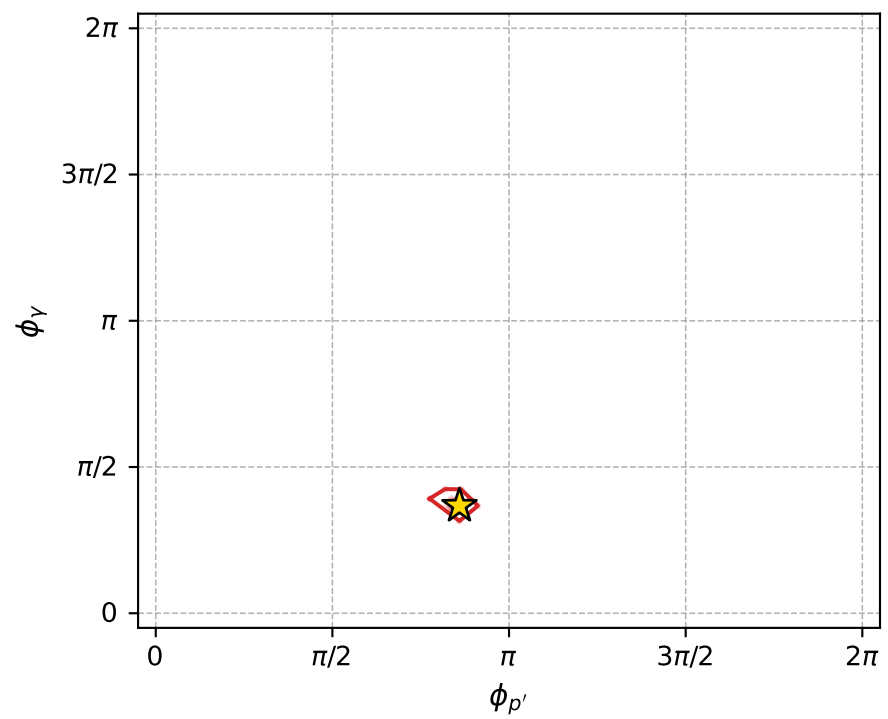
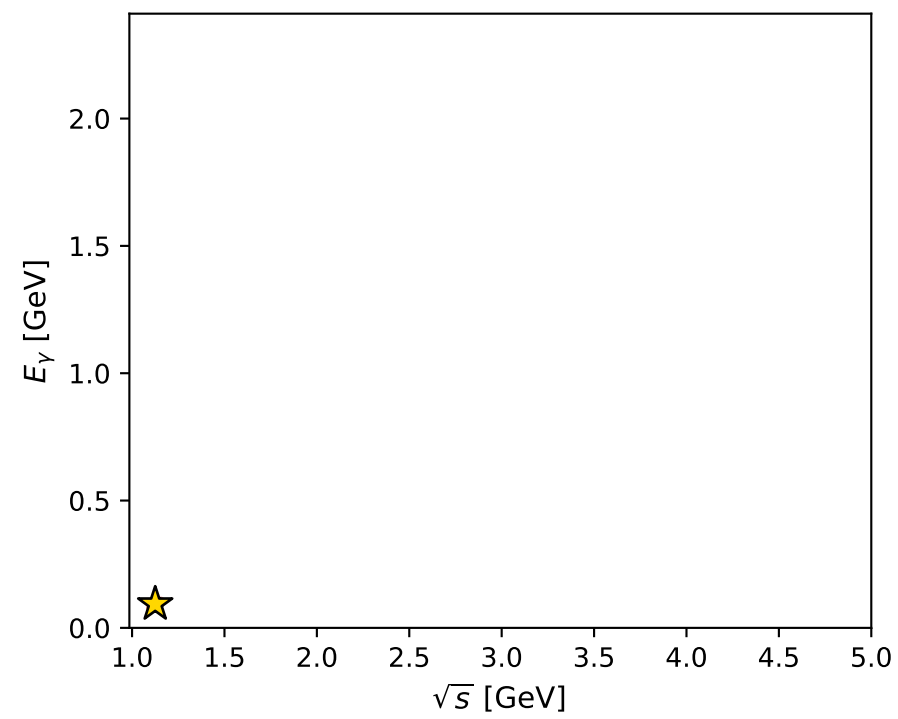
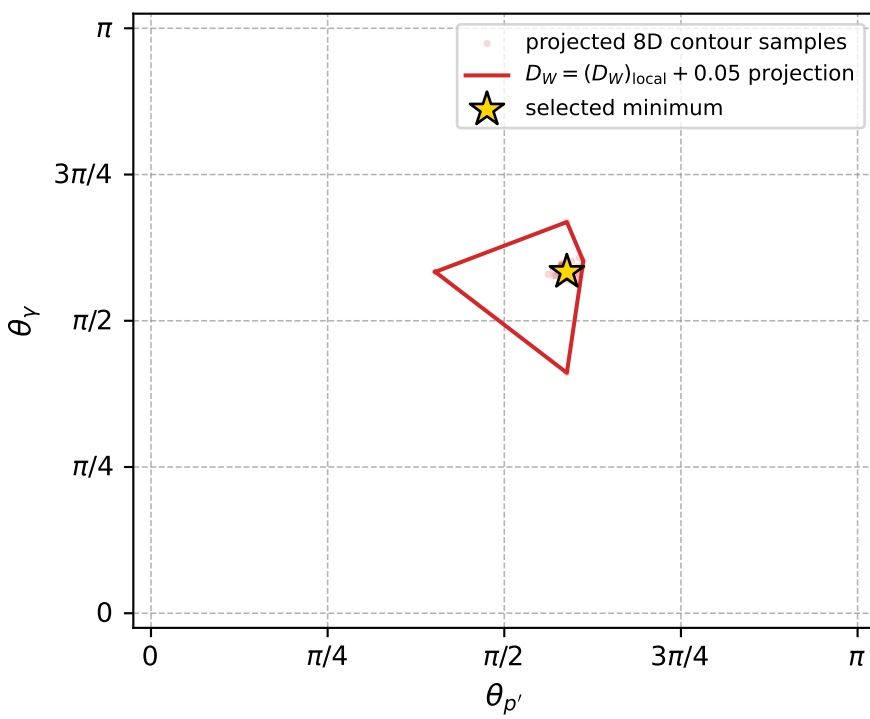
dw_mixing_angles_polarization_06_local_minimum_1078 D_W=0.0251826
 region: polarization_cluster_06_local_minimum_1078
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.359006$ rad
 incoming lepton: $-0.709092|+> +0.705116|->$
 proton mixing angle: $\alpha_p=1.5521592$ rad
 incoming proton: $0.0186361|+> +0.999826|->$

$s=1.26615$, $\sqrt{s}=1.12523$, $|\vec{P}|=0.171656$, $|\vec{P}'|=0.0011566$
 $\theta_{p'}=1.849$, $\theta_Y=1.83327$, $\phi_{p'}=2.70073$, $\phi_Y=1.15415$
 $E_Y=0.0935585$, $\theta_{\ell'}=1.30514$, $\phi_{\ell'}=4.30805$
 $Q^2=0.0406029$, $x_B=0.187891$, $t=-0.0293337$, $W^2=1.05534$, $y=0.559393$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1717$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1717$
 P $E= 0.9536$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1717$
 ℓ' $E= 0.0937$ $p_x= -0.0356$ $p_y= -0.0831$ $p_z= 0.0246$
 P' $E= 0.9380$ $p_x= -0.0010$ $p_y= 0.0005$ $p_z= -0.0003$
 q_Y $E= 0.0936$ $p_x= 0.0366$ $p_y= 0.0826$ $p_z= -0.0243$



electron: polarization cluster P6, local minimum 1078; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.025182648$
 $D_W \text{ contour} = 0.075182648$
 above global minimum = 0.025166216
 8D contour samples = 253

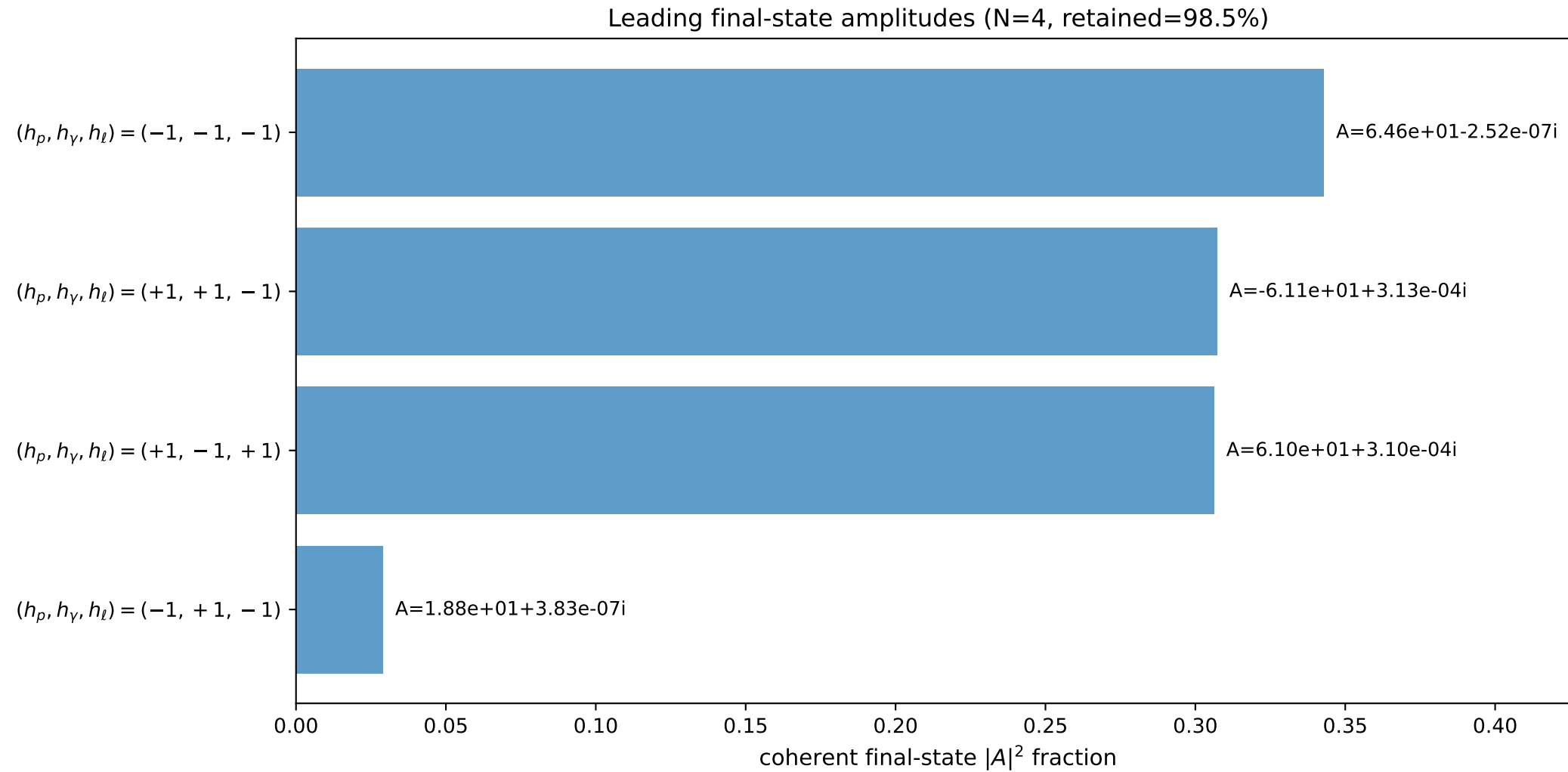
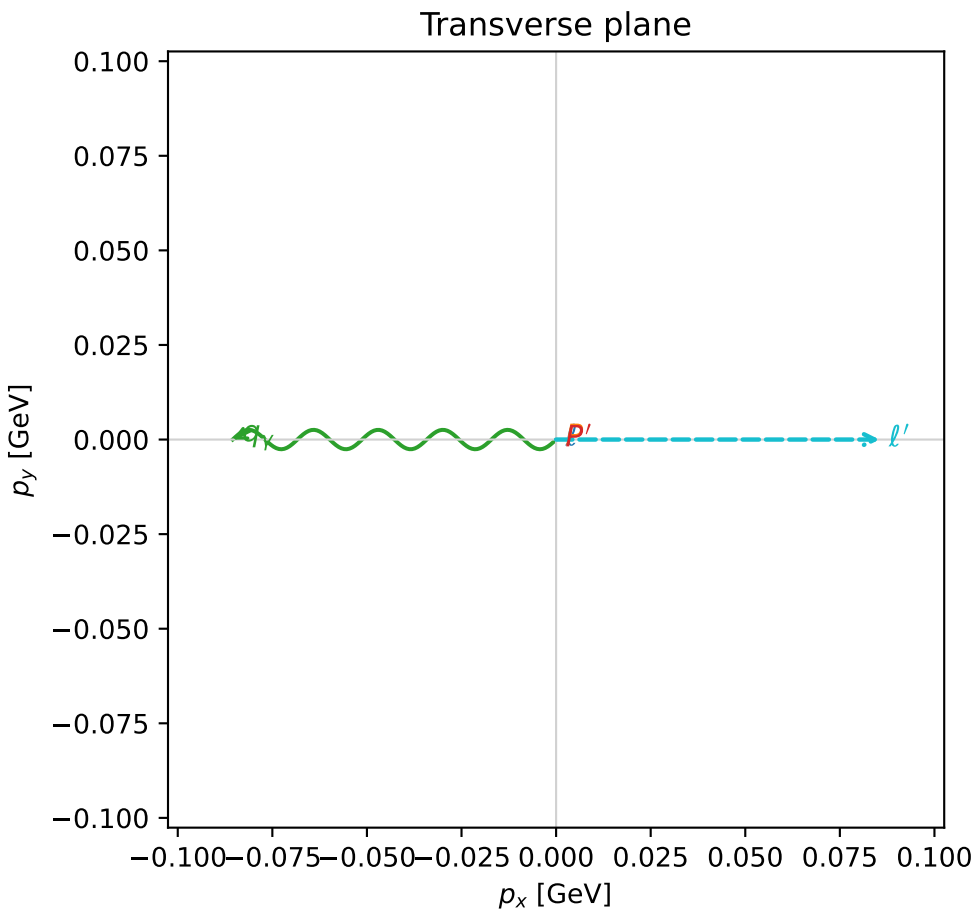
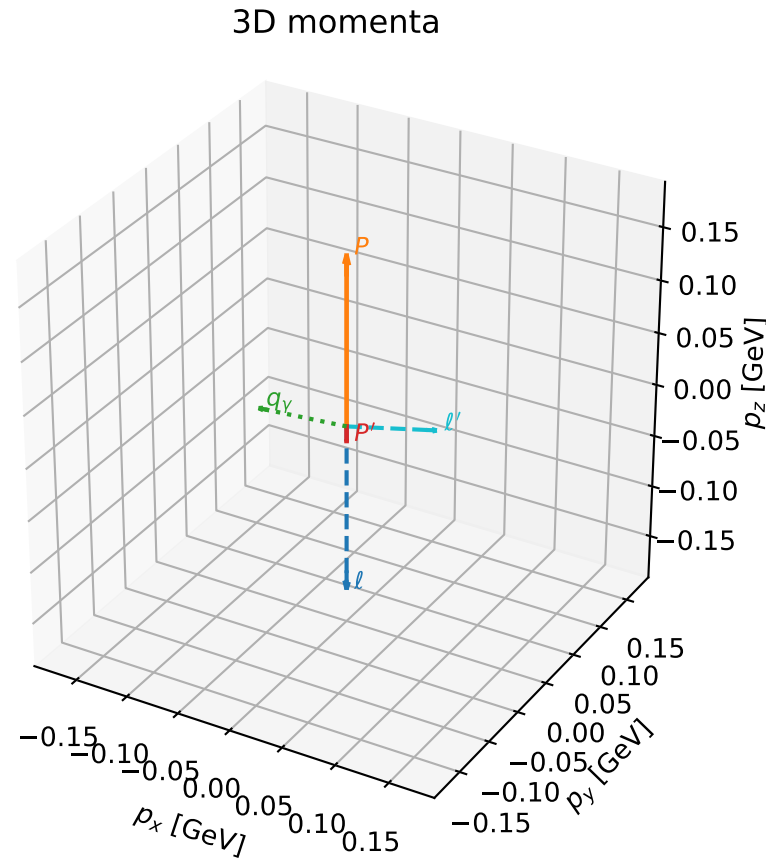
selected phase-space point:
 $\text{sqrt_s} = 1.125235$
 $\text{theta_p_out} = 1.849004$
 $\text{theta_gamma_out} = 1.833273$
 $\text{qOut} = 0.09355853$
 $\text{phi_p_out} = 2.700735$
 $\text{phi_gamma_out} = 1.154154$
 $\text{alpha_e} = 2.359006$
 $\text{alpha_p} = 1.552159$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

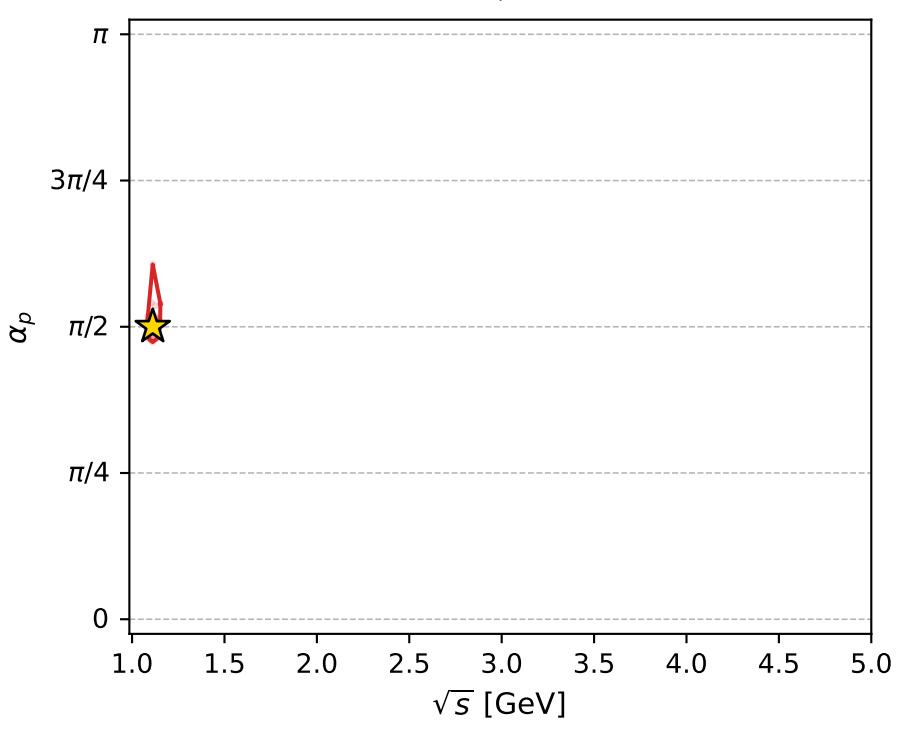
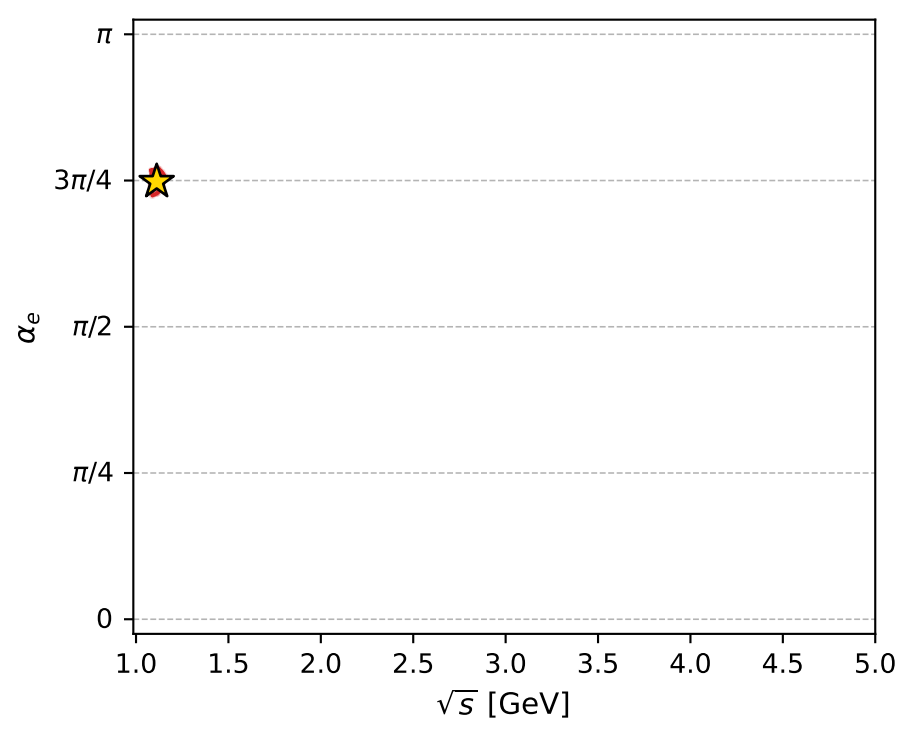
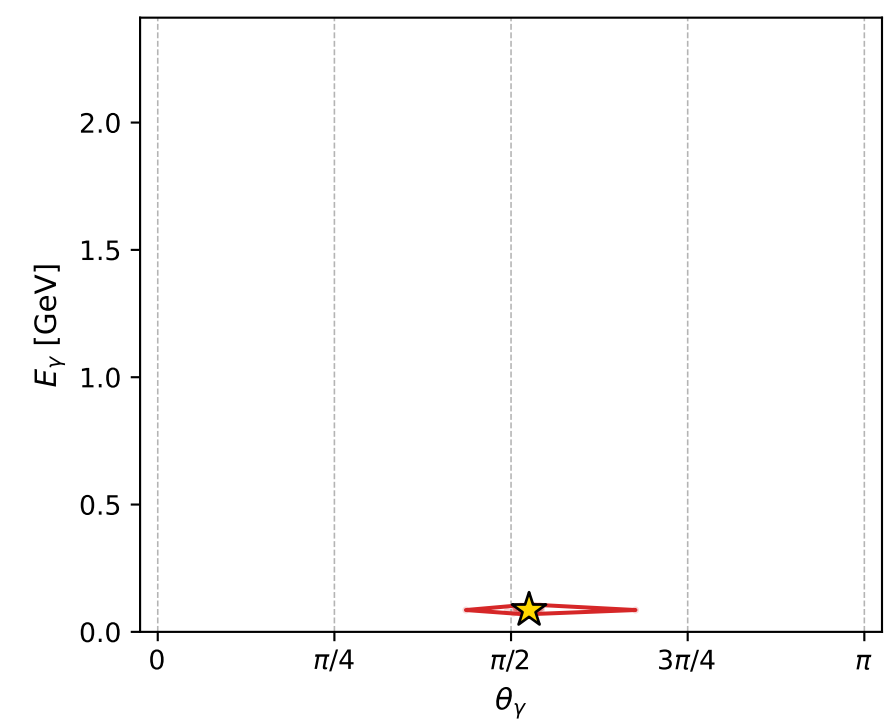
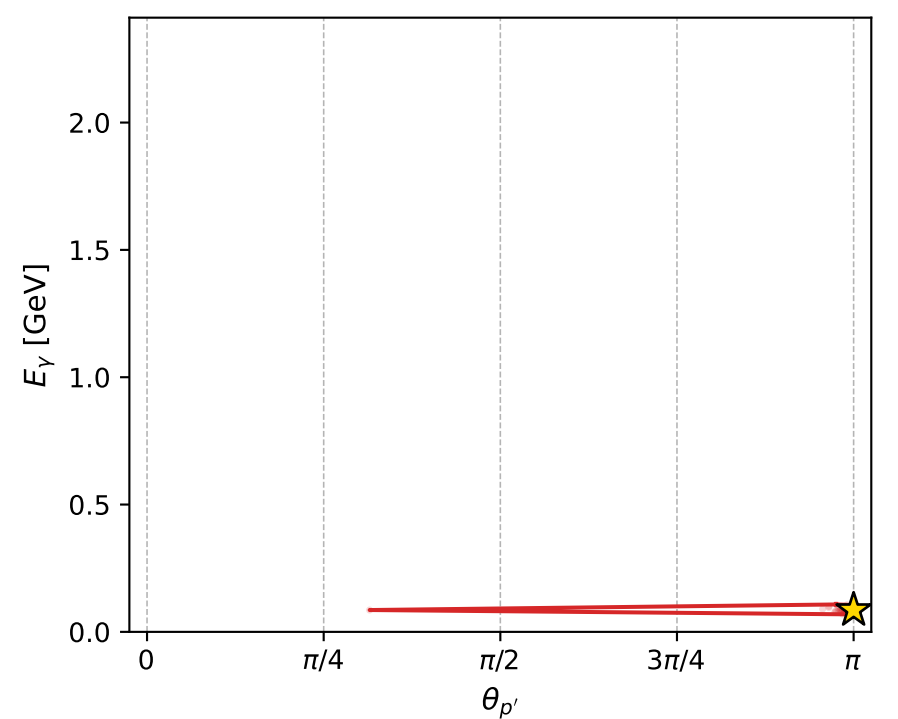
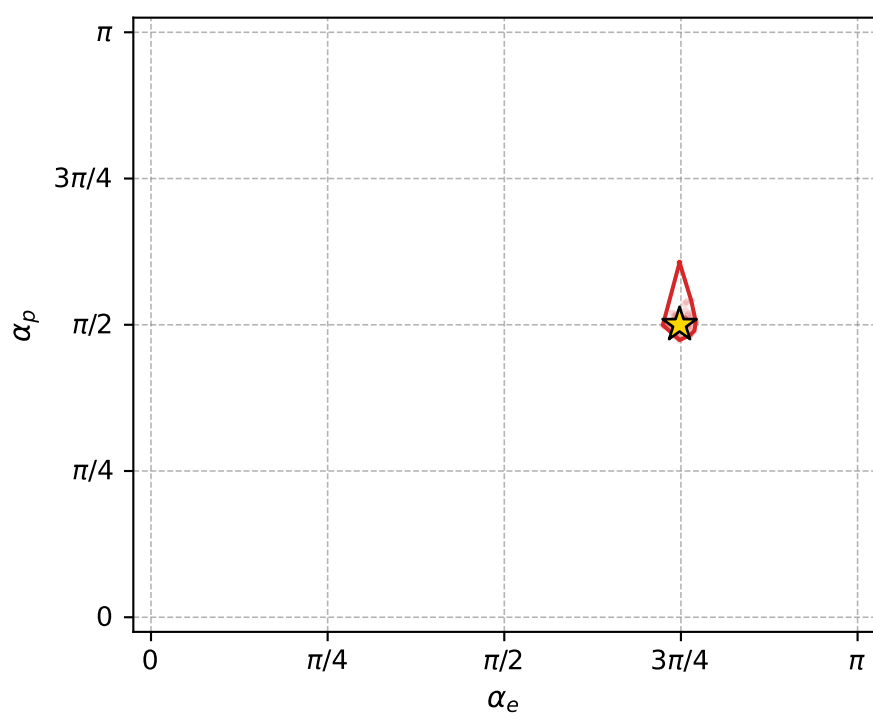
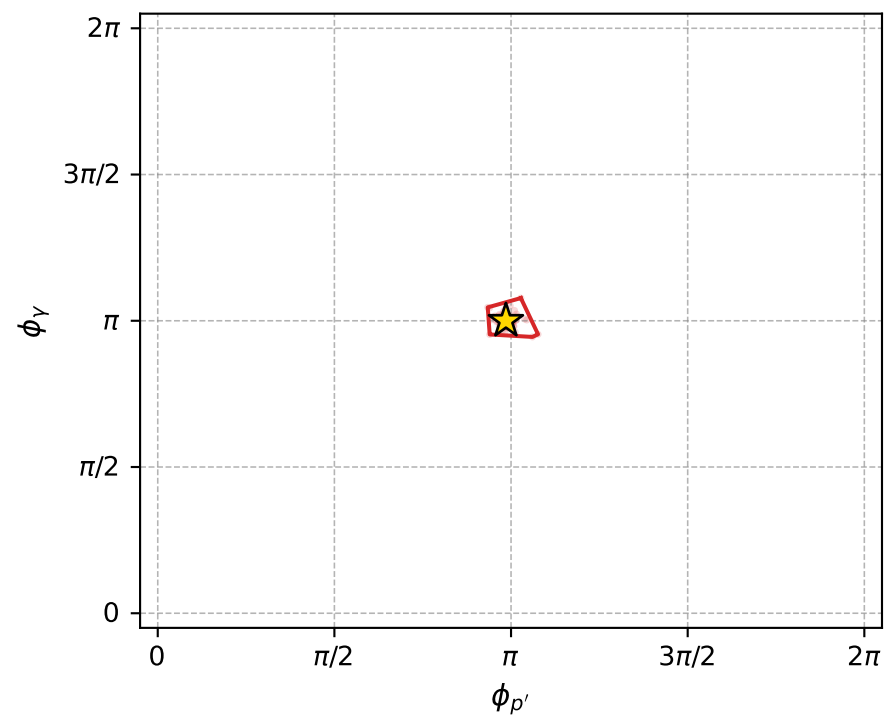
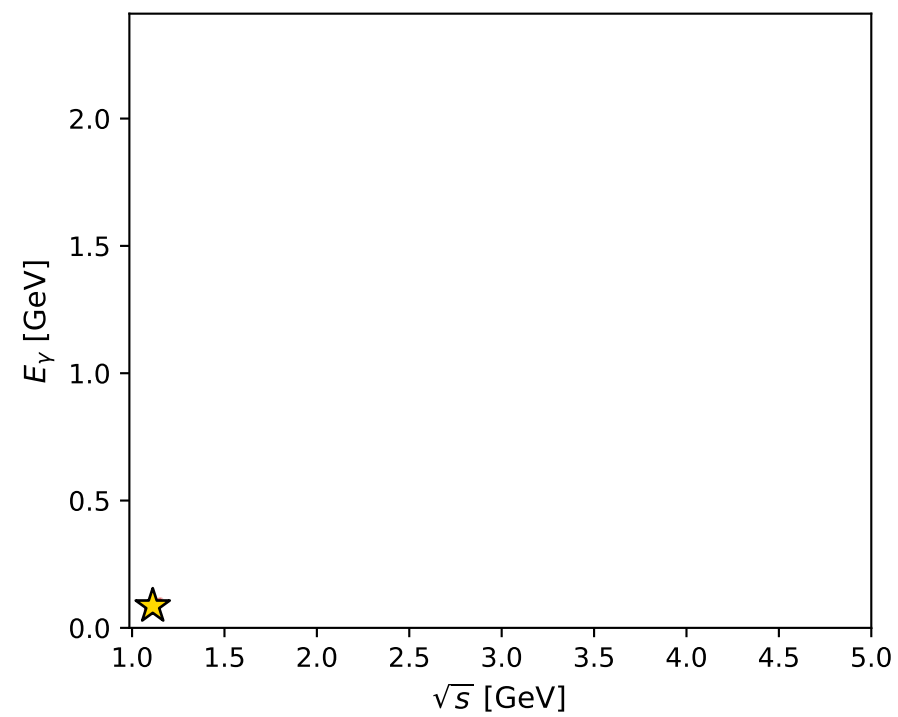
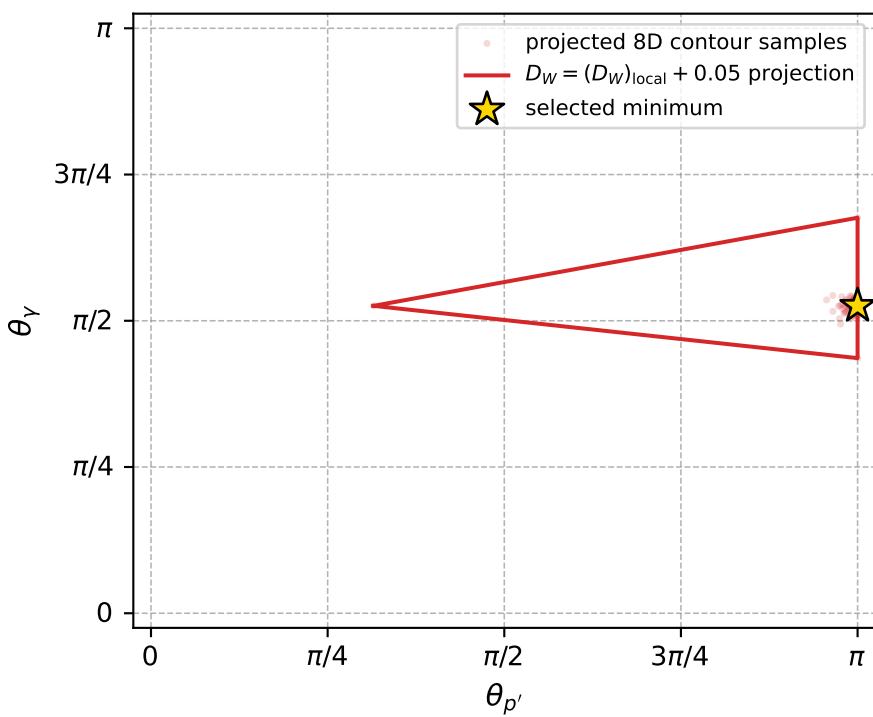
dw_mixing_angles_polarization_06_local_minimum_1086 D_W=0.0257855
 region: polarization_cluster_06_local_minimum_1086
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.350139$ rad
 incoming lepton: $-0.702812|+> +0.711376|->$
 proton mixing angle: $\alpha_p=1.5714981$ rad
 incoming proton: $-0.000701818|+> +1|->$

$s=1.23623$, $\sqrt{s}=1.11186$, $|\vec{P}|=0.160265$, $|\vec{P}'|=0.0140077$
 $\theta_{P'}=3.14159$, $\theta_Y=1.65091$, $\phi_{P'}=3.09606$, $\phi_Y=3.14159$
 $E_Y=0.0857585$, $\theta_{\ell'}=1.33133$, $\phi_{\ell'}=6.28318$
 $Q^2=0.0348948$, $x_B=0.178396$, $t=-0.0301892$, $W^2=1.04055$, $y=0.548853$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1603$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1603$
 P $E= 0.9516$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1603$
 ℓ' $E= 0.0880$ $p_x= 0.0855$ $p_y= -0.0000$ $p_z= 0.0209$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0140$
 q_Y $E= 0.0858$ $p_x= -0.0855$ $p_y= 0.0000$ $p_z= -0.0069$



electron: polarization cluster P6, local minimum 1086; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.025785465$
 $D_W \text{ contour} = 0.075785465$
 above global minimum = 0.025769032
 8D contour samples = 136

selected phase-space point:
 $\text{sqrt}_s = 1.111859$
 $\text{theta}_p_{\text{out}} = 3.141593$
 $\text{theta}_\gamma_{\text{out}} = 1.650912$
 $q_{\text{out}} = 0.08575855$
 $\text{phi}_p_{\text{out}} = 3.096061$
 $\text{phi}_\gamma_{\text{out}} = 3.141588$
 $\alpha_e = 2.350139$
 $\alpha_p = 1.571498$

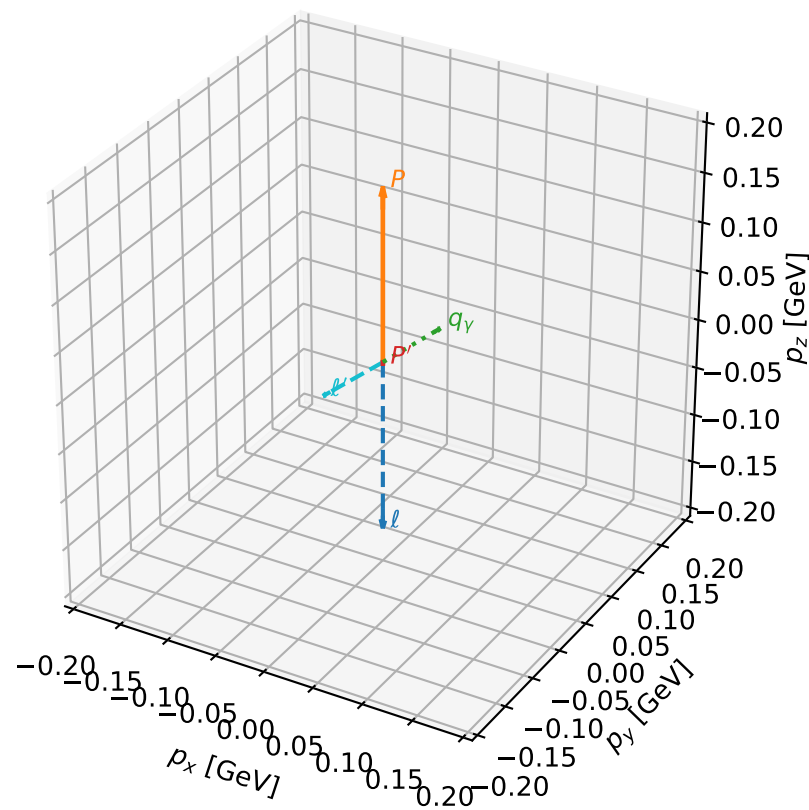
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_1088 D_W=0.0260274
 region: polarization_cluster_06_local_minimum_1088
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3547528$ rad
 incoming lepton: $-0.706087|+> +0.708126|->$
 proton mixing angle: $\alpha_p=1.5655727$ rad
 incoming proton: $0.00522358|+> +0.999986|->$

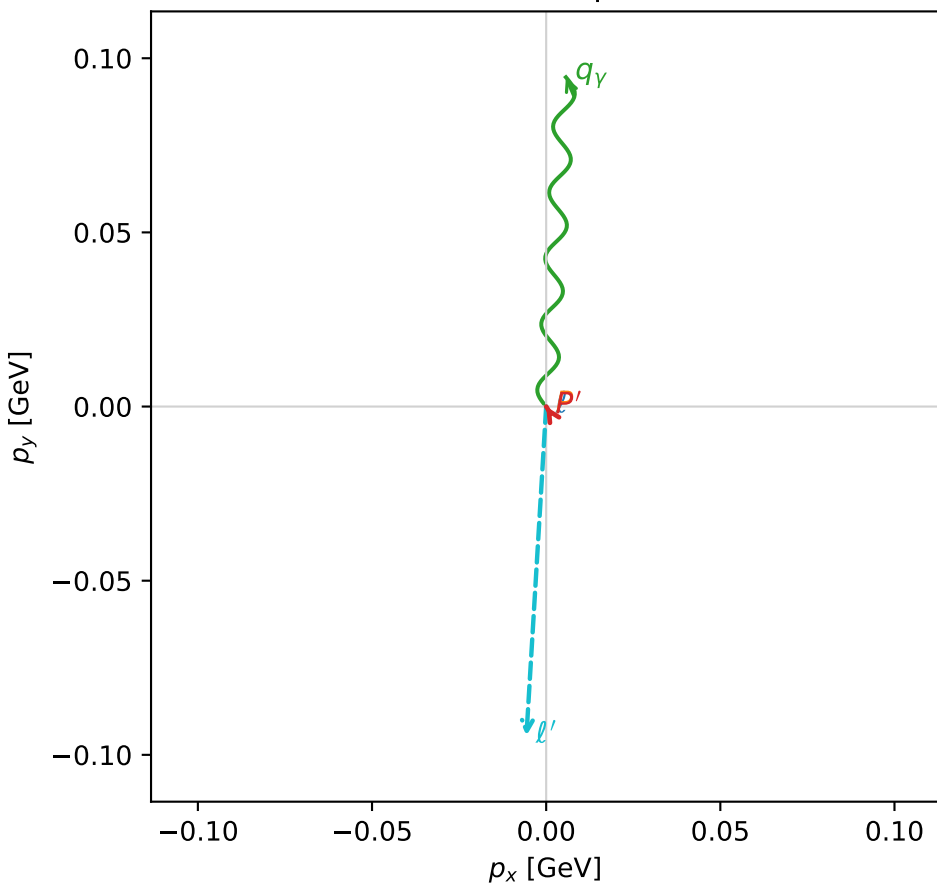
$s=1.27389$, $\sqrt{s}=1.12867$, $|\vec{P}|=0.17456$, $|\vec{P}'|=0.00146841$
 $\theta_{P'}=3.14139$, $\theta_V=1.67723$, $\phi_{P'}=2.12208$, $\phi_V=1.51094$
 $E_V=0.0952478$, $\theta_{\ell'}=1.44906$, $\phi_{\ell'}=4.65254$
 $Q^2=0.0373569$, $x_B=0.172939$, $t=-0.0307268$, $W^2=1.0585$, $y=0.548196$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1746$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1746$
 P $E= 0.9541$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1746$
 ℓ' $E= 0.0954$ $p_x= -0.0057$ $p_y= -0.0945$ $p_z= 0.0116$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0015$
 q_V $E= 0.0952$ $p_x= 0.0057$ $p_y= 0.0945$ $p_z= -0.0101$

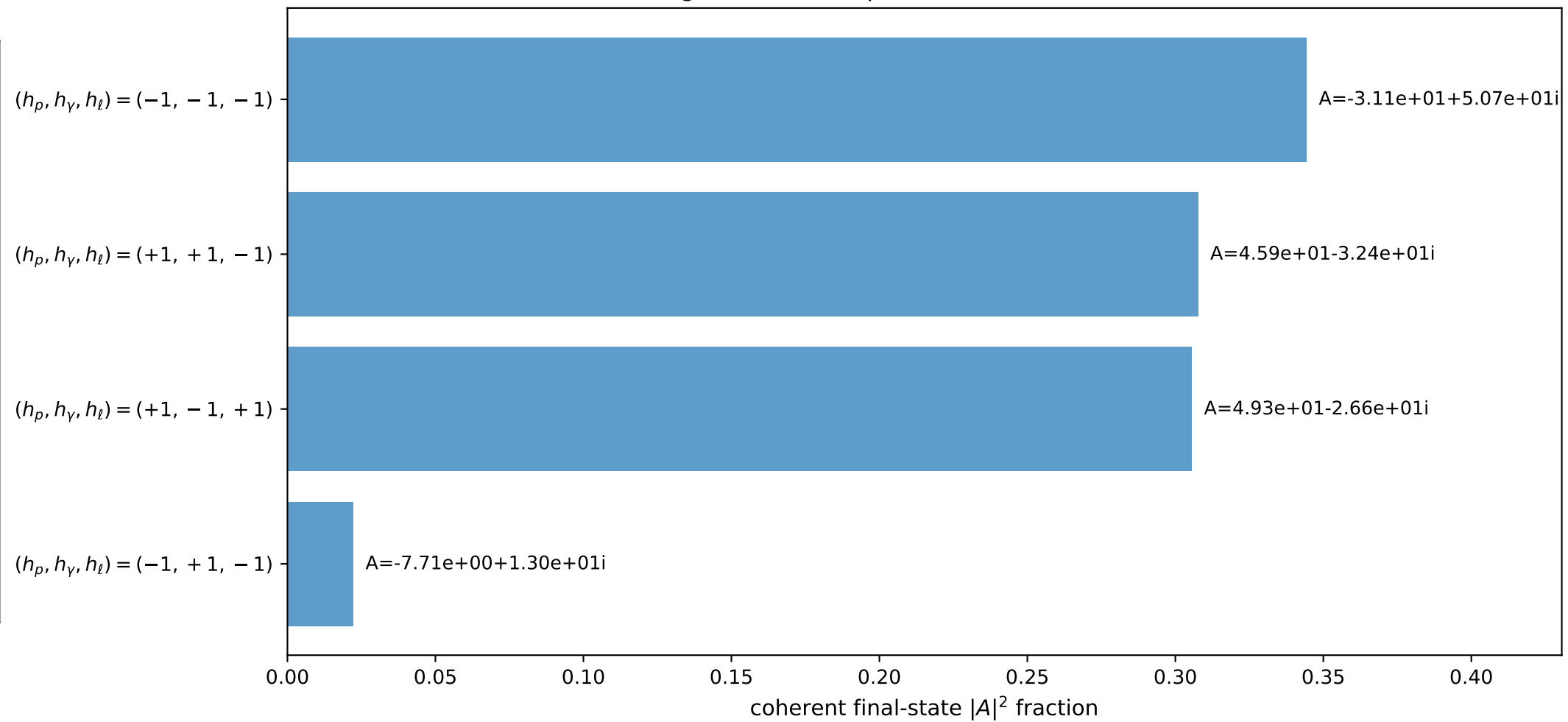
3D momenta



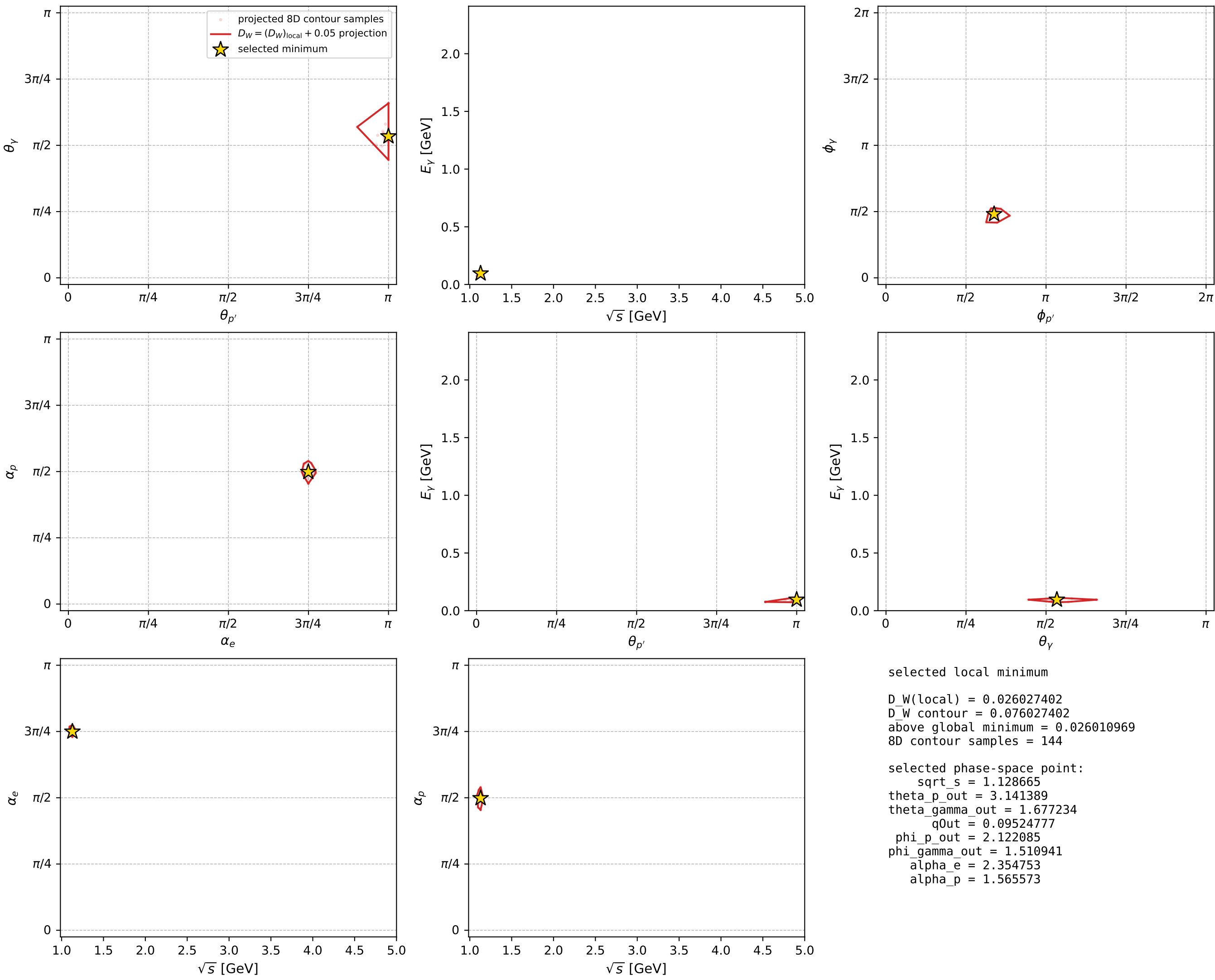
Transverse plane



Leading final-state amplitudes (N=4, retained=98.0%)



electron: polarization cluster P6, local minimum 1088; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

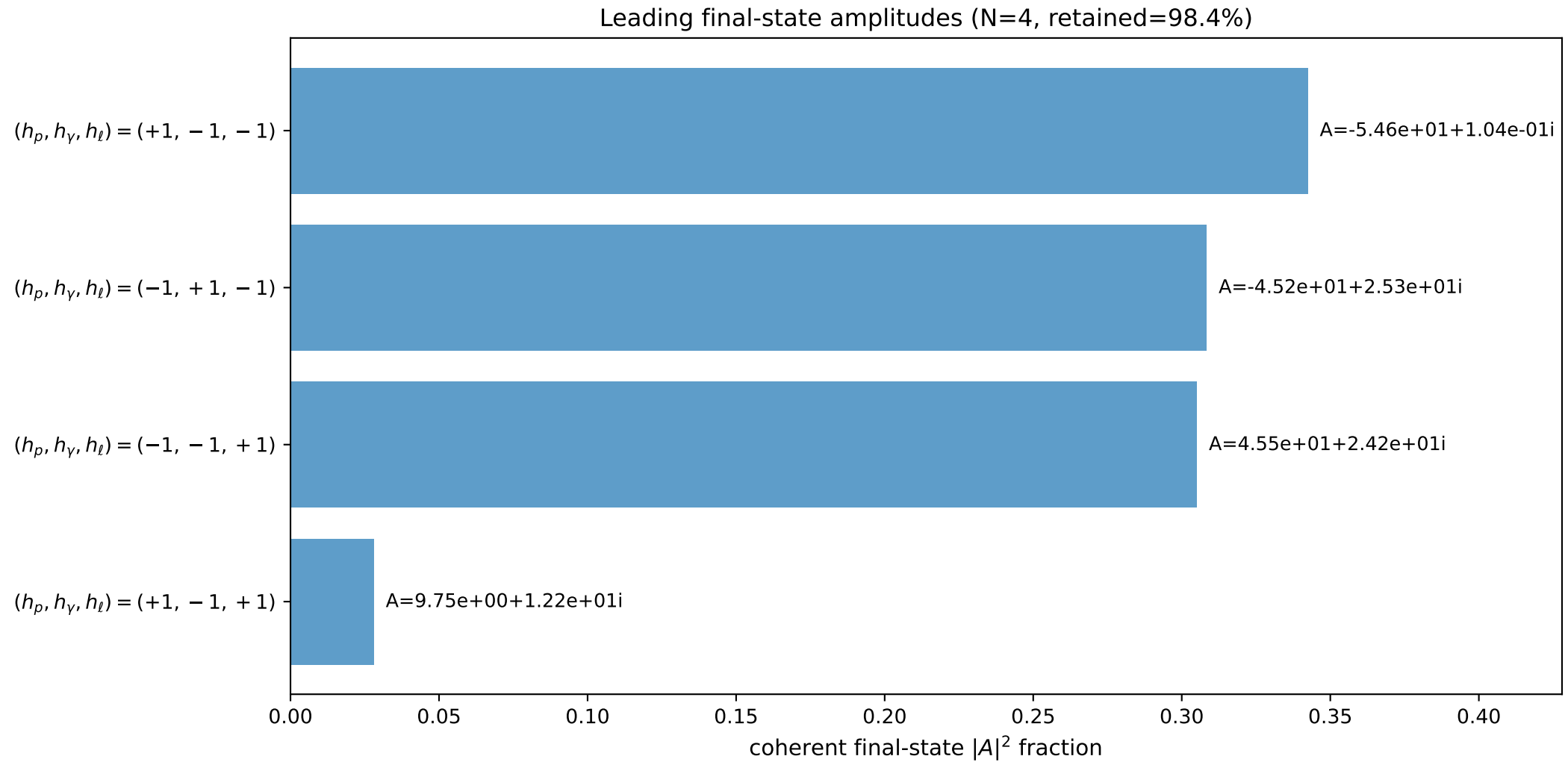
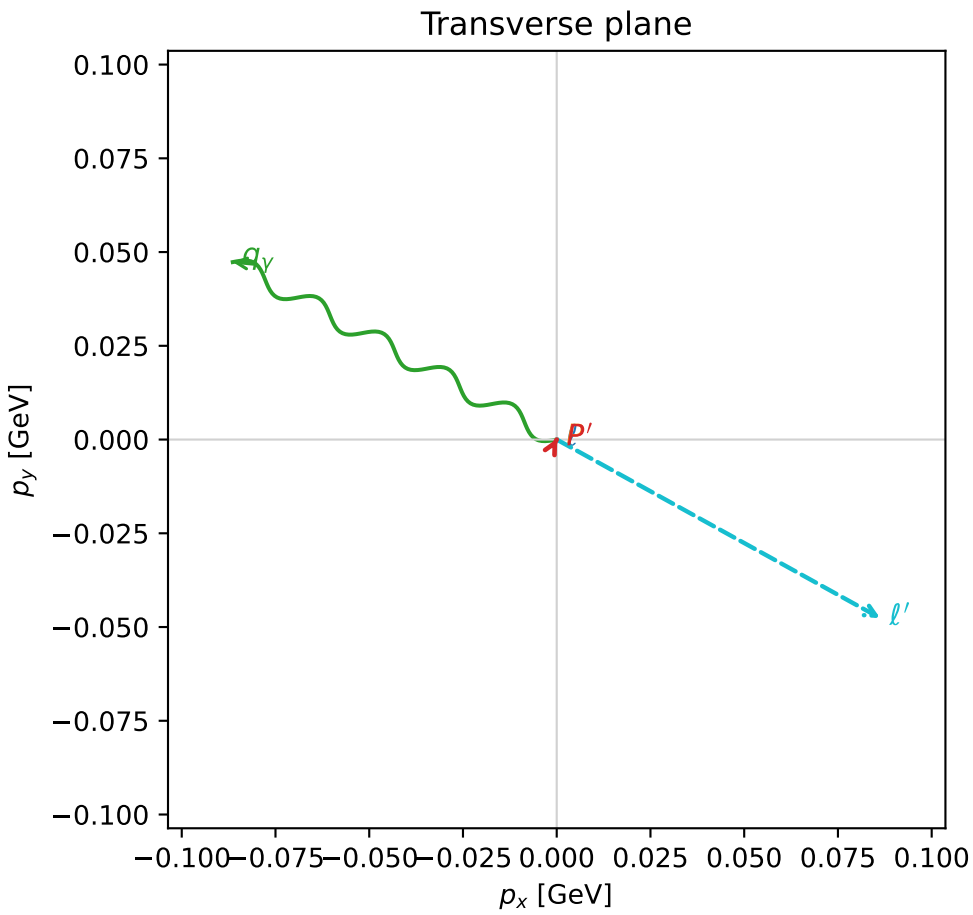
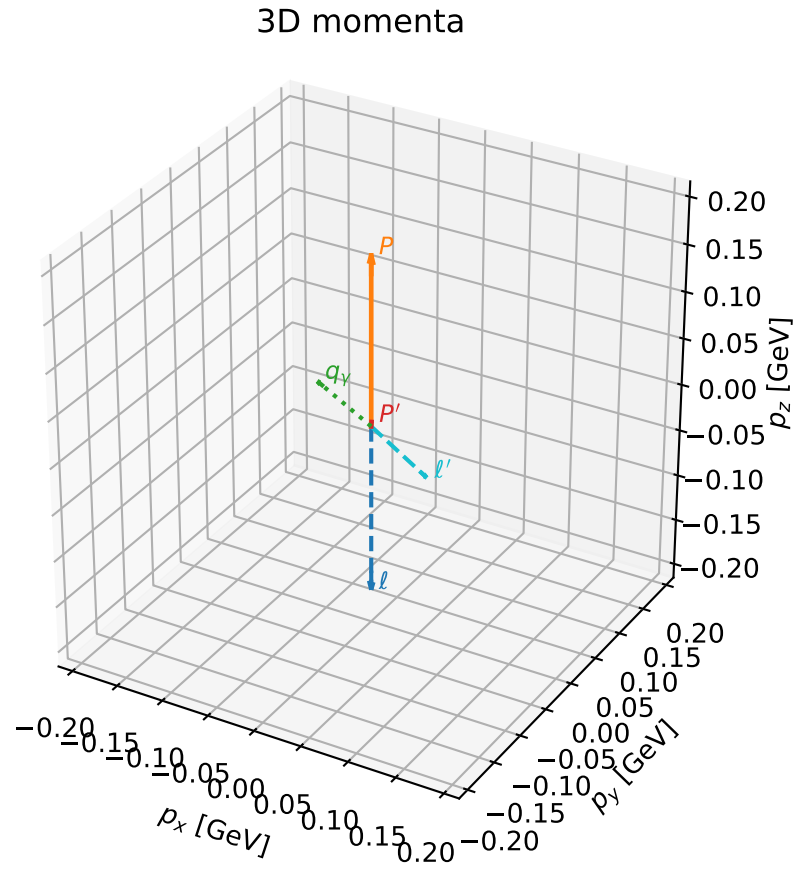


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

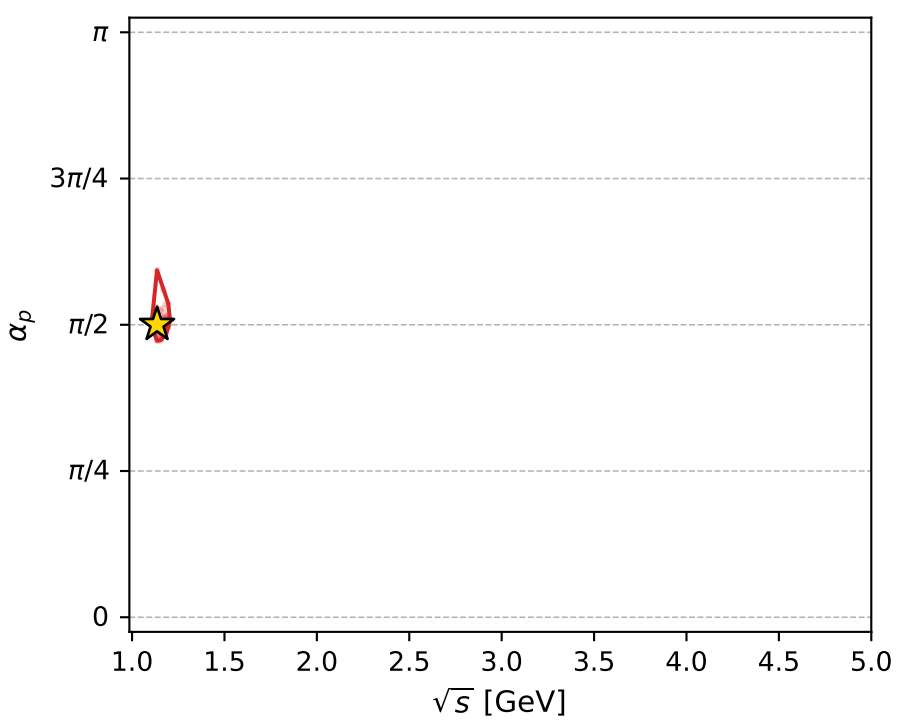
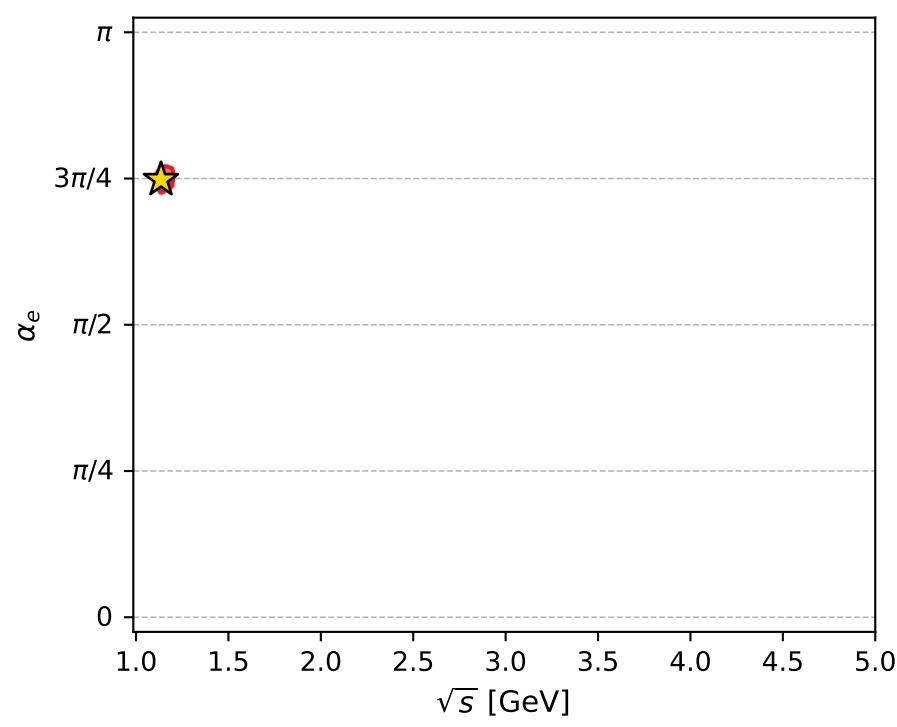
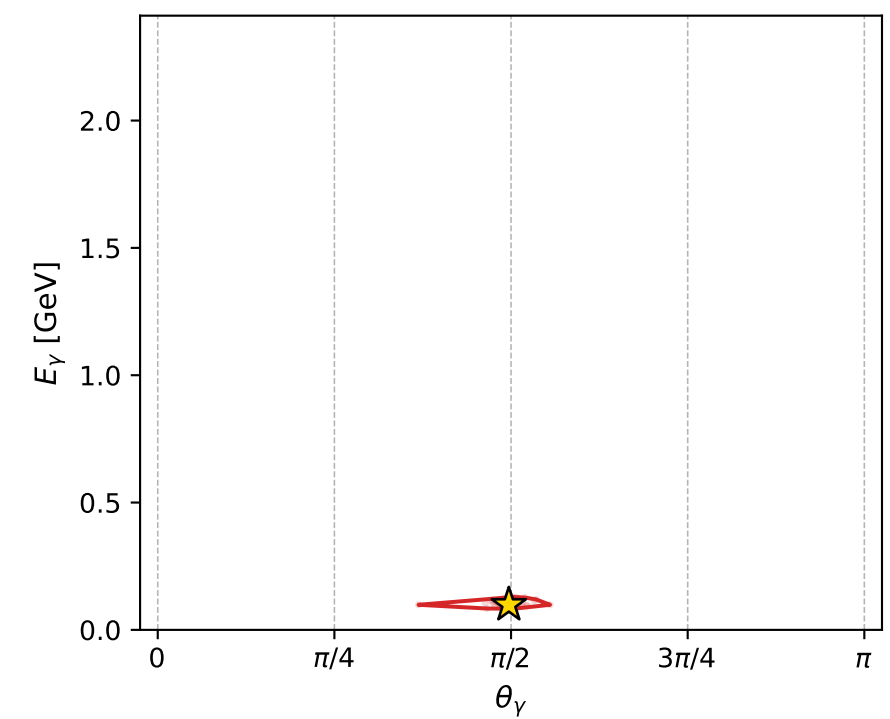
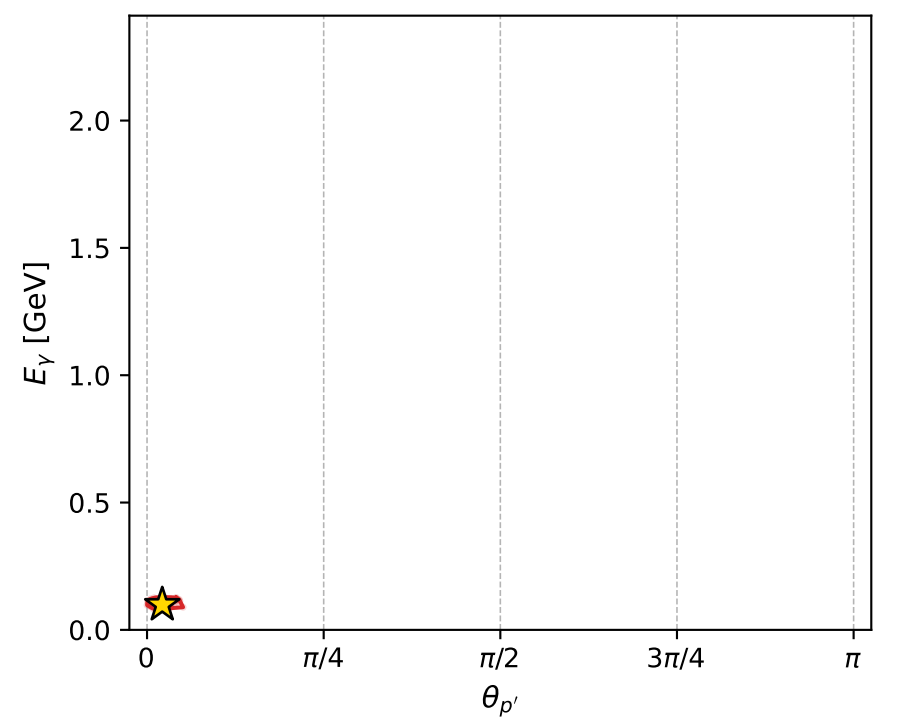
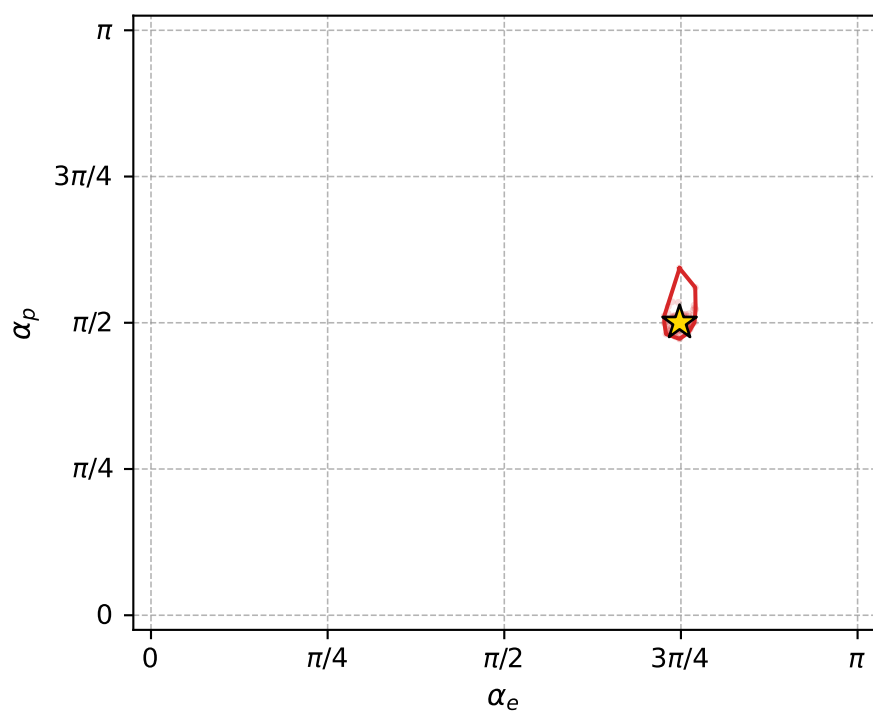
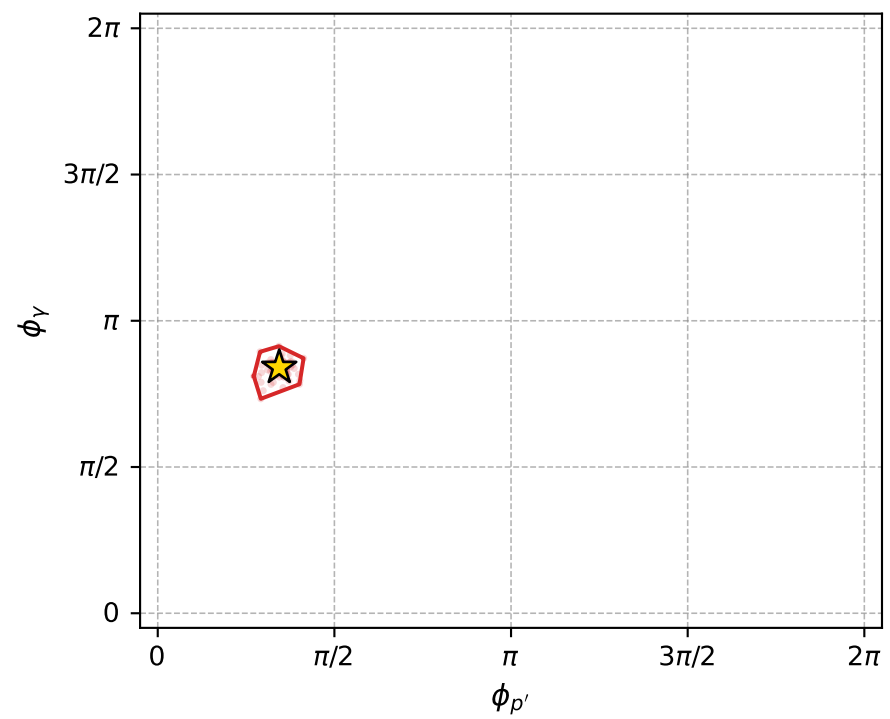
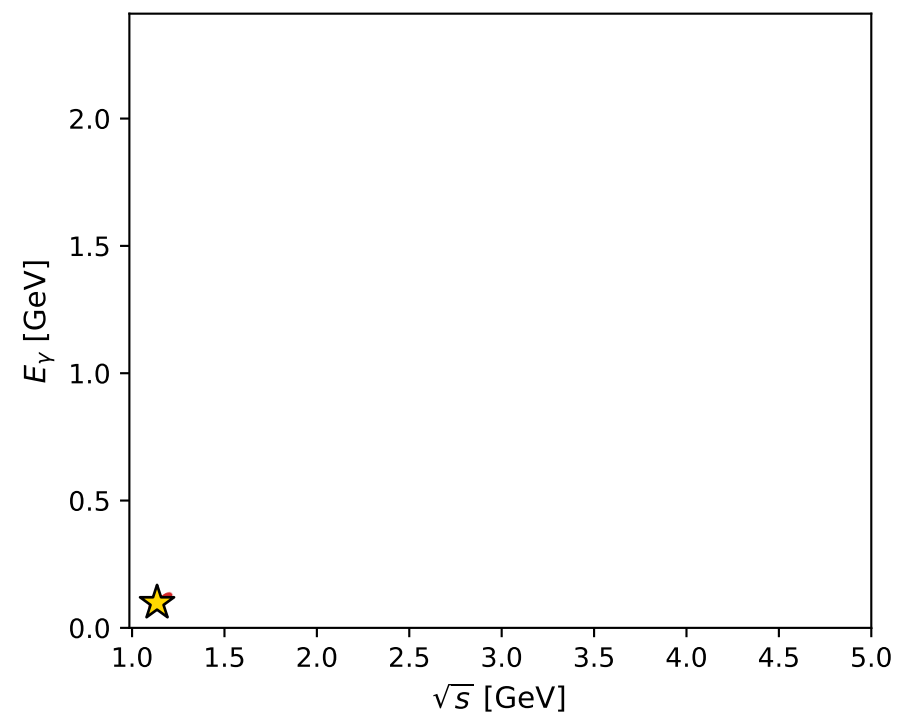
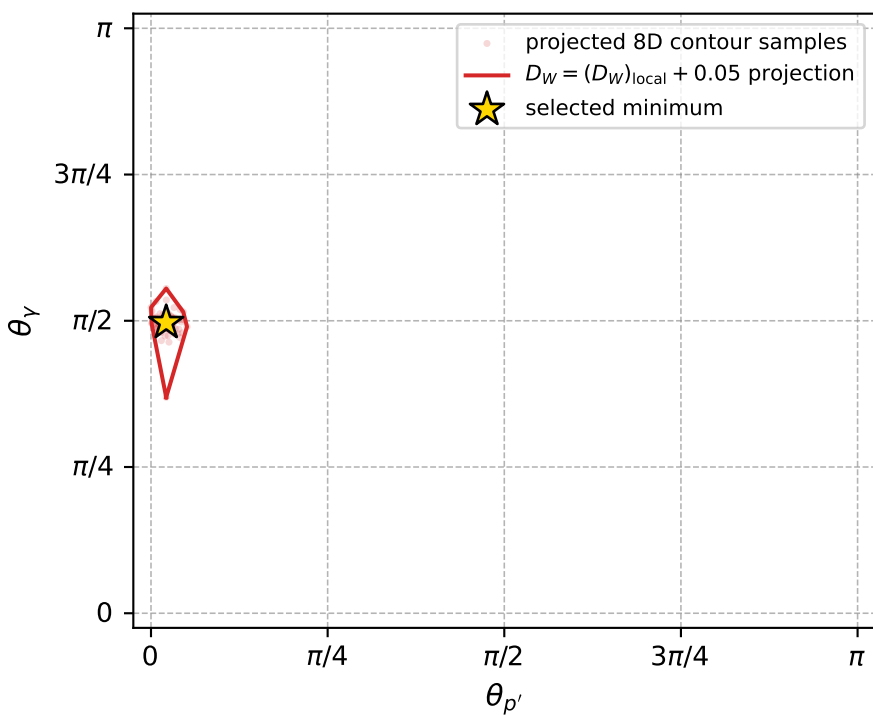
dw_mixing_angles_polarization_06_local_minimum_1090 D_W=0.0260761
 region: polarization_cluster_06_local_minimum_1090
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3500994$ rad
 incoming lepton: $-0.702784|+> +0.711403|->$
 proton mixing angle: $\alpha_p=1.5722119$ rad
 incoming proton: $-0.00141558|+> +0.999999|->$

$s=1.28864$, $\sqrt{s}=1.13518$, $|\vec{P}|=0.180058$, $|\vec{P}'|=0.00465505$
 $\theta_{p'}=0.0675842$, $\theta_Y=1.5611$, $\phi_{p'}=1.08019$, $\phi_Y=2.64044$
 $E_Y=0.0985065$, $\theta_{\ell'}=1.62758$, $\phi_{\ell'}=5.77884$
 $Q^2=0.0335143$, $x_B=0.153521$, $t=-0.0304771$, $W^2=1.06463$, $y=0.534017$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1801$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1801$
 P $E= 0.9551$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1801$
 ℓ' $E= 0.0987$ $p_x= 0.0862$ $p_y= -0.0476$ $p_z= -0.0056$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0003$ $p_z= 0.0046$
 q_Y $E= 0.0985$ $p_x= -0.0864$ $p_y= 0.0473$ $p_z= 0.0010$



electron: polarization cluster P6, local minimum 1090; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.026076144$
 $D_W \text{ contour} = 0.076076144$
 above global minimum = 0.026059711
 8D contour samples = 245

selected phase-space point:
 $\text{sqrt_s} = 1.135184$
 $\text{theta_p_out} = 0.06758423$
 $\text{theta_gamma_out} = 1.561101$
 $\text{qOut} = 0.0985065$
 $\text{phi_p_out} = 1.080189$
 $\text{phi_gamma_out} = 2.640439$
 $\text{alpha_e} = 2.350099$
 $\text{alpha_p} = 1.572212$

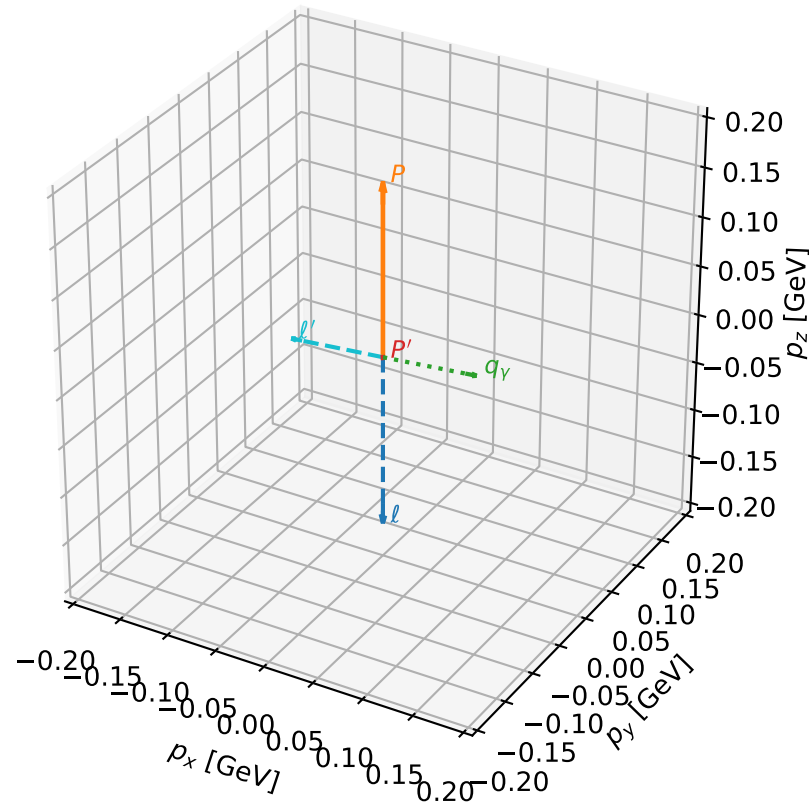
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_1092 D_W=0.0263125
region: polarization_cluster_06_local_minimum_1092
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3564245$ rad
incoming lepton: $-0.707269|+\rangle +0.706944|-\rangle$
proton mixing angle: $\alpha_p=1.5586349$ rad
incoming proton: $0.0121611|+\rangle +0.999926|-\rangle$

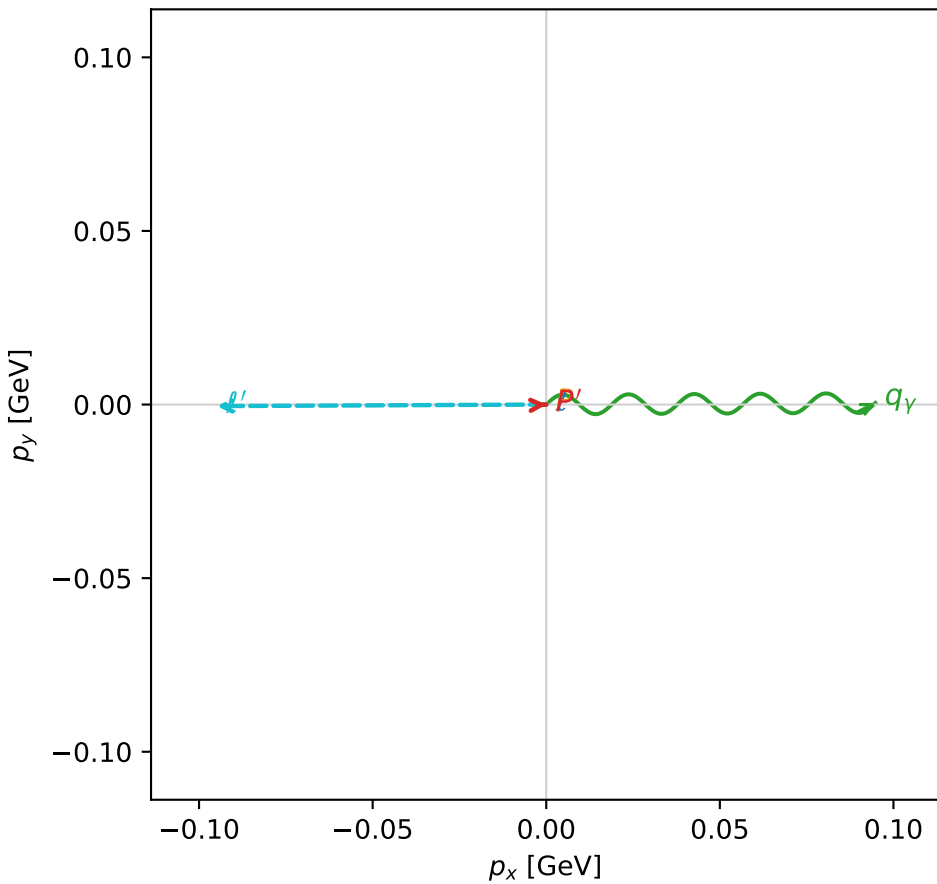
$s=1.27311$, $\sqrt{s}=1.12832$, $|\vec{P}|=0.174268$, $|\vec{P}'|=0.0006285$
 $\theta_{P'}=3.03001$, $\theta_V=1.48336$, $\phi_{P'}=0.0504662$, $\phi_V=0.00466219$
 $E_V=0.0951505$, $\theta_{\ell'}=1.65163$, $\phi_{\ell'}=3.14629$
 $Q^2=0.0304912$, $x_B=0.145897$, $t=-0.0303298$, $W^2=1.05834$, $y=0.53143$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1743$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1743$
 P $E=0.9541$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1743$
 ℓ' $E=0.0952$ $p_x=-0.0949$ $p_y=-0.0004$ $p_z=-0.0077$
 P' $E=0.9380$ $p_x=0.0001$ $p_y=0.0000$ $p_z=-0.0006$
 q_V $E=0.0952$ $p_x=0.0948$ $p_y=0.0004$ $p_z=0.0083$

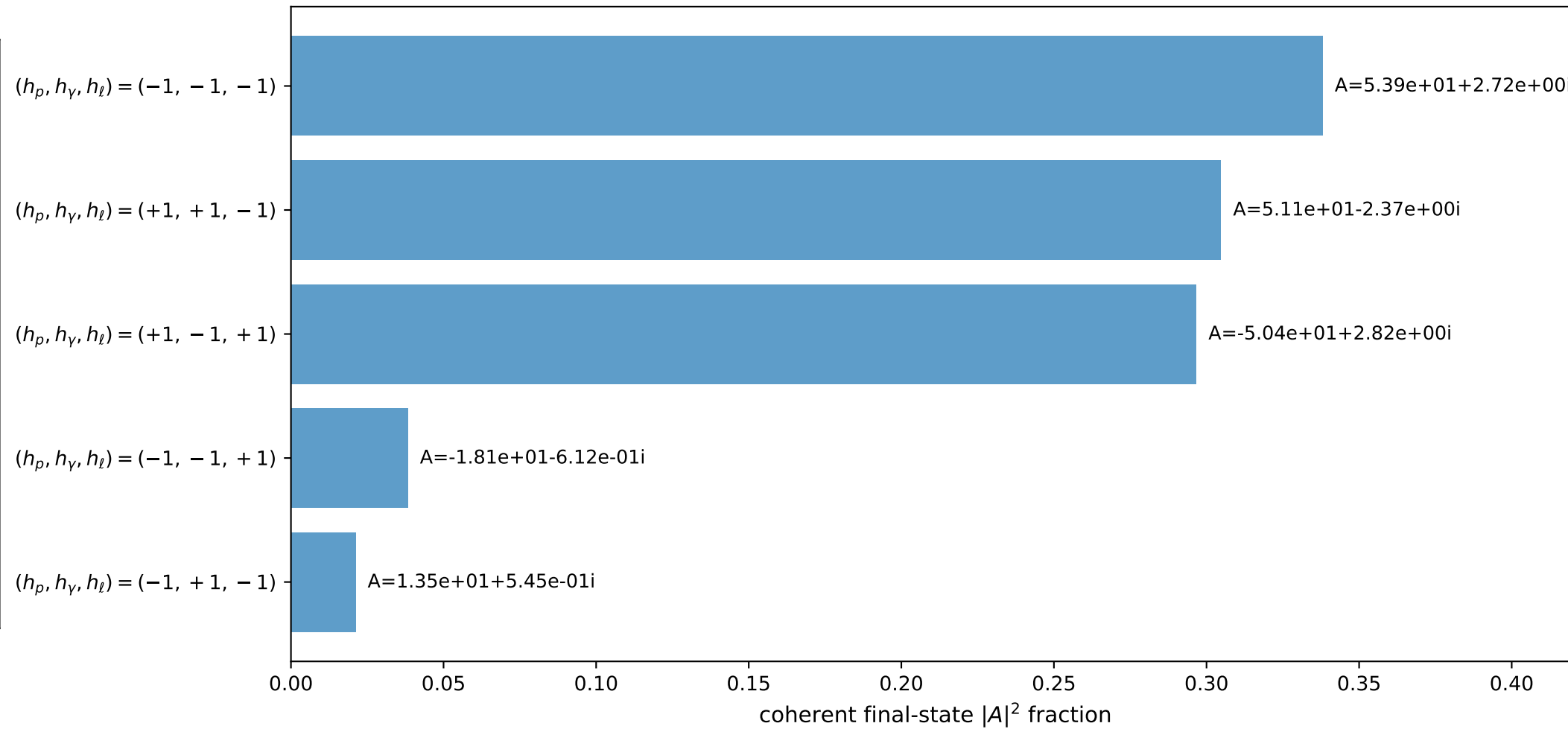
3D momenta



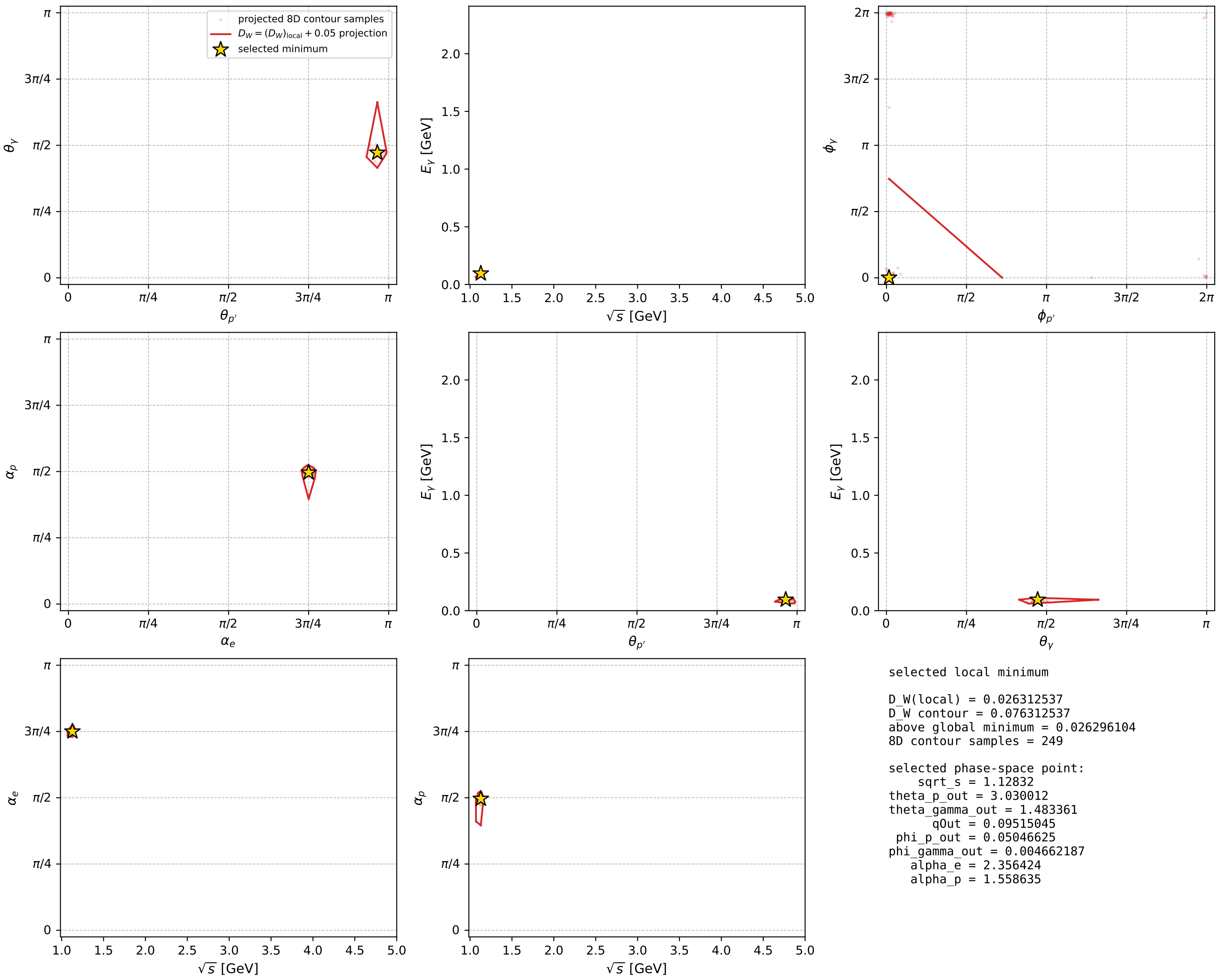
Transverse plane



Leading final-state amplitudes (N=5, retained=99.9%)



electron: polarization cluster P6, local minimum 1092; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

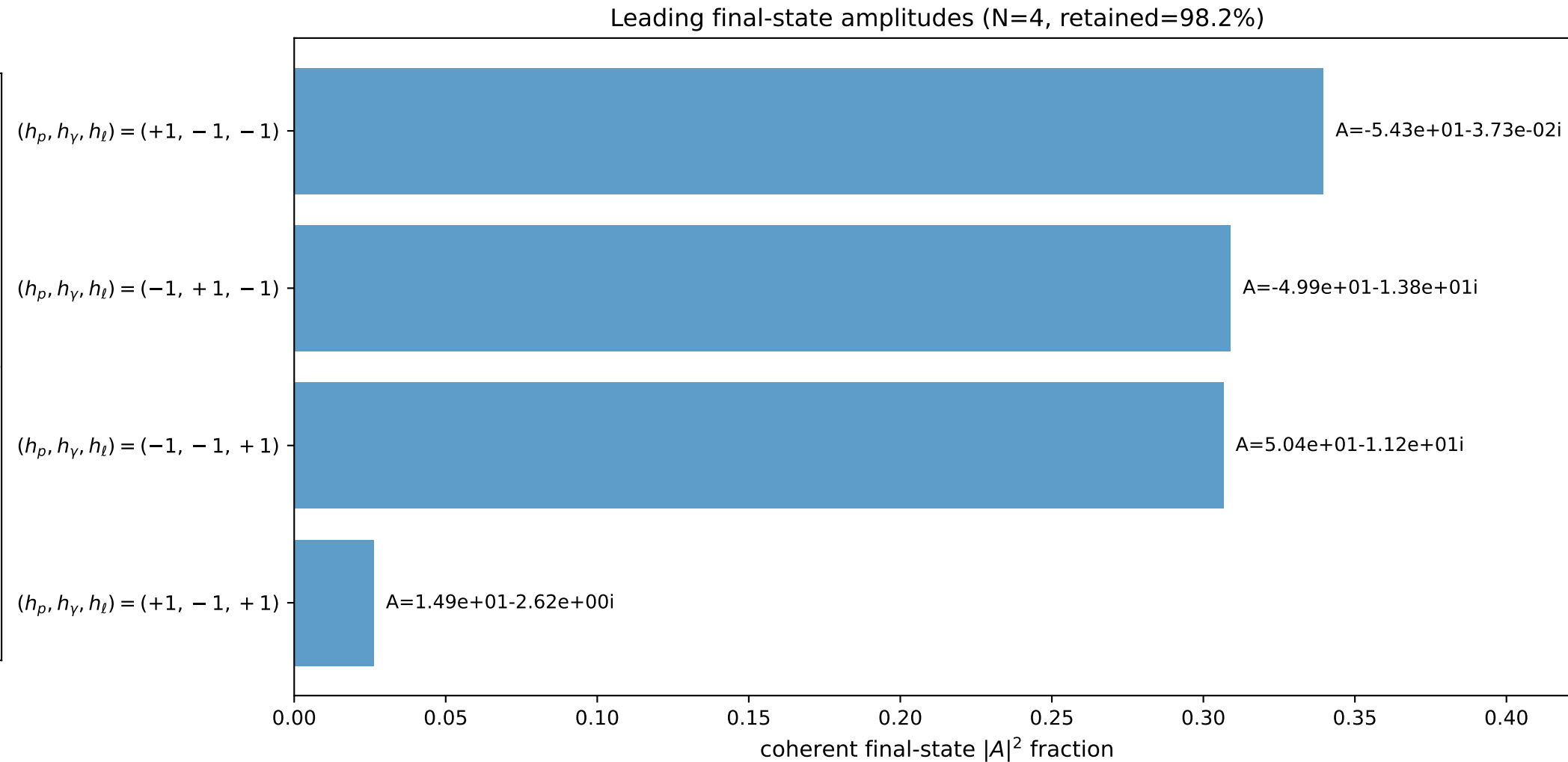
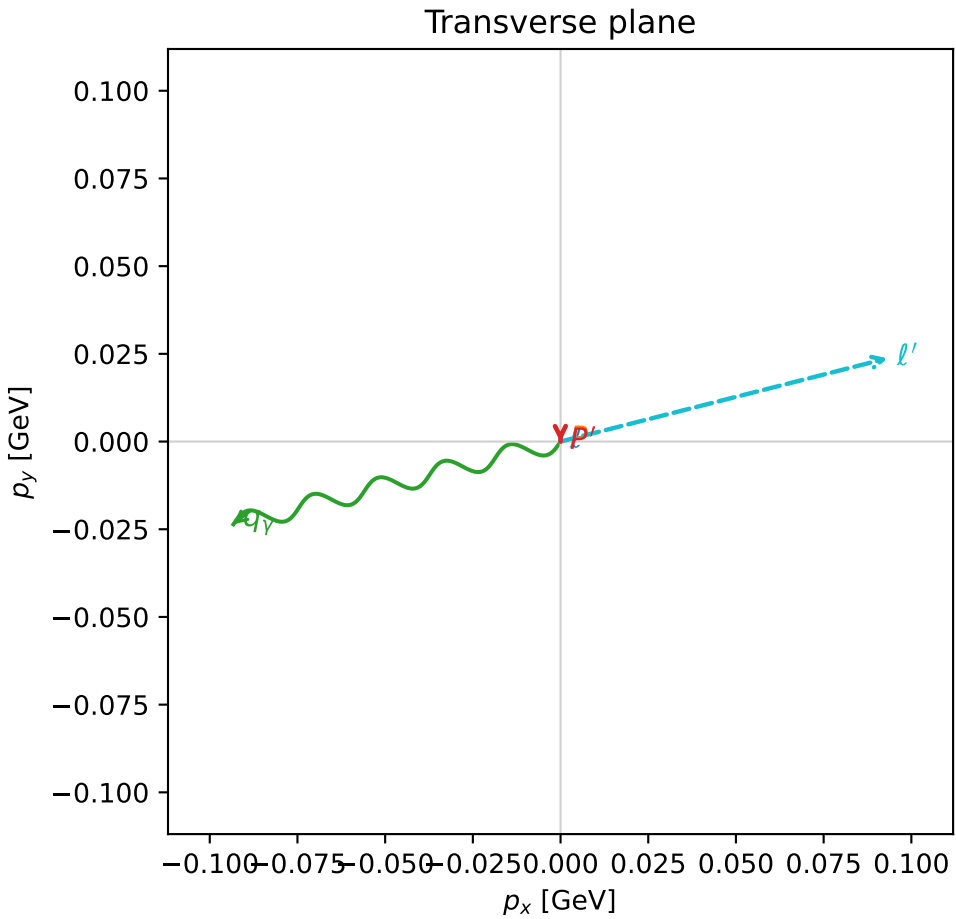
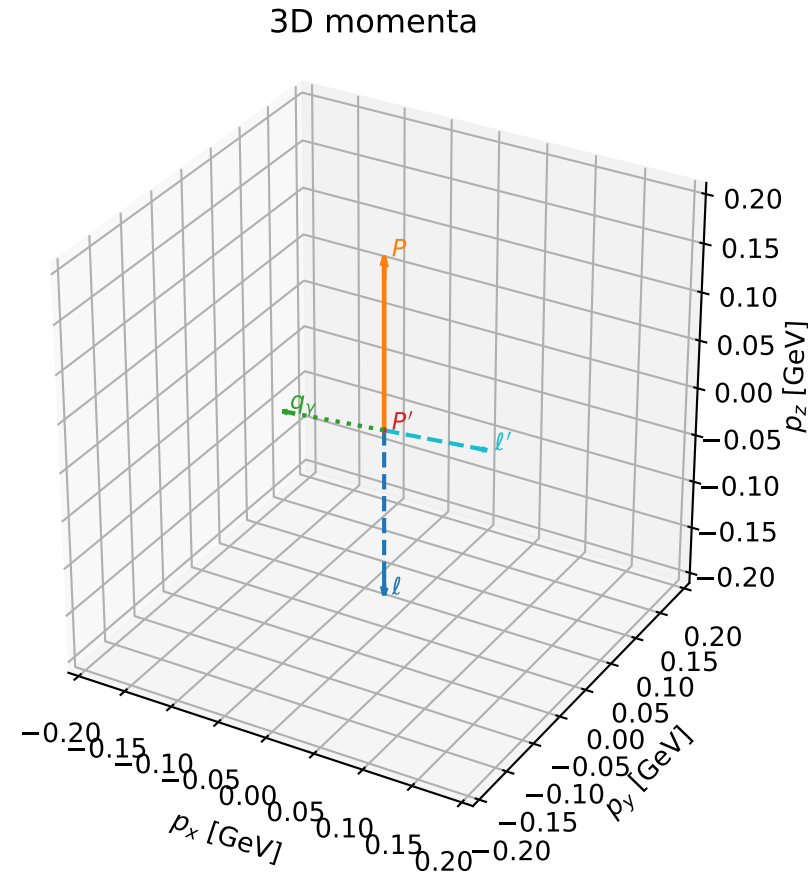


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_06_local_minimum_1093 D_W=0.0263428
 region: polarization_cluster_06_local_minimum_1093
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3495456$ rad
 incoming lepton: $-0.70239|+> +0.711793|->$
 proton mixing angle: $\alpha_p=1.5710648$ rad
 incoming proton: $-0.000268459|+> +1|->$

$s=1.27852$, $\sqrt{s}=1.13072$, $|\vec{P}|=0.176292$, $|\vec{P}'|=0.000723062$
 $\theta_{p'}=0.192171$, $\theta_Y=1.51936$, $\phi_{p'}=4.72333$, $\phi_Y=3.38948$
 $E_Y=0.096321$, $\theta_{\ell'}=1.62957$, $\phi_{\ell'}=0.249285$
 $Q^2=0.0319903$, $x_B=0.150419$, $t=-0.0305593$, $W^2=1.06053$, $y=0.533459$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1763$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1763$
 P $E= 0.9544$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1763$
 ℓ' $E= 0.0964$ $p_x= 0.0933$ $p_y= 0.0237$ $p_z= -0.0057$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0001$ $p_z= 0.0007$
 q_Y $E= 0.0963$ $p_x= -0.0933$ $p_y= -0.0236$ $p_z= 0.0050$



electron: polarization cluster P6, local minimum 1093; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

